

# Lecture Notes on Mathematical Analysis

Single and Multivariable Calculus

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# Preface

These notes are for students who know the main ideas of single-variable calculus and want to understand why its central statements are true. They develop rigorous real analysis before rebuilding calculus from limits, completeness, continuity, differentiation, and Riemann integration. The intended audience is an advanced undergraduate beginning proof-based mathematics. Familiar algebra and ordinary calculus are assumed, but the set-theoretic, logical, topological, and finite-dimensional linear-algebraic tools needed later are developed in the text.

Part I (Chapters 1–9) constructs the real number system and develops one-variable analysis through Riemann integration and the Fundamental Theorem of Calculus. Part II (Chapters 10–13) develops the finite-dimensional topology and linear algebra needed for multivariable differentiation and Riemann integration, and ends with differential forms and the general Stokes theorem. Measure theory, Lebesgue integration, complex and functional analysis, partial differential equations, and deeper differential geometry are outside the developed theory; Chapter 9 previews the Lebesgue FTC, and the final chapter describes further directions.

**AI-assisted preparation.** OpenAI Codex was used extensively in drafting, revising, proofreading, and preparing the  $\text{\LaTeX}$  source for these notes. The author determined the scope and mathematical organization, made the final decisions about retained content, and is responsible for approving and verifying the correctness of the published version.



# How to Read These Notes

These notes start from familiar calculus questions but use a different standard of explanation: every formal assertion has hypotheses, and every main claim is justified from material already established or from an explicitly identified external theorem. A definition fixes meaning; a theorem requires proof unless the prose immediately after its statement explains why the proof is omitted or deferred and, where useful, supplies a later or external reference. Proof status is not placed in a mathematical heading. Examples provide intuition, but not proof of a general assertion. Occasionally, an explicitly identified informal preview uses familiar notation to motivate a later formal development; such previews are not used as formal justification.

**Practice and hints.** Part I gives explicit proof strategies, ordinary calculations, and a few applications that require translating words into mathematical assumptions. The scaffolding becomes lighter in Part II as the reader gains proof maturity; its structural development expects greater independence. An asterisk  $*$  after an exercise number indicates that a hint is available in Appendix I; the symbol links directly to that hint, and the hint’s exercise number links back. The asterisk means *hint available*, not “difficult exercise.” Some theoretical exercises are deliberately left without hints because finding the idea is part of their purpose.

**Notation and reminders.** General functions and nonlinear maps use lowercase Latin letters; charts use lowercase Greek letters, curves use  $\gamma$ , and surface parametrizations usually use  $\sigma$ . Linear maps use capitals such as  $T, S, L$ . The Euclidean derivative is  $Df(a)$ ; the intrinsic differential and exterior derivative are  $df_p$  and  $d\omega$ . For a smooth Euclidean scalar function,  $df$  denotes the exact 1-form and  $(df)_p = Df(p)$  its value at  $p$ . Conventional symbols with a distinct role, such as the force or flux vector field  $F$  and an antiderivative  $F$  paired with its integrand  $f$ , are retained where they clarify that role. Less familiar structural notions reused after a gap receive a cross-reference; when central to the argument, they also receive a local reminder. Such reminders are selective rather than attached to every use of ordinary continuity or differentiation.

The chapters are designed to be read in order. Exercises are part of the course, but the main text contains the mathematics the reader is expected to learn. Exercises reinforce, apply, specialize, or locally reconstruct material already developed in the exposition; no major theorem, essential lemma, or concept needed later is established only in an exercise. They supply calculations, short corollaries, counterexamples, comparisons of hypotheses, and nearby applications. A reader should attempt them after the relevant section, consulting the stated hypotheses and earlier results rather than relying on calculus folklore. Results not proved in the notes are identified immediately after their statements. The bibliography also distinguishes the imported topological result from further-reading references. In particular, Chapter 9 states without proof the compatibility of proper Riemann and Lebesgue integration, the Lipschitz/a.e. Newton–Leibniz theorem, existence of the Cantor function, Lebesgue monotone and dominated convergence, and the Lebesgue Fundamental Theorem of Calculus. These previews clarify the limits of the Riemann theory but are

not prerequisites for later chapters. The optional analytic Lagrange inversion result in Chapter 11 and the Lebesgue change-of-variables preview in Chapter 12 are likewise stated without proof. Section 13.17 contains additional explicitly unproved preview theorems, none used as a prerequisite. The main route through these chapters is local differential structure, primitive substitution with finite Euclidean localization, and then locally finite manifold localization in Chapter 13.

Read a substantial proof twice with different aims. On the first pass, identify the problem, the main construction, and the purpose of each estimate. On the second, reconstruct the quantifiers and check where every hypothesis enters. Understanding a proof's architecture and reproducing all its details are related but distinct skills.

In Chapter 3, the construction of  $\mathbb{R}$  proves that the system we use exists. Difficulty reproducing every equivalence-class verification should not block further study. Retain its working consequences: ordered-field structure, the Archimedean property, rational density, completeness, Cauchy convergence, and the least-upper-bound property. Return to the implementation as these ideas become familiar. In Part II, watch these same tools reappear as norm estimates, compactness arguments, linear approximation, and local-to-global integration.

Chapter 10 assumes no previous topology or linear algebra course. It generalizes the real-line topology and compactness introduced in Chapter 5; its distance examples lead to metric continuity before abstract topology; its coordinate calculations lead to bases, direct sums, linear maps, and matrices. Appendices A and B collect foundational and structural language occasionally useful throughout the notes. Appendices C–H develop optional algebraic, topological, differential-geometric, and cohomological extensions. None is intended to be read before Chapter 1, and none is a prerequisite for the main analysis-to-Stokes route unless explicitly stated. The hierarchy is foundational language, then categorical language, then optional mathematical extensions; it is not a compulsory reading order.

Sections A.1–A.2 of Appendix A are a companion to Chapter 2. They review induction, recursive definitions and proofs, factorials, binomial coefficients, Pascal's identity, the finite binomial theorem, and the equivalence of induction and well-ordering of  $\mathbb{N}$ . Readers unfamiliar with these tools may consult those sections alongside Chapter 2. Elementary finite algebraic identities such as the binomial theorem may be used in the main text without proof; Appendix A gives proofs by induction for readers who want the foundational details.

Sections A.3–A.5 are fully optional foundations. Neither the equivalence of Choice, Zorn's Lemma, and the Well-Ordering Theorem nor its transfinite recursion and Hartogs arguments is needed for Chapters 3–9. For the main text, knowing the statement and a typical use of Zorn's Lemma is more than enough: if every chain in a nonempty partially ordered set has an upper bound in that set, then the poset has a maximal element. A standard use is the existence of a basis of an arbitrary vector space. The detailed foundational proofs may be read purely for interest.

Appendix B organizes constructions through maps and their compatibility. Appendix C develops determinants over a field; Appendix D gives a structural alternative to Chapter 13's concrete exterior algebra. Appendix E constructs manifolds by gluing. Appendix F connects the rank theorems to manifold geometry, and Appendix G uses that framework for Lie groups. Appendix H starts from forms and Stokes, preserves concrete computations, and then introduces the exact-sequence language connecting smooth analysis with topology. These are branching continuations, not one total dependency chain. Section 13.17 provides navigation, including differential topology, geometric topology, and geometric analysis; the appendices develop selected parts of that outlook.

At the first serious use of a proof method, look for the actual map, sequence, or estimate being constructed. Later uses refer back to that mechanism. In particular, existence of a map is a different task from proving equality of two maps, and choosing a finite index is a different task from subsequently taking a limit.

# Notation and Conventions

Most notation is introduced again when it first becomes mathematically relevant. The conventions collected here are those used repeatedly throughout the notes, especially where closely related meanings coexist.

**Numbers and maps.** We use  $\mathbb{N} = \{1, 2, 3, \dots\}$  and  $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ . The symbols  $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$  denote the integers, rationals, reals, and complex numbers. General maps usually use  $f, g, h$ ; linear maps and operators use  $T, S, L$ . Charts and coordinate maps use  $\varphi, \psi, \chi$ , curves use  $\gamma$ , and surface parametrizations usually use  $\sigma$ . Conventional symbols such as a force field  $F$  are retained. Composition is  $(g \circ f)(x) = g(f(x))$ ;  $f|_A$  is the restriction to  $A$ .

**Euclidean inputs and scalar volume.** In Part II the full Euclidean input is written

$$\mathbf{x} = (x_1, \dots, x_n), \quad f(\mathbf{x}), \quad \int_Q f(\mathbf{x}) \, d\mathbf{x}.$$

Components  $x_i$ , scalar coordinates, and abstract points retain ordinary type. The unsigned scalar volume symbol  $d\mathbf{x}$  is distinct from coordinate covectors  $dx_i$ , their oriented wedge product, and exterior differentiation. Established vector and manifold notation is retained.

**Four related differential notations.**

|                                       |                                                              |
|---------------------------------------|--------------------------------------------------------------|
| $Df(a)$                               | Euclidean/Fréchet derivative at $a$                          |
| $(df)_p$                              | value of the exact scalar-function 1-form at $p$             |
| $df_p : T_p M \rightarrow T_{f(p)} N$ | intrinsic differential of a smooth map $f : M \rightarrow N$ |
| $d\omega$                             | exterior derivative of a differential form                   |

For a Euclidean scalar function,  $(df)_p = Df(p)$  under the usual tangent-space identification. For a manifold scalar function, identify  $T_{f(p)}\mathbb{R}$  with  $\mathbb{R}$  to regard its differential as a covector.

|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| $D$ emphasizes Euclidean differentiation;<br>$d$ is used for intrinsic differentials and forms. |
|-------------------------------------------------------------------------------------------------|

**Bases and their measurements.** A basis  $e_1, \dots, e_n$  has dual basis  $e^1, \dots, e^n$ , with  $e^i(e_j) = \delta_j^i$ . The Kronecker delta  $\delta_j^i$  is 1 when  $i = j$  and 0 otherwise. In manifold coordinates the corresponding pair is

$$\left. \frac{\partial}{\partial x^i} \right|_p, \quad dx^i|_p.$$

The first is a tangent direction; the second measures its  $i$ th coordinate. A topological basis, by contrast, is a family of open sets, not vectors.

**Sets and boundaries.** The symbols  $\bar{A}$ ,  $A^\circ$ , and  $\partial A$  denote closure, interior, and topological boundary in the stated ambient space, with  $\partial A = \bar{A} \setminus A^\circ$ . The intrinsic manifold boundary  $\partial M$  is defined by half-space charts and need not equal the boundary of  $M$  as a subset of an ambient Euclidean space. For example, a circle has empty intrinsic boundary, while its topological boundary in  $\mathbb{R}^2$  is the entire circle. Context and the specified ambient space matter. The notation  $\text{supp } f$  or  $\text{supp } \omega$  means the closure of the set where the function or form is nonzero.

## Part I

# Single-Variable Calculus and Analysis

# Chapter 1

## Motivation and the Transition from Calculus to Analysis

### 1.1 Why Calculus Needs Foundations

Calculus describes change and accumulation. Its central calculations often begin with an informal phrase: “make the increment arbitrarily small,” “add infinitely many pieces,” or “approach a point.” These phrases are useful, but they conceal questions that computation alone cannot answer. What does it mean to approach? Why is a limiting value unique? Why may a limiting process be interchanged with addition, differentiation, or integration?

Mathematical analysis makes these meanings precise and proves the rules used in calculation. Its hypotheses tell us when a calculation is reliable and help locate the error in a plausible argument that fails.

**Example 1.1.1** (A number suggested by geometry). The diagonal of a square of side length 1 has a length customarily denoted by  $\sqrt{2}$ . If this length were a rational number

$$p/q$$

in lowest terms, then  $p^2 = 2q^2$ . Thus  $p^2$  is even. An odd integer has the form  $2k + 1$ , whose square is  $4k(k + 1) + 1$  and is odd; hence  $p$  is even. Write  $p = 2r$ . Then  $q^2 = 2r^2$ , so the same argument makes  $q$  even. This contradicts that

$$p/q$$

was in lowest terms. Hence no rational number has square 2.

The example does not yet construct  $\sqrt{2}$ ; rather, it points out why rational numbers are not sufficient for familiar geometry. Chapter 3 will construct a number system containing such limits and isolate the property that makes it adequate: completeness.

### 1.2 A First Look at Limits

Consider the values

$$1, \frac{1}{2}, \frac{1}{3}, \dots$$

Their terms become as small as we wish. A rigorous definition should express exactly that statement without appealing to a last term, because there is no last positive integer.

**Informal preview of convergence.** A sequence  $(a_n)_{n \in \mathbb{N}}$  of real numbers **converges to** a real number  $L$  when, for every positive tolerance  $\varepsilon$ , all sufficiently late terms  $a_n$  lie within distance  $\varepsilon$  of  $L$ . In symbols, the intended statement is

$$\text{for every } \varepsilon > 0 \text{ there is } N \in \mathbb{N} \text{ such that } n \geq N \implies |a_n - L| < \varepsilon.$$

This is not yet a formal definition: the quantifiers, real numbers, and natural numbers will be developed formally in Chapters 2–4.

For instance,

$$\frac{1}{n}$$

converges to 0: given  $\varepsilon > 0$ , choose a natural number

$$N > \frac{1}{\varepsilon}.$$

Then  $n \geq N$  implies

$$0 < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

The existence of such an  $N$  is an Archimedean property, which will be proved from the construction of the real numbers rather than silently assumed.

## 1.3 The Structural Ideas of Analysis

### Limits

A limit describes stable behavior at arbitrarily fine scales. The order of the quantifiers matters: a proposed tolerance is given first, and the required stage may depend on it.

### Completeness

A number system is complete when a process that should determine a number does not leave a gap. Two formulations are the least-upper-bound principle and the assertion that Cauchy sequences converge.

### Continuity

A function is continuous at a point when small changes in input force small changes in output. The exact definition uses the same quantifier pattern as a limit.

### Differentiation

The derivative is the best linear approximation to a function at a point. In one variable it is a number; in several variables it is a linear map.

### Integration

Integration formalizes the limiting process of approximating accumulated quantities by finite sums. The existence of an integral, and its relationship to derivatives, are the substance of the Fundamental Theorem of Calculus.

## 1.4 Warnings That Motivate Hypotheses

**Example 1.4.1** (Pointwise convergence need not preserve continuity). For  $n \in \mathbb{N}$ , define  $f_n : [0, 1] \rightarrow \mathbb{R}$  by  $f_n(x) = x^n$ . Every  $f_n$  is continuous. For  $0 \leq x < 1$ , the values  $x^n$  tend to 0, whereas  $f_n(1) = 1$  for every  $n$ . Thus the pointwise limiting function is

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1, \end{cases}$$

which is not continuous at 1. Chapter 6 will prove that **uniform** convergence supplies the missing hypothesis.

**Example 1.4.2** (A bounded set may lack a rational supremum). Let

$$S = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}.$$

The set is nonempty (for example  $1 \in S$ ) and bounded above in  $\mathbb{Q}$  (for example 2 is an upper bound). It has no least rational upper bound. Informally, any such bound would have to be  $\sqrt{2}$ , which is not rational. The least-upper-bound argument in Chapter 3 constructs positive square roots; its perturbation estimates also show why no rational candidate can be the least upper bound of this rational set.

## 1.5 Road Map

Chapter 2 establishes the language of proof. Chapters 3–4 build  $\mathbb{R}$  and its sequence theory. Chapters 5–9 reconstruct one-variable calculus from limits. The second part then enlarges the setting to finite-dimensional spaces, develops multivariable differentiation and integration, and studies work along curves and flux through surfaces. Green and parametrized-patch Stokes then motivate differential forms; the general Stokes theorem unifies these formulas on manifolds.

## Exercises

**Exercise 1.** Let  $(a_n)$  be a sequence of real numbers that converges to both  $L$  and  $M$ . Using the informal definition above, formulate a strategy for proving  $L = M$ . The complete proof appears after the necessary order facts have been proved.

**Exercise 2.** For each claim, identify an existence question and a separate computational question: finding a square root, summing an infinite series, and solving an equation for one variable. Explain why a plausible formula alone need not answer the existence question.

## Chapter 2

# Basic Set Theory, Logic, Functions, and Relations

Sections A.1–A.2 of Appendix A complement this chapter with induction, recursion, factorials, binomial coefficients, Pascal’s identity, and the finite binomial theorem. They also explain why induction and well-ordering of  $\mathbb{N}$  are equivalent. Elementary finite algebraic identities such as the binomial theorem are admitted background in the main text; the appendix supplies optional proofs by induction.

### 2.1 Sets

Mathematical analysis constantly describes collections: the possible inputs of a function, the points of an interval, or the terms of a sequence. The language of sets lets us describe such collections precisely and compare them without relying on a picture or an informal list.

We take

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

as the primitive discrete number system. Its usual arithmetic means that addition and multiplication have their familiar laws and keep natural numbers natural. Its induction property says that a statement true at 1, and inherited from each number to its successor, is true for every natural number. Its well-ordering property says that every nonempty collection of natural numbers has a smallest member. These principles will be used repeatedly, and induction is stated formally in Theorem 2.2.1. We write

$$\mathbb{N}_0 = \mathbb{N} \cup \{0\}.$$

The integers and rational numbers will be constructed in this chapter, and the real numbers in Chapter 3. A **set** is a collection of objects, called its elements. We write  $x \in A$  when  $x$  is an element of  $A$ .

**Definition 2.1.1** (Subsets and set equality). For sets  $A, B$ , write  $A \subseteq B$  if every element of  $A$  is an element of  $B$ . Define  $A = B$  to mean both  $A \subseteq B$  and  $B \subseteq A$ .

The empty set  $\emptyset$  has no elements, and is a subset of every set. Here “and” requires both stated conditions, “or” allows either condition (including both), and  $x \notin B$  means that  $x$  is not an element of  $B$ . For any sets  $A, B$ , define

$$\begin{aligned} A \cup B &= \{x : x \in A \text{ or } x \in B\}, & A \cap B &= \{x : x \in A \text{ and } x \in B\}, \\ A \setminus B &= \{x : x \in A \text{ and } x \notin B\}. \end{aligned}$$



If  $A \subseteq X$ , its complement is  $A^c = X \setminus A$ . The power set is  $\mathcal{P}(A) = \{E : E \subseteq A\}$ .

**Proposition 2.1.2** (De Morgan's laws). If  $A, B \subseteq X$ , then  $(A \cup B)^c = A^c \cap B^c$  and  $(A \cap B)^c = A^c \cup B^c$ .

*Proof.* For  $x \in X$ , membership in  $(A \cup B)^c$  means that it is false that  $x \in A$  or  $x \in B$ . This is equivalent to  $x \notin A$  and  $x \notin B$ , namely  $x \in A^c \cap B^c$ . The second identity follows identically with “and” and “or” exchanged.  $\square$

**Definition 2.1.3** (Cartesian product). For sets  $A, B$ , define  $A \times B = \{(a, b) : a \in A, b \in B\}$ , where  $(a, b) = (a', b')$  means  $a = a'$  and  $b = b'$ .

## 2.2 Logic and Proofs

Definitions specify the objects of mathematics, while proofs explain why their asserted properties follow. This section introduces the small amount of logical language needed to state definitions and organize proofs clearly.

A **proposition** is a statement that is either true or false. A **predicate** is a statement with a variable whose truth depends on the value of that variable. For example, let  $P(n)$  mean “ $n$  is even.” Then  $P(4)$  is a true proposition, whereas  $P(n)$  is not a complete proposition until  $n$  is assigned a value or is quantified.

If  $P$  and  $Q$  are propositions, then  $P$  and  $Q$  means that both are true, and  $P$  or  $Q$  means that at least one is true. This is the inclusive mathematical meaning of “or.” For instance,

$$n > 0 \text{ and } n < 5$$

requires both inequalities, while “ $n$  is even or divisible by 3” permits either property or both. The **negation** of  $P$ , written

$$\neg P,$$

means that  $P$  is false. Thus the negation of  $x > 3$  is  $x \leq 3$ . Negating a mathematical statement means negating its full meaning, not merely placing the word “not” at an arbitrary point.

The implication

$$P \Rightarrow Q$$

means “if  $P$ , then  $Q$ ”;  $P$  is its **hypothesis** (or antecedent), and  $Q$  is its **conclusion** (or consequent). For example,

$$n \text{ is divisible by } 4 \quad \Rightarrow \quad n \text{ is even.}$$

Its **contrapositive** is

$$\neg Q \Rightarrow \neg P,$$

and an implication is logically equivalent to its contrapositive. The **converse** is  $Q \Rightarrow P$ ; it is a different assertion and does not follow automatically. The notation

$$P \Longleftrightarrow Q$$

means “ $P$  if and only if  $Q$ ,” often abbreviated “iff”: both  $P \Rightarrow Q$  and  $Q \Rightarrow P$  hold. A proof of an iff statement normally proves the two directions separately.

Quantifiers turn predicates into propositions. The statement

$$\forall x \in A \, P(x)$$

means that  $P(x)$  holds for every  $x \in A$ , while

$$\exists x \in A \ P(x)$$

means that there is at least one  $x \in A$  for which  $P(x)$  holds. For example,

$$\forall n \in \mathbb{N} \quad n + 1 > n,$$

whereas

$$\exists n \in \mathbb{N} \quad n^2 = 4.$$

The first says something about every natural number; the second asserts the existence of a witness, here  $n = 2$ . After predicates, quantifiers, and negation have been introduced, their basic interaction can be written as

$$\neg(\forall x \in A \ P(x)) \iff \exists x \in A \ \neg P(x),$$

$$\neg(\exists x \in A \ P(x)) \iff \forall x \in A \ \neg P(x).$$

In words, “every student passed” is false exactly when there is a student who did not pass; “there is a student who passed” is false exactly when every student failed.

## Writing Proofs

Logic becomes useful here as a guide to proof writing. To prove a universal implication

$$\forall x \in A \quad P(x) \Rightarrow Q(x),$$

begin by letting  $x \in A$  be arbitrary, suppose  $P(x)$ , and show  $Q(x)$ . The word “arbitrary” matters: the argument must work for every allowed  $x$ , not only for a convenient example. To prove an existential statement, give a **witness** and verify that it has the required property. To prove that an object exists uniquely, first prove existence and then show that any two objects with the property are equal. This pattern will recur for limits, least upper bounds, inverse functions, and implicit functions.

Set equality is proved by showing both inclusions

$$A \subseteq B \quad \text{and} \quad B \subseteq A.$$

Similarly, two functions with the same domain and codomain are equal when their values agree at every point of the domain. A direct proof derives the conclusion from the hypotheses. A contrapositive proof establishes  $\neg Q \Rightarrow \neg P$  in order to prove  $P \Rightarrow Q$ . A proof by contradiction assumes the hypotheses and the negation of the desired conclusion, then derives an impossibility.

A proof should indicate why each nontrivial step follows: from a hypothesis, a definition, an earlier theorem, or an algebraic calculation. Calculations are part of a proof only when their purpose is clear, so short explanatory sentences around a nontrivial chain of formulas are often helpful. The aim is clarity and visible logical dependence, not rigid prose templates. The next theorem records the induction method: prove a base case, assume the statement at an arbitrary stage as the induction hypothesis, and then prove the induction step.

**Theorem 2.2.1** (Induction). Let  $P(n)$  be a proposition for each  $n \in \mathbb{N}$ . If  $P(1)$  is true and  $P(n) \Rightarrow P(n+1)$  for every  $n \in \mathbb{N}$ , then  $P(n)$  is true for all  $n \in \mathbb{N}$ .

*Proof.* This is the induction axiom for  $\mathbb{N}$ . Equivalently, it follows from the well-ordering principle that each nonempty subset of  $\mathbb{N}$  has a least element: a least  $n$  for which  $P(n)$  failed could be neither 1 nor a successor of a number for which  $P$  holds.  $\square$

## 2.3 Relations and Functions

Many mathematical statements compare two objects: two numbers may be equal, one number may divide another, or one set may be contained in another. A relation records exactly which ordered pairs of objects are being regarded as related.

**Definition 2.3.1** (Binary relation). Let  $A$  and  $B$  be sets. A **binary relation**  $R$  from  $A$  to  $B$  is a subset

$$R \subseteq A \times B.$$

If  $A = B$ , then  $R$  is a **binary relation on**  $A$ . When

$$(a, b) \in R,$$

we commonly write

$$a R b.$$

**Example 2.3.2** (Familiar relations). Equality, the usual numerical order, divisibility on  $\mathbb{N}$ , and inclusion between subsets of a fixed set are all binary relations. For example, on

$$A = \{1, 2, 3, 4\},$$

the divisibility relation is the subset

$$R = \{(a, b) \in A \times A : a \text{ divides } b\}.$$

It contains, among others,

$$(1, 2), \quad (1, 4), \quad (2, 4), \quad (3, 3),$$

but

$$(3, 4) \notin R.$$

Thus the statement that a relation is a set of ordered pairs has concrete content. Not every relation is an equivalence relation or an order: for example, on  $\mathbb{N}$ , the relation

$$m R n \iff n = m + 1$$

is neither, since it is not reflexive.

**Definition 2.3.3** (Equivalence relation). An **equivalence relation** is a binary relation  $\sim$  on  $A$  such that, for all  $a, b, c \in A$ , it is **reflexive**, **symmetric**, and **transitive**:

$$a \sim a, \quad a \sim b \implies b \sim a, \quad a \sim b \text{ and } b \sim c \implies a \sim c.$$

Its equivalence class at  $a$  is

$$[a] = \{b \in A : b \sim a\}.$$

Equivalence relations classify elements by grouping those regarded as the same for a specified purpose. Equality is the simplest example. Congruence modulo a natural number, used after the construction of  $\mathbb{Z}$  below, is a further familiar example.

**Proposition 2.3.4.** If  $\sim$  is an equivalence relation, then  $A$  is the union of its equivalence classes, and two classes are either equal or disjoint.

*Proof.* Reflexivity gives  $a \in [a]$  for every  $a$ . If  $x \in [a] \cap [b]$ , symmetry and transitivity imply  $a \sim b$ . Then  $y \sim a$  implies  $y \sim b$ , and conversely, so  $[a] = [b]$ .  $\square$

**Definition 2.3.5** (Partial order). An **order relation** or **partial order** on a set  $X$  is a binary relation  $\leq$  on  $X$  that is reflexive, antisymmetric, and transitive: for all  $x, y, z \in X$ ,

$$x \leq x,$$

$$x \leq y \text{ and } y \leq x \implies x = y, \quad x \leq y \text{ and } y \leq z \implies x \leq z.$$

A set equipped with a partial order is a **partially ordered set**, or **poset**. Antisymmetric does not mean “not symmetric”; it is the specific implication displayed above.

**Example 2.3.6** (Partial orders and incomparability). The power set  $\mathcal{P}(X)$ , ordered by inclusion, is a poset. If  $X$  contains 1 and 2, then

$$\{1\} \not\subseteq \{2\}, \quad \{2\} \not\subseteq \{1\},$$

so these two sets are not comparable. Divisibility is another partial order: on positive integers, define

$$m \leq_d n \iff m \text{ divides } n.$$

**Definition 2.3.7** (Total order and strict order). A **total order**, or **linear order**, is a partial order  $\leq$  on  $X$  such that every two elements are comparable:

$$x \leq y \text{ or } y \leq x \quad (x, y \in X).$$

For a total order, define its associated strict order by

$$x < y \quad \text{if} \quad x \leq y \text{ and } x \neq y.$$

We take  $\leq$  as the primitive relation. The usual order on  $\mathbb{N}$  is a total order, whereas inclusion on  $\mathcal{P}(X)$  is generally only a partial order.

Two different specializations of binary relations will recur: equivalence relations classify or group elements, whereas partial orders compare elements; total orders are the partial orders with no incomparability.

**Definition 2.3.8** (Function and inverse images). A function  $f : A \rightarrow B$  assigns exactly one  $f(a) \in B$  to every  $a \in A$ . The set  $A$  is its **domain**,  $B$  is its **codomain**, and its **range** is the subset  $f(A) \subseteq B$ . The codomain is part of the declared function and need not equal its range. Its graph

$$\{(a, f(a)) : a \in A\} \subseteq A \times B$$

is therefore a relation for which every  $a \in A$  occurs in exactly one ordered pair. For  $E \subseteq A$  and  $F \subseteq B$ , set  $f(E) = \{f(x) : x \in E\}$  and  $f^{-1}(F) = \{x \in A : f(x) \in F\}$ . It is **injective** if  $f(a) = f(a')$  implies  $a = a'$ , **surjective** if  $f(A) = B$ , and **bijective** if both.

**Definition 2.3.9** (Composition and identity functions). Let  $f : A \rightarrow B$  and  $g : B \rightarrow C$ . Their **composition** is the function  $g \circ f : A \rightarrow C$  defined by

$$(g \circ f)(a) = g(f(a)) \quad (a \in A).$$

For every set  $A$ , the **identity function**  $\text{id}_A : A \rightarrow A$  is defined by  $\text{id}_A(a) = a$ .

**Theorem 2.3.10** (Inverse function). If  $f : A \rightarrow B$  is bijective, there is a unique function  $f^{-1} : B \rightarrow A$  satisfying  $f^{-1}(f(a)) = a$  and  $f(f^{-1}(b)) = b$ .

*Proof.* For each  $b \in B$ , surjectivity supplies at least one  $a \in A$  with  $f(a) = b$ . If  $a'$  is another, injectivity gives  $a' = a$ . Thus there is exactly one such preimage; define  $g(b)$  to be it. This defines a function  $g : B \rightarrow A$ , not a choice depending on an arbitrary preimage. By its definition  $f(g(b)) = b$  for every  $b$ . For  $a \in A$ , the point  $a$  is a preimage of  $f(a)$ , and its uniqueness gives  $g(f(a)) = a$ . Hence  $fg = \text{id}_B$  and  $gf = \text{id}_A$ . If  $h : B \rightarrow A$  also satisfies the two identities, then

$$f(h(b)) = b = f(g(b)),$$

so injectivity gives  $h(b) = g(b)$  for every  $b$ . The inverse is unique, and we denote it  $f^{-1}$ . This inverse function is distinct from the inverse-image notation  $f^{-1}(E)$ , which is defined even when  $f$  is not bijective.  $\square$

**Proposition 2.3.11** (Basic laws of composition). Let  $f : A \rightarrow B$ ,  $g : B \rightarrow C$ , and  $h : C \rightarrow D$  be functions.

- (a)  $h \circ (g \circ f) = (h \circ g) \circ f$ .
- (b)  $f \circ \text{id}_A = f$  and  $\text{id}_B \circ f = f$ .
- (c) If  $f$  and  $g$  are injective, then  $g \circ f$  is injective; if  $f$  and  $g$  are surjective, then  $g \circ f$  is surjective. In particular, a composition of bijections is a bijection.
- (d) If  $f$  and  $g$  are bijections, then  $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ .

*Proof.* For  $a \in A$ , both functions in the first assertion have value  $h(g(f(a)))$ , proving equality. The identity assertions follow from  $\text{id}_A(a) = a$  and  $\text{id}_B(f(a)) = f(a)$ . If  $(g \circ f)(a) = (g \circ f)(a')$ , injectivity of  $g$  gives  $f(a) = f(a')$ , and injectivity of  $f$  gives  $a = a'$ . If  $c \in C$ , surjectivity of  $g$  gives  $b \in B$  with  $g(b) = c$ , and surjectivity of  $f$  gives  $a \in A$  with  $f(a) = b$ ; then  $(g \circ f)(a) = c$ . Finally,

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_B \circ f = \text{id}_A,$$

In the other order,

$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ \text{id}_B \circ g^{-1} = \text{id}_C.$$

A two-sided inverse is unique: if  $u$  and  $v$  are both two-sided inverses of a function  $w$ , then

$$u = u \circ (w \circ v) = (u \circ w) \circ v = v.$$

This proves the last assertion without using a later result.  $\square$

**Proposition 2.3.12** (Inverse-image rules). For  $f : A \rightarrow B$  and  $F, G \subseteq B$ , one has  $f^{-1}(F \cup G) = f^{-1}(F) \cup f^{-1}(G)$ ,  $f^{-1}(F \cap G) = f^{-1}(F) \cap f^{-1}(G)$ , and  $f^{-1}(B \setminus F) = A \setminus f^{-1}(F)$ .

*Proof.* For example,  $x \in f^{-1}(F \cap G)$  if and only if  $f(x) \in F$  and  $f(x) \in G$ , precisely when  $x \in f^{-1}(F) \cap f^{-1}(G)$ . The other two identities use the same membership calculation.  $\square$

**Proposition 2.3.13** (Image and inverse-image rules). Let  $f : A \rightarrow B$ , let  $E, H \subseteq A$ , and let  $F, G \subseteq B$ . Then

$$f(E \cup H) = f(E) \cup f(H), \quad f(E \cap H) \subseteq f(E) \cap f(H),$$

and equality holds in the second formula when  $f$  is injective. Moreover,

$$E \subseteq f^{-1}(f(E)), \quad f(f^{-1}(F)) \subseteq F,$$

and equality holds in the first inclusion when  $f$  is injective and in the second when  $f$  is surjective.

*Proof.* Each assertion follows by unpacking membership. For example, if

$$y \in f(E \cap H),$$

then  $y = f(x)$  for some  $x \in E \cap H$ , so  $y$  belongs to both  $f(E)$  and  $f(H)$ . Conversely, if  $f$  is injective and

$$y = f(e) = f(h)$$

with  $e \in E$  and  $h \in H$ , then  $e = h \in E \cap H$ . The remaining statements follow similarly; the equality cases use respectively injectivity to recover a unique preimage and surjectivity to supply one.  $\square$

## 2.4 The Natural Numbers, Integers, and Rational Numbers

The natural numbers are the starting point for the arithmetic used throughout the notes. A fully set-theoretic construction of  $\mathbb{N}$ , for example by finite ordinals, belongs to foundations rather than analysis. We therefore take  $\mathbb{N}$  and  $\mathbb{N}_0$  with their usual arithmetic, induction, and well-ordering structure as primitive. The next two quotient constructions explain the algebraic chain

$$\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q}$$

before Chapter 3 treats the analytic extension

$$\mathbb{Q} \longrightarrow \mathbb{R}.$$

Passing from  $\mathbb{N}_0$  to  $\mathbb{Z}$  permits subtraction, and passing from  $\mathbb{Z}$  to  $\mathbb{Q}$  permits division by nonzero elements.

**Definition 2.4.1** (The integers). On  $\mathbb{N}_0 \times \mathbb{N}_0$ , define

$$(a, b) \sim_{\mathbb{Z}} (c, d)$$

if

$$a + d = b + c.$$

The set of equivalence classes is denoted by

$$\mathbb{Z} = (\mathbb{N}_0 \times \mathbb{N}_0) / \sim_{\mathbb{Z}}.$$

Write  $[a, b]_{\mathbb{Z}}$  for the class of  $(a, b)$ , interpreted as the formal difference  $a - b$ . Define

$$\begin{aligned} [a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}} &= [a + c, b + d]_{\mathbb{Z}}, \\ [a, b]_{\mathbb{Z}} [c, d]_{\mathbb{Z}} &= [ac + bd, ad + bc]_{\mathbb{Z}}, \quad -[a, b]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}}. \end{aligned}$$

Set

$$0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}, \quad 1_{\mathbb{Z}} = [1, 0]_{\mathbb{Z}},$$

and define

$$\iota_{\mathbb{Z}} : \mathbb{N}_0 \longrightarrow \mathbb{Z}, \quad \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}.$$

Finally, define the order by

$$[a, b]_{\mathbb{Z}} < [c, d]_{\mathbb{Z}} \quad \text{if} \quad a + d < b + c.$$

**Proposition 2.4.2** (Integer arithmetic). The relation  $\sim_{\mathbb{Z}}$  is an equivalence relation, and the operations and order in the preceding definition are independent of representatives. The map  $\iota_{\mathbb{Z}}$  is injective and preserves addition, multiplication, and order. With these operations,  $\mathbb{Z}$  has commutative, associative, distributive addition and multiplication, identities  $0_{\mathbb{Z}}, 1_{\mathbb{Z}}$ , additive inverses, and no zero divisors. Its displayed order is total and is compatible with addition and multiplication by positive elements.

*Proof.* Reflexivity and symmetry of  $\sim_{\mathbb{Z}}$  are immediate. If

$$a + d = b + c, \quad c + f = d + e,$$

then

$$a + d + f = b + c + f = b + d + e.$$

Cancellation in  $\mathbb{N}_0$  gives  $a + f = b + e$ , proving transitivity.

An operation written with representatives gives an operation on equivalence classes only when changing representatives does not change the resulting class. This is what **well defined** means here. For the displayed sum, suppose

$$(a, b) \sim_{\mathbb{Z}} (a', b'), \quad (c, d) \sim_{\mathbb{Z}} (c', d').$$

Then

$$\begin{aligned} (a + c) + (b' + d') &= (a + b') + (c + d') \\ &= (b + a') + (d + c') \\ &= (b + d) + (a' + c'). \end{aligned}$$

Thus

$$(a + c, b + d) \sim_{\mathbb{Z}} (a' + c', b' + d'),$$

so addition is well defined. For multiplication we must avoid assuming integer subtraction before integers have been constructed. First change only  $(a, b)$ , using  $a + b' = b + a'$ . Multiply that equality by  $c$  and by  $d$ :

$$ac + b'c = bc + a'c, \quad ad + b'd = bd + a'd.$$

Adding these equalities in the required order gives

$$ac + bd + a'd + b'c = ad + bc + a'c + b'd.$$

This is precisely  $(ac+bd, ad+bc) \sim_{\mathbb{Z}} (a'c+b'd, a'd+b'c)$ . Next change only  $(c, d)$ , using  $c+d' = d+c'$ . Multiplication by  $a'$  and  $b'$  gives  $a'c + a'd' = a'd + a'c'$  and  $b'c + b'd' = b'd + b'c'$ . Thus

$$a'c + b'd + a'd' + b'c' = a'd + b'c + a'c' + b'd',$$

the equivalence between the new product and  $(a'c' + b'd', a'd' + b'c')$ . Transitivity proves independence when both representatives change. A formula on representatives was only a proposed operation; these checks turn it into an operation on classes. The comparison defining the order is likewise unchanged after adding the equalities associated with equivalent representatives. These checks show that all formulas descend to classes.

The associative, commutative, and distributive laws now follow from the corresponding arithmetic identities in  $\mathbb{N}_0$ . The formulas for  $0_{\mathbb{Z}}$ ,  $1_{\mathbb{Z}}$ , and  $-[a, b]_{\mathbb{Z}}$  give the required identities and additive inverses. The order says exactly that a formal difference is positive when its first entry exceeds its second; the order laws follow by adding inequalities and multiplying positive differences in  $\mathbb{N}_0$ . Consequently, a product can vanish only when one factor does. Finally,

$$[m, 0]_{\mathbb{Z}} + [n, 0]_{\mathbb{Z}} = [m + n, 0]_{\mathbb{Z}},$$

$$[m, 0]_{\mathbb{Z}}[n, 0]_{\mathbb{Z}} = [mn, 0]_{\mathbb{Z}},$$

and

$$[m, 0]_{\mathbb{Z}} < [n, 0]_{\mathbb{Z}} \iff m < n,$$

which proves the assertions about  $\iota_{\mathbb{Z}}$ . □

**Definition 2.4.3** (The rational numbers). On

$$\mathbb{Z} \times (\mathbb{Z} \setminus \{0_{\mathbb{Z}}\}),$$

define

$$(a, b) \sim_{\mathbb{Q}} (c, d)$$

if

$$ad = bc.$$

The set of equivalence classes is denoted by

$$\mathbb{Q} = (\mathbb{Z} \times (\mathbb{Z} \setminus \{0_{\mathbb{Z}}\})) / \sim_{\mathbb{Q}}.$$

Write  $[a, b]_{\mathbb{Q}}$ , or simply

$$a/b,$$

for the class of  $(a, b)$ . Define

$$[a, b]_{\mathbb{Q}} + [c, d]_{\mathbb{Q}} = [ad + bc, bd]_{\mathbb{Q}},$$

$$[a, b]_{\mathbb{Q}}[c, d]_{\mathbb{Q}} = [ac, bd]_{\mathbb{Q}},$$

and

$$\iota_{\mathbb{Q}} : \mathbb{Z} \longrightarrow \mathbb{Q}, \quad \iota_{\mathbb{Q}}(a) = [a, 1]_{\mathbb{Q}}.$$

The additive inverse of  $[a, b]_{\mathbb{Q}}$  is  $[-a, b]_{\mathbb{Q}}$ ; if  $a \neq 0_{\mathbb{Z}}$ , its multiplicative inverse is  $[b, a]_{\mathbb{Q}}$ . Define positivity by

$$[a, b]_{\mathbb{Q}} > 0 \quad \text{if} \quad ab > 0_{\mathbb{Z}}.$$

As in every ordered arithmetic system, define

$$r < s \quad \text{if} \quad s - r > 0.$$



**Proposition 2.4.4** (Rational arithmetic). The relation  $\sim_{\mathbb{Q}}$ , the operations, and the positivity relation in the preceding definition are well defined. The map  $\iota_{\mathbb{Q}}$  is an injective order-preserving arithmetic embedding. The resulting system  $\mathbb{Q}$  satisfies the ordered-field axioms stated in Chapter 3.

*Proof.* The only new point needed for the equivalence relation is cancellation of a nonzero integer. If

$$ad = bc, \quad cf = de,$$

then

$$adf = bcf = bde.$$

Cancelling the nonzero factor  $d$  gives  $af = be$ , proving transitivity. Again, well definedness asks whether a formula has the same output class after representatives are changed. For addition, suppose

$$ab' = a'b, \quad cd' = c'd.$$

Then

$$\begin{aligned} (ad + bc)b'd' &= ab'dd' + bb'cd' \\ &= a'bdd' + bb'c'd \\ &= (a'd' + b'c')bd. \end{aligned}$$

Hence

$$(ad + bc, bd) \sim_{\mathbb{Q}} (a'd' + b'c', b'd'),$$

so rational addition is well defined. For multiplication the proposed pairs are  $(ac, bd)$  and  $(a'c', b'd')$ . Their denominators are nonzero, and the hypotheses give

$$(ac)(b'd') = (ab')(cd') = (a'b)(c'd) = (a'c')(bd).$$

This is the defining equality of their rational classes. Thus multiplication too is independent of both representatives.

The formulas for additive and multiplicative inverses verify the inverse laws directly. If

$$[a, b]_{\mathbb{Q}} = [c, d]_{\mathbb{Q}},$$

then multiplying  $ad = bc$  by  $bd$  shows that  $ab$  and  $cd$  have the same sign, because nonzero squares are positive in  $\mathbb{Z}$ . Thus positivity is well defined. It is closed under addition and multiplication, and every nonzero rational is either positive or has positive negative; these facts follow by choosing the signs of numerator and denominator. Hence the order is compatible with the operations. The remaining field identities descend from the integer identities through the displayed fraction formulas. Finally, the definition of  $\iota_{\mathbb{Q}}$  immediately gives injectivity and preservation of arithmetic and order.  $\square$

## 2.5 Countability

An infinite set may still be exhausted by a list indexed by natural numbers. The question is whether every element appears at some finite stage, not whether the listing ever ends. Bijections distinguish a complete listing without repetitions from a list that misses elements.

**Definition 2.5.1** (Countability). Sets  $A, B$  have the same cardinality if a bijection  $A \rightarrow B$  exists. A set is **finite** if it is empty or bijective with  $\{1, \dots, n\}$  for some  $n \in \mathbb{N}$ ; it is **countably infinite** if it is bijective with  $\mathbb{N}$ ; and **countable** if finite or countably infinite.

**Proposition 2.5.2** (Elementary finite-set facts). Every subset of a finite set is finite. If  $A$  and  $B$  are finite, then

$$A \cup B, \quad A \times B$$

are finite. If  $A$  has  $m$  elements and  $B$  has  $n$  elements, then

$$A \times B$$

has  $mn$  elements.

*Proof.* Transport the question along bijections with initial segments of  $\mathbb{N}$ . A subset of

$$\{1, \dots, m\}$$

can be listed in increasing order and therefore has at most  $m$  elements. The union is listed by first listing  $A$  and then retaining the members of  $B$  not already listed. Finally, the ordered pairs

$$(i, j), \quad 1 \leq i \leq m, \quad 1 \leq j \leq n,$$

are enumerated by

$$(i, j) \mapsto n(i-1) + j,$$

which is a bijection onto  $\{1, \dots, mn\}$ . □

**Theorem 2.5.3** (Diagonal enumeration). The set  $\mathbb{N} \times \mathbb{N}$  is countable. If, for every  $n \in \mathbb{N}$ , a listing of a set  $A_n$  is given, then

$$\bigcup_{n=1}^{\infty} A_n$$

is countable.

*Proof.* For  $s \geq 2$ , the  $s$ th diagonal is the finite list of pairs  $(n, k)$  with  $n + k = s$ , ordered by increasing  $n$ . Every  $(n, k) \in \mathbb{N} \times \mathbb{N}$  belongs to the unique diagonal  $n + k$ , so concatenating these finite lists gives a listing of  $\mathbb{N} \times \mathbb{N}$  without omissions or repetitions.

Now let  $a_{n,1}, a_{n,2}, \dots$  be the given listing of  $A_n$ . Replace the pair  $(n, k)$  in the preceding diagonal list by  $a_{n,k}$ . If  $x$  belongs to the displayed union, then  $x \in A_n$  for some  $n$  and hence  $x = a_{n,k}$  for some  $k$ ; therefore this new sequence contains  $x$ . It can contain repetitions, either because a listing repeats an element or because two sets overlap. Read the sequence from left to right and retain an entry exactly when it has not occurred earlier. If infinitely many entries remain, their positions give a bijection from  $\mathbb{N}$  onto the union. If only finitely many remain, they give a finite listing. Thus the union is countable. □

**Corollary 2.5.4.** The rational numbers are countable.

*Proof.* The explicitly constructed integer classes are listed exactly once by

$$0, 1, -1, 2, -2, \dots,$$

so  $\mathbb{Z}$  is countable. Hence  $\mathbb{Z} \times \mathbb{N}$  is countable: compose its first-coordinate listing with the diagonal enumeration from [Theorem 2.5.3](#). The map

$$\mathbb{Z} \times \mathbb{N} \longrightarrow \mathbb{Q}, \quad (p, q) \longmapsto p/q,$$

is surjective because every rational class has a numerator in  $\mathbb{Z}$  and, after changing both signs if necessary, a positive denominator in  $\mathbb{N}$ . Applying this map to a listing of  $\mathbb{Z} \times \mathbb{N}$  gives a listing of  $\mathbb{Q}$ ; retaining the first occurrence of each value makes it a finite listing or a bijection with  $\mathbb{N}$ .  $\square$

**Theorem 2.5.5** (Cantor’s diagonal argument—informal preview). The interval  $(0, 1)$  is uncountable.

*Proof.* This familiar decimal argument is an informal preview: Chapter 3 constructs the real numbers and interprets decimals through shrinking intervals. Chapter 4 revisits that interpretation through limits of finite truncations. The argument is included here to display the diagonal idea before that analytic machinery is available. The formal result is [Theorem 4.1.4](#); later proofs use that result.

Suppose, for contradiction, that  $x_1, x_2, \dots$  listed  $(0, 1)$ . Give each  $x_n$  the decimal expansion that does not end in infinitely many 9s, and let  $d_n$  be its  $n$ th digit. Define  $y = 0.e_1e_2\dots$ , where  $e_n = 1$  if  $d_n \neq 1$ , and  $e_n = 2$  if  $d_n = 1$ . This decimal uses only 1 and 2, so it represents a number strictly between 0 and 1 and does not have an alternative expansion ending in infinitely many 9s. Its  $n$ th digit differs from  $d_n$ , so  $y \neq x_n$  for every  $n$ . This contradicts the assumed listing.  $\square$

## 2.6 Choice

A statement that each set contains an element does not itself specify one function selecting elements from all the sets at once. Finite selections can be made successively; an arbitrary family may have no given selection rule. The following axiom supplies that simultaneous existence.

**Definition 2.6.1** (Axiom of choice). The **axiom of choice** asserts that for every indexed family  $(A_i)_{i \in I}$  of nonempty sets, there is a function

$$c : I \rightarrow \bigcup_{i \in I} A_i$$

such that  $c(i) \in A_i$  for every  $i \in I$ . Such a function is a **choice function**.

A useful mental image is an unlabelled collection of nonempty boxes: a choice function selects one item from every box at once, without requiring an algorithm for the selections.

Choice allows simultaneous selections even when no selection rule has been given. In analysis, one application is the familiar unrestricted form of the countable-union theorem: for a sequence of nonempty countable sets, it permits one to select a listing for each set, after which the proof of [Theorem 2.5.3](#) applies. It also supports the standard existence of vector-space bases and, in more advanced analysis, the extension principles underlying Hahn–Banach and the existence of ultrafilters. Those later results are not prerequisites for this text.

A proof using choice may establish existence without describing an object or giving a procedure for finding one. In some foundational settings, very strong forms of choice also yield counterintuitive decompositions of sets. Accordingly, we distinguish an explicit construction from an existence proof using choice, and state the latter use when it occurs. The standard equivalents of the full axiom are Zorn’s Lemma and the Well-Ordering Theorem. Their equivalence is proved in [Appendix A](#), although the core development through [Chapters 3–9](#) does not require those proofs.

For the core development through Chapters 3–9, the proofs will be arranged to use explicit sequences and finite choices whenever possible. The construction of  $\mathbb{R}$  uses a specified rational representative at a time, and the Bolzano–Weierstrass proof in Chapter 4 selects indices recursively from explicitly nonempty sets. If a later chapter invokes a general basis, a nonconstructive global selection, or a result equivalent to choice, the dependence will be identified at that point.

## Exercises

**Exercise 1.\*** Prove that the set of all finite strings on a fixed finite alphabet is countable.

**Exercise 2.\*** Let  $f : A \rightarrow B$  and let  $E \subseteq A$ .

(a) Prove that

$$f(f^{-1}(f(E))) = f(E).$$

(b) Prove that

$$E \subseteq f^{-1}(f(E)).$$

(c) Prove that

$$f^{-1}(f(E)) = E$$

whenever  $f$  is injective.

(d) Give an explicit example showing that the equality in the preceding part can fail when  $f$  is not injective.

**Exercise 3.** Suppose that

$$(a, b) \sim_{\mathbb{Z}} (a', b') \quad \text{and} \quad (c, d) \sim_{\mathbb{Z}} (c', d').$$

Verify directly that

$$(a + c, b + d) \sim_{\mathbb{Z}} (a' + c', b' + d').$$

Thus the addition of integer classes is well defined.

**Exercise 4.** Prove directly from the equivalence relation defining  $\mathbb{Q}$  that

$$\iota_{\mathbb{Q}} : \mathbb{Z} \longrightarrow \mathbb{Q}, \quad a \longmapsto [a, 1_{\mathbb{Z}}]_{\mathbb{Q}},$$

is injective and preserves order.

**Exercise 5.\*** Use induction to prove

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad (n \in \mathbb{N}).$$

State clearly the base case and the induction step.

**Exercise 6.** Give an explicit bijection between

$$\mathbb{N}$$

and the set of even positive integers.

**Exercise 7.** Negate each statement, retaining the order of its quantifiers:

$$\forall x \in A \exists y \in B : f(x) = y, \quad \exists y \in B \forall x \in A : f(x) = y.$$

When  $A = B = \mathbb{N}$  and  $f(x) = x$ , decide which statements are true. Explain why interchanging the quantifiers changes the assertion.

## Chapter 3

# The Real Number System

### 3.1 Ordered Fields and the Problem to Be Solved

Chapter 2 constructs the arithmetic chain

$$\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q}.$$

The first extension permits subtraction, and the second permits division by a nonzero element. The rational numbers nevertheless have gaps: as Chapter 1 observed, no rational number has square 2. This chapter completes the chain

$$\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R}$$

by constructing a system without those analytic and order-theoretic gaps. We begin by isolating the algebraic and order structure that the final system must possess.

**Definition 3.1.1** (Field). A **field** is a set  $F$  with elements  $0, 1$  ( $0 \neq 1$ ) and operations  $+$  and  $\cdot$  such that, for all  $a, b, c \in F$ , addition and multiplication are associative and commutative, multiplication distributes over addition,  $0 + a = a$ ,  $1a = a$ , every  $a$  has an additive inverse  $-a$ , and every nonzero  $a$  has a multiplicative inverse  $a^{-1}$ .

**Definition 3.1.2** (Ordered field). An **ordered field** is a field  $F$  equipped with a total order  $\leq$ , in the sense of Chapter 2, that is compatible with its field operations: for all  $a, b, c \in F$ ,

$$a \leq b \implies a + c \leq b + c,$$

and

$$0 \leq a \text{ and } 0 \leq b \implies 0 \leq ab.$$

**Proposition 3.1.3** (Positive-cone characterization). Let  $F$  be a field. An ordered-field structure on  $F$  is equivalently specified by a subset

$$P = \{x \in F : x > 0\}$$

such that

$$P + P \subseteq P, \quad PP \subseteq P,$$

where

$$P + P = \{p + q : p, q \in P\}, \quad PP = \{pq : p, q \in P\},$$

and, for every  $x \in F$ , exactly one of

$$x = 0, \quad x \in P, \quad -x \in P$$

holds. Conversely, such a subset defines an ordered-field structure by

$$x \leq y \iff x = y \text{ or } y - x \in P.$$

*Proof.* Suppose first that  $F$  is an ordered field. If  $a, b > 0$ , compatibility with addition gives  $a + b \geq a > 0$ ; equality  $a + b = a$  would force  $b = 0$ . Compatibility with multiplication gives  $ab \geq 0$ . If  $ab = 0$ , multiplication by  $a^{-1}$  gives  $b = 0$ , contradicting  $b > 0$ ; hence  $ab > 0$ . Totality gives the displayed trichotomy.

Conversely, let  $P$  satisfy the stated conditions. The trichotomy for zero shows that  $0 \notin P$ . The displayed relation is reflexive. If both  $x \leq y$  and  $y \leq x$  hold and  $x \neq y$ , then both  $y - x$  and  $x - y$  belong to  $P$ , so closure under addition would put  $0$  in  $P$ , a contradiction; hence the relation is antisymmetric. Closure under addition also gives transitivity, and the trichotomy for  $y - x$  gives totality. Its compatibility with addition is immediate from

$$(y + c) - (x + c) = y - x.$$

Finally, if  $0 \leq a$  and  $0 \leq b$ , then either factor is zero or both belong to  $P$ , in which case  $ab \in P$ . Thus multiplication preserves nonnegativity, as required.  $\square$

**Definition 3.1.4** (Absolute value). In an ordered field, define  $|x| = x$  if  $x \geq 0$  and  $|x| = -x$  if  $x < 0$ .

**Proposition 3.1.5** (Basic order rules). In an ordered field,  $<$  is the strict order associated with a transitive total order. If  $a < b$ , then  $a + c < b + c$  for every  $c$ ; if  $a < b$  and  $0 < c$ , then  $ac < bc$ . Also  $a^2 \geq 0$  for every  $a$ , and

$$|a + b| \leq |a| + |b|, \quad |ab| = |a||b|$$

for all  $a, b$ .

*Proof.* The total-order axioms give totality and transitivity. Addition compatibility gives  $a + c \leq b + c$ , and equality would imply  $a = b$  after cancellation; hence  $a + c < b + c$ . If  $a < b$  and  $0 < c$ , then

$$0 \leq (b - a)c.$$

If this product were zero, multiplication by  $(b - a)^{-1}$  would give  $c = 0$ , a contradiction. Thus  $bc - ac > 0$  and  $ac < bc$ . Totality gives either  $a \geq 0$  or  $-a \geq 0$ , and multiplication compatibility then gives  $a^2 \geq 0$ . The four possible sign combinations for  $a$  and  $b$  prove the product formula for absolute values. Applying the order rules to

$$-|a| \leq a \leq |a|, \quad -|b| \leq b \leq |b|$$

gives

$$-|a| - |b| \leq a + b \leq |a| + |b|,$$

which is the triangle inequality.  $\square$

An upper bound is a ceiling, not necessarily a member of the set. For  $S = (0, 1)$  in  $\mathbb{R}$ , every number at least 1 is an upper bound. The smallest ceiling is 1, yet  $S$  has no largest member: between  $x < 1$  and 1 lies

$$(x + 1)/2.$$

For  $[0, 1]$  the same ceiling is also a member, so it is a maximum. A supremum is the least upper bound; a maximum is a member larger than or equal to all other members. Existence of a supremum does not assert that it is attained.

**Definition 3.1.6** (Upper bounds and completeness). Let  $S \subseteq F$ . An element  $u \in F$  is an **upper bound** of  $S$  if  $s \leq u$  for every  $s \in S$ . A **least upper bound** of  $S$  is an upper bound  $u$  with  $u \leq v$  for every upper bound  $v$  of  $S$ . An ordered field has the **least-upper-bound property** if every nonempty set having an upper bound has a least upper bound.

**Proposition 3.1.7** (Approaching the least upper bound from below). If  $s = \sup S$  for a nonempty set  $S$ , then for every  $\varepsilon > 0$  there is  $x \in S$  with  $s - \varepsilon < x \leq s$ .

*Proof.* Every  $x \in S$  satisfies  $x \leq s$ . If no member exceeded  $s - \varepsilon$ , that smaller number would be an upper bound, contradicting leastness.  $\square$

The estimate lets us choose elements arbitrarily close to the supremum. For

$$S = \{1 - 1/n : n \geq 1\},$$

choose

$$n > 1/\varepsilon$$

to obtain such an element. This choice previews the Archimedean property proved below; it is not part of the proof of the preceding proposition. The supremum is 1, even though no term is 1.

**Definition 3.1.8** (Complete ordered field). An ordered field having the least-upper-bound property is called a **complete ordered field**. This is a characterization, not a construction: it says what the real numbers are like, but does not yet exhibit them.

**Remark 3.1.9** (A first-reading route through this chapter). The detailed proofs in this chapter are important, but a first reading has a more modest goal. Before continuing, understand why  $\mathbb{Q}$  is inadequate for analysis and why the real numbers are characterized structurally as a complete ordered field. “Uniqueness of  $\mathbb{R}$ ” means that any two complete ordered fields are related by the unique order-preserving field isomorphism that fixes their canonical rational copies; it does not mean that two concrete constructions have literally the same underlying set.

Keep the architecture of the uniqueness argument in view:

|                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\begin{aligned} &\text{completeness} \implies \text{Archimedean property} \implies \mathbb{Q} \text{ dense} \\ &\implies \text{rational lower cuts determine points} \implies \text{canonical isomorphism} \end{aligned}$ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The existence argument has a parallel architecture:

|                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\begin{aligned} &\text{rational Cauchy sequences} \implies \text{equivalence classes} \\ &\implies \text{arithmetic and order} \implies \text{completeness} \end{aligned}$ |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Study the well-definedness, quotient, order, and completeness verifications, but do not let difficulty reproducing every implementation detail prevent progress to sequences and limits in Chapter 4. On a first reading, it is enough to understand what the real numbers are structurally, what uniqueness means, why an existence construction is needed, and how the two arguments are organized.



## 3.2 Rational Numbers, Completeness, and Uniqueness

The preceding axioms describe the target. Before constructing a model, we show that they determine its arithmetic and order uniquely. We begin with the copy of the rational numbers contained in every ordered field.

**Definition 3.2.1** (Field homomorphisms and isomorphisms). Let  $F$  and  $G$  be fields. A **field homomorphism** is a map

$$\phi : F \longrightarrow G$$

such that

$$\phi(1_F) = 1_G, \quad \phi(x + y) = \phi(x) + \phi(y), \quad \phi(xy) = \phi(x)\phi(y).$$

A bijective field homomorphism is a **field isomorphism**. Between ordered fields, a homomorphism is **order preserving** if

$$x < y \implies \phi(x) < \phi(y).$$

An order-preserving field isomorphism is called an **ordered-field isomorphism**.

**Theorem 3.2.2** (Canonical rational copy). Every ordered field  $F$  has a unique order-preserving field homomorphism

$$\iota_F : \mathbb{Q} \longrightarrow F.$$

It is injective, and its image is an ordered subfield of  $F$ . Thus we identify  $\mathbb{Q}$  with its image in  $F$ .

*Proof.* For  $n \in \mathbb{N}$ , let  $n_F$  be the sum of  $n$  copies of  $1_F$ . For negative integers, use additive inverses to define  $m_F$ . Induction gives  $n_F > 0$ , hence  $n_F \neq 0$ . Define

$$\iota_F(m/n) = m_F(n_F)^{-1} \quad (m \in \mathbb{Z}, n \in \mathbb{N}).$$

To check well definedness, suppose

$$\frac{m}{n} = \frac{p}{q}.$$

Then  $mq = np$ , so the corresponding integer elements of  $F$  satisfy

$$m_F q_F = n_F p_F.$$

Multiplying by  $(n_F)^{-1}(q_F)^{-1}$  gives

$$m_F(n_F)^{-1} = p_F(q_F)^{-1}.$$

Thus the definition does not depend on the chosen fraction. The field laws then directly show that  $\iota_F$  preserves addition and multiplication.

If  $u > 0$  in an ordered field, then  $u^{-1} > 0$ : it is nonzero, and if  $u^{-1} < 0$ , multiplication of this inequality by  $u > 0$  would give

$$1 = u^{-1}u < 0,$$

contrary to  $1 = 1^2 > 0$ . A positive rational has a representation

$$m/n$$

with  $m, n \in \mathbb{N}$ , so  $m_F > 0$  and  $(n_F)^{-1} > 0$ ; hence its image is positive. Therefore  $\iota_F$  is order preserving and thus injective. Any field homomorphism fixes 1, then every integer, and then every quotient of integers; uniqueness follows.  $\square$

**Definition 3.2.3** (Archimedean field). A field  $F$  is **Archimedean** if it is an ordered field and, for every  $x \in F$ , there exists  $n \in \mathbb{N}$  such that

$$x < n.$$

Equivalently,

$$F \text{ is Archimedean} \iff \forall \varepsilon > 0 \exists n \in \mathbb{N} : \frac{1}{n} < \varepsilon.$$

Indeed, the large-scale condition applied to

$$1/\varepsilon$$

gives

$$n > 1/\varepsilon,$$

hence

$$1/n < \varepsilon.$$

Conversely, assume the small-scale condition. If  $x < 1$ , take  $n = 1$ . If  $x \geq 1$ , choose  $n \in \mathbb{N}$  with

$$\frac{1}{n} < \frac{1}{x+1}.$$

Inversion reverses this inequality and gives  $n > x+1 > x$ . Thus natural numbers exceed every fixed size exactly when their reciprocals become arbitrarily small.

**Theorem 3.2.4** (Integer parts and rational density in an Archimedean field). Let  $F$  be an Archimedean ordered field, with its canonical copy of  $\mathbb{Q}$ . Then:

(a) for every  $x \in F$ , there is  $k \in \mathbb{Z}$  such that

$$k \leq x < k+1;$$

(b) if  $x < y$  in  $F$ , there is  $q \in \mathbb{Q}$  such that  $x < q < y$ .

*Proof.* For the first assertion, choose  $M \in \mathbb{N}$  with  $M > -x$ . The set

$$E = \{n \in \mathbb{N} : n > x + M\}$$

is nonempty, so it has a least member  $n$ . If  $n > 1$ , minimality gives  $n-1 \leq x+M < n$ ; if  $n = 1$ , the left inequality follows from  $x+M > 0$ . Thus  $k = n - M - 1$  has the required property.

For  $x < y$ , choose  $N \in \mathbb{N}$  with  $N(y-x) > 1$ , and apply the first assertion to  $Nx$ . If  $k \leq Nx < k+1$ , then

$$Nx < k+1 \leq Nx+1 < Ny.$$

Dividing by  $N > 0$  gives  $x < (k+1)/N < y$ . □

**Theorem 3.2.5** (Every complete ordered field is Archimedean). Every complete ordered field is Archimedean. Consequently, for every  $\varepsilon > 0$ , there is  $N \in \mathbb{N}$  with

$$1/N < \varepsilon.$$

*Proof.* Identify  $\mathbb{Q}$  with its canonical copy. If  $\mathbb{N}$  had an upper bound, let  $s = \sup \mathbb{N}$ . Since  $s - 1$  is not an upper bound, some  $n \in \mathbb{N}$  satisfies

$$s - 1 < n.$$

Then  $s < n + 1$ , contradicting that  $s$  is an upper bound. Thus  $\mathbb{N}$  is unbounded above, which is precisely the Archimedean property. The reciprocal statement follows from the equivalent small-scale form in [Definition 3.2.3](#).  $\square$

**Theorem 3.2.6** (Rational density). Let  $F$  be a complete ordered field and let  $x < y$  in  $F$ . Then there is  $q \in \mathbb{Q}$  such that

$$x < q < y.$$

*Proof.* By [Theorem 3.2.5](#),  $F$  is Archimedean, so the second assertion of [Theorem 3.2.4](#) applies.  $\square$

**Lemma 3.2.7** (Rational cuts determine points). If  $F$  is a complete ordered field and  $x, y \in F$  satisfy

$$\{q \in \mathbb{Q} : q < x\} = \{q \in \mathbb{Q} : q < y\},$$

then  $x = y$ .

*Proof.* If  $x < y$ , rational density gives a rational strictly between them, which belongs to exactly one of the two sets. The case  $y < x$  is symmetric.  $\square$

**Theorem 3.2.8** (Uniqueness of complete ordered fields). If  $F$  and  $G$  are complete ordered fields, there is a unique order-preserving field isomorphism

$$\Phi : F \longrightarrow G$$

which identifies their canonical rational copies.

*Proof.* Rational density means that rational numbers below a point determine that point. Match these lower cuts in the two fields, then use rational approximations to show that this matching preserves arithmetic. Any successful construction of a complete ordered field must therefore give this same system up to the unique canonical isomorphism.

Identify both rational copies with  $\mathbb{Q}$ . For  $x \in F$ , let

$$L_F(x) = \{q \in \mathbb{Q} : q < x\}.$$

The Archimedean property and rational density make this set nonempty and bounded above in  $G$ . Define

$$\Phi(x) = \sup_G L_F(x).$$

For every rational  $q$ ,

$$q < \Phi(x) \iff q < x. \tag{*}$$

Indeed, if  $q < x$ , choose  $r \in \mathbb{Q}$  with  $q < r < x$ , so  $q < r \leq \Phi(x)$ . Conversely, if  $x \leq q$ , then  $q$  is an upper bound of  $L_F(x)$ .

The equivalence (\*) shows that  $\Phi$  is strictly order preserving and fixes  $\mathbb{Q}$ . It also proves additivity: for a rational  $q$ ,

$$q < x + y$$

holds exactly when there are rationals  $p < x$  and  $r < y$  with

$$q = p + r.$$

For the nontrivial implication, choose  $p \in \mathbb{Q}$  between  $q - y$  and  $x$  and put  $r = q - p$ . Apply (\*) in both fields and then use the preceding lemma to conclude

$$\Phi(x + y) = \Phi(x) + \Phi(y).$$

For positive  $x, y$ , we next compare the lower cuts of the products. If

$$q < 0,$$

then automatically  $q < xy$ , and the same is true of

$$\Phi(x)\Phi(y).$$

Now let  $q \in \mathbb{Q}$  satisfy

$$q \geq 0.$$

We claim that

$$q < xy$$

if and only if there are positive rationals  $p, r$  such that

$$0 < p < x, \quad 0 < r < y, \quad q < pr. \quad (**)$$

The reverse implication follows from the order rules. Conversely, if

$$q < xy,$$

then

$$\frac{q}{y} < x.$$

Rational density gives a positive rational  $p$  with

$$\frac{q}{y} < p < x.$$

Because

$$p > q/y,$$

one has

$$q/p < y;$$

rational density again gives a positive rational  $r$  with

$$\frac{q}{p} < r < y.$$

Thus  $q < pr$ , proving (\*\*).

The equivalence (\*) transfers the inequalities

$$p < x, \quad r < y$$

exactly to

$$p < \Phi(x), \quad r < \Phi(y).$$

Applying (\*\*) in  $F$  and in  $G$  therefore shows that  $xy$  and  $\Phi(x)\Phi(y)$  have the same rational lower cut. By Lemma 3.2.7,

$$\Phi(xy) = \Phi(x)\Phi(y)$$

when  $x, y > 0$ . Additivity gives

$$\Phi(-x) = -\Phi(x).$$

If one factor is zero, multiplicativity is immediate. If exactly one factor is negative, apply the positive case to its negative and use this identity; if both are negative, apply it twice. Hence multiplicativity holds for all

$$x, y \in F.$$

Thus  $\Phi$  is an order-preserving field homomorphism.

Define the reverse map explicitly by

$$\Psi(y) = \sup_F \{q \in \mathbb{Q} : q < y \text{ in } G\}, \quad y \in G.$$

The rational set is nonempty and bounded by the Archimedean property, so completeness of  $F$  supplies its supremum. The proof of (\*) applied to this definition gives  $q < \Psi(y)$  if and only if  $q < y$ . Consequently for every rational  $q$  and  $x \in F$ ,

$$q < \Psi(\Phi(x)) \iff q < \Phi(x) \iff q < x.$$

The rational-cut lemma implies  $\Psi\Phi(x) = x$ . For  $y \in G$ ,

$$q < \Phi(\Psi(y)) \iff q < \Psi(y) \iff q < y,$$

so the same lemma gives  $\Phi\Psi(y) = y$ . Both composites are identities, and hence the already proved field homomorphism  $\Phi$  is an isomorphism. Finally, any order-preserving field isomorphism fixing  $\mathbb{Q}$  has property (\*), because it preserves rational lower cuts; therefore it equals  $\Phi$ .  $\square$

**Definition 3.2.9** (The real number system). The **real number system**, denoted by  $\mathbb{R}$ , is a complete ordered field. The uniqueness theorem makes this definition unambiguous up to the unique order-preserving field isomorphism fixing  $\mathbb{Q}$ . The construction below proves that such a field exists.

### 3.3 Cauchy Sequences of Rational Numbers

The preceding discussion gives the intended abstract meaning of  $\mathbb{R}$  and proves that such a field, if it exists, is unique. We now supply existence by constructing a concrete complete ordered field from rational Cauchy sequences. The uniqueness theorem will then identify this model with the real-number system just characterized.

We use the usual ordered-field properties of  $\mathbb{Q}$ , which Chapter 2 establishes from its construction as equivalence classes of integer pairs  $(p, q)$  with  $q \neq 0$ . No completeness property of  $\mathbb{Q}$  will be used.

### The construction in one picture

A real number will be an equivalence class of rational Cauchy approximation processes. Different processes represent the same number when their difference tends to zero. We can test whether approximations settle down without already knowing a limit. The Cauchy condition in  $\mathbb{Q}$  does just that. For example,

$$(1 + 1/n), \quad (1 - 1/n)$$

should represent the same number,

$$(1/n)$$

should represent zero, and the constant sequences (1) and (2) must remain different. Eventual equality would be much too restrictive: the first two sequences never agree, although their difference

$$2/n$$

becomes smaller than every positive rational tolerance.

The work has three layers. First identify equivalent processes and do arithmetic on their classes. Next build order, keeping nonzero classes separated from zero. Finally show that rational approximations converge to their class and that every real Cauchy sequence then has a limit.

**Definition 3.3.1** (Cauchy sequence in  $\mathbb{Q}$ ). A sequence  $(q_n)_{n \in \mathbb{N}}$  in  $\mathbb{Q}$  is **Cauchy** if, for every rational  $\varepsilon > 0$ , there is  $N \in \mathbb{N}$  such that  $m, n \geq N$  implies  $|q_m - q_n| < \varepsilon$ .

**Definition 3.3.2** (Null equivalence). For rational Cauchy sequences  $(q_n)$  and  $(r_n)$ , write  $(q_n) \sim (r_n)$  if, for every  $\varepsilon \in \mathbb{Q}$  with  $\varepsilon > 0$ , there is  $N$  such that  $n \geq N$  implies  $|q_n - r_n| < \varepsilon$ .

**Lemma 3.3.3.** The relation  $\sim$  is an equivalence relation on the set of rational Cauchy sequences.

*Proof.* Reflexivity and symmetry are immediate. If  $q \sim r$  and  $r \sim s$ , then for a given  $\varepsilon > 0$ , both

$$|q_n - r_n| < \varepsilon/2, \quad |r_n - s_n| < \varepsilon/2$$

eventually hold. The triangle inequality gives  $|q_n - s_n| < \varepsilon$ , proving transitivity.  $\square$

**Lemma 3.3.4** (Elementary Cauchy facts). Let  $(q_n)$  and  $(r_n)$  be Cauchy sequences in  $\mathbb{Q}$ .

- (a) Each sequence is bounded: there is  $M \in \mathbb{Q}$  with  $|q_n| \leq M$  for all  $n$ .
- (b) The sequences  $(q_n + r_n)$  and  $(q_n r_n)$  are Cauchy.
- (c) If  $(q_n) \sim (q'_n)$  and  $(r_n) \sim (r'_n)$ , then

$$(q_n + r_n) \sim (q'_n + r'_n) \quad \text{and} \quad (q_n r_n) \sim (q'_n r'_n).$$

*Proof.* Choose  $N$  such that  $m, n \geq N$  gives  $|q_m - q_n| < 1$ . Then  $|q_n| \leq |q_N| + 1$  for  $n \geq N$ ; enlarge this rational bound to cover the finitely many earlier terms. For sums, make each difference less than

$$\varepsilon/2$$

and apply the triangle inequality. For products, first bound the potentially large factors, then make the differences small. Take a common rational bound  $M \geq 1$  for both sequences and use

$$|q_m r_m - q_n r_n| \leq |q_m| |r_m - r_n| + |r_n| |q_m - q_n| \leq M(|r_m - r_n| + |q_m - q_n|).$$

Given  $\varepsilon > 0$ , choose a stage after which both differences are less than

$$\varepsilon/(2M).$$

The displayed bound is then less than  $\varepsilon$ , proving Cauchyness. For the final assertion, the triangle inequality gives the sum statement. Choose a common rational bound  $M \geq 1$  for the four sequences. The identity

$$q_n r_n - q'_n r'_n = q_n(r_n - r'_n) + r'_n(q_n - q'_n)$$

then gives

$$|q_n r_n - q'_n r'_n| \leq M|r_n - r'_n| + M|q_n - q'_n|,$$

which is less than  $\varepsilon$  once both null differences are less than

$$\varepsilon/(2M).$$

Thus the product class does not depend on the representatives.  $\square$

**Lemma 3.3.5** (Finite changes). Changing finitely many entries preserves Cauchyness and null-equivalence, and therefore does not change a represented class.

*Proof.* For each tolerance choose the original stage, then increase it beyond every changed index. All inequalities in the definitions are then unchanged.  $\square$

The next lemma solves the denominator problem. If a Cauchy process does not represent zero, one sufficiently late term stays a definite distance from zero; all later terms must stay near it, on the same side of zero.

**Lemma 3.3.6** (Eventual separation). Let  $(q_n)$  be a rational Cauchy sequence that is not null-equivalent to the zero sequence. Then there are  $\delta > 0$  in  $\mathbb{Q}$  and  $N \in \mathbb{N}$  such that either  $q_n \geq \delta$  for every  $n \geq N$ , or  $q_n \leq -\delta$  for every  $n \geq N$ .

*Proof.* Null-equivalence to zero means

$$\forall \eta \in \mathbb{Q}_{>0} \exists N \forall n \geq N : |q_n| < \eta.$$

Its negation is  $\exists \eta \in \mathbb{Q}_{>0} \forall N \exists n \geq N$  with  $|q_n| \geq \eta$ . Thus there is  $\eta > 0$  for which  $|q_n| \geq \eta$  for infinitely many  $n$ . Choose  $N$  so that

$$|q_m - q_n| < \eta/2$$

whenever  $m, n \geq N$ , and choose  $k \geq N$  with  $|q_k| \geq \eta$ . If  $q_k \geq \eta$ , then

$$q_n > \eta/2$$

for every  $n \geq N$ . If  $q_k \leq -\eta$ , then

$$q_n < -\eta/2$$

for every  $n \geq N$ . Take

$$\delta = \eta/2.$$

$\square$

**Construction 3.3.7** (The set of real numbers). Define  $\mathbb{R}$  to be the set of equivalence classes  $[(q_n)]$  under  $\sim$ . Define

$$[(q_n)] + [(r_n)] = [(q_n + r_n)], \quad [(q_n)] [(r_n)] = [(q_n r_n)], \quad -[(q_n)] = [(-q_n)].$$

For  $a \in \mathbb{Q}$ , define  $\iota(a) = [(a, a, a, \dots)]$ .

**Proposition 3.3.8** (Algebra of the construction). The operations in [construction 3.3.7](#) are well defined. With  $0 = \iota(0)$  and  $1 = \iota(1)$ , they make  $\mathbb{R}$  a field, and  $\iota : \mathbb{Q} \rightarrow \mathbb{R}$  is an injective field homomorphism.

*Proof.* The last part of [Lemma 3.3.4](#) proves well-definedness of addition and multiplication; it also proves that the displayed sequences are Cauchy. All associative, commutative, distributive, and identity laws hold term by term for rational sequences and therefore for their classes. Additive inverses are termwise. If  $x = [(q_n)] \neq 0$ , [Lemma 3.3.6](#) gives  $\delta > 0$  and  $N$  such that  $|q_n| \geq \delta$  for every  $n \geq N$ . Define

$$p_n = \begin{cases} 1, & n < N, \\ q_n^{-1}, & n \geq N. \end{cases}$$

The estimate

$$|p_m - p_n| = \frac{|q_m - q_n|}{|q_m q_n|} \leq \frac{1}{\delta^2} |q_m - q_n|$$

for  $m, n \geq N$  proves Cauchyness: given  $\varepsilon > 0$ , use the Cauchy tolerance  $\varepsilon \delta^2$  for  $(q_n)$  and a stage at least  $N$ . The finitely many initial choices are harmless by the finite-changes lemma. Since  $q_n p_n = 1$  for every  $n \geq N$ , its product class is 1; hence  $[(p_n)]$  is the multiplicative inverse of  $x$ . Finally, constant-sequence operations show that  $\iota$  is a homomorphism; if  $\iota(a) = \iota(b)$ , then the constant difference  $a - b$  is smaller in absolute value than every positive rational, which is possible only when  $a = b$ .  $\square$

### 3.4 Order and Completeness

Eventual positivity of the entries alone cannot define a positive class:

$$1/n > 0$$

at every stage, but

$$(1/n)$$

represents zero. We need a positive margin that survives arbitrarily accurate changes of representative.

**Definition 3.4.1** (Order on  $\mathbb{R}$ ). For  $x = [(q_n)] \in \mathbb{R}$ , write  $x > 0$  if there are a rational  $\delta > 0$  and  $N \in \mathbb{N}$  such that  $q_n > \delta$  for every  $n \geq N$ . Define  $x < y$  to mean  $y - x > 0$ .

**Lemma 3.4.2.** The preceding definition is independent of the chosen representative. It makes  $\mathbb{R}$  an ordered field, and  $\iota$  preserves order.

*Proof.* If  $q_n > \delta$  eventually and  $q \sim r$ , then

$$|q_n - r_n| < \delta/2$$



eventually, so

$$r_n > \delta/2$$

eventually. Thus positivity is well defined. Positive eventual lower bounds are closed under addition and multiplication. For a nonzero class, [Lemma 3.3.6](#) gives exactly one eventual sign. Therefore [Proposition 3.1.3](#) makes the resulting relation a compatible total order, and [Proposition 3.1.5](#) supplies its basic rules. Constant sequences visibly preserve rational order.  $\square$

**Theorem 3.4.3** (The constructed field is Archimedean). For every  $x \in \mathbb{R}$ , there is  $N \in \mathbb{N}$  with  $x < N$ . Consequently, for every  $\varepsilon > 0$ , there is  $N \in \mathbb{N}$  with

$$1/N < \varepsilon.$$

*Proof.* Let  $x = [(q_n)]$ . By [Lemma 3.3.4](#), the rational sequence  $(q_n)$  has a rational upper bound  $M$ . Choose a natural number  $N > M$ , using the elementary Chapter 2 fact that natural numbers are unbounded above in  $\mathbb{Q}$ . Then  $N - q_n \geq N - M > 0$  for all  $n$ , so the definition of positivity, with margin

$$(N - M)/2,$$

gives  $x < N$ . Thus the constructed field is Archimedean in the sense of [Definition 3.2.3](#). The reciprocal consequence follows from the equivalent small-scale form.  $\square$

**Lemma 3.4.4** (Integer-part lemma). For every  $x \in \mathbb{R}$ , there is an integer  $k \in \mathbb{Z}$  such that

$$k \leq x < k + 1.$$

*Proof.* The constructed field is Archimedean by [Theorem 3.4.3](#); apply the first assertion of [Theorem 3.2.4](#).  $\square$

**Theorem 3.4.5** (Density of the rational numbers). If  $x, y \in \mathbb{R}$  and  $x < y$ , then there is  $q \in \mathbb{Q}$  such that

$$x < \iota(q) < y.$$

*Proof.* Apply the second assertion of [Theorem 3.2.4](#), using [Theorem 3.4.3](#).  $\square$

**Definition 3.4.6** (Convergence of a real sequence). Let  $(x_n)$  be a sequence in  $\mathbb{R}$ , and let  $x \in \mathbb{R}$ . We say that  $(x_n)$  **converges to**  $x$ , and write

$$\lim_{n \rightarrow \infty} x_n = x,$$

if for every  $\varepsilon > 0$  there is  $N \in \mathbb{N}$  such that

$$n \geq N \implies |x_n - x| < \varepsilon.$$

**Definition 3.4.7** (Cauchy sequence in  $\mathbb{R}$ ). A real sequence  $(x_n)$  is **Cauchy** if, for every  $\varepsilon > 0$ , there is  $N \in \mathbb{N}$  such that  $m, n \geq N$  implies  $|x_m - x_n| < \varepsilon$ .

**Lemma 3.4.8** (Bounds for classes). If  $a \leq u_n \leq b$  eventually, with rational  $a, b$ , then

$$\iota(a) \leq [(u_n)] \leq \iota(b).$$

*Proof.* If  $[(u_n)] > \iota(b)$ , positivity would require  $u_n - b > \delta > 0$  eventually, contradicting the upper bound. The lower bound is identical.  $\square$

Strict bounds need not remain strict:

$$u_n = 1 - 1/(n+1) < 1$$

represents 1. To prove a strict conclusion, reserve a margin in the approximation.

**Lemma 3.4.9** (Rational approximation from a representative). Let  $x = [(q_n)] \in \mathbb{R}$ . Then

$$\lim_{n \rightarrow \infty} \iota(q_n) = x.$$

Consequently, for every  $\varepsilon > 0$  there is a rational  $q$  such that

$$|\iota(q) - x| < \varepsilon.$$

*Proof.* Let  $\varepsilon > 0$ . By [Theorem 3.4.5](#), choose  $e \in \mathbb{Q}$  with

$$0 < \iota(e) < \varepsilon.$$

Cauchyness of  $(q_n)$  supplies  $N$  such that

$$m, n \geq N \implies |q_m - q_n| < e/2.$$

For a fixed  $n \geq N$ , the real number  $\iota(q_n) - x$  is represented by the sequence  $(q_n - q_k)_{k \in \mathbb{N}}$ , whose tail has absolute value less than

$$e/2.$$

By [Lemma 3.4.8](#),

$$|\iota(q_n) - x| \leq \iota(e/2) < \iota(e) < \varepsilon.$$

This proves the convergence. Taking one sufficiently large  $n$  proves the final assertion.  $\square$

**Four levels of notation in the completeness argument.** The construction begins with rational sequences  $(q_k)$ , not sequences of already available real numbers. A constructed real is a class  $[(q_k)]$  of equivalent rational Cauchy sequences. A rational  $r$  is embedded as the class of the constant sequence  $(r, r, \dots)$ . Finally, the sequence whose completeness is now being proved is a sequence of these classes.

$$\begin{array}{ll} q_k \in \mathbb{Q} & \longrightarrow (q_k)_{k \geq 1} \text{ rational Cauchy sequence} \\ & (q_k) \longmapsto [(q_k)] \text{ one constructed real} \\ X_1, X_2, \dots & \text{a sequence of constructed reals.} \end{array}$$

To approximate its  $n$ th term  $X_n$ , choose one embedded rational  $r_n$  close to that class. The diagonal choice  $(r_n)$  is a new rational sequence, not a choice of one fixed representative sequence for all  $X_n$ . The proof checks it is Cauchy, takes its class as the candidate limit, and compares that class back to  $X_n$ . This avoids assuming the very completeness being established.

**Theorem 3.4.10** (Cauchy completeness of the construction). Every Cauchy sequence in  $\mathbb{R}$  converges to an element of  $\mathbb{R}$ .

*Proof.* For each  $j$ , choose a rational  $r_j$  within

$$1/j$$

of the  $j$ th real approximation. We show that  $(r_j)$  is rational Cauchy and then compare its class with  $x_j$ :

$$x_j \approx \iota(r_j) \longrightarrow [(r_j)].$$

No summable error is needed, only an error tending to zero.

By [Lemma 3.4.9](#), choose  $r_j \in \mathbb{Q}$  with

$$|x_j - \iota(r_j)| < 1/j.$$

This can be explicit: fix a listing of  $\mathbb{Q}$  and take its first entry satisfying the inequality for each  $j$ . Given rational  $e > 0$ , choose  $J$  so that

$$1/J < e/3, \quad |x_j - x_k| < \iota(e/3)$$

for  $j, k \geq J$ . Then

$$|\iota(r_j) - \iota(r_k)| \leq |\iota(r_j) - x_j| + |x_j - x_k| + |x_k - \iota(r_k)| < \iota(e).$$

Order preservation gives  $|r_j - r_k| < e$ , so  $(r_j)$  is rational Cauchy. Set  $x = [(r_j)]$ . The representative-approximation lemma gives  $\iota(r_j) \rightarrow x$ . Given real  $\varepsilon > 0$ , choose  $J$  so that, for  $j \geq J$ , both

$$1/j < \varepsilon/2, \quad |\iota(r_j) - x| < \varepsilon/2.$$

Hence

$$|x_j - x| \leq |x_j - \iota(r_j)| + |\iota(r_j) - x| < \varepsilon.$$

Thus  $x_j \rightarrow x$ , which is exactly what completing  $\mathbb{Q}$  was meant to achieve.  $\square$

**Theorem 3.4.11** (Completeness of the constructed field). The ordered field  $\mathbb{R}$  of [construction 3.3.7](#) has the least-upper-bound property.

*Proof.* We construct nested intervals whose left endpoints are not upper bounds and whose right endpoints are upper bounds. Bisection makes their lengths tend to zero; Cauchy completeness gives a common limit, which the endpoint properties identify as the least upper bound.

Let  $S \subseteq \mathbb{R}$  be nonempty and bounded above. Choose  $s_0 \in S$  and an upper bound  $u_0$ . If  $s_0$  is an upper bound, then it is a maximum of  $S$  and hence its least upper bound. Otherwise  $s_0$  is not an upper bound. Since every element of  $S$  is at most  $u_0$ , one has  $s_0 < u_0$ . Bisect  $[s_0, u_0]$  recursively. At stage  $n$  set

$$m_n = \frac{s_n + u_n}{2}, \quad (s_{n+1}, u_{n+1}) = \begin{cases} (s_n, m_n), & m_n \text{ is an upper bound of } S, \\ (m_n, u_n), & m_n \text{ is not an upper bound of } S. \end{cases}$$

The construction preserves the two roles:  $s_n$  is not an upper bound, whereas  $u_n$  is. The interval width is

$$u_n - s_n = 2^{-n}(u_0 - s_0).$$

The sequence  $(s_n)$  is Cauchy, because for  $m \geq n$ ,

$$0 \leq s_m - s_n \leq u_n - s_n = 2^{-n}(u_0 - s_0).$$

The elementary induction inequality  $2^n \geq n + 1$  gives

$$0 \leq 2^{-n} \leq \frac{1}{n+1} \longrightarrow 0$$

by [Theorem 3.4.3](#); hence the right side can be made arbitrarily small. By [Theorem 3.4.10](#), let  $s_n \rightarrow a$ . The upper endpoints have the same limit, since

$$|u_n - a| \leq |u_n - s_n| + |s_n - a| \longrightarrow 0.$$

Finally, for fixed  $n$ , the inequalities

$$s_n \leq s_m \leq u_m \leq u_n$$

for  $m \geq n$  pass to the limit: if, for example,  $s_n > a$ , then  $s_m$  cannot lie within

$$(s_n - a)/2$$

of  $a$  for any  $m \geq n$ . Hence  $s_n \leq a \leq u_n$ .

If  $t \in S$  and  $t > a$ , choose  $n$  with  $u_n - s_n < t - a$ ; then  $t > u_n$ , contradicting that  $u_n$  is an upper bound. Thus  $a$  is an upper bound. If  $v < a$ , choose  $n$  with  $u_n - s_n < a - v$ ; then  $v < s_n$ . Since  $s_n$  is not an upper bound, neither is  $v$ . Hence  $a$  is the least upper bound.  $\square$

The construction has now supplied the existence promised in [Definition 3.2.9](#). Therefore the ordered field constructed in [construction 3.3.7](#) is, up to the unique order-preserving field isomorphism fixing  $\mathbb{Q}$ , the real number system. The remaining results record useful equivalent forms and consequences of its completeness.

Other standard descriptions—Dedekind cuts, suitable decimal expansions, and nested-interval models—are not constructed in full here. The uniqueness theorem does not by itself prove that such descriptions are equivalent to this model: for each one, it remains necessary to prove the field, order, and completeness properties. Once that has been done, however, the uniqueness theorem supplies its canonical order-preserving field isomorphism with the Cauchy-sequence model and hence with  $\mathbb{R}$ .

**Proposition 3.4.12** (Positive square roots). For every  $a > 0$  there is a unique  $s > 0$  with  $s^2 = a$ , denoted  $\sqrt{a}$ .

*Proof.* Let  $S = \{x \geq 0 : x^2 < a\}$ . It contains a positive element, for example

$$\min\{1, a\}/2,$$

and is bounded above by  $a + 1$ . Let  $s = \sup S > 0$ . If  $s^2 < a$ , we seek a slightly larger member of  $S$ . The error is  $(s + h)^2 - s^2 = 2sh + h^2$ . First require  $h < 1$ , so this error is less than  $(2s + 1)h$ ; then require the latter to be smaller than the gap  $a - s^2$ . Accordingly choose

$$0 < h < \min\{1, (a - s^2)/(2s + 1)\}.$$

Then  $(s + h)^2 < s^2 + (2s + 1)h < a$ , contradicting that  $s$  is an upper bound. If  $s^2 > a$ , we instead seek a smaller upper bound. Since  $s^2 - (s - h)^2 = 2sh - h^2 \leq 2sh$ , it suffices to keep  $2sh$  smaller than  $s^2 - a$ , while  $h < s$  keeps  $s - h$  positive. Choose

$$0 < h < \min\{s, (s^2 - a)/(2s)\}.$$

Then  $(s - h)^2 > s^2 - 2sh > a$ . Any  $x \in S$  must satisfy  $x < s - h$ , since for nonnegative  $x \geq s - h$  we would have  $x^2 \geq (s - h)^2 > a$ . Thus  $s - h$  is a smaller upper bound, again a contradiction. Thus  $s^2 = a$ . For  $0 < u < v$ ,  $v^2 - u^2 = (v - u)(v + u) > 0$ , which proves uniqueness.  $\square$

In particular  $\sqrt{2}$  exists and is irrational by the initial parity argument. The same perturbation estimates, with rational  $s, h$ , show directly that  $\{q \in \mathbb{Q} : q > 0, q^2 < 2\}$  has no rational least upper bound.

**Proposition 3.4.13** (Density of the irrational numbers). Every nonempty open interval contains an irrational number.

*Proof.* Given  $a < b$ , rational density supplies  $q \in \mathbb{Q}$  with  $a - \sqrt{2} < q < b - \sqrt{2}$ . Then  $q + \sqrt{2} \in (a, b)$  is irrational, since otherwise subtraction of  $q$  would make  $\sqrt{2}$  rational.  $\square$

**Theorem 3.4.14** (Greatest-lower-bound property). Every nonempty subset of  $\mathbb{R}$  that is bounded below has a greatest lower bound.

*Proof.* Let  $S \subseteq \mathbb{R}$  be nonempty and bounded below. Define

$$-S = \{-s : s \in S\}.$$

Then  $-S$  is nonempty and bounded above. By [Theorem 3.4.11](#), let  $u = \sup(-S)$ , and put  $\ell = -u$ . For every  $s \in S$ , one has  $-s \leq u$ , hence  $\ell \leq s$ ; thus  $\ell$  is a lower bound. If  $v$  is any lower bound of  $S$ , then  $-s \leq -v$  for every  $s \in S$ , so  $-v$  is an upper bound of  $-S$ . Therefore  $u \leq -v$ , or  $v \leq \ell$ . Hence  $\ell$  is the greatest lower bound.  $\square$

**Theorem 3.4.15** (Nested-interval property). Let

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$$

be nested closed intervals in  $\mathbb{R}$ , and suppose

$$\lim_{n \rightarrow \infty} (b_n - a_n) = 0.$$

Then there is exactly one real number belonging to every interval.

*Proof.* The sequence  $(a_n)$  is increasing and bounded above by  $b_1$ , so its range has a least upper bound  $c$ . For every fixed  $n$ , nesting gives  $a_m \leq a_n \leq b_n$  when  $m < n$ , and  $a_m \leq b_m \leq b_n$  when  $m \geq n$ . Thus  $b_n$  is an upper bound of the whole range, so  $c \leq b_n$ . Also  $a_n \leq c$  by definition of upper bound. Hence  $c \in [a_n, b_n]$  for every  $n$ .

If  $d$  belongs to every interval, then

$$|c - d| \leq b_n - a_n \quad (n \in \mathbb{N}).$$

If  $|c - d| > 0$ , choose  $n$  so that  $b_n - a_n < |c - d|$ , contradicting this inequality. Thus  $c = d$ , and the common point is unique.  $\square$

## Decimal expansions of real numbers

The real number  $\sqrt{2}$  can be described by successively finer rational information. Squaring the positive endpoints verifies

$$1.4 < \sqrt{2} < 1.5, \quad 1.41 < \sqrt{2} < 1.42, \quad 1.414 < \sqrt{2} < 1.415.$$

Continuing this process gives the familiar notation  $\sqrt{2} = 1.41421356237\dots$ . The digits encode rational approximations, not a new kind of arithmetic operation.

For  $x \geq 0$ , [Lemma 3.4.4](#) gives a unique integer  $m \geq 0$  with  $m \leq x < m + 1$ . Divide this interval into ten equal half-open pieces, choose the one containing  $x$ , and repeat. If its successive digits are  $a_j \in \{0, \dots, 9\}$ , the finite truncation

$$t_n = m + \frac{a_1}{10} + \dots + \frac{a_n}{10^n} \quad \text{satisfies} \quad t_n \leq x < t_n + 10^{-n}.$$

Thus the schematic expression

$$x = m + a_1/10 + a_2/10^2 + \dots$$

means that these finite truncations approximate  $x$  with errors at most  $10^{-n}$ ; it invokes no infinite-series theory. Conversely, any prescribed digits give nested closed intervals  $[t_n, t_n + 10^{-n}]$ . Their lengths shrink to zero, so [Theorem 3.4.15](#) gives exactly one represented real number. Negative numbers are described by a minus sign and the digits of their absolute value. Chapter 4 develops convergence systematically, and Chapter 7 later interprets the same notation as a series.

**Two names for an endpoint.** For every finite  $n$ ,

$$0.\underbrace{99\dots 9}_{n \text{ digits}} = 1 - 10^{-n}.$$

This is a finite identity (also proved by induction). The corresponding closed intervals are  $[1 - 10^{-n}, 1]$ , whose unique common point is 1. Hence

$$1.000\dots = 0.999\dots = 0.\bar{9}, \quad 0.25000\dots = 0.24999\dots$$

The half-open selection rule above chooses the terminating version at an endpoint; the closed-interval interpretation admits both descriptions. These are the only ambiguities. Indeed, at the first differing place the two decimal subintervals have disjoint interiors. They can share the same represented point only when they are adjacent and that point is their common endpoint. Repeated subdivision then forces all remaining digits on the lower side to be 9 and those on the upper side to be 0. The same endpoint argument covers a change in the integer part.

Decimals can themselves serve as a construction of  $\mathbb{R}$ , after identifying these two representations of each terminating value. Order and completeness are comparatively transparent through nested decimal intervals. Arithmetic is less elegant: one must handle carries and show that operations respect the representation ambiguity. These notes do not develop that alternative construction.

**Remark 3.4.16** (Why the Cauchy construction?). The construction  $\mathbb{Q} \hookrightarrow \mathbb{R}$  is a prototype of completing a metric space  $X$  to a space  $\widehat{X}$  in which its Cauchy sequences converge. This is motivation, not a prerequisite or a claim that general completion theory has already been proved. Cauchy sequences depend on a metric, or more generally a uniform structure. A topology determines convergence but does not specify a canonical Cauchy notion. For example, on  $(0, \infty)$  the metrics

$$d(x, y) = |x - y|, \quad \rho(x, y) = |1/x - 1/y|$$

have the same open sets: reciprocal coordinates and their inverse are continuous. Yet

$$x_n = 1/n$$

is Cauchy for  $d$  and is not Cauchy for  $\rho$ , since  $\rho(x_n, x_{n+1}) = 1$ . The metric and topological language is developed in Chapter 10; uniform spaces are not needed here.

**Remark 3.4.17** (What to retain). Uniqueness identifies  $\mathbb{R}$  up to canonical isomorphism; the Cauchy construction proves existence. Subsequent analysis uses the ordered-field laws, Archimedean property, rational density, and completeness far more often than it uses equivalence classes. Understanding their construction matters; reproducing every verification on a first reading is a separate achievement.

### 3.5 Basic Inequalities

Exact computation is often unnecessary in analysis. The decisive step is usually an estimate that replaces a complicated quantity by one already known to be small or bounded.

**Proposition 3.5.1** (Basic absolute-value estimates). For  $x, y \in \mathbb{R}$ ,

$$|x + y| \leq |x| + |y|, \quad ||x| - |y|| \leq |x - y|.$$

Consequently,

$$|x_1 + \cdots + x_n| \leq \sum_{j=1}^n |x_j|.$$

*Proof.* The triangle inequality was established with the ordered-field absolute value. Applying it to  $x = (x - y) + y$  gives  $|x| - |y| \leq |x - y|$ ; exchange  $x$  and  $y$  for the reverse bound. The finite form follows by induction.  $\square$

**Proposition 3.5.2** (Quadratic and Bernoulli inequalities). For  $u, v \in \mathbb{R}$ ,

$$2uv \leq u^2 + v^2.$$

If  $t \geq -1$  and  $n \in \mathbb{N}$ , then

$$(1 + t)^n \geq 1 + nt.$$

*Proof.* The first inequality is  $(u - v)^2 \geq 0$ . For the second, induction begins at  $n = 1$ . If the assertion holds for  $n$ , multiplication by  $1 + t \geq 0$  gives

$$(1 + t)^{n+1} \geq 1 + (n + 1)t + nt^2 \geq 1 + (n + 1)t.$$

$\square$

**Remark 3.5.3** (How estimates are used). The triangle inequality splits an error into controllable pieces; the reverse triangle inequality keeps denominators away from zero; the quadratic estimate turns products into sums; and Bernoulli compares powers with linear growth. For nonnegative inputs, the quadratic estimate later yields the usual arithmetic–geometric mean inequality after roots are available. Part I obtains the finite scalar Cauchy–Schwarz inequality as the  $p = 2$  case of Hölder in [Theorem 8.9.7](#). Chapter 10 independently develops its geometric inner-product form and the norm comparisons

$$|x_i| \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty$$

for multivariable estimates. Later examples will repeatedly seek a sufficiently strong bound rather than an exact formula.

## Exercises

**Exercise 1.\*** Give two unequal rational Cauchy sequences representing 1, and a positive rational sequence representing zero. Explain why eventual equality and eventual positivity would give the wrong equivalence relation and order.

**Exercise 2.** Adapt the square-root construction to prove existence of a positive cube root of 2. First show that  $\{x \geq 0 : x^3 < 2\}$  is nonempty and bounded; then use  $(s+h)^3 - s^3 = 3s^2h + 3sh^2 + h^3$  to rule out either strict inequality for the cube of its supremum.

**Exercise 3.** Let  $F$  be a complete ordered field. Prove that every nonempty open interval in  $F$  contains a rational element.

**Exercise 4.\*** Suppose rational approximations satisfy

$$|x_j - r_j| < 1/j.$$

Explain why Cauchyness of  $(x_j)$  forces Cauchyness of  $(r_j)$ , and why replacing

$$1/j$$

by any positive error tending to zero leaves the argument valid.

**Exercise 5.** Give a bounded nonempty subset of  $\mathbb{R}$  whose supremum is not a maximum, and one whose supremum is a maximum. State the additional condition on  $\sup S$  that distinguishes the two cases. Identify where completeness enters the assertion that every nonempty bounded-above set has a supremum, and explain why the corresponding assertion fails in  $\mathbb{Q}$ .



## Chapter 4

# Sequences of Real Numbers

Threshold indices will usually be written explicitly in this chapter. A direct proof must exhibit which choices work together; later applications can invoke a proved theorem that already contains this bookkeeping. Discovering a proof and writing it are different tasks:

Discovery: reason backward from the desired error bound.

Proof: choose the threshold and verify forward.

For

$$\frac{1}{n} \rightarrow 0,$$

the target

$$\frac{1}{n} < \varepsilon$$

suggests

$$n > \frac{1}{\varepsilon}.$$

The proof starts instead by choosing a natural number

$$N > 1/\varepsilon$$

and then checks

$$n \geq N \Rightarrow \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

For  $|a_n - a| + |b_n - b| < \varepsilon$ , an *error budget* allocates

$$\varepsilon/2$$

to each summand. Different allocations are possible; each must be chosen before verifying the final estimate.

### 4.1 Convergence

We now study systematically the convergence used in Chapter 3, beginning by restating its precise error criterion. The error must eventually stay below each prescribed tolerance; merely returning close to a proposed limit infinitely often is not enough.

In this chapter the real numbers are the ordered field constructed in Chapter 3. A sequence in  $\mathbb{R}$  is a function  $a : \mathbb{N} \rightarrow \mathbb{R}$ ; its value at  $n$  is written  $a_n$ .

**Definition 4.1.1** (Limit of a sequence). Let  $(a_n)$  be a real sequence and let  $L \in \mathbb{R}$ . We say that  $(a_n)$  **converges to**  $L$ , and write

$$\lim_{n \rightarrow \infty} a_n = L,$$

or equivalently  $a_n \rightarrow L$ , if for every  $\varepsilon > 0$  there exists  $N \in \mathbb{N}$  such that, for every  $n \geq N$ , one has  $|a_n - L| < \varepsilon$ . A sequence that converges to some real number is **convergent**; otherwise it is **divergent**.

The number  $N$  may depend on  $\varepsilon$ , but it must work for every  $n \geq N$ . This order of quantifiers is essential.

### Revisiting decimals through sequence limits

Section 3.4 already interpreted decimals through finite truncations and shrinking intervals, including the terminating versus repeating-9 ambiguity. We now revisit those conclusions in the language of Definition 4.1.1. The same three questions become convergence of prescribed truncations, construction of digits for a given real number, and uniqueness apart from endpoint ambiguity. Finite tail estimates make the convergence explicit; no infinite-series theory is needed.

**Example 4.1.2** (Constructing digits by successive intervals). For

$$x = 1/7,$$

integer division successively gives  $10 = 1 \cdot 7 + 3$ ,  $30 = 4 \cdot 7 + 2$ ,  $20 = 2 \cdot 7 + 6$ ,  $60 = 8 \cdot 7 + 4$ ,  $40 = 5 \cdot 7 + 5$ , and  $50 = 7 \cdot 7 + 1$ . Thus the digits repeat 142857. The first three truncations are 0.1, 0.14, 0.142, and the  $n$ th truncation  $d_n$  satisfies

$$d_n \leq 1/7 < d_n + 10^{-n}.$$

The shrinking error, rather than the repeating pattern, gives

$$d_n \rightarrow 1/7$$

by the definition of convergence, recovering Chapter 3's interval interpretation.

**Proposition 4.1.3** (Decimal representations). For digits  $d_j \in \{0, \dots, 9\}$ , the truncations

$$s_n = \sum_{j=1}^n d_j 10^{-j}$$

converge to a real number in  $[0, 1]$ , denoted  $0.d_1 d_2 \dots$ . Every  $x \in [0, 1)$  has such an expansion which is not eventually 9. Two different digit strings represent the same number only when, at their first differing place, their digits differ by one and the smaller digit is followed entirely by 9's, the larger entirely by 0's.

*Proof.* For  $m > n$ , a finite geometric sum gives

$$0 \leq s_m - s_n \leq \sum_{j=n+1}^m 9 \cdot 10^{-j} = 10^{-n} - 10^{-m} < 10^{-n}.$$

Given  $\varepsilon > 0$ , choose  $N$  with  $10^{-N} < \varepsilon$  (use  $10^N \geq N + 1$ ). For  $m > n \geq N$ ,  $|s_m - s_n| < 10^{-n} \leq 10^{-N} < \varepsilon$ ; swapping  $m, n$  covers the other order, and equal indices give zero. The truncations are

Cauchy and converge by [Theorem 3.4.10](#). Their bounds  $0 \leq s_n \leq 1$  also bound their limit: a limit outside that interval would have a positive distance from every truncation.

To construct digits for  $x < 1$ , set  $r_0 = x$  and recursively choose the integer  $d_n$ , using [Lemma 3.4.4](#), with  $d_n \leq 10r_{n-1} < d_n + 1$ , putting  $r_n = 10r_{n-1} - d_n$ . Then  $0 \leq r_n < 1$ ,  $d_n \in \{0, \dots, 9\}$ , and  $x = s_n + 10^{-n}r_n$ . For justification set  $s_0 = 0$ ; the identity holds at  $n = 0$ . Substituting

$$r_{n-1} = (d_n + r_n)/10$$

into the identity at  $n-1$  proves it at  $n$ . For  $n \geq N$  as above,  $|x - s_n| = 10^{-n}r_n < 10^{-n} \leq 10^{-N} < \varepsilon$ , so  $s_n \rightarrow x$ . This expansion cannot have only 9's after position  $n$ : that tail would force  $r_n = 1$ .

For uniqueness, suppose two strings first differ at place  $k$ , with  $a_k < b_k$ . For every  $m > k$ , the maximum compensation from the next  $m-k$  digits is  $10^{-k} - 10^{-m}$  by the finite identity above. The limiting compensation is therefore at most  $10^{-k}$ . Equality of the represented numbers therefore forces  $b_k - a_k = 1$  and the maximal compensating tails. If even one later digit fails to be maximal, the compensation loses a fixed positive amount. Thus the tails must be respectively all 9's and all 0's.  $\square$

The endpoint example from [Section 3.4](#) now reads

$$0.\underbrace{99 \dots 9}_{n \text{ digits}} = 1 - 10^{-n}, \quad 0.999 \dots = 1.$$

Likewise  $0.24999 \dots = 0.25$ . A decimal expression represents a number; it is not the number itself. For a canonical representation use the one that is not eventually 9 (and the usual integer part for numbers outside  $[0, 1)$ ).

**Theorem 4.1.4** (Cantor's diagonal argument, now formal). The interval  $(0, 1)$ , and hence  $\mathbb{R}$ , is uncountable.

*Proof.* If  $(x_n)$  listed  $(0, 1)$ , write each entry in its canonical decimal expansion. Choose the  $n$ th digit  $c_n$  to be 2 if the  $n$ th digit of  $x_n$  is 1, and 1 otherwise. The proposition gives a real number  $c = 0.c_1c_2 \dots$  in  $(0, 1)$ . Its digits are only 1 and 2, so no terminating/repeating-9 ambiguity can identify it with an  $x_n$  whose  $n$ th digit differs. This contradicts the claimed listing.  $\square$

This completes the informal diagonal-argument preview in Chapter 2, using the decimal interpretation established in Chapter 3 and made explicit through sequence limits here.

**Lemma 4.1.5.** If  $x \in \mathbb{R}$  satisfies  $|x| < \varepsilon$  for every  $\varepsilon > 0$ , then  $x = 0$ .

*Proof.* If  $x \neq 0$ , then  $|x| > 0$ . Taking

$$\varepsilon = |x|/2$$

contradicts the hypothesis.  $\square$

**Theorem 4.1.6** (Uniqueness of limits). A convergent real sequence has exactly one limit.

*Proof.* Suppose  $a_n \rightarrow L$  and  $a_n \rightarrow M$ . Given  $\varepsilon > 0$ , choose  $N_1, N_2$  such that

$$|a_n - L| < \varepsilon/2$$

for  $n \geq N_1$  and

$$|a_n - M| < \varepsilon/2$$

for  $n \geq N_2$ . For  $n \geq \max\{N_1, N_2\}$ , the triangle inequality gives

$$|L - M| \leq |L - a_n| + |a_n - M| < \varepsilon.$$

By [Lemma 4.1.5](#),  $L - M = 0$ .  $\square$

**Example 4.1.7.** The sequence

$$a_n = 1/n$$

converges to 0. For a given  $\varepsilon > 0$ , [Theorem 3.4.3](#) provides

$$N > 1/\varepsilon,$$

and then  $n \geq N$  implies

$$0 < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

**Definition 4.1.8** (Divergence to infinity). We write

$$a_n \longrightarrow +\infty$$

if, for every  $M \in \mathbb{R}$ , there is  $N \in \mathbb{N}$  such that

$$n \geq N \implies a_n > M.$$

We write  $a_n \rightarrow -\infty$  if, for every  $M \in \mathbb{R}$ , there is  $N$  such that

$$n \geq N \implies a_n < M.$$

These statements describe two particular forms of divergence. Neither  $+\infty$  nor  $-\infty$  is a real number, and neither statement says that  $(a_n)$  converges in  $\mathbb{R}$ .

**Example 4.1.9** (Examples and nonexamples). The sequence  $n^2$  tends to  $+\infty$ : given  $M \in \mathbb{R}$ , choose  $N > \max\{M, 1\}$ . The sequence  $-n$  tends to  $-\infty$ . By contrast,  $(-1)^n n$  is unbounded in both directions but tends to neither infinity; for every proposed stage, later terms occur with each sign. Thus “unbounded” is much weaker than “eventually larger than every real bound.”

**Remark 4.1.10** (A useful proof pattern). To prove  $a_n \rightarrow +\infty$ , start with an arbitrary target height  $M$  and solve the inequality  $a_n > M$  for a sufficient condition on  $n$ . This is the infinite-limit analogue of solving  $|a_n - L| < \varepsilon$  in an ordinary  $\varepsilon$ - $N$  proof.

## 4.2 Algebra and Order of Limits

Once limits have been defined, the next question is whether familiar algebra survives passage to the limit. The following rules justify the calculations that will later occur inside functions, series, derivatives, and integrals.

**Theorem 4.2.1** (Algebra of limits). Let  $(a_n)$  and  $(b_n)$  be real sequences with  $a_n \rightarrow a$  and  $b_n \rightarrow b$ , and let  $c \in \mathbb{R}$ . Then

$$\lim_{n \rightarrow \infty} (a_n + b_n) = a + b, \quad \lim_{n \rightarrow \infty} ca_n = ca, \quad \lim_{n \rightarrow \infty} a_n b_n = ab.$$

If  $b \neq 0$ , then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{a}{b}$$

whenever  $b_n \neq 0$  for all sufficiently large  $n$ .

*Proof.* Fix  $\varepsilon > 0$ . For sums choose  $N_1, N_2$  so that

$$|a_n - a| < \varepsilon/2$$

for  $n \geq N_1$  and

$$|b_n - b| < \varepsilon/2$$

for  $n \geq N_2$ . Put  $N = \max\{N_1, N_2\}$ . For every  $n \geq N$ ,

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \varepsilon.$$

If  $c = 0$ , scalar multiplication has zero error. Otherwise choose  $N_c$  with

$$|a_n - a| < \varepsilon/|c|$$

for  $n \geq N_c$ ; then  $|ca_n - ca| = |c||a_n - a| < \varepsilon$  for every such  $n$ .

For products first choose  $N_0$  with  $|a_n - a| < 1$  for  $n \geq N_0$ . Set  $B = |a| + 1$ ; thus  $|a_n| < B$  on that tail. Choose  $N_1$  for

$$|b_n - b| < \varepsilon/(2B)$$

and  $N_2$  for

$$|a_n - a| < \varepsilon/(2(|b| + 1)).$$

For every  $n \geq N = \max\{N_0, N_1, N_2\}$ ,

$$|a_n b_n - ab| \leq |a_n||b_n - b| + |b||a_n - a| < \varepsilon.$$

The bound  $B$  was fixed before choosing the error thresholds.

For quotients choose  $N_0$  with

$$|b_n - b| < |b|/2$$

for  $n \geq N_0$ . Then

$$|b_n| \geq |b| - |b_n - b| > |b|/2 > 0,$$

so the quotient is defined. Choose  $N_1$  for

$$|a_n - a| < \varepsilon|b|/4$$

and  $N_2$  for

$$|b_n - b| < \varepsilon|b|^2/(4(|a| + 1)).$$

For every  $n \geq N = \max\{N_0, N_1, N_2\}$ ,

$$\left| \frac{a_n}{b_n} - \frac{a}{b} \right| \leq \frac{2}{|b|} |a_n - a| + \frac{2|a|}{|b|^2} |b_n - b| < \varepsilon.$$

Thus the denominator hypothesis on the tail follows from  $b \neq 0$  itself. □

**Theorem 4.2.2** (Order preservation). Let  $a_n \rightarrow a$  and  $b_n \rightarrow b$ . If  $a_n \leq b_n$  for every sufficiently large  $n$ , then  $a \leq b$ .

*Proof.* Let  $N_0$  be a stage from which  $a_n \leq b_n$ . If  $a > b$ , put

$$\varepsilon = (a - b)/3.$$

Choose  $N_1, N_2$  for the respective errors  $|a_n - a| < \varepsilon$  and  $|b_n - b| < \varepsilon$ . For every  $n \geq \max\{N_0, N_1, N_2\}$ ,  $a_n > a - \varepsilon = b + 2\varepsilon > b_n$ , a contradiction.  $\square$

**Corollary 4.2.3** (Squeeze theorem). If  $a_n \leq b_n \leq c_n$  eventually and  $a_n \rightarrow L$ ,  $c_n \rightarrow L$ , then  $b_n \rightarrow L$ .

*Proof.* Let  $N_0$  be a stage from which the ordering holds. Given  $\varepsilon > 0$ , choose  $N_1, N_2$  for  $|a_n - L| < \varepsilon$  and  $|c_n - L| < \varepsilon$ , respectively. Put  $N = \max\{N_0, N_1, N_2\}$ . For every  $n \geq N$ ,

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon,$$

so  $|b_n - L| < \varepsilon$ .  $\square$

**Example 4.2.4** (Integer powers and geometric sequences). A geometric sequence with initial term  $a$  and common ratio  $r$  is  $a, ar, ar^2, \dots$ . We first study powers of the ratio. For fixed  $n \in \mathbb{N}$ , the estimate

$$0 < \frac{1}{k^n} \leq \frac{1}{k}$$

holds for  $k \geq 1$ . Given  $\varepsilon > 0$ , choose

$$N > \max\{1, 1/\varepsilon\};$$

every  $k \geq N$  then has

$$\frac{1}{k^n} < \varepsilon.$$

Here  $n$  is fixed and  $k$  varies. For  $r > 1$ , Bernoulli's inequality supplies the growing lower bound

$$r^n = (1 + (r - 1))^n \geq 1 + n(r - 1).$$

Given  $M \in \mathbb{R}$ , choose

$$N > \max\{1, (M - 1)/(r - 1)\}.$$

Then  $n \geq N$  implies  $r^n > M$ , proving divergence to  $+\infty$ . If  $0 < |r| < 1$ , set

$$h = \frac{1}{|r|} - 1 > 0.$$

The same inequality gives

$$|r^n| \leq \frac{1}{1 + nh}.$$

Choosing

$$N > \max\{1, 1/(h\varepsilon)\}$$

now makes  $|r^n| < \varepsilon$  for every  $n \geq N$ . For  $r = 0$ , all positive powers are zero.

**Proposition 4.2.5** (Dominant powers in polynomial substitutions). Let  $f(x) = a_n x^n + \cdots + a_0$  and  $g(x) = b_m x^m + \cdots + b_0$  be nonzero polynomial expressions, with  $a_n \neq 0$ ,  $b_m \neq 0$ ,  $\deg f = n$  and  $\deg g = m$  (degrees may be zero). Here  $x$  is an indeterminate; substituting natural numbers gives the sequence

$$f(k)/g(k)$$

wherever its denominator is nonzero. From some stage onward this denominator is nonzero, and

$$\frac{f(k)}{g(k)} \rightarrow \begin{cases} 0, & n < m, \\ \frac{a_n}{b_m}, & n = m, \\ +\infty, & n > m, \frac{a_n}{b_m} > 0, \\ -\infty, & n > m, \frac{a_n}{b_m} < 0. \end{cases}$$

*Proof.* Divide by the leading powers, not by a guessed limit:

$$F_k = \frac{f(k)}{k^n} = a_n + \frac{a_{n-1}}{k} + \cdots + \frac{a_0}{k^n} \rightarrow a_n, \quad G_k = \frac{g(k)}{k^m} \rightarrow b_m.$$

For degree zero the corresponding expression is just its constant term. These limits follow from the integer-power example and limit arithmetic. Choose  $N_0$  such that

$$|G_k - b_m| < |b_m|/2$$

for every  $k \geq N_0$ . Then

$$|G_k| > |b_m|/2,$$

so  $g(k) \neq 0$  there, and

$$H_k = F_k/G_k \rightarrow c = a_n/b_m$$

by the quotient theorem. Now

$$f(k)/g(k) = k^{n-m} H_k.$$

For  $n < m$ , write this as

$$H_k/k^{m-n}$$

and use limit arithmetic; for  $n = m$  it is  $H_k$ . For  $n > m$  and  $c > 0$ , choose  $N_1$  such that

$$|H_k - c| < c/2$$

for  $k \geq N_1$ . Given  $M \in \mathbb{R}$ , choose a natural number

$$N \geq \max\{N_0, N_1\}, \quad N > \max\left\{1, \frac{2 \max\{M, 0\}}{c}\right\}.$$

Every  $k \geq N$  satisfies

$$k^{n-m} H_k > (c/2)k > M.$$

For  $c < 0$ , apply this positive argument to

$$-f(k)/g(k),$$

using target  $-M$  to obtain

$$f(k)/g(k) < M.$$

□

For example,

$$(2k + 1)/(k^2 + 1) \rightarrow 0,$$

$$(3k^2 - k)/(2k^2 + 5) \rightarrow 3/2, \quad (-k^3 + 2)/(k + 1) \rightarrow -\infty.$$

The leading coefficients decide the sign in the last case; the lower terms cannot reverse it on the controlled tail.

**Example 4.2.6** (Algebraic simplification and rationalization). The candidate limit in

$$a_n = \frac{3n^2 - n}{2n^2 + 5}$$

becomes visible after division by  $n^2$ ; the limit laws then give

$$a_n \rightarrow 3/2.$$

For

$$b_n = \sqrt{n^2 + n} - n,$$

direct substitution hides the answer. Multiplying by the conjugate gives

$$b_n = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \rightarrow \frac{1}{2}.$$

Here square roots exist by [Proposition 3.4.12](#); their needed limit follows directly from

$$0 \leq \sqrt{1 + \frac{1}{n}} - 1 = \frac{\frac{1}{n}}{\sqrt{1 + \frac{1}{n}} + 1} \leq \frac{1}{n}.$$

The algebra does not itself prove a limit; it rewrites the expression into terms to which already proved limit laws apply.

**Remark 4.2.7** (Strategy for sequence limits). Separate discovery from proof. First simplify, compute a few terms, or compare dominant terms to guess a candidate. Then prove it using the definition, the limit laws, a squeeze estimate, or monotonicity and boundedness. To prove that no finite limit exists, look for incompatible subsequences, failure of the Cauchy condition, or divergence to  $\pm\infty$ . A failed attempt to find a limit is not a proof of nonexistence.

**Example 4.2.8** (Translating repeated decay). A process removes half the remaining material at each step, starting with 8 grams. The varying quantity is the amount  $a_n$  after  $n$  steps; the assumption is that the same fraction is removed each time and none is added. Thus

$$a_0 = 8, \quad a_{n+1} = \frac{a_n}{2}.$$

Induction gives the model

$$a_n = 8 \left( \frac{1}{2} \right)^n.$$

The geometric-sequence result proves  $a_n \rightarrow 0$ : any positive tolerance in grams is met and retained after some step. For error below

$$1/10$$

gram,  $N = 7$  works, since

$$8/2^7 = 1/16 < 1/10$$

and the amounts decrease. Convergence does not mean the amount becomes exactly zero at a finite step.



**Disproving a proposed limit.** Negating  $a_n \rightarrow L$  gives precisely

$$\exists \varepsilon_0 > 0 \forall N \in \mathbb{N} \exists n \geq N : |a_n - L| \geq \varepsilon_0.$$

For

$$a_n = 1 + 1/n,$$

the candidate  $L = 0$  fails: take  $\varepsilon_0 = 1$  and  $n = N$ . This does not prove divergence; in fact  $a_n \rightarrow 1$ . To prove divergence one must exclude *every* real  $L$ . For  $(-1)^n$ , given  $L \geq 0$  choose any odd  $n \geq N$ ; given  $L < 0$  choose any even  $n \geq N$ . In either case  $|(-1)^n - L| \geq 1$ . The next subsection packages this obstruction into the particularly useful two-subsequence test.

### 4.3 Subsequences and Elementary Divergence

A subsequence samples an infinite process along a thinner infinite set of stages. It is especially useful for detecting divergence: incompatible long-term behaviors cannot belong to one convergent sequence.

**Definition 4.3.1** (Subsequence). Let  $(a_n)$  be a sequence. A **subsequence** is a sequence  $(a_{n_k})_{k \in \mathbb{N}}$ , where  $n_1 < n_2 < \dots$  are natural numbers.

**Proposition 4.3.2** (Subsequences preserve limits). If  $a_n \rightarrow L$ , then every subsequence of  $(a_n)$  converges to  $L$ .

*Proof.* For  $n_1 < n_2 < \dots$ , induction gives  $n_k \geq k$ . Given  $\varepsilon > 0$ , choose  $N$  such that  $n \geq N$  implies  $|a_n - L| < \varepsilon$ . Then  $k \geq N$  implies  $n_k \geq N$ , hence  $|a_{n_k} - L| < \varepsilon$ .  $\square$

**Corollary 4.3.3.** If a sequence has two subsequences converging to distinct limits, then it is divergent.

*Proof.* Otherwise [Proposition 4.3.2](#) would make both subsequences converge to the sequence's limit, contradicting [Theorem 4.1.6](#).  $\square$

**Example 4.3.4.** The sequence  $(-1)^n$  diverges. Its even-indexed subsequence is constantly 1, while its odd-indexed subsequence is constantly  $-1$ .

### 4.4 Completeness Consequences

The order completeness of the real numbers now becomes a working tool. It converts bounded monotone approximation into a limit, prevents Cauchy approximations from leaving a gap, and forces bounded sequences to have convergent subsequences.

**Definition 4.4.1** (Monotone sequence). A sequence  $(a_n)$  is **increasing** if  $a_n \leq a_{n+1}$  for all  $n$ , **decreasing** if  $a_{n+1} \leq a_n$  for all  $n$ , and **monotone** if it is increasing or decreasing.

We write  $a_n \uparrow L$  when  $(a_n)$  is increasing and  $a_n \rightarrow L$ , and  $a_n \downarrow L$  when  $(a_n)$  is decreasing and  $a_n \rightarrow L$ . Similarly,  $a_n \uparrow +\infty$  and  $a_n \downarrow -\infty$  record monotone divergence to the indicated infinite limit.

**Theorem 4.4.2** (Monotone convergence). An increasing sequence converges if and only if it is bounded above; in that event, its limit is  $\sup\{a_n : n \in \mathbb{N}\}$ . A decreasing sequence converges if and only if it is bounded below, and its limit is the infimum of its range.

*Proof.* If  $a_n \rightarrow L$ , choose  $N_0$  so that  $|a_n - L| < 1$  for  $n \geq N_0$ . Then  $B = \max\{|L| + 1, |a_1|, \dots, |a_{N_0}|\}$  bounds every  $|a_n|$ . Conversely, let  $s = \sup\{a_n : n \in \mathbb{N}\}$ . Given  $\varepsilon > 0$ ,  $s - \varepsilon$  is not an upper bound, so  $a_N > s - \varepsilon$  for some  $N$ . For  $n \geq N$ ,

$$s - \varepsilon < a_N \leq a_n \leq s,$$

and hence  $|a_n - s| < \varepsilon$ . The decreasing statement follows on applying the increasing statement to  $(-a_n)$ .  $\square$

The theorem immediately constructs one of the central constants of analysis. In elementary algebra the expression

$$(1 + 1/n)^n$$

appears in repeated growth and compound-interest calculations. Here it will do more: bounded monotone approximation will produce a real number before any exponential function has been defined.

**Theorem 4.4.3** (The sequential construction of  $e$ ). The sequence

$$a_n = \left(1 + \frac{1}{n}\right)^n \quad (n \in \mathbb{N})$$

is strictly increasing and bounded above. Consequently it converges.

*Proof. Mechanism.* Expand each finite power. For every fixed binomial term, increasing  $n$  increases each factor, while the next expansion also contains a new positive term.

The finite binomial identity (which follows by elementary induction) gives

$$\begin{aligned} a_n &= \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k} \\ &= \sum_{k=0}^n \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right), \end{aligned}$$

where the product for  $k = 0$  is 1. For  $0 \leq k \leq n$ , every factor in the  $k$ th term is nonnegative and is no smaller when  $n$  is replaced by  $n + 1$ . The  $k = 2$  term increases strictly, and the expansion of  $a_{n+1}$  has one additional positive term. Hence  $a_{n+1} > a_n$ .

Every factor in the product is at most 1, so

$$a_n \leq \sum_{k=0}^n \frac{1}{k!}.$$

For  $k \geq 2$ , one has  $k! \geq 2 \cdot 3^{k-2}$ . Therefore the finite geometric sum gives

$$a_n \leq 2 + \frac{1}{2} \sum_{j=0}^{n-2} \frac{1}{3^j} < \frac{11}{4} < 3.$$

Thus  $(a_n)$  is increasing and bounded above, so it converges by [Theorem 4.4.2](#).  $\square$

**Definition 4.4.4** (The number  $e$ ). Define

$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

In particular,  $2 < e < 3$ . This definition uses only finite powers and sequence completeness. Chapter 8 will construct the exponential function independently and prove that its value at 1 is this same number.

**Example 4.4.5** (Recognizing the defining sequence for  $e$ ). Write

$$a_n = \left(1 + \frac{1}{n}\right)^n,$$

so  $a_n \rightarrow e$ . Three altered expressions reduce to this sequence by operations already available:

$$\begin{aligned} \left(1 + \frac{1}{2n}\right)^{2n} &= a_{2n} \rightarrow e, \\ \left(1 + \frac{1}{n+1}\right)^n &= \frac{a_{n+1}}{1 + \frac{1}{n+1}} \rightarrow e, \\ \left(1 - \frac{1}{n}\right)^{-n} &= a_{n-1} \left(1 + \frac{1}{n-1}\right) \rightarrow e \quad (n \geq 2). \end{aligned}$$

The first is a subsequence; the second removes one factor after an index shift; the third rewrites a reciprocal and adds one factor. Thus recognition, index shifts, and limit algebra do the work, with only integer powers. Chapter 8 will define real powers and use logarithms to prove the broader principle

$$(1 + x/n)^n \rightarrow e^x$$

([Proposition 8.5.16](#)). That formula is a forward pointer, not a tool used here.

**Theorem 4.4.6** (Cauchy criterion). A real sequence converges if and only if it is Cauchy.

*Proof.* If  $a_n \rightarrow L$ , choose  $N$  with

$$|a_n - L| < \varepsilon/2$$

for  $n \geq N$  and, for all  $m, n \geq N$ , write  $|a_m - a_n| \leq |a_m - L| + |a_n - L| < \varepsilon$ . Conversely, the rational-Cauchy construction of  $\mathbb{R}$  proves exactly this assertion in [Theorem 3.4.10](#).  $\square$

**Definition 4.4.7** (Peak index). An index  $n \in \mathbb{N}$  is a **peak index** of  $(a_n)$  if  $a_n \geq a_m$  for every  $m > n$ .

**Theorem 4.4.8** (Monotone subsequence). Every real sequence has a monotone subsequence.

*Proof. Strategy.* Either infinitely many terms are never exceeded later, or eventually every term is exceeded by a later term. If there are infinitely many peak indices, list them as  $n_1 < n_2 < \dots$ . Then  $a_{n_k} \geq a_{n_{k+1}}$ , so this subsequence is nonincreasing. If there are only finitely many, choose  $N$  larger than all of them; if there are none, take  $N = 1$ . Put  $n_1 = N$  and choose  $n_{k+1}$  to be the least  $m > n_k$  with  $a_m > a_{n_k}$ . Such an index exists because  $n_k$  is not a peak. The resulting subsequence is strictly increasing. This argument uses only order and works in any ordered field, without completeness.  $\square$

**Theorem 4.4.9** (Bolzano–Weierstrass). Every bounded real sequence has a convergent subsequence.

*Proof.* By Theorem 4.4.8, the sequence has a monotone subsequence. That subsequence is still bounded, so it converges by the Monotone Convergence Theorem (Theorem 4.4.2). This is the useful habit of reusing a proved theorem rather than repeating its supremum argument.  $\square$

A bounded sequence need not settle at one value. We can still ask for upper and lower bounds on its eventual behavior by discarding an initial segment and bounding the whole remaining tail. These bounds change monotonically as the tail shrinks, so their limits exist even when the original sequence diverges.

**Definition 4.4.10** (Limsup and liminf). Let  $(a_n)$  be bounded. For  $n \in \mathbb{N}$ , put

$$s_n = \sup \{a_k : k \geq n\}, \quad i_n = \inf \{a_k : k \geq n\}.$$

Define

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} s_n, \quad \liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} i_n.$$

**Theorem 4.4.11** (Basic limsup and liminf facts). For every bounded sequence, limsup and liminf exist and

$$\liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n.$$

Moreover,  $a_n \rightarrow L$  if and only if the two displayed limits both equal  $L$ .

*Proof.* The decreasing sequence  $(s_n)$  and increasing sequence  $(i_n)$  are bounded by the same bounds as  $(a_n)$ , hence converge by monotone convergence. Since  $i_n \leq s_n$ , order preservation orders their limits. If  $a_n \rightarrow L$ , given  $\varepsilon > 0$  choose  $N$  with

$$|a_k - L| < \varepsilon/2$$

for  $k \geq N$ . For every  $n \geq N$ , all terms in the  $n$ th tail lie between

$$L - \varepsilon/2, \quad L + \varepsilon/2.$$

Thus

$$|i_n - L| \leq \varepsilon/2 < \varepsilon, \quad |s_n - L| \leq \varepsilon/2 < \varepsilon.$$

Conversely, if  $i_n, s_n \rightarrow L$ , choose  $N_1, N_2$  for their errors less than  $\varepsilon$ . For  $n \geq \max\{N_1, N_2\}$ ,

$$L - \varepsilon < i_n \leq a_n \leq s_n < L + \varepsilon.$$

$\square$

**Why boundedness remains in this definition.**

**Example 4.4.12** (Tail ceilings and floors). For

$$a_n = (-1)^n + 1/n,$$

the even subsequence tends to 1 and the odd subsequence to  $-1$ . Removing finitely many terms cannot remove either recurring behavior. The tail suprema decrease to 1, while the tail infima increase to  $-1$ . Thus  $\limsup a_n = 1$  and  $\liminf a_n = -1$ . For

$$b_n = 2 + (-1)^n/n,$$

both tail barriers tend to 2, so the sequence converges despite oscillating. The limsup is the eventual ceiling, not the largest early spike in the whole sequence.

For a bounded sequence, every subsequential limit lies between  $\liminf$  and  $\limsup$ : its late terms lie between every fixed pair of tail bounds. Each barrier itself is a subsequential limit. To reach the upper one  $s$ , make two choices at stage  $j$ . First choose a tail index  $N_j > k_{j-1}$  so late that

$$|s_{N_j} - s| < 1/j,$$

where  $k_0 = 0$ . Then the definition of supremum supplies  $k_j \geq N_j$  with

$$s_{N_j} - 1/j < a_{k_j} \leq s_{N_j}.$$

Thus  $k_j > k_{j-1}$  and

$$|a_{k_j} - s| < 2/j,$$

proving convergence. For the lower barrier  $i$ , first choose  $N_j > k_{j-1}$  with

$$|i_{N_j} - i| < 1/j,$$

then choose  $k_j \geq N_j$  with

$$i_{N_j} \leq a_{k_j} < i_{N_j} + 1/j.$$

Again the error is less than

$$2/j.$$

One can develop conventions for  $\limsup$  and  $\liminf$  of unbounded sequences, but they are unnecessary here. We retain real-valued suprema and infima and use [Definition 4.1.8](#) when a sequence eventually escapes every real bound.

## 4.5 Equivalent Completeness Principles

The least-upper-bound property is not the only possible formulation of completeness. The equivalences below explain why several apparently different principles may all serve as the decisive feature distinguishing the real numbers from the rationals.

**Lemma 4.5.1** (A Cauchy sequence with a convergent subsequence). If a Cauchy sequence has a subsequence converging to  $L$ , the entire sequence converges to  $L$ . This holds in every ordered field.

*Proof.* Given  $\varepsilon > 0$ , choose  $N$  so that

$$|a_m - a_n| < \varepsilon/2$$

for  $m, n \geq N$ . Choose  $K$  so that

$$|a_{n_k} - L| < \varepsilon/2$$

for  $k \geq K$ . Fix  $k = \max\{K, N\}$ ; then  $n_k \geq k \geq N$ . For all  $n \geq N$ , the triangle inequality gives  $|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \varepsilon$ .  $\square$

**Theorem 4.5.2** (Equivalence of standard completeness principles). Let  $F$  be an Archimedean field. The following assertions are equivalent.

- (a) Every nonempty subset of  $F$  having an upper bound has a least upper bound.
- (b) Every Cauchy sequence in  $F$  converges.
- (c) Every bounded sequence in  $F$  has a convergent subsequence.
- (d) Every increasing sequence in  $F$  that is bounded above converges.
- (e) Whenever

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots, \quad \lim_{n \rightarrow \infty} (b_n - a_n) = 0,$$

the intervals have a common point.

Consequently, each assertion may be used as a completeness axiom for an Archimedean field.

*Proof. Strategy.* Prove the single cycle  $(a) \Rightarrow (d) \Rightarrow (c) \Rightarrow (b) \Rightarrow (e) \Rightarrow (a)$ . The order-limit laws and the preceding lemma use only ordered-field laws, so they apply in  $F$  without assuming real completeness.

$(a) \Rightarrow (d)$ . For a bounded increasing sequence take the supremum  $s$  of its range. For  $\varepsilon > 0$  some  $a_N > s - \varepsilon$ , and

$$s - \varepsilon < a_N \leq a_n \leq s$$

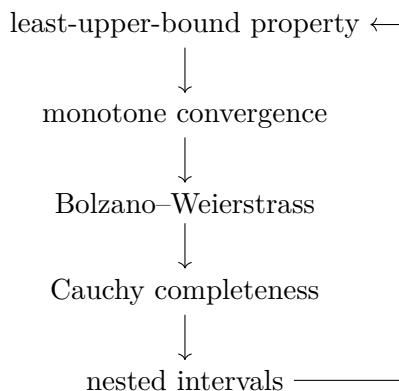
for  $n \geq N$ .

$(d) \Rightarrow (c)$ . The peak-index proof extracts a monotone subsequence in  $F$ . It remains bounded, so (d), applied to its negative if it is decreasing, gives convergence.

$(c) \Rightarrow (b)$ . Choose  $N_0$  with  $|a_m - a_n| < 1$  for  $m, n \geq N_0$ . For  $n \geq N_0$ ,  $|a_n| < |a_{N_0}| + 1$ , so  $\max\{|a_1|, \dots, |a_{N_0}| + 1\}$  bounds the whole sequence. By (c) it has a convergent subsequence. Apply [Lemma 4.5.1](#).

$(b) \Rightarrow (e)$ . For nested intervals with widths tending to zero, given  $\varepsilon > 0$  choose  $N$  with  $b_N - a_N < \varepsilon$ . For all  $m, n \geq N$ , both left endpoints lie in  $[a_N, b_N]$ , so  $|a_m - a_n| \leq b_N - a_N < \varepsilon$ . Thus  $(a_n)$  is Cauchy; let  $a_n \rightarrow x$ . For each fixed  $N$ , the late left endpoints belong to  $[a_N, b_N]$ . Order-limit preservation gives  $a_N \leq x \leq b_N$ .

$(e) \Rightarrow (a)$ . Let  $S \neq \emptyset$  be bounded above, choose  $s_0 \in S$ , and choose an upper bound  $u_0$ . If  $s_0$  is an upper bound it is the maximum. Otherwise bisect  $[s_0, u_0]$ , retaining the lower half when its midpoint is an upper bound of  $S$ , and the upper half otherwise. The resulting nested intervals  $[s_n, u_n]$  have  $s_n$  not an upper bound,  $u_n$  an upper bound, and width  $2^{-n}(u_0 - s_0)$ . The Archimedean property and  $2^n \geq n + 1$  imply that these widths tend to zero in  $F$ . By (e) choose a common point  $c$ . If  $t \in S$  and  $t > c$ , choose  $N_1$  with  $u_{N_1} - s_{N_1} < t - c$ ; then  $u_{N_1} \leq c + (u_{N_1} - s_{N_1}) < t$ , a contradiction. If  $v < c$ , choose  $N_2$  with  $u_{N_2} - s_{N_2} < c - v$ ; then  $s_{N_2} \geq c - (u_{N_2} - s_{N_2}) > v$ , and since  $s_{N_2}$  is not an upper bound, neither is  $v$ . Thus  $c = \sup S$ .  $\square$



Each arrow has one mechanism: a supremum, peak indices, the Cauchy estimate, shrinking widths, and finally bisection.

## Exercises

**Exercise 1.\*** Prove directly that

$$3/(n+2) \rightarrow 0, \quad (2n+1)/(n+1) \rightarrow 2.$$

For each, show the backward discovery of  $N$  separately from the forward proof.

**Exercise 2.\*** Evaluate

$$((4k^2 + 1)/(2k^2 - k)),$$

$$((k+1)/(k^3+2)),, \quad ((-2k^4+k)/(k^2+1))$$

as  $k \rightarrow \infty$ . Identify the degrees and leading coefficients and justify that denominators are nonzero on a tail.

**Exercise 3.\*** Give explicit thresholds proving

$$(3/4)^n \rightarrow 0, \quad (3/2)^n \rightarrow +\infty.$$

Determine whether

$$(-3/2)^n$$

has either infinite limit.

**Exercise 4.\*** For  $c > 0$ , find the limits of  $\sqrt{k^2 + ck} - k$  and  $\sqrt{k^2 + c} - k$ . Explain which part of rationalization determines the different answers.

**Exercise 5.\*** An approximation begins with error at most 6 units. Each correction reduces the error magnitude to at most one third of its preceding magnitude. Formulate a sequence inequality, prove convergence of the errors to zero, and find a number of corrections guaranteeing error below

$$1/100$$

unit.

**Exercise 6.\*** Disprove the proposed limit 1 for

$$a_n = 2 + 1/n$$

using the negated definition. Then prove that  $2 + (-1)^n$  has no limit. Explain why these are different tasks.

**Exercise 7.** Find the limsup and liminf of

$$(-1)^n + 2/n$$

and exhibit subsequences attaining them. Prove from the tail bounds that no other subsequential limit is possible.

**A common strategy for monotone recursive sequences.** First find a bound preserved by the recurrence, and prove it by induction. Then establish monotonicity and apply monotone convergence. Only after convergence is known should one pass to the recurrence, using the limit laws already established, to identify the limit. Later, continuity will often package this last step conveniently.

**Exercise 8.** Let  $a_1 = 0$  and

$$a_{n+1} = (a_n + 2)/3.$$

Prove monotonicity and boundedness, deduce convergence, and determine the limit. Which step uses completeness?

**Exercise 9.\*** Let  $(a_n)$  be increasing and bounded above, and let  $s$  be the supremum of its range. Adapt the proof of [Theorem 4.4.2](#) to show directly that

$$\lim_{n \rightarrow \infty} a_n = s.$$

Identify the exact use of the least-upper-bound property.

**Exercise 10.** Give an example of an unbounded sequence for which

$$\lim_{n \rightarrow \infty} \sup a_n$$

cannot be interpreted as a real number. Explain the purpose of the postponed extended convention.

**Exercise 11.\*** Assuming the least-upper-bound property, prove that every decreasing sequence that is bounded below converges, without appealing to the increasing case as a black box.

**Exercise 12.** Explain why changing the diagonal digits to only 0 and 9 would require an extra ambiguity check. Prove  $0.12999 \cdots = 0.13$  from truncations.



## Chapter 5

# Limits and Continuity of Functions

### 5.1 Limits at a Point

Sequence limits describe processes indexed by natural numbers. A function limit asks the same question when the input can approach a point through all nearby domain values, so the value assigned at the target itself must be kept logically separate.

**Definition 5.1.1** (Limit point). Let  $D \subseteq \mathbb{R}$  and  $a \in \mathbb{R}$ . The point  $a$  is a **limit point** of  $D$  if, for every  $\delta > 0$ , there is an  $x \in D$  such that  $0 < |x - a| < \delta$ .

There are points of  $D$  other than  $a$  arbitrarily close to  $a$ . Membership and approachability are different questions. For  $D = (0, 1)$ , the point 0 is a limit point although  $0 \notin D$ : for any  $\delta > 0$ ,

$$x = \min\{\delta/2, 1/2\}$$

belongs to  $D$  and satisfies  $0 < x < \delta$ . Thus a right-hand limit at this missing endpoint is meaningful. For  $D = \{0\} \cup [1, 2]$ , however,  $0 \in D$  is not a limit point: no point of  $D$  satisfies

$$0 < |x| < 1/2.$$

**Definition 5.1.2** (Isolated point). A point  $a \in D$  is **isolated in  $D$**  if some  $\delta > 0$  satisfies  $D \cap (a - \delta, a + \delta) = \{a\}$ . For  $a \in D$ , negating the limit-point definition shows that being isolated is equivalent to not being a limit point of  $D$ .

**Proposition 5.1.3** (Sequential test for a limit point). The point  $a$  is a limit point of  $D$  if and only if there is a sequence  $x_n \in D \setminus \{a\}$  with  $x_n \rightarrow a$ .

*Proof.* Choose

$$0 < |x_n - a| < 1/n$$

in one direction. Conversely, convergence puts some  $x_n \neq a$  in every prescribed punctured neighborhood of  $a$ .  $\square$

We can therefore test function limits by sequences; the limit-point hypothesis guarantees that there really are approaching inputs.

**Definition 5.1.4** (Function limit). Let  $f : D \rightarrow \mathbb{R}$ , let  $a$  be a limit point of  $D$ , and let  $L \in \mathbb{R}$ . We write

$$\lim_{x \rightarrow a} f(x) = L$$

if, for every  $\varepsilon > 0$ , there is  $\delta > 0$  such that every  $x \in D$  satisfying  $0 < |x - a| < \delta$  also satisfies  $|f(x) - L| < \varepsilon$ .

**Why require a limit point?** If  $a$  is not a limit point, some punctured neighborhood contains no domain points. The implication in the limit definition would then be true for *every* proposed  $L$ , because there is no admissible  $x$  to test. It could not determine a unique value. In the uniqueness proof below, the limit-point assumption is exactly what supplies an actual  $x \neq a$  close enough to compare both proposed limits. Notice that  $f(a)$  is not needed and, when defined, has no effect on the punctured limit.

**Discover backward; prove forward.** The target  $|f(x) - L| < \varepsilon$  suggests a sufficient restriction on  $|x - a|$ . During discovery we may start at the target and work backward. The proof must start with a positive  $\delta$  chosen from  $\varepsilon$  and then verify the implication for *every* admissible  $x$ . For  $3x - 1 \rightarrow 5$  at 2, discovery gives  $3|x - 2| < \varepsilon$ ; the forward proof chooses

$$\delta = \varepsilon/3$$

and checks  $0 < |x - 2| < \delta \Rightarrow |3x - 6| < 3\delta = \varepsilon$ . The next example adds a local bound before allocating the error.

**Example 5.1.5** (Localize before choosing the tolerance). To prove  $x^2 \rightarrow a^2$ , discovery begins with  $|x^2 - a^2| = |x - a||x + a|$ . The second factor is not fixed. First impose  $|x - a| < 1$ , which gives  $|x + a| < 1 + 2|a|$ . Now choose

$$\delta = \min \left\{ 1, \frac{\varepsilon}{1 + 2|a|} \right\}.$$

For  $0 < |x - a| < \delta$ , the product is less than  $\varepsilon$ . The pattern is to bound the uncontrolled factor locally and only then choose the accuracy.

**Example 5.1.6** (Controlling a denominator directly). To prove

$$1/x \rightarrow 1/2$$

as  $x \rightarrow 2$ , first require  $|x - 2| < 1$ . Then  $x > 1$  and

$$|1/x - 1/2| = |x - 2|/(2|x|) \leq |x - 2|/2.$$

Given  $\varepsilon > 0$ , choose  $\delta = \min\{1, 2\varepsilon\}$ . For every  $0 < |x - 2| < \delta$  the preceding estimate is strictly less than  $\varepsilon$ . A restriction that keeps a denominator away from zero is part of the proof, not an optional algebraic convenience.

**Theorem 5.1.7** (Uniqueness of function limits). Under the hypotheses of [Definition 5.1.4](#), there is at most one value  $L$  for which the displayed limit holds.

*Proof.* Suppose  $L$  and  $M$  both satisfy the definition. Given  $\varepsilon > 0$ , obtain positive numbers  $\delta_L, \delta_M$  for tolerance

$$\varepsilon/2.$$

Since  $a$  is a limit point, choose  $x \in D$  with

$$0 < |x - a| < \min\{\delta_L, \delta_M\}.$$

Then

$$|L - M| \leq |L - f(x)| + |f(x) - M| < \varepsilon.$$

By [Lemma 4.1.5](#),  $L = M$ . □

**Theorem 5.1.8** (Sequential criterion). Let  $f : D \rightarrow \mathbb{R}$  and let  $a$  be a limit point of  $D$ . Then

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if, for every sequence  $(x_n)$  in  $D \setminus \{a\}$  with  $x_n \rightarrow a$ , one has  $f(x_n) \rightarrow L$ .

*Proof.* Suppose first that the function limit exists. Given  $\varepsilon > 0$ , take  $\delta$  from [Definition 5.1.4](#). Choose  $N$  with  $|x_n - a| < \delta$  for all  $n \geq N$ . For every such  $n$ ,  $x_n \neq a$ , so  $|f(x_n) - L| < \varepsilon$ .

Conversely, suppose the sequential statement holds but the function limit fails. Negating the definition gives

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in D : 0 < |x - a| < \delta \quad \text{and} \quad |f(x) - L| \geq \varepsilon_0.$$

The order of quantifiers gives one fixed  $\varepsilon_0$  for all choices of  $\delta$ . For each  $n$ , set

$$\delta = 1/n$$

and choose the corresponding point  $x_n \in D$ , so

$$0 < |x_n - a| < \frac{1}{n} \quad \text{and} \quad |f(x_n) - L| \geq \varepsilon_0.$$

For any  $\eta > 0$ , choose

$$N > 1/\eta;$$

then  $n \geq N$  implies

$$|x_n - a| < 1/n \leq 1/N < \eta.$$

Hence  $x_n \rightarrow a$ , while the second prevents  $f(x_n)$  from converging to  $L$ , a contradiction. □

**Definition 5.1.9** (One-sided and infinite limits). Let  $f : D \rightarrow \mathbb{R}$ . For a right-hand limit assume that  $a$  is a limit point of  $D \cap (a, \infty)$ ; use  $D \cap (-\infty, a)$  for a left-hand limit. The assertion

$$\lim_{x \rightarrow a^+} f(x) = L$$

means that for every  $\varepsilon > 0$  there is  $\delta > 0$  such that

$$x \in D, \quad 0 < x - a < \delta \quad \implies \quad |f(x) - L| < \varepsilon.$$

The left-hand limit is defined with  $0 < a - x < \delta$ . We write  $f(x) \rightarrow +\infty$  as  $x \rightarrow a^+$  if for every  $M \in \mathbb{R}$  there is  $\delta > 0$  such that the same hypotheses imply  $f(x) > M$ ; replace the last inequality by  $f(x) < M$  for a limit of  $-\infty$ . The analogous two-sided definitions use  $0 < |x - a| < \delta$ .

At an unbounded end of the domain (assume  $D$  is unbounded in the stated direction),

$$\lim_{x \rightarrow +\infty} f(x) = L$$

means that for every  $\varepsilon > 0$  there is  $A \in \mathbb{R}$  such that  $x \in D$  and  $x > A$  imply  $|f(x) - L| < \varepsilon$ . Replacing this conclusion by  $f(x) > M$  defines  $f(x) \rightarrow +\infty$  as  $x \rightarrow +\infty$ ; changing the direction of the input or output inequalities gives the other cases.

**Remark 5.1.10** (Infinity is behavior, not a value). The notation  $f(x) \rightarrow +\infty$  does not assert convergence to a member of  $\mathbb{R}$ . It abbreviates an eventual-inequality statement. For example,

$$1/x \rightarrow +\infty$$

as  $x \rightarrow 0^+$ , but the two-sided limit at 0 does not tend to either sign of infinity because

$$1/x \rightarrow -\infty$$

from the left.

**Theorem 5.1.11** (Algebraic function-limit laws). Let  $a$  be a limit point of a common domain  $D$ , and suppose  $f(x) \rightarrow L$  and  $g(x) \rightarrow M$  as  $x \rightarrow a$  through  $D$ . For constants  $c, d$ ,  $cf + dg \rightarrow cL + dM$ ,  $fg \rightarrow LM$ , and if  $M \neq 0$ ,

$$f/g \rightarrow L/M$$

on a sufficiently small punctured domain.

*Proof.* For every  $x_n \in D \setminus \{a\}$  tending to  $a$ , transfer the sequence laws through the sequential criterion. The hypotheses give  $f(x_n) \rightarrow L$ ,  $g(x_n) \rightarrow M$ . The sequence sum and product laws give the asserted limits on every such sequence, hence the function limits. If  $M \neq 0$ , choose  $\delta_0 > 0$  such that  $0 < |x - a| < \delta_0$  implies

$$|g(x) - M| < |M|/2;$$

then

$$|g(x)| > |M|/2$$

and the quotient is defined locally. Apply the sequence quotient law and the same criterion on that punctured domain.  $\square$

**Example 5.1.12** (Direct proof and rationalization). To prove

$$\lim_{x \rightarrow 2} (3x - 1) = 5,$$

observe that  $|(3x - 1) - 5| = 3|x - 2|$  and choose

$$\delta = \varepsilon/3.$$

For a less direct example,

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x} + 1} = \frac{1}{2},$$

where the equality for  $x \neq 0$  follows by rationalization and the last step follows directly from an estimate, without invoking continuity. For

$$|x| < 1/2,$$

the positive root is defined by the real-number root construction, and

$$\left| \frac{1}{\sqrt{1+x}+1} - \frac{1}{2} \right| = \frac{|x|}{2(\sqrt{1+x}+1)^2} \leq \frac{|x|}{2}.$$

Choosing

$$\delta = \min\{1/2, 2\varepsilon\}$$

proves the limit.

**Example 5.1.13** (One-sided limits detect nonexistence). For

$$f(x) = x/|x|$$

on  $\mathbb{R} \setminus \{0\}$ ,

$$\lim_{x \rightarrow 0^-} f(x) = -1, \quad \lim_{x \rightarrow 0^+} f(x) = 1.$$

Hence the two-sided limit does not exist. Indeed, the sequences

$$x_n = -1/n, \quad y_n = 1/n$$

approach the same point while their images have different limits.

For one-sided approach,  $t \downarrow a$  means  $t \rightarrow a$  from above ( $t \rightarrow a^+$ ), while  $t \uparrow a$  means  $t \rightarrow a$  from below ( $t \rightarrow a^-$ ). This related notation concerns a variable approaching a point; it is grammatically distinct from the sequence notation  $a_n \uparrow L$ .

**Remark 5.1.14** (A working strategy for function limits). First discover a plausible answer by substitution, simplification, a graph, or comparison. Next prove it: use limit laws only after their hypotheses are met; cancel only on the punctured neighborhood; rationalize radicals; or bound the error by a simpler expression and squeeze. For nonexistence, compare one-sided limits or produce two approaching sequences with different image limits. The exact value  $f(a)$  is irrelevant to  $\lim_{x \rightarrow a} f(x)$ .

**Example 5.1.15** (A zero denominator is a question, not an answer). At  $x = 1$ , the quotient

$$(x^2 - 1)/(x - 1)$$

is undefined, but on the punctured domain it equals  $x + 1$  and its limit is 2. In contrast

$$1/x$$

has incompatible infinite limits at zero, while

$$x^2/x = x$$

has limit zero. In all three cases the denominator tends to zero, so the ordinary quotient theorem cannot decide the problem. For a direct disproof that

$$x/|x| \rightarrow 1$$

at zero, take  $\varepsilon_0 = 1$ . For every  $\delta > 0$ ,

$$x = -\delta/2$$

has  $0 < |x| < \delta$  and

$$|x/|x| - 1| = 2 \geq \varepsilon_0.$$

To exclude all proposed limits, use the two approaching sequences in the preceding example.

### Polynomial and rational limits

The identity function tends to  $a$  as  $x \rightarrow a$ , directly with  $\delta = \varepsilon$ . Repeated products and sums in the limit laws therefore give

$$P(x) = a_n x^n + \cdots + a_0 \implies \lim_{x \rightarrow a} P(x) = P(a).$$

If  $Q(a) \neq 0$ , denominator control and the quotient law give

$$P(x)/Q(x) \rightarrow P(a)/Q(a).$$

For example,

$$(x^2 + 1)/(x + 2) \rightarrow 5/4$$

as  $x \rightarrow 2$ .

The same algebraic laws hold at infinity. To justify this directly, in the sequence-limit proofs replace every tail condition  $n \geq N_i$  by  $x > A_i$  and use  $A = \max_i A_i$ ; the estimates do not otherwise change. The basic input is

$$1/x \rightarrow 0 :$$

choose

$$A = \max\{1, 1/\varepsilon\},$$

so  $x > A$  gives

$$|1/x| < \varepsilon.$$

Integer powers and limit arithmetic handle all lower-order terms.

For the expressions  $f(x) = a_n x^n + \cdots + a_0$  and  $g(x) = b_m x^m + \cdots + b_0$ , with nonzero leading coefficients and degrees  $n, m$ , divide by the dominant powers:

$$F(x) = \frac{f(x)}{x^n} \rightarrow a_n, \quad G(x) = \frac{g(x)}{x^m} \rightarrow b_m.$$

Choose  $A_0$  with

$$|G(x) - b_m| < |b_m|/2$$

for  $x > A_0$ . Then

$$|G(x)| > |b_m|/2, \quad H(x) = F(x)/G(x) \rightarrow c = a_n/b_m.$$

Consequently

$$f(x)/g(x) = x^{n-m} H(x)$$

tends to 0 if  $n < m$ , to  $c$  if  $n = m$ , and to the infinity with sign  $c$  if  $n > m$ . For completeness, when  $c > 0$  choose  $A_1$  with

$$H(x) > c/2$$

for  $x > A_1$ . For any target  $M$ , every

$$x > A = \max \left\{ A_0, A_1, 1, \frac{2 \max\{M, 0\}}{c} \right\}$$

satisfies

$$x^{n-m}H(x) > (c/2)x > M.$$

Negating handles  $c < 0$ . This is the mechanism of [Proposition 4.2.5](#), now for genuine function limits, not just natural-number substitutions.

For  $x \rightarrow -\infty$ , put  $t = -x \rightarrow +\infty$ . The leading coefficients become  $(-1)^n a_n$  and  $(-1)^m b_m$ . The limit is still zero for  $n < m$  and  $c$  for  $n = m$ ; for  $n > m$  its infinite sign is that of  $(-1)^{n-m}c$ . For example

$$(2x^3 + 1)/(x + 1)$$

tends to  $+\infty$  at both ends, whereas

$$(2x^2 + 1)/(x + 1)$$

tends to opposite infinities.

## 5.2 Asymptotic Notation

Limit arguments often need only a comparison of sizes rather than an exact formula. We use the following notation for two common kinds of comparison without replacing the underlying estimates.

**Definition 5.2.1** (Big-O and little-o). Let  $f, g$  be real-valued functions defined sufficiently near  $a$ . We write

$$f(x) = O(g(x)) \quad (x \rightarrow a)$$

if there are  $C > 0$  and a punctured neighborhood of  $a$  on which

$$|f(x)| \leq C|g(x)|.$$

We write

$$f(x) = o(g(x)) \quad (x \rightarrow a)$$

if  $g(x) \neq 0$  sufficiently near  $a$  and

$$\frac{f(x)}{g(x)} \rightarrow 0.$$

For sequences,  $a_n = O(b_n)$  as  $n \rightarrow \infty$  and  $a_n = o(b_n)$  as  $n \rightarrow \infty$  have the identical eventual-bound and quotient-limit meanings.

The limiting regime is part of the assertion. For example,  $x^2 = o(x)$  as  $x \rightarrow 0$ , but not as  $x \rightarrow \infty$ ; and  $n^{-2} = o(n^{-1})$  as  $n \rightarrow \infty$ . All rules below refer to one fixed regime, with comparison functions eventually nonzero when quotients occur.

**Translating the notation.** The assertion  $r = o(g)$  means  $r = \epsilon(x)g(x)$  with  $\epsilon(x) \rightarrow 0$ . Equivalently, for every  $\varepsilon > 0$ ,  $|r(x)| \leq \varepsilon|g(x)|$  sufficiently near the limit. The assertion  $r = O(g)$  uses one fixed constant  $C$ , rather than every positive tolerance. When  $g$  is eventually nonzero it is equivalent to  $r = C(x)g(x)$  with  $C(x)$  bounded. These are functions obeying estimates, not mysterious individual objects called  $O(g)$  or  $o(g)$ . Thus  $f = g + o(g)$  means that there exists a remainder  $r = o(g)$  for which  $f = g + r$ . Different occurrences of  $o(g)$  may denote different remainders.

**Proposition 5.2.2** (When does a remainder vanish?). One has  $r = o(1)$  exactly when  $r \rightarrow 0$ . The assertion  $r = O(1)$  says only that  $r$  is bounded. If  $g \rightarrow 0$ , either  $r = O(g)$  or  $r = o(g)$  implies  $r \rightarrow 0$ .

*Proof.* The first two statements translate the definitions with comparison scale 1. For big-O use  $|r| \leq C|g|$  and squeeze. Little-o implies big-O by choosing tolerance 1, so the same argument applies.  $\square$

For example  $\sin x = O(1)$  as  $x \rightarrow \infty$  but has no limit (use

$$x = 2\pi n + \pi/2, \quad x = 2\pi n + 3\pi/2$$

). As  $x \rightarrow \infty$ ,  $x = o(x^2)$  and  $\sqrt{x} = o(x)$ , although both remainders diverge. Little-o means small *relative to a scale*, not necessarily small in absolute size. For vector-valued remainders the same definitions use the norm of the remainder; Chapter 10 proves the norm estimates needed for composition.

Near  $h = 0$ , the three conditions are strictly ordered:

$$r(h) = o(|h|) \implies r(h) = O(|h|) \implies r(h) \rightarrow 0.$$

Neither converse holds:  $|h|$  is  $O(|h|)$  but not  $o(|h|)$ , while  $\sqrt{|h|} \rightarrow 0$  but is not  $O(|h|)$ . In contrast,  $|h|^2 = o(|h|)$ . Chapter 10 replaces absolute values by norms with the same examples.

**Proposition 5.2.3** (A calculus of remainder estimates). In a fixed regime,

$$\begin{aligned} O(g) + O(g) &= O(g), & o(g) + o(g) &= o(g), \\ O(f)O(g) &= O(fg), & o(f)O(g) &= o(fg), & o(f)o(g) &= o(fg). \end{aligned}$$

Also  $r = O(f)$ ,  $f = O(g)$  imply  $r = O(g)$ ; replacing either of these two hypotheses by little-o gives  $r = o(g)$ .

*Proof.* Translate each remainder into its comparison scale times a bounded or vanishing coefficient. For sums the coefficients add; for products they multiply. Sums of bounded coefficients are bounded, sums of vanishing coefficients vanish, and a vanishing coefficient times a bounded one vanishes. A vanishing coefficient is itself eventually bounded. For transitivity, multiply

$$r/f$$

by

$$f/g;$$

these same facts apply. For big-O sums even at zeros of  $g$ , the triangle inequality gives  $|r_1 + r_2| \leq (C_1 + C_2)|g|$ ; product bounds work in the same way.  $\square$

These displayed “equalities” assert estimates for arbitrary remainders; they do not license cancellation as if  $O(g)$  were a fixed number. In particular,  $O(g) - O(g)$  need not be zero, and the reciprocal of an  $O(1)$  function need not be  $O(1)$  without a bound away from zero.

For eventually nonzero  $g$ , write  $f \sim g$  when

$$f/g \rightarrow 1.$$

Equivalently  $f = g + o(g)$ , since

$$(f - g)/g = f/g - 1.$$

Once the elementary sine limit is established, it says  $\sin x = x + o(x)$  and  $\sin x \sim x$  as  $x \rightarrow 0$ .



**Preview: why differentiation uses little-o.** Chapter 8's derivative condition is equivalent to

$$f(a+h) = f(a) + f'(a)h + o(|h|) \quad (h \rightarrow 0).$$

Indeed, for  $r(h) = f(a+h) - f(a) - f'(a)h$ , the absolute error quotient is

$$|r(h)|/|h| = |(f(a+h) - f(a))/h - f'(a)|.$$

Thus the nonlinear error is negligible compared with the first-order scale, not merely a function tending to zero. The multivariable condition  $f(a+h) = f(a) + Df(a)h + o(\|h\|)$  has exactly the same meaning.

### 5.3 Basic Topology of the Real Line

The local language used throughout analysis can be organized with a small amount of topology. We develop here only the real-line notions needed in the remainder of Part I; Chapter 10 places them in the general setting of metric and topological spaces.

**Definition 5.3.1** (Euclidean balls). For  $x \in \mathbb{R}$  and  $r > 0$ , put

$$B(x, r) = \{y \in \mathbb{R} : |y - x| < r\} = (x - r, x + r).$$

**Definition 5.3.2** (Intervals, open sets, and closed sets). A set  $I \subseteq \mathbb{R}$  is an **interval** if  $x, y \in I$  and  $x < z < y$  imply  $z \in I$ . A set  $U \subseteq \mathbb{R}$  is **open** if

$$\forall x \in U \exists r > 0 : B(x, r) \subseteq U.$$

A set  $F \subseteq \mathbb{R}$  is **closed** if  $\mathbb{R} \setminus F$  is open.

An *open neighborhood* of  $x$  is an open subset of  $\mathbb{R}$  containing  $x$ . More generally, a *neighborhood* of  $x$  is a set containing an open neighborhood of  $x$ .

The intervals  $(a, b)$ ,  $(-\infty, a)$ , and  $(a, \infty)$  are open, while  $[a, b]$ ,  $(-\infty, a]$ , and  $[a, \infty)$  are closed. Both  $\emptyset$  and  $\mathbb{R}$  are open and closed: the defining condition is vacuous for the empty set, and every real point has a ball contained in  $\mathbb{R}$ .

**Proposition 5.3.3** (Elementary set operations). Arbitrary unions and finite intersections of open subsets of  $\mathbb{R}$  are open. Arbitrary intersections and finite unions of closed subsets of  $\mathbb{R}$  are closed.

*Proof.* If  $x \in \bigcup_{\alpha} U_{\alpha}$ , then  $x \in U_{\alpha}$  for at least one index, and a ball witnessing openness of that member lies in the union. If  $x \in U_1 \cap \cdots \cap U_n$ , choose radii  $r_i > 0$  with  $B(x, r_i) \subseteq U_i$  and set

$$r = \min\{r_1, \dots, r_n\} > 0.$$

Then  $B(x, r)$  lies in every  $U_i$ . The closed-set assertions follow by taking complements and using De Morgan's laws.  $\square$

Finiteness matters: for example,

$$\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\},$$

which is not open. Taking complements also gives an infinite union of closed sets that is not closed.

**Definition 5.3.4** (Relative openness and closedness). Let  $A \subseteq Y \subseteq \mathbb{R}$ . The set  $A$  is **relatively open in  $Y$**  if  $A = Y \cap U$  for some open  $U \subseteq \mathbb{R}$ . It is **relatively closed in  $Y$**  if  $A = Y \cap F$  for some closed  $F \subseteq \mathbb{R}$ ; equivalently,  $Y \setminus A$  is relatively open in  $Y$ .

For instance, if  $a < c < b$ , then

$$[a, c) = [a, b] \cap (-\infty, c)$$

is relatively open in  $[a, b]$  although it is not open in  $\mathbb{R}$ .

Limit points and isolated points were defined in [Proposition 5.1.3](#) and the paragraphs preceding it. Those notions record whether every punctured neighborhood meets a set; the sequential test there will remain our principal tool.

**Theorem 5.3.5** (Sequential characterization of closed sets). A set  $F \subseteq \mathbb{R}$  is closed if and only if every convergent sequence in  $F$  has its limit in  $F$ .

*Proof.* Suppose first that  $F$  is closed and  $x_n \in F$  converges to  $x$ . If  $x \notin F$ , the open set  $F^c$  contains a ball  $B(x, r)$ . Eventually  $x_n \in B(x, r)$ , a contradiction. Conversely, suppose the sequential condition holds and let  $x \in F^c$ . If no ball about  $x$  lay in  $F^c$ , then for every  $n$  one could choose  $x_n \in F$  with  $|x_n - x| < 1/n$ . This sequence converges to  $x$ , contradicting the hypothesis. Hence  $F^c$  is open.  $\square$

**Definition 5.3.6** (Dense subsets). For  $E \subseteq Y \subseteq \mathbb{R}$ , the set  $E$  is **dense in  $Y$**  if every nonempty relatively open subset of  $Y$  meets  $E$ . Equivalently, every neighborhood of a point of the subspace  $Y$  meets  $E$ .

By [Corollary 2.5.4](#) and [Theorem 3.4.5](#),  $\mathbb{Q} \cap [a, b]$  is countable and dense in  $[a, b]$ .

**Proposition 5.3.7** (Interval components of an open set). Every open subset of  $\mathbb{R}$  is a disjoint union of at most countably many open intervals.

*Proof.* Let  $G \subseteq \mathbb{R}$  be open. For each  $x \in G$ , take the union of all open intervals that contain  $x$  and lie in  $G$ . Overlapping intervals merge into an interval, so this union is a maximal open interval contained in  $G$ . Two such maximal intervals are either equal or disjoint, and together they cover  $G$ . Fix an enumeration  $\mathbb{Q} = \{q_1, q_2, \dots\}$ . Associate to each component interval the first  $q_n$  that it contains. Distinct disjoint components receive distinct rationals, so this gives an injection of the component intervals into  $\mathbb{N}$ .  $\square$

## 5.4 Continuity

Continuity compares nearby function values with the value at a domain point. At a limit point it removes the separation between the limiting value and the assigned value; isolated points require a separate explanation. The topology of [Section 5.3](#) gives a second language for this stability.

**Definition 5.4.1** (Continuity). Let  $f : D \rightarrow \mathbb{R}$  and let  $a \in D$ . The function  $f$  is **continuous at  $a$**  if, for every  $\varepsilon > 0$ , there is  $\delta > 0$  such that every  $x \in D$  satisfying  $|x - a| < \delta$  satisfies  $|f(x) - f(a)| < \varepsilon$ . It is **continuous on  $D$**  if it is continuous at every point of  $D$ .

**Theorem 5.4.2** (Open-set characterization of continuity). For  $f : D \rightarrow \mathbb{R}$ , the function  $f$  is continuous on  $D$  if and only if  $f^{-1}(U)$  is relatively open in  $D$  for every open  $U \subseteq \mathbb{R}$ . Equivalently, inverse images of closed sets are relatively closed in  $D$ .

This result is recorded to connect the  $\varepsilon$ - $\delta$  definition with the topological language used later. Its proof is not given here; Chapter 10 proves the corresponding statement for general topological spaces in [Theorem 10.4.3](#).

**Limits and continuity have different domain requirements.** A function limit requires  $a$  to be a limit point but need not have  $a \in D$ ; its test  $0 < |x - a| < \delta$  excludes  $x = a$ . Continuity requires  $a \in D$  because  $f(a)$  must exist, but need not require a limit point; its test  $|x - a| < \delta$  includes  $a$ .

For  $D = \{0\} \cup [1, 2]$ , choose

$$\delta = 1/2.$$

Any  $x \in D$  with  $|x| < \delta$  must equal 0. The only comparison required for continuity is therefore  $|f(0) - f(0)| = 0 < \varepsilon$ , for every  $\varepsilon > 0$ , regardless of the other values of  $f$ . Within that neighborhood the function is defined only at the isolated point itself. This differs from a punctured limit, where there would be no admissible points at all.

Consequently, if  $a \in D$  is a limit point,  $f$  is continuous at  $a$  if and only if

$$\lim_{x \rightarrow a} f(x) = f(a) :$$

the excluded case  $x = a$  adds only the comparison  $0 < \varepsilon$ . If  $a \in D$  is isolated, every function on  $D$  is continuous there, while its punctured function limit is not defined under the convention of these notes.

**Theorem 5.4.3** (Sequential criterion for continuity). Let  $f : D \rightarrow \mathbb{R}$  and  $a \in D$ . Then  $f$  is continuous at  $a$  if and only if every sequence  $(x_n)$  in  $D$  satisfying  $x_n \rightarrow a$  also satisfies  $f(x_n) \rightarrow f(a)$ .

*Proof.* The proof is the same as [Theorem 5.1.8](#), except that terms equal to  $a$  require no exclusion: for such a term,  $|f(x_n) - f(a)| = 0$ . Conversely, failure of continuity gives, for each  $n$ , an  $x_n \in D$  with

$$|x_n - a| < 1/n$$

and  $|f(x_n) - f(a)| \geq \varepsilon_0$ . □

**Theorem 5.4.4** (Algebra of continuous functions). Let  $f, g : D \rightarrow \mathbb{R}$  be continuous at  $a \in D$ , and let  $c \in \mathbb{R}$ . Then  $f + g$ ,  $cf$ , and  $fg$  are continuous at  $a$ . If  $g(a) \neq 0$ , then

$$f/g$$

is continuous at  $a$  on the set where it is defined.

*Proof.* For every sequence  $x_n \rightarrow a$  in  $D$ , continuity gives  $f(x_n) \rightarrow f(a)$  and  $g(x_n) \rightarrow g(a)$ . Apply the sequence limit laws of [Theorem 4.2.1](#) and then [Theorem 5.4.3](#). □

**Proposition 5.4.5** (Composition preserves continuity). Let  $D, E \subseteq \mathbb{R}$ , let  $f : D \rightarrow E$  and  $g : E \rightarrow \mathbb{R}$ , and let  $a \in D$ . If  $f$  is continuous at  $a$  and  $g$  is continuous at  $f(a)$ , then  $g \circ f$  is continuous at  $a$ .

*Proof.* For every sequence  $x_n \rightarrow a$  in  $D$ , the sequential criterion [Theorem 5.4.3](#) gives  $f(x_n) \rightarrow f(a)$  in  $E$  and then  $g(f(x_n)) \rightarrow g(f(a))$ . The same criterion proves the conclusion.  $\square$

**Example 5.4.6.** Every polynomial is continuous on  $\mathbb{R}$ : constant and identity functions are continuous directly from [Definition 5.4.1](#), and repeated application of [Theorem 5.4.4](#) proves the claim. Every rational function is continuous wherever its denominator is nonzero.

**Example 5.4.7** (Checking a junction and a removable value). Let  $f(x) = x + 1$  for  $x < 0$  and  $f(x) = x^2 + c$  for  $x \geq 0$ . Away from zero use polynomial continuity. At zero the left limit is 1 and the right limit and assigned value are  $c$ , so continuity holds exactly when  $c = 1$ . For a removable example, assign a value  $d$  at  $x = 1$  to

$$(x^2 - 1)/(x - 1).$$

The punctured limit is 2, so its extension is continuous exactly when  $d = 2$ . Altering the value cannot repair a jump whose two side limits disagree.

**Definition 5.4.8** (Bounded functions). A function  $f : D \rightarrow \mathbb{R}$  is **bounded** if

$$\exists M \geq 0 \forall x \in D : |f(x)| \leq M.$$

Equivalently, its range  $f(D)$  is a bounded subset of  $\mathbb{R}$ .

**Definition 5.4.9** (Piecewise continuity). A function  $f : [a, b] \rightarrow \mathbb{R}$  is **piecewise continuous** if there is a finite partition

$$a = x_0 < x_1 < \cdots < x_n = b$$

such that, on every  $(x_{i-1}, x_i)$ , the restriction of  $f$  extends continuously to  $[x_{i-1}, x_i]$ . Thus only finitely many breakpoints are allowed, and finite one-sided limits exist there.

## 5.5 Uniform Continuity

Ordinary continuity permits the required input tolerance to vary from point to point. Uniform continuity asks for one tolerance that works everywhere on the domain, precisely the stronger control needed for approximation and integration.

**Definition 5.5.1** (Uniform continuity). Let  $f : D \rightarrow \mathbb{R}$ . The function  $f$  is **uniformly continuous** on  $D$  if, for every  $\varepsilon > 0$ , there is  $\delta > 0$  such that, for all  $x, y \in D$ ,

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

The difference from ordinary continuity is that  $\delta$  may depend on  $\varepsilon$ , but not on the point of  $D$ . Schematically, the quantifiers are

$$\text{continuity: } \forall a \in D \forall \varepsilon > 0 \exists \delta = \delta(a, \varepsilon) > 0,$$

whereas

$$\text{uniform continuity: } \forall \varepsilon > 0 \exists \delta = \delta(\varepsilon) > 0 \forall x, y \in D.$$

Moving the point quantifiers past  $\exists \delta$  is the whole strengthening.

**Proposition 5.5.2** (Lipschitz continuity implies uniform continuity). Let  $f : D \rightarrow \mathbb{R}$ . If there is  $K \geq 0$  such that

$$|f(x) - f(y)| \leq K|x - y| \quad \text{for all } x, y \in D,$$

then  $f$  is uniformly continuous on  $D$ .

*Proof.* If  $K = 0$ , the function is constant. If  $K > 0$ , choose

$$\delta = \varepsilon/K.$$

Then the displayed inequality gives the conclusion of [Definition 5.5.1](#).  $\square$

A function satisfying the displayed inequality is **Lipschitz continuous**, or simply **Lipschitz**, on (D), and (K) is a **Lipschitz constant**. Thus the hierarchy is

$$\text{Lipschitz continuity} \implies \text{uniform continuity} \implies \text{continuity}.$$

Neither converse holds in general.

**Proposition 5.5.3** (Sequence pairs detect failure of uniform continuity). A function  $f : D \rightarrow \mathbb{R}$  is uniformly continuous if and only if every pair of sequences  $x_n, y_n \in D$  with  $|x_n - y_n| \rightarrow 0$  satisfies  $|f(x_n) - f(y_n)| \rightarrow 0$ .

*Proof.* For uniform continuity, given  $\varepsilon > 0$  choose its common  $\delta > 0$ . Choose  $N$  so that  $|x_n - y_n| < \delta$  for  $n \geq N$ . Then  $|f(x_n) - f(y_n)| < \varepsilon$  for every  $n \geq N$ . Conversely, failure gives  $\varepsilon_0 > 0$  such that for each  $n$  there are  $x_n, y_n \in D$  with

$$|x_n - y_n| < 1/n$$

but  $|f(x_n) - f(y_n)| \geq \varepsilon_0$ . These pairs contradict the stated test.  $\square$

**Example 5.5.4** (The three levels of control). The function  $x \mapsto x^2$  is not uniformly continuous on  $\mathbb{R}$ : with  $x_n = n$  and

$$y_n = n + 1/n,$$

the inputs approach one another while

$$y_n^2 - x_n^2 = 2 + \frac{1}{n^2}.$$

On  $[-A, A]$ , however,

$$|x^2 - y^2| \leq 2A|x - y|,$$

so it is Lipschitz there. The function  $\sqrt{x}$  on  $[0, 1]$  is uniformly continuous because

$$|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|},$$

as follows: if  $u = \sqrt{x} \geq v = \sqrt{y}$ , then

$$(u - v)^2 \leq (u - v)(u + v) = x - y;$$

interchange  $x, y$  for the other order. The uniform-continuity proof chooses  $\delta = \varepsilon^2$ . It is not Lipschitz: if a constant  $K$  worked, then  $\sqrt{x} \leq Kx$  for every  $x > 0$ , equivalently

$$1/\sqrt{x} \leq K,$$

which fails near 0.

**Example 5.5.5** (Continuity need not be uniform). The function  $f : (0, 1) \rightarrow \mathbb{R}$  given by

$$f(x) = \frac{1}{x}$$

is continuous by [Theorem 5.4.4](#), but not uniformly continuous. Indeed, put

$$x_n = \frac{1}{n+1}, \quad y_n = \frac{1}{n+2}.$$

Then  $|x_n - y_n| \rightarrow 0$ , whereas  $|f(x_n) - f(y_n)| = 1$  for every  $n$ .

On  $[a, \infty)$  with  $a > 0$ , the same function

$$1/x$$

is Lipschitz, since

$$\left| \frac{1}{x} - \frac{1}{y} \right| = \frac{|x - y|}{xy} \leq \frac{1}{a^2} |x - y|.$$

This contrast shows that uniform continuity depends on the domain as well as on the formula.

## 5.6 Intermediate Values

Continuity has a qualitative consequence beyond limiting estimates: it prevents a function from jumping over a value. Completeness supplies the point at which this intuitive crossing must occur.

**Theorem 5.6.1** (Intermediate value theorem). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, with  $a < b$ . If  $u$  lies between  $f(a)$  and  $f(b)$ , then there is  $c \in [a, b]$  such that  $f(c) = u$ .

*Proof.* It is enough to treat  $f(a) \leq u \leq f(b)$ . Let

$$S = \{x \in [a, b] : f(x) \leq u\}.$$

The set is nonempty and bounded above, so let  $c = \sup S$ . If  $f(c) < u$ , then  $c \neq b$  because

$$f(b) \geq u.$$

Continuity gives a positive  $\delta$  such that  $f(x) < u$  whenever

$$|x - c| < \delta.$$

Choose

$$0 < r < \min(\delta, b - c).$$

Then  $c + r \in S$ , contradicting that  $c$  is an upper bound of  $S$ . If  $f(c) > u$ , then  $c \neq a$  because

$$f(a) \leq u.$$

Continuity gives a positive  $\delta$  such that  $f(x) > u$  whenever

$$|x - c| < \delta.$$

By the defining property of the supremum, there is

$$x \in S$$

with

$$c - \delta < x \leq c,$$

which contradicts  $f(x) \leq u$ . Therefore  $f(c) = u$ . □

**Theorem 5.6.2** (Continuity of a monotone inverse). If  $f$  is continuous and strictly monotone on an interval  $I$ , then  $f : I \rightarrow f(I)$  is bijective and its inverse is continuous.

*Proof.* Strict monotonicity gives injectivity, while surjectivity onto  $f(I)$  is automatic. The IVT shows that  $f(I)$  is an interval. Suppose first that  $f$  is increasing, fix  $a \in I$ , and write  $y_0 = f(a)$ .

If  $a$  is an interior point of  $I$ , then for a given  $\varepsilon > 0$  choose

$$a - \varepsilon < \ell < a < r < a + \varepsilon, \quad \ell, r \in I.$$

The number

$$\delta = \min\{f(a) - f(\ell), f(r) - f(a)\} > 0$$

has the desired effect: if  $y \in f(I)$  and  $|y - y_0| < \delta$ , then

$$f(\ell) < y < f(r),$$

so monotonicity traps  $f^{-1}(y)$  between  $\ell$  and  $r$ , and hence within  $\varepsilon$  of  $a$ .

At a left endpoint  $a$ , choose only  $r \in I$  with  $a < r < a + \varepsilon$  and take  $\delta = f(r) - f(a)$ . Then  $a \leq f^{-1}(y) < r$  whenever  $y \in f(I)$  and  $|y - y_0| < \delta$ . The right-endpoint argument is symmetric; a singleton interval is immediate. Thus  $f^{-1}$  is continuous at every point of  $f(I)$ . If  $f$  is decreasing, apply the increasing case to  $-f$ . Since  $f^{-1}(y) = (-f)^{-1}(-y)$ , continuity follows by composition with negation.  $\square$

Integer powers are already defined. We now reverse the question: given  $a$ , can we solve  $x^n = a$ ? Elementary algebra uses radical notation for a chosen solution; analysis must justify its existence and uniqueness.

**Theorem 5.6.3** (Existence and uniqueness of roots). For every  $c \geq 0$  and  $n \in \mathbb{N}$ , there is a unique  $r \geq 0$  such that  $r^n = c$ .

*Proof.* The polynomial  $p(t) = t^n - c$  is continuous. On  $[0, \max\{1, c\}]$  its endpoint values have opposite weak signs, so the Intermediate Value Theorem gives a zero. If  $0 \leq r < s$ , then

$$s^n - r^n = (s - r)(s^{n-1} + s^{n-2}r + \cdots + r^{n-1}) > 0,$$

so  $t \mapsto t^n$  is strictly increasing on  $[0, \infty)$  and the root is unique.  $\square$

**Remark 5.6.4** (Roots of an equation and the principal radical). Saying that  $x$  is an  $n$ th root of  $a$  means  $x^n = a$ . The symbol  $\sqrt[n]{a}$  instead chooses the *principal real root*. For example, if  $a > 0$ , then

$$x^2 = a \iff x = \pm\sqrt{a},$$

whereas  $\sqrt{a}$  alone denotes the unique nonnegative solution. For even  $n$ , the principal root is nonnegative and exists exactly when  $a \geq 0$ ; for  $a > 0$  the equation also has its negative as a solution. For odd  $n$ , every real  $a$  has exactly one real root, negative when  $a < 0$ , given in that case by  $-\sqrt[n]{-a}$ . These assertions follow from the root theorem and the parity of integer powers. Analysis has now justified the familiar radicals, rather than assuming that writing a symbol produces a solution.

**Remark 5.6.5** (Continuity of roots). Positive square roots were constructed directly in [Proposition 3.4.12](#), in Chapter 3. The general positive  $n$ th roots established above using the IVT also vary continuously. The map  $t \mapsto t^n$  is continuous and strictly increasing on  $[0, \infty)$ . Its inverse  $x \mapsto \sqrt[n]{x}$  is therefore continuous by [Theorem 5.6.2](#). At the endpoint this also has the explicit tolerance

$$0 \leq x < \varepsilon^n \implies 0 \leq \sqrt[n]{x} < \varepsilon.$$

For odd roots on negative inputs, use the sign formula above and the same conclusion on each side of zero.

**Proposition 5.6.6** (Two-variable arithmetic–geometric mean inequality). For  $a, b \geq 0$ ,

$$\boxed{\sqrt{ab} \leq \frac{a+b}{2}.}$$

Equality holds exactly when  $a = b$ .

*Proof.* The quadratic inequality in [Proposition 3.5.2](#) says  $2uv \leq u^2 + v^2$ . Set  $u = \sqrt{a}$  and  $v = \sqrt{b}$ , whose existence has now been established. Then

$$2\sqrt{ab} = 2\sqrt{a}\sqrt{b} \leq a + b.$$

Equality in the quadratic inequality holds exactly when  $\sqrt{a} = \sqrt{b}$ , equivalently  $a = b$ . □

**Proposition 5.6.7** (Laws of real roots). Let  $n \in \mathbb{N}$ , let  $a, b \geq 0$ , and suppose  $b > 0$  in the quotient formula. Then

$$\sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}, \quad \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}.$$

For odd  $n$ , the product law also holds for arbitrary real  $a, b$  when the unique real odd roots are used.

*Proof.* Both sides of the product identity are nonnegative, and raising the right side to the  $n$ th power gives  $ab$ . Uniqueness in [Theorem 5.6.3](#) proves the equality. The quotient proof is identical: the displayed quotient is nonnegative and its  $n$ th power is

$$a/b.$$

For odd  $n$ , signs multiply in the same way, and uniqueness among all real odd roots completes the argument. □

To extend integer exponent laws, the value

$$a^{1/n}$$

for  $a > 0$  must satisfy

$$(a^{1/n})^n = a.$$

Requiring positive values therefore forces the positive  $n$ th root. The same laws then suggest

$$a^{1/n} = \sqrt[n]{a}, \quad a^{m/n} = (a^{1/n})^m = \sqrt[n]{a^m}.$$

The last equality follows by raising both positive expressions to the  $n$ th power and using uniqueness. Thus the progression is from integer powers to solving an equation, choosing its principal root, and extending the exponent. We still have to check that different fractions representing one rational number give the same value.



**Definition 5.6.8** (Positive rational powers). Let  $a > 0$  and  $r \in \mathbb{Q}$ . Choose  $m \in \mathbb{Z}$  and  $n \in \mathbb{N}$  with

$$r = \frac{m}{n},$$

and define

$$a^r = a^{m/n} := \sqrt[n]{a^m}.$$

Integer powers with negative exponent have their usual reciprocal meaning. The next theorem shows that the value is independent of the chosen fraction representing  $r$ .

**Theorem 5.6.9** (Rational-exponent laws). For  $a, b > 0$  and  $r, s \in \mathbb{Q}$ , the preceding definition is well defined and satisfies

$$\begin{aligned} a^{r+s} &= a^r a^s, & a^{r-s} &= \frac{a^r}{a^s}, \\ (a^r)^s &= a^{rs}, & (ab)^r &= a^r b^r. \end{aligned}$$

If  $0 < a < b$  and  $r > 0$ , then  $a^r < b^r$ . For fixed  $a > 1$ , the map  $r \mapsto a^r$  is strictly increasing on  $\mathbb{Q}$ ; for  $0 < a < 1$ , it is strictly decreasing.

*Proof.* Suppose

$$m/n = p/q,$$

with  $n, q > 0$ , and set  $u = \sqrt[q]{a^m}$ ,  $v = \sqrt[n]{a^p}$ . Both are positive, and

$$u^{nq} = a^{mq} = a^{pn} = v^{nq}.$$

Uniqueness of positive  $nq$ th roots gives  $u = v$ . This is the promised well-definedness: the notation depends on the rational number, not on how it is written as a fraction.

For addition, put the exponents over one positive denominator, say  $r = m/n$  and  $s = p/n$ . Both sides below are positive, and

$$(a^r a^s)^n = (a^r)^n (a^s)^n = a^m a^p = a^{m+p} = (a^{r+s})^n.$$

Uniqueness of the positive  $n$ th root therefore gives  $a^{r+s} = a^r a^s$ .

For the iterated-power law, write  $r = m/n$  and  $s = p/q$ , with  $n, q > 0$ . After raising both positive candidates to the common power  $nq$ , integer exponent laws give

$$((a^r)^s)^{nq} = a^{mp} = (a^{rs})^{nq}.$$

Uniqueness of the positive  $nq$ th root proves  $(a^r)^s = a^{rs}$ . The quotient law follows from the addition law applied to  $r - s + s = r$ , and the product law follows by raising both positive sides to a positive denominator of  $r$  and using [Proposition 5.6.7](#).

If  $0 < a < b$  and

$$r = m/n > 0,$$

then  $m, n \in \mathbb{N}$  may be chosen positive. Thus  $a^m < b^m$ . Were  $a^r \geq b^r$ , raising to the  $n$ th power would give  $a^m \geq b^m$ , contradicting  $a^m < b^m$ . Finally,

$$a^s / a^r = a^{s-r} > 1$$

when  $a > 1$  and  $s > r$ ; taking reciprocals gives the statement for  $0 < a < 1$ . □

**Remark 5.6.10** (Why the clean theory uses positive bases). The equality

$$\frac{1}{3} = \frac{2}{6}$$

exposes a representation problem for negative bases. Interpreting each fraction through a principal radical gives

$$\sqrt[3]{-8} = -2, \quad \sqrt[6]{(-8)^2} = 2.$$

Thus the same rational exponent would give incompatible answers under a naive representation-by-representation rule. Some negative-base expressions are meaningful: for example,

$$(-8)^{1/3} = -2,$$

and more generally odd roots of negative numbers exist. Even roots of negative numbers are not real, however, and a reduced rational exponent may impose a parity restriction on its denominator. To avoid hiding these domain issues, the general rational-power theory is formulated for positive bases. Chapter 8 will extend it to arbitrary real exponents, still for positive bases, by means of  $\exp$  and  $\log$ .

## 5.7 Compactness on the Real Line

Compactness is the mechanism that turns local information into finitely many choices and one common scale. We record both formulations used in Part I.

**Definition 5.7.1** (Covers and subcovers). Let  $K \subseteq \mathbb{R}$ . A family  $\{U_\alpha\}_{\alpha \in A}$  **covers**  $K$  if  $K \subseteq \bigcup_{\alpha \in A} U_\alpha$ . It is an **open cover** if every  $U_\alpha$  is open in  $\mathbb{R}$ . If  $B \subseteq A$  and the members indexed by  $B$  still cover  $K$ , they form a **subcover**; it is a **finite subcover** when  $B$  is finite.

A subcover selects members of the original family; it does not replace them by newly constructed sets. The cover condition is containment, not equality: the single interval  $(-1, 2)$  covers  $[0, 1]$ .

**Definition 5.7.2** (Compactness and sequential compactness). A set  $K \subseteq \mathbb{R}$  is **compact** if every open cover of  $K$  has a finite subcover. It is **sequentially compact** if every sequence in  $K$  has a subsequence converging to an element of  $K$ .

**Theorem 5.7.3** (Compactness and sequential compactness on the line). A subset of  $\mathbb{R}$  is compact if and only if it is sequentially compact.

The general equivalence is not proved here. Part I develops the two formulations directly in the concrete settings it needs; Chapter 10 proves the corresponding metric-space result in [Theorem 10.6.3](#).

**Theorem 5.7.4** (Sequential compactness of closed bounded intervals). If  $a \leq b$ , then  $[a, b]$  is sequentially compact.

*Proof.* Let  $(x_n)$  be a sequence in  $[a, b]$ . It is bounded, so [Theorem 4.4.9](#) gives a subsequence converging to some  $x \in \mathbb{R}$ . Since  $a \leq x_{n_k} \leq b$  for every  $k$ , [Theorem 4.2.2](#) gives  $a \leq x \leq b$ . Thus  $x \in [a, b]$ .  $\square$

For an interval  $J$ , write  $\ell(J)$  for its ordinary length.

**Lemma 5.7.5** (Common scale for an interval cover). For every open cover  $\mathcal{U}$  of  $[a, b]$ , there is  $\lambda > 0$  such that every subinterval of  $[a, b]$  having length less than  $\lambda$  is contained in one member of  $\mathcal{U}$ .

*Proof.* Suppose no such  $\lambda$  exists. For each  $n \in \mathbb{N}$ , choose an interval  $J_n \subseteq [a, b]$  of length less than  $1/n$  that lies in no member of  $\mathcal{U}$ , and choose  $x_n \in J_n$ . By [Theorem 5.7.4](#), some subsequence  $x_{n_k}$  converges to  $x \in [a, b]$ . Choose  $U \in \mathcal{U}$  containing  $x$  and  $r > 0$  with  $B(x, r) \subseteq U$ . For all sufficiently large  $k$ ,

$$|x_{n_k} - x| < r/2, \quad \ell(J_{n_k}) < 1/n_k < r/2.$$

Every  $y \in J_{n_k}$  then satisfies  $|y - x| \leq |y - x_{n_k}| + |x_{n_k} - x| < r$ , so  $J_{n_k} \subseteq U$ , a contradiction.  $\square$

**Theorem 5.7.6** (Compactness of closed intervals). Every closed bounded interval  $[a, b]$  is compact.

*Proof.* Given an open cover, take  $\lambda$  from [Lemma 5.7.5](#). Choose a finite partition of  $[a, b]$  whose cells have length less than  $\lambda$ . Each cell lies in one member of the cover, and selecting one such member for each of the finitely many cells gives a finite subcover.  $\square$

**Corollary 5.7.7** (Closed subsets of compact intervals). If  $K \subseteq [a, b]$  is relatively closed in  $[a, b]$ , then  $K$  is compact.

*Proof.* Let  $\mathcal{U}$  be an open cover of  $K$ . Relative closedness makes  $[a, b] \setminus K$  relatively open, so it equals  $[a, b] \cap W$  for some open  $W \subseteq \mathbb{R}$ . Then  $\mathcal{U} \cup \{W\}$  covers  $[a, b]$  and has a finite subcover by [Theorem 5.7.6](#). Discarding  $W$ , if it was selected, leaves finitely many original members that cover  $K$ .  $\square$

Boundedness of the domain supplies a convergent subsequence of trial points, closedness keeps its limit in the domain, and continuity transfers their function values to that limit. All three steps matter:  $x$  on  $(0, 1)$  approaches its supremum without attaining it, while

$$1/x$$

on  $(0, 1]$  is continuous but unbounded.

**Theorem 5.7.8** (Extreme value theorem). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, where  $a \leq b$ . Then there are points  $x_{\min}, x_{\max} \in [a, b]$  such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \text{for every } x \in [a, b].$$

*Proof.* Let  $s = \sup f([a, b])$ . To justify that this supremum exists, suppose first that  $f([a, b])$  were unbounded above. For each  $n$ , choose  $x_n \in [a, b]$  with  $f(x_n) > n$ . By [Theorem 5.7.4](#), a subsequence  $x_{n_k}$  converges to  $x \in [a, b]$ . Continuity gives  $f(x_{n_k}) \rightarrow f(x)$ , but  $f(x_{n_k}) \geq n_k \rightarrow \infty$ , an impossibility. Hence the image is bounded above, and it is nonempty. For each  $n$ , choose  $x_n \in [a, b]$  with

$$f(x_n) > s - 1/n.$$

A convergent subsequence has limit  $x_{\max} \in [a, b]$ . Continuity and the squeeze theorem give  $f(x_{\max}) = s$ . Applying the same argument to  $-f$  gives  $x_{\min}$ .  $\square$

**Theorem 5.7.9** (Heine–Cantor theorem). Every continuous function  $f : [a, b] \rightarrow \mathbb{R}$ , where  $a \leq b$ , is uniformly continuous.

*Proof.* Suppose not. Then some  $\varepsilon_0 > 0$  has the property that, for every  $n$ , there are  $x_n, y_n \in [a, b]$  such that

$$|x_n - y_n| < \frac{1}{n} \quad \text{and} \quad |f(x_n) - f(y_n)| \geq \varepsilon_0.$$

By [Theorem 5.7.4](#), a subsequence  $x_{n_k}$  converges to  $x \in [a, b]$ . Retain those same indices in the paired sequence. Then

$$|y_{n_k} - x| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - x| < \frac{1}{n_k} + |x_{n_k} - x| \rightarrow 0.$$

Thus  $y_{n_k} \rightarrow x$  as well. Choosing a different subsequence of the  $y_n$  would lose the paired estimate, which concerns  $x_n$  and  $y_n$  at one index. Continuity gives both  $f(x_{n_k}) \rightarrow f(x)$  and  $f(y_{n_k}) \rightarrow f(x)$ , so their difference tends to 0, contradicting the second inequality.  $\square$

### Roots, tolerances, and existence in use

The continuous roots constructed near [Theorem 5.6.3](#) now join our earlier continuity tools in concrete existence and tolerance problems.

**Example 5.7.10** (A threshold and a sensitivity guarantee). A tank starts empty and has volume  $V(t) = t + t^2$  litres after  $t$  minutes, for  $0 \leq t \leq 2$ ; this formula is the stated model. The varying input is time and the target is  $V(t) = 3$ . The polynomial  $V(t) - 3$  is continuous, negative at 1 and positive at 2. The IVT guarantees a filling time in  $(1, 2)$  minutes; it does not by itself compute that time. Since

$$V(y) - V(x) = (y - x)(1 + x + y) > 0$$

for  $0 \leq x < y \leq 2$ , it is unique. Moreover  $|V(t) - V(s)| \leq 5|t - s|$  on this operating interval. An input error below

$$\varepsilon/5$$

minutes therefore produces an output error below  $\varepsilon$  litres, uniformly across the interval. If only continuity were known, Heine–Cantor would still supply one tolerance, but without this numerical value. Chapter 8 will obtain Lipschitz constants from derivative bounds.

### Exercises

**Exercise 1.\*** Prove directly that  $5x + 2 \rightarrow 7$  at  $x = 1$ ,  $x^2 \rightarrow 9$  at  $x = 3$ , and

$$1/(x + 1) \rightarrow 1/2$$

at  $x = 1$ . Display all auxiliary bounds and positive  $\delta$  choices.

**Exercise 2.\*** Find the limits at both  $+\infty$  and  $-\infty$  of

$$(2x^2 + 1)/(x^3 - 1),$$

$$(3x^2 - x)/(x^2 + 2), \quad (-x^3 + 1)/(2x + 3).$$

For one infinite limit give the complete target-and-cutoff argument.

**Exercise 3.\*** Find all  $c$  making the function

$$(x^2 - 4)/(x - 2)$$

for  $x \neq 2$ , with value  $c$  at 2, continuous. Compare with

$$x/|x|$$

at zero: can any assigned value remove its discontinuity?

**Exercise 4.\*** Prove  $x^2$  is uniformly continuous on  $[0, 4]$  but not on  $[0, \infty)$ . Prove

$$1/x$$

is Lipschitz on  $[2, \infty)$  and determine whether it is uniformly continuous on  $(0, 2]$ .

**Exercise 5.\*** A continuous temperature reading rises from 18 degrees at time zero to 24 degrees after one hour. Must it equal 21 degrees at some time? Formulate the function, domain, and theorem. Must there be exactly one such time?

**Exercise 6.\*** A device maps an input length  $x \in [1, 3]$  centimetres to an output area  $x^2$  square centimetres. Find one input tolerance ensuring output error below 0.06 square centimetre for every pair of admissible inputs. Explain why the tolerance is uniform.

**Exercise 7.** Using only integer powers and root notation, prove continuity of  $x \mapsto \sqrt[3]{x}$  on  $[0, \infty)$ . Show that it is not Lipschitz near zero.

**Exercise 8.** Give continuous functions showing that the compact-interval hypotheses in the EVT and Heine–Cantor theorem cannot simply be dropped. Specify which conclusion fails in each example.

**Exercise 9.** For

$$D = \{0\} \cup \{1/n : n \in \mathbb{N}\} \cup [2, 3),$$

determine which of 0, 1, 2, 3 belong to  $D$ , are limit points of  $D$ , or are isolated in  $D$ . At which of these points is every function on  $D$  automatically continuous?

**Exercise 10.** As  $h \rightarrow 0$ , classify  $\sqrt{|h|}$ ,  $|h|$ ,  $|h|^2$  according to whether they are  $o(1)$ ,  $O(|h|)$ , or  $o(|h|)$ . Translate each answer into a bound or quotient limit. Repeat the relative-size comparison of  $x$  with  $x^2$  as  $x \rightarrow \infty$  and explain why the remainder does not vanish.

**Exercise 11.** Let  $r = o(g)$  and  $s = O(g)$  in one fixed regime. Prove  $r + s = O(g)$ , and give an example showing that  $r + s = o(g)$  need not hold.

**Exercise 12.** Use the sequential criterion to prove that

$$f(x) = x^2$$

is continuous at every real number.

**Exercise 13.\*** Let

$$f(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

Determine all points at which  $f$  is continuous, and explain your answer using sequences.

**Exercise 14.** Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous and suppose

$$f(a)f(b) < 0.$$

Use the Intermediate Value Theorem to prove that  $f$  has a zero in  $(a, b)$ .

## Chapter 6

# Sequences and Families of Functions

For a sequence of functions,  $f_k$  means the  $k$ th whole function, whereas  $f_k(x)$  is its scalar value at the fixed input  $x$ . Varying  $k$  with  $x$  fixed gives a sequence of numbers.

### 6.1 Pointwise and Uniform Convergence

For functions, convergence has two distinct quantifier patterns. Pointwise convergence lets the stage of approximation depend on the input; uniform convergence does not. This distinction explains exactly when a limit may inherit analytic properties from its approximants.

**Definition 6.1.1** (Pointwise convergence). Let  $D \subseteq \mathbb{R}$ , and let  $f_n : D \rightarrow \mathbb{R}$  and  $f : D \rightarrow \mathbb{R}$ . We say that  $(f_n)$  **converges pointwise** to  $f$  on  $D$  if, for every  $x \in D$ ,

$$\lim_{n \rightarrow \infty} f_n(x) = f(x).$$

**Definition 6.1.2** (Uniform convergence). Under the same hypotheses,  $(f_n)$  **converges uniformly** to  $f$  on  $D$  if, for every  $\varepsilon > 0$ , there is  $N \in \mathbb{N}$  such that, for every  $n \geq N$  and every  $x \in D$ ,

$$|f_n(x) - f(x)| < \varepsilon.$$

**Definition 6.1.3** (Sup norm). If  $h : D \rightarrow \mathbb{R}$  is bounded, define

$$\|h\|_\infty = \sup_{x \in D} |h(x)|.$$

Whenever  $f_n - f$  is bounded, uniform convergence is equivalently

$$\|f_n - f\|_\infty \longrightarrow 0.$$

Indeed, the inequality  $\|f_n - f\|_\infty < \varepsilon$  gives the same error bound at every  $x$ , while a uniform error bound of

$$\varepsilon/2$$

forces the supremum to be at most

$$\varepsilon/2 < \varepsilon.$$

The dependence of the index is worth displaying:

$$\text{pointwise: } \forall x \in D \forall \varepsilon > 0 \exists N = N(x, \varepsilon),$$

$$\text{uniform: } \forall \varepsilon > 0 \exists N = N(\varepsilon) \forall x \in D.$$

Uniform convergence implies pointwise convergence, because one may fix  $x$  after choosing the common stage  $N$ . The converse fails:  $f_n(x) = x^n$  on  $[0, 1]$  converges pointwise to the discontinuous function described in Chapter 1, and hence cannot converge uniformly by Theorem 6.2.1 below.

More directly, the pointwise limit of  $x^n$  on  $[0, 1]$  equals 0 on  $[0, 1)$  and 1 at 1. For every  $n$ , choosing

$$x_n = \sqrt[n]{1/2}$$

gives  $x_n < 1$  and hence  $f(x_n) = 0$ , but

$$|f_n(x_n) - f(x_n)| = \frac{1}{2}.$$

Thus the sup-norm error never tends to zero. In contrast,

$$g_n(x) = x/n$$

on  $[0, 1]$  converges uniformly to zero because

$$\|g_n\|_\infty = \frac{1}{n}.$$

**A working procedure.**

Fix  $x$  first; only then let  $n \rightarrow \infty$ .

Find the pointwise candidate, separating exceptional inputs. To prove uniform convergence, bound the error independently of  $x$ , or compute  $\sup_{x \in D} |f_n(x) - f(x)|$  and show it tends to zero. To disprove it, find *moving witnesses*  $x_n \in D$  and a fixed  $\varepsilon_0 > 0$  with  $|f_n(x_n) - f(x_n)| \geq \varepsilon_0$  for arbitrarily large  $n$ . The witnesses may depend on  $n$ ; the positive gap may not.

**Example 6.1.4** (Cases and domains). For

$$f_n(x) = 1/(1 + nx^2)$$

on  $\mathbb{R}$ , fixing  $x = 0$  gives the limit 1. For fixed  $x \neq 0$ ,

$$0 < f_n(x) \leq 1/(nx^2) \rightarrow 0.$$

Thus the limit has two cases. The witnesses

$$x_n = 1/n$$

have  $f(x_n) = 0$  but

$$f_n(x_n) = 1/(1 + 1/n) \geq 1/2,$$

so convergence is not uniform. On  $D = \{x : |x| \geq d\}$ , where  $d > 0$ ,

$$\sup_D |f_n| = 1/(1 + nd^2) \rightarrow 0 :$$

the same formula converges uniformly on this unbounded domain.

For

$$g_n(x) = x/n,$$



the pointwise limit is zero on  $\mathbb{R}$ . On a bounded interval the error has a common bound:

$$\|g_n\|_{\infty, [-A, A]} = \frac{A}{n}.$$

Choose

$$N > A/\varepsilon$$

for a uniform tolerance  $\varepsilon$ . On  $\mathbb{R}$ , however, the moving input  $x = n$  gives  $g_n(n) = 1$ , disproving uniform convergence. Boundedness of the domain often helps, but the preceding example shows that an unbounded domain does not force failure.

**Example 6.1.5** (A uniform approximation guarantee). Suppose a normalized input  $x \in [0, 1]$  should produce output  $f(x) = x^2$ , while approximation level  $n$  produces

$$f_n(x) = x^2 + x(1 - x)/n.$$

Since

$$0 \leq x(1 - x) = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2 \leq \frac{1}{4},$$

$$\|f_n - f\|_{\infty} = \frac{1}{4n}.$$

Choosing

$$N > 1/(4\varepsilon)$$

guarantees output error below  $\varepsilon$  for every input and every  $n \geq N$ . Testing only a few fixed inputs would not establish this simultaneous guarantee.

**Proposition 6.1.6** (Uniform Cauchy criterion). Let  $f_n : D \rightarrow \mathbb{R}$ . There is a function  $f : D \rightarrow \mathbb{R}$  to which  $f_n$  converges uniformly if and only if, for every  $\varepsilon > 0$ , there is  $N$  such that, for all  $m, n \geq N$  and all  $x \in D$ ,

$$|f_n(x) - f_m(x)| < \varepsilon.$$

*Proof.* If  $f_n \rightarrow f$  uniformly, choose  $N$  so that

$$|f_j(x) - f(x)| < \varepsilon/2$$

for every  $j \geq N$  and  $x \in D$ . For  $m, n \geq N$  the triangle inequality gives  $|f_n(x) - f_m(x)| < \varepsilon$  on all of  $D$ . Conversely, each  $(f_n(x))$  is a Cauchy real sequence, so define  $f(x) = \lim_n f_n(x)$  by completeness. Given  $\varepsilon > 0$ , choose  $N$  from the uniform Cauchy hypothesis with tolerance

$$\varepsilon/2.$$

For fixed  $n \geq N$  and  $x \in D$ , let  $m \rightarrow \infty$ ; order preservation gives

$$|f_n(x) - f(x)| \leq \varepsilon/2 < \varepsilon.$$

The same  $N$  works for every  $x$ , proving uniform convergence. □

## 6.2 Continuous Limits

The principal benefit of uniform convergence is that one approximation index works simultaneously near every point. The three-term estimate below makes this control visible and explains why pointwise convergence is insufficient.

**Theorem 6.2.1** (Uniform limit theorem). Let  $D \subseteq \mathbb{R}$ . If every  $f_n : D \rightarrow \mathbb{R}$  is continuous at  $a \in D$  and  $f_n \rightarrow f$  uniformly on  $D$ , then  $f$  is continuous at  $a$ .

*Proof.* Given  $\varepsilon > 0$ , choose  $N$  such that

$$|f_N(x) - f(x)| < \varepsilon/3$$

for every  $x \in D$ . Continuity of  $f_N$  at  $a$  supplies  $\delta > 0$  such that  $|x - a| < \delta$  implies

$$|f_N(x) - f_N(a)| < \varepsilon/3.$$

For such  $x$ ,

$$|f(x) - f(a)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(a)| + |f_N(a) - f(a)| < \varepsilon.$$

□

**Remark 6.2.2** (What uniform convergence permits). Uniform limits of continuous functions are continuous by the theorem above. Chapter 9 proves that a uniform limit may also be passed through a Riemann integral. Differentiation is more delicate: uniform convergence of  $f_n$  alone does not control the difference quotients, so Chapter 8 imposes uniform convergence on derivatives and fixes one anchor value. Each interchange requires an estimate suited to the operation: an integral uses a bound on function values, while a derivative requires control of difference quotients.

**Proposition 6.2.3** (Uniform convergence respects algebra). Let  $f_n \rightarrow f$  and  $g_n \rightarrow g$  uniformly on  $D$ . Then  $f_n + g_n \rightarrow f + g$  uniformly. If both  $(f_n)$  and  $(g_n)$  are uniformly bounded, then

$$f_n g_n \rightarrow f g$$

uniformly on  $D$ .

*Proof.* For the sum choose  $N_1, N_2$  for the uniform errors below

$$\varepsilon/2;$$

the triangle inequality works for every  $n \geq \max\{N_1, N_2\}$ . For the product let

$$|f_n(x)| \leq M, \quad |g_n(x)| \leq K \quad (n \in \mathbb{N}, x \in D).$$

Taking pointwise limits gives

$$|g(x)| \leq K \quad (x \in D).$$

Then

$$|f_n(x)g_n(x) - f(x)g(x)| \leq M|g_n(x) - g(x)| + K|f_n(x) - f(x)|.$$

Choose  $N_1$  with

$$\|g_n - g\|_\infty < \varepsilon/(2(M+1))$$

for  $n \geq N_1$  and  $N_2$  with

$$\|f_n - f\|_\infty < \varepsilon/(2(K+1))$$

for  $n \geq N_2$ . For every  $n \geq \max\{N_1, N_2\}$  the displayed error is below  $\varepsilon$ , simultaneously at every input. □

**Proposition 6.2.4** (Uniform limits preserve boundedness). Let  $f_n \rightarrow f$  uniformly on  $D$ . If every function  $f_n$  is bounded on  $D$ , then  $f$  is bounded on  $D$ .

*Proof.* Choose  $N$  so that

$$|f(x) - f_N(x)| < 1 \quad (x \in D).$$

If  $|f_N(x)| \leq M$  throughout  $D$ , then

$$|f(x)| \leq M + 1 \quad (x \in D).$$

□

No common bound on the approximating functions is required: only the chosen late function needs to supply a bound. An arbitrary early bounded term cannot do this. On  $\mathbb{R}$ , take  $f_1 = 0$  and  $f_n(x) = x$  for  $n \geq 2$ . Convergence to  $x$  is uniform, but the limit is unbounded.

**Example 6.2.5** (A continuous limit without uniform convergence). On  $[0, 1]$ , put

$$f_n(x) = \max\{1 - n|x - 1/n|, 0\}.$$

Each graph is a continuous triangular spike. At 0 its value is zero; for fixed  $x > 0$ , it is zero once

$$n > 2/x.$$

Thus  $f_n \rightarrow 0$  pointwise, with a continuous limit. Nevertheless

$$f_n(1/n) = 1,$$

so  $\|f_n\|_\infty = 1$  for every  $n$ . Continuity of the limit does not supply the missing uniform control.

**Example 6.2.6** (Pointwise limits need not preserve boundedness). On  $D = (0, 1]$ , define

$$f_n(x) = \begin{cases} n, & 0 < x \leq \frac{1}{n}, \\ \frac{1}{x}, & \frac{1}{n} < x \leq 1. \end{cases}$$

Each  $f_n$  is bounded by  $n$ , while for every  $x \in D$ ,

$$\lim_{n \rightarrow \infty} f_n(x) = \frac{1}{x}.$$

The pointwise limit is unbounded. Thus the uniform hypothesis in [Proposition 6.2.4](#) cannot be weakened to pointwise convergence.

**Example 6.2.7** (Two limits need not be interchanged). For

$$f_n(x) = x^n \quad (0 \leq x < 1),$$

one has

$$\lim_{n \rightarrow \infty} \left( \lim_{x \rightarrow 1^-} f_n(x) \right) = 1,$$

whereas

$$\lim_{x \rightarrow 1^-} \left( \lim_{n \rightarrow \infty} f_n(x) \right) = 0.$$

The convergence in  $n$  is pointwise but not uniform near  $x = 1$ . This is the basic obstruction behind unjustified interchanges of limits.

**Theorem 6.2.8** (Dini's theorem). Let  $f_n \in C([a, b])$ , suppose  $f_n \uparrow f$  pointwise on  $[a, b]$ , and suppose  $f \in C([a, b])$ . Then  $f_n \rightarrow f$  uniformly. The same conclusion holds when  $f_n \downarrow f$ .

Here  $f_n \uparrow f$  pointwise means that for every fixed  $x$ , the numerical sequence  $f_n(x)$  is increasing and converges to  $f(x)$ ; the decreasing notation has the analogous meaning.

*Proof.* Suppose the increasing case is not uniform. Then there are  $\varepsilon > 0$ , integers  $n_j$  that are strictly increasing with  $n_j \rightarrow \infty$ , and points  $x_j \in [a, b]$  such that

$$f(x_j) - f_{n_j}(x_j) \geq \varepsilon.$$

By sequential compactness, pass to a subsequence  $x_{j_k} \rightarrow x \in [a, b]$ . For any fixed  $N$ , once  $n_{j_k} \geq N$ , monotonicity gives

$$0 \leq f(x_{j_k}) - f_{n_{j_k}}(x_{j_k}) \leq f(x_{j_k}) - f_N(x_{j_k}).$$

Letting  $k \rightarrow \infty$  and using continuity of  $f$  and  $f_N$  shows that the right side tends to  $f(x) - f_N(x)$ . Pointwise convergence at the fixed point  $x$  supplies  $N$  with  $f(x) - f_N(x) < \varepsilon/2$ , contradicting the lower bound. For a decreasing sequence apply the increasing case to  $-f_n$ .  $\square$

## 6.3 Polynomial Approximation

Polynomials are among the simplest continuous functions: they can be added, multiplied, and differentiated exactly. How closely can they approximate other continuous functions? Write  $C([a, b])$  for the vector space of continuous real-valued functions on  $[a, b]$ . Every function in this space can be approximated on the whole interval by a polynomial, to any prescribed uniform tolerance.

### Reading the Bernstein weights

For degree two the approximation is

$$B_2(f)(x) = f(0)(1-x)^2 + 2f\left(\frac{1}{2}\right)x(1-x) + f(1)x^2.$$

At

$$x = 1/2$$

its weights are

$$1/4, 1/2, 1/4.$$

For  $f(t) = t^2$  this gives

$$B_2(f)(x) = (x + x^2)/2,$$

not  $x^2$  itself. The approximating polynomial samples nearby values heavily; exact reproduction of every quadratic is not the aim. The proof below establishes concentration using finite algebraic sums, with no probability prerequisite.

**Theorem 6.3.1** (Weierstrass approximation theorem). If  $f \in C([a, b])$ , then for every  $\varepsilon > 0$  there is a polynomial  $p$  such that

$$\|f - p\|_\infty < \varepsilon.$$

Equivalently, polynomials are uniformly dense in  $C([a, b])$ .

*Proof.* We first work on  $[0, 1]$ . For  $f \in C([0, 1])$ , define its  $n$ th Bernstein polynomial by

$$B_n(f)(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

It is a polynomial. Its nonnegative weights have sum 1 by the binomial theorem, so  $B_n(f)(x)$  is a weighted average of sampled values

$$f(k/n).$$

Before estimating a general function, check the samples  $1, t, t^2$ :

$$B_n(1) = 1, \quad B_n(t) = x, \quad B_n(t^2) = x^2 + \frac{x(1-x)}{n}.$$

For weights

$$w_{n,k} = \binom{n}{k} x^k (1-x)^{n-k},$$

the first moment is

$$\sum_{k=0}^n \frac{k}{n} w_{n,k} = x \sum_{k=1}^n \binom{n-1}{k-1} x^{k-1} (1-x)^{n-k} = x.$$

The factorial second moment is similarly

$$\sum_{k=0}^n k(k-1) w_{n,k} = n(n-1) x^2.$$

Since  $k^2 = k(k-1) + k$ , division by  $n^2$  gives the displayed quadratic identity.

These follow without differentiation: use

$$k \binom{n}{k} = n \binom{n-1}{k-1}, \quad k(k-1) \binom{n}{k} = n(n-1) \binom{n-2}{k-2},$$

sum with the binomial theorem (admitted elementary background; an optional induction proof is in [Theorem A.2.8](#)), and write  $k^2 = k(k-1) + k$  (the case  $n = 1$  is direct). Expanding the square now shows that the weights concentrate near

$$k/n = x :$$

$$\sum_{k=0}^n \left(\frac{k}{n} - x\right)^2 \binom{n}{k} x^k (1-x)^{n-k} = \frac{x(1-x)}{n} \leq \frac{1}{4n}.$$

Let  $M = \|f\|_\infty$ . Given  $\varepsilon > 0$ , uniform continuity of  $f$  supplies  $\delta > 0$  such that  $|s - t| < \delta$  implies

$$|f(s) - f(t)| < \varepsilon/2.$$

The mechanism is

near samples are accurate; far samples carry little total weight.

Indeed,

$$|B_n(f)(x) - f(x)| \leq \sum_{k=0}^n w_{n,k}(x) |f(k/n) - f(x)|.$$

Split this sum according as

$$|k/n - x| < \delta$$

or not. The near part is at most

$$\frac{\varepsilon}{2} \sum_{|k/n - x| < \delta} w_{n,k}(x) \leq \frac{\varepsilon}{2}.$$

On the far part, the error is at most  $2M$ , and therefore its contribution is at most

$$2M \sum_{|k/n - x| \geq \delta} w_{n,k}(x).$$

The second-moment estimate gives

$$\sum_{|k/n - x| \geq \delta} w_{n,k}(x) \leq \frac{1}{\delta^2} \sum_{k=0}^n \left(\frac{k}{n} - x\right)^2 w_{n,k}(x) \leq \frac{1}{4n\delta^2}.$$

Choose

$$N > \max\{1, M/(\varepsilon\delta^2)\}.$$

For every  $n \geq N$ , the second contribution is at most

$$M/(2n\delta^2) < \varepsilon/2,$$

uniformly in  $x$ . Hence  $B_n(f) \rightarrow f$  uniformly on  $[0, 1]$ .

For a general interval  $[a, b]$ , apply this result to  $F(t) = f(a + (b - a)t)$  and substitute

$$t = (x - a)/(b - a).$$

The resulting expression is a polynomial in  $x$  and has the required uniform error.  $\square$

The error estimate separates two reasons a contribution can be small:

| samples                 | function error    | total weight          |
|-------------------------|-------------------|-----------------------|
| $ k/n - x  < \delta$    | $< \varepsilon/2$ | $\leq 1$              |
| $ k/n - x  \geq \delta$ | $\leq 2M$         | $\leq 1/(4n\delta^2)$ |

The near samples are accurate because of uniform continuity. The far samples need not be accurate, but together carry little weight because each contributes at least  $\delta^2$  times its weight to the squared distance sum. First choose  $\delta$  from  $f$ , then choose  $N$  as above. Neither choice depends on the evaluation point  $x$ .

**Example 6.3.2** (Approximating a corner). The function  $f(x) = |x|$  on  $[-1, 1]$  is not a polynomial and is not differentiable at 0. Nevertheless, the theorem provides polynomials  $p_n$  with  $\|p_n - |x|\|_\infty \rightarrow 0$ . This does not contradict the fact that uniform limits of differentiable functions need not be differentiable: uniform convergence preserves continuity, but derivatives require the stronger hypotheses discussed in Chapter 8.

**Remark 6.3.3** (Optional perspective: Stone–Weierstrass). For a compact Hausdorff space  $X$ , the real Stone–Weierstrass theorem says that a subalgebra of  $C(X, \mathbb{R})$  which contains the constants and separates points is dense in the sup norm. On  $[a, b]$ , polynomials have these properties, so the classical theorem is its first example. The general result belongs naturally to functional analysis and is not used here; the term *Hausdorff* is defined in [Definition 10.5.5](#).

## 6.4 Equicontinuity

Uniform convergence controls variation in the sequence index. Equicontinuity instead controls variation in the input uniformly across a family; together with compactness it is a central compactness principle for function spaces.

**Definition 6.4.1** (Equicontinuity). Let  $\mathcal{F}$  be a family of functions  $D \rightarrow \mathbb{R}$ . It is **equicontinuous at**  $a \in D$  if, for every  $\varepsilon > 0$ , there is  $\delta > 0$  such that every  $f \in \mathcal{F}$  satisfies

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon \quad (x \in D).$$

It is equicontinuous on  $D$  if it is equicontinuous at every point of  $D$ .

**Proposition 6.4.2.** If  $f_n \rightarrow f$  pointwise on  $D$  and the family  $\{f_n : n \in \mathbb{N}\}$  is equicontinuous at  $a$ , then  $f$  is continuous at  $a$ .

*Proof.* Choose the common input neighborhood first, then take pointwise limits at the two fixed points. Equicontinuity gives  $\delta$  such that

$$|f_n(x) - f_n(a)| < \varepsilon/2$$

for every  $n$  whenever  $x \in D$  and  $|x - a| < \delta$ . Fix such  $x$  and let  $n \rightarrow \infty$ . The inequality becomes

$$|f(x) - f(a)| \leq \varepsilon/2 < \varepsilon.$$

This is continuity at  $a$ ; no uniform convergence was needed. □

**Example 6.4.3** (A shared Lipschitz bound and moving spikes). If  $|f(x) - f(y)| \leq L|x - y|$  for every member of a family, then

$$\delta = \varepsilon/(L + 1)$$

proves equicontinuity at every point. The functions  $x \mapsto c + sx$  with  $|s| \leq 3$  give an example even when their constants  $c$  are unbounded. Equicontinuity controls variation, not absolute heights. The triangular spikes introduced earlier are not equicontinuous at zero:  $f_n(0) = 0$ ,

$$f_n(1/n) = 1, \quad 1/n \rightarrow 0.$$

Every proposed common  $\delta$  fails for a late spike.

**Lemma 6.4.4** (Uniform equicontinuity on a compact interval). Let  $\mathcal{F}$  be an equicontinuous family of functions

$$[a, b] \longrightarrow \mathbb{R}.$$

Then, for every  $\varepsilon > 0$ , there is  $\delta > 0$  such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

for every  $f \in \mathcal{F}$  and every  $x, y \in [a, b]$ .

*Proof.* If not, there would be  $\varepsilon > 0$ , functions  $h_n \in \mathcal{F}$ , and points  $x_n, y_n \in [a, b]$  such that

$$|x_n - y_n| < \frac{1}{n}, \quad |h_n(x_n) - h_n(y_n)| \geq \varepsilon.$$

By compactness of  $[a, b]$ , pass to a subsequence with  $x_n \rightarrow x$ . Then also  $y_n \rightarrow x$ . Relabel this subsequence as  $(x_n, y_n, h_n)$ . Equicontinuity at  $x$ , with tolerance

$$\varepsilon/3,$$

supplies  $\delta > 0$  working for every  $h_n$ . Choose  $N_1, N_2$  so that  $|x_n - x| < \delta$  for  $n \geq N_1$  and  $|y_n - x| < \delta$  for  $n \geq N_2$ . For  $n \geq \max\{N_1, N_2\}$ , both  $|h_n(x_n) - h_n(x)|$  and  $|h_n(y_n) - h_n(x)|$  are below

$$\varepsilon/3.$$

The triangle inequality gives a contradiction. □

**Theorem 6.4.5** (Arzelà–Ascoli on an interval). Let  $(f_n)$  be an equicontinuous sequence in  $C([a, b])$  for which there is  $M > 0$  satisfying

$$|f_n(x)| \leq M \quad (n \in \mathbb{N}, x \in [a, b]).$$

Then it has a subsequence that converges uniformly on  $[a, b]$  to a continuous function.

*Proof.* The idea is to first obtain convergence on a countable dense set and then use equicontinuity to make that convergence uniform. Enumerate a countable dense subset of  $[a, b]$  as  $(q_j)$ . Uniform boundedness and Bolzano–Weierstrass allow us to choose successively a subsequence converging at  $q_1$ , then a subsequence of it converging at  $q_2$ , and so on. The diagonal subsequence can be made explicit. Write the nested increasing index lists as  $(n_k^{(j)})_{k \geq 1}$  and take  $g_k = f_{n_k^{(k)}}$ . Because list  $k+1$  is a subsequence of list  $k$ ,  $n_{k+1}^{(k+1)} \geq n_{k+1}^{(k)} > n_k^{(k)}$ . For fixed  $j$ , all entries with  $k \geq j$  belong to list  $j$ , in increasing order. Thus  $(g_k(q_j))$  converges. The remaining step upgrades these separate convergences to one uniform estimate by using only finitely many nearby test points.

The diagonal selection can be pictured as follows; each row is a subsequence of the preceding row, and the marked diagonal gives  $g_k$ :

|   | 1           | 2           | 3           | ... | convergence guaranteed at |
|---|-------------|-------------|-------------|-----|---------------------------|
| 1 | $n_1^{(1)}$ | $n_2^{(1)}$ | $n_3^{(1)}$ | ... | $q_1$                     |
| 2 | $n_1^{(2)}$ | $n_2^{(2)}$ | $n_3^{(2)}$ | ... | $q_1, q_2$                |
| 3 | $n_1^{(3)}$ | $n_2^{(3)}$ | $n_3^{(3)}$ | ... | $q_1, q_2, q_3$           |



Given  $\varepsilon > 0$ , [Lemma 6.4.4](#) supplies  $\delta > 0$  such that

$$|x - y| < \delta \implies |g_k(x) - g_k(y)| < \varepsilon/3 \quad (k \in \mathbb{N}).$$

Partition  $[a, b]$  into finitely many cells of length less than  $\delta/2$ . For each cell with nonempty relative interior, density supplies one  $q_j$  in that interior; include an endpoint separately if needed. Every  $x \in [a, b]$  then lies within less than  $\delta$  of one selected point. List them as  $q_{j_1}, \dots, q_{j_\ell}$ . At  $q_{j_i}$  choose a Cauchy stage  $N_i$  for tolerance

$$\varepsilon/3.$$

Put  $N = \max\{N_1, \dots, N_\ell\}$ . For  $r, s \geq N$  the values at every selected point differ by less than

$$\varepsilon/3.$$

For a nearby selected point  $q$ , the three-term estimate is

$$|g_r(x) - g_s(x)| \leq \underbrace{|g_r(x) - g_r(q)|}_{\text{equicontinuity}} + \underbrace{|g_r(q) - g_s(q)|}_{\text{finite test points}} + \underbrace{|g_s(q) - g_s(x)|}_{\text{equicontinuity}}.$$

There are only finitely many test points, so taking the maximum of their Cauchy stages gives one  $N$ . The triangle inequality then gives

$$|g_r(x) - g_s(x)| < \varepsilon \quad (x \in [a, b]).$$

Thus  $(g_k)$  is uniformly Cauchy, so [Proposition 6.1.6](#) gives a uniform limit. It is continuous by [Theorem 6.2.1](#).  $\square$

## Exercises

**Exercise 1.\*** Find the pointwise limit of  $x^n$  on  $[-1, 1]$  wherever it exists. Determine uniform convergence on  $[0, r]$  for  $0 < r < 1$ , on  $[0, 1)$ , and on  $[0, 1]$ .

**Exercise 2.** For

$$f_n(x) = x/(1 + nx)$$

on  $[0, \infty)$ , find the pointwise limit and an input-independent error bound. Does unboundedness of the domain prevent uniform convergence?

**Exercise 3.\*** On a normalized input interval  $[0, 1]$ , an approximation adds

$$x(1 - x)/n$$

to the desired output. Derive a worst-case error, and choose a level guaranteeing error below  $10^{-3}$  for every input.

**Exercise 4.** Prove directly that a uniform limit of functions with a shared Lipschitz constant  $K$  has the same Lipschitz constant. Can a uniformly convergent sequence have Lipschitz constants tending to infinity?

**Exercise 5.** On  $[0, 1]$ , let  $f_n(x) = x^n$ . Determine its pointwise limit and prove directly from Definition 6.1.2 that the convergence is not uniform.

**Exercise 6.** Let

$$f_n(x) = x/n$$

for  $x \in \mathbb{R}$ . Prove that  $f_n$  converges pointwise to zero but not uniformly on  $\mathbb{R}$ .

**Exercise 7.** Suppose that  $f_n \rightarrow f$  uniformly on  $D$  and that every  $f_n$  is bounded by the same number  $M$ . Prove that

$$|f(x)| \leq M \quad (x \in D).$$

**Exercise 8.\*** For

$$f_n(x) = nx/(1 + n^2x^2)$$

on  $[0, 1]$ , find the pointwise limit and a moving point preventing uniform convergence. Prove uniform convergence on  $[\delta, 1]$  for each  $\delta > 0$ .

**Exercise 9.** Compute  $B_2(f)$  for  $f(t) = |2t - 1|$ . Compare its error at

$$0, 1/2, 1.$$

Which part of the Bernstein argument depends on continuity, and which part only on the binomial identities?

**Exercise 10.** Show that pointwise convergence to  $f$  and equicontinuity on  $[a, b]$  imply uniform convergence there. Use the continuity of  $f$  proved above and a finite set of nearby test points. Contrast the moving-spike example.

**Exercise 11.** Give a uniformly bounded family that is not equicontinuous, and an equicontinuous family that is not uniformly bounded. Explain why neither alone supplies the hypotheses of Arzelà–Ascoli.

**Exercise 12.** Suppose  $f_n(0) = 0$  and  $|f_n(x) - f_n(y)| \leq 2|x - y|$  on  $[0, 1]$ . Use Arzelà–Ascoli and prove that every uniform subsequential limit also has this Lipschitz bound. Must the original sequence converge?

# Chapter 7

## Infinite Series

### 7.1 Series, Partial Sums, and the Cauchy Criterion

An infinite sum is never formed by adding infinitely many terms at once. Its mathematical meaning is the limit of finite partial sums, so all questions about series begin with the sequence theory already developed.

**Definition 7.1.1** (Series). Let  $(a_n)$  be a real sequence. Its  $N$ th **partial sum** is

$$s_N = \sum_{n=1}^N a_n.$$

The **series** with terms  $(a_n)$  is denoted by

$$\sum_{n=1}^{\infty} a_n.$$

It **converges** when  $(s_N)$  converges; its sum is then

$$\sum_{n=1}^{\infty} a_n = \lim_{N \rightarrow \infty} s_N.$$

**Theorem 7.1.2** (Cauchy criterion for series). The series

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if, for every  $\varepsilon > 0$ , there is  $N \in \mathbb{N}$  such that

$$m > n \geq N \implies \left| \sum_{k=n+1}^m a_k \right| < \varepsilon.$$

*Proof.* For  $m > n$ ,

$$s_m - s_n = \sum_{k=n+1}^m a_k.$$

Thus the stated condition says exactly that the partial sums are Cauchy. Apply [Theorem 4.4.6](#).  $\square$

**Proposition 7.1.3** (Finite changes and telescoping). Changing finitely many terms of a series does not affect convergence. If

$$a_n = b_n - b_{n+1},$$

then

$$\sum_{n=1}^N a_n = b_1 - b_{N+1}.$$

Consequently, this telescoping series converges if and only if  $(b_n)$  converges, and its sum is

$$b_1 - \lim_{n \rightarrow \infty} b_n.$$

*Proof.* Partial sums after a finite change differ by a constant from some point onward. The displayed finite identity follows by cancellation; its limit proves the last assertion.  $\square$

**Proposition 7.1.4** (Necessary condition). If a series converges, then its terms tend to zero.

*Proof.* If  $s_N \rightarrow s$ , then

$$a_N = s_N - s_{N-1} \quad (N \geq 2).$$

The algebra of limits gives  $a_N \rightarrow 0$ .  $\square$

**Remark 7.1.5** (Preliminary strategy for choosing a test). Before trying a sophisticated theorem, check the necessary condition  $a_n \rightarrow 0$ ; failure settles divergence, although success proves nothing by itself. Next look for exact geometric or telescoping structure. For nonnegative terms, compare with an already understood benchmark. Factorials and exponentials often favor the Ratio Test, while expressions raised to the  $n$ th power often favor the Root Test. Alternating signs suggest checking absolute convergence first and then the Alternating-Series Test. An inconclusive test is information about the test, not a verdict about the series.

## 7.2 Nonnegative Series and Comparison

When terms are nonnegative, partial sums can only increase. This lets order and completeness convert estimates by simpler benchmark series into decisive convergence tests.

**Theorem 7.2.1** (Comparison test). Let

$$0 \leq a_n \leq b_n \quad (n \in \mathbb{N}).$$

If

$$\sum_{n=1}^{\infty} b_n$$

converges, then

$$\sum_{n=1}^{\infty} a_n$$

converges. If the latter series diverges, then the former diverges.

*Proof.* The partial sums of  $(a_n)$  are increasing and bounded above by the sum of the series with terms  $b_n$ . Apply [Theorem 4.4.2](#). The second statement is the contrapositive.  $\square$

**Theorem 7.2.2** (Limit comparison test). Let  $a_n, b_n > 0$  eventually, and suppose

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$$

with

$$0 < L < \infty.$$

Then the series with terms  $a_n$  converges if and only if the series with terms  $b_n$  converges.

*Proof.* Choose  $N_0$  from which both sequences are positive, and  $N_1$  from the limit definition with tolerance

$$L/2.$$

For every  $n \geq N = \max\{N_0, N_1\}$ ,

$$\left| \frac{a_n}{b_n} - L \right| < \frac{L}{2}, \quad \frac{L}{2} < \frac{a_n}{b_n} < \frac{3L}{2}.$$

Multiplication by  $b_n > 0$  gives

$$\frac{L}{2}b_n < a_n < \frac{3L}{2}b_n.$$

The upper bound compares

$$\sum a_n$$

with a convergent multiple of

$$\sum b_n;$$

the lower bound can be rewritten

$$b_n < (2/L)a_n$$

for the reverse implication. Apply [Theorem 7.2.1](#) in both directions and [Proposition 7.1.3](#) to discard the finite initial segment. The particular constants do not matter; the tail is controlled by fixed positive multiples.  $\square$

**Theorem 7.2.3** (Geometric series). For  $r \in \mathbb{R}$ , the series with terms  $r^n$ , beginning at  $n = 0$ , converges if and only if

$$|r| < 1.$$

In that case,

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

*Proof.* For  $r \neq 1$ ,

$$\sum_{n=0}^N r^n = \frac{1 - r^{N+1}}{1 - r}.$$

If  $r = 0$ , the conclusion is immediate. If  $0 < r < 1$ , write

$$r = 1/(1+h)$$

with  $h > 0$ . Bernoulli's inequality gives

$$(1+h)^N \geq 1 + Nh,$$

so

$$0 \leq r^N \leq \frac{1}{1 + Nh} \rightarrow 0.$$

The same conclusion holds whenever  $|r| < 1$ , because

$$|r^N| = |r|^N.$$

Taking limits proves the formula. If  $|r| \geq 1$ , the terms do not tend to zero, so [Proposition 7.1.4](#) proves divergence.  $\square$

Multiplying by an initial coefficient  $a$  gives the usual general form:

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r} \quad (|r| < 1).$$

Here  $a$  is the first term and  $r$  the common ratio. If  $a = 0$ , the series is zero for every  $r$ ; the necessity of  $|r| < 1$  applies when  $a \neq 0$ .

**Example 7.2.4** (Two canonical positive series). The terms of both

$$\sum_{n=1}^{\infty} \frac{1}{n} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^2}$$

tend to zero, but only the second series converges.

For the square-reciprocal series, put

$$S_n = \sum_{k=1}^n \frac{1}{k^2}.$$

We claim by induction that

$$\boxed{S_n \leq 2 - \frac{1}{n}}.$$

The base case is  $S_1 = 1 = 2 - 1$ . If the estimate holds at  $n$ , then

$$S_{n+1} \leq 2 - \frac{1}{n} + \frac{1}{(n+1)^2} \leq 2 - \frac{1}{n} + \frac{1}{n(n+1)} = 2 - \frac{1}{n+1}.$$

Thus  $(S_n)$  is increasing and bounded above, so it converges by the monotone convergence theorem for sequences. Euler's solution of the Basel problem gives the deeper exact value

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}},$$

which is stated here without proof and is not needed for convergence.

For the harmonic series, group the terms dyadically:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots.$$

Each displayed block after the first is bounded below by  $1/2$ : for example,

$$\frac{1}{3} + \frac{1}{4} \geq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, \quad \frac{1}{5} + \cdots + \frac{1}{8} \geq 4 \cdot \frac{1}{8} = \frac{1}{2}.$$

In general, formalizing the same count gives

$$H_{2^m} = \sum_{n=1}^{2^m} \frac{1}{n} \geq 1 + \frac{m}{2} \longrightarrow +\infty.$$

Hence the harmonic series diverges. It is the first canonical example in which  $a_n \rightarrow 0$  but  $\sum a_n$  diverges; tending to zero is necessary, not sufficient. Cauchy condensation below abstracts this same dyadic grouping mechanism. The later rational  $p$ -series theorem recovers these as the model cases  $p = 1$  and  $p = 2$ .

The condensation test groups terms into blocks of doubling length:

$$\underbrace{b_1}_{1 \text{ term}}, \quad \underbrace{b_2 + b_3}_{2 \text{ terms}}, \quad \underbrace{b_4 + \cdots + b_7}_{4 \text{ terms}}, \quad \underbrace{b_8 + \cdots + b_{15}}_{8 \text{ terms}}.$$

On the  $k$ th block all  $2^k$  terms lie between the values at its ends. Comparing one block at a time gives the inequalities in the proof; it does not replace every term by an asymptotic guess.

**Theorem 7.2.5** (Cauchy condensation test). Let  $(b_n)$  be a nonincreasing sequence of nonnegative real numbers. Then

$$\sum_{n=1}^{\infty} b_n$$

converges if and only if

$$\sum_{k=0}^{\infty} 2^k b_{2^k}$$

converges.

*Proof.* For every  $k \geq 0$ ,

$$2^k b_{2^{k+1}} \leq \sum_{n=2^k}^{2^{k+1}-1} b_n \leq 2^k b_{2^k}.$$

Summing the right inequalities proves that convergence of the condensed series implies convergence of the original series. Summing the left inequalities gives

$$\frac{1}{2} \sum_{k=1}^{m+1} 2^k b_{2^k} \leq \sum_{n=1}^{2^{m+1}-1} b_n.$$

Hence boundedness of the original partial sums implies boundedness of the condensed partial sums. Both series have nonnegative terms, so monotone convergence completes the proof.  $\square$

**Example 7.2.6** (Recognizing a finite cancellation). For

$$a_n = 1/(n(n+1)),$$

write

$$a_n = 1/n - 1/(n+1).$$

The  $N$ th partial sum is

$$1 - 1/(N+1),$$

so the sum is 1. The finite identity must come before passage to an infinite sum. For comparison,

$$1/(n(n+1) + 1) \leq 1/(n(n+1))$$

for  $n \geq 1$ , so its series converges without an exact formula for its sum.

**Example 7.2.7** (Accumulating repeated contributions). A sequence of corrections adds

$$1/2$$

millimetre initially, and each subsequent correction adds half the preceding amount. After  $N$  corrections the total, in millimetres, is

$$s_N = \sum_{n=1}^N 2^{-n} = 1 - 2^{-N}.$$

The geometric-series theorem proves a limiting total of 1 millimetre, and the unperformed corrections total  $2^{-N}$  millimetres. Thus  $N = 10$  gives an error below

$$1/1000$$

millimetre. Writing an infinite sum only proposes the accumulated quantity; convergence proves that the proposed total is finite.

### 7.3 Absolute Convergence and Products

Signs permit cancellation, which is useful but dangerous: a series can look small only because large positive and negative tails offset each other. Absolute convergence rules out that instability and will justify changing orders and multiplying series.

**Definition 7.3.1** (Absolute and conditional convergence). A series with real terms  $a_n$  **converges absolutely** if

$$\sum_{n=1}^{\infty} |a_n|$$

converges. It **converges conditionally** if it converges but does not converge absolutely.

Absolute convergence controls the total size of the terms and will justify rearrangement. Conditional convergence is weaker, so changing the order of its terms requires separate care.

**Theorem 7.3.2** (Absolute convergence test). Absolute convergence implies convergence.

*Proof.* For  $m > n$ ,

$$\left| \sum_{k=n+1}^m a_k \right| \leq \sum_{k=n+1}^m |a_k|.$$

The Cauchy criterion for the absolute-value series and [Theorem 7.1.2](#) show that the original partial sums are Cauchy.  $\square$

**Theorem 7.3.3** (Weierstrass M-test). Let  $f_n : D \rightarrow \mathbb{R}$  be functions, and suppose that there are numbers  $M_n \geq 0$  such that

$$|f_n(x)| \leq M_n \quad (x \in D, n \in \mathbb{N}).$$

If

$$\sum_{n=1}^{\infty} M_n$$

converges, then

$$\sum_{n=1}^{\infty} f_n$$

converges uniformly and absolutely on  $D$ .



*Proof.* For  $q \geq p$ ,

$$\left| \sum_{n=p}^q f_n(x) \right| \leq \sum_{n=p}^q |f_n(x)| \leq \sum_{n=p}^q M_n.$$

Given  $\varepsilon > 0$ , choose  $N$  from the numerical Cauchy criterion so that

$$\sum_{n=p}^q M_n < \varepsilon$$

whenever  $q \geq p \geq N$ . The final bound is independent of  $x$  and works for all these indices. The uniform Cauchy criterion of Chapter 6 proves uniform convergence. The same estimate applied to  $|f_n|$  proves absolute convergence at every point.  $\square$

**Example 7.3.4** (A uniform majorant). On  $[-1, 1]$ ,

$$\left| \frac{x^n}{2^n} \right| \leq \frac{1}{2^n}.$$

Since

$$\sum_{n=1}^{\infty} 2^{-n}$$

converges, the M-test shows that

$$\sum_{n=1}^{\infty} \frac{x^n}{2^n}$$

converges uniformly and absolutely on  $[-1, 1]$ .

**Theorem 7.3.5** (Rearrangement of an absolutely convergent series). Let

$$\sum_{n=1}^{\infty} a_n$$

converge absolutely, and let  $(a_{\sigma(n)})$  be a rearrangement, where  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$  is a bijection. Then the rearranged series converges absolutely and has the same sum.

*Proof. Strategy.* Wait until every important early term has appeared; all remaining terms belong to a small absolute tail. Put

$$S = \sum_{n=1}^{\infty} a_n, \quad A = \sum_{n=1}^{\infty} |a_n|.$$

Every rearranged absolute partial sum is at most  $A$ , so monotone convergence proves absolute convergence of the rearranged series. Given  $\varepsilon > 0$ , choose  $N$  with

$$\sum_{r>N} |a_r| < \varepsilon/2.$$

Let  $M = \max\{\sigma^{-1}(1), \dots, \sigma^{-1}(N)\}$ . For every  $m \geq M$ ,  $I_m = \{\sigma(1), \dots, \sigma(m)\}$  contains  $\{1, \dots, N\}$ . Hence

$$\left| \sum_{j=1}^m a_{\sigma(j)} - S \right| \leq \left| \sum_{r \in I_m \setminus \{1, \dots, N\}} a_r \right| + \left| \sum_{r=1}^N a_r - S \right| \leq 2 \sum_{r>N} |a_r| < \varepsilon.$$

Thus the sums agree.  $\square$

**Theorem 7.3.6** (Riemann rearrangement phenomenon). If  $\sum a_n$  converges conditionally, then for every  $L \in \mathbb{R}$  there is a rearrangement of the series that converges to  $L$ .

*Proof.* Write  $a_n^+ = \max(a_n, 0)$  and  $a_n^- = \min(a_n, 0)$ .

**Stage 1: the two one-sign subseries diverge.** The positive subseries diverges to  $+\infty$  and the negative subseries to  $-\infty$ . If either had a finite sum, convergence of the original series would force the other to have a finite sum, and then  $\sum |a_n| = \sum a_n^+ - \sum a_n^-$  would converge, contradicting conditional convergence.

**Stage 2: alternate crossings.** Starting from zero, add unused positive terms in their original order until the running sum first exceeds  $L$ . Then add unused negative terms until it first falls below  $L$ , and repeat. Stage 1 guarantees every crossing.

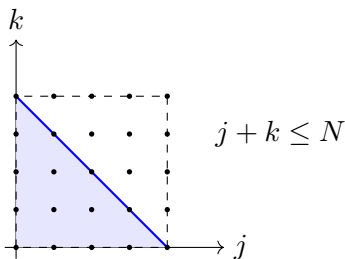
**Stage 3: no nonzero term is lost.** If a particular positive term were never used, only finitely many earlier positive terms could have been used, contradicting the infinitely many upper crossings. The same argument applies to negative terms. Thus every nonzero term eventually appears.

**Stage 4: crossing errors vanish.** At an upper crossing the overshoot is no larger than the last positive term used; at a lower crossing the undershoot is no larger than the absolute value of the last negative term used. Since convergence of the original series implies  $a_n \rightarrow 0$ , both errors tend to zero. Every intervening partial sum lies between the most recent upper and lower crossing values, so squeezing gives convergence to  $L$ .

After arranging all nonzero terms, insert zero terms at arbitrary unused positions; this changes neither convergence nor the sum and completes a rearrangement of the entire original series.  $\square$

Thus absolute convergence gives rearrangement invariance, whereas conditional convergence permits the sum to be changed arbitrarily.

A product of finite partial sums sums over a rectangle of index pairs  $(j, k)$ . A Cauchy product instead collects pairs along diagonals  $j + k = n$ ; its partial sums fill a triangle. Absolute convergence controls the terms in the difference of those two shapes.



**Theorem 7.3.7** (Cauchy product). Suppose that

$$\sum_{n=0}^{\infty} a_n \quad \text{and} \quad \sum_{n=0}^{\infty} b_n$$

converge absolutely. Define

$$c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Then

$$\sum_{n=0}^{\infty} c_n$$

converges absolutely and

$$\left(\sum_{n=0}^{\infty} a_n\right) \left(\sum_{n=0}^{\infty} b_n\right) = \sum_{n=0}^{\infty} c_n.$$

*Proof.* The proof follows the scheme

 absolute convergence  $\longrightarrow$  rectangle versus triangle  $\longrightarrow$  small missing corner.

The partial sums of

$$\sum_{n=0}^{\infty} |c_n|$$

are bounded by the product of the two absolute sums, because

$$\sum_{n=0}^N |c_n| \leq \sum_{\substack{j,k \geq 0 \\ j+k \leq N}} |a_j| |b_k|.$$

Thus the Cauchy product converges absolutely. Let  $\varepsilon > 0$ . Choose  $K$  so large that

$$\left(\sum_{j=K+1}^{\infty} |a_j|\right) \left(\sum_{k=0}^{\infty} |b_k|\right) + \left(\sum_{j=0}^{\infty} |a_j|\right) \left(\sum_{k=K+1}^{\infty} |b_k|\right) < \varepsilon.$$

Put

$$A_N = \sum_{j=0}^N a_j, \quad B_N = \sum_{k=0}^N b_k, \quad C_N = \sum_{n=0}^N c_n.$$

Then

$$A_N B_N = \sum_{0 \leq j, k \leq N} a_j b_k, \quad C_N = \sum_{j+k \leq N} a_j b_k,$$

and hence

$$A_N B_N - C_N = \sum_{\substack{0 \leq j, k \leq N \\ j+k > N}} a_j b_k.$$

For  $N \geq 2K$ , every pair  $(j, k)$  in this missing corner has

$$j > K \quad \text{or} \quad k > K,$$

because  $j \leq K$  and  $k \leq K$  would imply  $j + k \leq 2K \leq N$ . The absolute value of the right side is bounded by

$$\left(\sum_{j>K} |a_j|\right) \left(\sum_{k \geq 0} |b_k|\right) + \left(\sum_{j \geq 0} |a_j|\right) \left(\sum_{k>K} |b_k|\right) < \varepsilon.$$

The finite product tends to the product of the two original sums. Its difference from the Cauchy partial sum tends to zero, proving the formula. No infinite double series has been rearranged in this argument.  $\square$

## 7.4 Ratio, Root, and Alternating Tests

Many series are best compared not term by term with a fixed model, but by examining how consecutive terms or their roots behave. The following tests make that asymptotic comparison precise.

**Rational powers are already available.** Recall from [Definition 5.6.8](#) and [Theorem 5.6.9](#) in Chapter 5 that positive rational powers are well defined, independent of the fraction representing the exponent, and satisfy the usual exponent and order laws. We now use that established algebraic language in the Root Test and the rational (p)-series theorem; constructing it no longer interrupts the series narrative.

**Remark 7.4.1** (Informal calculus preview: the Integral Test). This paragraph recalls familiar calculus without using it as a proof prerequisite. Improper integrals have not yet been formally defined. In this preview only, the familiar notation for arbitrary real powers is allowed; Chapter 8 constructs those powers. The familiar rule is that

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges exactly when  $p > 1$ . The Integral Test says that for a positive continuous decreasing  $f$  on  $[1, \infty)$ , the series and improper integral

$$\sum_{n=1}^{\infty} f(n), \quad \int_1^{\infty} f(x) \, dx$$

have the same convergence behavior.

Recall the calculus procedure: truncate, evaluate the proper integral, then take the endpoint limit. For example,

$$\int_1^b \frac{dx}{x^2} = 1 - \frac{1}{b} \rightarrow 1, \quad \int_1^b \frac{dx}{\sqrt{x}} = 2(\sqrt{b} - 1) \rightarrow +\infty.$$

The preview predicts convergence of

$$\sum 1/n^2$$

and divergence of

$$\sum 1/\sqrt{n}.$$

At singular endpoints use the one-sided limits

$$\lim_{c \rightarrow a^+} \int_c^b f, \quad \lim_{c \rightarrow b^-} \int_a^c f.$$

Different improper pieces must converge separately: cancellation of divergent pieces is not allowed. Chapter 9, especially [Theorem 9.5.11](#), will define these limits and prove the test. The next theorem independently proves the rational- $p$  rule by condensation, with no integration.

**Theorem 7.4.2** (Rational  $p$ -series test). Let

$$p \in \mathbb{Q}.$$

Then

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if

$$p > 1.$$

*Proof.* Suppose first that

$$p > 0.$$

By [Theorem 5.6.9](#), the sequence

$$b_n = n^{-p}$$

is nonincreasing. The Cauchy Condensation Test gives convergence precisely when

$$\sum_{k=0}^{\infty} 2^k (2^k)^{-p} = \sum_{k=0}^{\infty} 2^{k(1-p)}$$

converges. This is a geometric series with positive ratio

$$2^{1-p};$$

by the order statement in [Theorem 5.6.9](#), that ratio is less than 1 exactly when

$$p > 1.$$

If

$$p \leq 0,$$

then

$$n^{-p} \geq 1 \quad (n \in \mathbb{N}),$$

so the terms do not tend to zero. The necessary condition in [Proposition 7.1.4](#) proves divergence.  $\square$

The cases  $p = 1$  and  $p = 2$  are precisely the harmonic and square-reciprocal series treated before condensation.

**Example 7.4.3** (Choosing a benchmark). For

$$a_n = (3n + 1)/(n^3 + 2),$$

compare with

$$b_n = 1/n^2.$$

Then

$$a_n/b_n = (3 + 1/n)/(1 + 2/n^3) \rightarrow 3,$$

so limit comparison and the rational  $p$ -series theorem give convergence. For

$$c_n = 1/\sqrt{n^2 + n},$$

comparison with

$$1/n$$

gives ratio

$$1/\sqrt{1 + 1/n} \rightarrow 1,$$

hence divergence. These conclusions use proved theorems, not the preceding calculus preview.

**Proposition 7.4.4** (Familiar root limits). For  $a > 0$ ,  $\sqrt[n]{a} \rightarrow 1$ , and  $\sqrt[n]{n} \rightarrow 1$ .

*Proof.* If  $a > 1$ , write  $u_n = \sqrt[n]{a} - 1 \geq 0$ . Bernoulli gives  $a = (1 + u_n)^n \geq 1 + nu_n$ , so

$$0 \leq u_n \leq (a - 1)/n.$$

Given  $\varepsilon > 0$ ,

$$N > \max\{1, (a - 1)/\varepsilon\}$$

works. For  $a = 1$  the sequence is constant; for  $0 < a < 1$  use the reciprocal of the root of

$$1/a$$

and the quotient limit law. For  $v_n = \sqrt[n]{n} - 1 \geq 0$  and  $n \geq 2$ , the binomial theorem (admitted elementary background; see [Theorem A.2.8](#) for an optional proof) gives

$$n = (1 + v_n)^n \geq \frac{n(n-1)}{2} v_n^2, \quad v_n \leq \sqrt{\frac{2}{n-1}}.$$

Choose

$$N > 1 + 2/\varepsilon^2;$$

every  $n \geq N$  has  $0 \leq v_n < \varepsilon$ . □

**Theorem 7.4.5** (Ratio test). Let  $a_n \neq 0$  eventually. If there are  $r < 1$  and  $N$  such that

$$\left| \frac{a_{n+1}}{a_n} \right| \leq r \quad (n \geq N),$$

then the series with terms  $a_n$  converges absolutely. If there are  $r > 1$  and  $N$  such that

$$\left| \frac{a_{n+1}}{a_n} \right| \geq r \quad (n \geq N),$$

then the series diverges.

*Proof.* The first inequality gives

$$|a_{N+k}| \leq |a_N| r^k,$$

so geometric comparison applies. The second makes the absolute values eventually increasing and nonzero, so the terms cannot tend to zero. □

The limit version requires one intermediate bound. If

$$|a_{n+1}/a_n| \rightarrow \rho < 1,$$

choose  $\rho < r < 1$ , for instance

$$r = (1 + \rho)/2.$$

The definition of the limit, with error  $r - \rho > 0$ , gives  $N_1$  such that

$$|a_{n+1}/a_n| < r$$

for every  $n \geq N_1$ . The theorem gives geometric comparison; merely knowing that each late ratio is less than 1 would not give this fixed geometric rate. If the limit is  $\rho > 1$ , choose  $1 < r < \rho$  and choose  $N_2$  for error  $\rho - r$ ; every  $n \geq N_2$  has ratio greater than  $r$ .

**Theorem 7.4.6** (Root test). Let  $(a_n)$  be a real sequence. If there are  $r$  and  $N$  such that

$$0 \leq r < 1$$

and

$$|a_n|^{\frac{1}{n}} \leq r \quad (n \geq N),$$

then the series with terms  $a_n$  converges absolutely. If there are  $r > 1$  and  $N$  such that

$$|a_n|^{\frac{1}{n}} \geq r \quad (n \geq N),$$

then the series diverges.

*Proof.* The first inequality gives

$$|a_n| \leq r^n \quad (n \geq N),$$

so geometric comparison applies. The second gives

$$|a_n| \geq r^n \geq 1$$

for every  $n \geq N$ , so the terms do not tend to zero. □

For the root limit

$$|a_n|^{1/n} \rightarrow \rho < 1,$$

again choose  $\rho < r < 1$ . With error  $r - \rho$ , choose  $N_1$  such that

$$|a_n|^{1/n} < r$$

for every  $n \geq N_1$ , hence  $|a_n| < r^n$ . This is the summable geometric majorant. If  $\rho > 1$ , choose  $1 < r < \rho$  and  $N_2$  for error  $\rho - r$ ; then  $|a_n| > r^n$  for every  $n \geq N_2$ . Neither limit test decides the case  $\rho = 1$ .

**Example 7.4.7** (When a test says nothing). For

$$a_n = 1/n^2,$$

$$\frac{a_{n+1}}{a_n} = \left( \frac{n}{n+1} \right)^2 \rightarrow 1,$$

so the common limit form of the Ratio Test is inconclusive. The series nevertheless converges by the rational  $p$ -series theorem. The same ratio limit is 1 for the divergent harmonic series. A value of 1 therefore cannot distinguish the two cases; comparison or the  $p$ -series theorem is the appropriate tool.

**Theorem 7.4.8** (Alternating-series test). Let  $b_n \geq 0$  decrease to 0. Then

$$\sum_{n=1}^{\infty} (-1)^{n+1} b_n$$

converges.

*Proof.* Write

$$s_N = \sum_{n=1}^N (-1)^{n+1} b_n.$$

The two-step differences are

$$s_{2k+2} - s_{2k} = b_{2k+1} - b_{2k+2} \geq 0, \quad s_{2k+3} - s_{2k+1} = -b_{2k+2} + b_{2k+3} \leq 0.$$

Pairing adjacent terms gives  $s_{2k} \geq 0$ , and

$$s_{2k} \leq s_{2k+1} \leq s_1 = b_1.$$

Thus both parity subsequences are bounded and monotone, and have limits by [Theorem 4.4.2](#). Since  $s_{2k+1} - s_{2k} = b_{2k+1} \rightarrow 0$ , those limits agree, say at  $s$ . For a given  $\varepsilon > 0$ , choose  $K_1, K_2$  for the even and odd subsequence errors below  $\varepsilon$ , respectively, and put  $K = \max\{K_1, K_2\}$ . Every  $N \geq 2K$  is either  $2k$  or  $2k+1$  with  $k \geq K$ , so  $|s_N - s| < \varepsilon$ .  $\square$

**Proposition 7.4.9** (Alternating remainder estimate). Let  $b_n \geq 0$  decrease to zero and set

$$S = \sum_{n=1}^{\infty} (-1)^{n-1} b_n.$$

Then  $|S - S_N| \leq b_{N+1}$ .

*Proof.* The even partial sums increase to  $S$  and the odd partial sums decrease to  $S$ , by the preceding proof. Consequently  $S$  lies between  $S_N$  and  $S_{N+1}$ , and

$$|S - S_N| \leq |S_{N+1} - S_N| = b_{N+1}.$$

$\square$

**Example 7.4.10** (Ratios, roots, and alternating signs). For

$$a_n = 3^n/n!,$$

$$a_{n+1}/a_n = 3/(n+1) \leq 1/2$$

for  $n \geq 5$ , so the series converges absolutely. For

$$b_n = ((n+1)/(3n))^n,$$

its  $n$ th root is

$$(n+1)/(3n) \rightarrow 1/3,$$

so the Root Test gives absolute convergence. The series

$$\sum (-1)^{n-1}/\sqrt{n}$$

converges by the Alternating-Series Test because

$$1/\sqrt{n}$$



decreases to zero, but is not absolutely convergent by the rational  $p$ -series test. Its error after  $N$  terms is at most

$$1/\sqrt{N+1};$$

choose

$$N+1 \geq 1/\varepsilon^2$$

for error at most  $\varepsilon$ .

The same ratio mechanism proves growth comparisons. For fixed  $m \in \mathbb{N}$  and  $a > 1$ , the positive terms

$$u_n = n^m/a^n$$

have

$$u_{n+1}/u_n = (1 + 1/n)^m/a \rightarrow 1/a < 1.$$

Choose  $r$  with

$$1/a < r < 1$$

and  $N_0$  with this ratio at most  $r$  for  $n \geq N_0$ . Induction gives  $u_{N_0+j} \leq u_{N_0}r^j$ . For any  $\varepsilon > 0$ , choose  $J$  with

$$r^j < \varepsilon/u_{N_0}$$

for  $j \geq J$ ; then  $N = N_0 + J$  proves  $u_n \rightarrow 0$ . For

$$a^n/n!$$

with  $a > 0$ , choose  $N_0$  so that

$$a/(n+1) \leq 1/2$$

for  $n \geq N_0$  and repeat the argument.

**Choosing among the tests.** First test  $a_n \rightarrow 0$ , then look for geometric or telescoping structure. For positive terms choose a comparison benchmark or compute a ratio to it. Factorials suggest the Ratio Test,  $n$ th powers the Root Test. For alternating signs check monotonicity and decay and separately decide absolute convergence. For power series find the radius and then test each endpoint as a numerical series. Record which theorem decided the answer. *An inconclusive test is not a conclusion about the series.*

**Forward reference: the Integral Test and real exponents.** The Integral Test compares a decreasing sequence of positive terms with an improper integral, so it belongs in Chapter 9. The rational  $p$ -series criterion above uses only Chapter 7 material. Arbitrary real powers have not yet been defined; after Chapter 8 constructs them from  $\exp$  and  $\log$ , Chapter 9 gives an analytic rederivation of the  $p$ -series criterion for every real exponent.

**Remark 7.4.11** (Choosing a convergence test). Once the full collection of tests is available, rational functions of  $n$  usually invite comparison with the power determined by their leading degrees. A positive decreasing term given by a manageable function invites the Integral Test. Factorials and exponentials invite ratios;  $n$ th powers invite roots; signs invite absolute convergence and then alternation. Limit comparison is often cleaner than forcing a termwise inequality. Always record the hypotheses (positivity, monotonicity, or eventual nonvanishing) before computing, and switch tests when the conclusion is inconclusive.

## 7.5 Power Series

Power series are infinite polynomials. Their special feature is a sharp radius inside which convergence is uniform on smaller intervals, allowing algebra and later differentiation to be recovered from finite polynomials.

**Definition 7.5.1** (Power series). For real coefficients  $(c_n)$  and center  $a \in \mathbb{R}$ , a **power series** is

$$\sum_{n=0}^{\infty} c_n (x - a)^n.$$

**Theorem 7.5.2** (Radius of convergence). Every power series has a number

$$R \in [0, \infty]$$

such that it converges absolutely for

$$|x - a| < R$$

and diverges for

$$|x - a| > R.$$

*Proof.* Let  $E$  be the set of  $t \geq 0$  for which

$$\sum_{n=0}^{\infty} |c_n| t^n$$

converges, with  $0^0 = 1$ . Then  $0 \in E$ . If  $t \in E$  and  $0 \leq u < t$ , comparison shows that  $u \in E$ . Let

$$R = \sup E,$$

with  $R = \infty$  if  $E$  is unbounded. Thus  $E$  is downward closed.

Suppose first that  $0 < R < \infty$ . If  $|x - a| < R$ , the defining property of the supremum supplies  $t \in E$  with

$$|x - a| < t;$$

comparison gives absolute convergence.

Conversely, suppose the series converges at a point with

$$z = |x - a| > 0.$$

The convergent terms  $|c_n|z^n$  tend to zero and are therefore bounded; choose  $M > 0$  such that  $|c_n|z^n \leq M$  for every  $n$ . Thus, for  $0 \leq t < z$  and every  $n$ ,

$$|c_n|t^n \leq M \left(\frac{t}{z}\right)^n.$$

Geometric comparison gives  $t \in E$ . If  $z > R$ , choosing  $t$  strictly between  $R$  and  $z$  contradicts the definition of  $R$ . Hence the series diverges outside its radius.

If  $R = \infty$ , then for every finite  $z = |x - a|$  there is  $t \in E$  with  $t > z$ , so comparison gives absolute convergence everywhere. If  $R = 0$ , there are no points with  $0 < |x - a| < R$ ; and convergence at any  $z > 0$  would, by the preceding geometric comparison, put every  $t < z$  in  $E$ , contradicting  $\sup E = 0$ . These cases complete the theorem.  $\square$

The theorem intentionally makes no assertion at points satisfying

$$|x - a| = R.$$

Endpoint behavior depends on the particular coefficients and must be examined separately.

**Theorem 7.5.3** (Uniform convergence inside the radius). Let a power series have radius  $R$ , and let

$$0 \leq \rho < R.$$

Then it converges uniformly on

$$[a - \rho, a + \rho].$$

*Proof.* Choose  $t$  with

$$\rho < t < R.$$

The series

$$\sum_{n=0}^{\infty} |c_n| t^n$$

converges. For  $|x - a| \leq \rho$ ,

$$|c_n(x - a)^n| \leq |c_n| t^n.$$

The Weierstrass M-test, [Theorem 7.3.3](#), proves the assertion.  $\square$

**Example 7.5.4** (The endpoints are separate problems). Consider

$$\sum_{n \geq 1} \frac{x^n}{n}.$$

For  $|x| < 1$ , comparison with the geometric series proves absolute convergence, and on  $[-r, r]$ ,  $r < 1$ , the M-test gives uniform convergence. For  $|x| > 1$  the terms do not tend to zero, so the radius is 1. At  $x = 1$  the harmonic series diverges; at  $x = -1$  the alternating-series test gives convergence, which is not absolute. Thus the convergence interval is  $[-1, 1)$ . The radius theorem settles neither endpoint. Uniform convergence on smaller closed intervals also does not assert uniform convergence on the entire interval  $(-1, 1)$ .

**Example 7.5.5** (Both endpoints, neither endpoint). For

$$\sum_{n \geq 1} \frac{x^n}{n^2},$$

the absolute ratio tends to  $|x|$ , so  $R = 1$ . At both  $x = 1$  and  $x = -1$ , absolute convergence follows from

$$\sum 1/n^2.$$

In contrast

$$\sum_{n \geq 0} x^n$$

has  $R = 1$  but diverges at both endpoints because its terms fail to tend to zero. Together with

$$\sum x^n/n$$

above, these show three possible endpoint patterns. A ratio or root limit of 1 at an endpoint settles none of them.

**Power-series operations deferred.** On every closed interval strictly inside its radius, a power series is uniformly convergent. Termwise differentiation and integration require additional theorems and are proved in Chapters 8 and 9.

## Exercises

**Exercise 1.\*** Choose and justify a test for each series:

$$\sum \frac{n}{n+1}, \quad \sum \frac{1}{n(n+2)}, \quad \sum \frac{2n+1}{n^3+1}, \quad \sum \frac{2^n}{n!}, \quad \sum \left( \frac{n+2}{4n} \right)^n.$$

Compute the sum only for the telescoping example.

**Exercise 2.** Classify

$$\sum \frac{(-1)^{n-1}}{n^2}, \quad \sum \frac{(-1)^{n-1}}{\sqrt{n}}$$

as absolutely convergent, conditionally convergent, or divergent. Give error bounds after  $N$  terms.

**Exercise 3.\*** A traveler walks 4 metres on the first move and on each later move walks one third as far as on the preceding move, always in the same direction. Formulate the distance after  $N$  moves, decide whether the total distance is finite, and bound the distance remaining after  $N$  moves.

**Exercise 4.\*** Using only the explicitly informal calculus preview, classify

$$\sum 1/n^2, \quad \sum 1/\sqrt{n}$$

by truncated integrals. Then replace each preview argument by the independent rational- $p$  theorem. Identify what Chapter 9 must prove to make the first method formal.

**Exercise 5.** Find the radius and test both endpoints of

$$\sum_{n \geq 1} \frac{(x-2)^n}{n2^n}, \quad \sum_{n \geq 1} \frac{(x+1)^n}{n^2 3^n}.$$

State the full intervals of convergence.

**Exercise 6.** Give two positive-term series whose ratio limits both equal 1, with one converging and the other diverging. Explain why an inconclusive test must be replaced, not interpreted as divergence.

**Exercise 7.** Reconstruct the induction proving  $S_n \leq 2 - 1/n$  for the square-reciprocal partial sums, and use Cauchy condensation to recover the harmonic-series conclusion from the main-text dyadic grouping.

**Exercise 8.** Use [Theorem 7.4.8](#) and [Theorem 7.4.2](#) to show that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

converges conditionally.

**Exercise 9.** Find the radius of convergence of

$$\sum_{n=0}^{\infty} 3^n (x - 2)^n.$$

What does [Theorem 7.5.2](#) leave undecided at the two endpoints?

**Exercise 10.** Let

$$\sum_{n=0}^{\infty} c_n (x - a)^n$$

have radius of convergence  $R > 0$ . Prove that it converges absolutely at every point satisfying

$$|x - a| < R.$$

Which step in the proof of [Theorem 7.5.2](#) gives this conclusion?

**Exercise 11.** How many terms of

$$\sum_{n \geq 1} (-1)^{n-1}/n$$

guarantee error at most  $10^{-3}$ ? Explain why a bound on the next term controls the entire tail.

**Exercise 12.\*** Compare the radius and the behavior at both endpoints for

$$\sum_{n \geq 1} x^n, \quad \sum_{n \geq 1} \frac{x^n}{n}, \quad \sum_{n \geq 1} \frac{x^n}{n^2}.$$

For the last numerical benchmark use condensation. Which series converge uniformly on the entire closed interval  $[-1, 1]$ ? Justify each answer.

## Chapter 8

# Differentiation

### 8.1 Derivative and Differentiability

The derivative formalizes instantaneous rate of change and the slope of the best local linear approximation. Its definition uses a function limit.

**Definition 8.1.1** (Derivative). Let  $f : D \rightarrow \mathbb{R}$  and let  $a$  be a limit point of  $D \cap \{x : x \neq a\}$ . Assume also  $a \in D$ . The function is **differentiable at**  $a$  if the following limit exists:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

The number  $f'(a)$  is the **derivative** of  $f$  at  $a$ .

**Definition 8.1.2** (One-sided derivatives). Where the domain permits, define

$$f'_+(a) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}, \quad f'_-(a) = \lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}.$$

At an interior point,  $f'(a)$  exists if and only if both one-sided derivatives exist and are equal; their common value is then  $f'(a)$ .

Differentiability says precisely that the function has a first-order linear approximation:

$$\boxed{f(a+h) = f(a) + f'(a)h + o(|h|) \quad (h \rightarrow 0, a+h \in D).}$$

Indeed, after subtracting the first two terms and dividing by  $|h|$ , the absolute value of the remainder quotient is

$$\left| \frac{f(a+h) - f(a)}{h} - f'(a) \right| \rightarrow 0.$$

Thus the linearization, or tangent-line approximation,

$$L_a(x) = f(a) + f'(a)(x - a)$$

satisfies  $f(x) - L_a(x) = o(|x - a|)$  as  $x \rightarrow a$  through  $D$ . This is the one-dimensional precursor of  $f(a+h) = f(a) + Df(a)h + o(\|h\|)$  in Chapter 11.

**Example 8.1.3** (A derivative from its definition). For  $f(x) = x^2$ , the difference quotient at  $a$  is

$$((a+h)^2 - a^2)/h = 2a + h$$

for  $h \neq 0$ . It tends to  $2a$ : given  $\varepsilon > 0$ , choose  $\delta = \varepsilon$  and  $0 < |h| < \delta$  makes its error exactly  $|h| < \varepsilon$ . More generally factor  $x^n - a^n$  to obtain

$$(x^n - a^n)/(x - a) = x^{n-1} + x^{n-2}a + \cdots + a^{n-1}.$$

Polynomial limits give  $(x^n)' = nx^{n-1}$  for integers  $n \geq 1$ ; constants have derivative zero directly.

**Definition 8.1.4** (Classes  $C^k$  and  $C^\infty$  in one variable). For an open interval  $I$ ,  $f \in C^k(I)$  means that derivatives through order  $k$  exist and are continuous (including  $f$  itself at order zero). For  $[a, b]$ , the derivatives on  $(a, b)$  extend continuously to  $[a, b]$ , with endpoint values given by the one-sided derivatives of Definition 8.1.2. The notation  $C^\infty$  means this holds for every finite order.

**Theorem 8.1.5** (Differentiability implies continuity). If  $f$  is differentiable at  $a$ , then it is continuous at  $a$ .

*Proof.* For  $x \neq a$ ,

$$f(x) - f(a) = (x - a) \frac{f(x) - f(a)}{x - a}.$$

The first factor tends to zero and the second tends to  $f'(a)$ , so the product tends to zero by the algebra of limits. Thus  $f(x) \rightarrow f(a)$ .  $\square$

**Theorem 8.1.6** (Derivative rules). Let  $f, g : D \rightarrow \mathbb{R}$  be differentiable at  $a$ , and let  $c \in \mathbb{R}$ . Then  $f + g$ ,  $cf$ , and  $fg$  are differentiable at  $a$ , with

$$\begin{aligned} (f + g)'(a) &= f'(a) + g'(a), & (cf)'(a) &= cf'(a), \\ (fg)'(a) &= f'(a)g(a) + f(a)g'(a). \end{aligned}$$

If  $g(a) \neq 0$ , then

$$f/g$$

is differentiable at  $a$  and

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

*Proof.* Expand each difference quotient. For products, use

$$f(x)g(x) - f(a)g(a) = f(x)(g(x) - g(a)) + g(a)(f(x) - f(a)).$$

Divide by  $x - a$ , then use continuity of differentiable functions and the algebra of limits. Continuity and  $g(a) \neq 0$  keep  $g(x)$  away from zero near  $a$ . Directly,

$$\frac{1/g(x) - 1/g(a)}{x - a} = -\frac{1}{g(x)g(a)} \frac{g(x) - g(a)}{x - a} \rightarrow -\frac{g'(a)}{g(a)^2}.$$

This establishes differentiability of

$$1/g$$

before any rule is applied to it. The product rule applied to

$$f(1/g)$$

now gives the quotient formula.  $\square$

**Example 8.1.7** (Exceptional points require the definition). Away from 0, the absolute-value function is given by a polynomial and has derivative 1 or  $-1$ . At 0, however,

$$\frac{|h| - |0|}{h} = \begin{cases} 1, & h > 0, \\ -1, & h < 0, \end{cases}$$

so the one-sided derivatives disagree and  $|x|$  is not differentiable there. It is nevertheless continuous. This shows both that continuity is necessary but insufficient and that a formula valid on separate pieces does not settle a junction point.

**Example 8.1.8** (A piecewise differentiability check). Let  $f(x) = x^2$  for  $x \geq 0$  and  $f(x) = -x^2$  for  $x < 0$ . At zero it is continuous since  $|f(x)| = x^2 \rightarrow 0$ . The quotient

$$f(h)/h$$

equals  $|h|$  on both sides, so  $f'(0) = 0$ . The derivative is  $2x$  for  $x > 0$  and  $-2x$  for  $x < 0$ . By contrast replacing the negative branch by  $x$  keeps continuity but makes the left quotient equal to 1 and the right quotient tend to zero. It then fails to be differentiable at the junction.

## 8.2 Chain Rule and Mean Value Theorems

Derivative rules explain how local linear approximations combine. The mean value theorems add a global principle: a net change across an interval is recorded by a derivative at some interior point.

**Theorem 8.2.1** (Chain rule). Let  $f : D \rightarrow \mathbb{R}$  be differentiable at  $a$ , let  $g : E \rightarrow \mathbb{R}$  be differentiable at  $f(a)$ , and suppose  $f(x) \in E$  for all  $x \in D$  sufficiently near  $a$ . Then  $g \circ f$  is differentiable at  $a$ , with

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

*Proof.* Put  $b = f(a)$  and define an auxiliary function on  $E$  by

$$H(y) = \begin{cases} \frac{g(y) - g(b)}{y - b}, & y \neq b, \\ g'(b), & y = b. \end{cases}$$

Differentiability of  $g$  at  $b$  says exactly that the first expression tends to the assigned value  $H(b)$ ; hence  $H$  is continuous at  $b$ . For every  $x$  sufficiently near  $a$ , including those with  $f(x) = b$ ,

$$g(f(x)) - g(b) = H(f(x))(f(x) - b).$$

The identity is valid in the equal-value case because both sides are zero. For  $x \neq a$  divide only by  $x - a$ . Continuity of  $f$  at  $a$  and [Proposition 5.4.5](#), applied after restricting  $D$  to a neighborhood of  $a$  mapped into  $E$ , give  $H(f(x)) \rightarrow H(b) = g'(b)$ , while

$$(f(x) - b)/(x - a) \rightarrow f'(a).$$

The product limit gives  $(g \circ f)'(a) = g'(b)f'(a)$ . The auxiliary function removes the division-by-zero problem before the limit is taken.  $\square$



**Implicit differentiation.** Suppose a differentiable function  $y = y(x)$  is already known to satisfy

$$F(x, y(x)) = 0.$$

One may differentiate this identity by the ordinary chain and product rules and solve for  $y'(x)$  where the resulting coefficient permits it. For example, from

$$x^2 + y(x)^2 = 1$$

one obtains

$$2x + 2y(x)y'(x) = 0, \quad y'(x) = -\frac{x}{y(x)}$$

where  $y(x) \neq 0$ . This calculation does not prove that a differentiable branch  $y(x)$  exists. The implicit function theorem [Theorem 11.5.8](#) supplies a local existence result under appropriate hypotheses in Chapter 11.

**Definition 8.2.2** (Local extrema and critical points). Let  $f : D \rightarrow \mathbb{R}$  and  $c \in D$ . The point  $c$  is a **local maximum** if some neighborhood  $U$  of  $c$  satisfies

$$f(x) \leq f(c) \quad (x \in D \cap U).$$

A **local minimum** reverses this inequality, and a **local extremum** is either one. Thus the definition is relative to the domain, and an endpoint can be a local extremum. If  $c$  is an interior point of  $D$ , it is a **critical point** when either  $f'(c) = 0$  or  $f'(c)$  does not exist.

The nondifferentiable alternative matters:  $f(x) = |x|$  has a strict local minimum at 0, where its derivative does not exist.

**Theorem 8.2.3** (Fermat's theorem on stationary points). If  $f$  is differentiable at an interior local extremum  $c$ , then

$$f'(c) = 0.$$

*Proof.* At a local maximum, for all sufficiently nearby points,

$$x < c \implies \frac{f(x) - f(c)}{x - c} \geq 0, \quad x > c \implies \frac{f(x) - f(c)}{x - c} \leq 0.$$

The two-sided derivative exists, so its left-hand value is nonnegative and its right-hand value is nonpositive; equality of the two forces  $f'(c) = 0$ . At a local minimum the signs are reversed.  $\square$

**Theorem 8.2.4** (Rolle's theorem). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ , where  $a < b$ . If  $f(a) = f(b)$ , then some  $c \in (a, b)$  satisfies  $f'(c) = 0$ .

*Proof.* By [Theorem 5.7.8](#),  $f$  attains a maximum and a minimum. If both values equal  $f(a)$ , then  $f$  is constant and every interior point works. Otherwise an extremum occurs at an interior point  $c$ , and [Theorem 8.2.3](#) gives  $f'(c) = 0$ .  $\square$

**Theorem 8.2.5** (Mean value theorem). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ , where  $a < b$ . Then some  $c \in (a, b)$  satisfies

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

*Proof.* Rolle's theorem requires equal endpoint values. Subtract the secant line joining  $(a, f(a))$  to  $(b, f(b))$ ; the resulting function vanishes at both endpoints, and its derivative is  $f'$  minus the average slope. Thus define

$$h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then  $h(a) = h(b) = 0$ . Apply [Theorem 8.2.4](#) to  $h$  and compute  $h'(c) = 0$ . □

A derivative may be discontinuous, but its discontinuities cannot be arbitrary jumps. The intermediate value property survives even without continuity of the derivative.

**Theorem 8.2.6** (Darboux's theorem for derivatives). If  $F$  is differentiable on an interval  $I$ , then  $F'$  has the intermediate value property. More precisely, if  $x < y$  lie in  $I$  and  $\lambda$  lies strictly between  $F'(x)$  and  $F'(y)$ , then some  $c \in (x, y)$  satisfies  $F'(c) = \lambda$ . At an included endpoint, differentiability is interpreted one-sidedly.

*Proof.* Suppose first that  $F'(x) < \lambda < F'(y)$ , and put  $G(t) = F(t) - \lambda t$ . Then  $G'(x) < 0 < G'(y)$ . For small  $h > 0$ ,

$$\frac{G(x+h) - G(x)}{h} < 0, \quad \frac{G(y-h) - G(y)}{-h} > 0.$$

Thus  $G(x+h) < G(x)$  and  $G(y-h) < G(y)$ , so neither endpoint minimizes  $G$  on  $[x, y]$ . Differentiability implies continuity, and the Extreme Value Theorem supplies a minimum at some  $c \in (x, y)$ . The derivative extremum argument used in Rolle's theorem gives  $G'(c) = 0$ , hence  $F'(c) = \lambda$ . If  $F'(x) > \lambda > F'(y)$ , apply the same argument to  $-F$  and  $-\lambda$ . □

**Definition 8.2.7** (Monotonicity). Let  $I \subseteq \mathbb{R}$  be an interval and  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is **increasing** (or **nondecreasing**) if

$$x < y \implies f(x) \leq f(y),$$

and **strictly increasing** if the final inequality is strict. It is **decreasing** (or **nonincreasing**) if

$$x < y \implies f(x) \geq f(y),$$

and **strictly decreasing** if the final inequality is strict. A function is **monotone** if it is increasing or decreasing, and **strictly monotone** if it is strictly increasing or strictly decreasing. Throughout these notes, “increasing” and “decreasing” are used in the non-strict sense; strict inequalities are indicated by “strictly increasing” and “strictly decreasing.”

**Theorem 8.2.8** (First derivative test). Suppose  $f$  is continuous at  $c$  and differentiable on a punctured neighborhood of  $c$ .

- (a) If some  $\delta > 0$  satisfies  $f'(x) > 0$  for  $c - \delta < x < c$  and  $f'(x) < 0$  for  $c < x < c + \delta$ , then  $c$  is a strict local maximum.
- (b) If the two signs are reversed on those same sides, then  $c$  is a strict local minimum.
- (c) If  $f'$  is positive on both sides or negative on both sides, then  $c$  is not a local extremum.

*Proof.* Take  $x \in (c - \delta, c)$ . Continuity at  $c$  and differentiability on  $(x, c)$  allow the Mean Value Theorem on  $[x, c]$ , so

$$f(c) - f(x) = f'(\xi)(c - x)$$

for some  $\xi \in (x, c)$ . Thus the left sign determines the strict comparison between  $f(x)$  and  $f(c)$ . Applying the same argument on  $[c, x]$  for  $x \in (c, c + \delta)$  gives the right comparison. The three conclusions now follow from their respective sign patterns.  $\square$

The equation  $f'(c) = 0$  alone is not sufficient. For  $f(x) = x^3$ ,  $f'(0) = 0$ , but  $f$  is increasing on both sides and 0 is not an extremum.

**Theorem 8.2.9** (Discontinuities of monotone functions). Let  $f : I \rightarrow \mathbb{R}$  be monotone on an interval. At every interior point it has finite one-sided limits, and its set of discontinuities is at most countable.

*Proof.* Suppose  $f$  is increasing and  $c$  is interior to  $I$ . Define

$$f(c^-) = \sup\{f(x) : x \in I, x < c\}, \quad f(c^+) = \inf\{f(x) : x \in I, x > c\}.$$

Choose  $u, v \in I$  with  $u < c < v$ . Then values to the left of  $c$  are bounded above by  $f(c)$  and below by  $f(u)$  once  $x \geq u$ ; the remaining farther left values are also at most  $f(u)$ . Similarly, values to the right lie between  $f(c)$  and  $f(v)$  locally. Hence the displayed supremum and infimum are finite. To see that the first is the left-hand limit, choose  $x_0 < c$  with  $f(x_0) > f(c^-) - \varepsilon$ . For  $x_0 < x < c$ ,

$$f(c^-) - \varepsilon < f(x) \leq f(c^-).$$

The right-hand argument is identical. Monotonicity also gives

$$f(c^-) \leq f(c) \leq f(c^+).$$

Thus  $f$  is continuous at  $c$  exactly when the two one-sided limits agree, and every discontinuity has a nonempty jump interval  $(f(c^-), f(c^+))$ .

Choose a rational number in each such interval. If  $c < d$ , monotonicity gives  $f(c^+) \leq f(d^-)$ , so the two jump intervals are disjoint. The chosen rationals are therefore distinct, giving an injection of the discontinuity set into  $\mathbb{Q}$ . Endpoint discontinuities, when the endpoints belong to  $I$ , add at most two points. For decreasing  $f$ , apply the increasing case to  $-f$ .  $\square$

Chapter 9 will use both this countability result and a direct Darboux proof that monotone functions are Riemann integrable; the latter does not depend on measure theory.

**Corollary 8.2.10** (Bounded derivative criterion for Lipschitz continuity). If  $f$  is continuous on an interval  $I$ , differentiable in its interior, and  $|f'| \leq M$  there, then

$$|f(y) - f(x)| \leq M|y - x| \quad (x, y \in I).$$

Hence  $f$  is Lipschitz, and therefore uniformly continuous, on  $I$ .

*Proof.* Apply the Mean Value Theorem on the closed interval with endpoints  $x$  and  $y$ ; endpoint cases follow by continuity.  $\square$

**Remark 8.2.11** (Three forms of mean-value control). For a continuous function on an interval, differentiable in its interior, the same increment formula gives

$$\begin{aligned} f' = 0 &\implies f \text{ is constant,} \\ f' \geq 0 &\implies f \text{ is increasing,} \\ |f'| \leq M &\implies |f(y) - f(x)| \leq M|y - x|. \end{aligned}$$

The sign controls direction; a bound on magnitude controls the amount of change. Each conclusion follows by applying the MVT between two points.

The ordinary Mean Value Theorem compares the net change of one function with one derivative. Cauchy's version compares the changes of two functions at once. It will allow L'Hôpital's Rule to relate a quotient of function increments to a quotient of derivatives.

**Theorem 8.2.12** (Cauchy's mean value theorem). Let  $f, g : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ , where  $a < b$ . Then some  $c \in (a, b)$  satisfies

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

*Proof.* Apply [Theorem 8.2.4](#) to

$$h(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x).$$

The endpoint values of  $h$  agree; setting  $h'(c) = 0$  gives the identity.  $\square$

**Infinite limits.** We use the eventual-inequality definitions from Chapter 5. In particular, expressions such as

$$\infty/\infty$$

name an **indeterminate limiting form**; they are not arithmetic quotients of numbers called infinity. The absolute-divergence formulation permits either sign for each function.

**Theorem 8.2.13** (L'Hôpital's rule: the  $0/0$  and  $\infty/\infty$  forms). Let  $f, g : (a, b) \rightarrow \mathbb{R}$  be differentiable, where  $a < b$ . Assume that

$$g'(x) \neq 0 \quad \text{for every } x \in (a, b),$$

and that one of the following alternatives holds:

(a)  $f$  and  $g$  extend continuously to  $a$  with  $f(a) = g(a) = 0$ ;

(b)

$$\lim_{x \rightarrow a^+} |f(x)| = \infty \quad \text{and} \quad \lim_{x \rightarrow a^+} |g(x)| = \infty.$$

If  $L \in \mathbb{R}$  and

$$\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L.$$

then the quotient is defined sufficiently close to  $a$  and

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

*Proof.* First suppose that  $f(a) = g(a) = 0$ . If  $g(x) = 0$  for some  $x > a$ , Rolle's theorem applied to  $g$  on  $[a, x]$  would give a point where  $g'$  vanishes. Thus the quotient is defined. Fix  $x \in (a, b)$ . Cauchy's theorem on  $[a, x]$  gives  $c_x \in (a, x)$  such that

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}.$$

As  $x \rightarrow a^+$ , one has  $a < c_x < x$ , hence  $c_x \rightarrow a^+$ . The assumed derivative-quotient limit therefore gives the conclusion.

Now suppose that both absolute values diverge. Let  $\varepsilon > 0$ . Choose  $\delta > 0$  so that

$$\left| \frac{f'(t)}{g'(t)} - L \right| < \frac{\varepsilon}{4} \quad \text{whenever} \quad a < t < a + \delta,$$

and choose  $c \in (a, a + \delta)$ . For  $x \in (a, c)$ , Rolle's theorem shows that  $g(x) \neq g(c)$ , and Cauchy's Mean Value Theorem on  $[x, c]$  gives a point  $\xi_x \in (x, c)$  for which

$$\frac{f(x) - f(c)}{g(x) - g(c)} = \frac{f'(\xi_x)}{g'(\xi_x)}.$$

Thus this quotient differs from  $L$  by less than

$$\varepsilon/4.$$

Since  $|g(x)| \rightarrow \infty$ , for all  $x$  sufficiently close to  $a$  we also have

$$\left| \frac{g(x) - g(c)}{g(x)} \right| \leq \frac{3}{2} \quad \text{and} \quad \left| \frac{f(c) - Lg(c)}{g(x)} \right| < \frac{\varepsilon}{2}.$$

Writing the desired difference as

$$\frac{f(x)}{g(x)} - L = \frac{g(x) - g(c)}{g(x)} \left( \frac{f(x) - f(c)}{g(x) - g(c)} - L \right) + \frac{f(c) - Lg(c)}{g(x)},$$

the triangle inequality proves that its absolute value is less than  $\varepsilon$ . □

**Infinite derivative-quotient variants.** Under the same hypotheses, if the derivative quotient tends to  $+\infty$  (respectively,  $-\infty$ ), then so does

$$f(x)/g(x).$$

Here is a proof using only finite thresholds. Suppose first that

$$f'/g' \rightarrow +\infty.$$

For the

$$0/0$$

form, let  $K > 0$ . Choose  $\delta > 0$  so that

$$f'(t)/g'(t) > K$$

for  $a < t < a + \delta$ . For  $a < x < a + \delta$ , the identity

$$f(x)/g(x) = f'(c_x)/g'(c_x)$$

with  $a < c_x < x$  gives

$$f(x)/g(x) > K.$$

For the

$$\infty/\infty$$

form, fix a finite target  $K > 0$ , and choose  $c \in (a, b)$  sufficiently close to  $a$  that

$$\frac{f'(t)}{g'(t)} > 2(K + 1) \quad (a < t < c).$$

For  $a < x < c$ , Rolle's theorem ensures  $g(x) \neq g(c)$ , and Cauchy's Mean Value Theorem on  $[x, c]$  gives

$$\frac{f(x) - f(c)}{g(x) - g(c)} > 2(K + 1).$$

For  $x$  sufficiently close to  $a$ ,  $g(x) \neq 0$  and the exact identity is

$$\frac{f(x)}{g(x)} = \frac{g(x) - g(c)}{g(x)} \frac{f(x) - f(c)}{g(x) - g(c)} + \frac{f(c)}{g(x)}.$$

Now freeze  $c$ . Since  $|g(x)| \rightarrow \infty$ , the first factor tends to 1 and the final term to 0. Choose  $x$  still closer to  $a$  so that

$$\frac{g(x) - g(c)}{g(x)} > \frac{1}{2}, \quad \left| \frac{f(c)}{g(x)} \right| < 1.$$

The identity yields

$$f(x)/g(x) > (K + 1) - 1 = K.$$

The order of choices is finite target  $K$ , then  $c$ , then  $x$  with  $c$  fixed. This proves the positive infinite limit. If

$$f'/g' \rightarrow -\infty,$$

apply the positive case to  $-f$  and  $g$ ; it gives

$$-f/g \rightarrow +\infty,$$

hence

$$f/g \rightarrow -\infty.$$

No infinite quantity has been substituted for the finite scalar  $L$ .

**Other domains and forms.** The same proof works for limits as  $x \rightarrow a^-$  and as  $x \rightarrow \pm\infty$ , after the evident changes of domain. For example,

$$\lim_{x \rightarrow \infty} \frac{3x^2 + x}{2x^2 - 1} = \lim_{x \rightarrow \infty} \frac{6x + 1}{4x} = \frac{3}{2}.$$

Here both the numerator and denominator have absolute value tending to infinity; their signs can affect the resulting quotient, but they require no separate theorem. L'Hôpital's Rule applies directly only to quotient forms of type

$$0/0, \quad \infty/\infty.$$

Forms such as  $0 \cdot \infty$ ,  $\infty - \infty$ ,  $1^\infty$ ,  $0^0$ , and  $\infty^0$  must first be transformed into a suitable quotient or logarithmic form.

Applications involving exponential and logarithmic functions follow their construction in [Section 8.5](#); applications involving trigonometric functions follow their construction in [Section 8.6](#). The theorem itself and the rational-function example above use only the results already proved.

### Calculating derivatives and using their meaning

The rules now replace repeated expansion of difference quotients. For  $u(x) = (2x - 1)(x^2 + 3)$ , the product rule gives  $u'(x) = 2(x^2 + 3) + (2x - 1)2x$ . For

$$v(x) = (x^2 + 1)/(x + 2)$$

on  $\mathbb{R} \setminus \{-2\}$ ,

$$v'(x) = \frac{2x(x + 2) - (x^2 + 1)}{(x + 2)^2}.$$

The chain rule gives  $((1 + x^2)^3)' = 6x(1 + x^2)^2$ . Sums and constant multiples combine these results term by term. At excluded points or piecewise junctions, return to the definition.

**Example 8.2.14** (Average and instantaneous speed). A position is modeled by  $s(t) = t^2 + 2t$  metres for  $0 \leq t \leq 2$  seconds. The displacement over this interval is  $s(2) - s(0) = 8$  metres, so the average velocity is 4 metres per second. The model is continuous on  $[0, 2]$  and differentiable inside. The MVT therefore guarantees  $c \in (0, 2)$  with  $s'(c) = 4$ ; since  $s'(t) = 2t + 2$ , this occurs at  $c = 1$  second. Also  $|s'(t)| \leq 6$  on the interval. Thus a time error less than  $\eta$  seconds changes the position by less than  $6\eta$  metres, by [Corollary 8.2.10](#).

**Example 8.2.15** (Constructing and justifying an optimum). A rectangle has perimeter 12 metres. Let  $x$  be one side in metres; the other is  $6 - x$ . The area objective is  $A(x) = x(6 - x)$ , with  $0 < x < 6$ . Extend it continuously to  $[0, 6]$  to include zero-area boundary candidates. The EVT guarantees a maximum there. The natural candidates for an absolute extremum on a closed interval are

interior critical points and endpoints.

Here the only interior critical point satisfies  $A'(x) = 6 - 2x = 0$ , hence  $x = 3$ . The endpoint areas are zero, while  $A(3) = 9$ ; alternatively the derivative is positive before 3 and negative after it. The optimum is therefore attained by a square with sides 3 metres and area 9 square metres. Constructing the domain, finding candidates, justifying the global extremum, and interpreting it are separate steps.

## 8.3 Differentiating Limits and Power Series

To differentiate a series term by term, we need uniform control of its derivative series and convergence at one point. The latter determines the additive constant that derivatives alone cannot detect.

**Theorem 8.3.1** (Differentiation of a sequence of functions). Let  $F_n$  be differentiable on  $I = [\alpha, \beta]$ , and suppose that  $F_n(x_0)$  converges at one  $x_0 \in I$  and  $F'_n \rightarrow g$  uniformly on  $I$ . Then  $F_n$  converges uniformly to a function  $f$ , differentiable in the interior, with  $f' = g$ .

*Proof.* Derivatives control variation but cannot detect additive constants. Convergence at one anchor fixes those constants; uniform Cauchy control of the derivatives then controls variation away from the anchor.

Given  $\varepsilon > 0$ , choose  $N_1$  from uniform Cauchyness of the derivatives with tolerance

$$\varepsilon/(2(1 + \beta - \alpha))$$

and  $N_2$  from Cauchyness at the anchor with tolerance

$$\varepsilon/2.$$

For  $m, n \geq N = \max\{N_1, N_2\}$ , the Mean Value Theorem implies

$$|(F_m(x) - F_n(x)) - (F_m(x_0) - F_n(x_0))| \leq \frac{\varepsilon}{2(1 + \beta - \alpha)}|x - x_0| < \varepsilon/2$$

throughout  $I$ . Adding

$$|F_m(x_0) - F_n(x_0)| < \varepsilon/2$$

proves that  $(F_n)$  is uniformly Cauchy, hence uniformly convergent to  $f$ .

Fix an interior point  $x$  and define, for  $h \neq 0$  with  $x + h \in I$ ,

$$Q_f(h) = \frac{f(x+h) - f(x)}{h}, \quad Q_N(h) = \frac{F_N(x+h) - F_N(x)}{h}.$$

Given  $\varepsilon > 0$ , choose  $N_1$  so that the uniform Cauchy error of the derivatives is below

$$\varepsilon/3$$

for indices at least  $N_1$ . Choose  $N_2$  so that

$$|F'_j(x) - g(x)| < \varepsilon/3$$

for  $j \geq N_2$ , and set  $N = \max\{N_1, N_2\}$ . Then

$$|F'_m(u) - F'_N(u)| < \varepsilon/3$$

for all  $m \geq N, u \in I$ , and

$$|F'_N(x) - g(x)| < \varepsilon/3.$$

The Mean Value Theorem for  $F_m - F_N$  gives

$$\left| \frac{F_m(x+h) - F_m(x)}{h} - Q_N(h) \right| < \varepsilon/3.$$

For each admissible  $h$ , let  $m \rightarrow \infty$ , using the already established convergence of the functions. We obtain

$$|Q_f(h) - Q_N(h)| \leq \varepsilon/3$$

for every such  $h$ , with the same  $N$ .



Now keep that  $N$  fixed. Differentiability of this one function  $F_N$  at  $x$  supplies  $\delta > 0$  such that  $0 < |h| < \delta$  implies

$$|Q_N(h) - F'_N(x)| < \varepsilon/3,$$

and shrink  $\delta$  so  $x + h \in I$ . Then

$$|Q_f(h) - g(x)| \leq |Q_f(h) - Q_N(h)| + |Q_N(h) - F'_N(x)| + |F'_N(x) - g(x)| < \varepsilon.$$

This directly proves  $Q_f(h) \rightarrow g(x)$ , so  $f'(x)$  exists and equals  $g(x)$ . The order is essential: choose  $N$  first, freeze it, then choose  $h$  small using differentiability of that fixed  $F_N$ .  $\square$

For instance  $F_n(x) = n$  has derivative zero for every  $n$ , yet the functions do not converge. Derivatives cannot detect additive constants; the anchor hypothesis is indispensable. The series version is now a direct application to its partial sums.

**Corollary 8.3.2** (Termwise differentiation). If each  $f_n$  is differentiable on  $I$ ,

$$\sum_{n=0}^{\infty} f_n(x_0)$$

converges at one point, and

$$\sum_{n=0}^{\infty} f'_n$$

converges uniformly to  $g$ , then

$$\sum_{n=0}^{\infty} f_n$$

converges uniformly to a differentiable sum  $f$  on the interior and  $f' = g$  there.

*Proof.* Set

$$F_N = \sum_{n=0}^N f_n.$$

The three hypotheses and the conclusions are exactly those of [Theorem 8.3.1](#) for  $(F_N)$ .  $\square$

**Theorem 8.3.3** (Power-series differentiation). Let

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$$

have radius of convergence  $R$ . Then, for

$$|x-a| < R,$$

the function  $f$  is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}.$$

The derivative series has the same radius of convergence. Repeated termwise differentiation is valid at every point strictly inside the radius.

*Proof.* Fix numbers  $\rho, t$  with

$$0 < \rho < t < R.$$

The numerical series

$$\sum_{n=0}^{\infty} |c_n| t^n$$

converges. Put

$$q = \rho/t < 1.$$

The positive terms  $nq^{n-1}$  have consecutive ratio

$$q(1 + 1/n) \rightarrow q < 1,$$

so the ratio test makes their series converge; in particular these terms have a finite bound  $C$ . For  $|x - a| \leq \rho$ ,

$$|nc_n(x - a)^{n-1}| \leq \frac{C}{t} |c_n| t^n.$$

The right side is a summable bound independent of  $x$ . The M-test gives uniform convergence of the derivative series on

$$[a - \rho, a + \rho].$$

The original series converges at  $a$ , so [Corollary 8.3.2](#) gives the formula on that interval. Since every point with distance less than  $R$  from  $a$  lies in such an interval, the formula holds throughout the open convergence interval. This proves that the derivative radius  $R'$  is at least  $R$ . For the reverse inequality, if  $0 < t < R'$ , then

$$\sum_{n \geq 1} |c_n| t^n \leq t \sum_{n \geq 1} n |c_n| t^{n-1} < \infty.$$

Thus every such  $t$  is within the original absolute-convergence range, so  $R \geq R'$ . The same implication handles  $R = 0$  or infinite radii. The radii are equal. Repeating the argument for the differentiated coefficients proves successive differentiation inside this common radius.  $\square$

This theorem is the analytic justification for formal power-series differentiation: the formal derivative represents the derivative of the summed function only after uniform convergence of the derivative series has been verified on smaller closed intervals.

## 8.4 Higher Derivatives and Taylor Approximation

The first derivative describes first-order change. To refine a local approximation beyond its tangent line, we ask how the derivative itself changes. We therefore differentiate the function  $x \mapsto f'(x)$ , not the already evaluated number  $f'(a)$ . Repeating this process provides the coefficients of Taylor polynomials.

**Definition 8.4.1** (Higher derivatives). If  $f'$  is differentiable, write  $f'' = (f')'$ . Inductively set

$$f^{(0)} = f, \quad f^{(n+1)} = (f^{(n)})'$$

whenever the indicated derivative exists.

In one variable each derivative is again a scalar-valued function of one real input, so the sum, product, quotient and chain rules apply repeatedly. For example, repeated use of the power rule gives

$$\frac{d^n}{dx^n} x^p = p(p-1) \cdots (p-n+1)x^{p-n}.$$

For integer powers use their domain of definition; for nonnegative integer  $p$  and  $n > p$  the derivative is identically zero, including at  $x = 0$ . For arbitrary real  $p$ , the formula on  $x > 0$  follows once [Theorem 8.5.15](#) is proved below.

**Proposition 8.4.2** (Higher product rule). For functions with derivatives through order  $n$ ,

$$(fg)^{(n)} = \sum_{j=0}^n \binom{n}{j} f^{(j)} g^{(n-j)}.$$

*Proof.* The order-zero identity is  $fg = fg$ . Suppose the formula holds at order  $n$ , and let  $f, g$  have derivatives through order  $n+1$ . Differentiating the finite sum by the ordinary product rule gives two sums:

$$\begin{aligned} (fg)^{(n+1)} &= \sum_{j=0}^n \binom{n}{j} f^{(j+1)} g^{(n-j)} + \sum_{j=0}^n \binom{n}{j} f^{(j)} g^{(n+1-j)} \\ &= \sum_{k=1}^{n+1} \binom{n}{k-1} f^{(k)} g^{(n+1-k)} + \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n+1-k)}. \end{aligned}$$

The first sum was reindexed with  $k = j + 1$ ; the second merely renamed its index. For  $1 \leq k \leq n$ , the two coefficients add to

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$$

by Pascal's identity. The terms  $k = 0$  and  $k = n+1$  each occur once with coefficient 1. This is the formula at order  $n+1$ , completing induction.  $\square$

For a composition, differentiating  $(g \circ f)' = g'(f)f'$  once more gives the second-order rule

$$(g \circ f)'' = g''(f)(f')^2 + g'(f)f''.$$

The first term measures curvature of  $g$  along the changing input; the second measures the change of speed of that input.

**Definition 8.4.3** (Taylor polynomial). If  $f$  has derivatives through order  $n$  at  $a$ , its Taylor polynomial of degree at most  $n$  centered at  $a$  is

$$P_{n,a}(x) = \sum_{j=0}^n \frac{f^{(j)}(a)}{j!} (x-a)^j.$$

In increment notation,  $T_n f(a; h) = P_{n,a}(a+h)$ .

The aim is to match the value, slope, and successive derivatives at the base point. The factorials are forced by that requirement.

**Proposition 8.4.4** (Derivative matching and uniqueness). The polynomial  $P_{n,a}$  is the unique polynomial of degree at most  $n$  such that  $P_{n,a}^{(j)}(a) = f^{(j)}(a)$  for  $0 \leq j \leq n$ .

*Proof.* The  $j$ th derivative of  $(x - a)^k$  at  $a$  is zero if  $k < j$  (the derivative vanishes identically) or  $k > j$  (a positive power remains). For  $k = j$  it is  $j!$ . Thus only term  $j$  contributes to  $P_{n,a}^{(j)}(a)$ , and the factorial cancels. Every polynomial of degree at most  $n$  can be written uniquely as

$$\sum_{j=0}^n c_j (x - a)^j,$$

by translating its variable. The same calculation forces

$$c_j = f^{(j)}(a)/j!,$$

proving uniqueness. □

Matching derivatives at a point defines a polynomial. It does not yet bound its error away from the point. Taylor's theorem supplies that additional information for this already determined approximation.

Write  $R_n(x) = f(x) - P_{n,a}(x)$  for the **remainder**. The Taylor polynomial is determined by derivative data at one point; Taylor's theorem controls the difference between the function and that polynomial. The first three orders reveal the link to the Mean Value Theorem:

$$\begin{aligned} f(x) &= f(a) + f'(\xi)(x - a), \\ f(x) &= f(a) + f'(a)(x - a) + \frac{f''(\xi)}{2}(x - a)^2, \\ f(x) &= f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2 + \frac{f^{(3)}(\xi)}{6}(x - a)^3. \end{aligned}$$

The intermediate point may differ from line to line; the corresponding smoothness hypotheses are specified in the theorem below. Order 0 is exactly the Mean Value Theorem in increment form. The MVT controls total change using  $f'$ ; Taylor controls successive approximation errors using higher derivatives.

**Theorem 8.4.5** (Taylor's theorem with Lagrange remainder). Let  $f : [a, b] \rightarrow \mathbb{R}$  have  $n + 1$  continuous derivatives on  $[a, b]$ , where  $n \geq 0$ . For every  $x \in [a, b]$  there is a number  $\xi$  between  $a$  and  $x$  such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - a)^{n+1}.$$

*Proof.* The case  $x = a$  is immediate. Fix  $x \neq a$  and write

$$P_n(t) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (t - a)^k.$$

Choose

$$C = \frac{f(x) - P_n(x)}{(x - a)^{n+1}}.$$

Here  $x$  is fixed;  $t$  will be the variable. The auxiliary function

$$H(t) = f(t) - \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (t - a)^k - C(t - a)^{n+1}$$

satisfies

$$H(a) = H(x) = 0.$$

By the definition of the Taylor polynomial,

$$H(a) = H'(a) = \cdots = H^{(n)}(a) = 0.$$

Rolle's theorem first gives a zero of  $H'$  between  $a$  and  $x$ . Together with  $H'(a) = 0$ , that gives two zeros of  $H'$ ; applying Rolle again gives a zero of  $H''$ . Continuing in this way is legitimate because each derivative through order  $n$  is continuous on the closed interval between  $a$  and  $x$  and differentiable in its interior. After  $n + 1$  applications, there is a point  $\xi$  between  $a$  and  $x$  for which

$$H^{(n+1)}(\xi) = 0.$$

Solving this identity for  $C$  gives

$$C = \frac{f^{(n+1)}(\xi)}{(n+1)!}.$$

The equation  $H(x) = 0$  is the required formula. □

In the remainder notation, the conclusion is

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}.$$

Thus an estimate on a higher derivative becomes an estimate on the approximation error, with the distance to the base point raised to the next power.

The same remainder proof works when  $x < a$ : apply Rolle's theorem on the closed interval with endpoints  $x, a$ . Thus a base point may lie inside the domain, and estimates apply on either side.

**Corollary 8.4.6** (Taylor error bound). If  $f$  has  $n + 1$  continuous derivatives on the segment from  $a$  to  $x$  and  $|f^{(n+1)}(t)| \leq M$  there, then

$$|R_n(x)| = |f(x) - P_{n,a}(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}.$$

*Proof.* Take absolute values in the Lagrange remainder and apply the bound at its intermediate point. At  $x = a$  the error is zero. □

**Corollary 8.4.7** (Taylor–Peano formula). If  $f$  is  $C^n$  near  $a$ , with  $n \geq 1$ , then

$$f(a+h) = T_n f(a; h) + o(|h|^n) \quad (h \rightarrow 0).$$

*Proof.* Apply [Theorem 8.4.5](#) at order  $n - 1$ . For  $h \neq 0$  sufficiently small, the remainder after subtracting also the degree- $n$  term is

$$f(a+h) - T_n f(a; h) = \frac{f^{(n)}(a+\theta h) - f^{(n)}(a)}{n!} h^n, \quad 0 < \theta < 1.$$

Continuity of  $f^{(n)}$  at  $a$  makes the coefficient tend to zero, because

$$|a+\theta h - a| = |\theta h| \leq |h| \rightarrow 0$$

for every possible  $0 < \theta < 1$ . Divide by  $|h|^n$ . □

This needs no  $(n + 1)$ st derivative. It prepares the finite-order Peano formula for vector increments in Chapter 11.

**Theorem 8.4.8** (Second derivative test). Suppose  $f'(c) = 0$  and  $f''(c)$  exists. If  $f''(c) > 0$ , then  $c$  is a strict local minimum; if  $f''(c) < 0$ , then  $c$  is a strict local maximum.

*Proof.* Taylor–Peano at  $c$  gives

$$f(c+h) = f(c) + \frac{1}{2}f''(c)h^2 + o(h^2).$$

For sufficiently small nonzero  $h$ , the remainder has absolute value less than  $|f''(c)|h^2/4$ . Since  $h^2 > 0$ , the sign of  $f(c+h) - f(c)$  is then the sign of  $f''(c)$ .  $\square$

The case  $f''(c) = 0$  is inconclusive. At zero,  $x^3$  has no local extremum, whereas  $x^4$  has a strict local minimum.

**Remark 8.4.9** (The multilinear notation hidden in one variable). For  $f : \mathbb{R} \rightarrow \mathbb{R}$ , Chapter 11 will write the  $k$ th derivative as

$$D^k f(a)[h_1, \dots, h_k] = f^{(k)}(a)h_1 \cdots h_k.$$

The base point  $a$  is fixed; the  $k$  increments are independent inputs, and the output is a scalar. This rule is linear in each input and unchanged by permuting them. Setting all inputs equal gives  $D^k f(a)[h, \dots, h] = f^{(k)}(a)h^k$ , which becomes the  $k$ th term in the Taylor polynomial after division by  $k!$ . In several variables the scalar coefficient is replaced by a symmetric  $k$ -linear map.

## 8.5 Exponential, Logarithm, and Real Powers

Integer and rational powers are available, but arbitrary real powers still need a definition. We first construct a function from the behavior we want, prove its laws, and identify its value at 1 with Chapter 4's constant  $e$ . Only then do we introduce its inverse and use that inverse to define real powers. Convergence and termwise differentiation justify the construction.

### Reverse-engineering the exponential

From algebra we expect an exponential  $E$  to satisfy

$$E(x+y) = E(x)E(y), \quad E(0) = 1,$$

and calculus suggests the normalization  $E' = E$ . If a power series

$$E(x) = \sum_{n=0}^{\infty} a_n x^n$$

has this behavior, formal differentiation forces

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{a_0}{n!}, \quad a_0 = E(0) = 1.$$

This motivates the coefficients. It is not yet a definition or a proof: convergence and legitimate differentiation come next.

**Proposition 8.5.1** (Convergence of the exponential series). The series

$$\sum_{n=0}^{\infty} \frac{x^n}{n!}$$

converges absolutely for every real  $x$ .

*Proof.* At zero this is immediate. Otherwise the ratio of consecutive absolute terms is

$$\frac{|x|}{n+1},$$

which tends to zero; apply the Ratio Test. □

**Definition 8.5.2** (Analytic exponential). For  $x \in \mathbb{R}$ , define

$$\exp x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

**Theorem 8.5.3** (Exponential derivative and addition law). For all real  $x, y$ ,

$$\exp'(x) = \exp(x), \quad \exp(0) = 1, \quad \exp(x+y) = \exp(x) \exp(y).$$

*Proof.* The series has infinite radius of convergence, so [Theorem 8.3.3](#) gives the derivative formula. The constant term gives  $\exp(0) = 1$ . For the addition law, absolute convergence permits the Cauchy product and the regrouping by total degree. Hence

$$\begin{aligned} \exp(x) \exp(y) &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{x^j y^k}{j! k!} = \sum_{n=0}^{\infty} \sum_{j=0}^n \frac{x^j y^{n-j}}{j! (n-j)!} \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{j=0}^n \binom{n}{j} x^j y^{n-j} = \sum_{n=0}^{\infty} \frac{(x+y)^n}{n!} = \exp(x+y), \end{aligned}$$

where the penultimate equality is the finite binomial theorem, proved by elementary induction. □

**Theorem 8.5.4** (The series value at 1 is the sequential constant). For the number  $e$  constructed in [Definition 4.4.4](#),

$$\boxed{\exp(1) = e.}$$

*Proof.* The binomial expansion used in [Theorem 4.4.3](#) gives

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

For each fixed  $k$ , the finite product tends to 1 as  $n \rightarrow \infty$ . Moreover it lies between 0 and 1. Given  $\varepsilon > 0$ , choose  $K$  so that

$$\sum_{k=K+1}^{\infty} \frac{1}{k!} < \frac{\varepsilon}{2};$$

this is possible because the exponential series converges at 1. For the finitely many indices  $0 \leq k \leq K$ , choose  $N$  so large that the sum of the coefficient errors is less than

$$\varepsilon/2$$

whenever  $n \geq N$ . The remaining terms, whether they occur in the binomial sum or only in the infinite exponential series, contribute at most the displayed tail. Hence

$$\left| \left(1 + \frac{1}{n}\right)^n - \exp(1) \right| < \varepsilon$$

for all sufficiently large  $n$ . The sequential definition of  $e$  now gives  $\exp(1) = e$ .  $\square$

**Definition 8.5.5** (The base- $e$  exponential). Only now, after identifying the independently constructed constant and function, define

$$e^x := \exp(x) \quad (x \in \mathbb{R}).$$

Thus  $e^x$  is not notation assumed in advance: it is the real-exponent extension associated with the Chapter 4 constant  $e$ .

**Theorem 8.5.6** (Exponential bijection). The function  $\exp$  is a strictly increasing bijection from  $\mathbb{R}$  onto

$$(0, \infty).$$

*Proof.* The exponential identity gives

$$\exp\left(\frac{x}{2}\right) \exp\left(-\frac{x}{2}\right) = \exp(0) = 1.$$

Thus

$$\exp(x/2) \neq 0,$$

and consequently

$$\exp(x) = \exp\left(\frac{x}{2}\right)^2 > 0.$$

Its derivative is positive, so the Mean Value Theorem makes it strictly increasing. Its series gives

$$\exp(n) \geq 1 + n \quad (n \in \mathbb{N}),$$

so for any  $M > 0$ , choose an integer  $n > M$ ; then  $x \geq n$  implies  $\exp(x) \geq \exp(n) \geq 1 + n > M$ . Hence  $\exp(x) \rightarrow \infty$  as  $x \rightarrow \infty$ . For  $t > 0$ , the addition law gives

$$\exp(-t) = \frac{1}{\exp(t)}.$$

Given  $\varepsilon > 0$ , choose  $T$  with  $\exp(T) > 1/\varepsilon$ . Then  $x < -T$  gives  $0 < \exp(x) < \varepsilon$ , so  $\exp(x) \rightarrow 0$  as  $x \rightarrow -\infty$ . If  $y > 0$ , the two limits provide real numbers  $A < B$  such that

$$\exp(A) < y < \exp(B).$$

Continuity and the Intermediate Value Theorem then give an  $x \in (A, B)$  with  $\exp(x) = y$ . Positivity shows that no other values occur, proving the asserted range.  $\square$



**Definition 8.5.7** (Natural logarithm). Define

$$\log : (0, \infty) \rightarrow \mathbb{R}, \quad \log = (\exp)^{-1}.$$

The symbol  $\log$  is the primary notation in these notes. The alternative symbol  $\ln$  is widespread and denotes exactly the same natural logarithm.

Strict monotonicity gives one inverse value, but does not guarantee a finite derivative of the inverse. A nonzero slope permits division in the inverse difference quotient. For example,  $x^3$  is strictly increasing, but its inverse has quotient

$$y^{1/3}/y = y^{-2/3}$$

for  $y > 0$ , which is unbounded as  $y \rightarrow 0^+$ .

**Theorem 8.5.8** (Inverse derivative theorem). Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f : I \rightarrow \mathbb{R}$  be continuously differentiable and strictly monotone. If

$$f'(x) \neq 0 \quad (x \in I),$$

then its inverse on  $f(I)$  is differentiable, with

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

In particular,

$$\log'(x) = \frac{1}{x} \quad (x > 0).$$

*Proof. Strategy.* Take the reciprocal of the original difference quotient. Put  $g = f^{-1}$  and  $y_0 = f(x_0)$ . By [Theorem 5.6.2](#),  $g$  is continuous. For  $y \neq y_0$  in  $f(I)$ , put  $x = g(y)$ . Injectivity gives  $x \neq x_0$ , and

$$\frac{g(y) - g(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)} = \left( \frac{f(x) - f(x_0)}{x - x_0} \right)^{-1}.$$

As  $y \rightarrow y_0$ , continuity of  $g$  gives  $x \rightarrow x_0$ . The last expression tends to

$$1/f'(x_0),$$

since  $f'(x_0) \neq 0$ . This proves differentiability before any differentiation of  $g \circ f = \text{id}$  is legitimate. Applying the formula to  $f = \exp$  gives

$$\log'(x) = \frac{1}{\exp(\log x)} = \frac{1}{x}.$$

□

**Proposition 8.5.9** (Arithmetic laws of the natural logarithm). For  $a, b > 0$ ,

$$\log 1 = 0, \quad \log(ab) = \log a + \log b,$$

$$\log\left(\frac{a}{b}\right) = \log a - \log b, \quad \log(a^{-1}) = -\log a.$$

*Proof.* Because  $\exp 0 = 1$  and  $\log$  is the inverse of  $\exp$ ,  $\log 1 = 0$ . Write  $a = \exp(\log a)$  and  $b = \exp(\log b)$ . The exponential addition law gives

$$ab = \exp(\log a + \log b).$$

Applying the inverse function  $\log$  proves the product law. Since

$$a/b = ab^{-1},$$

the identity  $\exp(-\log b) = b^{-1}$  and the product law give the remaining two formulas.  $\square$

Logarithms turn products, not sums, into sums. In general

$$\log(a + b) \neq \log a + \log b;$$

for instance,

$$\log(1 + 1) = \log 2 \neq 0 = \log 1 + \log 1.$$

Chapter 5 defined  $a^q$  for rational  $q$ . What should  $a^x$  mean when  $x$  is irrational? This is an extension problem: the new values must agree with the old ones and preserve the exponent laws. Logarithms turn multiplication into addition, since the exponential addition law and injectivity give  $\log(ab) = \log a + \log b$ . Thus the logarithm of a power should be the exponent times  $\log a$ . Applying  $\exp$  suggests the following definition.

**Definition 8.5.10** (Real powers of positive numbers). For  $a > 0$  and  $x \in \mathbb{R}$ , define

$$a^x := e^{x \log a}.$$

The definition agrees with the rational powers constructed in Definition 5.6.8. Indeed, for

$$r = m/n,$$

$n > 0$ , set

$$y = e^{(m/n) \log a} > 0.$$

The exponential addition law gives  $y^n = e^{m \log a} = a^m$ , also for negative  $m$  by the reciprocal identity. Uniqueness of a positive  $n$ th root therefore identifies  $y$  with the Chapter 5 value

$$a^{m/n}.$$

**Proposition 8.5.11** (Real-exponent laws). For  $a, b > 0$  and  $x, y \in \mathbb{R}$ ,

$$\begin{aligned} a^{x+y} &= a^x a^y, & a^{x-y} &= \frac{a^x}{a^y}, \\ (a^x)^y &= a^{xy}, & (ab)^x &= a^x b^x. \end{aligned}$$

Moreover,

$$\log(a^x) = x \log a.$$

The general definition requires a positive base. Separately, one may set  $0^x = 0$  for  $x > 0$ ; it does not define  $0^0$  or negative powers of zero. A negative base has no real-valued power  $a^x$  for every real  $x$ .

*Proof.* The first two identities follow immediately from the exponential addition law. The inverse identity

$$\log(a^x) = \log(e^{x \log a}) = x \log a$$

proves the displayed logarithm law, and the third exponent law follows from it and the definition. The identity  $\log(ab) = \log a + \log b$ , obtained by applying the inverse of  $\exp$  to its addition law, proves the fourth.  $\square$

**Theorem 8.5.12** (Exponential functions with arbitrary positive base). Fix  $a > 0$  and define

$$E_a : \mathbb{R} \rightarrow (0, \infty), \quad E_a(x) = a^x = e^{x \log a}.$$

Then

$$E'_a(x) = a^x \log a.$$

If  $a > 1$ , then  $E_a$  is strictly increasing; if  $0 < a < 1$ , then  $E_a$  is strictly decreasing; and if  $a = 1$ , then  $E_a(x) = 1$  for every  $x$ . For  $a \neq 1$ , the function  $E_a$  is a bijection from  $\mathbb{R}$  onto  $(0, \infty)$ .

*Proof. Strategy.* Differentiate the defining composite, then let the sign of  $\log a$  control monotonicity. Surjectivity will come from solving the equation  $a^x = y$  explicitly with the already constructed natural logarithm. The chain rule gives

$$E'_a(x) = e^{x \log a} \log a = a^x \log a.$$

The natural logarithm is strictly increasing because it is the inverse of the strictly increasing exponential, and  $\log 1 = 0$ . Thus  $\log a$  is positive when  $a > 1$  and negative when  $0 < a < 1$ . Since  $a^x > 0$ , the Mean Value Theorem gives the stated strict monotonicity. If  $a = 1$ , the definition gives  $1^x = e^{x \log 1} = 1$ .

Now suppose  $a \neq 1$ . Strict monotonicity gives injectivity. For any  $y > 0$ ,  $\log a \neq 0$ , and

$$x = \frac{\log y}{\log a}$$

satisfies

$$a^x = e^{\frac{\log y}{\log a} \log a} = e^{\log y} = y.$$

Hence the actual range is all of  $(0, \infty)$ , proving bijectivity with the displayed codomain.  $\square$

**Definition 8.5.13** (Logarithm to base  $a$ ). Let  $a > 0$  and  $a \neq 1$ . After restricting to such a base, [Theorem 8.5.12](#) proves that  $E_a : \mathbb{R} \rightarrow (0, \infty)$  is a bijection. Define its inverse

$$\log_a : (0, \infty) \rightarrow \mathbb{R}.$$

Equivalently,

$$\log_a x = y \iff a^y = x.$$

The base  $a = 1$  is excluded because  $1^x$  is constant and therefore is not invertible onto  $(0, \infty)$ . No logarithm with a nonpositive base is being defined here.

By the two inverse identities,

$$\log_a(a^x) = x \quad (x \in \mathbb{R}), \quad a^{\log_a x} = x \quad (x > 0).$$

Taking the natural logarithm of  $a^y = x$  gives  $y \log a = \log x$ . Therefore

$$\boxed{\log_a x = \frac{\log x}{\log a}} \quad (a > 0, a \neq 1, x > 0).$$

This change-of-base formula and [Propositions 8.5.9](#) and [8.5.11](#) give, for  $x, y > 0$  and  $r \in \mathbb{R}$ ,

$$\log_a(xy) = \log_a x + \log_a y, \quad \log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y,$$

$$\log_a(x^r) = r \log_a x.$$

Differentiating the change-of-base formula yields

$$\boxed{(\log_a x)' = \frac{1}{x \log a}}.$$

Thus  $\log_a$  is increasing when  $a > 1$  and decreasing when  $0 < a < 1$ , as is also forced by its being the inverse of the corresponding increasing or decreasing exponential.

The notation used in these notes is

$$\boxed{\log x := \log_e x}, \quad \ln x = \log x, \quad \boxed{\lg x := \log_{10} x}.$$

The symbol  $\lg$  for the common, or decimal, logarithm is familiar but not universal, so its meaning is fixed explicitly here. Elementary values include

$$\log e = 1, \quad \lg 10 = 1, \quad \lg 100 = 2, \quad \lg 1000 = 3.$$

**Example 8.5.14** (Solving an exponential equation). The equation  $2^x = 5$  has exactly one solution because  $2^x$  is strictly increasing. Applying the inverse, or using change of base, gives

$$x = \log_2 5 = \frac{\log 5}{\log 2}.$$

Thus arbitrary-base logarithms recover an unknown exponent even when the base is not  $e$ .

**Theorem 8.5.15** (Derivative of a real power). For every  $p \in \mathbb{R}$ , the function

$$x \mapsto x^p$$

is differentiable on  $(0, \infty)$  and

$$\frac{d}{dx} x^p = p x^{p-1}.$$

*Proof.* The chain rule and [Theorem 8.5.8](#) give

$$\frac{d}{dx} e^{p \log x} = e^{p \log x} \frac{p}{x} = p x^{p-1}.$$

□

**Proposition 8.5.16** (A familiar compound-growth limit). For every fixed  $x \in \mathbb{R}$ ,

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x.$$

The base is positive for all sufficiently large  $n$ .

*Proof.* Choose  $N$  so that

$$1 + x/n > 0$$

for  $n \geq N$ . Then

$$\log \left(1 + \frac{x}{n}\right)^n = n \log \left(1 + \frac{x}{n}\right) = x \frac{\log(1 + x/n)}{x/n} \rightarrow x,$$

with the case  $x = 0$  immediate. The logarithmic quotient tends to 1 because it is the derivative of  $\log t$  at  $t = 1$ , after writing

$$t = 1 + x/n.$$

Continuity of  $\exp$  and [Theorem 8.5.4](#) give the result.  $\square$

## Recognizing hidden exponential limits

Logarithms turn a changing exponent into a multiplier:

$$u_n^{v_n} = e^{v_n \log u_n} \quad (u_n > 0).$$

The derivative of  $\log$  at 1 says that

$$\frac{\log(1+h)}{h} \rightarrow 1$$

as  $h \rightarrow 0$ . Start with concrete cases. If

$$u_n = 1 + 3/n,$$

then

$$n \log u_n = 3 \frac{\log(1 + 3/n)}{3/n} \rightarrow 3, \quad u_n^n \rightarrow e^3.$$

For a quotient, first expose the small increment:

$$\frac{n+3}{n+1} = 1 + \frac{2}{n+1}, \quad n \log \left(\frac{n+3}{n+1}\right) = \frac{2n}{n+1} \frac{\log(1 + 2/(n+1))}{2/(n+1)} \rightarrow 2.$$

Exponentiating gives the limit  $e^2$ . The same computation, for fixed real  $a, b, c$  and eventually positive bases, gives

$$\left(\frac{n+a}{n+b}\right)^{cn} \rightarrow e^{c(a-b)} : \quad h_n = \frac{a-b}{n+b}, \quad cnh_n \rightarrow c(a-b).$$

If  $a = b$  the expression is identically 1 wherever defined. For a less obvious rational function, subtract 1 before taking the limit:

$$\begin{aligned} \frac{n^2 + 3n + 1}{n^2 - n + 4} &= 1 + h_n, & h_n &= \frac{4n - 3}{n^2 - n + 4}, \\ h_n &\rightarrow 0, & nh_n &\rightarrow 4. \end{aligned}$$

Consequently its  $n$ th power tends to  $e^4$ . Comparing dominant degrees only shows that the base tends to 1; the rate of approach determines the answer when the exponent grows.

These computations reveal the common mechanism. If  $h_n \rightarrow 0$ ,  $1 + h_n > 0$  eventually, and  $v_n h_n \rightarrow L \in \mathbb{R}$ , then

$$v_n \log(1 + h_n) = v_n h_n \frac{\log(1 + h_n)}{h_n} \rightarrow L, \quad (1 + h_n)^{v_n} \rightarrow e^L.$$

At indices with  $h_n = 0$ , assign the quotient its continuous extension 1; the identity still holds. The assumptions  $u_n \rightarrow 1$  and  $v_n \rightarrow \infty$  alone do not determine  $u_n^{v_n}$ . With exponent  $n$ , the bases 1,

$$1 + 1/n, \quad 1 + 2/n$$

already give three different limits.

The function-limit version uses the same normalization. For fixed real  $a, b$ , both  $1 + ax$  and  $1 + bx$  are positive near zero, and

$$\frac{1}{x} \log \left( \frac{1 + ax}{1 + bx} \right) = \frac{\log(1 + ax)}{x} - \frac{\log(1 + bx)}{x} \rightarrow a - b.$$

(The term is zero if its parameter is zero.) Hence

$$\lim_{x \rightarrow 0} \left( \frac{1 + ax}{1 + bx} \right)^{1/x} = e^{a-b}, \quad \lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$$

This generalizes the index-shift and reciprocal examples after [Definition 4.4.4](#). Algebra reveals the hidden  $1^\infty$  form; logarithms reveal the rate that controls it.

**Example 8.5.17** (Taylor's theorem resolves a second-order limit). The exponential is now constructed and satisfies  $(\exp)'' = \exp$ . Taylor's theorem at 0 gives, for  $x \neq 0$  and some  $\xi$  between 0 and  $x$ ,

$$\frac{e^x - 1 - x}{x^2} = \frac{e^\xi}{2} \rightarrow \frac{1}{2}.$$

The remainder identifies the first contribution left after the constant and linear terms cancel; continuity of  $\exp$  controls the intermediate point.

**Example 8.5.18** (A numerical Taylor approximation with certified error). The cubic approximation to  $e^{0.1}$  is

$$1 + 0.1 + \frac{0.1^2}{2} + \frac{0.1^3}{6} = 1.105166666 \dots$$

On  $[0, 0.1]$ , comparison of the exponential and geometric series gives

$$e^x \leq \frac{10}{9}.$$

Thus [Corollary 8.4.6](#) certifies

$$|R_3(0.1)| \leq \frac{10}{9} \frac{0.1^4}{24} < 0.00000463.$$

This is a bound on the actual error, not just a formal series expression.

## Hyperbolic functions

The exponential also produces a pair of functions adapted to the hyperbola:

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad \sinh x = \frac{e^x - e^{-x}}{2}.$$

Their definitions immediately give

$$\cosh(-x) = \cosh x, \quad \sinh(-x) = -\sinh x,$$

and differentiation gives

$$(\sinh x)' = \cosh x, \quad (\cosh x)' = \sinh x.$$

Expanding the squares yields the fundamental identity

$$\boxed{\cosh^2 x - \sinh^2 x = 1.}$$

Since  $\cosh x = (e^x + e^{-x})/2 > 0$ , define

$$\tanh x = \frac{\sinh x}{\cosh x}.$$

The quotient rule and the boxed identity give

$$(\tanh x)' = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = 1 - \tanh^2 x.$$

Structurally,

$$(\sin x)'' = -\sin x, \quad (\cos x)'' = -\cos x, \quad (\sinh x)'' = \sinh x, \quad (\cosh x)'' = \cosh x.$$

The strictly increasing function  $\sinh : \mathbb{R} \rightarrow \mathbb{R}$  has inverse  $\operatorname{arsinh}$ . To solve  $x = \sinh y$ , put  $u = e^y > 0$ . Then

$$x = \frac{u - u^{-1}}{2} \implies u^2 - 2xu - 1 = 0.$$

The positive root is  $u = x + \sqrt{x^2 + 1}$ , and therefore

$$\operatorname{arsinh} x = \log(x + \sqrt{x^2 + 1}), \quad (\operatorname{arsinh} x)' = \frac{1}{\sqrt{1 + x^2}}.$$

Similarly, solving  $x = (u + u^{-1})/2$  gives  $u^2 - 2xu + 1 = 0$ ; on  $y \geq 0$  the appropriate root is  $u = x + \sqrt{x^2 - 1}$ . Thus the restriction of  $\cosh$  to  $[0, \infty)$  has inverse

$$\operatorname{arcosh} x = \log(x + \sqrt{x^2 - 1}), \quad x \geq 1,$$

with

$$(\operatorname{arcosh} x)' = \frac{1}{\sqrt{x^2 - 1}}, \quad x > 1.$$

The derivative formulas follow either by differentiating the logarithmic expressions or by the inverse-derivative theorem.

## 8.6 Trigonometric and Inverse Trigonometric Functions

### Geometry suggests the construction

Recall first the right-triangle ratios for an acute angle  $\theta$ , as motivation rather than definitions:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Figure 8.1 restores the familiar school-level picture before analysis begins.

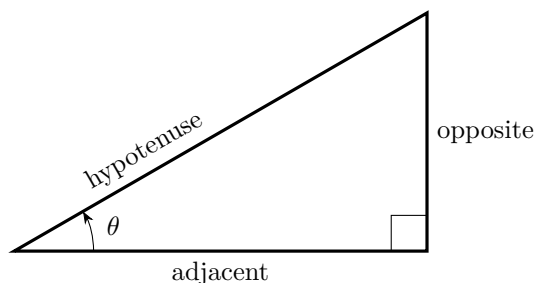


Figure 8.1: The familiar right-triangle meanings of sine, cosine, and tangent. At this point the picture motivates the analytic definitions; it does not replace them.

The two special triangles in Figure 8.2 similarly remind us of familiar side ratios. We deliberately use degree notation here:  $30^\circ$ ,  $45^\circ$ , and  $60^\circ$  are geometric targets, not yet expressions involving an analytically constructed constant  $\pi$ .

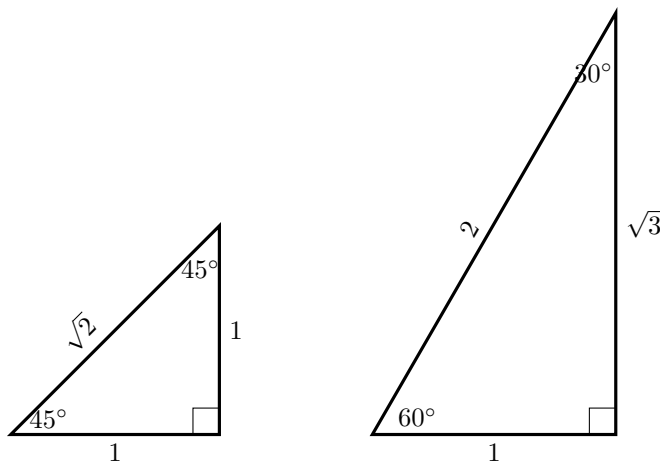


Figure 8.2: The  $45^\circ$ – $45^\circ$ – $90^\circ$  and  $30^\circ$ – $60^\circ$ – $90^\circ$  triangles. Their side ratios are familiar geometric targets that the analytic construction will later justify.

The desired analytic functions should reproduce those ratios and the familiar geometric identities. Calculus also suggests the derivative pattern

$$s' = c, \quad c' = -s, \quad s(0) = 0, \quad c(0) = 1.$$

These are still targets, not premises. They motivate equations that force the power-series coefficients: if

$$s(x) = \sum a_n x^n, \quad c(x) = \sum b_n x^n,$$



then formally

$$(n+1)a_{n+1} = b_n, \quad (n+1)b_{n+1} = -a_n.$$

Starting from  $a_0 = 0, b_0 = 1$  produces alternating odd and even series. The construction below will first define sine and cosine by those series, then prove their derivative and addition laws, then derive the Pythagorean identity. Only after that analytic work will we recover the global circle picture and identify the familiar degree angles with radian values.

**Proposition 8.6.1** (Convergence of the trigonometric series). For every  $x \in \mathbb{R}$ , the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

converge absolutely.

*Proof.* For  $x \neq 0$  the ratios of consecutive absolute terms are respectively

$$\frac{|x|^2}{(2n+2)(2n+3)}, \quad \frac{|x|^2}{(2n+1)(2n+2)}.$$

Both tend to zero, so apply the Ratio Test. At  $x = 0$  convergence is immediate. □

**Definition 8.6.2** (Analytic sine and cosine). For  $x \in \mathbb{R}$ , define

$$\begin{aligned} \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}. \end{aligned}$$

These convergent series define functions; their geometric properties remain to be proved.

**Theorem 8.6.3** (Derivatives of sine and cosine). The analytic functions satisfy  $\sin' = \cos$  and  $\cos' = -\sin$ .

*Proof.* Their series have infinite radius of convergence. Apply [Theorem 8.3.3](#). In full for sine,

$$\frac{d}{dx} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = \cos x.$$

For cosine, the constant term disappears and reindexing gives

$$\cos'(x) = \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{(2n-1)!} = -\sin x.$$

□

**Theorem 8.6.4** (Trigonometric addition and fundamental identities). For all  $x, y \in \mathbb{R}$ ,

$$\begin{aligned}\sin(x + y) &= \sin x \cos y + \cos x \sin y, \\ \cos(x + y) &= \cos x \cos y - \sin x \sin y, \\ \sin^2 x + \cos^2 x &= 1.\end{aligned}$$

The series also give oddness of sine and evenness of cosine.

*Proof.* Fix  $y$  and define

$$\begin{aligned}h(x) &= \sin(x + y) - \sin x \cos y - \cos x \sin y, \\ k(x) &= \cos(x + y) - \cos x \cos y + \sin x \sin y.\end{aligned}$$

By [Theorem 8.6.3](#),

$$h' = k, \quad k' = -h.$$

The product rule now gives

$$(h^2 + k^2)' = 2hk + 2k(-h) = 0.$$

The Mean Value Theorem makes  $h^2 + k^2$  constant. At  $x = 0$ , both differences vanish, because

$$\sin 0 = 0, \quad \cos 0 = 1,$$

as follows from their series. Hence both differences vanish identically. Finally, the derivative of

$$\sin^2 x + \cos^2 x$$

is zero, and its value at 0 is 1; the Mean Value Theorem proves the last identity.  $\square$

The Pythagorean identity now makes the promised geometric interpretation dependency-safe. Define the algebraic locus

$$C = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 = 1\}.$$

For every real  $t$ ,

$$(\cos t, \sin t) \in C.$$

In familiar coordinate geometry,  $C$  is the circle of radius 1 centered at the origin. We use that picture here as an interpretation; Chapter 10 formally develops Euclidean distance in  $\mathbb{R}^2$ . The coordinate labels in [Figure 8.3](#) also reconnect this global picture with the right-triangle ratios. The figure explains the identity; it is not part of its proof.

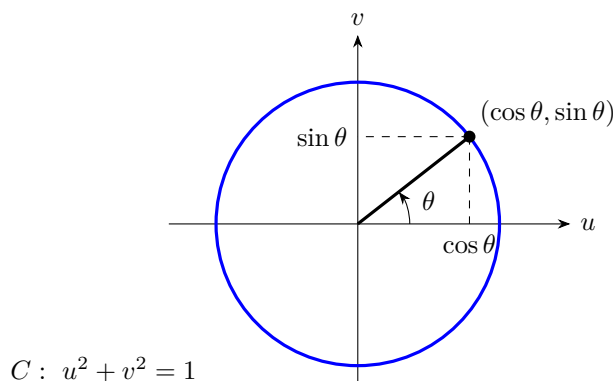


Figure 8.3: After the Pythagorean identity is proved, the pair  $(\cos \theta, \sin \theta)$  lies on the algebraic locus  $C$ . Familiar coordinate geometry interprets  $C$  as the unit circle.

The elementary derivative formulas proved in [Theorems 8.5.3](#) and [8.6.3](#) give  $(\exp)^{(n)} = \exp$  and the cycles

$$\sin \mapsto \cos \mapsto -\sin \mapsto -\cos \mapsto \sin,$$

with the cosine cycle starting at the second entry. These are repeated uses of ordinary rules, not new differentiation operations.

**Example 8.6.5** (Taylor's theorem as an inequality). Since  $|\sin t|, |\cos t| \leq 1$ , the remainder estimate in [Corollary 8.4.6](#) gives, for every real  $x$ ,

$$|\sin x - x| \leq \frac{|x|^3}{6}, \quad \left| \cos x - \left(1 - \frac{x^2}{2}\right) \right| \leq \frac{|x|^4}{24}.$$

Use degree 2 for sine and degree 3 for cosine: their missing even or odd coefficients vanish. The bounds hold on the segment from 0 to  $x$ , so the formulas are rigorous inequalities on either side of zero.

**Theorem 8.6.6** (Construction of  $\pi$  and the trigonometric ranges). There is a least positive zero  $c$  of  $\cos$ . Put

$$\pi = 2c.$$

Then

$$\cos c = 0, \quad \sin c = 1, \quad \cos \pi = -1, \quad \sin \pi = 0, \quad \cos(x + 2\pi) = \cos x, \quad \sin(x + 2\pi) = \sin x.$$

Moreover,  $\cos x > 0$  for  $-c < x < c$ , and  $\sin$  maps  $[-c, c]$  continuously and strictly increasingly onto  $[-1, 1]$ .

*Proof.* Taylor's theorem applied to  $\cos$  at 0 with degree 3 gives, for some  $\xi$  between 0 and 3,

$$\cos 3 = 1 - \frac{3^2}{2} + \frac{\cos \xi}{4!} 3^4.$$

Since the identity in [Theorem 8.6.4](#) gives  $|\cos \xi| \leq 1$ , the right side is at most

$$1 - \frac{9}{2} + \frac{81}{24} < 0.$$

Thus  $\cos$  has a zero in  $(0, 3)$  by the Intermediate Value Theorem. Since  $\cos 0 = 1$ , continuity gives a number

$$\delta > 0$$

such that

$$\cos x > 0 \quad (0 \leq x \leq \delta).$$

Consequently the set

$$Z = \{x \in [\delta, 3] : \cos x = 0\}$$

is nonempty: the zero just obtained cannot lie in  $[0, \delta]$ . Let

$$c = \inf Z.$$

For every  $n \in \mathbb{N}$ , the defining property of the infimum gives a point

$$z_n \in Z$$

with

$$c \leq z_n < c + \frac{1}{n}.$$

Thus  $z_n \rightarrow c$ , and continuity gives

$$\cos c = \lim_{n \rightarrow \infty} \cos z_n = 0.$$

Also  $c \geq \delta > 0$ . Every positive zero of cosine is at least  $c$ : a zero larger than 3 is certainly so, and a zero in  $(0, 3]$  belongs to  $Z$ . Thus  $c$  is the least positive zero of cosine.

If  $0 \leq x < c$  and  $\cos x \leq 0$ , the Intermediate Value Theorem applied on  $[0, x]$  would give a zero smaller than  $c$ . Thus  $\cos x > 0$  on  $[0, c)$ . The power-series definitions give that cosine is even and sine is odd. Hence cosine is positive on  $(-c, c)$ . By the Mean Value Theorem, sine is strictly increasing on  $[-c, c]$ . The identity

$$\sin^2 c + \cos^2 c = 1$$

and monotonicity give  $\sin c = 1$ . The addition formulas now give

$$\sin(2c) = 0, \quad \cos(2c) = -1,$$

and, after one further doubling,

$$\sin(4c) = 0, \quad \cos(4c) = 1.$$

The addition formulas therefore give the stated  $2\pi = 4c$  periodicity. Finally, oddness and the endpoint values show that the continuous strictly increasing function sine maps  $[-c, c]$  onto  $[-1, 1]$ .  $\square$

**Proposition 8.6.7** (Zeros, quadrant signs, and special angles). The zero sets are

$$\sin x = 0 \iff x = k\pi, \quad \cos x = 0 \iff x = \frac{\pi}{2} + k\pi \quad (k \in \mathbb{Z}).$$

The familiar first-quadrant values are

| $x$      | 0 | $\frac{\pi}{6}$      | $\frac{\pi}{4}$      | $\frac{\pi}{3}$      | $\frac{\pi}{2}$ |
|----------|---|----------------------|----------------------|----------------------|-----------------|
| $\sin x$ | 0 | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1               |
| $\cos x$ | 1 | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0               |

Parity, periodicity, and the addition formulas determine the corresponding values in every quadrant; for example,

$$\sin \frac{3\pi}{4} = \frac{1}{\sqrt{2}}, \quad \cos \frac{3\pi}{4} = -\frac{1}{\sqrt{2}}.$$

*Proof.* The addition formulas and the values at  $\pi$  give

$$\sin(x + \pi) = -\sin x, \quad \cos(x + \pi) = -\cos x.$$

On

$$(-\pi/2, \pi/2),$$

cosine is positive, and sine is strictly increasing with its only zero at 0. For every real  $x$ , the integer-part lemma (Lemma 3.4.4), applied after division by  $\pi > 0$ , places  $x - k\pi$  in a half-open

interval of length  $\pi$  for some  $k \in \mathbb{Z}$ . Translating that interval by the identities above therefore gives exactly the displayed zero sets and the usual alternating quadrant signs.

At

$$x = \pi/4,$$

the double-angle identity gives

$$0 = \cos(\pi/2) = \cos^2 x - \sin^2 x.$$

Both functions are positive there, so they are equal; the Pythagorean identity gives both values as

$$\frac{1}{\sqrt{2}}.$$

Next the addition formulas give

$$\cos(3t) = 4\cos^3 t - 3\cos t.$$

For

$$t = \pi/3,$$

let  $u = \cos t > 0$ . Since  $\cos \pi = -1$ ,

$$0 = 4u^3 - 3u + 1 = (2u - 1)^2(u + 1),$$

so

$$u = \frac{1}{2},$$

and positivity plus the Pythagorean identity gives

$$\sin(\pi/3) = \frac{\sqrt{3}}{2}.$$

Finally,

$$(\sin(\pi/2 - t), \cos(\pi/2 - t)) = (\cos t, \sin t)$$

supplies the

$$\pi/6$$

values. Taking

$$3\pi/4 = \pi - \pi/4$$

proves the last pair. □

Before defining inverses, recall the distinction from [Definition 2.3.8](#): for  $f : A \rightarrow B$ , the domain is  $A$ , the codomain is  $B$ , and the actual range is  $f(A) \subseteq B$ . For example,  $x \mapsto x^2$  may be declared from  $\mathbb{R}$  to  $\mathbb{R}$ , although its range is only  $[0, \infty)$ . We first regard  $\sin, \cos : \mathbb{R} \rightarrow \mathbb{R}$ ; [Theorem 8.6.6](#) and periodicity prove that each has actual range  $[-1, 1]$ .

**Definition 8.6.8** (Tangent). Define

$$\tan x = \frac{\sin x}{\cos x}$$

on its actual domain

$$\mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi : k \in \mathbb{Z} \right\}.$$

**Theorem 8.6.9** (Trigonometric restrictions). The function

$$\sin : \left[ -\frac{\pi}{2}, \frac{\pi}{2} \right] \rightarrow [-1, 1]$$

is continuous, strictly increasing, and onto. The function

$$\cos : [0, \pi] \rightarrow [-1, 1]$$

is continuous, strictly decreasing, and onto. Tangent is continuous, strictly increasing, and maps

$$\left( -\frac{\pi}{2}, \frac{\pi}{2} \right)$$

onto  $\mathbb{R}$ .

*Proof.* The sine assertion is part of [Theorem 8.6.6](#). For cosine on  $(0, \pi)$ , sine is positive: on

$$\left( 0, \frac{\pi}{2} \right)$$

this follows from strict increase, and on

$$\left( \frac{\pi}{2}, \pi \right)$$

it follows from

$$\sin x = \sin(\pi - x).$$

Hence

$$\cos' = -\sin < 0,$$

so the endpoint values and the Intermediate Value Theorem give the cosine assertion.

[Theorem 8.6.6](#) gives positivity of cosine on the tangent domain, and hence the quotient rule gives

$$\tan'(x) = \frac{1}{\cos^2 x} > 0.$$

Thus tangent is strictly increasing. As

$$x \rightarrow (\pi/2)^-,$$

continuity gives  $\sin x \rightarrow 1$  and  $\cos x \rightarrow 0^+$ . Given  $M > 0$ , choose  $x$  sufficiently close to

$$\pi/2$$

that

$$\sin x > \frac{1}{2}, \quad 0 < \cos x < \frac{1}{2M};$$

then  $\tan x > M$ . Oddness gives the corresponding limit  $-\infty$  at

$$-\pi/2.$$

The Intermediate Value Theorem proves that tangent is onto  $\mathbb{R}$ . □

The identity  $\cos^2 t + \sin^2 t = 1$  puts the series-defined pair  $(\cos t, \sin t)$  on the algebraic locus  $C$ . The proved special values, signs, and periodicity show the coordinate progression

$$(1, 0) \rightarrow (0, 1) \rightarrow (-1, 0) \rightarrow (0, -1) \rightarrow (1, 0)$$

as

$$t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi.$$

In the familiar orientation of the coordinate plane, this is the counterclockwise direction. Formal coordinatewise differentiation, velocity, and speed are postponed to [Section 8.7](#), where those terms are defined.

Solving  $\sin \theta = y$  means finding an angle with a prescribed vertical coordinate. Periodicity gives many solutions, so a global inverse cannot exist. The right half of the unit circle, parametrized by

$$[-\pi/2, \pi/2],$$

has each vertical coordinate exactly once. Similarly, the upper half, parametrized by  $[0, \pi]$ , has each horizontal coordinate exactly once. These observations explain the principal choices for arcsine and arccosine. Tangent measures slope on the open right semicircle, where every real slope occurs once. The preceding injectivity and range proofs make these geometric choices rigorous.

**Definition 8.6.10** (Inverse trigonometric functions). Periodicity prevents sine, cosine, and tangent from being globally injective. The restrictions in [Theorem 8.6.9](#) first prove injectivity, then determine the actual range, and only then use that range as the codomain of a bijection. Define the inverse maps

$$\begin{aligned} \arcsin : [-1, 1] &\longrightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \\ \arccos : [-1, 1] &\longrightarrow [0, \pi], \\ \arctan : \mathbb{R} &\longrightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right). \end{aligned}$$

They invert, respectively, sine on

$$[-\pi/2, \pi/2],$$

cosine on  $[0, \pi]$ , and tangent on

$$(-\pi/2, \pi/2).$$

These target intervals are the **principal ranges**. The notations  $\sin^{-1}, \cos^{-1}, \tan^{-1}$  are also widespread, but here  $\sin^{-1} x$  means inverse sine, not

$$1/\sin x.$$

The two possible orders of composition must be distinguished. For example,

$$\sin(\arcsin x) = x \quad (-1 \leq x \leq 1),$$

whereas

$$\arcsin(\sin x) = x$$

only for

$$x \in [-\pi/2, \pi/2].$$

Analogous statements hold for cosine on  $[0, \pi]$  and tangent on

$$(-\pi/2, \pi/2).$$

From Proposition 8.6.7,

$$\begin{aligned} \arcsin \frac{1}{2} &= \frac{\pi}{6}, & \arcsin \frac{1}{\sqrt{2}} &= \frac{\pi}{4}, \\ \arccos \frac{1}{2} &= \frac{\pi}{3}, & \arctan 1 &= \frac{\pi}{4}. \end{aligned}$$

**Theorem 8.6.11** (Derivatives of the inverse trigonometric functions). The derivative formulas are

$$(\arcsin)'(x) = \frac{1}{\sqrt{1-x^2}}, \quad (\arccos)'(x) = -\frac{1}{\sqrt{1-x^2}} \quad (-1 < x < 1),$$

and

$$(\arctan)'(x) = \frac{1}{1+x^2} \quad (x \in \mathbb{R}).$$

*Proof.* The inverse-derivative theorem and

$$\cos(\arcsin x) > 0, \quad \sin(\arccos x) > 0$$

give the first two formulas using  $\sin^2 + \cos^2 = 1$ . In particular, if

$$y = \arcsin x \in (-\pi/2, \pi/2),$$

then cosine is positive there, so

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2};$$

the positive square root is forced by the principal range. The tangent formula follows from the inverse-derivative theorem and

$$\tan'(x) = 1 + \tan^2 x.$$

□

**Example 8.6.12** (An oscillating piecewise function revisited). The elementary functions and the bound  $|\sin t| \leq 1$  are now proved, so we can treat

$$f(x) = x^2 \sin(1/x)$$

for  $x \neq 0$ , with  $f(0) = 0$ , without a calculus preview. For  $h \neq 0$ ,

$$\left| \frac{f(h) - f(0)}{h} \right| = |h \sin(1/h)| \leq |h|.$$

Given  $\varepsilon > 0$ , choose  $\delta = \varepsilon$ ; every  $0 < |h| < \delta$  makes this quotient's absolute value less than  $\varepsilon$ . Thus  $f'(0) = 0$ . Away from zero the rules give

$$f'(x) = 2x \sin(1/x) - \cos(1/x).$$

Rules handle each open piece, while the junction requires its own limit.



## Elementary derivatives and familiar limits

Now that the elementary functions have been constructed and differentiated, the chain and product rules give, on their stated domains,

$$\begin{aligned}\frac{d}{dx}e^{x^2} &= 2xe^{x^2}, & \frac{d}{dx}\log(1+x^2) &= \frac{2x}{1+x^2}, \\ \frac{d}{dx}(x \sin x) &= \sin x + x \cos x, & \frac{d}{dx}\sqrt{x} &= \frac{1}{2\sqrt{x}} \quad (x > 0).\end{aligned}$$

The last identity follows from the real-power derivative theorem and the identification of positive rational powers with their roots.

**Derivative quotients and local equivalents.** The base point in a derivative quotient determines which derivative is being used. For sine and the exponential, the base point is zero:

$$\begin{aligned}\frac{\sin x}{x} &= \frac{\sin x - \sin 0}{x - 0} \longrightarrow (\sin)'(0) = 1, \\ \frac{e^x - 1}{x} &= \frac{e^x - e^0}{x - 0} \longrightarrow (e^x)'|_{x=0} = 1.\end{aligned}$$

For the logarithm put  $t = 1 + x$ . Then  $t \rightarrow 1$ , with  $t > 0$  locally, and

$$\frac{\log(1+x)}{x} = \frac{\log t - \log 1}{t - 1} \longrightarrow (\log)'(1) = 1.$$

Continuity of cosine and the quotient limit law now give

$$\frac{\tan x}{x} = \frac{\sin x}{x} \frac{1}{\cos x} \longrightarrow 1.$$

The first-order cosine limit is another derivative quotient:

$$\frac{\cos x - \cos 0}{x - 0} \longrightarrow -\sin 0 = 0, \quad \frac{1 - \cos x}{x} \longrightarrow 0.$$

The second-order limit instead uses Taylor–Peano ([Corollary 8.4.7](#)):

$$\begin{aligned}\cos x &= 1 - \frac{x^2}{2} + o(x^2), \\ 1 - \cos x &= \frac{x^2}{2} + o(x^2), \\ \frac{1 - \cos x}{x^2} &= \frac{1}{2} + \frac{o(x^2)}{x^2} \longrightarrow \frac{1}{2}.\end{aligned}$$

For parameters, the chain rule for the exponential and the real-power derivative at  $t = 1$  give, respectively,

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{a^x - 1}{x} &= \log a \quad (a > 0), \\ \lim_{x \rightarrow 0} \frac{(1+x)^p - 1}{x} &= \lim_{t \rightarrow 1} \frac{t^p - 1}{t - 1} = p \quad (p \in \mathbb{R}).\end{aligned}$$

In the second line  $1 + x > 0$  locally. These are derivatives, rather than additional limit rules to memorize. For example,

$$\frac{\sin(3x)}{x} = 3 \frac{\sin(3x)}{3x} \longrightarrow 3, \quad \frac{e^{2x} - 1}{\sin x} = 2 \frac{e^{2x} - 1}{2x} \frac{x}{\sin x} \longrightarrow 2.$$

The normalized denominator tends to one, so the quotient law applies.

**Inverse trigonometric limits and principal ranges.** The derivatives already established, and the zero values of arcsine and arctangent at zero, give

$$\lim_{x \rightarrow 0} \frac{\arcsin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\arctan x}{x} = 1.$$

There is also an exact identity, including the endpoints:

$$\arccos x = \frac{\pi}{2} - \arcsin x \quad (-1 \leq x \leq 1).$$

Indeed, put  $y = \arcsin x$ . The principal range gives

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, \quad 0 \leq \frac{\pi}{2} - y \leq \pi, \quad \cos\left(\frac{\pi}{2} - y\right) = \sin y = x.$$

Uniqueness in the principal range of arccosine proves the identity. Hence

$$\lim_{x \rightarrow 0} \frac{\arccos x - \pi/2}{x} = -\lim_{x \rightarrow 0} \frac{\arcsin x}{x} = -1.$$

At the endpoint put  $t = \arccos x$ . Continuity of the inverse gives  $t \rightarrow 0^+$  as  $x \rightarrow 1^-$ , and  $x = \cos t$ . Thus

$$\frac{\arccos x}{\sqrt{2(1-x)}} = \left(2 \frac{1 - \cos t}{t^2}\right)^{-1/2} \rightarrow 1.$$

The identity  $\arccos(-x) = \pi - \arccos x$ , proved by the same principal-range argument, gives its symmetric companion:

$$\lim_{x \rightarrow -1^+} \frac{\pi - \arccos x}{\sqrt{2(1+x)}} = 1.$$

This square-root behavior describes the endpoints at which

$$(\arccos)'(x) = -\frac{1}{\sqrt{1-x^2}}$$

becomes unbounded.

**The continuous-variable exponential limit.** For  $x \neq 0$  sufficiently near zero, real powers give

$$(1+x)^{1/x} = \exp\left(\frac{\log(1+x)}{x}\right) \rightarrow e$$

by the logarithmic limit and continuity of the exponential. Taking

$$x = \frac{1}{n}$$

recovers the Chapter 4 sequence limit

$$\left(1 + \frac{1}{n}\right)^n \rightarrow e.$$

The same logarithm-then-exponential argument underlies the exponential limit family; no new differentiation rule is needed.

**Local asymptotic equivalents.** Here  $f(x) \sim g(x)$  means

$$\frac{f(x)}{g(x)} \longrightarrow 1.$$

As  $x \rightarrow 0$ , the preceding calculations give

$$\begin{aligned} \sin x &\sim x, & \tan x &\sim x, & e^x - 1 &\sim x, & \log(1+x) &\sim x, \\ \arcsin x &\sim x, & \arctan x &\sim x, \\ \arccos x - \frac{\pi}{2} &\sim -x, & 1 - \cos x &\sim \frac{x^2}{2}. \end{aligned}$$

Scaling, substitution, reciprocals, and derivative quotients give the useful variants

$$\begin{aligned} \frac{\sin(ax)}{x} &\rightarrow a, & \frac{\tan(ax)}{x} &\rightarrow a, & \frac{e^{ax} - 1}{x} &\rightarrow a, & \frac{\log(1+ax)}{x} &\rightarrow a, \\ \frac{(1+ax)^p - 1}{x} &\rightarrow ap, & \frac{1 - \cos(ax)}{x^2} &\rightarrow \frac{a^2}{2}, \\ \frac{x}{\sin x} &\rightarrow 1, & \frac{x}{e^x - 1} &\rightarrow 1, & \frac{x}{\log(1+x)} &\rightarrow 1. \end{aligned}$$

Here  $a, p \in \mathbb{R}$ , with the expressions considered where defined. The logarithm-then-exponential argument also gives

$$\lim_{x \rightarrow 0} (1+ax)^{b/x} = e^{ab}$$

whenever the base is positive near the limiting regime. These are consequences of the canonical limits, not additional formulas to memorize.

**Example 8.6.13** (Two legitimate applications). With the elementary-function derivatives now established, L'Hôpital's Rule gives

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x}{1} = 1$$

from the

$$0/0$$

form, and

$$\lim_{x \rightarrow +\infty} \frac{\log x}{x} = \lim_{x \rightarrow +\infty} \frac{1/x}{1} = 0$$

from the

$$\infty/\infty$$

form. In each calculation the equal sign relates two limits under the theorem's hypotheses; it is not the pointwise identity

$$f/g = f'/g'.$$

**Remark 8.6.14** (When not to apply the rule). The quotient

$$(x^2 + 1)/x$$

tends to  $+\infty$  as  $x \rightarrow 0^+$ , but it is not a

$$0/0 \quad \text{or} \quad \infty/\infty$$

form, so differentiating numerator and denominator would incorrectly suggest the limit 0. For

$$x^2 \sin(1/x)/x$$

at 0, the derivative quotient is more complicated than the original expression, while the estimate

$$|x \sin(1/x)| \leq |x|$$

settles the limit immediately. Finally, if

$$f'/g'$$

has no limit, the theorem gives no conclusion; the original quotient may still have a limit by another method. Always identify the form and verify the hypotheses before differentiating.

## 8.7 Parametrized Curves and Polar Coordinates

The pair  $(\cos t, \sin t)$  suggests motion described by two ordinary one-variable functions. This section develops that idea without importing multivariable differentiation or a previously defined Euclidean norm. The formal statements are coordinate formulas; familiar plane geometry supplies interpretation and will be formalized in Chapters 10–11.

**Definition 8.7.1** (Parametrized plane curve, velocity, and speed). A **parametrized plane curve** on an interval  $I$  is a pair of ordinary scalar functions  $x, y : I \rightarrow \mathbb{R}$ , written

$$\gamma(t) = (x(t), y(t)).$$

When both coordinates are differentiable, define the coordinatewise **velocity** and the **speed** by

$$\gamma'(t) := (x'(t), y'(t)), \quad \text{speed}(t) = \sqrt{x'(t)^2 + y'(t)^2}.$$

The curve is  $C^k$  when both coordinate functions are  $C^k$ , and is smooth when both are smooth. Endpoint derivatives, when used, are one-sided.

Geometrically,  $\gamma(t)$  represents the position of a moving point in the plane,  $\gamma'(t)$  represents instantaneous motion, and the displayed nonnegative number is its speed. Chapter 10 will identify the square-root expression with Euclidean length, and Chapter 11 will show that coordinatewise differentiation agrees with the Fréchet derivative of  $\gamma : I \rightarrow \mathbb{R}^2$ . Thus “velocity vector” is useful familiar language, while the present definition needs only ordered pairs and one-variable derivatives.

For  $\gamma(t) = (\cos t, \sin t)$ , coordinatewise differentiation gives

$$\gamma'(t) = (-\sin t, \cos t), \quad \text{speed}(t) = 1.$$

The coordinate progression proved in the preceding section gives the familiar geometric interpretation that, on  $0 \leq t \leq 2\pi$ , the curve starts at  $(1, 0)$  and moves once counterclockwise around its algebraic locus; on the same interval,  $(\cos 2t, \sin 2t)$  has speed two and appears to make two traversals. The exact multiplicities follow from angular nonuniqueness in [Proposition 8.7.4](#). A parametrization therefore records speed and multiplicity as well as its geometric image.

**Standard parametrizations.** For a graph over  $[a, b]$ , use  $\gamma(t) = (t, f(t))$ ,  $a \leq t \leq b$ ; it moves from  $(a, f(a))$  to  $(b, f(b))$  in increasing horizontal coordinate. For two points  $(x_0, y_0)$  and  $(x_1, y_1)$ , first define the set

$$S = \{((1-t)x_0 + tx_1, (1-t)y_0 + ty_1) : 0 \leq t \leq 1\}.$$

The displayed formula is the formal definition of  $S$ ; familiar coordinate geometry calls it the line segment from  $(x_0, y_0)$  to  $(x_1, y_1)$ . Its displayed parameter runs from the first endpoint to the second.

For  $R > 0$ , let  $\gamma(t) = (R \cos t, R \sin t)$ . Its coordinates satisfy

$$x(t)^2 + y(t)^2 = R^2(\cos^2 t + \sin^2 t) = R^2.$$

Elementary coordinate geometry recognizes this locus as the circle of radius  $R$ . Coordinate-wise differentiation gives velocity  $(-R \sin t, R \cos t)$  and speed  $R$ . Likewise, for  $a, b > 0$ ,  $\gamma(t) = (a \cos t, b \sin t)$  satisfies

$$\frac{x(t)^2}{a^2} + \frac{y(t)^2}{b^2} = 1,$$

the familiar ellipse equation. On  $[0, 2\pi]$  the coordinate progression

$$(1, 0) \rightarrow (0, 1) \rightarrow (-1, 0) \rightarrow (0, -1) \rightarrow (1, 0)$$

at

$$t = 0, \pi/2, \pi, 3\pi/2, 2\pi$$

shows that both parametrizations move counterclockwise in the familiar orientation.

The hyperbolic identity gives the parallel parametrization

$$x = a \cosh t, \quad y = b \sinh t,$$

for which

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.}$$

Because  $\cosh t > 0$ , this parametrizes the right branch of the hyperbola; the left branch is obtained from  $x = -a \cosh t$ . Thus

$$\cos^2 t + \sin^2 t = 1 \quad \text{and} \quad \cosh^2 t - \sinh^2 t = 1$$

govern, respectively, circle/ellipse geometry and hyperbola geometry. This comparison motivates the term *hyperbolic functions*.

**Definition 8.7.2** (Polar coordinates). A **polar representation** of an ordered pair  $(x, y)$  is a pair  $(r, \theta)$  satisfying

$$\boxed{x = r \cos \theta, \quad y = r \sin \theta, \quad r \geq 0.}$$

The Pythagorean identity then gives

$$x^2 + y^2 = r^2, \quad \boxed{r = \sqrt{x^2 + y^2}}.$$

Geometrically,  $r$  is the distance from the origin and  $\theta$  is a direction angle. Chapter 10 will formalize the distance interpretation through the Euclidean metric. In the same familiar interpretation, fixed  $r > 0$  gives a circle and fixed  $\theta$  with varying  $r \geq 0$  gives a ray; neither picture is a prerequisite for the algebraic definition.

**Proposition 8.7.3** (Existence of polar representations). Every ordered pair  $(x, y)$  has a polar representation. If  $(x, y) \neq (0, 0)$ , one can choose  $r > 0$  and  $0 \leq \theta < 2\pi$ .

*Proof.* The origin is represented by  $r = 0$  and any angle. Suppose  $(x, y) \neq (0, 0)$  and set

$$r = \sqrt{x^2 + y^2} > 0, \quad u = \frac{x}{r}, \quad v = \frac{y}{r}.$$

Then  $u^2 + v^2 = 1$ . Let

$$\alpha = \arccos u \in [0, \pi], \quad \theta = \begin{cases} \alpha, & v \geq 0, \\ 2\pi - \alpha, & v < 0. \end{cases}$$

Because  $\cos \alpha = u$  and  $\sin \alpha \geq 0$ , the Pythagorean identity gives  $\sin \alpha = \sqrt{1 - u^2} = |v|$ . If  $v \geq 0$ , this is  $v$ . If  $v < 0$ , the addition formulas and periodicity give

$$\cos(2\pi - \alpha) = \cos \alpha = u, \quad \sin(2\pi - \alpha) = -\sin \alpha = v.$$

Thus  $x = r \cos \theta$  and  $y = r \sin \theta$ . In the second case  $v < 0$  rules out  $\alpha = 0, \pi$ , so  $0 < 2\pi - \alpha < 2\pi$ ; the first case also has  $0 \leq \theta \leq \pi$ . Hence the stated angular range is valid.  $\square$

**Proposition 8.7.4** (Exact angular nonuniqueness). If  $r > 0$ , then

$$(r \cos \theta, r \sin \theta) = (r \cos \phi, r \sin \phi)$$

if and only if

$$\theta - \phi \in 2\pi\mathbb{Z}.$$

At  $r = 0$ , every angle represents the origin.

*Proof.* When  $r > 0$ , equality of the ordered pairs gives equality of the two cosines and of the two sines. The addition formulas then yield

$$\cos(\theta - \phi) = \cos \theta \cos \phi + \sin \theta \sin \phi = \cos^2 \phi + \sin^2 \phi = 1.$$

Put  $s = \theta - \phi$ . If  $\cos s = 1$ , the Pythagorean identity gives  $\sin s = 0$ . The zero-set result in [Proposition 8.6.7](#) then gives  $s = k\pi$  for some  $k \in \mathbb{Z}$ . Since

$$\cos(k\pi) = (-1)^k = 1,$$

the integer  $k$  is even, so  $s \in 2\pi\mathbb{Z}$ . Conversely,  $2\pi$ -periodicity gives equality whenever the angles differ by a multiple of  $2\pi$ . For  $r = 0$ , both coordinates vanish for every angle.  $\square$

Thus polar coordinates are not globally one-to-one. The quadrant information cannot be discarded. For example  $(-1, 1)$  has  $r = \sqrt{2}$  and

$$\theta = 3\pi/4,$$

since

$$\cos(3\pi/4) = -1/\sqrt{2}, \quad \sin(3\pi/4) = 1/\sqrt{2}.$$

The value

$$\arctan(y/x) = \arctan(-1) = -\pi/4$$

would select the wrong quadrant. For  $r > 0$ , the chosen angle must satisfy *both*

$$\cos \theta = x/r, \quad \sin \theta = y/r.$$

If radius is prescribed by  $r = \rho(\theta) \geq 0$ , the resulting polar curve is

$$\gamma(\theta) = (\rho(\theta) \cos \theta, \rho(\theta) \sin \theta).$$

For example  $\rho(\theta) = \theta$  on  $[0, 2\pi]$  moves outward from the origin through one turn, ending at  $(2\pi, 0)$ ; “spiral” is the familiar geometric interpretation of this coordinate formula.

**Proposition 8.7.5** (Polar motion). For differentiable  $r(t) \geq 0$  and  $\theta(t)$ , set

$$\gamma(t) = (r(t) \cos \theta(t), r(t) \sin \theta(t)).$$

Then

$$\begin{aligned} \gamma'(t) &= (r' \cos \theta - r\theta' \sin \theta, r' \sin \theta + r\theta' \cos \theta), \\ \text{speed}(t) &= \sqrt{(r')^2 + r^2(\theta')^2}. \end{aligned}$$

All quantities on the right are evaluated at  $t$ .

*Proof.* Apply the ordinary product and chain rules to each coordinate. Squaring and adding cancels the cross terms:

$$\begin{aligned} x'^2 + y'^2 &= (r')^2(\cos^2 \theta + \sin^2 \theta) + r^2(\theta')^2(\sin^2 \theta + \cos^2 \theta) \\ &\quad - 2rr'\theta' \cos \theta \sin \theta + 2rr'\theta' \sin \theta \cos \theta \\ &= (r')^2 + r^2(\theta')^2. \end{aligned}$$

Taking the nonnegative square root gives speed. □

Informally,  $r'$  measures radial motion toward or away from the origin, while  $r\theta'$  measures motion around the origin. Chapters 10–11 will supply the formal Euclidean vector interpretation. For a circle  $r = R$ , the scalar speed formula gives  $R|\theta'|$ ; for  $\theta = t$  it gives  $R$ . The derivation used only one-variable differentiation and trigonometric identities: no Jacobian, norm, inner product, or multivariable derivative was invoked.

**Remark 8.7.6** (Polar coordinates can fail at the origin). The smooth coordinatewise curve  $\gamma(t) = (t, 0)$  passes through the origin. A nonnegative polar radius must satisfy  $r(t)^2 = t^2$ , hence  $r(t) = |t|$ , which is not differentiable at 0; no unique angle is determined there. Thus a smooth coordinatewise curve need not have differentiable polar coordinates through the origin. The preceding motion formulas apply where differentiable choices of  $r, \theta$  have actually been made.

## 8.8 A Smooth Cutoff and the Limits of Taylor Series

If every derivative exists, every Taylor coefficient is available. Must those coefficients reconstruct the function nearby? Analyticity names the reconstruction property; the smooth cutoff below will show that having all derivatives is insufficient.

**Definition 8.8.1** (Analytic at a point). A function is **real analytic** at  $a$  if on some open interval about  $a$  it equals a convergent power series in  $x - a$ . It is real analytic on an open interval if this holds at every point there.

Repeated termwise differentiation on smaller intervals (Theorem 8.3.3) shows that analytic functions are smooth and that the coefficients at  $a$  must be

$$f^{(j)}(a)/j!.$$

Recall that  $C^k$  means continuous derivatives through order  $k$ , while  $C^\infty$  means continuous derivatives of every finite order. Thus

$$\text{real analytic} \implies C^\infty \implies C^k \implies \dots \implies C^1 \quad (k \geq 1).$$

The reverse implications fail in general. For example, for an integer  $k \geq 1$ , the function  $x \mapsto (x_+)^{k+1}$ , where  $x_+ = \max(x, 0)$ , is  $C^k$  but not  $C^{k+1}$ : its  $k$ th derivative is  $(k+1)!x_+$ , which is continuous but not differentiable at zero. The next theorem supplies a smooth function that is not analytic.

For every fixed  $N \geq 1$ , Corollary 8.4.7 gives a  $C^\infty$  function the approximation

$$f(a+h) = \sum_{j=0}^N \frac{f^{(j)}(a)}{j!} h^j + o(|h|^N).$$

It does not assert

$$f(a+h) = \sum_{j=0}^{\infty} \frac{f^{(j)}(a)}{j!} h^j.$$

The first statement lets  $h \rightarrow 0$  with  $N$  fixed. The second holds a nearby  $h$  fixed and lets  $N \rightarrow \infty$ . A remainder estimate for each fixed order provides no common control as the order grows.

**Theorem 8.8.2** (The smooth cutoff building block). The function

$$\theta(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0 \end{cases}$$

is  $C^\infty$  on  $\mathbb{R}$ , with  $\theta^{(k)}(0) = 0$  for every  $k \geq 0$ .

*Proof.* Each differentiation creates powers of

$$1/t,$$

but exponential decay dominates every fixed power, even after the extra division by  $t$  needed for a derivative at zero. For an integer  $m \geq 0$  and  $u > 0$ , the exponential series gives

$$e^u \geq \frac{u^{m+1}}{(m+1)!}, \quad 0 \leq u^m e^{-u} \leq \frac{(m+1)!}{u} \longrightarrow 0.$$

The first derivatives on  $t > 0$  are

$$\theta'(t) = t^{-2}e^{-1/t}, \quad \theta''(t) = (t^{-4} - 2t^{-3})e^{-1/t}.$$

Each is a polynomial in

$$1/t$$



times the same decaying exponential. On  $t > 0$ , induction gives

$$\theta^{(k)}(t) = P_k(1/t)e^{-1/t},$$

where  $P_0 = 1$  and  $P_{k+1}(u) = u^2(P_k(u) - P'_k(u))$ . Define  $g_k$  by this formula on  $t > 0$  and by zero on  $t \leq 0$ . The estimate makes each  $g_k$  continuous at zero. Moreover

$$g_k(t)/t \rightarrow 0$$

as  $t \rightarrow 0^+$  by the same estimate with one extra power, while the quotient is zero on the negative side. Thus  $g'_k(0) = 0$ ; away from zero  $g'_k = g_{k+1}$ . Inductively  $\theta^{(k)} = g_k$  everywhere.  $\square$

Every Taylor polynomial of  $\theta$  at zero is zero, yet  $\theta(t) > 0$  for  $t > 0$ . This smooth function is not analytic at zero. Thus knowing every derivative of a  $C^\infty$  function at one point need not determine the function nearby:  $\theta$  and the zero function have the same derivatives at zero. For an analytic function, the Taylor coefficients do determine it locally, by its convergent-series representation. Smoothness permits the flexibility of cutoffs; analyticity is a stronger reconstruction property. Chapter 11 uses this building block to construct Euclidean cutoffs equal to one near compact sets and finite partitions of unity (Lemma 11.6.18 and Theorem 11.6.19). These will localize the change-of-variables proof before manifolds appear. The cutoff is also useful:

$$b(x) = \theta(1 - \|x\|_2^2)$$

will be a smooth function that vanishes outside the closed unit ball once the multivariable chain rule is available. Chapter 13 uses translations and rescalings of this same building block to localize forms.

**Exercise 1.** Show that

$$\theta(t)/(\theta(t) + \theta(1-t))$$

is smooth, equals zero for  $t \leq 0$ , and equals one for  $t \geq 1$ . Explain why its denominator never vanishes. This constructs a smooth transition between two constant values.

## 8.9 Convexity, Concavity, and Inequalities

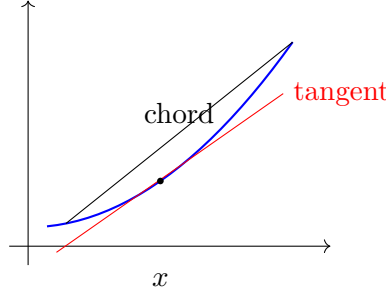
Derivatives describe local slope and curvature; convexity turns inequalities between those slopes into a statement about the whole graph. Supporting tangents and monotone derivatives relate the local differential tests to global shape. The definition first makes sense even when derivatives are unavailable.

If  $I \subseteq \mathbb{R}$  is an interval,  $x, y \in I$ , and  $0 \leq t \leq 1$ , then  $(1-t)x + ty$  lies between  $x$  and  $y$ , and therefore belongs to  $I$ . Thus the expression in the following definition remains in the domain.

**Definition 8.9.1** (Convex function). Let  $I$  be an interval. A function  $f : I \rightarrow \mathbb{R}$  is **convex** if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

for all  $x, y \in I$  and  $0 \leq t \leq 1$ . Geometrically, the graph lies below each of its chords.



For  $x < z < y$ , the chord slope across  $[x, y]$  is the weighted average of the two smaller chord slopes, with weights

$$(z - x)/(y - x), \quad (y - z)/(y - x).$$

Convexity orders the smaller slopes; their weighted average lies between them. Passing to a tangent means taking a limit of these ordered secant slopes.

**Theorem 8.9.2** (Derivative criteria for convexity). Let  $f$  be differentiable on an open interval  $I$ . Then  $f$  is convex if and only if  $f'$  is nondecreasing. In that case every tangent line supports the graph:

$$f(y) \geq f(x) + f'(x)(y - x) \quad (x, y \in I).$$

If  $f''$  exists throughout  $I$  and  $f'' \geq 0$ , then  $f$  is convex.

*Proof.* For  $x < z < y$ , convexity at  $z$  rearranges to

$$\frac{f(z) - f(x)}{z - x} \leq \frac{f(y) - f(x)}{y - x} \leq \frac{f(y) - f(z)}{y - z}.$$

Keep  $x, y$  fixed. Let  $z \downarrow x$  in the left inequality and  $z \uparrow y$  in the right one to get

$$f'(x) \leq \frac{f(y) - f(x)}{y - x} \leq f'(y).$$

In particular  $f'$  is nondecreasing. The first bound gives  $f(y) \geq f(x) + f'(x)(y - x)$  for  $y > x$ . For  $y < x$ , apply the second bound to the ordered pair  $y < x$  and rearrange to the same tangent inequality.

Conversely suppose  $f'$  is nondecreasing. For  $x < y$  and  $0 < t < 1$ , put  $z = tx + (1 - t)y$ , so  $x < z < y$ . The Mean Value Theorem on  $[x, z]$  and  $[z, y]$  supplies  $\xi_1 \in (x, z)$  and  $\xi_2 \in (z, y)$  with

$$\frac{f(z) - f(x)}{z - x} = f'(\xi_1) \leq f'(\xi_2) = \frac{f(y) - f(z)}{y - z}.$$

Multiply by the positive denominators and collect the  $f(z)$  terms:  $(y - x)f(z) \leq (y - z)f(x) + (z - x)f(y)$ . Since

$$(y - z)/(y - x) = t, \quad (z - x)/(y - x) = 1 - t,$$

this is the chord inequality. The cases  $t = 0, 1$  are equalities; reversing  $x, y$  covers the other order. Finally, if  $f'' \geq 0$ , applying the Mean Value Theorem to  $f'$  shows that  $f'$  is nondecreasing, so the result applies.  $\square$

Thus  $x^2$ ,  $e^x$ , and  $-\log x$  are convex on their natural domains, by their nonnegative second derivatives. The function  $-x^2$  is not convex; at the midpoint of  $-1$  and  $1$  its value  $0$  lies above the chord value  $-1$ . Applying the tangent inequality to  $e^x$  at  $0$  gives the useful estimate

$$e^y \geq 1 + y.$$

**Definition 8.9.3** (Concavity and inflection points). A function  $f$  is **concave** on an interval when  $-f$  is convex. An interior point  $c$  is an **inflection point** when the function changes concavity at  $c$ .

Applying [Theorem 8.9.2](#) to  $-f$  shows that a differentiable function is concave exactly when  $f'$  is nonincreasing, and  $f'' \leq 0$  implies concavity. A sign change of  $f''$  across  $c$  is a useful sufficient condition for an inflection point when the relevant derivatives exist. The equation  $f''(c) = 0$  alone is not sufficient:  $x^3$  changes concavity at zero, while  $x^4$  does not.

**Theorem 8.9.4** (Finite Jensen inequality). Let  $f$  be convex on an interval  $I$ . If  $x_i \in I$ ,  $\lambda_i \geq 0$ , and

$$\sum_{i=1}^n \lambda_i = 1,$$

then

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i).$$

For concave  $f$ , the inequality is reversed.

*Proof.* For  $n = 1$  the assertion is equality. The assertion for  $n = 2$  is the definition of convexity. Assume it for  $n - 1$ . If  $\lambda_n = 1$ , the result is equality. Otherwise put

$$\Lambda = 1 - \lambda_n, \quad y = \sum_{i=1}^{n-1} \frac{\lambda_i}{\Lambda} x_i.$$

The induction hypothesis gives

$$f(y) \leq \sum_{i=1}^{n-1} \frac{\lambda_i}{\Lambda} f(x_i).$$

Apply two-point convexity to  $\Lambda y + \lambda_n x_n$ , then substitute this bound. The concave form follows by applying the result to  $-f$ .  $\square$

**Corollary 8.9.5** (Weighted arithmetic–geometric mean inequality). If  $x_i > 0$ ,  $\lambda_i \geq 0$ , and  $\sum_i \lambda_i = 1$ , then

$$\prod_{i=1}^n x_i^{\lambda_i} \leq \sum_{i=1}^n \lambda_i x_i.$$

If every weight is positive, equality holds exactly when all  $x_i$  are equal; zero-weight terms are irrelevant. The conclusion extends to nonnegative  $x_i$  by treating a positive-weight zero factor separately.

*Proof.* The logarithm is strictly concave because

$$(\log x)'' = -1/x^2 < 0.$$

The concave form of Jensen gives

$$\log\left(\sum_i \lambda_i x_i\right) \geq \sum_i \lambda_i \log x_i = \log\left(\prod_i x_i^{\lambda_i}\right).$$

Apply the increasing exponential. Strict concavity gives the equality condition.  $\square$

For equal weights this becomes

$$\sqrt[n]{x_1 \cdots x_n} \leq \frac{x_1 + \cdots + x_n}{n}.$$

Appendix A gives an independent proof of this equal-weight case by Cauchy forward-backward induction, starting from the elementary two-variable result [Proposition 5.6.6](#).

**Corollary 8.9.6** (Young's inequality). If  $p, q > 1$  and  $1/p + 1/q = 1$ , then for  $a, b \geq 0$ ,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

*Proof.* Apply weighted AM–GM to  $a^p, b^q$  with weights  $1/p, 1/q$ . □

**Theorem 8.9.7** (Finite Hölder inequality). For conjugate exponents  $p, q > 1$  and real numbers  $a_i, b_i$ ,

$$\sum_{i=1}^n |a_i b_i| \leq \left( \sum_{i=1}^n |a_i|^p \right)^{1/p} \left( \sum_{i=1}^n |b_i|^q \right)^{1/q}.$$

*Proof.* Put

$$A = \left( \sum_i |a_i|^p \right)^{1/p}, \quad B = \left( \sum_i |b_i|^q \right)^{1/q}.$$

If  $A = 0$  or  $B = 0$ , every term on the corresponding side is zero. Otherwise apply Young's inequality [Corollary 8.9.6](#) to

$$\frac{|a_i|}{A}, \quad \frac{|b_i|}{B},$$

and sum:

$$\sum_i \frac{|a_i b_i|}{AB} \leq \frac{1}{p} \sum_i \frac{|a_i|^p}{A^p} \frac{1}{q} \sum_i \frac{|b_i|^q}{B^q} = \frac{1}{p} + \frac{1}{q} = 1.$$

Multiplication by  $AB$  proves the result. □

For  $p = q = 2$ , Hölder first gives

$$\sum_{i=1}^n |a_i b_i| \leq \left( \sum_i a_i^2 \right)^{1/2} \left( \sum_i b_i^2 \right)^{1/2}.$$

Together with

$$\left| \sum_i a_i b_i \right| \leq \sum_i |a_i b_i|,$$

this is the familiar finite scalar Cauchy–Schwarz inequality:

$$\left| \sum_{i=1}^n a_i b_i \right| \leq \left( \sum_i a_i^2 \right)^{1/2} \left( \sum_i b_i^2 \right)^{1/2}.$$

Here it appears as the  $p = 2$  case of Hölder. Chapter 10 retains an independent proof and develops its geometric interpretation through inner products.

**Example 8.9.8** (A continuous trigonometric series). For  $0 < a < 1$  and any positive integer  $b$ , the series

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(\pi b^n x)$$

converges uniformly on  $\mathbb{R}$  by the bound  $|a^n \cos(\pi b^n x)| \leq a^n$ . Its sum is therefore continuous. This argument establishes only continuity; it proves no assertion about differentiability of this family. The formal derivative terms have bounds  $\pi(ab)^n$ , which supply no convergent geometric majorant when  $ab \geq 1$ . Failure of this bound is not itself a proof of nondifferentiability. To separate continuity from differentiability within results proved here, recall  $|x|$ : its one-sided slopes at zero are  $-1$  and  $1$ , so it is continuous there but not differentiable.

## Exercises

**Exercise 2.\*** Prove from the difference quotient that

$$(1/x)' = -1/x^2$$

for  $x \neq 0$ . Give denominator control explicitly. For the function equal to  $x^2$  on  $x \geq 0$  and  $cx$  on  $x < 0$ , determine all  $c$  for differentiability at zero.

**Exercise 3.** Differentiate  $(x^2 + 1)e^x$ ,  $\sin(x^2)$ ,  $\log(1 + x^2)$ , and

$$(x + 1)/(x^2 + 1).$$

State domains and identify the rule used at each step.

**Exercise 4.** Evaluate

$$\sin(2x)/x,$$

$$(e^{3x} - 1)/x,$$

$$\log(1 + 2x)/x, \quad (1 - \cos(2x))/x^2$$

as  $x \rightarrow 0$ , using the established elementary limits.

**Exercise 5.\*** A rectangle is made with 20 metres of boundary. Determine its largest area. Formulate the variable, objective, and domain; identify all candidates and justify a global maximum.

**Exercise 6.** A position is  $s(t) = t^3$  metres for  $0 \leq t \leq 2$  seconds. Find its average velocity and a time when the instantaneous velocity equals it. State the theorem hypotheses and give a uniform sensitivity bound for position errors caused by time errors.

**Exercise 7.** Use Taylor's theorem to approximate  $e^{0.1}$  by

$$1 + 0.1 + 0.1^2/2$$

and prove an error bound using  $e^t \leq e$  on  $[0, 0.1]$ .

**Exercise 8.** Use the Mean Value Theorem to prove that if

$$|f'(x)| \leq M \quad (x \in (a, b)),$$

where  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , then

$$|f(y) - f(x)| \leq M|y - x| \quad (x, y \in [a, b]).$$

**Exercise 9.** Use [Theorem 8.3.3](#) to differentiate

$$\sum_{n=0}^{\infty} (x-1)^n$$

on its interval of convergence. Explain why this is an analytic theorem rather than merely a formal manipulation of symbols.

**Exercise 10.** Prove from the definitions that

$$\arcsin(-x) = -\arcsin(x) \quad (-1 \leq x \leq 1).$$

**Exercise 11.** For

$$F_n(x) = x + x^2/n$$

on  $[-1, 1]$ , verify the sequence differentiation theorem including the anchor. Then explain why  $F_n(x) = n + x$  fails it despite uniformly convergent derivatives.

**Exercise 12.\*** Use the degree-two Taylor polynomial of  $\log(1+x)$  at zero to approximate  $\log(1.1)$  with an explicit third-derivative error bound.

**Exercise 13.** If all derivatives of a smooth function vanish at zero, must it vanish near zero? Give a counterexample. What changes if it is analytic there?

**Exercise 14.** Use a supporting tangent line to prove  $\log x \leq x - 1$  for  $x > 0$ . State precisely which function is convex or concave in your argument.

## Chapter 9

# Riemann Integration

This chapter answers four related questions:

When does an integral exist?  
How do we compute it?  
Why do the familiar calculus rules work?  
Where does the Riemann theory begin to break down?

Most functions in ordinary calculus are assembled from elementary functions by algebraic operations and composition. On their natural domains they are usually very regular—often  $C^\infty$ , and therefore continuous—so the existence hypotheses for proper integrals can become nearly invisible. Continuity on a compact interval, not continuous differentiability, is already enough for Riemann integrability. At the same time,

elementary formula  $\neq$  automatic global regularity.

The functions

$$1/x,$$

$\log x$ ,  $\sqrt{x}$ ,  $\tan x$ , and

$$1/\sqrt{1-x^2}$$

all demand attention to their domains and singularities. Accordingly, existence is checked before computation throughout the chapter.

### 9.1 Partitions and Riemann Sums

Integration formalizes approximation of accumulated quantities by finite sums. Darboux upper and lower sums avoid choosing sample points: they record the best guaranteed under- and over-approximations on every subinterval.

**Definition 9.1.1** (Partition). Let  $a < b$ . A **partition** of  $[a, b]$  is a finite ordered set

$$P = \{x_0, x_1, \dots, x_n\}$$

with  $a = x_0 < x_1 < \dots < x_n = b$ . Its mesh is

$$\text{mesh}(P) = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

Thus  $\text{mesh}(P) < \delta$  means that every cell has width less than  $\delta$ . Refinement cannot increase mesh. Equal partitions  $P_N$  satisfy

$$\text{mesh}(P_N) = \frac{b-a}{N} \rightarrow 0,$$

so arbitrarily fine partitions exist.

**Definition 9.1.2** (Interval oscillation). For a bounded function  $f$  on a nonempty interval  $I$ , define

$$\text{osc}_I(f) = \sup_I f - \inf_I f.$$

This is the largest possible vertical variation of  $f$  on  $I$ :

$$\boxed{\text{osc}_I(f) = \sup_{x, y \in I} |f(x) - f(y)|.}$$

Indeed, every difference is at most  $\sup_I f - \inf_I f$ . Conversely, for each  $\eta > 0$ , choose  $x, y \in I$  with

$$f(x) > \sup_I f - \eta/2, \quad f(y) < \inf_I f + \eta/2.$$

Then  $|f(x) - f(y)| > \text{osc}_I(f) - \eta$ , so taking the supremum proves the reverse inequality without assuming extrema are attained. Restriction can only reduce the available pairs, and hence

$$J \subseteq I \implies \text{osc}_J(f) \leq \text{osc}_I(f).$$

Also  $\text{osc}_I(f) = 0$  exactly when  $f$  is constant on  $I$ .

**Definition 9.1.3** (Upper and lower sums). Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded and let  $P$  be a partition. Put

$$I_i = [x_{i-1}, x_i], \quad \Delta x_i = x_i - x_{i-1},$$

and

$$M_i = \sup \{f(x) : x \in [x_{i-1}, x_i]\}, \quad m_i = \inf \{f(x) : x \in [x_{i-1}, x_i]\}.$$

The **upper sum** and **lower sum** are

$$U(f, P) = \sum_{i=1}^n M_i \Delta x_i, \quad L(f, P) = \sum_{i=1}^n m_i \Delta x_i.$$

The two panels of Figure 9.1 use the same function and partition. The excess height on cell  $I_i$  is its oscillation  $M_i - m_i$ , so, writing  $\Delta x_i = x_i - x_{i-1}$ ,

$$\boxed{U(f, P) - L(f, P) = \sum_i \underbrace{\text{osc}_{I_i}(f)}_{\text{vertical uncertainty}} \underbrace{\Delta x_i}_{\text{horizontal width}}.}$$

This cellwise quantity is distinct from the pointwise oscillation introduced in section 9.6. It organizes the next several estimates: uniform continuity makes every cell's oscillation small when the partition intervals are short.



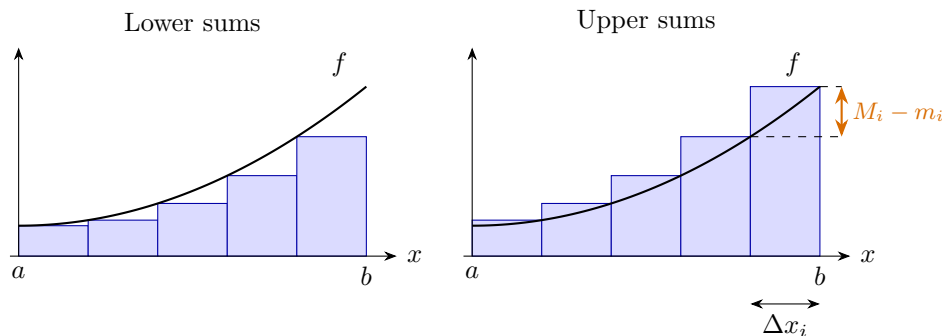


Figure 9.1: The same increasing function and partition give lower rectangles on the left and upper rectangles on the right. On the marked cell, the extra area is  $(M_i - m_i)\Delta x_i$ . Summing these cellwise gaps gives  $U(f, P) - L(f, P)$ ; small oscillation makes the total gap small.

**Definition 9.1.4** (Riemann integrability). Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. It is **Riemann integrable** if

$$\inf_P U(f, P) = \sup_P L(f, P),$$

where the infimum and supremum range over all partitions of  $[a, b]$ . Their common value is the **Riemann integral** of  $f$  and is denoted by

$$\int_a^b f(x) \, dx.$$

**Remark 9.1.5** (The integration variable is bound). The letter  $x$  in the integral is a bound variable:

$$\int_a^b f(x) \, dx = \int_a^b f(t) \, dt.$$

Renaming the variable changes neither the function being integrated nor the integral. Here  $dx$  is part of standard Riemann-integration notation; no infinitesimal numbers are assumed. Later coordinate differentials will give related notation a precise linear-map meaning, without changing this Darboux definition.

**Example 9.1.6** (All the Darboux sums for a simple function). For  $f(x) = x$  on  $[0, 1]$ , use

$$x_j = j/n.$$

On cell  $j$ , the minimum is

$$(j-1)/n,$$

the maximum

$$j/n,$$

and the width

$$1/n.$$

Therefore

$$L(f, P_n) = \frac{1}{n^2} \sum_{j=1}^n (j-1) = \frac{n-1}{2n}, \quad U(f, P_n) = \frac{1}{n^2} \sum_{j=1}^n j = \frac{n+1}{2n}.$$

The gap is

$$1/n$$

and both sums tend to

$$1/2.$$

Every tagged sum for this partition lies between them, regardless of the tags. The Darboux criterion asks whether such gaps can be made arbitrarily small.

The Darboux criterion below is therefore the default integrability tool in this chapter:

to prove Riemann integrability, make  $U(f, P) - L(f, P)$  arbitrarily small.

**Refinement narrows the Darboux gap.** A refinement inserts division points and removes none. To see explicitly what happens to one old cell  $[u, v]$ , insert

$$u = t_0 < t_1 < \cdots < t_k = v, \quad J_j = [t_{j-1}, t_j].$$

Write  $m = \inf_{[u,v]} f$ ,  $M = \sup_{[u,v]} f$  and  $m_j = \inf_{J_j} f$ ,  $M_j = \sup_{J_j} f$ . Since  $J_j \subseteq [u, v]$ ,

$$m \leq m_j \leq M_j \leq M.$$

The widths add by cancellation of the inserted endpoints:

$$(t_1 - u) + (t_2 - t_1) + \cdots + (v - t_{k-1}) = v - u.$$

Multiplying each bound by its corresponding positive width yields

$$\begin{aligned} m(v - u) &= m(t_1 - u) + m(t_2 - t_1) + \cdots + m(v - t_{k-1}) \\ &\leq m_1(t_1 - u) + m_2(t_2 - t_1) + \cdots + m_k(v - t_{k-1}) \\ &\leq M_1(t_1 - u) + M_2(t_2 - t_1) + \cdots + M_k(v - t_{k-1}) \\ &\leq M(t_1 - u) + M(t_2 - t_1) + \cdots + M(v - t_{k-1}) \\ &= M(v - u). \end{aligned}$$

For the original cells  $I_i = [x_{i-1}, x_i]$ , write their inserted points as  $t_{i,0} = x_{i-1}, \dots, t_{i,k_i} = x_i$  and put  $J_{i,j} = [t_{i,j-1}, t_{i,j}]$ . Adding the preceding inequalities gives

$$\begin{aligned} L(f, P) &= \sum_{i=1}^n m_i(x_i - x_{i-1}) \\ &\leq \sum_{i=1}^n \sum_{j=1}^{k_i} (\inf_{J_{i,j}} f)(t_{i,j} - t_{i,j-1}) = L(f, R) \\ &\leq U(f, R) = \sum_{i=1}^n \sum_{j=1}^{k_i} (\sup_{J_{i,j}} f)(t_{i,j} - t_{i,j-1}) \\ &\leq \sum_{i=1}^n M_i(x_i - x_{i-1}) = U(f, P). \end{aligned}$$

Figure 9.2 shows the resulting reduction in uncertainty.

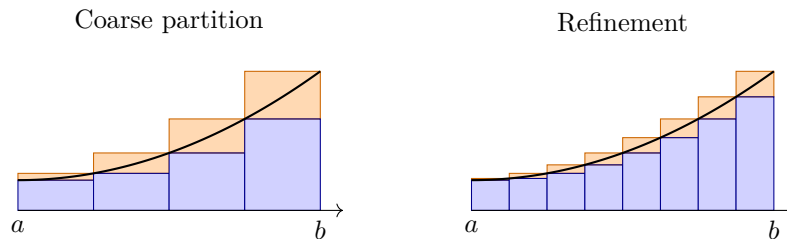


Figure 9.2: Blue rectangles give the lower sum; adding the orange bands gives the upper sum. Inserting cuts raises the lower sum and lowers the upper sum. The orange area, the Darboux gap, decreases.

For two partitions use their common refinement

$$R = P \cup Q.$$

Then

$$L(f, P) \leq L(f, R) \leq U(f, R) \leq U(f, Q).$$

Taking the supremum over  $P$  and then the infimum over  $Q$  proves

$$\sup_P L(f, P) \leq \inf_Q U(f, Q).$$

**Proposition 9.1.7** (Darboux criterion). A bounded function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable if and only if, for every  $\varepsilon > 0$ , there is a partition  $P$  with

$$U(f, P) - L(f, P) < \varepsilon.$$

*Proof.* Put  $L^* = \sup_P L(f, P)$  and  $U^* = \inf_P U(f, P)$ . If  $f$  is integrable, then  $L^* = U^* = I$ . Given  $\varepsilon > 0$ , choose partitions  $P, Q$  such that

$$U(f, P) < I + \frac{\varepsilon}{2}, \quad L(f, Q) > I - \frac{\varepsilon}{2}.$$

For their common refinement  $R$ ,

$$L(f, R) \geq L(f, Q) > I - \frac{\varepsilon}{2}, \quad U(f, R) \leq U(f, P) < I + \frac{\varepsilon}{2}.$$

Consequently  $U(f, R) - L(f, R) < \varepsilon$ .

Conversely, suppose that for every  $\varepsilon > 0$  there is a partition  $P$  with  $U(f, P) - L(f, P) < \varepsilon$ . Then

$$L(f, P) \leq L^* \leq U^* \leq U(f, P),$$

and hence

$$0 \leq U^* - L^* \leq U(f, P) - L(f, P) < \varepsilon.$$

Since this holds for every  $\varepsilon > 0$ , [Lemma 4.1.5](#) gives  $U^* - L^* = 0$ . Thus  $L^* = U^*$ , so  $f$  is Riemann integrable.  $\square$

## 9.2 Continuous Functions Are Integrable

Continuity prevents large oscillation on sufficiently short intervals. Uniform continuity on a compact interval makes this local control uniform, which is exactly what the Darboux criterion requires.

**Theorem 9.2.1** (Integrability of continuous functions). Every continuous function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable.

*Proof.* By Theorem 5.7.9,  $f$  is uniformly continuous, and by the extreme value theorem it is bounded. Given  $\varepsilon > 0$ , choose  $\eta > 0$  with

$$\eta(b - a) < \varepsilon.$$

Choose  $\delta > 0$  so that

$$|x - y| < \delta \implies |f(x) - f(y)| < \eta.$$

Choose, for example, an equal partition  $P$  with  $\text{mesh}(P) < \delta$ . Then every cell  $I_i$  has  $\text{osc}_{I_i}(f) < \eta$ . Consequently

$$U(f, P) - L(f, P) = \sum_i \text{osc}_{I_i}(f) \Delta x_i < \eta \sum_i \Delta x_i = \eta(b - a) < \varepsilon.$$

The Darboux criterion proves integrability. Figure 9.3 depicts the cellwise height control used in this estimate.  $\square$

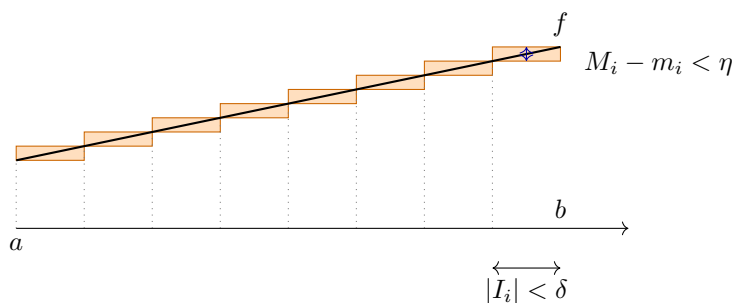


Figure 9.3: Uniform continuity makes the vertical oscillation bands narrow on every sufficiently short cell. Their total area is at most the common height bound times the full interval length. The strict cell bound shown here holds for continuous functions since extrema are attained; the proof needs only the weak bound.

The elementary oscillation inequalities used below are

$$\begin{aligned} \text{osc}_I(f + g) &\leq \text{osc}_I(f) + \text{osc}_I(g), \\ \text{osc}_I(cf) &= |c| \text{osc}_I(f), \\ \text{osc}_I(|f|) &\leq \text{osc}_I(f). \end{aligned}$$

They follow from the triangle inequality, homogeneity of absolute value, and the reverse triangle inequality, applied to pairs of values in  $I$ .

**Theorem 9.2.2** (Basic properties of the integral). If  $f, g : [a, b] \rightarrow \mathbb{R}$  are Riemann integrable and  $c \in \mathbb{R}$ , then  $f + g$  and  $cf$  are Riemann integrable, and

$$\int_a^b (f + g) \, dx = \int_a^b f \, dx + \int_a^b g \, dx, \quad \int_a^b cf \, dx = c \int_a^b f \, dx.$$

Moreover,  $f \leq g$  implies that their integrals have the same order.

*Proof. Strategy.* Use one common partition to squeeze both candidate integral values into the same short interval.

Fix  $\varepsilon > 0$ . By the Darboux criterion, choose partitions  $P_f$  and  $P_g$  such that

$$U(f, P_f) - L(f, P_f) < \frac{\varepsilon}{2}, \quad U(g, P_g) - L(g, P_g) < \frac{\varepsilon}{2}.$$

Let  $P$  be their common refinement. The oscillation inequality gives

$$\begin{aligned} U(f + g, P) - L(f + g, P) &= \sum_i \operatorname{osc}_{I_i}(f + g) |I_i| \\ &\leq [U(f, P) - L(f, P)] + [U(g, P) - L(g, P)] < \varepsilon. \end{aligned}$$

Thus  $f + g$  is integrable. Cellwise infimum and supremum inequalities give

$$L(f, P) + L(g, P) \leq L(f + g, P) \leq \int_a^b (f + g) \, dx \leq U(f + g, P) \leq U(f, P) + U(g, P),$$

while separately

$$L(f, P) + L(g, P) \leq \int_a^b f \, dx + \int_a^b g \, dx \leq U(f, P) + U(g, P).$$

The common enclosing interval has length  $[U(f, P) - L(f, P)] + [U(g, P) - L(g, P)] < \varepsilon$ . Since  $\varepsilon$  is arbitrary, the two values agree.

The scalar oscillation identity proves integrability of  $cf$  by the Darboux criterion (with a sufficiently small gap for  $f$  when  $c \neq 0$ ). To identify its integral, for  $c > 0$  and every interval  $I$ ,

$$\sup_I(cf) = c \sup_I f, \quad \inf_I(cf) = c \inf_I f.$$

Thus  $U(cf, P) = cU(f, P)$  and  $L(cf, P) = cL(f, P)$ , which prove both integrability and the scalar formula by the Darboux criterion. If  $c < 0$ , multiplication reverses order, so

$$\sup_I(cf) = c \inf_I f, \quad \inf_I(cf) = c \sup_I f.$$

Hence  $U(cf, P) = cL(f, P)$  and  $L(cf, P) = cU(f, P)$ , and the same conclusion follows; the case  $c = 0$  is immediate. Finally, if  $f \leq g$ , then on every partition interval  $I$ ,

$$\sup_I f \leq \sup_I g.$$

Therefore  $U(f, P) \leq U(g, P)$  for every partition  $P$ . Taking infima and using integrability gives

$$\int_a^b f \, dx \leq \int_a^b g \, dx.$$

□

**Theorem 9.2.3** (Additivity and absolute-value estimates). Let  $a < c < b$ .

- (a) If  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable, then its restrictions to  $[a, c]$  and  $[c, b]$  are Riemann integrable and

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx.$$

- (b) If  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable, then  $|f|$  is Riemann integrable and

$$\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx.$$

- (c) If  $f$  is Riemann integrable and

$$|f(x)| \leq M \quad (x \in [a, b]),$$

then

$$\left| \int_a^b f(x) \, dx \right| \leq M(b - a).$$

*Proof.* For (a), let  $\varepsilon > 0$  and choose a partition  $P$  of  $[a, b]$  with  $U(f, P) - L(f, P) < \varepsilon$ . Refine  $P$  by inserting  $c$ ; the Darboux gap cannot increase under refinement. Let  $P_{\text{left}}$  and  $P_{\text{right}}$  be the induced partitions of  $[a, c]$  and  $[c, b]$ . The sums split as

$$U_{[a,b]}(f, P) = U_{[a,c]}(f, P_{\text{left}}) + U_{[c,b]}(f, P_{\text{right}})$$

and

$$L_{[a,b]}(f, P) = L_{[a,c]}(f, P_{\text{left}}) + L_{[c,b]}(f, P_{\text{right}}).$$

Consequently,

$$U_{[a,b]} - L_{[a,b]} = (U_{[a,c]} - L_{[a,c]}) + (U_{[c,b]} - L_{[c,b]}).$$

Both terms on the right are nonnegative, so each is less than  $\varepsilon$ . The Darboux criterion proves that both restrictions are integrable.

Write

$$I = \int_a^b f, \quad I_{\text{left}} = \int_a^c f, \quad I_{\text{right}} = \int_c^b f.$$

For every partition containing  $c$ , the split lower and upper sums give

$$L_{[a,b]}(f, P) \leq I_{\text{left}} + I_{\text{right}} \leq U_{[a,b]}(f, P).$$

Choose such a partition with  $U_{[a,b]}(f, P) - L_{[a,b]}(f, P) < \varepsilon$ . Since  $L_{[a,b]}(f, P) \leq I \leq U_{[a,b]}(f, P)$ , it follows that

$$I - \varepsilon < I_{\text{left}} + I_{\text{right}} < I + \varepsilon.$$

As this holds for every  $\varepsilon > 0$ , the desired additivity identity follows.

For (b), the inequality

$$||u| - |v|| \leq |u - v|$$

implies that, for  $x, y$  in a partition interval  $I_i$ ,

$$||f(x)| - |f(y)|| \leq |f(x) - f(y)|.$$

Thus

$$\operatorname{osc}_{I_i}(|f|) \leq \operatorname{osc}_{I_i}(f), \quad U(|f|, P) - L(|f|, P) \leq U(f, P) - L(f, P).$$

The Darboux criterion proves integrability of  $|f|$ . Since

$$-|f| \leq f \leq |f|,$$

[Theorem 9.2.2](#) gives

$$-\int_a^b |f| \, dx \leq \int_a^b f \, dx \leq \int_a^b |f| \, dx,$$

which yields the stated inequality. For (c), order preservation applied to

$$0 \leq |f| \leq M.$$

gives

$$\int_a^b |f| \, dx \leq \int_a^b M \, dx = M(b-a).$$

Combine this with (b). □

**Theorem 9.2.4** (Integral Hölder and Cauchy–Schwarz inequalities). Let  $p, q > 1$  be conjugate exponents, and let  $f, g$  be continuous on  $[a, b]$ . Then

$$\int_a^b |f(x)g(x)| \, dx \leq \left( \int_a^b |f(x)|^p \, dx \right)^{1/p} \left( \int_a^b |g(x)|^q \, dx \right)^{1/q}.$$

*Proof.* Set

$$A = \left( \int_a^b |f|^p \right)^{1/p}, \quad B = \left( \int_a^b |g|^q \right)^{1/q}.$$

If  $A = 0$ , continuity and nonnegativity imply  $f = 0$  throughout the interval: a nonzero value would persist on a short interval and force a positive integral. The same applies when  $B = 0$ . Otherwise Young's inequality [Corollary 8.9.6](#), applied pointwise to  $|f|/A$  and  $|g|/B$ , gives

$$\frac{|fg|}{AB} \leq \frac{|f|^p}{pA^p} + \frac{|g|^q}{qB^q}.$$

Integrate and use  $1/p + 1/q = 1$ , then multiply by  $AB$ . □

For  $p = q = 2$ , the absolute-value estimate for integrals gives the **integral Cauchy–Schwarz inequality**

$$\left| \int_a^b f(x)g(x) \, dx \right| \leq \left( \int_a^b |f(x)|^2 \, dx \right)^{1/2} \left( \int_a^b |g(x)|^2 \, dx \right)^{1/2}.$$

Taking  $g \equiv 1$  in Hölder also gives

$$\int_a^b |f(x)| \, dx \leq (b-a)^{1/q} \left( \int_a^b |f(x)|^p \, dx \right)^{1/p}.$$

These are Riemann-integral consequences of Young's inequality; no  $L^p$  theory is being introduced. Chapter [10](#) later interprets the  $p = 2$  case through an integral inner product.

**Theorem 9.2.5** (Integrability of monotone functions). Every monotone function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable.

*Proof.* It is enough to consider an increasing function; the decreasing case follows by applying the result to  $-f$ . For a partition  $P$ , monotonicity gives

$$M_i = f(x_i), \quad m_i = f(x_{i-1}).$$

Thus

$$U(f, P) - L(f, P) = \sum_{i=1}^n (f(x_i) - f(x_{i-1}))(x_i - x_{i-1}) \leq \text{mesh}(P) \sum_{i=1}^n (f(x_i) - f(x_{i-1})).$$

Here  $x_i - x_{i-1} \leq \text{mesh}(P)$  and each difference  $f(x_i) - f(x_{i-1})$  is nonnegative. The sum telescopes:

$$\sum_{i=1}^n (f(x_i) - f(x_{i-1})) = f(b) - f(a).$$

Therefore

$$U(f, P) - L(f, P) \leq \text{mesh}(P)(f(b) - f(a)).$$

Choose a partition with norm less than

$$\frac{\varepsilon}{f(b) - f(a)}$$

when  $f(b) > f(a)$ ; if  $f(b) = f(a)$ , the function is constant and the upper and lower sums are equal. The Darboux criterion proves integrability.  $\square$

### Tagged sums: the calculus meaning of the integral

Darboux sums enclose an accumulated quantity without sampling the function. Calculus computations usually sample once in each cell instead. We must show that every sufficiently fine sampling gives the same answer, regardless of the tags and without requiring nested partitions.

Fix a coarse partition first. A fine cell lying inside one old cell inherits its oscillation bound. A *crossing cell* has an old interior division point in its interior; it needs only the global height bound. Each old point lies in the interior of at most one fine cell. Thus there are at most as many crossing cells as old interior division points. Figure 9.4 displays these two contributions.

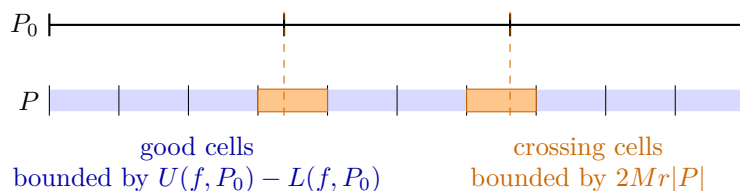


Figure 9.4: A fine partition need not refine the coarse one. Blue cells inherit the coarse oscillation bounds; orange cells cross old division points. Their total width is at most  $r|P|$ , so their gap contribution is at most  $2Mr|P|$ .

For a partition  $P : a = x_0 < \cdots < x_m = b$ , choose tags  $\xi_i \in [x_{i-1}, x_i]$ . Its tagged sum is

$$S(f, P, \xi) = \sum_i f(\xi_i)(x_i - x_{i-1});$$

its mesh is  $\text{mesh}(P) = \max_i (x_i - x_{i-1})$ .



**Theorem 9.2.6** (Fine-mesh characterization). A bounded function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable with integral  $I$  if and only if for every  $\varepsilon > 0$  there is  $\delta > 0$  such that every tagged partition of mesh less than  $\delta$  satisfies  $|S(f, P, \xi) - I| < \varepsilon$ .

*Proof.* Suppose  $f$  is integrable and  $|f| \leq M$ . Choose a coarse partition  $P_0$  such that

$$U(f, P_0) - L(f, P_0) < \frac{\varepsilon}{2},$$

and let  $r$  be its number of interior division points. For any partition  $P$ , the good cells inside a fixed coarse cell have total length at most that coarse cell's length, and their oscillations are no larger. More explicitly, if  $I_j$  is a coarse cell and  $\mathcal{G}_j$  is the family of fine cells contained in it, then

$$\sum_{J \in \mathcal{G}_j} \text{osc}_J(f) \ell(J) \leq \text{osc}_{I_j}(f) \sum_{J \in \mathcal{G}_j} \ell(J) \leq \text{osc}_{I_j}(f) \ell(I_j).$$

Summing over  $j$  bounds all good cells by the coarse Darboux gap. There are at most  $r$  crossing cells, each of length at most  $\text{mesh}(P)$  and oscillation at most  $2M$ . Therefore

$$U(f, P) - L(f, P) < \frac{\varepsilon}{2} + 2Mr \text{mesh}(P).$$

If  $Mr > 0$ , choose  $\delta > 0$  so small that

$$2Mr\delta < \frac{\varepsilon}{2}.$$

If  $Mr = 0$ , the crossing contribution is already zero and any  $\delta > 0$  works. On each fine cell,  $m_i \leq f(\xi_i) \leq M_i$ ; multiplying by  $\Delta x_i$  and summing gives

$$L(f, P) \leq S(f, P, \xi) \leq U(f, P).$$

For  $\text{mesh}(P) < \delta$ , both  $I$  and every tagged sum therefore lie between  $L(f, P)$  and  $U(f, P)$ , whose gap is less than  $\varepsilon$ .

Conversely, fix  $\varepsilon > 0$ , use the tagged-sum hypothesis with tolerance

$$\frac{\varepsilon}{4},$$

and choose a partition  $P$  with sufficiently small mesh. Choose  $\eta > 0$  once so that

$$\eta(b-a) < \frac{\varepsilon}{4}.$$

On each cell choose near-supremum and near-infimum tags satisfying

$$f(\xi_i) > M_i - \eta, \quad f(\zeta_i) < m_i + \eta.$$

No attained extrema are required. Summing gives

$$\begin{aligned} 0 &\leq U(f, P) - S(f, P, \xi) < \eta(b-a) < \frac{\varepsilon}{4}, \\ 0 &\leq S(f, P, \zeta) - L(f, P) < \eta(b-a) < \frac{\varepsilon}{4}. \end{aligned}$$

Both tagged sums are within the chosen tolerance of  $I$ , so

$$U(f, P) < I + \frac{\varepsilon}{2}, \quad L(f, P) > I - \frac{\varepsilon}{2}.$$

The gap is less than  $\varepsilon$ , proving integrability by the Darboux criterion. Its integral  $J$  lies between the same sums; hence

$$|J - I| < \frac{\varepsilon}{2}.$$

Since  $\varepsilon > 0$  was arbitrary,  $J = I$ . □

Thus sampled values times interval widths approximate the integral, with an error controlled by the mesh independently of the sample choices.

### Deciding integrability in ordinary problems

For a function on  $[a, b]$ , the available tests depend on its regularity.

1. Check boundedness. Without it, the proper Riemann integral is not defined.
2. Check continuity on the whole compact interval; if it holds, [Theorem 9.2.1](#) settles the question.
3. If continuity fails, check monotonicity.
4. Check whether the function is bounded and piecewise continuous with only finitely many jumps.
5. If no theorem applies immediately, construct a partition for which  $U(f, P) - L(f, P)$  is small.
6. To disprove integrability, look instead for a fixed positive lower bound on  $U(f, P) - L(f, P)$  that survives every partition.
7. After [Theorem 9.6.7](#) is available, use the Lebesgue criterion when the discontinuity set is easier to analyze than partitions are.

Continuity and monotonicity supply the first two positive tests just proved.

**Theorem 9.2.7** (Finitely many discontinuities). A bounded function on  $[a, b]$  that is continuous except at finitely many points is Riemann integrable. In particular, every piecewise-continuous function in the sense of [Definition 5.4.9](#) is integrable.

*Proof.* The mechanism is

$$\boxed{\text{one bad point} \longrightarrow \text{one short bad interval} \longrightarrow \text{finitely many bad points.}}$$

Suppose first that only  $c$  is exceptional and choose  $M$  with  $|f| \leq M$ . Enclose  $c$  in an interval of length less than  $\varepsilon/[4(M+1)]$ . On the two remaining closed pieces, continuity and [Theorem 9.2.1](#) give partitions whose combined Darboux gap is less than  $\varepsilon/2$ . Insert the endpoints of the short bad interval. Its oscillation is at most  $2M$ , so its contribution is less than  $\varepsilon/2$ .

For finitely many exceptional points, choose their surrounding intervals with total length

$$\ell < \frac{\varepsilon}{4(M+1)}.$$

Repeat the preceding construction on the finitely many complementary closed pieces. With  $\mathcal{G}$  and  $\mathcal{E}$  denoting the good and bad cells of the resulting common partition,

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{I \in \mathcal{G}} \text{osc}_I(f) \ell(I) + \sum_{I \in \mathcal{E}} \text{osc}_I(f) \ell(I), \\ \sum_{I \in \mathcal{E}} \text{osc}_I(f) \ell(I) &\leq 2M \sum_{I \in \mathcal{E}} \ell(I) \leq 2M\ell < \varepsilon/2. \end{aligned}$$

The good contribution was chosen below  $\varepsilon/2$ , so the Darboux criterion proves the result.  $\square$

**Example 9.2.8** (A direct Darboux calculation). For  $f(x) = x^2$  on  $[0, 1]$ , the equal partition  $P_n$  has

$$U(f, P_n) - L(f, P_n) = \frac{1}{n} \sum_{k=1}^n \left( \left( \frac{k}{n} \right)^2 - \left( \frac{k-1}{n} \right)^2 \right) = \frac{1}{n}.$$

Given  $\varepsilon > 0$ , choose

$$n > 1/\varepsilon;$$

the Darboux criterion proves integrability directly. For the step function that is zero on

$$[0, 1/2)$$

and one on

$$[1/2, 1],$$

isolate the jump in

$$[1/2 - \eta, 1/2 + \eta],$$

where

$$0 < \eta < \min\{1/2, \varepsilon/2\}.$$

Every other cell has zero oscillation; the gap is at most  $2\eta < \varepsilon$ .

To prove nonintegrability it suffices to find  $c > 0$  such that  $U(f, P) - L(f, P) \geq c$  for *every* partition  $P$ . For  $\mathbf{1}_{\mathbb{Q}}$  on  $[0, 1]$ , rational and irrational density make every cell's supremum 1 and infimum 0. Thus the gap is always 1. The later [Examples 9.6.9](#) and [9.6.10](#) contrast a fixed gap on every interval with a function whose large oscillations can be confined to intervals of arbitrarily small total length. Together with the continuous, monotone, and finite-jump cases above, they show why choosing an efficient integrability test is part of the problem.

### 9.3 Uniform Limits and Integration

An integral is itself a limit, so a second limiting process must be handled with care. In the Riemann theory, uniform convergence is the clean condition that permits a limit to pass through an integral: one error bound works on the whole interval and hence controls the accumulated error.

**Theorem 9.3.1** (Uniform interchange of limits and Riemann integrals). Let

$$f_n : [a, b] \longrightarrow \mathbb{R}$$

be Riemann integrable for every  $n$ , and suppose that  $f_n$  converges uniformly on  $[a, b]$  to a function  $f$ . Then  $f$  is Riemann integrable and

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) \, dx = \int_a^b f(x) \, dx.$$

In fact,

$$\lim_{n \rightarrow \infty} \int_a^b |f_n(x) - f(x)| \, dx = 0.$$

*Proof. Stage 1: integrability of the limit.* Put  $d_n = \|f - f_n\|_\infty$ . For  $x, y$  in any interval  $I$ ,

$$|f(x) - f(y)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)|,$$

so

$$\text{osc}_I(f) \leq \text{osc}_I(f_n) + 2d_n.$$

Consequently, for every partition  $P$ ,

$$U(f, P) - L(f, P) \leq U(f_n, P) - L(f_n, P) + 2d_n(b - a).$$

Given  $\varepsilon > 0$ , first choose  $n$  so that  $2d_n(b - a) < \varepsilon/2$ , and then choose  $P$  so that the Darboux gap of  $f_n$  is below  $\varepsilon/2$ . The Darboux criterion proves that  $f$  is integrable.

**Stage 2: convergence of the integrals.** The basic integral estimates give the sup-norm bound

$$\left| \int_a^b f_n \, dx - \int_a^b f \, dx \right| \leq \int_a^b |f_n - f| \, dx \leq (b - a) \sup_{x \in [a, b]} |f_n(x) - f(x)|.$$

The right-hand side tends to zero. The same final estimate proves the last assertion.  $\square$

**Corollary: integrating a uniformly convergent series.** If

$$\sum_{n=1}^{\infty} f_n$$

converges uniformly on  $[a, b]$  and each  $f_n$  is Riemann integrable, then its sum is Riemann integrable and

$$\int_a^b \left( \sum_{n=1}^{\infty} f_n(x) \right) \, dx = \sum_{n=1}^{\infty} \int_a^b f_n(x) \, dx.$$

Apply [Theorem 9.3.1](#) to the partial sums and use linearity for each finite sum.

**Pointwise bounded convergence requires a different argument**

Uniform convergence provides a single bound on all pointwise errors. Pointwise convergence does not: a bad location may move with  $n$ . Nevertheless bounded heights and an integrable limit suffice for integral convergence. We need an elementary fact about intervals. We prove it explicitly, without introducing Lebesgue integration.

The open-cover, relative-topology, compactness, finite-subcover, and common-scale terminology used below is established in [Sections 5.3](#) and [5.7](#).

For an open subset  $G$  of  $(a, b)$ , use the component-interval decomposition of [Proposition 5.3.7](#) and define

$$\ell(G) = \sum_j \ell(I_j), \quad \ell((u, v)) = v - u,$$

where  $G = \sqcup_j I_j$ . Every finite sum of their lengths is at most  $b - a$ , so the full sum exists. This definition is only a tool for open interval covers.

**Lemma 9.3.2** (Two elementary facts about open-set lengths). If an open  $G$  is contained in

$$\bigcup_j G_j$$

with the  $G_j$  open, then

$$\ell(G) \leq \sum_j \ell(G_j).$$

Also, for every  $\eta > 0$  an open  $G \subset (a, b)$  contains a finite union  $K$  of compact intervals such that  $\ell(G \setminus K) < \eta$ .

*Proof.* **Claim 1: inner approximation.** If  $\ell(K)$  denotes the combined length of a finite union of disjoint closed intervals, then

$$\ell(G) = \sup\{\ell(K) : K \subset G, K \text{ a finite union of closed intervals}\}.$$

Indeed, retain finitely many component intervals of  $G$  and shorten their endpoints slightly. These finite inner approximations approach the defining component sum from below.

**Claim 2: finite interval covers.** If finitely many open intervals cover a finite union  $K$  of disjoint closed intervals, their total length is at least  $\ell(K)$ . Sort all endpoints and subdivide at them. Every subinterval belonging to  $K$  is counted at least once among the covering lengths, which proves the inequality.

**Claim 3: reduce to the finite case.** The component intervals of all  $G_j$  form an open cover of any such  $K$ . Compactness from [Corollary 5.7.7](#) selects a finite subcover, and Claim 2 gives

$$\ell(K) \leq \sum_j \ell(G_j).$$

Taking the supremum over  $K$  in Claim 1 proves subadditivity.

**Claim 4: finite approximation with controlled loss.** Retain finitely many components of  $G$  whose omitted tail has length less than

$$\eta/2;$$

then shorten those retained intervals by a combined length less than

$$\eta/2.$$

Their compact union is  $K$ , and the components of  $G \setminus K$  consist of the discarded tail and endpoint strips. Their total length is less than  $\eta$ .  $\square$

**Lemma 9.3.3** (Nested open sets of uniformly positive length). If  $G_n \subset (a, b)$  are decreasing open sets and  $\ell(G_n) \geq c > 0$  for every  $n$ , then

$$\bigcap_n G_n \neq \emptyset.$$

*Proof.* The roadmap is

$$G_n \supset K_n \longrightarrow H_m = K_1 \cap \cdots \cap K_m \longrightarrow H_m \neq \emptyset \longrightarrow \bigcap_m H_m \neq \emptyset.$$

The sets  $G_n$  are open, so the nested-closed-set argument does not apply directly. Choose finite unions of closed bounded intervals  $K_n \subset G_n$  with

$$\ell(G_n \setminus K_n) < c/2^{n+1}.$$

The approximations  $K_n$  need not be nested: shortening and retaining different intervals at each stage makes independent choices. Set

$$H_m = K_1 \cap \cdots \cap K_m$$

instead. It is a finite union of closed bounded intervals (possibly degenerate), hence satisfies the local finite-subcover result just proved, and  $H_{m+1} \subseteq H_m$ . We must still show nonemptiness; the summable errors were chosen for exactly this purpose. If  $H_m$  were empty, then every  $x \in G_m$  belongs to  $G_i$  for  $i \leq m$  by nestedness, but failure of membership in  $H_m$  means that  $x \notin K_i$  for at least one such  $i$ . Hence

$$G_m \subset \bigcup_{i=1}^m (G_i \setminus K_i), \quad \ell(G_m) < \sum_{i=1}^m c/2^{i+1} < c/2,$$

contradicting its lower bound. The nonempty sets  $H_m$ , each a finite union of closed bounded intervals, decrease inside  $[a, b]$ , so their intersection is nonempty. Explicitly, otherwise the open complements of the  $H_m$  would finitely cover  $H_1$ , forcing one late  $H_m$  to be empty. A point in every  $H_m$  belongs to every  $G_n$ .  $\square$

**Theorem 9.3.4** (Arzelà bounded convergence). Let  $f_n : [a, b] \rightarrow \mathbb{R}$  be Riemann integrable,  $f_n \rightarrow f$  pointwise, and  $|f_n(x)| \leq M$  for all  $n, x$ . If  $f$  is Riemann integrable, then

$$\int_a^b |f_n - f| \longrightarrow 0, \quad \int_a^b f_n \longrightarrow \int_a^b f.$$

*Proof.* The proof follows

large error integral  $\Rightarrow$  large bad set  $\Rightarrow$  one fixed point is bad infinitely often  $\Rightarrow$  contradiction.

Let  $g_n = |f_n - f|$ . These are integrable by the integral algebra already proved, satisfy  $0 \leq g_n \leq 2M$ , and tend pointwise to zero. The cases  $a = b$  or  $M = 0$  are immediate. If the integrals do not tend to zero, pass to a subsequence and choose  $\eta > 0$  such that

$$\int_a^b g_n \geq \eta$$

for every term of the subsequence. Put

$$h = \frac{\eta}{2(b-a)}.$$

For each  $n$ , integrability supplies a lower Darboux sum greater than  $3\eta/4$ . Call a cell low when its lower height is at most  $h$  and high otherwise. The low cells contribute at most  $h(b-a) = \eta/2$ , so the high cells contribute more than  $\eta/4$ . Because all heights are at most  $2M$ , their combined length is at least

$$c = \frac{\eta}{8M} > 0.$$

Let  $U_n$  be the union of the interiors of the high cells. Endpoints do not change interval length, so  $\ell(U_n) \geq c$ , and  $g_n > h$  on  $U_n$ . The open sets

$$G_N = \bigcup_{n \geq N} U_n$$

decrease and each has length at least  $c$ . By [Lemma 9.3.3](#), some  $x$  belongs to every  $G_N$ . Unpacking the union and intersection,

$$x \in \bigcap_N G_N \iff \text{for every } N \text{ there exists } n \geq N \text{ with } x \in U_n.$$

It is one fixed point  $x$  lying in arbitrarily late bad sets, so  $g_n(x) > h$  at arbitrarily large indices. Pointwise convergence at this fixed  $x$  says all sufficiently late errors are less than  $h$ , a contradiction. The point cannot vary with  $N$  in this last step. This proves convergence of the absolute error integrals. Finally

$$\left| \int f_n - \int f \right| \leq \int |f_n - f|$$

proves the second assertion. □

**Example 9.3.5** (The limit's integrability cannot be dropped). Enumerate the rationals in  $[0, 1]$  as  $q_1, q_2, \dots$  and set  $f_n = \mathbf{1}_{\{q_1, \dots, q_n\}}$ . A finite set is covered by finitely many intervals of arbitrarily small total length, so each  $f_n$  has integral zero. Also  $0 \leq f_n \leq 1$  and  $f_n$  converges pointwise to  $\mathbf{1}_{\mathbb{Q}}$ . Every nondegenerate interval contains both rational and irrational points, so the limit has lower sum 0 and upper sum 1 for every partition. It is not Riemann integrable.

Uniform Riemann interchange automatically supplies an integrable limit. Arzelà's theorem permits pointwise convergence but separately assumes that integrability. Neither theorem permits arbitrary unbounded moving spikes. Bartle [\[1\]](#) is a classical companion for comparing these convergence hypotheses; the interval-cover proof above is internal.

Measure-theoretic convergence theorems weaken these hypotheses substantially; they are stated and compared with the present Riemann results in [Section 9.7](#). Dini's theorem itself now appears in [Chapter 6](#).

## 9.4 The Fundamental Theorem of Calculus

Differentiation measures local change, while integration accumulates local change. The Fundamental Theorem identifies these two operations under the continuity hypotheses that make both constructions stable. In the familiar elementary setting one often has  $F \in C^1$ , in the one-variable sense of [Definition 8.1.4](#), so  $F'$  is continuous. The results below first work in that comfortable setting and then identify exactly which hypotheses can be weakened.

**Proposition 9.4.1** (Accumulation at continuity points). For Riemann-integrable  $f$  on  $[a, b]$ , the accumulation function

$$F(x) = \int_a^x f(t) \, dt$$

is continuous. Here  $x$  is the free endpoint parameter and  $t$  is a bound integration variable. Renaming  $t$  does not change the function  $F$ . At every interior point  $c$  where  $f$  is continuous,  $F'(c) = f(c)$ .

*Proof.* If  $|f| \leq M$ , additivity and the integral bound give  $|F(y) - F(x)| \leq M|y - x|$ , so  $F$  is continuous. For  $h > 0$  with  $c + h \in [a, b]$ , additivity first gives the increment identity

$$\frac{F(c+h) - F(c)}{h} - f(c) = \frac{1}{h} \int_c^{c+h} (f(t) - f(c)) \, dt.$$

For  $h < 0$  use only forward-endpoint integrals at this stage:

$$\frac{F(c+h) - F(c)}{h} - f(c) = \frac{1}{-h} \int_{c+h}^c (f(t) - f(c)) \, dt.$$

In either case the absolute error is at most  $\sup\{|f(t) - f(c)| : |t - c| \leq |h|, t \in [a, b]\}$ . Given  $\varepsilon > 0$ , continuity at  $c$  supplies a neighborhood on which

$$|f(t) - f(c)| < \varepsilon/2.$$

Choose  $|h|$  small enough that the whole interval of integration lies there. The supremum is at most

$$\varepsilon/2 < \varepsilon,$$

so the difference quotient tends to  $f(c)$ . □

**Corollary 9.4.2** (Fundamental theorem of calculus, first part). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous and define

$$F(x) = \int_a^x f(t) \, dt.$$

Then  $F$  is differentiable on  $(a, b)$  and  $F'(x) = f(x)$ .

*Proof.* Continuity makes  $f$  Riemann integrable and makes every interior point a continuity point. Apply [Proposition 9.4.1](#) at each such point. □

**Theorem 9.4.3** (Fundamental theorem of calculus, second part). Let  $F : [a, b] \rightarrow \mathbb{R}$  be differentiable on  $(a, b)$  and continuous on  $[a, b]$ . Suppose that its derivative extends to a continuous function  $f : [a, b] \rightarrow \mathbb{R}$ , so that

$$F'(x) = f(x) \quad \text{for every } x \in (a, b).$$

Then

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

*Proof.* Define

$$G(x) = \int_a^x f(t) \, dt.$$

By [Corollary 9.4.2](#),  $G'(x) = f(x)$  for  $x \in (a, b)$ . Therefore the function  $H = F - G$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and satisfies  $H'(x) = 0$  throughout  $(a, b)$ . For  $x = a$ , the identity

$$H(x) = H(a)$$



is immediate. If  $x \in (a, b]$ , the Mean Value Theorem applied on  $[a, x]$  gives a point  $c \in (a, x)$  for which

$$H(x) - H(a) = H'(c)(x - a) = 0.$$

Thus  $H$  is constant. Since  $G(a) = 0$ , evaluating at  $b$  gives

$$F(b) - F(a) = G(b) = \int_a^b f(x) \, dx.$$

□

## Antiderivatives and indefinite integrals

**Definition 9.4.4** (Antiderivative). Let  $f : I \rightarrow \mathbb{R}$  be defined on an interval  $I$ . A differentiable function  $F : I \rightarrow \mathbb{R}$  is an **antiderivative** of  $f$  on  $I$  if

$$F'(x) = f(x) \quad (x \in I),$$

with one-sided derivatives at included endpoints when appropriate.

**Proposition 9.4.5** (Antiderivatives differ by a constant). If  $F$  and  $G$  are antiderivatives of the same function on an interval  $I$ , then there is a constant  $C$  such that

$$F(x) - G(x) = C \quad (x \in I).$$

*Proof.* Put  $H = F - G$ . Then  $H' = 0$  on the interior of  $I$ . For any  $x < y$  in  $I$ , the Mean Value Theorem on  $[x, y]$  gives a point  $c \in (x, y)$  such that

$$H(y) - H(x) = H'(c)(y - x) = 0.$$

Thus  $H$  has one value throughout  $I$ . □

This proposition is the rigorous reason for the arbitrary constant in

$$\int f(x) \, dx = F(x) + C.$$

The expression on the left is **indefinite-integral notation**: it denotes the family of all antiderivatives on the interval under discussion. It is not a definite Riemann integral with its endpoints deleted. The distinction is one of type:

$$\begin{array}{l|l} \text{definite integral } \int_a^b f(x) \, dx & \text{a number} \\ \text{indefinite integral } \int f(x) \, dx & \text{a family of functions } F + C. \end{array}$$

The interval hypothesis matters. On  $\mathbb{R} \setminus \{0\}$ , two antiderivatives of

$$1/x$$

may differ by one constant on  $(-\infty, 0)$  and by another on  $(0, \infty)$ . Thus

$$\int \frac{dx}{x} = \log |x| + C$$

is understood on one interval avoiding zero. More generally, one constant is available on each maximal interval contained in the domain.

In the first line below,  $F$  is the accumulation function just defined. The two FTC statements read in opposite directions:

FTC I:  $f$  continuous  $\implies F'(x) = f(x), \quad x \in (a, b),$

FTC II:  $F' = f \implies \int_a^b f(x) \, dx = F(b) - F(a) \quad \text{under the stated continuity hypotheses.}$

FTC I says that integrating a continuous rate produces an antiderivative. FTC II, or the Newton–Leibniz formula, is the direction most often used in elementary calculations: recognize  $F$ , verify  $F' = f$ , and evaluate the endpoints. Also,

having an antiderivative  $\neq$  having an elementary closed-form antiderivative.

For every continuous  $f$ , FTC I supplies the perfectly valid antiderivative

$$x \mapsto \int_a^x f(t) \, dt,$$

even when no familiar elementary formula is available.

**Example 9.4.6** (The two directions of the FTC). The integral

$$A(x) = \int_0^x (1 + t^2) \, dt$$

for  $x > 0$  defines an antiderivative: the first part of the FTC gives  $A'(x) = 1 + x^2$  directly from continuity of the integrand. Conversely, once

$$B(x) = x + x^3/3$$

is known to satisfy  $B'(x) = 1 + x^2$ , the second part evaluates

$$\int_0^2 (1 + x^2) \, dx = [x + x^3/3]_0^2 = 2 + 8/3 = 14/3.$$

The first use constructs an antiderivative from an integral; the second uses an antiderivative to evaluate a definite integral.

**Example 9.4.7** (A basic antiderivative library). Each entry below is justified by differentiating the proposed antiderivative and checking its domain:

| integrand         | antiderivative                                   |
|-------------------|--------------------------------------------------|
| $x^p$             | $\frac{x^{p+1}}{p+1} + C \quad (p \neq -1)$      |
| $1/x$             | $\log  x  + C$                                   |
| $e^x$             | $e^x + C$                                        |
| $a^x$             | $\frac{a^x}{\log a} + C \quad (a > 0, a \neq 1)$ |
| $\sin x$          | $-\cos x + C$                                    |
| $\cos x$          | $\sin x + C$                                     |
| $1/\cos^2 x$      | $\tan x + C$                                     |
| $1/(1+x^2)$       | $\arctan x + C$                                  |
| $1/\sqrt{1-x^2}$  | $\arcsin x + C$                                  |
| $-1/\sqrt{1-x^2}$ | $\arccos x + C.$                                 |

For arbitrary real  $p$ , the power formula holds on intervals contained in  $(0, \infty)$ , where Chapter 8 defines  $x^p$ . It also holds on any larger real domain on which the relevant power is defined and the derivative formula is valid. The logarithmic formula is used only on an interval avoiding zero. The tangent formula requires  $\cos x \neq 0$ , and the arcsine formula is used on intervals contained in  $(-1, 1)$ .

Representative checks reveal the mechanisms behind the table. For real powers on the positive half-line,

$$\frac{d}{dx} \left( \frac{x^{p+1}}{p+1} \right) = \frac{p+1}{p+1} x^p = x^p \quad (p \neq -1).$$

The sine, cosine, exponential, tangent, and arctangent entries follow by the same reversal of their established derivatives. The arcsine entry uses

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}} \quad (-1 < x < 1).$$

By the principal-range identity from Chapter 8,

$$\arcsin x + \arccos x = \frac{\pi}{2}.$$

Thus the same antiderivative family can equally be written

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C_1 = -\arccos x + C_2 \quad (-1 < x < 1),$$

where the arbitrary constants are adjusted by the fixed difference.

**Why the logarithm contains an absolute value.** The real logarithm retains its domain

$$\log : (0, \infty) \longrightarrow \mathbb{R}.$$

For  $x > 0$  its derivative is the reciprocal. For  $x < 0$ , the positive argument  $-x$  is available, and the chain rule gives

$$\frac{d}{dx} \log(-x) = \frac{1}{-x}(-1) = \frac{1}{x}.$$

Since  $|x| = x$  on the positive half-line and  $|x| = -x$  on the negative half-line, these two checks prove

$$\int \frac{dx}{x} = \log |x| + C$$

on each interval avoiding zero. No logarithm of a negative number is used. For a differentiable  $g$  that is nonzero throughout an interval, continuity and the intermediate value theorem make its sign constant. Applying the same two cases and the chain rule gives

$$\frac{d}{dx} \log |g(x)| = \frac{g'(x)}{g(x)}, \quad \int \frac{g'(x)}{g(x)} dx = \log |g(x)| + C.$$

For example,

$$\int \frac{2x}{1+x^2} dx = \log(1+x^2) + C, \quad \int \frac{dx}{x-a} = \log |x-a| + C.$$

The first absolute value disappears because  $1+x^2 > 0$ ; the second formula is used on an interval on one side of  $a$ .

The same recognition pattern gives

$$\boxed{\int \tan x \, dx = -\log |\cos x| + C}$$

on intervals avoiding zeros of  $\cos x$ , because

$$\tan x = -\frac{(\cos x)'}{\cos x}.$$

All affine variants follow from one chain-rule template. If  $a \neq 0$ ,

$$\int F'(ax + b) \, dx = \frac{1}{a} F(ax + b) + C.$$

Consequently,

$$\begin{aligned} \int e^{ax+b} \, dx &= \frac{1}{a} e^{ax+b} + C, & \int \sin(ax + b) \, dx &= -\frac{1}{a} \cos(ax + b) + C, \\ \int \cos(ax + b) \, dx &= \frac{1}{a} \sin(ax + b) + C, & \int \frac{dx}{ax + b} &= \frac{1}{a} \log |ax + b| + C. \end{aligned}$$

These are not additional rules to memorize; they follow from the canonical antiderivatives by the chain rule or substitution. For  $a > 0$ ,

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C \quad (|x| < a),$$

and

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C \quad (x \in \mathbb{R}).$$

The scaled arcsine family also equals

$$-\arccos\left(\frac{x}{a}\right) + C \quad (|x| < a).$$

The chain rule checks the scale factors explicitly:

$$\begin{aligned} \frac{d}{dx} \arcsin\left(\frac{x}{a}\right) &= \frac{1/a}{\sqrt{1 - x^2/a^2}} = \frac{1}{\sqrt{a^2 - x^2}}, \\ \frac{d}{dx} \left[ \frac{1}{a} \arctan\left(\frac{x}{a}\right) \right] &= \frac{1}{a^2} \frac{1}{1 + x^2/a^2} = \frac{1}{a^2 + x^2}. \end{aligned}$$

Here  $a > 0$  ensures the stated simplification of the square root. The inverse-hyperbolic derivative in [Section 8.5](#) similarly gives

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \operatorname{arsinh}(x/a) + C = \log(x + \sqrt{x^2 + a^2}) + C_1, \quad a > 0,$$

where the two constants differ by  $\log a$ . These formulas are recognition tools, not permissions to ignore singularities or cross disconnected components.

**Definition 9.4.8** (Oriented interval integrals). If  $f$  is Riemann integrable on the closed interval with endpoints  $r$  and  $s$ , retain the preceding Riemann integral when

$$r \leq s,$$

and, when

$$s < r,$$

define

$$\int_r^s f(x) \, dx = - \int_s^r f(x) \, dx.$$

In particular,

$$\int_r^r f(x) \, dx = 0.$$

Thus reversing endpoints reverses the orientation of an integral.

**Integrating power series.** For a power series

$$\sum_{n=0}^{\infty} c_n(t-a)^n$$

of radius  $R > 0$ , [Theorem 7.5.3](#) gives uniform convergence on every closed interval strictly inside its radius. Applying the uniform-series corollary of section 9.3 on the segment joining  $a$  and  $x$  gives

$$\int_a^x \left( \sum_{n=0}^{\infty} c_n(t-a)^n \right) dt = \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1}, \quad |x-a| < R.$$

For  $x < a$  use the orientation convention of [Definition 9.4.8](#). The polynomial integrals follow from the FTC. This is the integration counterpart to [Theorem 8.3.3](#).

For example, integrate the geometric series

$$\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n \quad (|t| < 1)$$

from 0 to  $x$ . The FTC and the derivative of  $-\log(1-t)$  then give

$$-\log(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n}, \quad |x| < 1.$$

Uniform convergence on the segment from 0 to  $x$  justifies the interchange.

## Substitution

**Theorem 9.4.9** (Substitution). Let

$$\phi : [\alpha, \beta] \longrightarrow \mathbb{R}$$

be of class  $C^1$ , and let  $f$  be continuous on an open interval containing

$$\phi([\alpha, \beta]).$$

Then

$$\int_{\alpha}^{\beta} f(\phi(t))\phi'(t) \, dt = \int_{\phi(\alpha)}^{\phi(\beta)} f(u) \, du.$$

*Proof.* Choose  $c$  in the open interval on which  $f$  is defined and set

$$F(x) = \int_c^x f(u) \, du,$$

using [Definition 9.4.8](#). The first part of the Fundamental Theorem of Calculus, applied locally on closed subintervals, gives

$$F'(x) = f(x).$$

The chain rule therefore gives

$$(F \circ \phi)'(t) = f(\phi(t))\phi'(t).$$

Applying [Theorem 9.4.3](#) to  $F \circ \phi$  yields

$$\int_{\alpha}^{\beta} f(\phi(t))\phi'(t) \, dt = F(\phi(\beta)) - F(\phi(\alpha)) = \int_{\phi(\alpha)}^{\phi(\beta)} f(u) \, du.$$

□

**Substitution is the reverse chain rule.** The pattern to recognize is

$$f(g(x))g'(x).$$

Choose an inner expression  $u = g(x)$  and check whether its derivative is present, perhaps up to a constant. Rewrite the entire integrand in  $u$ ; for a definite integral, replace the bounds by  $g(a)$  and  $g(b)$ . The substitution is useful when the transformed integral is simpler. For instance,

$$\int 2xe^{x^2} \, dx = e^{x^2} + C, \quad \int \frac{2x}{1+x^2} \, dx = \log(1+x^2) + C.$$

For the definite version, continuity first gives existence, and then

$$\int_0^1 2xe^{x^2} \, dx = \int_0^1 e^u \, du = e - 1.$$

Here  $u = x^2$  sends the old endpoints 0, 1 to 0, 1. Likewise,

$$\int_0^1 \frac{2x}{1+x^2} \, dx = \int_1^2 \frac{du}{u} = \log 2.$$

A substitution may be legal without being useful. Setting  $u = x^2$  in

$$\int e^x \, dx$$

introduces  $x = \sqrt{u}$  and a factor

$$1/(2\sqrt{u});$$

it obscures an integral already in the basic table.

**A representative trigonometric substitution.** For  $a > 0$ , the radical in

$$\int \sqrt{a^2 - x^2} \, dx$$

suggests  $x = a \sin \theta$ . On a range where  $-\pi/2 \leq \theta \leq \pi/2$ , one has

$$dx = a \cos \theta \, d\theta, \quad \sqrt{a^2 - x^2} = a\sqrt{1 - \sin^2 \theta} = a \cos \theta.$$

Hence

$$\int \sqrt{a^2 - x^2} \, dx = a^2 \int \cos^2 \theta \, d\theta = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin(x/a) + C.$$

The recognition principle is to match a radical with an identity:  $\sqrt{a^2 - x^2}$  suggests sine,  $\sqrt{a^2 + x^2}$  suggests a hyperbolic sine (or tangent), and  $\sqrt{x^2 - a^2}$  suggests a hyperbolic cosine (or secant). One carefully chosen substitution is usually more useful than a catalogue.

**Remark 9.4.10** (Substitution, Jacobians, and pullbacks). In one variable, substitution is a short consequence of the chain rule and the Fundamental Theorem of Calculus; no injectivity of  $\phi$  is required. In several variables, the analogous transformation factor is

$$\phi'(t) \longrightarrow \det D\Phi(x),$$

where these are signed derivative factors, interpreted as oriented factors in Chapter 13. The ordinary scalar multiple-integral theorem uses  $|\det D\Phi|$ , since scalar volume is unsigned. Its rigorous Riemann/Jordan treatment is the genuinely nontrivial change-of-variables theorem of Chapter 12. Chapter 13 then interprets both formulas as pullback: one-variable substitution leads to multivariable change of variables and then to the pullback of differential forms.

## Integration by parts

**Theorem 9.4.11** (Integration by parts). Let

$$u, v : [a, b] \longrightarrow \mathbb{R}$$

be of class  $C^1$ . Then

$$\int_a^b u(x)v'(x) \, dx = [u(x)v(x)]_{x=a}^{x=b} - \int_a^b u'(x)v(x) \, dx.$$

*Proof.* The product rule gives

$$(uv)' = u'v + uv'.$$

Integrating and applying [Theorem 9.4.3](#) to  $uv$  gives

$$u(b)v(b) - u(a)v(a) = \int_a^b u'(x)v(x) \, dx + \int_a^b u(x)v'(x) \, dx.$$

Rearrangement proves the formula. The mnemonic notation

$$\int u \, dv = uv - \int v \, du$$

records this identity but is not its formal statement. □

**Example 9.4.12** (A short integration-by-parts computation). Taking

$$u(x) = x, \quad v(x) = e^x$$

on  $[0, 1]$  gives

$$\int_0^1 x e^x \, dx = [x e^x]_{x=0}^{x=1} - \int_0^1 e^x \, dx = 1.$$

**Translate differential shorthand into derivatives.** Here  $du$  abbreviates  $u'(x) \, dx$  and  $dv$  abbreviates  $v'(x) \, dx$ . Choosing  $dv = dx$  means choosing  $v'(x) = 1$ , for example  $v = x$ . This shorthand records the already proved [Theorem 9.4.11](#); it is not a proof by manipulating infinitesimal numbers. Chapter 11 will give  $dx$  and  $df$  their covector interpretation.

**Example 9.4.13** (Choosing which factor to differentiate). To compute the integral below, choose

$$u = \log x, \quad dv = dx, \quad du = \frac{1}{x} \, dx, \quad v = x,$$

because

$$u'(x) = 1/x$$

and  $v'(x) = 1$ . This choice simplifies the remaining product to  $v \, du = dx$ :

$$\int_1^e \log x \, dx = [x \log x]_1^e - \int_1^e 1 \, dx = e - (e - 1) = 1.$$

Differentiating the logarithm removed it; integrating the constant factor was easy.

**Choosing the parts.** Integration by parts reverses the product rule. Choose  $u$  so that differentiating it simplifies it, and choose  $dv$  so that an antiderivative  $v$  is easy. The mnemonic LIATE (logarithmic, inverse trigonometric, algebraic, trigonometric, exponential) can suggest a choice of  $u$ , but it is a heuristic, not a theorem. After one step, verify that the new integral is simpler. For the product  $x e^x$ , choosing

$$u = x, \quad dv = e^x \, dx$$

removes the polynomial degree:

$$\int x e^x \, dx = x e^x - e^x + C.$$

The reversed choice asks for an antiderivative of  $x$  but differentiates  $e^x$ , producing

$$\frac{1}{2} x^2 e^x - \frac{1}{2} \int x^2 e^x \, dx,$$

a harder integral than the original. A formally valid choice can therefore be a poor strategy.

**Example 9.4.14** (Integration-by-parts patterns). Polynomial–trigonometric products behave like the polynomial–exponential case:

$$\int x \sin x \, dx = -x \cos x + \sin x + C, \quad \int x \cos x \, dx = x \sin x + \cos x + C.$$

Hidden products use a factor of 1. Thus

$$\int \log x \, dx = x \log x - x + C \quad (x > 0),$$



and

$$\int \arctan x \, dx = x \arctan x - \frac{1}{2} \log(1 + x^2) + C.$$

For the arctangent formula, the hidden product is  $1 \cdot \arctan x$ ; choose  $u = \arctan x$  and  $v = x$ . The remaining integral is

$$\int \frac{x}{1 + x^2} \, dx = \frac{1}{2} \log(1 + x^2) + C.$$

The same choice for arcsine gives, on  $(-1, 1)$ ,

$$\begin{aligned} \int \arcsin x \, dx &= x \arcsin x - \int \frac{x}{\sqrt{1 - x^2}} \, dx \\ &= x \arcsin x + \sqrt{1 - x^2} + C, \end{aligned}$$

where the last step follows by substituting  $u = 1 - x^2$ , or differentiating the square root. Subtract this identity from the integral of the constant principal-range sum to obtain

$$\int \arccos x \, dx = x \arccos x - \sqrt{1 - x^2} + C.$$

As a check, the two formulas add to

$$\frac{\pi}{2}x + C;$$

the square-root terms cancel, as the exact arcsine–arccosine identity requires. Repeated integration by parts gives

$$\int x^2 e^x \, dx = e^x(x^2 - 2x + 2) + C.$$

For a cyclic example, write

$$I = \int e^x \cos x \, dx.$$

Two applications give

$$I = e^x \cos x + e^x \sin x - I,$$

so

$$\int e^x \cos x \, dx = \frac{1}{2} e^x (\sin x + \cos x) + C.$$

Finally, continuity gives existence of the proper integral, and the definite formula keeps the boundary term visible:

$$\int_0^\pi x \sin x \, dx = [-x \cos x]_0^\pi + \int_0^\pi \cos x \, dx = \pi.$$

## Partial fractions for rational functions

For a rational function

$$P(x)/Q(x),$$

partial fractions turn algebraic factorization into logarithmic, power, and arctangent antiderivatives. We use the method only when the needed real factorization of  $Q$  is supplied or easy to obtain; no general polynomial-factorization theorem is assumed.

1.  $\deg P \geq \deg Q$  ? Perform polynomial division first.
2. Factor the remaining denominator as needed over  $\mathbb{R}$ .
3. Write one term for every required factor and multiplicity.
4. Solve for the coefficients.
5. Integrate the elementary terms.
6. Check poles and the domain of every logarithm.

Distinct linear factors use

$$\frac{1}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b},$$

whereas a repeated factor  $(x-a)^m$  requires all terms

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_m}{(x-a)^m}.$$

An irreducible quadratic receives a linear numerator

$$\frac{Ax+B}{x^2+px+q}.$$

If an irreducible quadratic  $q(x)$  occurs to power  $m$ , include every term

$$\frac{A_1x+B_1}{q(x)} + \frac{A_2x+B_2}{q(x)^2} + \cdots + \frac{A_mx+B_m}{q(x)^m}.$$

Rewriting  $Ax+B$  as a multiple of  $2x+p$  plus a remainder separates a logarithmic term from an arctangent-type term after completing the square.

**Example 9.4.15** (Four partial-fraction patterns). For distinct linear factors,

$$\frac{1}{(x-1)(x+2)} = \frac{1}{3} \frac{1}{x-1} - \frac{1}{3} \frac{1}{x+2},$$

so on any interval avoiding both poles,

$$\int \frac{dx}{(x-1)(x+2)} = \frac{1}{3} \log|x-1| - \frac{1}{3} \log|x+2| + C.$$

For a repeated factor,

$$\frac{1}{x(x-1)^2} = \frac{1}{x} - \frac{1}{x-1} + \frac{1}{(x-1)^2},$$

and therefore

$$\int \frac{dx}{x(x-1)^2} = \log|x| - \log|x-1| - \frac{1}{x-1} + C.$$

Polynomial division comes first in

$$\frac{x^2+1}{x-1} = x+1 + \frac{2}{x-1},$$

giving

$$\int \frac{x^2+1}{x-1} dx = \frac{x^2}{2} + x + 2 \log|x-1| + C.$$

Finally,

$$\frac{3x+5}{x^2+2x+5} = \frac{3}{2} \frac{2x+2}{x^2+2x+5} + 2 \frac{1}{(x+1)^2+4},$$

so

$$\int \frac{3x+5}{x^2+2x+5} dx = \frac{3}{2} \log(x^2+2x+5) + \arctan\left(\frac{x+1}{2}\right) + C.$$

**Remark 9.4.16** (Algebra does not remove a pole). The decomposition may be correct wherever the original rational function is defined, but it does not make an integral across a pole proper. For example,

$$1/(x^2 - 1)$$

has a pole at  $x = 1$ , so

$$\int_0^2 1/(x^2 - 1) \, dx$$

is not a proper integral and must later be split into independent one-sided improper integrals. In particular,

algebraic manipulation does not override domain or convergence questions.

**Definition 9.4.17** (Average value). For an integrable function  $f : [a, b] \rightarrow \mathbb{R}$ , its **average value** is

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) \, dx.$$

**Theorem 9.4.18** (Mean value theorem for integrals). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, where  $a < b$ . There is a point

$$c \in [a, b]$$

such that

$$\int_a^b f(x) \, dx = f(c)(b-a).$$

Equivalently, a continuous function on  $[a, b]$  attains its average value:  $f(c) = f_{\text{avg}}$ .

*Proof.* By the Extreme Value Theorem,  $f$  has a minimum  $m$  and maximum  $M$ . Order preservation of the integral gives

$$m(b-a) \leq \int_a^b f(x) \, dx \leq M(b-a).$$

Dividing by  $b-a > 0$  gives a number between  $m$  and  $M$ . The Intermediate Value Theorem supplies  $c \in [a, b]$  at which  $f$  takes that value.  $\square$

## Using the FTC and formulating accumulations

The two parts of the FTC answer different questions. A continuous  $f$  defines an antiderivative

$$F(x) = \int_a^x f(t) \, dt;$$

conversely, a known antiderivative  $G$  evaluates

$$\int_a^b f(t) \, dt = G(b) - G(a).$$

Once the hypotheses are checked, neither computation requires reconstructing Riemann sums.

For ordinary integrals, let the structure of the integrand suggest the first move:

| integrand structure                                 | first method to consider        |
|-----------------------------------------------------|---------------------------------|
| recognizable derivative formula                     | basic antiderivative            |
| $f(g(x))g'(x)$                                      | substitution                    |
| product where differentiation simplifies one factor | integration by parts            |
| $P(x)/Q(x)$                                         | division and partial fractions. |

Real problems may combine methods. Before selecting one, split at changes of formula or sign when necessary, reverse orientation if useful, exploit odd or even symmetry on symmetric intervals, and check the domain of every formula.

For a definite integral, keep three questions separate:

1. Does the proper integral exist?
2. Which antiderivative or technique is appropriate?
3. How does the FTC evaluate the result?

A complete calculation therefore often follows

check continuity and domain  $\rightarrow$  select a method  $\rightarrow$  find or transform an antiderivative  
 $\rightarrow$  apply the FTC  $\rightarrow$  interpret or check the answer.

**Example 9.4.19** (Evaluation, orientation, and splitting). For a polynomial integrand,

$$\int_0^2 (3x^2 - 2x + 1) \, dx = [x^3 - x^2 + x]_0^2 = 6.$$

Reversing the bounds gives  $-6$ . For an absolute value first split where its formula changes:

$$\int_0^2 |x - 1| \, dx = \int_0^1 (1 - x) \, dx + \int_1^2 (x - 1) \, dx = 1.$$

Substitution  $u = 1 + x^2$  gives

$$\int_0^1 \frac{2x}{1 + x^2} \, dx = \int_1^2 \frac{du}{u} = \log 2;$$

the new bounds are the substituted endpoint values. For an integrable odd function, substitution  $x = -t$  shows that the two integrals on  $[-a, 0]$  and  $[0, a]$  cancel. Thus

$$\int_{-1}^1 (x^3 + x) \, dx = 0.$$

For an even function the same argument doubles the integral on  $[0, a]$ .

**From local contribution to total amount.** When  $q(x)$  measures amount per unit  $x$ , a short interval contributes approximately its width times a representative rate or density:

$$\Delta A_i \approx q(\xi_i) \Delta x_i, \quad A \approx \sum_i q(\xi_i) \Delta x_i.$$

When the model defines the total as the limit of these sums and  $q$  is Riemann integrable, [Theorem 9.2.6](#) identifies it as

$$A = \int_a^b q(x) \, dx.$$

This makes the modeling assumption visible before the FTC is used to compute. For a time rate the units satisfy

$$(\text{amount}/\text{time})(\text{time}) = \text{amount}.$$

**Example 9.4.20** (From a rate to an amount). A container receives water at rate  $r(t) = 2 + t$  litres per minute during the first 3 minutes, with no outflow. The variable is time; the desired quantity is the volume added, not the rate at one instant. On a short interval the constant-rate approximation is  $r(\xi)\Delta t$  litres. Summing and taking the tagged-sum limit gives

$$\Delta V = \int_0^3 (2 + t) \, dt = [2t + t^2/2]_0^3 = 21/2 \quad \text{litres.}$$

Continuity ensures integrability; the FTC evaluates the accumulation. Equivalently, if a volume function has derivative  $r$ , the FTC gives exactly its total change. The average rate, in litres per minute, is

$$\frac{1}{3} \int_0^3 r(t) \, dt = \frac{7}{2}.$$

**Example 9.4.21** (Displacement and distance). A particle has velocity  $v(t) = t - 1$  metres per second on  $[0, 2]$ . If  $s'(t) = v(t)$ , the FTC identifies the signed change in position:

$$s(b) - s(a) = \int_a^b v(t) \, dt.$$

On short intervals the magnitude of motion is approximately

$$|\Delta s| \approx |v(\xi)|\Delta t,$$

so the accumulation of speed gives total distance. For this particle, the displacement, in metres, is

$$\int_0^2 v(t) \, dt = 0.$$

The distance traveled is the accumulation of speed  $|v(t)|$ ; splitting at the reversal time  $t = 1$  gives 1 metre, by the absolute-value computation above:

$$\text{distance} = \int_0^2 |v(t)| \, dt = 1 \text{ metre.}$$

The negative and positive velocity contributions cancel in displacement; their magnitudes add in distance. Figure 9.5 shows precisely which contributions change sign.

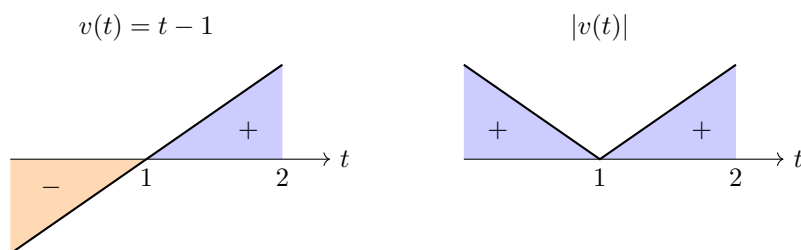


Figure 9.5: Displacement adds signed contributions (left), giving zero here. Distance adds their magnitudes (right), giving one metre.

**Example 9.4.22** (Mass, average value, and accumulated deviation). If a thin rod occupying  $[0, L]$  has continuous linear density  $\rho(x)$  kilograms per metre, a short piece near  $\xi$  has mass

$$\Delta M \approx \rho(\xi)\Delta x.$$

Adding the pieces and taking the tagged-sum limit gives total mass

$$M = \int_0^L \rho(x) \, dx.$$

The dimensional check is

$$(\text{mass/length})(\text{length}) = \text{mass}.$$

For continuous  $f$  on  $[a, b]$ , its average is the constant  $c$  producing the same total accumulation:

$$\int_a^b c \, dx = \int_a^b f(x) \, dx, \quad c(b-a) = \int_a^b f(x) \, dx.$$

Dividing by interval length gives

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) \, dx;$$

the mean value theorem for integrals says this average is attained at some point. For two continuous profiles  $f, g$ , the local magnitude of discrepancy is approximately

$$|f(\xi_i) - g(\xi_i)|\Delta x_i.$$

Thus the nonnegative quantity

$$\int_a^b |f(x) - g(x)| \, dx$$

measures total accumulated vertical deviation, whereas

$$\int_a^b (f - g)$$

permits positive and negative discrepancies to cancel. In each application, translate the assumptions, choose the integrand and bounds, check continuity, compute, and interpret the units.

**Standard and extended Newton–Leibniz hypotheses.** The familiar theorem and the next proposition differ only in the regularity required of the derivative:

| standard continuous-derivative setting | integrable-derivative extension |
|----------------------------------------|---------------------------------|
| $F'$ continuous                        | $F'$ merely Riemann integrable  |
| $F$ essentially $C^1$                  | $F$ merely differentiable       |
| Newton–Leibniz holds                   | Newton–Leibniz still holds.     |

Thus continuous differentiability is a convenient sufficient hypothesis, not the boundary of the theorem.

**Proposition 9.4.23** (Newton–Leibniz with an integrable derivative). If  $F$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and  $F'$  extends to a Riemann-integrable function  $g$  on  $[a, b]$ , then

$$F(b) - F(a) = \int_a^b g.$$

*Proof.* On each partition interval the mean value theorem supplies an interior tag  $\xi_i$  with  $F(x_i) - F(x_{i-1}) = g(\xi_i)(x_i - x_{i-1})$ . These tags may depend on the partition. Their tagged sum is exactly

$$S(g, P, \xi) = \sum_i g(\xi_i)(x_i - x_{i-1}) = \sum_i (F(x_i) - F(x_{i-1})) = F(b) - F(a).$$

Take, for example, equal subdivisions with mesh

$$(b - a)/N \rightarrow 0.$$

By [Theorem 9.2.6](#), these particular tagged sums converge to

$$\int_a^b g,$$

since  $g$  is integrable and the theorem allows every choice of tags. They all equal  $F(b) - F(a)$ , proving the identity.  $\square$

After the measure-zero theory and Lebesgue criterion in section 9.6, we return to Newton–Leibniz when the derivative identity has a small exceptional set; see “[How Far Can Newton–Leibniz Be Extended?](#)”

## 9.5 Improper Integrals

For a nonnegative time rate  $r$ , first distinguish the amount accumulated by a finite time from a proposed infinite-time total:

$$A(T) = \int_0^T r(t) \, dt, \quad A(\infty) = \int_0^\infty r(t) \, dt.$$

The second expression will mean the limit of the first, provided it is finite. When it is finite, the amount remaining after time  $T$  is the tail

$$A(\infty) - A(T) = \int_T^\infty r(t) \, dt.$$

The finite truncation is the modeled amount before any improper limit is taken.

Finite-interval Riemann integration does not directly cover infinite domains or unbounded behavior. Improper integrals retain the finite theory and add an explicit limiting process whose convergence must be established.

**Definition 9.5.1** (Improper Riemann integral). Let  $f : [a, \infty) \rightarrow \mathbb{R}$  be Riemann integrable on every finite interval  $[a, b]$ . The **improper integral** of  $f$  converges if

$$\int_a^\infty f(x) \, dx = \lim_{b \rightarrow \infty} \int_a^b f(x) \, dx$$

exists as a finite real number. Define an improper integral at a finite endpoint analogously by a one-sided limit.

Thus an improper integral is not a new kind of finite-interval integral: it is a limit of ordinary Riemann integrals whose existence must be proved.

**Theorem 9.5.2** (Cauchy criterion for improper integrals). The improper integral  $\int_a^\infty f$  converges if and only if, for every  $\varepsilon > 0$ , there is  $A > a$  such that

$$d > c > A \implies \left| \int_c^d f(x) \, dx \right| < \varepsilon.$$

At a finite singular endpoint,  $\int_a^b f$  converges from the right if and only if, for every  $\varepsilon > 0$ , some  $\delta > 0$  satisfies

$$a < c < d < a + \delta \implies \left| \int_c^d f(x) \, dx \right| < \varepsilon.$$

*Proof.* For the infinite endpoint, put  $F(t) = \int_a^t f$ . Additivity gives

$$F(d) - F(c) = \int_c^d f.$$

Thus the displayed condition is exactly the Cauchy criterion for the real limit of  $F(t)$  as  $t \rightarrow \infty$ . The finite-endpoint statement is the same argument for the truncation function  $F(c) = \int_c^b f$  as  $c \rightarrow a^+$ .  $\square$

This is the integral analogue of the series tail criterion:

$$\sum_{n=N+1}^M a_n \longleftrightarrow \int_c^d f.$$

At a singular left endpoint, the definition is

$$\int_a^b f(x) \, dx = \lim_{c \rightarrow a^+} \int_c^b f(x) \, dx,$$

when this limit exists finitely; a right-endpoint singularity is analogous. An interior singularity at  $c$  requires two separately convergent integrals,

$$\int_a^c f + \int_c^b f,$$

and cancellation between divergent sides is not allowed. Likewise,

$$\int_{-\infty}^\infty f = \int_{-\infty}^c f + \int_c^\infty f$$

means that both one-sided improper integrals converge, independently of the chosen finite splitting point  $c$ . Infinity is never an endpoint in  $\mathbb{R}$ .



**Example 9.5.3** ( $p$ -integrals at infinity and at zero). For real  $p$ , direct evaluation of the truncated proper integrals gives

$$\int_1^\infty x^{-p} dx \text{ converges exactly when } p > 1,$$

whereas

$$\int_0^1 x^{-p} dx \text{ converges exactly when } p < 1.$$

In particular,

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{c \rightarrow 0^+} 2(1 - \sqrt{c}) = 2,$$

while

$$\int_0^1 x^{-1} dx, \quad \int_1^\infty x^{-1} dx$$

diverge. The opposite thresholds are a useful reminder to identify which endpoint causes the improper behavior.

The  $p$ -integral classifications follow from the explicit truncations

$$\int_1^b x^{-p} dx = \begin{cases} \frac{b^{1-p} - 1}{1-p}, & p \neq 1, \\ \log b, & p = 1, \end{cases}$$

$$\int_c^1 x^{-p} dx = \begin{cases} \frac{1 - c^{1-p}}{1-p}, & p \neq 1, \\ -\log c, & p = 1. \end{cases}$$

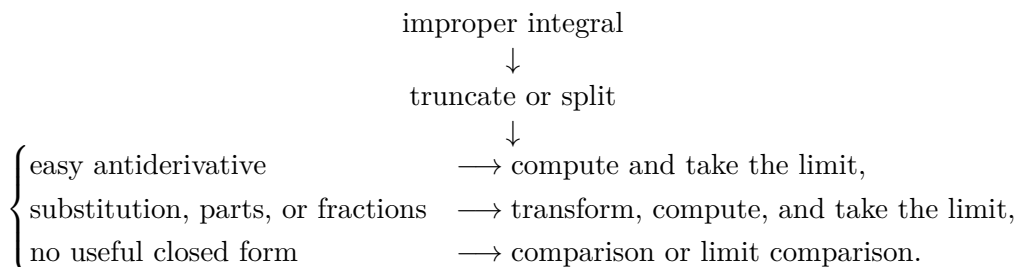
For  $p > 1$  the first limit is

$$1/(p-1);$$

for  $p < 1$  the second is

$$1/(1-p).$$

The other cases diverge to  $+\infty$ . Here real powers are available from Chapter 8. Locate each source of impropriety, split at interior singularities, and truncate each piece. Evaluate the resulting proper integrals and take their limits independently; convergence requires all of those limits to exist. The computational choice occurs only after truncation:



**Example 9.5.4** (Ordinary techniques inside improper integrals). Substitution evaluates

$$\int_0^\infty x e^{-x^2} dx = \lim_{b \rightarrow \infty} \int_0^b x e^{-x^2} dx = \frac{1}{2} \lim_{b \rightarrow \infty} \int_0^{b^2} e^{-u} du = \frac{1}{2}.$$

Integration by parts is applied before the limit:

$$\int_1^\infty x e^{-x} dx = \lim_{b \rightarrow \infty} \left( [-x e^{-x}]_1^b + \int_1^b e^{-x} dx \right) = \frac{2}{e}.$$

Partial fractions give

$$\int_1^\infty \frac{dx}{x(x+1)} = \lim_{b \rightarrow \infty} [\log x - \log(x+1)]_1^b = \log 2.$$

In each calculation the finite truncation is a proper integral and only its evaluated value is sent to the improper endpoint.

**Definition 9.5.5** (Absolute convergence of an improper integral). An improper integral of  $f$  is **absolutely convergent** if the corresponding improper integral of  $|f|$  converges.

**Proposition 9.5.6** (Absolute convergence implies convergence). If  $f$  is integrable on every finite truncation and its improper integral converges absolutely, then its improper integral converges.

*Proof.* By the Cauchy criterion applied to  $|f|$ , for every  $\varepsilon > 0$  there is  $A$  such that  $d > c > A$  implies

$$\left| \int_c^d f \right| \leq \int_c^d |f| < \varepsilon.$$

The criterion for  $f$  now proves convergence. Finite-endpoint and multiple improper-endpoint cases are handled one side at a time in the same way.  $\square$

**Theorem 9.5.7** (Comparison for improper integrals). Let  $0 \leq f \leq g$  on  $[a, \infty)$ , with both functions Riemann integrable on finite intervals. If the improper integral of  $g$  converges, then that of  $f$  converges.

*Proof.* Put

$$H(b) = \int_a^b f.$$

Nonnegativity makes  $H$  increasing in its real endpoint  $b$ , and comparison bounds it by the finite improper integral of  $g$ . Set  $L = \sup\{H(b) : b \geq a\}$ . Given  $\varepsilon > 0$ , choose  $c$  with  $H(c) > L - \varepsilon$ . For every real  $b \geq c$ ,

$$L - \varepsilon < H(c) \leq H(b) \leq L.$$

This is  $H(b) \rightarrow L$  as  $b \rightarrow \infty$ ; the monotone-convergence argument applies, now with a real endpoint rather than an integer index.  $\square$

**Corollary 9.5.8** (Limit comparison for improper integrals). Let  $f, g : [a, \infty) \rightarrow \mathbb{R}$  be Riemann integrable on every finite interval  $[a, b]$ . Suppose  $f, g > 0$  eventually and

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L, \quad 0 < L < \infty.$$

Then their improper integrals have the same convergence behavior.

*Proof.* Choose  $A_0$  from which  $f, g > 0$ , and  $A_1$  such that

$$|f(x)/g(x) - L| < L/2$$

for  $x > A_1$ . For every  $x > A = \max\{A_0, A_1, a\}$ ,

$$(L/2)g(x) < f(x) < (3L/2)g(x).$$

Apply comparison on that tail in both directions; changing a finite initial interval does not affect convergence.  $\square$

**Example 9.5.9** (Comparison instead of an antiderivative). On  $[1, \infty)$ ,

$$0 \leq 1/(1+x^2) \leq 1/x^2,$$

so its improper integral converges. For

$$(2x+1)/(x^3+1),$$

the ratio to

$$1/x^2$$

tends to 2; positive limit comparison gives convergence. For

$$1/\sqrt{x^2+1},$$

the ratio to

$$1/x$$

tends to 1; its integral diverges. All integrands are continuous on finite truncations. These tests decide existence without requiring a formula for the accumulated value.

**Example 9.5.10** (An infinite-time accumulation). A device releases material at rate

$$r(t) = 2/(1+t)^2$$

grams per hour for  $t \geq 0$ . The assumptions specify the rate at every nonnegative time. The amount released by time  $b$  is

$$A(b) = \int_0^b \frac{2}{(1+t)^2} dt = 2 - \frac{2}{1+b} \quad \text{grams.}$$

Therefore the infinite-time total is 2 grams, defined as the improper limit. The amount yet to be released after time  $b$  is

$$2/(1+b)$$

grams, below a tolerance  $\eta > 0$  when

$$b > 2/\eta.$$

The formula yields both a finite limiting total and a usable error bound.

We now close the logical gap explicitly marked in [Remark 7.4.1](#): proper integrals, their improper limits, and the needed comparison principles have all been established. The preview's rectangle argument can now be a proof.

**Theorem 9.5.11** (Integral test). Let  $f : [1, \infty) \rightarrow \mathbb{R}$  be continuous, nonnegative, and decreasing, and put  $a_n = f(n)$ . Then

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if the improper integral

$$\int_1^{\infty} f(x) \, dx$$

converges. More precisely, for every  $N \in \mathbb{N}$ ,

$$\int_1^{N+1} f(x) \, dx \leq \sum_{n=1}^N f(n) \leq f(1) + \int_1^N f(x) \, dx.$$

*Proof.* For  $x \in [n, n+1]$ , monotonicity gives

$$f(n+1) \leq f(x) \leq f(n).$$

Integrating these inequalities over the unit intervals and summing gives the displayed bounds. The partial sums and truncated integrals are increasing, so each is bounded above if and only if the other is bounded above. For real endpoints, if  $b \geq 1$ , choose an integer  $N \geq b$ ; then

$$\int_1^b f \leq \int_1^N f.$$

Thus boundedness at integer endpoints is boundedness at all endpoints. The supremum argument in [Theorem 9.5.7](#) proves convergence of these increasing truncations; monotone convergence proves convergence of the partial sums. This proves the equivalence for the genuine improper limit.  $\square$

**Theorem 9.5.12** ( $p$ -series test: analytic rederivation). For  $p \in \mathbb{R}$ , the series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if  $p > 1$ .

*Proof.* For rational  $p$ , this conclusion was already proved without integration in [Theorem 7.4.2](#). We now give an analytic proof that extends the criterion to every real exponent, using the real powers constructed in [Chapter 8](#).

For  $p > 0$ , the function

$$f(x) = x^{-p}$$

is continuous, nonnegative, and decreasing on  $[1, \infty)$  by [Theorem 8.5.15](#). If  $p \neq 1$ , that theorem gives

$$\frac{d}{dx} \left( \frac{x^{1-p}}{1-p} \right) = x^{-p};$$

if  $p = 1$ , [Theorem 8.5.8](#) gives

$$\frac{d}{dx} \log x = \frac{1}{x}.$$

Thus [Theorem 9.4.3](#) gives

$$\int_1^b x^{-p} \, dx = \begin{cases} \frac{b^{1-p} - 1}{1-p}, & p \neq 1, \\ \log b, & p = 1. \end{cases}$$

Thus the improper integral converges exactly when  $p > 1$ , and [Theorem 9.5.11](#) gives the conclusion. If  $p \leq 0$ , the terms do not tend to zero, so the series diverges by [Proposition 7.1.4](#).  $\square$

**Example 9.5.13** (Cancellation at a singularity). The two one-sided integrals of

$$1/x$$

across zero diverge: the negative side gives  $\log \delta \rightarrow -\infty$ , the positive side gives  $-\log \eta \rightarrow +\infty$ . Choosing  $\delta = \eta$  cancels them at each stage but does not define an improper integral. That linked-cutoff limit is a Cauchy principal value. Choosing  $\eta = \delta^2$  instead makes the sum  $-\log \delta$  diverge. Independent one-sided convergence is essential.

## 9.6 The Lebesgue Criterion for Riemann Integrability

Continuity is sufficient for Riemann integrability, but it is not necessary. The natural question is how large a discontinuity set Riemann integration can tolerate. Darboux sums measure oscillation on intervals; the criterion below answers the question by asking whether discontinuities fit inside interval covers of arbitrarily small total length. This will make the Dirichlet–Thomae comparison decisive: density alone does not decide integrability. We use only countable interval covers, not a formal theory of Lebesgue measure.

For comparison, [Theorem 8.2.9](#) already proves that a monotone function has at most countably many discontinuities. The direct Darboux proof of monotone integrability earlier in this chapter remains logically independent of the criterion developed here.

Ordinary interval length is the starting point for Lebesgue measure  $\mu$ , which assigns a generalized length to a large class of subsets of  $\mathbb{R}$ , called measurable sets. Not every subset of  $\mathbb{R}$  is measurable. Ordinary intervals are measurable, and for  $a < b$ ,

$$\boxed{\mu([a, b]) = \mu((a, b)) = \mu([a, b)) = \mu((a, b]) = b - a.}$$

Indeed,  $\mu(\{a\}) = \mu(\{b\}) = 0$ , so adding or deleting finitely many endpoints does not change measure. The central property, stated here without proof, is countable additivity: for pairwise disjoint measurable sets  $E_j$ ,

$$\mu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \mu(E_j).$$

Lebesgue measure is also *translation invariant*, a fact stated without proof: for every measurable  $E \subseteq \mathbb{R}$  and  $t \in \mathbb{R}$ , the translate is measurable and

$$E + t = \{x + t : x \in E\}, \quad \mu(E + t) = \mu(E).$$

Translation preserves length; it does not remove overlap. For example,

$$E = [0, 1], \quad F = E + \frac{1}{2},$$

have equal measure by translation invariance, but

$$\mu(E) + \mu(F) = 2, \quad \mu(E \cup F) = \frac{3}{2}.$$

The overlap has been counted twice in the sum. Pairwise disjointness in countable additivity ensures that no portion is counted more than once. A standard consequence of countable additivity, also used here without proof, is countable subadditivity:

$$\mu\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} \mu(E_j).$$

Thus, if a measurable set  $E$  is covered by open intervals  $I_j$  of total length less than  $\varepsilon$ , then

$$\begin{aligned}\mu(E) &\leq \mu\left(\bigcup_{j=1}^{\infty} I_j\right) \leq \sum_{j=1}^{\infty} \mu(I_j) \\ &= \sum_{j=1}^{\infty} \ell(I_j) < \varepsilon.\end{aligned}$$

If such a cover exists for every  $\varepsilon > 0$ , then  $\mu(E) = 0$ . A theorem of Lebesgue measure theory, also stated without proof, says that *all* subsets satisfying this interval-cover condition are measurable and have measure zero, and conversely every measure-zero set has such covers. We can therefore use the following self-contained covering definition. Only this property is used in this chapter; neither a construction of Lebesgue measure nor Lebesgue integration is required.

A discontinuity need not be a jump. To measure all possible local failures of continuity at once, compare every pair of nearby values and then shrink the neighborhood. The resulting oscillation records the variation that cannot be removed by choosing a smaller neighborhood.

**Definition 9.6.1** (Oscillation and measure-zero sets). Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. For each  $x \in [a, b]$ , define

$$N_\delta(x) = [a, b] \cap (x - \delta, x + \delta), \quad \omega_f(x) = \inf_{\delta > 0} \operatorname{osc}_{N_\delta(x)}(f).$$

Shrinking the neighborhood can only decrease its oscillation by restriction monotonicity. Thus  $\omega_f(x)$  is the variation that cannot be removed by zooming in at  $x$ . A set  $E \subseteq [a, b]$  has **Lebesgue measure zero** if, for every  $\eta > 0$ , there are open intervals  $I_1, I_2, \dots$  such that

$$E \subseteq \bigcup_{j=1}^{\infty} I_j \quad \text{and} \quad \sum_{j=1}^{\infty} \ell(I_j) < \eta.$$

Finite covers are allowed as a special case.

**Remark 9.6.2** (Null sets and almost everywhere). A measure-zero set is also called a *null set*. A property holds *almost everywhere* (a.e.) if it holds outside a set of Lebesgue measure zero.

**Proposition 9.6.3** (Oscillation and continuity). Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. Then

$$\omega_f(x) = 0$$

if and only if  $f$  is continuous at  $x$ .

*Proof.* If  $f$  is continuous at  $x$ , then, for a given  $\varepsilon > 0$ , all values in a sufficiently small relative neighborhood of  $x$  lie within

$$\varepsilon/2$$

of  $f(x)$ . Their pairwise differences are therefore less than  $\varepsilon$ , since

$$|f(y) - f(z)| \leq |f(y) - f(x)| + |f(z) - f(x)|.$$

Thus the defining infimum is zero. Conversely, if  $\omega_f(x) = 0$ , then the defining infimum supplies a  $\delta > 0$  for which the pairwise oscillation in the relative  $\delta$ -neighborhood of  $x$  is less than  $\varepsilon$ . Taking one of the two points to be  $x$  gives

$$|f(y) - f(x)| < \varepsilon$$

whenever  $|y - x| < \delta$ . □

**Proposition 9.6.4** (Closed oscillation sets). Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded and let  $\varepsilon > 0$ . Then

$$D_\varepsilon = \{x \in [a, b] : \omega_f(x) \geq \varepsilon\}$$

is closed in  $[a, b]$ .

*Proof.* If  $\omega_f(x) < \varepsilon$ , choose  $r > 0$  with

$$\text{osc}_{N_r(x)}(f) < \varepsilon.$$

If  $|y - x| < r/2$ , then  $N_{r/2}(y) \subseteq N_r(x)$ , and hence

$$\omega_f(y) \leq \text{osc}_{N_{r/2}(y)}(f) \leq \text{osc}_{N_r(x)}(f) < \varepsilon.$$

Hence the complement of  $D_\varepsilon$  is open. □

**Proposition 9.6.5** (Countable sets have measure zero). Every countable subset of  $[a, b]$  has measure zero.

*Proof.* Let

$$E \subseteq \{x_1, x_2, \dots\},$$

where repetitions are allowed if  $E$  is finite. Given  $\eta > 0$ , cover  $x_n$  by an open interval of length

$$\frac{\eta}{2^{n+1}}.$$

These intervals cover  $E$ , and their total length is less than  $\eta$ . □

**Lemma 9.6.6** (Countable unions of measure-zero sets). If

$$E_1, E_2, \dots \subseteq [a, b]$$

have measure zero, then

$$\bigcup_{m=1}^{\infty} E_m$$

has measure zero.

*Proof.* Given  $\eta > 0$ , cover  $E_m$  by open intervals having total length less than

$$\frac{\eta}{2^{m+1}}.$$

The combined doubly indexed family is countable by the countability result of Chapter 2, covers the displayed union, and has total length less than  $\eta$ . □

The criterion uses two error accounts. On good intervals the oscillation is small, even if their total length is large. On bad intervals the oscillation may be large, but their total length must be small.

$$\underbrace{\sum_{\text{good}} \text{osc}_I(f)|I|}_{\leq \eta(b-a)} + \underbrace{\sum_{\text{bad}} \text{osc}_I(f)|I|}_{\leq 2M\ell_{\text{bad}}}.$$

With  $|f| \leq M$ , choose

$$\eta(b-a) < \varepsilon/2, \quad 2M\ell_{\text{bad}} < \varepsilon/2.$$

Large oscillation is acceptable when confined to intervals of sufficiently small combined length.

good: small oscillation

---

bad: short total length

**Theorem 9.6.7** (Lebesgue criterion for Riemann integrability). Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. Then

$$f \text{ is Riemann integrable} \iff \text{Disc}(f) \text{ has Lebesgue measure zero.}$$

Here  $\text{Disc}(f)$  denotes the discontinuity set. For the proof, we use the following equivalent formulations.

(a)  $f$  is Riemann integrable.

(b) For every  $\varepsilon > 0$ , the set

$$D_\varepsilon = \{x \in [a, b] : \omega_f(x) \geq \varepsilon\}$$

has measure zero.

(c) The discontinuity set of  $f$  has measure zero.

**Remark 9.6.8** (Terminology). This result is commonly called **Lebesgue's criterion for Riemann integrability**. Some sources use a name such as the **Riemann–Lebesgue criterion**; it must not be confused with the Riemann–Lebesgue lemma from Fourier analysis.

*Proof.* **(a) $\Rightarrow$ (b): small gap forces short large-oscillation cells.** Fix an oscillation threshold  $\alpha > 0$  and a cover budget  $\eta > 0$ . By the Darboux criterion choose  $P$  with  $U(f, P) - L(f, P) < \alpha\eta/4$ . Then

$$\alpha \sum_{\text{osc}_{I_i}(f) \geq \alpha} \ell(I_i) \leq \sum_{\text{osc}_{I_i}(f) \geq \alpha} \text{osc}_{I_i}(f) \ell(I_i) \leq U(f, P) - L(f, P).$$

Thus the large-oscillation cells have combined length below  $\eta/4$ . Every non-partition point  $x \in D_\alpha$  lies in the interior of one such cell: otherwise a small relative neighborhood of  $x$  inside its cell would have oscillation below  $\alpha$ , contradicting  $\omega_f(x) \geq \alpha$ . Enlarge the finitely many large cells slightly to open intervals of total length below  $\eta/2$ , and cover the finitely many partition points separately with total length below  $\eta/2$ . This covers  $D_\alpha$  within budget  $\eta$ .

**(b) $\Rightarrow$ (a): four stages.** Fix  $\varepsilon > 0$  and choose  $M \geq 1$  with  $|f| \leq M$ .

*Stage 1: choose the vertical threshold.* Select  $\alpha > 0$  so that

$$\alpha(b - a) < \varepsilon/2.$$

*Stage 2: make the bad cover short and finite.* Cover  $D_\alpha$  by open intervals with total length below  $\varepsilon/(4M)$ . The set  $D_\alpha$  is relatively closed by [Proposition 9.6.4](#), hence compact by [Corollary 5.7.7](#); retain a finite subcover and let  $O$  be its union.

*Stage 3: obtain one common horizontal scale.* For each  $x \in [a, b] \setminus O$ , the inequality  $\omega_f(x) < \alpha$  supplies an open interval  $V_x$  on which  $\text{osc}(f) < \alpha$ . The finitely many bad intervals together with the  $V_x$  cover  $[a, b]$ . Apply [Lemma 5.7.5](#) and choose  $P$  with  $\text{mesh}(P) < \lambda$ .

*Stage 4: split the Darboux gap.* A cell is bad if it lies in one of the selected bad intervals; every other cell lies in some  $V_x$  and is good. Assigning each bad cell to one selected interval shows that their combined length is below  $\varepsilon/(4M)$ . Therefore

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{\text{good } I} \text{osc}_I(f) \ell(I) + \sum_{\text{bad } I} \text{osc}_I(f) \ell(I) \\ &< \alpha(b - a) + 2M \frac{\varepsilon}{4M} < \varepsilon. \end{aligned}$$

The Darboux criterion proves (a).



Finally, [Proposition 9.6.3](#) and the Archimedean property give

$$\text{Disc}(f) = \bigcup_{m=1}^{\infty} D_{1/m}.$$

Indeed, discontinuity is equivalent to  $\omega_f(x) > 0$ , and then some  $m$  satisfies  $1/m \leq \omega_f(x)$ . Countable-union stability proves (b) $\Rightarrow$ (c), while  $D_\alpha \subseteq \text{Disc}(f)$  proves (c) $\Rightarrow$ (b).  $\square$

**Example 9.6.9** (The Dirichlet function). For

$$\mathbf{1}_{\mathbb{Q}} : [0, 1] \rightarrow \mathbb{R},$$

every neighborhood contains rational and irrational points. Hence

$$\omega_{\mathbf{1}_{\mathbb{Q}}}(x) = 1 \quad (x \in [0, 1]),$$

so

$$D_{1/2} = [0, 1]$$

does not have measure zero.

**Example 9.6.10** (Thomae's function). For  $x \in [0, 1]$ , define

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ is rational and written in lowest terms,} \\ 0, & x \text{ is irrational.} \end{cases}$$

This is commonly called **Thomae's function**, or the **popcorn function**; some texts call it Riemann's function, a name best avoided because it is ambiguous in analysis. The function  $T$  is continuous at every irrational point and discontinuous at every rational point. Consequently,

$$\text{Disc}(T) = \mathbb{Q} \cap [0, 1],$$

$T$  is Riemann integrable, and

$$\int_0^1 T(x) \, dx = 0.$$

*Proof.* Let  $x$  be irrational, and fix  $\varepsilon > 0$ . There are only finitely many rational numbers in  $[0, 1]$  that, in lowest terms, have denominator

$$q \leq \frac{1}{\varepsilon};$$

indeed, there are only finitely many possible numerators for each such denominator. Let  $F_\varepsilon$  be this finite set. Since

$$x \notin F_\varepsilon,$$

there is a neighborhood of  $x$  avoiding  $F_\varepsilon$ . At every point  $y$  in this neighborhood, either  $y$  is irrational and  $T(y) = 0$ , or its reduced denominator satisfies

$$q > \frac{1}{\varepsilon},$$

so

$$T(y) = \frac{1}{q} < \varepsilon.$$

Since  $T(x) = 0$ , this proves that  $T$  is continuous at  $x$ .

If

$$x = p/q \in [0, 1]$$

is rational in lowest terms, irrational density (Proposition 3.4.13) supplies an irrational point in every relative open neighborhood of  $x$ . There  $T$  is zero, whereas

$$T(x) = 1/q.$$

The continuity test with

$$\varepsilon = 1/(2q)$$

therefore fails at  $x$ , including at the endpoints. This proves the stated discontinuity set.

The set

$$\mathbb{Q} \cap [0, 1]$$

is countable and therefore has measure zero by Proposition 9.6.5. The criterion proves that  $T$  is integrable. Finally, every nondegenerate subinterval of  $[0, 1]$  contains an irrational point by Proposition 3.4.13. The infimum of  $T$  on it is therefore zero. Every lower Darboux sum is zero; since  $T$  is integrable, its integral is zero.  $\square$

**Remark 9.6.11** (The Dirichlet–Thomae contrast). Earlier integrability proofs constructed partitions and estimated

$$U(f, P) - L(f, P).$$

The criterion often saves that work: identify the discontinuity set, show it has measure zero, and conclude integrability. A direct Darboux proof for Thomae’s function requires separating the finitely many large values from the small ones. Once continuity has been analyzed, the criterion gives

$$\text{Disc}(T) = \mathbb{Q} \cap [0, 1],$$

a countable set, hence a set of measure zero. Thus  $T$  is Riemann integrable. For the Dirichlet function,

$$\text{Disc}(\mathbf{1}_{\mathbb{Q}}) = [0, 1], \quad \mu([0, 1]) = 1,$$

so it is not Riemann integrable. Density alone does not determine integrability; the measure-theoretic size of the discontinuity set does.

## How Far Can Newton–Leibniz Be Extended?

Three separate assertions enter the Fundamental Theorem of Calculus:

|                                                                                     |
|-------------------------------------------------------------------------------------|
| $F \text{ is differentiable,} \quad f \text{ is Riemann integrable,} \quad F' = f.$ |
|-------------------------------------------------------------------------------------|

The first concerns the existence of a local rate of change, the second the convergence of sums, and the third identifies two functions. Neither of the first two assertions alone gives the other assertions. In the finite- and countable-exception theorems below,  $F'$  exists at every interior point; only the identity  $F' = f$  is allowed to fail on the stated exceptional set. This differs from the later Lipschitz/a.e. theorem, where differentiability itself may fail on a null set.

**Example 9.6.12** (Differentiable need not mean  $C^1$ ). Set

$$F(0) = 0, \quad F(x) = x^2 \sin(1/x) \quad (x \neq 0).$$

Then

$$|F(h)/h| \leq |h|,$$

so  $F'(0) = 0$ , while for  $x \neq 0$ ,

$$F'(x) = 2x \sin(1/x) - \cos(1/x).$$

This derivative is bounded near zero but has no limit there. Hence  $F$  is differentiable and is not  $C^1$ . On every compact interval,  $F'$  is bounded and continuous except at zero, so the finite-discontinuity argument following [Theorem 9.2.6](#) makes it Riemann integrable. Consequently,

$$F' \text{ Riemann integrable} \not\Rightarrow F' \text{ continuous},$$

and [Proposition 9.4.23](#) is genuinely stronger than the continuous-derivative version of the FTC.

**Example 9.6.13** (A Riemann-integrable jump is not a derivative). Let  $j(x) = 0$  for  $x < 0$  and  $j(x) = 1$  for  $x \geq 0$  on  $[-1, 1]$ . This bounded monotone function is Riemann integrable. It cannot equal  $F'$  everywhere for any differentiable  $F$ , because derivatives have the intermediate value property by [Theorem 8.2.6](#), while  $j$  omits every value strictly between 0 and 1. Thus

$$\text{Riemann integrable} \not\Rightarrow \text{is a derivative}.$$

**Example 9.6.14** (A differentiable function with an unbounded derivative). Set  $F(0) = 0$  and

$$F(x) = x^2 \sin(1/x^2)$$

for  $x \neq 0$ . Since

$$\left| \frac{F(h) - F(0)}{h} \right| = |h \sin(1/h^2)| \leq |h| \rightarrow 0,$$

we have  $F'(0) = 0$ . At other points,

$$F'(x) = 2x \sin(1/x^2) - \frac{2}{x} \cos(1/x^2).$$

For

$$x_n = (2\pi n)^{-1/2},$$

one has  $F'(x_n) = -2\sqrt{2\pi n}$ . The sequence  $-x_n$  gives the corresponding positive values. Thus  $F'$  is unbounded on every nondegenerate compact interval containing zero, even if zero is an endpoint, and is not Riemann integrable there:

$$F \text{ differentiable} \not\Rightarrow F' \text{ Riemann integrable}.$$

**Example 9.6.15** (An accumulation function with a corner). On  $[-1, 1]$  let  $f(x) = 0$  for  $x < 0$  and  $f(x) = 1$  for  $x \geq 0$ . This monotone function is Riemann integrable, and direct integration gives

$$A(x) = \int_{-1}^x f(t) \, dt = \begin{cases} 0, & -1 \leq x \leq 0, \\ x, & 0 \leq x \leq 1. \end{cases}$$

The left derivative of  $A$  at zero is 0, while the right derivative is 1. The accumulation-function theorem guarantees  $A' = f$  at continuity points of  $f$ ; it makes no such promise at its jump. In particular,

$$f \text{ Riemann integrable} \not\Rightarrow \left( \int_a^x f \right)' = f \text{ everywhere.}$$

**Lemma 9.6.16** (Finite modifications). If bounded functions  $f, g : [a, b] \rightarrow \mathbb{R}$  differ only at finitely many points and  $f$  is Riemann integrable, then  $g$  is Riemann integrable and

$$\int_a^b g = \int_a^b f.$$

*Proof.* Put  $u = g - f$ . Then  $u = 0$  off a finite set  $E$  and  $|u| \leq B$  for some  $B$ . Surround the points of  $E$  by intervals of arbitrarily small total length and insert their endpoints in a partition. Every other cell has zero oscillation, while each bad cell has oscillation at most  $2B$ . Consequently the Darboux gap for  $u$  can be made arbitrarily small, so  $u$  is integrable. Moreover both  $|L(u, P)|$  and  $|U(u, P)|$  are at most  $B$  times the total bad length; squeezing gives  $\int u = 0$ . Linearity yields the conclusion. Endpoint modifications are included.  $\square$

**Theorem 9.6.17** (Newton–Leibniz with finitely many exceptions). Let  $F$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ , and let  $f : [a, b] \rightarrow \mathbb{R}$  be Riemann integrable. If  $F'(x) = f(x)$  except at finitely many points of  $(a, b)$ , then

$$F(b) - F(a) = \int_a^b f(t) \, dt.$$

*Proof.* Replace  $f(x)$  by  $F'(x)$  at the exceptional interior points, retaining the endpoint values of  $f$ . The resulting function is a Riemann-integrable extension of  $F'$  with the same integral as  $f$ . Apply [Proposition 9.4.23](#).  $\square$

**Theorem 9.6.18** (Newton–Leibniz with countably many exceptions). Let  $F$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ , and let  $f : [a, b] \rightarrow \mathbb{R}$  be Riemann integrable. Suppose  $E \subset (a, b)$  is countable and  $F'(x) = f(x)$  for  $x \in (a, b) \setminus E$ . Then

$$F(b) - F(a) = \int_a^b f(t) \, dt.$$

*Proof. Strategy.* Transfer local bounds on  $f$  to  $F'$  by Darboux's theorem, then squeeze increments of  $F$  between the Darboux sums of  $f$ .

**Stage 1: transfer local bounds from  $f$  to  $F'$ .** First let  $J \subset (a, b)$  be a nonempty open interval on which  $\alpha \leq f \leq \beta$ . We claim that the same bounds hold for  $F'$ . If  $F'(t_0) > \beta$ , choose  $s \in J \setminus E$ , possible because every nondegenerate interval is uncountable. Then  $F'(s) = f(s) \leq \beta$ . By [Theorem 8.2.6](#), every value in  $(\beta, F'(t_0))$  is attained between  $s$  and  $t_0$ , hence inside  $J$ . All such values must be attained at exceptional points. This puts an uncountable interval inside the

countable image  $F'(E \cap J)$ , a contradiction. Apply the same argument to  $-F, -f$  for the lower bound. Thus

$$\alpha \leq F'(t) \leq \beta \quad (t \in J).$$

**Stage 2: compare increments with Darboux sums.** For a partition  $a = x_0 < \cdots < x_n = b$ , put  $m_i = \inf_{[x_{i-1}, x_i]} f$  and  $M_i = \sup_{[x_{i-1}, x_i]} f$ . The local bound applies on  $(x_{i-1}, x_i)$ . The Mean Value Theorem, using continuity of  $F$  at the endpoints and differentiability at every interior point, gives

$$m_i \Delta x_i \leq F(x_i) - F(x_{i-1}) \leq M_i \Delta x_i.$$

Summing yields

$$L(f, P) \leq F(b) - F(a) \leq U(f, P).$$

The integral lies in this same interval; Riemann integrability makes its length arbitrarily small, proving the equality.  $\square$

Finite changes preserve integrability without further assumptions. Countable changes need not: changing the zero function to 1 at rational points produces the Dirichlet function. In the theorem, the changed function is a derivative. Darboux's property prevents its exceptional values from escaping the bounds forced by nearby nonexceptional values. Countability excludes an interval of such values. The local bounds then place each increment of  $F$  between the corresponding lower and upper Darboux contributions of  $f$ , so integrability of  $f$  completes the proof.

**Theorem 9.6.19** (Lipschitz, almost-everywhere Riemann form). Let  $F : [a, b] \rightarrow \mathbb{R}$  be Lipschitz and let  $f : [a, b] \rightarrow \mathbb{R}$  be Riemann integrable. If  $F' = f$  almost everywhere on  $(a, b)$  (in particular,  $F'$  exists outside a set of Lebesgue measure zero), then

$$F(b) - F(a) = \int_a^b f(t) \, dt.$$

Here *almost everywhere*, abbreviated a.e., means outside a set of Lebesgue measure zero. This is the strongest form developed here whose right-hand side remains an ordinary Riemann integral. The proof is not developed here because it requires the measure-theoretic structure of Lipschitz functions. For the needed results, see for example [9, 16]. For

$$A(x) = \int_a^x f(t) \, dt,$$

the integral estimate makes  $A$  Lipschitz, and the accumulation theorem and Lebesgue criterion give  $A' = f$  a.e. Thus  $F - A$  is Lipschitz with derivative zero a.e. To conclude that it is constant requires a deeper theorem about Lipschitz functions; its proof belongs to measure-theoretic real analysis and is deferred here.

**Remark 9.6.20** (The Cantor-function warning). There exists a continuous nondecreasing function  $C : [0, 1] \rightarrow [0, 1]$ , called the Cantor function, with  $C(0) = 0$ ,  $C(1) = 1$ , and  $C' = 0$  almost everywhere. Its construction is not needed here; see [7, 9] for standard treatments. For  $f = 0$  it gives

$$C' = f \text{ a.e.}, \quad \int_0^1 f = 0, \quad C(1) - C(0) = 1.$$

Continuity together with an a.e. derivative identity therefore does not suffice for Newton–Leibniz. The Cantor function is not Lipschitz. This does not show that Lipschitz continuity is the weakest possible sufficient condition: absolute continuity, defined next, is weaker and is the condition that characterizes indefinite Lebesgue integrals.

## 9.7 Limits of the Riemann Theory

Section 9.6 has already encountered the simplest important measure-theoretic notion of size: sets of Lebesgue measure zero. Full measure theory extends this from recognizing null sets to assigning generalized length systematically and defining a new integral. This leads to the relation between absolute continuity and integration described below.

The Dirichlet function differs from zero only on the null set  $\mathbb{Q} \cap [0, 1]$ , yet it is not Riemann integrable. Thus changing a function on a null set can destroy Riemann integrability. Thomae's function, whose discontinuity set is the same dense countable set, shows that density of the discontinuity set alone is not the obstruction. Improper integrals also force one to treat unbounded behavior and infinite domains by separate limiting conventions.

The resulting progression is:

|                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $  \begin{aligned}  &\text{Riemann integrability} \longrightarrow \text{measure-zero exceptional sets} \\  &\quad \longrightarrow \text{Lebesgue measure and integration} \\  &\quad \quad \longrightarrow \text{MCT and DCT} \\  &\quad \quad \longrightarrow \text{absolute continuity and the Lebesgue FTC}  \end{aligned}  $ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### An informal working framework for Lebesgue integration

A notion of size is generally assigned not to every subset of a set  $X$ , but to a specified collection on which it behaves consistently. Some measures, including counting measure and the zero measure, can be defined on all of  $\mathcal{P}(X)$ . The obstruction is instead geometric: ordinary length on  $\mathbb{R}$  cannot be extended to every subset while remaining countably additive, translation invariant, and consistent with interval length, for example  $\mu([0, 1]) = 1$ . Lebesgue measure is therefore defined on a large family of Lebesgue-measurable sets, not on all of  $\mathcal{P}(\mathbb{R})$ .

**Definition 9.7.1** ( $\sigma$ -algebra and measurable space). For a set  $X$ , a  **$\sigma$ -algebra** is a family  $\mathcal{A} \subseteq \mathcal{P}(X)$  such that

$$X \in \mathcal{A}, \quad E \in \mathcal{A} \implies X \setminus E \in \mathcal{A},$$

and

$$E_1, E_2, \dots \in \mathcal{A} \implies \bigcup_{n=1}^{\infty} E_n \in \mathcal{A}.$$

Its members are **measurable sets**, and  $(X, \mathcal{A})$  is a **measurable space**. Complements and countable unions repeatedly arise in limiting arguments, so the admissible sets must remain stable under them.

**Definition 9.7.2** (Measure). A **measure** on  $(X, \mathcal{A})$  is a function

$$\mu : \mathcal{A} \rightarrow [0, \infty]$$

such that  $\mu(\emptyset) = 0$  and, for pairwise disjoint measurable sets,

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n).$$

The *Borel  $\sigma$ -algebra* on the line is

$$\mathcal{B}(\mathbb{R}) = \sigma(\{\text{open subsets of } \mathbb{R}\}).$$

Here “generated by” means

$\mathcal{B}(\mathbb{R})$  is the smallest  $\sigma$ -algebra containing every open subset of  $\mathbb{R}$ .

Starting with the open sets, one repeatedly closes the family under complements and countable unions. De Morgan’s law also supplies countable intersections:

$$\bigcap_{n=1}^{\infty} E_n = \left( \bigcup_{n=1}^{\infty} E_n^c \right)^c.$$

The word *repeatedly* matters: one round of these operations does not describe all Borel sets. By [Proposition 5.3.7](#), open intervals generate the same  $\sigma$ -algebra. The Lebesgue  $\sigma$ -algebra is larger: it contains all Borel sets and all subsets of Lebesgue-null sets.

**Definition 9.7.3** (Measurable maps). A map

$$f : (X, \mathcal{A}) \rightarrow (Y, \mathcal{B})$$

is **measurable** if  $f^{-1}(B) \in \mathcal{A}$  for every  $B \in \mathcal{B}$ . A real-valued measurable function has target  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ .

**Proposition 9.7.4** (Threshold criterion for real-valued measurability). A real-valued function  $f$  on  $(X, \mathcal{A})$  is measurable if and only if

$$\{x \in X : f(x) > \alpha\} \in \mathcal{A} \quad \text{for every } \alpha \in \mathbb{R}.$$

The proof is not developed here. The threshold form says concretely that the region on which  $f$  lies above each prescribed level is measurable. If  $X$  is a topological space equipped with its Borel  $\sigma$ -algebra, every continuous map  $f : X \rightarrow \mathbb{R}$  is Borel measurable. In particular, a continuous real-valued function on an interval is Lebesgue measurable.

**Definition 9.7.5** (Simple functions and their integrals). Let  $E \in \mathcal{A}$ . A nonnegative measurable **simple function**  $s : E \rightarrow [0, \infty)$  has the form

$$s = \sum_{k=1}^m a_k \mathbf{1}_{E_k}, \quad a_k \geq 0,$$

where the measurable sets  $E_k \subseteq E$  may be taken pairwise disjoint. Define

$$\int_E s \, d\mu = \sum_{k=1}^m a_k \mu(E_k).$$

Each value is weighted by the measure of the set on which it occurs.

This highlights two organizing ideas:

Riemann: partition the domain into short intervals;

Lebesgue: measure sets on which values lie in specified ranges.

The second box is a conceptual contrast, not a claim that every later calculation literally uses level sets.

For a nonnegative measurable function, the defining idea is approximation from below:

$$\int_E f \, d\mu = \sup \left\{ \int_E s \, d\mu : 0 \leq s \leq f, s \text{ nonnegative measurable simple} \right\}.$$

For a real-valued measurable function define

$$f^+ = \max\{f, 0\}, \quad f^- = \max\{-f, 0\}.$$

Then  $f = f^+ - f^-$  and  $|f| = f^+ + f^-$ . The function  $f$  is *Lebesgue integrable* on  $E$  when

$$\int_E |f| \, d\mu < \infty;$$

in that case

$$\int_E f \, d\mu = \int_E f^+ \, d\mu - \int_E f^- \, d\mu.$$

Changing a measurable function on a set of Lebesgue measure zero does not change its Lebesgue integral. This null-set invariance is not proved here; it explains why equality and convergence almost everywhere are natural in the convergence theorems and the Lebesgue FTC.

For an open  $G = \bigsqcup_j I_j \subset (a, b)$ , countable additivity bridges the elementary length of Section 9.3 with Lebesgue measure:

$$\mu(G) = \sum_j \mu(I_j) = \sum_j \ell(I_j) = \ell(G).$$

The conceptual progression is

$$\sigma\text{-algebras} \longrightarrow \text{measures} \longrightarrow \text{measurable functions} \longrightarrow \text{simple functions} \longrightarrow \text{Lebesgue integral}.$$

Because nonnegative integration is built from approximation from below, monotone increasing limits fit naturally. An integrable majorant controls the absolute size in dominated convergence, and null-set invariance makes almost-everywhere identities sufficient in the Lebesgue FTC.

This subsection supplies only the working framework needed for the theorem statements below. A systematic construction of Lebesgue measure and development of the Lebesgue integral belongs to a subsequent real-analysis course.

**Theorem 9.7.6** (Compatibility of Riemann and Lebesgue integration). If  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable, then  $f$  is Lebesgue measurable and Lebesgue integrable, and

$$\int_a^b f(x) \, dx = \int_{[a,b]} f \, d\mu.$$

This concerns the proper Riemann integral on a bounded closed interval. Its proof belongs to the systematic measure-theoretic construction omitted from these notes. Conceptually, Lebesgue integration extends ordinary proper Riemann integration without changing its value. The converse fails:  $\mathbf{1}_{\mathbb{Q} \cap [0,1]}$  is not Riemann integrable, but it is Lebesgue integrable and

$$\int_{[0,1]} \mathbf{1}_{\mathbb{Q}} \, d\mu = 0$$

because it vanishes almost everywhere. A conditionally convergent improper Riemann integral need not define a Lebesgue-integrable function with  $\int |f| \, d\mu < \infty$ .



**Theorem 9.7.7** (Lebesgue monotone convergence theorem). Let  $E \subseteq \mathbb{R}$  be measurable. If

$$0 \leq f_1 \leq f_2 \leq \cdots$$

are measurable and  $f_n(x) \rightarrow f(x)$  almost everywhere on  $E$ , then

$$\boxed{\int_E f \, d\mu = \lim_{n \rightarrow \infty} \int_E f_n \, d\mu,}$$

allowing the value  $+\infty$ . Equivalently,

$$\int_E f_n \, d\mu \uparrow \int_E f \, d\mu.$$

Its proof is not given here because it depends on the systematic development of the Lebesgue integral just placed outside our scope. The arrow in the displayed equivalent form means that the numerical sequence of integrals is increasing and converges to the displayed limit.

**Theorem 9.7.8** (Lebesgue dominated convergence theorem). Let  $E \subseteq \mathbb{R}$  be measurable. Suppose  $f_n$  are measurable,

$$f_n \rightarrow f \quad \text{almost everywhere,}$$

and there is a Lebesgue-integrable function  $g$  such that

$$|f_n| \leq g \quad \text{almost everywhere for every } n.$$

Then  $f$  is Lebesgue integrable and

$$\boxed{\int_E |f_n - f| \, d\mu \rightarrow 0.}$$

Consequently,

$$\boxed{\int_E f_n \, d\mu \rightarrow \int_E f \, d\mu.}$$

Its proof is likewise deferred to measure-theoretic real analysis.

**Dini and monotone convergence.** Dini's theorem [Theorem 6.2.8](#) says that

$$f_n \uparrow f, \quad f_n, f \in C([a, b])$$

implies uniform convergence. The Riemann uniform-interchange theorem then gives convergence of the integrals:

$$\text{Dini} \implies \text{uniform convergence} \implies \text{Riemann integral convergence}.$$

Lebesgue MCT instead obtains integral convergence directly from nonnegative monotone convergence of measurable functions, without continuity or a compactness-based uniform-convergence step. Dini is a useful Riemann-side precursor, not literally a Riemann MCT.

**Arzelà bounded convergence and dominated convergence.** The Riemann theorem [Theorem 9.3.4](#) assumes pointwise convergence, one constant bound  $|f_n| \leq M$ , and Riemann integrability of all relevant functions; it concludes

$$\int_a^b |f_n - f| \, dx \rightarrow 0.$$

Lebesgue DCT allows almost-everywhere convergence and a variable integrable dominator  $g(x)$ , and it supplies integrability of the limit. The same absolute-error integral conclusion makes the analogy visible, but the hypotheses and proof mechanisms are genuinely different.

For continuous functions, integration and differentiation are cleanly inverse operations. Under weaker Riemann hypotheses, one must separately track differentiability of  $F$ , integrability of  $f$ , the identity  $F' = f$ , and exceptional sets on which equality or differentiability may fail. This fragmentation reflects real limitations of the framework, not merely inefficient notation.

**Remark 9.7.9** (Riemann–Stieltjes integration). Another classical extension replaces the ordinary Riemann sums

$$\sum_i f(t_i)(x_i - x_{i-1})$$

by sums of the form

$$\sum_i f(t_i)(\alpha(x_i) - \alpha(x_{i-1})),$$

whose limiting value, when it exists, is denoted by

$$\int_a^b f \, d\alpha.$$

The ordinary Riemann integral is recovered when

$$\alpha(x) = x.$$

This Riemann–Stieltjes integral records accumulation relative to changes in another function: increasing or step-function integrators can encode weighted or discrete contributions, and the theory leads naturally to functions of bounded variation. Under suitable regularity hypotheses, it has the later integration-by-parts identity

$$\int_a^b f \, d\alpha + \int_a^b \alpha \, df = f(b)\alpha(b) - f(a)\alpha(a),$$

which generalizes [Theorem 9.4.11](#). No Riemann–Stieltjes theory is developed here. The measure-theoretic viewpoint later subsumes many such integrals by associating suitable measures to increasing functions; it gives the broader framework previewed again in [Section 13.17](#).

Measure theory introduces a systematic notion of size fine enough to distinguish a dense null set from an interval. The definitions above explain the objects in the preview theorems; their construction and full theory are not developed here. The Lebesgue criterion relates Riemann integrability to the size of the discontinuity set, while Lebesgue theory uses set size in the definition of the integral itself.

## Absolute continuity and the Lebesgue FTC

Uniform continuity controls the change over one short interval. Integration also requires control of the total change over a collection of intervals whose total length is small.

**Definition 9.7.10** (Absolute continuity). A function  $F : [a, b] \rightarrow \mathbb{R}$  is **absolutely continuous** if for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that any finite collection of pairwise disjoint intervals  $(a_k, b_k) \subset [a, b]$ ,  $1 \leq k \leq N$ , satisfies

$$\sum_{k=1}^N (b_k - a_k) < \delta \implies \sum_{k=1}^N |F(b_k) - F(a_k)| < \varepsilon.$$

The definition gives the hierarchy

$$\text{Lipschitz} \implies \text{absolutely continuous} \implies \text{uniformly continuous}.$$

For a Lipschitz constant  $L$ , sum  $|F(b_k) - F(a_k)| \leq L(b_k - a_k)$  and choose

$$\delta = \varepsilon / (L + 1).$$

For the second implication, use just one interval.

**Theorem 9.7.11** (Lebesgue FTC). If  $F : [a, b] \rightarrow \mathbb{R}$  is absolutely continuous, then  $F'$  exists almost everywhere, is Lebesgue integrable, and

$$F(x) = F(a) + \int_a^x F'(t) \, d\mu(t) \quad (x \in [a, b]).$$

Conversely, if  $f$  is Lebesgue integrable and

$$F(x) = c + \int_a^x f(t) \, d\mu(t),$$

then  $F$  is absolutely continuous and  $F' = f$  almost everywhere.

Lebesgue integrability means integrability in the later measure-theoretic sense, with finite integral of the absolute value. Values assigned to  $F'$  where it does not exist do not affect this integral. Almost everywhere has the same measure-zero meaning as in section 9.6. No construction of the Lebesgue integral or proof of this theorem is attempted in these notes; the omission reflects the measure-theoretic machinery required. Standard proofs may be found, for example, in [9, 16]. The two directions characterize the same functions:

|                                                                           |
|---------------------------------------------------------------------------|
| $\text{absolute continuity} \iff \text{indefinite Lebesgue integration}.$ |
|---------------------------------------------------------------------------|

The additive constant is free, and an integrand is recovered only up to equality almost everywhere. Absolute continuity is precisely the class on which differentiation and integration become inverse operations in the Lebesgue setting.

**The hierarchy of FTC statements.** In the first four lines below,  $F$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . In the exceptional-set and Lipschitz lines,  $f$  is Riemann integrable. Endpoint extensions are understood in the integrable-derivative line.

$$F \in C^1 \implies F' \text{ continuous and Newton-Leibniz},$$

$$F' \text{ Riemann integrable} \implies \text{Newton-Leibniz},$$

$$F' = f \text{ with finite exceptions} \implies F(b) - F(a) = \int_a^b f,$$

$$F' = f \text{ with countable exceptions} \implies F(b) - F(a) = \int_a^b f,$$

$$F \text{ Lipschitz, } F' = f \text{ a.e.} \implies F(b) - F(a) = \int_a^b f,$$

$$F \text{ absolutely continuous} \iff \text{indefinite Lebesgue integration}.$$

The first four statements have been proved in this chapter. The last two are forward-looking results. The successive extensions pass from classical FTC to an integrable derivative, then to small exceptional sets, a.e. identities, and finally absolute continuity.

## Exercises

### Integrability, computation, and improper integrals

**Exercise 1.\*** Use equal partitions to prove directly that  $f(x) = 2x + 1$  is Riemann integrable on  $[0, 2]$ . For  $\mathbf{1}_{\mathbb{Q}}$  on this interval show a fixed positive Darboux gap for every partition.

**Exercise 2.** Evaluate

$$\int_2^0 (x^2 + 1) \, dx, \quad \int_{-2}^2 x^3 \, dx, \quad \int_0^3 |x - 1| \, dx,$$

$$\int_0^1 2x \cos(x^2) \, dx, \quad \int_1^e \log x \, dx.$$

Name the operation that simplifies each.

**Exercise 3.\*** Water enters an initially empty container at a rate of  $3 + 2t$  litres per minute for  $0 \leq t \leq 4$ , with no outflow. Determine the volume at four minutes and the average inflow rate. Formulate the integrand and bounds before evaluating. Explain why rate times a short time interval approximates the volume added, check the units, and explain why average rate is divided by elapsed time.

**Exercise 4.** A particle has velocity  $2t - 2$  metres per second from  $t = 0$  to  $t = 3$ . Formulate and calculate its displacement and total distance. Explain the split needed for one of them, and why cancellation is meaningful for displacement but not for distance.

**Exercise 5.** Classify

$$\int_0^1 x^{-2/3} \, dx, \quad \int_1^\infty x^{-2/3} \, dx, \quad \int_{-1}^1 (1/x) \, dx.$$

Write every required one-sided truncation; do not combine divergent pieces.

**Exercise 6.** Use comparison or limit comparison to classify

$$\int_1^\infty \frac{3x + 1}{x^3 + 2} \, dx, \quad \int_1^\infty \frac{1}{\sqrt{x^2 + 4}} \, dx.$$

Verify finite-truncation integrability.

**Exercise 7.** A source releases material at

$$3/(1 + t)^2$$

grams per hour for all  $t \geq 0$ . Determine its infinite-time total and a time after which less than 0.01 gram remains to be released. State the improper limit that represents the total.

**Exercise 8.** Now justify formally the Integral-Test calculations requested in Chapter 7, Exercise 4. Check continuity, positivity, and monotonicity, and identify the theorem that closes the preview.

### Integrability criteria, convergence, and the FTC

**Exercise 9.** Use Theorem 9.6.7 to prove that a bounded function with finitely many discontinuities is Riemann integrable.

**Exercise 10.** Let  $f : [a, b] \rightarrow \mathbb{R}$  be monotone. Show that, for each  $\varepsilon > 0$ , the oscillation set  $D_\varepsilon$  from Proposition 9.6.4 is finite.

**Exercise 11.** Prove directly from the Darboux criterion that a function having only finitely many discontinuities on  $[a, b]$  and otherwise constant on finitely many intervals is Riemann integrable.

**Exercise 12.** Determine whether

$$\int_1^\infty \frac{1}{x^2} dx, \quad \int_1^\infty \frac{1}{x} dx$$

converge. Use Theorem 9.4.3, then compare the conclusion with the two corresponding series.

**Exercise 13.** Use Theorem 9.4.9 to evaluate

$$\int_0^1 2te^{t^2} dt.$$

State the substitution and identify the endpoint values.

**Exercise 14.** Use Theorem 9.4.11 to evaluate

$$\int_0^1 x \sin x dx.$$

**Exercise 15.** Explain why the function  $\mathbf{1}_{\mathbb{Q}}$  on  $[0, 1]$  has the same upper and lower sums for no partition. Which density facts from earlier chapters are being used?

**Exercise 16.** For  $f_n(x) = x^n$  on  $[0, 1]$ , apply Arzelà bounded convergence to the pointwise limit and verify the integral conclusion directly. Explain why the uniform convergence theorem does not apply on this interval.

**Exercise 17.\*** Construct integrable nonnegative triangular spikes on  $[0, 1]$  tending pointwise to zero with integrals equal to one. Identify exactly which Arzelà hypothesis fails.

**Exercise 18.** Give a direct tagged-sum proof that changing finitely many values of a Riemann-integrable function preserves its integrability and integral. Bound the contribution of cells whose tags belong to the exceptional set in terms of the mesh. Deduce Theorem 9.6.17.

**Exercise 19.** Compare the zero function and  $\mathbf{1}_{\mathbb{Q}}$  on  $[0, 1]$ . They agree outside a countable set: why does this not contradict Theorem 9.6.18? Explain where its proof would fail if an arbitrary function agreeing with  $f$  outside a countable set were substituted for  $F'$ . In particular, check the intermediate value property and integrability of  $\mathbf{1}_{\mathbb{Q}}$ .

### Computational techniques and applications

**Exercise 20.** For each integral, decide whether direct use of a basic antiderivative, substitution, integration by parts, partial fractions, or symmetry is the most efficient first move. Explain the choice before computing:

$$\int e^{4x+1} dx, \quad \int x \cos(x^2) dx, \quad \int x^2 e^x dx, \quad \int \frac{dx}{x^2 - 4}, \quad \int_{-2}^2 x^5 e^{x^2} dx.$$

State the interval on which each indefinite answer is valid.

**Exercise 21.** Evaluate

$$\int \cos(5x - 2) dx, \quad \int \frac{3x^2}{1 + x^3} dx, \quad \int_0^2 x e^{x^2} dx.$$

For the definite integral, check existence and transform the endpoint values. Then explain why  $u = x^2$  is not an efficient substitution for

$$\int e^x dx.$$

**Exercise 22.** Use integration by parts to compute

$$\int x \cos x dx, \quad \int \arctan x dx, \quad \int x^2 e^x dx.$$

For the first integral, compare the choice  $u = x$  with the reversed choice and identify which produces the simpler remaining integral.

**Exercise 23.** Apply integration by parts twice and solve algebraically for the original integral to evaluate

$$\int e^x \sin x dx.$$

Differentiate the answer to check it.

**Exercise 24.** Evaluate

$$\int_0^\pi x \sin x dx$$

by integration by parts. State why the proper integral exists before using the FTC, and interpret why the answer is positive.

**Exercise 25.** Decompose and integrate each rational function on an interval avoiding its poles:

$$\frac{1}{(x-2)(x+1)}, \quad \frac{1}{x(x+1)^2}, \quad \frac{x^2+3}{x+1}, \quad \frac{2x+7}{x^2+4x+8}.$$

Name the distinct-factor, repeated-factor, division, or irreducible-quadratic pattern used in each case.

**Exercise 26.** Before attempting to evaluate

$$\int_0^3 \frac{dx}{(x-1)(x+2)},$$

decide whether it is proper. Write the independent one-sided truncations required by the definition and classify them. Explain why a partial-fraction antiderivative cannot authorize evaluation straight across the pole.

**Exercise 27.** Let a rod on  $[0, 2]$  have density  $\rho(x) = 1 + x$  kilograms per metre. Explain the approximate mass of a short piece. Find its mass and its average density, with units, deriving the average from equal total mass. Then compute the total accumulated deviation of  $\rho$  from its average,

$$\int_0^2 |\rho(x) - \rho_{\text{avg}}| dx.$$

Identify the split point before integrating.

**Exercise 28.** For each improper integral, first locate the impropriety and write a proper truncation. Then select the indicated ordinary method and classify or evaluate the integral:

$$\int_0^\infty x e^{-x^2} dx \quad (\text{substitution}),$$

$$\int_1^\infty x e^{-x} dx \quad (\text{integration by parts}), \quad \int_1^\infty \frac{dx}{x(x+1)} \quad (\text{partial fractions}).$$

### Extended FTC and uniform integration

**Exercise 29.** Let  $F(0) = 0$  and

$$F(x) = x^2 \sin(1/x)$$

for  $x \neq 0$ . Prove from the definition that  $F'(0) = 0$ , compute  $F'(x)$  for  $x \neq 0$ , and show that  $F'$  is bounded but discontinuous at zero. Which version of Newton–Leibniz applies on  $[-1, 1]$ , and why does the standard  $C^1$  version not apply?

**Exercise 30.** Give one example for each strict failure:

differentiable  $\not\Rightarrow C^1$ ,

differentiable  $\not\Rightarrow$  Riemann-integrable derivative,

Riemann integrable  $\not\Rightarrow$  is a derivative,

Riemann integrable  $f \not\Rightarrow \left(\int_a^x f\right)' = f$  everywhere.

For each, identify the exact failed hypothesis rather than merely naming the example.

**Exercise 31.** Suppose  $f_n \rightarrow f$  uniformly on  $[a, b]$ . Prove directly that

$$\left| \int_a^b f_n - \int_a^b f \right| \leq (b-a) \|f_n - f\|_\infty.$$

Then give a pointwise-convergent sequence from this chapter for which the uniform theorem does not apply, and state which stronger special result, if any, does apply.

## Part II

# Multivariable Calculus and Analysis



## Chapter 10

# Topology and Linear Algebra

In one variable,  $|x - y|$  measures distance, one number  $f'(a)$  describes the best linear approximation, and substitution rescales a short interval. In several variables these three familiar operations separate into several ideas. A point is now a vector, and proximity must make sense without privileging one coordinate. A derivative sends an input displacement to an output displacement: it is a linear transformation. Its Jacobian matrix is the table representing that transformation in chosen coordinates, not the transformation itself. Substitution must measure the volume distortion of a small box; the absolute determinant gives its magnitude, while the sign records orientation. Finally, forms evaluate several tangent directions at once, requiring dual spaces and alternating multilinear maps.

Part I introduced the real-line versions of open and closed sets, relative topology, compactness, and sequential compactness in [Sections 5.3](#) and [5.7](#). This chapter now generalizes those notions systematically to metric and topological spaces. It is not trying to reproduce full courses in topology and linear algebra. It develops the parts of those subjects needed for Chapters 11–13, but develops those parts carefully enough that they do not become hidden prerequisites. Our roadmap is

distance and neighborhoods  $\longrightarrow$  compactness and continuity  
 $\longrightarrow$  vectors and linear maps  $\longrightarrow$  matrices and coordinates  
 $\longrightarrow$  inner products and norms  $\longrightarrow$  multilinearity  
 $\longrightarrow$  determinants and orientation.

Keep asking two questions: what is the object, and which numbers describe it? A coordinate column, a matrix, and a covector row will answer different questions even when all three are arrays of real numbers.

**Euclidean input convention.** When the input is a full Euclidean coordinate tuple, we write

$$\mathbf{x} = (x_1, \dots, x_n), \quad f(\mathbf{x}), \quad Df(\mathbf{x}).$$

The components  $x_i$  and individual scalar coordinates retain ordinary type. Points of an abstract metric space and established vector letters such as  $v \in V$  retain their existing notation. For a sequence of Euclidean inputs,  $\mathbf{x}_k$  denotes its  $k$ th point, whereas  $x_i$  denotes a component.

# 10.1 Distances, Balls, and Neighborhoods

On  $\mathbb{R}$ , distance is  $|x - y|$ . In the plane the Pythagorean theorem gives

$$d_2(\mathbf{x}, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

In  $\mathbb{R}^n$  define the *Euclidean length*

$$\|\mathbf{x}\|_2 = \left( \sum_{j=1}^n x_j^2 \right)^{1/2}, \quad d_2(\mathbf{x}, y) = \|\mathbf{x} - y\|_2.$$

We call this a length for the moment: the triangle inequality, and hence the fact that it is a norm, will follow from Cauchy–Schwarz below. Which properties of distance actually enter convergence and continuity arguments? Zero distance identifies a point, reversing the two points changes nothing, and an intermediate stop cannot shorten the trip. We abstract these properties, then verify that the Euclidean formula has them.

**Definition 10.1.1** (Metric space). A metric on  $X$  is a map  $d : X \times X \rightarrow [0, \infty)$  satisfying, for all  $x, y, z \in X$ ,

$$\begin{aligned} d(x, y) = 0 &\iff x = y && \text{(identity of indiscernibles),} \\ d(x, y) &= d(y, x) && \text{(symmetry),} \\ d(x, z) &\leq d(x, y) + d(y, z) && \text{(triangle inequality).} \end{aligned}$$

The pair  $(X, d)$  is a metric space.

**Theorem 10.1.2** (Cauchy–Schwarz inequality). Writing

$$\mathbf{x} \cdot y = \sum_j x_j y_j, \quad \|\mathbf{x}\|_2 = \sqrt{\mathbf{x} \cdot \mathbf{x}},$$

we have

$$|\mathbf{x} \cdot y| \leq \|\mathbf{x}\|_2 \|y\|_2.$$

Consequently  $\|\cdot\|_2$  is a norm and  $d_2(\mathbf{x}, y) = \|\mathbf{x} - y\|_2$  is its associated metric.

*Proof.* The triangle inequality for Euclidean length contains the cross term  $2u \cdot v$ . We therefore establish Cauchy–Schwarz first; it is exactly the estimate that controls this term. If  $y = 0$  the inequality reads  $0 \leq 0$ . Otherwise substitute

$$t = (\mathbf{x} \cdot y) / \|y\|_2^2$$

into a sum of squares. This value is chosen because

$$\|\mathbf{x} - ty\|_2^2 = \|\mathbf{x}\|_2^2 - 2t(\mathbf{x} \cdot y) + t^2\|y\|_2^2$$

is a nonnegative quadratic in  $t$ , minimized when

$$t = (\mathbf{x} \cdot y) / \|y\|_2^2$$

(equivalently, this choice completes the square). At its minimum,

$$0 \leq \sum_j (x_j - ty_j)^2 = \|\mathbf{x}\|_2^2 - \frac{(\mathbf{x} \cdot y)^2}{\|y\|_2^2}.$$

Multiplication by  $\|y\|_2^2$  and square roots give the assertion. For  $u = \mathbf{x} - y$  and  $v = y - z$ , it follows that

$$\|u + v\|_2^2 = \|u\|_2^2 + 2u \cdot v + \|v\|_2^2 \leq (\|u\|_2 + \|v\|_2)^2.$$

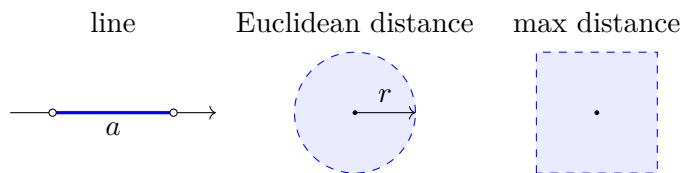
Taking square roots proves the triangle inequality. A sum of nonnegative squares is zero exactly when every coordinate is zero, and  $\|\alpha \mathbf{x}\|_2 = |\alpha| \|\mathbf{x}\|_2$ . Thus Euclidean length satisfies all three norm axioms. Applying definiteness to  $\mathbf{x} - y$  and observing that  $\|\mathbf{x} - y\|_2 = \|y - \mathbf{x}\|_2$  verify the remaining metric axioms for  $d_2$ .  $\square$

From now on, unless another norm is explicitly indicated, the standing convention on Euclidean space is

$$\|\mathbf{x}\| := \|\mathbf{x}\|_2 \quad (\mathbf{x} \in \mathbb{R}^n).$$

The later abstract treatment of norms generalizes this notation; it does not change the present Euclidean convention.

The points within distance  $r > 0$  of  $a$  form  $(a - r, a + r)$  on the line and an open disk in the plane. The limiting endpoints or circle are excluded.



**Definition 10.1.3** (Balls and neighborhoods). The open ball of center  $a$  and radius  $r > 0$  is

$$B_d(a, r) = \{x \in X : d(x, a) < r\}.$$

A **closed ball** is denoted by

$$\overline{B}_d(a, r) := \{x \in X : d(x, a) \leq r\}.$$

The bar is part of the closed-ball notation; it does not assert that this set is the closure of  $B_d(a, r)$ . The two sets can differ: for the discrete metric and  $r = 1$ , one has  $B(a, 1) = \{a\}$  but  $\overline{B}(a, 1) = X$ . When the metric is clear, write  $B(a, r)$  and  $\overline{B}(a, r)$ . A neighborhood of  $a$  is any set containing some  $B_d(a, r)$ . The neighborhood itself need not be a ball, or even an open set.

**Example 10.1.4** (A ball need not be round). On  $\mathbb{R}^n$  set  $d_\infty(\mathbf{x}, y) = \max_j |x_j - y_j|$ . Each coordinate distance is at most  $d_\infty(\mathbf{x}, z) + d_\infty(z, y)$ , proving the triangle inequality after taking the maximum. The other axioms follow coordinatewise. Its balls in  $\mathbb{R}^2$  are squares with sides parallel to the axes. Moreover,

$$d_\infty(\mathbf{x}, y) \leq d_2(\mathbf{x}, y) \leq \sqrt{n} d_\infty(\mathbf{x}, y).$$

Thus a sufficiently small ball for either distance fits into any ball for the other. Neighborhood structure, rather than the roundness of balls, is what the topology will retain.

## 10.2 Open and Closed Sets in an Ambient Space

An interior point of an interval can move a little in either direction without leaving it. An interior point of a disk can move a little in every direction. In  $\mathbb{R}$ , the sets  $(0, 1)$ ,  $(2, \infty)$ , and  $(0, 1) \cup (2, 3)$  all have this property at every one of their points. For  $x \in (0, 1)$ , for instance, the radius

$$\min(x, 1 - x)/2$$

keeps the ball inside the interval. The set  $[0, 1)$  fails this test at 0 when the ambient space is  $\mathbb{R}$ .

**Definition 10.2.1** (Open and closed). A set  $U \subseteq X$  is open if

$$\text{for every } x \in U \text{ there exists } r > 0 \text{ with } B_d(x, r) \subseteq U.$$

A set  $F \subseteq X$  is closed if its complement  $X \setminus F$  is open.

The set  $[0, 1]$  is closed in  $\mathbb{R}$ , since its complement is  $(-\infty, 0) \cup (1, \infty)$ . Neither word is the negation of the other:  $[0, 1]$  is neither open nor closed in  $\mathbb{R}$ , while  $\emptyset$  and  $X$  are both.

**Proposition 10.2.2** (Metric balls and open/closed sets). For every metric space  $(X, d)$ , point  $a \in X$ , and  $r > 0$ , the open ball  $B(a, r)$  is open and the closed ball  $\overline{B}(a, r)$  is closed.

*Proof.* If  $x \in B(a, r)$ , put  $\rho = r - d(a, x) > 0$  and choose  $0 < s < \rho$ . For  $y \in B(x, s)$ , the triangle inequality gives

$$d(a, y) \leq d(a, x) + d(x, y) < d(a, x) + \rho = r.$$

Thus  $B(x, s) \subseteq B(a, r)$ , so open balls are open.

If  $x \notin \overline{B}(a, r)$ , put  $\rho = d(a, x) - r > 0$  and choose  $0 < s < \rho$ . For  $y \in B(x, s)$ , the reverse triangle estimate obtained directly from the metric triangle inequality gives

$$d(a, y) \geq d(a, x) - d(x, y) > d(a, x) - \rho = r.$$

Hence  $B(x, s) \subseteq X \setminus \overline{B}(a, r)$ . The complement of the closed ball is open, so the closed ball is closed.  $\square$

**Proposition 10.2.3** (Metric open sets). The empty set and  $X$  are open. Arbitrary unions and finite intersections of open sets are open. Arbitrary intersections and finite unions of closed sets are closed.

*Proof.* There is no point to test in the empty set; every ball in  $X$  lies in  $X$ . If

$$x \in \bigcup_{\alpha} U_{\alpha},$$

choose a member containing  $x$  and a ball inside that member. That same ball lies in the union. If

$$x \in U_1 \cap \cdots \cap U_m,$$

choose  $r_i > 0$  with  $B(x, r_i) \subseteq U_i$ . The minimum of these finitely many positive radii is positive, and its ball lies in every  $U_i$ . The closed-set assertions follow by taking complements and applying De Morgan's laws. Finiteness matters:

$$\bigcap_{m \geq 1} (-1/m, 1/m) = \{0\}$$

is not open in  $\mathbb{R}$ .  $\square$

### Subspaces: change the universe before testing openness

Restrict a metric on  $X$  to  $Y \subseteq X$ . Then  $B_Y(y, r) = Y \cap B_X(y, r)$ : only points allowed in the new universe remain. For  $Y = [0, 1]$ , a ball about 0 is  $[0, r)$  when  $0 < r \leq 1$ . In particular

$$[0, 1/2)$$

is open in  $Y$ : it equals

$$Y \cap (-1, 1/2).$$

It is not open in  $\mathbb{R}$ , where every ball about 0 includes negative points.

**Proposition 10.2.4** (Subspace open sets). A set  $A \subseteq Y$  is open for the restricted metric if and only if  $A = Y \cap U$  for some open  $U \subseteq X$ .

*Proof.* If  $A = Y \cap U$ , each  $a \in A$  has an  $X$ -ball in  $U$ ; intersecting with  $Y$  gives a  $Y$ -ball in  $A$ . Conversely, for each  $a \in A$  choose  $r_a > 0$  with  $B_Y(a, r_a) \subseteq A$ . Put

$$U = \bigcup_{a \in A} B_X(a, r_a),$$

which is open. Every  $a \in A$  belongs to  $Y \cap U$ , and each point of  $Y \cap U$  belongs to one of the chosen  $Y$ -balls, hence to  $A$ . Thus  $Y \cap U = A$ .  $\square$

Closedness also depends on the ambient space:  $[0, 1]$  is both open and closed in itself;  $(0, 1)$  is closed in the space  $(0, 1)$  but not in  $\mathbb{R}$ .

Figure 10.1 gives a planar version: an open segment  $A$  in a line  $Y$  contains small relative balls, although it contains no planar ball. Fix the ambient space before asking whether a set is open, closed, interior, or boundary.

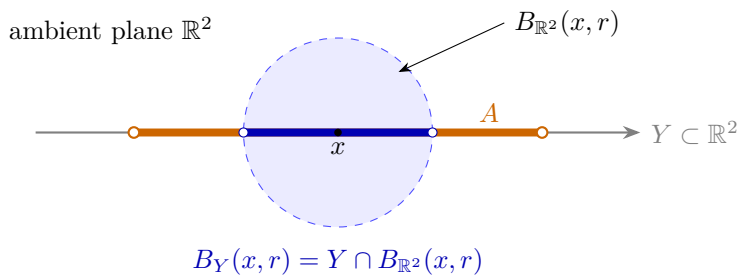


Figure 10.1: A relative ball stays on the line and fits inside the open segment  $A$ . Every planar ball about  $x$  also contains points off the line, so  $A$  is open in  $Y$  but not in  $\mathbb{R}^2$ . Endpoints of the colored open intervals are excluded. The ambient space determines the meaning of openness.

### Interior, closure, and boundary

Three geometric questions give three different sets. Which points have a whole neighborhood inside  $E$ ? Which points can be approximated from  $E$ ? Which points are at the interface with the complement?

**Definition 10.2.5** (Interior, closure, boundary). For  $E \subseteq X$  define, with all balls taken in  $X$ ,

$$\begin{aligned} E^\circ &= \{x : \text{some } B(x, r) \text{ lies in } E\}, \\ \overline{E} &= \{x : \text{every } B(x, r) \text{ meets } E\}, \\ \partial E &= \{x : \text{every } B(x, r) \text{ meets both } E \text{ and } X \setminus E\}. \end{aligned}$$

These are the interior, closure, and boundary relative to  $X$ .

**Proposition 10.2.6** (Interior, closure, and boundary identities).  $E^\circ$  is the largest open subset of  $E$ , and  $\overline{E}$  is the smallest closed superset of  $E$ . Moreover,

$$E^\circ \subseteq E \subseteq \overline{E}, \quad \partial E = \overline{E} \setminus E^\circ, \quad \boxed{\overline{E} = E^\circ \cup \partial E}.$$

*Proof.*  $E^\circ$  is the union of all open balls contained in  $E$ , so it is open; every open subset of  $E$  is included by the ball test. If  $x \notin \overline{E}$ , some ball about  $x$  misses  $E$ . Each point in that ball has a smaller ball still missing  $E$ , so  $X \setminus \overline{E}$  is open. The closure contains  $E$ , because a ball about a point of  $E$  meets it at its center. If a closed  $F$  contains  $E$  and  $x \notin F$ , openness of  $X \setminus F$  gives a ball missing  $E$ ; hence  $x \notin \overline{E}$ . This proves minimality. Finally, failure of the interior test means every ball meets the complement. Combining that test with the closure test proves  $\partial E = \overline{E} \setminus E^\circ$ . Since  $E^\circ \subseteq E \subseteq \overline{E}$ , adjoining  $E^\circ$  to this difference recovers all of  $\overline{E}$  and proves the boxed identity.  $\square$

**Proposition 10.2.7** (A closed ball inside an open set). If  $U \subseteq X$  and  $x \in U^\circ$ , then there is  $r > 0$  such that

$$x \in \overline{B}(x, r) \subseteq U.$$

*Proof.* Choose  $R > 0$  with  $B(x, R) \subseteq U$ , and take any  $r$  with  $0 < r < R$ . If  $z \in \overline{B}(x, r)$ , then  $d(z, x) \leq r < R$ , so  $z \in B(x, R) \subseteq U$ .  $\square$

**Definition 10.2.8** (Limit points and isolated points). A limit point of  $E$  is a point  $a$  such that

$$\text{for every } r > 0 \text{ there exists } x \in E \text{ with } 0 < d(x, a) < r.$$

The set of limit points is denoted by  $E'$ . A limit point is also called an **accumulation point**. An isolated point is  $a \in E$  for which  $B(a, r) \cap E = \{a\}$  for some  $r > 0$ . These are exactly the Chapter 5 tests with  $d(x, a)$  replacing  $|x - a|$ . Closure allows the point itself; the limit-point test excludes it.

**Proposition 10.2.9** (Closure from points and limit points). For every subset  $E$  of a metric space,

$$\boxed{\overline{E} = E \cup E'}.$$

*Proof.* Every point of  $E$  belongs to  $\overline{E}$ . Every limit point belongs to  $\overline{E}$  because each of its balls meets  $E$ . Conversely, if  $x \in \overline{E} \setminus E$ , every ball about  $x$  meets  $E$ , and any point of intersection differs from  $x$ . Hence  $x \in E'$ .  $\square$

**Example 10.2.10** (An isolated point in a closed set). In  $\mathbb{R}$  let

$$E = \{0\} \cup [1, 2].$$

Then  $0 \in E$  is isolated, so  $0 \notin E'$ , while  $E' = [1, 2]$  and  $\overline{E} = E$ . Also  $E^\circ = (1, 2)$  and  $\partial E = \{0, 1, 2\}$ . Thus a point of a set need not be a limit point, and even in a metric space a boundary point may fail to be a limit point of the set. Conversely, a limit point may be an interior point. The two closed-set tests below concern containment of these sets, not equality between boundary points and limit points. Closure keeps the points already in  $E$  and adds precisely those forced by arbitrarily close points of  $E$ .

**Theorem 10.2.11** (Characterizations of closed subsets of a metric space). Let  $(X, d)$  be a metric space and  $E \subseteq X$ . The following are equivalent:

- (1)  $E$  is closed;
- (2)  $X \setminus E$  is open;
- (3)  $E = \overline{E}$ ;
- (4)  $\partial E \subseteq E$ ;
- (5)  $E' \subseteq E$ .

Equivalently, closedness has the metric neighborhood form

$$E \text{ closed} \iff \forall x \notin E \exists r > 0 : B(x, r) \cap E = \emptyset.$$

*Proof.* The equivalence (1)  $\iff$  (2) is the definition of closedness. By the extremal characterization of closure in [Proposition 10.2.6](#),  $E$  is closed if and only if its smallest closed superset is itself; hence (1)  $\iff$  (3).

For (3)  $\iff$  (4), use  $\overline{E} = E^\circ \cup \partial E$  and  $E^\circ \subseteq E \subseteq \overline{E}$ . Equality  $E = \overline{E}$  clearly implies  $\partial E \subseteq E$ ; conversely, if  $\partial E \subseteq E$ , then  $\overline{E} = E^\circ \cup \partial E \subseteq E$ , and the reverse inclusion always holds. For (3)  $\iff$  (5), use  $\overline{E} = E \cup E'$  from [Proposition 10.2.9](#).

Finally,  $X \setminus E$  is open exactly when every  $x \notin E$  is the center of a ball contained in  $X \setminus E$ , which is exactly the displayed disjoint-ball condition. It is simply openness of the complement written in metric language.  $\square$

### 10.3 Convergence, Completeness, and Metric Continuity

The sequence definitions on  $\mathbb{R}$  used distance rather than multiplication or order. Replacing  $|x - y|$  by  $d(x, y)$  preserves their quantifier pattern. Completeness still asks whether the limit belongs to the space under consideration, so the choice of ambient space matters.

**Definition 10.3.1** (Metric convergence). A sequence  $(x_n)$  in  $X$  converges to  $x \in X$  if

$$\forall \varepsilon > 0 \exists N \forall n \geq N : d(x_n, x) < \varepsilon.$$

It is Cauchy if

$$\forall \varepsilon > 0 \exists N \forall m, n \geq N : d(x_m, x_n) < \varepsilon.$$

The space is complete if every Cauchy sequence has a limit in  $X$ .

**Proposition 10.3.2** (Basic laws of metric convergence). In a metric space, limits are unique; every subsequence of a convergent sequence converges to the same limit; and every convergent sequence is Cauchy.

*Proof.* If  $x_n \rightarrow x$  and  $x_n \rightarrow y$ , then eventually both

$$d(x_n, x) < \varepsilon/2, \quad d(x_n, y) < \varepsilon/2.$$

Hence  $d(x, y) < \varepsilon$  for every  $\varepsilon > 0$ , so  $x = y$ . If  $x_n \rightarrow x$  and  $n_k$  is strictly increasing, then  $n_k \geq k$ ; every sufficiently late  $x_{n_k}$  is therefore among the terms already within any prescribed ball about  $x$ . Thus  $x_{n_k} \rightarrow x$ . Finally, if  $x_n \rightarrow x$ , put two late terms within

$$\varepsilon/2$$

of  $x$  and use the triangle inequality to obtain  $d(x_m, x_n) < \varepsilon$ .  $\square$

**Example 10.3.3** (Where must the limit live?). The sequence

$$1/n$$

is Cauchy in  $(0, 1]$ , since

$$|1/m - 1/n| \leq 1/m + 1/n,$$

but its only possible real limit, 0, is missing. Thus  $(0, 1]$  is not complete. The space  $\mathbb{R}^n$  is complete: each coordinate of a Cauchy sequence is real Cauchy; the coordinate limits give a vector, and  $\|\mathbf{x}_n - \mathbf{x}\|_2 \leq \sqrt{n} \max_j |x_{n,j} - x_j| \rightarrow 0$ . In the discrete metric  $d(x, y) = 1$  for  $x \neq y$ , a Cauchy sequence is eventually constant by the test

$$\varepsilon = 1/2,$$

so any such space is complete.

**Definition 10.3.4** (Boundedness and diameter).  $E \subseteq X$  is bounded if it is empty or lies in some ball  $B(a, R)$ . Its diameter is  $\sup\{d(x, y) : x, y \in E\}$ , possibly  $+\infty$ ; the empty set has diameter zero. Boundedness alone gives no convergence:  $(-1)^n$  is bounded but not Cauchy.

**Theorem 10.3.5** (Sequential closure and limit-point criteria). Let  $E$  be a subset of a metric space. Then

$$x \in \overline{E} \iff \text{there exists } (x_n) \subseteq E \text{ with } x_n \rightarrow x,$$

and

$$x \in E' \iff \text{there exists } x_n \in E \setminus \{x\} \text{ with } x_n \rightarrow x.$$

*Proof.* If  $x \in \overline{E}$ , each

$$E \cap B(x, 1/n)$$

is nonempty. Choose  $x_n$  in that intersection. Then

$$d(x_n, x) < 1/n \rightarrow 0.$$

Conversely, if  $x_n \in E$  and  $x_n \rightarrow x$ , every ball about  $x$  contains an eventual term and therefore meets  $E$ . Thus  $x \in \overline{E}$ .

If  $x \in E'$ , choose

$$x_n \in (E \setminus \{x\}) \cap B(x, 1/n);$$

then  $x_n \rightarrow x$ . Conversely, if such a sequence exists, every ball about  $x$  contains an eventual term from  $E$  different from  $x$ , so  $x \in E'$ .  $\square$

When  $x$  is a limit point, the sequence may be chosen with distinct terms. Indeed, after choosing  $x_1, \dots, x_{n-1}$ , take a ball about  $x$  small enough to exclude these finitely many nonzero distances and then choose  $x_n$  from that punctured ball.

**Theorem 10.3.6** (Sequential characterization of closed sets). A subset  $E$  of a metric space is closed if and only if it contains the limit of every convergent sequence of its points:

$$\boxed{E \text{ is closed} \iff [x_n \in E, x_n \rightarrow x] \implies x \in E.}$$



*Proof.* If  $E$  is closed and  $x_n \in E$  converges to  $x$ , the first part of [Theorem 10.3.5](#) gives  $x \in \overline{E} = E$ . Conversely, if every such limit belongs to  $E$  and  $x \in \overline{E}$ , the same theorem supplies a sequence in  $E$  converging to  $x$ , so  $x \in E$ . Hence  $\overline{E} \subseteq E$ ; the reverse inclusion always holds, and [Theorem 10.2.11](#) gives closedness.  $\square$

Together, the topological and sequential forms give the reusable metric-space picture

$$\begin{aligned} E \text{ closed} &\iff X \setminus E \text{ open} \iff E = \overline{E} \\ &\iff \partial E \subseteq E \iff E' \subseteq E \\ &\iff E \text{ contains every limit of every convergent sequence in } E. \end{aligned}$$

**Proposition 10.3.7** (Closed subsets of complete spaces). If  $X$  is complete and  $F \subseteq X$  is closed, then  $F$ , with the restricted metric, is complete.

*Proof.* A Cauchy sequence in  $F$  is also Cauchy in  $X$ , so completeness of  $X$  gives an ambient limit  $x \in X$ . Because  $F$  is closed, [Theorem 10.3.6](#) forces  $x \in F$ . Thus the sequence converges in the restricted metric on  $F$ .  $\square$

**Corollary 10.3.8** (Closed balls in complete spaces). If  $X$  is complete, then every  $\overline{B}(a, r) \subseteq X$  is complete. In particular, every Euclidean closed ball in  $\mathbb{R}^n$  is complete.

*Proof.* Closed balls are closed by [Proposition 10.2.2](#); apply [Proposition 10.3.7](#). Euclidean space is complete by the preceding coordinate argument.  $\square$

The contrast with open balls is essential. The open ball  $(-1, 1)$  in  $\mathbb{R}$  is not complete: the Cauchy sequence

$$1 - 1/n$$

converges in  $\mathbb{R}$  to the missing boundary point 1. The successful logical chain is

$$\begin{aligned} &\overline{B}(a, r) \text{ closed} \\ &\implies \overline{B}(a, r) \text{ contains all its limit points} \\ &\implies \overline{B}(a, r) \text{ is complete when } X \text{ is complete.} \end{aligned}$$

**Definition 10.3.9** (Metric continuity). For metric spaces  $X, Y$ , a map  $f : X \rightarrow Y$  is continuous at  $a \in X$  if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in X : d_X(x, a) < \delta \implies d_Y(f(x), f(a)) < \varepsilon.$$

It is continuous if this holds at every point of  $X$ .

The output tolerance is prescribed first; the input tolerance may depend on both it and the base point. For  $f(x, y) = x + 2y$ ,  $|f(x, y) - f(a, b)| \leq 3\|(x - a, y - b)\|_2$ , so

$$\delta = \varepsilon/3$$

works. For any metric,  $|d(x, c) - d(a, c)| \leq d(x, a)$  by two applications of the triangle inequality. Hence distance to a fixed point is continuous.

**Definition 10.3.10** (Limit of a map). If  $a$  is a limit point of  $D \subseteq X$ , then  $\lim_{x \rightarrow a} f(x) = L$  for  $f : D \rightarrow Y$  means

$$\forall \varepsilon > 0 \exists \delta > 0 : \quad x \in D, \quad 0 < d_X(x, a) < \delta \implies d_Y(f(x), L) < \varepsilon.$$

**Theorem 10.3.11** (Sequential criterion for limits of maps). Let  $f : D \subseteq X \rightarrow Y$ , where  $X, Y$  are metric spaces, and let  $a$  be a limit point of  $D$ . Then

$$\lim_{x \rightarrow a} f(x) = L \iff x_n \in D \setminus \{a\}, \quad x_n \rightarrow a \implies f(x_n) \rightarrow L.$$

*Proof.* If the metric  $\varepsilon$ - $\delta$  limit holds, apply its  $\delta$  to the eventual terms of any such sequence. Conversely, if the limit fails, then for some  $\varepsilon_0 > 0$  and every  $n$  there is  $x_n \in D \setminus \{a\}$  with

$$d_X(x_n, a) < 1/n, \quad d_Y(f(x_n), L) \geq \varepsilon_0.$$

Then  $x_n \rightarrow a$  but  $f(x_n) \not\rightarrow L$ , contradicting the sequential condition.  $\square$

**Example 10.3.12** (Paths disprove, estimates prove). The function

$$xy/(x^2 + y^2)$$

has values

$$1/2$$

along  $y = x \neq 0$  and

$$-1/2$$

along  $y = -x \neq 0$ , so has no limit at the origin. In contrast,

$$\left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y| \leq \|(x, y)\|_2$$

proves a zero limit in every direction at once. Agreement along a few paths would not establish that all approaches are controlled.

**Theorem 10.3.13** (Sequential criterion for continuity).  $f$  is continuous at  $a$  if and only if  $x_n \rightarrow a$  implies  $f(x_n) \rightarrow f(a)$ .

*Proof.* For the forward direction choose  $\delta$  for the prescribed  $\varepsilon$ ; eventually  $d_X(x_n, a) < \delta$ , so the output error is less than  $\varepsilon$ . For the converse the negation of continuity is

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in X : \quad d_X(x, a) < \delta, \quad d_Y(f(x), f(a)) \geq \varepsilon_0.$$

One fixed  $\varepsilon_0$  works at every scale. Set

$$\delta = 1/n$$

and choose such an  $x_n$ . The inputs converge to  $a$ , but the output errors never fall below  $\varepsilon_0$ , contradicting the sequential assertion.  $\square$

## 10.4 Topological Spaces and Finite Products

The metric open-set rules allow us to retain neighborhoods without retaining a distance formula. This is useful for charts: two different coordinate descriptions can have the same open sets.

**Definition 10.4.1** (Topological space). A topology  $\mathcal{T}$  on  $X$  is a collection of subsets satisfying

$$\begin{aligned} \emptyset, X &\in \mathcal{T}, \\ U_\alpha \in \mathcal{T} \ (\alpha \in I) &\implies \bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}, \\ U_1, \dots, U_m \in \mathcal{T} &\implies \bigcap_{i=1}^m U_i \in \mathcal{T}. \end{aligned}$$

Its members are called open. A topological space is a pair  $(X, \mathcal{T})$ . Closed means open complement. Given a topological space  $X$  and  $x \in X$ , an **open neighborhood**  $U$  of  $x$  is an open subset  $U \subseteq X$  containing  $x$ . A **neighborhood** of  $x$  more generally is a set containing an open neighborhood of  $x$ ; it need not itself be open. This agrees with our metric convention. We say **open neighborhood** whenever openness is required. The subspace topology on  $Y \subseteq X$  consists of  $Y \cap U$  with  $U$  open in  $X$ .

[Propositions 10.2.3](#) and [10.2.4](#) explain why these definitions recover the metric notions. The subspace rules hold because intersections with  $Y$  commute with unions and finite intersections. In any topology define interior as the union of open subsets of  $E$ , closure as the intersection of closed supersets, and boundary as  $\overline{E} \setminus E^\circ$ . The proof of [Proposition 10.2.6](#), replacing balls by open neighborhoods, gives the same extremal and neighborhood characterizations. The sequential criterion is asserted here only for metric spaces.

**Definition 10.4.2** (Topological continuity). A map  $f : X \rightarrow Y$  is continuous if  $f^{-1}(U)$  is open in  $X$  for every open  $U \subseteq Y$ .

**Theorem 10.4.3** (The two definitions agree). For metric spaces this is equivalent to metric continuity at every point.

*Proof.* Suppose first that  $f$  is metric continuous. Given open  $U \subseteq Y$ , test a point  $a \in f^{-1}(U)$ . Openness gives  $\varepsilon > 0$  with  $B_Y(f(a), \varepsilon) \subseteq U$ . Continuity gives  $\delta > 0$  sending  $B_X(a, \delta)$  into that ball. Thus  $B_X(a, \delta) \subseteq f^{-1}(U)$ . Every point of the inverse image passes the open-set test.

Conversely assume inverse images of open sets are open. Fix  $a$  and  $\varepsilon > 0$ . The open ball  $B_Y(f(a), \varepsilon)$  has an open inverse image containing  $a$ , since its center belongs to it. Therefore some  $B_X(a, \delta)$  lies in this inverse image. Unpacking membership gives exactly the metric continuity implication at  $a$ .  $\square$

Composition is continuous: for open  $U$ , first  $g^{-1}(U)$  is open and then  $f^{-1}(g^{-1}(U)) = (g \circ f)^{-1}(U)$  is open.

**Definition 10.4.4** (Homeomorphisms). A homeomorphism  $h : X \rightarrow Y$  is a bijection such that  $h$  and  $h^{-1}$  are continuous. It identifies the open-set structures of the spaces.

For example

$$x \mapsto \tan\left(\pi\left(x - \frac{1}{2}\right)\right)$$

gives a homeomorphism  $(0, 1) \rightarrow \mathbb{R}$  with inverse

$$y \mapsto \frac{1}{2} + \frac{\arctan y}{\pi}.$$

It does not preserve lengths or boundedness.

**Bases: describing all open sets from a smaller family**

To verify a local property at  $x$ , we often need only a smaller family of basic open neighborhoods: every open neighborhood must contain one of them. This is the abstract version of fitting a small ball inside an open set.

**Definition 10.4.5** (Topological basis). A **basis** for a topology  $\mathcal{T}$  on  $X$  is a family  $\mathcal{B} \subseteq \mathcal{T}$  such that every open set is a union of members of  $\mathcal{B}$ , allowing the empty union. Equivalently,

$$U \in \mathcal{T}, \quad x \in U \implies \text{there is } B \in \mathcal{B} \text{ with } x \in B \subseteq U.$$

If  $U$  is a union of basic sets, choose a member containing  $x$ . Conversely the local criterion gives

$$U = \bigcup \{B \in \mathcal{B} : B \subseteq U\} :$$

every point of  $U$  is covered, and nothing outside  $U$  is included. It is this local criterion that most basis arguments actually use.

**Proposition 10.4.6** (Basis criterion). A family  $\mathcal{B}$  of subsets of  $X$  is a basis for a topology if:

- (a) every  $x \in X$  belongs to some  $B \in \mathcal{B}$ ;
- (b) if  $x \in B_1 \cap B_2$ , with  $B_1, B_2 \in \mathcal{B}$ , there is  $B_3 \in \mathcal{B}$  with  $x \in B_3 \subseteq B_1 \cap B_2$ .

The generated topology  $\mathcal{T}_{\mathcal{B}}$  consists of all unions of members of  $\mathcal{B}$ .

*Proof.* The empty union gives  $\emptyset$ ; the covering condition gives  $X$ . An arbitrary union of unions of basic sets is again such a union. For  $x \in U \cap V$ , choose basic sets  $x \in B_1 \subseteq U$  and  $x \in B_2 \subseteq V$ . The second axiom refines their intersection to  $x \in B_3 \subseteq U \cap V$ . Taking all such basic sets expresses the intersection as a union. Induction gives finite intersections, so this is a topology. Each basic set is open, and by construction every open set is a union of basic sets. Conversely any basis satisfies the two axioms by applying its local criterion to  $X$  and to the open set  $B_1 \cap B_2$ .  $\square$

**Working rule.** To construct a topology, check the two basis axioms. To prove  $U$  open in that topology, find for each  $x \in U$  a basic set  $x \in B \subseteq U$ . To prove  $f : X \rightarrow Y$  continuous, it suffices to check that  $f^{-1}(B)$  is open for each member of a basis of  $Y$ : inverse images preserve unions. Here a *topological basis* is a family of open sets. Later a *vector-space basis* will be a family of vectors giving unique linear coordinates. These are different uses of the word.

**Example 10.4.7** (Metric and rational bases). All open balls form a basis of a metric space. They cover it, and if  $z \in B(x, r) \cap B(y, s)$ , choose

$$0 < \varepsilon < \min\{r - d(x, z), s - d(y, z)\}.$$

The triangle inequality puts  $B(z, \varepsilon)$  inside both balls. The generated topology is exactly the metric topology by the ball test. On  $\mathbb{R}$ , rational-endpoint intervals form a countable basis: inside any open interval about  $x$ , density supplies rationals  $a < x < b$  still inside it. On  $\mathbb{R}^n$ , a countable basis is

$$\mathcal{B}_{\mathbb{Q}} = \{B(q, r) : q \in \mathbb{Q}^n, r \in \mathbb{Q}, r > 0\}.$$

The indexing set  $\mathbb{Q}^n \times \mathbb{Q}_{>0}$  is countable by [Corollary 2.5.4](#) and [Theorem 2.5.3](#), iterating the countable-product argument finitely many times. Given  $\mathbf{x} \in U$  open, choose  $B(\mathbf{x}, \varepsilon) \subseteq U$ . For  $n \geq 1$ , choose rational  $q_i$  with

$$|q_i - x_i| < \varepsilon/(3\sqrt{n})$$

for each  $i$ . Then

$$\|q - \mathbf{x}\|^2 = \sum_i |q_i - x_i|^2 < \varepsilon^2/9.$$

Choose a rational radius satisfying

$$\|q - \mathbf{x}\| < r < \varepsilon - \|q - \mathbf{x}\|.$$

Density of the rationals supplies these choices. Then  $\mathbf{x} \in B(q, r) \subseteq B(\mathbf{x}, \varepsilon) \subseteq U$  by the triangle inequality. This verifies the local criterion, not just countability. For  $n = 0$  the one-point space has the one-member basis  $\{\mathbb{R}^0\}$ .

**Example 10.4.8** (An algebraic basis: the  $I$ -adic topology). First consider integers: for a fixed prime  $p$ , the sets  $a + p^n\mathbb{Z}$  consist of integers congruent to  $a$  modulo  $p^n$ . Larger  $n$  demands agreement modulo a larger power, so gives a smaller neighborhood. If  $x$  belongs to two such sets, the neighborhood  $x + p^{\max(m,n)}\mathbb{Z}$  lies in both. This is nearness by congruence, not by the usual distance between integers.

The following generalization is optional and assumes familiarity with rings and ideals. Let  $R$  be a ring and  $I$  an ideal (two-sided if the ring is not commutative). Its powers satisfy  $I^{n+1} \subseteq I^n$ . Consider

$$\mathcal{B}_I = \{x + I^n : x \in R, n \geq 1\}.$$

Every  $x$  lies in  $x + I$ , so these sets cover  $R$ . If  $x \in (a + I^m) \cap (b + I^n)$ , put  $r = \max(m, n)$ . For  $z \in I^r$ , both  $x - a + z \in I^m$  and  $x - b + z \in I^n$ , whence

$$x \in x + I^r \subseteq (a + I^m) \cap (b + I^n).$$

The basis criterion (Proposition 10.4.6) therefore gives a topology, called the  **$I$ -adic topology**. The powers  $I^n$  act as successively smaller basic neighborhoods of 0; their translates give basic neighborhoods of all points. If  $I = (0)$ , the basic sets are singletons and the topology is discrete. This example illustrates that bases also organize algebraic notions of nearness, without a metric.

### Why rectangles suffice

To control a pair  $(x, y)$ , control each component. The family of open rectangles  $U \times V$  is a basis:  $X \times Y$  itself covers every point, and

$$(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$$

is again an open rectangle. By Proposition 10.4.6 it generates the **product topology**. Thus  $G$  is product-open exactly when every  $(x, y) \in G$  lies in an open rectangle contained in  $G$ .

**Proposition 10.4.9** (Products of bases). If  $\mathcal{B}_X, \mathcal{B}_Y$  are bases, then  $\{B \times C : B \in \mathcal{B}_X, C \in \mathcal{B}_Y\}$  is a basis for the product topology.

*Proof.* Inside an open rectangle  $U \times V$  about  $(x, y)$ , choose  $x \in B \subseteq U$  and  $y \in C \subseteq V$  using the two local basis criteria. Then  $(x, y) \in B \times C \subseteq U \times V$ . Every product-open neighborhood contains such a rectangle, so Definition 10.4.5 proves the assertion.  $\square$

Figure 10.2 illustrates this local test in the plane.

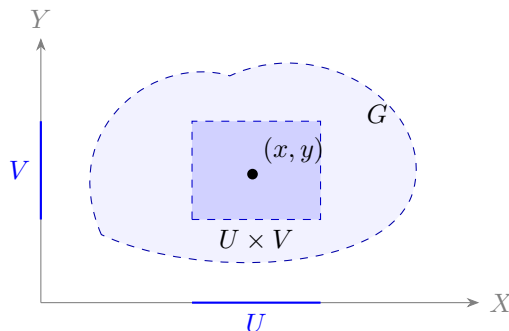


Figure 10.2: The local basis test:  $(x, y) \in U \times V \subseteq G$ . The planar drawing illustrates a product-open neighborhood; in general the factors need not be Euclidean spaces. Dashed edges are excluded.

**Definition 10.4.10** (Second countability). A topological space is **second countable** if it has a countable basis. These countably many open sets must generate **every** open set, or equivalently provide a basic neighborhood inside every open neighborhood.

**Proposition 10.4.11** (Subspaces and finite products). Every subspace of a second-countable space is second countable, and every finite product of second-countable spaces is second countable. In particular every subspace of  $\mathbb{R}^n$  is second countable.

*Proof.* For  $Y \subseteq X$  and a countable basis  $\mathcal{B}$  of  $X$ , the family

$$\{B \cap Y : B \in \mathcal{B}, B \cap Y \neq \emptyset\}$$

is countable and relatively open. If  $y \in U \cap Y$  with  $U$  open in  $X$ , choose  $y \in B \subseteq U$ ; then  $y \in B \cap Y \subseteq U \cap Y$ . This proves the basis property for arbitrary subspaces, not only open ones. For two factors use [Proposition 10.4.9](#); the product of the two countable indexing sets is countable. Induct on the number of factors. The Euclidean assertion follows from [Example 10.4.7](#).  $\square$

The projections are  $\pi_X(x, y) = x$  and  $\pi_Y(x, y) = y$ .

## Limits defined by neighborhoods

**Definition 10.4.12** (Topological limits). In a topological space,  $x_n \rightarrow x$  means that for every neighborhood  $U$  of  $x$  there is  $N$  such that  $x_n \in U$  for all  $n \geq N$ . For  $f : A \rightarrow Y$ , where  $A \subseteq X$ , assume  $a \in X$  is an accumulation point of  $A$ : every neighborhood of  $a$  meets  $A \setminus \{a\}$ . We write  $\lim_{x \rightarrow a, x \in A} f(x) = L$  if for every neighborhood  $V$  of  $L$  there is a neighborhood  $U$  of  $a$  with

$$f((U \cap A) \setminus \{a\}) \subseteq V.$$

It suffices in both definitions to use open neighborhoods. In metric spaces, testing balls gives exactly the earlier  $\varepsilon$ - $N$  and  $\varepsilon$ - $\delta$  definitions. A function limit omits the point  $a$ ; continuity at a point  $a \in A$  instead asks that every neighborhood of  $f(a)$  contain the image of some neighborhood of  $a$  in  $A$ , including its value at  $a$ . This pointwise condition at every point is equivalent to the inverse-image definition of topological continuity: an inverse image of an open set is open precisely when it is a neighborhood of each of its points. At an accumulation point  $a \in A$ , continuity is therefore equivalent to the displayed limit with  $L = f(a)$ . Limits need not be unique until a separation hypothesis is imposed; [Proposition 10.5.6](#) supplies that hypothesis.

**Definition 10.4.13** (Sequential continuity and sequential closedness). A map  $f : X \rightarrow Y$  is **sequentially continuous** if  $x_n \rightarrow x$  implies  $f(x_n) \rightarrow f(x)$ . A subset  $E \subseteq X$  is **sequentially closed** if every limit in  $X$  of a sequence in  $E$  belongs to  $E$ .

**Proposition 10.4.14.** Every continuous map is sequentially continuous, and every closed subset is sequentially closed, in arbitrary topological spaces.

*Proof.* If  $V$  is an open neighborhood of  $f(x)$ , its inverse image is an open neighborhood of  $x$ . Eventually  $x_n$  belongs to that inverse image, so eventually  $f(x_n) \in V$ . If a sequence in a closed set  $E$  converged to  $x \notin E$ , the open neighborhood  $X \setminus E$  would eventually contain its terms, a contradiction.  $\square$

**Definition 10.4.15** (First countability). A countable local neighborhood basis at  $x$  is a family  $(U_n)_{n \geq 1}$  of neighborhoods of  $x$  such that every neighborhood of  $x$  contains some  $U_n$ . A space is **first countable** if each point has such a family. We may take the  $U_n$  open, by replacing them with contained open neighborhoods.

Metric spaces are first countable: take

$$B(x, 1/n).$$

Second countability (Definition 10.4.10) implies first countability by taking the members of a countable topological basis that contain the point. First countability is local; it does not require one countable family to generate all open sets of the space.

**Theorem 10.4.16** (When sequences detect topology). If  $X$  is first countable, then  $f : X \rightarrow Y$  is continuous if and only if it is sequentially continuous, for any topological space  $Y$ . Also  $E \subseteq X$  is closed if and only if it is sequentially closed.

*Proof.* The forward implications were proved in Proposition 10.4.14. At a point  $x$ , replace a countable open local basis by

$$W_n = U_1 \cap \cdots \cap U_n$$

to make it decreasing. If  $f$  is not continuous at  $x$ , there is an open neighborhood  $V$  of  $f(x)$  such that no  $W_n$  maps into  $V$ . Choose  $x_n \in W_n$  with  $f(x_n) \notin V$ . The local basis property gives  $x_n \rightarrow x$ , but the images do not converge to  $f(x)$ . For the set assertion, if  $x \in \overline{E} \setminus E$ , every  $W_n$  meets  $E$ . Choosing  $x_n \in E \cap W_n$  gives a sequence in  $E$  converging outside  $E$ , contrary to sequential closedness.  $\square$

The same argument characterizes punctured function limits by sequences when the ambient domain space is first countable at  $a$ : require  $f(x_n) \rightarrow L$  for every sequence in  $A \setminus \{a\}$  converging to  $a$ . Failure of the neighborhood limit produces  $x_n \in W_n \cap A$  with  $x_n \neq a$  and  $f(x_n) \notin V$  for one fixed neighborhood  $V$  of  $L$ .

**Example 10.4.17** (Why the converse needs a hypothesis). On the uncountable set  $\mathbb{R}$ , let the open sets be  $\emptyset$  and all sets with countable complement. They form a topology. A sequence converges to  $x$  in this topology exactly when it is eventually equal to  $x$ : the neighborhood obtained by removing all its terms different from  $x$  forces eventual equality. Consequently the identity map from this space to discrete  $\mathbb{R}$  is sequentially continuous. It is not continuous, since a singleton is open in the target but not in the source. Every subset is sequentially closed in the source, although, for example,  $[0, \infty)$  is not closed there. Thus sequences can miss both discontinuity and nonclosedness.

Nets and filters allow genuinely topological versions of sequence tests without first countability. We will not develop them. In any topological space, sequential compactness means that every sequence has a subsequence converging to a point of the space. It must not be substituted for open-cover compactness in arbitrary topological spaces. The equivalence in [Theorem 10.6.3](#) uses the metric essentially.

**Proposition 10.4.18** (Product continuity).  $f : Z \rightarrow X \times Y$  is continuous if and only if both components  $\pi_X \circ f$  and  $\pi_Y \circ f$  are continuous.

*Proof.*  $\pi_X^{-1}(U) = U \times Y$  and  $\pi_Y^{-1}(V) = X \times V$  are open, so the projections are continuous; composition proves one direction. Conversely write an arbitrary product-open set as

$$G = \bigcup_{\alpha} (U_{\alpha} \times V_{\alpha}).$$

Then

$$f^{-1}(G) = \bigcup_{\alpha} \left[ (\pi_X \circ f)^{-1}(U_{\alpha}) \cap (\pi_Y \circ f)^{-1}(V_{\alpha}) \right].$$

Each bracket is an intersection of two open sets, and the union is open. Checking rectangles therefore suffices for arbitrary open sets.  $\square$

For a finite product use rectangles  $U_1 \times \cdots \times U_r$ . The metric  $d(x, y) = \max_j d_j(x_j, y_j)$  has balls equal to products of equal-radius balls. A rectangle about  $x$  contains such a ball: choose one radius for each component and take their positive minimum. Thus it induces the product topology.

**Proposition 10.4.19** (Convergence in finite products). For metric spaces  $(X_j, d_j)$ ,  $1 \leq j \leq k$ , equipped with the maximum product metric,

$$(x_n^{(1)}, \dots, x_n^{(k)}) \longrightarrow (x^{(1)}, \dots, x^{(k)}) \iff x_n^{(j)} \longrightarrow x^{(j)} \text{ for every } j.$$

Consequently,

$$v_n \rightarrow v \text{ in } \mathbb{R}^m \iff (v_n)_j \rightarrow v_j \quad (j = 1, \dots, m).$$

*Proof.* Convergence in the maximum metric forces  $d_j(x_n^{(j)}, x^{(j)})$  to zero for each  $j$ . Conversely, for a prescribed  $\varepsilon > 0$ , choose one eventual index for each of the finitely many components and take their maximum. Then all component distances, and hence their maximum, are less than  $\varepsilon$ .

For Euclidean space use the quantitative comparisons

$$|v_j| \leq \|v\|_2 \leq \sqrt{m} \max_{1 \leq j \leq m} |v_j|.$$

They compare the Euclidean and maximum metrics, so the just-proved product criterion gives the Euclidean statement.  $\square$

The displayed Euclidean inequalities are often useful estimates in their own right, beyond their role in comparing metrics. The same product argument and [Proposition 10.4.18](#) show that a map to  $\mathbb{R}^m$  is continuous exactly when each coordinate function is continuous.



## 10.5 Compactness: From Local Pieces to Global Control

A local estimate often comes with a neighborhood depending on its point. To obtain one global estimate we want only finitely many of those neighborhoods, so that their constants have a maximum and their positive scales have a positive minimum. Compactness solves this finite-selection problem:

$$\text{local information} \longrightarrow \text{finitely many local pieces} \longrightarrow \text{uniform control}.$$

For example, the relative intervals

$$U_t = [0, 1] \cap (t - 1/4, t + 1/4)$$

for  $t \in [0, 1]$  cover  $[0, 1]$ . The five with centers

$$0, 1/4, 1/2, 3/4, 1$$

already cover it. This verifies one cover, whereas compactness makes a claim about every open cover. In contrast

$$(0, 1) = \bigcup_{n \geq 3} (1/n, 1 - 1/n)$$

has no finite subcover: any finite selection lies in its largest member, leaving points near zero. Likewise

$$\mathbb{R} = \bigcup_{n \geq 1} (-n, n)$$

has no finite subcover, since every finite selection is bounded.

**Definition 10.5.1** (Covers and compact subsets). Let  $Y \subseteq X$ , where  $X$  is a topological space. A family  $\{U_i\}_{i \in I}$  of subsets of  $X$  **covers**  $Y$  if

$$Y \subseteq \bigcup_{i \in I} U_i.$$

It is an **open cover of  $Y$  in  $X$**  if each  $U_i$  is open in  $X$ . A subfamily indexed by  $J \subseteq I$  is a **subcover** if

$$Y \subseteq \bigcup_{j \in J} U_j.$$

It is finite if  $J$  is finite. The subset  $Y$  is **compact** if every open cover of  $Y$  in  $X$  has a finite subcover.

Containment is essential: the open neighborhood  $(-1, 2)$  covers  $[0, 1]$  although its union is larger than  $[0, 1]$ . When  $Y = X$ , containment becomes equality automatically, since every cover member is already a subset of  $X$ .

**Proposition 10.5.2** (Ambient and subspace compactness). A subset  $Y \subseteq X$  is compact in the preceding sense if and only if  $Y$ , with its subspace topology, is a compact space.

*Proof.* Intersect an ambient open cover with  $Y$  to obtain a relatively open cover; a finite selection covers the same points of  $Y$ . Conversely, write each member of a relatively open cover as  $V_i = Y \cap U_i$ , with  $U_i$  open in  $X$ . The  $U_i$  cover  $Y$ , and a finite selection of them gives the same finite selection of the  $V_i$ . These two translations prove both directions.  $\square$

The empty space is compact: the empty index subset supplies a finite subcover. The interval  $[0, 1]$  is compact by [Theorem 5.7.6](#); the general Euclidean criterion will be proved below. The bounded interval  $(0, 1)$  shows that boundedness is not a substitute for compactness.

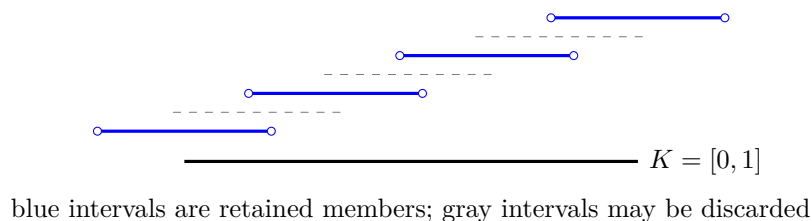


Figure 10.3: A finite subcover retains members of the original cover. The vertical offsets separate the intervals visually; all intervals live on the same horizontal line. This schematic illustrates one cover, whereas compactness controls every open cover.

**Theorem 10.5.3** (Closed subsets of compact spaces). If  $X$  is compact and  $F \subseteq X$  is closed, then  $F$  is compact.

*Proof.* Start with an open cover  $(U_\alpha)$  of  $F$  by open subsets of  $X$ . Adjoin the open set  $X \setminus F$  to cover  $X$ , and select a finite subcover. The selected original members still cover  $F$ , because no point of  $F$  lies in  $X \setminus F$ . This proves compactness in the ambient sense, hence also in the subspace topology.  $\square$

**Theorem 10.5.4** (Continuous images of compact spaces). If  $X$  is compact and  $f : X \rightarrow Y$  is continuous, then  $f(X)$  is compact.

*Proof.* Let  $(U_\alpha)$  cover  $f(X)$  by open subsets of  $Y$ . Continuity makes  $f^{-1}(U_\alpha)$  an open cover of  $X$ . A finite subcover of  $X$  selects finitely many  $U_\alpha$  covering every value  $f(x)$ , hence all of  $f(X)$ . [Proposition 10.5.2](#) identifies this with subspace compactness.  $\square$

**Definition 10.5.5** (Hausdorff space).  $X$  is Hausdorff if distinct points admit disjoint open neighborhoods. Every metric space is Hausdorff: balls of radius

$$d(x, y)/3$$

about distinct  $x, y$  are disjoint by the triangle inequality.

Hausdorffness isolates a familiar feature of analysis: distinct points can be enclosed in disjoint open neighborhoods. Metrics provide this separation automatically, but arbitrary topologies need not: on a set with two points, the topology consisting only of the empty set and the whole set cannot separate them. The following proposition makes precise how separation ensures uniqueness of limits. Compact subsets will also behave as they do in Euclidean analysis: they are closed. These well-separated-point properties explain why Hausdorffness will be required in the definition of a manifold.

**Proposition 10.5.6** (Uniqueness of limits in Hausdorff spaces). In a Hausdorff space every convergent sequence has a unique limit. A function limit at an accumulation point is likewise unique if its target is Hausdorff.

*Proof.* If  $x_n \rightarrow x$  and  $x_n \rightarrow y$  with  $x \neq y$ , choose disjoint open neighborhoods  $U, V$  of  $x, y$  by Definition 10.5.5. For all sufficiently large  $n$ , convergence (Definition 10.4.12) puts  $x_n$  in both  $U$  and  $V$ , a contradiction. For two proposed function limits, choose disjoint target neighborhoods. Their corresponding source neighborhoods intersect in a neighborhood of the accumulation point; this contains a domain point other than that point, whose image would have to belong to both target neighborhoods.  $\square$

Hausdorffness makes limits unambiguous.

Only this implication is asserted: uniqueness of sequence limits alone does not characterize Hausdorffness in arbitrary spaces.

**Proposition 10.5.7** (Closed points in a Hausdorff space). If  $X$  is Hausdorff, every singleton  $\{x\}$  is closed. Consequently every finite subset of  $X$  is closed.

*Proof.* For each  $y \neq x$ , choose disjoint open neighborhoods of  $x$  and  $y$ . The neighborhood of  $y$  lies in  $X \setminus \{x\}$ . Thus this complement is open, being a union of such neighborhoods, and  $\{x\}$  is closed. Finite unions of closed sets are closed, proving the second assertion.  $\square$

**Example 10.5.8** (The Zariski affine line). Let  $k$  be an infinite field. On  $\mathbb{A}^1(k) = k$ , the Zariski closed sets are common zero sets  $V(\mathcal{S})$  of collections  $\mathcal{S} \subseteq k[t]$  of polynomials. If the collection contains a nonzero polynomial, its common zero set is finite, since that polynomial has finitely many roots. If every polynomial is zero, the zero set is  $k$ . Conversely

$$\{a_1, \dots, a_r\} = V\left(\prod_{j=1}^r (t - a_j)\right);$$

also  $\emptyset = V(1)$ . Thus the closed sets are exactly

$$\emptyset, \quad k, \quad \text{finite subsets of } k.$$

These sets do form the closed sets of a topology: they are closed under arbitrary intersections and finite unions. Every singleton is closed,  $\{a\} = V(t - a)$ , but two nonempty open sets are cofinite and intersect because  $k$  is infinite. Hence this space is not Hausdorff:

closed singletons do not imply Hausdorff separation.

The Zariski topology is fundamental in algebraic geometry; its large open sets encode algebraic rather than metric geometry.

There is a structural way to express the same separation property in a product space: equality should be a closed condition.

**Proposition 10.5.9** (The diagonal criterion). For a topological space  $X$ , let

$$\Delta_X = \{(x, x) : x \in X\} \subseteq X \times X.$$

Then  $X$  is Hausdorff if and only if  $\Delta_X$  is closed in the product topology.

*Proof.* Suppose first that  $X$  is Hausdorff. If  $(x, y) \notin \Delta_X$ , then  $x \neq y$ , so there are disjoint open neighborhoods  $U$  of  $x$  and  $V$  of  $y$ . The product  $U \times V$  is an open neighborhood of  $(x, y)$  which misses  $\Delta_X$ : a diagonal point in it would belong to both  $U$  and  $V$ . Thus every point of the complement has an open neighborhood in that complement. The complement is open, so  $\Delta_X$  is closed.

Conversely, suppose  $\Delta_X$  is closed and take  $x \neq y$ . The open complement contains  $(x, y)$ . By the product-topology basis of [Proposition 10.4.9](#), there are open  $U, V \subseteq X$  such that

$$(x, y) \in U \times V \subseteq (X \times X) \setminus \Delta_X.$$

If  $z \in U \cap V$ , then  $(z, z) \in U \times V \cap \Delta_X$ , a contradiction. Hence  $U$  and  $V$  are disjoint open neighborhoods of  $x$  and  $y$ , proving Hausdorffness.  $\square$

Thus Hausdorffness means that the relation  $x = y$  is closed in  $X \times X$ . The product formulation records separation as a property of a set, rather than as a separate choice of neighborhoods for every pair of points.

**Theorem 10.5.10** (Compact subsets are closed in Hausdorff spaces). A compact subset  $K$  of a Hausdorff space  $X$  is closed in  $X$ .

*Proof.* Fix  $y \notin K$ . For each  $x \in K$ , the Hausdorff hypothesis supplies open sets  $x \in U_x$  and  $y \in V_x$  with  $U_x \cap V_x = \emptyset$ . The  $U_x$  form an open cover of  $K$ , so compactness supplies  $x_1, \dots, x_m$  with

$$K \subseteq U_{x_1} \cup \dots \cup U_{x_m}.$$

Put

$$V = V_{x_1} \cap \dots \cap V_{x_m}.$$

It is open because the intersection is finite. It contains  $y$  because every factor does. It misses  $K$ : any  $z \in K$  belongs to some  $U_{x_i}$ , whereas a point of  $V$  belongs to  $V_{x_i}$ , which is disjoint from that  $U_{x_i}$ . Thus every  $y \notin K$  has an open neighborhood outside  $K$ . If  $K$  is empty use  $V = X$ . The complement is open, so  $K$  is closed.  $\square$

This proof converts neighborhoods depending on  $x$  into one neighborhood of  $y$ . An infinite intersection of the  $V_x$  would not necessarily be open; the finite selection is the precise job of compactness.

**Proposition 10.5.11** (A uniform neighborhood of a compact set). If  $K$  is compact in a metric space  $X$  and  $K \subseteq U$  with  $U$  open, there exists  $\delta > 0$  such that

$$B(K, \delta) := \bigcup_{x \in K} B(x, \delta) \subseteq U.$$

Consequently, if nonempty  $K$  is compact, nonempty  $F$  is closed, and  $K \cap F = \emptyset$ , then  $d(K, F) := \inf\{d(x, y) : x \in K, y \in F\} > 0$ .

*Proof.* We first leave room inside each local neighborhood, then take the smallest of finitely many radii. For each  $x \in K$  choose  $r_x > 0$  with  $B(x, 2r_x) \subseteq U$ . Choose a finite subcover

$$K \subseteq \bigcup_{i=1}^m B(x_i, r_i)$$

and put  $\delta = \min_i r_i > 0$ . If  $y \in B(K, \delta)$ , choose  $x \in K$  with  $d(y, x) < \delta$ , and then  $i$  with  $d(x, x_i) < r_i$ . Thus

$$d(y, x_i) \leq d(y, x) + d(x, x_i) < \delta + r_i \leq 2r_i,$$

so  $y \in U$ . If  $K = \emptyset$ , any positive  $\delta$  works. For the consequence take  $U = X \setminus F$ . No  $y \in F$  can satisfy  $d(x, y) < \delta$  for any  $x \in K$ , so  $d(K, F) \geq \delta > 0$ . Conversely, this positive-distance assertion applied to  $X \setminus U$  gives the neighborhood assertion, with any smaller positive radius; if the complement is empty there is nothing to check.  $\square$

This is a reusable form of compactness: point-dependent room inside an open set becomes one uniform margin. In Chapter 13 it will let a function supported on  $K$  extend smoothly by zero across the artificial edge of  $U$ .

## 10.6 Sequential Compactness, Total Boundedness, and a Common Scale

On intervals, convergent subsequences supplied global control. We now connect that sequence property with the open-cover definition of compactness. The intermediate tasks are to find finitely many nearby sample points and one positive radius small enough for the given cover.

**Definition 10.6.1** (Sequential compactness and total boundedness). A metric space is sequentially compact if every sequence has a subsequence converging to a point of that space. It is **totally bounded** if, for every  $\varepsilon > 0$ , it can be covered by finitely many open balls of radius  $\varepsilon$  with centers in the space.

Total boundedness says that at each resolution only finitely many small regions are needed. It does not yet make those balls fit inside a given open cover. A common radius for that cover supplies the second ingredient.

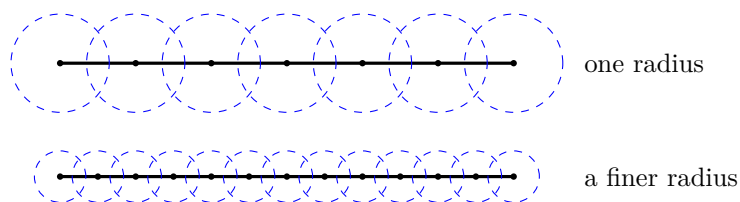


Figure 10.4: Total boundedness requires a finite cover by equal-radius balls at every positive radius. Finer resolution may require more balls, but still only finitely many. The balls in the pictured interval are relative balls.

**Lemma 10.6.2** (Two consequences of sequential compactness). A sequentially compact metric space is totally bounded. For every open cover there is  $\lambda > 0$  such that each  $B(x, \lambda)$  is contained in some member of that cover.

*Proof.* If finitely many radius- $\varepsilon$  balls cannot cover the space, choose  $x_1$ , then choose

$$x_{n+1} \notin \bigcup_{i=1}^n B(x_i, \varepsilon).$$

This choice is possible because the previous balls do not cover the space. Distinct selected points have distance at least  $\varepsilon$ . A convergent subsequence would be Cauchy, contradicting this separation.

If no common cover radius exists, for each  $n$  choose  $x_n$  such that

$$B(x_n, 1/n)$$

lies in no cover member. Sequential compactness gives  $x_{n_j} \rightarrow x$ , where  $n_j$  increases strictly. Choose a cover member  $U$  containing  $x$  and  $r > 0$  with  $B(x, r) \subseteq U$ . Eventually both

$$d(x_{n_j}, x) < r/2, \quad 1/n_j < r/2.$$

For any

$$z \in B(x_{n_j}, 1/n_j),$$

$$d(z, x) \leq d(z, x_{n_j}) + d(x_{n_j}, x) < 1/n_j + r/2 < r.$$

Thus that entire ball lies in  $U$ , contradicting its selection. The empty space satisfies both assertions without choosing points.  $\square$

**Theorem 10.6.3** (Metric compactness). For a metric space  $X$ ,

|                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------|
| $\begin{aligned} X \text{ compact} &\iff X \text{ sequentially compact} \\ &\iff X \text{ complete and totally bounded.} \end{aligned}$ |
|-----------------------------------------------------------------------------------------------------------------------------------------|

*Proof. Strategy.* A finite cover forces one point to see infinitely many sequence indices; shrinking balls then select a convergent subsequence. Let  $(x_n)$  lie in compact  $X$ . Suppose every  $p \in X$  has an open neighborhood containing  $x_n$  for only finitely many indices  $n$ . A finite subcover would then contain terms for only finitely many indices, although it covers all of  $X$ , a contradiction. Thus some  $p \in X$  has infinitely many sequence indices in every open neighborhood. Starting with  $n_0 = 0$ , choose  $n_k > n_{k-1}$  with

$$x_{n_k} \in B(p, 1/k).$$

Then  $x_{n_k} \rightarrow p$ . Counting indices handles repeated values correctly.

Conversely, for a given open cover choose  $\lambda$  from the preceding lemma, then cover the space by finitely many radius-

$$\lambda/2$$

balls with centers  $a_1, \dots, a_m$ . For each center choose a cover member  $U_i$  containing  $B(a_i, \lambda)$ . Every point is within

$$\lambda/2$$

of a center, so lies in one of these  $U_i$ . These are finitely many members of the original cover. For the empty space all three properties hold directly.

Suppose next that  $X$  is sequentially compact. Total boundedness was proved in [Lemma 10.6.2](#). If  $(x_n)$  is Cauchy, choose a subsequence  $x_{n_j} \rightarrow x \in X$ . Given  $\varepsilon > 0$ , choose  $N$  so

$$d(x_n, x_m) < \varepsilon/2$$

for  $n, m \geq N$ , and choose  $j$  with  $n_j \geq N$  and

$$d(x_{n_j}, x) < \varepsilon/2.$$

For every  $n \geq N$ ,

$$d(x_n, x) \leq d(x_n, x_{n_j}) + d(x_{n_j}, x) < \varepsilon.$$

Thus the whole sequence converges in  $X$ , proving completeness.

Finally assume completeness and total boundedness. We will select a subsequence whose tail is trapped in one ball at each successively finer scale. Cover  $X$  by finitely many balls of radius  $2^{-r}$

for  $r = 0, 1, 2, \dots$ . Starting with  $N_{-1} = \mathbb{N}$ , choose at stage  $r$  one ball containing  $x_n$  for infinitely many indices  $n \in N_{r-1}$ , and let  $N_r$  be those indices. A finite union of finite index sets could not exhaust the infinite  $N_{r-1}$ , so the choice is possible. Then

$$N_0 \supseteq N_1 \supseteq N_2 \supseteq \dots, \quad n_r \in N_r, \quad n_r > n_{r-1}.$$

For  $s, t \geq r$ , both  $x_{n_s}, x_{n_t}$  belong to the stage- $r$  ball, and hence  $d(x_{n_s}, x_{n_t}) < 2^{1-r}$ . Given  $\varepsilon > 0$ , choose  $r$  with  $2^{1-r} < \varepsilon$ . This proves the subsequence is Cauchy; completeness supplies its limit in  $X$ . We have proved sequential compactness.  $\square$

The selected index sets, not necessarily the balls themselves, are nested. The two hypotheses have different jobs: total boundedness produces Cauchy approximations; completeness prevents those approximations from escaping.

**Example 10.6.4** (Which compactness ingredient fails?). All subspaces of  $\mathbb{R}$  in the table use the usual distance. In the last row,  $D$  is infinite and  $d(x, y) = 1$  for  $x \neq y$ , with  $d(x, x) = 0$ .

| $X$                      | bounded | complete | totally bounded | compact |
|--------------------------|---------|----------|-----------------|---------|
| $[0, 1]$                 | yes     | yes      | yes             | yes     |
| $(0, 1)$                 | yes     | no       | yes             | no      |
| $\mathbb{Q} \cap [0, 1]$ | yes     | no       | yes             | no      |
| $\mathbb{R}$             | no      | yes      | no              | no      |
| $D$                      | yes     | yes      | no              | no      |

Finite subdivisions of  $[0, 1]$  into intervals shorter than  $\varepsilon$  give finite ball covers: choose a center in each interval meeting the subspace. The sequence

$$1/n$$

escapes  $(0, 1)$ ; rational approximations to

$$\sqrt{2}/2$$

escape  $\mathbb{Q} \cap [0, 1]$ . Both are Cauchy. Finitely many balls of fixed radius in  $\mathbb{R}$  have bounded union and cannot cover  $\mathbb{R}$ . In  $D$ , every Cauchy sequence is eventually constant, but each ball of radius

$$1/2$$

is a singleton. Thus  $D$  is bounded and complete but not totally bounded. These examples distinguish boundedness from total boundedness, and total boundedness from compactness without completeness.

**Lemma 10.6.5** (Lebesgue number). For an open cover of a nonempty compact metric space there is  $\lambda > 0$  such that every subset of diameter less than  $\lambda$  lies in one cover member.

*Proof.* Use the ball radius just proved. If  $E$  is nonempty, choose  $a \in E$ . For every  $z \in E$ ,  $d(z, a) \leq \text{diam}(E) < \lambda$ , so  $E \subseteq B(a, \lambda)$ , which lies in a cover member. The empty set lies in any member; the cover has a member because the space is nonempty.  $\square$

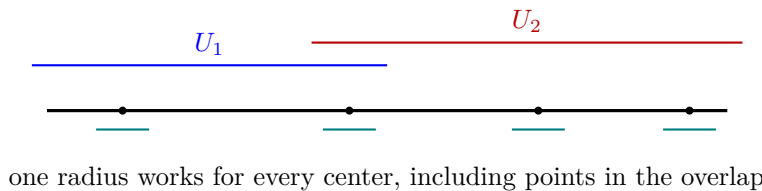


Figure 10.5: A common scale for a prescribed cover. Each sufficiently small relative ball fits in one cover member; the containing member may depend on the center. A Lebesgue number extends this control to all subsets of sufficiently small diameter.

This is the scale needed when a fine grid must fit its cells into local neighborhoods. Total boundedness gives finite ball covers; a Lebesgue number makes sufficiently small pieces fit inside the prescribed cover.

**Theorem 10.6.6** (Compact extreme-value theorem). A continuous  $f : K \rightarrow \mathbb{R}$  on a nonempty compact space attains its extrema.

*Proof.* By Theorems 10.5.4 and 10.5.10, the image is compact and closed in  $\mathbb{R}$ . Cover it by the intervals  $(-m, m)$ ,  $m \in \mathbb{N}$ . Choose a finite subcover; its largest selected radius bounds the image. Let  $s$  be its supremum. For each  $n$  choose an image value

$$y_n > s - 1/n,$$

with  $y_n \leq s$ . Then  $y_n \rightarrow s$ , and closedness places  $s$  in the image. Applying the same argument to  $-f$  gives its minimum.  $\square$

**Theorem 10.6.7** (Uniform continuity on compact metric spaces). A continuous map from a compact metric space to a metric space is uniformly continuous.

*Proof.* Failure would give one  $\varepsilon_0 > 0$  and pairs  $x_n, y_n$  with

$$d_X(x_n, y_n) < 1/n$$

but  $d_Y(f(x_n), f(y_n)) \geq \varepsilon_0$ . Extract  $x_{n_j} \rightarrow x$  by metric compactness. Retain the same indices for the paired points, because only those pairs have the small-distance estimate:

$$d_X(y_{n_j}, x) \leq d_X(y_{n_j}, x_{n_j}) + d_X(x_{n_j}, x) < 1/n_j + d_X(x_{n_j}, x) \rightarrow 0.$$

Both image subsequences tend to  $f(x)$  by continuity, so their mutual distance tends to zero by the triangle inequality, contradicting  $\varepsilon_0$ .  $\square$

**Theorem 10.6.8** (Heine–Borel). A subset of Euclidean  $\mathbb{R}^n$  is compact exactly when it is closed and bounded.

*Proof.* If  $K$  is compact, it is closed because Euclidean space is Hausdorff (Theorem 10.5.10). For boundedness use the increasing open cover

$$\mathbb{R}^n = \bigcup_{m \geq 1} B(0, m).$$

A finite subcover of  $K$  has a largest selected radius  $M$ , so  $K \subseteq B(0, M)$ .



Conversely, by [Theorem 10.6.3](#) it suffices to prove sequential compactness. Boundedness lets us extract convergence coordinate by coordinate; finite dimensionality makes the extraction terminate; closedness keeps the resulting limit inside  $K$ .

First see the mechanism in  $\mathbb{R}^2$ . For a sequence  $\mathbf{x}_j = (x_j^1, x_j^2)$  in bounded  $K$ , Bolzano–Weierstrass gives indices  $j_k$  with  $x_{j_k}^1 \rightarrow a_1$ . Apply it again to the bounded scalar sequence  $(x_{j_k}^2)$  to obtain  $k_\ell$  with  $x_{j_{k_\ell}}^2 \rightarrow a_2$ . The first coordinates still converge to  $a_1$ : a subsequence of a convergent sequence has the same limit ([Proposition 10.3.2](#)). The same selected points now work in both coordinates.

In  $\mathbb{R}^n$  make the finite sequence of extractions

$$(\mathbf{x}_j) \supset (\mathbf{x}_{j_k^{(1)}}) \supset \cdots \supset (\mathbf{x}_{j_k^{(n)}}).$$

Each extraction retains convergence of all previously treated coordinates. Write the final indices simply as  $j_k$  and their coordinate limits as  $a_i$ . Given  $\varepsilon > 0$ , for each  $i$  choose  $K_i$  with

$$|x_{j_k}^i - a_i| < \varepsilon/\sqrt{n}$$

for  $k \geq K_i$ . The single index  $k \geq \max_{1 \leq i \leq n} K_i$  then works for every coordinate, and

$$\|\mathbf{x}_{j_k} - \mathbf{a}\|_2 \leq \sqrt{n} \max_i |x_{j_k}^i - a_i| < \varepsilon.$$

Closedness and [Theorem 10.3.6](#) give  $\mathbf{a} \in K$ . The empty set and the case  $n = 0$  are immediate. Thus

boundedness produces a convergent subsequence;  
closedness prevents the limit from escaping.

□

**A second viewpoint.** A closed subset of complete  $\mathbb{R}^n$  is complete by [Proposition 10.3.7](#). A bounded subset lies in a bounded box. Subdivide that box into finitely many cubes of side

$$a < \varepsilon/\sqrt{n}$$

and choose a point of the subset in each cube that meets it. Every point in that cube is less than  $\varepsilon$  from the chosen point, since the cube has diameter  $a\sqrt{n}$ . Thus the subset is totally bounded. The general metric compactness theorem gives the same conclusion, while the coordinate proof above explains explicitly why finite dimension is decisive.

In particular the unit sphere is compact: it is bounded, and the reverse length estimate  $\|\mathbf{x}\|_2 - \|y\|_2 \leq \|\mathbf{x} - y\|_2$  makes it closed. A continuous positive function there has a positive minimum: attainment places the minimum at a point where the function is strictly positive. This observation will supply lower bounds in norm equivalence.

## 10.7 Connectedness and Paths

Can a space be split into two separated nonempty open pieces? The space  $[0, 1] \cup [2, 3]$  can: each displayed interval is relatively open, using ambient open intervals that stop in the gap. A single interval has no such gap. A disk also has a geometric feature suggesting connectedness: one can travel along a straight segment between any two of its points without leaving it.

**Definition 10.7.1** (Connectedness). A space is connected if it cannot be written  $X = A \cup B$  with  $A, B$  disjoint, nonempty, and open in  $X$ . Such a pair is called a separation.

**Proposition 10.7.2** (Clopen sets and locally constant functions). A space  $X$  is connected if and only if its only subsets that are both open and closed (**clopen**) are  $\emptyset$  and  $X$ . A function is **locally constant** if every point has an open neighborhood on which the function is constant. Every locally constant function on a nonempty connected space is constant.

*Proof.* A nontrivial clopen subset  $A$  gives the separation  $A, X \setminus A$ . Conversely either part of a separation is nontrivial and clopen. For a locally constant function  $f$ , each fiber  $f^{-1}(\{c\})$  is open: every point of it has a constant neighborhood contained in that fiber. Its complement is the union of all the other fibers and is also open. Choose  $x \in X$ ; its nonempty clopen fiber must be  $X$ , so  $f = f(x)$ .  $\square$

This observation will turn locally constant orientation signs into one sign on each connected overlap. No topology on the function's target is needed for the locally constant assertion.

**Definition 10.7.3** (Paths and convexity). A path from  $x$  to  $y$  is a continuous  $\gamma : [0, 1] \rightarrow X$  with  $\gamma(0) = x$  and  $\gamma(1) = y$ . A space is path connected if every two points can be joined by a path. A set  $C \subseteq \mathbb{R}^n$  is convex if  $(1 - t)\mathbf{x} + ty \in C$  for  $\mathbf{x}, y \in C$  and  $0 \leq t \leq 1$ . Thus convex sets have the straight-line paths  $\gamma(t) = (1 - t)\mathbf{x} + ty$ .

**Proposition 10.7.4** (Connectedness). Continuous images of connected spaces are connected. Intervals in  $\mathbb{R}$  are connected, every path-connected space is connected, and every connected subset of  $\mathbb{R}$  is an interval (including the empty set and singletons).

*Proof.* Suppose  $A, B$  separate  $f(X)$ . Their inverse images under the continuous map into  $f(X)$  are open in  $X$ . They cover  $X$ , are disjoint, and are both nonempty because every point of  $f(X)$  has a preimage. This contradicts connectedness.

If an interval  $I$  had a separation  $A, B$ , define  $h = 0$  on  $A$  and  $h = 1$  on  $B$ . The relative openness of the pieces makes  $h$  locally constant, hence continuous. Choose one point from each piece. The closed interval between them lies in  $I$ , so the Intermediate Value Theorem would force the value

$$1/2,$$

impossible.

If a path-connected  $X$  had a separation, choose  $x \in A, y \in B$  and a path joining them. The inverse images of  $A, B$  under the path would be a separation of  $[0, 1]$ : they are open, disjoint, cover, and contain  $0, 1$  respectively. The preceding paragraph excludes this.

Finally, a set  $E \subseteq \mathbb{R}$  that is not an interval has  $a < c < b$  with  $a, b \in E$  and  $c \notin E$ . The nonempty relatively open pieces  $E \cap (-\infty, c)$  and  $E \cap (c, \infty)$  separate it.  $\square$

**Corollary 10.7.5** (Intermediate values on a connected space). If  $X$  is connected and  $f : X \rightarrow \mathbb{R}$  is continuous, then  $f(X)$  is an interval. In particular,  $f(x) < c < f(y)$  implies  $f(z) = c$  for some  $z \in X$ .

*Proof.* The image is connected by [Proposition 10.7.4](#); a connected subset of  $\mathbb{R}$  is an interval by the same proposition. Membership of  $c$  in this interval means precisely that such a preimage  $z$  exists.  $\square$

**Corollary 10.7.6** (Constant sign). A continuous nonvanishing real function on a nonempty connected space has one constant sign.

*Proof.* Its connected image is an interval avoiding zero. If it contained both a negative and a positive value, the interval property would put zero in the image. Thus all its values have the same sign.  $\square$

Later, continuity of a determinant, connectedness of a domain, and nonvanishing of that determinant will play these three separate roles.

### Three increasingly flexible ways to stay in one piece

$$\boxed{\text{convex} \implies \text{path connected} \implies \text{connected}.}$$

The straight segment proves the first implication; [Proposition 10.7.4](#) proves the second. The circle  $S^1$  is path connected: choose angles for two given points and interpolate the angles continuously. It is not convex, since the midpoint of opposite points is its missing center.

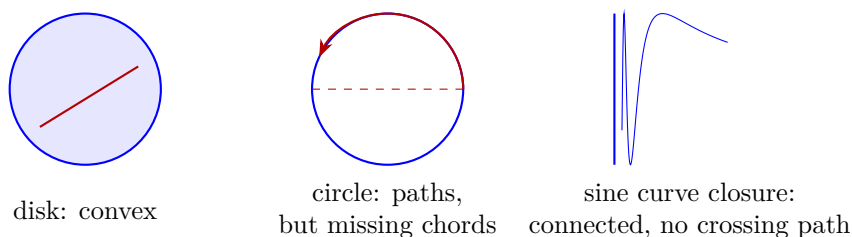


Figure 10.6: The implications convex implies path connected implies connected cannot be reversed. Dashed chords need not stay in the circle; the sine curve has a stronger obstruction to paths.

**Proposition 10.7.7** (Closure preserves connectedness). If  $A$  is connected, its closure  $\overline{A}$  is connected.

*Proof.* If two nonempty relatively open sets separated  $\overline{A}$ , each would meet  $A$ : write one as  $\overline{A} \cap U$  with  $U$  ambient open, choose  $x$  in it, and use the closure neighborhood test to find  $U \cap A \neq \emptyset$ . Their intersections with  $A$  would therefore separate  $A$ , a contradiction.  $\square$

**Example 10.7.8** (The topologist's sine curve). Let

$$G = \{(x, \sin(1/x)) : 0 < x \leq 1\}, \quad S = G \cup (\{0\} \times [-1, 1]).$$

The graph  $G$  is a continuous image of the interval  $(0, 1]$ , so is connected. Its closure is exactly  $S$ . Indeed limits with positive first coordinate remain on the graph by continuity; limits with first coordinate zero have second coordinate in  $[-1, 1]$ . Conversely, for every  $y \in [-1, 1]$  choose  $\theta$  with  $\sin \theta = y$ . For all sufficiently large integers  $m$ ,

$$x_m = 1/(\theta + 2\pi m)$$

lies in  $(0, 1]$ , and  $(x_m, y) \rightarrow (0, y)$ . Thus [Proposition 10.7.7](#) proves  $S$  connected.

To rule out a path from the vertical segment to the graph, we force opposite oscillations arbitrarily close to its last contact with that segment. Suppose  $\gamma(t) = (u(t), v(t))$ ,  $0 \leq t \leq 1$ , were

such a path, with  $u(0) = 0$  and  $u(1) > 0$ . The closed zero set of  $u$  has a largest point  $a < 1$ . Thus  $u(a) = 0$  and  $u(t) > 0$  for  $a < t \leq 1$ . Choose  $b_m \downarrow a$  with  $b_m > a$ . Since  $u(b_m) > 0$ , choose numbers

$$0 < r_m^+ = \frac{1}{\pi/2 + 2\pi N_m} < u(b_m), \quad 0 < r_m^- = \frac{1}{3\pi/2 + 2\pi N_m} < u(b_m)$$

by taking  $N_m$  sufficiently large. The intermediate value theorem on  $[a, b_m]$  supplies  $t_m^+, t_m^- \in (a, b_m)$  with  $u(t_m^\pm) = r_m^\pm$ . These points are on  $G$ , so  $v(t_m^+) = 1$  and  $v(t_m^-) = -1$ . But both sequences of times converge to  $a$ , contradicting continuity of  $v$  at  $a$ . Hence  $S$  is connected but not path connected.

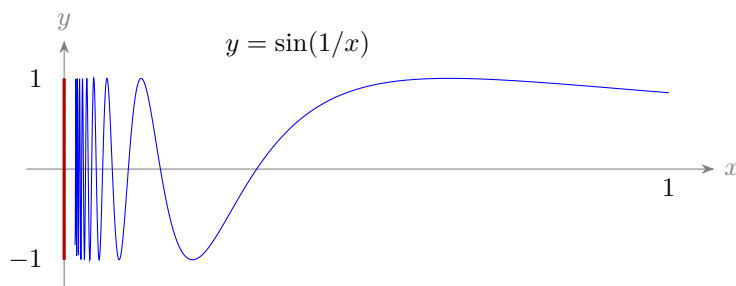


Figure 10.7: The vertical segment belongs to the closure of the oscillating graph. The drawing truncates the oscillations near zero; the proof uses all of them. A path leaving the segment would have to attain both heights arbitrarily close to one time.

## Connected components and total disconnectedness

**Lemma 10.7.9** (Connected sets sharing a point). A union of connected subsets with a common point is connected.

*Proof.* In a proposed separation of the union, the common point lies in one part, say  $A$ . Every connected member must lie wholly in  $A$ : if it met the other part, the two relative intersections would separate it. Thus the other part of the union is empty, a contradiction.  $\square$

**Definition 10.7.10** (Connected component). For  $x \in X$ , the **connected component**  $C(x)$  is the union of all connected subsets of  $X$  containing  $x$ .

**Proposition 10.7.11** (Components partition a space). The components are maximal connected subsets, partition  $X$ , and are closed.

*Proof.* The singleton  $\{x\}$  is connected and all the sets defining  $C(x)$  share  $x$ , so [Lemma 10.7.9](#) makes  $C(x)$  connected. Any connected superset of it contains  $x$  and is already included in its defining union; this proves maximality. If  $C(x)$  and  $C(y)$  intersect, their union is connected by the same lemma, so maximality makes both components equal. Every point belongs to its component, proving the partition assertion. Finally  $\overline{C(x)}$  is connected by [Proposition 10.7.7](#). Maximality gives  $\overline{C(x)} = C(x)$ , proving closedness.  $\square$

**Definition 10.7.12** (Totally disconnected). A space is **totally disconnected** if all its connected components are singletons.

**Example 10.7.13** (Totally disconnected need not mean discrete). Let

$$E = \{0\} \cup \{1/n : n \in \mathbb{N}\} \subset \mathbb{R}.$$

A connected subset of  $\mathbb{R}$  is an interval ([Proposition 10.7.4](#)); an interval between distinct points of  $E$  contains an irrational number, hence cannot lie in  $E$ . Thus all connected subsets of  $E$  are singletons or empty. Yet 0 is not isolated: every relative neighborhood contains

$$1/n$$

for large  $n$ . Therefore  $E$  is totally disconnected but not discrete. Likewise, if rational  $p < q$  belong to a subset of  $\mathbb{Q}$ , choose irrational  $c \in (p, q)$ . Intersecting that subset with  $(-\infty, c)$  and  $(c, \infty)$  separates it. Thus  $\mathbb{Q}$  is totally disconnected; no rational is isolated, since

$$p + 1/n \rightarrow p$$

with distinct rational terms. Total disconnectedness concerns connected subsets, whereas discreteness concerns open singletons.

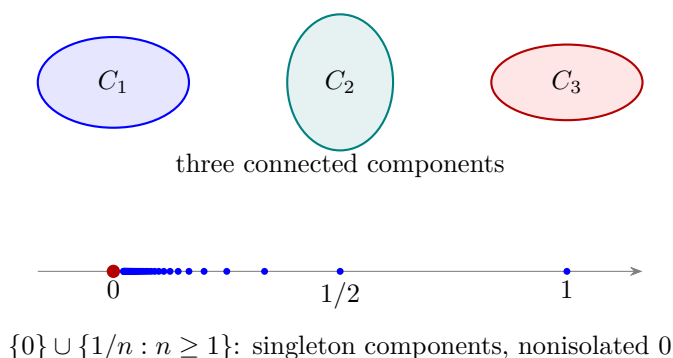


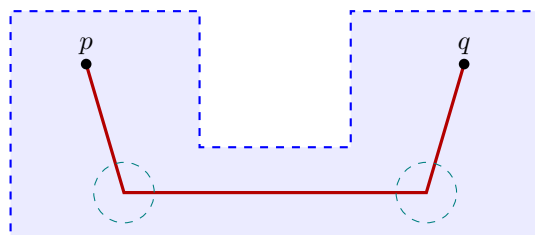
Figure 10.8: Components are maximal connected pieces. In the accumulating sequence below, every component is a point, but the component consisting of zero is not open.

**Theorem 10.7.14** (Open connected Euclidean sets admit polygonal paths). Every nonempty open connected  $U \subseteq \mathbb{R}^n$  is polygonally path connected: any two points can be joined by finitely many line segments in  $U$ .

*Proof.* Fix  $p \in U$  and let  $A$  consist of the points reachable from  $p$  by such paths, including the constant path. Small balls will show that reachability and its failure are both open conditions. For  $\mathbf{x} \in U$ , choose  $B(\mathbf{x}, r) \subseteq U$ . If  $\mathbf{x} \in A$ , append the segment from  $\mathbf{x}$  to any  $y \in B(\mathbf{x}, r)$  to a path reaching  $\mathbf{x}$ ; convexity of the ball gives  $B(\mathbf{x}, r) \subseteq A$ . If  $\mathbf{x} \notin A$ , no  $y \in B(\mathbf{x}, r)$  can belong to  $A$ , because appending the reverse segment would reach  $\mathbf{x}$ . Thus  $B(\mathbf{x}, r) \subseteq U \setminus A$ . Both sets are open in  $U$ . Since  $p \in A$ , connectedness forces  $A = U$ .  $\square$

**Corollary 10.7.15** (Components of Euclidean open sets). Every connected component of an open subset of  $\mathbb{R}^n$  is open and polygonally path connected.

*Proof.* For  $\mathbf{x}$  in a component  $C$ , a small ball  $B(\mathbf{x}, r) \subseteq U$  is connected. Its union with  $C$  is connected by Lemma 10.7.9, so maximality puts the whole ball in  $C$ . Thus  $C$  is open; apply the preceding theorem.  $\square$



small balls allow one more straight segment

Figure 10.9: A polygonal path can go around a gap in an open connected domain. The proof uses small convex balls to show that the reachable set and its complement are relatively open.

## 10.8 Vector Spaces, Bases, and Dimension

Vectors describe directions; linear maps will describe first-order approximations. Coordinates let us calculate with these intrinsic objects. We develop the finite-dimensional real theory needed for that purpose.

### What can be added and rescaled?

In  $\mathbb{R}^2$ ,  $(2, 1) + (1, -3) = (3, -2)$  and  $2(2, 1) = (4, 2)$ . In  $\mathbb{R}^3$  the same operations describe displacements in three directions. But coordinate tuples are not essential: the polynomials  $1 + t$  and  $t^2$  can also be added and rescaled, as can continuous functions on  $[a, b]$ . Their sums remain objects of the same kind. The vector-space definition records the arithmetic laws common to these examples, without requiring that an object look like an arrow. Scalars are numbers; vectors are whatever objects the space contains. In the polynomial example, the zero vector is the zero polynomial, not a coordinate location.

**Definition 10.8.1** (Real vector space). A real vector space is a set  $V$  with operations  $+: V \times V \rightarrow V$  and  $\mathbb{R} \times V \rightarrow V$ ,  $(a, v) \mapsto av$ . For all vectors  $u, v, w$  and scalars  $a, b$ , they satisfy

$$\begin{array}{ll}
 (u + v) + w = u + (v + w) & \text{(associativity of addition),} \\
 u + v = v + u & \text{(commutativity),} \\
 v + 0 = v & \text{(a zero vector exists),} \\
 v + (-v) = 0 & \text{(each vector has an additive inverse),} \\
 a(u + v) = au + av & \text{(distribution over vector addition),} \\
 (a + b)v = av + bv & \text{(distribution over scalar addition),} \\
 a(bv) = (ab)v & \text{(compatibility of scalar multiplication),} \\
 1v = v & \text{(scalar identity).}
 \end{array}$$

For another concrete example, the functions  $a \cos t + b \sin t$  form a vector space under pointwise operations: addition adds the two coefficients and scalar multiplication rescales them. These are functions, not pairs; coefficients are a convenient description. More generally  $P_n$  denotes polynomials of degree at most  $n$ , including zero.

The model is  $\mathbb{R}^n$  with componentwise operations. Another example is  $P_2$ , the real polynomials of degree at most two (including the zero polynomial), with ordinary addition and scalar multiplication.

## Subspaces

The plane  $x + y + z = 0$  in  $\mathbb{R}^3$  contains zero, and sums and multiples of solutions remain solutions. The parallel plane  $x + y + z = 1$  lacks zero and is not a vector space under these operations. Polynomials with zero constant term form a subspace of  $P_n$ ; polynomials of degree exactly  $n$  do not, since zero is missing. The union of the two coordinate axes is not a subspace either:  $(1, 0)$  and  $(0, 1)$  belong to it but their sum does not.

**Definition 10.8.2** (Subspace). A subspace  $U$  of  $V$  is a subset that is itself a vector space with the operations inherited from  $V$ .

**Proposition 10.8.3** (Subspace criterion). A nonempty  $U \subseteq V$  is a subspace exactly when  $au + bv \in U$  for all  $u, v \in U$  and all scalars  $a, b$ .

*Proof.* A subspace is closed under scaling and addition, giving the condition. Conversely choose  $u \in U$ . Taking  $a = b = 0$  puts  $0$  in  $U$ ; taking  $a = b = 1$  gives closure under addition, and taking  $b = 0$  gives scalar closure. In particular  $-u \in U$ . All other axioms are identities already true in  $V$ , so they remain true for its subset under the inherited operations.  $\square$

## Linear combinations and span

Can  $(3, 1)$  be produced from  $u = (1, 1)$  and  $v = (1, -1)$ ? The equation  $au + bv = (3, 1)$  asks for  $a + b = 3$  and  $a - b = 1$ , so  $a = 2, b = 1$ . For a general  $(x, y)$  the same equations give

$$a = (x + y)/2,$$

$$b = (x - y)/2.$$

Thus this pair produces every vector in the plane.

**Definition 10.8.4** (Linear combination). A linear combination of  $v_1, \dots, v_r$  is a vector  $a_1v_1 + \dots + a_rv_r$ , where the  $a_i$  are scalars. The empty sum is zero.

**Definition 10.8.5** (Span). For  $S \subseteq V$ , define

$$\text{span}(S) = \left\{ \sum_{j=1}^r a_j v_j : r \geq 0, v_j \in S, a_j \in \mathbb{R} \right\}.$$

Only finite sums are used, even if  $S$  is infinite.

**Proposition 10.8.6** (What span constructs).  $\text{span}(S)$  is a subspace, contains  $S$ , and is contained in every subspace containing  $S$ .

*Proof.* The empty sum gives zero. If

$$x = \sum_{i=1}^r a_i u_i, \quad y = \sum_{j=1}^s b_j v_j$$

belong to the span, then

$$cx + dy = \sum_i (ca_i)u_i + \sum_j (db_j)v_j$$

is another finite combination of members of  $S$ . The subspace criterion applies. Each  $v \in S$  is the one-term combination  $1v$ , so  $S$  is contained. If  $U$  is any subspace containing  $S$ , scalar closure puts each  $a_j v_j$  in  $U$  and repeated addition puts their finite sum in  $U$ . Hence the whole span lies in  $U$ .  $\square$

**Example 10.8.7** (Three kinds of span). In  $\mathbb{R}^2$ ,  $\text{span}(1, 2)$  is the line  $(a, 2a)$ . In  $\mathbb{R}^3$ , combinations of  $u = (1, 0, 1)$  and  $v = (0, 1, 1)$  are  $(a, b, a+b)$ , exactly the plane  $z = x+y$ . Thus  $(2, -1, 1) = 2u - v$  lies in the span, while  $(0, 0, 1)$  does not. In  $P_2$ , combinations of  $1+t$  and  $t+t^2$  have coefficients  $(a, a+b, b)$ , so  $c+dt+et^2$  is in the span exactly when  $d=c+e$ .

**Definition 10.8.8** (Spanning set). A set spans or generates  $V$  if its span is  $V$ . The question is whether every vector can be produced, regardless of whether there are redundant generators. A space is finite dimensional if it has a finite spanning set.

For example  $(e_1, e_2, e_1+e_2)$  spans  $\mathbb{R}^2$ , although the third vector adds nothing. In  $P_2$ ,  $(1, t, t^2, 1+t)$  is a redundant spanning list.

### Independence means no redundancy

The list  $u, v, u+v$  produces no more vectors than  $u, v$ . The relation  $u+v-(u+v)=0$  detects its redundancy with coefficients  $1, 1, -1$ . More generally a relation with some nonzero coefficient lets us solve for the corresponding vector and delete it from a generating list. The definition below says that such deletion is impossible. A basis must meet both requirements: it spans everything and contains no redundancy.



**Example 10.8.9** (Solve the zero relation first). For  $u = (1, 1)$  and  $v = (1, -1)$ , the relation  $au + bv = 0$  gives  $a + b = 0$ ,  $a - b = 0$ , hence  $a = b = 0$ . For the three vectors  $(1, 0, 1)$ ,  $(0, 1, 1)$ ,  $(1, 1, 0)$  in  $\mathbb{R}^3$ , a zero relation gives

$$a + c = 0, \quad b + c = 0, \quad a + b = 0.$$

The first two imply  $a = b = -c$ , so the third gives  $-2c = 0$ ; all three coefficients vanish. By contrast, replacing the third vector with  $(1, 1, 2)$  gives the nontrivial relation  $(1, 0, 1) + (0, 1, 1) - (1, 1, 2) = 0$ . Whenever

$$\sum_i a_i v_i = 0$$

has  $a_j \neq 0$ , solving gives

$$v_j = - \sum_{i \neq j} (a_i / a_j) v_i.$$

Thus this vector lies in the span of the others.

**Definition 10.8.10** (Linear independence). A list  $(v_1, \dots, v_r)$  is linearly independent if

$$\sum_{j=1}^r a_j v_j = 0 \implies a_1 = \dots = a_r = 0.$$

Otherwise it is dependent. A set is independent if every finite list of distinct members is independent.

**Definition 10.8.11** (Basis). A basis is an independent spanning set: it produces every vector and has no redundancy.

Spanning provides an expression for every vector; independence makes it unique, since subtracting two expressions gives a relation equal to zero. For instance  $(1, t, t^2)$  is a basis of  $P_2$ , since polynomial coefficients are unique. The empty list is a basis of the zero space.

**Example 10.8.12** (Polynomial coordinates). The list  $(1, 1 + t, t^2)$  is a basis of  $P_2$ . Indeed  $a + b(1 + t) + ct^2 = (a + b) + bt + ct^2$ , so a zero combination has  $c = b = a = 0$ . Also  $2 + 3t - t^2 = -1 \cdot 1 + 3(1 + t) - t^2$ . Every  $c + dt + et^2$  is  $(c - d) \cdot 1 + d(1 + t) + et^2$ , proving spanning as well. The same polynomial has columns  $(2, 3, -1)^T$  in  $(1, t, t^2)$  and  $(-1, 3, -1)^T$  in  $(1, 1 + t, t^2)$ . For contrast  $(1 + t, t + t^2, 1 - t^2)$  is dependent because its first vector minus its second equals its third. Looking at three polynomials is not enough to conclude that they form a basis of a three-coordinate space.

## Coordinates: existence and uniqueness

Fix an ordered basis  $\mathcal{B} = (v_1, \dots, v_n)$ . Spanning provides

$$v = \sum_j x_j v_j.$$

If also

$$v = \sum_j y_j v_j,$$

subtraction gives

$$\sum_j (x_j - y_j)v_j = 0;$$

independence gives  $x_j = y_j$  for every  $j$ . Define  $[v]_{\mathcal{B}} = (x_1, \dots, x_n)^T$ , its coordinate column. For  $\mathcal{B} = ((1, 1), (1, -1))$ ,  $[(3, 1)]_{\mathcal{B}} = (2, 1)^T$ . For  $\mathcal{C} = ((1, 0, 1), (0, 1, 1), (0, 0, 1))$ ,

$$(x, y, z) = x(1, 0, 1) + y(0, 1, 1) + (z - x - y)(0, 0, 1),$$

so  $[(2, -1, 4)]_{\mathcal{C}} = (2, -1, 3)^T$ . The polynomial example above supplies a third, non-geometric coordinate system.

**Proposition 10.8.13** (Coordinate and synthesis maps). The maps

$$\kappa_{\mathcal{B}} : V \rightarrow \mathbb{R}^n, \quad v \mapsto [v]_{\mathcal{B}}, \quad s_{\mathcal{B}} : \mathbb{R}^n \rightarrow V, \quad \mathbf{x} \mapsto \sum_j x_j v_j$$

preserve addition and scalar multiplication (they are linear), and are mutual inverses.

*Proof.* If

$$u = \sum_j x_j v_j, \quad v = \sum_j y_j v_j,$$

then

$$au + bv = \sum_j (ax_j + by_j)v_j.$$

Uniqueness of coordinates proves  $\kappa_{\mathcal{B}}(au + bv) = a\kappa_{\mathcal{B}}(u) + b\kappa_{\mathcal{B}}(v)$ . Distributing a finite sum proves  $s_{\mathcal{B}}(a\mathbf{x} + b\mathbf{y}) = as_{\mathcal{B}}(\mathbf{x}) + bs_{\mathcal{B}}(\mathbf{y})$ . For any  $\mathbf{x} \in \mathbb{R}^n$ , the coefficients of  $s_{\mathcal{B}}(\mathbf{x})$  are precisely  $\mathbf{x}$ , so  $\kappa_{\mathcal{B}}s_{\mathcal{B}}(\mathbf{x}) = \mathbf{x}$ . For any  $v \in V$ , its coordinate expansion gives  $s_{\mathcal{B}}\kappa_{\mathcal{B}}(v) = v$ . Thus both composites are identities.  $\square$

These explicit inverse maps will be called an isomorphism after linear maps are introduced formally. No dimension theorem was needed to construct them.

### Why dimension is a theorem before it is a number

We would like to call the number of basis vectors the dimension. But that would be ambiguous if another basis could have a different size. The next three arguments remove that ambiguity. Reduction deletes redundant generators; exchange compares an independent list with a generating list; extension fills the missing directions.

**Lemma 10.8.14** (Basis reduction). Every finite spanning set contains a basis.

*Proof.* Remove one redundant vector at each step. If  $a_1v_1 + \dots + a_rv_r = 0$  and  $a_j \neq 0$ , then

$$v_j = -\sum_{i \neq j} \frac{a_i}{a_j} v_i.$$

In any combination of the original list, substitute this expression for  $v_j$ . The result uses only the remaining vectors, so deletion preserves span. Each deletion shortens the finite list by one. When no deletion is possible the list is independent and still spans. In the zero space the process ends with the empty list.  $\square$

**Example 10.8.15** (One replacement by hand). Insert  $u = (2, 1)$  into the spanning list  $e_1, e_2$  of  $\mathbb{R}^2$ . Since  $u = 2e_1 + e_2$ , the nonzero coefficient 2 allows us to solve

$$e_1 = (u - e_2)/2.$$

Thus  $u, e_2$  still produces both old generators, and every new generator was already in the old span. The spans coincide. For example

$$(3, 4) = (3/2)u + (5/2)e_2.$$

In the general proof we must remove an old generator, never one of the independent vectors already inserted.

**Theorem 10.8.16** (Replacement and exchange). If  $(u_1, \dots, u_m)$  is independent and  $(v_1, \dots, v_n)$  spans  $V$ , then  $m \leq n$ , and  $m$  members of the spanning list can be replaced by the  $u_i$  while preserving span. In particular, all finite bases of  $V$  have the same size.

*Proof.* Insert independent vectors one at a time, solving each time for a still-unused generator. The induction invariant is that the current list spans  $V$  and retains every previously inserted  $u_i$ . After inserting  $u_1, \dots, u_{k-1}$ , write  $u_k$  in the current spanning list. Some coefficient of a remaining  $v_j$  must be nonzero, or  $u_k$  would lie in the span of its predecessors. Solving for that  $v_j$  lets us replace it by  $u_k$  while preserving span. There must be a remaining generator at every step, so  $m \leq n$ . Applying this inequality in both directions to two bases proves equality. More explicitly, at step  $k$  write

$$u_k = \sum_{i < k} a_i u_i + \sum_{j \in J} b_j v_j.$$

If all  $b_j$  vanished this would be a dependence of the  $u$ -list. Choose  $b_{j_0} \neq 0$  and solve

$$v_{j_0} = b_{j_0}^{-1} \left( u_k - \sum_{i < k} a_i u_i - \sum_{j \in J \setminus \{j_0\}} b_j v_j \right).$$

The equation preserves the old span. Since no previously inserted  $u_i$  is removed, the induction can continue. An infinite basis is impossible: it would contain an independent list of length  $n + 1$ .  $\square$

**Definition 10.8.17** (Dimension). The common size of the finite bases of  $V$ , now proved independent of the basis by exchange, is the dimension  $\dim V$ . In particular  $\dim \mathbb{R}^n = n$  and  $\dim P_n = n + 1$ .

**Lemma 10.8.18** (Subspaces and extensions). Every subspace of a finite-dimensional space has a finite basis. Every independent list extends to a basis.

*Proof.* Adjoin vectors outside the current span. The enlarged list remains independent: a relation with nonzero coefficient on the new vector would put it in the old span. Exchange bounds its length by  $\dim V$ , so the process stops with a spanning list. For a subspace start with the empty list and choose all new vectors in the subspace. Any independent set is already finite by exchange, so the extension assertion applies to sets too.  $\square$

**Example 10.8.19** (Extending a basis by finding a missing direction). Start in  $\mathbb{R}^3$  with  $u = (1, 0, 1)$  and  $v = (0, 1, 1)$ . Their span is  $z = x + y$ , so  $e_3$  lies outside it. The formula above shows that  $(u, v, e_3)$  spans; the zero-combination equations  $a = b = a + b + c = 0$  show independence. Extension keeps the original independent vectors. Reduction of  $(u, v, u + v, e_3)$  instead deletes the redundant third vector. Both procedures give a basis, but answer different starting questions.

## 10.9 Finite Direct Sums

Sometimes a vector has several independent kinds of components. A polynomial in  $P_2$  has a constant part and a part with zero constant term. A point in  $\mathbb{R}^3$  has a component on the horizontal plane and one on the vertical axis. We need to distinguish building tuples from adding subspaces inside one space.

### External sums: keep the components in separate positions

To combine a vector of  $V_1$  with a vector of  $V_2$ , we can retain both as a tuple rather than trying to add objects from different spaces. Slot injections put them into one common vector space; projections recover the original data.

**Definition 10.9.1** (External direct sum). For finitely many vector spaces  $V_1, \dots, V_r$ , their external direct sum is

$$V_1 \oplus \cdots \oplus V_r = \{(v_1, \dots, v_r) : v_i \in V_i\},$$

with addition and scalar multiplication performed componentwise. The injection  $\iota_i : V_i \rightarrow \bigoplus_j V_j$  places a vector in slot  $i$  and zeros in the other slots; the projection  $\pi_i : \bigoplus_j V_j \rightarrow V_i$  reads slot  $i$ .

The vector-space axioms hold in each component. Each injection and projection preserves both operations. Every tuple has the decomposition

$$(v_1, \dots, v_r) = \sum_i \iota_i(v_i), \quad \pi_i \iota_j = \begin{cases} \text{id}_{V_i} & i = j, \\ 0 & i \neq j. \end{cases}$$

Here the zero map when  $i \neq j$  has source  $V_j$  and target  $V_i$ .

**Proposition 10.9.2** (Basis of an external sum). If  $\mathcal{B}_i$  is a basis of  $V_i$ , the vectors  $\iota_i(b)$  with  $b \in \mathcal{B}_i$  form a basis of the external sum. Consequently

$$\dim \bigoplus_{i=1}^r V_i = \sum_{i=1}^r \dim V_i.$$

*Proof.* Expand each  $v_i$  in  $\mathcal{B}_i$  and insert these expansions into

$$(v_1, \dots, v_r) = \sum_i \iota_i(v_i).$$

This proves spanning. For a zero relation among the proposed basis vectors, apply  $\pi_i$ . All terms from other slots vanish, leaving a zero relation in  $\mathcal{B}_i$ . Its coefficients are zero. Doing this for each  $i$  proves independence. Counting the basis vectors gives the dimension formula.  $\square$

**Proposition 10.9.3** (Assembling maps from the components). Given maps  $L_i : V_i \rightarrow W$  preserving addition and scalar multiplication, there is a unique such map  $L : \bigoplus_i V_i \rightarrow W$  with  $L\iota_i = L_i$ :

$$L(v_1, \dots, v_r) = \sum_i L_i(v_i).$$

*Proof.* The formula defines a map on tuples and preserves both operations by the corresponding identities for each  $L_i$ . On a tuple supported in slot  $i$  it gives  $L_i(v_i)$ , because  $L_j(0) = 0$ . Any other map with the prescribed restrictions must send

$$\sum_i \iota_i(v_i)$$

to

$$\sum_i L_i(v_i)$$

by additivity. Thus it equals  $L$ . This is the direct-sum construction and uniqueness rule.  $\square$

### Internal sums: uniqueness is an extra requirement

Inside one ambient vector space we can add the component vectors directly. The resulting vector may lose information about which component supplied which part. Directness is exactly the condition that permits recovery of all components from their sum.

**Definition 10.9.4** (Internal sum and directness). For subspaces  $U_i \subseteq V$ , define

$$U_1 + \dots + U_r = \{u_1 + \dots + u_r : u_i \in U_i\}.$$

This sum is direct if each of its vectors has exactly one such representation. We then write  $U_1 \oplus \dots \oplus U_r$  for the internal sum. Writing  $V = \bigoplus_i U_i$  asserts also that the sum equals  $V$ .

The sum is a subspace by combining terms componentwise and applying the subspace criterion. The notation  $\oplus$  asserts uniqueness; it does not make an arbitrary sum direct.

**Theorem 10.9.5** (Tests for directness). The following conditions are equivalent:

(a) The sum of the  $U_i$  is direct.

(b) If  $u_i \in U_i$  and

$$\sum_i u_i = 0,$$

then every  $u_i = 0$ .

(c) For each  $i$ ,

$$U_i \cap \sum_{j \neq i} U_j = \{0\}.$$

In particular  $V = U \oplus W$  exactly when  $V = U + W$  and  $U \cap W = \{0\}$ .

*Proof.* Uniqueness applied to zero gives (b), since the all-zero representation already exists. If (b) holds and

$$\sum_i u_i = \sum_i u'_i,$$

then

$$\sum_i (u_i - u'_i) = 0,$$

so each difference is zero; this proves (a). For (b) $\Rightarrow$ (c), write a point in the intersection as

$$u_i = \sum_{j \neq i} u_j.$$

Move the right side to the left and use (b). For (c) $\Rightarrow$ (b), a zero sum gives

$$u_i = - \sum_{j \neq i} u_j$$

in that intersection, so  $u_i = 0$  for each  $i$ . With two subspaces, (c) is just the trivial-intersection condition.  $\square$

**Example 10.9.6** (A plane and a complementary line). Let  $U = \{(x, y, z) : z = x + y\}$  and  $W = \text{span}(0, 0, 1)$  in  $\mathbb{R}^3$ . The defining equation is preserved by linear combinations, so  $U$  is a subspace;  $W$  is a subspace by its span description. For every vector,

$$(x, y, z) = (x, y, x + y) + (0, 0, z - x - y).$$

The first term satisfies the equation of  $U$ , and the second is a multiple of  $(0, 0, 1)$ . This proves existence. For uniqueness, any decomposition  $(x, y, z) = (a, b, a + b) + (0, 0, c)$  forces  $a = x$ ,  $b = y$ , and  $c = z - x - y$ . Thus  $\mathbb{R}^3 = U \oplus W$ .

The component maps are the linear maps

$$P_U(x, y, z) = (x, y, x + y), \quad P_W(x, y, z) = (0, 0, z - x - y).$$

Their sum is the identity. The external summation map and its proposed inverse are

$$\Sigma : U \oplus_{\text{ext}} W \rightarrow \mathbb{R}^3, \quad (u, w) \mapsto u + w, \quad K(v) = (P_U v, P_W v).$$

Both maps are linear by their formulas. We have  $\Sigma K(v) = v$ . Conversely, for  $u = (a, b, a + b)$  and  $w = (0, 0, c)$ ,

$$K \Sigma(u, w) = K(a, b, a + b + c) = ((a, b, a + b), (0, 0, c)) = (u, w).$$

These are the two inverse identities. Directness means uniqueness of components. It imposes no condition on dot products:  $(1, 0, 1) \in U$  has dot product 1 with  $(0, 0, 1) \in W$ .

Figure 10.10 contrasts unique components with a three-subspace obstruction.

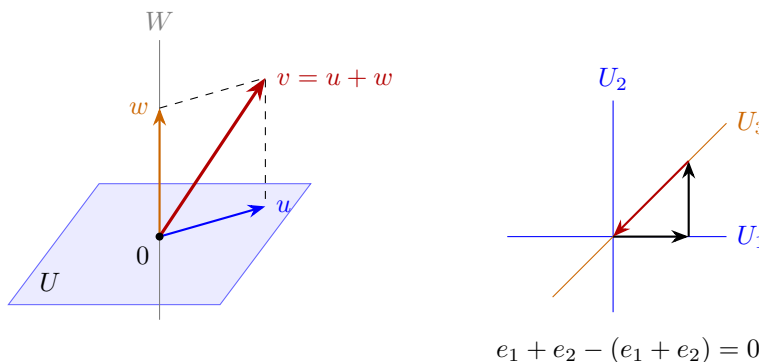


Figure 10.10: Left: a complementary plane and line give unique components; the drawing is oblique and does not assert orthogonality. Right: three lines can meet pairwise only at zero while admitting a nonzero relation.

**Example 10.9.7** (Pairwise intersections do not suffice). In  $\mathbb{R}^2$  let  $U_1 = \text{span}(e_1)$ ,  $U_2 = \text{span}(e_2)$ , and  $U_3 = \text{span}(e_1 + e_2)$ . Any two meet only at zero. Nevertheless  $e_1 + e_2 - (e_1 + e_2) = 0$  is a nonzero relation with one term in each space. Thus the sum of three subspaces is not direct. The correct test compares one subspace with the sum of all the others, not with each separately.

The summation map connects the two constructions:

$$\Sigma : U_1 \oplus_{\text{ext}} \cdots \oplus_{\text{ext}} U_r \rightarrow V, \quad \Sigma(u_1, \dots, u_r) = \sum_i u_i.$$

It preserves addition and scalar multiplication, reaches exactly the internal sum, and is injective exactly when the representation is unique. When the sum is all of  $V$  and direct, its reverse map sends a vector to its unique component tuple. Summing that tuple returns the vector, and decomposing a sum returns the original tuple. Thus both composites are identities.

**Proposition 10.9.8** (Dimension of a sum). For finite-dimensional subspaces,

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

*Proof.* Choose a basis  $(z_1, \dots, z_k)$  of  $U \cap W$ , extend it by  $u_1, \dots, u_a$  to a basis of  $U$ , and by  $w_1, \dots, w_b$  to a basis of  $W$ . The combined list  $(z_i, u_j, w_\ell)$  spans  $U + W$ . If

$$\sum c_i z_i + \sum d_j u_j + \sum e_\ell w_\ell = 0,$$

then

$$\sum d_j u_j = -\sum c_i z_i - \sum e_\ell w_\ell$$

lies in  $U \cap W$ . Express it in the  $z_i$  and use independence of the basis of  $U$  to get all  $d_j = 0$ . The remaining relation is in the basis of  $W$ , so all  $c_i, e_\ell$  vanish. The combined list is a basis. Its size is  $k + a + b = (k + a) + (k + b) - k$ , proving the formula.  $\square$

## 10.10 Linear Transformations, Kernels, and Images

A linear map preserves the two operations that define a vector space. The map  $(x, y) \mapsto (x, 0)$  projects the plane onto its horizontal axis;  $(x, y, z) \mapsto (x - y, y - z)$  records two differences; and

differentiation  $p \mapsto p'$  maps  $P_2$  to  $P_1$ . Each respects sums and scalar multiples. In contrast a translation  $v \mapsto v + c$  with  $c \neq 0$  cannot be linear, since a linear map sends zero to zero. The map  $x \mapsto x^2$  sends zero to zero but fails additivity:  $(1 + 1)^2 \neq 1^2 + 1^2$ . Thus the zero test is a useful obstruction, not a sufficient test.

**Definition 10.10.1** (Linear transformation). A map  $T : V \rightarrow W$  is linear if

$$\begin{aligned} T(u + v) &= T(u) + T(v) && \text{(addition),} \\ T(av) &= aT(v) && \text{(scalar multiplication).} \end{aligned}$$

Equivalently  $T(au + bv) = aT(u) + bT(v)$  for all vectors and scalars.

Indeed the two separate rules give the combined one; in the reverse direction take  $a = b = 1$  for addition and  $b = 0$  for scalar multiplication. For a linear map  $T(0) = T(0 + 0) = T(0) + T(0)$ , so subtraction gives  $T(0) = 0$ . Also  $0 = T(v + (-v)) = T(v) + T(-v)$ , giving  $T(-v) = -T(v)$ .

**Definition 10.10.2** (Linear-map notation). For real vector spaces  $V, W$ , write

$$\mathcal{L}(V, W) := \{T : V \rightarrow W : T \text{ is linear}\}.$$

When source and target agree, our standing convention is

$$\boxed{\mathcal{L}(V) := \mathcal{L}(V, V).}$$

Thus endomorphisms of  $V$  belong to  $\mathcal{L}(V)$ , whereas  $\mathcal{L}(V, W)$  retains both spaces when they differ.

**Definition 10.10.3** (Affine map). Let  $V, W$  be real vector spaces. A map  $F : V \rightarrow W$  is **affine** if

$$F(v) = L(v) + b$$

for some  $L \in \mathcal{L}(V, W)$  and  $b \in W$ . The map  $L$  is the **linear part** of  $F$ , and  $b = F(0)$  is its translation part. For all  $v, w \in V$ ,

$$F(v) - F(w) = L(v - w).$$

Consequently an affine map is linear exactly when  $F(0) = 0$ , equivalently when  $b = 0$ .

For a fixed affine map  $F(x) = Lx + b$ , choose  $x_0 \in V$  and put  $y_0 = F(x_0)$ . Then

$$F(x) = y_0 + L(x - x_0),$$

because  $y_0 - Lx_0 = b$ . Compositions of affine maps are affine. If  $F(x) = Ax + b$  and  $A$  is invertible, then  $F$  is an affine bijection and

$$F^{-1}(y) = A^{-1}(y - b).$$

**Theorem 10.10.4** (Constructing a linear map from basis values). Given a basis  $(v_1, \dots, v_n)$  of  $V$  and arbitrary  $w_1, \dots, w_n \in W$ , there exists exactly one linear map  $T : V \rightarrow W$  with  $T(v_i) = w_i$ .

*Proof.* For the unique coordinate expansion

$$v = \sum_i x_i v_i,$$

define



$$T(v) = \sum_i x_i w_i.$$

Uniqueness of the input coordinates makes this a function; the  $w_i$  need not be independent. Coordinates of  $au + bv$  are the corresponding combinations of the coordinates of  $u, v$ , so distributing the finite output sum proves linearity. Substitution of a basis vector gives  $T(v_i) = w_i$ . If  $L$  is any other linear map with those basis values, then

$$L(v) = \sum_i x_i L(v_i) = \sum_i x_i w_i = T(v)$$

for every  $v$ . This proves uniqueness after existence has been established.  $\square$

**Definition 10.10.5** (Kernel). The kernel  $\ker T = \{v \in V : T(v) = 0\}$  is a subset of the source. It records the inputs whose information is lost.

**Definition 10.10.6** (Image). The image  $\operatorname{im} T = T(V) = \{T(v) : v \in V\}$  is a subset of the target. It records the outputs that can be reached.

Both are subspaces. The kernel contains zero, and if  $T(u) = T(v) = 0$ , then  $T(au + bv) = 0$ . The image contains zero, and a combination of image elements  $aT(u) + bT(v)$  equals  $T(au + bv)$ , another image element.

**Proposition 10.10.7** (Injectivity and surjectivity). A linear map is injective exactly when  $\ker T = \{0\}$ . It is surjective exactly when  $\operatorname{im} T = W$ .

*Proof.* If  $T$  is injective,  $T(v) = 0 = T(0)$  forces  $v = 0$ . Conversely  $T(u) = T(v)$  gives  $T(u - v) = 0$ ; a zero kernel forces  $u - v = 0$ . The image equality is the assertion that every element of the target has a preimage.  $\square$

**Definition 10.10.8** (Rank and nullity). For a finite-dimensional source, the nullity is  $\dim \ker T$  and the rank is  $\dim \operatorname{im} T$ . The image is finite dimensional because the images of a finite source basis span it.

**Example 10.10.9** (A complete kernel and image calculation). For  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ ,  $T(x, y, z) = (x - y, y - z)$ , the kernel equations are  $x - y = 0$  and  $y - z = 0$ . Their solutions are  $(t, t, t)$ , so a kernel basis is  $(1, 1, 1)$  and the nullity is 1. Given any target  $(a, b)$ , the source vector  $(a, 0, -b)$  maps to it. Hence the image is all  $\mathbb{R}^2$ , with basis  $(1, 0), (0, 1)$  and rank 2. Numerically  $3 = 1 + 2$ , anticipating rank-nullity. The map is surjective but not injective.

**Example 10.10.10** (Differentiation loses one coefficient). For  $D : P_2 \rightarrow P_1$ ,  $D(a + bt + ct^2) = b + 2ct$ . The kernel consists of constants, with basis  $(1)$ , and every target  $u + vt$  is the derivative of

$$ut + (v/2)t^2.$$

Thus  $D$  is surjective but not injective. Its nullity is 1, its rank is 2, and their sum is 3. A map is injective when each output has at most one preimage, surjective when every target has a preimage, and bijective when both hold.

**Theorem 10.10.11** (Rank–nullity). For a linear map  $T : V \rightarrow W$  with  $V$  finite dimensional,  $\dim V = \text{nullity } T + \text{rank } T$ .

*Proof.* Choose a basis  $(u_1, \dots, u_k)$  of  $\ker T$  and extend it to a basis  $(u_1, \dots, u_k, v_1, \dots, v_r)$  of  $V$  using Lemma 10.8.18. We prove that  $T(v_1), \dots, T(v_r)$  is a basis of the image in two steps.

*Spanning.* For  $y \in \text{im } T$ , choose  $x \in V$  with  $T(x) = y$ . Write

$$x = \sum_i b_i u_i + \sum_j a_j v_j.$$

Applying  $T$  gives

$$y = \sum_j a_j T(v_j),$$

because each  $T(u_i) = 0$ .

*Independence.* Suppose

$$\sum_j a_j T(v_j) = 0.$$

Then

$$\sum_j a_j v_j \in \ker T,$$

so

$$\sum_j a_j v_j = \sum_i b_i u_i$$

for some  $b_i$ . Consequently

$$\sum_j a_j v_j - \sum_i b_i u_i = 0.$$

Independence of the full source basis forces every  $a_j$  and  $b_i$  to be zero. The image basis therefore has  $r$  elements. Since  $k + r$  is the size of the source basis,  $\dim V = k + r = \dim \ker T + \dim \text{im } T$ .  $\square$

**Corollary 10.10.12** (Linear bijectivity). For  $\dim V = \dim W < \infty$ , a linear  $T : V \rightarrow W$  is injective if and only if it is surjective, hence if and only if it is bijective.

*Proof.* Rank–nullity makes zero nullity equivalent to  $\text{rank } T = \dim W$ . A subspace of  $W$  of this dimension is  $W$ , since its basis is already an independent list of maximal possible size.  $\square$

**Definition 10.10.13** (Isomorphism and endomorphism). A bijective linear map is an isomorphism. A linear map from  $V$  to itself is an endomorphism; it need not be bijective.

**Proposition 10.10.14** (The inverse is linear). If  $T : V \rightarrow W$  is a bijective linear map, its inverse  $S : W \rightarrow V$  is linear.

*Proof.* For  $y, z \in W$  and scalars  $a, b$ , compute

$$T(S(ay + bz)) = ay + bz, \quad T(aS(y) + bS(z)) = aT(S(y)) + bT(S(z)) = ay + bz.$$

Injectivity of  $T$  gives  $S(ay + bz) = aS(y) + bS(z)$ . This method proves equality of two vectors by applying an injective map to both of them.  $\square$

The coordinate and synthesis maps of [Proposition 10.8.13](#) are therefore an explicitly constructed isomorphism. For the summation map of [Section 10.9](#), its image is the internal sum and its kernel is exactly the tuples  $(u_1, \dots, u_r)$  with

$$\sum_i u_i = 0.$$

These are the linear relations among the components. Thus a direct internal sum is equivalent to injectivity of that map; if the sum is  $V$ , it is an isomorphism.

An isomorphism gives a reversible translation between vector spaces. The coordinate map  $p \mapsto [p]_{(1,t,t^2)}$  is an isomorphism  $P_2 \rightarrow \mathbb{R}^3$ , although a polynomial and a column are different objects.

## 10.11 Matrices as Coordinate Representations

**Example 10.11.1** (A square table from basis images). Take  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(x, y) = (2x - y, x + 3y)$ , with source and target bases both  $((1, 0), (0, 1))$ . The basis images  $T(1, 0) = (2, 1)$  and  $T(0, 1) = (-1, 3)$  have target columns

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

Assemble  $A = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$ . Directly  $T(3, 2) = (4, 9)$ ; the column calculation

$$3 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$$

gives the same result.

Before using indices, consider differentiation  $D : P_2 \rightarrow P_1$  in  $\mathcal{B} = (1, t, t^2)$  and  $\mathcal{C} = (1, t)$ . Its three basis images are  $0, 1, 2t$ , whose target columns are  $(0, 0)^T$ ,  $(1, 0)^T$ , and  $(0, 2)^T$ . Store them in that order:

$$[D]_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

For  $p = 2 + 3t - t^2$ , direct differentiation gives  $3 - 2t$ . Its output column is

$$\begin{pmatrix} 3 \\ -2 \end{pmatrix},$$

and the table acting on  $(2, 3, -1)^T$  gives

$$2 \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}.$$

There are two rows because outputs need two coordinates, and three columns because there are three input basis vectors. The matrix of  $T(x, y, z) = (x - y, y - z)$  in standard bases is similarly  $\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$ . Here the source basis is  $(e_1, e_2, e_3)$  and the target basis is  $(e_1, e_2)$ . The basis images are  $(1, 0), (-1, 1), (0, -1)$ , already in target coordinates. For  $(2, -1, 3)$ , direct evaluation gives  $(3, -4)$ ; the column sum  $2(1, 0) - (-1, 1) + 3(0, -1) = (3, -4)$  agrees. These examples suggest

the following general rule. Choose ordered bases  $\mathcal{B} = (v_1, \dots, v_n)$  of  $V$  and  $\mathcal{C} = (w_1, \dots, w_m)$  of  $W$ . To know  $T$  it suffices to know  $T(v_j)$ : linearity then determines

$$T\left(\sum_j x_j v_j\right).$$

Conversely, any prescribed  $m$ -dimensional vectors  $T(v_j)$  define one linear map by this formula, because the coordinates  $x_j$  are unique. Write each basis image in the target basis:

$$T(v_j) = \sum_{i=1}^m a_{ij} w_i.$$

Here  $j$  labels the input basis vector, hence the column. For that fixed  $j$ ,  $i$  runs through the target basis and labels the row containing its coefficient. Thus  $a_{ij}$  means “coefficient of  $w_i$  in  $T(v_j)$ ,” not the coefficient of the  $j$ th output in the  $i$ th input. An  $m$ -by- $n$  **matrix** is a rectangular array of numbers; their space is denoted  $M_{m,n}(\mathbb{R})$ , or  $M_n(\mathbb{R})$  when square. Define the representing matrix by its columns:

$$A = [T]_{\mathcal{C} \leftarrow \mathcal{B}} = \left( \begin{array}{c|ccc|c} & & & & \\ & [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} & \\ & & & & \end{array} \right).$$

The notation  $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$  says: input coordinates are read in  $\mathcal{B}$ ; output coordinates are written in  $\mathcal{C}$ . For

$$v = \sum_j x_j v_j,$$

apply linearity first and then collect coefficients:

$$T(v) = \sum_j x_j T(v_j) = \sum_i \left( \sum_j a_{ij} x_j \right) w_i, \quad [T(v)]_{\mathcal{C}} = \sum_j x_j [T(v_j)]_{\mathcal{C}}.$$

Thus the compatible definition of matrix-vector multiplication is

$$(A\mathbf{x})_i = \sum_j a_{ij} x_j.$$

Matrix-vector multiplication is not an arbitrary convention. It is the unique coordinate rule compatible with linearity and the fact that the columns record the images of basis vectors.

**Example 10.11.2** (A rectangular multiplication, two views). Let  $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$  and  $\mathbf{x} = (2, -1, 3)^T$ . By columns,

$$Ax = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}.$$

By rows,  $(Ax)_1 = 1(2) + (-1)(-1) + 0(3) = 3$  and  $(Ax)_2 = 0(2) + 1(-1) + (-1)(3) = -4$ . The column view combines basis images; the row view gives one formula for each output coordinate. They are the same calculation grouped differently.

**Example 10.11.3** (Recovering a map with a nonstandard target basis). Suppose a map  $H : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  has matrix

$$[H]_{\mathcal{C} \leftarrow \text{std}} = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}, \quad \mathcal{C} = ((1, 1), (1, -1)).$$

The columns are target coordinates, not the standard output vectors. They say  $H(e_1) = (1, 1) - (1, -1) = (0, 2)$  and  $H(e_2) = 2(1, 1) + (1, -1) = (3, 1)$ . The basis-value theorem therefore constructs  $H(x, y) = (3y, 2x + y)$ . For input  $(2, -1)$ , matrix multiplication gives the  $\mathcal{C}$ -column  $(0, -3)^\top$ . Synthesizing the output gives  $0(1, 1) - 3(1, -1) = (-3, 3)$ , agreeing with the intrinsic calculation.

**Example 10.11.4** (Coordinates through a nonstandard input basis). Let  $\mathcal{B} = ((1, 1), (1, -1))$  and consider the map  $H(x, y) = (x + y, y, 2x)$ . The two basis images are  $(2, 1, 2)$  and  $(0, -1, 2)$ , so

$$[H]_{\text{std} \leftarrow \mathcal{B}} = \begin{pmatrix} 2 & 0 \\ 1 & -1 \\ 2 & 2 \end{pmatrix}.$$

The vector  $(3, 1)$  has  $\mathcal{B}$ -column  $(2, 1)^\top$ , not  $(3, 1)^\top$ . Multiplying the correct column gives  $2(2, 1, 2) + (0, -1, 2) = (4, 1, 6) = H(3, 1)$ .

## 10.12 Matrix Multiplication Comes from Composition

First compose two concrete maps. For  $T(x, y) = (x + y, y)$  and  $S(u, v) = (2u - v, u)$ , substitution gives  $S(T(x, y)) = (2x + y, x + y)$ . The composite sends  $e_1$  to  $(2, 1)$  and  $e_2$  to  $(1, 1)$ , so its table is  $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ . The tables of  $T$  and  $S$  are respectively  $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $B = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$ . Applying  $B$  to each column of  $A$  produces exactly those composite columns. Now let  $S : W \rightarrow X$  have matrix  $B = [S]_{\mathcal{D} \leftarrow \mathcal{C}}$ . The middle coordinates must match, since

$$[S(T(v))]_{\mathcal{D}} = B[T(v)]_{\mathcal{C}} = B(A[v]_{\mathcal{B}}).$$

The  $j$ th column of the composite is  $B$  applied to the  $j$ th column of  $A$ , and therefore has entries

$$\sum_k b_{ik} a_{kj}.$$

This forces:

**Proposition 10.12.1** (Coordinate rules and composition). For the preceding choices,

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}, \quad [S \circ T]_{\mathcal{D} \leftarrow \mathcal{B}} = BA, \quad (BA)_{ij} = \sum_{k=1}^m b_{ik} a_{kj}.$$

The types can be read from

$$\mathbb{R}^n \xrightarrow{A \ (m \times n)} \mathbb{R}^m \xrightarrow{B \ (\ell \times m)} \mathbb{R}^\ell.$$

Here  $A$  is  $m$ -by- $n$ ,  $B$  is  $\ell$ -by- $m$ , and  $BA$  is  $\ell$ -by- $n$ .

*Proof.* The preceding derivation computes every output coordinate and every basis image of the composite, which uniquely determine its matrix.  $\square$

Matrix multiplication is the unique multiplication rule compatible with composition of linear maps. The number of columns of  $B$  must equal the number of rows of  $A$ , the dimension of the intermediate space. “First  $A$ , then  $B$ ” gives  $BA$ , just as  $S \circ T$  applies  $T$  first. Associativity follows from associativity of composition and uniqueness of representing matrices. The identity map has the matrix  $I_n$  with diagonal entries 1 and all other entries zero. An inverse matrix represents the inverse linear map and satisfies  $A^{-1}A = AA^{-1} = I_n$ .

**Example 10.12.2** (Why entrywise multiplication fails). Recall the opening maps  $T(x, y) = (x + y, y)$  and  $S(u, v) = (2u - v, u)$ . Their composite sends  $e_2$  to  $(1, 1)$ . Multiplying the entries in their two tables position by position would instead give

$$\begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix} e_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

This fails the basis-image test. Matrix multiplication must sum over the intermediate coordinate; multiplying matching entries loses that information.

**Example 10.12.3** (A rectangular composition). Let  $T(x, y, z) = (x - y, y - z)$  and  $S(u, v) = (u + v, 2u)$ . Then

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 0 & -1 \\ 2 & -2 & 0 \end{pmatrix}.$$

For instance entry  $(2, 2)$  is  $2(-1) + 0(1) = -2$ : the second input basis vector goes to  $(-1, 1)$ , and the second coordinate of its image under  $S$  is  $-2$ . Directly,  $S(T(x, y, z)) = (x - z, 2x - 2y)$ .  $AB$  is not defined:  $S$  produces two coordinates, while  $T$  needs three.

## Solving Linear Systems and Row Reduction

For  $A \in M_{m \times n}(\mathbb{R})$ , the equation  $Ax = b$  asks for an input with prescribed output. Its augmented matrix  $[A \mid b]$  records the coefficients and right-hand side together. Three row operations preserve the solution set: interchange two equations, multiply an equation by a nonzero scalar, and add a scalar multiple of another equation to an equation. Each is reversible, so equivalence holds in both directions.

A matrix is in *row echelon form* if zero rows are last, each nonzero row begins farther to the right than the row above, and entries below each leading entry are zero. A leading entry is a *pivot*. In *reduced row echelon form*, pivots are 1 and are the only nonzero entries in their columns. Gaussian elimination makes echelon form by choosing a nonzero pivot, interchanging rows if needed, and eliminating below it, then continuing on the remaining rows and columns. Back substitution solves the resulting triangular equations. Gauss–Jordan elimination also scales pivots to 1 and eliminates above them.

**Example 10.12.4** (A three-variable system). Solve  $x + y + z = 6$ ,  $2x + 3y + z = 11$ ,  $x + 2y + 3z = 14$ . Apply  $R_2 \leftarrow R_2 - 2R_1$ ,  $R_3 \leftarrow R_3 - R_1$ , then  $R_3 \leftarrow R_3 - R_2$ :

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 11 \\ 1 & 2 & 3 & 14 \end{array} \right] \longrightarrow \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & 2 & 8 \end{array} \right] \longrightarrow \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 3 & 9 \end{array} \right].$$

Back substitution gives  $z = 3$ ,  $y = -1 + z = 2$ ,  $x = 6 - y - z = 1$ . To continue to reduced form, divide row 3 by 3, add row 3 to row 2, subtract row 3 from row 1, and finally subtract row 2 from row 1. The result is  $[I_3 \mid (1, 2, 3)^T]$ . Substitution in the three original equations verifies the answer.

An equation  $0 = c$  with  $c \neq 0$  signals inconsistency. Otherwise columns of the coefficient matrix without pivots correspond to free variables; pivot variables are determined by them. Thus a consistent system has a unique solution exactly when every variable has a pivot. For example,

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 4 \end{array} \right] \longrightarrow \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

gives  $(x, y, z) = (2 - s - t, s, t)$  for arbitrary  $s, t \in \mathbb{R}$ . Changing the last right-hand entry from 4 to 5 instead produces  $[0 \ 0 \ 0 \mid 1]$  and no solution. The free parameters describe the kernel directions already understood through rank-nullity.

**Elementary matrices and inversion.** Applying one row operation to  $I_m$  produces an *elementary matrix*  $E$ . The multiplication rule shows that  $EA$  performs that same row operation on  $A$ : each new row is the specified linear combination of old rows. Reversing the operation gives  $E^{-1}$ . Thus a sequence of row operations is multiplication by  $E_s \cdots E_1$ , in that order. For a square matrix, reducing the left block of  $[A \mid I_n]$  to  $I_n$  produces  $[I_n \mid E_s \cdots E_1]$ . The identity  $(E_s \cdots E_1)A = I_n$  shows that the right block is  $A^{-1}$ , since the square map is injective and hence invertible. If a pivot is missing, the homogeneous system has a free variable and  $A$  is not invertible. For example, successively subtract twice row 1 from row 2, subtract row 2 from row 1:

$$\left[ \begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array} \right] \longrightarrow \left[ \begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right] \longrightarrow \left[ \begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -2 & 1 \end{array} \right].$$

Hence the inverse is  $\begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}$ . These algorithms compute with coordinate tables; the structural results about linear maps explain why their outputs have the claimed meanings.

### 10.13 Changing Coordinates while Keeping the Map

Before introducing a conversion matrix, solve a conversion. For  $\mathcal{B} = ((1, 1), (1, -1))$  and the standard basis  $\mathcal{C}$ , a column  $(a, b)^T$  in  $\mathcal{B}$  describes  $(a + b, a - b)$ . Thus the conversion is  $(a, b) \mapsto (a + b, a - b)$ , with inverse

$$(x, y) \mapsto ((x + y)/2, (x - y)/2).$$

Both preserve linear combinations. For example  $(2, 1)^T$  converts to  $(3, 1)^T$ . Coordinate conversion is itself a linear map on columns. Figure 10.11 shows the two descriptions of this vector.

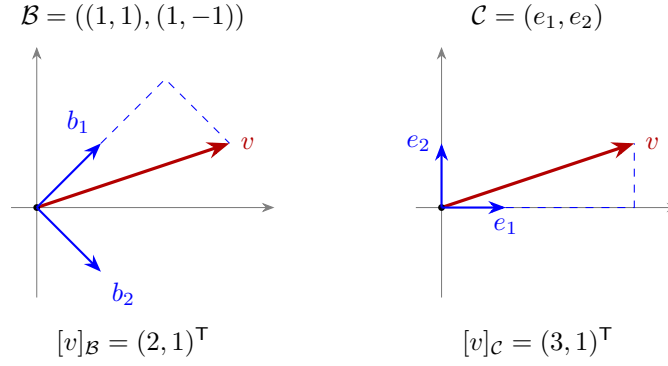


Figure 10.11: One vector, two coordinate columns. The dashed routes build  $v = 2b_1 + b_2 = 3e_1 + e_2$ ; the red vector stays fixed.

The vector  $v$  is one object. The columns  $[v]_{\mathcal{B}}$  and  $[v]_{\mathcal{C}}$  are two descriptions of it. To translate between descriptions, the intrinsic operation is simply  $v \mapsto v$ ; only its coordinate table changes. An identity map can therefore have a nonidentity matrix when its source and target bases differ. For two bases of the same space, define

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = [\text{id}_V]_{\mathcal{C} \leftarrow \mathcal{B}}, \quad [v]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}}[v]_{\mathcal{B}}.$$

The columns are the old basis vectors expressed in the new basis. Composing the identity coordinate changes in both directions gives  $P_{\mathcal{B} \leftarrow \mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}}^{-1}$ . For  $T : V \rightarrow W$ , change the input coordinates, apply the old matrix, then change the output coordinates. For a fixed input  $v$ , follow

$$[v]_{\mathcal{B}'} \mapsto [v]_{\mathcal{B}} \mapsto [Tv]_{\mathcal{C}} \mapsto [Tv]_{\mathcal{C}'}.$$

The three arrows apply  $P_{\mathcal{B} \leftarrow \mathcal{B}'}$ , then  $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ , then  $P_{\mathcal{C}' \leftarrow \mathcal{C}}$ . Their matrices multiply in the reverse written order because the rightmost matrix acts first. The linear map does not change; only its coordinate table changes. The composition rule yields

$$[T]_{\mathcal{C}' \leftarrow \mathcal{B}'} = P_{\mathcal{C}' \leftarrow \mathcal{C}}[T]_{\mathcal{C} \leftarrow \mathcal{B}}P_{\mathcal{B} \leftarrow \mathcal{B}'}.$$

For an endomorphism using the same basis on both sides, abbreviate its matrix by  $[T]_{\mathcal{B}}$ . Setting  $P = P_{\mathcal{B} \leftarrow \mathcal{B}'}$  gives  $[T]_{\mathcal{B}'} = P^{-1}[T]_{\mathcal{B}}P$ .

**Example 10.13.1** (A complete change of basis). Take  $T(x, y) = (2x + y, y)$ ,  $\mathcal{B} = (e_1, e_2)$ , and  $\mathcal{B}' = ((1, 1), (0, 1))$ . Then

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \quad P^{-1}AP = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}.$$

To check the columns directly,  $T(1, 1) = (3, 1) = 3(1, 1) - 2(0, 1)$ , and  $T(0, 1) = (1, 1)$ . For  $v = (2, 3)$  the old coordinates are  $(2, 3)$ , the new ones are  $P^{-1}(2, 3)^T = (2, 1)^T$ , and the new matrix gives  $(7, -4)^T$ . Converting back gives  $P(7, -4)^T = (7, 3)^T = T(v)$ .

The linear map is intrinsic; its matrix depends on the coordinate systems. In Chapter 11,  $Df(a)$  will be the derivative itself and the Jacobian its matrix in standard coordinates.



$$\begin{array}{ccccc}
\mathbb{R}^n & \xrightarrow{P_{\mathcal{B} \leftarrow \mathcal{B}'}} & \mathbb{R}^n & \xrightarrow{[T]_{\mathcal{C} \leftarrow \mathcal{B}}} & \mathbb{R}^m \\
\downarrow [T]_{\mathcal{C}' \leftarrow \mathcal{B}'} & & & & \downarrow P_{\mathcal{C}' \leftarrow \mathcal{C}} \\
\mathbb{R}^m & \xlongequal{\hspace{1.5cm}} & & & \mathbb{R}^m
\end{array}$$

Read the top route in order: translate the input, perform the old calculation, translate the output. The equality of routes derives the transformation law. The expression  $P^{-1}AP$  is called a similarity transformation here only in this coordinate sense.

**Example 10.13.2** (A second plane calculation, with different bases on each side). Let  $T(x, y) = (x - y, 2y)$ ,  $\mathcal{B} = ((1, 1), (0, 1))$ , and  $\mathcal{C} = ((1, 0), (1, 1))$ . The standard-coordinate matrices of these bases and the map are

$$P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}.$$

Thus  $[T]_{\mathcal{C} \leftarrow \mathcal{B}} = Q^{-1}AP = \begin{pmatrix} -2 & -3 \\ 2 & 2 \end{pmatrix}$ . Directly  $T(1, 1) = (0, 2) = -2(1, 0) + 2(1, 1)$  and  $T(0, 1) = (-1, 2) = -3(1, 0) + 2(1, 1)$ , verifying both columns. For the identity map the transition is  $P_{\mathcal{C} \leftarrow \mathcal{B}} = Q^{-1}P = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ , whose inverse is  $\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$ .

**Example 10.13.3** (Three-dimensional coordinates). Take  $T(x, y, z) = (x + y, y + z, z)$  and  $\mathcal{B}' = ((1, 0, 0), (1, 1, 0), (0, 0, 1))$ . Relative to the standard basis,

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplication gives

$$P^{-1}AP = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

The images of the new basis vectors are  $(1, 0, 0), (2, 1, 0), (0, 1, 1)$ . Writing

$$(x, y, z) = (x - y)(1, 0, 0) + y(1, 1, 0) + z(0, 0, 1)$$

gives their columns  $(1, 0, 0), (1, 1, 0), (-1, 1, 1)$ , exactly as above.

Matrices have entrywise vector-space operations:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = ca_{ij}.$$

The zero matrix and entrywise negatives satisfy the vector-space axioms because real numbers do. For linear maps  $S, T : V \rightarrow W$ , the  $j$ th column of the matrix of  $aS + bT$  records  $(aS + bT)(v_j) = aS(v_j) + bT(v_j)$ . Linearity of target coordinates gives

$$[aS + bT]_{\mathcal{C} \leftarrow \mathcal{B}} = a[S]_{\mathcal{C} \leftarrow \mathcal{B}} + b[T]_{\mathcal{C} \leftarrow \mathcal{B}}.$$

This proves the linearity in the next identification.

**Proposition 10.13.4** (The vector space of linear maps). The set  $\mathcal{L}(V, W)$  is a vector space under

$$(S + T)(v) = S(v) + T(v), \quad (cT)(v) = cT(v).$$

If  $\dim V = n$  and  $\dim W = m$ , then

$$\dim \mathcal{L}(V, W) = mn.$$

*Proof.* The vector-space axioms hold pointwise. Choosing bases identifies  $\mathcal{L}(V, W)$  linearly and bijectively with  $M_{m,n}(\mathbb{R})$ ; the matrices with one entry 1 and all others 0 form a basis.  $\square$

The scalar-target case  $\mathcal{L}(V, \mathbb{R})$  receives its own terminology and interpretation in [Definition 10.16.1](#), after the norm estimates. In particular, the standard basis identifies  $\mathcal{L}(\mathbb{R}^n) \cong M_n(\mathbb{R})$ , although the notation on the left keeps the underlying operators visible.

## 10.14 Inner Products and Norms

The dot product measures how two displacements interact. On polynomials, another meaningful interaction is the average product over an interval. Both lead to a length by pairing a vector with itself. Near the start of this chapter,  $\|\mathbf{x}\|$  was fixed as the standard Euclidean norm on  $\mathbb{R}^n$ . We now generalize the same notation: on an arbitrary normed space it denotes the norm supplied as part of that space's structure.

**Definition 10.14.1** (Real inner product). An inner product is a map  $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$  satisfying, for all vectors and real scalars,

$$\begin{aligned} \langle au + bv, w \rangle &= a\langle u, w \rangle + b\langle v, w \rangle && \text{(linearity),} \\ \langle u, v \rangle &= \langle v, u \rangle && \text{(symmetry),} \\ \langle v, v \rangle &\geq 0, \quad \langle v, v \rangle = 0 \iff v = 0 && \text{(positive definiteness).} \end{aligned}$$

Symmetry also gives linearity in the second variable:  $\langle u, av + bw \rangle = a\langle u, v \rangle + b\langle u, w \rangle$ . Set  $\|v\| = \sqrt{\langle v, v \rangle}$ ; the next theorem proves its length laws.

**Example 10.14.2** (Two inner products). On  $\mathbb{R}^n$  and  $P_2$ , respectively, use

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_i x_i y_i, \quad \langle p, q \rangle = \int_0^1 p(t)q(t) \, dt.$$

Linearity of the integral proves bilinearity, and  $pq = qp$  proves symmetry. If  $p$  is nonzero, it cannot vanish at every point of  $[0, 1]$ ; a polynomial of degree at most two with three distinct roots is zero by successive factorization. At a point where  $p \neq 0$ , continuity supplies a subinterval of positive length on which  $p^2$  is bounded below by a positive constant. Hence

$$\int_0^1 p^2 > 0.$$

For example,

$$\langle 1, t \rangle = \frac{1}{2}, \quad \|t\| = \frac{1}{\sqrt{3}}, \quad \|1 + t\|^2 = \int_0^1 (1 + 2t + t^2) \, dt = \frac{7}{3}.$$

The goal is to show that the induced length satisfies the following three axioms, which also make sense without an inner product.

**Definition 10.14.3** (Norm). A **norm** on a real vector space  $V$  is a function

$$\|\cdot\| : V \longrightarrow [0, \infty)$$

such that

$$\begin{aligned} \|v\| = 0 &\iff v = 0 && \text{(definiteness),} \\ \|av\| &= |a|\|v\| && \text{(absolute homogeneity),} \\ \|v + w\| &\leq \|v\| + \|w\| && \text{(triangle inequality).} \end{aligned}$$

A vector space equipped with a norm is a **normed vector space**. The associated metric is

$$d(v, w) = \|v - w\|.$$

**Standard coordinate norms.** For  $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ ,  $n \geq 1$ , the standard formulas are

$$\begin{aligned} \|\mathbf{x}\|_1 &= \sum_i |x_i| && \text{(sum norm or 1-norm),} \\ \|\mathbf{x}\|_2 &= \left( \sum_i x_i^2 \right)^{1/2} && \text{(Euclidean norm or 2-norm),} \\ \|\mathbf{x}\|_3 &= \left( \sum_i |x_i|^3 \right)^{1/3}, \quad \|\mathbf{x}\|_4 = \left( \sum_i |x_i|^4 \right)^{1/4}, \\ \|\mathbf{x}\|_p &= \left( \sum_i |x_i|^p \right)^{1/p} && (1 \leq p < \infty), \\ \|\mathbf{x}\|_\infty &= \max_i |x_i| && \text{(maximum or sup norm).} \end{aligned}$$

For the 1-norm, summing  $|x_i + y_i| \leq |x_i| + |y_i|$  proves the triangle inequality. For the maximum norm, each  $|x_i + y_i|$  is at most  $\|\mathbf{x}\|_\infty + \|y\|_\infty$ ; take the maximum. For the Euclidean norm, [Theorem 10.1.2](#) already proved the triangle inequality by expanding the square. Definiteness and homogeneity follow directly in all three cases. The general  $p$ -norm triangle inequality is Minkowski's inequality; this standard result is not proved or needed here. Later arguments use only the three norms just verified, or specified abstract norms.

The useful elementary comparisons are

$$\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 \leq \sqrt{n}\|\mathbf{x}\|_2, \quad \|\mathbf{x}\|_2 \leq \sqrt{n}\|\mathbf{x}\|_\infty, \quad |x_i| \leq \|\mathbf{x}\|_2.$$

A coordinate square is at most the sum of squares. Also

$$\sum_i |x_i|^2 \leq \left( \sum_i |x_i| \right)^2,$$

and Cauchy-Schwarz applied to  $(|x_i|)$  and  $(1)$  gives the 1-to-2 comparison. Finally each square is at most  $\|\mathbf{x}\|_\infty^2$ ; summing gives the last norm comparison.

**Remark 10.14.4** (Default norm convention). Recall the convention introduced with Euclidean distance: on  $\mathbb{R}^n$ , unless another norm is explicitly indicated,

$$\|\mathbf{x}\| := \|\mathbf{x}\|_2.$$

This convention governs Chapters 11–13. On an abstract normed space  $V$ ,  $\|v\|$  means its specified norm, not an implicitly Euclidean one. Write  $\|\cdot\|_V$  and  $\|\cdot\|_W$  when two norms must be distinguished. The operator norm introduced next remains  $\|L\|_{\text{op}}$ , or  $\|L\|_{V \rightarrow W}$  when its dependence on those norms matters.

For bounded functions on a nonempty set  $D$ , the notation from Chapter 6 is  $\|f\|_\infty = \sup_{x \in D} |f(x)|$ . This and  $\|\mathbf{x}\|_\infty = \max_i |x_i|$  express the same largest-magnitude idea on different vector spaces; context specifies the space.

**Theorem 10.14.5** (Inner-product Cauchy–Schwarz). For a real inner product,  $|\langle u, v \rangle| \leq \|u\| \|v\|$ . The induced length is a norm.

*Proof.* Use the nonnegative squared length of  $u - tv$  and choose the scalar that removes the linear term. If  $v = 0$  the inequality is immediate. Otherwise set

$$t = \langle u, v \rangle / \langle v, v \rangle.$$

Bilinearity and symmetry give

$$0 \leq \langle u - tv, u - tv \rangle = \|u\|^2 - \frac{\langle u, v \rangle^2}{\|v\|^2}.$$

Multiply by  $\|v\|^2$  and take square roots. Consequently

$$\|u + v\|^2 = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \leq (\|u\| + \|v\|)^2,$$

which proves the triangle inequality. Positive definiteness gives zero length only at zero, and bilinearity gives  $\|av\| = |a|\|v\|$ .  $\square$

Thus an inner product gives a norm, and a norm gives a metric. A general norm need only preserve the length properties needed for estimates; it need not come from an inner product. Norms quantify vector size and approximation errors. Operator norms measure the largest amplification of those errors by a linear map. Compactness will make these bounds finite in the spaces used throughout Part II.

Norms measure vector size and therefore quantify errors in linear approximations. On  $\mathbb{R}^2$ , the unit balls of  $\|\mathbf{x}\|_1 = |x_1| + |x_2|$ ,  $\|\mathbf{x}\|_2$ , and  $\|\mathbf{x}\|_\infty$  are, respectively, a diamond, a circle, and a square. They look different, but in finite dimensions they lead to the same convergence and continuity theory; the next theorem explains why this is important before derivatives are defined.

For a map  $r$  with values in a normed space, the notation

$$r(h) = o(\|h\|) \quad (h \rightarrow 0)$$

means precisely

$$\frac{\|r(h)\|}{\|h\|} \longrightarrow 0.$$

Likewise,  $r(h) = O(\|h\|)$  means that  $\|r(h)\| \leq C\|h\|$  for all sufficiently small  $h$ . This is the norm-valued version of the asymptotic notation introduced in Chapter 5; it will describe approximation errors in Chapter 11.

**Proposition 10.14.6** (Reverse triangle inequality). Every norm satisfies

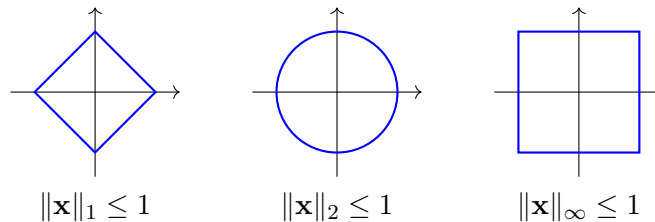
$$|\|v\| - \|w\|| \leq \|v - w\|.$$

Hence its norm map is continuous.

*Proof.* The triangle inequality gives

$$\|v\| \leq \|v - w\| + \|w\|.$$

Interchange  $v, w$  and combine the two inequalities. □



Different unit balls do not mean different convergent sequences. The coordinate comparisons above already show this for the 1-, 2-, and maximum norms. The following proof follows the same pattern for any norm: coordinate upper bound, continuity, compact sphere, positive minimum, then homogeneous rescaling to obtain a lower bound.

**Theorem 10.14.7** (Equivalence of norms). If  $V$  is finite dimensional and

$$\|\cdot\|_a, \quad \|\cdot\|_b$$

are norms on  $V$ , then there are  $c, C > 0$  such that

$$c\|v\|_a \leq \|v\|_b \leq C\|v\|_a$$

for every  $v \in V$ . Thus the two norms have the same open sets, convergent and Cauchy sequences, continuous maps, and compact subsets.

*Proof.* First bound a norm above in coordinates, then obtain a uniform lower bound on the compact Euclidean sphere. If  $V = \{0\}$  any  $c, C > 0$  work. Otherwise choose a basis and identify  $V$  with  $\mathbb{R}^n$ . For any norm on  $\mathbb{R}^n$ ,

$$\|\mathbf{x}\|_b \leq \sum_{j=1}^n |x_j| \|e_j\|_b \leq C \|\mathbf{x}\|_2$$

by Cauchy–Schwarz. Together with [Proposition 10.14.6](#), this gives

$$|\|\mathbf{x}\|_b - \|y\|_b| \leq C \|\mathbf{x} - y\|_2,$$

so  $\mathbf{x} \mapsto \|\mathbf{x}\|_b$  is Euclidean-continuous. It has a positive minimum  $c$  on the Euclidean unit sphere, which is compact by [Theorem 10.6.8](#); [Theorem 10.6.6](#) supplies the attained minimum. It is positive because every sphere point is nonzero. For  $\mathbf{x} \neq 0$ , the vector

$$u = \mathbf{x} / \|\mathbf{x}\|_2$$

is on that sphere, so homogeneity gives  $\|\mathbf{x}\|_b = \|\mathbf{x}\|_2 \|u\|_b \geq c \|\mathbf{x}\|_2$ ; the same inequality holds at zero. Apply these bounds to each given norm, writing  $c_a \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_a \leq C_a \|\mathbf{x}\|_2$  and  $c_b \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_b \leq C_b \|\mathbf{x}\|_2$ . Then

$$\frac{c_b}{C_a} \|\mathbf{x}\|_a \leq \|\mathbf{x}\|_b \leq \frac{C_b}{c_a} \|\mathbf{x}\|_a.$$

Denote the resulting constants by  $c, C$ . Their ball inclusions are

$$B_a(\mathbf{x}, \varepsilon/C) \subseteq B_b(\mathbf{x}, \varepsilon)$$

and  $B_b(\mathbf{x}, c\varepsilon) \subseteq B_a(\mathbf{x}, \varepsilon)$ , so the open sets agree. Applying the same inequalities to  $\mathbf{x}_j - \mathbf{x}$  and  $\mathbf{x}_j - \mathbf{x}_k$  proves the convergence and Cauchy assertions. Continuity and compactness agree because their definitions use only open sets.  $\square$

**Corollary 10.14.8** (Completeness in finite dimensions). Every finite-dimensional normed vector space is complete, and its compact subsets are exactly its closed bounded subsets.

*Proof.* Coordinates reduce the assertion to Euclidean space, where completeness follows coordinatewise from [Theorem 3.4.10](#) and compactness from [Theorem 10.6.8](#); use [Theorem 10.14.7](#).  $\square$

Completeness ensures that Cauchy approximations have limits. Chapter 11 uses it to obtain fixed points and solve nonlinear equations.

## 10.15 Operator Norms and Finite-Dimensional Estimates

A derivative will be a linear map, and its approximation error must be controlled in every direction at once. We first bound the output by a constant times the input norm. The operator norm will then record the smallest such constant and let us measure differences between linear maps.

**Proposition 10.15.1** (Finite-dimensional linear maps are Lipschitz). Every linear map  $L : V \rightarrow W$  between finite-dimensional normed spaces is Lipschitz, hence continuous.

*Proof.* A bound on the finitely many basis images controls every linear combination. If  $V = \{0\}$  there is nothing to prove. Otherwise choose a basis  $e_1, \dots, e_n$  of  $V$ . For

$$v = \sum_i a_i e_i$$

put

$$\|v\|_c = \sum_i |a_i|.$$

Norm equivalence gives  $\|v\|_c \leq K \|v\|_V$ . With  $M = \max_i \|L(e_i)\|_W$ , the triangle inequality gives

$$\|Lv\|_W \leq \sum_i |a_i| \|L(e_i)\|_W \leq M \|v\|_c \leq MK \|v\|_V.$$

Apply this to  $v = x - y$  to obtain  $\|Lx - Ly\|_W \leq MK \|x - y\|_V$ . No target basis is needed. Finite dimensionality enters through the finite basis and the norm-equivalence estimate.  $\square$

The following characterization does not require finite dimensionality; it explains which linear maps admit an operator norm. A linear map between normed spaces is called **bounded** if  $\|Lv\|_W \leq C \|v\|_V$  for some  $C \geq 0$  and every  $v \in V$ .

**Theorem 10.15.2** (Bounded linear maps). For a linear map  $L : V \rightarrow W$  between normed spaces, the following are equivalent:

$$\begin{aligned} &L \text{ is continuous}; \quad L \text{ is continuous at } 0; \\ &\exists C \geq 0 : \|Lv\|_W \leq C\|v\|_V \quad (v \in V); \quad L \text{ is Lipschitz.} \end{aligned}$$

In finite dimensions these conditions hold for every linear map by [Proposition 10.15.1](#).

*Proof.* Continuity implies continuity at zero. If  $L$  is continuous at zero, choose  $\delta > 0$  such that  $\|u\|_V < \delta$  implies  $\|Lu\|_W < 1$ . For  $v \neq 0$ , apply this to

$$u = \delta v / (2\|v\|_V)$$

and use linearity to obtain

$$\|Lv\|_W < \frac{2}{\delta} \|v\|_V.$$

Consequently the uniform weak inequality

$$\|Lv\|_W \leq \frac{2}{\delta} \|v\|_V \quad (v \in V)$$

also includes  $v = 0$ . Such a bound applied to  $v = x - y$  is the Lipschitz estimate  $\|Lx - Ly\|_W \leq C\|x - y\|_V$ , and every Lipschitz map is continuous.  $\square$

**Definition 10.15.3** (Operator norm). For a bounded linear map  $L : V \rightarrow W$ , define

$$\|L\|_{\text{op}} = \sup \{ \|Lv\|_W : \|v\|_V \leq 1 \}.$$

Both norms matter;  $\|L\|_{V \rightarrow W}$  makes their roles visible.

**Theorem 10.15.4** (Characterizations of the operator norm). Let  $V \neq \{0\}$  and let  $L : V \rightarrow W$  be bounded and linear. Then

$$\begin{aligned} \|L\|_{\text{op}} &= \sup_{\|v\|_V \leq 1} \|Lv\|_W = \sup_{\|v\|_V = 1} \|Lv\|_W \\ &= \sup_{v \neq 0} \frac{\|Lv\|_W}{\|v\|_V} = \sup_{x \neq y} \frac{\|Lx - Ly\|_W}{\|x - y\|_V} \\ &= \inf \{ C \geq 0 : \|Lv\|_W \leq C\|v\|_V \text{ for every } v \in V \}. \end{aligned}$$

Thus the operator norm is the optimal Lipschitz constant of a linear map. If  $V$  is finite dimensional, then

$$\|L\|_{\text{op}} = \max_{\|v\|_V = 1} \|Lv\|_W.$$

*Proof.* For every  $v \neq 0$ , normalize it by

$$u = \frac{v}{\|v\|_V}, \quad \|u\|_V = 1.$$

Linearity gives

$$\frac{\|Lv\|_W}{\|v\|_V} = \|Lu\|_W.$$

This proves equality of the unit-sphere and nonzero-vector suprema. It also shows that every vector in the closed unit ball contributes at most the unit-sphere supremum, while the sphere is contained in that ball. Hence the first three suprema agree and

$$\|Lv\|_W \leq \|L\|_{\text{op}} \|v\|_V.$$

Conversely, every constant  $C$  satisfying this inequality bounds the unit-ball supremum, so  $\|L\|_{\text{op}} \leq C$ . The operator norm itself is one such bound, proving the infimum formula.

Since  $Lx - Ly = L(x - y)$ , every difference quotient is one of the nonzero-vector ratios. Conversely every nonzero  $v$  occurs as  $x - y$  by taking  $x = v$  and  $y = 0$ . This proves the Lipschitz formula and its optimality.

In finite dimensions the unit sphere is closed and bounded, hence compact by [Corollary 10.14.8](#). The function  $v \mapsto \|Lv\|_W$  is continuous by [Theorem 10.15.2](#) and [Proposition 10.14.6](#); therefore [Theorem 10.6.6](#) turns its supremum into a maximum.  $\square$

An upper estimate proves only  $\|L\| \leq C$ ; an input attaining  $C$  proves equality. The coefficient bound

$$\left(\sum_{i,j} a_{ij}^2\right)^{1/2}$$

is generally only an upper bound: for the Euclidean identity on  $\mathbb{R}^2$  it is  $\sqrt{2}$ , whereas the operator norm is 1.

**Proposition 10.15.5** (Operator-norm estimates). The operator norm is a norm on the vector space of bounded linear maps  $V \rightarrow W$ . In finite dimensions every linear map is bounded, so it is a norm on  $\mathcal{L}(V, W)$ . For bounded maps  $L : V \rightarrow W$  and  $S : W \rightarrow Z$ ,

$$\|S \circ L\|_{\text{op}} \leq \|S\|_{\text{op}} \|L\|_{\text{op}}.$$

*Proof.* Homogeneity and the triangle inequality follow pointwise and then by taking suprema. If the norm is zero, the displayed bound gives  $Lv = 0$  for every  $v$ , so  $L = 0$ . Finally  $\|(S \circ L)v\| \leq \|S\|_{\text{op}} \|L\|_{\text{op}} \|v\|$ ; take the supremum over  $\|v\| \leq 1$ .  $\square$

The estimate  $\|T(v)\| \leq C\|v\|$  says quantitatively that a linear map cannot amplify a small input error without bound. Every finite-dimensional linear map has such a bound; this is the prototype for the operator-norm estimates on derivatives used later.

**Example 10.15.6** (A diagonal operator norm). For  $L(x, y) = (3x, -2y)$  in Euclidean norm,

$$\|L(x, y)\|_2^2 = 9x^2 + 4y^2 \leq 9(x^2 + y^2).$$

Equality on  $(1, 0)$  proves  $\|L\|_{\text{op}} = 3$ . More generally a diagonal map has Euclidean operator norm  $\max_i |a_{ii}|$ , by the same bound and a maximizing coordinate vector.

**Example 10.15.7** (An operator norm depends on the chosen lengths). Let  $T(x, y) = (x + 2y, 0)$ . With the max norm on both spaces,

$$\|T(x, y)\|_{\infty} = |x + 2y| \leq 3\|(x, y)\|_{\infty}.$$

The max-unit vector  $(1, 1)$  attains 3, so  $\|T\|_{\infty \rightarrow \infty} = 3$ . With the sum norm on both spaces,

$$\|T(x, y)\|_1 = |x + 2y| \leq |x| + 2|y| \leq 2\|(x, y)\|_1.$$

The sum-unit vector  $(0, 1)$  attains 2, so  $\|T\|_{1 \rightarrow 1} = 2$ . The map is unchanged but the exact norms differ. An upper bound alone is not automatically the exact operator norm: equality requires an input attaining the bound, or inputs approaching it.



**Example 10.15.8** (An estimate transferred to a polynomial space). On  $P_2$  use the coefficient norm  $\|a + bt + ct^2\|_c = |a| + |b| + |c|$ , and on  $P_1$  use the analogous norm. Differentiation satisfies  $\|D(a + bt + ct^2)\|_c = |b| + 2|c| \leq 2\|a + bt + ct^2\|_c$ . Equality at  $t^2$  gives operator norm 2. If the spaces instead use the integral inner-product norms, this number need not remain 2, but norm equivalence still guarantees a finite bound. The intrinsic map has not changed, while a numerical error-amplification constant depends on how the two spaces measure error.

Composition means successive amplification:  $L$  amplifies by at most  $\|L\|$ , then  $S$  by at most  $\|S\|$ . This explains the product bound. In Chapter 11 the approximation  $f(a + h) = f(a) + Df(a)h + o(\|h\|)$  uses  $\|Df(a)h\| \leq \|Df(a)\|_{\text{op}}\|h\|$ . For  $\lambda(h) = c \cdot h$  on Euclidean  $\mathbb{R}^n$ , with absolute value on the scalar target,  $\|\lambda\|_{\text{op}} = \|c\|_2$ : Cauchy–Schwarz gives the upper bound and

$$h = c/\|c\|_2$$

attains it when  $c \neq 0$ . The zero case is immediate. Chapter 11 will apply this to scalar derivatives. For example  $\lambda(x, y) = 3x - 4y$  has norm 5, attained at

$$(3/5, -4/5).$$

## Series of vectors and operators

A series

$$\sum_{k=0}^{\infty} v_k$$

in a finite-dimensional normed space  $V$  means the limit of its partial sums. Norm equivalence shows that convergence is equivalent to convergence of every coordinate series in any basis. If

$$\sum_k \|v_k\| < \infty,$$

the estimate

$$\left\| \sum_{k=p}^q v_k \right\| \leq \sum_{k=p}^q \|v_k\|$$

makes partial sums Cauchy; finite-dimensional completeness supplies their limit. For a linear map  $L : V \rightarrow W$ , continuity applied to partial sums gives

$$L\left(\sum_k v_k\right) = \sum_k L(v_k).$$

Scalar convergence tests need no redevelopment.

**Proposition 10.15.9** (Operator geometric series). For  $T \in \mathcal{L}(V)$  with  $\|T\|_{\text{op}} < 1$ ,

$$(I - T)^{-1} = \sum_{k=0}^{\infty} T^k$$

with convergence in operator norm.

*Proof.* The space  $\mathcal{L}(V)$  is finite dimensional (its basis coordinates are the finitely many matrix entries). Since  $\|T^k\| \leq \|T\|^k$  for  $k \geq 1$ , absolute convergence follows from the scalar geometric series. For

$$S_N = \sum_{k=0}^N T^k,$$

direct cancellation gives  $(I-T)S_N = S_N(I-T) = I - T^{N+1}$ . Pass to the limit using the composition estimate; both products equal  $I$ . This also covers the zero space.  $\square$

## 10.16 Dual Spaces and Multilinear Maps

A scalar-valued derivative measures how a direction changes a scalar output. This calls for a linear functional, rather than another direction.

**Definition 10.16.1** (Covectors and the dual space). A **linear functional**, or **covector**, on a real vector space  $V$  is a linear map  $\lambda : V \rightarrow \mathbb{R}$ . The **dual space** is  $V^* = \mathcal{L}(V, \mathbb{R})$ , with pointwise addition and scalar multiplication.

$$v \in V \quad (\text{vector}), \quad \lambda \in V^* \quad (\text{covector}), \quad \lambda(v) \in \mathbb{R} \quad (\text{scalar}).$$

A vector represents a direction or displacement; a covector evaluates on a vector and returns a scalar. Thus scalar-valued derivatives naturally produce covectors.

**Definition 10.16.2** (Kronecker delta and dual basis). The **Kronecker delta** is

$$\delta_j^i = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Thus  $(\delta_j^i)$  is the identity matrix written entrywise. For an ordered basis  $\mathcal{B} = (e_1, \dots, e_n)$  of  $V$ , its **dual basis** is the unique family  $\mathcal{B}^* = (e^1, \dots, e^n)$  in  $V^*$  satisfying  $e^i(e_j) = \delta_j^i$ .

Here are existence, uniqueness, and the reason it is a basis. Define

$$e^i\left(\sum_j v^j e_j\right) = v^i.$$

Unique vector coordinates make this well defined; coordinates of sums and scalar multiples show it is linear. Any covector with the prescribed values on  $e_j$  must have these same values on every linear combination, proving uniqueness. For any  $\lambda \in V^*$ ,

$$v = \sum_i e^i(v) e_i, \quad \lambda = \sum_i \lambda(e_i) e^i.$$

The second equality holds on each  $e_j$ , hence on every vector by linearity. Evaluating a relation

$$\sum_i c_i e^i = 0$$

on  $e_j$  gives  $c_j = 0$ . Thus the  $e^i$  span  $V^*$  independently, and  $\dim V^* = n$ . The  $e_i$  build vectors from coordinates; the  $e^i$  extract coordinates. Subscripts label vectors and superscripts label covectors, not powers. Chapter 11 identifies the standard dual basis with coordinate differentials  $dx_i$  by

differentiating coordinate projections. For example,  $\lambda = 2e^1 - e^2$  evaluates on  $(3, 4)$  as 2. Its coordinate row is  $(2, -1)$ , whereas a vector's coordinate column has a different role. A column and a row represent these objects after bases are chosen; they do not intrinsically identify a vector with a covector. For scalar-valued  $f$  in Chapter 11,  $Df(a) \in V^*$ . Its Euclidean vector representative is  $\nabla f(a)$ , determined by  $Df(a)h = \nabla f(a) \cdot h$ .

**Example 10.16.3** (Classical planes and covectors). For  $0 \neq \lambda \in V^*$ , the equation  $\lambda(x) = c$  defines an **affine hyperplane**. Choose  $p$  with  $\lambda(p) = c$ : if  $\lambda(v) \neq 0$ , take

$$p = cv/\lambda(v).$$

Linearity gives

$$\{x : \lambda(x) = c\} = p + \ker \lambda.$$

The direction space is  $\ker \lambda$ , of codimension one by rank-nullity; the affine set itself need not contain zero. In  $\mathbb{R}^3$  this connects a Cartesian equation  $ax + by + cz = d$  with  $p + \text{span}(u, v)$ . For example, for  $2x - y + 3z = 5$  take

$$p = (1, 0, 1), \quad u = (1, 2, 0), \quad v = (0, 3, 1), \quad \lambda = 2 \, dx - dy + 3 \, dz.$$

Here  $dx, dy, dz$  denote the standard coordinate covectors, and  $\ker \lambda = \{(u_1, u_2, u_3) : 2u_1 - u_2 + 3u_3 = 0\}$ . With the Euclidean inner product the normal is  $n = (2, -1, 3)$ , since  $\lambda(w) = n \cdot w$ . The covector is defined without an inner product; its representing normal vector depends on that additional choice.

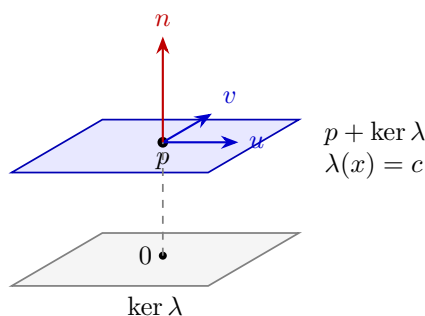


Figure 10.12: An affine plane and its parallel direction plane. The arrows  $u, v$  span the direction space; a Euclidean normal  $n$  represents the covector. Orthogonality is spatial, not an angle measured on this oblique drawing.

**Example 10.16.4** (Finding a dual basis, rather than guessing it). For  $b_1 = (1, 1), b_2 = (1, -1)$ , seek  $b^1(x, y) = ax + by$  satisfying  $a + b = 1$  and  $a - b = 0$ . Thus

$$b^1(x, y) = (x + y)/2.$$

Similarly

$$b^2(x, y) = (x - y)/2.$$

Each extracts the coefficient of its own basis vector:  $(x, y) = b^1(x, y)b_1 + b^2(x, y)b_2$ . For  $\lambda(x, y) = 3x - y$ , its values on  $b_1, b_2$  are 2, 4, so  $\lambda = 2b^1 + 4b^2$ . The coordinate row changes from  $(3, -1)$  to  $(2, 4)$  while the functional stays fixed.

The gradient is an inner-product representative, not the covector itself. For example the functional  $\lambda(x, y) = x$  is represented by  $(1, 0)$  for the usual dot product, but by

$$(1/2, 0)$$

for  $\langle u, v \rangle = 2u_1v_1 + u_2v_2$ . Both pairings return  $x$ . No such choice is needed to evaluate  $\lambda$  or to pull it back.

**Definition 10.16.5** (Pullback of a covector). For linear  $T : V \rightarrow W$ , define  $T^* : W^* \rightarrow V^*$  by  $T^*(\lambda) = \lambda \circ T$ . Vectors are sent forward by  $T$ ; covectors are pulled back by composition. This map is linear because composition distributes over sums and scalar multiples of functionals.

For  $T(x, y) = (x + 2y, 3y)$  and  $\lambda(u, v) = 2u - v$ , one obtains  $T^*\lambda(x, y) = 2x + y$ . In coordinates the row  $(2, -1)$  is multiplied on the right by the matrix of  $T$ .

**Definition 10.16.6** (Multilinear maps). A map  $A : V_1 \times \cdots \times V_k \rightarrow W$  is  $k$ -multilinear if each slot  $r$  satisfies the following conditions, with all other inputs fixed:

$$\begin{aligned} A(\dots, u + v, \dots) &= A(\dots, u, \dots) + A(\dots, v, \dots), \\ A(\dots, au, \dots) &= aA(\dots, u, \dots). \end{aligned}$$

The vectors  $u, v$  belong to  $V_r$  and  $a \in \mathbb{R}$ . For  $k = 2$  the map is called bilinear. A single scaling of the whole input tuple is not the condition; one input varies at a time.

Examples are  $x \cdot y$ ,  $x^\top Ay$  for a fixed matrix  $A$ , and the signed-area expression  $v_1w_2 - v_2w_1$  in the plane. A bilinear map is generally *not* linear on the product vector space:  $B(2x, 2y) = 4B(x, y)$ , whereas product-space linearity would require  $2B(x, y)$ . Ordinary multiplication  $B(x, y) = xy$  already demonstrates this.

Expanding one argument at a time shows that a multilinear map is determined by its values on tuples of basis vectors. In the bilinear case,

$$B\left(\sum_i x_i e_i, \sum_j y_j f_j\right) = \sum_i \sum_j x_i y_j B(e_i, f_j).$$

Repeating this for  $k$  slots gives a sum of products of  $k$  coordinates times the corresponding basis-tuple value. If all  $V_j$  and  $W$  are finite dimensional, this identifies the map space with a finite collection of  $W$ -valued basis data, so the map space is finite dimensional. The logical chain is: basis values determine the map, the finite expansion gives a norm estimate, and the estimate gives continuity. In finite-dimensional normed spaces every multilinear map is bounded: coordinate bounds and the triangle inequality give a constant  $C$  such that

$$\|A(v_1, \dots, v_k)\| \leq C \prod_j \|v_j\|.$$

For bilinear maps, for example, bound each  $|x_i|$  and  $|y_j|$  by fixed multiples of  $\|x\|$  and  $\|y\|$  and sum the finitely many  $\|B(e_i, f_j)\|$ . This bound also proves continuity, by telescoping the difference of two evaluations one slot at a time.

**Example 10.16.7** (Basis values and a complete bilinear estimate). On  $\mathbb{R}^2$ , suppose  $B(e_1, e_1) = 1$ ,  $B(e_1, e_2) = 2$ ,  $B(e_2, e_1) = -1$ , and  $B(e_2, e_2) = 3$ . Then

$$B(\mathbf{x}, y) = x_1y_1 + 2x_1y_2 - x_2y_1 + 3x_2y_2.$$

The four basis values force this expression. Since every coordinate magnitude is at most the Euclidean norm,  $|B(\mathbf{x}, y)| \leq 7\|\mathbf{x}\|_2\|y\|_2$ . This is an upper bound, not a claim that 7 is optimal. To compare nearby input pairs, write the exact two-slot telescoping identity

$$B(x, y) - B(x', y') = B(x - x', y) + B(x', y - y').$$

Near a fixed pair the factors  $\|y\|$  and  $\|x'\|$  stay bounded, so the displayed estimate proves continuity. For more slots replace one argument at a time and add the resulting differences. We will use this estimate for higher derivatives.

**Definition 10.16.8** (Alternating maps). An alternating  $k$ -multilinear map  $A : V^k \rightarrow W$  is one that vanishes whenever two arguments are equal:

$$v_i = v_j \text{ for some } i \neq j \implies A(v_1, \dots, v_k) = 0.$$

The repeated-argument-zero condition is the definition; a sign rule will be a consequence.

**Proposition 10.16.9** (Alternation). Swapping two arguments of an alternating map changes its sign. An alternating map vanishes on linearly dependent arguments.

*Proof.* Put  $u + v$  in the two slots and expand the value 0. The terms with two  $u$ 's or two  $v$ 's vanish, leaving  $A(\dots, u, \dots, v, \dots) + A(\dots, v, \dots, u, \dots) = 0$ . For dependent arguments, express one as a combination of the others; every term then repeats an argument. A zero argument gives zero directly by linearity, including the one-slot case.  $\square$

**Proposition 10.16.10** (Norm-valued remainder rules). As  $h \rightarrow 0$ , sums of  $O(\|h\|)$  terms remain  $O(\|h\|)$  and sums of  $o(\|h\|)$  terms remain  $o(\|h\|)$ . If  $L$  is bounded linear and  $r(h) = o(\|h\|)$ , then  $Lr(h) = o(\|h\|)$ . If  $k(h) = O(\|h\|)$  and  $r(k) = o(\|k\|)$  with  $r(0) = 0$ , then  $r(k(h)) = o(\|h\|)$ , whenever the composition is defined near 0. For bounded bilinear  $B$ , if  $u(h), v(h) = O(\|h\|)$ , then  $B(u(h), v(h)) = O(\|h\|^2) = o(\|h\|)$ .

*Proof.* Replace the symbols by norm inequalities, including the case of a zero intermediate increment. Triangle inequalities prove the sum rules. The bound  $\|Lr(h)\| \leq \|L\|_{\text{op}}\|r(h)\|$  proves the linear rule after division by  $\|h\|$ . For composition, choose  $C \geq 1$  with  $\|k(h)\| \leq C\|h\|$ . Given  $\varepsilon > 0$ , the estimate

$$\|r(k)\| \leq (\varepsilon/C)\|k\|$$

holds for sufficiently small nonzero  $k$ , and for  $k = 0$  because  $r(0) = 0$ . Since  $k(h) \rightarrow 0$ , it gives  $\|r(k(h))\| \leq \varepsilon\|h\|$  eventually. Finally the bilinear bound gives  $\|B(u(h), v(h))\| \leq C_B C_u C_v \|h\|^2$ ; division by  $\|h\|$  makes this tend to zero.  $\square$

The condition  $r(0) = 0$  is essential if  $k(h)$  sometimes vanishes; the punctured little-o statement alone imposes no value at zero. Derivative remainders have this value by their definition. These estimates justify the product and chain-rule manipulations in Chapter 11.

## 10.17 Determinants and Orientation

In  $\mathbb{R}^2$ ,  $v_1w_2 - v_2w_1$  measures the signed area of the ordered parallelogram spanned by  $v, w$ . Doubling one side doubles it; replacing a side by a sum adds the signed contributions; equal sides give zero. The standard ordered unit square has value 1. These suggest three properties of signed volume in higher dimensions: multilinearity, alternation, and normalization on the standard basis. Volume magnitude and orientation sign are related but distinct; Chapter 12 proves the precise volume-scaling theorem.

**Definition 10.17.1** (Normalized determinant form). A normalized determinant form on  $\mathbb{R}^n$  is an alternating  $n$ -linear scalar-valued map whose value on  $(e_1, \dots, e_n)$  is 1.

The sign-change lemma is [Proposition 10.16.9](#): putting  $u + v$  in two slots and expanding its four terms proves the minus sign for an interchange. This lemma supplies the effect of a swap on a form; the elementary parity facts supply how the signs of successive swaps combine. It remains to show that the structural requirements are consistent and determine exactly one form.

### The permutation signs needed here

Multilinear expansion will produce terms indexed by reorderings of basis vectors. Alternation requires a sign for each reordering. Counting inverted pairs gives a sign directly from the permutation, without choosing a possibly nonunique sequence of swaps.

A permutation is a bijection of  $\{1, \dots, n\}$  to itself;  $S_n$  is their set, with  $(\sigma\tau)(j) = \sigma(\tau(j))$ . Define

$$\text{inv}(\sigma) = \#\{(i, j) : i < j, \sigma(i) > \sigma(j)\}, \quad \text{sgn}(\sigma) = (-1)^{\text{inv}(\sigma)}.$$

**Proposition 10.17.2** (Sign rules). Swapping two entries reverses sign, and

$$\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau), \quad \text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma).$$

*Proof.* Swapping adjacent entries changes only their mutual inversion: comparisons with each outside entry have the same total before and after. Thus it reverses sign. Swapping positions  $p < q$  can be done by moving the first entry right through  $q - p$  adjacent swaps, then the other left through  $q - p - 1$  swaps. The odd total  $2(q - p) - 1$  reverses sign. Repeatedly swap an adjacent inverted pair. Each step reduces the inversion number by one; a list with none is the increasing identity list. Reversing these steps expresses  $\tau$  by  $\text{inv}(\tau)$  adjacent swaps. Apply the same position swaps to the list for  $\sigma$ . The final list is  $(\sigma(\tau(1)), \dots, \sigma(\tau(n)))$ , so its sign changes by  $(-1)^{\text{inv}(\tau)}$ . This proves multiplicativity. Apply it to  $\sigma\sigma^{-1}$  to prove the inverse identity.  $\square$

**Theorem 10.17.3** (The normalized determinant form). There is a unique alternating  $n$ -linear map  $\det : (\mathbb{R}^n)^n \rightarrow \mathbb{R}$  with  $\det(e_1, \dots, e_n) = 1$ . More generally, every alternating  $n$ -linear form  $F$  on  $\mathbb{R}^n$  is determined by its one value  $c = F(e_1, \dots, e_n)$  and satisfies  $F = c \det$ .

*Proof.* Expand basis coordinates one slot at a time. Alternation removes repeated choices; the remaining basis tuples can be reordered to the standard one. We separate uniqueness from existence. Write

$$x_j = \sum_i a_{ij} e_i.$$

The slot number  $j$  specifies which input is being expanded. The index  $i$  specifies which standard basis vector occurs in that input. Thus  $a_{ij}$  lies in row  $i$ , column  $j$  of the matrix of input columns.

The separate indices  $i_1, i_2, \dots$  below are independent choices:  $i_2$  is not constrained by the choice already made for  $i_1$ . Expanding the first slot gives

$$F(x_1, x_2, \dots, x_n) = \sum_{i_1=1}^n a_{i_1,1} F(e_{i_1}, x_2, \dots, x_n).$$

For  $n \geq 2$ , expand the second slot of *each* resulting term:

$$F(x_1, \dots, x_n) = \sum_{i_1=1}^n \sum_{i_2=1}^n a_{i_1,1} a_{i_2,2} F(e_{i_1}, e_{i_2}, x_3, \dots, x_n).$$

Continuing through all slots gives

$$F(x_1, \dots, x_n) = \sum_{i_1, \dots, i_n=1}^n a_{i_1,1} \cdots a_{i_n,n} F(e_{i_1}, \dots, e_{i_n}).$$

Each term chooses one basis vector from each input, with the product of the chosen coefficients. There are  $n^n$  choices. Terms with repeated indices vanish. In each remaining term every index occurs once, and a sequence of swaps puts the arguments into  $(e_1, \dots, e_n)$ , determining its value from  $c$ . Thus two forms with the same  $c$  agree on every tuple. A multi-index  $(i_1, \dots, i_n)$  records one such choice for every slot. If  $i_r = i_s$ , the basis tuple repeats a vector, so alternation kills that whole term even if its coefficient is nonzero. Without repetitions,  $n$  choices from  $n$  labels use every label exactly once: these are exactly the  $n!$  permutations. Multilinearity produces all choices; alternation removes repeated choices; permutations are exactly the choices that remain.

A surviving tuple defines the permutation  $\sigma(j) = i_j$ . Therefore

$$F(e_{i_1}, \dots, e_{i_n}) = F(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = \text{sgn}(\sigma) F(e_1, \dots, e_n).$$

Reordering each surviving tuple contributes its sign, so necessarily

$$F(x_1, \dots, x_n) = c \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j),j}.$$

For existence define the determinant by this Leibniz sum with  $c = 1$ . The constructed expression is linear in each column of coordinates. When two inputs agree, say columns  $r, s$ , pair the permutations by

$$\sigma \longleftrightarrow \sigma \circ (r \ s).$$

Applying this pairing twice returns  $\sigma$ , and it has no fixed point because  $\sigma(r) \neq \sigma(s)$ . Equal columns give explicitly

$$a_{\sigma(r),r} a_{\sigma(s),s} = a_{\sigma(r),s} a_{\sigma(s),r}, \quad \text{sgn}(\sigma \circ (r \ s)) = -\text{sgn}(\sigma).$$

All remaining factors are unchanged. Each pair therefore sums to zero. Thus the expression is alternating. On the standard basis only the identity choice remains and has value 1. This proves existence. The same uniqueness argument applied to  $F$  and  $c \det$  proves the final assertion, including  $c = 0$ .  $\square$

**Remark 10.17.4** (A fuller permutation discussion). The treatment here is deliberately streamlined toward the determinant and is self-contained. Appendix C develops cycles, transpositions, adjacent swaps, inversion counting, parity, and permutation matrices more slowly. Readers wanting concrete practice may read that supplementary material before continuing.

**Remark 10.17.5** (Why this proof has so many indices). This first complete multilinear expansion is unusually index-heavy because every input is simultaneously expressed in a basis. Later arguments may say “expand by multilinearity” after identifying the slots being expanded; they usually work directly with the given vectors and do not need this full bookkeeping. The underlying operation is always the same distribution of one sum at a time.

**Example 10.17.6** (From 27 choices to six). For  $n = 3$ , expand in terms  $a_{i1}a_{j2}a_{k3}F(e_i, e_j, e_k)$  for  $1 \leq i, j, k \leq 3$ . The term  $(1, 1, 2)$  vanishes. The choices  $(1, 2, 3)$  and  $(2, 1, 3)$  contribute  $a_{11}a_{22}a_{33}c$  and  $-a_{21}a_{12}a_{33}c$ . Of the 27 formal terms, only the six orderings of  $(1, 2, 3)$  can survive; some of these may still have zero coefficients. The calculation uses multilinearity and alternation alone.

For a square matrix  $A$  with columns  $A_1, \dots, A_n$ , define  $\det A = \det(A_1, \dots, A_n)$ . The preceding proof adapts the slot-by-slot argument in the author’s algebra notes, Theorem 5.12.3.

**Theorem 10.17.7** (Determinant laws). Swapping columns negates the determinant, scaling a column scales it by the same factor, and adding a multiple of another column leaves it unchanged. Dependent columns give zero. Moreover

$$\det I_n = 1, \quad \det(AB) = \det A \det B, \quad \det A \neq 0 \iff A \text{ is invertible.}$$

An upper or lower triangular matrix has determinant

$$\prod_i a_{ii}.$$

*Proof.* Use alternation for column operations and uniqueness for composition. Scaling is multilinearity; swaps and dependent columns were proved in Proposition 10.16.9. Adding  $cA_i$  to a different column adds a determinant with two equal columns, which is zero. To measure what a fixed map  $A$  does to every parallelepiped, feed its transformed edge vectors into the volume form. Define

$$F(v_1, \dots, v_n) = \det(Av_1, \dots, Av_n).$$

For any slot,  $A(av + bw) = aAv + bAw$  followed by determinant multilinearity proves linearity there; equal inputs have equal images, proving alternation. Its value at the standard basis is  $\det A$ . The normalized determinant theorem therefore compares the two alternating forms  $F$  and  $(\det A) \det$ , which have the same value on that basis, and gives

$$\det(Av_1, \dots, Av_n) = (\det A) \det(v_1, \dots, v_n).$$

Matrix multiplication gives  $AB = [AB_1 \cdots AB_n]$ , so

$$\begin{aligned} \det(AB) &= \det(AB_1, \dots, AB_n) \\ &= (\det A) \det(B_1, \dots, B_n) = (\det A)(\det B). \end{aligned}$$

If  $A$  is invertible,  $1 = \det(AA^{-1}) = \det A \det A^{-1}$ . If it is not, its columns are dependent by the basis and bijectivity criteria, so its determinant is zero. For upper triangular  $A$ , the first column



is  $a_{11}e_1$ . In the expansion of column two only  $a_{22}e_2$  can survive, since the  $e_1$  choice repeats. For  $n \geq 2$  the first two stages are

$$\begin{aligned}\det A &= a_{11} \det(e_1, a_{12}e_1 + a_{22}e_2, A_3, \dots, A_n) \\ &= a_{11}a_{12} \det(e_1, e_1, A_3, \dots, A_n) + a_{11}a_{22} \det(e_1, e_2, A_3, \dots, A_n) \\ &= a_{11}a_{22} \det(e_1, e_2, A_3, \dots, A_n).\end{aligned}$$

At stage  $j$ , the choices  $e_1, \dots, e_{j-1}$  all repeat a previous input; only  $a_{jj}e_j$  survives. Induction yields

$$\det A = \left( \prod_j a_{jj} \right) \det(e_1, \dots, e_n) = \prod_j a_{jj}.$$

The case  $n = 1$  is just scaling. For lower triangular matrices run this argument from the last column to the first.  $\square$

**Proposition 10.17.8** (Rank and nonzero minors). An  $r \times r$  **minor** of an  $m \times n$  matrix  $A$  is the determinant of the square submatrix selected by  $r$  rows and  $r$  columns, in their original order. For  $0 \leq r \leq \min(m, n)$ ,

$$\text{rank } A \geq r \iff \text{some } r \times r \text{ minor is nonzero.}$$

Thus the rank is the largest size of a nonzero minor, with the empty  $0 \times 0$  determinant taken to be 1.

*Proof.* The assertion for  $r = 0$  follows from the empty-determinant convention. If the rank is at least  $r > 0$ , choose  $r$  independent columns and form the corresponding  $m \times r$  matrix  $B$ . Its rows span  $\mathbb{R}^r$ : otherwise a basis of their span has  $s < r$  elements, and the map taking a vector to its scalar products with these  $s$  rows has a nonzero kernel by [Theorem 10.10.11](#). A nonzero vector in that kernel is annihilated by every row of  $B$ , contrary to the independence of its columns. Choose  $r$  independent rows of  $B$ . The resulting square matrix has zero kernel, since its rows span  $\mathbb{R}^r$ , and hence is invertible. By [Theorem 10.17.7](#) its determinant is nonzero. Conversely, a nonzero  $r \times r$  minor gives an invertible submatrix. Any relation among the corresponding columns of  $A$ , restricted to the selected rows, therefore has all coefficients zero. Those columns are independent, so  $\text{rank } A \geq r$ .  $\square$

### The coordinate formula follows from the structure

The determinant proof now reads as the forced **Leibniz formula**:

$$\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j),j}.$$

Multilinearity produces all choices; alternation removes repeated choices; permutations are exactly the choices that remain. Each product uses one entry from each column and each row. For

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \text{ the six terms are}$$

$$aei + bfg + cdh - ceg - bdi - afh.$$

**Example 10.17.9** (The six Leibniz terms). In the table, the list is  $(\sigma(1), \sigma(2), \sigma(3))$ . It selects one row in column one, one in column two, and one in column three. Use the matrix

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}.$$

| list | parity/sign | selected positions     | product | contribution |
|------|-------------|------------------------|---------|--------------|
| 123  | even/+      | (1, 1), (2, 2), (3, 3) | $aei$   | $+aei$       |
| 132  | odd/-       | (1, 1), (3, 2), (2, 3) | $ahf$   | $-ahf$       |
| 213  | odd/-       | (2, 1), (1, 2), (3, 3) | $dbi$   | $-dbi$       |
| 231  | even/+      | (2, 1), (3, 2), (1, 3) | $dhc$   | $+dhc$       |
| 312  | even/+      | (3, 1), (1, 2), (2, 3) | $gbf$   | $+gbf$       |
| 321  | odd/-       | (3, 1), (2, 2), (1, 3) | $gec$   | $-gec$       |

For  $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 1 & 3 & 6 \end{pmatrix}$  the six contributions in this order are 30, -24, -24, 18, 16, -15. Their sum is 1. Only for three-by-three matrices does this short six-term mnemonic apply; a four-by-four determinant has 24 terms.

**Proposition 10.17.10** (Transpose and intrinsic determinant). The transpose  $A^T$ , obtained by exchanging rows and columns, has the same determinant as  $A$ . For an endomorphism  $T : V \rightarrow V$ ,  $\det[T]_{\mathcal{B}}$  is independent of the basis  $\mathcal{B}$ ; we may therefore denote this scalar by  $\det T$ .

*Proof.* The reindexing  $i = \sigma(j)$  gives

$$\begin{aligned} \det(A^T) &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_j a_{j, \sigma(j)} \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma^{-1}) \prod_i a_{\sigma^{-1}(i), i} = \det A. \end{aligned}$$

Inversion is a bijection of  $S_n$  preserving sign, so the last sum is exactly the Leibniz sum. If  $P$  expresses the new basis in the old one, the new matrix is  $P^{-1}[T]_{\mathcal{B}}P$ . Therefore

$$\det(P^{-1}[T]_{\mathcal{B}}P) = (\det P)^{-1} \det[T]_{\mathcal{B}} \det P = \det[T]_{\mathcal{B}}.$$

This proves that the scalar is independent of coordinates.  $\square$

Transpose invariance gives all the column rules for rows as well. In particular, interchanging two rows changes the determinant's sign, multiplying a row by  $c$  multiplies the determinant by  $c$ , and adding  $c$  times a different row leaves it unchanged. For the last assertion, row multilinearity splits the new determinant into the original one and  $c$  times a determinant with two equal rows; the latter vanishes. These are consequences of the determinant's construction and transpose invariance, so row reduction can now be used to compute it.

For maps between different spaces a determinant requires choices of volume normalizations; a scalar independent of both bases is not asserted. Appendix D, Theorem D.9.10, gives a retrospective fourth characterization: the induced action on the top exterior power multiplies by this same scalar. Its four-viewpoint synthesis connects normalized alternation, Leibniz, recursive Laplace expansion, and top exterior action without replacing the proofs here.

# Minors, cofactors, and efficient computation

The structural laws are best for theory. Jacobians also require actual numbers, and cofactor expansion reduces a determinant to smaller ones.

**Definition 10.17.11** (Minors and cofactors). For  $A \in M_n(\mathbb{R})$ , let  $M_{ij}(A)$  be the determinant obtained by deleting row  $i$  and column  $j$ , preserving the remaining order. The **cofactor** is  $C_{ij}(A) = (-1)^{i+j} M_{ij}(A)$ . The empty 0-by-0 determinant is 1, so these conventions cover  $n = 1$ .

**Theorem 10.17.12** (Laplace expansion). Along any fixed column  $j$  or fixed row  $i$ , respectively,

$$\det A = \sum_{i=1}^n a_{ij} C_{ij}(A), \quad \det A = \sum_{j=1}^n a_{ij} C_{ij}(A).$$

*Proof.* Expand a column in the standard basis, and identify the coefficient of each entry by moving that basis vector to the corner. Multilinearity in column  $j$  gives

$$\det A = \sum_i a_{ij} \det(A_1, \dots, e_i, \dots, A_n).$$

Move this column to the first position using  $j - 1$  swaps, then move row  $i$  to the first position using  $i - 1$  swaps. The resulting first column is  $e_1$ ; the remaining rows and columns have their original relative order. The matrix now has block form

$$\begin{pmatrix} 1 & * \\ 0 & B \end{pmatrix},$$

where  $B$  is the deleted-row, deleted-column matrix. Write its other columns as  $b_\ell e_1 + \iota(B_\ell)$ , where  $\iota : \mathbb{R}^{n-1} \rightarrow \mathbb{R}^n$  inserts an initial zero. Expanding those columns, every  $e_1$  choice repeats the first argument, so

$$\det \begin{pmatrix} 1 & * \\ 0 & B \end{pmatrix} = \det(e_1, \iota(B_1), \dots, \iota(B_{n-1})).$$

The map  $H(z_1, \dots, z_{n-1}) = \det(e_1, \iota(z_1), \dots, \iota(z_{n-1}))$  is alternating multilinear and takes value  $\det(e_1, e_2, \dots, e_n) = 1$  on the standard basis. The normalized determinant theorem in dimension  $n - 1$  therefore gives  $H(B_1, \dots, B_{n-1}) = \det B = M_{ij}$ ; for  $n = 1$  this uses the empty determinant convention. Undoing the swaps gives  $(-1)^{i+j-2} M_{ij} = C_{ij}$ . This proves column expansion; applying it to  $A^\top$  proves row expansion, since deleting its row  $j$  and column  $i$  transposes the minor.  $\square$

**Example 10.17.13** (Choose the zeros first). For

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 4 \\ 5 & 6 & 7 \end{pmatrix},$$

column two has only one nonzero entry. Expanding there gives

$$\det A = 6(-1)^{3+2} \det \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} = -6(8 - 3) = -30.$$

As a check, row one gives  $2(0 \cdot 7 - 4 \cdot 6) + (3 \cdot 6 - 0 \cdot 5) = -48 + 18 = -30$ , requiring two minors instead of one. For an even greater saving take

$$D = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 5 & 0 \\ 2 & 1 & 0 & 3 \\ 4 & 0 & 1 & 2 \end{pmatrix}.$$

Row two leaves a single minor:

$$\begin{aligned} \det D &= -5 \det \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 0 & 2 \end{pmatrix} \\ &= -5\{1(2) - 2(4 - 12) + 4(0 - 4)\} = -10. \end{aligned}$$

Expanding the original first row would require four 3-by-3 minors.

A determinant may be expanded along *any row or any column*. For hand computation normally choose the one with the most zeros, since each zero entry removes a term. Cofactors are useful for small matrices, sparse rows or columns, and recursive proofs. For a dense matrix, row reduction to triangular form often takes fewer operations. The determinant of a triangular matrix is its diagonal product: in the Leibniz formula only the identity permutation can contribute. Track every swap and row scaling; row additions require no correction factor.

**Example 10.17.14** (Triangularize without changing the determinant). The coefficient matrix of our three-variable system reduces by row additions:

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \end{pmatrix}.$$

Its determinant is  $1 \cdot 1 \cdot 3 = 3$ . This also confirms invertibility. If one scales the last row by

$$1/3$$

to obtain reduced form, the new determinant is 1; recovering the original requires the factor 3.

The geometric transformations below supply the structural factorization needed for volume.

The determinant exists algebraically without order; its *sign* uses the ordered real field. Over  $\mathbb{R}$ ,  $|\det|$  records volume magnitude, while sign distinguishes the two orientations. These facts lead to the following definition using the coordinate changes already constructed.

In the plane, an ordered coordinate pair can be counterclockwise or clockwise; in three dimensions this is the familiar distinction between right-handed and left-handed coordinates. In arbitrary finite dimension, the sign of a change-of-basis determinant is the algebraic version of this handedness. Orientation matters because scalar volume forgets the sign via  $|\det|$ , while oriented integration and Stokes' theorem retain it.

**Definition 10.17.15** (Orientation). For  $n \geq 1$ , two ordered bases of an  $n$ -dimensional real vector space are **equivalently oriented** if the change-of-coordinate matrix between them has positive determinant. An **orientation** is one of the two equivalence classes of ordered bases. The class of the standard basis is the standard orientation of  $\mathbb{R}^n$ .

In  $\mathbb{R}^2$ , the ordered basis

$$(e_1, e_2)$$

has the standard orientation, whereas

$$(e_2, e_1)$$

has the opposite orientation because the change-of-basis determinant is  $-1$ . The basis  $(e_1 + e_2, e_2)$  has the standard orientation because its change-of-basis determinant is positive.

**Proposition 10.17.16** (Orientation). Equivalent orientation is an equivalence relation. The sign of the determinant of an invertible map between oriented spaces does not depend on the positively oriented bases chosen.

*Proof.* Fix one ordered basis. Every other basis has either positive or negative transition determinant relative to it. Multiplicativity shows bases of the same sign are equivalent, and bases of opposite sign are not. Both signs occur by negating the first basis vector, so there are exactly two classes. The identity has determinant 1, inverses of positive-determinant matrices have positive determinant, and products of such matrices have positive determinant. A change to another positively oriented basis multiplies the matrix of a map on the left or right by a positive-determinant matrix, so multiplicativity preserves its sign.  $\square$

**Lemma 10.17.17** (Geometric factorization). Every invertible  $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is a composition of coordinate permutations, nonzero coordinate scalings, and shears  $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_i + cx_j, \dots, x_n)$ ,  $i \neq j$ . Their determinants are respectively the permutation sign, the scale, and 1.

*Proof.* Fix one basis direction, separate it from the others, and repeat on the remaining space. The determinant laws give the stated values; each transformation has an inverse of the same type. Since  $Le_1 \neq 0$ , a target permutation  $P$  puts a nonzero component first. A target scaling  $C$  makes that component one; target shears  $R$  subtract the remaining components times the first coordinate. Set  $A = R \circ C \circ P$ , so  $ALe_1 = e_1$ . For  $j > 1$  write  $ALe_j = a_j e_1 + v_j$ , with  $v_j \in E = \text{span}(e_2, \dots, e_n)$ . Now use a map on the *source*: let  $Be_1 = e_1$  and  $Be_j = e_j - a_j e_1$ . It is a composition of source shears. Thus

$$(A \circ L \circ B)e_1 = e_1, \quad (A \circ L \circ B)e_j = AL(e_j - a_j e_1) = v_j.$$

Consequently  $A \circ L \circ B = I_{\mathbb{R}e_1} \oplus L'$  with  $L' : E \rightarrow E$  invertible: a vector in its kernel would be in the kernel of the full invertible composite, and equal dimensions give surjectivity. Induct on  $n$ , starting with a single nonzero scaling for  $n = 1$ . Extend each factor of  $L'$  to fix  $e_1$ . Finally the exact order is

$$L = A^{-1} \circ (I_{\mathbb{R}e_1} \oplus L') \circ B^{-1}, \quad A^{-1} = P^{-1} \circ C^{-1} \circ R^{-1}.$$

The rightmost factor acts first. This undoes the source and target changes on their respective sides and completes the factorization.  $\square$

**Example 10.17.18** (Stretch, shear, reflection). The maps  $(x, y) \mapsto (3x, y)$ ,  $(x, y) \mapsto (x + 2y, y)$ , and  $(x, y) \mapsto (-x, y)$  have determinants 3, 1,  $-1$ . The first stretches a unit square, the second tilts its sides into a parallelogram, and the third reflects it. The determinant sign records orientation; its absolute value will be proved to give area scaling in Chapter 12.

Figure 10.13 separates unsigned area from the orientation encoded by the determinant sign.

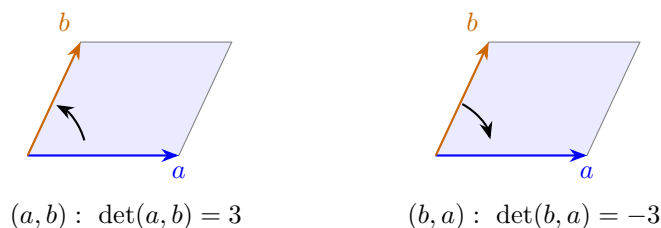


Figure 10.13: Swapping the ordered edges reverses orientation while preserving area. Here  $a = (2, 0)$  and  $b = (0.7, 1.5)$ ; both parallelograms have area 3.

## 10.18 Perspective and Transition

Before continuing, check that you can move between a linear map and its matrix, multiply a matrix by a vector, and compute composition both ways. Use  $\|h\|$  to quantify an input error and explain how compactness turns local estimates into uniform ones. Distinguish  $\det L$ ,  $|\det L|$ , and the sign of  $\det L$ : they answer related but different questions.

**Exercise 1.** For  $L(x, y) = (2x - y, y)$  and  $S(u, v) = (u, u + v)$ , compute  $SL$  as maps and matrices. Bound  $\|Lh\|_2$  by a constant times  $\|h\|_2$ , and determine whether each map preserves orientation.

The derivative in Chapter 11 will be a linear map between finite-dimensional normed spaces. The equivalence of norms makes differentiability independent of a coordinate norm. Compactness supplies finite local control, while determinants and orientation distinguish the two local senses of direction needed in change of variables and Stokes' theorem.

## Exercises

**Exercise 2.** Prove that every open ball in a metric space is open and every singleton is closed.

**Exercise 3.** Prove that, for  $E \subseteq X$ ,

$$\partial E = \overline{E} \cap \overline{X \setminus E}.$$

**Exercise 4.** Which of the sets

$$[0, 1/2), \quad (0, 1/2), \quad [0, 1/2]$$

are open in the subspace  $[0, 1] \subseteq \mathbb{R}$ ?

**Exercise 5.** Prove directly from epsilon–delta definitions that a composition of continuous maps between metric spaces is continuous.

**Exercise 6.** For  $f(t) = (t^2, \sin t)$ , use coordinate continuity to prove continuity into  $\mathbb{R}^2$ . Explain why an estimate for the max metric also controls Euclidean continuity.

**Exercise 7.** Use [Theorem 10.6.3](#) to prove that a compact metric space is complete.

**Exercise 8.** Which of the punctured plane, two disjoint open disks, and a closed triangle are connected? Give paths or a separation as appropriate.

**Exercise 9.** Show that the open unit ball in  $\mathbb{R}^n$  is not compact.

**Exercise 10.** Extend  $(1, 1, 0), (0, 1, 1)$  to a basis of  $\mathbb{R}^3$  using standard basis vectors. Identify a generator that can be replaced when inserting each given vector into the standard spanning list.

**Exercise 11.** For

$$T(x, y, z) = (x + 2y - z, 2x + 4y - 2z),$$

find bases of the kernel and image, and verify rank–nullity. Explain why the explicit target–lift argument for the earlier surjective example cannot work for an arbitrary target here.

**Exercise 12.** Let  $T : V \rightarrow V$  be a linear endomorphism of a finite-dimensional real vector space and suppose that

$$T^2 = T$$

Show that

$$\ker T \cap T(V) = \{0\},$$

and that every  $v \in V$  can be written uniquely in the form

$$v = u + w, \quad u \in \ker T, \quad w \in T(V).$$

**Exercise 13.** Give maps between finite-dimensional spaces that are injective but not surjective, and surjective but not injective.

**Exercise 14.** For

$$\|\mathbf{x}\|_1 = \sum_{j=1}^n |x_j|, \quad \|\mathbf{x}\|_\infty = \max_{1 \leq j \leq n} |x_j|,$$

prove

$$\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 \leq \sqrt{n} \|\mathbf{x}\|_2.$$

**Exercise 15.** Give an incomplete normed vector space, for example by considering polynomials on  $[0, 1]$  with the supremum norm.

**Exercise 16.** Suppose square matrices  $A, B$  satisfy  $AB = I$ . First use linear bijectivity to prove  $BA = I$ , then determine  $\det B$  from  $\det A$ . Explain why the first implication can fail for rectangular matrices, and give an example.

**Exercise 17.** Show that

$$(x_1, \dots, x_n) \mapsto (-x_1, x_2, \dots, x_n)$$

reverses the standard orientation of  $\mathbb{R}^n$ .

**Exercise 18.** In  $P_2$ , decide whether  $(1+t, t+t^2, 1-t^2)$  is independent and find a basis of its span. Give a polynomial outside that span.

**Exercise 19.** Let  $\mathcal{B} = ((1, 1), (1, -1))$  and let  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$  satisfy  $T(1, 1) = (1, 2, 0)$ ,  $T(1, -1) = (0, 1, 3)$ . Find its matrix from  $\mathcal{B}$  to the standard basis. Compute  $T(3, 1)$  first by expressing the input in  $\mathcal{B}$  and then by the corresponding column sum.

**Exercise 20.** For  $T(x, y) = (x + 2y, y)$  and  $S(u, v) = (u - v, 2u)$ , compute the matrix of  $S \circ T$  and explain each column using basis images. Compare  $T \circ S$ .

**Exercise 21.** For  $\mathcal{B} = ((1, 1), (0, 1))$  and  $\mathcal{C} = ((1, 0), (1, 1))$ , find both change-of-coordinate matrices. Transform the standard matrix of  $T(x, y) = (x - y, 2y)$  to  $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$  and verify its two columns directly. Then compute  $[T]_{\mathcal{B}}$ .

**Exercise 22.** For  $\lambda = 3e^1 - 2e^2$  in the standard dual basis and  $T(x, y) = (x + y, 2x)$ , compute  $\lambda(2, 1)$  and  $T^*\lambda$ . Distinguish the input vector, the covector, and their coordinate column and row.

**Exercise 23.** Show  $B(\mathbf{x}, y) = x_1y_2 + x_2y_1$  is bilinear and symmetric but not alternating. Show it is not linear as a function on  $\mathbb{R}^2 \times \mathbb{R}^2$ .

**Exercise 24.** Compute the determinant of  $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 1 & 3 & 6 \end{pmatrix}$  using row additions and a triangular matrix.

Record which operations preserve the determinant.

**Exercise 25.** Compute  $\det \begin{pmatrix} 2 & 0 & 3 \\ 1 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$  by two different cofactor expansions. Which row or column gives the shortest calculation, and why?

**Exercise 26.** If  $A(t)$  is a continuous square-matrix-valued function on an interval and every  $A(t)$  is invertible, prove that  $\det A(t)$  has constant sign. State exactly where continuity, connectedness, and invertibility are used.

**Exercise 27.** For

$$E = \{1/n : n \geq 1\} \cup \{2\}$$

in  $\mathbb{R}$ , compute interior, closure, boundary, isolated points, and accumulation points. Repeat with ambient space  $\overline{E}$ , identifying which answers change.

**Exercise 28.** On  $P_1$  with

$$\langle p, q \rangle = \int_0^1 pq,$$

compute  $\|1 - t\|$  and  $\langle 1 + t, 1 - t \rangle$ . Verify Cauchy-Schwarz for these two polynomials. Does the coordinate sum norm arise from this pairing?



**Exercise 29.** Find the dual basis of  $((1, 0, 1), (0, 1, 1), (0, 0, 1))$  in  $\mathbb{R}^3$ . Express  $2e^1 - e^2 + 3e^3$  (using the standard dual basis) in that basis and check its value on  $(1, 2, 3)$  using both descriptions.

**Exercise 30.** For  $T(x, y) = (x + y, x - y)$  find its operator norm from the sum norm to the max norm, then from the max norm to the sum norm. Give an upper bound and a maximizing input in each case.

**Exercise 31.** A student identifies the matrix of the identity in two different bases with  $I$ . Supply a concrete counterexample and explain when the claim is valid. Then explain why  $[T]_{\mathcal{B}'} = P^{-1}[T]_{\mathcal{B}}P$  preserves the determinant but need not preserve any particular entry.

**Exercise 32.** Reconstruct the proof that a compact subset of a Hausdorff space is closed. Label the step using Hausdorff separation, the finite selection using compactness, and the step that would fail for an infinite intersection. Explain why boundedness alone supplies none of that finite selection.

**Exercise 33.** Negate metric continuity at  $a$  with all quantifiers displayed. Construct a sequence witnessing the failure, and explain why its output error uses one fixed positive number, not a number depending on the sequence index.

**Exercise 34.** For subspaces  $U_1, U_2, U_3$ , prove that uniqueness of a representation in  $U_1 + U_2 + U_3$  is equivalent to the zero-relation criterion. Test both the criterion and pairwise intersections on three distinct lines in  $\mathbb{R}^2$ .

**Exercise 35.** Keep  $U = \{z = x + y\} \subset \mathbb{R}^3$  but replace the complementary line in [Example 10.9.6](#) by  $W_t = \text{span}(1, 0, t)$ . For which real  $t$  is  $\mathbb{R}^3 = U \oplus W_t$ ? In those cases construct both component maps. At the exceptional value, identify separately what fails for existence and for uniqueness.

**Exercise 36.** Let  $\mathcal{C} = ((1, 1), (1, -1))$ , and suppose the matrices of  $S, T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  from the standard basis to  $\mathcal{C}$  are  $A = \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$  and  $B = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$ . Construct  $2S - T$  from its basis images and compare its matrix with  $2A - B$ . Explain why adding a matrix with a different target basis would not be valid until its output coordinates had been converted.

## Chapter 11

# Differentiation in Finite-Dimensional Spaces

### 11.1 Limits, Continuity, and Families of Maps

Before differentiating, separate three questions: how one map behaves near a point, how whole maps approach a limit, and when a family has a convergent subsequence. Chapter 10 already develops convergence, limits of maps, continuity, closure, completeness, and their sequential criteria for arbitrary metric spaces; see [Theorems 10.2.11, 10.3.5, 10.3.6, 10.3.11](#) and [10.3.13](#). We use those results here rather than rebuilding them for Euclidean space. Chapter 6 supplies the one-variable models. The new work is finite-dimensional: coordinate descriptions, norm estimates, standard multivariable limit methods, and convergence of families of vector-valued maps. We use the Euclidean default  $\|\mathbf{x}\| = \|\mathbf{x}\|_2$  established near the beginning of Chapter 10, retaining subscripts when comparing norms.

#### Limits and continuity in finite dimensions

For  $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ , apply the metric definitions and sequential criteria of Chapter 10 with Euclidean distance. What is special to the target  $\mathbb{R}^m$  is the componentwise description. By [Proposition 10.4.19](#),

$$f(\mathbf{x}) \rightarrow b \iff f_j(\mathbf{x}) \rightarrow b_j \quad (j = 1, \dots, m),$$

and the same equivalence holds for continuity. The quantitative estimates

$$|v_j| \leq \|v\|_2 \leq \sqrt{m} \max_j |v_j|$$

let component estimates be converted into one norm estimate and vice versa. The sequential criterion used below is precisely [Theorem 10.3.11](#): the limit is  $b$  if and only if  $f(\mathbf{x}_k) \rightarrow b$  for every sequence  $\mathbf{x}_k \in D \setminus \{a\}$  tending to  $a$ .

**Start with continuity and algebra.** Try substitution before estimating. For example,

$$(x^2 + y)/(1 + x^2 + y^2) \rightarrow 2/3$$

as  $(x, y) \rightarrow (1, 1)$ , by the algebra of continuous functions and the nonzero denominator. At a singular formula, use the limit laws and squeeze theorem of Chapter 5: factor, cancel only where

the cancelled factor is nonzero, or rationalize. Thus, near the origin,

$$\frac{\sqrt{1+x^2+y^2}-1}{x^2+y^2} = \frac{1}{\sqrt{1+x^2+y^2}+1} \longrightarrow \frac{1}{2}.$$

The equality is needed only on the punctured neighborhood where the original formula is defined. Assigning a value at the origin cannot change its limit.

**Recall polar coordinates for limits.** Recall from [Definition 8.7.2](#) that

$$(x, y) = (r \cos \theta, r \sin \theta), \quad r = \sqrt{x^2 + y^2}.$$

We now use these coordinates for genuinely two-variable limits. The circle/ray geometry and nonuniqueness are as in Chapter 8. Since  $\|\sigma(r, \theta)\| = r$  for  $\sigma(r, \theta) = (r \cos \theta, r \sin \theta)$ , approaching the origin is equivalent to  $r \rightarrow 0$ , while  $\theta$  may vary. A bound

$$|f(r \cos \theta, r \sin \theta) - L| \leq g(r), \quad g(r) \longrightarrow 0,$$

proves the limit if it is uniform in  $\theta$ . Convergence at every fixed angle is insufficient: the angle may depend on  $r$ .

**Turn the distance into a bound.** A particularly useful target is  $|f(\mathbf{x}) - L| \leq C\|\mathbf{x} - a\|^\alpha$  with  $C > 0$  and  $\alpha > 0$ . Then

$$\delta = (\varepsilon/C)^{1/\alpha}$$

works. In two variables put  $r = \sqrt{x^2 + y^2}$ . The estimates  $|x|, |y| \leq r$  and  $2|xy| \leq r^2$  often remove the directional dependence. For the Chapter 10 example,

$$\left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y| \leq r.$$

Given  $\varepsilon > 0$ , choose  $\delta = \varepsilon$ . Whenever  $0 < r < \delta$ , the absolute value is less than  $\varepsilon$ , proving the zero limit in every direction. Equivalently, substitute  $x = r \cos \theta$ ,  $y = r \sin \theta$  to obtain  $r \cos^2 \theta \sin \theta$ . Its angular factor is bounded by 1 *uniformly in*  $\theta$ . A bound for each fixed angle alone is not enough: the angle can vary while  $r \rightarrow 0$ . This is a substitution for estimating a function, with no integration formula involved.

**Use paths to test a suspected failure.** Begin with the axes, then lines  $y = mx$ . For

$$f(x, y) = xy/(x^2 + y^2),$$

both axes give zero, but  $y = x$  gives

$$1/2$$

and  $y = -x$  gives

$$-1/2.$$

In radial coordinates this is  $\cos \theta \sin \theta$ , whose directional dependence remains with no decaying radial factor. The two paths yield the sequences

$$(1/k, 1/k), \quad (1/k, -1/k)$$

with different limiting outputs, so the sequential criterion disproves a joint limit. The two iterated limits are nevertheless both zero: for each fixed  $y$ , the limit as  $x \rightarrow 0$  is zero, and the reversed order gives the same result. Iterated limits test successive one-variable processes, not all simultaneous approaches to the point.

Even agreement on every line is insufficient. Consider

$$q(x, y) = \frac{x^2 y}{x^4 + y^2}, \quad (x, y) \neq (0, 0).$$

On  $y = mx$  with  $m \neq 0$  this is

$$mx/(x^2 + m^2) \rightarrow 0;$$

both axes also give zero. To choose a curved path, balance the denominator terms:  $y^2$  is comparable to  $x^4$  when  $y$  is of order  $x^2$ . On  $y = x^2$ ,

$$q = 1/2.$$

Thus

$$(1/k, 0), \quad (1/k, 1/k^2)$$

disprove the zero limit suggested by all straight lines. Successful path tests are evidence for a candidate, never a proof of a joint limit.

**Limit checklist.** (1) Try continuity and substitution. (2) Simplify on the actual punctured domain. (3) Seek a bound in  $\|\mathbf{x} - a\|$ . (4) Try radial coordinates with a uniform angular bound. (5) Test axes and lines to disprove a candidate. (6) If lines agree, balance competing terms to select curved paths.

Here  $\mathbf{x}_k \rightarrow a$  varies the input of one fixed function. In  $f_k \rightarrow f$ , we vary entire functions, as follows.

## Pointwise and uniform convergence

The distinction from Chapter 6 persists for vector-valued maps: at a fixed input we take a sequence of outputs, whereas uniform convergence controls every input with one stage. Norms replace absolute values in the errors.

For maps  $f_k, f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ , convergence means

$$\text{pointwise: } \forall \mathbf{x} \in D \forall \varepsilon > 0 \exists N = N(\mathbf{x}, \varepsilon) \forall k \geq N : \|f_k(\mathbf{x}) - f(\mathbf{x})\| < \varepsilon,$$

$$\text{uniform: } \forall \varepsilon > 0 \exists N = N(\varepsilon) \forall \mathbf{x} \in D \forall k \geq N : \|f_k(\mathbf{x}) - f(\mathbf{x})\| < \varepsilon.$$

For pointwise convergence, fix *all* input coordinates first; only  $k$  varies. On  $D = [0, 1]^2$ , the sequence  $f_k(x, y) = x^k$  therefore has limit

$$f(x, y) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

The unused coordinate  $y$  is still part of the fixed input.

Once the pointwise limit is known, form the error  $E_k(\mathbf{x}) = \|f_k(\mathbf{x}) - f(\mathbf{x})\|$ . Uniform convergence is exactly  $\sup_{\mathbf{x} \in D} E_k(\mathbf{x}) \rightarrow 0$ . One may compute this supremum or bound it by a number  $a_k \rightarrow 0$  independent of the input. To disprove uniform convergence, find  $\varepsilon_0 > 0$  and moving inputs  $\mathbf{x}_k \in D$  with  $E_k(\mathbf{x}_k) \geq \varepsilon_0$  for arbitrarily large  $k$ . For  $x^k$ , choose

$$(2^{-1/k}, 0).$$

Its first coordinate is less than 1, so its limit-function value is zero and its error is exactly

$$1/2.$$

The input changes with  $k$ : this contradicts uniform convergence without contradicting convergence at any fixed input. In fact the error supremum is 1 for every  $k$ , approached as  $x \uparrow 1$  but not attained. Consequently

$$\limsup_k \sup_D E_k = 1, \quad \sup_D \lim_k E_k = 0.$$

Interchanging a limit and a supremum would erase precisely the moving bad points that uniform convergence must control.

**Example 11.1.1** (The domain determines the uniform estimate). For

$$f_k(x, y) = (x/k, y/k)$$

on  $[-1, 1]^2$ , the pointwise limit is zero and

$$E_k(x, y) = \frac{\sqrt{x^2 + y^2}}{k} \leq \frac{\sqrt{2}}{k}.$$

Equality holds at a corner, so this is the exact supremum. Any bounded rectangle admits the same argument with its own fixed bound on  $\sqrt{x^2 + y^2}$ . On all of  $\mathbb{R}^2$ , the same formula converges pointwise but the error supremum is infinite for every  $k$ , so convergence is not uniform.

An estimate such as

$$E_k(x, y) \leq (x^2 + y^2)/k$$

gives an index depending on  $x, y, \varepsilon$  and hence only proves pointwise convergence until the domain supplies a common bound on  $x^2 + y^2$ . Inspect the dependencies of the chosen  $N$ , not merely the last inequality in the calculation. For finitely many output components, pointwise (respectively uniform) convergence is equivalent to pointwise (respectively uniform) convergence of each component. Indeed each component error is bounded by  $E_k$ , and  $E_k$  is bounded by  $\sqrt{m}$  times the largest component error; for uniform convergence take the largest of the finitely many component indices.

**Sequence checklist.** Fix the input and find the pointwise limit. Subtract that limit and compute the error. Seek a supremum or a bound independent of the input; otherwise try moving inputs with a fixed positive error. Check that the proposed index depends only on  $\varepsilon$  and fixed domain bounds before claiming uniform convergence.

**Theorem 11.1.2** (Uniform limits preserve continuity). A uniform limit of continuous maps  $D \rightarrow \mathbb{R}^m$  is continuous on  $D$ .

*Proof.* Choose one sufficiently late continuous map. Given  $a \in D$  and  $\varepsilon > 0$ , choose  $k$  with

$$\|f - f_k\|_\infty < \varepsilon/3,$$

then choose  $\delta$  by continuity of this fixed  $f_k$  at  $a$ . For  $\mathbf{x} \in D$  within  $\delta$  of  $a$ ,

$$\|f(\mathbf{x}) - f(a)\| \leq \|f(\mathbf{x}) - f_k(\mathbf{x})\| + \|f_k(\mathbf{x}) - f_k(a)\| + \|f_k(a) - f(a)\| < \varepsilon.$$

□

Uniform convergence does not preserve differentiability. The smooth maps

$$f_k(x, y) = \sqrt{x^2 + 1/k}$$

satisfy

$$0 \leq f_k(x, y) - |x| \leq 1/\sqrt{k}$$

everywhere, but  $|x|$  is not differentiable on  $x = 0$ . Differentiability will require stronger control of the linear approximations, not just the function values.

## Equicontinuity and compactness of families

Uniform convergence concerns an already chosen sequence of maps. Equicontinuity instead bounds input variation across an entire family, so that nearby sample values control every member. Compactness will let us reduce that control to finitely many sample points.

Let  $K \subseteq \mathbb{R}^n$  be compact and let  $C(K, \mathbb{R}^m)$  denote the continuous maps from  $K$  to  $\mathbb{R}^m$ . A family  $\mathcal{F} \subseteq C(K, \mathbb{R}^m)$  is *equicontinuous* if for every  $a \in K$  and  $\varepsilon > 0$  there is  $\delta = \delta(a, \varepsilon) > 0$  such that

$$\mathbf{x} \in K, \|\mathbf{x} - a\| < \delta \implies \|g(\mathbf{x}) - g(a)\| < \varepsilon \quad \text{for every } g \in \mathcal{F}.$$

It is uniformly equicontinuous if one  $\delta(\varepsilon)$  works for every  $a \in K$  as well. The common scale across all maps is essential.

**Lemma 11.1.3.** An equicontinuous family on compact  $K$  is uniformly equicontinuous.

*Proof.* Otherwise choose  $\mathbf{x}_j, y_j \in K$  and  $g_j \in \mathcal{F}$  with

$$\|\mathbf{x}_j - y_j\| < 1/j$$

and  $\|g_j(\mathbf{x}_j) - g_j(y_j)\| \geq \varepsilon_0 > 0$ . Pass to a subsequence  $\mathbf{x}_j \rightarrow a \in K$ ; then  $y_j \rightarrow a$  too. Equicontinuity at this one  $a$  makes both  $\|g_j(\mathbf{x}_j) - g_j(a)\|$  and  $\|g_j(y_j) - g_j(a)\|$  less than

$$\varepsilon_0/2$$

eventually, contradicting the triangle inequality.  $\square$

**Theorem 11.1.4** (Finite-dimensional Arzelà–Ascoli). An equicontinuous sequence  $f_j : K \rightarrow \mathbb{R}^m$  on compact  $K \subseteq \mathbb{R}^n$  with  $\sup_{j, \mathbf{x}} \|f_j(\mathbf{x})\| < \infty$  has a uniformly convergent subsequence. Its limit is continuous.

*Proof.* Select convergence on countably many sample points, then use a common continuity scale to control the gaps between samples. The empty domain is immediate. For each positive integer  $r$ , compactness and total boundedness ([Theorem 10.6.3](#)) give a finite cover of  $K$  by radius-

$$1/r$$

balls centered in  $K$ . The union of their finite sets of centers is a countable dense set  $\{d_1, d_2, \dots\}$  (repeat points if it is finite). Boundedness and Bolzano–Weierstrass in  $\mathbb{R}^m$  select a subsequence whose values converge at  $d_1$ . From it select one converging at  $d_2$ , and continue with nested subsequences. Taking the  $j$ th index of the  $j$ th subsequence gives increasing indices  $k_j$ ; each tail lies in every earlier selection. Thus  $f_{k_j}(d_i)$  converges for each fixed  $i$ .

Given  $\varepsilon > 0$ , choose the uniform equicontinuity scale for

$$\varepsilon/3.$$

The balls of that radius centered at the dense points  $d_i$  cover  $K$ ; compactness selects finitely many. Thus every point is less than that scale from one of the finitely many selected centers. At all these finitely many points, sufficiently late  $f_{k_p}, f_{k_q}$  differ by less than

$$\varepsilon/3.$$

For any  $\mathbf{x} \in K$  choose one such point  $d$  and write

$$\|f_{k_p}(\mathbf{x}) - f_{k_q}(\mathbf{x})\| \leq \|f_{k_p}(\mathbf{x}) - f_{k_p}(d)\| + \|f_{k_p}(d) - f_{k_q}(d)\| + \|f_{k_q}(d) - f_{k_q}(\mathbf{x})\| < \varepsilon.$$

This is uniform Cauchyness. Completeness of  $\mathbb{R}^m$  supplies pointwise limits, and passing  $q \rightarrow \infty$  in this common estimate proves uniform convergence. The uniform-limit theorem proves continuity.  $\square$

This generalizes the interval theorem of Chapter 6, [Theorem 6.4.5](#). Compactness and finite ball covers replace one-dimensional order. Its hypotheses require a compact domain  $K$ , one bound on every value of every map, and equicontinuity with a scale common to the family. A common Lipschitz constant supplies the third condition; later a common derivative bound on a neighborhood containing each connecting segment will supply such a constant. The conclusion concerns a uniformly convergent *subsequence*, not the entire original sequence.

## 11.2 Differentiability and the Derivative

With the little-o notation introduced in Chapter 5 and extended to norm-valued remainders in Chapter 10, one-variable differentiability at  $a$  may be written as

$$f(a+h) = f(a) + f'(a)h + o(|h|).$$

The remainder is negligible relative to the increment: after subtracting the best linear approximation, its size divided by  $|h|$  tends to zero. Multiplication by the scalar  $f'(a)$  is already a linear map of the increment  $h$ . For a map between higher-dimensional spaces, the correct replacement is a linear map  $Df(a)$ , and the derivative remains the best linear approximation to the change in  $f$ .

Here  $a$  is fixed,  $h \in V$  is an input increment, and  $f(a+h) - f(a) \in W$  is an output increment. We cannot divide one vector by another. Instead propose a linear map  $L : V \rightarrow W$ , subtract  $Lh$ , and compare the scalar sizes of the error and the input increment. Openness of  $U$  makes all sufficiently small increments admissible in every direction. The map  $h \mapsto Df(a)h$  approximates the output *change*;  $x \mapsto f(a) + Df(a)(x-a)$  is the base-point form of an affine map from [Definition 10.10.3](#), and is the affine approximation to the function itself. “Best” means first-order asymptotic approximation, not least-squares fitting.

We formulate the ordinary ambient Fréchet derivative at interior points of the domain. Assuming  $U$  open makes this setup available at every point and guarantees that sufficiently small increments are admissible in every ambient direction. This also makes the derivative unique: the proof tests  $h = tv$  for arbitrary  $v$ , with  $a + tv \in U$  for all sufficiently small  $t$ .

Without an interior condition, an ambient linear approximation need not be determined. For

$$U = \{(x, 0) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2, \quad f \equiv 0,$$

every linear map  $L_c(x, y) = cy$  vanishes on all admissible increments in  $U$ . Thus the data on  $U$  cannot distinguish the maps  $L_c$ . For maps on smooth submanifolds, the natural derivative acts on the tangent space; that theory is developed in Chapter 13. Other relative notions of differentiability on arbitrary subsets are possible, but are not developed here.

**Definition 11.2.1** (Fréchet derivative). Let  $V, W$  be finite-dimensional normed real vector spaces, let  $U \subseteq V$  be open, and let

$$f : U \longrightarrow W.$$

The function  $f$  is **differentiable at**  $a \in U$  if there is a linear map

$$L : V \longrightarrow W$$

such that

$$\lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - Lh\|_W}{\|h\|_V} = 0.$$

The map  $L$ , if it exists, is denoted by

$$Df(a)$$

and is called the **derivative** of  $f$  at  $a$ .

The limit in the definition means explicitly that for every  $\varepsilon > 0$  there is  $\delta > 0$  such that

$$0 < \|h\|_V < \delta \implies \|f(a+h) - f(a) - Lh\|_W < \varepsilon \|h\|_V.$$

**Example 11.2.2** (Affine maps). For  $f(x) = Ax + b$ , with  $A : V \rightarrow W$  linear and  $b \in W$ ,  $f(a+h) - f(a) = Ah$ . Thus  $Df(a) = A$  with zero remainder.

**Example 11.2.3** (A derivative with its error visible). For  $f(x, y) = (x^2 + y, xy)$  at  $a = (1, 1)$ , put  $h = (u, v)$ . Then

$$f(a+h) - f(a) = (2u + v + u^2, u + v + uv) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + (u^2, uv).$$

The remainder satisfies  $\|(u^2, uv)\| = |u|\sqrt{u^2 + v^2} \leq \|h\|^2$ . For  $h \neq 0$  its relative size is at most  $\|h\|$ , which tends to zero. Thus the displayed matrix represents  $Df(1, 1)$ ; it is the Jacobian (formally defined in [Definition 11.2.9](#) below), and the estimate proves the approximation for every small increment at once.

**Proposition 11.2.4** (The one-dimensional case). For a function  $f : U \subseteq \mathbb{R} \rightarrow \mathbb{R}$ , Fréchet differentiability at  $a \in U$  is equivalent to differentiability in the sense of Chapter 8. Under this equivalence,

$$Df(a)h = f'(a)h.$$

*Proof.* Every linear map  $L : \mathbb{R} \rightarrow \mathbb{R}$  has the form

$$Lh = ch$$

with  $c = L(1)$ . The Fréchet remainder condition is therefore exactly

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = c,$$

after separating the harmless case  $h = 0$ . This is the ordinary derivative definition, and the asserted formula follows.  $\square$



**Remark 11.2.5.** Equivalently,

$$f(a+h) = f(a) + Df(a)h + r(h), \quad \frac{\|r(h)\|_W}{\|h\|_V} \longrightarrow 0.$$

By [Theorem 10.14.7](#), this definition is independent of the norms chosen on finite-dimensional spaces.

**Proposition 11.2.6** (Uniqueness and continuity). If  $f$  is differentiable at  $a$ , then  $Df(a)$  is unique and  $f$  is continuous at  $a$ .

*Proof.* If  $L, M$  both have the required remainder property, then, for  $h \neq 0$ ,

$$\frac{\|(L-M)h\|_W}{\|h\|_V} \leq \frac{\|f(a+h) - f(a) - Lh\|_W}{\|h\|_V} + \frac{\|f(a+h) - f(a) - Mh\|_W}{\|h\|_V}.$$

Fix any nonzero  $v \in V$  and put  $h = tv$ . For  $t \neq 0$ , linearity gives

$$\frac{\|(L-M)(tv)\|_W}{\|tv\|_V} = \frac{\|(L-M)v\|_W}{\|v\|_V}.$$

The vector  $v$  need not be small: since  $a$  is an interior point, scaling by sufficiently small  $t$  makes  $a+tv \in U$  and brings this fixed direction into the small-increment regime. The left side is bounded by a quantity tending to zero as  $t \rightarrow 0$ , whereas the right side is independent of  $t$ . It must therefore vanish. This gives

$$(L-M)v = 0,$$

so  $L = M$ . Write  $f(a+h) - f(a) = Df(a)h + r(h)$ . Boundedness of the linear map from [Proposition 10.15.1](#) gives

$$\|f(a+h) - f(a)\|_W \leq \|Df(a)\|_{\text{op}}\|h\|_V + \|r(h)\|_W \rightarrow 0,$$

which proves continuity. □

**Proposition 11.2.7** (Algebra and the chain rule). Let  $f, g : U \rightarrow W$  be differentiable at  $a$  and let  $c \in \mathbb{R}$ . Then

$$D(f+g)(a) = Df(a) + Dg(a), \quad D(cf)(a) = cDf(a).$$

For real-valued  $f, g$ ,

$$D(fg)(a) = f(a)Dg(a) + g(a)Df(a).$$

If  $g(a) \neq 0$ , then

$$D\left(\frac{f}{g}\right)(a) = \frac{g(a)Df(a) - f(a)Dg(a)}{g(a)^2}.$$

If  $f : U \rightarrow W$  is differentiable at  $a$  and  $g : Z \rightarrow X$  is differentiable at  $f(a)$ , with  $f(a) \in Z$  and  $Z$  open, then

$$D(g \circ f)(a) = Dg(f(a)) \circ Df(a).$$

*Proof.* Adding and scaling the remainder formulas proves the first two identities. We use the norm-valued estimates of [Proposition 10.16.10](#); each derivative remainder is defined to be zero at increment zero. For products, expand

$$f(a+h)g(a+h) - f(a)g(a)$$

by adding and subtracting

$$f(a)g(a+h).$$

The product of the two increments is

$$o(\|h\|_V)$$

because each differentiable increment is  $O(\|h\|)$ . To establish the reciprocal, set  $c = g(a) \neq 0$  and  $\Delta = g(a+h) - c = Dg(a)h + o(\|h\|)$ . Continuity bounds  $|c + \Delta|$  away from zero. The exact identity

$$\frac{1}{c + \Delta} - \frac{1}{c} = -\frac{\Delta}{c^2} + \frac{\Delta^2}{c^2(c + \Delta)}$$

has final term  $O(\|h\|^2)$ , so the reciprocal has derivative

$$-Dg(a)/c^2.$$

Applying the already proved product rule now gives the quotient.

For the chain rule set  $A = Df(a)$  and  $B = Dg(f(a))$ . Write both remainder formulas with their own input variables:

$$f(a+h) = f(a) + Ah + r(h), \quad g(f(a)+k) = g(f(a)) + Bk + s(k),$$

where  $r(h) = o(\|h\|)$  and  $s(k) = o(\|k\|)$ . Substituting  $k = Ah + r(h)$  gives the complete expression

$$g(f(a+h)) = g(f(a)) + BAh + Br(h) + s(Ah + r(h)).$$

The candidate derivative is the linear map  $BA$ . Its remainder is  $Br(h) + s(Ah + r(h))$ . Boundedness of  $A$  and the first remainder bound give  $\|Ah + r(h)\| = O(\|h\|)$ . Boundedness of  $B$  gives  $Br(h) = o(\|h\|)$ , and the composition remainder rule of [Proposition 10.16.10](#) gives  $s(Ah + r(h)) = o(\|h\|)$ . Their sum is therefore  $o(\|h\|)$ , proving the asserted derivative.  $\square$

**Remark 11.2.8.** The chain rule says that local linear approximations compose:

$$f(a+h) \approx f(a) + Df(a)h$$

and then

$$g(f(a)+k) \approx g(f(a)) + Dg(f(a))k.$$

Thus the linear part of  $g \circ f$  is the composite

$$Dg(f(a))Df(a).$$

After coordinates are chosen, composition of linear maps is matrix multiplication, which is why that operation appears in the Jacobian formula.

Fix  $v \in \mathbb{R}^n$  and restrict  $f$  to the line through  $a$ :  $g_v(t) = f(a + tv)$ , defined for scalar  $t$  in an open interval about zero, with values in  $\mathbb{R}^m$ . Then  $D_v f(a) = g'_v(0)$ . The vector  $v$  stays fixed while  $t$  varies. It specifies both direction and speed; a unit vector measures rate per unit distance. The one-variable limit is taken componentwise when the output is a vector.

**Definition 11.2.9** (Directional derivatives and Jacobian). Let  $U \subseteq \mathbb{R}^n$  be open and write  $f = (f_1, \dots, f_m) : U \rightarrow \mathbb{R}^m$ . The **directional derivative** at  $a$  in direction  $v$  is

$$D_v f(a) = \lim_{t \rightarrow 0} \frac{f(a + tv) - f(a)}{t},$$

when this limit exists. The partial derivative with respect to  $x_j$  is

$$\frac{\partial f}{\partial x_j}(a) = D_{e_j} f(a).$$

Whenever all scalar partial derivatives  $\partial_j f_i(a)$  exist, the **Jacobian matrix** is the  $m$ -by- $n$  matrix

$$J_f(a) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(a) & \cdots & \frac{\partial f_1}{\partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(a) & \cdots & \frac{\partial f_m}{\partial x_n}(a) \end{pmatrix}.$$

If  $f$  is differentiable at  $a$ , the next proposition proves  $J_f(a) = [Df(a)]_{\text{std}}$ . Existence of  $J_f(a)$  alone does not imply existence of  $Df(a)$ : its entries give only a candidate linear map.

**Compute a partial derivative on a coordinate slice.** Fix  $a = (a_1, \dots, a_n)$  and a coordinate  $j$ , and define

$$g_j(t) = f(a_1, \dots, a_{j-1}, t, a_{j+1}, \dots, a_n) = f(a + (t - a_j)e_j).$$

Here  $t$  varies near  $a_j$  and every other coordinate stays fixed. Substitution in the definition gives  $\partial_j f(a) = g'_j(a_j)$ . For scalar output this is an ordinary real derivative. For vector output, differentiate each scalar component:

$$\partial_j f(a) = \begin{pmatrix} \partial_j f_1(a) \\ \vdots \\ \partial_j f_m(a) \end{pmatrix} \in \mathbb{R}^m.$$

Thus to compute a partial, freeze *every other coordinate* and use the one-variable rules. The next result explains how this calculation identifies a basis image once total differentiability is known.

**Proposition 11.2.10** (Derivative and directional derivatives). If  $f$  is differentiable at  $a$ , then

$$D_v f(a) = Df(a)v$$

for every  $v \in \mathbb{R}^n$ . In particular, its Jacobian is the matrix displayed above.

*Proof.* For  $v = 0$  the parameter function is constant and both sides are zero. For fixed  $v \neq 0$ , write  $f(a + h) - f(a) = Df(a)h + r(h)$ . Put  $h = tv$ . For  $t \neq 0$ ,

$$\frac{\|r(tv)\|}{|t|} = \frac{\|r(tv)\|}{\|tv\|} \|v\| \longrightarrow 0.$$

Divide the remainder formula by  $t$  and take the limit. Taking  $v = e_j$  identifies the columns of the matrix.  $\square$

**Read the Jacobian as a table of basis images.** When  $f$  is differentiable, its derivative is the linear map  $Df(a) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ ; its standard matrix is  $J_f(a) = [Df(a)]_{\text{std}}$ . Taking  $v = e_j$  in [Proposition 11.2.10](#) gives

$$Df(a)e_j = \partial_j f(a) = \begin{pmatrix} \partial_j f_1(a) \\ \vdots \\ \partial_j f_m(a) \end{pmatrix}.$$

Thus column  $j$  records the response of *every output* to the  $j$ th input direction. Row  $i$  records the response of the scalar output  $f_i$  to all input directions. There are  $m$  rows because outputs have  $m$  coordinates, and  $n$  columns because the input basis has  $n$  vectors. For

$$h = \sum_j h_j e_j,$$

linearity now explains matrix multiplication:

$$Df(a)h = \sum_{j=1}^n h_j Df(a)e_j = \sum_{j=1}^n h_j \partial_j f(a) = J_f(a)h.$$

The computational chain is therefore: choose the varying coordinate, freeze the others, differentiate the slice, and place the resulting vector in column  $j$ , since it is  $Df(a)e_j$ . The columns are weighted by the coordinates of the increment. Accordingly,

$$f(a+h) = f(a) + J_f(a)h + o(\|h\|).$$

The matrix acts on  $h$  to approximate the *change*; adding  $f(a)$  produces the affine prediction for the function value.

$Df(a)$  is the intrinsic linear map.  $J_f(a)$  is its table in the chosen standard bases. Changing bases changes the table, by Chapter 10's coordinate rules, while retaining the same derivative map.

## Coordinate projections and differentials

**Remark: familiar notation, new structure.** Earlier chapters used

$$d/dx, \quad \int f(x) \, dx$$

operationally: we could differentiate and integrate correctly before studying  $dx$  as an object. We now begin to explain the linear structure behind this notation. This does not make every earlier integration symbol a covector; the  $dx$  in a scalar Riemann or Lebesgue integral specifies the variable or measure. The tangent/cotangent viewpoint in [Section 13.11](#) will explain the geometric differential. The later theory explains familiar valid procedures rather than correcting them.

**Definition 11.2.11** (Coordinate differentials and the scalar differential). Let  $\pi_i : \mathbb{R}^n \rightarrow \mathbb{R}$  be the coordinate projection  $\pi_i(x_1, \dots, x_n) = x_i$ . Since  $\pi_i$  is linear,  $D\pi_i(a)v = \pi_i(v) = v_i$  for every base point  $a$ . Define

$$(dx_i)_a := d\pi_i|_a := D\pi_i(a), \quad (dx_i)_a(v) = v_i.$$

This derivative is independent of  $a$ , so in Euclidean coordinates we suppress the base point and write  $dx_i = d\pi_i = e^i$ , the standard dual basis from [Definition 10.16.2](#). For a scalar function  $f$

differentiable at  $a$ , write  $Df(a) \in (\mathbb{R}^n)^*$ . When  $f$  is differentiable on an open set, write  $df$  for the covector field with value  $(df)_a = Df(a)$ . For smooth  $f$  this is its smooth exact 1-form. This notation anticipates Chapter 13;  $D$  denotes the Euclidean derivative and  $d$  the operation forming its exact differential form.

The symbol  $dx_i$  denotes a covector  $\mathbb{R}^n \rightarrow \mathbb{R}$ , not a small number or an infinitesimal displacement. Likewise  $(df)_a$  is the same Fréchet derivative, with its covector type emphasized, not a new derivative. Expanding in the dual basis gives

$$(df)_a = \sum_{i=1}^n Df(a)e_i \, dx_i = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) \, dx_i.$$

The partial derivatives are exactly the coordinates of this covector.

**Example 11.2.12** (A differential and an actual change). For  $f(x, y) = x^2 + 2y^2$  and  $a = (1, 1)$ ,

$$(df)_a = 2 \, dx + 4 \, dy.$$

If  $v = (1, 2)$ , then  $dx(v) = 1$ ,  $dy(v) = 2$ , and  $(df)_a(v) = 10$ . For a real increment parameter  $t$ , direct expansion gives

$$f(a + tv) - f(a) = (1 + t)^2 + 2(1 + 2t)^2 - 3 = 10t + 9t^2, \quad (df)_a(tv) = 10t.$$

Here  $(df)_a$  is a covector,  $(df)_a(v)$  a scalar evaluation, and  $f(a + tv) - f(a)$  an actual finite change. The differential captures its first-order part, not generally the whole change.

For the line,  $\pi(x) = x$  and  $dx = d\pi$  is the standard basis of  $\mathbb{R}^*$ . Thus  $(df)_x = f'(x) \, dx$ . The notation

$$df/dx = f'(x)$$

says that  $f'(x)$  is the coefficient of  $df$  relative to  $dx$ ; it is not an ordinary quotient of real numbers and does not justify arbitrary fraction manipulations.

Coordinate differentials recur because they provide the coordinate basis for scalar derivatives. Chapter 13 will build alternating forms from their wedges and explain their transformation by pullback. Schematically,

$$dx_i \longrightarrow df \longrightarrow dx_i \wedge dx_j \longrightarrow dx_1 \wedge \cdots \wedge dx_n \longrightarrow \text{pullback and integration.}$$

The top wedge measures oriented coordinate volume; its transformation produces a determinant. Line, surface, and manifold integrals will pair scalar coefficient functions with these coordinate forms. This is a preview, not yet a definition of form integration.

**Definition 11.2.13** (Euclidean gradient). Let  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  be differentiable at  $a$ . Its **gradient** is the coordinate vector

$$\nabla f(a) = \left( \frac{\partial f}{\partial x_1}(a), \dots, \frac{\partial f}{\partial x_n}(a) \right).$$

Thus the invariant derivative has the coordinate form

$$Df(a)h = \nabla f(a) \cdot h.$$

For a unit vector  $v$ ,  $D_v f(a) = \nabla f(a) \cdot v \leq \|\nabla f(a)\|$  by Cauchy–Schwarz. If the gradient is nonzero, equality holds for

$$v = \nabla f(a) / \|\nabla f(a)\|.$$

Thus it points toward greatest first-order increase, and its norm is the maximal unit directional derivative. This representation depends on the Euclidean inner product; the intrinsic derivative remains the covector  $Df(a)$ .

The derivative  $Df(a)$  is a linear functional, independent of coordinates. Its Jacobian is a row matrix after bases are chosen. The gradient is a vector representing that functional through the Euclidean inner product:  $Df(a)h = \langle \nabla f(a), h \rangle$ . Changing the inner product changes this vector representation, not the underlying functional. This distinction was introduced through dual spaces in Chapter 10 and will govern differential forms in Chapter 13.

### The chain rule along a parametrized curve

**Definition 11.2.14** (Differentiable curve and velocity). Recall the plane curves of Definition 8.7.1. For an interval  $I$ , the same coordinate description extends to a map  $\gamma : I \rightarrow \mathbb{R}^n$ . With differentiable coordinates, write

$$\gamma(t) = (x_1(t), \dots, x_n(t)), \quad \gamma'(t) = (x'_1(t), \dots, x'_n(t)).$$

Each coordinate remainder is  $o(|h|)$ ; the coordinate norm estimates combine them to give  $\gamma(t+h) - \gamma(t) = h\gamma'(t) + o(|h|)$ . Thus coordinatewise differentiation agrees with the Fréchet derivative:  $\gamma'(t) = D\gamma(t)1$  is the velocity. Statements at endpoints use one-sided derivatives when these exist.

**Proposition 11.2.15** (Chain rule along a curve). Let  $U \subseteq \mathbb{R}^n$  be open,  $f : U \rightarrow \mathbb{R}^m$  differentiable, and  $\gamma : I \rightarrow U$  a differentiable curve. Then

$$(f \circ \gamma)'(t) = Df(\gamma(t))\gamma'(t).$$

*Proof.* The chain rule gives  $D(f \circ \gamma)(t) = Df(\gamma(t))D\gamma(t)$ . Evaluate on  $1 \in \mathbb{R}$ . □

At fixed time  $t$ , the derivative transports input velocity to output velocity:

$$\underbrace{\gamma'(t) \in \mathbb{R}^n}_{\text{input velocity}} \xrightarrow{Df(\gamma(t))} \underbrace{(f \circ \gamma)'(t) \in \mathbb{R}^m}_{\text{output velocity}}.$$

For scalar output, Definition 11.2.11 and the Euclidean gradient representation give

$$\frac{d}{dt}f(\gamma(t)) = (df)_{\gamma(t)}(\gamma'(t)) = \nabla f(\gamma(t)) \cdot \gamma'(t) = \sum_i \partial_i f(\gamma(t))x'_i(t),$$

since  $dx_i(\gamma'(t)) = x'_i(t)$ . For the function  $f(x, y) = x^2 + 2y^2$ , take  $\gamma(t) = (t, t^2)$ . The formula gives  $2t + 4t^2(2t) = 2t + 8t^3$ , agreeing with direct differentiation of  $t^2 + 2t^4$ .

In two variables, the same chain rule reads

$$\frac{d}{dt}f(x(t), y(t)) = f_x(x(t), y(t))x'(t) + f_y(x(t), y(t))y'(t).$$

The partial derivatives measure change in the coordinate directions; the velocity selects the combination followed by this curve. Writing

$$df = f_x dx + f_y dy, \quad dx = x'(t) dt, \quad dy = y'(t) dt \quad \text{along the curve}$$

is compatible with that calculation: substitution gives  $(f_x x' + f_y y') dt$ , with the partials evaluated at  $\gamma(t)$ . Here the last two equations are shorthand for transporting differentials to the parameter interval, not identities between objects on different domains. Chapter 13 makes this precise using pullbacks in Sections 13.7 and 13.12.

**Why a gradient is normal to a level curve.** Let  $f$  be differentiable and let a differentiable curve  $\gamma$  lie in the level set  $f(\gamma(t)) = c$ . By Proposition 11.2.15,

$$0 = (f \circ \gamma)'(0) = Df(\gamma(0))\gamma'(0) = \nabla f(\gamma(0)) \cdot \gamma'(0).$$

At every parameter value,  $\gamma'(t) \in \ker Df(\gamma(t))$ : the derivative covector annihilates tangent velocities to the level set. Thus every tangent velocity is orthogonal to the gradient. At a regular point, where the gradient is nonzero, it supplies a normal direction. Equivalently the directional derivative along the tangent velocity is zero. This uses velocities of curves, without introducing tangent-space theory.

**Example 11.2.16** (A rectangular Jacobian). For  $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ ,  $f(x, y, z) = (xy + z, x^2 - yz)$ , computing partials gives the candidate matrix and linear map

$$J_f(a, b, c) = \begin{pmatrix} b & a & 1 \\ 2a & -c & -b \end{pmatrix}, \quad L \begin{pmatrix} h \\ k \\ \ell \end{pmatrix} = \begin{pmatrix} bh + ak + \ell \\ 2ah - ck - b\ell \end{pmatrix}.$$

To justify it directly, put  $u = (h, k, \ell)$  and expand:

$$f((a, b, c) + u) - f(a, b, c) - J_f(a, b, c)u = \begin{pmatrix} hk \\ h^2 - k\ell \end{pmatrix}.$$

Since

$$|hk| \leq \|u\|^2/2$$

and  $|h^2 - k\ell| \leq \|u\|^2$ , the remainder norm is at most  $2\|u\|^2$ . Dividing by  $\|u\|$  gives zero in the limit. Thus  $Df(a, b, c) = L$  is the total derivative, without needing a continuous-partial theorem in advance. The third column  $(1, -b)^T$  measures both outputs' response to the third input. A 2-by-3 table is exactly what converts three input increment coordinates into two output increment coordinates.

For  $c \neq 0$ , substitution  $s = ct$  in the directional limit gives  $D_{cv}f(a) = cD_vf(a)$  whenever  $D_vf(a)$  exists. For  $c = 0$  both sides are zero. Homogeneity alone does not imply additivity in  $v$ .

**Example 11.2.17** (A polynomial along a line). For  $f(x, y) = x^2 + xy$  at  $a = (1, 2)$  and fixed  $v = (r, s)$ ,  $g_v(t) = 3 + t(4r + s) + t^2(r^2 + rs)$ . Hence  $D_vf(a) = 4r + s$ , the value of the linear functional  $Df(a)$  on  $v$ .

**Proposition 11.2.18** (Componentwise differentiability). For  $f = (f_1, \dots, f_m) : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ ,  $f$  is differentiable at  $a$  if and only if every  $f_i$  is differentiable there. In that case  $Df(a)h = (Df_1(a)h, \dots, Df_m(a)h)$ .

*Proof.* If the vector remainder is  $r(h) = o(\|h\|)$ , then  $|r_i(h)| \leq \|r(h)\|$ , so coordinate projection yields each scalar derivative. Conversely assemble the scalar derivatives into the linear map  $Lh = (Df_i(a)h)_i$ . For its remainder,  $\|r(h)\| \leq \sqrt{m} \max_i |r_i(h)| = o(\|h\|)$ , using a common neighborhood for the finitely many components.  $\square$

Reducing *output components* is not the same as testing *input coordinates*: each  $f_i$  must still be differentiable in all  $n$  variables.

**Example 11.2.19** (The routine continuous-partials calculation). For  $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $f(x, y) = (xe^y, x^2 + y^2)$ , the partials are

$$\partial_x f = (e^y, 2x), \quad \partial_y f = (xe^y, 2y).$$

All four scalar entries are continuous. The continuous-partials theorem proved below therefore gives differentiability everywhere, with

$$J_f(a, b) = \begin{pmatrix} e^b & ae^b \\ 2a & 2b \end{pmatrix}, \quad Df(a, b)(h, k) = \begin{pmatrix} e^b h + ae^b k \\ 2ah + 2bk \end{pmatrix}.$$

Its columns are  $(e^b, 2a)^\top$  and  $(ae^b, 2b)^\top$ . The displayed output increment is  $h$  times the first column plus  $k$  times the second. The affine prediction is therefore

$$f(a + h, b + k) = \begin{pmatrix} ae^b \\ a^2 + b^2 \end{pmatrix} + h \begin{pmatrix} e^b \\ 2a \end{pmatrix} + k \begin{pmatrix} ae^b \\ 2b \end{pmatrix} + o(\sqrt{h^2 + k^2}).$$

For example at  $(a, b) = (1, 0)$  this predicts  $f(1 + h, k) \approx (1 + h + k, 1 + 2h)$ . The base value and the linear response have distinct roles in that prediction.

**Composition and determinant.** For differentiable  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$  and  $g : \mathbb{R}^m \rightarrow \mathbb{R}^p$  (or maps on appropriate open subsets), the chain rule in standard coordinates is

$$\underbrace{J_{g \circ f}(a)}_{p \times n} = \underbrace{J_g(f(a))}_{p \times m} \underbrace{J_f(a)}_{m \times n}.$$

This is the coordinate form of  $D(g \circ f)(a) = Dg(f(a)) \circ Df(a)$ : the input increment passes through  $Df(a)$  first. Matching the middle dimension and evaluating  $J_g$  at  $f(a)$  catch two common errors. Every differentiable map has a Jacobian matrix, including rectangular ones. Only when  $m = n$  can we form its *Jacobian determinant*  $\det J_f(a)$ . Some books call either object “the Jacobian”; here we specify which is meant. The progression into integration will be the derivative map, its matrix, its determinant, and the absolute determinant as the local volume factor.

**Example 11.2.20** (The polar Jacobian and first-order area). Return to the polar map  $\sigma(r, \theta) = (r \cos \theta, r \sin \theta)$  of Chapter 8. For  $r > 0$ , the columns are

$$\partial_r \sigma = (\cos \theta, \sin \theta), \quad \partial_\theta \sigma = (-r \sin \theta, r \cos \theta), \quad J_\sigma = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}.$$

With our Euclidean convention,

$$\|\partial_r \sigma\| = 1, \quad \|\partial_\theta \sigma\| = r, \quad \partial_r \sigma \cdot \partial_\theta \sigma = 0.$$

Chapter 8’s [Proposition 8.7.5](#) already separated radial and angular velocity contributions. They are now the two coordinate-direction derivatives of a map of two variables. The trigonometric derivative rules and the continuous-partials theorem proved below justify differentiability. Its linear approximation sends a small coordinate rectangle of sides  $\Delta r, \Delta \theta$  to a rectangle of sides  $\Delta r, r \Delta \theta$ , of area  $r \Delta r \Delta \theta$ . This is first-order geometric motivation; Chapter 12 proves the integral change-of-variables theorem using  $|\det D\sigma| = r$ . In the order  $(r, \theta)$ ,  $\det D\sigma = r$ ; swapping to  $(\theta, r)$  swaps columns and gives  $-r$ . Orientation changes, but the scalar area factor remains  $r$ .



**Jacobian checklist.** (1) Identify input and output dimensions. (2) List the scalar components. (3) Compute  $\partial_j f_i$  in row  $i$ , column  $j$ . (4) Evaluate at the stated base point. (5) Multiply by an input increment. (6) Add the base value for an affine prediction. (7) For composition, match dimensions and evaluate at the intermediate point. (8) Take a determinant only for a square matrix. These computations represent a derivative only after differentiability has been justified, for example by continuous partials.

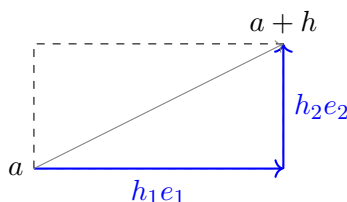
**Example 11.2.21** (Partial derivatives need not give differentiability). Define

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Both partial derivatives at the origin exist and equal zero, since  $f$  vanishes on both coordinate axes. But along the line  $y = x$ ,

$$f(x, x) = \frac{1}{2}$$

for  $x \neq 0$ ; hence  $f$  is not even continuous at the origin and cannot be differentiable there. Continuous partial derivatives, as in the next theorem, are sufficient because they control all nearby coordinate increments.



The continuous-partial proof walks from  $a$  to  $a+h$  one coordinate at a time. Each segment admits the one-variable mean value theorem. All intermediate points remain within  $\|h\|$  of  $a$ , so continuity of the finitely many partials supplies one error bound for all segments. That common neighborhood is what upgrades coordinate information to a uniform first-order approximation.

**Theorem 11.2.22** (Continuous partial derivatives). Let  $U \subseteq \mathbb{R}^n$  be open. If all partial derivatives of

$$f : U \longrightarrow \mathbb{R}^m$$

exist on a neighborhood of  $a$  and are continuous at  $a$ , then  $f$  is differentiable at  $a$  and its derivative is represented in standard coordinates by  $J_f(a)$ ; explicitly  $Df(a)h = J_f(a)h$ .

*Proof.* By Proposition 11.2.18, it is enough to consider a real-valued component. Choose  $r > 0$  so that  $B(a, r)$  lies in the neighborhood where all partial derivatives exist. Telescope the increment from  $a$  to  $a+h$  along the coordinate polygonal path. Put

$$p_0 = a, \quad p_j = a + h_1e_1 + \cdots + h_je_j \quad (1 \leq j \leq n).$$

If

$$\|h\|_2 < r/\sqrt{n},$$

every point  $q$  on one of these segments satisfies

$$\|q - a\|_2 \leq |h_1| + \cdots + |h_n| \leq \sqrt{n} \|h\|_2 < r.$$

Thus the entire coordinate polygonal path lies in  $B(a, r)$ . If  $h_j = 0$ , that segment contributes zero; set  $a_j = p_{j-1}$  and do not apply the Mean Value Theorem. For  $h_j \neq 0$ , applying the one-variable Mean Value Theorem to the function that varies only its  $j$ th coordinate gives a point

$$a_j = p_{j-1} + t_j h_j e_j \quad (0 < t_j < 1)$$

such that

$$f(a + h) - f(a) = \sum_{j=1}^n \frac{\partial f}{\partial x_j}(a_j) h_j.$$

Each  $a_j$  tends to  $a$  with  $h$ . Subtract the same sum with  $a_j$  replaced by  $a$ . Given  $\varepsilon > 0$ , continuity of the partial derivatives makes the absolute value of the difference at most

$$\varepsilon \sum_{j=1}^n |h_j| \leq \varepsilon \sqrt{n} \|h\|.$$

This is the required remainder estimate. Apply it componentwise.  $\square$

**Definition 11.2.23** (Class  $C^1$ ). A differentiable map on an open set is of class

$$C^1$$

when its derivative is continuous as a map into the finite-dimensional space of linear maps. This extends the one-variable terminology of [Definition 8.1.4](#): ordinary derivatives are replaced by Fréchet derivatives between finite-dimensional spaces.

Here  $Df : U \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$  is itself a function, with linear maps as values. Its continuity at  $a$  means  $\|Df(\mathbf{x}) - Df(a)\|_{\text{op}} \rightarrow 0$ .

**Proposition 11.2.24** ( $C^1$  and continuous partials). A map  $f : U \rightarrow \mathbb{R}^m$  is  $C^1$  if and only if all its first scalar partial derivatives exist and are continuous on  $U$ .

*Proof.* Continuous partials give differentiability by [Theorem 11.2.22](#). Continuity of every matrix entry is equivalent to operator-norm continuity because matrix space is finite dimensional. Conversely a continuous derivative has continuous entries  $\partial_j f_i$ , by coordinate evaluation.  $\square$

**Example 11.2.25** (Differentiable need not mean  $C^1$ ). Reuse

$$\phi(t) = t^2 \sin(1/t)$$

for  $t \neq 0$ ,  $\phi(0) = 0$ , from Chapter 8, and put  $f(x, y) = \phi(x)$ . At zero,  $|f(h)| \leq h_1^2 \leq \|h\|^2$ , so  $Df(0, 0) = 0$ . Away from  $x = 0$  the usual rules apply; at every  $(0, y)$  the same remainder bound proves differentiability. But for  $x \neq 0$ ,

$$\partial_x f = 2x \sin(1/x) - \cos(1/x),$$

which does not approach zero. Thus  $f$  is differentiable everywhere and is not  $C^1$  near the origin.

If a total derivative exists, [Proposition 11.2.10](#) determines its values on every basis vector. A linear map is determined by those values, so the partial derivatives supply the *only possible* derivative  $L(h) = J_f(a)h$ . Computing this candidate is not a proof of differentiability.

**How to find the total derivative.**

1. Compute the partial derivative functions and evaluate them at  $a$ .
2. Assemble their columns into the candidate matrix  $J_f(a)$ .
3. Justify differentiability: use nearby partials continuous at  $a$  (Theorem 11.2.22), the derivative rules and chain rule, or a direct estimate of the Fréchet remainder at an exceptional point.
4. Only then conclude  $Df(a)h = J_f(a)h$ .

For scalar output this reads  $Df(a)h = \nabla f(a) \cdot h$ . The gradient is the Euclidean vector representing the derivative covector; the intrinsic derivative is the linear functional, not that vector. In the earlier computation of  $(xe^y, x^2 + y^2)$ , the two partial functions produce the columns and their continuity justifies the derivative.

**Example 11.2.26** (Continuous with partials, but not differentiable). Set  $f(0, 0) = 0$  and

$$f(x, y) = x^3/(x^2 + y^2)$$

elsewhere. The bound

$$|f(x, y)| \leq |x| \leq \sqrt{x^2 + y^2}$$

proves continuity at zero. On the axes  $f(t, 0) = t$  and  $f(0, t) = 0$ , so  $\partial_x f(0) = 1$ ,  $\partial_y f(0) = 0$ . The only candidate is  $L(h, k) = h$ . But

$$f(t, t) = t/2$$

for  $t \neq 0$ , and hence

$$\frac{|f(t, t) - L(t, t)|}{\|(t, t)\|} = \frac{1}{2\sqrt{2}}.$$

The normalized error does not tend to zero. Without absolute values the quotient is

$$-1/(2\sqrt{2})$$

for  $t > 0$  and has the opposite sign for  $t < 0$ ; either one-sided approach disproves differentiability.

The earlier example

$$xy/(x^2 + y^2)$$

showed that partials alone need not even imply continuity. This example shows that adding continuity still does not prove total differentiability. By contrast, nearby partials continuous at the point do suffice. Partial test coordinate directions; differentiability requires one linear approximation controlling all sufficiently small increments at once.

**Example 11.2.27** (A direct remainder estimate). Define  $f(0, 0) = 0$  and, away from the origin,

$$f(x, y) = \frac{x^2 y^2}{x^2 + y^2}.$$

The candidate derivative at the origin is zero. Since  $4x^2 y^2 \leq (x^2 + y^2)^2$ ,

$$\frac{|f(x, y)|}{\|(x, y)\|} \leq \frac{1}{4} \|(x, y)\| \longrightarrow 0.$$

Hence  $f$  is differentiable at the origin by the definition, even without first proving continuity of its nearby partial derivatives.

The counterexamples form a hierarchy of increasingly demanding tests:

$$\begin{aligned} & \text{partials} < \text{all directional derivatives} \\ & < \text{linear directional data} < \text{uniform linear approximation.} \end{aligned}$$

Here each step means a stronger requirement, not a numerical inequality. Partial derivatives test coordinate lines only. Directional derivatives test each fixed line, but their values might not depend linearly on direction. Even linear directional data can fail to control directions that change with the size of the increment. Differentiability requires one remainder estimate valid for all sufficiently small increments simultaneously.

**Remark 11.2.28** (The same example tests all directions). For the function in [Example 11.2.26](#) and  $v \neq 0$ ,

$$D_v f(0) = \frac{v_1^3}{v_1^2 + v_2^2}.$$

All directional derivatives exist, but they are not linear in  $v$ : the values on  $e_1, e_2, e_1 + e_2$  are

$$1, 0, 1/2.$$

They cannot be the values of one linear map  $Df(0)$ .

**Example 11.2.29** (Even linear directional data can fail). Let  $h(0, 0) = 0$  and

$$h(x, y) = x^3 y / (x^4 + y^2)$$

elsewhere. Since  $2|x^2 y| \leq x^4 + y^2$ ,

$$|h(x, y)| \leq |x|/2,$$

so  $h$  is continuous at zero. For  $v = (a, b)$  with  $b \neq 0$ ,

$$\frac{h(ta, tb)}{t} = \frac{ta^3 b}{t^2 a^4 + b^2} \rightarrow 0;$$

for  $b = 0$  it is identically zero. Every directional derivative is therefore zero; hence  $v \mapsto D_v h(0)$  is the zero linear functional. But

$$h(x, x^2) = x/2$$

for  $x \neq 0$ , and

$$|h(x, x^2)| / \|(x, x^2)\| \rightarrow 1/2.$$

Total differentiability requires uniform first-order control over all increments, including curved approaches.

In the preceding example  $h_t = (t, t^2)$  tends to zero while its direction changes with  $t$ . Fixed-line tests do not control this approach. For a proposed linear map  $L$ , directional information may give

$$[f(a + tv) - f(a)]/t \rightarrow Lv$$

for each fixed unit  $v$ , with the allowed size of  $t$  depending on  $v$ . Differentiability requires a single neighborhood:

$$\begin{aligned} & \forall \varepsilon > 0 \exists \delta > 0 \forall \|v\| = 1 \forall 0 < |t| < \delta : \\ & \left\| \frac{f(a + tv) - f(a)}{t} - Lv \right\| < \varepsilon. \end{aligned}$$

Indeed every nonzero increment has the form  $h = tv$  with  $v$  unit; this is exactly the normalized remainder condition.

For a scalar graph  $z = f(x, y)$  at  $a = (a_1, a_2)$ , Fréchet differentiability reads  $f(a + h) = f(a) + Df(a)h + o(\|h\|)$ . Its linear map is  $Df(a)$ ; the affine graph of  $h \mapsto f(a) + Df(a)h$  is the tangent plane

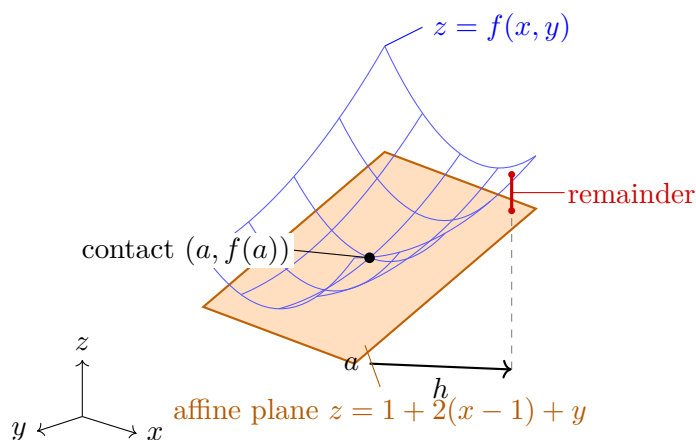
$$z - f(a_1, a_2) = f_x(a_1, a_2)(x - a_1) + f_y(a_1, a_2)(y - a_2).$$

The derivative is a map, whereas this plane is a set of points in  $\mathbb{R}^3$ .

**Example 11.2.30** (A surface and its affine approximation). Let  $f(x, y) = x^2 + xy + 2y^2$  and  $a = (1, 0)$ . Then  $f(a) = 1$  and  $Df(a)(h, k) = 2h + k$ . The tangent plane is the affine graph  $z = 1 + 2(x - 1) + y$ . The exact remainder is

$$f(1 + h, k) - 1 - (2h + k) = h^2 + hk + 2k^2, \quad |h^2 + hk + 2k^2| \leq \frac{5}{2}(h^2 + k^2).$$

At  $(h, k) = (0.01, 0.01)$  the linear increment is 0.03 and the remainder is 0.0004, compared with  $\|(h, k)\| \approx 0.01414$ . Dividing the bound by the increment norm gives a quantity tending to zero.



The orange plane and blue graph meet at the marked contact point. At a nearby input  $a + h$ , the red vertical segment measures their discrepancy,  $h_1^2 + h_1h_2 + 2h_2^2$ ; scaling  $h$  by  $t$  scales this error by  $t^2$ . The blue mesh therefore bends away from the plane quadratically.  $Df(a)$  is the linear part acting on  $(h, k)$ ; adding  $f(a)$  produces the affine graph.

For a Euclidean input the preceding notation specializes to

$$Df(\mathbf{x})h, \quad \nabla f(\mathbf{x}), \quad D^2f(\mathbf{x})[h, h], \quad h = \mathbf{x} - a.$$

The increment  $h$  and base point  $a$  keep their established notation.

### 11.3 Higher Derivatives and Local Estimates

The one-variable Mean Value and Taylor theorems describe a function by its successive local derivatives. In  $\mathbb{R}^n$ , restrict the function to a line segment and apply the Part I theorem. The resulting coefficients are evaluations of multilinear derivatives on the segment direction.

**Definition 11.3.1** (Convex subset). Extending the Euclidean definition in [Definition 10.7.3](#), a subset  $C$  of a real vector space  $V$  is **convex** if

$$(1-t)x + ty \in C$$

for every  $x, y \in C$  and every  $0 \leq t \leq 1$ . The interval property used in Chapter 8 is exactly the one-dimensional special case.

**Proposition 11.3.2** (Normed-space balls are convex). Every open or closed ball in a normed real vector space is convex.

*Proof.* If  $x, y \in \overline{B}(a, r)$  and  $0 \leq t \leq 1$ , then

$$\begin{aligned} \|(1-t)x + ty - a\| &= \|(1-t)(x-a) + t(y-a)\| \\ &\leq (1-t)\|x-a\| + t\|y-a\| \\ &\leq r. \end{aligned}$$

Thus  $\overline{B}(a, r)$  is convex. If  $x, y \in B(a, r)$ , the last bound is strict because both endpoint distances are strictly less than  $r$ , proving the open-ball case.  $\square$

**Theorem 11.3.3** (Scalar mean-value equality). For differentiable  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  on open  $U$  and  $[\mathbf{x}, y] \subset U$ , there is  $t \in (0, 1)$  such that

$$f(y) - f(\mathbf{x}) = Df(\mathbf{x} + t(y - \mathbf{x}))(y - \mathbf{x}).$$

*Proof.* Apply the ordinary mean value theorem to  $g(t) = f(\mathbf{x} + t(y - \mathbf{x}))$  on  $[0, 1]$ ; the chain rule gives  $g'(t) = Df(\mathbf{x} + t(y - \mathbf{x}))(y - \mathbf{x})$ .  $\square$

Scalar output admits this exact intermediate-point equality. For vector-valued functions there need not be a single intermediate point giving the scalar mean-value equality. For  $c(t) = (\cos t, \sin t)$  on  $[0, 2\pi]$ , the endpoint difference is zero but  $c'(t)$  is never zero. A norm estimate still bounds the change between endpoints.

**Theorem 11.3.4** (Mean-value estimate). Let  $U \subseteq \mathbb{R}^n$  be open, let  $f : U \rightarrow \mathbb{R}^m$  be differentiable, and suppose the segment from  $\mathbf{x}$  to  $y$  lies in  $U$ . If

$$\|Df((1-t)\mathbf{x} + ty)\|_{\text{op}} \leq M \quad (0 \leq t \leq 1),$$

then

$$\|f(y) - f(\mathbf{x})\| \leq M\|y - \mathbf{x}\|.$$

*Proof.* If  $f(y) \neq f(\mathbf{x})$ , put

$$u = \frac{f(y) - f(\mathbf{x})}{\|f(y) - f(\mathbf{x})\|}$$

The choice of this unit vector is made so that

$$u \cdot (f(y) - f(\mathbf{x})) = \frac{\|f(y) - f(\mathbf{x})\|^2}{\|f(y) - f(\mathbf{x})\|} = \|f(y) - f(\mathbf{x})\|.$$

Thus one scalar projection captures the full length of the difference. Keep  $u$  fixed while applying the one-variable Mean Value Theorem to

$$t \mapsto u \cdot f((1-t)\mathbf{x} + ty).$$

For some  $c \in (0, 1)$ ,

$$\|f(y) - f(\mathbf{x})\| = u \cdot Df((1-c)\mathbf{x} + cy)(y - \mathbf{x}).$$

Cauchy–Schwarz and the operator-norm bound prove the assertion. The zero case is immediate.  $\square$

**Corollary 11.3.5** (Local Lipschitz continuity). If  $U$  is convex and  $\|Df(\mathbf{x})\|_{\text{op}} \leq M$  on  $U$ , then  $f$  is  $M$ -Lipschitz on  $U$ . Every  $C^1$  map is locally Lipschitz.

*Proof.* Convexity keeps the segment between any  $\mathbf{x}, y \in U$  in  $U$ , so [Theorem 11.3.4](#) applies. For a  $C^1$  map, continuity of  $Df$  at any  $a$  supplies a ball in  $U$  where  $\|Df(\mathbf{x})\|_{\text{op}} \leq \|Df(a)\|_{\text{op}} + 1$ . Apply the first assertion on that convex ball, using [Proposition 11.3.2](#). A bound outside the connecting segment would not suffice.  $\square$

**Proposition 11.3.6** (Differentiating limits of maps). Let  $f_k : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$  be  $C^1$  on open  $U$ , with  $\overline{B}(a, r) \subset U$ ,  $r > 0$ . Let  $A : \overline{B}(a, r) \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ . Suppose  $f_k(a)$  converges and  $Df_k \rightarrow A$  uniformly in operator norm on this closed ball. Then  $f_k$  converges uniformly there to a map  $f$ , and  $Df = A$  on  $B(a, r)$ .

*Proof.* Anchor the functions at  $a$  and control their increments by the derivatives. The closed ball is convex by [Proposition 11.3.2](#), so every segment from  $a$  to  $\mathbf{x} \in \overline{B}(a, r)$  stays in the ball. The mean-value estimate gives

$$\|f_k(\mathbf{x}) - f_l(\mathbf{x})\| \leq \|f_k(a) - f_l(a)\| + r \sup_{z \in \overline{B}(a, r)} \|Df_k(z) - Df_l(z)\|.$$

Thus the functions are uniformly Cauchy; completeness gives a uniform limit. The map  $A : \overline{B}(a, r) \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$  is continuous by the uniform-limit theorem, using finite-dimensional coordinates on the target. Fix  $\mathbf{x} \in B(a, r)$  and put  $\eta = r - \|\mathbf{x} - a\| > 0$ . If  $\|h\| < \eta$ , then  $\|\mathbf{x} + th - a\| < r$  for  $0 \leq t \leq 1$ , so the segment stays in the ball. Apply the mean-value estimate to  $z \mapsto f_k(z) - A(\mathbf{x})z$  to obtain

$$\|f_k(\mathbf{x} + h) - f_k(\mathbf{x}) - A(\mathbf{x})h\| \leq \sup_{0 \leq t \leq 1} \|Df_k(\mathbf{x} + th) - A(\mathbf{x})\| \|h\|.$$

Set

$$e_k = \sup_{z \in \overline{B}(a, r)} \|Df_k(z) - A(z)\|_{\text{op}} \longrightarrow 0.$$

The triangle inequality gives the uniform comparison

$$\sup_{0 \leq t \leq 1} \|Df_k(\mathbf{x} + th) - A(\mathbf{x})\|_{\text{op}} \leq e_k + \sup_{0 \leq t \leq 1} \|A(\mathbf{x} + th) - A(\mathbf{x})\|_{\text{op}}.$$

For fixed  $h$ , letting  $k \rightarrow \infty$  therefore gives

$$\|f(\mathbf{x} + h) - f(\mathbf{x}) - A(\mathbf{x})h\| \leq \sup_{0 \leq t \leq 1} \|A(\mathbf{x} + th) - A(\mathbf{x})\|_{\text{op}} \|h\|.$$

Now let  $h \rightarrow 0$ . Since  $\|th\| \leq \|h\|$ , continuity of  $A$  at  $\mathbf{x}$  makes the remaining supremum tend to zero. Hence  $Df(\mathbf{x}) = A(\mathbf{x})$ .  $\square$

A common derivative bound on a convex neighborhood of a compact set gives a common Lipschitz bound on that set, hence equicontinuity. With uniform boundedness, [Theorem 11.1.4](#) then gives subsequence compactness.

**Definition 11.3.7** (Higher derivatives). Let  $V, W$  be finite-dimensional real normed spaces,  $U \subseteq V$  open, and  $f : U \rightarrow W$  differentiable. Define inductively

$$D^0 f = f, \quad D^{k+1} f(a) = D(D^k f)(a),$$

when the right side exists. For  $V = \mathbb{R}^n$  and  $W = \mathbb{R}^m$ , the first two derivatives initially have the types

$$Df(a) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m), \quad D(Df)(a) \in \mathcal{L}(\mathbb{R}^n, \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)).$$

The canonical identification sends a linear map  $A : V \rightarrow \mathcal{L}(V, W)$  to the bilinear map  $(h, k) \mapsto (Ah)(k)$ ; its inverse sends a bilinear map  $B$  to  $h \mapsto B(h, \cdot)$ . Neither construction chooses a basis. Recall  $\mathcal{L}(V, W)$  from [Definition 10.10.2](#). Since  $Df : U \rightarrow \mathcal{L}(V, W)$ , its derivative satisfies

$$D(Df)(a) : V \rightarrow \mathcal{L}(V, W), \quad D^2 f(a)[h, k] = (D(Df)(a)h)(k).$$

This is linear in  $h$  because  $D(Df)(a)$  is linear, and in  $k$  because each output is itself a linear map. Thus it is bilinear in the sense of [Section 10.16](#). Inductively, evaluate a linear-map-valued derivative on its first argument and then on the remaining arguments to obtain  $D^k f(a) : V^k \rightarrow W$ . Its coefficients in bases form a finite array; continuity and differentiation of that array use the equivalent finite-dimensional norms. Extending the convention of [Definition 8.1.4](#), the map is of class  $C^k$  when all Fréchet derivatives through order  $k$  exist and are continuous.

**Differentiate the base point, not the increment.** For fixed  $a$ , the map  $L_a(h) = Df(a)h$  is linear in  $h$ , and indeed  $DL_a(h) = L_a$ . But the second derivative differentiates a different map:

$$g = Df : U \rightarrow \mathcal{L}(V, W), \quad x \mapsto Df(x).$$

It varies the base point  $x$ . Its derivative  $Dg(a) : V \rightarrow \mathcal{L}(V, W)$  measures how the first derivative changes as that point moves. After evaluation,

$$D^2 f(a)[h, k] = (Dg(a)h)(k).$$

Here  $h$  moves the base point and  $k$  is the vector on which the changed first derivative is evaluated. In particular,

$$Df(a) \in \mathcal{L}(V, W), \quad D^2 f(a) \in \mathcal{L}^2(V; W),$$

where  $\mathcal{L}^2(V; W)$  denotes the space of bilinear maps  $V \times V \rightarrow W$ . For scalar-valued  $f$ ,  $Df(a)$  is a linear functional on  $V$ , whereas  $D^2 f(a)$  is a bilinear form on  $V \times V$ . These are different object types.

**Example 11.3.8** (A constant derivative and a changing derivative). If  $f(x) = Lx$ , then  $Df(a) = L$  at every base point. The map  $a \mapsto Df(a)$  is constant, so  $D^2 f(a) = 0$ , even when  $L \neq 0$ . For  $f(x) = x^2$ ,  $Df(a)h = 2ah$  is linear in  $h$  at fixed  $a$ , but

$$Df(a+k)h - Df(a)h = 2kh, \quad D^2 f(a)[k, h] = 2kh.$$

The two independent slots record base-point motion and evaluation.

In general,  $D^{k+1} f$  differentiates the function  $x \mapsto D^k f(x)$  of the base point. Each differentiation introduces one new independent slot.

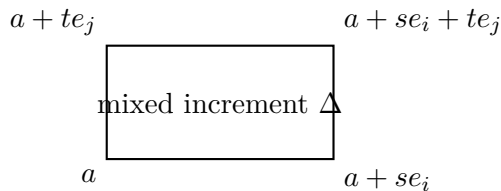
For a  $k$ -linear map  $B : V^k \rightarrow W$  we use the multilinear operator norm

$$\|B\| = \sup_{\|v_1\| \leq 1, \dots, \|v_k\| \leq 1} \|B(v_1, \dots, v_k)\|, \quad \|B(v_1, \dots, v_k)\| \leq \|B\| \prod_j \|v_j\|.$$



Finite-dimensional boundedness follows from the coefficient estimates for multilinear maps in Chapter 10; the inequality follows by scaling each nonzero input to a unit vector, with zero inputs handled by multilinearity.

The mixed-increment proof below visualizes the two independent directions.



**Theorem 11.3.9** (Equality of mixed partials). Let  $U \subseteq \mathbb{R}^n$  be open and let  $f : U \rightarrow \mathbb{R}$  have continuous second partial derivatives. Then

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(a) = \frac{\partial^2 f}{\partial x_j \partial x_i}(a).$$

*Proof.* Compute the same rectangular increment in both orders. For  $i = j$  there is nothing to prove. Otherwise choose small positive  $s, t$  so the entire coordinate rectangle is in  $U$ , and write

$$\Delta = f(a + se_i + te_j) - f(a + se_i) - f(a + te_j) + f(a).$$

Apply the mean value theorem to  $u \mapsto f(a + ue_i + te_j) - f(a + ue_i)$  on  $[0, s]$ . For some  $\xi \in (0, s)$ ,

$$\Delta = s\{\partial_i f(a + \xi e_i + te_j) - \partial_i f(a + \xi e_i)\}.$$

Apply it again, now to  $v \mapsto \partial_i f(a + \xi e_i + ve_j)$  on  $[0, t]$ . For some  $\eta \in (0, t)$  this yields

$$\Delta/(st) = \partial_j \partial_i f(a + \xi e_i + \eta e_j).$$

Reversing the two initial differences instead gives, for possibly different  $\xi', \eta'$ ,

$$\Delta/(st) = \partial_i \partial_j f(a + \xi' e_i + \eta' e_j).$$

Both points tend to  $a$  as  $s, t \rightarrow 0^+$ . Continuity of the two mixed partials equates their limits. A single choice of positive shrinking rectangles suffices to compare their values at  $a$ .  $\square$

**Definition 11.3.10** (Hessian and critical points). Let  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  be differentiable at  $a \in U$ , with  $U$  open. If  $f$  is of class  $C^2$ , its **Hessian** at  $a$  is the matrix

$$h_f(a) = \left( \frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right)_{i,j=1}^n.$$

Under the standard coordinate identification,

$$D^2 f(a)[h, k] = h^\top h_f(a) k.$$

By [Theorem 11.3.9](#), the Hessian is symmetric. A point  $a \in U$  is a **critical point** of a differentiable scalar-valued function when

$$Df(a) = 0,$$

or equivalently when  $\nabla f(a) = 0$ .

**Proposition 11.3.11** (First-order necessary condition). Let  $U \subseteq \mathbb{R}^n$  be open and  $f : U \rightarrow \mathbb{R}$  differentiable at  $a \in U$ . If  $a$  is a local maximum or minimum, then  $Df(a) = 0$ .

*Proof.* Fix  $v \in \mathbb{R}^n$ . For sufficiently small  $t$  of both signs,  $a + tv \in U$ . The scalar function  $g_v(t) = f(a + tv)$  has a local extremum at the interior point 0. The one-variable necessary condition gives  $0 = g'_v(0) = Df(a)v$ . Since  $v$  was arbitrary,  $Df(a) = 0$ .  $\square$

Interior matters because both signs of  $t$  must be available. Optimization therefore begins with critical points, but they are only candidates. The definite second-order tests below are sufficient for classification; semidefinite cases can be inconclusive.

**Compute the Hessian before evaluating its two inputs.** For scalar  $f$ , first compute the partial derivative functions  $\partial_j f$ . Differentiate each of these functions again, freezing all other coordinates, to obtain  $h_f(a) = (\partial_i \partial_j f(a))_{i,j}$ . Continuous second partials justify  $C^2$  regularity by applying the continuous-partial criterion to  $f$  and to its derivative's entries. Once twice differentiability is justified, expanding both inputs in the standard basis gives

$$D^2 f(a)[h, k] = \sum_{i,j} h_i k_j D^2 f(a)[e_i, e_j] = \sum_{i,j} h_i (\partial_i \partial_j f)(a) k_j = h^\top h_f(a) k.$$

There are two independently variable input slots. The diagonal value  $h^\top h_f(a) h$  is a quadratic expression obtained by repeating one input; it is a use of the bilinear map, not its full two-slot formula.

For a  $C^2$  map  $f = (f_1, \dots, f_m) : U \rightarrow \mathbb{R}^m$ , compute one scalar Hessian per output component. Then

$$D^2 f(a)[h, k] = \begin{pmatrix} h^\top h_{f_1}(a) k \\ \vdots \\ h^\top h_{f_m}(a) k \end{pmatrix} \in \mathbb{R}^m.$$

There is no single ordinary Hessian matrix representing all of this vector-valued bilinear map; there are  $m$  two-index coefficient arrays.

**Example 11.3.12** (Two directions versus a repeated direction). For  $f(x, y) = x^2 + 3xy$  and  $a = (a_1, a_2)$ , the first derivative is the linear map

$$Df(a)h = (2a_1 + 3a_2)h_1 + 3a_1h_2.$$

Its coefficient functions are  $2x + 3y$  and  $3x$ . Differentiating again gives the constant Hessian  $h_f = \begin{pmatrix} 2 & 3 \\ 3 & 0 \end{pmatrix}$ . The polynomial rules justify smoothness, and the preceding calculation gives the second derivative at every  $a$ :

$$D^2 f(a)[h, k] = 2h_1k_1 + 3h_1k_2 + 3h_2k_1.$$

This is bilinear and symmetric: interchanging  $h, k$  preserves its value. For  $h = (1, 0)$  and  $k = (0, 1)$  it gives 3, whereas  $D^2 f(a)[h, h] = 2$ . The diagonal specialization is the quadratic expression  $2h_1^2 + 6h_1h_2$  in one direction; it is not the two-input bilinear map.

**Example 11.3.13** (A vector-valued second derivative). For  $f(x, y) = (x^2 + 3xy, y^2)$ , the first derivative sends  $h$  to

$$Df(a)h = ((2a_1 + 3a_2)h_1 + 3a_1h_2, 2a_2h_2).$$

The second derivative is the bilinear map  $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$  given by

$$D^2f(a)[h, k] = (2h_1k_1 + 3h_1k_2 + 3h_2k_1, 2h_2k_2).$$

Its two components have Hessians  $\begin{pmatrix} 2 & 3 \\ 3 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$ . Repeating the input gives  $D^2f(a)[h, h] = (2h_1^2 + 6h_1h_2, 2h_2^2)$ , a vector rather than a scalar.

**Distinguishing the second-order objects.** For scalar  $C^2$  functions, keep the following distinctions:

- $D^2f(a)$  is a bilinear map;  $h_f(a)$  is its matrix in the standard basis, with entries  $\partial_i \partial_j f(a)$ .
- $D^2f(a)[h, k]$  is a scalar obtained by evaluation on two directions.
- $D^2f(a)[h, h] = h^\top h_f(a)h$  is quadratic as a function of  $h$ . For fixed  $h$ , it is the second derivative at zero of  $t \mapsto f(a + th)$ , as the chain rule below verifies.

The second directional derivative is therefore an evaluation of the second differential when  $f$  is  $C^2$ . Existence of separate directional derivatives alone does not justify a bilinear second differential.

**Higher orders: arrays represent multilinear maps.** The hierarchy of types is

$$Df(a) : \text{linear}, \quad D^2f(a) : \text{bilinear}, \quad D^3f(a) : \text{trilinear}, \quad D^k f(a) : k\text{-linear}.$$

For scalar  $f \in C^k(U)$ , evaluation on basis vectors and repeated coordinate differentiation give

$$D^k f(a)[h^{(1)}, \dots, h^{(k)}] = \sum_{i_1, \dots, i_k=1}^n (\partial_{i_1} \cdots \partial_{i_k} f)(a) \prod_{j=1}^k h_{i_j}^{(j)}.$$

Indeed the coefficient on  $(e_{i_1}, \dots, e_{i_k})$  is the displayed iterated partial by the recursive definition; multilinearity expands arbitrary inputs in those basis vectors. The mixed-partial theorem applied to lower derivatives permits adjacent differentiation orders to be interchanged. Adjacent swaps generate all permutations, so  $D^k f(a)$  is symmetric: for every permutation  $\sigma$  of  $\{1, \dots, k\}$ ,

$$D^k f(a)[h_{\sigma(1)}, \dots, h_{\sigma(k)}] = D^k f(a)[h_1, \dots, h_k].$$

Apply the formula componentwise for vector-valued outputs. At order two, equality of mixed partials expresses symmetry of the bilinear map in a basis. The Hessian represents that map in coordinates; it does not define the second derivative independently of  $D(Df)$ .

Order two has two coefficient indices, hence the ordinary Hessian matrix. Order three has three indices, order four has four, and so on. For  $k \geq 3$  there is no preferred ordinary matrix unless indices are grouped artificially; the symmetric  $k$ -linear map keeps the input roles visible. Complete higher-order arrays are rarely computed by hand beyond low orders. Applications usually need their symmetry, a norm bound, or values on a repeated direction.

**Example 11.3.14** (Three input slots). For  $f(x, y) = x^3 + x^2y$ , the nonzero third partials are  $f_{xxx} = 6$  and  $f_{xxy} = f_{xyx} = f_{yxx} = 2$ . Thus at every  $a$ ,

$$\begin{aligned} D^3 f(a)[u, v, w] &= 6u_1v_1w_1 \\ &\quad + 2(u_1v_1w_2 + u_1v_2w_1 + u_2v_1w_1). \end{aligned}$$

Each input appears once in every term, and permutation of the three inputs leaves the value unchanged. On a repeated input the value is  $6h_1^3 + 6h_1^2h_2$ ; dividing by  $3!$  recovers the cubic polynomial.

**Remark 11.3.15** (Symmetric and alternating multilinearity). Higher derivatives and differential forms share the multilinear language of Chapter 10, but impose different symmetry. For scalar  $C^k$  functions,  $D^k f(a)$  is symmetric and describes higher-order local variation. Chapter 13's differential  $k$ -form  $\omega_a$  is alternating and describes oriented  $k$ -dimensional measurement. Permuting inputs leaves the former unchanged; exchanging two inputs changes the latter's sign. For degree at least two an alternating map vanishes on a repeated-input tuple, whereas  $D^k f(a)[h, \dots, h]$  gives the Taylor term.

**Proposition 11.3.16** (Bilinear differentiation). Let  $V, W, Z$  be finite-dimensional normed spaces and  $B : V \times W \rightarrow Z$  bilinear with  $\|B(u, v)\| \leq C\|u\|\|v\|$ . Then

$$DB(u, v)[p, q] = B(p, v) + B(u, q).$$

If  $f : O \rightarrow V$  and  $g : O \rightarrow W$  are differentiable on an open subset of a finite-dimensional normed space, then

$$D(B(f, g))(a)h = B(Df(a)h, g(a)) + B(f(a), Dg(a)h).$$

*Proof.* On  $V \times W$  use  $\|(p, q)\| = \sqrt{\|p\|^2 + \|q\|^2}$ ; the triangle inequality follows from those in the factors and the Euclidean triangle inequality in  $\mathbb{R}^2$ . Expand

$$B(u + p, v + q) - B(u, v) = B(p, v) + B(u, q) + B(p, q).$$

The first two terms form a bounded linear map of  $(p, q)$ , with bound  $(C\|v\| + C\|u\|)\|(p, q)\|$ . The remainder satisfies

$$\|B(p, q)\| \leq C\|p\|\|q\| \leq \frac{C}{2}(\|p\|^2 + \|q\|^2) = o(\|(p, q)\|).$$

This proves differentiability. The pair map  $(f, g)$  has derivative  $h \mapsto (Df(a)h, Dg(a)h)$  by its two remainder estimates. Apply the chain rule.  $\square$

In particular evaluation  $E : \mathcal{L}(W, Z) \times W \rightarrow Z$ ,  $E(A, w) = Aw$ , is bounded bilinear since  $\|Aw\| \leq \|A\|_{\text{op}}\|w\|$ .

**Proposition 11.3.17** (Second-order chain rule). Let  $V, W, Z$  be finite-dimensional real normed spaces. For  $C^2$  maps  $f : U \subseteq V \rightarrow W$  and  $g : U_1 \subseteq W \rightarrow Z$ , where  $U, U_1$  are open and  $f(U) \subseteq U_1$ , we have for  $h, k \in V$

$$\begin{aligned} D^2(g \circ f)(a)[h, k] &= D^2g(f(a))[Df(a)h, Df(a)k] \\ &\quad + Dg(f(a))[D^2f(a)[h, k]]. \end{aligned}$$

*Proof.* For fixed  $k \in V$ , differentiate  $D(g \circ f)(x)k = Dg(f(x))(Df(x)k)$  in direction  $h \in V$ . Apply [Proposition 11.3.16](#) to the bounded evaluation map  $E(A, w) = Aw$ . Its rule differentiates each factor once. The first derivative of the first factor is  $D^2g(f(a))[Df(a)h, \cdot]$ ; that of the second is  $D^2f(a)[h, k]$ . Their two contributions are exactly the formula.  $\square$

Here  $Df(a) : V \rightarrow W$  sends both directions to  $W$ ;  $D^2g(f(a)) : W \times W \rightarrow Z$  is evaluated on those two images. The other term first uses  $D^2f(a) : V \times V \rightarrow W$ , then  $Dg(f(a)) : W \rightarrow Z$ . Both outputs belong to  $Z$ . For scalar one-variable maps and  $h = k = 1$ , this is  $(g \circ f)'' = g''(f)(f')^2 + g'(f)f''$ .

Unlike the first-order rule, the second-order rule has two contributions. The first comes from curvature of the outer map  $g$ , tested on the two directions transported by  $Df(a)$ . The second comes from nonlinear variation of  $f$ , carried to the target by  $Dg(f(a))$ . If  $f$  is affine, the second term vanishes; if  $g$  is affine, the first vanishes. At higher orders, differentiating a term either adds an input to an outer derivative or differentiates one of its inner factors. The resulting sums are organized by partitions of the input directions into groups: each group enters one derivative of  $f$ , and their outputs enter a derivative of  $g$ . We will not need the general combinatorial formula.

**Corollary 11.3.18** (Second derivative along a curved path). If  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  and  $\gamma : I \rightarrow U$  are  $C^2$ , then

$$(f \circ \gamma)''(t) = D^2f(\gamma(t))[\gamma'(t), \gamma'(t)] + Df(\gamma(t))\gamma''(t).$$

*Proof.* Apply [Proposition 11.3.17](#) to  $f \circ \gamma$  and evaluate on  $1, 1 \in \mathbb{R}$ . □

The first term measures second-order variation of  $f$  along current velocity; the second pairs its first derivative with the path's acceleration. A line  $\gamma(t) = a + tv$  has  $\gamma'' = 0$ , explaining the simpler identity  $g''(t) = D^2f(a + tv)[v, v]$  for  $g(t) = f(a + tv)$ . For  $f(x, y) = x^2 + y^2$  and  $\gamma(t) = (\cos t, \sin t)$ ,

$$D^2f(\gamma)[\gamma', \gamma'] = 2, \quad Df(\gamma)\gamma'' = 2\gamma \cdot (-\gamma) = -2.$$

Their sum is zero, as it must be since  $f \circ \gamma = 1$ . A curved path generally contributes more than the Hessian term alone.

The *Taylor polynomial in the increment  $h$*  is

$$T_k f(a; h) = \sum_{j=0}^k \frac{1}{j!} D^j f(a)[h, \dots, h],$$

where the  $j = 0$  term means  $f(a)$ . Each  $D^j f(a)$  is a symmetric  $j$ -linear map; evaluating on  $j$  copies of  $h$  gives a homogeneous polynomial of degree  $j$  (scaling  $h$  by  $t$  scales the term by  $t^j$ ). For scalar output, its first two coordinate terms are

$$Df(a)h = \sum_i f_{x_i}(a)h_i, \quad D^2f(a)[h, h] = h^T h_f(a)h = \sum_{i,j} f_{x_i x_j}(a)h_i h_j.$$

Thus in two variables the intrinsic formula reads

$$\begin{aligned} f(a + h) &= f(a) + f_x(a)h_1 + f_y(a)h_2 \\ &\quad + \frac{1}{2}(f_{xx}(a)h_1^2 + 2f_{xy}(a)h_1 h_2 + f_{yy}(a)h_2^2) + \text{remainder}. \end{aligned}$$

For a fixed direction  $v$  and  $g(t) = f(a + tv)$ , the chain rule repeated  $j$  times gives

$$g^{(j)}(t) = D^j f(a + tv)[v, \dots, v], \quad g^{(j)}(0) = D^j f(a)[v, \dots, v].$$

Only  $t$  varies; the direction remains fixed. Each differentiation inserts one more copy of  $v$ , since the line has constant velocity and no higher derivatives. This generalizes the one-variable formula in the subsection “[Higher Derivatives and Taylor Approximation](#).” The following theorem explains the remainder by taking  $v = h$  and applying one-variable Taylor to  $g$ .

**Theorem 11.3.19** (Taylor's theorem along a segment). Let  $k \geq 0$  be an integer. Let  $U \subseteq \mathbb{R}^n$  be open, let  $f : U \rightarrow \mathbb{R}^m$  be of class  $C^{k+1}$ , and suppose the segment from  $a$  to  $a + h$  is in  $U$ . If  $m = 1$ , there is

$$\theta \in (0, 1)$$

such that

$$f(a + h) = \sum_{j=0}^k \frac{1}{j!} D^j f(a)[h, \dots, h] + \frac{1}{(k+1)!} D^{k+1} f(a + \theta h)[h, \dots, h].$$

For vector-valued  $f$ , this formula holds componentwise, possibly with a different value of  $\theta$  for each component.

*Proof.* For a scalar component define

$$g(t) = f(a + th).$$

Keep  $h$  fixed. Always  $g'(t) = Df(a + th)h$ . If  $k \geq 1$ , the hypothesis also supplies the second derivative

$$g''(t) = D^2 f(a + th)[h, h].$$

For  $k = 0$  only  $g'$  is needed. Each differentiation inserts the same segment direction once more. Repeated application of the chain rule gives

$$g^{(j)}(t) = D^j f(a + th)[h, \dots, h].$$

Apply the one-variable Taylor theorem [Theorem 8.4.5](#) to  $g$  at zero and evaluate at one. Do this for each component.  $\square$

**Corollary 11.3.20** (Taylor error bound and general Peano form). If  $f$  is  $C^{k+1}$  on a neighborhood of a segment and  $\|D^{k+1}f\| \leq M$  there, then, in Euclidean target norm,

$$\|f(a + h) - T_k f(a; h)\| \leq \frac{M}{(k+1)!} \|h\|^{k+1}.$$

If  $f$  is only  $C^k$  near  $a$ ,  $k \geq 1$ , then  $f(a + h) = T_k f(a; h) + o(\|h\|^k)$ .

*Proof.* For a nonzero remainder vector  $R$ , fix

$$u = R/\|R\|$$

and apply the scalar segment theorem to  $u \cdot f$ . Its remainder is  $u \cdot R = \|R\|$ , and  $|u \cdot D^{k+1} f[h, \dots, h]| \leq M \|h\|^{k+1}$ . The zero case is immediate; no common intermediate point for all components is asserted. For the Peano assertion follow the proof of [Corollary 8.4.7](#): apply the same projection argument to  $R = f(a + h) - T_k f(a; h)$ , using the segment theorem of order  $k - 1$ . After subtracting the degree- $k$  term, it bounds  $\|R\|$  by

$$\frac{\|h\|^k}{k!} \sup_{0 \leq t \leq 1} \|D^k f(a + th) - D^k f(a)\|.$$

Continuity at  $a$  makes the supremum tend to zero.  $\square$

The  $C^{k+1}$  hypothesis gives a scalar Lagrange remainder along the segment and a vector norm estimate. The weaker  $C^k$  hypothesis suffices for the Peano error.

**Corollary 11.3.21** (Second-order Taylor–Peano formula). For  $f : U \rightarrow \mathbb{R}^m$  of class  $C^2$  near  $a$ ,

$$f(a+h) = f(a) + Df(a)h + \frac{1}{2}D^2f(a)[h, h] + o(\|h\|^2).$$

*Proof.* Apply the  $C^k$  Peano part of [Corollary 11.3.20](#) with  $k = 2$ . No third derivative is required.  $\square$

**Example 11.3.22** (Computing a quadratic Taylor polynomial). For  $f(x, y) = e^x \cos y$ , first compute the gradient and Hessian:

$$\nabla f = (e^x \cos y, -e^x \sin y), \quad h_f = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ -e^x \sin y & -e^x \cos y \end{pmatrix}.$$

At  $a = (0, 0)$  these are  $(1, 0)$  and  $\text{diag}(1, -1)$ . For  $h = (u, v)$ , the coordinate quadratic expression is  $h^\top h_f(0)h = u^2 - v^2$ . Hence

$$T_2f(0; h) = 1 + u + \frac{1}{2}(u^2 - v^2), \quad e^u \cos v = T_2f(0; h) + O(\|h\|^3).$$

Indeed the third derivative is continuous and therefore bounded by some  $M$  on a fixed compact ball around zero; the error is at most

$$M\|h\|^3/6$$

there. With only  $C^2$  information the justified conclusion would instead be  $o(\|h\|^2)$ . The bilinear Hessian has two input slots; the polynomial repeats the increment in both.

**Coordinate expansion of the Taylor polynomial.** The polynomial is defined by evaluating multilinear derivatives on the increment. Multi-indices abbreviate its coordinate expansion. A multi-index is a tuple  $\alpha = (\alpha_1, \dots, \alpha_n)$  of nonnegative integers, with

$$|\alpha| = \sum_i \alpha_i, \quad \alpha! = \prod_i \alpha_i!, \quad h^\alpha = \prod_i h_i^{\alpha_i}, \quad \partial^\alpha = \partial_1^{\alpha_1} \dots \partial_n^{\alpha_n}.$$

Expanding  $D^j f(a)[h, \dots, h]$  by multilinearity gives one term for each ordered choice of  $j$  coordinate indices. If index  $i$  occurs  $\alpha_i$  times, the same monomial  $h^\alpha$  occurs

$$j!/\alpha!$$

times. Symmetry makes their derivative coefficients equal. Consequently

$$\frac{1}{j!}D^j f(a)[h, \dots, h] = \sum_{|\alpha|=j} \frac{\partial^\alpha f(a)}{\alpha!} h^\alpha.$$

This is a finite identity for  $C^j$  maps, componentwise for vector output; it does not require analyticity. The Taylor polynomial specifies the coefficients, the Peano remainder asserts local approximation as  $h \rightarrow 0$ , and the quantitative bound above uses one more bounded derivative.

### Smoothness and real analyticity

The distinction in [Definition 8.8.1](#) and [Theorem 8.8.2](#) remains essential in several variables. Here  $C^\infty$  means  $C^k$  for every finite  $k$ , not equality to an infinite Taylor expansion.

**Definition 11.3.23** (Real-analytic map). A scalar function on an open subset of  $\mathbb{R}^n$  is real analytic at  $a$  if on a neighborhood of  $a$  it is represented by a convergent power series in the coordinates of  $h = \mathbf{x} - a$ . It is real analytic on its domain if this holds at every point. Vector-valued maps are real analytic when each scalar component is.

Here convergent power series means absolute convergence on a sufficiently small coordinate box. Termwise differentiation is valid on smaller boxes: choose  $0 < r < R$  with the closed box of radius  $R$  inside it. For a fixed multi-index  $\beta$  of order  $q = |\beta|$  and  $\alpha \geq \beta$  coordinatewise, put  $m = |\alpha|$ . Then

$$\begin{aligned} \sup_{\|h\|_\infty \leq r} |\partial^\beta(c_\alpha h^\alpha)| &\leq |c_\alpha| m^q r^{m-q} \\ &= r^{-q} m^q (r/R)^m |c_\alpha| R^m \leq C_{\beta,r,R} |c_\alpha| R^{|\alpha|}, \end{aligned}$$

where

$$C_{\beta,r,R} = r^{-q} \sup_{m \geq 0} m^q (r/R)^m < \infty.$$

For  $q = 0$  take  $m^0 = 1$ ; if  $\alpha \not\geq \beta$ , the derivative is zero. The supremum is finite because geometric decay dominates fixed polynomial growth. Absolute convergence gives

$$\sum_{\alpha} |c_\alpha| R^{|\alpha|} < \infty,$$

so the M-test proves uniform convergence of each derivative series on the smaller box. Apply the differentiating-limits theorem successively in each coordinate. As in Chapter 8, this proves smoothness and the forced coefficients

$$\partial^\alpha f(a)/\alpha!.$$

Thus, for sufficiently small  $h$ ,

$$f(a+h) = \sum_{\alpha} \frac{\partial^\alpha f(a)}{\alpha!} h^\alpha = \sum_{j=0}^{\infty} \frac{1}{j!} D^j f(a)[h, \dots, h].$$

Group terms by  $|\alpha| = j$  and use the finite coordinate identity above. Absolute convergence justifies this grouping of the infinite series.

Analytic implies  $C^\infty$ , which implies  $C^k$  for each finite  $k$ . The reverse implications fail, as the one-variable examples composed with a coordinate projection show. For each *fixed*  $k$ , [Corollary 11.3.20](#) gives  $T_k f(a; h) + o(\|h\|^k)$  as  $h \rightarrow 0$ . It supplies no estimate allowing  $k \rightarrow \infty$  with  $h$  fixed.

**Example 11.3.24** (A smooth map not recovered by its Taylor series). Let  $f(x_1, \dots, x_n) = \theta(x_1)$  with  $\theta$  from [Theorem 8.8.2](#). The chain rule with coordinate projection makes  $f$  smooth. A derivative involving another coordinate is zero; the derivatives using only  $x_1$  are derivatives of  $\theta$ . Hence  $f(0) = 0$  and  $D^k f(0) = 0$  for every  $k \geq 1$ . Its Taylor series at zero is the zero series, although  $f(\mathbf{x}) > 0$  whenever  $x_1 > 0$ . All derivatives at a point therefore need not determine a smooth map nearby.

The inverse and implicit function arguments below preserve  $C^r$  regularity, including  $r = \infty$ . They do not establish analyticity: existence of derivatives of every order is not a proof that a Taylor series converges to the inverse or implicit function.



**Higher derivatives of local inverses and implicit graphs.** The inverse theorem below first constructs a  $C^1$  inverse and then gains derivatives from its first-derivative formula. Writing  $g = f^{-1}$ , the identity  $g \circ f = \text{id}$  gives

$$Dg(f(\mathbf{x})) = [Df(\mathbf{x})]^{-1}, \quad Dg = \text{Inv} \circ Df \circ g,$$

where  $\text{Inv}(A) = A^{-1}$  on  $GL(\mathbb{R}^n)$ . Smoothness of operator inversion, proved below, makes the available regularity visible:

$$f \in C^k \implies Df \in C^{k-1} \implies (\mathbf{x} \mapsto [Df(\mathbf{x})]^{-1}) \in C^{k-1}.$$

Composition with  $g$  must be justified inductively. Once  $g \in C^j$  for  $j < k$ , the formula for  $Dg$  gives  $Dg \in C^j$  and hence  $g \in C^{j+1}$ . Repeating this reaches  $Dg \in C^{k-1}$  and  $g \in C^k$ . Here  $[Df(\mathbf{x})]^{-1}$  means inversion of a linear map, not differentiation of an inverse function. The implicit graph is constructed from the local inverse of  $h(x, y) = (x, f(x, y))$  by restricting to  $(x, 0)$  and projecting onto the remaining coordinates. These operations preserve  $C^k$  regularity, so no separate higher-order existence argument is needed.

### Using the quadratic term at an extremum

**Corollary 11.3.25** (Second-order test). Let  $f : U \rightarrow \mathbb{R}$  be of class  $C^2$  and satisfy

$$Df(a) = 0.$$

If

$$D^2f(a)[h, h] > 0 \quad (h \neq 0),$$

then  $a$  is a strict local minimum. If the displayed inequality is reversed, then  $a$  is a strict local maximum.

*Proof.* On the Euclidean unit sphere, the continuous quadratic form has a positive minimum: compactness follows from [Theorem 10.6.8](#), and attainment from [Theorem 10.6.6](#). Hence

$$D^2f(a)[h, h] \geq c\|h\|^2$$

for some  $c > 0$ . The sphere minimum is attained at a nonzero vector, so it is strictly positive; pointwise positivity alone would not yield this uniform constant without compactness. By [Corollary 11.3.21](#), choose  $\|h\|$  small enough that the absolute remainder is at most

$$c\|h\|^2/4.$$

Then

$$f(a + h) - f(a) \geq (c/2 - c/4)\|h\|^2 > 0$$

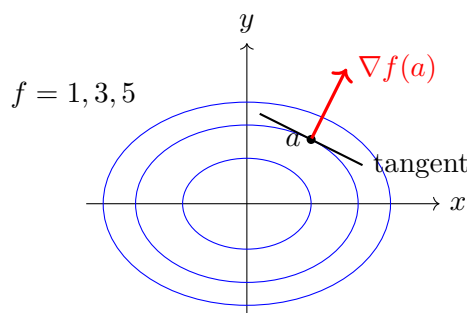
for  $h \neq 0$ . Apply this to  $-f$  for the maximum assertion. □

**Remark 11.3.26** (Saddles and inconclusive Hessians). At a critical point, an indefinite Hessian gives directions  $u, v$  with  $D^2f(a)[u, u] > 0 > D^2f(a)[v, v]$ . Taylor expansion along  $tu$  and  $tv$  then gives values above and below  $f(a)$  arbitrarily nearby: a saddle. For example  $x^2 - y^2$  has this behavior at zero. A semidefinite Hessian is generally inconclusive: both  $x^4 + y^4$  and  $x^4 - y^4$  have zero Hessian at zero, but the first has a strict minimum and the second a saddle.

**Example 11.3.27** (Level ellipses and the gradient). For  $f(x, y) = x^2 + 2y^2$ , take  $a = (1, 1)$  on the level  $f = 3$ . Then  $\nabla f(a) = (2, 4)$ , and for unit directions

$$D_{(1,0)}f(a) = 2, \quad D_{(2,-1)/\sqrt{5}}f(a) = 0, \quad D_{(1,2)/\sqrt{5}}f(a) = 2\sqrt{5}.$$

The middle direction is tangent to the ellipse; the last is the gradient direction, which maximizes first-order increase by Cauchy–Schwarz.



More generally, for  $a, b > 0$  the ellipse

$$x^2/a^2 + y^2/b^2 = 1$$

has normal

$$(2x/a^2, 2y/b^2)$$

at  $(x, y)$ . At  $(x_0, y_0)$  its tangent line is

$$x_0x/a^2 + y_0y/b^2 = 1.$$

The hyperbola

$$x^2/a^2 - y^2/b^2 = 1$$

similarly has nonzero normal

$$(2x/a^2, -2y/b^2)$$

everywhere, although it has two components, distinguished by  $x > 0$  and  $x < 0$ .

## Convexity and global minima

The line-segment method also generalizes Chapter 8's convexity theory. For an open convex  $U \subseteq \mathbb{R}^n$ , a function  $f : U \rightarrow \mathbb{R}$  is convex if

$$f((1-t)\mathbf{x} + ty) \leq (1-t)f(\mathbf{x}) + tf(y) \quad (\mathbf{x}, y \in U, 0 \leq t \leq 1).$$

It is strictly convex if the inequality is strict for  $\mathbf{x} \neq y$  and  $0 < t < 1$ . The tangent affine function will now replace the tangent line.

**Proposition 11.3.28** (Criteria for convexity). For differentiable  $f$ , convexity is equivalent to

$$f(y) \geq f(\mathbf{x}) + Df(\mathbf{x})(y - \mathbf{x}) \quad (\mathbf{x}, y \in U).$$

For  $C^2$  functions it is equivalent to  $D^2f(\mathbf{x})[h, h] \geq 0$  for all  $\mathbf{x} \in U$ ,  $h \in \mathbb{R}^n$ ; in coordinates, the Hessian is positive semidefinite everywhere.

*Proof.* Restrict to  $g(t) = f(\mathbf{x} + t(y - \mathbf{x}))$ . Convexity gives

$$(g(t) - g(0))/t \leq g(1) - g(0)$$

for  $0 < t < 1$ ; let  $t \downarrow 0$  to obtain the supporting inequality. Conversely put  $z = (1 - t)\mathbf{x} + ty$ , apply that inequality based at  $z$  to  $\mathbf{x}$  and  $y$ , and multiply by  $1 - t$  and  $t$ . The derivative terms sum to  $Df(z)((1 - t)\mathbf{x} + ty - z) = 0$ , proving convexity. For  $C^2$  functions the restriction has  $g''(t) = D^2f(\mathbf{x} + t(y - \mathbf{x}))[y - \mathbf{x}, y - \mathbf{x}]$ ; the one-variable second-derivative criterion proves sufficiency. Necessity follows by restricting a convex  $f$  to a small line interval  $\mathbf{x} + th$  within the open domain and testing at zero.  $\square$

If  $Df(a) = 0$ , the supporting inequality gives  $f(y) \geq f(a)$  everywhere:  $a$  is a global minimum. For strictly convex  $f$  there cannot be two minimizers, since their midpoint would have a smaller value. For example  $f(x, y) = (x - 1)^2 + 2(y + 1)^2$  has Hessian  $\text{diag}(2, 4)$  everywhere and its unique critical point  $(1, -1)$  is its unique global minimum. A positive definite Hessian at one *critical* point gives the earlier local minimum test; a positive semidefinite Hessian everywhere on a convex domain gives global convexity.

## 11.4 The Contraction Principle

To solve an inverse or implicit equation, we will rearrange it as  $x = T(x)$  and look for a fixed point. Repeatedly applying  $T$  should improve an approximate solution. A contraction gives existence, uniqueness, and convergence of these iterates together; completeness keeps their limit inside the space. This is the tool behind the next section's local inverse and implicit function theorems.

**Definition 11.4.1** (Contraction). A map

$$T : X \longrightarrow X$$

on a metric space is a **contraction** if

$$d(Tx, Ty) \leq qd(x, y)$$

for all  $x, y \in X$  and some

$$0 \leq q < 1.$$

**Theorem 11.4.2** (Contraction mapping theorem). A contraction of a nonempty complete metric space has a unique fixed point. For every  $x_0$ , the sequence

$$x_{n+1} = T(x_n)$$

converges to it.

*Proof.* The contraction inequality gives

$$d(x_{n+1}, x_n) \leq q^n d(x_1, x_0).$$

Thus, for  $m > n$ ,

$$d(x_m, x_n) \leq \sum_{j=n}^{m-1} d(x_{j+1}, x_j) \leq d(x_1, x_0) \sum_{j=n}^{m-1} q^j \leq \frac{q^n}{1-q} d(x_1, x_0).$$

The last bound is independent of  $m > n$ , proving the Cauchy property. Completeness gives a limit  $x$ . Continuity of  $T$  implies

$$T(x) = T(\lim_n x_n) = \lim_n T(x_n) = \lim_n x_{n+1} = x.$$

If  $x, y$  are fixed, then

$$d(x, y) \leq qd(x, y),$$

so  $x = y$ . □

**Corollary 11.4.3** (Contractions on a Euclidean open ball). Let  $r > 0$ , and suppose

$$T : B_2(0, r) \longrightarrow \mathbb{R}^n$$

is Lipschitz with constant  $0 \leq q < 1$  and

$$\|T(0)\| < (1 - q)r.$$

Then  $T$  maps  $B_2(0, r)$  into itself, has a unique fixed point  $p$  there, and

$$\|p\| \leq \frac{\|T(0)\|}{1 - q} < r.$$

For every  $\mathbf{x}_0 \in B_2(0, r)$ , the iteration  $\mathbf{x}_{j+1} = T(\mathbf{x}_j)$  converges to  $p$ .

*Proof.* Put

$$c = \|T(0)\|/(1 - q) < r.$$

If  $\|\mathbf{x}\| < r$ , then

$$\|T(\mathbf{x})\| \leq \|T(\mathbf{x}) - T(0)\| + \|T(0)\| \leq q\|\mathbf{x}\| + (1 - q)c < r,$$

so  $T$  is a self-map of the open ball. That ball need not be complete, so we do not apply the contraction theorem to it. Choose  $s$  with  $c < s < r$ . For  $\|\mathbf{x}\| \leq s$ ,

$$\|T(\mathbf{x})\| \leq qs + (1 - q)c < s.$$

Thus the closed ball  $\overline{B}_2(0, s)$  is invariant and is complete by [Corollary 10.3.8](#), so [Theorem 11.4.2](#) gives a fixed point  $p$  in it. Any fixed point in the open ball is unique by the contraction inequality, and

$$(1 - q)\|p\| \leq \|T(0)\|$$

gives the asserted bound. Given an arbitrary  $\mathbf{x}_0$  in the open ball, choose instead  $s$  with  $\max\{c, \|\mathbf{x}_0\|\} < s < r$ . The same invariant closed-ball argument then proves convergence of its iterates to  $p$ . □

The strict inequality is sharp. On  $(-1, 1)$ , the map

$$T(x) = (x + 1)/2$$

has

$$q = 1/2, \quad |T(0)| = (1 - q)r = 1/2,$$

but its only fixed point is the boundary point 1.

**Example 11.4.4** (Solving  $x = \cos x$  by iteration). On

$$X = [1/2, 1]$$

with the usual metric,  $T(x) = \cos x$  maps  $X$  into itself. Indeed, the elementary-function theory gives

$$1/2 < \cos 1 \leq \cos x < 1$$

there. The Mean Value Theorem and  $|\sin x| \leq \sin 1 < 1$  give

$$|T(x) - T(y)| \leq (\sin 1)|x - y|.$$

The closed interval is complete, so the contraction theorem gives a unique solution in  $X$ , and the numerical iteration  $x_{n+1} = \cos x_n$  converges to it from every starting point in  $X$ . Completeness is used precisely when the Cauchy iterates are asserted to have a limit. Similar successive approximations will give existence results for differential equations, integral equations, and problems in functional analysis.

**What each contraction hypothesis contributes.** The self-map hypothesis keeps every iterate in the space. The factor  $q < 1$  makes successive steps shrink geometrically. Completeness places the Cauchy limit in that same space. For

$$T(x) = x/2$$

on  $(0, 1)$ , the first two properties hold but the fixed point 0 is missing, so completeness cannot be omitted. On  $\mathbb{R}$ ,  $T(x) = x + 1$  shows why a Lipschitz factor 1 is insufficient even in a complete space.

The geometric tail gives usable stopping bounds. For  $x_n = T^n x_0$  and the fixed point  $p$ ,

$$d(x_n, p) \leq \frac{q^n}{1 - q} d(x_1, x_0), \quad d(x_n, p) \leq \frac{q}{1 - q} d(x_n, x_{n-1}) \quad (n \geq 1).$$

Indeed every future step is bounded by a successive power of  $q$  times the first remaining step; sum that geometric series and let the far endpoint tend to  $p$ . The second bound uses the observed last step, so it can certify the accuracy during an iteration.

**Proposition 11.4.5** (Perturbation of an invertible map). Let  $A, B \in \mathcal{L}(\mathbb{R}^n)$  and let  $A$  be invertible. If

$$\|I - A^{-1}B\|_{\text{op}} < 1,$$

then  $B$  is invertible.

*Proof.* Set  $T = I - A^{-1}B$ . By [Proposition 10.15.9](#),  $I - T$  has inverse

$$S = \sum_{k=0}^{\infty} T^k.$$

Since  $B = A(I - T)$ , its inverse is  $SA^{-1}$ , in that order. □

## 11.5 Inverse and Implicit Functions

There are two related questions. An inverse problem solves  $y = f(\mathbf{x})$  for  $\mathbf{x}$  as a function of  $y$ . An implicit problem solves  $f(\mathbf{x}, y) = 0$  for one block of variables, say  $y$ , in terms of the independent Euclidean block  $\mathbf{x}$ . In each case we first ask whether the linear approximation can solve for the desired unknowns.

For an invertible operator  $T \in \mathcal{L}(\mathbb{R}^n)$ , invertibility is equivalent to  $\det T \neq 0$ . In one variable, the corresponding hypothesis  $f'(a) \neq 0$  says that the linear approximation  $h \mapsto f'(a)h$  can be inverted. For a map

$$\mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

the exact replacement is that  $Df(a)$  is an invertible linear map. Since

$$f(a + h) = f(a) + Df(a)h + o(\|h\|),$$

the natural question is whether an invertible best linear approximation forces  $f$  itself to be invertible nearby. The inverse function theorem answers yes, locally: close to  $a$ , the map neither folds space nor collapses a direction, and nearby points can be recovered uniquely from their images.

Regard  $\mathcal{L}(\mathbb{R}^n)$  as a finite-dimensional normed vector space with the operator norm. Any equivalent norm gives the same topology and the same finite-dimensional  $C^k$  notion; the operator norm is natural here because it controls composition and perturbations. Write

$$\mathrm{GL}(\mathbb{R}^n) = \{A \in \mathcal{L}(\mathbb{R}^n) : A \text{ is invertible}\}.$$

Thus  $\mathrm{GL}(\mathbb{R}^n) \subseteq \mathcal{L}(\mathbb{R}^n)$ ; after a basis is chosen these operators are represented by the usual invertible matrices.

**Lemma 11.5.1** (Smooth inversion of operators). The set  $\mathrm{GL}(\mathbb{R}^n)$  is open in  $\mathcal{L}(\mathbb{R}^n)$ , and

$$\mathrm{Inv} : \mathrm{GL}(\mathbb{R}^n) \longrightarrow \mathcal{L}(\mathbb{R}^n), \quad \mathrm{Inv}(A) = A^{-1},$$

is smooth. Its derivative is the linear map

$$D\mathrm{Inv}(A) : \mathcal{L}(\mathbb{R}^n) \longrightarrow \mathcal{L}(\mathbb{R}^n), \quad D\mathrm{Inv}(A)[h] = -A^{-1}hA^{-1}.$$

*Proof.* Fix  $A \in \mathrm{GL}(\mathbb{R}^n)$ . The estimate

$$\|A^{-1}(B - A)\|_{\mathrm{op}} \leq \|A^{-1}\|_{\mathrm{op}}\|B - A\|_{\mathrm{op}}$$

shows that

$$\|B - A\|_{\mathrm{op}} < \frac{1}{2\|A^{-1}\|_{\mathrm{op}}} \implies \|A^{-1}(B - A)\|_{\mathrm{op}} < \frac{1}{2} < 1.$$

The perturbation result [Proposition 11.4.5](#) then makes  $B$  invertible, so  $\mathrm{GL}(\mathbb{R}^n)$  is open.

We first establish continuity and the local bound used in the derivative estimate. If

$$\|A^{-1}(B - A)\|_{\mathrm{op}} < 1/2,$$

the geometric series gives

$$\|B^{-1}\|_{\mathrm{op}} \leq 2\|A^{-1}\|_{\mathrm{op}}.$$

The exact resolvent identity

$$B^{-1} - A^{-1} = B^{-1}(A - B)A^{-1}$$

then proves  $B^{-1} \rightarrow A^{-1}$  when  $B \rightarrow A$ .

For

$$\|h\|_{\text{op}} < 1/(2\|A^{-1}\|_{\text{op}}),$$

the preceding argument makes  $A + h$  invertible. Rearranging  $(A + h)^{-1}(A + h) = I$  first gives

$$(A + h)^{-1} - A^{-1} = -(A + h)^{-1}HA^{-1}.$$

Combining the same identity in the opposite order gives the exact remainder identity

$$(A + h)^{-1} - A^{-1} + A^{-1}HA^{-1} = A^{-1}HA^{-1}h(A + h)^{-1}.$$

The local bound for  $(A + h)^{-1}$  makes the right side  $O(\|h\|_{\text{op}}^2)$ . This proves the displayed derivative formula. Operator composition  $(R, S) \mapsto RS$  is bilinear, hence smooth, so

$$(A, h) \mapsto -\text{Inv}(A)h\text{Inv}(A)$$

is as differentiable in  $A$  as  $\text{Inv}$  itself. Starting from  $C^1$  and applying this observation inductively proves  $C^j$  regularity for every finite  $j$ , and therefore smoothness.  $\square$

**Definition 11.5.2** (Diffeomorphisms). Let  $U \subseteq \mathbb{R}^n$  and  $V \subseteq \mathbb{R}^m$  be open. A bijection  $f : U \rightarrow V$  is a  $C^k$  **diffeomorphism** if both  $f$  and  $f^{-1}$  are of class  $C^k$ . When  $k = \infty$ , it is a **smooth diffeomorphism**. A map is a **local  $C^k$  diffeomorphism at  $a$**  if its restriction to some open neighborhood of  $a$  is a  $C^k$  diffeomorphism onto an open set.

The correction map  $T_y(\mathbf{x}) = \mathbf{x} - Df(a)^{-1}(f(\mathbf{x}) - y)$  comes from solving the affine model. If  $f(\mathbf{x}) = f(a) + B(\mathbf{x} - a)$  with  $B$  invertible, then

$$T_y(\mathbf{x}) = a + B^{-1}(y - f(a)),$$

independent of  $\mathbf{x}$  and exactly the solution of  $f(\mathbf{x}) = y$ . For a nonlinear map, source and target translations move  $a$  and  $f(a)$  to zero, while multiplication by  $B^{-1}$  reduces the linear part to the identity. The remaining nonlinear deviation then has small derivative and generates a contraction.

**Theorem 11.5.3** (Inverse Function Theorem). Let  $U \subseteq \mathbb{R}^n$  be open and let

$$f_0 : U \longrightarrow \mathbb{R}^n$$

be  $C^k$ ,  $1 \leq k \leq \infty$ . If  $\mathbf{x}_0 \in U$  and  $Df_0(\mathbf{x}_0)$  is invertible, then there are open neighborhoods  $U_0$  of  $\mathbf{x}_0$  and  $V_0$  of  $f_0(\mathbf{x}_0)$  such that

$$f_0 : U_0 \longrightarrow V_0$$

is a  $C^k$  diffeomorphism, and

$$D(f_0^{-1})(f_0(\mathbf{x})) = (Df_0(\mathbf{x}))^{-1} \quad (\mathbf{x} \in U_0).$$

*Proof. 1. Normalization.* Let

$$A = Df_0(\mathbf{x}_0).$$

For  $a \in \mathbb{R}^n$ , let  $\tau_a(\mathbf{x}) = \mathbf{x} + a$ . Directly,

$$\tau_a(\mathbf{x} + h) - \tau_a(\mathbf{x}) - Ih = (\mathbf{x} + h + a) - (\mathbf{x} + a) - h = 0.$$

Hence  $D\tau_a(\mathbf{x}) = I$  at every  $\mathbf{x}$ , with zero remainder. The derivative map is constant, so  $D^j\tau_a = 0$  for  $j \geq 2$ ; thus  $\tau_a \in C^\infty$ . Its inverse is  $\tau_{-a}$ , making  $\tau_a$  a global smooth diffeomorphism.

Define

$$g_0(h) = f_0(\mathbf{x}_0 + h) - f_0(\mathbf{x}_0).$$

Then

$$g_0 = \tau_{-f_0(\mathbf{x}_0)} \circ f_0 \circ \tau_{\mathbf{x}_0}.$$

Consequently  $g_0$  is  $C^k$  on the open set  $\tilde{U} = U - \mathbf{x}_0$ , and the chain rule gives

$$Dg_0(h) = Df_0(\mathbf{x}_0 + h), \quad Dg_0(0) = Df_0(\mathbf{x}_0) = A.$$

Thus translations change the choice of origin without changing the derivative. Apply the invertible linear normalization

$$\tilde{f}(h) = A^{-1}g_0(h) = A^{-1}(f_0(\mathbf{x}_0 + h) - f_0(\mathbf{x}_0)).$$

The chain rule yields

$$D\tilde{f}(h) = A^{-1}Df_0(\mathbf{x}_0 + h), \quad \tilde{f}(0) = 0, \quad D\tilde{f}(0) = I.$$

It is therefore enough to prove the theorem under the normalized assumptions

$$0 \in U, \quad f(0) = 0, \quad Df(0) = I.$$

We now write  $U, f$  for the normalized domain and map.

**2. The contraction estimate.** For each  $y \in \mathbb{R}^n$ , define

$$g_y(\mathbf{x}) = \mathbf{x} - f(\mathbf{x}) + y.$$

Then

$$f(\mathbf{x}) = y \iff g_y(\mathbf{x}) = \mathbf{x}.$$

Set

$$g_0(\mathbf{x}) = \mathbf{x} - f(\mathbf{x}).$$



Thus  $g_y(\mathbf{x}) = g_0(\mathbf{x}) + y$  and

$$Dg_0(\mathbf{x}) = I - Df(\mathbf{x}).$$

The derivative is a map

$$Df : U \longrightarrow \mathcal{L}(\mathbb{R}^n),$$

where the target has the operator norm. Therefore continuity of  $Df$  at 0 means

$$\|Df(\mathbf{x}) - Df(0)\|_{\text{op}} \longrightarrow 0.$$

Since  $Df(0) = I$ , use

$$\varepsilon = 1/2$$

in this continuity statement. There is  $\delta > 0$  such that

$$\|\mathbf{x}\| < \delta \implies \|Df(\mathbf{x}) - I\|_{\text{op}} < \frac{1}{2}.$$

Separately, because  $0 \in U$  and  $U$  is open, [Proposition 10.2.7](#) gives  $\rho > 0$  such that

$$\overline{B}(0, \rho) \subseteq U.$$

Choose

$$0 < r < \min\{\rho, \delta\}.$$

Then

$$\overline{B}(0, r) \subseteq U, \quad \|Df(\mathbf{x}) - I\|_{\text{op}} < \frac{1}{2} \quad (\mathbf{x} \in \overline{B}(0, r)).$$

By [Theorem 10.15.4](#),

$$\|(Df(\mathbf{x}) - I)v\| \leq \|Df(\mathbf{x}) - I\|_{\text{op}}\|v\| \leq \frac{1}{2}\|v\| \quad (v \in \mathbb{R}^n).$$

Since  $Dg_0(\mathbf{x}) = I - Df(\mathbf{x})$ ,

$$\|Dg_0(\mathbf{x})\|_{\text{op}} < \frac{1}{2} \quad (\mathbf{x} \in \overline{B}(0, r)).$$

If  $\mathbf{x}_1, \mathbf{x}_2 \in \overline{B}(0, r)$  and  $0 \leq t \leq 1$ , then

$$\|(1-t)\mathbf{x}_1 + t\mathbf{x}_2\| \leq (1-t)\|\mathbf{x}_1\| + t\|\mathbf{x}_2\| \leq r.$$

Thus the connecting segment lies in  $\overline{B}(0, r) \subseteq U$ . The mean-value estimate [Theorem 11.3.4](#) therefore applies and gives

$$\|g_0(\mathbf{x}_1) - g_0(\mathbf{x}_2)\| \leq \frac{1}{2}\|\mathbf{x}_1 - \mathbf{x}_2\|.$$

Translation by  $y$  cancels in differences, so every  $g_y$  has contraction constant

$$q = 1/2 :$$

$$\|g_y(\mathbf{x}_1) - g_y(\mathbf{x}_2)\| \leq \frac{1}{2}\|\mathbf{x}_1 - \mathbf{x}_2\|.$$

**3. Solving nearby equations.** Set

$$V_0 = B(0, r/2).$$

Because  $g_0(0) = 0$ , for  $\mathbf{x} \in \overline{B}(0, r)$  and  $y \in V_0$ ,

$$\begin{aligned}\|g_y(\mathbf{x})\| &\leq \|g_0(\mathbf{x}) - g_0(0)\| + \|y\| \\ &\leq \frac{1}{2}\|\mathbf{x}\| + \|y\| \\ &< \frac{r}{2} + \frac{r}{2} = r.\end{aligned}$$

Therefore

$$g_y(\overline{B}(0, r)) \subseteq B(0, r) \subseteq \overline{B}(0, r);$$

Thus  $g_y$  maps the closed ball into itself. By [Corollary 10.3.8](#), this closed ball is complete because it is a closed subset of the complete metric space  $\mathbb{R}^n$ . The contraction mapping theorem [Theorem 11.4.2](#) now gives, for every  $y \in V_0$ , a unique fixed point  $\mathbf{x} \in \overline{B}(0, r)$ . The strict self-map estimate places that fixed point in  $B(0, r)$ . Hence each  $y \in V_0$  has exactly one solution of  $f(\mathbf{x}) = y$  inside  $B(0, r)$ .

Define

$$U_0 = f^{-1}(V_0) \cap B(0, r).$$

One has  $0 \in U_0$  because  $f(0) = 0$ . The map  $f$  is continuous, so  $f^{-1}(V_0)$  is open in the open domain  $U$ ; hence it is open in  $\mathbb{R}^n$ , and intersection with the open ball shows that  $U_0$  is open. The fixed-point argument gives every  $y \in V_0$  a preimage in  $B(0, r)$ , and that preimage belongs to  $U_0$  by definition. It is unique there by uniqueness in the closed ball. Thus

$$f : U_0 \longrightarrow V_0$$

is bijective.

**4. The inverse Lipschitz estimate.** Since  $f(\mathbf{x}) = \mathbf{x} - g_0(\mathbf{x})$ , for  $\mathbf{x}, z \in U_0$ ,

$$\begin{aligned}\|f(\mathbf{x}) - f(z)\| &= \|(\mathbf{x} - z) - (g_0(\mathbf{x}) - g_0(z))\| \\ &\geq \|\mathbf{x} - z\| - \|g_0(\mathbf{x}) - g_0(z)\| \\ &\geq \frac{1}{2}\|\mathbf{x} - z\|.\end{aligned}$$

Therefore the inverse satisfies

$$\|f^{-1}(y) - f^{-1}(w)\| \leq 2\|y - w\| \quad (y, w \in V_0).$$

In particular, the inverse is continuous. The same estimate will provide the  $O(\|h\|)$  comparison required when differentiating it.

**5. Differentiability of the inverse.** For every  $\mathbf{x} \in \overline{B}(0, r)$ ,

$$\|I - Df(\mathbf{x})\|_{\text{op}} < \frac{1}{2} < 1.$$

Apply [Proposition 11.4.5](#) with the invertible map  $A = I$  and  $B = Df(\mathbf{x})$ . It follows that  $Df(\mathbf{x})$  is invertible throughout the closed ball.

Let  $g = f^{-1}$ . Fix  $y = f(\mathbf{x}) \in V_0$  and, for small  $h$  with  $y + h \in V_0$ , set

$$\Delta\mathbf{x} = g(y + h) - g(y).$$

Since  $f(\mathbf{x} + \Delta\mathbf{x}) - f(\mathbf{x}) = h$ , differentiability of  $f$  at  $\mathbf{x}$  gives

$$h = Df(\mathbf{x})\Delta\mathbf{x} + R(\Delta\mathbf{x}), \quad R(\Delta\mathbf{x}) = o(\|\Delta\mathbf{x}\|).$$

The inverse Lipschitz estimate yields

$$\|\Delta \mathbf{x}\| \leq 2\|h\|,$$

so  $\Delta \mathbf{x} = O(\|h\|)$  and in particular  $\Delta \mathbf{x} \rightarrow 0$  as  $h \rightarrow 0$ . For  $h \neq 0$ , injectivity of  $g$  gives  $\Delta \mathbf{x} \neq 0$ , and

$$\frac{\|R(\Delta \mathbf{x})\|}{\|h\|} = \frac{\|R(\Delta \mathbf{x})\|}{\|\Delta \mathbf{x}\|} \frac{\|\Delta \mathbf{x}\|}{\|h\|} \rightarrow 0.$$

Thus  $R(\Delta \mathbf{x}) = o(\|h\|)$ . Multiplication by the fixed bounded linear map  $Df(\mathbf{x})^{-1}$  preserves this estimate, because

$$\|Df(\mathbf{x})^{-1}R(\Delta \mathbf{x})\| \leq \|Df(\mathbf{x})^{-1}\|_{\text{op}}\|R(\Delta \mathbf{x})\|.$$

Solving the remainder equation gives

$$\Delta \mathbf{x} = Df(\mathbf{x})^{-1}h + o(\|h\|),$$

and therefore

$$Dg(y) = Df(\mathbf{x})^{-1}.$$

Equivalently,

$$D(f^{-1})(f(\mathbf{x})) = (Df(\mathbf{x}))^{-1}.$$

The remainder calculation establishes differentiability before the identity  $f^{-1} \circ f = \text{id}$  is differentiated.

**6. Higher regularity.** The derivative formula can be written

$$Dg = \text{Inv} \circ Df \circ g.$$

The map  $g$  is continuous by the Lipschitz estimate,  $Df$  is continuous because  $f$  is  $C^1$ , and  $\text{Inv} : \text{GL}(\mathbb{R}^n) \rightarrow \text{GL}(\mathbb{R}^n)$  is continuous, indeed smooth, by [Lemma 11.5.1](#). Hence  $Dg$  is continuous and  $g \in C^1$ .

If  $g \in C^j$  with  $j < k$ , then  $Df \in C^{k-1} \subseteq C^j$ , so  $Dg = \text{Inv} \circ Df \circ g$  is  $C^j$ . Therefore  $g \in C^{j+1}$ . Iteration gives  $g \in C^k$ ; when  $k = \infty$ , repeat the argument for every finite  $j$ .

**7. Returning to the original coordinates.** Return to the original map  $f_0$ , the point  $\mathbf{x}_0$ , and  $A = Df_0(\mathbf{x}_0)$ . The normalized relation was

$$f(h) = A^{-1}(f_0(\mathbf{x}_0 + h) - f_0(\mathbf{x}_0)).$$

Thus the normalized bijection  $f : U_0 \rightarrow V_0$  corresponds to

$$f_0 : \mathbf{x}_0 + U_0 \longrightarrow f_0(\mathbf{x}_0) + AV_0.$$

The two displayed sets are open neighborhoods of  $\mathbf{x}_0$  and  $f_0(\mathbf{x}_0)$ . With  $g = f^{-1}$ , the inverse in the original coordinates is

$$f_0^{-1}(y) = \mathbf{x}_0 + g(A^{-1}(y - f_0(\mathbf{x}_0))).$$

These affine coordinate changes preserve the inverse identities and the  $C^k$  regularity. Moreover,

$$Df(h) = A^{-1}Df_0(\mathbf{x}_0 + h),$$

and hence

$$[Df(h)]^{-1} = [Df_0(\mathbf{x}_0 + h)]^{-1}A.$$

The chain rule for the displayed formula for  $f_0^{-1}$  gives

$$\begin{aligned} D(f_0^{-1})(f_0(\mathbf{x}_0 + h)) &= [Df(h)]^{-1}A^{-1} \\ &= [Df_0(\mathbf{x}_0 + h)]^{-1}AA^{-1} \\ &= [Df_0(\mathbf{x}_0 + h)]^{-1}. \end{aligned}$$

Writing  $\mathbf{x} = \mathbf{x}_0 + h$  proves  $D(f_0^{-1})(f_0(\mathbf{x})) = (Df_0(\mathbf{x}))^{-1}$  on the original neighborhood. Renaming the translated neighborhoods as  $U_0, V_0$  gives the theorem.  $\square$

**Example 11.5.4** (Local is not global). The map

$$P(u, v) = (e^u \cos v, e^u \sin v)$$

has  $\det DP(u, v) = e^{2u} > 0$  at every point. The theorem therefore makes  $P$  a local smooth diffeomorphism everywhere, but  $P$  is not globally one-to-one because  $P(u, v) = P(u, v + 2\pi)$ . An invertible derivative, even at every point, gives a local conclusion unless additional global information is supplied.

**Example 11.5.5** (Using the inverse function theorem). Let

$$f(u, v) = (u + v^2, u^2 + v).$$

Its Jacobian is

$$J_f(u, v) = \begin{pmatrix} 1 & 2v \\ 2u & 1 \end{pmatrix}, \quad \det J_f(u, v) = 1 - 4uv.$$

At  $(0, 0)$  the determinant is 1, so the theorem supplies neighborhoods of  $(0, 0)$  on which  $f$  has a unique smooth inverse. In ordinary language, every output sufficiently close to  $(0, 0)$  comes from exactly one nearby input, and that input varies differentiably with the output. The theorem does not make a global claim; on the curve

$$uv = 1/4$$

the derivative is singular, so the Inverse Function Theorem gives no conclusion about local invertibility there.

Formal differentiation predicts the derivative formula: from  $f^{-1} \circ f = \text{id}$ , the chain rule suggests

$$D(f^{-1})(f(a))Df(a) = I,$$

and hence  $D(f^{-1})(f(a)) = Df(a)^{-1}$ . The theorem proves that this formal calculation applies to an actual local inverse.

At the nonzero source point  $a = (1, 0)$  in the preceding example,  $f(a) = (1, 1)$  and the standard matrices are

$$J_f(1, 0) = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad J_{f^{-1}}(1, 1) = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}.$$

The inverse derivative is evaluated at the corresponding *target* point. Invertibility of the derivative is sufficient for a differentiable local inverse;  $x \mapsto x^3$  is locally bijective at zero but its inverse is not differentiable there.

The graph mechanism behind implicit functions is already present in a linear equation: an invertible block determines the dependent variables.

**Theorem 11.5.6** (Linear implicit function theorem). Let  $V = V_1 \oplus V_2$  and let  $T : V \rightarrow W$  be linear. Write  $T_i = T|_{V_i}$ . If  $T_2 : V_2 \rightarrow W$  is invertible, put  $U = -T_2^{-1}T_1 : V_1 \rightarrow V_2$ . Then

$$\ker T = \{v_1 + U(v_1) : v_1 \in V_1\} = \text{im } \Gamma_U, \quad \Gamma_U : V_1 \rightarrow V, \quad \Gamma_U(v_1) = v_1 + U(v_1).$$

*Proof.* Every  $v \in V$  has a unique decomposition  $v = v_1 + v_2$ , and

$$T(v_1 + v_2) = 0 \iff T_1v_1 + T_2v_2 = 0 \iff v_2 = -T_2^{-1}T_1v_1.$$

This proves both inclusions in the graph identity. Invertibility of  $T_2$  already implies surjectivity of  $T$ .  $\square$

For  $V = \mathbb{R}^n \times \mathbb{R}^m$  and  $W = \mathbb{R}^m$ , write  $T_x, T_y$  for the restrictions to the two coordinate subspaces. The conclusion becomes

$$T(x, y) = 0 \iff y = -T_y^{-1}T_x x.$$

Thus an invertible dependent-variable block turns the solution space into a graph over the independent variables. The nonlinear theorem keeps this mechanism, with a graph that can bend.

**Remark 11.5.7** (From an equation to a local graph). The circle

$$x^2 + y^2 = 1$$

is not the graph of one function  $y = g(x)$  globally: most  $x$  values have two points on the circle. Yet near  $(0, 1)$ , the equation

$$f(x, y) = x^2 + y^2 - 1 = 0$$

can be solved for  $y$  because  $f_y(0, 1) = 2 \neq 0$ . Near  $(1, 0)$  it can instead be solved for  $x$ , since  $f_x(1, 0) = 2 \neq 0$ . The Implicit Function Theorem is the general existence-and-local-parametrization principle behind this familiar calculus picture: it solves selected variables locally even when there is no useful explicit algebraic formula.

For  $f : U \subseteq \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ , define the block derivatives by

$$\begin{aligned} D_x f(a, b)u &= Df(a, b)(u, 0), & D_x f(a, b) : \mathbb{R}^n &\rightarrow \mathbb{R}^m, \\ D_y f(a, b)v &= Df(a, b)(0, v), & D_y f(a, b) : \mathbb{R}^m &\rightarrow \mathbb{R}^m. \end{aligned}$$

Linearity gives  $Df(a, b)(u, v) = D_x f(a, b)u + D_y f(a, b)v$ . Here  $\mathbf{x}, y$  are blocks of variables and  $f(\mathbf{x}, y) = 0$  means  $m$  scalar equations;  $0$  is the zero vector of  $\mathbb{R}^m$ . Thus only the square dependent-variable block has the invertibility type required below. The auxiliary map

$$h : U \rightarrow \mathbb{R}^{n+m}, \quad h(\mathbf{x}, y) = (\mathbf{x}, f(\mathbf{x}, y))$$

retains the  $n$  independent coordinates beside the  $m$  equations, so source and target have the same dimension. Its block derivative is invertible when  $D_y f$  is; a local inverse then recovers the dependent variables.

The linear graph formula in [Theorem 11.5.6](#) now has the following nonlinear counterpart:

| linear                        | nonlinear near $(a, b)$                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------|
| $T(\mathbf{x}, y) = 0$        | $f(\mathbf{x}, y) = 0$                                                                  |
| $T_y$ invertible              | $D_y f(a, b)$ invertible                                                                |
| $y = -T_y^{-1}T_x \mathbf{x}$ | $y = \varphi(\mathbf{x})$                                                               |
| $-T_y^{-1}T_x$                | $-(D_y f(\mathbf{x}, \varphi(\mathbf{x})))^{-1}D_x f(\mathbf{x}, \varphi(\mathbf{x})).$ |

The last row solves the differential of the nonlinear equation by exactly the linear solution formula.

**Theorem 11.5.8** (Implicit Function Theorem). Let  $U \subseteq \mathbb{R}^n \times \mathbb{R}^m$  be open, let  $1 \leq k \leq \infty$ , and let  $f : U \rightarrow \mathbb{R}^m$  be of class  $C^k$ . Suppose

$$f(a, b) = 0$$

and  $D_y f(a, b) : \mathbb{R}^m \rightarrow \mathbb{R}^m$  is invertible. Then there are open neighborhoods  $A \subseteq \mathbb{R}^n$  of  $a$  and  $B \subseteq \mathbb{R}^m$  of  $b$ , with  $A \times B \subseteq U$ , and a unique  $C^k$  map  $\varphi : A \rightarrow B$  such that

$$\varphi(a) = b, \quad f(\mathbf{x}, \varphi(\mathbf{x})) = 0,$$

and

$$\{(\mathbf{x}, y) \in A \times B : f(\mathbf{x}, y) = 0\} = \{(\mathbf{x}, \varphi(\mathbf{x})) : \mathbf{x} \in A\}.$$

Moreover,

$$D\varphi(\mathbf{x}) = -(D_y f(\mathbf{x}, \varphi(\mathbf{x})))^{-1} D_x f(\mathbf{x}, \varphi(\mathbf{x})).$$

*Proof. 1. The straightening map.* The construction straightens the zero set to the coordinate plane  $\mathbb{R}^n \times \{0\}$ . Define

$$h(\mathbf{x}, y) = (\mathbf{x}, f(\mathbf{x}, y)).$$

It keeps the independent variable  $\mathbf{x}$  and replaces  $y$  by the residual  $f(\mathbf{x}, y)$ , so  $f(\mathbf{x}, y) = 0$  exactly when  $h(\mathbf{x}, y) = (\mathbf{x}, 0)$ .

**2. The block derivative.** Its derivative at  $(a, b)$  is the block operator

$$Dh(a, b)(u, v) = (u, D_x f(a, b)u + D_y f(a, b)v).$$

Given  $(p, q) \in \mathbb{R}^n \times \mathbb{R}^m$ , the explicit inverse sends it to

$$\left(p, (D_y f(a, b))^{-1}(q - D_x f(a, b)p)\right).$$

Thus  $Dh(a, b)$  is invertible.

**3. Local inversion of  $h$ .** The Inverse Function Theorem gives open neighborhoods  $W$  of  $(a, b)$  and  $Z$  of  $(a, 0)$  such that  $h : W \rightarrow Z$  is a  $C^k$  diffeomorphism.

**4. The graph function.** Choose an open neighborhood  $A_0$  of  $a$  with  $A_0 \times \{0\} \subseteq Z$ , and immediately define

$$\varphi_0(\mathbf{x}) = \pi_2(h^{-1}(\mathbf{x}, 0)) \quad (\mathbf{x} \in A_0).$$

If  $h^{-1}(\mathbf{x}, 0) = (\mathbf{x}', y')$ , then  $h(\mathbf{x}', y') = (\mathbf{x}, 0)$ , and the first component of  $h$  forces  $\mathbf{x}' = \mathbf{x}$ . Hence

$$h^{-1}(\mathbf{x}, 0) = (\mathbf{x}, \varphi_0(\mathbf{x})), \quad f(\mathbf{x}, \varphi_0(\mathbf{x})) = 0.$$

Also  $\varphi_0(a) = b$ , and  $\varphi_0$  is  $C^k$ .

**5. Graph uniqueness.** Because  $W$  is open at  $(a, b)$ , choose open neighborhoods  $A_1$  of  $a$  and  $B$  of  $b$  with  $A_1 \times B \subseteq W$ . Continuity of  $\varphi_0$  at  $a$  permits shrinking  $A_1$  inside  $A_0$  so that  $\varphi_0(A_1) \subseteq B$ . Set  $A = A_1$  and  $\varphi = \varphi_0|_A$ . If  $(\mathbf{x}, y) \in A \times B$  and  $f(\mathbf{x}, y) = 0$ , then

$$h(\mathbf{x}, y) = (\mathbf{x}, 0) = h(\mathbf{x}, \varphi(\mathbf{x})).$$

Both points lie in  $W$ , so injectivity of  $h|_W$  gives  $y = \varphi(\mathbf{x})$ . This proves the local graph equality and uniqueness.

**6. Differentiating the graph equation.** The operator in the derivative formula is invertible throughout the graph neighborhood. Indeed, since  $h|_W$  is a diffeomorphism, the chain rule for  $h^{-1} \circ h$  shows that  $Dh(\mathbf{x}, y)$  is invertible at every  $(\mathbf{x}, y) \in W$ . At a graph point, if  $D_y f(\mathbf{x}, \varphi(\mathbf{x}))v = 0$ , then

$$Dh(\mathbf{x}, \varphi(\mathbf{x}))(0, v) = (0, 0).$$

Thus  $v = 0$ . The endomorphism  $D_y f(\mathbf{x}, \varphi(\mathbf{x})) : \mathbb{R}^m \rightarrow \mathbb{R}^m$  is injective and hence, in finite dimension, invertible.

The graph function  $\varphi$  has now been constructed. Differentiating  $f(\mathbf{x}, \varphi(\mathbf{x})) = 0$  as an identity of maps from  $\mathbb{R}^n$  to  $\mathbb{R}^m$  gives

$$D_x f(\mathbf{x}, \varphi(\mathbf{x})) + D_y f(\mathbf{x}, \varphi(\mathbf{x}))D\varphi(\mathbf{x}) = 0.$$

Here

$$D_x f : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad D_y f : \mathbb{R}^m \rightarrow \mathbb{R}^m, \quad D\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

Multiplication on the left by  $D_y f(\mathbf{x}, \varphi(\mathbf{x}))^{-1}$  proves

$$D\varphi(\mathbf{x}) = -D_y f(\mathbf{x}, \varphi(\mathbf{x}))^{-1} D_x f(\mathbf{x}, \varphi(\mathbf{x})).$$

□

**Example 11.5.9** (An equation without an explicit solution). For  $f(x, y) = y + \sin y - x$ ,  $f(0, 0) = 0$  and  $f_y(0, 0) = 2$ . Thus near zero there is a unique  $y = \varphi(x)$  with  $\varphi(x) + \sin \varphi(x) = x$ , and

$$\varphi'(x) = 1/(1 + \cos \varphi(x)),$$

in particular

$$\varphi'(0) = 1/2.$$

The theorem supplies existence and derivatives without an explicit inverse formula.

For a local graph  $f(x, y(x)) = 0$ , the chain rule gives

$$0 = \frac{d}{dx}f(x, y(x)) = f_x + f_y y', \quad y' = -\frac{f_x}{f_y} \quad (f_y \neq 0),$$

where the partials are evaluated on the graph. In differential notation,  $df = f_x dx + f_y dy$  vanishes on its velocity  $(1, y')$ . Thus  $f_x dx + f_y dy = 0$  *along the level curve* means zero change in its tangent directions, not that  $df$  is the zero functional on the ambient plane. We return to this kernel interpretation in [Section 13.12](#).

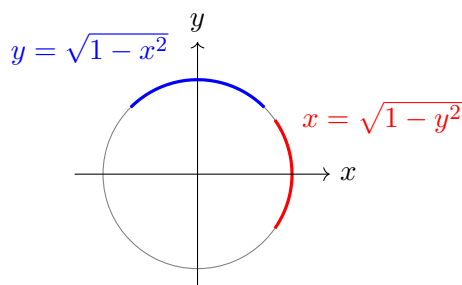
**Example 11.5.10** (Implicit differentiation justified). For  $f(x, y) = x^2 + y^2 - 1$ , take  $(a, b) = (0, 1)$ . Then

$$f(0, 1) = 0, \quad f_y(0, 1) = 2 \neq 0.$$

The Implicit Function Theorem therefore gives a unique differentiable function  $y = g(x)$  near 0 with  $g(0) = 1$  and  $x^2 + g(x)^2 = 1$ . Its derivative is

$$g'(x) = -\frac{f_x(x, g(x))}{f_y(x, g(x))} = -\frac{x}{g(x)}.$$

This is the familiar implicit-differentiation rule from calculus, now backed by a theorem that first supplies the local function being differentiated. At  $(0, -1)$  one obtains the lower local graph; at  $(\pm 1, 0)$  one must solve locally for  $x$  as a function of  $y$  instead.



Near the north pole the circle is a graph over  $x$ ; near the rightmost point it is a graph over  $y$ . Neither graph covers the whole circle. The same choice of a nonzero partial derivative gives local graph descriptions of the regular conics above.

**Example 11.5.11** (Two dependent variables in an implicit system). Consider

$$f(x, y, z) = (y + z - x, y - z + x^2).$$

At  $(0, 0, 0)$  the matrix of derivatives in  $(y, z)$  is  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ , of determinant  $-2$ . Thus both  $y$  and  $z$  can be solved as smooth functions of the one independent variable  $x$ . Directly adding and subtracting the equations gives

$$y(x) = (x - x^2)/2,$$

$$z(x) = (x + x^2)/2.$$

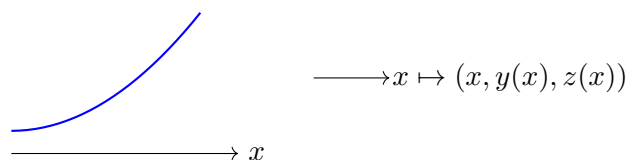
At zero the implicit derivative formula yields

$$(y'(0), z'(0))^T = (1/2, 1/2)^T,$$

which agrees with the explicit solution. The invertible object is the entire dependent-variable block, not one selected partial derivative.



constraint locally becomes a graph



### Analytic refinement: Lagrange inversion (optional)

The implicit theorem gives a local solution. When the data are analytic, Lagrange inversion gives its Taylor coefficients explicitly. Analyticity is essential here: the Implicit Function Theorem alone does not assert it.

**Theorem 11.5.12** (Analytic Lagrange inversion; stated result). Let  $\phi$  be real analytic near 0, with  $\phi(0) \neq 0$ . There is a unique real analytic solution  $w(z)$  near 0, with  $w(0) = 0$ , of  $w = z\phi(w)$ . If  $h$  is analytic near 0, then

$$[z^n]h(w(z)) = \frac{1}{n}[t^{n-1}]h'(t)\phi(t)^n \quad (n \geq 1).$$

Here  $[t^j]$  denotes the coefficient of  $t^j$  in a power series. In particular,

$$w(z) = \sum_{n \geq 1} \frac{z^n}{n!} \left[ \frac{d^{n-1}}{dt^{n-1}} \phi(t)^n \right]_{t=0}.$$

This optional analytic result, including convergence, is used without proof; it is not needed below. See [19, §2.1 and §2.6] for the coefficient formulations and [20] for analytic implicit solutions. No complex integration is needed for its use here. The smooth implicit theorem applied to  $f(z, w) = w - z\phi(w)$ , whose  $w$  derivative at  $(0, 0)$  is 1, already supplies the unique smooth local solution. The additional analytic theorem identifies that solution with a convergent series; it is this extra conclusion that licenses Taylor reconstruction.

**Example 11.5.13** (A computable implicit series). Take  $\phi(t) = 1 + t^2$  and  $h(t) = t$ . The coefficient rule gives zero in even degrees, and in degree  $2j + 1$  gives

$$\binom{2j+1}{j} / (2j+1).$$

Hence

$$w = z + z^3 + 2z^5 + 5z^7 + \cdots.$$

The first coefficients can also be derived directly: substituting  $w = a_1z + a_2z^2 + \cdots$  into  $w = z + zw^2$  gives  $a_1 = 1$ ,  $a_2 = 0$ ,  $a_3 = 1$ ,  $a_4 = 0$ , and  $a_5 = 2$ . Convergence also follows from an elementary coefficient bound. Write

$$c_j = \frac{1}{2j+1} \binom{2j+1}{j} = \frac{1}{j+1} \binom{2j}{j}.$$

The finite binomial theorem gives

$$0 \leq c_j \leq \binom{2j}{j} \leq 2^{2j} = 4^j.$$

Thus, for

$$|z| < 1/2,$$

$$\sum_{j=0}^{\infty} c_j |z|^{2j+1} \leq |z| \sum_{j=0}^{\infty} (4|z|^2)^j = \frac{|z|}{1-4|z|^2}.$$

The algebraic solution

$$w = (1 - \sqrt{1 - 4z^2})/(2z),$$

with its removable value 0 at zero, provides an independent consistency check.

## 11.6 Rank and Local Normal Forms

An invertible derivative preserves all directions. More generally its rank counts how many independent first-order output directions remain. Constant rank lets us turn this linear information into local coordinates.

**Definition 11.6.1** (Rank of a map). For a  $C^1$  map  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$  on an open set, define

$$\text{rank}_a f = \text{rank } Df(a) = \dim \text{im } Df(a) = \text{rank } J_f(a).$$

The map has **constant rank**  $r$  on a set if this number equals  $r$  at every point of that set.

**Lemma 11.6.2** (Persistence of a nonzero minor). If  $\text{rank } Df(a) = r$ , then  $\text{rank } Df(\mathbf{x}) \geq r$  on a neighborhood of  $a$ .

*Proof.* For  $r = 0$  the claim is automatic. Otherwise [Proposition 10.17.8](#) supplies an  $r \times r$  minor with determinant  $d \neq 0$ . Its determinant is continuous in  $\mathbf{x}$ , since the entries of  $Df$  are continuous and a determinant is a polynomial in them. On a neighborhood where its difference from  $d$  is less than

$$|d|/2,$$

that minor is nonzero, so the rank is at least  $r$ . □

**Theorem 11.6.3** (Constant Rank Theorem). Let  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$  be  $C^k$ ,  $1 \leq k \leq \infty$ , and suppose its rank is constantly  $r$  near  $a$ . There are local  $C^k$  diffeomorphisms  $\phi$  near  $a$  and  $\psi$  near  $f(a)$ , taking these points to zero, for which

$$(\psi \circ f \circ \phi^{-1})(u_1, \dots, u_n) = (u_1, \dots, u_r, 0, \dots, 0)$$

on a neighborhood of zero.

*Proof.* We first use  $r$  outputs as source coordinates, then remove the dependence of the remaining outputs on those coordinates. If  $r = 0$ , the derivative vanishes on a ball about  $a$ ; the mean-value estimate makes  $f$  constant there. Translations in source and target give the assertion. Assume  $r > 0$ . Translate  $a$  and  $f(a)$  to zero and permute coordinates so that the first  $r$  rows and columns form an invertible submatrix. Write  $\mathbf{x} = (s, t)$ ,  $s \in \mathbb{R}^r$ ,  $t \in \mathbb{R}^{n-r}$ , and set

$$A(s, t) = (f_1(s, t), \dots, f_r(s, t), t).$$

Its derivative at zero is block triangular with the chosen submatrix and  $I_{n-r}$  on its diagonal. The inverse function theorem gives a local  $C^k$  diffeomorphism  $A$ . Restrict its image to a product of open boxes  $B \times D$  centered at zero, and its source to the inverse image of that product. Then

$$f \circ A^{-1}(u, v) = (u, g(u, v)), \quad D(f \circ A^{-1}) = \begin{pmatrix} I_r & 0 \\ D_u g & D_v g \end{pmatrix}.$$

Right multiplication by the invertible  $DA^{-1}$  preserves rank, so this matrix has rank  $r$ . Its first  $r$  columns are independent. Every other column has zero first block and must be their linear combination; the top block of that combination is exactly its coefficient vector. That vector must therefore be zero. The entire column vanishes, proving  $D_v g = 0$ . For fixed  $u$ , the segment between any two points of  $D$  stays in  $D$ ; the mean-value estimate gives  $g(u, v) = g(u, 0) = h(u)$ . If there are no  $v$  coordinates this assertion simply defines  $h = g$ .

On the open target set  $B \times \mathbb{R}^{m-r}$  the map

$$\psi_0(y, z) = (y, z - h(y))$$

is a  $C^k$  diffeomorphism, with inverse  $(y, z) \mapsto (y, z + h(y))$ . It sends  $(u, h(u))$  to  $(u, 0)$ . Taking  $\phi = A$  and  $\psi = \psi_0$  in these translated and permuted coordinates gives the desired formula. Composing with the original translations and permutations gives actual coordinate diffeomorphisms for the original map and base points. Empty blocks when  $r = m$  are omitted.  $\square$

The principal special cases are

$$\begin{aligned} r = n = m &\Rightarrow \text{inverse-function normal form,} \\ r = m \leq n &\Rightarrow \text{projection normal form,} \\ r = n \leq m &\Rightarrow \text{inclusion normal form.} \end{aligned}$$

**Definition 11.6.4** (Immersion and submersion). A  $C^1$  map  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$  is an **immersion at**  $a$  if  $Df(a)$  is injective, equivalently  $\text{rank}_a f = n$ . It is a **submersion at**  $a$  if  $Df(a)$  is surjective, equivalently  $\text{rank}_a f = m$ . These require  $n \leq m$  and  $m \leq n$ , respectively. It is an immersion or submersion on  $U$  if the corresponding condition holds at every point.

The persistence lemma makes either full-rank condition hold on a neighborhood, so the Constant Rank Theorem applies. Projection  $(x, y) \mapsto x$  is a submersion, and inclusion  $x \mapsto (x, 0)$  is an immersion. The terminology is Euclidean here. Chapter 13 constructs the intrinsic differential needed to formulate the manifold notions, developed in Appendix F.

**Example 11.6.5** (A normal form and a rank drop). The map  $f(x, y) = (x + y, (x + y)^2)$  has rank one everywhere. The coordinate changes  $\phi(x, y) = (x + y, y)$  and  $\psi(u, v) = (u, v - u^2)$  give  $\psi \phi^{-1}(u, v) = (u, 0)$ . In contrast,  $(x, y) \mapsto (x, xy)$  has determinant  $x$  and rank drops from two to one on  $x = 0$ ; there is no constant-rank neighborhood there.

## Submersions, Regular Level Sets, and Multipliers

The submersion normal form describes level sets as coordinate planes. The implicit graph description also identifies their tangent directions directly.

**Corollary 11.6.6** (Regular level sets and tangent directions). Let  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$  be  $C^1$  on open  $U$ ,  $m \leq n$ , and suppose  $f(a) = c$ ,  $\text{rank } Df(a) = m$ . Near  $a$ , the level set  $f^{-1}(c)$  is a  $C^1$  graph of  $n - m$  variables. Its tangent directions, defined as velocities at  $a$  of  $C^1$  curves in the level set, are exactly  $\ker Df(a)$ .

*Proof.* The images of the coordinate basis vectors span  $\mathbb{R}^m$ ; select  $m$  of them forming a basis and reorder these as dependent coordinates  $y$ . The remaining coordinates are  $\mathbf{x} \in \mathbb{R}^{n-m}$ , and  $D_y f(a)$  is invertible. Apply the implicit theorem to  $f(\mathbf{x}, y) - c$  to obtain  $y = \phi(\mathbf{x})$  locally. A curve  $\gamma$  in the level set satisfies  $Df(a)\gamma'(0) = 0$  by the chain rule. Conversely write a kernel vector as  $(u, v)$ . The equation  $D_x f(a)u + D_y f(a)v = 0$  gives  $v = D\phi(a_x)u$ . The curve  $(a_x + tu, \phi(a_x + tu))$  has exactly this velocity and stays in the level set for small  $t$ . If  $m = n$ , the graph has zero free variables and the local level set is a single point; its only velocity is zero.  $\square$

Writing  $T_a f^{-1}(c)$  for these curve velocities, the conclusion is

$$T_a f^{-1}(c) = \ker Df(a).$$

Thus  $m$  independent equations leave local dimension  $n - m$ . For scalar  $f$ ,  $\nabla f(a) \neq 0$  gives an  $(n - 1)$ -dimensional graph and  $\ker Df(a) = \{v : \nabla f(a) \cdot v = 0\}$ . This makes the gradient normal to the constraint and prepares the multiplier condition below.

**Remark 11.6.7** (Equations, graphs, and parametrizations). The classical descriptions  $y = f(x)$ ,  $f(x, y) = 0$ , and  $(\mathbf{x}, y) = \gamma(t)$  become, in higher dimensions, a graph  $y = \varphi(\mathbf{x})$ , a level set  $f(\mathbf{x}) = c$ , and a parametrization  $\mathbf{x} = \sigma(u)$ . The implicit theorem turns a regular equation locally into a graph; the constant-rank theorem gives its coordinate normal form. In Chapter 13, the inverse of a chart will provide the corresponding local parametrization. For a regular scalar level surface in  $\mathbb{R}^3$ , the affine tangent plane is

$$\nabla f(a) \cdot (\mathbf{x} - a) = 0, \quad T_a(f^{-1}(c)) = \ker Df(a).$$

The gradient is the Euclidean representative of the derivative covector. For  $a, b, c, R, p > 0$ , useful familiar models are

|                                   |                      |
|-----------------------------------|----------------------|
| $x^2/a^2 + y^2/b^2 = 1$           | ellipse,             |
| $x^2/a^2 - y^2/b^2 = 1$           | hyperbola,           |
| $y = x^2/(4p)$                    | parabola,            |
| $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ | ellipsoid,           |
| $x^2 + y^2 = R^2$                 | cylinder,            |
| $x^2 + y^2 = z^2$                 | cone,                |
| $z = x^2 + y^2$                   | elliptic paraboloid. |

Their defining gradients are nonzero on these sets, except at the cone's vertex. Thus the regular-level argument applies away from that vertex. Scaling turns the ellipsoid into a sphere; cylindrical coordinates simplify the cylinder and cone. These coordinate choices will also simplify integrals.

The auxiliary map already used in the proof,  $h(\mathbf{x}, y) = (\mathbf{x}, f(\mathbf{x}, y))$ , is a local diffeomorphism when  $D_y f(a)$  is invertible. In coordinates  $(\mathbf{x}, z) = h(\mathbf{x}, y)$ , the equation  $f(h^{-1}(\mathbf{x}, z)) = z$  says that  $f$  becomes projection onto the second block. This is the local submersion form supplied by the same proof.

If  $f : U \rightarrow \mathbb{R}^n$  is  $C^1$  with invertible derivative everywhere, the inverse theorem makes it a local diffeomorphism at every point. It is an open map: for any open  $V \subseteq U$ , restrict a local inverse neighborhood at each  $\mathbf{x} \in V$ ; its intersection with  $V$  has open image under that local

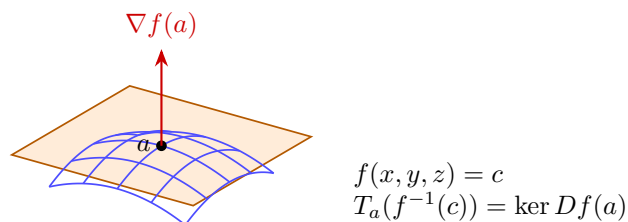


Figure 11.1: The regular level surface has tangent direction space  $\ker Df(a)$ . Its affine tangent plane passes through  $a$ ; the gradient is normal to it.

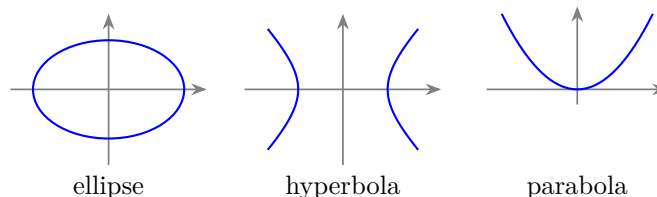


Figure 11.2: Three regular conics. Their equations become local graphs wherever the corresponding partial derivative is nonzero; a vertical tangent requires solving for the other variable.

homeomorphism, and these images cover  $f(V)$ . This does not give global injectivity, as the earlier periodic map  $(x, y) \mapsto (e^x \cos y, e^x \sin y)$  demonstrates:  $y$  and  $y + 2\pi$  have the same image.

Continuity of the determinant also shows that  $\det Df(a) \neq 0$  has constant nonzero sign on a sufficiently small neighborhood of  $a$ . Positive sign means locally orientation preserving; negative sign means locally orientation reversing. Chapters 12–13 use its magnitude and sign for their different integration purposes.

Finally the implicit formula is a statement of smooth dependence on parameters: if  $f(t, y) = 0$  and  $D_y f(t_0, y_0)$  is invertible, the nearby solution  $y = \phi(t)$  has the regularity of  $f$  and  $D\phi(t) = -(D_y f(t, \phi(t)))^{-1} D_t f(t, \phi(t))$ . For  $f(t, y) = y^2 - t$  at  $(1, 1)$  this selects the positive branch  $\phi(t) = \sqrt{t}$  near 1, and

$$\phi'(t) = 1/(2\phi(t)).$$

The negative branch illustrates why the conclusion is local to the chosen solution.

**Theorem 11.6.8** (Lagrange multipliers for one constraint). Let  $f, g : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  be of class  $C^1$ . Suppose that  $a$  is a local extremum of  $f$  subject to

$$g(\mathbf{x}) = c,$$

with  $g(a) = c$ , and that  $\nabla g(a) \neq 0$ . Then there is  $\lambda \in \mathbb{R}$  such that

$$\nabla f(a) = \lambda \nabla g(a).$$

*Proof.* By Corollary 11.6.6, the level set  $g = c$  is locally a  $C^1$  graph with tangent space  $T = \ker Dg(a)$ . Every tangent vector is the velocity of a curve in that graph. The restriction of  $f$  has a local extremum, so the one-variable derivative along each such curve is zero. The chain rule gives  $Df(a)v = 0$  for every  $v \in T$ . Thus  $Df(a)$  vanishes on  $\ker Dg(a)$ . Write  $\ell = Dg(a)$  and  $m = Df(a)$ . Choose  $w$  with  $\ell(w) = 1$ ; this is possible by rescaling a vector where the nonzero functional  $\ell$  does not vanish. Every vector splits as

$$v = (v - \ell(v)w) + \ell(v)w, \quad v - \ell(v)w \in \ker \ell.$$

Applying  $m$  kills the first summand, giving  $m(v) = m(w)\ell(v)$ . Thus  $m = m(w)\ell$  as functionals. Geometrically the allowed tangent directions lie in the constraint kernel; a constrained extremum has no first-order change in any such direction. With the Euclidean inner product the two representing gradients are parallel. Thus  $\lambda = m(w)$  gives the stated gradient equation.  $\square$

**Example 11.6.9** (Why the defining equation must be regular). Let  $g(x, y) = y^2$  and  $f(x, y) = x^2 + y$ . On  $g = 0$  we have  $y = 0$  and  $f(x, 0) = x^2$ , so zero is a strict constrained minimum. But  $\nabla g(0) = 0$  while  $\nabla f(0) = (0, 1)$ ; no multiplier equation is possible. The equivalent defining equation  $y = 0$  has nonzero gradient and gives multiplier 1. Regularity concerns the chosen equation, not merely its geometric zero set.

**Example 11.6.10** (Extrema of  $xy$  on the unit circle). To maximize or minimize

$$f(x, y) = xy$$

subject to  $g(x, y) = x^2 + y^2 = 1$ , observe that

$$\nabla f = (y, x), \quad \nabla g = (2x, 2y).$$

The Lagrange equations are

$$(y, x) = \lambda(2x, 2y), \quad x^2 + y^2 = 1.$$

They give

$$\lambda = 1/2$$

and  $y = x$ , or

$$\lambda = -1/2$$

and  $y = -x$ . Thus

$$f = 1/2$$

at

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right),$$

and

$$f = -1/2$$

at the two points with  $y = -x$ . These are respectively the maximum and minimum, as also follows from  $2|xy| \leq x^2 + y^2 = 1$ .

**Remark 11.6.11** (How to recognize which theorem to use). For a map  $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ , a question about a local inverse calls for computing  $Df(a)$  and checking that it is invertible (equivalently, that its determinant is nonzero). For equations  $f(x, y) = 0$  and a request to solve for the  $y$  variables, compute the block  $D_y f(a, b)$  instead. In either case, the result concerns neighborhoods of the specified point, not a global inverse or a global parametrization.

## Dimension consequences

The normal form detects lost source directions. A different covering argument will detect when a source has too few dimensions to fill an open target.

**Theorem 11.6.12** (More source dimensions prevent injectivity). If  $U \subseteq \mathbb{R}^n$  and  $V \subseteq \mathbb{R}^m$  are nonempty open sets,  $n > m$ , and  $f : U \rightarrow V$  is  $C^1$ , then  $f$  is not injective.

*Proof.* The set of attained ranks is a nonempty subset of the finite set  $\{0, \dots, m\}$ . It therefore has a largest member  $r$ , attained at some  $a \in U$ . By Lemma 11.6.2 the rank is at least  $r$  near  $a$ ; maximality makes it equal to  $r$  there. The Constant Rank Theorem supplies coordinates in which  $f$  is  $(u_1, \dots, u_n) \mapsto (u_1, \dots, u_r, 0, \dots, 0)$ . Since  $r \leq m < n$ , a small change in coordinate  $u_{r+1}$  stays in the open coordinate neighborhood and leaves the image unchanged. The inverse source coordinate map gives two distinct points with the same  $f$  value, because the target coordinate map is injective.  $\square$

**Theorem 11.6.13** (Fewer source dimensions prevent surjectivity). Under the same open-set and  $C^1$  hypotheses, if  $n < m$ , then  $f$  is not surjective.

The proof is deferred to Theorem 12.3.18 in Chapter 12. It uses small-box volume bounds and compactness, not the Constant Rank Theorem. In particular, a countable union of Jordan-null sets need not be Jordan-null; that inference will not be used.

**Proposition 11.6.14** (Diffeomorphisms preserve dimension). A  $C^1$  diffeomorphism between nonempty open subsets of  $\mathbb{R}^n$  and  $\mathbb{R}^m$  requires  $n = m$ .

*Proof.* Writing  $g = f^{-1}$ , the chain rule at  $a$  and  $f(a)$  gives

$$Dg(f(a))Df(a) = I_{\mathbb{R}^n}, \quad Df(a)Dg(f(a)) = I_{\mathbb{R}^m}.$$

Thus  $Df(a) : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is a linear isomorphism, so  $n = m$ .  $\square$

## Primitive diffeomorphisms and local factorization

Elementary coordinate operations have nonlinear counterparts: one output coordinate changes while the others remain available as parameters.

**Definition 11.6.15** (Primitive map). A  $C^k$  map on an open subset of  $\mathbb{R}^n$  is **primitive** if for some  $i$  it has the form

$$q(\mathbf{x}) = (x_1, \dots, x_{i-1}, p(\mathbf{x}), x_{i+1}, \dots, x_n).$$

A primitive diffeomorphism is such a map that is a diffeomorphism onto an open image. Coordinate permutations are treated separately.

Expanding its determinant along the unchanged rows gives  $\det Dq = \partial_i p$ . Thus  $\partial_i p(a) \neq 0$  makes  $q$  a local  $C^k$  diffeomorphism at  $a$  by the inverse theorem. To see why one product box suffices, write the changed coordinate last:  $q(\mathbf{x}', t) = (\mathbf{x}', p(\mathbf{x}', t))$ . Continuity keeps  $\partial_t p$  of one sign on a sufficiently small open product box  $B$  around  $a$ . For fixed  $\mathbf{x}'$ , the mean value theorem makes  $t \mapsto p(\mathbf{x}', t)$  strictly monotone. If  $q(\mathbf{x}', t) = q(\mathbf{y}', s)$ , the unchanged coordinates give  $\mathbf{x}' = \mathbf{y}'$ , and strict monotonicity then gives  $t = s$ . Thus  $q$  is injective on all of  $B$ . At every point of  $B$  its derivative is invertible, so the inverse function theorem gives an open image neighborhood and a  $C^k$  local inverse. These image neighborhoods cover  $q(B)$ , making it open. On overlaps the local inverses agree because injectivity permits only one preimage in  $B$ . They therefore define a  $C^k$  inverse  $q^{-1} : q(B) \rightarrow B$ . Solving a primitive map requires solving only one scalar equation.

**Theorem 11.6.16** (Local primitive factorization). Let  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$  be  $C^k$ ,  $k \geq 1$ , with  $Df(a)$  invertible. There is a source neighborhood  $N$  on which, after a permutation of output coordinates,  $f$  is a composition of  $n$  primitive  $C^k$  diffeomorphisms between open sets. This is a local assertion.

The coordinate replacements can be chosen successively. Once the first  $j-1$  outputs equal their input coordinates, invertibility forces some remaining output to detect the  $j$ th source direction. Using that output as a new coordinate makes one more coordinate exact.

*Proof.* By the inverse function theorem, restrict  $f_0 = f$  to a local  $C^k$  diffeomorphism near  $a_0 = a$ . Suppose  $f_{j-1}$  is a local diffeomorphism near  $a_{j-1}$  whose first  $j-1$  components are  $x_1, \dots, x_{j-1}$ . Invertibility gives

$$Df_{j-1}(a_{j-1})e_j \neq 0.$$

Its first  $j-1$  components are zero, since  $\partial_j x_l = 0$  for  $l < j$ . Thus some  $i \geq j$  satisfies  $\partial_j f_{j-1,i}(a_{j-1}) \neq 0$ . Choose an output permutation  $\tau_j$  fixing coordinates  $1, \dots, j-1$  and bringing this component to position  $j$ . Put  $g_j = \tau_j \circ f_{j-1}$  and define

$$q_j(\mathbf{x}) = (x_1, \dots, x_{j-1}, g_{j,j}(\mathbf{x}), x_{j+1}, \dots, x_n).$$

The scalar function  $g_{j,j}$  is component  $j$  of the permuted residual map. Since  $\det Dq_j(a_{j-1}) = \partial_j g_{j,j}(a_{j-1}) \neq 0$ , the inverse function theorem makes  $q_j$  a primitive  $C^k$  diffeomorphism on a smaller neighborhood. Set

$$a_j = q_j(a_{j-1}), \quad f_j = g_j \circ q_j^{-1} = \tau_j \circ f_{j-1} \circ q_j^{-1}.$$

For  $y = q_j(\mathbf{x})$ , the first  $j-1$  components of  $g_j(\mathbf{x})$  are  $x_l = y_l$ , and its  $j$ th component is  $g_{j,j}(\mathbf{x}) = y_j$ . Consequently

$$f_j(y) = (y_1, \dots, y_j, f_{j,j+1}(y), \dots, f_{j,n}(y)).$$

It is again a local diffeomorphism, now near  $a_j$ , so induction continues.

All these constructions take place on open neighborhoods. At each stage restrict the current neighborhood so that the next  $q_j$  and its inverse are defined, and pull that restriction back through the preceding primitive factors to a neighborhood of  $a$ . Previously constructed maps remain diffeomorphisms when restricted to these open subsets. There are only finitely many stages, so this gives one final source neighborhood  $N$  on which every factor and inverse in the composite is defined.

At the last stage  $f_n = \text{id}$ . Each  $\tau_j$  fixes the coordinates already made exact, and their composite is the single permutation  $\sigma = \tau_n \circ \dots \circ \tau_1$ . Iterating the defining identities on the final neighborhoods gives

$$\text{id} = f_n = \sigma \circ f \circ q_1^{-1} \circ \dots \circ q_n^{-1}.$$

Composing on the right with the primitive factors in reverse order yields

$$\sigma \circ f|_N = q_n \circ \dots \circ q_1.$$

□



**Example 11.6.17** (Two successive coordinate replacements). For  $f(x, y) = (x + y, y + (x + y)^2)$ , take  $q_1(x, y) = (x + y, y)$  and  $q_2(u, v) = (u, v + u^2)$ . Then  $f = q_2 \circ q_1$  and both primitive determinants equal one.

Chapter 12 will reduce a local change of variables to repeated one-coordinate substitutions through [Theorem 11.6.16](#); compare the structural treatments in [\[10, 13, 16\]](#).

### Finite Euclidean localization

Compact support makes finitely many local formulas enough. We now build smooth weights that assemble them without counting overlaps twice. For a function on an open set  $G$ , its support is the closure in  $G$  of its nonzero set;  $C_c^\infty(G)$  denotes smooth functions whose support is compact in  $G$ . Such functions extend smoothly by zero to  $\mathbb{R}^n$ .

**Lemma 11.6.18** (Euclidean cutoff). If  $K \subset G \subseteq \mathbb{R}^n$ ,  $K$  is compact, and  $G$  is open, there is  $\chi \in C_c^\infty(G)$  with  $0 \leq \chi \leq 1$  and  $\chi = 1$  on a neighborhood of  $K$ .

*Proof.* For empty  $K$  use zero. From [Theorem 8.8.2](#) form

$$s(t) = \frac{\theta(t)}{\theta(t) + \theta(1-t)}.$$

The denominator is positive everywhere. Thus  $s$  is smooth, lies in  $[0, 1]$ , equals zero for  $t \leq 0$ , and equals one for  $t \geq 1$ . For each  $a \in K$  choose  $r_a > 0$  with  $\bar{B}(a, 2r_a) \subset G$ . Finitely many  $B(a_j, r_j)$  cover  $K$ . Define

$$b_j(\mathbf{x}) = s\left(\frac{4r_j^2 - \|\mathbf{x} - a_j\|^2}{3r_j^2}\right), \quad \chi(\mathbf{x}) = 1 - \prod_j (1 - b_j(\mathbf{x})).$$

The function  $b_j$  is one on  $\bar{B}(a_j, r_j)$  and supported in  $\bar{B}(a_j, 2r_j)$ . Hence  $\chi$  has the required range, equals one on the union of the smaller open balls, and has compact support in the finite union of the larger closed balls inside  $G$ .  $\square$

**Theorem 11.6.19** (Finite Euclidean partition of unity). If a compact  $K \subseteq \mathbb{R}^n$  is covered by open sets  $U_1, \dots, U_N$ , there are smooth  $\rho_i : \mathbb{R}^n \rightarrow [0, 1]$  with compact support in  $U_i$  and

$$\sum_i \rho_i = 1$$

on a neighborhood of  $K$ .

*Proof. Strategy.* Assign each cutoff the portion not already assigned; the finite product identity telescopes.

For each point of  $K$  choose a ball about it whose doubled closed ball lies in one of the  $U_i$ , and choose a finite subcover by the smaller balls. Let  $K_i$  be the union of the smaller closed balls assigned to  $U_i$  (possibly empty). Then  $K_i$  is compact in  $U_i$ , and their interiors cover  $K$ . Choose  $\chi_i$  by [Lemma 11.6.18](#), equal to one near  $K_i$  and compactly supported in  $U_i$ . Define globally

$$\rho_i = \chi_i \prod_{j < i} (1 - \chi_j), \quad p_0 = 1, \quad p_i = \prod_{j=1}^i (1 - \chi_j).$$

Then  $\rho_i = p_{i-1} - p_i$ , so

$$\sum_{i=1}^N \rho_i = 1 - \prod_{i=1}^N (1 - \chi_i).$$

Near every point of  $K$ , at least one  $\chi_i$  equals one. The sum is therefore one on an open neighborhood of  $K$ . Each  $\rho_i$  is globally smooth,  $0 \leq \rho_i \leq 1$ , and  $\text{supp } \rho_i \subseteq \text{supp } \chi_i \subset U_i$  is compact. Each successive cutoff receives the portion not already assigned by earlier cutoffs. The order dependence is harmless for existence.  $\square$

The finite Euclidean construction is all that Chapter 12 needs. Chapter 13 replaces the finite compact situation by locally finite covers of manifolds.

## Exercises

The first exercises practice the computational tests and conic geometry; the remaining problems range across the chapter.

**Exercise 1.** For  $f(x, y) = xy + y^2$ ,  $a = (1, 2)$ , and  $v = (2, -1)$ , compute  $(df)_a$ , express it in  $dx, dy$ , and evaluate  $(df)_a(v)$ . For  $\gamma(t) = (1 + 2t, 2 - t)$ , compare this value with  $(f \circ \gamma)'(0)$  computed directly and by the chain rule.

**Exercise 2.** Find polar coordinates for  $(-\sqrt{3}, -1)$  with  $0 \leq \theta < 2\pi$ . Explain why

$$\arctan(y/x)$$

alone selects the wrong quadrant.

**Exercise 3.** For

$$f(x, y) = xy/\sqrt{x^2 + y^2},$$

extended by zero, decide which of continuity, coordinate partials, all two-sided directional derivatives, and differentiability hold at zero. Explain why the direction  $(1, 1)$  is decisive.

**Exercise 4.** Estimate directly the normalized remainder of  $x^2 + xy + 2y^2$  at an arbitrary point. Find the tangent line and a unit normal to

$$x^2/4 + y^2 = 1$$

at

$$(\sqrt{2}, 1/\sqrt{2}).$$

**Exercise 5.** For

$$f(x, y) = (x^2y, \sin(x + y)),$$

compute  $Df(x, y)$  and  $Df(x, y)(u, v)$ .

**Exercise 6.** Give a function from  $\mathbb{R}^2$  to  $\mathbb{R}$  with both partial derivatives at the origin but which is not differentiable there.

**Exercise 7.** Prove that a linear map between finite-dimensional normed spaces is differentiable everywhere and is its own derivative.

**Exercise 8.** Prove directly from the definition that differentiability at  $a$  implies continuity at  $a$ .

**Exercise 9.** Use [Theorem 11.3.4](#) to prove that a  $C^1$  map with zero derivative on a convex open set is constant.

**Exercise 10.** Compute the Hessian of

$$x^3 + xy^2 - 3x$$

and classify its critical points.

**Exercise 11.** Apply Taylor's theorem to find the quadratic approximation of

$$e^x \sin y$$

at the origin.

**Exercise 12.** Give a function with unequal mixed partial derivatives at the origin and explain why [Theorem 11.3.9](#) does not apply.

**Exercise 13.** Use the contraction mapping theorem to prove that

$$x = \cos x$$

has a unique solution in

$$[0, 1].$$

**Exercise 14.** Let  $A$  be a matrix with

$$\|I - A\|_{\text{op}} < 1.$$

Show that

$$\mathbf{x}_{k+1} = (I - A)\mathbf{x}_k + b$$

converges to the unique solution of

$$Ax = b.$$

**Exercise 15.** Apply the inverse function theorem to

$$f(x, y) = (e^x \cos y, e^x \sin y)$$

at  $(0, 0)$  and compute the derivative of its local inverse at  $(1, 0)$ .

**Exercise 16.** Near  $(0, 1)$ , use the implicit function theorem on

$$x^2 + y^2 + xy = 1$$

to solve for  $y$  as a  $C^1$  function of  $x$  and find its derivative at zero.

**Exercise 17.** Explain why

$$y^2 = x^3$$

has a full local zero set that is not the graph of one function on an open interval containing zero. Separately show that its one-sided branches

$$y = \pm x^{3/2},$$

$x \geq 0$ , are  $C^1$  up to the origin.

**Exercise 18.** For  $f(x, y) = \log(1 + x) + y^2$  near zero, compute  $Df(0)$  and  $D^2f(0)$  and write the second-order Peano formula. Distinguish its quadratic term from the bilinear second derivative evaluated on two different vectors.

**Exercise 19.** Iterate

$$T(x) = \cos x/2$$

on  $[0, 1]$ . Check self-map, contraction, and completeness, then give an a priori error bound after ten steps starting at zero. Which bound uses the last observed step?

**Exercise 20.** For  $f(x, y, z) = (y + z - x, y - z + x^2)$  calculate the implicit derivative at  $x = 1$  from the block formula and the explicit solution. Identify its source dimension and target dimension.

**Exercise 21.** At the origin compare the second-order tests for  $x^2 + 2y^2$ ,  $x^2 - y^2$ , and  $x^2 + y^4$ . In the last case, explain why the theorem is inconclusive even though the definition decides the minimum.

**Exercise 22.** For  $T(x, y, z) = (x + 2y + z, y - z)$  choose  $x$  independent and  $(y, z)$  dependent. Compute  $-T_y^{-1}T_x$  and parametrize the kernel by its graph map.

**Exercise 23.** Find the ranks and rank-drop sets of  $(x, y) \mapsto (x, xy)$  and  $(x, y, z) \mapsto (x + y, z)$ . Identify the submersions and immersions.

**Exercise 24.** For  $f(x, y) = (x + y, y + (x + y)^2)$  compute the two primitive inverses and verify the determinant product. Put  $(x + y, (x + y)^2)$  in constant-rank form.

**Exercise 25.** Use the transition  $s$  from the cutoff proof to construct two smooth weights subordinate to  $(-\infty, 2)$  and  $(-1, \infty)$  that sum to one. For  $w = z(1 + w)^2$ , use Lagrange inversion to compute through degree four.

## 11.7 Perspective and Transition

Chapter 12 combines primitive factorization, one-variable substitution, Fubini, and finite Euclidean localization to transform integrals. Rank and its normal forms explain the underlying local geometry. In Chapter 13 derivatives transport tangent directions, while pullbacks transport covectors and forms. Exterior differentiation and integration then lead to Stokes' theorem.

## Chapter 12

# Multivariable Riemann Integration

### 12.1 Rectangles, Partitions, and Darboux Sums

This is the direct extension of Chapter 9: intervals become rectangles, length becomes volume, and one partitions each coordinate interval at once. The Darboux definition is unchanged in spirit. Chapter 10 supplies the compactness and uniform-continuity facts that make continuous functions integrable on these higher-dimensional compact rectangles.

We begin with *scalar-valued* functions  $f : Q \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ . The scalar outputs are ordered real numbers, so on every nonempty cell  $R$  a bounded  $f$  has a finite infimum and supremum by completeness:

$$\inf_{\mathbf{x} \in R} f(\mathbf{x}), \quad \sup_{\mathbf{x} \in R} f(\mathbf{x}).$$

| Chapter 9            | Chapter 12              |
|----------------------|-------------------------|
| subinterval          | subrectangle            |
| length               | $n$ -dimensional volume |
| $\inf_I f, \sup_I f$ | $\inf_R f, \sup_R f$    |

**Definition 12.1.1** (Rectangles and volume). A **rectangle** in  $\mathbb{R}^n$  is a product

$$Q = [a_1, b_1] \times \cdots \times [a_n, b_n],$$

where

$$a_j < b_j \quad (1 \leq j \leq n).$$

Its **volume** is

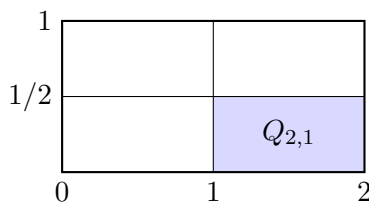
$$\text{vol}(Q) = \prod_{j=1}^n (b_j - a_j).$$

**A grid you can see.** On  $Q = [0, 2] \times [0, 1]$ , choose horizontal-coordinate cuts  $0 = x_0 < x_1 = 1 < x_2 = 2$  and vertical-coordinate cuts

$$0 = y_0 < y_1 = 1/2 < y_2 = 1.$$

A typical cell is  $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$ , with area  $(x_i - x_{i-1})(y_j - y_{j-1})$ . There are four cells, each of area

$$1/2.$$



Higher dimensions add no new idea: partition every coordinate interval and take products. In general choose

$$a_j = x_{j,0} < x_{j,1} < \cdots < x_{j,m_j} = b_j \quad (1 \leq j \leq n).$$

The resulting closed cells are

$$Q_{i_1, \dots, i_n} = [x_{1,i_1-1}, x_{1,i_1}] \times \cdots \times [x_{n,i_n-1}, x_{n,i_n}], \quad 1 \leq i_j \leq m_j.$$

There are

$$r = \prod_j m_j$$

cells. From now on, when only this finite collection and its volumes matter, suppress the multi-indices and relabel them  $Q_1, \dots, Q_r$ . Then explicitly

$$Q = \bigcup_{i=1}^r Q_i, \quad \text{vol}(Q) = \sum_{i=1}^r \text{vol}(Q_i).$$

The interiors are pairwise disjoint; shared faces are allowed. The volume identity follows by expanding the product of the sums of coordinate lengths. This is the same convention as the closed subintervals in Chapter 9: shared endpoints caused no problem for Darboux sums, and shared faces cause none here. We are adding cell volumes, not claiming disjointness of closed sets. Half-open cells will enter only when an actual step function needs one value on a shared face.

**Definition 12.1.2** (Grid partitions and Darboux sums). Let  $Q$  be a rectangle. A **grid partition** of  $Q$  is obtained by choosing a finite partition of each interval  $[a_j, b_j]$  and taking all products of the resulting subintervals. Its members are **subrectangles**. For bounded

$$f : Q \longrightarrow \mathbb{R}$$

and a grid partition  $\mathcal{P} = \{Q_1, \dots, Q_r\}$ , put  $m_i = \inf_{Q_i} f$  and  $M_i = \sup_{Q_i} f$ . Define

$$U(f, \mathcal{P}) = \sum_{i=1}^r M_i \text{vol}(Q_i), \quad L(f, \mathcal{P}) = \sum_{i=1}^r m_i \text{vol}(Q_i).$$

The function  $f$  is **Riemann integrable** on  $Q$  if

$$\inf_{\mathcal{P}} U(f, \mathcal{P}) = \sup_{\mathcal{P}} L(f, \mathcal{P}).$$

We may subsequently write the same finite sums as

$$\sum_{R \in \mathcal{P}}$$

when individual cell numbers do not matter. The common value is the multiple Riemann integral (a double integral for  $n = 2$ , a triple integral for  $n = 3$ ), denoted by

$$\int_Q f(x_1, \dots, x_n) \, d(x_1, \dots, x_n).$$

In scalar-integral notation we use

$$\mathbf{x} = (x_1, \dots, x_n), \quad \int_Q f(\mathbf{x}) \, d\mathbf{x}.$$

Here  $d\mathbf{x}$  is shorthand for unsigned scalar coordinate-volume integration in the variables  $x_1, \dots, x_n$ . It is not an exterior derivative applied to the vector  $\mathbf{x}$ . In particular, it is distinct from the individual coordinate covectors  $dx_i$  of [Definition 11.2.11](#) and from the oriented top-degree form

$$dx_1 \wedge \dots \wedge dx_n$$

introduced in Chapter 13. That form retains orientation information that scalar volume does not. We retain the useful geometric notations

$$\iint_Q f(x, y) \, dA, \quad \iiint_Q f(x, y, z) \, dV,$$

as well as explicit coordinate lists in scalar integrals.

**Refinement puts approximations on the same grid.** A refinement adds coordinate cuts and removes none. Every old cell becomes a finite union of smaller cells. If the coordinate-cut sets of two grids are  $P_j$  and  $P'_j$ , their common refinement uses  $P_j \cup P'_j$  for every  $j$ . It therefore refines both grids. For one coarse cell write the operation out:

$$R = R_1 \cup \dots \cup R_s, \quad \text{vol}(R) = \sum_{k=1}^s \text{vol}(R_k).$$

Since every  $R_k \subseteq R$ ,

$$\inf_R f \leq \inf_{R_k} f \leq \sup_{R_k} f \leq \sup_R f.$$

Multiply by the nonnegative cell volumes and add to get

$$\begin{aligned} (\inf_R f) \text{vol}(R) &\leq \sum_{k=1}^s (\inf_{R_k} f) \text{vol}(R_k), \\ \sum_{k=1}^s (\sup_{R_k} f) \text{vol}(R_k) &\leq (\sup_R f) \text{vol}(R). \end{aligned}$$

Summing these inequalities over the coarse cells  $Q_1, \dots, Q_r$  proves the rule we may now abbreviate:

$$\mathcal{P}' \text{ refines } \mathcal{P} \implies L(f, \mathcal{P}) \leq L(f, \mathcal{P}') \leq U(f, \mathcal{P}') \leq U(f, \mathcal{P}).$$

In particular a common refinement shows that every lower sum is at most every upper sum, even when they originally use different grids.

The definition permits the best lower and upper approximations to come from different grids. The Darboux criterion says one grid can make them arbitrarily close: a common refinement puts two separately good approximations onto one grid. This follows the proof of [Proposition 9.1.7](#). The new geometric step is taking the union of the cut sets in *each* coordinate direction; the supremum–infimum argument that follows is unchanged.

**Proposition 12.1.3** (Darboux criterion). A bounded function  $f : Q \rightarrow \mathbb{R}$  is Riemann integrable if and only if, for every  $\varepsilon > 0$ , there is a grid partition  $\mathcal{P}$  such that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon.$$

*Proof.* Refining a grid partition decreases upper sums and increases lower sums. Thus every lower sum is at most every upper sum. If the two extremal values are equal, choose partitions whose upper and lower sums are within

$$\varepsilon/2$$

of that common value and take their common refinement. Conversely, the displayed condition forces the difference of the two extremal values to be smaller than every positive  $\varepsilon$ , hence zero by [Lemma 4.1.5](#).  $\square$

Choose one tag  $\xi_i \in Q_i$  per cell. The familiar tagged sum is

$$S(f, \mathcal{P}, \xi) = \sum_{i=1}^r f(\xi_i) \operatorname{vol}(Q_i).$$

Each sampled height lies between its cell infimum and supremum, so  $L(f, \mathcal{P}) \leq S(f, \mathcal{P}, \xi) \leq U(f, \mathcal{P})$ . Lower sums approximate from below and upper sums from above. Refinement narrows their uncertainty; when both reach the same number, the integral is determined.

**Example 12.1.4** (A four-cell calculation). For  $f(x, y) = x + y$  on  $[0, 1]^2$ , split each axis at

$$1/2.$$

| cell (lower left corner) | area | inf $f$ | sup $f$ |
|--------------------------|------|---------|---------|
| (0, 0)                   | 1/4  | 0       | 1       |
| (1/2, 0)                 | 1/4  | 1/2     | 3/2     |
| (0, 1/2)                 | 1/4  | 1/2     | 3/2     |
| (1/2, 1/2)               | 1/4  | 1       | 2       |

Hence

$$L = \frac{0 + \frac{1}{2} + \frac{1}{2} + 1}{4} = \frac{1}{2}, \quad U = \frac{1 + \frac{3}{2} + \frac{3}{2} + 2}{4} = \frac{3}{2}.$$

On an  $m$ -by- $m$  equal grid the heights on cell  $(i, j)$  are

$$(i + j)/m, \quad (i + j + 2)/m,$$



$0 \leq i, j < m$ . Finite summation gives

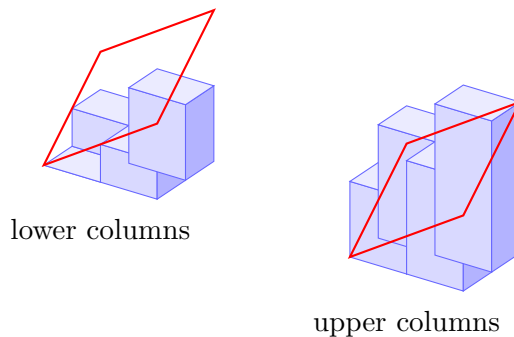
$$L = 1 - 1/m,$$

$$U = 1 + 1/m.$$

Refinement narrows the gap to

$$2/m$$

and squeezes the integral to 1, using the definition without Fubini.



The red outline is  $z = x + y$  on  $[0, 1]^2$ . On each of the four cells, the lower column stops at the minimum and the upper column at the maximum. Each base has area

$$1/4;$$

the total column volumes are

$$1/2, \quad 3/2.$$

Define the mesh  $|\mathcal{P}| = \max_{R \in \mathcal{P}} \text{diam}(R)$ . Subtracting the finite sums displays their gap:

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) = \sum_{R \in \mathcal{P}} \text{osc}_R(f) \text{vol}(R).$$

Here  $\text{osc}_R(f) = \sup_R f - \inf_R f$ . Small oscillations on all cells make the total gap small. For later estimates distinguish the cell oscillation  $\text{osc}_R(f) = \sup_R f - \inf_R f$  from the pointwise oscillation  $\omega_f(a)$  of Chapter 9. For a continuous function on a compact domain  $K$ , its modulus of continuity is instead

$$\mu_f(\delta) = \sup\{|f(\mathbf{x}) - f(y)| : \mathbf{x}, y \in K, \|\mathbf{x} - y\| \leq \delta\}.$$

Uniform continuity gives  $\mu_f(\delta) \rightarrow 0$  as  $\delta \rightarrow 0$ , and  $\text{osc}_R(f) \leq \mu_f(\text{diam } R)$  for  $R \subseteq K$ . The inputs to these three notions are respectively a cell, a point, and a nonnegative scalar distance.

The proof of [Theorem 9.2.1](#) now has a precise translation: interval length becomes  $\text{vol}(Q)$ , and interval mesh becomes maximum cell diameter. Uniform continuity still supplies one oscillation bound for all cells; summing their volumes gives the global gap.

**Theorem 12.1.5** (Continuous functions are integrable). Every continuous function

$$f : Q \longrightarrow \mathbb{R}$$

is Riemann integrable.

*Proof.* Small cells give small oscillation on every cell, hence a small total Darboux gap. Compactness makes  $f$  bounded and uniformly continuous, so  $\mu_f(\delta) \rightarrow 0$ . For every cell,

$$\text{osc}_R(f) \leq \mu_f(\text{diam } R) \leq \mu_f(|\mathcal{P}|).$$

The gap identity and finite volume additivity give

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) \leq \mu_f(|\mathcal{P}|) \text{vol}(Q).$$

Equal subdivisions in each coordinate make the mesh arbitrarily small. The right side tends to zero, so the Darboux criterion applies.  $\square$

**Example 12.1.6** (Continuity succeeds; boundedness alone fails). On  $Q = [0, 1]^2$ , the function  $f(x, y) = x^2 + y^2$  is integrable by [Theorem 12.1.5](#). In contrast, define

$$g(x, y) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Every nondegenerate subrectangle contains points of both kinds, so its oscillation is 1. Consequently every grid partition satisfies

$$U(g, \mathcal{P}) - L(g, \mathcal{P}) = \text{vol}(Q) = 1,$$

and  $g$  is not Riemann integrable. The example separates boundedness from integrability in several variables just as the Dirichlet function does in one variable.

**Actual bounding functions: owned cells.** The basic Darboux construction is now in place. Later proofs need actual bounding functions, not only numerical sums. If two neighboring cells carry different constants, a shared face would receive conflicting values. Only for this purpose do we assign every face to one cell. Partition each coordinate interval at finitely many points. For bookkeeping, assign each partition point to the interval on its right, except the final endpoint, which belongs to the last interval. Products of these half-open intervals (with their final endpoints included) give genuinely disjoint *owned cells* covering  $Q$ . Their volumes are the products of their side lengths, the same as those of their closures. A rectangular step function  $s : Q \rightarrow \mathbb{R}$  takes a constant value  $c_R$  on each owned cell. Define its step integral by

$$\int_Q s = \sum_R c_R \text{vol}(R).$$

This is height times length in one dimension, height times area in two, and height times  $n$ -dimensional volume in general. Refining a cell gives

$$c \text{vol}(R) = \sum_j c \text{vol}(R_j),$$

so refinement does not change the answer. Two representations have a common refinement; on each owned refined cell their values agree. Thus this definition is independent of representation, without any theory of null sets.

For bounded  $f$ , use on each owned cell the infimum and supremum over its *closed* rectangle to define  $s_{\mathcal{P}}^-$  and  $s_{\mathcal{P}}^+$ . Then  $s_{\mathcal{P}}^- \leq f \leq s_{\mathcal{P}}^+$ , and their step integrals are exactly the lower and upper sums defined above. Closed rectangles in Darboux sums retain their shared faces; the ownership rule is used to make the step functions single-valued.

**Proposition 12.1.7** (Compatibility with preliminary step integration). Every owned-cell rectangular step function is Darboux integrable, with integral equal to its preliminary step sum.

*Proof.* The ownership convention changes values only on finitely many jump planes. Thin strips around those planes make their possible Darboux contribution arbitrarily small. Let  $I_s$  be the preliminary sum and  $M$  a bound for  $|s|$ . Given any grid  $\mathcal{P}$ , refine it together with the step grid using the union of their coordinate cuts. Each refined owned cell belongs to one owned cell of each grid. Its constant value lies between the infimum and supremum on the closed  $\mathcal{P}$ -cell containing it. Multiply by its volume and add. Common owned-cell refinement preserves  $I_s$ , so

$$L(s, \mathcal{P}) \leq I_s \leq U(s, \mathcal{P}).$$

Now include the finitely many jump planes and nearby parallel planes in a grid. The closed cells touching jump planes lie in thin grid strips of total volume  $\delta$ , as small as desired: each strip volume is its width times a fixed product of side lengths. Elsewhere  $s$  is constant on each closed cell. Therefore  $U - L \leq 2M\delta$ . The Darboux criterion proves integrability. The preceding squeeze for these grids identifies its integral with  $I_s$ . If  $M = 0$  the conclusion is immediate.  $\square$

**Theorem 12.1.8** (Basic properties). If  $f, g : Q \rightarrow \mathbb{R}$  are integrable and  $c \in \mathbb{R}$ , then

$$\int_Q (f + g) \, d\mathbf{x} = \int_Q f \, d\mathbf{x} + \int_Q g \, d\mathbf{x}, \quad \int_Q cf \, d\mathbf{x} = c \int_Q f \, d\mathbf{x}.$$

If

$$f \leq g,$$

then their integrals have the same order. Also,

$$\left| \int_Q f \, d\mathbf{x} \right| \leq \int_Q |f| \, d\mathbf{x},$$

and

$$\left| \int_Q f \, d\mathbf{x} \right| \leq M \operatorname{vol}(Q)$$

whenever

$$|f(\mathbf{x})| \leq M \quad (\mathbf{x} \in Q).$$

*Proof. Strategy.* Place both candidate sums of integrals between common Darboux bounds, then shrink the gap.

The upper and lower sum inequalities used in the proof of [Theorem 9.2.2](#) hold verbatim for subrectangles:

$$U(f + g, \mathcal{P}) \leq U(f, \mathcal{P}) + U(g, \mathcal{P}),$$

$$L(f + g, \mathcal{P}) \geq L(f, \mathcal{P}) + L(g, \mathcal{P}).$$

Taking a common refinement of partitions that make the two individual gaps small proves integrability of the sum. Both

$$\int_Q (f + g), \quad \int_Q f + \int_Q g$$

lie in

$$[L(f, \mathcal{P}) + L(g, \mathcal{P}), U(f, \mathcal{P}) + U(g, \mathcal{P})].$$

The length is the sum of the two individual Darboux gaps and can be made arbitrarily small on one common refinement. This proves equality. Scalar multiplication and order preservation are identical to the one-variable arguments.

For absolute values and the two bounds, follow the proof of [Theorem 9.2.3](#), with volume in place of length. The inequality

$$||u| - |v|| \leq |u - v|$$

shows from a common partition that  $|f|$  is integrable. Since

$$-|f| \leq f \leq |f|,$$

order preservation gives the first estimate. The second follows by integrating the constant functions  $-M$  and  $M$ .  $\square$

### Rectangular restriction, gluing, and translation

The basic multiple Riemann integral has now been constructed. The next lemmas are bookkeeping tools for cutting a rectangle into finitely many pieces, enlarging its container, or translating it. Later they support Jordan regions and change of variables. The gluing argument generalizes the interval additivity in [Theorem 9.2.3](#): inserting a division point becomes inserting subdivision planes.

**Lemma 12.1.9** (Grid restriction and finite gluing). Let  $Q_1, \dots, Q_N$  be the closed subrectangles of a finite grid in  $Q$ . A bounded  $f : Q \rightarrow \mathbb{R}$  is integrable exactly when each restriction is, and then

$$\int_Q f = \sum_{i=1}^N \int_{Q_i} f.$$

*Proof.* Write

$$Q = Q_1 \cup \dots \cup Q_N,$$

with disjoint interiors and possible shared faces. Insert all subdivision planes into any testing grid. Each refined cell belongs to one  $Q_i$ , although their closed faces can be shared. Upper and lower sums are exactly the sums of the corresponding local sums: explicitly, for the induced local grids  $\mathcal{P}_i$ ,

$$L(f, \mathcal{P}) = \sum_{i=1}^N L(f|_{Q_i}, \mathcal{P}_i), \quad U(f, \mathcal{P}) = \sum_{i=1}^N U(f|_{Q_i}, \mathcal{P}_i).$$

Every small cell contributes its height times its volume to exactly one local sum; shared closed faces do not add any extra cell volume. If  $f$  is integrable, make its total nonnegative gap small; each local gap is at most that total, proving restriction. Conversely choose a grid on each  $Q_i$  with gap below

$$\varepsilon/N$$

and extend all its coordinate cuts through  $Q$ . Refinement decreases local gaps, so the total is below  $\varepsilon$ . The criterion gives gluing. The sum of the local integrals and the global integral both lie between the same total lower and upper sums with arbitrarily small gap, proving additivity of the integral, not merely volume.  $\square$

**Lemma 12.1.10** (Face strips and zero extension). A bounded function supported on finitely many coordinate hyperplane sections of a rectangle is integrable with integral zero. If  $Q \subseteq P$  are rectangles and  $f$  is integrable on  $Q$ , its extension by zero to  $P$  is integrable and has the same integral.

*Proof.* For the first assertion, put every supporting plane and neighboring parallel cuts into a grid. All cells with a nonzero value lie in strips of total volume  $\delta \rightarrow 0$ . With bound  $M$ , the lower and upper sums are between  $-M\delta$  and  $M\delta$ , and their gap is at most  $2M\delta$ . The criterion and squeeze prove the assertion directly from grids. For extension, subdivide  $P$  at the faces of  $Q$ . On the central rectangle the restriction is  $f$ . On every other subrectangle it is zero except possibly on shared faces of  $Q$ . The first assertion gives integral zero there, even though the function need not vanish on the closed shared face. Grid gluing and additivity now prove the claim. For two unrelated containing rectangles, enlarge both to a common rectangle.  $\square$

**Lemma 12.1.11** (Translation of rectangular integrals). For  $a \in \mathbb{R}^n$ , set  $Q + a = \{\mathbf{x} + a : \mathbf{x} \in Q\}$  and  $g(y) = f(y - a)$  there. Then  $g$  is integrable exactly when  $f$  is, and

$$\int_{Q+a} g = \int_Q f.$$

*Proof.* Translate every coordinate cut by the corresponding coordinate of  $a$ . This gives a bijection of grids preserving side lengths and cell volumes. The value set of  $g$  on  $R + a$  equals that of  $f$  on  $R$ , so the lower and upper sums agree exactly. Taking their extrema proves the assertion.  $\square$

**Definition 12.1.12** (Vector-valued Riemann integrals). For  $f = (f_1, \dots, f_m) : Q \rightarrow \mathbb{R}^m$ , say  $f$  is integrable when every scalar component is integrable, and define

$$\int_Q f = \left( \int_Q f_1, \dots, \int_Q f_m \right) \in \mathbb{R}^m.$$

There is no natural total order on  $\mathbb{R}^m$  corresponding to the scalar order used by Darboux sums; this is a componentwise extension of the already constructed scalar integral.

**Proposition 12.1.13** (Linearity of vector integration). For integrable  $f, g : Q \rightarrow \mathbb{R}^m$ , scalars  $a, b$ , and a linear map  $T : \mathbb{R}^m \rightarrow \mathbb{R}^k$ , the functions  $af + bg$  and  $T \circ f$  are integrable and

$$\int_Q (af + bg) = a \int_Q f + b \int_Q g, \quad \int_Q T(f(\mathbf{x})) \, d\mathbf{x} = T \left( \int_Q f(\mathbf{x}) \, d\mathbf{x} \right).$$

Continuous vector-valued functions on  $Q$  are integrable.

*Proof.* Apply scalar linearity to each component. If  $[T] = (t_{ij})$ , the  $i$ th component of  $T \circ f$  is

$$\sum_j t_{ij} f_j,$$

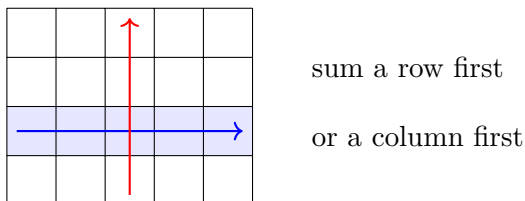
whose integral is

$$\sum_j t_{ij} \int_Q f_j.$$

Continuity also passes to each component, so [Theorem 12.1.5](#) proves the last assertion.  $\square$

## 12.2 Iterated Integration

The equality of an integral over a rectangle with repeated one-variable integrals is not a new definition. It is a theorem saying that the same finite rectangular approximations can be grouped in either order. The continuity hypothesis keeps the oscillation on every sufficiently small product rectangle under control.



A finite rectangular grid has the same total whether its terms are added row by row or column by column. Riemann–Fubini passes this finite regrouping to a limit; the estimates ensure both limiting operations describe the same integral. A *multiple integral* is the directly defined Riemann integral over a multidimensional rectangle. An *iterated integral* performs successive integrations. On  $Q = [a, b] \times [c, d]$ ,

$$\iint_Q f \, dA$$

is defined by two-dimensional grids, whereas

$$\int_a^b \left( \int_c^d f(x, y) \, dy \right) dx$$

is a later computational representation. Fubini proves their equality under its hypotheses; it does not define the multiple integral.

For  $A \subseteq \mathbb{R}^p$ ,  $B \subseteq \mathbb{R}^q$ , write  $\mathbf{x} = (x_1, \dots, x_p)$  and  $y = (y_1, \dots, y_q)$ . With  $\mathbf{x}$  fixed,  $f(\mathbf{x}, \cdot) : B \rightarrow \mathbb{R}$  is a scalar function of  $q$  variables. Its integral

$$g(\mathbf{x}) = \int_B f(\mathbf{x}, y) \, dy$$

is one real number, defining  $g : A \rightarrow \mathbb{R}$ . The integral

$$\int_A g(\mathbf{x}) \, d\mathbf{x}$$

handles the remaining  $p$  variables. In order, these operations are

$$f(\mathbf{x}, y) \xrightarrow{\text{integrate in } y \text{ with } \mathbf{x} \text{ fixed}} g(\mathbf{x}) \xrightarrow{\text{integrate in } \mathbf{x}} \text{a scalar.}$$

For general  $p, q$ , either stage can itself be a multiple integral.

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| $a_{03}$ | $a_{13}$ | $a_{23}$ | $a_{33}$ | $a_{43}$ |
| $a_{02}$ | $a_{12}$ | $a_{22}$ | $a_{32}$ | $a_{42}$ |
| $a_{01}$ | $a_{11}$ | $a_{21}$ | $a_{31}$ | $a_{41}$ |
| $a_{00}$ | $a_{10}$ | $a_{20}$ | $a_{30}$ | $a_{40}$ |

vertical groups

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| $a_{03}$ | $a_{13}$ | $a_{23}$ | $a_{33}$ | $a_{43}$ |
| $a_{02}$ | $a_{12}$ | $a_{22}$ | $a_{32}$ | $a_{42}$ |
| $a_{01}$ | $a_{11}$ | $a_{21}$ | $a_{31}$ | $a_{41}$ |
| $a_{00}$ | $a_{10}$ | $a_{20}$ | $a_{30}$ | $a_{40}$ |

horizontal groups

Both pictures contain exactly the same twenty contributions:

$$\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}.$$

With  $a_{ij}$  equal to a tagged function value times cell volume, this is the finite model behind Riemann–Fubini. The theorem justifies passing from that identity to integrals.

**Theorem 12.2.1** (Riemann–Fubini theorem for continuous functions). Let

$$Q = A \times B,$$

where  $A \subseteq \mathbb{R}^p$  and  $B \subseteq \mathbb{R}^q$  are rectangles, and let

$$f : Q \longrightarrow \mathbb{R}$$

be continuous. Then the functions

$$\mathbf{x} \longmapsto \int_B f(\mathbf{x}, y) \, dy, \quad y \longmapsto \int_A f(\mathbf{x}, y) \, d\mathbf{x}$$

are continuous, and

$$\int_{A \times B} f(\mathbf{x}, y) \, d(\mathbf{x}, y) = \int_A \left( \int_B f(\mathbf{x}, y) \, dy \right) d\mathbf{x} = \int_B \left( \int_A f(\mathbf{x}, y) \, d\mathbf{x} \right) dy.$$

*Proof.* Every section is continuous on its rectangle, so its integral exists. Write  $a = \text{vol}(A) > 0$ ,  $b = \text{vol}(B) > 0$ , and

$$G(\mathbf{x}) = \int_B f(\mathbf{x}, y) \, dy.$$

The bounded-integral estimate gives

$$|G(\mathbf{x}') - G(\mathbf{x})| \leq b \sup_{y \in B} |f(\mathbf{x}', y) - f(\mathbf{x}, y)|.$$

Uniform continuity of  $f$  makes the supremum less than

$$\varepsilon/b$$

whenever  $\mathbf{x}'$  is sufficiently close to  $\mathbf{x}$ , proving continuity of  $G$ . Thus its outer integral also exists.

**Strategy.** The section integral and the multiple integral lie between the same product Darboux sums. For a product cell  $R \times S$  put  $m_{R,S} = \inf_{R \times S} f$  and  $M_{R,S} = \sup_{R \times S} f$ . For  $\mathbf{x} \in R$ , integration over the  $S$  cells gives

$$\sum_S m_{R,S} \text{vol}(S) \leq G(\mathbf{x}) \leq \sum_S M_{R,S} \text{vol}(S).$$

Integrate over  $R$  and add, using rectangular restriction and gluing:

$$L(f, \mathcal{P}_A \times \mathcal{P}_B) \leq \int_A G \leq U(f, \mathcal{P}_A \times \mathcal{P}_B).$$

The multiple integral lies in this same interval. Uniform continuity of  $f$  gives arbitrarily small product Darboux gaps, so the two integrals agree. Reverse the coordinate blocks to prove continuity of the other section-integral function and its integral identity.  $\square$

**Example 12.2.2** (A double integral on a rectangle). For the continuous function  $f(x, y) = x + 2y$  on  $[0, 1] \times [0, 2]$ , Riemann–Fubini gives

$$\begin{aligned} \int_{[0,1] \times [0,2]} (x + 2y) \, d(x, y) &= \int_0^1 \left( \int_0^2 (x + 2y) \, dy \right) dx \\ &= \int_0^1 (2x + 4) \, dx = 5. \end{aligned}$$

Here the rectangle makes iterated integration natural; no coordinate change is needed.

### Beyond continuous integrands

Continuity is sufficient, but not necessary, for the equality of multiple and iterated integrals. It buys us two conveniences: every section is integrable, and its integral varies continuously with the parameter. For a bounded  $f : A \times B \rightarrow \mathbb{R}$  that is merely Riemann integrable, a section  $f(\mathbf{x}, \cdot) : B \rightarrow \mathbb{R}$  need not be integrable for every  $\mathbf{x}$ . Thus

$$\mathbf{x} \mapsto \int_B f(\mathbf{x}, y) \, dy$$

may not even be defined everywhere. The multiple integral still makes sense; the difficulty lies in requiring an ordinary Riemann integral on every slice.

**Definition 12.2.3** (Lower and upper section integrals). Let  $A \subseteq \mathbb{R}^p$  and  $B \subseteq \mathbb{R}^q$  be compact rectangles and  $f : A \times B \rightarrow \mathbb{R}$  bounded. Define functions on  $A$  by

$$\underline{I}_B(\mathbf{x}) = \int_B f(\mathbf{x}, y) \, dy := \sup_{\mathcal{P}_B} L(f(\mathbf{x}, \cdot), \mathcal{P}_B), \quad \bar{I}_B(\mathbf{x}) = \overline{\int_B f(\mathbf{x}, y) \, dy} := \inf_{\mathcal{P}_B} U(f(\mathbf{x}, \cdot), \mathcal{P}_B).$$

Define functions on  $B$  by reversing the roles:

$$\underline{I}_A(y) = \sup_{\mathcal{P}_A} L(f(\cdot, y), \mathcal{P}_A), \quad \bar{I}_A(y) = \inf_{\mathcal{P}_A} U(f(\cdot, y), \mathcal{P}_A).$$

All partitions here are grid partitions of the indicated rectangle. These are Darboux integrals, whether or not the sections are integrable.

If  $|f| \leq K$ , both functions on  $A$  lie between  $-K \operatorname{vol}(B)$  and  $K \operatorname{vol}(B)$ . Common refinement gives  $\underline{I}_B \leq \bar{I}_B$ ; equality at  $\mathbf{x}$  holds exactly when  $f(\mathbf{x}, \cdot)$  is Riemann integrable. The same statements hold with  $A, B$  interchanged.

**Theorem 12.2.4** (Darboux–Fubini theorem for a Riemann-integrable function). Let  $f : A \times B \rightarrow \mathbb{R}$  be Riemann integrable on a product of compact rectangles. All four lower and upper section-integral functions are Riemann integrable on their parameter rectangles, and

$$\begin{aligned} \int_A \underline{I}_B(\mathbf{x}) \, d\mathbf{x} &= \int_A \bar{I}_B(\mathbf{x}) \, d\mathbf{x} = \int_{A \times B} f, \\ \int_B \underline{I}_A(y) \, dy &= \int_B \bar{I}_A(y) \, dy = \int_{A \times B} f. \end{aligned}$$

No hypothesis of section integrability is imposed.



*Proof.* Take grid partitions  $\mathcal{P}_A, \mathcal{P}_B$ . For their product cells put

$$m_{R,S} = \inf_{R \times S} f, \quad M_{R,S} = \sup_{R \times S} f.$$

For each  $\mathbf{x} \in R$ , comparison with the section sums gives

$$\sum_S m_{R,S} \text{vol}(S) \leq \underline{I}_B(\mathbf{x}) \leq \bar{I}_B(\mathbf{x}) \leq \sum_S M_{R,S} \text{vol}(S).$$

Indeed the left sum is at most the section's lower sum on  $\mathcal{P}_B$ , which is at most its lower integral; the upper comparison is the reverse. For either  $h = \underline{I}_B$  or  $h = \bar{I}_B$ , take the infimum and supremum over  $\mathbf{x} \in R$ , multiply by  $\text{vol}(R)$ , and sum:

$$L(f, \mathcal{P}_A \times \mathcal{P}_B) \leq L(h, \mathcal{P}_A) \leq U(h, \mathcal{P}_A) \leq U(f, \mathcal{P}_A \times \mathcal{P}_B).$$

The cross bounds also give

$$L(f, \mathcal{P}_A \times \mathcal{P}_B) \leq L(\underline{I}_B, \mathcal{P}_A) \leq U(\bar{I}_B, \mathcal{P}_A) \leq U(f, \mathcal{P}_A \times \mathcal{P}_B).$$

Every grid partition of  $A \times B$  is such a product: separate its first  $p$  coordinate-cut lists from its last  $q$  lists. Integrability of  $f$  therefore supplies product grids with gap smaller than any prescribed  $\varepsilon > 0$ . The first chain bounds the outer Darboux gap for each  $h$  by this gap, proving its integrability by [Proposition 12.1.3](#). Both

$$\int_A h, \quad \int_{A \times B} f$$

lie between the same product lower and upper sums. Their difference has absolute value less than  $\varepsilon$ , so they agree. For the other order, hold  $y \in S$  fixed and use

$$\sum_R m_{R,S} \text{vol}(R) \leq \underline{I}_A(y) \leq \bar{I}_A(y) \leq \sum_R M_{R,S} \text{vol}(R).$$

Taking extrema on  $S$  and summing with weights  $\text{vol}(S)$  proves integrability and the same equality on  $B$ . Only finite Darboux sums and the Darboux criterion have been used.  $\square$

**Corollary 12.2.5** (Ordinary iterated integrals). Under the hypotheses of [Theorem 12.2.4](#), if every section  $f(\mathbf{x}, \cdot)$  is Riemann integrable, then

$$G(\mathbf{x}) = \int_B f(\mathbf{x}, y) \, dy$$

is Riemann integrable on  $A$  and

$$\int_{A \times B} f = \int_A G(\mathbf{x}) \, d\mathbf{x}.$$

The corresponding conclusion holds in the reverse order. If every section in both directions is integrable, both iterated integrals exist and equal the multiple integral.

*Proof.* Section integrability says  $G = \underline{I}_B = \bar{I}_B$ . Apply the theorem, and interchange the coordinate blocks for the other order.  $\square$

The continuous theorem remains the main computational case: continuity automatically supplies every section integral, and even continuity of  $G$ .

**Example 12.2.6** (A dense family of nonintegrable sections). Recall Thomae's function from [Example 9.6.10](#), writing it as  $t$  on  $[0, 1]$ :  $t(x) = 0$  at irrational  $x$  and

$$t(a/b) = 1/b$$

for a reduced rational with  $b > 0$  (including

$$0/1$$

). It is Riemann integrable with integral zero. Set

$$f(x, y) = t(x)\mathbf{1}_{\mathbb{Q}}(y), \quad (x, y) \in [0, 1]^2.$$

On any product cell  $I \times J$ , the infimum is zero because  $J$  contains irrationals; the supremum is  $\sup_I t$  because  $J$  contains rationals. Thus every lower sum is zero and

$$U(f, \mathcal{P}_x \times \mathcal{P}_y) = \sum_I (\sup_I t) |I| \sum_J |J| = U(t, \mathcal{P}_x).$$

The one-dimensional upper sums can be made arbitrarily small, so the two-dimensional Darboux criterion gives

$$\int_{[0,1]^2} f = 0.$$

For irrational  $x$  the vertical section is zero. For rational  $x$  it is a positive multiple of the Dirichlet function: every lower sum is zero and every upper sum is  $t(x) > 0$ . It is not Riemann integrable. Bad section parameters are therefore infinite and dense. Nevertheless, on all of  $[0, 1]$ ,

$$\underline{I}_B = 0, \quad \bar{I}_B = t.$$

Both outer functions are Riemann integrable with integral zero, as the theorem predicts. In the reverse order every section is either  $t$  or zero, so every inner integral is zero and the ordinary iterated integral exists.

**Remark 12.2.7** (Undefined section values cannot be filled arbitrarily). In the preceding example the ordinary vertical-section integral is defined and zero only at irrational parameters. Giving it the value 1 at all rational parameters produces  $\mathbf{1}_{\mathbb{Q}}$ , which is not Riemann integrable. The canonical lower and upper section integrals avoid this problem; arbitrary assignments need not preserve outer integrability. This is a concrete reason for the later almost-everywhere language of Lebesgue theory, not an invocation of that theory here.

**Remark 12.2.8** (Riemann–Fubini and the later Fubini–Tonelli theorems). The continuous theorem gives the simplest computational case; the Darboux theorem above also treats Riemann-integrable functions with bad sections. In Lebesgue integration on Euclidean product spaces, Tonelli applies to nonnegative measurable functions: the product integral and iterated integrals agree, possibly with common value  $+\infty$ . In the standard absolute-integrability formulation, Fubini assumes

$$\int |f| < \infty.$$

Almost every section is then integrable, and both iterated integrals are finite and equal to the product integral. Exceptional sections are ignored in the almost-everywhere sense of that later theory.

Fubini is not a license to interchange arbitrary iterated integrals. Product Riemann integrability supplies the Darboux bounds in the general theorem. Continuity is a convenient sufficient condition that additionally guarantees integrability of every section and continuous variation of the section integral. Later measurability and integral-size hypotheses provide greater flexibility; those measure-theoretic results are not proved here.

### Coordinatewise calculus on rectangles

**Corollary 12.2.9** (Coordinatewise FTC on a rectangle). Let  $F$  be  $C^1$  on a neighborhood of

$$Q = \prod_{i=1}^n [a_i, b_i].$$

Write  $\mathbf{x}_{\hat{j}}$  for the tuple omitting coordinate  $j$ , and

$$Q_{\hat{j}} = \prod_{i \neq j} [a_i, b_i].$$

Then

$$\int_Q \partial_j F(\mathbf{x}) \, d\mathbf{x} = \int_{Q_{\hat{j}}} [F(\mathbf{x})]_{x_j=a_j}^{x_j=b_j} d\mathbf{x}_{\hat{j}}.$$

For  $n = 1$ , integration over the remaining empty tuple means evaluation.

*Proof.* Hold the other coordinates fixed, apply the one-variable FTC ([Theorem 9.4.3](#)) in coordinate  $j$ , then integrate over those other coordinates using Fubini. All integrands and face restrictions are continuous on their compact rectangles.  $\square$

In one variable there are two endpoints. Here fixing  $x_j$  at its endpoints leaves functions on two opposite  $(n-1)$ -dimensional faces, which must themselves be integrated. This is a coordinatewise FTC; Chapter 13's Stokes theorem will give the intrinsic generalization. For example, with  $F(x, y) = x^2 y$  on  $[0, 1]^2$ ,

$$\iint_{[0,1]^2} 2xy \, dA = \int_0^1 [F(1, y) - F(0, y)] \, dy = \int_0^1 y \, dy = \frac{1}{2}.$$

Integrating  $2xy$  first in  $x$  likewise gives  $[x^2 y]_0^1 = y$ .

For  $Q = [a, b] \times [c, d]$ , coordinatewise integration by parts reads

$$\iint_Q uv_x \, dA = \int_c^d [u(b, y)v(b, y) - u(a, y)v(a, y)] \, dy - \iint_Q vu_x \, dA.$$

Evaluating at  $x = a, b$  removes the  $x$  variable but leaves a function of  $y$ . In  $n$  dimensions, write  $\mathbf{x}_{\hat{j}} = (x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)$ . The hat means that the indicated coordinate is omitted. The rectangle  $Q_{\hat{j}}$  contains these remaining coordinates, and  $d\mathbf{x}_{\hat{j}}$  is their scalar coordinate volume. The face term below is an  $(n-1)$ -dimensional integral. For  $n = 1$  the remaining zero-dimensional integral is evaluation at the single empty tuple, so the formula is the ordinary one-variable integration-by-parts identity.

**Theorem 12.2.10** (Coordinatewise integration by parts). Let

$$Q = [a_1, b_1] \times \cdots \times [a_n, b_n],$$

let  $u, v$  be of class  $C^1$  on an open neighborhood of  $Q$ , and fix

$$j \in \{1, \dots, n\}.$$

Write

$$Q_{\hat{j}} = \prod_{\substack{1 \leq i \leq n \\ i \neq j}} [a_i, b_i], \quad d\mathbf{x}_{\hat{j}} = dx_1 \cdots dx_{j-1} dx_{j+1} \cdots dx_n.$$

Then

$$\int_Q u(\mathbf{x}) \frac{\partial v}{\partial x_j}(\mathbf{x}) \, d\mathbf{x} = \int_{Q_j^+} [u(\mathbf{x})v(\mathbf{x})]_{x_j=a_j}^{x_j=b_j} \, d\mathbf{x}_j - \int_Q v(\mathbf{x}) \frac{\partial u}{\partial x_j}(\mathbf{x}) \, d\mathbf{x}.$$

In the middle integral, the coordinates other than  $x_j$  are held fixed while the bracket is evaluated at the two endpoints.

*Proof.* For fixed  $\mathbf{x}_j \in Q_j^+$ , apply [Theorem 9.4.11](#) to the  $C^1$  functions of  $x_j$  obtained from  $u$  and  $v$ . The resulting identity has continuous dependence on  $\mathbf{x}_j$ . Integrating it over  $Q_j^+$  and using [Theorem 12.2.1](#) gives the displayed formula.  $\square$

**Example 12.2.11** (Integration by parts in one coordinate). On  $Q = [0, 1]^2$ , take  $u(x, y) = x + y$  and  $v(x, y) = xy$ . We have  $v_x = y$  and  $u_x = 1$ . The formula in the  $x$  coordinate gives

$$\iint_Q (x + y)y \, dA = \int_0^1 [(x + y)xy]_{x=0}^{x=1} \, dy - \iint_Q xy \, dA.$$

The face term is

$$\int_0^1 (y + y^2) \, dy = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}.$$

The remaining integral is

$$\int_0^1 [x^2 y/2]_{x=0}^{x=1} \, dy = \int_0^1 y/2 \, dy = 1/4.$$

Thus the answer is

$$\frac{5}{6} - \frac{1}{4} = \frac{7}{12}.$$

Direct integration of  $xy + y^2$  checks it:

$$\frac{1}{4} + \frac{1}{3} = \frac{7}{12}.$$

The endpoint term has become a face integral.

|                   | One variable  | Rectangle in $\mathbb{R}^n$                  |
|-------------------|---------------|----------------------------------------------|
| FTC               | $F(b) - F(a)$ | integral of $F$ 's opposite-face difference  |
| IBP boundary term | $[uv]_a^b$    | integral of $uv$ 's opposite-face difference |

**Remark 12.2.12** (Rectangular and intrinsic integration by parts). The preceding formula uses only one-variable integration by parts and Fubini, so it is naturally a statement about rectangles and their coordinate faces. At this point no intrinsic integral over a general boundary has been developed. Chapter 13 replaces the coordinate-face term by an integral over the oriented boundary of a manifold, obtained from the graded Leibniz rule and Stokes' theorem.

## 12.3 Jordan Sets and Integration over Bounded Regions

A triangular region is not a rectangle, so our existing definition does not yet integrate over it. Place it inside a rectangle and extend the integrand by zero outside the region. The ordinary rectangular integral then counts exactly the desired portion. The new question is whether the introduced jump along the boundary obstructs Riemann integrability. Jordan measurability isolates the case where it does not. For area itself the integrand is 1 inside, 0 outside: the indicator function.

**Definition 12.3.1** (Jordan measurability). Let  $E$  be a bounded subset of  $\mathbb{R}^n$  and choose a rectangle  $Q$  containing  $E$ . The set  $E$  is **Jordan measurable** if its indicator

$$\mathbf{1}_E(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in E, \\ 0, & \mathbf{x} \notin E \end{cases}$$

is Riemann integrable on  $Q$ . Its **Jordan volume** is

$$\text{vol}(E) = \int_Q \mathbf{1}_E(\mathbf{x}) \, d\mathbf{x}.$$

**Definition 12.3.2** (Jordan-null set). A bounded set  $N \subseteq \mathbb{R}^n$  is **Jordan-null** if, for every

$$\varepsilon > 0,$$

it can be covered by finitely many rectangles whose total volume is less than  $\varepsilon$ .

Chapter 9 uses Lebesgue measure-zero (null) sets, allowing countably many covering intervals; Jordan-nullness requires finitely many rectangles. The rationals in  $[0, 1]$  have Lebesgue measure zero, by Proposition 9.6.5, but are not Jordan-null. A finite union of closed intervals covering this dense set is closed, so it covers its closure  $[0, 1]$  and has total length at least one. Countably many small intervals need not have this property.

**Lemma 12.3.3** (Fine grids near a compact set). Let  $K$  be compact and let  $G$  be an open neighborhood of  $K$  in  $\mathbb{R}^n$ . There is  $d > 0$  such that every closed grid cell of diameter less than  $d$  meeting  $K$  lies in  $G$ . If  $G$  is a finite union of open rectangles of combined volume  $v$ , the total volume of such cells in any fixed finite grid is at most  $v$ .

*Proof.* For each  $\mathbf{x} \in K$  choose  $r_{\mathbf{x}} > 0$  with  $B(\mathbf{x}, 2r_{\mathbf{x}}) \subset G$ . Finitely many balls  $B(\mathbf{x}, r_{\mathbf{x}})$  cover  $K$ ; let  $d$  be the minimum of their radii. If a cell contains  $y \in K \cap B(\mathbf{x}, r_{\mathbf{x}})$ , then for every point  $z$  in the cell,  $|z - \mathbf{x}| < d + r_{\mathbf{x}} \leq 2r_{\mathbf{x}}$ , so  $z \in G$ . For the volume bound, subdivide at all grid and covering-rectangle faces. Each resulting cell is counted at most once in the grid union and at least once among the covering rectangles. Adding its elementary volume gives the inequality. This is a finite volume comparison, not an assumption that arbitrary neighborhoods can be refined by their faces.  $\square$

**Proposition 12.3.4** (Boundary criterion for Jordan measurability). A bounded set  $E \subseteq \mathbb{R}^n$  is Jordan measurable if and only if its boundary is Jordan-null.

*Proof.* Put all cells meeting the boundary inside an open cover of small volume, using its positive margin. Choose  $Q$  with  $\overline{E}$  in its interior. The boundary is compact because it is closed and bounded. If it is Jordan-null, enlarge a finite rectangular cover slightly to make it open while keeping total volume below  $\varepsilon$ . By Lemma 12.3.3, a sufficiently fine grid has all its boundary cells inside this cover. To see why every other cell  $C$  has constant indicator, write

$$C = (C \cap E^\circ) \cup (C \cap (\mathbb{R}^n \setminus E)^\circ).$$

This equality holds because  $C$  misses  $\partial E$ : each of its points has a neighborhood either in  $E$  or in its complement. The two displayed pieces are disjoint and relatively open in  $C$ . If both were nonempty, they would separate  $C$ . But  $C$  is convex: the segment between any two points remains in  $C$ , giving a path and hence connectedness by Proposition 10.7.4. One piece must therefore be

empty, so the indicator is constant on  $C$ . Its Darboux gap is consequently at most the combined volume of the boundary cells, less than  $\varepsilon$ .

Conversely choose a partition whose indicator gap is less than

$$\varepsilon/2.$$

Its mixed cells, where the indicator takes both values, have combined volume less than

$$\varepsilon/2.$$

Any boundary point in the interior of a grid cell makes that cell mixed by the definition of boundary. The other boundary points lie on finitely many grid faces, coverable by thin boxes of combined volume less than

$$\varepsilon/2.$$

Together these rectangles cover the boundary with total volume less than  $\varepsilon$ .  $\square$

Grid faces are now Jordan-null: cover each bounded face by a rectangle with arbitrarily small thickness in its normal coordinate. The null-set modification result below therefore makes the ownership of grid faces irrelevant to integrals, although it was essential to defining our step functions unambiguously before Jordan theory.

**Proposition 12.3.5** (Elementary algebra of Jordan sets). If  $E, F \subseteq \mathbb{R}^n$  are Jordan measurable, so are  $E \cup F$ ,  $E \cap F$ , and  $E \setminus F$ .

*Proof.* All three sets are bounded. If  $\mathbf{x} \notin \partial E \cup \partial F$ , choose one ball about  $\mathbf{x}$  on which membership in  $E$  is constant and another on which membership in  $F$  is constant. On the smaller ball, membership in each of the three displayed sets is constant. Therefore

$$\partial(E \cup F), \partial(E \cap F), \partial(E \setminus F) \subseteq \partial E \cup \partial F.$$

The right side is Jordan-null: combine finite rectangle covers of its two parts with total volumes below

$$\varepsilon/2$$

each. Each boundary on the left is thus Jordan-null, and [Proposition 12.3.4](#) proves the claims.  $\square$

For  $f : E \rightarrow \mathbb{R}$  and  $E \subseteq Q$ , its zero extension is the new map

$$\tilde{f} : Q \rightarrow \mathbb{R}, \quad \tilde{f}(\mathbf{x}) = \begin{cases} f(\mathbf{x}), & \mathbf{x} \in E, \\ 0, & \mathbf{x} \in Q \setminus E. \end{cases}$$

Thus  $\tilde{f}|_E = f$  and  $\tilde{f}|_{Q \setminus E} = 0$ . They have different domains and are not literally the same function. Writing  $f\mathbf{1}_E$  on  $Q$  would require first defining  $f$  outside  $E$ .

**Proposition 12.3.6** (Continuous functions on Jordan sets). Let  $E \subseteq \mathbb{R}^n$  be bounded with Jordan-null boundary, and let

$$f : E \rightarrow \mathbb{R}$$

be bounded and continuous for the relative topology on  $E$ . Then its zero extension to a containing rectangle is Riemann integrable.

*Proof. Strategy.* A single open cover separates small boundary volume from small oscillation. Let  $\tilde{f}$  be the zero extension to  $Q$  and choose  $M \geq 1$  with  $|\tilde{f}| \leq M$ . Given  $\varepsilon > 0$ , cover  $\partial E$  by a finite union  $G$  of open boxes with total volume less than

$$\varepsilon/(4M).$$

At every  $\mathbf{x} \in Q \setminus \partial E$ , the extension is continuous: locally the point is inside  $E^\circ$  or outside  $\overline{E}$ . Choose a relatively open neighborhood  $U_{\mathbf{x}}$  in  $Q$  with  $\text{osc}_{U_{\mathbf{x}}} \tilde{f} < \eta$ , where

$$\eta = \varepsilon/[2(1 + \text{vol } Q)].$$

The open cover consisting of  $G \cap Q$  and these  $U_{\mathbf{x}}$  has a Lebesgue number by [Lemma 10.6.5](#). Choose a grid with cell diameters below that number. Assign every cell contained in  $G$  to the bad collection; every remaining cell lies in some  $U_{\mathbf{x}}$ . The bad cells have disjoint interiors and total volume at most the sum of the covering-box volumes (subdivide along all box faces to see this). Their Darboux contribution is less than

$$2M\varepsilon/(4M) = \varepsilon/2.$$

All other cells contribute at most

$$\eta \text{vol } Q < \varepsilon/2.$$

The Darboux criterion proves integrability. No uniform continuity on the possibly noncompact set  $Q \setminus \partial E$  was assumed.  $\square$

Translation preserves Jordan measurability and volume: the indicator of  $E + a$  is the translate of the indicator of  $E$ , so [Lemma 12.1.11](#) applies before any change-of-variables theorem.

**Proposition 12.3.7** (Independence and elementary Jordan sets). The Jordan volume is independent of the containing rectangle. Every rectangle, and every finite union of rectangles with pairwise disjoint interiors, is Jordan measurable; its volume is the sum of the volumes of its rectangles.

*Proof.* Apply [Lemma 12.1.10](#) to the indicator on its containing rectangle. The lemma explicitly accounts for nonzero shared face values. Enlarging any two containing rectangles to a common one proves both independence of integrability and equality of the integrals.

The indicator of a rectangle is constant except on its finitely many faces. Given

$$\varepsilon > 0,$$

cover these faces by finitely many thin rectangles of total volume less than

$$\varepsilon.$$

Outside that cover, use a grid refining the rectangle's faces; there the upper and lower sums agree. The Darboux criterion proves integrability. For a finite union with pairwise disjoint interiors, the union indicator differs from the sum of the individual indicators only on their finitely many faces. The face-strip lemma makes that difference integrable with integral zero. Scalar linearity and rectangular zero extension therefore give the asserted volume formula before general Jordan null modification.  $\square$

**Proposition 12.3.8** (Null modifications and finite additivity). Let  $N \subseteq \mathbb{R}^n$  be bounded and Jordan-null.

- (a) The set  $N$  is Jordan measurable and has volume zero.
- (b) A bounded function supported on  $N$ , extended by zero to a containing rectangle, is Riemann integrable with integral zero. Consequently, changing a bounded Riemann-integrable function on  $N$  does not change its integral.
- (c) Let  $E_1, \dots, E_m$  be Jordan measurable and suppose every  $E_i \cap E_j$ ,  $i \neq j$ , is Jordan-null. Then their union is Jordan measurable and

$$\text{vol} \left( \bigcup_{i=1}^m E_i \right) = \sum_{i=1}^m \text{vol}(E_i).$$

More generally, if  $h$  is bounded on the union and the zero extension of  $h|_{E_i}$  is integrable for every  $i$ , then the zero extension of  $h$  from the union is integrable and

$$\int_{\bigcup_i E_i} h = \sum_{i=1}^m \int_{E_i} h.$$

*Proof.* Given  $\varepsilon > 0$ , cover  $N$  by finitely many closed rectangles of combined volume less than

$$\varepsilon/2.$$

Their union also contains  $\overline{N}$ . Enlarge them to open rectangles with combined volume less than  $\varepsilon$ . The compact set  $\overline{N}$  has a positive margin inside this open union, so [Lemma 12.3.3](#) puts every sufficiently fine grid cell meeting  $N$  inside it. All other cells have indicator zero. The upper sum is at most  $\varepsilon$ , and lower sums are nonnegative. Hence the indicator is integrable with integral zero.

If  $|h| \leq M$  and  $h$  is supported on  $N$ , then

$$|h| \leq M \mathbf{1}_N.$$

The same grid argument proves integrability, and order preservation gives zero integral. The difference of two functions which agree off  $N$  is of this form, proving the modification assertion.

Put

$$N_0 = \bigcup_{i < j} (E_i \cap E_j).$$

If  $m \leq 1$ , there is nothing to prove. If  $m \geq 2$ , this finite union is Jordan-null: combine finite covers whose total volumes are each less than

$$\varepsilon / \binom{m}{2}.$$

Away from  $N_0$ ,

$$\mathbf{1}_{\bigcup_i E_i} = \sum_i \mathbf{1}_{E_i}.$$

The right side is integrable, so the null-modification result proves integrability of the left side and the volume formula. The same argument applied to the zero extensions of  $h|_{E_i}$  proves the final assertion.  $\square$



**Definition 12.3.9** (Integration over a Jordan set). Let  $E \subseteq \mathbb{R}^n$  be Jordan measurable and let  $f : E \rightarrow \mathbb{R}$  be bounded. If the extension

$$\tilde{f}(\mathbf{x}) = \begin{cases} f(\mathbf{x}), & \mathbf{x} \in E, \\ 0, & \mathbf{x} \notin E \end{cases}$$

is integrable on a containing rectangle  $Q$ , define

$$\int_E f(\mathbf{x}) \, d\mathbf{x} = \int_Q \tilde{f}(\mathbf{x}) \, d\mathbf{x}.$$

**Remark 12.3.10.** This definition is intentionally more restrictive than later Lebesgue integration. A bounded function on a Jordan-measurable set need not have an integrable zero extension, so this condition must be checked separately.

### Simple planar regions and changing the order

Extending by zero lets us integrate over a curved region, but usually introduces a discontinuity at its boundary. The continuous Fubini theorem therefore does not directly apply. The next lemma restates [Corollary 12.2.5](#) in the form needed for zero extensions. We retain its direct finite-sum proof here. Product integrability and integrability of every section are separate hypotheses; the latter is needed even to define the ordinary outer integrand.

**Lemma 12.3.11** (Riemann–Fubini with integrable sections). Let  $g$  be Riemann integrable on  $A \times B$ , and suppose every section  $y \mapsto g(\mathbf{x}, y)$  is integrable on  $B$ . Then

$$G(\mathbf{x}) = \int_B g(\mathbf{x}, y) \, dy$$

is integrable on  $A$  and

$$\int_A G = \int_{A \times B} g.$$

*Proof.* Apply [Corollary 12.2.5](#) to  $g$ . Its product integrability and the separately assumed integrability of every indicated section are exactly the hypotheses of that corollary.  $\square$

**Example 12.3.12** (An integrable product with a nonintegrable section). On  $[0, 1]^2$  let  $g(x, y) = 1$  when  $x = 0$  and  $y \in \mathbb{Q}$ , and let  $g(x, y) = 0$  otherwise. This bounded function differs from zero only on the Jordan-null face  $\{0\} \times [0, 1]$ . By the null-modification rule, it is Riemann integrable with integral zero. Its section at  $x = 0$  is  $\mathbf{1}_{\mathbb{Q}}$  on  $[0, 1]$ : every interval has infimum 0 and supremum 1, so this section is not Riemann integrable. Thus the section hypothesis in the lemma is separate from product integrability.

**Theorem 12.3.13** (Integration between continuous graphs). Let  $\alpha, \beta : [a, b] \rightarrow \mathbb{R}$  be continuous,  $\alpha \leq \beta$ , and

$$D = \{(x, y) : a \leq x \leq b, \alpha(x) \leq y \leq \beta(x)\}.$$

Then  $D$  is Jordan measurable, and for continuous  $f : D \rightarrow \mathbb{R}$ ,

$$\int_D f = \int_a^b \left( \int_{\alpha(x)}^{\beta(x)} f(x, y) \, dy \right) dx.$$

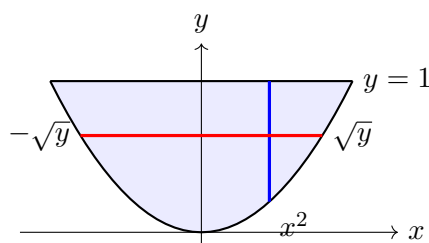
The analogous assertion holds with  $x$  and  $y$  interchanged.

*Proof.* The boundary of  $D$  lies in the two graphs and the two vertical end segments. To cover a continuous graph, partition  $[a, b]$  so that its oscillation in each cell is less than  $\eta$ . Rectangles of height at most  $2\eta$  cover these pieces, with total area at most  $2\eta(b - a)$ ; end segments have covers of arbitrarily small area. Uniform continuity and the boundary criterion give Jordan measurability. Compactness of  $D$  and [Proposition 12.3.6](#) make the zero extension of  $f$  integrable on a containing rectangle. Each vertical section is continuous on  $[\alpha(x), \beta(x)]$  and zero outside, with at most two junction discontinuities, hence integrable by the one-variable Darboux criterion. Apply [Lemma 12.3.11](#) to this extension. We have not applied the continuous-rectangle theorem to a discontinuous function.  $\square$

**Drawing is part of finding the bounds.** For fixed  $x$ , ask which  $y$  values occur: this gives  $a \leq x \leq b$ ,  $\alpha(x) \leq y \leq \beta(x)$ . For fixed  $y$ , ask which  $x$  values occur: this gives  $c \leq y \leq d$ ,  $\lambda(y) \leq x \leq \mu(y)$ . Changing integration order changes how the same geometric set is sliced.

**Example 12.3.14** (Between a parabola and a line). The region  $E = \{(x, y) : x^2 \leq y \leq 1\}$  has the two descriptions

$$-1 \leq x \leq 1, x^2 \leq y \leq 1; \quad 0 \leq y \leq 1, -\sqrt{y} \leq x \leq \sqrt{y}.$$



The blue vertical slice starts at  $x^2$ ; the red horizontal slice ends at the two square roots. Consequently

$$\text{area}(E) = \int_{-1}^1 (1 - x^2) \, dx = \int_0^1 2\sqrt{y} \, dy = \frac{4}{3}.$$

For a nonconstant integrand the same bounds give

$$\int_E y = \int_0^1 2y^{3/2} \, dy = 4/5.$$

**Example 12.3.15** (The same triangle, two orders). For  $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$ ,

$$\int_D x = \int_0^1 \int_0^{1-x} x \, dy \, dx = \frac{1}{6} = \int_0^1 \int_0^{1-y} x \, dx \, dy.$$

Reversing the order means describing the same points with horizontal sections; it does not mean merely exchanging two differential symbols.

**Example 12.3.16** (A triangular zero extension and a split order). For  $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x\}$  and  $f = x + y$ , the zero extension on  $[0, 1]^2$  has jumps only along the three boundary segments. The graph-region theorem gives

$$\int_D (x + y) = \int_0^1 \int_0^x (x + y) \, dy \, dx = \int_0^1 \frac{3}{2}x^2 \, dx = \frac{1}{2}.$$

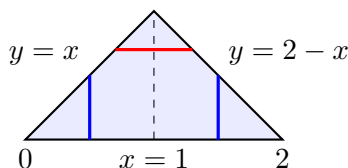
Reversing the order gives

$$\int_0^1 \int_y^1 (x+y) \, dx \, dy = 1/2.$$

For a region requiring a split, take  $E = \{(x, y) : 0 \leq x \leq 2, 0 \leq y \leq \min(x, 2-x)\}$ . Horizontal slices have  $0 \leq y \leq 1, y \leq x \leq 2-y$ ; vertical slices require  $[0, 1]$  with upper edge  $x$  and  $[1, 2]$  with upper edge  $2-x$ . Thus its area is both

$$\int_0^1 (2-2y) \, dy = 1 \quad \text{and} \quad \int_0^1 x \, dx + \int_1^2 (2-x) \, dx = 1.$$

The switch in the upper boundary at  $x = 1$  is why one iterated description must be split.



In the preceding split-order example, vertical slices change their upper edge at the dashed line; every horizontal slice has endpoints  $y, 2-y$ .

**Lemma 12.3.17** (Lipschitz images of boxes and faces). If  $F$  is  $L$ -Lipschitz on a subset of an  $n$ -cube of side  $h$ , its image fits in a box of side  $2L\sqrt{n}h$  (or is a singleton when  $L = 0$ ). The image of a compact  $(n-1)$ -dimensional rectangular face under a Lipschitz map into  $\mathbb{R}^n$  is Jordan-null.

*Proof.* Choose one point of the subset. Every other point has distance at most  $\sqrt{n}h$  from it, so every image coordinate differs by at most  $L\sqrt{n}h$ . This proves the box bound. Partition a face into at most  $Ch^{-(n-1)}$  pieces of side at most  $h$ . The covering image boxes have total volume at most  $C(2L\sqrt{n})^n h$ , which tends to zero. For  $n = 1$  a face is a point and a small interval covers its image. The same argument handles a finite union of faces.  $\square$

**Theorem 12.3.18** (Lower-dimensional images cannot contain an open set). Let  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$  be  $C^1$  on an open set, with  $n < m$ . The image of every closed cube  $Q \subset U$  is compact and Jordan-null in  $\mathbb{R}^m$ . The full image  $f(U)$  has empty interior. Thus a map into a nonempty open  $V \subseteq \mathbb{R}^m$  cannot be surjective.

*Proof.* On  $Q$ , bound  $\|Df\|$  by  $L > 0$ . Its convexity makes  $f|_Q$   $L$ -Lipschitz. Subdivide a cube of side  $s$  into  $k^n$  cubes of side

$$s/k.$$

Each image lies in an  $m$ -box of side

$$2L\sqrt{n}s/k.$$

Their total  $m$ -volume is at most

$$(2L\sqrt{n}s)^m k^{n-m} \rightarrow 0.$$

Enlarging these finitely many boxes slightly makes them open while keeping arbitrarily small total volume. Compactness of  $f(Q)$  follows from continuity. This proves the first assertion.

There are countably many closed cubes with rational center and positive rational side length contained in  $U$ , and their interiors cover  $U$ : a sufficiently small such cube around any point lies

in an open ball contained in  $U$ . Enumerate them as  $Q_j$ . Suppose  $f(U)$  contained a closed target cube  $B$  of positive volume. For a fixed  $\varepsilon > 0$ , cover each  $f(Q_j)$  by finitely many open boxes of total volume less than  $\varepsilon 2^{-j}$ . Together these form an open cover of the compact cube  $B$ , so finitely many of the boxes cover  $B$ . Finite grid comparison then gives  $\text{vol}(B) < \varepsilon$ . Choosing  $\varepsilon < \text{vol}(B)$  is a contradiction. Every nonempty open set contains such a cube, proving empty interior. For  $n = 0$  the source contains at most one point and the conclusion is immediate.  $\square$

This completes the deferred proof of [Theorem 11.6.13](#). The countable family was used only to construct an open cover of one compact target cube and extract a finite subcover. No assertion that  $f(U)$  itself is Jordan-null, and no countable additivity of Jordan content, is needed.

## Volumes in three dimensions by cross-sections

Sketch the solid and choose a slicing direction. Describe a typical two-dimensional cross-section by inequalities, then integrate its contribution. Alternatively project onto the  $xy$ -plane and write  $z_-(x, y) \leq z \leq z_+(x, y)$ ; projections onto the  $xz$ - or  $yz$ -plane instead describe intervals in  $y$  or  $x$ .

**What Fubini does for a solid.** Let  $\Omega$  lie between heights  $a$  and  $b$ , and define its horizontal section

$$\Omega_z = \{(x, y) : (x, y, z) \in \Omega\}, \quad A(z) = \int_{\Omega_z} 1 \, d(x, y).$$

When  $\Omega$  and all these sections are Jordan measurable, apply [Lemma 12.3.11](#) to the zero-extended indicator:

$$\begin{aligned} \text{vol}(\Omega) &= \int_{\Omega} 1 \, d(x, y, z) \\ &= \int_a^b \left( \int_{\Omega_z} 1 \, d(x, y) \right) dz = \int_a^b A(z) \, dz. \end{aligned}$$

Fixing  $z$  leaves a planar region of admissible  $(x, y)$  points. The inner integral of 1 gives its area; the outer integral accumulates those areas.

**Checking the geometry needed by the theorem.** For  $R, h > 0$ , the cylinder and cone sides are images of  $[0, 2\pi] \times [0, h]$  under, respectively,

$$\begin{aligned} (\theta, z) &\longmapsto (R \cos \theta, R \sin \theta, z), \\ (\theta, z) &\longmapsto (R(1 - z/h) \cos \theta, R(1 - z/h) \sin \theta, z). \end{aligned}$$

Their derivatives are bounded on these compact rectangles, so the maps are Lipschitz. Their images are therefore Jordan-null by [Lemma 12.3.17](#). No regularity of the cone as a surface at its apex is needed for this argument. The flat disk pieces lie in planes and are also null in three dimensions. For the sphere use the six coordinate graph pieces where one coordinate has magnitude at least

$$R/\sqrt{3};$$

the corresponding square-root graphs have bounded derivatives on their compact disk domains. The same covering argument applies to these pieces. Thus each solid has Jordan-null boundary. All horizontal sections are disks, whose circular boundaries are Lipschitz images of an interval and hence null in the plane. The required indicators and sections are therefore integrable.

The disk area used below follows from Cartesian slices and the one-variable substitution  $x = r \sin t$ :

$$\int_{-r}^r 2\sqrt{r^2 - x^2} \, dx = 2r^2 \int_{-\pi/2}^{\pi/2} \cos^2 t \, dt = \pi r^2.$$

The case  $r = 0$  is zero. These examples do not presuppose polar change of variables.

**Example 12.3.19** (Cylinder and cone). Let  $R, h > 0$ . For the cylinder

$$C = \{(x, y, z) : x^2 + y^2 \leq R^2, 0 \leq z \leq h\},$$

each height  $z \in [0, h]$  gives the same section  $C_z = \{(x, y) : x^2 + y^2 \leq R^2\}$ . Its area is  $A(z) = \pi R^2$ . The cross-sectional Fubini formula gives

$$\text{vol}(C) = \int_0^h A(z) \, dz = \int_0^h \pi R^2 \, dz = \pi R^2 [z]_0^h = \pi R^2 h.$$

Thus “base area times height” is itself an iterated-integral calculation.

For the cone

$$K = \{(x, y, z) : 0 \leq z \leq h, x^2 + y^2 \leq R^2(1 - z/h)^2\},$$

fixing  $z$  leaves a disk with radius

$$r(z) = R(1 - z/h),$$

since

$$1 - z/h \geq 0.$$

Consequently

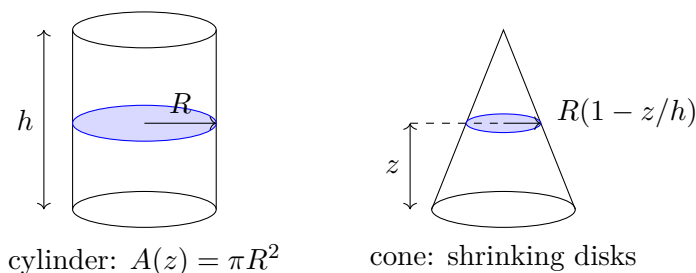
$$A(z) = \pi r(z)^2 = \pi R^2(1 - z/h)^2.$$

The radius decreases linearly with height, so area decreases quadratically. Integrating this quadratic profile is what produces the factor

$$1/3 :$$

$$\begin{aligned} \text{vol}(K) &= \int_0^h \pi R^2(1 - z/h)^2 \, dz \\ &= \pi R^2 \int_0^h \left(1 - \frac{2z}{h} + \frac{z^2}{h^2}\right) \, dz \\ &= \pi R^2 \left[ z - \frac{z^2}{h} + \frac{z^3}{3h^2} \right]_0^h \\ &= \pi R^2(h - h + h/3) = \frac{1}{3}\pi R^2 h. \end{aligned}$$

Compared with the cylinder of the same base and height, the cone has one third of the volume because its sections shrink according to this profile.



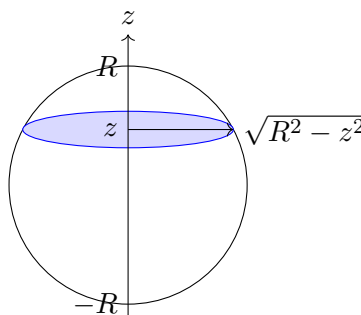
**Example 12.3.20** (A ball from horizontal disks). For  $B_R = \{(x, y, z) : x^2 + y^2 + z^2 \leq R^2\}$ , the height range is  $[-R, R]$ . At a fixed height,

$$(B_R)_z = \{(x, y) : x^2 + y^2 \leq R^2 - z^2\}, \quad A(z) = \pi(R^2 - z^2).$$

The radius is  $\sqrt{R^2 - z^2}$ , zero at the poles and  $R$  at the equator. Applying the same cross-sectional formula gives

$$\begin{aligned} \text{vol}(B_R) &= \int_{-R}^R \pi(R^2 - z^2) \, dz = \pi \left[ R^2 z - \frac{z^3}{3} \right]_{-R}^R \\ &= \pi \left( \frac{2}{3} R^3 - \left( -\frac{2}{3} R^3 \right) \right) = \frac{4}{3} \pi R^3. \end{aligned}$$

This is a genuine triple integral reduced to one variable by first understanding the geometry and area of every two-dimensional section.



Once the disk geometry is understood, the three-dimensional calculation reduces to an elementary one-variable integral.

## 12.4 Linear Changes of Variables

The one-variable substitution rule says that  $|\phi'(t)|$  measures the local stretching of length. In several variables, the corresponding linear fact comes first: an invertible linear map  $L$  sends a small box to a parallelepiped and scales its volume by  $|\det L|$ . A coordinate scaling by  $c$  has factor  $|c|$ , a diagonal map has the product of its coordinate scalings, a shear has factor 1, and a coordinate reflection preserves volume. Thus the determinant is the natural volume-scaling number, not a formal correction inserted after the fact.

**Example 12.4.1** (A rectangle becomes a parallelogram). Let

$$L(s, t) = (2s + t, s + 3t).$$

The unit square is sent to the parallelogram spanned by  $(2, 1)$  and  $(1, 3)$ . Since

$$\det L = \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 5,$$

its area is 5. More generally, the image of a rectangle of area  $A$  has area  $5A$ . The linear change-of-variables theorem below turns this geometric calculation into an integral formula.

**Lemma 12.4.2** (Volume under an invertible linear map). Let

$$L : \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

be invertible and let  $Q$  be a rectangle. Then  $L(Q)$  is Jordan measurable and

$$\text{vol}(L(Q)) = |\det L| \text{vol}(Q).$$

More generally, if  $E \subseteq \mathbb{R}^n$  is Jordan measurable, then  $L(E)$  is Jordan measurable and

$$\text{vol}(L(E)) = |\det L| \text{vol}(E).$$

*Proof.* We first record the effect of the three simple invertible linear maps on Jordan volume. If a rectangle has a zero side length, it lies in a rectangular face. Its image under any linear map is Jordan-null by Lemma 12.3.17, since linear maps are Lipschitz. Both volumes are then zero. For the rectangle calculations below assume all side lengths are positive. A coordinate permutation merely permutes the side intervals of a rectangle and preserves its volume. Multiplication of one coordinate by  $c \neq 0$  takes a rectangle to a rectangle and multiplies its volume by  $|c|$ .

**Strategy for a shear.** A shear translates every fiber without changing its length; first prove Jordan measurability, then apply Fubini. After relabeling coordinates, write  $R = A \times [a, b]$  and  $S(z, t) = (z, t + cz_j)$ . As  $S$  is a homeomorphism,  $\partial S(R) = S(\partial R)$ . The boundary of  $R$  is a finite union of rectangular faces, whose Lipschitz images are Jordan-null by Lemma 12.3.17. The boundary criterion (Proposition 12.3.4) therefore proves that  $S(R)$  is Jordan measurable before Fubini is invoked. Choose a compact interval  $J$  containing all translated fibers. For each  $z \in A$ , the fiber is  $[a + cz_j, b + cz_j]$ , so

$$\int_J \mathbf{1}_{S(R)}(z, t) \, dt = b - a.$$

All these sections are integrable. By Lemma 12.3.11,

$$\text{vol}(S(R)) = \int_A \left( \int_J \mathbf{1}_{S(R)}(z, t) \, dt \right) dz = (b - a) \text{vol}(A) = \text{vol}(R).$$

These conclusions extend from rectangles to every Jordan-measurable set. Indeed, for such a set  $E$ , the Darboux criterion applied to  $\mathbf{1}_E$  gives finite unions  $I$  and  $O$  of grid rectangles with pairwise disjoint interiors such that

$$I \subseteq E \subseteq O, \quad \text{vol}(O) - \text{vol}(I) < \varepsilon.$$

Images of distinct grid rectangles under a simple map overlap only in images of their common faces. By Lemma 12.3.17, those images are Jordan-null. Thus these overlaps are Jordan-null, and Proposition 12.3.8 shows that the volumes of the images of  $I$  and  $O$  are the sums of the volumes of their pieces. The inclusions

$$S(I) \subseteq S(E) \subseteq S(O)$$

then make the upper and lower Jordan integrals of  $\mathbf{1}_{S(E)}$  differ by less than  $\varepsilon$ . The same argument applies to coordinate permutations and coordinate scalings. Consequently, each of these simple maps sends Jordan-measurable sets to Jordan-measurable sets and multiplies their volumes by the absolute value of its determinant.

Lemma 10.17.17 gives geometric maps  $G_1, \dots, G_k$  with  $L = G_k \circ \dots \circ G_1$ , each a permutation, scaling, or shear. The rightmost map acts first. Start with  $A_0 = E$  and put  $A_j = G_j(A_{j-1})$  for  $j = 1, \dots, k$ . The result just proved first makes  $A_j$  Jordan measurable and then gives

$$\text{vol}(A_j) = |\det G_j| \text{vol}(A_{j-1}).$$

Thus  $A_k = L(E)$  is Jordan measurable before its volume is used. Multiplying these finite identities in their order of application gives

$$\text{vol}(L(E)) = \left( \prod_{j=1}^k |\det G_j| \right) \text{vol}(E) = |\det L| \text{vol}(E).$$

For  $L = I$  the empty composition has factor 1. Taking  $E = [0, 1]^n$  gives volume  $|\det L|$ ; taking  $E = Q$  gives the rectangle assertion directly, including degenerate rectangles.

For completeness, measurability can also be read directly from the boundary. Since  $L$  is a homeomorphism,

$$\partial L(Q) = L(\partial Q).$$

Each face of  $Q$  can be covered by at most  $C_0 h^{-(n-1)}$  face-cubes of side at most  $h$ . If  $M = \|L\|_{\text{op}}$ , the image of each such piece has diameter at most  $M\sqrt{n}h$ , and hence lies in an axis-parallel  $n$ -box of side  $2M\sqrt{n}h$ . The total volume of these boxes is at most

$$C_0(2M\sqrt{n})^n h,$$

which tends to zero. A finite union over the faces proves that  $\partial L(Q)$  is Jordan-null, in agreement with [Proposition 12.3.4](#).  $\square$

**Theorem 12.4.3** (Linear change of variables). Let  $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be invertible, let  $E$  be Jordan measurable, and let

$$f : L(E) \rightarrow \mathbb{R}$$

have an integrable zero extension. Then  $L(E)$  is Jordan measurable and

$$\int_{L(E)} f(y) \, dy = \int_E f(L\mathbf{x}) |\det L| \, d\mathbf{x}.$$

*Proof.* Let  $R$  be a rectangle and let

$$s = \sum_{i=1}^N c_i \mathbf{1}_{R_i}$$

be a finite sum for the closed cells of a grid partition of  $R$ . On shared faces this formula sums all incident terms. This is harmless for its integral, but we do not use those values as pointwise bounds for another function. This enlarges the initial owned-cell convention: arbitrary bounded values on the finitely many grid faces change no integral by the established null-modification result. More generally we call a bounded function a grid step function if it is constant on each cell interior, allowing arbitrary bounded values on the union  $G$  of grid faces. Such a function differs from the displayed sum only on the Jordan-null set  $G$ , so has the same integral by [Proposition 12.3.8](#). Its transform differs only on  $L(G)$ , also Jordan-null by [Lemma 12.3.17](#). The sets  $L(R_i)$  are Jordan measurable by [Lemma 12.4.2](#). Two of them intersect only in the image of a common grid face, which is Jordan-null by [Lemma 12.3.17](#). Hence [Proposition 12.3.8](#) gives

$$\int_{L(R)} s(L^{-1}y) \, dy = \sum_{i=1}^N c_i \text{vol}(L(R_i)) = |\det L| \int_R s(\mathbf{x}) \, d\mathbf{x}.$$

Thus the formula is established first for rectangular step functions; their transforms are finite sums of indicators of the Jordan sets  $L(R_i)$ , not rectangular step functions in the  $y$ -coordinates.



Now let  $h : R \rightarrow \mathbb{R}$  be Riemann integrable. For every  $\varepsilon > 0$ , choose a grid whose Darboux gap is less than  $\varepsilon$ . On the interior of its cell  $R_i$ , put  $\ell = \inf_{R_i} h$  and  $u = \sup_{R_i} h$ . On the entire face set  $G$ , instead put  $\ell = -M$  and  $u = M$ , where  $M \geq \sup_R |h|$ . These are grid step functions in the preceding sense. The bounds now hold even on shared and exterior faces. Null modification leaves their integrals equal to the lower and upper Darboux sums, so

$$\ell \leq h \leq u, \quad \int_R (u - \ell) < \varepsilon.$$

Their transforms are integrable on  $L(R)$ , and

$$\ell(L^{-1}y) \leq h(L^{-1}y) \leq u(L^{-1}y).$$

The gap between the transformed upper and lower integrals is

$$|\det L| \int_R (u - \ell) < |\det L| \varepsilon.$$

To make the integrability conclusion explicit, choose Darboux partitions for the two transformed integrable step functions whose upper–lower gaps have sum less than an arbitrary  $\delta > 0$ , and pass to a common refinement. On that refinement, the upper sum of  $h \circ L^{-1}$  is at most the upper sum of  $u \circ L^{-1}$ , while its lower sum is at least the lower sum of  $\ell \circ L^{-1}$ . Its Darboux gap is consequently less than

$$|\det L| \varepsilon + \delta.$$

Since  $\varepsilon$  and  $\delta$  are arbitrary,  $h \circ L^{-1}$  is integrable on  $L(R)$ . Squeezing its integral between the two transformed step-function integrals proves

$$\int_{L(R)} h(L^{-1}y) \, dy = |\det L| \int_R h(\mathbf{x}) \, d\mathbf{x}.$$

All integrations here may be read through zero extensions to containing rectangles, as in the definition of integration over a Jordan set.

Apply the same lower–upper construction to  $\mathbf{1}_E$  on a rectangle containing  $E$ . The transformed indicators squeeze  $\mathbf{1}_{L(E)}$ , with integral gap multiplied by  $|\det L|$ . Hence  $L(E)$  is Jordan measurable before any integral over it is used.

Finally, choose a rectangle  $P$  containing  $L(E)$ , and let  $\tilde{f}$  be the integrable zero extension of  $f$  to  $P$ . Apply the preceding argument to  $L^{-1}$  and  $\tilde{f}$ . It proves that

$$\mathbf{x} \mapsto \tilde{f}(L\mathbf{x})$$

has an integrable zero extension from  $L^{-1}(P)$ , and

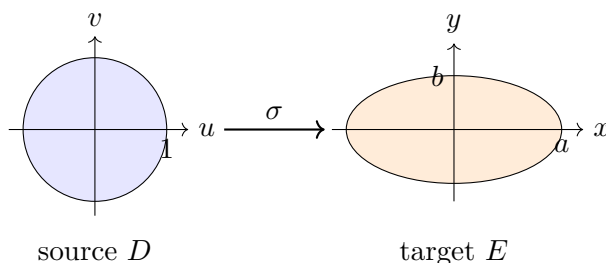
$$\int_P \tilde{f}(y) \, dy = |\det L| \int_{L^{-1}(P)} \tilde{f}(L\mathbf{x}) \, d\mathbf{x}.$$

The integrands vanish, respectively, outside  $L(E)$  and outside  $E$ . Using the definition of integration over Jordan sets therefore turns this identity into the asserted formula.  $\square$

**Example 12.4.4** (An ellipse and an ellipsoid by scaling). Let  $a, b, c > 0$ . The map  $\sigma(u, v) = (au, bv)$  takes the unit disk  $D$  onto

$$E = \{x^2/a^2 + y^2/b^2 \leq 1\}.$$

Its derivative is  $\text{diag}(a, b)$ , so  $|\det D\sigma| = ab$  and  $\text{area}(E) = ab \text{area}(D) = \pi ab$ .



Likewise  $(u, v, w) \mapsto (au, bv, cw)$  has determinant  $abc$  and takes the unit ball to

$$x^2/a^2 + y^2/b^2 + z^2/c^2 \leq 1.$$

Hence the ellipsoid volume is

$$4\pi abc/3.$$

Slicing found the ball's volume; linear scaling now transfers it to a new solid.

## 12.5 Nonlinear Changes of Variables

The local structure from Chapter 11 turns a multivariable substitution into one-variable substitutions with parameters. Our main route is

$$\boxed{\text{primitive} \longrightarrow \text{composition} \longrightarrow \text{local} \longrightarrow \text{finite localization.}}$$

Jordan boundaries will be handled separately after the integral identity. The determinant measures local volume distortion, but establishing that geometric interpretation need not precede the structural proof.

### Primitive substitution and composition

For a continuous function  $f$  on an open set  $V$ , write  $f \in C_c(V)$  if its support in  $V$  is compact. Recall that support is the closure of the nonzero set (Section 11.6). Such a support has positive distance from the complement of  $V$  when that complement is nonempty: finitely many doubled balls inside  $V$  cover it by their smaller balls. Therefore zero extension of  $f$  is continuous on  $\mathbb{R}^n$ . Its integral over  $V$  means the ordinary Riemann integral of that extension on any rectangle containing its support in the interior. Restriction and gluing show independence of this rectangle. No improper integral is being introduced.

**Lemma 12.5.1** (Primitive change of variables). Let  $q : U \rightarrow V$  be a primitive  $C^1$  diffeomorphism of open sets. For every  $f \in C_c(V)$ ,

$$\int_V f(y) \, dy = \int_U f(q(\mathbf{x})) |\det Dq(\mathbf{x})| \, d\mathbf{x}.$$

*Proof.* First suppose the changed coordinate is the last one, so  $q(\mathbf{x}', t) = (\mathbf{x}', p(\mathbf{x}', t))$  and  $\det Dq = \partial_t p$ . Both  $f$  and  $g = (f \circ q)|\partial_t p|$  have continuous zero extensions with compact support: the support of  $g$  is contained in  $q^{-1}(\text{supp } f)$ , which is compact by continuity of  $q^{-1}$ . Choose enclosing rectangles for these two supports.

Fix  $\mathbf{x}'$ . The source slice  $U_{\mathbf{x}'} = \{t : (\mathbf{x}', t) \in U\}$  is open in  $\mathbb{R}$ , and its components are open intervals. On each such interval  $\partial_t p$  is continuous and never zero, so has constant sign; the mean value theorem makes  $p(\mathbf{x}', \cdot)$  strictly monotone there. Injectivity of  $q$  makes the images of distinct component intervals disjoint, and surjectivity onto  $V$  makes their union  $V_{\mathbf{x}'}$ .

Only finitely many of these intervals meet the compact slice of  $q^{-1}(\text{supp } f)$ . Indeed, every point of that slice has an interval neighborhood contained in  $U_{\mathbf{x}'}$ , and a finite subcover meets only finitely many components. Within each relevant component choose a closed subinterval  $[\alpha, \beta]$  with that component's compact support slice in its interior. Define

$$\begin{aligned} s_- &= \min\{p(\mathbf{x}', \alpha), p(\mathbf{x}', \beta)\}, \\ s_+ &= \max\{p(\mathbf{x}', \alpha), p(\mathbf{x}', \beta)\}. \end{aligned}$$

One-variable substitution, [Theorem 9.4.9](#), gives

$$\int_{\alpha}^{\beta} f(\mathbf{x}', p(\mathbf{x}', t)) |\partial_t p(\mathbf{x}', t)| dt = \int_{s_-}^{s_+} f(\mathbf{x}', s) ds.$$

For an increasing fiber map this is substitution directly. For a decreasing map,  $|\partial_t p| = -\partial_t p$ ; the negative sign reverses the endpoint order in the substituted integral. The minimum and maximum therefore give the same formula in both cases. Outside these finitely many subintervals and their images the slice integrands are zero. Adding yields equality of the integrals of the full zero-extended slices. This uses only finitely many ordinary interval integrals for each  $\mathbf{x}'$ , even if the open domain has infinitely many slice components. Fubini for the continuous zero extensions now integrates that equality over  $\mathbf{x}'$ . For  $n = 1$  there are no parameter coordinates. Any other changed coordinate is moved to the last position by coordinate permutations, which preserve integrals by Fubini and have absolute determinant one.  $\square$

Thus a primitive multivariable substitution is ordinary one-variable substitution with parameters. The absolute value accounts for decreasing fibers as well as increasing ones.

**Lemma 12.5.2** (Stability under composition). Suppose the compact-support formula holds for  $C^1$  diffeomorphisms  $\sigma : U \rightarrow V$  and  $\tau : V \rightarrow W$ . It then holds for  $\tau \circ \sigma$ . It also holds for restrictions of any such map to open subsets.

*Proof.* For  $f \in C_c(W)$  the function  $h = (f \circ \tau)|\det D\tau|$  lies in  $C_c(V)$ , since its support lies in the compact set  $\tau^{-1}(\text{supp } f)$ . Applying the two formulas gives

$$\begin{aligned} \int_W f &= \int_V (f \circ \tau) |\det D\tau| \\ &= \int_U (f \circ \tau \circ \sigma) |\det D\tau(\sigma(\mathbf{x}))| |\det D\sigma(\mathbf{x})| d\mathbf{x}. \end{aligned}$$

The chain rule and multiplicativity of the determinant identify the product as  $|\det D(\tau \circ \sigma)(\mathbf{x})|$ . For a restriction  $\sigma|_{U_0}$ ,  $U_0 \subset U$  open, extend a function in  $C_c(\sigma(U_0))$  by zero to  $V$  and use the original formula. Its pullback vanishes outside  $U_0$ , giving exactly the restricted identity.  $\square$

**Corollary 12.5.3** (Local change of variables). If  $D\sigma(a)$  is invertible for a  $C^1$  map  $\sigma$  on an open set, there is a neighborhood  $N$  of  $a$  on which the compact-support change-of-variables formula holds for  $\sigma : N \rightarrow \sigma(N)$ .

*Proof.* Use [Theorem 11.6.16](#). Each primitive factor satisfies [Lemma 12.5.1](#); coordinate permutations satisfy the formula by Fubini, with absolute determinant one. Successive applications of [Lemma 12.5.2](#) give the assertion.  $\square$

### The compact-support theorem

**Theorem 12.5.4** (Change of variables on open sets with compact support). Let  $\sigma : U \rightarrow V$  be a  $C^1$  diffeomorphism between open subsets of  $\mathbb{R}^n$ . For every  $f \in C_c(V)$ ,

$$\int_V f(y) \, dy = \int_U f(\sigma(\mathbf{x})) |\det D\sigma(\mathbf{x})| \, d\mathbf{x}.$$

If  $\det D\sigma > 0$  the absolute value may be omitted. If it is negative everywhere, omitting it reverses the sign.

*Proof.* Let  $K = \text{supp } f$  and  $L = \sigma^{-1}(K)$ . The latter set is compact as the continuous image of  $K$  under  $\sigma^{-1}$ . If  $K$  is empty the identity is zero equals zero. Otherwise cover  $L$  by finitely many neighborhoods  $N_i \subset U$  from [Corollary 12.5.3](#). Choose the finite Euclidean weights  $\rho_i$  of [Theorem 11.6.19](#), compactly supported in  $N_i$ , with sum one near  $L$ . On  $\sigma(N_i)$  put

$$f_i(y) = \rho_i(\sigma^{-1}(y))f(y).$$

Its support is contained in the compact subset  $\sigma(\text{supp } \rho_i) \subset \sigma(N_i)$ , so  $f_i$  belongs to  $C_c(\sigma(N_i))$  and extends continuously by zero to  $V$ . Where  $f(y) \neq 0$ ,  $\sigma^{-1}(y) \in L$  and the weights sum to one; where  $f(y) = 0$ , all  $f_i$  vanish. Thus

$$\sum_i f_i = f$$

on  $V$ . The local formula says

$$\int_{\sigma(N_i)} f_i = \int_{N_i} \rho_i(\mathbf{x}) f(\sigma(\mathbf{x})) |\det D\sigma(\mathbf{x})| \, d\mathbf{x}.$$

All these functions have compact support in their displayed domains. Their zero extensions permit a finite sum on containing rectangles. On  $U$  the sum of the source integrands is  $f(\sigma(\mathbf{x})) |\det D\sigma(\mathbf{x})|$ , because that function vanishes outside  $L$ . Summing gives the claimed formula. The sign assertions follow from the absolute determinant.  $\square$

### Jordan boundaries and compact regions

The preceding theorem establishes an integral identity for continuous compactly supported functions. The next estimate supplies the separate set-theoretic fact needed to integrate on compact Jordan regions.

**Lemma 12.5.5** ( $C^1$  maps preserve compact Jordan-null sets). Let  $\sigma : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$  be  $C^1$ , with  $U$  open. The image of a compact Jordan-null set  $A \subset U$  is Jordan-null. If  $\sigma : U \rightarrow V$  is a diffeomorphism and  $E \subset U$  is compact and Jordan measurable, then  $\sigma(E)$  is Jordan measurable and  $\partial\sigma(E) = \sigma(\partial E)$ .

*Proof.* Choose  $\delta > 0$  so that the closed  $\delta$ -neighborhood of  $A$  is compact in  $U$ , using a finite cover by doubled balls in  $U$ . Bound  $\|D\sigma\|$  there by a positive constant  $L$ . Each cube of diameter less than  $\delta$  meeting  $A$  lies in that neighborhood. Segments in that cube remain there, so the mean-value estimate bounds image distances by  $L$  times source distances. A cube of side  $h$  maps into a box of side  $2L\sqrt{n}h$  by Lemma 12.3.17.

For any  $\varepsilon > 0$ , cover  $A$  by finitely many open boxes of total volume below  $\varepsilon$ . Their union contains a positive neighborhood of  $A$  by compactness. A sufficiently fine grid has all cells meeting  $A$  inside that union; their total volume is at most  $\varepsilon$  by finite grid comparison. Their image boxes have total volume at most  $(2L\sqrt{n})^n \varepsilon$ . This proves nullity. A homeomorphism of open sets preserves relative boundary. The compact sets  $E, \sigma(E)$  lie inside their respective open domains, so their relative and ambient boundaries coincide. Consequently  $\partial\sigma(E) = \sigma(\partial E)$  is null, and the Jordan boundary criterion proves the final assertion.  $\square$

**Corollary 12.5.6** (Change of variables for compact Jordan regions). Let  $\sigma : U \rightarrow V$  be a  $C^1$  diffeomorphism of open sets,  $E \subset U$  compact and Jordan measurable, and  $f$  continuous on  $\sigma(E)$ . Then

$$\int_{\sigma(E)} f = \int_E f(\sigma(\mathbf{x})) |\det D\sigma(\mathbf{x})| \, d\mathbf{x}.$$

*Proof.* Both regions are Jordan measurable by Lemma 12.5.5. Put  $M = \max_{\sigma(E)} |f|$  and  $J_0 = \max_E |\det D\sigma|$ ; the empty case is immediate. Fine-grid cells meeting  $\partial E$  have total volume  $v_h \rightarrow 0$ . Work in a fixed compact neighborhood of  $E$  inside  $U$  and bound  $\|D\sigma\|$  there by  $L > 0$ . The image of any union of these small cells has outer volume at most  $Cv_h$ ,  $C = (2L\sqrt{n})^n$ , by the preceding cube estimate.

Let  $O_h$  be a bounded open union of slightly enlarged boundary cells. The enlargements can be chosen so their total volume is at most  $2v_h$ , all stay in the fixed neighborhood, and their image boxes have total volume at most  $2Cv_h$ . The set  $K_h = E \setminus O_h$  is compact in  $E^\circ$ . Choose  $\eta_h \in C_c^\infty(E^\circ)$  equal to one near  $K_h$ , with values in  $[0, 1]$ . Define on  $V$

$$f_h(y) = \begin{cases} \eta_h(\sigma^{-1}(y))f(y), & y \in \sigma(E), \\ 0, & y \notin \sigma(E). \end{cases}$$

Although  $f$  is given only on  $\sigma(E)$ , this extension is continuous: the cutoff has compact support in  $E^\circ$  and vanishes near the boundary. Its image support is compact in  $\sigma(E^\circ)$ , so  $f_h \in C_c(V)$ . Apply Theorem 12.5.4. The source and target errors are confined to  $E \cap O_h$  and its image, respectively. All involved restricted integrands are integrable by the compact Jordan theory, and domination by the finite box covers gives

$$\left| \int_{\sigma(E)} f - \int_V f_h \right| \leq 2MCv_h, \quad \left| \int_E (f \circ \sigma) |\det D\sigma| - \int_E \eta_h(f \circ \sigma) |\det D\sigma| \right| \leq 2MJ_0v_h.$$

Letting the grid size tend to zero proves the assertion. No extension of  $f$  beyond its given region has been assumed.  $\square$

**Theorem 12.5.7** (Change of variables on a rectangle). Let  $Q \subseteq \mathbb{R}^n$  be a rectangle,  $U$  an open neighborhood of  $Q$ , and  $\sigma : U \rightarrow \mathbb{R}^n$  be  $C^1$ . Suppose  $\sigma$  is injective on  $Q$  and  $\det D\sigma(\mathbf{x}) \neq 0$  for  $\mathbf{x} \in Q$ . Then  $\sigma(Q)$  is Jordan measurable and, for continuous  $f$  on  $\sigma(Q)$ ,

$$\int_{\sigma(Q)} f(y) \, dy = \int_Q f(\sigma(\mathbf{x})) |\det D\sigma(\mathbf{x})| \, d\mathbf{x}.$$

*Proof.* There is an open neighborhood of  $Q$  on which  $\sigma$  is injective. Otherwise, in neighborhoods of distance less than

$$1/j$$

from  $Q$  inside a fixed compact neighborhood in  $U$ , there would be distinct  $\mathbf{x}_j, y_j$  with  $\sigma(\mathbf{x}_j) = \sigma(y_j)$ . Subsequences converge to  $\mathbf{x}, y \in Q$ . Continuity and injectivity on  $Q$  give  $\mathbf{x} = y$ . The inverse theorem gives a neighborhood of this common limit where  $\sigma$  is injective; eventually both sequences lie there, a contradiction. Continuity of the determinant and compactness also give a neighborhood of  $Q$  where it never vanishes. On the intersection  $\sigma$  is an injective local diffeomorphism, hence a diffeomorphism onto an open image: the local smooth inverses agree by injectivity. Apply [Corollary 12.5.6](#) with  $E = Q$ .  $\square$

### Optional geometric refinement: local volume distortion

The structural proof is complete. The following independent local comparison retains the geometric insight of a direct Riemann proof. Its inner estimate uses the inverse already supplied by Chapter 11.

**The maximum-norm segment estimate.** Let  $H : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$  be differentiable on an open neighborhood of  $[\mathbf{x}, y]$  and suppose  $\|DH(z)\|_{\infty \rightarrow \infty} \leq M$  along that segment. Choose a coordinate  $i$  with  $|H_i(y) - H_i(\mathbf{x})| = \|H(y) - H(\mathbf{x})\|_{\infty}$ . The scalar mean value theorem applied to  $t \mapsto H_i(\mathbf{x} + t(y - \mathbf{x}))$  on  $[0, 1]$  gives some  $z \in [\mathbf{x}, y]$  with

$$\begin{aligned} \|H(y) - H(\mathbf{x})\|_{\infty} &= |DH_i(z)(y - \mathbf{x})| \\ &\leq \|DH(z)(y - \mathbf{x})\|_{\infty} \leq M\|y - \mathbf{x}\|_{\infty}. \end{aligned}$$

For  $\mathbf{x} = y$  the conclusion is immediate. In particular the estimate applies on a convex region with this derivative bound, with the exact same constant.

**Proposition 12.5.8** (Local Jacobian volume distortion). Let  $\sigma$  be  $C^1$  near  $a$ , with  $D\sigma(a)$  invertible. For every  $\varepsilon > 0$  there is a neighborhood  $N$  of  $a$  such that every nondegenerate closed cube  $C \subset N$  satisfies

$$\left| \frac{\text{vol } \sigma(C)}{\text{vol } C} - |\det D\sigma(a)| \right| < \varepsilon.$$

In particular this ratio tends to  $|\det D\sigma(a)|$  as the cubes shrink to  $a$ .

*Proof.* Put  $A = D\sigma(a)$  and  $T(\mathbf{x}) = a + A^{-1}(\sigma(\mathbf{x}) - \sigma(a))$ . Then  $T(a) = a$ ,  $DT(a) = I$ , and the inverse theorem provides a local inverse  $S$  with  $DS(a) = I$ . Use the maximum norm and its operator norm. Choose a convex ball about  $a$  in the source and a convex ball about  $a$  in the image on which respectively

$$\|DT - I\|_{\text{op}} \leq \delta, \quad \|DS - I\|_{\text{op}} \leq \delta, \quad 0 < \delta < 1.$$

Continuity of  $DT, DS$  allows this for any  $\delta > 0$ . Shrink  $N$  so that cubes  $C \subset N$  stay in the source ball and all points within their half side length of  $T(c)$  stay in the image ball, where  $c$  is the cube center. Concretely, if the image ball has radius  $R$ , take  $N = B_\infty(a, r)$  with

$$r < R/4,$$

inside the source ball, and

$$T(N) \subset B_\infty(a, R/2).$$

A cube in  $N$  has side  $h \leq 2r$ , so these target points lie in

$$B_\infty(a, 3R/4).$$

The maximum-norm segment estimate just proved, applied to  $T-I$  and  $S-I$  on their respective convex balls, gives

$$\|T(\mathbf{x}) - T(c) - (\mathbf{x} - c)\|_\infty \leq \delta \|\mathbf{x} - c\|_\infty \quad (\mathbf{x} \in C).$$

Thus  $T(C)$  lies in the cube centered at  $T(c)$  of side  $(1 + \delta)h$ . Conversely, if

$$\|y - T(c)\|_\infty \leq h/(2(1 + \delta)),$$

the segment from  $T(c)$  to  $y$  is in the chosen image ball and

$$\|S(y) - c\|_\infty \leq (1 + \delta)\|y - T(c)\|_\infty \leq h/2.$$

So the cube of side

$$h/(1 + \delta)$$

about  $T(c)$  lies in  $T(C)$ . The image is Jordan measurable by [Lemma 12.5.5](#), independently of the integral formula. Monotonicity of volume and linear volume scaling yield

$$|\det A|(1 + \delta)^{-n} \leq \frac{\text{vol } \sigma(C)}{\text{vol } C} \leq |\det A|(1 + \delta)^n.$$

Choose  $\delta$  so that both endpoints differ from  $|\det A|$  by less than  $\varepsilon$ . This proves the uniform neighborhood assertion.  $\square$

**Schwartz's 1954 perspective.** An alternative direct Riemann/Jordan proof is Schwartz's [\[21\]](#). Its proof principle is to obtain the easy one-sided volume or integral bound from local derivative estimates, then apply that bound to the inverse map. For example, once the upper bound

$$\int_{\sigma(E)} q \leq \int_E (q \circ \sigma) J_\sigma$$

is known for nonnegative continuous  $q$ , apply it to  $\sigma^{-1}$  and  $h = (q \circ \sigma) J_\sigma$ . Since  $J_{\sigma^{-1}}(\sigma(\mathbf{x})) J_\sigma(\mathbf{x}) = 1$ , this gives the reverse inequality. This avoids solving a hard inner-containment problem directly:

$\text{one inequality for } \sigma + \text{the same inequality for } \sigma^{-1} \implies \text{equality.}$

This is a proof perspective, not a second full proof. The local geometric viewpoint can also be compared with [\[9, 1\]](#).

**Lebesgue perspective (unproved preview).** In Lebesgue theory the theorem first describes transformed measure:

$$m(\sigma(E)) = \int_E |\det D\sigma(\mathbf{x})| \, d\mathbf{x}$$

for measurable  $E \subset U$ . More precisely, the measure on  $U$  defined by  $\nu(E) = m(\sigma(E))$  is  $(\sigma^{-1})_* m|_U$ ; its Radon–Nikodym density relative to  $m|_U$  is  $|\det D\sigma|$ . Integration with respect to this transformed measure gives the function formula, first for nonnegative measurable functions and then for integrable ones. The present Riemann/Jordan proof uses finite smooth localization; Lebesgue theory permits measurable localization and countable additivity. These measure-theoretic statements are a non-prerequisite preview, not results proved here; see [10].

**Lemma 12.5.9** (Derivatives on half-space domains). Let  $O$  be relatively open in  $\mathbb{H}^n = \{x_n \geq 0\}$  and let  $F : O \rightarrow \mathbb{R}^m$  be  $C^1$  in the local-extension sense. Define  $DF(p)$  using any local  $C^1$  Euclidean extension. This is independent of the extension, including when  $p_n = 0$ , and compositions obey the chain rule. The same convention applies to smooth maps.

*Proof.* Two extensions agree on the Euclidean-open portion  $O \cap \{x_n > 0\}$  of their common neighborhood, so their derivatives agree there. Approach each boundary point by points in that portion. Continuity of derivatives makes their boundary values equal. This uses no boundary-preservation claim. For  $F : O \rightarrow P$  and  $G : P \rightarrow \mathbb{R}^r$ , where  $P$  is relatively open in a half-space, choose ambient extensions near  $p$  and  $F(p)$ . By continuity shrink the first neighborhood so its extension maps into the second. The ordinary chain rule then gives  $D(G \circ F)(p) = DG(F(p))DF(p)$ , independent of extensions by the first part. In particular, for inverse smooth half-space maps  $\chi, \chi^{-1}$ ,

$$D\chi^{-1}(\chi(p))D\chi(p) = I, \quad D\chi(p)D\chi^{-1}(\chi(p)) = I.$$

These are derivatives of the identity compositions, so both derivatives are invertible even at boundary points.  $\square$

**Corollary 12.5.10** (The half-space version). The same formula holds for a  $C^1$  diffeomorphism between relatively open subsets of  $\mathbb{H}^n = \{x_n \geq 0\}$ , smooth across the boundary locally, with smooth inverse and preserving the boundary face. A continuous compactly supported integrand is integrated over a containing half-rectangle, extending by zero only across artificial chart boundaries.

*Proof.* Such an extension is continuous within the half-space, so its integral over a half-rectangle exists. For  $\delta > 0$  let

$$\chi_\delta(t) = \min\{1, \max\{0, (t - \delta)/\delta\}\}.$$

Apply [Theorem 12.5.4](#) on the interiors to the target function  $f(y)\chi_\delta((\sigma^{-1}(y))_n)$ . It has compact support away from the boundary. The source error is supported in a fixed bounded box intersected with  $0 \leq x_n \leq 2\delta$ , and is bounded, hence its integral tends to zero. The target error is likewise bounded on  $K = \text{supp } f$ . By compactness of  $\sigma^{-1}(K)$  and boundary preservation, its support lies in  $0 \leq y_n \leq \eta(\delta)$  with  $\eta(\delta) \rightarrow 0$ . Otherwise there would be  $\varepsilon_0 > 0$ , numbers  $\delta_j \rightarrow 0$ , and  $y_j \in K$  such that, with  $\mathbf{x}_j = \sigma^{-1}(y_j)$ ,

$$0 \leq (\mathbf{x}_j)_n \leq 2\delta_j, \quad (y_j)_n \geq \varepsilon_0.$$

A subsequence of  $(\mathbf{x}_j)$  converges in the compact set  $\sigma^{-1}(K)$  to  $\mathbf{x}$  with  $x_n = 0$ . Continuity gives  $y_j = \sigma(\mathbf{x}_j) \rightarrow \sigma(\mathbf{x})$ , while boundary preservation gives  $(\sigma(\mathbf{x}))_n = 0$ , a contradiction. In a fixed containing box, a strip of height  $\eta(\delta)$  has volume at most  $C\eta(\delta)$ . If the target integrand is bounded there by  $M$ , its error is at most  $MC\eta(\delta) \rightarrow 0$ . Pass to the limit. No zero extension across the genuine boundary is required to be continuous.  $\square$



### Alternative Viewpoint: Pullbacks and Manifold Integration

The compact-support theorem was obtained by primitive substitution and finite Euclidean localization. Its geometric interpretation is supplied by [Proposition 12.5.8](#):  $|\det D\sigma|$  is infinitesimal unsigned volume distortion. Differential forms retain the sign as well.

For this preview take a smooth diffeomorphism  $\sigma : U \rightarrow V$  between open sets and a smooth compactly supported scalar  $f : V \rightarrow \mathbb{R}$ . The compact-support theorem ensures that both scalar integrals exist. Accordingly, the scalar formula just proved has the schematic form

$$\int_V f(y) \, dy = \int_U f(\sigma(\mathbf{x})) |\det D\sigma(\mathbf{x})| \, d\mathbf{x}.$$

The absolute value occurs because ordinary scalar volume is unsigned. Chapter 13 gives a complementary interpretation. For a top-degree form on  $V$ ,

$$\omega = f(y) \, dy_1 \wedge \cdots \wedge dy_n,$$

the pullback by a smooth map  $\sigma : U \rightarrow V$  is

$$\sigma^*\omega = f(\sigma(\mathbf{x})) \det D\sigma(\mathbf{x}) \, dx_1 \wedge \cdots \wedge dx_n.$$

Indeed,

$$\sigma^*(dy_j) = \sum_{i=1}^n \frac{\partial \sigma^j}{\partial x_i} \, dx_i,$$

and the alternating multilinearity of the wedge product produces the determinant. In this language, the determinant appears automatically when a top-degree form is transported to the parameter domain.

The sign is now meaningful. For an orientation-preserving diffeomorphism, the oriented formula is

$$\int_V \omega = \int_U \sigma^*\omega;$$

an orientation-reversing diffeomorphism changes the sign. This distinction between  $\det D\sigma$  and  $|\det D\sigma|$  is why orientation is part of the natural language of integration on manifolds. One can formulate unsigned integration intrinsically using densities, but that theory is beyond the scope of these notes.

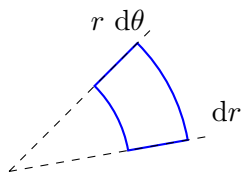
This pullback identity is the form of change of variables that survives on a smooth manifold, where no single global coordinate system need exist. Chapter 13 develops pullbacks and top-degree forms in [Section 13.7](#), and its change-of-parametrization result [Proposition 13.8.6](#) uses the present theorem to establish coordinate independence. The logical order is essential:

$$\begin{aligned} & \text{Euclidean change of variables} \\ \implies & \text{coordinate invariance of form integration} \\ \implies & \text{well-defined integration on manifolds.} \end{aligned}$$

Thus the differential-form discussion is a conceptual, coordinate-invariant reformulation, not an independent proof of the Euclidean analytic theorem.

The unproved Lebesgue perspective above points toward the later measure theory described in [Section 13.17](#).

**The geometric source of the polar factor.** For the parameter order  $(r, \theta)$ , the map  $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$  takes a tiny parameter rectangle to an annular sector. The radial side has length about  $dr$  and the angular side about  $r d\theta$ , so its area is about  $r dr d\theta$ . This picture motivates the factor  $r$ ; the change-of-variables theorem now justifies it. The calculation below removes the origin and an angular seam and estimates the omitted pieces.



**Example 12.5.11** (Polar coordinates and the unit disk). Consider

$$\sigma(\theta, r) = (r \cos \theta, r \sin \theta).$$

Its derivative is represented by the Jacobian

$$J_\sigma(\theta, r) = \begin{pmatrix} -r \sin \theta & \cos \theta \\ r \cos \theta & \sin \theta \end{pmatrix}, \quad \det D\sigma(\theta, r) = -r, \quad |\det D\sigma(\theta, r)| = r.$$

The parameter order  $(\theta, r)$  specifies the two Jacobian columns. Compared with the order  $(r, \theta)$  in [Example 11.2.20](#), this swaps columns: the determinant changes from  $+r$  to  $-r$ , while its absolute value remains  $r$ . The nested notation  $dr d\theta$  instead instructs us to integrate in  $r$  first, then  $\theta$ . Exchanging legitimate scalar iterated integrals has no minus sign; exchanging wedge factors does. The order  $(\theta, r)$  reverses orientation, while the absolute determinant for scalar area remains positive. The geometric preview above used actual increments: an angular change  $\Delta\theta$  at radius  $r$  sweeps an arc of length approximately  $r\Delta\theta$ , while a radial change has length  $\Delta r$ . The present theorem supplies the rigorous integral statement.

To use the theorem without hiding its hypotheses, first take

$$Q_{\varepsilon, \eta} = [\eta, 2\pi - \eta] \times [\varepsilon, 1]$$

with  $\varepsilon, \eta > 0$ . On this rectangle the map is injective and its Jacobian determinant never vanishes. Its image is an annular sector. For example, since  $x^2 + y^2 = r^2$ ,

$$\begin{aligned} \int_{\sigma(Q_{\varepsilon, \eta})} (x^2 + y^2) d(x, y) &= \int_{\eta}^{2\pi - \eta} \int_{\varepsilon}^1 r^2 r dr d\theta \\ &= \frac{(2\pi - 2\eta)(1 - \varepsilon^4)}{4}. \end{aligned}$$

Letting  $\varepsilon, \eta \rightarrow 0^+$  gives

$$\int_{x^2 + y^2 \leq 1} (x^2 + y^2) d(x, y) = \frac{\pi}{2}.$$

The following discarded-area bounds are deliberately independent of the disk area already computed by Cartesian slices. The central disk lies in  $[-\varepsilon, \varepsilon]^2$ , so its area is at most  $4\varepsilon^2$ . For

$$0 < \eta < \pi/2,$$

the omitted angular seam has  $0 \leq x \leq 1$  and  $|y| \leq \sin \eta \leq \eta$ ; it lies in  $[0, 1] \times [-\eta, \eta]$ , of area  $2\eta$ . Their combined outer area is at most  $4\varepsilon^2 + 2\eta \rightarrow 0$ . The integrand is bounded by 1, so the

discarded integrals obey the same bound. The unit circle is Jordan-null independently, by the Lipschitz image-of-an-interval estimate applied to  $(\cos t, \sin t)$  on  $[0, 2\pi]$ ; thus the disk's Jordan measurability also precedes its area calculation. The same calculation with integrand 1 recovers the disk-area formula by a second method:

$$\int_{x^2+y^2 \leq 1} 1 \, d(x, y) = \pi.$$

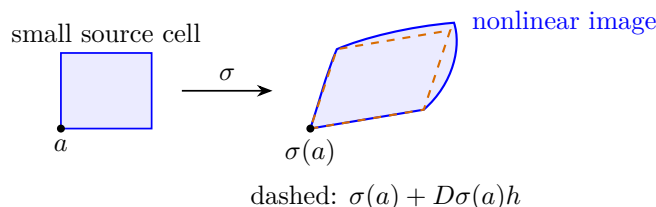


Figure 12.1: The derivative gives a first-order parallelogram model, not an exact image. Its area is  $|\det D\sigma(a)|$  times the source area; the nonlinear error is controlled as the cell shrinks.

## Worked changes of variables

**Computational checklist.** Choose coordinates and write the source-to-target map  $\sigma$ ; describe the source domain; compute  $D\sigma$  and  $|\det D\sigma|$ ; substitute in the integrand; check the hypotheses or controlled exceptions; evaluate.

Classical analytic geometry asks which coordinates make a curve, surface, or region's equation simplest. Change of variables asks the same question simultaneously for the domain and the integrand. A disk suggests polar coordinates, an ellipse suggests scaling, the linear expressions  $x + y, x - y$  suggest oblique coordinates, and a cone suggests cylindrical coordinates.

**Example 12.5.12** (Two substitutions for an ellipse). Use [Example 12.4.4](#). For

$$f(x, y) = x^2/a^2 + y^2/b^2,$$

$f(\sigma(u, v)) = u^2 + v^2$  and  $|\det D\sigma| = ab$ . On the source disk use  $(u, v) = (r \cos \theta, r \sin \theta)$ , now with parameter order  $(r, \theta)$  and determinant  $r$ . The preceding polar truncation justifies the seam and central-point exceptions. Thus

$$\int_E f \, d(x, y) = ab \int_0^{2\pi} \int_0^1 r^2 r \, dr \, d\theta = \frac{\pi ab}{2}.$$

The two absolute Jacobian determinant factors describe two successive changes of coordinates.

**Example 12.5.13** (Linear expressions as coordinates). On  $E = \{|x + y| \leq 1, |x - y| \leq 1\}$  put  $u = x + y, v = x - y$ . Solving for the old variables gives the required source-to-target map

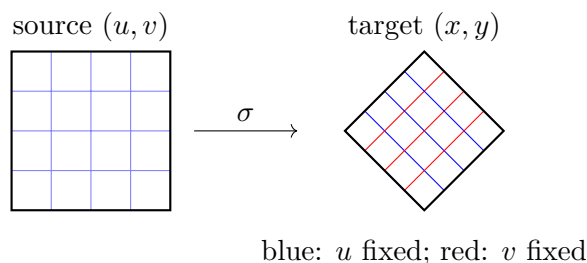
$$\sigma(u, v) = ((u + v)/2, (u - v)/2), \quad (u, v) \in [-1, 1]^2, \quad D\sigma = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

It is globally invertible and

$$|\det D\sigma| = 1/2,$$

whereas the inverse coordinate map has absolute determinant 2. For the adapted integrand,

$$\int_E ((x+y)^2 + (x-y)^2) \, d(x,y) = \frac{1}{2} \int_{-1}^1 \int_{-1}^1 (u^2 + v^2) \, dv \, du = \frac{4}{3}.$$



Good coordinates simplify both the domain and the integrand: the oblique lines become coordinate lines on the source square.

**Example 12.5.14** (A rectangle becomes a trapezoid). Take  $\sigma(u, v) = (u, (1+u)v)$  on  $Q = [0, 1]^2$ . Its image is  $E = \{0 \leq x \leq 1, 0 \leq y \leq 1+x\}$ , and

$$D\sigma = \begin{pmatrix} 1 & 0 \\ v & 1+u \end{pmatrix}, \quad \det D\sigma = 1+u.$$

The inverse is

$$(x, y) \mapsto (x, y/(1+x)),$$

smooth on  $x > -1$ . For  $f(x, y) = y$ , there are two separate transformations:

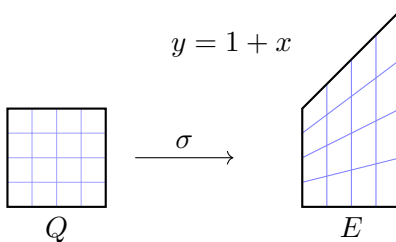
$$f(\sigma(u, v)) = (1+u)v, \quad d(x, y) = (1+u) \, d(u, v).$$

Therefore

$$\int_E y \, d(x, y) = \int_0^1 \int_0^1 (1+u)^2 v \, dv \, du = \frac{1}{2} \int_0^1 (1+u)^2 \, du = \frac{7}{6}.$$

With integrand 1 the area is

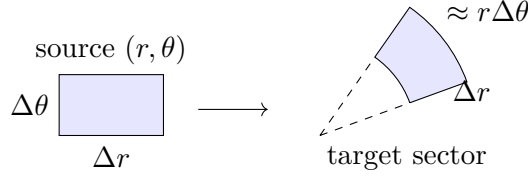
$$\int_0^1 (1+u) \, du = 3/2.$$



Horizontal coordinate lines  $v = \text{constant}$  become sloping lines.

**Example 12.5.15** (A radial integrand). For  $R > 0$ , polar coordinates transform  $x^2 + y^2 \leq R^2$  into  $0 \leq r \leq R$ ,  $0 \leq \theta \leq 2\pi$ , subject to the same controlled seam and center exceptions as before. Since  $(x^2 + y^2)^2 = r^4$ ,

$$\int_{x^2+y^2 \leq R^2} (x^2 + y^2)^2 \, d(x, y) = \int_0^{2\pi} \int_0^R r^5 \, dr \, d\theta = \frac{\pi R^6}{3}.$$



The two small side lengths explain the area factor  $r$ ; the theorem turns this first-order picture into an exact integral formula.

**Example 12.5.16** (Cylindrical coordinates revisit the cone). Use  $\sigma(r, \theta, z) = (r \cos \theta, r \sin \theta, z)$ . With this parameter order,

$$D\sigma = \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \det D\sigma = r.$$

The cone in [Example 12.3.19](#) has source bounds  $0 \leq z \leq h$ ,  $0 \leq \theta \leq 2\pi$ ,

$$0 \leq r \leq R(1 - z/h).$$

On  $r > 0$ ,  $0 < \theta < 2\pi$  the map is a diffeomorphism onto the slit space. For  $0 < \varepsilon < R$  and

$$0 < \eta < \pi/2,$$

retain the compact source

$$E_{\varepsilon, \eta} = \{(r, \theta, z) : 0 \leq z \leq h(1 - \varepsilon/R), \quad \eta \leq \theta \leq 2\pi - \eta, \\ \varepsilon \leq r \leq R(1 - z/h)\}.$$

The shortened height range follows by solving

$$\varepsilon \leq R(1 - z/h)$$

for  $z$ ; it prevents reversed radial bounds near the apex. This Jordan region lies inside the diffeomorphism domain, so [Corollary 12.5.6](#) applies. The omitted target lies in a central box of volume  $4\varepsilon^2 h$  and a seam box of volume at most  $2R^2 \eta h$ . Their volumes tend to zero. The nonnegative omitted source integral, including the small apex portion, is at most

$$\int_0^h \int_0^{2\pi} \int_0^\varepsilon r \, dr \, d\theta \, dz + 2\eta \int_0^h \int_0^R r \, dr \, dz = \pi h \varepsilon^2 + \eta h R^2 \longrightarrow 0.$$

Hence

$$\text{vol}(K) = \int_0^h \int_0^{2\pi} \int_0^{R(1-z/h)} r \, dr \, d\theta \, dz = \pi R^2 \int_0^h (1 - z/h)^2 \, dz = \frac{1}{3} \pi R^2 h.$$

The inner two integrals recover the disk area at height  $z$ . This is the same solid and the same answer as cross-sectional Fubini. Geometrically, a small cylindrical cell has side lengths  $\Delta r$ , approximately  $r\Delta\theta$ , and  $\Delta z$ , explaining its volume factor  $r$ .

**Example 12.5.17** (Spherical coordinates and the ball). Use the parameter order  $(\rho, \varphi, \theta)$  and the convention

$$\sigma(\rho, \varphi, \theta) = (\rho \sin \varphi \cos \theta, \rho \sin \varphi \sin \theta, \rho \cos \varphi).$$

Here  $\rho \geq 0$  is distance from the origin,  $0 \leq \varphi \leq \pi$  is the angle from the positive  $z$ -axis, and  $0 \leq \theta < 2\pi$  is the azimuthal angle in the  $xy$ -plane. Differentiating in this order gives

$$D\sigma = \begin{pmatrix} \sin \varphi \cos \theta & \rho \cos \varphi \cos \theta & -\rho \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & \rho \cos \varphi \sin \theta & \rho \sin \varphi \cos \theta \\ \cos \varphi & -\rho \sin \varphi & 0 \end{pmatrix}.$$

Expansion of the determinant gives  $\det D\sigma = \rho^2 \sin \varphi$ , which is nonnegative on the stated parameter ranges. For  $R > 0$  the resulting formula is

$$\text{vol}(B_R) = \int_0^R \int_0^\pi \int_0^{2\pi} \rho^2 \sin \varphi \, d\theta \, d\varphi \, d\rho = \frac{4}{3}\pi R^3.$$

Here is the justification at the coordinate exceptions. The map is a diffeomorphism on  $\rho > 0$ ,  $0 < \varphi < \pi$ ,  $0 < \theta < 2\pi$  onto the complement of the polar axis and angular seam: recover  $\rho$  as the norm,

$$\varphi = \arccos(z/\rho),$$

and the unique azimuth in that interval. These inverses are smooth locally. At the origin all angles coincide; on the polar axis azimuth is undetermined; and the closed angular box repeats the seam at  $\theta = 0, 2\pi$ . Thus the full box is not a global diffeomorphism domain.

Apply [Corollary 12.5.6](#) first on  $[\delta, R] \times [\eta, \pi - \eta] \times [\eta, 2\pi - \eta]$ , where  $0 < \delta < R$  and

$$0 < \eta < \pi/2.$$

The omitted target lies in a central cube of volume  $8\delta^3$ , an axis box of volume at most  $8R^3 \sin^2 \eta$ , and a seam slab of volume at most  $4R^3 \sin \eta$ . These tend to zero. The omitted nonnegative source integral is bounded by

$$\frac{4\pi}{3}\delta^3 + \frac{4\pi R^3}{3}(1 - \cos \eta) + \frac{4\eta R^3}{3},$$

which also tends to zero. The sphere bounding the ball is Jordan-null: it is the Lipschitz image under  $\sigma(R, \cdot, \cdot)$  of a parameter rectangle, by the earlier lower-dimensional image theorem. Hence the ball is Jordan measurable and its open and closed versions have equal volume. This completes the same truncation argument used for polar and cylindrical coordinates, without discarding a coordinate exception silently.

**Change-of-variables warnings.** Transform the integrand *and* the domain and include the absolute Jacobian determinant. Do not retain old bounds after substitution, or omit the absolute Jacobian determinant after changing bounds. For  $\sigma$  from new coordinates to old, the factor is  $|\det D\sigma|$ , not the determinant of the inverse map. Ordinary scalar area and volume use an absolute value. Verify injectivity and control singular-coordinate exceptions. Oriented forms instead keep a sign.

## 12.6 Convergence and Multiple Integration

The same principle from Chapter 9 survives unchanged on a rectangle: a uniform error in the integrand is at most that error times the volume of the domain after integration. Pointwise convergence alone is not the Riemann theory's general interchange principle.

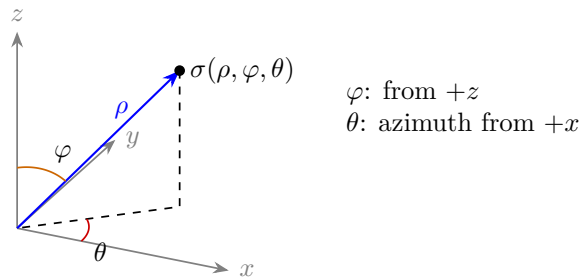


Figure 12.2: Our spherical-coordinate convention. The dashed vertical segment projects the point to the  $xy$ -plane. The polar angle is measured from the positive  $z$ -axis.

**Theorem 12.6.1** (Uniform interchange for multiple Riemann integrals). Let  $Q \subseteq \mathbb{R}^n$  be a rectangle. Suppose that every function

$$f_k : Q \longrightarrow \mathbb{R}$$

is Riemann integrable and that  $f_k$  converges uniformly to  $f$  on  $Q$ . Then  $f$  is Riemann integrable and

$$\lim_{k \rightarrow \infty} \int_Q f_k(\mathbf{x}) \, d\mathbf{x} = \int_Q f(\mathbf{x}) \, d\mathbf{x}.$$

Moreover,

$$\lim_{k \rightarrow \infty} \int_Q |f_k(\mathbf{x}) - f(\mathbf{x})| \, d\mathbf{x} = 0.$$

*Proof.* The oscillation argument in the proof of [Theorem 9.3.1](#) applies to grid subrectangles, using [Proposition 12.1.3](#). Explicitly the bound is  $U(f, \mathcal{P}) - L(f, \mathcal{P}) \leq U(f_k, \mathcal{P}) - L(f_k, \mathcal{P}) + 2\|f - f_k\|_\infty \text{vol}(Q)$ : the interval length in Chapter 9 has become volume. Once integrability of  $f$  is known, [Theorem 12.1.8](#) gives

$$\left| \int_Q f_k \, d\mathbf{x} - \int_Q f \, d\mathbf{x} \right| \leq \int_Q |f_k - f| \, d\mathbf{x} \leq \text{vol}(Q) \sup_{\mathbf{x} \in Q} |f_k(\mathbf{x}) - f(\mathbf{x})|.$$

The right-hand side tends to zero, and the same estimate gives the final claim.  $\square$

The uniform theorem gives

$$\int_Q |f_k - f| \leq \text{vol}(Q) \|f_k - f\|_\infty \longrightarrow 0.$$

Thus uniform convergence gives  $L^1$  convergence (convergence of the integral of the absolute error), then convergence of integrals. The same result holds on a Jordan region for integrable functions: extend all maps by zero to one containing rectangle. Their uniform errors remain bounded by the original error, so the rectangular theorem applies.

### Arzelà bounded convergence on rectangles

The proof of [Theorem 9.3.4](#) used open interval lengths and compact inner approximations. Its replacement dictionary is

**Chapter 9: intervals**

 Open-set length  $\ell(G)$ 

Finite unions of closed intervals

Nested open sets of uniformly positive length

**Here: rectangles**

 Open-box content  $c(G)$ 

Finite unions of closed boxes

Nested open sets of uniformly positive content

The following lemma proves the new box geometry. Once it is available, the bounded-height, large-error-region contradiction is the same one. For open  $G \subset Q^\circ$ , let  $c(G)$  be the supremum of the total volumes of finite families of closed boxes contained in  $G$  with pairwise disjoint interiors; include the empty family, of volume zero. Every such total is at most  $\text{vol}(Q)$ , by a common grid.

**Lemma 12.6.2** (Open-box content tools). For open subsets of  $Q^\circ$ , the following hold:

(1) If

$$G \subseteq \bigcup_j G_j,$$

then

$$c(G) \leq \sum_j c(G_j).$$

(2) Given  $\eta > 0$ , there is a finite union  $K$  of closed boxes in  $G$  with  $c(G \setminus K) < \eta$ .

(3) If open  $G_N$  decrease and  $c(G_N) \geq c_0 > 0$  for every  $N$ , then

$$\bigcap_N G_N$$

is nonempty.

*Proof.* Compare only finite box families, and use compactness to turn open covers into finite subdivisions. For (1), take any admissible compact box union  $K \subset G$ . A finite subcover from the  $G_j$  covers  $K$ . A Lebesgue number for this compact cover (Lemma 10.6.5) permits subdivision of its boxes into closed cells, each contained in one chosen  $G_j$ . The common-scale argument is proved in Lemma 10.6.2. Assign each cell to one such  $j$ . The cells assigned to a given  $j$  have disjoint interiors and lie in  $G_j$ , so their total is at most  $c(G_j)$ . Sum and take the supremum over  $K$ . For (2), choose an admissible  $K$  of volume greater than  $c(G) - \eta$ . Any admissible box family in  $G \setminus K$  can be joined to the boxes of  $K$  and still lies in  $G$  with disjoint interiors. Its total is at most  $c(G) - \text{vol}(K) < \eta$ ; the same bound holds for its supremum. Here volume of a finite box union means its finite grid volume.

For (3), choose  $K_i \subset G_i$  as in (2) with error

$$c_0/2^{i+1}.$$

The compact sets

$$H_N = K_1 \cap \cdots \cap K_N$$

decrease. If  $H_N$  were empty, nesting would imply

$$G_N \subseteq \bigcup_{i=1}^N (G_i \setminus K_i);$$

(1) would give



$$c(G_N) < c_0/2,$$

a contradiction. Thus all  $H_N$  are nonempty and compact; their intersection is nonempty by compactness, and lies in every  $G_N$ .  $\square$

For a finite collection of boxes with disjoint interiors, the content of the union of their interiors equals the sum of their volumes: shrink each box slightly for the lower bound, and subdivide a finite test family with the box faces for the upper bound. This identifies the numerical content needed in the next proof; it does not assert Jordan measurability of arbitrary open sets.

**Theorem 12.6.3** (Multivariable Arzelà bounded convergence). Let  $Q$  be a compact rectangle, let  $f_k : Q \rightarrow \mathbb{R}$  be Riemann integrable, and suppose  $f_k \rightarrow f$  pointwise and  $|f_k(\mathbf{x})| \leq M$  for all  $k, \mathbf{x}$ . Assume separately that  $f$  is Riemann integrable. Then

$$\int_Q |f_k - f| \rightarrow 0, \quad \int_Q f_k \rightarrow \int_Q f.$$

*Proof.* A large integral with bounded heights forces large open error regions; compact inner approximations force one point into arbitrarily late regions, contradicting pointwise convergence. Set  $g_k = |f_k - f|$ . The integral algebra makes  $g_k$  integrable and  $0 \leq g_k \leq 2M$ , since the pointwise limit also satisfies  $|f| \leq M$ . The case  $M = 0$  is immediate. Otherwise replace  $g_k$  by

$$g_k/(2M),$$

so  $0 \leq g_k \leq 1$ . Write  $V = \text{vol}(Q) > 0$ . If its integrals do not tend to zero, select a subsequence with

$$\int_Q g_k > 2\varepsilon V$$

for one fixed  $\varepsilon > 0$ . Choose a lower sum exceeding  $2\varepsilon V$ . Cells of lower height at most  $\varepsilon$  contribute at most  $\varepsilon V$ ; the remaining cells therefore have combined volume greater than  $\varepsilon V$ , since their heights are at most one. Let  $U_k$  be the union of their interiors. Then  $c(U_k) > \varepsilon V$  and  $g_k(\mathbf{x}) > \varepsilon$  throughout  $U_k$ . The open sets

$$G_N = \bigcup_{k \geq N} U_k$$

decrease and each has content at least  $\varepsilon V$ . By [Lemma 12.6.2](#), there is a fixed

$$\mathbf{x} \in \bigcap_N G_N.$$

For every  $N$  some  $k \geq N$  then satisfies  $g_k(\mathbf{x}) > \varepsilon$ , contrary to  $g_k(\mathbf{x}) \rightarrow 0$ . Undoing the height normalization proves the first limit. Finally

$$\left| \int_Q f_k - \int_Q f \right| \leq \int_Q |f_k - f|$$

proves the second.  $\square$

Uniform convergence controls the whole domain by one error bound. Arzelà bounded convergence uses compactness to prevent regions of large error and substantial content from moving forever without revisiting a fixed point. The integrability of the limit is still a separate assumption. For example  $f_k(x, y) = x^k$  on  $[0, 1]^2$  has a limit supported on the face  $x = 1$ , hence integrable with integral zero by the face-strip lemma. The convergence is not uniform, but this theorem gives integral convergence; direct calculation yields

$$\int f_k = 1/(k+1).$$

As in one variable, pointwise convergence alone does not control a Riemann integral. On a compact rectangle, continuity and monotonicity provide the extra structure that can force uniform convergence. The proof of [Theorem 6.2.8](#) applies: continuity makes the good sets relatively open, monotonicity makes them nested, and compactness reduces their cover to one uniform index. No ordering of the domain is used.

**Theorem 12.6.4** (Dini's theorem). Let  $Q \subseteq \mathbb{R}^n$  be a rectangle, and let

$$f_k : Q \longrightarrow \mathbb{R}$$

be continuous functions satisfying

$$f_k(\mathbf{x}) \leq f_{k+1}(\mathbf{x}) \quad (\mathbf{x} \in Q, k \in \mathbb{N}).$$

If  $f_k$  converges pointwise on  $Q$  to a continuous function  $f$ , then the convergence is uniform and

$$\lim_{k \rightarrow \infty} \int_Q f_k(\mathbf{x}) \, d\mathbf{x} = \int_Q f(\mathbf{x}) \, d\mathbf{x}.$$

The same conclusion holds for a pointwise decreasing sequence.

*Proof. Strategy.* Nested open good sets turn a finite subcover into one uniform index. In the increasing case set  $h_n = f - f_n \geq 0$  on  $K = Q$ . For  $\varepsilon > 0$ , the relatively open sets

$$U_n = \{\mathbf{x} \in K : h_n(\mathbf{x}) < \varepsilon/2\}$$

are increasing, and pointwise convergence gives

$$K = \bigcup_n U_n.$$

Compactness of  $K$  supplies a finite subcover. Its largest-index member  $U_N$  already covers  $K$ . For  $n \geq N$  and every  $\mathbf{x} \in K$ ,

$$0 \leq h_n(\mathbf{x}) < \varepsilon/2,$$

hence

$$\sup_K |f - f_n| \leq \varepsilon/2 < \varepsilon.$$

This proves uniform convergence, and [Theorem 12.6.1](#) gives the integral formula. Apply the increasing case to  $-f_n$  for a decreasing sequence.  $\square$

**Remark 12.6.5** (Which convergence theorem applies?). Uniform interchange assumes integrable approximants and uniform convergence; it proves integrability of the limit and bounds the absolute-error integral by volume times the uniform error. Arzelà assumes pointwise convergence, a common bound, and separately an integrable limit; compact inner box approximations give convergence of absolute-error integrals. Dini assumes continuous approximants, a continuous limit, monotonicity, and compactness; these first give uniform convergence, then integral convergence. The later Lebesgue dominated convergence theorem instead uses measurable functions, almost-everywhere convergence, and an integrable dominating function; in that different theory it also supplies integrability of the limit. It is not the uniform theorem under another name and is not proved here.

## Exercises

**Exercise 1.** For  $P(x, t) = (x, e^x t)$  and  $f \in C_c(\mathbb{R}^2)$ , prove the substitution formula directly by integrating in  $t$  first. Explain the support of the pulled-back integrand and the factor  $e^x$ .

**Exercise 2.** Prove that replacing a grid partition by a refinement decreases its upper sum and increases its lower sum.

**Exercise 3.** Compute

$$\int_{\{0 \leq y \leq 1, y \leq x \leq y+2\}} (x + 2y) \, d(x, y)$$

directly from iterated one-variable integrals.

**Exercise 4.** Use [Theorem 12.2.10](#) with  $j = 1$  to evaluate

$$\int_{[0,1]^2} x \frac{\partial}{\partial x} (x^2 y) \, d(x, y).$$

**Exercise 5.** Prove that a constant function  $c$  on  $Q$  has integral

$$c \operatorname{vol}(Q).$$

**Exercise 6.** Show that a finite set in  $\mathbb{R}^n$  has Jordan volume zero.

**Exercise 7.** Prove that the open unit ball in  $\mathbb{R}^n$  is Jordan measurable, assuming that its boundary can be covered by finitely many boxes of arbitrarily small total volume.

**Exercise 8.** Give a bounded subset of  $[0, 1]$  whose indicator is not Riemann integrable.

**Exercise 9.** Use [Theorem 12.4.3](#) to compute the area of the parallelogram spanned by two vectors in  $\mathbb{R}^2$ .

**Exercise 10.** Verify the determinant computation in the polar-coordinate example.

**Exercise 11.** For  $a, b > 0$ , use a linear map and polar coordinates to compute

$$\int_{\{x^2/a^2 + y^2/b^2 \leq 1\}} x^2 \, d(x, y).$$

Transform the domain and both scaling factors. Justify the exceptional origin and angular seam by truncation and bounds, as in the disk example.

## 12.7 Strategy for Multiple Integrals: Geometry, Coordinates, and Computation

Existence and evaluation are different tasks. The chapter's continuity, Darboux, and Jordan results establish existence; the following procedure helps choose an efficient evaluation using those results.

1. **Identify dimension and meaning.** Is this a double integral over  $E \subset \mathbb{R}^2$  or a triple integral over  $\Omega \subset \mathbb{R}^3$ ? Integrand 1 computes area or volume; a density computes total mass.
2. **Draw and label the set.** Mark boundaries, intersection points, axes, symmetries, and a representative slice. For a solid, include the projection region that will carry the outer integral.
3. **Try slicing.** Compare vertical and horizontal slices in the plane. In space compare horizontal cross-sections with projections onto the  $xy$ ,  $xz$ , and  $yz$  planes. Solve the inequalities for the inner variable; the picture helps detect missing bounds.
4. **Split where a boundary changes.** The roof-shaped triangle requires two vertical descriptions but only one horizontal description. Two simple integrals can be clearer than one complicated bound.
5. **Use symmetry precisely.** Name a transformation preserving the domain and check how the integrand changes. For example if  $(x, y) \mapsto (-x, y)$  preserves  $E$  and  $f(-x, y) = -f(x, y)$ , its absolute Jacobian determinant is 1 and substitution gives

$$\int_E f = - \int_E f = 0.$$

6. **Look for adapted coordinates.** Expressions  $x^2 + y^2$  suggest polar or cylindrical coordinates; ellipses suggest scaling;  $x + y$  and  $x - y$  boundaries suggest using those expressions as coordinates. The best substitution simplifies the domain, the integrand, or ideally both.
7. **Check the substitution.** Write  $\sigma$ , the parameter domain,  $D\sigma$ ,  $|\det D\sigma|$ , the transformed integrand, and the hypotheses or controlled exceptions before integrating.
8. **Check the answer.** Area and volume are nonnegative and no larger than those of containing boxes. If  $0 \leq f \leq M$ , then

$$0 \leq \int_E f \leq M \operatorname{vol}(E).$$

Scaling every length by  $c > 0$  scales  $n$ -dimensional volume by  $c^n$ .

For the cone, cross-sections immediately give

$$A(z) = \pi R^2(1 - z/h)^2.$$

Cylindrical coordinates instead compute that area as the inner two integrals and accommodate densities depending on  $r, \theta, z$  naturally. Neither method is universally superior. Choose the description in which the geometry and the resulting integral are easiest to handle.

**Exercise 12** (Choose and justify a method). (a) Reverse the order in the picture of [Example 12.3.14](#) and compute

$$\int_E x^2.$$

Label a representative slice of each kind.

(b) Compute the volume of  $0 \leq z \leq h$ ,

$$x^2 + y^2 \leq R^2 z/h$$

by cross-sections, and independently by cylindrical coordinates.

(c) Find the volume of

$$x^2/4 + y^2/9 + z^2/16 \leq 1$$

by scaling.

(d) On  $|x+y| \leq 2$ ,  $|x-y| \leq 1$ , integrate  $(x+y)^2$  using these two linear expressions as coordinates.

(e) Compare Cartesian, polar, and symmetry methods for

$$\int_{x^2+y^2 \leq 1} x$$

and for

$$\int_{x^2+y^2 \leq 1} (x^2 + y^2).$$

Explain which steps each choice saves.

## 12.8 Limits of the Riemann Theory

The Riemann integral handles continuous functions and geometrically regular domains very effectively. Its limitations become sharper in several variables: indicators of sets with complicated boundaries may fail to be integrable, iterated integrals need not be available for arbitrary bounded functions, and the present change-of-variables theorem requires continuity and a globally injective parametrization. Lebesgue measure and integration replace Jordan volume by a much more flexible notion of size and later permit theorems under substantially weaker hypotheses. Those extensions are not developed in these notes.

## Chapter 13

# Differential Forms, Stokes' Theorem, and Calculus on Manifolds

### 13.1 Parametrized Curves and Line Integrals

Integration can accumulate mass along a wire or work along a path. The same geometric curve supports both questions, but they require different integrands: mass uses length, while work records direction. We start with these familiar quantities before introducing the objects that unify their boundary formulas.

**Recall and choose a parametrization.** Chapter 8 introduced plane curves in “[Parametrized Curves and Polar Coordinates](#)”; [Definition 11.2.14](#) identified coordinatewise velocity with the Fréchet derivative. Recalling those choices, now also using vector notation,

|                                |                                                                          |
|--------------------------------|--------------------------------------------------------------------------|
| segment from $p$ to $q$        | $\gamma(t) = (1-t)p + tq, \quad 0 \leq t \leq 1,$                        |
| graph $y = f(x)$               | $\gamma(t) = (t, f(t)), \quad t \text{ in the graph's interval,}$        |
| circle of radius $R > 0$       | $\gamma(t) = (R \cos t, R \sin t), \quad 0 \leq t \leq 2\pi,$            |
| ellipse, $a, b > 0$            | $\gamma(t) = (a \cos t, b \sin t), \quad 0 \leq t \leq 2\pi,$            |
| polar curve $r = \rho(\theta)$ | $\gamma(\theta) = (\rho(\theta) \cos \theta, \rho(\theta) \sin \theta).$ |

For the polar curve choose the desired angular interval and  $\rho \geq 0$ . Start by identifying the set and the intended traversal. Choose one scalar parameter, express every coordinate in it, and select an interval covering exactly the desired portion. Check endpoints, orientation, and multiplicity; then differentiate to obtain velocity and speed. The geometric image does not determine the traversal: speed, direction, and the number of passes can all change while the image stays fixed.

By [Proposition 11.2.15](#),

$$(f \circ \gamma)'(t) = Df(\gamma(t))\gamma'(t).$$

It transports tangent velocities through maps. For scalar  $f$  it pairs the derivative covector with velocity; this will explain work for gradient fields, the fundamental theorem for line integrals, and curve pullback.

#### A curve is a traversal, not just a set

The maps  $t \mapsto (\cos t, \sin t)$  on  $[0, 2\pi]$  and  $t \mapsto (\cos 2t, \sin 2t)$  on the same interval have the same image, but the second traverses it twice. Integrals can distinguish these maps. The velocity  $\gamma'(t)$  is

a tangent vector; its norm is speed. For a short time  $\Delta t$ , the displacement is approximately  $\gamma'(t)\Delta t$  and the distance traveled is approximately  $\|\gamma'(t)\|\Delta t$ . This suggests length as accumulated speed.

For a scalar density  $f$ , multiply each small length by  $f$  to obtain a scalar line integral. For a force field  $F$ , multiply displacement by the component of force in its direction:  $F \cdot \gamma'\Delta t$ . This gives work. These two constructions use different information, although both are computed by a one-variable integral.

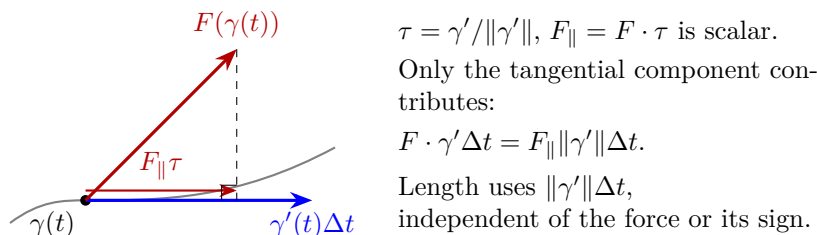


Figure 13.1: The dot product extracts the force component along displacement. The blue vector follows the velocity  $\gamma'(t)$  and approximates the actual short displacement along the curve.

**Definition 13.1.1** (Parametrized curve and velocity). For the recalled parametrized curves, our integration convention is a map  $\gamma : [a, b] \rightarrow \mathbb{R}^n$  of class  $C^1$ , with continuous derivative, using one-sided derivatives at endpoints. Its velocity at time  $t$  is  $\gamma'(t)$ , and it is **regular** if this vector is never zero. Its orientation is the direction of increasing parameter. Its image is the set  $\gamma([a, b])$ , which does not record the speed or multiplicity of traversal.

For  $\gamma(t) = (t, t^2)$  the velocity is  $(1, 2t)$  and the speed  $\sqrt{1 + 4t^2}$  varies with time. A tangent vector is a displacement direction; speed is a nonnegative scalar.

**Definition 13.1.2** (Smooth-curve length). We define the length of a  $C^1$  curve by

$$\ell(\gamma) = \int_a^b \|\gamma'(t)\| \, dt.$$

For continuous piecewise  $C^1$  curves, add the lengths of the finitely many smooth pieces. Refining that subdivision does not change the sum, by additivity of the one-variable integral. No equivalence with a polygonal-supremum definition is assumed here.

To integrate a scalar function along the curve, the small length is weighted by the scalar value at the visited point. For the fixed curve  $\gamma : [a, b] \rightarrow \mathbb{R}^n$ , the field types are  $f : \gamma([a, b]) \rightarrow \mathbb{R}$  and  $F : \gamma([a, b]) \rightarrow \mathbb{R}^n$ . Assume these fields are continuous. The actual integrands on the parameter interval are the scalar functions

$$g(t) = f(\gamma(t))\|\gamma'(t)\|, \quad h(t) = F(\gamma(t)) \cdot \gamma'(t).$$

Both are continuous on each smooth piece, so Chapter 9 supplies their integrals. Thus the following definitions are the ordinary integrals

$$\int_a^b g(t) \, dt, \quad \int_a^b h(t) \, dt.$$

The curve lies in  $\mathbb{R}^n$  but the integration domain is one-dimensional. Work is scalar because the dot product is taken *before* integration; it differs from the componentwise vector integral

$$\int_a^b F(\gamma(t)) \, dt.$$

**Definition 13.1.3** (Scalar line integral). For continuous  $f$  along the image of  $\gamma$ , set

$$\int_{\gamma} f \, ds = \int_a^b f(\gamma(t)) \|\gamma'(t)\| \, dt.$$

Taking  $f = 1$  recovers length. Negative values of  $f$  contribute negatively.

**Arc length is not a coordinate differential.** Coordinate covectors are linear:  $dx(-v) = -dx(v)$ . Length instead satisfies  $\| -v \| = \| v \|$ . Thus the parametrization shorthand  $ds = \|\gamma'(t)\| \, dt$  records a speed factor in an ordinary integral, not evaluation of a linear 1-form analogous to  $dx$ . This explains why scalar arc-length integrals are independent of orientation, whereas work and 1-form integrals change sign under reversal. For the counterclockwise unit circle on  $[0, 2\pi]$ ,

$$\int_{\gamma} dx = \int_0^{2\pi} (-\sin t) \, dt = 0, \quad \int_{\gamma} ds = \int_0^{2\pi} 1 \, dt = 2\pi.$$

Here the first integral means accumulating the coordinate covector on velocity; the work-integral definition below includes this case.

Work instead pairs a vector field with the directed displacement.

**Definition 13.1.4** (Work integral). For a continuous vector field  $F$  along the image, set

$$\int_{\gamma} F \cdot dr = \int_a^b F(\gamma(t)) \cdot \gamma'(t) \, dt.$$

Both kinds of line integral extend to piecewise  $C^1$  curves by finite addition over their smooth pieces.

**Proposition 13.1.5** (Reparametrization and reversal). Let  $h : [c, d] \rightarrow [a, b]$  be a  $C^1$  diffeomorphism and set  $\tilde{\gamma} = \gamma \circ h$ . The scalar line integral and length are unchanged. The work integral is unchanged if  $h$  is increasing and changes sign if  $h$  is decreasing.

*Proof.* The chain rule gives  $\tilde{\gamma}' = \gamma'(h)h'$ , and hence  $\|\tilde{\gamma}'\| = \|\gamma'(h)\||h'|$ . If  $h$  is increasing, substitution in Chapter 9 gives both assertions. If it is decreasing, substitution with  $h'$  changes the limits from  $b$  to  $a$ . For work this contributes a minus sign. For the scalar integral,  $|h'| = -h'$  contributes a second minus sign, cancelling the first.  $\square$

Thus

$$\int_C f \, ds$$

is independent of orientation, but it need not be positive: if  $f = -1$  on a nonconstant curve, it equals minus the length. In contrast,

$$\int_C F \cdot dr$$

is signed. The notation  $C$  presupposes a specified traversal up to these reparametrizations: tracing a curve twice generally doubles its integral and is not a diffeomorphic reparametrization.



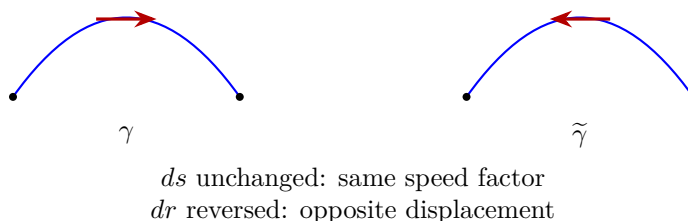


Figure 13.2: The same arc with opposite traversals. Scalar arc-length integrals are unchanged; work and 1-form integrals change sign.

**Example 13.1.6** (Changing speed and reversing direction). Let  $\gamma(t) = (t, t^2)$  for  $0 \leq t \leq 1$  and

$$h(s) = (s + s^2)/2$$

on  $[0, 1]$ . Then

$$h'(s) = (1 + 2s)/2 > 0,$$

so  $h$  is an increasing diffeomorphism with the same endpoint values. The new velocity is  $(h'(s), 2h(s)h'(s))$ ; the traversal speeds differ, but substitution shows both lengths equal

$$\int_0^1 \sqrt{1 + 4t^2} \, dt.$$

For  $F = (1, 0)$ , both work integrals are 1: in the new parameter the integral is

$$\int_0^1 h'(s) \, ds = h(1) - h(0).$$

Using  $h(s) = 1 - s$  instead reverses direction and gives work  $-1$ . The scalar integral of  $f = -1$  stays the same negative length.

For a planar curve  $\gamma(t) = (x(t), y(t))$ , the notation means

$$\int_{\gamma} P \, dx + Q \, dy = \int_a^b [P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)] \, dt.$$

As previewed after [Proposition 11.2.15](#), along the parametrization  $dx$  becomes  $x'(t) \, dt$  and  $dy$  becomes  $y'(t) \, dt$ . These are evaluations of the coordinate covectors on velocity; later we formalize the resulting parameter-line covectors as pullbacks. These differential factors are not the Chapter 12 area notation  $d(x, y)$ .

**Example 13.1.7** (One running circulation). For  $F(x, y) = (-y, x)$  and the counterclockwise unit circle  $\gamma(t) = (\cos t, \sin t)$ ,  $0 \leq t \leq 2\pi$ , the velocity is  $F(\gamma(t))$ . Thus its length and circulation are both  $2\pi$ . Reversal keeps the length and negates the circulation. The expression  $x \, dy - y \, dx$  will reappear as a differential form later in this chapter.

**Theorem 13.1.8** (Fundamental theorem for line integrals). If  $f$  is  $C^1$  near a piecewise  $C^1$  curve  $\gamma$ , then

$$\int_{\gamma} \nabla f \cdot dr = f(\gamma(b)) - f(\gamma(a)).$$

*Proof.* On each smooth piece the chain rule gives  $(f \circ \gamma)' = Df(\gamma)\gamma' = \nabla f(\gamma) \cdot \gamma'$ . Integrate by the one-variable FTC. Intermediate endpoint values cancel when the pieces are added.  $\square$

Reparametrization invariance compares the same traversal at different speeds or coordinates. Path independence compares genuinely different curves with the same endpoints. The latter requires a special hypothesis: the theorem shows that work for  $F = \nabla f$  depends only on endpoints, so any two such curves have the same work.

This is our first boundary/interior formula: integrating a derivative along the interior leaves only the oriented endpoint values.

**Exercise 1.** Integrate  $F = (2x, y)$  along the segment from  $(0, 0)$  to  $(1, 2)$ , directly and using a potential. Compute

$$\int_C y \, ds$$

as well and explain how the two answers change under reversal.

## Computing line integrals

**Working rule.** Write  $\gamma(t)$ , differentiate to obtain  $\gamma'(t)$ , and substitute along the curve. For scalar density integrate  $f(\gamma(t))\|\gamma'(t)\|$ ; for work integrate  $F(\gamma(t)) \cdot \gamma'(t)$ . The curve can live in  $\mathbb{R}^n$ , but the parameter is one-dimensional. For work the dot product is taken *before* the ordinary scalar integral.

**Example 13.1.9** (Mass of a quarter-circle wire). Let  $R > 0$  and  $\gamma(t) = (R \cos t, R \sin t)$  for

$$0 \leq t \leq \pi/2.$$

For density  $\rho(x, y) = x + y$ , we have  $\gamma' = (-R \sin t, R \cos t)$  and  $\|\gamma'\| = R$ . Thus the mass is

$$\int_{\gamma} \rho \, ds = R^2 \int_0^{\pi/2} (\cos t + \sin t) \, dt = 2R^2.$$

The density is nonnegative on this wire; reversing traversal leaves its mass unchanged.

**Example 13.1.10** (Work along a segment). For  $\gamma(t) = (t, 2t)$ ,  $0 \leq t \leq 1$ , and  $F(x, y) = (y, x)$ ,

$$F(\gamma(t)) = (2t, t), \quad \gamma'(t) = (1, 2), \quad F(\gamma(t)) \cdot \gamma'(t) = 4t.$$

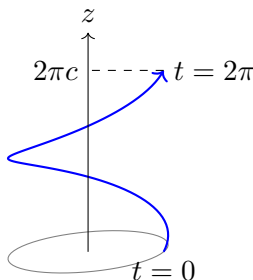
Consequently

$$\int_{\gamma} F \cdot dr = \int_0^1 4t \, dt = 2.$$

The reverse traversal has work  $-2$ .

**Example 13.1.11** (A helix). For  $R > 0$ ,  $c > 0$ , let  $\gamma(t) = (R \cos t, R \sin t, ct)$ ,  $0 \leq t \leq 2\pi$ . Its derivative is  $(-R \sin t, R \cos t, c)$ , so

$$\text{length}(\gamma) = 2\pi\sqrt{R^2 + c^2}, \quad \int_{\gamma} z \, ds = \int_0^{2\pi} ct\sqrt{R^2 + c^2} \, dt = 2\pi^2 c\sqrt{R^2 + c^2}.$$



The arrow shows one traversal: one turn around the axis and rise  $2\pi c$ .

## 13.2 Green's Theorem in the Plane

The next formula turns circulation around a planar boundary into accumulated local rotation. We first prove it for a precise elementary class of regions.

Integrating a vertical derivative first subtracts its lower boundary value from its upper value. Positive boundary traversal runs rightward on the lower edge and leftward on the upper edge, explaining the minus sign in the  $P$  term. The horizontal calculation explains the  $Q$  term. After subdivision, a shared edge is traversed in opposite directions by its two adjacent pieces, leaving only the outside boundary.

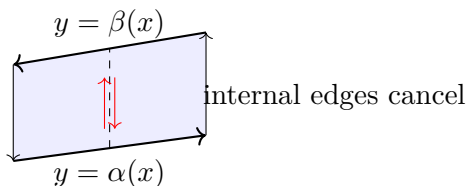
**Theorem 13.2.1** (Green for elementary planar regions). Suppose  $D$  has both a vertical and a horizontal description

$$\begin{aligned} D &= \{(x, y) : a \leq x \leq b, \alpha(x) \leq y \leq \beta(x)\}, \\ D &= \{(x, y) : c \leq y \leq d, \lambda(y) \leq x \leq \mu(y)\}, \end{aligned}$$

where the four bounding functions are continuous and piecewise  $C^1$ , with strict lower/upper inequalities in the interiors of their intervals. Give its piecewise  $C^1$  boundary the positive direction, with the region on the left. If  $P, Q$  are  $C^1$  on a neighborhood of  $D$ , then

$$\int_{\partial D} P \, dx + Q \, dy = \int_D (Q_x - P_y) \, d(x, y).$$

The formula also holds for a finite union of such regions with disjoint interiors and shared boundary arcs that cancel in opposite directions.



*Proof.* Integrate  $P_y$  vertically and  $Q_x$  horizontally; the FTC produces exactly the upper/lower and right/left boundary contributions. By [Theorem 12.3.13](#) and the FTC,

$$-\int_D P_y = \int_a^b (P(x, \alpha(x)) - P(x, \beta(x))) \, dx.$$

The lower boundary goes left to right and the upper one right to left; vertical boundary segments contribute nothing to  $P \, dx$ . This is therefore

$$\int_{\partial D} P \, dx.$$

Similarly,

$$\int_D Q_x = \int_c^d (Q(\mu(y), y) - Q(\lambda(y), y)) \, dy = \int_{\partial D} Q \, dy.$$

The right boundary goes upward and the left downward. Add the identities. For a finite decomposition, each  $C^1$  arc on a compact interval has bounded derivative, hence is Lipschitz by the mean-value estimate. Its image is Jordan-null by [Lemma 12.3.17](#) (a one-dimensional face mapped into the plane). A piecewise  $C^1$  arc is a finite union of these null images. Overlaps of the regions lie in these boundary arcs, so area integrals add by [Proposition 12.3.8](#). Each shared arc is traversed twice in opposite directions, so its line integrals cancel.  $\square$

Rectangles and triangles belong to this class. The theorem does not assert a decomposition for every conceivable piecewise smooth boundary. Later manifold Stokes will supply the global smooth-boundary result.

**Example 13.2.2** (Area from circulation). For

$$P = -y/2,$$

$$Q = x/2,$$

one has  $Q_x - P_y = 1$ . Thus for these regions

$$\text{area}(D) = \frac{1}{2} \int_{\partial D} (x \, dy - y \, dx).$$

On the triangle with vertices  $(0, 0), (1, 0), (0, 1)$ , the two coordinate edges contribute zero. On  $(x, y) = (1 - t, t)$ ,  $0 \leq t \leq 1$ , the integrand is  $dt$ , yielding area

$$1/2.$$

Internal-edge cancellation is already a concrete model of the later passage from local formulas to a global one.

**Exercise 2.** Verify Green for  $P = xy$ ,  $Q = x^2$  on the unit square by computing all four edge integrals. Then split the square along its diagonal and identify the two cancelling contributions.

### 13.3 Parametrized Surfaces and Surface Integrals

A surface needs two independent tangent directions. A parametrization turns a small parameter rectangle into an approximately planar parallelogram; its area factor is the two-dimensional counterpart of  $\|\gamma'\|$ .

**Definition 13.3.1** (Cross product). For  $a, b \in \mathbb{R}^3$ , define

$$a \times b = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1).$$

**Proposition 13.3.2** (Cross-product identities). The cross product is linear in each slot, and

$$a \times b = -b \times a, \quad a \times a = 0.$$

For  $a, b, c \in \mathbb{R}^3$ ,

$$(a \times b) \cdot c = \det(a, b, c), \quad \|a \times b\|^2 = \|a\|^2\|b\|^2 - (a \cdot b)^2.$$

The cross product is orthogonal to  $a, b$  (meaning that its Euclidean dot product with each is zero). It is nonzero exactly when  $a, b$  are independent, and its norm is their parallelogram area.

*Proof.* Each coordinate  $a_i b_j - a_j b_i$  is linear separately in  $a$  and  $b$ . For example, the first coordinate of  $(a + a') \times b$  is

$$(a_2 + a'_2)b_3 - (a_3 + a'_3)b_2 = (a_2b_3 - a_3b_2) + (a'_2b_3 - a'_3b_2).$$

The other two coordinates and scalar multiplication give  $(a + a') \times b = a \times b + a' \times b$  and  $(ta) \times b = t(a \times b)$ ; the same calculation applies to the second slot. Interchanging  $a, b$  negates every coordinate; equal inputs give zero. The dot product expands as  $(a_2b_3 - a_3b_2)c_1 + (a_3b_1 - a_1b_3)c_2 + (a_1b_2 - a_2b_1)c_3$ , the determinant expansion in the third column. Taking  $c = a$  or  $c = b$  gives zero by alternation. For the squared norm, expansion gives

$$\sum_{i < j} (a_i b_j - a_j b_i)^2 = \sum_{i \neq j} a_i^2 b_j^2 - 2 \sum_{i < j} a_i a_j b_i b_j = \|a\|^2 \|b\|^2 - (a \cdot b)^2.$$

If  $a \neq 0$ , write  $b = ta + h$  with

$$t = (a \cdot b) / \|a\|^2$$

and  $a \cdot h = 0$ . Substitution in the last identity gives  $\|a \times b\|^2 = \|a\|^2 \|h\|^2$ . It vanishes exactly when  $b$  is a multiple of  $a$ ; otherwise  $\|a\|$  and  $\|h\|$  are the base and perpendicular height. If  $a = 0$ , both the cross product and the area are zero, and the pair is dependent.  $\square$

**Definition 13.3.3** (Regular surface patch and its orientation). Let  $D \subset \mathbb{R}^2$  be a compact Jordan region equal to the closure of its interior. A regular  $C^1$  surface patch is a  $C^1$  map  $\sigma$  on an open neighborhood of  $D$ , injective on  $D$ , with independent derivatives  $\sigma_u, \sigma_v$  throughout  $D$ . Its tangent plane at  $\sigma(u, v)$  is the span of these vectors, translated to that point. The parameter order  $(u, v)$  chooses the unit normal

$$n = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|}.$$

The opposite normal gives the reverse orientation.

### A small parameter rectangle becomes a tangent parallelogram

At  $(u, v)$  the two edge increments are approximately  $\sigma_u \Delta u$  and  $\sigma_v \Delta v$ . Their parallelogram has area  $\|\sigma_u \times \sigma_v\| \Delta u \Delta v$ . The scalar  $\|\sigma_u \times \sigma_v\|$  measures area per unit parameter area. The vector  $\sigma_u \times \sigma_v$  contains the same magnitude together with an oriented normal direction. Taking its norm loses precisely the information that flux needs.

For a regular patch  $\sigma : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ ,  $D\sigma(u, v) : \mathbb{R}^2 \rightarrow \mathbb{R}^3$  has a  $3 \times 2$  matrix with columns  $\sigma_u, \sigma_v \in \mathbb{R}^3$ ;  $\det D\sigma$  is not defined. Put  $a_\sigma = \|\sigma_u \times \sigma_v\| > 0$  and

$$n = (\sigma_u \times \sigma_v) / a_\sigma.$$

The area factor  $a_\sigma$  is a scalar,  $n$  a unit vector, and for  $f : S \rightarrow \mathbb{R}$ ,  $F : S \rightarrow \mathbb{R}^3$ , both  $f(\sigma)a_\sigma$  and  $(F(\sigma) \cdot n)a_\sigma$  are scalars on  $D$ . In particular,

$$(F(\sigma) \cdot n)a_\sigma = F(\sigma) \cdot \left( \frac{\sigma_u \times \sigma_v}{a_\sigma} a_\sigma \right) = F(\sigma) \cdot (\sigma_u \times \sigma_v).$$

Here  $dS$  denotes surface area and  $d(u, v)$  parameter-plane area. All these are double integrals: the ambient space  $\mathbb{R}^3$  does not make the parameter domain three-dimensional. Assume the fields  $f, F$  are continuous along the patch. Under the patch hypotheses, the scalar coefficients are continuous on compact Jordan  $D$ , so their integrals exist by Chapter 12.

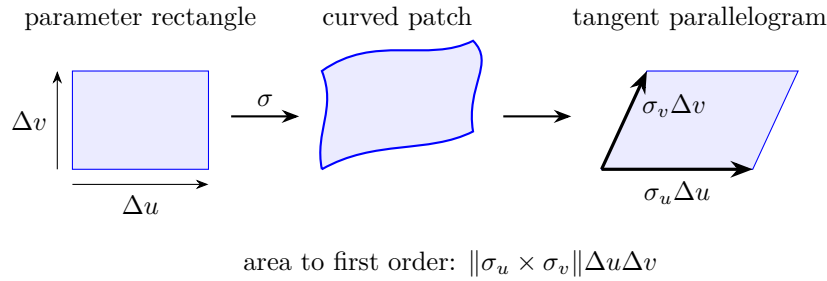


Figure 13.3: The derivative replaces a small curved patch by its tangent parallelogram. The cross-product norm gives the local scalar area factor.

**Definition 13.3.4** (Area, scalar surface integral, and flux). For  $S = \sigma(D)$  and continuous  $f, F$  along  $S$ , define

$$\begin{aligned} \text{area}(S) &= \int_D \|\sigma_u \times \sigma_v\| \, d(u, v), & \int_S f \, dS &= \int_D f(\sigma) \|\sigma_u \times \sigma_v\| \, d(u, v), \\ \int_S F \cdot n \, dS &= \int_D F(\sigma) \cdot (\sigma_u \times \sigma_v) \, d(u, v). \end{aligned}$$

These are classical smooth parametrized-surface integrals. We are not constructing an independent general surface-measure theory.

**Proposition 13.3.5** (Surface reparametrization). Let  $\tilde{\psi} : U' \rightarrow U$  be a  $C^1$  diffeomorphism between open neighborhoods of  $D'$  and  $D$ , with  $\tilde{\psi}(D') = D$ , and put  $\psi = \tilde{\psi}|_{D'}$ . Then scalar surface integrals are unchanged. Flux is unchanged for a positive determinant and changes sign for a negative determinant.

*Proof.* First  $D' = \tilde{\psi}^{-1}(D)$  is compact, and Lemma 12.5.5 applied to the inverse diffeomorphism makes it Jordan measurable. A homeomorphism of neighborhoods preserves interiors and closures, so  $D' = \overline{(D')^\circ}$  as well. Thus it is an admissible parameter region before change of variables is used. Write  $\psi(s, t) = (u(s, t), v(s, t))$ . With  $\sigma_u, \sigma_v$  evaluated at  $\psi(s, t)$ , the chain rule gives

$$(\sigma \circ \psi)_s = \sigma_u u_s + \sigma_v v_s, \quad (\sigma \circ \psi)_t = \sigma_u u_t + \sigma_v v_t.$$

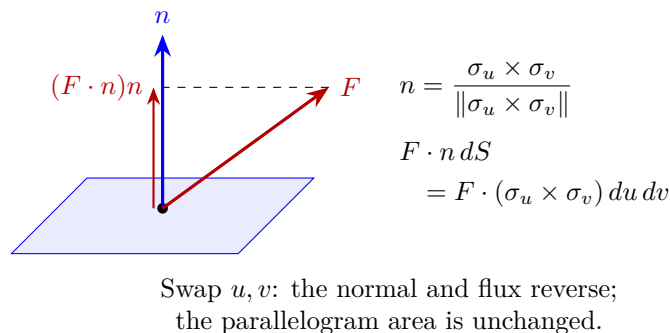


Figure 13.4: Flux measures the normal component of the field. Tangential motion contributes no flux through the patch. The sketch uses a positively directed normal component.

Expanding their cross product gives zero from the two equal-vector terms; the remaining terms are  $u_s v_t (\sigma_u \times \sigma_v)$  and  $v_s u_t (\sigma_v \times \sigma_u) = -v_s u_t (\sigma_u \times \sigma_v)$ . Their sum has coefficient  $u_s v_t - v_s u_t = \det D\psi$ . This determinant belongs to the  $2 \times 2$  change of parameters; the original  $3 \times 2$  surface derivative has no determinant. Thus

$$(\sigma \circ \psi)_s \times (\sigma \circ \psi)_t = (\det D\psi)(\sigma_u \times \sigma_v) \circ \psi.$$

Taking norms produces  $|\det D\psi|$  for scalar integrals; retaining the vector produces  $\det D\psi$  for flux. Apply [Corollary 12.5.6](#) to obtain both assertions.  $\square$

At this stage these are integrals of a chosen parametrized patch, invariant under the admissible reparametrizations just proved. Full chart-independence for smooth surfaces follows later from manifold integration.

### Computing surface area and flux

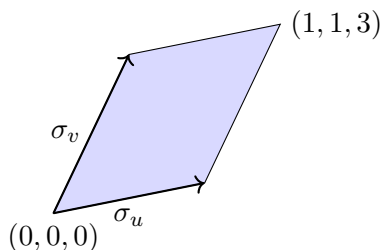
**Working rule.** Write  $\sigma(u, v)$ , compute  $\sigma_u, \sigma_v$ , then  $\sigma_u \times \sigma_v$ . For scalar surface integration use  $f(\sigma) \|\sigma_u \times \sigma_v\|$ ; for flux use  $F(\sigma) \cdot (\sigma_u \times \sigma_v)$ , and take the ordinary double integral. Here  $D\sigma$  is 3 by 2, so  $\det D\sigma$  is undefined.

**Example 13.3.6** (A plane patch). For  $\sigma(u, v) = (u, v, u + 2v)$  on  $[0, 1]^2$ ,

$$\sigma_u = (1, 0, 1), \quad \sigma_v = (0, 1, 2), \quad \sigma_u \times \sigma_v = (-1, -2, 1).$$

Thus

$$\text{area}(S) = \int_0^1 \int_0^1 \sqrt{6} \, dv \, du = \sqrt{6}.$$



The unit parameter square becomes this parallelogram with area  $\sqrt{6}$ . For the nonconstant scalar field  $f(x, y, z) = z$ , substitute  $z = u + 2v$  and use the same unsigned area factor:

$$\begin{aligned}\int_S z \, dS &= \sqrt{6} \int_0^1 \int_0^1 (u + 2v) \, dv \, du \\ &= \sqrt{6} \int_0^1 [uv + v^2]_{v=0}^{v=1} \, du = \sqrt{6} \int_0^1 (u + 1) \, du = \frac{3}{2}\sqrt{6}.\end{aligned}$$

For  $F = (0, 0, z)$ , the upward cross product instead gives  $F(\sigma) \cdot (\sigma_u \times \sigma_v) = u + 2v$ . Thus

$$\int_S F \cdot n \, dS = \int_0^1 \int_0^1 (u + 2v) \, dv \, du = \int_0^1 (u + 1) \, du = \frac{3}{2}.$$

Area  $\sqrt{6}$ , scalar-weighted area

$$\frac{3\sqrt{6}}{2},$$

and flux

$$\frac{3}{2}$$

are three different integrals on the same surface. The scalar density weights unsigned area; flux measures the field's oriented normal component.

**Example 13.3.7** (The side of a cylinder). For  $R, h > 0$ , parametrize the side by

$$\sigma(\theta, z) = (R \cos \theta, R \sin \theta, z), \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq h.$$

Use two half-cylinder patches to avoid the repeated seam; their common edges contribute zero parameter area. We obtain

$$\sigma_\theta = (-R \sin \theta, R \cos \theta, 0), \quad \sigma_z = (0, 0, 1), \quad \sigma_\theta \times \sigma_z = (R \cos \theta, R \sin \theta, 0).$$

This points outward and has norm  $R$ , so the side area is

$$\text{area}(S) = \int_0^{2\pi} \int_0^h R \, dz \, d\theta = \int_0^{2\pi} Rh \, d\theta = 2\pi Rh.$$

For  $F(x, y, z) = (x, y, 0)$ , first take the dot product:

$$F(\sigma) \cdot (\sigma_\theta \times \sigma_z) = R^2 \cos^2 \theta + R^2 \sin^2 \theta = R^2.$$

Hence the outward flux is

$$\int_S F \cdot n \, dS = \int_0^{2\pi} \int_0^h R^2 \, dz \, d\theta = \int_0^{2\pi} R^2 h \, d\theta = 2\pi R^2 h.$$

The compact angular notation represents the sum over the two injective half-cylinder patches already specified. Reversing orientation negates flux and preserves area. Chapter 12 computed the solid's volume  $\pi R^2 h$ ; here we integrate over its two-dimensional side, excluding the caps.



**Example 13.3.8** (A graph and a simple flux). For  $\sigma(u, v) = (u, v, h(u, v))$ ,

$$\sigma_u = (1, 0, h_u), \quad \sigma_v = (0, 1, h_v), \quad \sigma_u \times \sigma_v = (-h_u, -h_v, 1).$$

The upward area factor is  $\sqrt{1 + h_u^2 + h_v^2}$ . The upward flux of  $(0, 0, 1)$  is

$$\int_D 1 = \text{area}(D),$$

although the graph's surface area is generally larger. Reversing orientation negates the flux and leaves the scalar area unchanged.

**Exercise 3.** For  $\sigma(u, v) = (u, v, u + 2v)$  on  $[0, 1]^2$ , use the computed area factor to find the scalar surface integral of  $x^2$  and upward flux of  $F = (0, 0, y^2)$ . Explain the effect of interchanging the parameter order.

**Example 13.3.9** (Area and flux on a curved paraboloid). Let  $\sigma(u, v) = (u, v, u^2 + v^2)$  for  $u^2 + v^2 \leq 1$ . The map is injective because its first two coordinates recover  $(u, v)$ ; its cross product  $(-2u, -2v, 1)$  never vanishes and points upward. Thus the area is

$$\int_0^{2\pi} \int_0^1 \sqrt{1 + 4r^2} r \, dr \, d\theta = \frac{\pi}{6}(5^{3/2} - 1).$$

Here substitution  $s = 1 + 4r^2$ ,  $ds = 8r \, dr$  evaluates the inner integral. For  $F = (0, 0, z)$  the upward flux is instead

$$\int_{u^2+v^2 \leq 1} (u^2 + v^2) \, d(u, v) = 2\pi \int_0^1 r^3 \, dr = \frac{\pi}{2}.$$

The normalizing factor cancels against the scalar area factor in the flux formula. Reversing the surface orientation negates only the flux.

## 13.4 Classical Stokes for a Parametrized Surface Patch

A parameter-plane velocity  $(u', v')$  becomes  $\sigma_u u' + \sigma_v v'$  on the surface. Taking its dot product with  $F(\sigma)$  gives  $Au' + Bv'$  with  $A = F(\sigma) \cdot \sigma_u$  and  $B = F(\sigma) \cdot \sigma_v$ . These coefficients are forced by the work integrand. Differentiating them produces two kinds of terms: derivatives of  $F$ , through the chain rule, and second derivatives of  $\sigma$ . In  $B_u - A_v$  the latter cancel because  $\sigma_{vu} = \sigma_{uv}$ . The surviving force-field derivatives combine into the curl. This is the plan behind the component calculation.

All four classical integrals have real outputs:

| Integral                | Field input | Output      |
|-------------------------|-------------|-------------|
| Scalar line integral    | scalar      | real number |
| Work                    | vector      | real number |
| Scalar surface integral | scalar      | real number |
| Flux                    | vector      | real number |

The parameter integrand is scalar in every case, unlike componentwise vector integration, whose output is a vector.

$$\begin{array}{ccc}
\partial D & \xrightarrow{\sigma|_{\partial D}} & \partial S \\
\downarrow & & \downarrow \\
D \subset \mathbb{R}^2 & \xrightarrow{\sigma} & S = \sigma(D)
\end{array}$$

$A du + B dv$  on  $\partial D$                       work on  $\partial S$   
 $B_u - A_v$  in  $D$                       curl flux through  $S$

$$B_u - A_v = (\nabla \times F)(\sigma) \cdot (\sigma_u \times \sigma_v)$$

Figure 13.5: Patch Stokes is Green's theorem in parameters. The same parametrization transfers both the oriented boundary and its interior integrand.

**Theorem 13.4.1** (Parametrized-patch Stokes theorem). Let  $D$  satisfy Theorem 13.2.1, let  $\sigma$  be a regular  $C^2$  patch on  $D$ , and let  $F = (F_1, F_2, F_3)$  be  $C^1$  near  $S = \sigma(D)$ . Orient  $S$  by  $\sigma_u \times \sigma_v$ , and traverse its boundary by the image of the positive boundary of  $D$ . Here  $\partial S$  means the patch boundary  $\sigma(\partial D)$ , not the ambient topological boundary in  $\mathbb{R}^3$ . Then

$$\int_{\partial S} F \cdot dr = \int_S (\nabla \times F) \cdot n \, dS,$$

where  $\nabla \times F = (\partial_y F_3 - \partial_z F_2, \partial_z F_1 - \partial_x F_3, \partial_x F_2 - \partial_y F_1)$ .

*Proof.* Transport boundary work to the parameter plane, differentiate there, and identify the interior integrand before applying Green. Put  $A = F(\sigma) \cdot \sigma_u$  and  $B = F(\sigma) \cdot \sigma_v$ . The open set where  $\sigma$  is defined and takes values in the domain of  $F$  contains  $D$ . On it  $F \circ \sigma$  is  $C^1$  and the first derivatives of  $\sigma$  are  $C^1$ , so  $A, B$  are  $C^1$  on an open neighborhood of  $D$ . For each boundary piece  $c(t) = (u(t), v(t))$ , the chain rule gives

$$(\sigma \circ c)' = \sigma_u(c)u' + \sigma_v(c)v', \quad F(\sigma(c)) \cdot (\sigma \circ c)' = A(c)u' + B(c)v'.$$

Integrating and adding boundary pieces yields

$$\int_{\partial S} F \cdot dr = \int_{\partial D} A \, du + B \, dv.$$

The product and chain rules next give

$$\begin{aligned}
B_u &= DF(\sigma)[\sigma_u] \cdot \sigma_v + F(\sigma) \cdot \sigma_{vu}, \\
A_v &= DF(\sigma)[\sigma_v] \cdot \sigma_u + F(\sigma) \cdot \sigma_{uv}.
\end{aligned}$$

Since  $\sigma$  is  $C^2$ , its mixed derivatives agree and the last terms cancel. Write  $a = \sigma_u$ ,  $b = \sigma_v$ . The remaining expression, with derivatives evaluated at  $\sigma$ , is

$$\begin{aligned}
B_u - A_v &= \sum_{i,j} (\partial_j F_i)(a_j b_i - b_j a_i) \\
&= (\partial_2 F_3 - \partial_3 F_2)(a_2 b_3 - a_3 b_2) \\
&\quad + (\partial_3 F_1 - \partial_1 F_3)(a_3 b_1 - a_1 b_3) \\
&\quad + (\partial_1 F_2 - \partial_2 F_1)(a_1 b_2 - a_2 b_1) \\
&= (\nabla \times F)(\sigma) \cdot (\sigma_u \times \sigma_v).
\end{aligned}$$

Green gives the desired surface integral. Our boundary orientation is precisely the one induced by the ordered tangent pair; reversing it and the surface orientation negates both sides.  $\square$

This proves Stokes for one patch. An arbitrary compact oriented surface may require several charts, and proving the global statement is the purpose of the manifold machinery below.

**Example 13.4.2** (Stokes on a tilted patch). Take  $\sigma(u, v) = (u, v, u + v)$  on the unit square and

$$F = (-y/2, x/2, 0).$$

Its curl is  $(0, 0, 1)$ , so the flux is 1. Boundary work pulls back to

$$(-v \, du + u \, dv)/2;$$

the bottom and left edges contribute zero and the other two contribute

$$1/2$$

each.

**Proposition 13.4.3** (Divergence on a box). For a  $C^1$  vector field near a box  $Q \subset \mathbb{R}^3$ ,

$$\int_Q \operatorname{div} F = \int_{\partial Q} F \cdot n \, dS,$$

where the right side is the sum of the outward fluxes across its six faces and  $\operatorname{div} F = \partial_x F_1 + \partial_y F_2 + \partial_z F_3$ .

*Proof.* Integrate each  $\partial_j F_j$  first in the  $j$ th coordinate by FTC and then in the other two by Fubini. The endpoint difference is exactly the flux through the positive and negative  $j$ -faces. Add the three identities.  $\square$

The curved-boundary divergence theorem will follow from general Stokes.

## 13.5 Beyond One Parametrization: Why Manifolds?

A single parametrization describes only one coordinate region. A sphere, torus, or more complicated smooth space generally requires several overlapping local descriptions. We therefore need a language for doing ordinary Euclidean calculus locally and proving that the local answers fit together globally. This is a vocabulary tour, not a request to memorize formal definitions; those and their proofs come after local Stokes.

### Manifolds: local Euclidean geometry

**What and why?** A manifold looks locally like Euclidean space, so the calculus we already know can be used in small neighborhoods. Its intrinsic dimension counts local coordinates, not ambient coordinates. **Examples.** Circle arcs look like intervals; sphere and torus patches look two-dimensional. Our formal torus will be  $T^2 = S^1 \times S^1$ ; quotient pictures are developed in [Appendix E](#).

## Charts: coordinates versus parametrizations

**What?** A chart assigns Euclidean coordinates to geometric points. Its inverse builds geometric points from coordinates. **Why?** It lets us transfer a local geometric problem to an ordinary open set.

$$U \subset M \begin{array}{c} \xrightarrow{\varphi \text{ gives coordinates}} \\ \xleftarrow{\varphi^{-1} \text{ parametrizes}} \end{array} V \subset \mathbb{R}^n$$

On the upper open semicircle,  $\varphi(x, y) = x$  has image  $(-1, 1)$  and inverse  $x \mapsto (x, \sqrt{1-x^2})$ . Near  $(1, 0)$  use the right semicircle chart  $\psi(x, y) = y$ , whose inverse is  $y \mapsto (\sqrt{1-y^2}, y)$ . The map  $t \mapsto (\cos t, \sin t)$  on all of  $\mathbb{R}$  repeats points and is not an inverse chart. The entire circle cannot be one chart: it is compact, whereas a nonempty open subset of  $\mathbb{R}$  is not compact ([Theorems 10.5.4](#) and [10.6.8](#)).

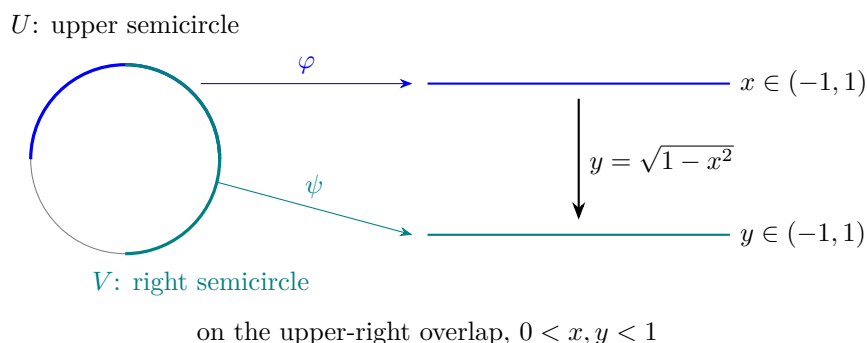


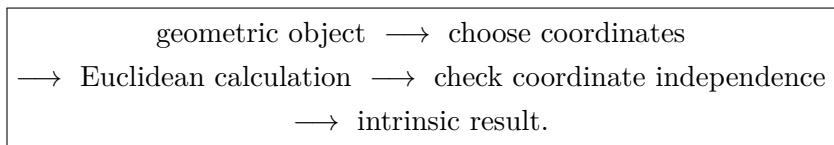
Figure 13.6: Two ways to flatten circle neighborhoods into intervals. The transition compares coordinates of the same point on their overlap; the inverse chart returns a coordinate to the circle.

## Transitions: compare two descriptions of one point

On an overlap of chart neighborhoods  $U, V$ , the coordinate change is

$$\varphi(U \cap V) \xrightarrow{\psi \circ \varphi^{-1}} \psi(U \cap V).$$

For the upper-right circle overlap it is  $x \mapsto \sqrt{1-x^2}$  on  $(0, 1)$ , with smooth inverse  $y \mapsto \sqrt{1-y^2}$ . Transition maps compare coordinate descriptions of the same geometry. Their job is compatibility:



## Tangent and cotangent spaces: directions and measurements

**What?** Velocities of curves through  $p$  describe tangent directions. Their space  $T_p M$  is linear even though  $M$  is curved. **Why?** Derivatives act on directions, so we need a vector space at each point. On a surface in  $\mathbb{R}^3$ , curve velocities lie in the tangent plane. A cotangent measurement is simply a linear map

$$\alpha_p : T_p M \rightarrow \mathbb{R}, \quad v \mapsto \alpha_p(v).$$

It could, for example, measure the instantaneous change of a scalar along a velocity. This is only the pointwise idea; fields of measurements come later.

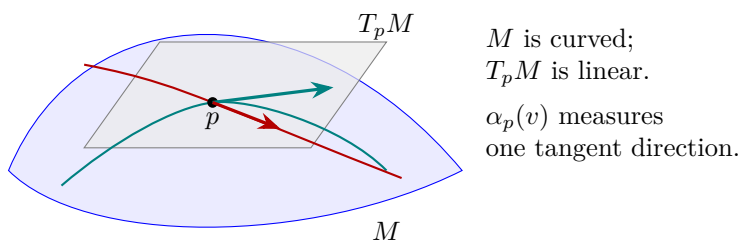


Figure 13.7: Curves through a point supply velocities in a common tangent plane. The plane is drawn translated to the point; its vectors form a linear space, on which a cotangent measurement is linear.

### Pullbacks: bring measurements back

**What?** For a smooth map  $f : M \rightarrow N$ , a covector or form  $\omega$  on  $N$  gives a corresponding measurement  $f^*\omega$  on  $M$ . **Why?** A direction at the starting point first moves forward under the differential; the measurement at the image point then reads it. Thus

directions move forward, measurements move backward.

This is a preview of the later pullback construction, not an additional definition required for the visual tour.

### Boundary: an intrinsic edge

**What and why?** Boundary points use a half-space local model  $H^n = \{x_n \geq 0\}$  at its edge; interior points use open Euclidean neighborhoods. This is the boundary that enters Stokes, not the ambient boundary of a set. For the circle and closed disk,

$$\partial_{\text{man}} S^1 = \emptyset, \quad \partial_{\mathbb{R}^2} S^1 = S^1, \quad \partial_{\text{man}} \overline{D} = S^1.$$

Thus an embedded circle has no intrinsic edge even though every one of its points is on its boundary as a subset of the plane.

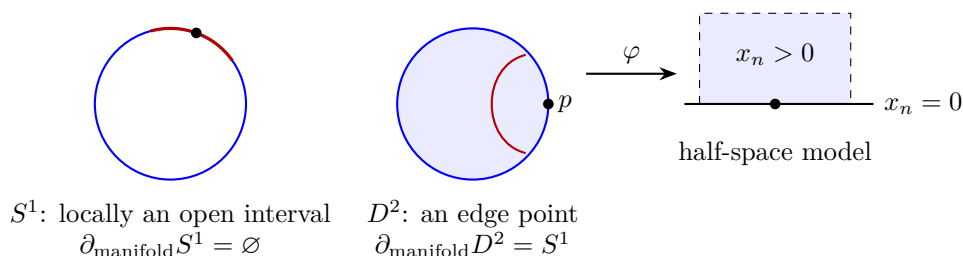
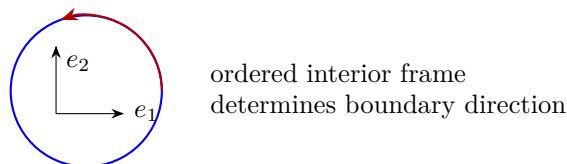


Figure 13.8: Manifold boundary is detected by local models, not by an ambient drawing. The whole circle is its boundary as a subset of the plane, but it has no manifold boundary. A disk edge flattens to the genuine face of a half-space.

### Orientation: choose consistent signs

**What?** Orientation consistently chooses which ordered tangent frames are positive, extending [Definition 10.17.15](#). **Why?** Oriented integration requires consistent signs across coordinate systems. For a disk with its usual ordered axes, the positive boundary traversal is counterclockwise. Reversing the orientation reverses the integral's sign.



### Partitions of unity: smoothly assemble local answers

**What?** Nonnegative smooth weights  $\rho_i$ , supported in their assigned chart neighborhoods, add to one. Recall from [Section 11.6](#) that support is the closure of the nonzero set. The manifold version and local finiteness ([Definition 13.14.1](#)) will be made precise before the construction. **Why?** Abruptly cutting a smooth quantity off at a chart edge can destroy smoothness. Fading weights avoid that defect. For a three-set cover, the bookkeeping is

$$\rho_1 + \rho_2 + \rho_3 = 1, \quad \omega = \rho_1\omega + \rho_2\omega + \rho_3\omega.$$

Here  $\omega$  merely denotes the quantity we wish to integrate; its precise form-valued meaning will be developed below.

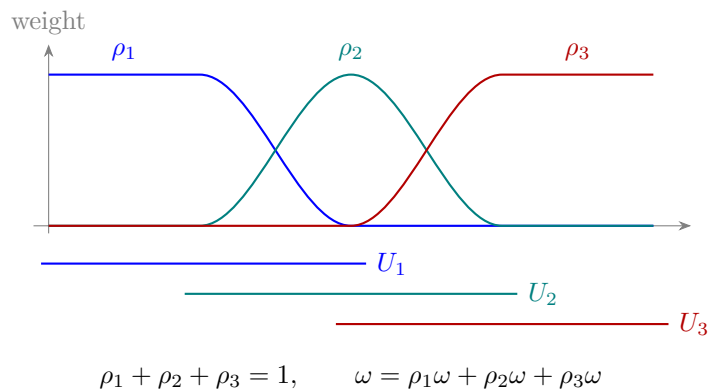


Figure 13.9: Schematic profiles for a smooth partition of unity: nonnegative weights fade out within their assigned open sets and add to one. Smooth profiles replace the abrupt jump of a hard cutoff; the exact smooth construction comes later.

### Global integration and Stokes: the two tasks

Integrate each weighted piece in its chart and add. Transition rules must make a change of chart harmless, and orientation must keep signs consistent. Applying local Stokes to the pieces will then produce global Stokes.

compatibility on overlaps  $\rightarrow$  transitions and pullbacks,  
 assembling local calculations  $\rightarrow$  orientation and smooth weights.

The guiding equation is

|                                                                                                   |
|---------------------------------------------------------------------------------------------------|
| local Euclidean calculus + compatibility + smooth localization<br>= global calculus on manifolds. |
|---------------------------------------------------------------------------------------------------|

First we need algebra that remembers orientation at one vector space. After that we will let its measurements vary smoothly with position.

## 13.6 Alternating Multilinear Algebra

|                                                           |
|-----------------------------------------------------------|
| What can consume $k$ directions and remember orientation? |
|-----------------------------------------------------------|

This section constructs pointwise algebra at one vector space  $V$ ; nothing varies with position yet. We proceed from covectors to alternating  $k$ -covectors, their elementary wedges and basis, and finally wedge products. Later a differential form will be a smoothly varying assignment  $p \mapsto \omega_p$ . Keep  $\Lambda^k(V^*)$  (one vector space's algebra) distinct from that field of algebraic objects. Recall from [Definition 10.16.1](#) that a covector is a linear map  $V \rightarrow \mathbb{R}$ , and  $V^*$  is the dual space. Recall also multilinear maps and alternation from [Section 10.16](#). The determinant is our prototype: an alternating form measures an oriented volume from an ordered list of directions. We now organize all degrees of such forms without using tensor products. The symmetric higher derivatives in [Remark 11.3.15](#) describe local variation; these alternating forms describe oriented measurement. The progression is multilinear maps, symmetric higher derivatives, then alternating covectors. They share linearity in each input, but have different behavior under interchange. In standard coordinates, [Definition 11.2.11](#) already identified  $dx_i = d\pi_i = e^i$  by differentiating the coordinate projection. Thus  $dx_i(v) = v_i$ . Ordinary covectors are precisely the degree-one case of the following hierarchy: linear maps  $V \rightarrow \mathbb{R}$ , alternating bilinear maps  $V \times V \rightarrow \mathbb{R}$ , and alternating  $k$ -linear maps  $V^k \rightarrow \mathbb{R}$ .

For complementary viewpoints, Hubbard and Hubbard [\[12\]](#) emphasize geometric measurements, and Tu [\[15\]](#) connects covector fields with general forms. Munkres [\[13\]](#) provides the analysis-to-Stokes comparison; Lee [\[5\]](#) is a structural reference. Spivak [\[14\]](#) gives a compact correctness benchmark. The constructions and proofs below are self-contained.

**First reading.** Understand covectors, oriented area measurements,  $k$ -covectors, the determinant meaning of elementary wedges, the low-degree wedge rules, and the coordinate wedge basis. These are the working tools for pullbacks and exterior derivatives. On a first reading one may admit the full general basis proof, shuffle construction, and general algebra laws temporarily while using their stated conclusions. Their proofs are given here in dependency order. Appendix [D](#), especially [Proposition D.10.9](#), gives a structural continuation: Readers who prefer a more structural introduction to multilinear algebra may consult its universal-property and tensor constructions ([Definition D.2.2](#) and [Section D.6](#)), exterior powers ([Section D.9](#)), and exterior algebra ([Sections D.9](#) and [D.10](#)). The canonical agreement in [Proposition D.10.9](#) connects that route to the concrete alternating covectors here. Appendix [D](#) may make the algebra conceptually cleaner for some readers, but tensor products and quotient constructions are not prerequisites for the main analysis-to-Stokes route.

| degree | vector inputs | geometric role                        |
|--------|---------------|---------------------------------------|
| 0      | 0             | scalar                                |
| 1      | 1             | directed linear measurement           |
| 2      | 2             | oriented area measurement             |
| 3      | 3             | oriented volume measurement           |
| $k$    | $k$           | oriented $k$ -dimensional measurement |

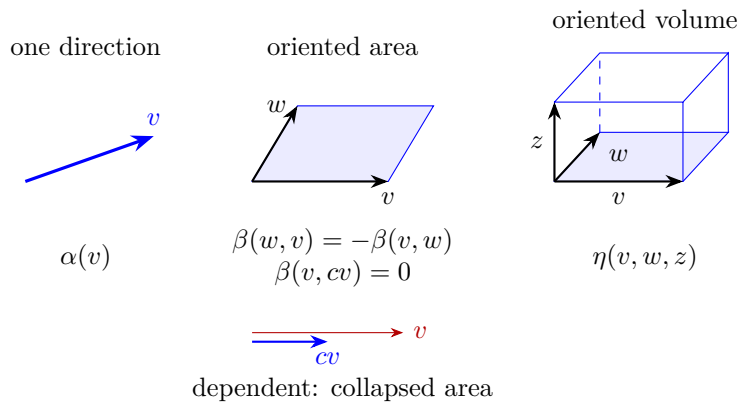


Figure 13.10: Alternating measurements progress from a vector to an ordered parallelogram to an ordered parallelepiped. Reordering two inputs reverses the sign; dependent inputs give zero.

**Definition 13.6.1** (Alternating  $k$ -covectors and exterior powers). Fix a finite-dimensional real vector space  $V$ , with  $\dim V = n$ . For  $k \geq 1$ , a  **$k$ -covector** is an alternating  $k$ -linear map  $A : V^k \rightarrow \mathbb{R}$ . Denote the space of these maps by

$$\Lambda^k(V^*) = \text{Alt}^k(V; \mathbb{R}).$$

Pointwise addition and scalar multiplication preserve multilinearity and alternation, and the vector-space axioms hold on each input tuple. The notation is neither an ordinary numerical power nor the Cartesian product  $(V^*)^k$ . An element is a scalar-valued map on  $k$  vectors, not a list of  $k$  covectors, and need not be a single elementary wedge. By convention  $\Lambda^0(V^*) = \mathbb{R}$ , and  $\Lambda^1(V^*) = V^*$ . If  $k > \dim V$ , every list is dependent, so  $\Lambda^k(V^*) = \{0\}$  by [Proposition 10.16.9](#).

A 1-covector accepts one vector and returns a scalar, as work does. A 2-covector accepts two vectors; on  $\mathbb{R}^2$  the rule

$$(v, w) \mapsto v_1 w_2 - v_2 w_1$$

measures signed area, is bilinear, and vanishes on equal inputs. Thus  $\Lambda^2(V^*)$  consists of alternating bilinear measurements. The degree-zero convention is a scalar with no vector inputs.

**Definition 13.6.2** (Elementary wedges). For covectors  $\alpha_1, \dots, \alpha_k : V \rightarrow \mathbb{R}$ , their **elementary wedge** is the map  $\alpha_1 \wedge \dots \wedge \alpha_k : V^k \rightarrow \mathbb{R}$  defined by

$$(\alpha_1 \wedge \dots \wedge \alpha_k)(v_1, \dots, v_k) = \det \begin{pmatrix} \alpha_1(v_1) & \dots & \alpha_1(v_k) \\ \vdots & & \vdots \\ \alpha_k(v_1) & \dots & \alpha_k(v_k) \end{pmatrix}.$$

Rows correspond to covectors; columns correspond to vector inputs. The determinant combines scalar measurements into one oriented  $k$ -dimensional scalar. At this stage  $\wedge$  denotes this actual map, not a formal quotient product from [Appendix D](#). The empty elementary wedge is the scalar 1.



**Proposition 13.6.3** (Elementary-wedge identities). An elementary wedge is alternating and multilinear in its vector inputs, so belongs to  $\Lambda^k(V^*)$ . It is also multilinear in its covector factors. Repeated factors give zero, and for any permutation  $\sigma$ ,

$$\alpha_{\sigma(1)} \wedge \cdots \wedge \alpha_{\sigma(k)} = \text{sgn}(\sigma) \alpha_1 \wedge \cdots \wedge \alpha_k.$$

*Proof.* A sum or scalar multiple in one vector slot changes the corresponding column in the evaluation matrix; column multilinearity of the determinant proves input multilinearity. Equal inputs give equal columns, hence zero. The same argument using rows proves multilinearity in the covectors and zero for repeated covectors. Swapping factors swaps rows, negating the determinant. Decomposing a permutation into swaps gives its sign by the determinant sign rules of Chapter 10. These are identities of maps because the evaluations agree on every input tuple.  $\square$

An elementary wedge is a special  $k$ -covector. A general  $k$ -covector may be a sum of such measurements without being one elementary wedge itself. The basis theorem for alternating covectors (Theorem 13.6.4) will establish that these sums exhaust the space; the dimension-four example after the product laws will show why a single elementary wedge need not suffice.

For increasing  $I = (i_1, \dots, i_k)$ , evaluation gives

$$(dx_{i_1} \wedge \cdots \wedge dx_{i_k})(v_1, \dots, v_k) = \det((v_j)_{i_r})_{r,j=1}^k.$$

In degree two this reads

$$(dx_i \wedge dx_j)(v, w) = v_i w_j - v_j w_i.$$

This is alternating coordinate-area measurement, not ordinary symbolic multiplication. Repeated coordinate covectors give repeated rows, so  $dx_i \wedge dx_i = 0$ ; repeated input directions give repeated columns and zero oriented area. Swapping coordinate covectors reverses the row orientation, and swapping input directions reverses the column orientation. In particular,  $dx_i \wedge dx_j = -dx_j \wedge dx_i$ . These rules encode oriented measurement. In particular,  $(dx_1 \wedge \cdots \wedge dx_n)(v_1, \dots, v_n) = \det[v_1 \cdots v_n]$ . For two covectors,  $(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v)$ . Thus  $dx \wedge dy$  is signed area, and

$$\begin{aligned} (dx + 2dy) \wedge (3dx - dy) &= 3dx \wedge dx - dx \wedge dy \\ &\quad + 6dy \wedge dx - 2dy \wedge dy \\ &= -7dx \wedge dy. \end{aligned}$$

For example  $dx \wedge dy$  evaluates on  $((1, 2), (3, 1))$  as  $-5$ .

**The basis mechanism in  $\mathbb{R}^3$ , degree two.** Let  $A$  be any alternating bilinear form and write

$$\begin{aligned} v &= \sum_i v_i e_i, \\ w &= \sum_j w_j e_j. \end{aligned}$$

Expanding both slots gives

$$\begin{aligned} A(v, w) &= \sum_{i,j=1}^3 v_i w_j A(e_i, e_j) \\ &= A(e_1, e_2)(v_1 w_2 - v_2 w_1) \\ &\quad + A(e_1, e_3)(v_1 w_3 - v_3 w_1) + A(e_2, e_3)(v_2 w_3 - v_3 w_2). \end{aligned}$$

The diagonal terms vanish; the  $(j, i)$  term combines with the  $(i, j)$  term using its opposite sign. Each parenthesis is one coordinate wedge, so as an identity of maps,

$$A = A(e_1, e_2) dx \wedge dy + A(e_1, e_3) dx \wedge dz + A(e_2, e_3) dy \wedge dz.$$

Only three oriented coordinate-area measurements remain. The general proof uses exactly this expansion-and-sorting mechanism.

**What information determines an alternating  $k$ -covector?** Expand on basis tuples, discard repeated indices, and sort the surviving tuples. Only increasing tuples remain. Which forms extract their coefficients? The coordinate elementary wedges do: the next proof shows that each is 1 on its own increasing tuple and 0 on the others.

**Theorem 13.6.4** (Basis of alternating covectors). For the dual basis  $(e^1, \dots, e^n)$  of [Definition 10.16.2](#), the elementary wedges

$$e^{i_1} \wedge \dots \wedge e^{i_k}, \quad i_1 < \dots < i_k,$$

form a basis of  $\Lambda^k(V^*)$ . Consequently its dimension is

$$\binom{n}{k}$$

for  $0 \leq k \leq n$ .

*Proof.* For increasing lists  $I = (i_1 < \dots < i_k)$  and  $J = (j_1 < \dots < j_k)$ , write  $e^I = e^{i_1} \wedge \dots \wedge e^{i_k}$ . The dual-basis rule gives

$$e^I(e_{j_1}, \dots, e_{j_k}) = \det(\delta_{j_s}^{i_r})_{r,s=1}^k = \begin{cases} 1, & I = J, \\ 0, & I \neq J. \end{cases}$$

If the lists match, the matrix is the identity. Otherwise choose  $j_s \in J \setminus I$ . Then  $e^{i_r}(e_{j_s}) = 0$  for every  $r$ , so column  $s$  is zero. Evaluation on increasing basis tuples therefore extracts the corresponding coefficient.

*Spanning.* For arbitrary  $A \in \Lambda^k(V^*)$ , construct

$$B = \sum_{i_1 < \dots < i_k} A(e_{i_1}, \dots, e_{i_k}) e^I.$$

First, the displayed evaluations show that  $A$  and  $B$  agree on every increasing basis tuple. Second, these tuples determine an alternating multilinear map everywhere: expand each input in the basis. Terms with repeated indices vanish. Sort each remaining tuple increasingly; each swap contributes a minus sign by [Proposition 10.16.9](#). Thus all basis-tuple values agree. Explicitly, for

$$v_s = \sum_i a_{is} e_i,$$

$$(A - B)(v_1, \dots, v_k) = \sum_{i_1, \dots, i_k} a_{i_1,1} \dots a_{i_k,k} (A - B)(e_{i_1}, \dots, e_{i_k}) = 0.$$

Therefore  $A = B$ .

*Independence.* If

$$\sum_I c_I e^I = 0,$$

evaluate on the increasing tuple indexed by  $J$  to obtain

$$0 = \sum_I c_I \delta_{IJ} = c_J.$$

Consequently the displayed family is a basis, and counting increasing  $k$ -subsets gives dimension

$$\binom{n}{k}.$$

For  $k = 0$  the basis is  $\{1\}$ . □

Write an arbitrary  $k$ -covector as

$$A = \sum_I a_I e^I$$

over increasing  $I$ . Then  $a_I = A(e_{i_1}, \dots, e_{i_k})$ . In dimension three the two-covector basis is  $dx \wedge dy, dx \wedge dz, dy \wedge dz$ . Thus the flux convention

$$A \, dy \wedge dz + B \, dz \wedge dx + C \, dx \wedge dy = C \, dx \wedge dy - B \, dx \wedge dz + A \, dy \wedge dz$$

has increasing-order coefficients  $(C, -B, A)$ . Elementary wedges span, but a general covector of degree  $k$  need not itself be one elementary wedge. A concrete degree-two example will follow the product laws, where it can be checked in one calculation.

For degrees  $(1, 1)$  the product must recover  $(\alpha \wedge \beta)(v_1, v_2) = \alpha(v_1)\beta(v_2) - \alpha(v_2)\beta(v_1)$ . The general mechanism is to choose which inputs go to the first form, preserve their internal order, send the remaining inputs to the second, and attach the sign of this rearrangement. Before general shuffle notation, take a one-covector  $\alpha$  and an alternating two-covector  $\beta$ . On three vectors there are exactly three ways to assign an input to  $\alpha$ , keeping the other two in their original order:

$$\begin{aligned} (\alpha \wedge \beta)(v_1, v_2, v_3) &= \alpha(v_1)\beta(v_2, v_3) - \alpha(v_2)\beta(v_1, v_3) \\ &\quad + \alpha(v_3)\beta(v_1, v_2). \end{aligned}$$

The corresponding lists are 123, 213, 312, with signs  $+, -, +$ . For  $\alpha = dx$ ,  $\beta = dy \wedge dz$ , and  $v_1 = (1, 1, 0)$ ,  $v_2 = (2, 0, 1)$ ,  $v_3 = (0, 1, 1)$ , the three contributions are  $-1, -2, 0$ , totaling  $-3$ . This is also the determinant of the three vector columns. A shuffle is precisely this selection of two internally ordered blocks, generalized to degrees  $p$  and  $q$ .

**Definition 13.6.5** (Wedge product by shuffles). For  $\alpha \in \Lambda^p(V^*)$ ,  $\beta \in \Lambda^q(V^*)$ , define

$$\begin{aligned} (\alpha \wedge \beta)(v_1, \dots, v_{p+q}) &= \sum_{\sigma \in \text{Sh}(p, q)} \text{sgn}(\sigma) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \\ &\quad \cdot \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}). \end{aligned}$$

Here a  $(p, q)$ -**shuffle** is a permutation with  $\sigma(1) < \dots < \sigma(p)$  and  $\sigma(p+1) < \dots < \sigma(p+q)$ : choose the first increasing block, and the remaining inputs form the second increasing block. Degree-zero factors act by scalar multiplication.

**Remark 13.6.6** (Equivalent full-permutation formula). Once the input-selection rule is understood, it can also be written over  $\mathbb{R}$  as the normalized full-permutation formula

$$(\alpha \wedge \beta)(v_1, \dots, v_{p+q}) = \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \text{sgn}(\sigma) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \cdot \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}).$$

**Proposition 13.6.7** (The shuffle product is an alternating covector). The two formulas agree and define an element of  $\Lambda^{p+q}(V^*)$ .

*Proof.* In each summand, any fixed vector input occurs exactly once, in one of the two factors. That factor is linear in it and the other is independent of it. Thus the finite sum is linear in every vector input. Write each permutation uniquely as

$$\pi = \sigma \circ (\rho \oplus \eta), \quad \sigma \in \text{Sh}(p, q), \quad \rho \in S_p, \quad \eta \in S_q.$$

To recover these factors, sort the set  $\{\pi(1), \dots, \pi(p)\}$  into the first block of  $\sigma$  and sort its complement into the second. Then  $\rho(i)$  is the position of  $\pi(i)$  in the first sorted block, and  $\eta(j)$  is the position of  $\pi(p+j)$  in the second. Here  $\rho \oplus \eta$  acts as  $\rho$  on the first block and sends  $p+j$  to  $p+\eta(j)$ . Thus

$$\text{sgn}(\pi) = \text{sgn}(\sigma) \text{sgn}(\rho) \text{sgn}(\eta).$$

Alternation in the two factors makes the full-sum term equal to

$$\begin{aligned} & \text{sgn}(\sigma) \text{sgn}(\rho)^2 \text{sgn}(\eta)^2 \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}) \\ &= \text{sgn}(\sigma) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}). \end{aligned}$$

Exactly  $p!q!$  pairs  $(\rho, \eta)$  give this value for each shuffle. Dividing the full sum by  $p!q!$  therefore gives the shuffle formula. In the full sum, interchanging inputs in positions  $a, b$  replaces each permutation  $\sigma$  by  $(a \ b) \circ \sigma$ , a bijective reindexing that negates its sign. The resulting evaluation is therefore the negative of the original one. Equal inputs make the value its own negative, hence zero over  $\mathbb{R}$ . Together with input multilinearity this proves membership in  $\Lambda^{p+q}(V^*)$ . Degree-zero factors give scalar multiplication.  $\square$

**Proposition 13.6.8** (Compatibility with elementary wedges). The shuffle product of elementary wedges is their concatenated wedge:

$$(\alpha_1 \wedge \dots \wedge \alpha_p) \wedge (\beta_1 \wedge \dots \wedge \beta_q) = \alpha_1 \wedge \dots \wedge \alpha_p \wedge \beta_1 \wedge \dots \wedge \beta_q.$$

*Proof.* This is a block form of Laplace expansion: choose columns for the first determinant, retain their internal order, and attach the shuffle sign. Evaluate on  $v_1, \dots, v_{p+q}$ . A shuffle  $\sigma$  chooses the columns for the first  $p$  rows of the concatenated evaluation matrix, in increasing order; its remaining columns belong to the last  $q$  rows. Expand the two smaller determinants by the Leibniz formula. A permutation  $\rho \in S_p$  assigns column  $\sigma(\rho(i))$  to row  $i$  of the first block, and  $\eta \in S_q$  assigns column  $\sigma(p+\eta(j))$  to row  $p+j$ . The resulting full permutation is  $\tau = \sigma \circ (\rho \oplus \eta)$ , where  $\rho \oplus \eta$  acts as  $\rho$  on the first block and as the shifted permutation  $\eta$  on the second block. Its sign is

$$\text{sgn}(\tau) = \text{sgn}(\sigma) \text{sgn}(\rho) \text{sgn}(\eta).$$

Conversely, the set of columns assigned by  $\tau$  to the first block determines  $\sigma$  uniquely by sorting the two blocks, and then determines  $\rho, \eta$  uniquely. Thus each Leibniz term of the large determinant occurs once, with its correct sign. The evaluated maps agree everywhere. Empty blocks reduce to the scalar unit convention.  $\square$

Only now have we justified using the same wedge symbol for the general product and the earlier elementary determinant maps.

**Why the sign depends on both degrees.** An element of  $\Lambda^p(V^*)$  has degree  $p$ ; such a fixed-degree element is called homogeneous. First take  $\alpha = \alpha_1 \wedge \alpha_2$  and  $\beta = \beta_1 \wedge \beta_2 \wedge \beta_3$ . By the elementary concatenation rule just proved, moving the entire  $\beta$  block to the front means the following successive reorderings:

$$\begin{aligned} \alpha_1 \alpha_2 \beta_1 \beta_2 \beta_3 &\longrightarrow \beta_1 \alpha_1 \alpha_2 \beta_2 \beta_3 && \text{two swaps,} \\ &\longrightarrow \beta_1 \beta_2 \alpha_1 \alpha_2 \beta_3 && \text{two more,} \\ &\longrightarrow \beta_1 \beta_2 \beta_3 \alpha_1 \alpha_2 && \text{two more.} \end{aligned}$$

Here juxtaposition abbreviates the ordered list of wedge factors, not ordinary multiplication. There are  $2 + 2 + 2 = 6 = 2 \cdot 3$  swaps, hence  $\text{sign } (-1)^6 = +1$ . In general each of the  $q$  factors crosses all  $p$  factors:  $pq$  swaps each contribute  $-1$ . The following proposition extends this mechanism from elementary wedges to arbitrary homogeneous elements.

**Proposition 13.6.9** (Exterior-algebra rules). The wedge product is bilinear, associative, and graded commutative:  $\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$  for degrees  $p, q$ .

*Proof.* Bilinearity follows by distributing each summand. Both parenthesizations of a triple product are therefore already-defined trilinear maps. By Proposition 13.6.8, on elementary wedges both concatenate the same three covector lists. Elementary wedges span each factor, so expanding each of the three arguments proves the two maps equal everywhere. For graded commutativity, moving  $p$  covectors past  $q$  covectors makes  $pq$  interchanges, giving  $(-1)^{pq}$ ; bilinearity extends the identity from elementary wedges. The shuffle picture gives the same explanation: each grouping lists the same three internally increasing blocks, with the same permutation sign.  $\square$

| $\deg \alpha$ | $\deg \beta$ | $(-1)^{pq}$ |
|---------------|--------------|-------------|
| even          | even         | +1          |
| even          | odd          | +1          |
| odd           | even         | +1          |
| odd           | odd          | -1          |

The sign is negative exactly when both degrees are odd. Thus even-degree elements commute with all homogeneous elements; odd-degree elements anticommute with odd-degree elements. For example,

$$dx \wedge dy = -dy \wedge dx, \quad dx \wedge (dy \wedge dz) = (dy \wedge dz) \wedge dx.$$

The exterior algebra is *graded commutative*, not simply anticommutative. If  $\alpha$  has odd degree, the rule gives  $\alpha \wedge \alpha = -\alpha \wedge \alpha$ , so  $2\alpha \wedge \alpha = 0$  and  $\alpha \wedge \alpha = 0$  over  $\mathbb{R}$ . For a 1-covector  $\alpha$ ,  $\alpha \wedge \alpha = 0$ , also directly from the determinant with repeated rows. For even degree the square need not vanish. In dimension four, put  $\omega = e^1 \wedge e^2 + e^3 \wedge e^4$ . Then

$$\omega \wedge \omega = 2e^1 \wedge e^2 \wedge e^3 \wedge e^4 \neq 0.$$

If  $\omega$  were one elementary wedge  $\alpha \wedge \beta$ , its square would have repeated factors and be zero. Thus this also proves that a general 2-covector need not be elementary.

For a three-dimensional  $V$ , the basis theorem for alternating covectors (Theorem 13.6.4) gives the complete degree table:

| $k$ | basis of $\Lambda^k(V^*)$                        | dim |
|-----|--------------------------------------------------|-----|
| 0   | 1                                                | 1   |
| 1   | $e^1, e^2, e^3$                                  | 3   |
| 2   | $e^1 \wedge e^2, e^1 \wedge e^3, e^2 \wedge e^3$ | 3   |
| 3   | $e^1 \wedge e^2 \wedge e^3$                      | 1   |

We have related vector spaces, one for each degree. The exterior algebra collects all degrees into one graded vector space and lets wedge multiplication move between them.

**Three related types.**  $\Lambda^k(V^*)$  is a vector space;  $\alpha_1 \wedge \cdots \wedge \alpha_k$  is one elementary wedge;  $\alpha \wedge \beta$  is the product of arbitrary homogeneous elements. A general  $k$ -covector need not be elementary, as the preceding dimension-four example proves.

**Definition 13.6.10** (The exterior algebra). Using the finite direct sum constructed in [Section 10.9](#), set

$$\Lambda^*(V^*) = \bigoplus_{k=0}^n \Lambda^k(V^*).$$

Every element has a unique component in each degree. Distribute the wedge product over these components: the degree- $r$  part is

$$\sum_{p+q=r} \alpha_p \wedge \beta_q.$$

An element supported in degree  $p$  is homogeneous of degree  $p$ . The direct-sum basis and dimension facts are those already proved in Chapter 10.

The direct-sum construction gives uniqueness of the homogeneous decomposition. Multiplication is a finite distribution across degrees; the degree- $r$  parts of the two triple products are respectively

$$\sum_{p+q+s=r} (\alpha_p \wedge \beta_q) \wedge \gamma_s, \quad \sum_{p+q+s=r} \alpha_p \wedge (\beta_q \wedge \gamma_s).$$

They agree termwise by [Proposition 13.6.9](#); degrees above  $n$  vanish. The degree-zero element 1 acts identically on each component, so is a unit. Here “algebra” means a vector space with bilinear associative multiplication and a unit, the degree-zero scalar 1. Mixed-degree elements have several nonzero components. In dimension two, for scalars  $a, b$ , 1-covectors  $u, v$ , and 2-covectors  $\omega_2, \eta_2$ ,

$$(a + u + \omega_2) \wedge (b + v + \eta_2) = ab + (av + bu) + (a\eta_2 + b\omega_2 + u \wedge v),$$

with groups in degrees 0, 1, 2; higher terms vanish. The graded sign rule uses specified homogeneous degrees, not an invented degree for a mixed sum. A covector is one fixed rule; a differential form lets such a rule vary smoothly with its base point. The tensor construction in [Appendix D](#) gives an algebraic explanation of these spaces; none of the following analysis requires it.

## 13.7 Differential Forms, Pullbacks, and Exterior Derivatives

### Why differential forms?

The Euclidean derivative  $Df(p)$  already measures first-order change in one direction. Work along a curve needs more general covector fields, since not every force comes from a potential. Surface flux needs a measurement of two tangent directions that remembers their order; higher-dimensional integration needs the corresponding measurement of  $k$  directions:

$$f \longrightarrow df \longrightarrow \text{general 1-form} \longrightarrow \text{2-form} \longrightarrow k\text{-form}.$$

Forms put these directed integrands in one language. We first learn the one-direction case, then use the preceding alternating algebra to extend it.

**From scalar differentials to directed integrands.** Start with a smooth scalar function  $f$  on an open set  $U \subseteq \mathbb{R}^n$ . Its Euclidean derivative  $Df(p) : \mathbb{R}^n \rightarrow \mathbb{R}$  is a covector. As  $p$  varies, these covectors define the exact 1-form  $df$ , with  $(df)_p = Df(p)$  (Definition 11.2.11). The field  $df$  and its value  $(df)_p$  are different types of object. Here tangent directions at  $p$  are ordinary vectors in  $\mathbb{R}^n$ ; we may write  $T_p U = \mathbb{R}^n$  to remember their base point.

**Definition 13.7.1** (A Euclidean 1-form). A smooth 1-form on  $U$  is a field of covectors  $p \mapsto \omega_p \in (\mathbb{R}^n)^*$  with smooth coordinate coefficients. In coordinates this means

$$\omega = \sum_{i=1}^n a_i dx^i$$

with smooth functions  $a_i$  and

$$\omega_p(v) = \sum_i a_i(p) v^i.$$

For example on a planar open set,

$$\omega = P dx + Q dy, \quad \omega_p : T_p U \rightarrow \mathbb{R}, \quad \omega_p(v) = P(p)v_1 + Q(p)v_2.$$

The coordinate covectors form the dual basis of Definition 10.16.2:  $dx(v) = v_1$ ,  $dy(v) = v_2$ .

Keep three objects separate:  $\omega$  is the field,  $\omega_p$  is one covector, and  $\omega_p(v)$  is a real number. The differential  $df = f_x dx + f_y dy$  is one special 1-form. A general 1-form need not be  $df$ , even globally on the plane:  $x dy$  would require  $f_x = 0$  and  $f_y = x$ , contradicting equality of mixed partials for smooth  $f$ .

A covector is one linear measuring device at one point; a differential 1-form is a smoothly varying field of such measuring devices.

| object     | type                                  | output                       |
|------------|---------------------------------------|------------------------------|
| $f$        | $U \rightarrow \mathbb{R}$            | $f(p)$ is a scalar           |
| $Df(p)$    | $\mathbb{R}^n \rightarrow \mathbb{R}$ | Euclidean derivative         |
| $(df)_p$   | $T_p U \rightarrow \mathbb{R}$        | value of the exact form $df$ |
| $\omega_p$ | $T_p U \rightarrow \mathbb{R}$        | linear measurement           |
| $\omega$   | $p \mapsto \omega_p$                  | a covector at each point     |

For example, at  $p = (1, 2)$  and  $v = (3, -1)$ ,  $dx_p(v) = 3$ ,  $dy_p(v) = -1$ . For  $f(x, y) = xy$ ,  $(df)_p = 2dx + dy$  and  $(df)_p(v) = 5$ . For the different form  $\omega = x dy - y dx$ , we obtain  $\omega_p(v) = -7$ . If  $X$  is a smooth vector field, the function  $p \mapsto \omega_p(X_p)$  is an ordinary smooth scalar function: its coordinate formula is a sum of products of smooth coefficients.

1-form + tangent direction = scalar.

For a smooth curve  $\gamma : I \rightarrow U$ , with  $[a, b] \subset I$ , define its pulled-back 1-form by composing the covector with  $D\gamma(t) : \mathbb{R} \rightarrow \mathbb{R}^n$ . On a scalar direction  $h$ , this gives  $\omega_{\gamma(t)}(h\gamma'(t))$ , so

$$(\gamma^* \omega)_t = \omega_{\gamma(t)}(\gamma'(t)) dt, \quad \int_{\gamma} \omega := \int_a^b \omega_{\gamma(t)}(\gamma'(t)) dt.$$

For  $\gamma(t) = (x(t), y(t))$ , evaluate first on a scalar direction  $h$ :

$$(\gamma^* dx)_t(h) = dx_{\gamma(t)}(h\gamma'(t)) = hx'(t).$$

Since  $dt(h) = h$ , this proves  $\gamma^*dx = x'(t) dt$ ; the same calculation gives  $\gamma^*dy = y'(t) dt$ . Therefore

$$\gamma^*(P dx + Q dy) = (P(\gamma(t))x'(t) + Q(\gamma(t))y'(t)) dt.$$

Integrating its coefficient gives precisely

$$\int_{\gamma} P dx + Q dy.$$

For piecewise smooth curves apply the rule on each piece and add. Pullback changes the domain to the parameter interval while preserving the scalar measurement along the traversal.

**Exact forms and general forms.** A form of the shape  $df$  is called **exact**. The chain rule gives

$$\gamma^*(df) = d(f \circ \gamma), \quad \int_{\gamma} df = f(\gamma(b)) - f(\gamma(a))$$

by [Theorem 13.1.8](#). Passing from  $f$  to  $df$  turns a scalar potential into a measuring field, and its integral remembers only the endpoints. In contrast  $\omega_0 = x dy - y dx$  cannot be  $df$  on the plane:  $f_x = -y$  and  $f_y = x$  would give  $f_{xy} = -1$  and  $f_{yx} = 1$ , contradicting equality of mixed partials. On the counterclockwise circle, the preceding pullback rule gives  $\gamma^*\omega_0 = dt$ , so

$$\int_{S^1} \omega_0 = 2\pi.$$

An exact form would integrate to zero on this closed curve. General 1-forms therefore describe more than changes of potential.

**From directions to oriented area.** A 1-form measures one tangent direction. Surface flux depends on two independent tangent directions. What should measure an ordered pair? It should be linear in each input, negate its value when their order is reversed, and vanish on dependent inputs. For covectors  $\alpha, \beta$ , recall the two-factor elementary wedge of [Definition 13.6.2](#):

$$(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v).$$

Distributing verifies linearity in each vector; exchanging  $v, w$  negates the formula, and substituting  $w = cv$  makes it zero (as does a zero input). For the coordinate measurements this becomes

$$(dx \wedge dy)(v, w) = v_1w_2 - v_2w_1,$$

the signed area of the coordinate parallelogram. Surface flux makes this two-direction measurement unavoidable: in  $\mathbb{R}^3$  the rule

$$\beta_F = F_1 dy \wedge dz + F_2 dz \wedge dx + F_3 dx \wedge dy, \quad (\beta_F)_p(v, w) = F(p) \cdot (v \times w)$$

is exactly the earlier oriented flux factor. Indeed, the explicit expansion is

$$\begin{aligned} F(p) \cdot (v \times w) &= F_1(p)(v_2w_3 - v_3w_2) + F_2(p)(v_3w_1 - v_1w_3) \\ &\quad + F_3(p)(v_1w_2 - v_2w_1), \end{aligned}$$

which is the evaluation of the three displayed coordinate wedges. A smoothly varying such alternating bilinear rule is a 2-form. We will now define the same idea for  $k$  directions using the basis already proved in [Theorem 13.6.4](#). No tensor products are needed for this route.



# General forms: smoothly varying alternating measurements

For a smooth function  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ , Chapter 11 gives

$$Df(\mathbf{x}) : \mathbb{R}^n \longrightarrow \mathbb{R}.$$

Thus  $Df(\mathbf{x})$  is a 1-covector. Recall from [Definition 11.2.11](#) that  $(df)_{\mathbf{x}} = Df(\mathbf{x})$  is the same derivative viewed intrinsically as a covector, independent of a chosen basis. We write this intrinsic differential as  $df$ . In one coordinate, its coordinate expression is

$$df = f'(x) \, dx$$

and in standard coordinates on  $\mathbb{R}^n$  the same covector field has the expansion

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \, dx_i.$$

This is the direct chain from the Part I derivative, through the Fréchet derivative of Chapter 11, to a differential 1-form.

There are two sorts of inputs to keep separate. A form  $\omega : p \mapsto \omega_p$  varies with its base point  $p$ . At fixed  $p$ , the evaluation  $\omega_p(v_1, \dots, v_k)$  is multilinear and alternating in  $k$  tangent vectors, and outputs a scalar. For example  $\omega = x^2 \, dy$  has degree one although its coefficient is quadratic. At  $p = (a, b)$ , for the *single vector*  $v = (v_1, v_2)$ ,

$$\omega_p(v) = a^2 v_2.$$

Degree counts tangent-vector inputs, not polynomial powers or coordinate variables. A two-form is not merely a function of two variables. Superscripts in  $e^i$ ,  $h^i$ , and  $dx^I$  label covectors, components, or increasing index tuples; in these uses they are not powers. The space  $\Lambda^k((\mathbb{R}^n)^*)$  describes finite-dimensional covector data at one point;  $\Omega^k(U)$  describes smoothly varying coefficient *functions* on  $U$ .

**Definition 13.7.2** (Differential forms). Let  $U \subseteq \mathbb{R}^n$  be open and let

$$0 \leq k \leq n.$$

A **smooth  $k$ -form** on  $U$  is a function that assigns to each

$$\mathbf{x} \in U$$

an element of

$$\Lambda^k((\mathbb{R}^n)^*)$$

whose coordinate coefficients are smooth. By [Theorem 13.6.4](#), it has a unique expression

$$\omega = \sum_{1 \leq i_1 < \dots < i_k \leq n} a_{i_1 \dots i_k}(\mathbf{x}) \, dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

Here the sum is over the increasing multi-indices

$$I = (i_1, \dots, i_k), \quad 1 \leq i_1 < \dots < i_k \leq n,$$

and we abbreviate

$$dx_I = dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

By convention,

$$\Lambda^0((\mathbb{R}^n)^*) = \mathbb{R},$$

so a smooth 0-form is simply a smooth real-valued function. The integer

$$k$$

is the **degree** of a  $k$ -form. Write  $\Omega^k(U)$  for the vector space of smooth  $k$ -forms, with pointwise operations. For  $k > n$  set  $\Omega^k(U) = \{0\}$ , consistent with the empty increasing-index basis. Thus the derivative of a top-degree form and a two-form pulled back to a curve have a specified zero target.

In

$$\omega = \sum_I a_I(\mathbf{x}) \, dx_I,$$

each  $dx_I$  is a fixed coordinate  $k$ -covector basis element at every point;  $a_I(\mathbf{x})$  is a scalar coefficient varying with position. It is their combination  $\omega_x$  that is a smoothly varying  $k$ -covector. Every differential of a smooth scalar function is a 1-form, but a general 1-form need not globally equal  $df$  for one scalar function. A coordinate expansion alone makes no such assertion.

For forms of degrees  $p, q$ , define their wedge pointwise by  $(\alpha \wedge \beta)_x = \alpha_x \wedge \beta_x$ . Expanding in the coordinate wedge bases, its coefficients are finite sums of signed products of smooth coefficients. They are therefore smooth, so this operation maps  $\Omega^p(U) \times \Omega^q(U)$  into  $\Omega^{p+q}(U)$ . The algebra rules of [Proposition 13.6.9](#) hold at every point and hence as identities of forms. This includes multiplication by smooth 0-forms.

**Regularity convention.** A continuous  $k$ -form is defined in the same way as a smooth  $k$ -form, except that its coefficient functions are continuous. Exterior differentiation below remains restricted to the stated smoothness class. Integration will also be defined for continuous top-degree coefficients.

**Proposition 13.7.3** (Differential of a function). For a smooth function  $f : U \rightarrow \mathbb{R}$ ,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \, dx_i.$$

At each  $\mathbf{x} \in U$ , the form  $(df)_{\mathbf{x}}$  is the derivative  $Df(\mathbf{x})$ .

*Proof.* The covector identity is [Definition 11.2.11](#). For smooth  $f$ , its partial derivatives are smooth coefficient functions, so the pointwise covectors form the asserted smooth 1-form.  $\square$

**Definition 13.7.4** (Pullback). Let

$$h : U \longrightarrow V$$

be a smooth map, where

$$U \subseteq \mathbb{R}^n, \quad V \subseteq \mathbb{R}^m$$

are open. For a  $k$ -form  $\omega$  on  $V$ , its **pullback** is the smooth  $k$ -form

$$h^*\omega$$

on  $U$  defined by

$$(h^*\omega)_{\mathbf{x}}(v_1, \dots, v_k) = \omega_{h(\mathbf{x})}(Dh(\mathbf{x})v_1, \dots, Dh(\mathbf{x})v_k).$$

Composition in each input with the linear map  $Dh(\mathbf{x})$  preserves multilinearity and alternation, so the value is a  $k$ -covector. Its coefficient on an increasing source basis tuple is its evaluation on that tuple by [Theorem 13.6.4](#). Expanding the vectors  $Dh(\mathbf{x})e_i$  in the target basis expresses this coefficient as a finite sum of products of coefficients of  $\omega$  composed with  $h$  and entries of  $Dh$ . Hence all coefficients are smooth. For  $k = 0$  the definition is composition of functions; for  $k > n$  input dependence makes the pullback zero.

For target coordinates  $y^j$  and  $h = (h^1, \dots, h^m)$ , pullback of a function is composition, so  $h^*y^j = h^j$ . Evaluation on a source tangent vector  $v$  proves the differential rule:

$$\begin{aligned}(h^*dy^j)_u(v) &= dy^j_{h(u)}(Dh(u)v) \\ &= (Dh(u)v)_j = Dh^j(u)v.\end{aligned}$$

Thus  $h^*(dy^j) = d(h^j)$ , rather than an unproved substitution rule. Pullback transforms *both* coefficient functions and differential factors. It exists even for noninvertible maps; invariance of integrals under reparametrization needs stronger hypotheses. For a  $C^1$  map and continuous form the same formula defines a continuous pullback: its coefficients are finite sums of products of continuous coefficients composed with  $h$  and first derivatives of  $h$ .

Writing source coordinates as  $x_j$ , the rule just proved is

$$h^*(dy^i) = dh^i = \sum_{j=1}^n \partial_j h^i \, dx_j.$$

These coefficients are row  $i$  of  $J_h$ . This is the coordinate rule for covector pullback from [Definition 10.16.5](#), applied at each base point to the linear map  $Dh$ . For a smooth curve  $\gamma(t) = (x_1(t), \dots, x_n(t))$ , it yields

$$\gamma^*(dx_i) = d(x_i(t)) = x'_i(t) \, dt, \quad \gamma^*(df) = d(f \circ \gamma).$$

Indeed the coefficient of the latter pullback is  $(df)_{\gamma(t)}(\gamma'(t)) = (f \circ \gamma)'(t)$  by [Proposition 11.2.15](#). The familiar writing  $dx_i = x'_i(t) \, dt$  along a curve abbreviates this pullback identity, not an equality between infinitesimal real numbers.

**Example 13.7.5** (Pullback as substitution and directions). For  $\omega = y \, dx + x \, dy$  and  $\gamma(t) = (t, t^2)$ ,

$$\gamma^*x = t, \quad \gamma^*y = t^2, \quad \gamma^*(dx) = dt, \quad \gamma^*(dy) = 2t \, dt.$$

Substitution into the coefficients and application of  $D\gamma$  to directions therefore give

$$\gamma^*\omega = t^2 \, dt + t(2t \, dt) = 3t^2 \, dt, \quad \int_0^1 3t^2 \, dt = 1.$$

The output is a 1-form on the parameter line, not a form on the plane. For the running form  $\omega_0 = x \, dy - y \, dx$ , the circle gives  $\gamma^*\omega_0 = dt$ . If using injective curve patches, split the circle into the two semicircles  $[0, \pi]$  and  $[\pi, 2\pi]$ , each injective, and add their integrals to obtain  $2\pi$ .

**Exercise 4.** If  $h : \mathbb{R}^2 \rightarrow \mathbb{R}^3$  is smooth and  $\eta$  is a 2-form on  $\mathbb{R}^3$ , what are the degree and domain of  $h^*\eta$ ? What happens for a 3-form? Compute the pullback of the flux form of  $(0, 0, z)$  under  $h(u, v) = (u, v, u + 2v)$  and compare with the worked plane-patch flux calculation.

**Remark 13.7.6.** Pullback is the form-theoretic version of substitution. Let  $h : U \rightarrow V$  be smooth, with  $U, V \subseteq \mathbb{R}^n$  open. For the standard volume form,

$$h^*(dy_1 \wedge \cdots \wedge dy_n) = dh_1 \wedge \cdots \wedge dh_n = \det Dh(\mathbf{x}) \, dx_1 \wedge \cdots \wedge dx_n.$$

To prove this directly, evaluate the pullback on  $v_1, \dots, v_n$ . The result is

$$\det[Dh(\mathbf{x})v_1 \cdots Dh(\mathbf{x})v_n] = \det Dh(\mathbf{x}) \det[v_1 \cdots v_n].$$

The middle wedge gives the same evaluation. Thus the determinant is the coefficient produced by pulling oriented coordinate volume back to the parameter domain. This interprets the Chapter 12 change-of-variables theorem; it does not replace its Riemann/Jordan proof. Scalar volume uses  $|\det Dh|$ , whereas the oriented form retains  $\det Dh$  and its sign.

**Proposition 13.7.7** (Pullback rules). For smooth maps

$$U \xrightarrow{h} V \xrightarrow{g} W,$$

and forms of compatible degree,

$$(g \circ h)^* = h^* \circ g^*, \quad h^*(\alpha \wedge \beta) = h^*\alpha \wedge h^*\beta.$$

*Proof.* The chain rule from [Proposition 11.2.7](#) gives

$$D(g \circ h)(\mathbf{x}) = Dg(h(\mathbf{x}))Dh(\mathbf{x}).$$

For an arbitrary  $k$ -form  $\omega$  on  $W$ , evaluate both proposed pullbacks at  $\mathbf{x} \in U$  on  $v_1, \dots, v_k$ . The left side gives  $\omega_{g(h(\mathbf{x}))}(D(g \circ h)(\mathbf{x})v_1, \dots, D(g \circ h)(\mathbf{x})v_k)$ . The chain rule makes this the same scalar as the right side:

$$\omega_{g(h(\mathbf{x}))}(Dg(h(\mathbf{x}))Dh(\mathbf{x})v_1, \dots, Dg(h(\mathbf{x}))Dh(\mathbf{x})v_k).$$

Both forms already exist by the pullback definition; equality on every tuple proves their equality. The reversal of order comes from first evaluating  $g^*\omega$  on the vectors  $Dh(\mathbf{x})v_i$ . For the wedge rule, its alternating finite sum has the same terms on both sides after each argument is replaced by  $Dh(\mathbf{x})v_i$ , proving the second identity.  $\square$

**Example 13.7.8** (Polar pullback: area and circulation). Use the familiar map  $h : (0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}^2$ ,  $h(r, \theta) = (r \cos \theta, r \sin \theta)$ . The coordinate rule gives

$$h^*(dx) = \cos \theta \, dr - r \sin \theta \, d\theta,$$

$$h^*(dy) = \sin \theta \, dr + r \cos \theta \, d\theta.$$

Pullback respects wedges by [Proposition 13.7.7](#). Expanding once,

$$\begin{aligned} h^*(dx \wedge dy) &= \cos \theta \sin \theta \, dr \wedge dr + r \cos^2 \theta \, dr \wedge d\theta \\ &\quad - r \sin^2 \theta \, d\theta \wedge dr - r^2 \sin \theta \cos \theta \, d\theta \wedge d\theta \\ &= r(\cos^2 \theta + \sin^2 \theta) \, dr \wedge d\theta = r \, dr \wedge d\theta. \end{aligned}$$

Repeated factors vanish and reversing the middle wedge changes its sign. This is the oriented-form version of the polar Jacobian factor in [Example 11.2.20](#) and Chapter 12. For the running circulation form  $\omega_0 = x \, dy - y \, dx$ , transform the coefficient functions as well as the differential factors:

$$\begin{aligned} h^*\omega_0 &= r \cos \theta (\sin \theta \, dr + r \cos \theta \, d\theta) \\ &\quad - r \sin \theta (\cos \theta \, dr - r \sin \theta \, d\theta) = r^2 \, d\theta. \end{aligned}$$

For  $R > 0$ , pulling this back further by  $\theta \mapsto (R, \theta)$  gives  $R^2 \, d\theta$ . The classical line-integral formula therefore yields

$$\int_{\gamma} \omega_0 = \int_0^{2\pi} R^2 \, d\theta = 2\pi R^2$$

for one counterclockwise circle, recovering the running example at  $R = 1$ . Pullback exists despite the polar map's lack of global injectivity. Identifying region integrals by Chapter 12's change-of-variables theorem has separate domain and injectivity hypotheses; this computation does not dispense with them.

### Why differentiate forms?

The familiar operation  $f \mapsto df$  already produces a 1-form. Once general covector fields such as  $\omega = P \, dx + Q \, dy$  are admitted, how should we differentiate an object that already measures a direction? Functions live on points, 1-forms are integrated along curves, and 2-forms over oriented regions or surfaces. A boundary has one fewer dimension than its interior, so a derivative relating their integrals must raise degree:

$$\Omega^0(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \xrightarrow{d} \dots$$

Green's theorem suggests the planar rule

$$d(P \, dx + Q \, dy) = (Q_x - P_y) \, dx \wedge dy.$$

Its coefficient is the local circulation quantity already proved to convert boundary work into an interior integral. We seek the higher-degree operation that makes

$$\int_{\partial M} \omega = \int_M d\omega$$

type-correct: an  $(n-1)$ -form on the boundary corresponds to an  $n$ -form in the interior. This motivates the following definition; the product and coordinate-change rules still require proof below.

**Definition 13.7.9** (Exterior derivative). For a smooth  $k$ -form

$$\omega = \sum_I a_I \, dx_I$$

on an open subset of  $\mathbb{R}^n$ , define its **exterior derivative** by

$$d\omega = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x_j} dx_j \wedge dx_I.$$

**Remark 13.7.10.** The exterior derivative raises degree by one:

$$\text{functions} \longrightarrow 1\text{-forms} \longrightarrow 2\text{-forms} \longrightarrow \cdots .$$

It generalizes ordinary differentiation while retaining the product rule. The identities  $d^2 = 0$  and pullback-commutation below are the two algebraic facts that allow a local integral identity to pass through coordinates and then to global manifolds.

For our running example,

$$d(x \, dy - y \, dx) = dx \wedge dy - dy \wedge dx = 2 \, dx \wedge dy.$$

Green's area formula is thus already the statement that the boundary integral of  $\omega_0$  equals the interior integral of  $d\omega_0$ .

**Theorem 13.7.11** (Fundamental exterior-derivative rules). For smooth forms  $\alpha, \beta$  of fixed degrees,

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge d\beta,$$

$$d(d\alpha) = 0, \quad h^*(d\omega) = d(h^*\omega).$$

*Proof.* For elementary coordinate forms  $\alpha = a \, dx_I$  and  $\beta = b \, dx_J$  with  $|I| = p$ , the coefficient product rule gives

$$d(\alpha \wedge \beta) = d(ab) \wedge dx_I \wedge dx_J = b \, da \wedge dx_I \wedge dx_J + a \, db \wedge dx_I \wedge dx_J.$$

The first term is  $d\alpha \wedge \beta$ . To identify the second, move the one-form  $db$  past the  $p$  one-forms in  $dx_I$ :  $dx_I \wedge db = (-1)^p db \wedge dx_I$ . Thus  $(-1)^p \alpha \wedge d\beta$  is precisely the second term, since  $(-1)^{2p} = 1$ . If indices repeat both expressions still obey the same alternating identities. Every form is a finite sum of elementary coordinate forms; bilinearity and linearity of  $d$  extend the rule to all forms. For the second identity write

$$\alpha = \sum_I a_I \, dx_I.$$

Applying the definition twice gives the double sum

$$d(d\alpha) = \sum_I \sum_{j,k} \frac{\partial^2 a_I}{\partial x_k \partial x_j} \, dx_k \wedge dx_j \wedge dx_I.$$

If a differential is repeated, the corresponding term vanishes because

$$dx_j \wedge dx_j = 0.$$

For distinct indices, the terms indexed by  $(j, k)$  and  $(k, j)$  pair: their coefficients agree by [Theorem 11.3.9](#), while

$$dx_k \wedge dx_j = -dx_j \wedge dx_k.$$

They therefore cancel, proving  $d^2 = 0$ .

For a smooth function  $f$ , the multivariable chain rule gives

$$h^*(df) = d(f \circ h).$$

In particular, pullback commutes with  $d$  on coordinate functions. For an elementary term  $\omega = f \, dy_{i_1} \wedge \cdots \wedge dy_{i_k}$ , its pullback is  $(f \circ h) \, dh^{i_1} \wedge \cdots \wedge dh^{i_k}$ . The Leibniz rule and  $d^2 h^i = 0$  show that its derivative is  $d(f \circ h) \wedge dh^{i_1} \wedge \cdots \wedge dh^{i_k}$ , exactly the pullback of  $d\omega$ . Add the finitely many elementary terms.  $\square$

**Remark 13.7.12** (The running form is not a global differential). For  $\omega_0 = x \, dy - y \, dx$  on  $\mathbb{R}^2$ ,

$$d\omega_0 = 2 \, dx \wedge dy \neq 0.$$

Indeed its value on  $(e_1, e_2)$  is 2. If  $\omega_0 = df$  for a smooth global scalar function, [Theorem 13.7.11](#) would give  $d\omega_0 = d(df) = 0$ , a contradiction. Thus the earlier warning about general 1-forms has a concrete example.

**Example 13.7.13.** In  $\mathbb{R}^3$ , write

$$\omega = P \, dx + Q \, dy + R \, dz.$$

Then

$$d\omega = \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy + \left( \frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) dx \wedge dz + \left( \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy \wedge dz.$$

The three coefficients encode the usual curl.

**Distinguish  $D^2f$  from  $d^2f$ .** For a smooth scalar function  $f$ ,  $D^2f$  is the second Fréchet derivative, represented by the Hessian, and is generally nonzero. By contrast,  $d^2f = d(df) = 0$ . In coordinates,

$$df = \sum_i \partial_i f \, dx_i, \quad d(df) = \sum_{i,j} \partial_j \partial_i f \, dx_j \wedge dx_i.$$

Terms with  $i = j$  vanish; terms with  $i \neq j$  cancel in pairs because the second partial coefficients are symmetric and the wedge is alternating. Thus symmetric coefficients combined with alternating basis elements produce cancellation, not a vanishing Hessian.

**Example 13.7.14** (The second derivative and  $d(df)$  have different types). For  $f(x, y) = x^2 + y^2$ ,  $D^2f(p)[u, v] = 2(u_1v_1 + u_2v_2)$  is a nonzero symmetric bilinear map. But  $df = 2x \, dx + 2y \, dy$ , and

$$d(df) = 2 \, dx \wedge dx + 2 \, dy \wedge dy = 0.$$

The exterior derivative antisymmetrizes coefficient derivatives. Equality of mixed partials together with alternation forces  $d^2 = 0$ ; it does not assert that the symmetric second derivative vanishes.

### One form through all its types

Keep  $\omega = x \, dy - y \, dx$  as a running example. The form  $\omega$  is a field of covectors on the plane. At  $p = (2, 3)$ , its value is the covector  $\omega_p = 2 \, dy|_p - 3 \, dx|_p$ . On  $v = (4, -1) \in \mathbb{R}^2$ , viewed as a displacement based at  $p$ , it returns the scalar  $\omega_p(v) = 2(-1) - 3(4) = -14$ . Along  $\gamma(t) = (\cos t, \sin t)$ ,

$$(\gamma^*\omega)_t = \omega_{\gamma(t)}(\gamma'(t)) \, dt = (\cos^2 t + \sin^2 t) \, dt = dt.$$

Its exterior derivative is  $d\omega = dx \wedge dy - dy \wedge dx = 2 \, dx \wedge dy$ . Thus  $\omega$ ,  $\omega_p$ , and  $\omega_p(v)$  are respectively a form, a covector at a point, and a number. Exterior differentiation acts on the varying form, not on an isolated numerical evaluation.

## Computing pullbacks on a surface

For  $h : U \rightarrow V$ , the evaluation rule reads

$$\begin{array}{ccc} (h_1, \dots, h_k) \text{ at } u & \xrightarrow{Dh(u)} & (Dh(u)h_1, \dots, Dh(u)h_k) \text{ at } h(u) \\ & & \downarrow \omega_{h(u)} \\ & & (h^*\omega)_u(h_1, \dots, h_k) \in \mathbb{R} \end{array}$$

First move the source tangent vectors forward, then evaluate the target form. The resulting rule is a covector on the source. The degree  $k$  is preserved and the base domain changes. Invertibility is unnecessary. If the source dimension is smaller than  $k$ , all  $k$  input vectors are dependent and the pulled-back  $k$ -form is zero.

**Example 13.7.15** (Pulling a two-form onto a graph). Let  $h(u, v) = (u, v, u^2 + v^2)$  and take  $\eta = z \, dx \wedge dy + x \, dy \wedge dz$  on  $\mathbb{R}^3$ . Substitute both coefficients and covectors:

$$h^*z = u^2 + v^2, \quad h^*x = u, \quad h^*dx = du, \quad h^*dy = dv, \quad h^*dz = 2u \, du + 2v \, dv.$$

Consequently

$$h^*\eta = (u^2 + v^2) \, du \wedge dv + u \, dv \wedge (2u \, du + 2v \, dv) = (v^2 - u^2) \, du \wedge dv.$$

Over  $[0, 1]^2$  its integral is

$$1/3 - 1/3 = 0.$$

The map from a plane into space is not invertible, yet every step of the pullback is defined. As a second example, pulling  $dx \wedge dy$  onto any curve gives zero, a degree-two form on a one-dimensional source.

## 13.8 Integration and the Local Stokes Theorem

A top-degree form on a coordinate region has one coefficient, which can be integrated by the existing multiple Riemann integral. The boundary has one fewer dimension. We first establish the boundary identity on a box and a half-space, where the Fundamental Theorem acts in one coordinate at a time; dimension zero needs the signed-sum convention below.

For a signed finite set, use the convention

$$\int_{\{p_1, \dots, p_r\}} f = \sum_{j=1}^r \epsilon_j f(p_j), \quad \epsilon_j \in \{+1, -1\}.$$

With endpoint signs  $\partial[a, b] = \{b\} - \{a\}$  this is  $f(b) - f(a)$ . The manifold interpretation and orientation signs will be justified later.

Before the higher-dimensional statement, recall its one-dimensional model. For a smooth function on an oriented interval,

$$\int_{[a, b]} df = f(b) - f(a) = \int_{\partial[a, b]} f.$$

The local theorem is the same assertion on a rectangle: its proof applies the Part I Fundamental Theorem of Calculus one coordinate at a time.



**Definition 13.8.1** (Integration of top-degree forms). Let  $Q \subseteq \mathbb{R}^n$  be a rectangle with the standard orientation. For a continuous  $n$ -form

$$\omega = f \, dx_1 \wedge \cdots \wedge dx_n,$$

define

$$\int_Q \omega = \int_Q f(\mathbf{x}) \, d\mathbf{x}.$$

Reversing the orientation changes the sign of the integral.

Thus for standard orientation, integrating  $f(\mathbf{x}) \, dx_1 \wedge \cdots \wedge dx_n$  corresponds to the scalar Riemann integral

$$\int_Q f(\mathbf{x}) \, d(x_1, \dots, x_n)$$

of Chapter 12.

For a smooth parametrized curve  $\gamma : [a, b] \rightarrow \mathbb{R}^n$  and a smooth 1-form

$$\omega = \sum_i a_i \, dx_i$$

near its image, this becomes

$$\begin{aligned} \gamma^* \omega &= \sum_i a_i(\gamma(t)) x'_i(t) \, dt, \\ \int_\gamma \omega &:= \int_{[a,b]} \gamma^* \omega = \int_a^b \sum_i a_i(\gamma(t)) x'_i(t) \, dt. \end{aligned}$$

Thus integration of 1-forms recovers the earlier work integral, including its dependence on traversal and orientation. The scalar notation specifies unsigned volume; the wedge specifies an oriented covector. Reversing orientation negates the form integral, not the scalar volume integral. Consequently scalar substitution uses an absolute determinant while oriented pullback retains its sign.

**Integration over an open set.** Let  $V \subseteq \mathbb{R}^n$  be open, and let

$$\omega = f \, dx_1 \wedge \cdots \wedge dx_n$$

be a continuous top-degree form whose support

$$K = \text{supp}(\omega) = \overline{\{\mathbf{x} \in V : f(\mathbf{x}) \neq 0\}}$$

is compact and contained in  $V$ , where the closure is taken in  $V$ . By [Proposition 10.5.11](#), a compact subset of the open set  $V$  is separated from  $\mathbb{R}^n \setminus V$  (by a positive distance when  $K \neq \emptyset$ ), there is a neighborhood of  $\mathbb{R}^n \setminus V$  disjoint from  $K$ . The function  $f$  vanishes on the part of that neighborhood lying in  $V$ . Consequently, extending  $f$  by zero outside  $V$  gives a continuous function  $\tilde{f}$ . Choose any rectangle  $Q$  whose interior contains  $K$ , and define

$$\int_V \omega = \int_Q \tilde{f}(\mathbf{x}) \, d\mathbf{x}.$$

This is independent of  $Q$ . Indeed, after enlarging two such rectangles to a common containing rectangle, finite additivity of the Riemann integral and the vanishing of  $\tilde{f}$  off  $K$  show that both integrals equal the integral over the larger rectangle. If  $\omega$  is smooth, then the extended coefficients are identically zero on a neighborhood of the artificial boundary of  $V$ , so the same zero extension is smooth.

**Definition 13.8.2** (Integration over a relatively open half-space set). For  $O \subseteq \mathbb{H}^n$  relatively open and a continuous top form  $\eta = f \, dx_1 \wedge \cdots \wedge dx_n$  compactly supported in  $O$ , extend  $f$  by zero within  $\mathbb{H}^n$  outside  $O$ . This is continuous because the support is compact and avoids artificial edges. Define

$$\int_O \eta = \int_{[a_1, b_1] \times \cdots \times [a_{n-1}, b_{n-1}] \times [0, b_n]} \tilde{f}(\mathbf{x}) \, d\mathbf{x},$$

where the half-rectangle contains the support in its relative interior. The scalar construction and independence are those of [Corollary 12.5.10](#), using rectangular zero extension and gluing. No zero extension across the genuine boundary  $x_n = 0$  is used. For smooth coefficients local extensions across that boundary need not vanish.

**Definition 13.8.3** (Boundary orientation). The boundary of the standard rectangle is oriented by the **outward-normal-first rule**: a basis of a boundary face is positive when adjoining an outward normal before it gives the orientation of the rectangle. The integral over the boundary is the signed sum over its faces. Restriction means pullback by inclusion: if  $\iota : \partial Q \rightarrow Q$  is the inclusion,

$$\int_{\partial Q} \omega$$

abbreviates

$$\int_{\partial Q} \iota^* \omega,$$

interpreted face by face here.

For the standard square the outward-normal-first calculation gives: bottom edge  $(-e_2, e_1)$  is positive, right edge  $(e_1, e_2)$  is positive, top edge  $(e_2, -e_1)$  is positive, and left edge  $(-e_1, -e_2)$  is positive. Thus the boundary goes right, up, left, down: counterclockwise. Rectangles and cubes have corners. The theorem below is a facewise result; they are not automatically smooth-boundary examples under our later definitions.

Restriction to a face is pullback along its inclusion. For  $i(t) = (t, 0)$  and  $\omega = P(x, y) \, dx + Q(x, y) \, dy$ ,

$$i^* \omega = P(t, 0) \, dt, \quad i^*(dy) = d(0) = 0.$$

It evaluates coefficients at boundary points *and* restricts covectors to boundary tangent directions. Merely setting  $y = 0$  while retaining all differential symbols would be wrong. More generally a term containing  $dx_j$  pulls back to zero on an  $x_j$ -constant face.

**Theorem 13.8.4** (Stokes theorem on a rectangle). Let  $Q \subseteq \mathbb{R}^n$  be a rectangle and let  $\omega$  be a smooth

$(n - 1)$ -form

on an open neighborhood of  $Q$ . Then

$$\int_Q d\omega = \int_{\partial Q} \omega.$$

*Proof.* Write

$$\omega = \sum_{j=1}^n (-1)^{j-1} f_j \, dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_n.$$

Only

$$\frac{\partial f_j}{\partial x_j}$$

contributes to the coefficient of

$$dx_1 \wedge \cdots \wedge dx_n$$

in  $d\omega$ . Apply the one-variable Fundamental Theorem of Calculus in the  $x_j$  variable and then [Theorem 12.2.1](#) to integrate the remaining variables. To verify the signs, put  $E_j = (e_1, \dots, \widehat{e}_j, \dots, e_n)$ , the standard ordered face frame. The ordered frame  $(e_j, E_j)$  has ambient sign  $(-1)^{j-1}$ , since moving  $e_j$  into position  $j$  uses  $j-1$  swaps. On the upper face  $x_j = b_j$  the outward normal is  $e_j$ , so its induced face orientation has sign  $(-1)^{j-1}$  relative to  $E_j$ . The coefficient of the  $j$ th term of  $\omega$  has this same sign; multiplying gives  $+f_j$ . On the lower face  $x_j = a_j$  the outward normal is  $-e_j$ , reversing the induced face sign to  $(-1)^j$ . Multiplication by the coefficient  $(-1)^{j-1}$  gives  $-f_j$ . Every other term restricts to zero on these faces because it contains  $dx_j$ . The paired face integral is therefore

$$\int_{\prod_{i \neq j} [a_i, b_i]} (f_j(\dots, b_j, \dots) - f_j(\dots, a_j, \dots)) dx_1 \cdots \widehat{dx_j} \cdots dx_n,$$

exactly the endpoint difference from the FTC. Summing over  $j$  proves the formula.  $\square$

**Definition 13.8.5** (Parametrized integration). Let  $D \subseteq \mathbb{R}^k$  be a compact rectangle and let

$$\sigma : D \longrightarrow \mathbb{R}^N$$

be a smooth injective parametrization with injective derivative on the interior. For a continuous  $k$ -form  $\omega$  on a neighborhood of  $\sigma(D)$ , define

$$\int_{\sigma(D)} \omega = \int_D \sigma^* \omega,$$

where the image carries the orientation induced by  $\sigma$ .

To see the scalar integral inside this definition, keep parameter dimension  $k$  and ambient dimension  $N$  fixed. Every top-degree form on  $D$  is a scalar coefficient times  $du_1 \wedge \cdots \wedge du_k$ . Evaluate  $\sigma^* \omega$  on  $(e_1, \dots, e_k)$ ; since  $D\sigma(u)e_j = \partial_j \sigma(u)$ , its coefficient is

$$g(u) = \omega_{\sigma(u)}(\partial_1 \sigma(u), \dots, \partial_k \sigma(u)).$$

Consequently

$$\sigma^* \omega = g(u) du_1 \wedge \cdots \wedge du_k, \quad \int_{\sigma(D)} \omega = \int_D g(u) d(u_1, \dots, u_k).$$

The coefficient is continuous, so its compact-rectangle Riemann integral exists. At each parameter point, compute the geometric point and tangent vectors, evaluate the form on them, and integrate the resulting scalar function. For a graph  $\sigma(u, v) = (u, v, h(u, v))$  with its parameter orientation,  $\sigma^*(dx \wedge dy) = du \wedge dv$ , so

$$\int_{\sigma(D)} dx \wedge dy = \text{area}(D), \quad \text{area}(\sigma(D)) = \int_D \sqrt{1 + h_u^2 + h_v^2} d(u, v).$$

A two-form therefore does not automatically represent unsigned surface area.

**Proposition 13.8.6** (Change of parametrization). Keep  $\sigma : D \rightarrow \mathbb{R}^N$  and the continuous  $k$ -form  $\omega$  as above. Let  $D, D' \subseteq \mathbb{R}^k$  be compact rectangles, and suppose that  $\psi : D' \rightarrow D$  is the restriction of a smooth diffeomorphism between open neighborhoods of these rectangles, with  $\psi(D') = D$ . Replacing  $\sigma$  by  $\sigma \circ \psi$  leaves the integral unchanged when  $\det D\psi > 0$  on  $D'$ , and changes its sign when  $\det D\psi < 0$  on  $D'$ .

*Proof.* Pullback functoriality gives

$$(\sigma \circ \psi)^*\omega = \psi^*(\sigma^*\omega).$$

Write  $\sigma^*\omega = g(\mathbf{x}) \, dx_1 \wedge \cdots \wedge dx_k$ . Then

$$\psi^*(\sigma^*\omega) = g(\psi(u)) \det D\psi(u) \, du_1 \wedge \cdots \wedge du_k.$$

Because  $D'$  is connected and the determinant is continuous and nowhere zero, its sign is constant. The scalar change-of-variables theorem [Theorem 12.5.7](#) therefore gives

$$\int_{D'} g(\psi(u)) \det D\psi(u) \, du = \begin{cases} \int_D g(\mathbf{x}) \, d\mathbf{x}, & \det D\psi > 0, \\ -\int_D g(\mathbf{x}) \, d\mathbf{x}, & \det D\psi < 0. \end{cases}$$

These are precisely the asserted orientation-preserving and orientation-reversing cases.  $\square$

**Proposition 13.8.7** (Invariance on chart overlaps). For an orientation-preserving smooth diffeomorphism  $\psi : V' \rightarrow V$  and a continuous compactly supported top-degree form  $\eta$  on  $V$ ,

$$\int_{V'} \psi^*\eta = \int_V \eta.$$

This holds both for open Euclidean sets and for relatively open half-space sets with boundary-preserving transitions.

*Proof.* Write  $\eta = f \, dx_1 \wedge \cdots \wedge dx_n$ . The pullback coefficient is  $f(\psi) \det D\psi$ , and the determinant is positive. Apply [Theorem 12.5.4](#) or [Corollary 12.5.10](#). These statements cover arbitrary compact support on an overlap, unlike the rectangular reparametrization proposition by itself.  $\square$

## 13.9 Charts, Atlases, and Smooth Manifolds

The local Stokes theorem works inside one coordinate region. To pass to a whole space, we first specify coordinates and then compare them on overlaps.

**The local-to-global roadmap.** A manifold may have complicated global shape and no single coordinate system covering it. Every point nevertheless has a neighborhood modeled on Euclidean space. We can do familiar calculus there; the additional work is to prove that the answers agree on overlaps. This is why smooth manifolds let calculus reach spaces with complicated global geometry:

$$\text{local coordinates} \rightarrow \text{Euclidean calculus} \rightarrow \text{overlap compatibility} \rightarrow \text{global geometry}.$$

Charts supply coordinates; transition maps compare them. Tangent spaces record directions independently of coordinates, and forms measure those directions. Orientation supplies a consistent sign. Partitions of unity reduce global problems to chartwise ones. Integration is defined locally and glued globally; Stokes is proved locally and glued globally.

**First reading.** Follow the conceptual route

$$\text{chart} \rightarrow T_p M \rightarrow T_p^* M \rightarrow 1\text{-form} \rightarrow k\text{-form} \rightarrow \text{orientation} \rightarrow \text{integration} \rightarrow \text{Stokes}.$$

The refinement, extension, and independence-of-choices proofs provide essential supporting infrastructure. Read their statements and their role now; their technical details may take a slower second pass. This separates the conceptual structure from its supporting proofs without making the main theorems optional.

**Definition 13.9.1** (Locally Euclidean space). A topological space  $M$  is called **locally Euclidean of dimension**  $n$  if for every  $x \in M$ , there is an open neighborhood  $U \subseteq M$  of  $x$ , an open subset  $V \subseteq \mathbb{R}^n$ , and a homeomorphism  $\varphi : U \rightarrow V$ .

Each point therefore has a Euclidean-type open neighborhood; the whole space need not be globally Euclidean. The neighborhood convention is [Definition 10.4.1](#). The local criterion of [Definition 10.4.5](#) also explains chart restriction: inside any open neighborhood of  $p$ , intersect with a chart domain, take a small Euclidean ball around  $\varphi(p)$  in its image, and pull the ball back. Such smaller coordinate neighborhoods form a topological basis. Here  $U$  is an open neighborhood in the space, whereas  $V$  is its Euclidean coordinate image. The map  $\varphi$  assigns coordinates; its inverse parametrizes the neighborhood. Openness of  $U$  refers to  $M$ 's topology, not to an ambient space in which  $M$  might be drawn.

**Definition 13.9.2** (Coordinate chart). An  $n$ -dimensional **coordinate chart** is a pair

$$(U, \varphi), \quad \varphi : U \xrightarrow{\cong} \varphi(U) \subseteq \mathbb{R}^n,$$

where  $U$  is open in  $M$ ,  $\varphi(U)$  is open in  $\mathbb{R}^n$ , and  $\varphi$  is a homeomorphism. Its **coordinate functions** are

$$x^i = \pi_i \circ \varphi : U \rightarrow \mathbb{R}, \quad \varphi(p) = (x^1(p), \dots, x^n(p)), \quad 1 \leq i \leq n.$$

The projection  $\pi_i$  selects the  $i$ th coordinate; superscripts label coordinates, not powers. Keep the two directions distinct:

$$\text{geometry} \xrightarrow{\varphi} \text{coordinates}, \quad \text{coordinates} \xrightarrow{\varphi^{-1}} \text{geometry}.$$

For example,  $(\mathbb{R}^n, \text{id})$  is a chart; so is  $(U, \text{id}_U)$  for every open  $U \subseteq \mathbb{R}^n$ .

## Overlaps are changes of coordinate system

Two charts can describe the same point with different tuples of numbers.

**Definition 13.9.3** (Transition maps and smooth compatibility). For charts  $(U, \varphi)$  and  $(V, \psi)$  with  $U \cap V \neq \emptyset$ , their **transition maps** are the inverse homeomorphisms

$$\boxed{\psi \circ \varphi^{-1} : \varphi(U \cap V) \longrightarrow \psi(U \cap V)},$$

$$\boxed{\varphi \circ \psi^{-1} : \psi(U \cap V) \longrightarrow \varphi(U \cap V)}.$$

Define **smooth compatibility** by

$$(U, \varphi) \sim_{\text{sm}} (V, \psi) \quad \text{if} \quad \begin{cases} \psi \circ \varphi^{-1} \text{ is a smooth diffeomorphism,} & U \cap V \neq \emptyset, \\ \text{automatically,} & U \cap V = \emptyset. \end{cases}$$

The intersection is open in both chart domains. Homeomorphisms take it to open subsets of the open coordinate images, so both overlap images are open in  $\mathbb{R}^n$ . Smooth diffeomorphism here has its Euclidean meaning: the transition and its inverse are  $C^\infty$ .

$$\begin{array}{ccc}
 & U \cap V \subseteq M & \\
 \varphi \swarrow & & \searrow \psi \\
 \varphi(U \cap V) & \xrightarrow{\psi \circ \varphi^{-1}} & \psi(U \cap V)
 \end{array}$$

The two routes send the same point  $p$  to its two coordinate descriptions; the bottom arrow changes the description, not the point.

**Remark 13.9.4** (Empty overlaps). If  $U \cap V = \emptyset$ , there are no common points whose coordinates need comparison. Equivalently, the transition is the unique map  $\emptyset \rightarrow \emptyset$ . The empty set is open, and this map and its inverse are smooth vacuously, exactly as the definition requires.

**Definition 13.9.5** (Atlas and smooth atlas). An  $n$ -**atlas** is a family of  $n$ -dimensional charts satisfying

$$\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}, \quad M = \bigcup_{\alpha \in A} U_\alpha.$$

It is a **smooth atlas** if, for every  $\alpha, \beta \in A$ ,

$$U_\alpha \cap U_\beta = \emptyset \quad \text{or} \quad \varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta) \text{ is smooth.}$$

Both ordered pairs  $(\alpha, \beta)$  and  $(\beta, \alpha)$  occur, so the condition includes smoothness of the inverse. An atlas supplies coverage; a smooth atlas additionally permits calculus to pass between its charts.

**Definition 13.9.6** (Maximal smooth atlas and smooth structure). A smooth atlas  $\mathcal{A}$  is **maximal** if, for every chart  $(U, \varphi)$ ,

$$[ \forall (V, \psi) \in \mathcal{A}, \quad (U, \varphi) \sim_{\text{sm}} (V, \psi) ] \implies (U, \varphi) \in \mathcal{A}.$$

A **smooth structure** is a chosen maximal smooth atlas:

$$\boxed{\text{smooth structure} = \text{maximal smooth atlas.}}$$

Maximality adds no points or geometry. It makes every coordinate system compatible with the chosen structure available for use. Every smooth atlas determines exactly one maximal one. To check this, adjoin all compatible charts. If two added charts meet at  $p$ , choose an original chart at  $p$ . On a smaller overlap the transition between the added charts factors through that original chart, so is smooth in both directions. Smoothness is local, hence the added charts are pairwise compatible. Any chart compatible with this enlarged atlas was already compatible with the original one and has already been added. This proves maximality and uniqueness of the maximal extension.

Recall that a space is **second countable** if its topology has a countable basis (Definition 10.4.10); a basis means that every open neighborhood of a point contains a basic neighborhood of that point (Definition 10.4.5). Rational balls give such a basis for  $\mathbb{R}^n$  (Example 10.4.7). Recall also that Hausdorff separation means distinct points have disjoint open neighborhoods (Definition 10.5.5).

**Definition 13.9.7** (Topological manifold). A topological space  $M$  is a **topological  $n$ -manifold** if

- (i)  $M$  is Hausdorff,
- (ii)  $M$  is second countable,
- (iii)  $M$  is locally Euclidean of dimension  $n$ .

Hausdorff means that distinct points have disjoint open neighborhoods; it ensures uniqueness of limits. The three axioms have distinct jobs:

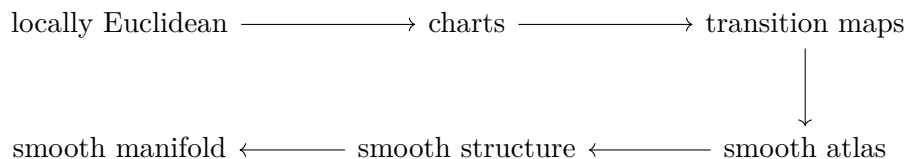
|                   |                                          |
|-------------------|------------------------------------------|
| Hausdorff         | separates distinct points,               |
| second countable  | controls global topological size,        |
| locally Euclidean | supplies local coordinates for calculus. |

**Definition 13.9.8** (Smooth manifold). A **smooth  $n$ -manifold** is a pair  $(M, \mathcal{A})$  with

$$\boxed{M \text{ a topological } n\text{-manifold,} \quad \mathcal{A} \text{ a smooth structure on } M.}$$

Its charts mean the charts belonging to  $\mathcal{A}$ .

The topology specifies continuity; the smooth structure specifies coordinates for differentiation. For  $n = 0$ ,  $\mathbb{R}^0$  is one point, and the manifolds are discrete second-countable spaces.



The last step also requires the Hausdorff and second-countability conditions. The arrows summarize the order of definitions, not implications from local Euclidean structure alone.

## Standard structures and first examples

**Definition 13.9.9** (Standard Euclidean smooth structure). For an open  $U \subseteq \mathbb{R}^n$ , define its **standard smooth structure** by

$$\mathcal{A}_{\text{std}}(U) = \text{the maximal smooth atlas generated by } \{(U, \text{id}_U)\}.$$

The chart  $(U, \text{id}_U)$  is one generating chart; the atlas  $\mathcal{A}_{\text{std}}(U)$  is the smooth structure. The smooth manifold is  $(U, \mathcal{A}_{\text{std}}(U))$ .

In particular the chart  $(\mathbb{R}^n, \text{id}_{\mathbb{R}^n})$  generates  $\mathcal{A}_{\text{std}}(\mathbb{R}^n)$ , and the standard Euclidean manifold is  $(\mathbb{R}^n, \mathcal{A}_{\text{std}}(\mathbb{R}^n))$ . Translations and many other compatible coordinate systems also belong to this maximal atlas.

**Proposition 13.9.10** (Open subsets inherit smooth structures). If  $W \subseteq M$  is open and  $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$  is a smooth atlas for  $M$ , then

$$\mathcal{A}|_W = \{(U_\alpha \cap W, \varphi_\alpha|_{U_\alpha \cap W}) : U_\alpha \cap W \neq \emptyset\}$$

is a smooth atlas on  $W$ . Its maximal extension is the canonical **inherited smooth structure** on  $W$ .

*Proof.* The restricted domains cover  $W$ , and their images are open because each original chart is a homeomorphism. Their transitions are restrictions of the original smooth transitions. Hausdorffness passes to subspaces by restricting separating open neighborhoods; second countability passes by Proposition 10.4.11. Replacing the generating atlas by another for the same structure gives compatible restricted charts, hence the same maximal extension.  $\square$

Thus open subsets of  $\mathbb{R}^n$  are manifolds with their standard structures; the identity restrictions generate exactly those structures.

**Example 13.9.11** (Graphs connect to Chapter 11). For smooth  $f : U \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^q$ ,  $U$  open, set

$$\Gamma_f = \{(\mathbf{x}, f(\mathbf{x})) : \mathbf{x} \in U\}, \quad \varphi : \Gamma_f \rightarrow U, \quad \varphi(\mathbf{x}, f(\mathbf{x})) = \mathbf{x}.$$

With the subspace topology this is a homeomorphism, with inverse

$$\varphi^{-1} : U \rightarrow \Gamma_f, \quad \mathbf{x} \mapsto (\mathbf{x}, f(\mathbf{x})).$$

Subspaces of Euclidean space are Hausdorff; they are second countable by Example 10.4.7 and Proposition 10.4.11. Thus this chart generates a smooth  $m$ -manifold structure. For  $m = 2, q = 1$ , its inverse has derivative columns  $(1, 0, f_{x^1})$ ,  $(0, 1, f_{x^2})$ , the directions already found for graph surfaces. Chapter 11's implicit-function theorem supplies this local model for regular level sets.

**Example 13.9.12** (An atlas for the circle). The upper and lower open semicircles use  $x$  as coordinate; the right and left open semicircles use  $y$ . Their images are  $(-1, 1)$ , with inverses  $u \mapsto (u, \pm\sqrt{1-u^2})$  or  $u \mapsto (\pm\sqrt{1-u^2}, u)$ . These four charts cover  $S^1$ . Nonempty adjacent overlaps have smooth signed square-root transitions away from the endpoints, so they generate a smooth structure. Opposite semicircles are disjoint and compatible.

**Example 13.9.13** (Computing an actual chart transition). On  $U = \{(x, y) \in S^1 : y > 0\}$  use  $\varphi(x, y) = x$ ; on  $V = \{(x, y) \in S^1 : x > 0\}$  use  $\psi(x, y) = y$ . Both full images are  $(-1, 1)$ , but the overlap is the first-quadrant arc. Consequently

$$\psi \circ \varphi^{-1} : (0, 1) \rightarrow (0, 1), \quad u \mapsto \sqrt{1-u^2}.$$

Its derivative is

$$-u/\sqrt{1-u^2},$$

nonzero on  $(0, 1)$ . The inverse has the same square-root formula, with its input renamed. Both are smooth on the overlap images. A negative derivative changes coordinate orientation but does not prevent smooth compatibility.

**Example 13.9.14** (Six charts for the sphere). Give  $S^2 = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = 1\}$  its subspace topology. For  $i = 1, 2, 3$  and  $\epsilon \in \{+1, -1\}$  let

$$U_i^\epsilon = \{\mathbf{x} \in S^2 : \epsilon x_i > 0\}, \quad \varphi_i^\epsilon : U_i^\epsilon \rightarrow B^2, \quad B^2 = \{u \in \mathbb{R}^2 : \|u\| < 1\},$$

where  $\varphi_i^\epsilon$  deletes the  $i$ th coordinate, keeping the other two in increasing order. Its inverse inserts

$$x_i = \epsilon \sqrt{1 - \sum_{j \neq i} x_j^2}.$$

Both maps are continuous and inverse. Each  $U_i^\epsilon$  is open in  $S^2$ , and the six sets cover it because a unit vector has a nonzero coordinate.



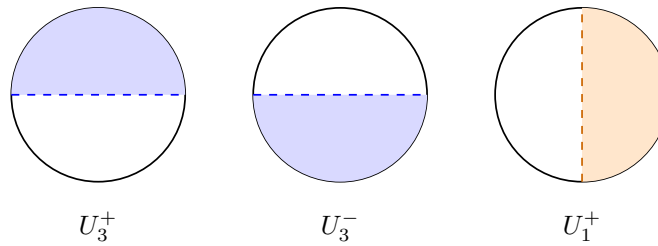


Figure 13.11: Front-view silhouettes of three hemisphere domains; dashed great-circle boundaries are excluded. Upper and lower hemispheres miss the equator, so side hemispheres are needed too. The six domains  $U_i^\pm$  cover the sphere and overlap.

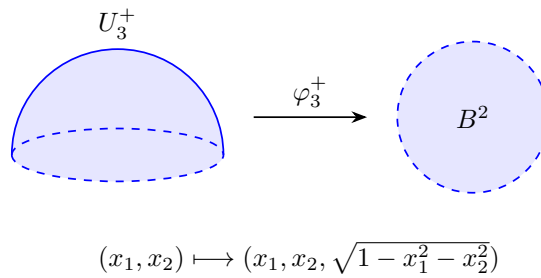


Figure 13.12: Deleting the third coordinate maps the open upper hemisphere to the open unit disk. The formula below reconstructs the hemisphere; both boundary circles are excluded.

For a concrete transition, use the upper hemisphere  $U_3^+$  and the right hemisphere  $U_1^+$ . In  $\varphi_3^+$  coordinates write  $u = (a, b) = (x_1, x_2)$ . Then

$$\varphi_1^+ \circ (\varphi_3^+)^{-1}(a, b) = (b, \sqrt{1 - a^2 - b^2}), \quad a > 0, \quad a^2 + b^2 < 1.$$

Its image is  $\{(b, c) : c > 0, \quad b^2 + c^2 < 1\}$ , and its inverse is

$$(b, c) \mapsto (\sqrt{1 - b^2 - c^2}, b).$$

The radicands are strictly positive throughout these open domains, so both maps are smooth. All other nonempty transitions similarly select components of an inverse chart, using a square root with positive radicand; reversing the pair proves inverse smoothness. Thus these charts form a smooth atlas beyond the one-dimensional circle example.

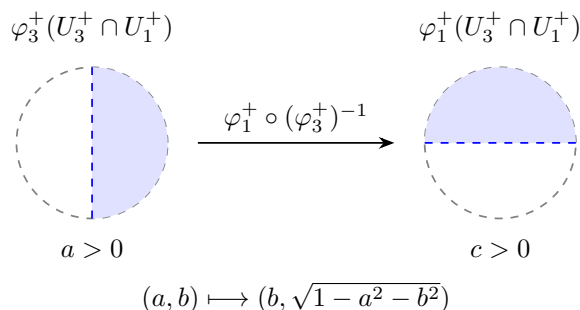


Figure 13.13: The same sphere overlap has two different coordinate images: a right half-disk and an upper half-disk. All displayed boundaries are excluded.

**Example 13.9.15** (Products and tori). For manifolds  $M^m, N^n$  without boundary, product charts are

$$\varphi \times \psi : U \times V \rightarrow \varphi(U) \times \psi(V), \quad (p, q) \mapsto (\varphi(p), \psi(q)).$$

The maximal atlas generated by these product charts is the product smooth structure. Their transitions are products of smooth transitions. [Propositions 10.4.9](#) and [10.4.11](#) supplies the countable product basis, and separating open neighborhoods in either factor separate distinct pairs. Thus the product topology and these charts give  $\dim(M \times N) = m + n$ . In particular  $T^n = (S^1)^n$  is a smooth  $n$ -manifold. This assertion concerns factors without boundary; products of boundary models can have corners.

Another fundamental family is real projective space,  $\mathbb{RP}^n$ , the space of lines through the origin, equivalently

$$S^n / (\mathbf{x} \sim -\mathbf{x}).$$

[Appendix E, Section E.7](#), constructs its topology rigorously and gives the standard affine charts. No projective-space theory is needed here.

**Example 13.9.16** (The two circular directions of a torus). Formally  $T^2 = S^1 \times S^1$ , with the product charts just constructed. For  $R > r > 0$  its familiar realization in  $\mathbb{R}^3$  is

$$\sigma(\theta, \varphi) = ((R + r \cos \varphi) \cos \theta, (R + r \cos \varphi) \sin \theta, r \sin \varphi).$$

This formula descends to the product of circles: the first circle records  $(\cos \theta, \sin \theta)$  and the second records  $(\cos \varphi, \sin \varphi)$ . On the image, put  $s = \sqrt{x^2 + y^2} > 0$ . The two circle points are recovered by

$$(x/s, y/s), \quad ((s - R)/r, z/r).$$

These smooth formulas identify the image with the product construction. The periodic angle formula repeats values at opposite sides of a parameter square and is not a global chart. Restricting both angles to sufficiently small open intervals gives a local parametrization; its inverse is an ordinary local coordinate chart. Equivalently, the torus admits the quotient description

$$T^2 \cong \mathbb{R}^2 / \mathbb{Z}^2, \quad (x, y) \sim (x + m, y + n) \quad (m, n \in \mathbb{Z}).$$

A unit square is a fundamental domain: identify left with right and bottom with top, with all four corners representing one point. In angular coordinates this is  $\mathbb{R}^2 / (2\pi\mathbb{Z})^2$ , with  $(\theta, \varphi) \sim (\theta +$

$2\pi n, \varphi + 2\pi n$ ); dividing both coordinates by  $2\pi$  gives the unit-square convention. The quotient topology and a rigorous proof of these identifications are given in [Section E.4](#) of Appendix E. The formal smooth-manifold construction in the main development remains  $S^1 \times S^1$ ; the appendix is not a prerequisite for Stokes' theorem. Figure 13.14 shows the two circular directions, and Figure 13.15 explains the successive gluings. The entire fundamental square is not a single global manifold chart.

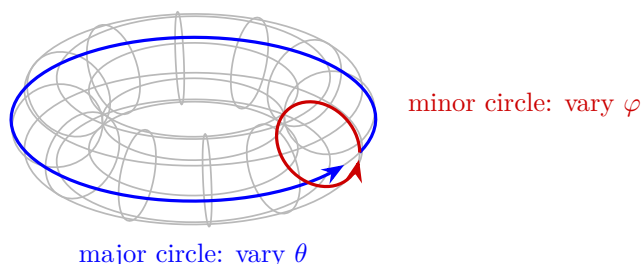


Figure 13.14: A wireframe of the embedded torus shows both independent circular directions. Hidden curves remain visible in this transparent schematic; periodic parameters are not a single global chart.

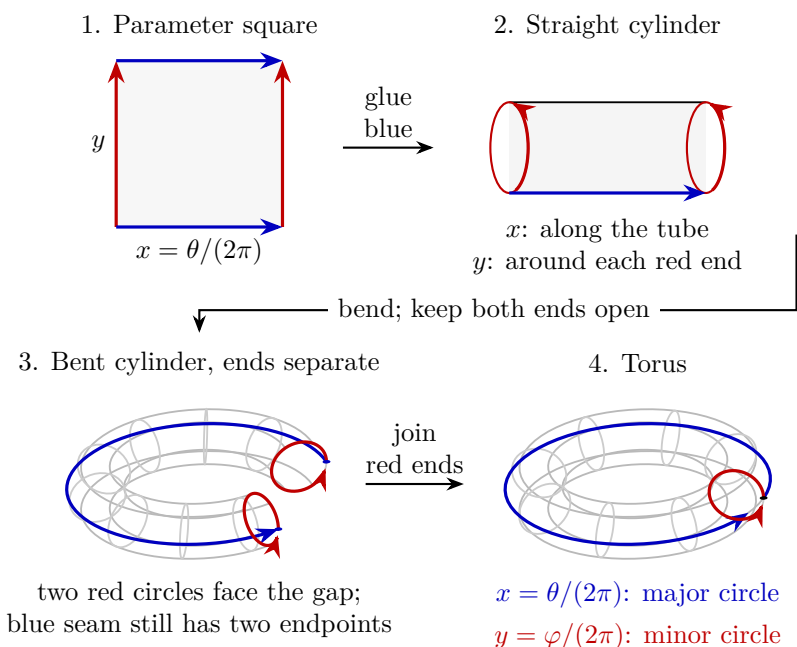


Figure 13.15: Square, straight cylinder, bent cylinder, and torus. Glue the blue edges first, making  $y$  circular while  $x$  runs along the tube. Bending preserves these directions. The two red end circles remain separate in panel 3; joining corresponding points closes the blue  $x$ -curve in panel 4. Either pair can be glued first, but this order agrees throughout with the angles in Figure 13.14. The square is a fundamental domain, not a global chart; see [Section E.4](#).

### An edge-identification gallery

The arrow convention in the following pictures is literal: corresponding points are matched by moving in the direction of the arrows on both edges. Red and blue arrows distinguish the paired

edges. The torus figure glues blue edges first to keep  $x$  along its major circle; the other gallery figures begin with the red pair. Thus parallel arrows on opposite edges prescribe a translation, whereas opposite arrows prescribe a reflection. These are identification arrows, not the induced boundary orientations of an oriented rectangle. The latter run around its perimeter and serve a different purpose.

**Cylinder.** On  $[0, 1] \times [-1, 1]$ , identify  $(0, t) \sim (1, t)$ . Bring the two red edges together without a twist. The remaining edges  $t = 1$  and  $t = -1$  close separately, giving two boundary circles (Figure 13.16).

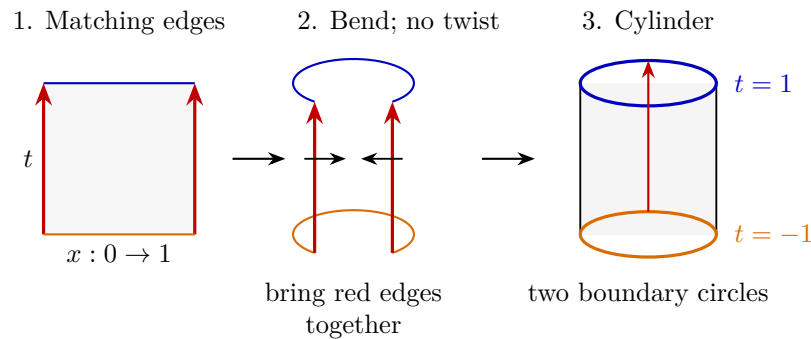


Figure 13.16: The cylinder glues  $(0, t)$  to  $(1, t)$ . The red arrows match without reversal; bending brings these edges together. The two unglued long edges close separately into the highlighted boundary circles.

**Möbius strip.** Change the seam rule to  $(0, t) \sim (1, -t)$ . The opposite red arrows demand a half-twist before the ends meet. The top edge now continues into the bottom edge; following both returns to the starting point. There is one boundary circle, traced in Figure 13.17.

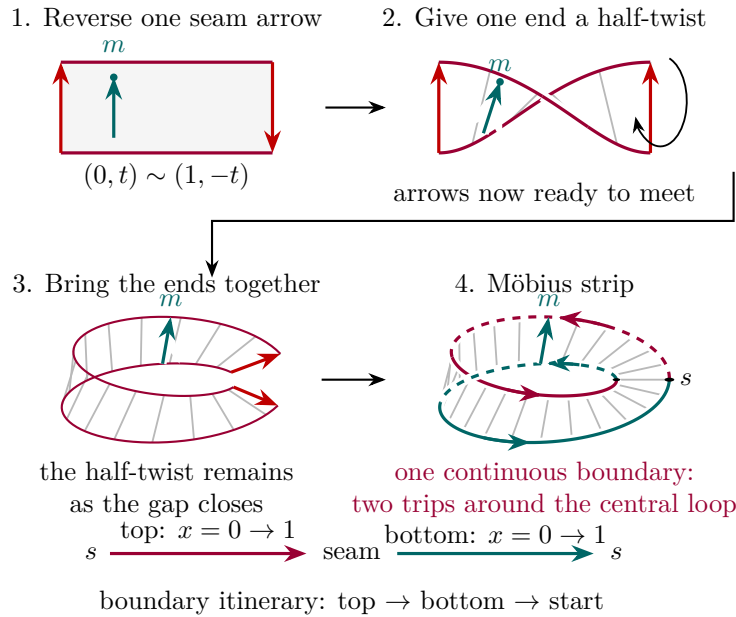


Figure 13.17: A half-twist implements the reversed seam rule. In the second panel the band is viewed obliquely through the twist; crossing strokes are projection effects. The marker  $m$  follows the same transverse segment. Dashed rear arcs and white gaps distinguish depth at projected crossings. Starting at  $s$ , the purple and teal portions trace one boundary: the edge that starts on top returns along the bottom before closing. See [Section E.5](#) for the local charts.

**Torus.** On  $[0, 1]^2$ , use  $(0, y) \sim (1, y)$  and  $(x, 0) \sim (x, 1)$ . The first gluing gives a cylinder; the second joins its boundary circles in matching directions. There is no remaining boundary. The two directions in [Figure 13.15](#) correspond to

$$x = \theta/(2\pi), \quad y = \varphi/(2\pi).$$

**Klein bottle.** Keep  $(0, y) \sim (1, y)$  but replace the other rule by  $(x, 0) \sim (1 - x, 1)$ . First form a cylinder, then identify its remaining circular boundaries with the reversal  $x \mapsto 1 - x$ . [Figure 13.18](#) separates this reversal from the familiar bottle-like drawing. The standard drawing of the Klein bottle in  $\mathbb{R}^3$  self-intersects. The abstract Klein bottle itself has no self-intersection; the crossing is an artifact of representing it in three-dimensional Euclidean space.

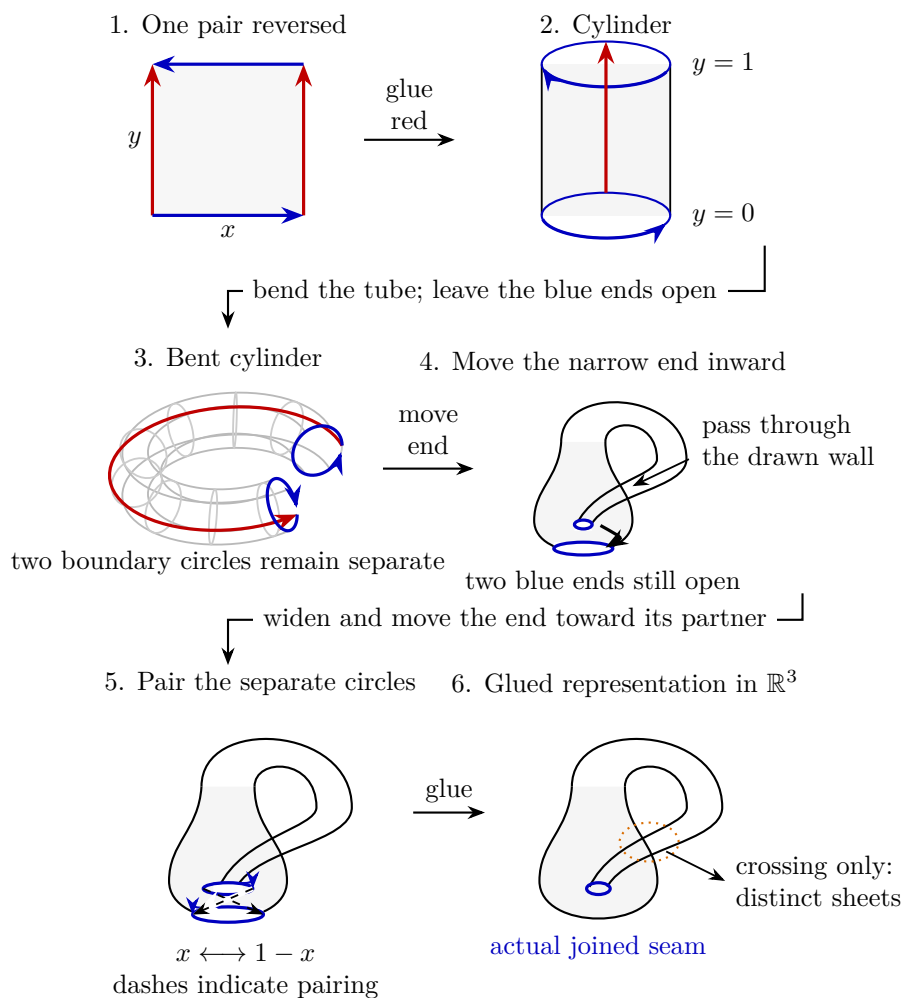


Figure 13.18: Klein-bottle gluing:  $(0, y) \sim (1, y)$  and  $(x, 0) \sim (1 - x, 1)$ . The bent cylinder keeps two boundary circles separate. Panels 4–5 move one end through the drawn wall, widen it, and display the reversed matching before identification in panel 6. The crossing in the  $\mathbb{R}^3$  picture is not an additional identification. The abstract Klein bottle has ordinary disk neighborhoods and no self-intersection. Figure E.7 verifies the corner neighborhood.

These quotient constructions, including seam and corner neighborhoods, are made rigorous in [Sections E.5 and E.6](#). They enrich the geometric examples without entering the prerequisite chain of this chapter. The comparison below previews orientability; [Section 13.13](#) makes it precise using positive tangent bases and determinant signs of coordinate changes.

| Surface      | Edge identification        | Boundary    | Orientability (preview) |
|--------------|----------------------------|-------------|-------------------------|
| Cylinder     | one pair, same direction   | two circles | orientable              |
| Möbius strip | one pair, reversed         | one circle  | non-orientable          |
| Torus        | both pairs, same direction | none        | orientable              |
| Klein bottle | one same, one reversed     | none        | non-orientable          |

**Remark 13.9.17** (The analytic-geometry dictionary: a preview). A classical local parametrization is an inverse chart, and a change of parameters is a transition map. For the embedded surfaces already studied, the classical tangent direction plane will realize the intrinsic space  $T_p M$  constructed in [Section 13.11](#), while the affine tangent plane passes through  $p$ . A normal vector represents a linear measurement vanishing on that direction plane. This previews how the familiar local geometry will acquire intrinsic language; it assumes no embedded-submanifold theory or already constructed intrinsic tangent space.

## 13.10 Manifolds with Boundary and Smooth Maps

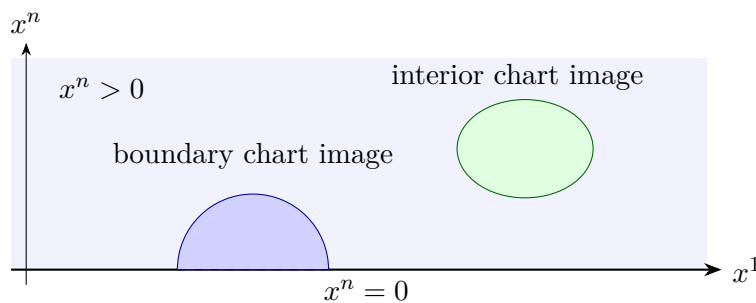
See [Figure 13.8](#) for the distinction between intrinsic and ambient boundary. An endpoint of an interval and an edge point of a disk do not have open-ball neighborhoods in their own dimensions. A half-space allows coordinates right up to that genuine boundary.

**Definition 13.10.1** (Half-space model). For  $n \geq 1$ , define

$$\mathbb{H}^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n : x^n \geq 0\}.$$

Its interior and boundary model are

$$(\mathbb{H}^n)^\circ = \{x^n > 0\}, \quad \partial\mathbb{H}^n = \{x^n = 0\}.$$



In this two-dimensional schematic, the half-disk includes its flat edge and excludes its curved edge; both chart images are relatively open.

**Definition 13.10.2** (Relative openness and smoothness on a half-space). For  $V \subseteq \mathbb{H}^n$ , define

$$V \text{ relatively open} \iff \exists \tilde{V} \subseteq \mathbb{R}^n \text{ open} : V = \tilde{V} \cap \mathbb{H}^n.$$

For such  $V$  and  $h : V \rightarrow \mathbb{R}^m$ , define  $h \in C^\infty(V, \mathbb{R}^m)$  by

$$\begin{aligned} \forall z \in V, \quad \exists O \subseteq \mathbb{R}^n \text{ open}, \quad \exists \tilde{h} \in C^\infty(O, \mathbb{R}^m), \\ z \in O, \quad \tilde{h}|_{O \cap V} = h|_{O \cap V}. \end{aligned}$$

A map into a half-space is smooth when its coordinate components satisfy this condition.

The ambient extension need not take values in the target half-space outside  $V$ . For example  $[0, 1)$  is open relative to  $[0, \infty)$  but not to  $\mathbb{R}$ . The function  $t \mapsto t^2$  on  $[0, 1)$  is smooth because the same formula extends across zero. By [Lemma 12.5.9](#), derivatives of extensions are unique on the half-space and obey the chain rule. Equality on interior points, followed by continuity to the face, is the mechanism.

**Definition 13.10.3** (Smooth manifold with boundary). A **smooth  $n$ -manifold with boundary**,  $n \geq 1$ , is a Hausdorff, second-countable space  $M$  equipped with a maximal smooth atlas of charts

$$\varphi : U \xrightarrow{\cong} V \subseteq \mathbb{H}^n, \quad U \text{ open in } M, \quad V \text{ relatively open in } \mathbb{H}^n.$$

The chart domains cover  $M$ : every  $p$  has an open neighborhood  $U$  modeled on a relatively open half-space set. Second countability has the meaning of [Definition 10.4.10](#); relative model bases are supplied by [Proposition 10.4.11](#). On every nonempty overlap, the transitions and their inverses are smooth in the extension sense; disjoint charts are compatible automatically. The candidate designations are

$$p \text{ boundary if } \varphi(p)^n = 0, \quad p \text{ interior if } \varphi(p)^n > 0.$$

**Definition 13.10.4** (Standard half-space smooth structure). For relatively open  $V \subseteq \mathbb{H}^n$ , the **standard smooth structure** is the maximal atlas generated by

$$\{(V, \text{id}_V)\}.$$

The restricted-chart proof of [Proposition 13.9.10](#) applies verbatim to open subsets of manifolds with boundary, using relative openness of their chart images.

These are candidate designations until we check that another chart gives the same answer. Ordinary interior charts can be regarded as half-space charts after restriction and translation. Manifolds without boundary are included in this terminology; a zero-dimensional manifold has empty boundary.

**Proposition 13.10.5** (Boundary status is independent of chart). Whether the last coordinate of a point is zero in a half-space chart is independent of the chart.

*Proof.* Suppose a transition sends a boundary model point to an interior model point. Its inverse is smooth on an open ball about the latter, takes values in  $\mathbb{H}^n$ , and has last component zero at the center. That component has a local minimum, so [Proposition 11.3.11](#) makes its derivative zero there. The inverse derivative has a zero row. But transition and inverse derivatives are inverse linear maps by [Lemma 12.5.9](#), whose proof did not assume boundary preservation. This is a contradiction. Reversing the transition also excludes an interior point mapping to a boundary point.  $\square$



**Definition 13.10.6** (Manifold interior and boundary). For a manifold with boundary, define the chart-independent subsets

$$M^\circ = \{p \in M : \varphi(p)^n > 0\}, \quad \partial M = \{p \in M : \varphi(p)^n = 0\},$$

where  $\varphi$  is any half-space chart at  $p$ .

**Example 13.10.7** (Flattening the closed disk near its north pole). On the part of the closed unit disk with

$$|x| < 1/2,$$

$$y > 1/2,$$

put

$$\varphi(x, y) = (x, \sqrt{1 - x^2} - y) = (u, t).$$

Its image is the relatively open half-space set

$$|u| < 1/2,$$

$$0 \leq t < \sqrt{1 - u^2} - 1/2.$$

The inverse  $(u, t) \mapsto (u, \sqrt{1 - u^2} - t)$  is smooth, as is  $\varphi$ , with determinant  $-1$ . The second coordinate is zero on the circular boundary and positive in the disk's interior. Thus this chart exhibits the local half-space model directly.

### Smoothness is tested in local coordinates

A map between manifolds is smooth when choosing source and target coordinates turns it into an ordinary smooth Euclidean map.

$$\begin{array}{ccc} U \subseteq M & \xrightarrow{f} & V \subseteq N \\ \varphi \downarrow & & \downarrow \psi \\ \varphi(U) \subseteq \mathbb{R}^m & \xrightarrow{\psi \circ f \circ \varphi^{-1}} & \psi(V) \subseteq \mathbb{R}^n \end{array}$$

**Definition 13.10.8** (Smooth manifold map). For manifolds  $M^m, N^n$ , a map  $f : M \rightarrow N$  is **smooth at  $p$**  if there exist charts  $\varphi : U \rightarrow \varphi(U)$ ,  $\psi : V \rightarrow \psi(V)$  with

$$p \in U, \quad f(p) \in V, \quad f(U) \subseteq V,$$

for which

$$\boxed{\psi \circ f \circ \varphi^{-1} : \varphi(U) \longrightarrow \psi(V)}$$

is smooth. The chart images lie in  $\mathbb{R}^m, \mathbb{R}^n$ , or in the corresponding half-spaces, where smooth means locally smoothly extendible. Define

$$f \text{ smooth} \quad \text{if} \quad \forall p \in M, \ f \text{ smooth at } p.$$

This condition implies continuity: on each  $U$  the map is  $\psi^{-1} \circ (\psi f \varphi^{-1}) \circ \varphi$ , a composition of continuous maps, and these neighborhoods cover  $M$ . For other charts  $\varphi' : U' \rightarrow \varphi'(U')$  and  $\psi' : V' \rightarrow \psi'(V')$  at the same points, restrict to the open neighborhood  $U \cap U' \cap f^{-1}(V \cap V')$ . The new coordinate representative is

$$\psi' f(\varphi')^{-1} = (\psi' \psi^{-1}) \circ (\psi f \varphi^{-1}) \circ (\varphi(\varphi')^{-1})$$

on the image of that neighborhood under  $\varphi'$ . Smooth transition maps and composition preserve smoothness, including local extensions at boundary points. Thus the definition does not privilege the initial charts.

**Definition 13.10.9** (Diffeomorphism and local diffeomorphism). For  $f : M \rightarrow N$ , define

$f$  a diffeomorphism if  $f$  bijective and  $f, f^{-1}$  smooth.

It is a **local diffeomorphism** if

$$\begin{aligned} \forall p \in M, \quad \exists U \text{ an open neighborhood of } p \text{ in } M : \quad & f(U) \text{ open in } N, \\ & f|_U : U \longrightarrow f(U) \text{ a diffeomorphism.} \end{aligned}$$

In Euclidean charts these are the notions of Chapter 11. The Inverse Function Theorem supplies local diffeomorphisms at interior points with invertible coordinate derivative; boundary domains require the stated half-space mapping condition as well.

**Proposition 13.10.10** (A chart is a diffeomorphism). Every chart  $\varphi : U \rightarrow \varphi(U)$  is a diffeomorphism from  $U$  with its inherited structure to its coordinate image with the standard structure.

*Proof.* Use  $\varphi$  as coordinates on  $U$  and the identity on  $\varphi(U)$ . The coordinate representatives of  $\varphi$  and  $\varphi^{-1}$  are both identity maps. They are smooth, including in the half-space extension sense.  $\square$

For the graph chart, this says that projection to the independent variables and the graph parametrization are inverse smooth manifold maps. Chart pullbacks below are therefore pullbacks along actual diffeomorphisms.

**Proposition 13.10.11** (Basic boundary structure). For a smooth  $n$ -manifold with boundary,  $n \geq 1$ , the interior  $M^\circ$  is open,  $\partial M$  is closed, and  $\partial M$  is naturally a smooth  $(n-1)$ -manifold without boundary. Its inclusion  $\iota : \partial M \rightarrow M$  is smooth and, in boundary coordinates, has injective derivative. The intrinsic differential will be defined in [Section 13.12](#).

*Proof.* In every chart the interior is the preimage of  $\{x_n > 0\}$ , relatively open in the chart image. These open preimages cover  $M^\circ$ , so its complement  $\partial M$  is closed. Restrict each boundary chart to  $U \cap \partial M$  and omit its last zero coordinate. The resulting image is open in  $\mathbb{R}^{n-1}$ : it is the intersection of an ambient open set with the hyperplane, identified with  $\mathbb{R}^{n-1}$ . The induced transitions are the first  $n-1$  components of the original smooth transitions evaluated at  $(u, 0)$ ; they and their inverses are smooth. Hausdorffness passes to the subspace by intersecting separating open sets; second countability passes to  $\partial M$  by the subspace-basis construction of [Proposition 10.4.11](#). Thus these charts define the asserted manifold structure. The inclusion in these coordinates is  $u \mapsto (u, 0)$ , smooth with injective derivative. For  $n = 1$  these charts have image the one-point space  $\mathbb{R}^0$ .  $\square$

At a boundary point a boundary-preserving transition  $\chi$  has derivative

$$[D\chi] = \begin{pmatrix} A & b \\ 0 & c \end{pmatrix}.$$

The zero block follows by differentiating the last component along the boundary, where it vanishes. Invertibility gives  $\det A \neq 0$  and  $c \neq 0$ . For small  $t > 0$ , the last component of  $\chi(p + te_n)$  is nonnegative and that of  $\chi(p)$  is zero; dividing by  $t$  and taking the limit gives  $c \geq 0$ , hence  $c > 0$ . Thus inward and outward transverse directions are preserved, explaining the consistency of the induced orientation below.

**Remark 13.10.12** (Two different boundaries). The circle in  $\mathbb{R}^2$  is its own ambient topological boundary, but its manifold boundary is empty: its local coordinates are open intervals. The closed disk has the circle as its manifold boundary. Boundary here refers to the intrinsic half-space model, not to an arbitrary embedding.

## 13.11 Tangent and Cotangent Spaces

A tangent line records a regular curve's first-order direction; a tangent plane records all first-order directions along a regular surface. For a graph, the familiar approximation

$$f(a + h, b + k) \approx f(a, b) + f_x(a, b)h + f_y(a, b)k$$

gives the direction plane spanned by  $(1, 0, f_x(a, b))$  and  $(0, 1, f_y(a, b))$ . Translating it to  $(a, b, f(a, b))$  gives the affine tangent plane. The progression is

$$\text{tangent line} \longrightarrow \text{tangent plane} \longrightarrow T_p M.$$

The tangent space is the higher-dimensional abstraction of these familiar objects: it captures the first-order linear behavior of a smooth space near a point. Charts let us construct it without an ambient embedding.

### Deriving the tangent-coordinate rule from a moving point

At an interior point  $p$ , let  $\gamma(0) = p$  and suppose two charts  $\varphi, \psi$  are available. Put  $v = (\varphi \circ \gamma)'(0)$  and  $w = (\psi \circ \gamma)'(0)$ . Since  $\psi \circ \gamma = (\psi \circ \varphi^{-1}) \circ (\varphi \circ \gamma)$ , the chain rule forces

$$w = D(\psi \circ \varphi^{-1})(\varphi(p))v.$$

The geometric motion has not changed, only its velocity coordinates. This equation suggests declaring chart-vector pairs equivalent when they represent the same direction by exactly this rule. The equivalence-class construction below preserves that interpretation without choosing an ambient Euclidean embedding. It also gives a full vector space at boundary points, where two-sided curves inside the manifold would be too restrictive. We construct tangent vectors by declaring when two coordinate vectors mean the same direction. For charts  $\varphi, \psi$  at  $p$ , define the linear map

$$C_{\psi\varphi}(p) = D(\psi \circ \varphi^{-1})(\varphi(p)) \in GL_n(\mathbb{R}).$$

The rightmost chart is the input coordinate system; the leftmost is the output. At boundary points derivatives are those of local smooth extensions. They are unique: extensions agree on interior points, hence their derivatives agree at the boundary by continuity. The chain rule identities below therefore hold at boundary points as well.

**Lemma 13.11.1** (Transition identities). For charts at a common point,

$$C_{\varphi\varphi} = I, \quad C_{\varphi\psi} = C_{\psi\varphi}^{-1}, \quad C_{\chi\varphi} = C_{\chi\psi}C_{\psi\varphi}.$$

*Proof.* The first transition is the identity in coordinates, so its derivative is  $I$ . The maps  $\varphi \circ \psi^{-1}$  and  $\psi \circ \varphi^{-1}$  are inverse on the overlap. Differentiating their two identity composites gives  $C_{\varphi\psi}C_{\psi\varphi} = I$  and  $C_{\psi\varphi}C_{\varphi\psi} = I$ , proving the inverse assertion. Finally factor the actual transition as  $\chi \circ \varphi^{-1} = (\chi \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1})$ . The intermediate evaluation point is  $\psi(p)$ ; the chain rule at  $\varphi(p)$  gives  $C_{\chi\varphi} = C_{\chi\psi}C_{\psi\varphi}$ .  $\square$

**Definition 13.11.2** (Tangent-vector relation). For chart-vector pairs  $(\varphi, v)$  with  $v \in \mathbb{R}^n$ , declare

$$(\varphi, v) \sim_p (\psi, w) \quad \text{if} \quad w = C_{\psi\varphi}(p)v.$$

**Proposition 13.11.3** (The relation is an equivalence relation). The relation  $\sim_p$  is reflexive, symmetric, and transitive.

*Proof.* Reflexivity follows from  $C_{\varphi\varphi}v = v$ . If  $w = C_{\psi\varphi}v$ , applying the inverse gives  $v = C_{\varphi\psi}w$ , proving symmetry. If also  $z = C_{\chi\psi}w$ , then  $z = C_{\chi\psi}C_{\psi\varphi}v = C_{\chi\varphi}v$ , proving transitivity. These are precisely the three identities of the lemma.  $\square$

**Definition 13.11.4** (Tangent vector and tangent space). The set of equivalence classes is the **tangent space**

$$T_pM = \{(\varphi, v) : \varphi \text{ a chart at } p, v \in \mathbb{R}^n\} / \sim_p.$$

Its elements are **tangent vectors at  $p$** . A tangent vector is the direction represented by all coordinate columns related through transition derivatives, rather than one distinguished column in  $\mathbb{R}^n$ .

The set becomes a vector space only after we define operations and check that they do not depend on which coordinates represent a direction.

**Theorem 13.11.5** (Vector-space structure of the tangent space). Fix one chart  $\varphi$  at  $p$ . Represent two classes in it as  $[\varphi, v]$  and  $[\varphi, w]$ , and define

$$[\varphi, v] + [\varphi, w] = [\varphi, v + w], \quad a[\varphi, v] = [\varphi, av].$$

These operations are independent of chart and make  $T_pM$  an  $n$ -dimensional real vector space.

*Proof.* Every class has a representative in  $\varphi$ : replace  $(\psi, z)$  by  $(\varphi, C_{\varphi\psi}z)$ . It has only one representative in that chart, since  $[\varphi, v] = [\varphi, v']$  forces  $v' = Iv = v$ . Thus the displayed operations specify outputs once  $\varphi$  is fixed. If we use  $\psi$  instead, the input representatives become  $C_{\psi\varphi}v$  and  $C_{\psi\varphi}w$ . Linearity gives

$$C_{\psi\varphi}(v + w) = C_{\psi\varphi}v + C_{\psi\varphi}w, \quad C_{\psi\varphi}(av) = aC_{\psi\varphi}v.$$

The outputs from the two charts are therefore equivalent pairs. This proves well-definedness before any vector-space conclusion is asserted. The zero vector is  $[\varphi, 0]$ , and the negative of  $[\varphi, v]$  is  $[\varphi, -v]$ . All vector-space axioms can now be checked in this one chart, where they are exactly those of  $\mathbb{R}^n$ .

Explicitly the coordinate map  $K_\varphi : T_pM \rightarrow \mathbb{R}^n$  sends a class to its unique  $\varphi$  representative vector; its reverse is  $S_\varphi(v) = [\varphi, v]$ . The definitions of operations make both maps linear. We have  $K_\varphi S_\varphi(v) = v$  and  $S_\varphi K_\varphi(\xi) = \xi$ . They are inverse isomorphisms, so the classes  $[\varphi, e_i]$  form a basis and  $\dim T_pM = n$ .  $\square$

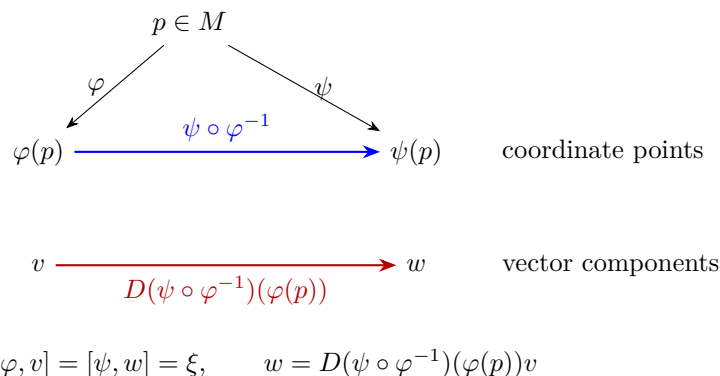


Figure 13.19: One intrinsic tangent direction, two representatives. The transition acts on coordinate points; its derivative acts on vector components. These are different maps with different inputs.

**Definition 13.11.6** (Coordinate tangent basis). For a chart  $\varphi$  with coordinate functions  $x^1, \dots, x^n$ , set

$$\left. \frac{\partial}{\partial x^i} \right|_p = [\varphi, e_i], \quad 1 \leq i \leq n.$$

These are the **coordinate tangent basis**: under the isomorphism  $K_\varphi : T_p M \rightarrow \mathbb{R}^n$  they become the standard basis.

**Example 13.11.7** (One circle direction in two coordinate systems). On the first-quadrant arc use  $u = x$  and  $v = y = \sqrt{1 - u^2}$ . At

$$p = (3/5, 4/5)$$

the transition derivative is

$$-3/4.$$

Hence

$$[\varphi, 1] = [\psi, -3/4], \quad \left. \frac{\partial}{\partial u} \right|_p = -\frac{3}{4} \left. \frac{\partial}{\partial v} \right|_p.$$

The first graph parametrization has velocity

$$(1, -3/4);$$

the second has velocity

$$(-4/3, 1)$$

per unit  $v$ . Multiplying the latter by

$$-3/4$$

gives

$$(1, -3/4)$$

again. The different columns describe the same direction.

We now have an intrinsic notion of direction. The next step is to construct objects that measure those directions. Tangent vectors represent infinitesimal directions. A scalar function responds to such a direction by a first-order change, which depends linearly on the direction. Cotangent vectors are these linear measurements: for smooth  $f$ , we will construct the differential  $df_p : T_p M \rightarrow \mathbb{R}$  in the next section and prove that it measures change intrinsically. We first revisit Chapter 11's differential notation ([Definition 11.2.11](#)) from this tangent/cotangent viewpoint.

**Definition 13.11.8** (Cotangent space). The **cotangent space** at  $p$  is

$$T_p^*M = (T_pM)^* = \mathcal{L}(T_pM, \mathbb{R}).$$

For chart coordinates  $x^1, \dots, x^n$ , define the **coordinate covectors** as the dual basis of [Definition 10.16.2](#), specified using the Kronecker delta by

$$dx^i|_p \left( \frac{\partial}{\partial x^j} \Big|_p \right) = \delta_j^i, \quad 1 \leq i, j \leq n.$$

Equivalently  $dx^i|_p([\varphi, v]) = v_i$  in the unique representative of the fixed chart.

Linearity follows from the tangent operations. Direct evaluation gives

$$dx^i|_p \left( \frac{\partial}{\partial x^j} \Big|_p \right) = (e_j)_i = \delta_j^i.$$

Every covector  $\lambda$  therefore equals

$$\sum_i \lambda((\partial/\partial x^i)|_p) dx^i|_p;$$

these covectors are precisely the dual basis. Directions lie in  $T_pM$ , while their linear scalar measurements lie in  $T_p^*M$ . The notation  $dx^i$  will also be justified as the differential of the coordinate function. Specifically,

$$\partial/\partial x^i$$

represents motion in the  $x^i$  direction, whereas  $dx^i$  measures that motion's  $x^i$  component:

$$v = \sum_j v^j \frac{\partial}{\partial x^j} \Big|_p \implies dx^i|_p(v) = v^i.$$

Here superscripts label coordinates and components; they are not powers.

## 13.12 Differentials and Differential Forms on Manifolds

A smooth map transports a direction at a source point into a direction at its image. The coordinate derivative describes this transport.

**Definition 13.12.1** (Differential of a smooth map). For smooth  $f : M \rightarrow N$ , define

$$df_p : T_pM \rightarrow T_{f(p)}N, \quad df_p[\varphi, v] = [\psi, D(\psi \circ f \circ \varphi^{-1})(\varphi(p))v].$$

Here the charts are at  $p, f(p)$  and the source neighborhood is restricted so that its image lies in the target chart. The following theorem verifies that this rule descends to equivalence classes.

For the proof, abbreviate the coordinate matrix by

$$A_{\psi\varphi} = D(\psi \circ f \circ \varphi^{-1})(\varphi(p)) : \mathbb{R}^{\dim M} \rightarrow \mathbb{R}^{\dim N}.$$

**Theorem 13.12.2** (Well-defined differential). The rule  $df_p[\varphi, v] = [\psi, A_{\psi\varphi}v]$  defines one intrinsic linear map  $df_p : T_pM \rightarrow T_{f(p)}N$ , independent of both charts.

*Proof.* For other charts  $\varphi', \psi'$ , the actual coordinate maps factor as

$$\psi' f(\varphi')^{-1} = (\psi' \psi^{-1}) \circ (\psi f \varphi^{-1}) \circ (\varphi(\varphi')^{-1}).$$

Differentiating at  $\varphi'(p)$  gives

$$A_{\psi'\varphi'} = C_{\psi'\psi}(f(p))A_{\psi\varphi}C_{\varphi\varphi'}(p).$$

If  $v' = C_{\varphi'\varphi}(p)v$  is the new representative of the same input tangent vector, then

$$A_{\psi'\varphi'}v' = C_{\psi'\psi}A_{\psi\varphi}C_{\varphi\varphi'}C_{\varphi'\varphi}v = C_{\psi'\psi}A_{\psi\varphi}v.$$

These are exactly equivalent output tangent representatives. Thus the rule is well defined as a map on classes, with target  $T_{f(p)}N$ . Fixing charts, linearity of  $A_{\psi\varphi}$  proves addition and scalar preservation; the already proved chart independence makes this intrinsic linearity.  $\square$

$$\begin{array}{ccc} p \in M & \xrightarrow{f} & f(p) \in N \\ \text{map of points} & \longrightarrow & \text{map of directions} \\ T_pM & \xrightarrow{df_p} & T_{f(p)}N \\ K_\varphi \downarrow & & \downarrow K_\psi \\ \mathbb{R}^m & \xrightarrow{D(\psi \circ f \circ \varphi^{-1})(\varphi(p))} & \mathbb{R}^n \end{array}$$

Figure 13.20: The intrinsic differential is represented by the Euclidean Jacobian. Here  $m = \dim M$ ,  $n = \dim N$ , and  $K_\varphi, K_\psi$  send tangent classes to their coordinate columns. The lower square commutes.

This is the Chapter 10 change-of-basis formula with transition derivatives as coordinate conversions. The map  $df_p$  stays fixed while the source and target coordinate descriptions change. In those coordinates the chain rule gives, by differentiating the composite coordinate maps,

$$d(g \circ f)_p = dg_{f(p)} \circ df_p, \quad d(\text{id}_M)_p = \text{id}_{T_pM}.$$

The representatives agree by the Euclidean chain rule, so chart independence turns these into equalities of intrinsic linear maps.

In particular, the boundary inclusion from Proposition 13.10.11 has  $d\iota_p[\varphi_\partial, v] = [\varphi, (v, 0)]$  in induced boundary and ambient charts. This intrinsic differential is injective, as the earlier coordinate calculation promised.

For a chart coordinate function  $u^i$ , identify the tangent space of its real-valued target with  $\mathbb{R}$  through the standard coordinate. Its coordinate expression is the projection  $\pi_i$ , so the differential just constructed gives

$$d(u^i)_p[\varphi, v] = D\pi_i(\varphi(p))v = v_i = du_p^i([\varphi, v]).$$

Thus the earlier dual-basis notation really is the differential of the coordinate function, with

$$du_p^i((\partial/\partial u^j)|_p) = \delta_j^i.$$

This is the local manifold version of  $dx_i = d\pi_i$  in Definition 11.2.11. For a smooth scalar function  $f$ , evaluation on each coordinate basis vector therefore gives

$$df_p = \sum_i \frac{\partial(f \circ \varphi^{-1})}{\partial x^i}(\varphi(p)) \, dx^i|_p, \quad \text{usually written} \quad df = \sum_i \frac{\partial f}{\partial x^i} \, dx^i.$$

This is the expansion of a cotangent vector in its dual basis, not merely a formal manipulation of symbols.

**Remark: velocities at the boundary.** At an interior point, every coordinate vector  $v$  is the velocity of  $t \mapsto \varphi^{-1}(\varphi(p) + tv)$  for small positive and negative  $t$ . At a boundary point this interpretation needs care. A two-sided curve into  $[0, \infty)$  through zero has derivative zero there, since zero is a minimum. Yet  $T_0[0, \infty) = \mathbb{R}$ : the tangent space contains inward and outward vectors. Coordinate vectors, rather than two-sided curves constrained to remain inside the manifold, give the correct full tangent space. Derivatives of local smooth extensions in half-space charts are unique: they agree on interior points and hence at boundary points by continuity.

**Example: the tangent plane of a graph.** For the embedded graph  $M = \{(x, y, h(x, y))\}$ , use its graph coordinates and inclusion  $i : M \rightarrow \mathbb{R}^3$ . Fix  $p = (a, b, h(a, b))$ ; all partials below are evaluated at  $(a, b)$ . The differential just constructed gives

$$di_p(u, v) = (u, v, h_x u + h_y v).$$

Its image is the tangent vector plane spanned by  $(1, 0, h_x)$  and  $(0, 1, h_y)$  from the earlier surface calculation. Here  $(u, v)$  denotes the coordinate representative of a tangent vector in  $T_p M$ . This image is a linear plane through zero in  $\mathbb{R}^3$ ; the affine tangent plane through the graph point is  $p + di_p(T_p M)$ .

**Example: implicit differentiation as a kernel condition.** Let  $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  be smooth, and let  $p \in f^{-1}(c)$  be regular:  $Df(p) \neq 0$ . By Corollary 11.6.6, the level set  $M = f^{-1}(c)$  is a smooth hypersurface near  $p$ . Identifying its tangent space with its image under the inclusion into  $\mathbb{R}^n$ , we have

$$T_p M = \ker Df(p).$$

Indeed differentiating  $f \circ i = c$  gives the inclusion in this kernel; both spaces have dimension  $n - 1$ , so they are equal. In the plane,

$$v = v^x \partial_x + v^y \partial_y \in T_p M \iff f_x(p)v^x + f_y(p)v^y = 0.$$

For a graph  $y = y(x)$  its velocity is  $(1, y')$ , recovering  $f_x + f_y y' = 0$  and, when  $f_y \neq 0$ ,

$$y' = -f_x/f_y.$$

Thus the implicit differentiation of Example 11.5.10 says that the cotangent vector  $Df(p)$  annihilates precisely the allowed directions.



**Manifold 1-forms first.** In a chart  $(x^1, \dots, x^n)$ , recall from Definition 13.11.8 that  $dx^i|_p$  is dual to

$$(\partial/\partial x^i)|_p.$$

A smooth 1-form is a field of linear measurements, locally

$$\omega = \sum_i a_i dx^i, \quad \omega_p \left( \sum_i v^i \frac{\partial}{\partial x^i} \Big|_p \right) = \sum_i a_i(p) v^i,$$

where the  $a_i$  are smooth. The form  $df$  is one example. The types are  $\omega \mapsto \omega_p \in T_p^*M \mapsto \omega_p(v) \in \mathbb{R}$ , with  $v \in T_pM$ .

On a circle arc, let  $\theta$  be an angle coordinate and  $s = 2\theta$ . The same tangent vector has representatives  $[\theta, a] = [s, 2a]$ . Since  $ds = 2d\theta$ , the form written  $d\theta$  in the first chart is

$$\frac{1}{2}ds$$

in the second, and

$$d\theta_p([\theta, a]) = a = \frac{1}{2}ds_p([s, 2a]).$$

The coordinate formula changed; the covector did not. This calculation is local and assumes no global angle coordinate on the circle. The analogous smoothly varying alternating measurement of  $k$  tangent directions is a  $k$ -form, using the already constructed exterior powers.

**Definition 13.12.3** (Differential form on a manifold). For  $0 \leq k \leq n = \dim M$ , a **smooth  $k$ -form** is a rule

$$\omega : p \mapsto \omega_p, \quad \omega_p \in \Lambda^k(T_p^*M)$$

such that in every chart  $(U, x^1, \dots, x^n)$  there are smooth functions  $a_I : U \rightarrow \mathbb{R}$  with

$$\omega|_U = \sum_{|I|=k} a_I dx^I, \quad dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_k}, \quad i_1 < \dots < i_k.$$

Smoothness of  $a_I$  means that  $a_I \circ \varphi^{-1}$  is smooth on  $\varphi(U)$ , using extensions at boundary points. Set

$$\Omega^k(M) = \{\text{smooth } k\text{-forms on } M\}.$$

Operations are pointwise;  $\Omega^0(M) = C^\infty(M, \mathbb{R})$  and  $\Omega^k(M) = \{0\}$  for  $k > n$ .

$$\begin{array}{ccccc} (\varphi, v) & \xrightarrow{\text{quotient}} & T_p M & \xrightarrow{\text{dual}} & T_p^* M \\ & & & & \downarrow \text{alternating} \\ & & \Omega^k(M) & \xleftarrow{\text{smooth variation}} & \Lambda^k(T_p^* M) \end{array}$$

This is the dependency order: first a direction, then its linear measurements, then alternating measurements varying smoothly over  $M$ .

There are three levels to keep distinct:

$$p \mapsto \omega_p \mapsto \omega_p(v_1, \dots, v_k) \in \mathbb{R}, \quad v_j \in T_p M.$$

The form is the assignment, its value at a point is an alternating multilinear map  $(T_p M)^k \rightarrow \mathbb{R}$ , and evaluation on directions produces a number. The circle calculation above illustrates this distinction in degree one.

Smooth transition derivatives convert the coordinate bases by smooth invertible matrices; the induced coefficient changes are finite sums of products of their entries. Thus smoothness in covering charts gives smoothness in every compatible chart. The following overlap calculation makes the corresponding rule for form operations explicit.

### Pullbacks, wedge products, and exterior derivatives on manifolds

The preceding definition specifies the values of a form. We now construct the operations used in manifold integration and Stokes, starting from the Euclidean operations in [Section 13.7](#).

For a smooth  $f : M \rightarrow N$  and a 1-form  $\omega$  on  $N$ , start with

$$(f^*\omega)_p(v) = \omega_{f(p)}(df_p v), \quad v \in T_p M.$$

Push the tangent vector forward, then let the target form measure it. The differential  $df_p : T_p M \rightarrow T_{f(p)} N$  is already intrinsic by [Theorem 13.12.2](#); composing it with a target covector therefore gives a source covector. For several inputs use the same rule on each direction. Define

$$f^* : \Omega^k(N) \rightarrow \Omega^k(M), \quad (f^*\omega)_p(v_1, \dots, v_k) = \omega_{f(p)}(df_p v_1, \dots, df_p v_k).$$

Initially this is a pointwise rule. The intrinsic linear map  $df_p$  from [Theorem 13.12.2](#) makes it an alternating covector at  $p$ , independent of tangent representatives. For  $k = 0$  the rule is  $f^*h = h \circ f$ .

To check smoothness, choose charts with  $f(U) \subseteq V$  and write  $h = \psi \circ f \circ \varphi^{-1}$ . If  $\omega_\psi$  is the expression of  $\omega$  in target coordinates, evaluation on coordinate vectors gives

$$(f^*\omega)_\varphi = h^*\omega_\psi.$$

Indeed both sides apply  $\omega_\psi(h(\mathbf{x}))$  to  $Dh(\mathbf{x})a_1, \dots, Dh(\mathbf{x})a_k$ . The right side is the already constructed Euclidean pullback and has smooth coefficients. This verifies the claimed target space, and proves agreement with Euclidean pullback. Linearity follows by evaluation at every point and tuple. The chain rule similarly gives  $(g \circ f)^* = f^*g^*$  and  $\text{id}^* = \text{id}$ .

For a scalar function  $f$  on a planar open set and the curve  $\gamma(t) = (x(t), y(t))$ , the rule precisely explains Chapter 11's shorthand:

$$\begin{aligned} \gamma^*(dx) &= x'(t) dt, & \gamma^*(dy) &= y'(t) dt, \\ \gamma^*(df) &= ((f_x \circ \gamma)x' + (f_y \circ \gamma)y') dt = d(f \circ \gamma). \end{aligned}$$

Equivalently,  $df_{\gamma(t)}(\gamma'(t)) = (f \circ \gamma)'(t)$ : the covector measures change along the velocity selected by the curve.

For  $\alpha \in \Omega^p(M)$  and  $\beta \in \Omega^q(M)$ , define

$$(\alpha \wedge \beta)_x = \alpha_x \wedge \beta_x.$$

The right side is the alternating-covector wedge already defined in [Section 13.6](#). In coordinates its coefficients are finite sums of signed products of the coefficients of  $\alpha$  and  $\beta$ ; they are smooth. Thus this defines a bilinear map into  $\Omega^{p+q}(M)$ . Pointwise pullback preserves it by the Euclidean shuffle formula.

Write a form in local Euclidean coordinates, differentiate it using the already-defined Euclidean exterior derivative, and prove that overlapping charts give the same intrinsic result. The dependency is

$$\boxed{\text{Euclidean } d + \text{pullback commutation} \implies \text{intrinsic manifold } d.}$$

Exterior differentiation extends this familiar scalar differential to all degrees. Locally it differentiates coefficients and alternates the resulting covectors:

$$\omega = \sum_I a_I dx^I \implies d\omega = \sum_{I,j} \frac{\partial a_I}{\partial x^j} dx^j \wedge dx^I.$$

For degree zero this is exactly the formula for  $df$ .

**Proposition 13.12.4** (The manifold exterior derivative). There is a unique operator  $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$  whose expression in every chart is the Euclidean exterior derivative.

*Proof. Strategy.* Differentiate in each chart, then use Euclidean pullback compatibility to show that the local answers agree on overlaps.

For  $\varphi : U \rightarrow \varphi(U)$  write  $\omega_\varphi = (\varphi^{-1})^*(\omega|_U)$ ; this notation is now legitimate by the pullback construction. On an overlap with  $\psi$ , put  $\chi = \psi\varphi^{-1}$ . The identity  $\varphi^{-1} = \psi^{-1} \circ \chi$  and pullback composition give

$$\omega_\varphi = \chi^* \omega_\psi.$$

Both sides have the same values on every tuple of coordinate vectors. Apply the already proved Euclidean identity in [Theorem 13.7.11](#) to obtain

$$d\omega_\varphi = d(\chi^* \omega_\psi) = \chi^*(d\omega_\psi).$$

Define the candidate on  $U$  as  $\varphi^*(d\omega_\varphi)$ . On the overlap the displayed identity gives

$$\varphi^*(d\omega_\varphi) = \varphi^* \chi^*(d\omega_\psi) = \psi^*(d\omega_\psi).$$

The local candidates therefore define a single global form with smooth coefficients. This is existence and chart independence. Any operator with the prescribed coordinate rule has these same local values, so is unique.

For a relatively open half-space domain, use local smooth extensions of the coefficients. Two extensions agree on interior points; all their derivatives agree there and then agree at boundary points by continuity. Thus the derivative is independent of the extension, and the Euclidean pullback identity above extends to the boundary by the same argument. The first-derivative and chain-rule foundation is [Lemma 12.5.9](#); the same interior-limit argument here applies to all coefficient derivatives.  $\square$

Equality of forms can be checked in charts. Consequently [Theorem 13.7.11](#) gives the intrinsic identities

$$d^2 = 0, \quad d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta, \quad f^*(d\omega) = d(f^*\omega).$$

The last identity uses the local representative  $h = \psi \circ f \circ \varphi^{-1}$  and the Euclidean identity for  $h$ , so it applies between different manifolds as well. Degrees above the dimension use the zero-space convention.

**Definition 13.12.5** (Support of a differential form). For a differential form  $\omega$  on  $M$ , define

$$\text{supp}(\omega) = \overline{\{p \in M : \omega_p \neq 0\}}.$$

The form has **compact support** when this closed set is compact.

**Lemma 13.12.6** (Support under differentiation and boundary pullback). For a smooth form  $\omega$  on  $M$ ,

$$\text{supp}(d\omega) \subseteq \text{supp}(\omega), \quad \text{supp}_{\partial M}(\iota^*\omega) \subseteq \text{supp}_M(\omega) \cap \partial M.$$

If  $\omega$  has compact support, both forms on the left have compact support.

*Proof.* Off the support of  $\omega$  it vanishes on a neighborhood. Its coordinate coefficients and their derivatives vanish there, so  $d\omega$  vanishes there too (at a boundary point use interior limits of derivatives). Likewise the pullback vanishes on the intersection of that neighborhood with  $\partial M$ . This proves both inclusions. The first support is a closed subset of a compact set. For the second, Proposition 13.10.11 makes  $\partial M$  closed in  $M$ , so the right side is compact, and its closed subset on the left is compact in  $\partial M$ .  $\square$

Compact support is not preserved by arbitrary pullback: a constant map  $\mathbb{R} \rightarrow \mathbb{R}$  with value at a point where a compactly supported function is nonzero pulls it back to a nonzero constant. The boundary inclusion works because its source is the closed subspace  $\partial M$ .

We now have the correct local integrands. To integrate them globally we still need a consistent sign and a way to combine coordinate patches.

## 13.13 Orientation and Boundary Orientation

Forms record signed volume. To integrate their local coefficients with consistent signs, we must decide which ordered tangent bases count as positive.

**Definition 13.13.1** (Orientation of a vector space). For ordered bases  $\mathcal{B}, \mathcal{C}$  of a real vector space  $V$ ,  $\dim V = n \geq 1$ , define

$$\mathcal{B} \sim_{\text{or}} \mathcal{C} \quad \text{if} \quad \det[\text{id}_V]_{\mathcal{C} \leftarrow \mathcal{B}} > 0.$$

An **orientation** chooses one of the two equivalence classes as positive.

The determinant product and inverse rules prove reflexivity, symmetry, and transitivity. There are exactly two classes because every invertible matrix has either positive or negative determinant; reversing one basis vector exchanges the classes. No lengths or angles enter this choice. In dimension zero there is only the empty basis; for integration we use a separate sign  $+1$  or  $-1$  at each point.

### Orientations specified by charts

On an overlap let  $C_{\psi\varphi} = D(\psi \circ \varphi^{-1})$ . This matrix converts the  $\varphi$ -components of a tangent vector into its  $\psi$ -components. Thus the columns of  $C_{\psi\varphi}$  are the  $\psi$ -components of the  $\varphi$  coordinate basis. The two coordinate bases have the same orientation precisely when  $\det C_{\psi\varphi} > 0$ . Its determinant is continuous and nonzero, so its sign is constant on each connected component of the overlap (Definition 10.7.10). Indeed the sign is locally constant by continuity and nonvanishing, and Proposition 10.7.2 makes it constant on a connected set. Distinct components can have different signs.

**Definition 13.13.2** (Oriented atlas and orientation). For a manifold without boundary of positive dimension, an **oriented atlas** is a smooth atlas all of whose transition determinants satisfy, at every point of every nonempty overlap,

$$\det D(\varphi_\beta \circ \varphi_\alpha^{-1}) > 0.$$

Two such atlases determine the same **orientation** if their union is oriented. A manifold is **orientable** if it admits an oriented atlas. After an orientation is chosen it is an **oriented manifold**; a **positive chart** is a chart compatible in this determinant sense with the chosen atlas.

At  $p$ , declare the coordinate basis of a positive chart positive in  $T_p M$ . The transition test makes the declaration independent of that chart. The upper and right circle graph charts have a negative transition derivative. Replacing the right coordinate  $y$  by  $-y$  makes the two charts agree in orientation on their first-quadrant overlap.

For a boundary model fixed as  $\mathbb{H}^n$ , we must allow an assigned orientation of that model. This matters already for the two endpoints of an interval: both cannot have positive derivative in coordinates increasing into the same half-line.

**Definition 13.13.3** (Oriented boundary atlas). An oriented atlas for a manifold with boundary consists of covering charts and a locally constant sign  $s_\varphi \in \{+1, -1\}$  on each chart, subject on overlaps to

$$\operatorname{sgn} \det D(\psi \circ \varphi^{-1}) = s_\psi s_\varphi.$$

Atlases are equivalent if their union satisfies this rule. An orientation is such an equivalence class; orientable means that one exists. A chart with  $s_\varphi = +1$  is positive relative to the standard model orientation.

Here is the rule's meaning and derivation. A basis with component matrix  $B$  in  $\varphi$  has matrix  $C_{\psi\varphi}B$  in  $\psi$ , and

$$s_\psi \det(C_{\psi\varphi}B) = s_\psi \det C_{\psi\varphi} \det B.$$

Its sign agrees with that of  $s_\varphi \det B$  for every invertible  $B$  exactly when  $s_\psi \operatorname{sgn} \det C_{\psi\varphi} = s_\varphi$ . Thus  $s_\varphi$  is precisely the sign of the coordinate basis relative to the tangent orientation. Shrink chart components if needed to make each sign constant. On a manifold without boundary one can reverse a coordinate to obtain positive charts; the definition then reduces to the preceding positive-transition definition. For zero-dimensional manifolds an orientation is a sign at each point.

## Returning to the gluing gallery

The examples in [Section 13.9](#) distinguish a local sign change from a global obstruction. On a lifted rectangle use the local top form  $dx \wedge dy$ . For the cylinder, the seam map  $f(x, y) = (x + 1, y)$  has derivative  $I$  and  $f^*(dx \wedge dy) = dx \wedge dy$ . Thus the local positive tangent bases fit together consistently. For the torus every lattice translation  $(x, y) \mapsto (x + m, y + n)$  has identity derivative, so these local forms likewise agree, giving a nowhere-vanishing global 2-form. This also has a direct product-circle description, independent of the quotient appendix: on  $S^1 = \{(u, v) : u^2 + v^2 = 1\}$  the form  $\alpha = u \, dv - v \, du$  takes the value 1 on the tangent vector  $(-v, u)$ , and in angle coordinates equals  $d\theta$ . If  $p_1, p_2 : T^2 \rightarrow S^1$  are the projections, then

$$\frac{1}{(2\pi)^2} p_1^* \alpha \wedge p_2^* \alpha$$

is nowhere vanishing and equals  $dx \wedge dy$  in unit coordinates. This illustrates [Theorem 13.13.7](#).

For the Möbius strip, the seam map is  $f(x, y) = (x + 1, -y)$ , with

$$Df = \text{diag}(1, -1), \quad f^*(dx \wedge dy) = -dx \wedge dy.$$

A negative determinant in one chosen transition alone does not prove non-orientability, since an individual chart can be reversed. Here the obstruction is global: follow the central loop represented by  $(x, 0)$ ,  $0 \leq x \leq 1$ . Along this connected lifted path a continuous choice of orientation has a constant sign relative to its coordinate frame. On returning through the seam, the frame is reflected, requiring the opposite sign at the same point. Reversing any chart merely relocates the sign change and cannot remove this contradiction around the loop.

For the Klein bottle our exact convention is  $A(x, y) = (x + 1, y)$  and  $B(x, y) = (1 - x, y + 1)$ . Their derivatives have determinants  $+1$  and  $-1$ , respectively, and  $B^*(dx \wedge dy) = -dx \wedge dy$ . The loop represented by

$$(1/2, y),$$

$0 \leq y \leq 1$ , closes under  $B$ . A continuous orientation along its lift has constant coordinate sign, but the closing identification reverses it. This is the same global obstruction, now on a surface without boundary. The quotient appendix supplies the explicit charts behind these computations.

|                  | Orientable | Non-orientable |
|------------------|------------|----------------|
| With boundary    | cylinder   | Möbius strip   |
| Without boundary | torus      | Klein bottle   |

Having boundary does not cause non-orientability, and being boundaryless does not imply orientability. Orientability asks whether local choices of positive tangent bases can be made globally consistent.

### The induced boundary orientation

The inclusion identifies  $T_p\partial M$  with the subspace of  $T_pM$  whose last coordinate in a boundary chart is zero.

**Definition 13.13.4** (Outward transverse vector). At  $p \in \partial M$ , let  $K_\varphi : T_pM \rightarrow \mathbb{R}^n$  give the components in a half-space chart. Define

$$\nu \text{ outward transverse} \quad \text{if} \quad \nu \in T_pM, \quad (K_\varphi(\nu))_n < 0.$$

Such a vector is not in  $T_p\partial M$ . The boundary-transition block calculation in [Section 13.10](#) has positive last diagonal entry, so the inequality is independent of the boundary chart.

**Definition 13.13.5** (Outward-normal-first boundary orientation). For an oriented  $n$ -manifold with boundary,  $n \geq 2$ , and  $p \in \partial M$ , define positivity of a basis of  $T_p\partial M$  by

$$(v_1, \dots, v_{n-1}) \text{ positive in } T_p\partial M \\ \text{if } \boxed{(\nu, v_1, \dots, v_{n-1}) \text{ positive in } T_pM},$$

where  $\nu$  is any outward transverse vector. For  $n = 1$ , the boundary point has sign  $+1$  or  $-1$  according as an outward vector is positive or negative in  $T_pM$ .

No metric or unit normal is needed: “normal” here means transverse and outward. For another outward vector  $\nu'$ , write  $\nu' = a\nu + w$  with  $a > 0$  and  $w \in T_p\partial M$ , by comparing last components. Replacing the first column by  $a\nu + w$  multiplies the determinant by  $a$ : the contribution from  $w$  is zero because it is a combination of the remaining columns. Explicitly,

$$\det(\nu', v_1, \dots, v_{n-1}) = a \det(\nu, v_1, \dots, v_{n-1}), \quad a > 0,$$

in any fixed coordinate basis. Hence the orientation is independent of the outward vector.

In coordinates one can use  $-e_n$ ; its determinant sign relative to the boundary coordinate basis is fixed locally. Thus the induced boundary orientations vary consistently in boundary charts.

**Example 13.13.6** (Why the oriented interval needs chart signs). Orient  $[a, b]$  by increasing  $x$ . Near  $a$  use  $u = x - a$ , whose coordinate basis is  $+\partial_x$ , so  $s_u = +1$ . Near  $b$  use  $v = b - x$ , whose coordinate basis is  $-\partial_x$ , so  $s_v = -1$ . The outward vectors are  $-\partial_x$  at  $a$  and  $+\partial_x$  at  $b$ . Therefore

$$\partial[a, b] = \{b\} - \{a\}, \quad \int_{\partial[a, b]} f = f(b) - f(a).$$

Requiring both endpoint charts into the fixed half-line to be positive would exclude this consistently oriented interval.

For a standard oriented annulus, outward-first gives counterclockwise traversal on the outer circle and clockwise traversal on the inner circle. These signs agree with the classical boundary traversals already used. Figure 13.21 shows why positive boundary orientation does not mean counterclockwise on every component: at the inner circle, outward points into the hole.

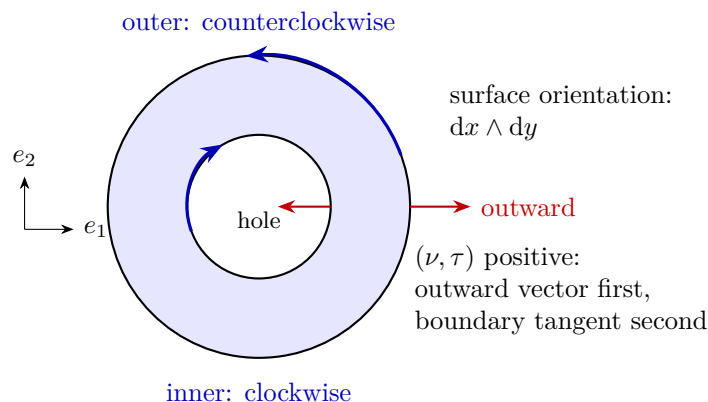


Figure 13.21: Outward-first on a standard oriented annulus. At the rightmost outer point, outward is right and the positive tangent is up. At the rightmost inner point, outward is left into the hole and the positive tangent is down. Thus the two boundary components run in opposite senses.

An oriented cylinder likewise induces orientations on both boundary circles. The Möbius strip has a boundary circle, and this circle is orientable as a one-dimensional manifold. But no orientation of the strip exists to induce a global boundary orientation by the outward-first rule. Choosing an orientation of that circle on its own is different from inducing one from an oriented surface.

**Theorem 13.13.7** (Orientation and top forms). A smooth  $n$ -manifold, possibly with boundary, is orientable if and only if it admits a nowhere-vanishing smooth  $n$ -form.

*Proof of the top-form-to-orientation direction.* Let  $\omega_p \neq 0$  for every  $p$ . For  $n \geq 1$  declare a tangent basis  $(v_1, \dots, v_n)$  positive when  $\omega_p(v_1, \dots, v_n) > 0$ . Alternation and the determinant rule give exactly one orientation class. In a chart write  $\omega = a(\mathbf{x}) d\mathbf{x}^1 \wedge \dots \wedge d\mathbf{x}^n$ . The smooth coefficient never vanishes, so its sign is locally constant. Take  $s_\varphi = \operatorname{sgn} a$ ; the change-of-coordinates formula gives the signed-atlas compatibility rule above. In interior charts we can reverse a coordinate when necessary to get an ordinary positive atlas. For  $n = 0$ ,  $\omega$  is a nowhere-zero function on a discrete space; its sign specifies the point orientations.  $\square$

The converse needs a way to combine local positive top forms without cancellation. We prove it immediately after the partition-of-unity theorem in [Section 13.14](#). Thus it is not used to construct partitions of unity. A top form both detects orientation and supplies the type of object that can be integrated over the whole manifold.

## 13.14 Partitions of Unity and Integration on Manifolds

We now know what “positive” means. The remaining difficulty is global: one chart rarely covers everything we want to integrate. Let us first solve the integration problem in stages, before constructing its machinery.

**Stage 1: support inside one chart.** Suppose a smooth top form  $\omega$  has compact support contained in one positive chart  $\varphi : U \rightarrow V$ . Recall that support is the closure of the nonzero set ([Definition 13.12.5](#)), so this assumption includes its limiting points. Write

$$(\varphi^{-1})^*\omega = g(u) du^1 \wedge \dots \wedge du^n, \quad I_\varphi(\omega) := \int_V g(u) d(u_1, \dots, u_n).$$

This is an ordinary scalar integral of one coordinate coefficient. For an interior chart, extend by zero to a containing rectangle; for a boundary chart use the half-space integral of [Definition 13.8.2](#). Compact support stays away from artificial chart edges, but may meet the genuine face. For a chart of sign  $s_\varphi$ , multiply this candidate by  $s_\varphi$ .

**Stage 2: why positive coordinates give the same answer.** What if another compatible oriented chart is used? Put  $\chi = \psi \circ \varphi^{-1}$  and write the other coefficient as  $h$ . On the overlap, the top-form pullback rule gives

$$g(u) = h(\chi(u)) \det D\chi(u).$$

For positive coordinates,  $\det D\chi > 0$ , so the signed determinant is exactly the absolute Jacobian in ordinary change of variables. Hence  $I_\varphi(\omega) = I_\psi(\omega)$  by [Proposition 13.8.7](#) and [Corollary 12.5.10](#). For signed charts the relation  $\operatorname{sgn} \det D\chi = s_\psi s_\varphi$  gives the same equality of signed candidates. Orientation supplies precisely the sign needed to make the coordinate answer intrinsic.

**Stage 3: what if one chart is not enough?** We seek smooth weights  $\rho_\alpha$  supported in chart domains and summing to 1. Then

$$\omega = \left( \sum_\alpha \rho_\alpha \right) \omega = \sum_\alpha \rho_\alpha \omega.$$



Each piece is supported in one chart. The intended definition is the sum of its Stage 1 integrals. The remaining obligations are to construct the weights, control the sum, and show independence of the decomposition.

For a toy picture, cover the counterclockwise circle by the two angle charts omitting  $(-1, 0)$  and  $(1, 0)$  respectively. Smooth weights with  $\rho_1 + \rho_2 = 1$  and supports in these arcs give  $\omega = \rho_1\omega + \rho_2\omega$ : two ordinary coordinate integrals will replace the unavailable single global angle chart. The theorem below supplies such weights; the worked circle example after the independence proof computes the resulting integral.

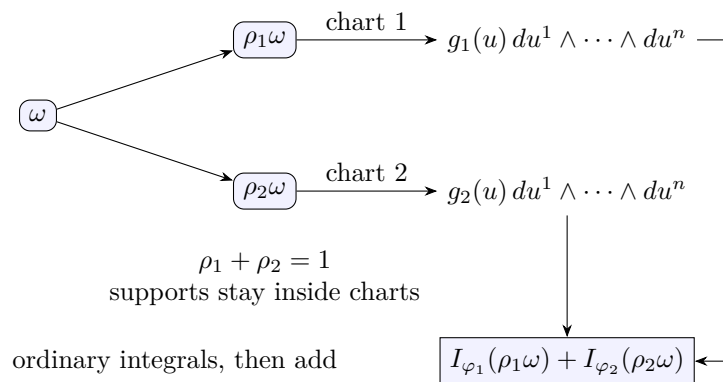


Figure 13.22: Two-chart integration: multiply by smooth weights, express each piece in coordinates, integrate its scalar coefficient, and add. The weights fade to zero near artificial chart edges. The same architecture works with any locally finite chart cover.

**First-reading signpost.** Conceptually, localize to charts, integrate the coordinate top forms by ordinary Euclidean integration, and prove independence of choices. The refinement and extension lemmas below supply the technical infrastructure, especially near genuine manifold boundaries. We now introduce the terminology needed to make this construction rigorous.

**Definition 13.14.1** (Covers for localization). A family  $(A_i)_{i \in I}$  in  $M$  is **locally finite** if

$$\forall p \in M, \quad \exists U \text{ an open neighborhood of } p : \quad \#\{i \in I : U \cap A_i \neq \emptyset\} < \infty.$$

A cover  $(V_j)_{j \in J}$  **refines** a cover  $(U_\alpha)_{\alpha \in A}$ , written  $(V_j) \prec (U_\alpha)$ , if

$$\forall j \in J, \quad \exists \alpha \in A : V_j \subseteq U_\alpha.$$

**Definition 13.14.2** (Precompactness and paracompactness). For a subset  $E \subseteq M$  and a topological space  $M$ , define

$$E \text{ precompact if } \overline{E} \text{ compact,}$$

$$M \text{ paracompact if every open cover has a locally finite open refinement.}$$

Local finiteness controls a whole neighborhood of each point. Refinement makes cover members smaller; precompactness gives compact closures. All closures here are in  $M$  unless another ambient space is specified.

Recall the Euclidean support convention of [Section 11.6](#); the same closure-based definition works on any topological space.

**Definition 13.14.3** (Support). For a function  $f : X \rightarrow \mathbb{R}$  on a topological space, define

$$\text{supp } f = \overline{\{x \in X : f(x) \neq 0\}}, \quad f \text{ compactly supported if } \text{supp } f \text{ compact.}$$

Support includes limiting points of the nonzero set, not just points where the value is nonzero. This closure controls smooth extension by zero.

A sharp subdivision by indicators of chart regions generally makes  $\mathbf{1}_U \omega$  discontinuous at artificial interfaces, destroying smoothness. Instead use smooth weights with

$$\sum_{\alpha} \rho_{\alpha} = 1,$$

so

$$\omega = \sum_{\alpha} \rho_{\alpha} \omega$$

and each localized form is smooth. Although differentiation introduces extra terms,

$$d(\rho_{\alpha} \omega) = d\rho_{\alpha} \wedge \omega + \rho_{\alpha} d\omega, \quad \sum_{\alpha} d\rho_{\alpha} = d\left(\sum_{\alpha} \rho_{\alpha}\right) = 0,$$

they cancel in the sum. This is the smooth analogue of partitioning a domain. We need local finiteness, not merely finitely many nonzero values at each point: a whole neighborhood must meet only finitely many supports so ordinary finite-sum differentiation applies there.

$$\begin{array}{ccc} \omega \text{ on } U & \xrightarrow{\text{multiply by } b} & b\omega \text{ on } U \\ & & \downarrow \text{extend by zero} \\ & & \widetilde{b\omega} \text{ on } M \end{array}$$

Here  $b$  is a smooth bump with compact support  $K \subset U$ . Outside  $K$ , the product is already zero on a neighborhood, so the zero extension is smooth across the artificial edge of  $U$ . At a genuine manifold boundary, smoothness still uses half-space extensions. This is the mechanism used in the bump construction and the Stokes localization.

**Lemma 13.14.4** (Working with local finiteness). For a locally finite family of subsets  $(A_i)$ , every compact set meets only finitely many members. A locally finite family of smooth functions (meaning their supports form such a family) has a smooth pointwise sum. A locally finite union of closed subsets is closed.

*Proof.* For each  $x$  of a compact set  $K$ , choose an open neighborhood  $N_x$  meeting only finitely many  $A_i$ . Select finitely many  $N_{x_1}, \dots, N_{x_m}$  covering  $K$ . Any  $A_i$  meeting  $K$  meets one selected neighborhood. Writing

$$I_j = \{i : A_i \cap N_{x_j} \neq \emptyset\}, \quad \{i : A_i \cap K \neq \emptyset\} \subseteq \bigcup_{j=1}^m I_j,$$

we see that the right side is a finite union of finite sets.

Near any point, only finitely many supports of the smooth functions meet one chosen neighborhood. All other functions vanish on that neighborhood, so the sum there is an ordinary finite smooth sum, and is smooth.

For closed  $A_i$ , let

$$x \notin \bigcup_i A_i.$$

Choose an open neighborhood  $N$  of  $x$  meeting only  $A_{i_1}, \dots, A_{i_m}$ . The set

$$N \setminus (A_{i_1} \cup \dots \cup A_{i_m})$$

is open, contains  $x$ , and misses every  $A_i$ . Thus the complement of the full union is open.  $\square$

**Example 13.14.5** (Pointwise finiteness does not control a neighborhood). Let

$$r_n = 1/(4n(n+1))$$

for  $n \geq 1$ , and use the smooth function  $\theta$  of [Theorem 8.8.2](#) to define

$$\chi_n(x) = \frac{\theta(1 - ((x - 1/n)/r_n)^2)}{\theta(1)}.$$

This is a smooth bump of height 1, with support

$$I_n = [1/n - r_n, 1/n + r_n].$$

These intervals avoid zero and are disjoint: the distance between neighboring centers is

$$1/(n(n+1)),$$

larger than  $r_n + r_{n+1}$ . Thus at each point at most one bump is nonzero. Nevertheless

$$S(x) = \sum_{n \geq 1} \chi_n(x)$$

satisfies  $S(0) = 0$  and

$$S(1/n) = 1,$$

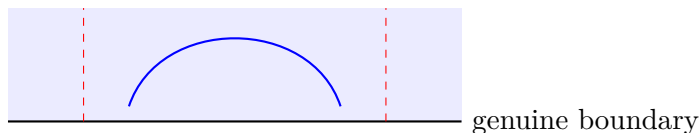
so it is not even continuous at zero. Every neighborhood of zero meets infinitely many  $I_n$ . Pointwise finiteness lets us evaluate the sum, but local finiteness would give one finite list valid throughout a neighborhood, which is what the preceding smoothness proof needs.

With local finiteness established ([Definition 13.14.1](#) and [Lemma 13.14.4](#)), we can construct the smooth weights. The next theorem is our external topological input. The subsequent construction of smooth weights uses the Euclidean cutoff in [Lemma 11.6.18](#).

Both this construction and [Theorem 11.6.19](#) use local bumps. The finite Euclidean proof assigns successive remainders by a telescoping product; here a locally finite family is naturally normalized by its positive sum. The new issues are the external refinement, infinitely many chart domains, local finiteness of supports, smooth local sums, and regrouping by the original cover. Those issues, rather than the quotient formula itself, are the work of the next proof.

As in the finite Euclidean construction, a chart-supported function can vanish near its artificial edges while remaining nonzero on the genuine boundary. Zero extension across the former is smooth; zero extension across the latter generally is not. Local finiteness in the general construction makes the normalization arithmetic finite in a neighborhood of each point.

support stays away from artificial chart edges



**Theorem 13.14.6** (External refinement theorem). Here second countability means a countable topological basis (Definition 10.4.10); precompact means compact closure (Definition 13.14.2). Every second-countable Hausdorff smooth manifold, possibly with boundary, is paracompact. Given an open cover of  $M$ , there are countable locally finite families of open coordinate neighborhoods  $V_j$  and  $W_j$ , with the  $V_j$  covering  $M$ , such that  $\overline{V_j}$  is compact and

$$\overline{V_j} \subseteq W_j \subseteq U_{\alpha(j)}$$

for a suitable index  $\alpha(j) \in I$ .

This is an external topological input. See Lee's second edition, [5, Theorem 1.15 and the coordinate-ball refinement in the proof of Theorem 2.23, pp. 43–44; boundary adaptation in Exercise 2.24]. The nested coordinate-neighborhood version stated here uses the same shrinking of coordinate balls (half-balls at boundary points). We import that refinement, not the smooth partition theorem: the bump construction, normalization, and regrouping are proved below using Chapter 11.

*Why the refinement can be nested.* First choose a countable locally finite cover  $(O_i)$  by precompact coordinate neighborhoods refining the given cover. Apply that statement again using a basis of smaller coordinate neighborhoods whose compact closures lie in some  $O_i$ . Call the resulting cover  $(V_j)$  and choose

$$\overline{V_j} \subset W_j = O_{i(j)}.$$

Local finiteness and compactness give

$$\#\{j : V_j \cap \overline{O_i} \neq \emptyset\} < \infty.$$

Consequently each  $O_i$  is repeated only finitely often in  $(W_j)$ . Local finiteness of  $(O_i)$  then proves local finiteness of  $(W_j)$ . Small coordinate balls or relative half-balls provide the required bases: inside a prescribed open neighborhood, intersect with a chart and choose a model ball of sufficiently small radius, with compact closure still inside the chart image. Pulling it back gives the local refinement required by Definition 10.4.5; compactness of its closure is preserved by the inverse chart. This explains the nested formulation without importing a partition of unity.

**Theorem 13.14.7** (Partition of unity). Let  $M$  be a smooth manifold, possibly with boundary, and let  $(U_\alpha)_{\alpha \in I}$  be an open cover. There are smooth functions  $\rho_\alpha : M \rightarrow [0, 1]$  whose supports form a locally finite family and satisfy

$$\text{supp}(\rho_\alpha) \subseteq U_\alpha \quad (\alpha \in I), \quad \sum_{\alpha \in I} \rho_\alpha = 1.$$

Such a partition of unity is **subordinate** to the cover: each weight has its full closed support inside its assigned open set.

*Proof. Strategy.* Build bumps on the refined charts, normalize by their positive sum, then regroup by the original cover.

*Local bumps.* Apply the external refinement theorem Theorem 13.14.6 to obtain

$$\overline{V_j} \subseteq W_j \subseteq U_{\alpha(j)}.$$

For a manifold with boundary, use boundary charts in this construction; the smooth-extension convention in the definition of a manifold with boundary makes the same Euclidean bump-function construction applicable.

For each  $j$ , let  $\varphi_j : W_j \rightarrow O_j$  be its chart, with  $O_j$  open in  $\mathbb{R}^n$  or relatively open in  $\mathbb{H}^n$ . Write  $O_j = \tilde{O}_j \cap \mathbb{H}^n$  in the boundary case, with  $\tilde{O}_j$  open in  $\mathbb{R}^n$ ; in the interior case use  $\tilde{O}_j = O_j$ . The coordinate set  $K_j = \varphi_j(\overline{V_j})$  is compact in  $\tilde{O}_j$ . By [Lemma 11.6.18](#), choose a smooth  $b_j$  compactly supported there, with  $0 \leq b_j \leq 1$  and  $b_j = 1$  near  $K_j$ . Restrict to  $O_j$ , transfer by  $\varphi_j$ , and extend by zero outside  $W_j$  in  $M$ , obtaining  $\chi_j$ . Its support is compact inside  $W_j$ , so this extension is smooth across artificial chart boundaries. At the genuine boundary we retain the smooth Euclidean bump extension, which need not be zero on the other side. Thus

$$\chi_j \geq 0, \quad \chi_j > 0 \text{ on } V_j, \quad \text{supp } \chi_j \subset W_j.$$

*Normalization.* Because the family  $(W_j)$  is locally finite and  $\text{supp}(\chi_j) \subseteq W_j$ , the family of supports  $(\text{supp}(\chi_j))$  is locally finite. Therefore the sum

$$\chi = \sum_{j=1}^{\infty} \chi_j$$

is smooth and well defined by [Lemma 13.14.4](#); it is positive because the  $V_j$  cover  $M$ . Set

$$\rho_j = \frac{\chi_j}{\chi}.$$

The denominator is positive, and local finiteness makes  $\chi$  smooth and each quotient smooth. The family  $(\rho_j)$  is locally finite, satisfies

$$\text{supp}(\rho_j) \subseteq \text{supp}(\chi_j) \subseteq W_j \subseteq U_{\alpha(j)},$$

and sums to one.

*Regrouping.* For each original index  $\alpha$ , define

$$\rho_\alpha = \sum_{\{j: \alpha(j)=\alpha\}} \rho_j.$$

These sums are locally finite, so the  $\rho_\alpha$  are smooth and the regrouped family remains locally finite. Moreover, by [Lemma 13.14.4](#), a locally finite union of closed sets is closed, and therefore

$$\text{supp}(\rho_\alpha) \subseteq \bigcup_{\{j: \alpha(j)=\alpha\}} \text{supp}(\rho_j) \subseteq U_\alpha.$$

Regrouping the locally finite sum also gives

$$\sum_{\alpha} \rho_\alpha = 1.$$

Thus the partition is subordinate to the original cover. □

We can now reduce global questions to chart-supported local questions. Local finiteness makes smooth sums legitimate on a whole neighborhood; compact support makes only finitely many pieces relevant to an integral.

*Completion of Theorem 13.13.7.* Suppose  $M$  is oriented and  $n \geq 1$ . Choose a covering signed oriented atlas, with constant sign on each chart domain  $U_\alpha$ . Its local positive top form is

$$\omega_\alpha = s_\alpha \, dx_\alpha^1 \wedge \cdots \wedge dx_\alpha^n.$$

On a positive chart  $s_\alpha = 1$ , so this is the usual coordinate volume form. The signs allow the same argument at both ends of an interval. Choose a subordinate partition  $(\rho_\alpha)$  by [Theorem 13.14.7](#). Each  $\rho_\alpha \omega_\alpha$  extends smoothly by zero outside  $U_\alpha$ : every point outside that chart has a neighborhood disjoint from the closed support of  $\rho_\alpha$ . Inside a boundary chart its coefficients retain their smooth extensions across the genuine face. Form the locally finite smooth sum

$$\omega = \sum_{\alpha} \rho_{\alpha} \omega_{\alpha}.$$

At  $p$ , evaluate on a positive tangent basis  $\mathcal{B}_p$ :

$$\omega_p(\mathcal{B}_p) = \sum_{\alpha: \rho_{\alpha}(p) > 0} \rho_{\alpha}(p) (\omega_{\alpha})_p(\mathcal{B}_p) > 0.$$

Every displayed summand is positive, and at least one occurs because the weights sum to one. Thus  $\omega_p \neq 0$ . For  $n = 0$ , the function taking the chosen sign at each point is smooth on the discrete manifold and nowhere zero. This proves the converse.  $\square$

Orientation has now produced a global top form. Integration will apply the same localization idea to an arbitrary compactly supported top form, using coordinate changes to prove that the resulting number is intrinsic.

The Euclidean compact cutoff used below is already available as [Lemma 11.6.18](#); it was proved directly in Chapter 11.

**Lemma 13.14.8** (One extension near a compact coordinate set). Let  $O$  be relatively open in  $\mathbb{H}^n$ , let  $h$  be smooth on  $O$  in the local-extension sense, and let  $K \subset O$  be compact. There is one smooth ambient extension of  $h$  on a Euclidean neighborhood of  $K$ . The assertion applies coefficient by coefficient to differential forms.

*Proof. Strategy.* Average finitely many local extensions with smooth weights; on the half-space they all agree with  $h$ .

For each point of  $K$  choose an ambient open neighborhood  $V$  with  $V \cap \mathbb{H}^n \subset O$  on which a smooth extension exists. Choose smaller balls with closed balls inside these neighborhoods and extract a finite cover of  $K$ . The Euclidean cutoff construction in [Lemma 11.6.18](#) gives smooth nonnegative functions  $b_1, \dots, b_r$ , compactly supported in the respective neighborhoods  $V_i$ , whose sum  $B$  is positive on  $K$ . If  $h_i$  is the corresponding local extension, extend  $b_i h_i$  by zero outside  $V_i$ ; compact support makes this smooth. On the open set  $G = \{B > 0\}$  define

$$\tilde{h} = \frac{\sum_{i=1}^r b_i h_i}{B}.$$

This is smooth and  $G$  contains  $K$ . On  $G \cap \mathbb{H}^n$ ,

$$\tilde{h} = \frac{\sum_i b_i h}{B} = h,$$

because each active local extension equals  $h$  there. For a form, first intersect the extension neighborhoods for its finitely many coefficients, then use the same weights for all of them.  $\square$

**Integrating in a boundary chart.** Use the half-space form integral already defined in [Definition 13.8.2](#). The compact boundary-extension lemma supplies smooth ambient coefficients when needed for local Stokes; zero extension still occurs only at artificial chart edges.

For an oriented zero-dimensional manifold, charts into  $\mathbb{R}^0$  make singletons open. A compact support is therefore finite, by its singleton open cover. The signed finite-set convention above consequently defines every compactly supported zero-dimensional integral.

Integration on manifolds is built locally from the familiar integration on Euclidean space. With standard orientation and compactly supported continuous coefficient  $f$  on an open  $V \subseteq \mathbb{R}^n$ ,

$$\int_V f(u) du^1 \wedge \cdots \wedge du^n = \int_V f(u) d(u^1, \dots, u^n).$$

The right side is the scalar Riemann integral, defined by zero extension to a containing rectangle; its volume notation is distinct from the wedge product on the left. If  $\sigma : V \rightarrow U \subset M$  is an orientation-preserving chart parametrization and  $\omega$  has compact support in  $U$ , then

$$\int_U \omega = \int_V \sigma^* \omega.$$

In the equal-dimensional Euclidean special case, let  $\sigma : V \rightarrow U$  be a smooth change of variables between open subsets of  $\mathbb{R}^n$ . Its derivative is square. In coordinates  $x^1, \dots, x^n$ , multilinearity and alternation give

$$\begin{aligned} \sigma^*(dx^1 \wedge \cdots \wedge dx^n) &= d\sigma^1 \wedge \cdots \wedge d\sigma^n \\ &= \det D\sigma du^1 \wedge \cdots \wedge du^n. \end{aligned}$$

The determinant measures the change in oriented volume. Forms retain its sign; the unoriented scalar change-of-variables formula uses  $|\det D\sigma|$ . An orientation-reversing parametrization therefore requires a minus sign in the displayed integral identity when  $U$  keeps its fixed orientation. The chart signs in the following definition record exactly this.

**Definition 13.14.9** (Integration on an oriented manifold). Let  $M$  be an oriented  $n$ -manifold and let  $\omega$  be a smooth  $n$ -form with compact support. Choose a partition of unity subordinate to oriented coordinate charts. First localize:  $\eta_\alpha = \rho_\alpha \omega$ . In coordinates extract the scalar coefficient  $g_\alpha$  by

$$(\varphi_\alpha^{-1})^* \eta_\alpha = g_\alpha(u) du^1 \wedge \cdots \wedge du^n.$$

Define

$$\int_M \omega = \sum_\alpha s_\alpha \int_{\varphi_\alpha(U_\alpha)} g_\alpha(u) d(u_1, \dots, u_n).$$

The sequence is: localize, express in coordinates, extract the scalar top-degree coefficient, integrate, apply the orientation sign, and sum. Equivalently, in compact form,

$$\int_M \omega = \sum_\alpha s_\alpha \int_{\varphi_\alpha(U_\alpha)} (\varphi_\alpha^{-1})^*(\rho_\alpha \omega).$$

Here  $s_\alpha$  is the coordinate-frame sign defined above; each written coordinate integral uses the standard orientation of its Euclidean or half-space model. Equivalently integrate without the explicit sign in the correspondingly oriented model. Only finitely many supports  $\text{supp}(\rho_\alpha)$  meet the compact set  $\text{supp}(\omega)$ . Furthermore,

$$\text{supp}(\rho_\alpha \omega) \subseteq \text{supp}(\rho_\alpha) \cap \text{supp}(\omega) \subseteq U_\alpha.$$

This support is a closed subset of the compact set  $\text{supp}(\omega)$ , so it is compact. Its image under  $\varphi_\alpha$  is therefore compact and contained in the chart image, open in Euclidean space or relatively open in the half-space. Thus every nonzero summand is the appropriate compactly supported coordinate integral defined above, and the displayed sum is finite.

Two global decompositions are difficult to compare directly. Intersect their localized chart pieces: each smaller piece lies in one chart from each construction. The independence proof therefore follows the route

two choices  $\rightarrow$  common refinement  $\rightarrow$  overlap comparison  $\rightarrow$  coordinate invariance  $\rightarrow$  reassembly.

**Proposition 13.14.10** (Well-definedness). The preceding integral is independent of the oriented charts and subordinate partition of unity.

*Proof.* To compare two choices, compare both to a common refinement. Let  $(\rho_\alpha)$  and  $(\sigma_\beta)$  be partitions subordinate to chart families  $(U_\alpha, \varphi_\alpha)$  and  $(V_\beta, \psi_\beta)$ . Let  $K = \text{supp}(\omega)$ , which is compact. By Lemma 13.14.4, only finitely many supports in either family meet  $K$ . Thus only finitely many products below can be nonzero. Define  $\eta_{\alpha\beta} = \rho_\alpha \sigma_\beta \omega$ . Its support is closed and is contained in

$$K \cap \text{supp}(\rho_\alpha) \cap \text{supp}(\sigma_\beta) \subseteq U_\alpha \cap V_\beta.$$

It is therefore compact and lies within the chart overlap, so either chart can integrate this same localized form.

For a chart-supported form  $\theta$ , write its signed coordinate integral as

$$I_\alpha(\theta) = s_\alpha \int_{\varphi_\alpha(U_\alpha)} (\varphi_\alpha^{-1})^* \theta, \quad J_\beta(\theta) = s_\beta \int_{\psi_\beta(V_\beta)} (\psi_\beta^{-1})^* \theta.$$

On the overlap the transition determinant has sign  $s_\beta s_\alpha$ . The signed change-of-coordinates formula (Proposition 13.8.7 and Corollary 12.5.10) therefore gives  $I_\alpha(\eta_{\alpha\beta}) = J_\beta(\eta_{\alpha\beta})$ , including opposite coordinate signs. Consequently,

$$\begin{aligned} \sum_\alpha I_\alpha(\rho_\alpha \omega) &= \sum_{\alpha, \beta} I_\alpha(\eta_{\alpha\beta}) \\ &= \sum_{\alpha, \beta} J_\beta(\eta_{\alpha\beta}) \\ &= \sum_\beta J_\beta(\sigma_\beta \omega). \end{aligned}$$

The first equality uses

$$\sum_\beta \sigma_\beta = 1$$

and linearity in a fixed chart. The middle equality is the overlap formula just verified. The last uses

$$\sum_\alpha \rho_\alpha = 1.$$

Compact support and local finiteness make this a finite double sum, so no infinite rearrangement is involved. Both choices give the same integral.  $\square$



**Example 13.14.11** (The circle integral using two chart-supported pieces). Orient  $S^1$  counterclockwise and restrict  $\omega = -y \, dx + x \, dy$  from the plane. The angle parametrization  $\gamma(t) = (\cos t, \sin t)$  has  $\gamma^*\omega = dt$ , but no single angle chart covers the circle. Use  $U_+ = S^1 \setminus \{(-1, 0)\}$  with angles  $t \in (-\pi, \pi)$  and  $U_- = S^1 \setminus \{(1, 0)\}$  with angles  $t \in (0, 2\pi)$ . Both angle coordinate frames have positive orientation.

Let  $\theta$  be the smooth function from Chapter 8 that is zero on  $(-\infty, 0]$  and

$$e^{-1/s}$$

for  $s > 0$ . For  $(x, y) \in S^1$ , define

$$\rho_+(x, y) = \frac{\theta(x + 1/2)}{\theta(x + 1/2) + \theta(1/2 - x)}, \quad \rho_- = 1 - \rho_+.$$

The denominator is positive: the two arguments cannot both be nonpositive. These are smooth nonnegative weights summing to one. The support of  $\rho_+$  lies in

$$\{x \geq -1/2\} \subset U_+,$$

and that of  $\rho_-$  lies in

$$\{x \leq 1/2\} \subset U_-.$$

Both supports are compact. Thus this is an explicit subordinate partition of unity. Put  $r_\pm(t) = \rho_\pm(\gamma(t))$ , which are  $2\pi$ -periodic. The definition of manifold integration gives the two localized coordinate integrals

$$\int_{S^1} \omega = \int_{-\pi}^{\pi} r_+(t) \, dt + \int_0^{2\pi} r_-(t) \, dt.$$

Their integrands vanish near the artificial endpoints of their respective charts, as required by chart-supported integration. Split the first integral at zero and substitute  $s = t + 2\pi$  on  $[-\pi, 0]$ . Periodicity changes it to

$$\int_0^{2\pi} r_+(s) \, ds.$$

Consequently

$$\int_{S^1} \omega = \int_0^{2\pi} (r_+ + r_-)(t) \, dt = \int_0^{2\pi} 1 \, dt = 2\pi.$$

The global definition reproduces the earlier line integral by adding two chart-supported pieces; it never assumes a global angle coordinate.

## Smooth Euclidean Domains and the Riemann Integral

The manifold integral should recover the ordinary multiple integral on a smooth bounded Euclidean domain. To prove this, we must relate its boundary charts to the Jordan boundary criterion and then compare the two integrals after localization. We specify the class of domains first.

For the classical corollaries, a **compact smooth domain**  $\Omega \subset \mathbb{R}^n$  ( $n \geq 1$ ) carries the smooth structure induced by its Euclidean embedding. Interior points have identity coordinates. At each boundary point there is a smooth ambient diffeomorphism  $\varphi : W \rightarrow O$  between open subsets of  $\mathbb{R}^n$  with

$$\varphi(W \cap \Omega) = O \cap \mathbb{H}^n, \quad \varphi(W \cap \partial\Omega) = O \cap \{x_n = 0\}.$$

Its restriction is a boundary chart. Here  $\partial\Omega$  is the ordinary Euclidean boundary, which these charts identify with the manifold boundary. We use the standard Euclidean orientation. This describes only the concrete domains needed below; it assumes no theorem about the region enclosed by an arbitrary simple closed curve.

**Proposition 13.14.12** (Smooth compact domains are Jordan measurable). Every compact smooth Euclidean domain is Jordan measurable.

*Proof.* Its boundary is closed and bounded, hence compact. At each boundary point choose a straightening chart as above and a small closed coordinate rectangle  $R = R' \times [-\delta, \delta]$  inside  $O$ , centered at that point's coordinate on  $x_n = 0$ . By the boundary-straightening identity, the inverse image under  $\varphi$  of the relative interior of the central face  $R' \times \{0\}$  is a relative neighborhood of the boundary point. On the compact rectangle  $R$ ,  $D\psi$  is bounded, where  $\psi = \varphi^{-1}$ . Since  $R$  is convex, the mean-value estimate on its segments makes  $\psi$  Lipschitz there. The image of its central face is Jordan-null by Lemma 12.3.17. Choose finitely many of these boundary neighborhoods covering  $\partial\Omega$ ; their associated null face images cover it. Thus  $\partial\Omega$  is Jordan-null, and Proposition 12.3.4 proves the assertion. In dimension one the faces are points, already covered by the same lemma.  $\square$

**Proposition 13.14.13** (Agreement with the Euclidean Riemann integral). Let  $\Omega \subset \mathbb{R}^n$  be a compact smooth domain with the standard orientation, and let  $f$  be smooth on a neighborhood of  $\Omega$ . Then

$$\int_{\Omega}^{\text{manifold}} f \, dx_1 \wedge \cdots \wedge dx_n = \int_{\Omega}^{\text{Riemann}} f(\mathbf{x}) \, d\mathbf{x}.$$

*Proof.* Localize to the concrete ambient charts, use Chapter 12 to compare each pair of integrals, and add the finitely many identities. No form of Stokes is needed. The scalar integral exists by Propositions 12.3.6 and 13.14.12.

1. *Choose finitely many coordinate neighborhoods and rectangle unions.* Choose a partition of unity  $\rho_{\alpha}$  subordinate to a finite cover by the interior or straightening charts just described, shrinking the charts to connected coordinate neighborhoods if necessary. Put  $h_{\alpha} = \rho_{\alpha}f$ . Each  $h_{\alpha}$  is continuous on  $\Omega$  and has compact support inside its chart, so its scalar integral exists too. Write  $\psi_{\alpha}$  for the inverse ambient coordinate map and  $s_{\alpha}$  for the sign of its determinant. This determinant is nonzero and has constant sign on the chosen neighborhood. Standard orientation makes  $s_{\alpha}$  precisely the chart-frame sign in manifold integration.

For one nonzero localized term, its compact coordinate support lies in a finite union  $E_{\alpha}$  of small closed grid rectangles inside the ambient coordinate domain, contained in  $\mathbb{H}^n$  in a boundary chart. To obtain them, choose for each support point a ball of doubled radius inside the coordinate domain, extract finitely many balls of the original radii, and take a grid whose cell diameter is smaller than the least selected radius. Every cell meeting the support then lies in one of the doubled balls. Only finitely many such cells occur in their bounded union; in a boundary chart, align the grid with  $x_n = 0$ . They cover a relative neighborhood of the coordinate support. Their union is compact and Jordan measurable, and  $\psi_{\alpha}(E_{\alpha}) \subset \Omega$ .

2. *Compare the localized integrals.* Corollary 12.5.6 first gives measurability of this image and then, for the continuous function  $h_{\alpha}$  on it, gives

$$\int_{\psi_{\alpha}(E_{\alpha})} h_{\alpha}(\mathbf{x}) \, d\mathbf{x} = \int_{E_{\alpha}} h_{\alpha}(\psi_{\alpha}(u)) |\det D\psi_{\alpha}(u)| \, du.$$

The function vanishes outside its support. The scalar integral on the left is therefore

$$\int_{\Omega} h_{\alpha} :$$

the omitted difference is Jordan measurable by Proposition 12.3.5 and has zero integrand. The right side is also the coordinate integral over the whole chart, using zero extension only across artificial edges. Indeed the coordinate integrand vanishes outside  $E_{\alpha}$ . Meanwhile the pullback formula gives

$$\psi_{\alpha}^*(h_{\alpha} \, dx_1 \wedge \cdots \wedge dx_n) = h_{\alpha}(\psi_{\alpha}(u)) \det D\psi_{\alpha}(u) \, du_1 \wedge \cdots \wedge du_n.$$

The orientation sign gives, pointwise,

$$s_\alpha \det D\psi_\alpha = |\det D\psi_\alpha|.$$

Thus the localized manifold integral equals the localized scalar integral.

3. *Sum the localized identities.* Since

$$\sum_\alpha \rho_\alpha = 1$$

on  $\Omega$ ,

$$\sum_\alpha h_\alpha = f.$$

Sum the finitely many integral identities. Linearity and the definition of manifold integration prove the proposition.  $\square$

## 13.15 The General Stokes Theorem

The local formula is now available in each chart. Smooth weights let us add these formulas, and their derivative terms cancel because the weights sum to one. Compact support will leave only finitely many contributions, even if the manifold needs infinitely many charts.

For the boundary inclusion  $\iota : \partial M \rightarrow M$ , the notation

$$\int_{\partial M} \omega$$

always means

$$\int_{\partial M} \iota^* \omega.$$

The pullback restricts the form to tangent directions of the boundary.

The general theorem unifies the Fundamental Theorem of Calculus, Green's theorem, classical Stokes, and the divergence theorem: differentiating an integrand over a region corresponds to integrating it over the oriented boundary. The first three have already appeared in this chapter; the following section recovers all four from the single identity below.

Recall that the integral is independent of charts and smooth weights by [Definition 13.14.9](#) and [Proposition 13.14.10](#). Before asserting equality, both integrals must exist. By [Lemma 13.12.6](#), compact support of a smooth  $(n-1)$ -form  $\omega$  gives compact support of the smooth top-degree form  $d\omega$  on  $M$  and of the smooth top-degree form  $\iota^*\omega$  on  $\partial M$ . The latter manifold and smooth inclusion were constructed in [Proposition 13.10.11](#). Thus the preceding integration definitions (including dimension zero) supply both sides of the theorem.

**Theorem 13.15.1** (General Stokes theorem). Let  $M$  be an oriented smooth  $n$ -manifold with boundary,  $n \geq 1$ , and let  $\omega$  be a smooth  $(n-1)$ -form with compact support. Then

$$\int_M d\omega = \int_{\partial M} \omega.$$

*Proof.* The proof follows the same local-to-global pattern used to define manifold integration:

localize  $\rightarrow$  coordinates  $\rightarrow$  local Euclidean Stokes  $\rightarrow$  reassemble.

All ingredients are now in place. The new payoff is that derivatives of the smooth localization weights cancel, leaving the original derivative. Choose a partition of unity subordinate to oriented manifold-with-boundary charts that carry  $M$  locally into  $\mathbb{H}^n$  and carry  $\partial M$  into  $\{x_n = 0\}$ . Compactness of  $\text{supp}(\omega)$  and local finiteness show that only finitely many forms

$$\rho_\alpha \omega$$

are nonzero. Each has compact support contained in the corresponding chart, because  $\text{supp}(\rho_\alpha \omega) \subseteq \text{supp}(\rho_\alpha) \cap \text{supp}(\omega)$ . By the product rule,

$$d(\rho_\alpha \omega) = d\rho_\alpha \wedge \omega + \rho_\alpha d\omega.$$

After summing, the first terms cancel because

$$\sum_\alpha d\rho_\alpha = d\left(\sum_\alpha \rho_\alpha\right) = d1 = 0.$$

This identity is legitimate because the partition is locally finite: near each point both sums are finite, so exterior differentiation commutes with the displayed sum. In each local Stokes calculation multiply both sides by the chart sign  $s_\alpha$ . Outward-normal-first transfers the same sign to its genuine boundary integral, so the local formula remains valid for either orientation of the coordinate model. It is therefore enough to prove the formula for a form supported in one chart, where “supported in” means that its closure-based support lies in the chart domain.

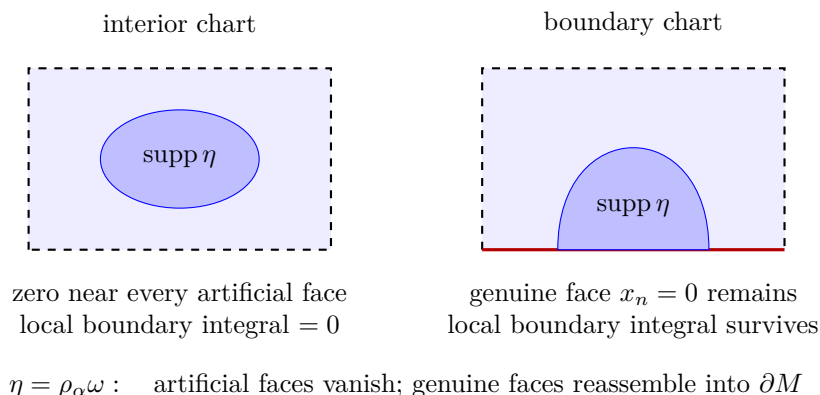


Figure 13.23: Local Stokes applied to a chart-supported form. Compact support removes artificial boundary terms. Smooth extension across the genuine face need not be zero, and this is exactly where the manifold boundary integral comes from.

*Interior charts.* In an interior chart, the coordinate support is compact in the open chart image. Hence the coordinate representative vanishes on a neighborhood of the artificial boundary of that image and extends smoothly by zero to  $\mathbb{R}^n$ . Choose a rectangle whose interior contains the extended support. The extension vanishes near the rectangle boundary, so [Theorem 13.8.4](#) gives the formula with no manifold boundary contribution.

*Boundary charts.* In a boundary chart, [Lemma 13.14.8](#) combines the local smooth extensions into one smooth (not necessarily zero) extension near the compact coordinate support, across

$$x_n = 0$$

to a neighborhood in  $\mathbb{R}^n$ . Compact coordinate support permits zero extension across the remaining artificial boundary of the coordinate image; no zero extension is made across the genuine face  $x_n = 0$ . Choose a rectangle

$$[a_1, b_1] \times \cdots \times [a_{n-1}, b_{n-1}] \times [0, b_n]$$

whose relative interior contains its coordinate support and whose artificial faces are disjoint from it. Using [Lemma 11.6.18](#) in the ambient extension neighborhood to choose a cutoff equal to one near that support, retain the smooth extension across  $x_n = 0$  and extend by zero only across the artificial faces; the resulting form vanishes near all of those faces. Apply [Theorem 13.8.4](#) to the smooth extension. Its only surviving boundary term is the face  $x_n = 0$ , which is the coordinate representative of the genuine manifold boundary. The outward-normal-first convention gives exactly the induced boundary sign. Pullback compatibility from [Proposition 13.12.4](#) transfers each coordinate formula back to the chart.

*Reassembly.* Linearity and the cancellation above now give

$$\begin{aligned} \int_M d\omega &= \sum_{\alpha} \int_M d(\rho_{\alpha}\omega) \\ &= \sum_{\alpha} \int_{\partial M} \iota^*(\rho_{\alpha}\omega) = \int_{\partial M} \iota^*\omega. \end{aligned}$$

All nonzero contributions belong to a finite list, so the reassembly uses only finite sums. □

**Corollary 13.15.2** (Integration by parts for differential forms). Let  $M$  be an oriented smooth  $n$ -manifold with boundary, let

$$0 \leq p \leq n-1,$$

let  $\alpha$  be a smooth  $p$ -form with compact support, and let  $\beta$  be a smooth

$$(n-p-1)\text{-form}.$$

Then

$$\int_M d\alpha \wedge \beta = \int_{\partial M} \alpha \wedge \beta - (-1)^p \int_M \alpha \wedge d\beta.$$

*Proof.* The form

$$\alpha \wedge \beta$$

has degree  $n-1$  and compact support, so [Theorem 13.15.1](#) applies. The graded Leibniz rule from [Theorem 13.7.11](#) gives

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta.$$

Integrate this identity over  $M$ , apply Stokes to its left side, and rearrange. □

**Remark 13.15.3** (Three levels of integration by parts). The same structural identity appears at three increasing levels of generality:

$$\begin{aligned} &\text{product rule plus the Fundamental Theorem of Calculus} \\ &\implies \text{one-variable integration by parts,} \\ &\text{one-variable integration by parts plus Fubini} \\ &\implies \text{the coordinatewise rectangular formula,} \\ &\quad \text{graded Leibniz rule plus Stokes} \\ &\implies \text{integration by parts for forms.} \end{aligned}$$

The final formula is coordinate-free. Green's identities and weak formulations of partial differential equations are important later descendants, but they require additional analytic machinery.

## 13.16 Classical Theorems Revisited

These are now instances of one theorem. The form degree specifies the dimension of integration; the exterior derivative specifies the interior quantity. In three dimensions the flux form is the one already calculated on surface patches. The general theorem additionally joins the chart pieces consistently and handles arbitrary compact smooth oriented surfaces.

Choose the form and check its degree before applying Stokes:

| Formula        | $\dim M$ | degree of $\omega$ | degree of $d\omega$ |
|----------------|----------|--------------------|---------------------|
| FTC            | 1        | 0                  | 1                   |
| Green          | 2        | 1                  | 2                   |
| Surface Stokes | 2        | 1                  | 2                   |
| Divergence     | 3        | 2                  | 3                   |

A surface in  $\mathbb{R}^3$  is still a *two*-dimensional integration domain. For  $F = (F_1, F_2, F_3)$  the work and flux forms are different objects:

$$\begin{aligned}\alpha_F &= F_1 \, dx + F_2 \, dy + F_3 \, dz, \\ \beta_F &= F_1 \, dy \wedge dz + F_2 \, dz \wedge dx + F_3 \, dx \wedge dy.\end{aligned}$$

Here  $d\alpha_F$  encodes curl, while  $d\beta_F$  encodes divergence. Use the work one-form for circulation and the flux two-form for flux.

**Corollary 13.16.1** (Fundamental theorem of calculus). For a smooth function  $f$  on  $[a, b]$ ,

$$\int_a^b f'(x) \, dx = f(b) - f(a).$$

*Proof.* Apply [Theorem 13.15.1](#) to the oriented interval and the 0-form  $f$ . The interior integral is the ordinary Riemann integral by [Proposition 13.14.13](#); the boundary integral has the two endpoint signs.  $\square$

**Corollary 13.16.2** (Green's theorem). Let  $D \subseteq \mathbb{R}^2$  be a compact smooth Euclidean domain as in “[Smooth Euclidean Domains and the Riemann Integral](#),” with the standard orientation, and let  $P, Q$  be smooth on a neighborhood of  $D$ . The induced boundary orientation is the usual positive orientation. Then

$$\int_{\partial D} P \, dx + Q \, dy = \int_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy.$$

*Proof.* Apply [Theorem 13.15.1](#) to

$$\omega = P \, dx + Q \, dy.$$

Its derivative is

$$(\partial Q / \partial x - \partial P / \partial y) \, dx \wedge dy.$$

By [Proposition 13.14.13](#), the manifold integral of this form is the displayed scalar Riemann integral.  $\square$

**Corollary 13.16.3** (Classical Stokes theorem). Let  $S$  be an oriented smooth compact surface with boundary in  $\mathbb{R}^3$ , carrying the smooth structure induced by its Euclidean embedding, and let

$$F = (P, Q, R)$$

be smooth near  $S$ . Then

$$\int_{\partial S} P \, dx + Q \, dy + R \, dz = \int_S (\nabla \times F) \cdot n \, dS.$$

*Proof.* Let  $i : S \rightarrow \mathbb{R}^3$  and  $j : \partial S \rightarrow S$  be the inclusions and let  $\alpha = P \, dx + Q \, dy + R \, dz$  on the ambient neighborhood. The smooth form on  $S$  is  $i^*\alpha$ . It has compact support because  $S$  is compact. Apply [Theorem 13.15.1](#) to obtain

$$\int_{\partial S} j^* i^* \alpha = \int_S d(i^* \alpha) = \int_S i^*(d\alpha).$$

The last equality uses commutation of pullback with  $d$ . Evaluation on boundary velocities identifies the first integral with boundary work; evaluation of  $d\alpha$  on each oriented surface tangent pair gives  $(\nabla \times F) \cdot (\sigma_u \times \sigma_v)$ , identifying the last integral with curl flux by the parametrized formula.  $\square$

**Definition 13.16.4** (Flux in arbitrary dimension). For a smooth vector field  $F = (F_1, \dots, F_n)$  on a Euclidean open set, its flux form is the  $(n-1)$ -form

$$\omega_F = \sum_{j=1}^n (-1)^{j-1} F_j \, dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_n.$$

For an oriented boundary with inclusion  $\iota$ , the notation

$$\int_{\partial \Omega} F \cdot n \, dS \quad \text{means} \quad \int_{\partial \Omega} \iota^* \omega_F.$$

This defines the shorthand through form integration; no separate hypersurface-measure construction is required here.

For  $n = 3$  and an oriented patch  $\sigma(u, v)$ , evaluation gives  $\omega_F(\sigma_u, \sigma_v) = \det(F, \sigma_u, \sigma_v) = F \cdot (\sigma_u \times \sigma_v)$  by [Proposition 13.3.2](#). Thus the definition agrees with the earlier unit-normal flux formula. For  $n = 1$ , the flux form is the function  $F_1$  and the signed endpoint integral gives its value at the right endpoint minus its value at the left.

**Corollary 13.16.5** (Divergence theorem). Let  $\Omega \subseteq \mathbb{R}^n$  be a compact smooth Euclidean domain as in “[Smooth Euclidean Domains and the Riemann Integral](#),” with the standard orientation, and let

$$F = (F_1, \dots, F_n)$$

be smooth near  $\Omega$ . The induced boundary orientation is the usual outward orientation. Then

$$\int_{\partial \Omega} F \cdot n \, dS = \int_{\Omega} \left( \sum_{j=1}^n \frac{\partial F_j}{\partial x_j} \right) d\mathbf{x}.$$

*Proof.* Apply [Theorem 13.15.1](#) to the flux form  $\omega_F$  of [Definition 13.16.4](#). Its exterior derivative is

$$\left( \sum_j \partial F_j / \partial x_j \right) dx_1 \wedge \cdots \wedge dx_n :$$

only the  $j$ th derivative of  $F_j$  survives, and moving  $dx_j$  past  $j-1$  factors cancels the coefficient sign. Its boundary integral is the outward flux by definition. Finally, [Proposition 13.14.13](#) identifies the manifold integral of its derivative with the scalar Riemann integral on the right.  $\square$

**Computational practice****Exercise 5.** Compute

$$\int_{\gamma} x^2 \, ds$$

on the quarter circle of radius  $R > 0$ . Compute the work of  $(y, x)$  along  $(t, 3t)$ ,  $0 \leq t \leq 1$ , and its reverse.

**Exercise 6.** On the cylinder side of radius  $R$  and height  $h$ , compute the scalar surface integral of  $z$  and the outward flux of  $(2x, 2y, z)$ . Explain which changes sign if orientation is reversed.

**Exercises****Exercise 7.** Prove that

$$\dim \Lambda^k((\mathbb{R}^n)^*) = \binom{n}{k}.$$

**Exercise 8.** Let  $\alpha, \beta$  be 1-covectors. Prove

$$\alpha \wedge \beta = -\beta \wedge \alpha$$

and give an example in which

$$\alpha \wedge \beta \neq 0.$$

**Exercise 9.** For  $\omega = x \, dy + y \, dz + z \, dx$ , compute  $d\omega$ .**Exercise 10.** Let

$$\sigma(u, v) = (u^2 - v^2, 2uv).$$

Compute

$$\sigma^*(x \, dy - y \, dx).$$

**Exercise 11.** Use [Theorem 13.8.4](#) in dimension one to recover the Fundamental Theorem of Calculus.

**Exercise 12.** Apply [Corollary 13.15.2](#) to the oriented interval  $[a, b]$  and the forms  $\alpha = u$  and  $\beta = v$  to recover the one-variable integration-by-parts formula for smooth functions  $u$  and  $v$ .

**Exercise 13.** For the unit square with its standard orientation, evaluate

$$\int_{\partial Q} x \, dy$$

and compare it with

$$\int_Q d(x \, dy).$$



**Exercise 14.** Let  $D \subset \mathbb{R}^2$  be a compact smooth planar domain as in Corollary 13.16.2, with positively oriented boundary. Use that theorem to prove

$$\text{area}(D) = \frac{1}{2} \int_{\partial D} (x \, dy - y \, dx).$$

**Exercise 15.** Use the divergence theorem to compute the outward flux of

$$F(x, y, z) = (x, y, z)$$

across the unit sphere.

**Exercise 16.** Explain why

$$d^2 = 0$$

is compatible with the boundary-of-boundary principle reflected in Stokes' theorem. Distinguish this compatibility from a chain-level theorem, which has not been developed here.

**Exercise 17** (Hypotheses and synthesis). Test the scope of the preceding constructions.

- (a) Why does the full circular parametrization on  $[0, 2\pi]$  fail the injective-patch hypothesis? Describe a valid subdivision.
- (b) Why can a two-sided curve in  $[0, \infty)$  through zero not represent the outward tangent vector there?
- (c) If  $\rho_1 + \rho_2 = 1$  near the support of  $\omega$ , expand  $d(\rho_1\omega) + d(\rho_2\omega)$  and identify the cancelling terms.
- (d) For  $\sigma(u, v) = (u, v, u^2 + v^2)$  on a triangle and  $F = (-y, x, 0)$ , verify patch Stokes by pulling back work to the triangle and computing curl flux. State exactly why this does not by itself prove Stokes on a sphere.

**Exercise 18.** For  $\gamma(t) = (t, t^2)$  on  $[0, 1]$ , compute the work of  $F = (y, x)$ . Repeat with

$$h(s) = (s + s^2)/2$$

by substitution and under reversal. Explain why doubling the traversal is a different operation.

**Exercise 19.** Pull back  $z \, dx \wedge dy$  by  $(u, v) \mapsto (u, v, uv)$  and integrate over  $[0, 1]^2$ . Interpret the result as an oriented flux.

**Exercise 20.** For  $\omega = x \, dy - y \, dx$ , compute  $\omega_{(1,2)}(3, 4)$ , its pullback by  $t \mapsto (t, t^2)$ , and  $d\omega$ . State the type of each answer and why a two-form pulled back to this curve is zero.

**Exercise 21.** Use the upper- and right-semicircle charts to transform a tangent coordinate at

$$(1/\sqrt{2}, 1/\sqrt{2}).$$

Check the answer by differentiating the corresponding parametrizations in the ambient plane.

**Exercise 22.** For a positively oriented interval verify its two endpoint chart signs and Stokes for  $f(x) = x^2$  on  $[1, 3]$ . Explain why the endpoint at 3 can have a negative chart sign and a positive boundary sign.

**Exercise 23.** Construct two smooth weights on an overlapping cover of an interval by normalizing cutoff functions. Check positivity of the denominator and show directly that  $d\rho_1 + d\rho_2 = 0$ .

**Exercise 24.** For a disk covered by two overlapping surface charts, explain which chart edges disappear in the Stokes proof and which boundary remains. Distinguish cancellation of derivatives of weights from zero extension near an artificial chart edge.

**Exercise 25.** For two circle angle charts differing by  $2\pi$  on an overlap component, compute their transition derivative and check the tangent-vector equivalence relation. Then replace an angle coordinate by twice that coordinate and compute how a tangent-vector component changes without changing the vector.

**Exercise 26.** Use the coordinate transformation formula for  $df_p$  to compare two source and two target charts for  $f : S^1 \rightarrow S^1$ ,  $f(\cos t, \sin t) = (\cos 2t, \sin 2t)$  locally. Check explicitly that equivalent source representatives give equivalent outputs.

**Exercise 27.** Reconstruct the common-refinement proof for manifold integration. Identify which support assumption permits each overlap integral and which hypothesis makes the double sum finite.

**Exercise 28.** Let  $f : S^1 \rightarrow S^1$  be  $f(x, y) = (x^2 - y^2, 2xy)$ , and let  $h(x, y) = x$  on  $S^1$ . In angle coordinates compute  $f^*(dh)$  and  $d(f^*h)$ . Repeat near the same point with source coordinate  $s = 2t$  and explain why the two coordinate answers describe the same one-form.

**Exercise 29.** For the inclusion  $\iota : S^1 \rightarrow \mathbb{R}^2$  and  $\alpha = x \, dy$ , compute  $\iota^*\alpha$  in angle coordinates and verify  $d(\iota^*\alpha) = \iota^*(d\alpha) = 0$ . Explain the role of  $\Omega^2(S^1) = \{0\}$ . Then reconstruct the overlap identity that makes the manifold exterior derivative well defined.

**Exercise 30** (Charts, coordinate functions, and tangent columns). On  $S^1$  let  $U = \{y > 0\}$ ,  $V = \{x > 0\}$ ,  $\varphi(x, y) = x$ , and  $\psi(x, y) = y$ . Write the full domains and codomains of both charts, their inverses, and both overlap transitions. For a general chart  $\chi : W \rightarrow \mathbb{R}^n$ , write its coordinate functions using projections. At

$$(\sqrt{3}/2, 1/2)$$

convert  $[\varphi, 2]$  to its  $\psi$  representative and verify the equality of the two parametrized velocities in  $\mathbb{R}^2$ .

**Exercise 31** (Dual bases and a graph differential). For  $M = \{(x, y, x^2 + y^2) : (x, y) \in \mathbb{R}^2\}$ , use graph coordinates to construct the coordinate tangent and dual bases at  $(1, 2, 5)$ . Verify all four duality evaluations. Compute the differential of the inclusion  $M \rightarrow \mathbb{R}^3$  on  $[\varphi, (2, -1)]$  and explain which space contains its input and which contains its output.

**Exercise 32** (Orientation from charts and forms). Use the circle transition above to test the signs of the two coordinate bases. Compute the coefficient of  $x \, dy - y \, dx$  in each chart on the first-quadrant overlap and check that it gives the same tangent orientation. Verify the endpoint chart signs and boundary signs for  $[a, b]$  oriented by decreasing  $x$ . Finally explain, using [Theorem 13.13.7](#), why a nonorientable manifold cannot admit a nowhere-zero top form; no example of nonorientability is required.

## 13.17 Perspective and Further Directions

Differentiation, orientation, and integration meet in Stokes' theorem. The subjects below develop different parts of that meeting. This section is a map for further study; Appendices [E–H](#) provide selected developed continuations. Named later theorems in this section are unproved previews, not prerequisites for the principal Chapters 1–13 route.

There are three overlapping starting points: topology leads toward general, algebraic, and geometric topology; smooth manifolds lead toward differential topology, differential geometry, Lie theory, and de Rham theory; analysis leads toward function spaces, PDE, complex analysis, probability, and geometric analysis. These are connections, not rigid disciplinary partitions.

| Starting mechanism    | A developed bridge and later direction                   |
|-----------------------|----------------------------------------------------------|
| Quotients and gluing  | <a href="#">Appendix E</a> → geometric topology          |
| Constant-rank ideas   | <a href="#">Appendix F</a> → differential topology       |
| Manifolds and algebra | <a href="#">Appendix G</a> → Lie theory                  |
| Forms and Stokes      | <a href="#">Appendix H</a> → de Rham theory and topology |
| Analysis and geometry | PDE and variational methods → geometric analysis         |

### General Topology

The open-set language of Chapter 10 also describes spaces obtained by identifying points. [Appendix E](#), *Quotient Spaces, Gluing, and Classical Manifolds*, develops quotient topology and its universal property, then applies them to circles, tori, the cylinder, the Möbius strip, the Klein bottle, and real projective spaces. This developed continuation makes the chapter's gluing pictures rigorous while keeping the main analysis-to-Stokes development independent of quotient topology.

The external refinement theorem used for partitions of unity is another point where topology does work that coordinate calculations alone cannot. A topology course develops product and quotient spaces, separation axioms, compactness, connectedness, paracompactness, and criteria for metrizability. That broader theory explains why certain global constructions are possible.

General topology supplies the language for constructing and comparing spaces. Algebraic topology asks what global information can be extracted from them by assigning algebraic invariants.

### Algebraic Topology

Algebraic topology assigns algebraic invariants to topological spaces and continuous maps, transferring geometric questions to algebra, where computation and comparison are often more tractable. This deliberately loses information: distinct spaces may have the same invariant. The art is to forget enough structure to obtain something computable while retaining enough to detect the phenomenon of interest. Fundamental groups, homology, cohomology, and Euler characteristics are useful invariants, not complete classifications. The working pattern is to construct an invariant

that respects continuous deformation, compute it for model spaces, and then deform or compare a new problem until those computations apply:

spaces and continuous maps  $\longrightarrow$  groups, chain complexes, and cohomology groups.

The definitions below make this philosophy concrete; all landmark results are unproved previews and remain outside the analysis-to-Stokes route.

**Definition 13.17.1** (Homotopy and homotopy equivalence). For continuous  $f, g : X \rightarrow Y$ , a **homotopy** from  $f$  to  $g$  is a continuous map  $H : X \times [0, 1] \rightarrow Y$  with

$$H(x, 0) = f(x), \quad H(x, 1) = g(x) \quad (x \in X).$$

Write  $f \simeq g$ . Spaces  $X, Y$  are **homotopy equivalent**, written  $X \simeq Y$ , if continuous maps  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$  satisfy

$$g \circ f \simeq \text{id}_X, \quad f \circ g \simeq \text{id}_Y.$$

Homotopy formalizes continuous deformation of maps. Homotopy equivalence is weaker than homeomorphism: for instance  $\mathbb{R}^n$  contracts to a point by  $H(\mathbf{x}, t) = (1 - t)\mathbf{x}$ , but is not homeomorphic to a point for  $n \geq 1$ . It forgets some local structure while retaining global shape information.

**Definition 13.17.2** (Based loops and the fundamental group). A **path** is a continuous map  $\gamma : [0, 1] \rightarrow X$ ; its endpoints are  $\gamma(0)$  and  $\gamma(1)$ . Path homotopies keep both endpoints fixed. Fix  $x_0 \in X$ . A **based loop** is a path with  $\gamma(0) = \gamma(1) = x_0$ . A **based homotopy** from  $\gamma_0$  to  $\gamma_1$  is a continuous  $H : [0, 1]^2 \rightarrow X$  satisfying

$$\begin{aligned} H(s, 0) &= \gamma_0(s), & H(s, 1) &= \gamma_1(s), \\ H(0, t) &= H(1, t) = x_0 & \text{for every } s, t \in [0, 1]. \end{aligned}$$

Thus the basepoint stays fixed throughout the deformation. Define

$$(\alpha * \beta)(s) = \begin{cases} \alpha(2s), & 0 \leq s \leq 1/2, \\ \beta(2s - 1), & 1/2 \leq s \leq 1, \end{cases} \quad e_{x_0}(s) = x_0, \quad \alpha^{-1}(s) = \alpha(1 - s).$$

Concatenation traverses  $\alpha$  first and then  $\beta$ . The **fundamental group** is

$$\pi_1(X, x_0) = \{\text{based loops at } x_0\} / \text{based homotopy}, \quad [\alpha][\beta] = [\alpha * \beta].$$

**Group structure (unproved preview).** Based homotopy is an equivalence relation, this product is well defined, and it makes  $\pi_1(X, x_0)$  a group with identity  $[e_{x_0}]$  and inverse  $[\gamma]^{-1} = [\gamma^{-1}]$ . The loops  $(\alpha * \beta) * \gamma$  and  $\alpha * (\beta * \gamma)$  generally have different parametrizations; they are based-homotopic, which gives associativity on classes. The symbol  $\pi_1$  denotes a group, whereas Appendix E's  $\pi$  denotes a projection map.

Continuous pointed maps induce group homomorphisms by  $[\gamma] \mapsto [f \circ \gamma]$ . Composing a based homotopy with  $f$  proves representative independence, and composition respects concatenation.

**Landmark examples (unproved previews).** For  $n \geq 1$  and any basepoint in the indicated spaces,

$$\pi_1(\mathbb{R}^n, x_0) = 0, \quad \pi_1(S^1, x_0) \cong \mathbb{Z}, \quad \pi_1(T^2, x_0) \cong \mathbb{Z}^2.$$

Euclidean loops contract to a point. On the circle the integer counts winding; on the torus the two integers count winding in the circular directions of Figure 13.14. In a path-connected space, a chosen connecting path gives an isomorphism between groups at different basepoints (unproved preview); that isomorphism need not be canonical without the path choice.

**Building a space one dimension at a time.** Appendix E glues edges of simple domains. Cell attachment extends this idea to disks in successive dimensions.

**Definition 13.17.3** (Cells, skeleta, and CW complexes). Put  $D^n = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \leq 1\}$  and, for  $n \geq 1$ ,  $S^{n-1} = \partial D^n$ . An  $n$ -cell is a copy of the open ball  $\text{int } D^n$ ; a 0-cell is a point. Start with a discrete set  $X^0$  of 0-cells. Given continuous **attaching maps**  $\phi_\alpha : S^{n-1} \rightarrow X^{n-1}$ , form

$$X^n = \left( X^{n-1} \sqcup \coprod_\alpha D_\alpha^n \right) / \sim, \quad z \sim \phi_\alpha(z) \quad (z \in \partial D_\alpha^n).$$

The disjoint union carries the topology in which a subset is open exactly when its intersection with each summand is open; the displayed identification is then given the quotient topology. The canonical maps from the attached disks into  $X$  are the **characteristic maps**  $\varphi_\alpha^n : D_\alpha^n \rightarrow X$ . Each maps its interior homeomorphically onto its cell  $e_\alpha^n$ ; these cells partition  $X$ . Here “open cell” means a copy of an open ball, not necessarily an open subset of  $X$ . The  $n$ -**skeleton**  $X^n$  is the union of all cells of dimension at most  $n$ , giving  $X^0 \subseteq X^1 \subseteq \cdots$ . A **CW complex** is the resulting Hausdorff space

$$X = \bigcup_{n \geq 0} X^n$$

with these conditions:

- (a) **Closure finiteness (C)**: the closure of each cell meets only finitely many cells.
- (b) **Weak topology (W)**:  $A \subseteq X$  is closed exactly when its inverse image under every characteristic map  $D_\alpha^n \rightarrow X$  (the map from the attached disk) is closed in that disk.

It is **finite** if it has finitely many cells in total.

The attaching map may identify different boundary points; a closed cell need not itself be a disk. In the finite examples below the topology is precisely that obtained from the displayed attachments.

**Example 13.17.4** (Cell structures for familiar quotients). The following standard cell descriptions connect to Appendix E; their general existence identifications are previews, not new prerequisites.

- $S^1$  has one vertex and one 1-cell: attach both endpoints of an interval to that vertex, as in [Section E.3](#).
- For  $n \geq 1$ ,

$$S^n \cong D^n / \partial D^n$$

by [Section E.3](#) (Figure E.2), so one 0-cell and one  $n$ -cell suffice. Separately,  $S^0$  has two 0-cells.

- For  $T^2$ , the square has one vertex class, two edge classes  $a, b$ , and one 2-cell. Let  $a$  follow increasing  $x$  on horizontal edges and  $b$  increasing  $y$  on vertical edges. Counterclockwise traversal from  $(0,0)$  reads  $aba^{-1}b^{-1}$ . Thus there is one 0-cell, two 1-cells, and one 2-cell, agreeing with [Section E.4](#). The relation to the fundamental group is an *unproved preview*:

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z}^2.$$

Here the presentation means that the generators commute, with no further relations; its derivation requires later tools.

- $\mathbb{RP}^n$  has one cell in every dimension  $0, \dots, n$  (unproved preview). This cell structure can be understood from the filtration

$$\mathbb{RP}^0 \subset \mathbb{RP}^1 \subset \dots \subset \mathbb{RP}^n$$

together with the antipodal sphere model illustrated in Figure E.8. Geometrically, build  $S^k$  from its equatorial  $S^{k-1}$  and two open hemispheres, starting with the two points of  $S^0$ . Antipodal cells occur in pairs in every dimension; identifying each pair leaves one cell in each dimension  $0, \dots, n$ . The complement of  $\mathbb{RP}^{k-1}$  in  $\mathbb{RP}^k$  is its affine  $k$ -cell.

- The figure eight  $S^1 \vee S^1$ , formed by identifying one point of each circle, has one vertex and two 1-cells. At the vertex four branches meet; it is not a 1-manifold. CW complexes extend well beyond manifolds.

**Definition 13.17.5** (Euler characteristic). For a finite CW complex with  $c_k$  cells of dimension  $k$ , define

$$\chi(X) = \sum_{k \geq 0} (-1)^k c_k.$$

For a finite two-dimensional CW complex put  $V = c_0$ ,  $E = c_1$ , and  $F = c_2$ . Then the definition specializes to

$$\chi(X) = V - E + F.$$

This recovers the classical Euler formula for finite cell decompositions of surfaces:  $\chi(S^2) = 2$  whereas  $\chi(T^2) = 0$ ; the value is not always 2. The cell structures just displayed give

$$\begin{aligned} \chi(S^1) &= 1 - 1 = 0, & \chi(S^2) &= 1 + 1 = 2, \\ \chi(S^n) &= 1 + (-1)^n, & \chi(T^2) &= 1 - 2 + 1 = 0, \\ \chi(\mathbb{RP}^n) &= \sum_{k=0}^n (-1)^k = \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases} \end{aligned}$$

The Klein square likewise has one vertex, two edges, and one face, so  $\chi(K) = 1 - 2 + 1 = 0$ . **Invariance (unproved preview).** If a space admits two finite CW decompositions, their alternating cell counts agree. Thus  $\chi$  is an invariant of the space rather than its chosen cell structure. In particular,  $\chi(S^2) \neq \chi(T^2)$  rules out a homeomorphism. But  $\chi(S^1) = \chi(T^2) = 0$ : useful information need not be complete information.

**Singular homology: cycles modulo boundaries.** We now choose real coefficients throughout, to prepare for the comparison with differential forms. The integral version instead uses free abelian groups.

**Definition 13.17.6** (Singular chains and homology). The standard  $k$ -simplex is

$$\Delta^k = \{(t_0, \dots, t_k) \in \mathbb{R}^{k+1} : t_i \geq 0, \sum_i t_i = 1\}.$$

For  $k \geq 1$  and  $0 \leq i \leq k$ , the **face map**  $\delta_i : \Delta^{k-1} \rightarrow \Delta^k$  inserts zero as coordinate  $i$ . A **singular  $k$ -simplex** in  $X$  is a continuous map  $\sigma : \Delta^k \rightarrow X$ . Let  $C_k(X; \mathbb{R})$  be the free real vector space

on these maps: its elements are finite formal linear combinations of them. For  $k \geq 1$ , define the boundary on generators and extend linearly:

$$\partial_k : C_k(X; \mathbb{R}) \rightarrow C_{k-1}(X; \mathbb{R}), \quad \partial_k \sigma = \sum_{i=0}^k (-1)^i \sigma \circ \delta_i.$$

Put  $C_{-1} = 0$  and  $\partial_0 = 0$ . The **boundary identity (unproved preview)** is  $\partial_{k-1} \partial_k = 0$ . Thus boundaries are cycles, and the following quotient is defined:

$$\begin{aligned} Z_k(X; \mathbb{R}) &= \ker \partial_k, & B_k(X; \mathbb{R}) &= \operatorname{im} \partial_{k+1}, \\ H_k(X; \mathbb{R}) &= Z_k(X; \mathbb{R}) / B_k(X; \mathbb{R}). \end{aligned}$$

The inclusion  $B_k \subseteq Z_k$  follows from the boundary identity. Cycles have no boundary; boundaries are cycles coming from one dimension higher. Homology records cycles modulo those already bounding a chain. A singular simplex need not be injective, and these chain spaces generally have many more generators than a cell structure.

**Continuous maps induce homology maps.** For continuous  $f : X \rightarrow Y$ , put  $f_*(\sigma) = f \circ \sigma$  on singular simplices and extend linearly to chains. The boundary formula gives  $\partial f_* = f_* \partial$ , so cycles map to cycles and boundaries to boundaries. Hence there is an induced linear map

$$f_* : H_k(X; \mathbb{R}) \rightarrow H_k(Y; \mathbb{R}), \quad [c] \mapsto [f_* c].$$

Again  $(g \circ f)_* = g_* \circ f_*$  and  $(\operatorname{id}_X)_* = \operatorname{id}$ . The passage from continuous maps to linear maps on chains, then to linear maps on homology, realizes the opening philosophy concretely.

**Definition 13.17.7** (Real singular cohomology). Put  $C^k(X; \mathbb{R}) = \operatorname{Hom}_{\mathbb{R}}(C_k(X; \mathbb{R}), \mathbb{R})$  and

$$\delta^k : C^k \rightarrow C^{k+1}, \quad (\delta^k \lambda)(c) = \lambda(\partial_{k+1} c).$$

Then  $\delta^{k+1} \delta^k = 0$  by the boundary identity. With  $C^{-1} = 0$ , define

$$H_{\text{sing}}^k(X; \mathbb{R}) = \ker \delta^k / \operatorname{im} \delta^{k-1}.$$

**Invariance (unproved preview).** Homotopy equivalent spaces have isomorphic singular homology and cohomology groups. These definitions therefore supply topological invariants, not just algebra attached to a presentation.

For a finite CW complex, the further identity

$$\chi(X) = \sum_k (-1)^k \dim H_k(X; \mathbb{R})$$

is an unproved preview. Cellular homology makes this computationally useful: it uses one generator per cell and boundary maps determined by the attachments, not just by cell counts. A later course proves agreement with singular homology. Hatcher [22] provides further reading for these optional previews. Xiao's lecture notes [23] supply a supplementary pedagogical benchmark for the progression through maps, homotopy, chains, and CW examples; the full course is not imported here. The later “[De Rham Theory](#)” discussion compares real singular cohomology with the differential forms already developed.

## Geometric Topology

Geometric topology studies manifolds and related spaces through their geometric realization, decomposition, and global topological structure. Its working pictures involve cutting, gluing, decomposition, embedding, and isotopy, meaning a deformation through embeddings. The surfaces in Appendix E are an accessible starting point: one can ask how a cut exposes a handle, or whether two placements of a curve can be deformed into one another without crossings. Dimensions three and four contain especially rich phenomena beyond this introductory picture.

Chapter 10 supplies the topological language; Appendix E supplies actual quotient and gluing constructions; algebraic topology supplies invariants with which to test them. Algebraic topology emphasizes extracting algebraic invariants, while geometric topology emphasizes understanding the spaces through their geometric and topological structure. These emphases overlap: an invariant can obstruct an embedding or distinguish two constructions. No classification theorem for surfaces or higher-dimensional manifolds is assumed here.

## Differential Topology

Differential topology studies smooth manifolds and smooth maps through their qualitative and global behavior, emphasizing properties that persist under appropriate smooth deformations. Its entry point is

local differential information  $\longrightarrow$  global topology of smooth maps and manifolds.

Appendix F supplies immersions, submersions, embeddings, regular values, and the constant-rank theorem. Later questions include: how typical are regular values; when do two submanifolds meet in the expected dimension; and how does the topology of a level set change as its level crosses a critical value?

Sard's theorem, transversality and intersection theory, degree theory, indices of vector fields, Morse theory, and cobordism develop these questions beyond Appendix F's scope. Differential geometry often adds metrics and curvature, while differential topology emphasizes smooth structure and smooth maps at a more topological level. This is a useful distinction of emphasis, with substantial overlap between the subjects.

## Differential Geometry

Tangent spaces tell us what directions are available, but do not assign lengths or angles. A **Riemannian metric** on a smooth manifold  $M$  is a family

$$g_p : T_p M \times T_p M \rightarrow \mathbb{R} \quad (p \in M)$$

of positive-definite inner products, varying smoothly: in each chart the functions

$$g_{ij}(p) = g_p((\partial/\partial x^i)|_p, (\partial/\partial x^j)|_p)$$

are smooth. For a smooth curve  $\gamma$ , write

$$\dot{\gamma}(t) = d\gamma_t(\partial/\partial t).$$

Its speed and length are

$$\|\dot{\gamma}(t)\|_g = \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))}, \quad L(\gamma) = \int_a^b \|\dot{\gamma}(t)\|_g dt.$$



Piecewise smooth curves use the sum of the lengths of their pieces.

On  $S^2$ , the inclusion  $i : S^2 \rightarrow \mathbb{R}^3$  gives  $g_p(v, w) = di_p(v) \cdot di_p(w)$ , recovering the usual surface metric. For example  $\gamma(t) = (\cos t, \sin t, 0)$  has unit speed, so one full equatorial circuit has length  $2\pi$ . The sphere's nonconstant geodesics are great circles with constant-speed parametrizations (unproved preview). Geodesics are curves whose motion has no intrinsic acceleration; the shorter great-circle arcs locally minimize length, though a whole great circle does not minimize between its repeated endpoints.

Orientation and Stokes required no Riemannian metric; a metric adds lengths and angles. A later course follows the route

$$\text{connections} \longrightarrow \text{covariant differentiation} \longrightarrow \text{geodesics} \longrightarrow \text{curvature}.$$

These notions describe how directions at nearby points can be compared and how intrinsic geometry differs from that of Euclidean space.

Appendix F, *Submanifolds*, develops the smooth geometric consequences of inverse, implicit, and constant-rank theory before any metric is imposed. Appendix G, *Lie Groups and Lie Algebras*, gives another setting for geometry, in which smooth structure coexists with group multiplication. Lie theory has its own algebraic questions, as the next direction explains.

## Lie Theory

The route smooth manifolds + groups  $\rightarrow$  Lie groups  $\rightarrow$  Lie algebras makes continuous symmetry accessible to linear algebra. Appendix G, *Lie Groups and Lie Algebras*, develops translations, the tangent space at the identity, the bracket, and concrete matrix exponentials. The circle and the additive line already show why local tangent information and global group structure are different questions. Further study leads to representation theory, symmetry in differential equations, and group actions in geometry. These connections extend well beyond Riemannian geometry.

## De Rham Theory

The equation  $d^2 = 0$  gives  $\text{im } d \subseteq \ker d$ . How far is that inclusion from equality? The quotient

$$H_{\text{dR}}^k(M) = \ker d / \text{im } d,$$

with the maps in degrees  $k$  and  $k - 1$ , records closed forms modulo exact forms. The angular form on the punctured plane is locally a derivative but has nonzero integral around the missing point, so it cannot be globally exact. Appendix H, *de Rham Cohomology*, develops this obstruction, proves local exactness, and computes the circle and the two degree-one torus directions. Integration and Stokes connect these classes to topology. The de Rham theorem, stated there without proof, identifies them with real singular cohomology. This continuation needs forms and Stokes, not a Riemannian metric.

## Measure Theory and Lebesgue Integration

Chapters 9 and 12 controlled integrals by finite partitions and Jordan content. Limits of functions and countable decompositions suggest a more flexible collection of sets on which size can be assigned.

A  $\sigma$ -**algebra** on a set  $X$  is a family  $\mathcal{A} \subseteq \mathcal{P}(X)$  satisfying

$$X \in \mathcal{A}, \quad E \in \mathcal{A} \implies X \setminus E \in \mathcal{A}, \quad (E_j)_{j \geq 1} \subseteq \mathcal{A} \implies \bigcup_{j=1}^{\infty} E_j \in \mathcal{A}.$$

A **measure** on this family is a map

$$\mu : \mathcal{A} \rightarrow [0, \infty], \quad \mu(\emptyset) = 0,$$

with countable additivity: for pairwise disjoint  $E_j \in \mathcal{A}$ ,

$$\mu \left( \bigsqcup_{j=1}^{\infty} E_j \right) = \sum_{j=1}^{\infty} \mu(E_j).$$

The triple  $(X, \mathcal{A}, \mu)$  is the setting for Lebesgue integration. Countable additivity on a  $\sigma$ -algebra supplies the flexibility absent from Jordan content: the allowed sets remain allowed under countable unions. Lebesgue measure on  $\mathbb{R}^n$  extends the usual volumes of boxes.

Here is one landmark result, **Dominated Convergence (unproved preview)**. For measurable real functions  $f_j, f$  on a measure space, suppose

$$f_j \rightarrow f \quad \text{almost everywhere}, \quad |f_j| \leq g \quad \text{almost everywhere for every } j, \quad \int_X g \, d\mu < \infty,$$

where  $g \geq 0$  is measurable. Then

$$\int_X |f| \, d\mu < \infty, \quad \lim_{j \rightarrow \infty} \int_X f_j \, d\mu = \int_X f \, d\mu.$$

The vocabulary in this preview has precise later meanings: measurable real functions have measurable inverse images of open sets; “almost everywhere” allows failure on a set of measure zero; and  $L^1$  denotes integrable functions, with functions equal almost everywhere identified.

Uniform convergence in the Riemann theory controls errors at every point. The one-dimensional Arzelà theorem proved earlier also allows uniformly bounded pointwise convergence, but assumes integrability of the limit. The displayed Lebesgue theorem instead derives integrability of the limit from measurable convergence and an integrable dominating function.

For a concrete preview, the rationals in  $[0, 1]$  form a countable set of Lebesgue measure zero. Thus

$$\int_0^1 \mathbf{1}_{\mathbb{Q}}(x) \, dx = 0 \quad (\text{Lebesgue integral}),$$

although every nondegenerate interval contains rational and irrational points, so its lower and upper Riemann sums remain 0 and 1. A later course constructs this integral and proves its convergence theorems; see Knapp [10, Theorem 5.30] (digital second edition).

## Functional Analysis

Chapter 10 studied norms on finite-dimensional spaces, and Chapter 6 studied limits of functions. Functional analysis treats whole function spaces as geometric spaces on which linear operators act.

$$\text{Banach space} = \text{complete normed vector space},$$

$$\text{Hilbert space} = \text{complete inner-product space}.$$

Completeness refers to the metric induced by the norm; for a Hilbert space the norm is  $\|x\| = \sqrt{\langle x, x \rangle}$ . A central real example is

$$C([a, b]), \quad \|f\|_{\infty} = \max_{x \in [a, b]} |f(x)|.$$

This is Banach: a Cauchy sequence in the maximum norm has pointwise limits by completeness of  $\mathbb{R}$ , the same Cauchy estimate makes convergence uniform, and the uniform limit is continuous. Its vectors are functions, so its dimension is infinite when  $a < b$ .

Finite-dimensional intuition has limits. Norms need not be equivalent: on  $C([0, 1])$ ,

$$\int_0^1 |f|$$

is another norm, but the continuous functions  $f_j(x) = \max(1 - jx, 0)$  have maximum norm 1 and integral norm

$$1/(2j).$$

Closed bounded sets need not be compact, either. Continuous unit-height bumps with disjoint supports in  $(0, 1)$  have pairwise maximum-norm distance 1, so finitely many open balls of radius

$$1/3$$

cannot cover the closed unit ball.

The **Riesz representation theorem for Hilbert spaces (unproved preview)** gives a striking survival of finite-dimensional duality. For a real Hilbert space  $H$ , let  $H^*$  mean its *continuous* linear functionals, not the unrestricted algebraic dual. Then

$$\forall \ell \in H^*, \quad \exists! y \in H : \quad \boxed{\ell(x) = \langle x, y \rangle \quad (x \in H).}$$

This extends the finite-dimensional identification of a covector with a gradient once an inner product is chosen. A later course studies bounded operators, spectral theory, and weak convergence in such spaces; see Knapp [10, Theorem 12.5] (digital second edition).

The function-space viewpoint also prepares the variational problems described under Geometric Analysis below.

## Complex Analysis

The real power-series and differentiation chapters suggest replacing  $\mathbb{R}$  by  $\mathbb{C}$ . For an open  $U \subseteq \mathbb{C}$  and  $f : U \rightarrow \mathbb{C}$ , complex differentiability at  $z \in U$  means existence of

$$f'(z) = \lim_{h \rightarrow 0, h \in \mathbb{C}} \frac{f(z+h) - f(z)}{h}.$$

The increment approaches zero through *every* complex direction. A function complex differentiable at every point of  $U$  is called **holomorphic** on  $U$ .

For  $f = u + iv$  with  $u, v$  continuously real differentiable on  $U$ , the Cauchy–Riemann criterion (preview) says that holomorphicity is equivalent to

$$u_x = v_y, \quad u_y = -v_x \quad \text{throughout } U.$$

These equations say the real derivative matrix is multiplication by one complex number. For  $f(z) = \bar{z}$ , the difference quotient is 1 along real increments and  $-1$  along imaginary increments; this real smooth map is not complex differentiable.

**Cauchy's integral formula (unproved preview)** states: if  $f$  is holomorphic on an open neighborhood of the closed disk  $\bar{B}(a, R)$ , and  $\gamma$  is its positively oriented boundary circle, then for  $|z - a| < R$ ,

$$\boxed{f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta.}$$

Here the complex contour integral is defined by parametrization, integrating real and imaginary parts. The formula recovers interior values from boundary values. One consequence, also left to a later course, is that holomorphic functions are analytic: near every point they equal convergent power series. This is much stronger than real smoothness. Complex analysis develops this rigidity through contour integration, analytic continuation, and residues. For the formula, see Knapp [10, Appendix B, §B4] (digital second edition).

## Partial Differential Equations

The derivatives and integration-by-parts identities of multivariable calculus lead to equations for unknown functions. For a twice differentiable function on an open set in  $\mathbb{R}^n$ , the **Laplacian** is

$$\Delta u = \sum_{j=1}^n \frac{\partial^2 u}{\partial x_j^2}.$$

Representative equations are

$$\begin{aligned} \Delta u &= 0 & (\text{Laplace}), & & -\Delta u &= f & (\text{Poisson}), \\ u_t - \Delta u &= 0 & (\text{heat}), & & u_{tt} - \Delta u &= 0 & (\text{wave}). \end{aligned}$$

In the last two,  $\Delta$  acts on spatial variables and  $t$  is time. Boundary data, and for evolution equations initial data, are part of a well-posed problem; an equation alone need not determine a solution.

For Poisson's equation on  $\Omega \subseteq \mathbb{R}^n$ , multiply by a smooth test function  $\varphi$  with compact support in  $\Omega$ . Integration by parts formally removes one derivative from the unknown:

$$\boxed{\int_{\Omega} \nabla u \cdot \nabla \varphi = \int_{\Omega} f \varphi.}$$

For smooth  $u$ , this follows from  $\operatorname{div}(\varphi \nabla u) = \nabla \varphi \cdot \nabla u + \varphi \Delta u$  and vanishing of the boundary contribution. It also suggests a definition of solution when classical second derivatives do not exist. This suggestion is motivational: no weak-solution existence theorem is asserted here. Sobolev spaces provide the later rigorous setting for weak derivatives, weak solutions, and variational methods.

One landmark is the **maximum principle (unproved preview)**: if  $\Omega$  is bounded and connected,  $u \in C^2(\Omega) \cap C(\overline{\Omega})$ , and  $\Delta u = 0$ , then an interior maximum forces  $u$  to be constant. Consequently its maximum and minimum on  $\overline{\Omega}$  occur on the boundary, and two such harmonic functions with the same boundary values coincide. PDE develops existence, uniqueness, and regularity beyond these initial identities.

When the unknown represents a metric, a surface, or a map between manifolds, these methods lead naturally to the Geometric Analysis direction below.

## Geometric Analysis

Geometric analysis uses differential equations, variational methods, inequalities, and functional analysis to answer geometric questions:

$$\boxed{\text{geometry} + \text{analysis/PDE} \longrightarrow \text{geometric information.}}$$

Multivariable differentiation measures change, integration defines energies and volumes, and manifolds and metrics specify the spaces on which these quantities live. Geodesics can be studied as critical points of an energy or length functional; minimal surfaces arise from area variation;

harmonic functions and maps connect energy to differential equations. Curvature equations and heat-type or curvature flows provide further examples.

The Functional Analysis direction supplies spaces of candidate functions or maps and ways to take limits. The PDE direction supplies equations and estimates for their solutions. Differential Geometry supplies the geometric quantities those estimates control. The productive point of contact is that an analytic inequality or convergence theorem can imply a geometric or topological conclusion. The theories of these examples are left to later courses; they illustrate why the analytic core of this book remains useful far into geometry.

## Probability

The measure-theoretic preview has a second interpretation: size can measure the likelihood of events. A **probability space** is a measure space

$$(\Omega, \mathcal{F}, \mathbb{P}), \quad \mathbb{P}(\Omega) = 1,$$

where  $\mathcal{F}$  is a  $\sigma$ -algebra of events and  $\mathbb{P}$  is a countably additive measure. A real **random variable** is a measurable map  $X : \Omega \rightarrow \mathbb{R}$ , meaning

$$X^{-1}(U) \in \mathcal{F} \quad \text{for every open } U \subseteq \mathbb{R}.$$

Its expectation, when

$$\int_{\Omega} |X| d\mathbb{P} < \infty,$$

is the integral

$$\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P}.$$

For a finite set of equally likely outcomes this is the familiar arithmetic average of the values of  $X$ . Measure theory extends that calculation to continuous distributions and infinite outcome spaces.

Chapter 6 distinguished pointwise from uniform convergence. Probability introduces different controls on exceptional outcomes: almost sure convergence means pointwise convergence outside one probability-zero set; convergence in probability means  $\mathbb{P}(|X_j - X| > \varepsilon) \rightarrow 0$  for every  $\varepsilon > 0$ ; and convergence in  $L^p$ ,  $1 \leq p < \infty$ , means  $\mathbb{E}[|X_j - X|^p] \rightarrow 0$  for variables with finite  $p$ th moments. These conditions control different aspects of the error and must not be silently interchanged.

The **strong law of large numbers (unproved preview)** states that independent identically distributed real random variables  $(X_j)_{j \geq 1}$  with  $\mathbb{E}|X_1| < \infty$  satisfy

$$\frac{X_1 + \cdots + X_n}{n} \longrightarrow \mathbb{E}[X_1] \quad \text{almost surely.}$$

Independence means probabilities of finite joint events factor into their individual probabilities; identical distribution means the variables give the same probabilities to inverse images of real Borel sets. These terms are only preview vocabulary here. A later probability course supplies the measure-theoretic foundations and explains how such limit laws turn repeated observations into stable averages.

# Appendix A

## Foundations: Natural Numbers, Induction, and Choice

### Introduction

Chapter 2 uses natural numbers, induction, and finite recursive constructions directly. This optional appendix separates their logical roles and gives worked guidance in induction before turning to the genuinely different issue of choice. Throughout,

$$\mathbb{N} = \{1, 2, 3, \dots\}, \quad 0 \notin \mathbb{N}, \quad \mathbb{N}_0 = \mathbb{N} \cup \{0\}.$$

No particular set-theoretic realization of the natural numbers is needed.

Sections A.1–A.2 of Appendix A are a companion to Chapter 2. They review induction, recursive definitions and proofs, factorials, binomial coefficients, Pascal’s identity, the finite binomial theorem, and the equivalence of induction and well-ordering of  $\mathbb{N}$ . Readers unfamiliar with these tools may consult those sections alongside Chapter 2. Elementary finite algebraic identities such as the binomial theorem may be used in the main text without proof; Appendix A gives proofs by induction for readers who want the foundational details.

Sections A.3–A.5 are fully optional foundations. Neither the equivalence of Choice, Zorn’s Lemma, and the Well-Ordering Theorem nor its transfinite recursion and Hartogs arguments is needed for Chapters 3–9. For the main text, knowing the statement and a typical use of Zorn’s Lemma is more than enough: if every chain in a nonempty partially ordered set has an upper bound in that set, then the poset has a maximal element. A standard use is the existence of a basis of an arbitrary vector space. The detailed foundational proofs may be read purely for interest.

### A.1 The Natural Numbers and the Peano Axioms

The **Peano axioms** express the elementary structure of  $\mathbb{N}$ . At this stage  $n+1$  denotes the successor of  $n$ ; the notation does not yet presuppose that the usual binary addition operation has already been defined. They assert:

- (a)  $1 \in \mathbb{N}$ ;
- (b) if  $n \in \mathbb{N}$ , then  $n+1 \in \mathbb{N}$ ;
- (c)  $n+1 \neq 1$  for every  $n \in \mathbb{N}$ ;

(d) if  $m + 1 = n + 1$ , then  $m = n$ ;

(e) if  $A \subseteq \mathbb{N}$  satisfies

$$1 \in A, \quad n \in A \implies n + 1 \in A \quad (n \in \mathbb{N}),$$

then  $A = \mathbb{N}$ .

The final assertion is induction. We need neither model-theoretic qualifications nor a separate successor function for these analysis notes.

## A.2 Addition, Induction, and Well-Ordering

### Addition by recursion

The recursion principle permits a value at the next stage to be specified from values already constructed.

**Theorem A.2.1** (Recursion principle). Let  $X$  be a set, let  $x_0 \in X$ , and let

$$\Phi : \mathbb{N}_0 \times X \longrightarrow X.$$

There is a unique function  $f : \mathbb{N}_0 \rightarrow X$  such that

$$f(0) = x_0, \quad f(n + 1) = \Phi(n, f(n)) \quad (n \in \mathbb{N}_0).$$

The proof is not included here because this appendix uses the recursion principle as a foundational construction theorem.

Before using induction from zero, record the required form explicitly.

**Proposition A.2.2** (Induction on  $\mathbb{N}_0$ ). If  $P(0)$  holds and

$$P(n) \implies P(n + 1) \quad (n \in \mathbb{N}_0),$$

then  $P(n)$  holds for every  $n \in \mathbb{N}_0$ .

*Proof.* Apply ordinary induction on  $\mathbb{N}$  to the assertion  $Q(k) = P(k - 1)$ . Its base case is  $P(0)$ , and its induction step is exactly the displayed implication with  $n = k - 1$ . Thus  $P(n)$  holds for every  $n \geq 0$ .  $\square$

For each fixed  $m \in \mathbb{N}_0$ , apply recursion with  $X = \mathbb{N}_0$ , initial value  $m$ , and update rule  $\Phi(n, x) = x + 1$ , where  $+1$  still means successor. This produces the unique map  $n \mapsto m + n$  satisfying

$$m + 0 = m, \quad m + (n + 1) = (m + n) + 1.$$

The final  $+1$  on the right initially means taking the successor. Once this construction is made for every  $m$ , it defines a binary operation

$$+ : \mathbb{N}_0 \times \mathbb{N}_0 \longrightarrow \mathbb{N}_0.$$

The earlier successor notation is then compatible with ordinary addition. Induction now proves properties of the operation. For example,  $0 + n = n$ : it holds at  $n = 0$ , and

$$0 + (n + 1) = (0 + n) + 1 = n + 1.$$

Thus

|                                           |
|-------------------------------------------|
| recursion defines;      induction proves. |
|-------------------------------------------|

**Proposition A.2.3** (Elementary addition laws). For all  $m, n, r \in \mathbb{N}_0$ ,

$$0 + n = n, \quad m + 0 = m, \quad (m + 1) + n = (m + n) + 1,$$

and

$$(m + n) + r = m + (n + r), \quad m + n = n + m.$$

*Proof.* The first identity was just proved; the second is part of the recursive definition. Fix  $m$  and induct on  $n$  for successor compatibility. At  $n = 0$ , both sides equal  $m + 1$ . If the identity holds at  $n$ , then

$$(m + 1) + (n + 1) = ((m + 1) + n) + 1 = ((m + n) + 1) + 1 = (m + (n + 1)) + 1.$$

For associativity, fix  $m, n$  and induct on  $r$ . The case  $r = 0$  is immediate. If the identity holds at  $r$ , then

$$\begin{aligned} (m + n) + (r + 1) &= ((m + n) + r) + 1 \\ &= (m + (n + r)) + 1 \\ &= m + (n + (r + 1)). \end{aligned}$$

Finally fix  $m$  and induct on  $n$  for commutativity. The case  $n = 0$  uses  $0 + m = m = m + 0$ . If  $m + n = n + m$ , then successor compatibility gives

$$m + (n + 1) = (m + n) + 1 = (n + m) + 1 = (n + 1) + m.$$

Each proof uses the previously established law and does not smuggle the desired arithmetic into the recursion principle.  $\square$

Further arithmetic—multiplication, order,  $\mathbb{Z}$ , and  $\mathbb{Q}$ —is developed in Chapter 2 rather than duplicated here.

## How an induction proof works

To prove  $P(n)$  for every  $n \in \mathbb{N}$ , organize the argument as

|                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------|
| state $P(n) \rightarrow$ base case $\rightarrow$ induction hypothesis $\rightarrow$ prove $P(n + 1) \rightarrow$ conclude. |
|----------------------------------------------------------------------------------------------------------------------------|

The induction hypothesis is not an assumption that the theorem is already true. It is the temporary assumption that  $P(n)$  holds at one arbitrary stage  $n$ , used to prove the next stage.

**Example A.2.4** (A finite-sum identity). For every  $n \in \mathbb{N}$ ,

$$1 + 2 + \cdots + n = \frac{n(n + 1)}{2}.$$

**Base case.** At  $n = 1$ , both sides equal 1.

**Induction hypothesis.** Fix arbitrary  $n \in \mathbb{N}$  and suppose

$$1 + 2 + \cdots + n = \frac{n(n + 1)}{2}.$$

**Induction step.** Add the next term and use the hypothesis:

$$\begin{aligned} 1 + 2 + \cdots + n + (n + 1) &= \frac{n(n + 1)}{2} + (n + 1) \\ &= \frac{(n + 1)(n + 2)}{2}. \end{aligned}$$

**Conclusion.** By induction, the identity holds for every  $n \in \mathbb{N}$ .



**Proposition A.2.5** (Bernoulli's inequality). If  $x \in \mathbb{R}$  and  $x \geq -1$ , then, for every  $n \in \mathbb{N}_0$ ,

$$(1+x)^n \geq 1+nx.$$

*Proof.* The assertion is equality at  $n = 0$ . If it holds at  $n$ , then

$$\begin{aligned} (1+x)^{n+1} &= (1+x)^n(1+x) \\ &\geq (1+nx)(1+x) \\ &= 1+(n+1)x+nx^2 \\ &\geq 1+(n+1)x. \end{aligned}$$

Here  $1+x \geq 0$  justifies multiplying the induction inequality. □

### Common variants

**Shifted induction.** If  $P(n_0)$  holds and

$$P(n) \implies P(n+1) \quad (n \geq n_0),$$

then  $P(n)$  holds for every  $n \geq n_0$ . This is ordinary induction with a shifted start and is useful when a statement is meaningful only after an initial index.

**Strong induction.** Suppose  $P(1)$  holds and, for every  $n \in \mathbb{N}$ ,

$$P(1), P(2), \dots, P(n) \implies P(n+1).$$

Then  $P(n)$  holds for every  $n \in \mathbb{N}$ . This is not a stronger axiom. Define

$$Q(n) : \quad P(1), P(2), \dots, P(n) \text{ all hold.}$$

The initial case gives  $Q(1)$ . If  $Q(n)$  holds, the strong-induction hypothesis gives  $P(n+1)$ , hence  $Q(n+1)$ . Ordinary induction proves every  $Q(n)$ , and therefore every  $P(n)$ . Thus

when the next step needs more information, strengthen the induction statement.

**Theorem A.2.6** (Prime factorization: existence). Every integer  $n \geq 2$  can be written as a product of primes.

*Proof.* Use strong induction beginning at 2. The integer 2 is prime. Fix  $n \geq 2$  and suppose every integer  $k$  with  $2 \leq k \leq n$  is a product of primes. If  $n+1$  is prime, there is nothing to prove. If it is composite, then

$$n+1 = ab, \quad 2 \leq a, b < n+1.$$

The induction hypothesis applies to both smaller factors. Writing each as a product of primes and multiplying gives the required representation. □

Knowing the assertion only at  $n$  would not address the possibly much smaller factors  $a$  and  $b$ , which is why strong induction is natural.

**Two-step induction.** If

$$P(n_0), \quad P(n_0 + 1)$$

hold, and

$$P(n) \text{ and } P(n + 1) \implies P(n + 2) \quad (n \geq n_0),$$

then  $P(n)$  holds for every  $n \geq n_0$ . Reduce this to ordinary induction by defining

$$Q(n) : \quad P(n) \text{ and } P(n + 1).$$

The two initial cases give  $Q(n_0)$ , and the two-step hypothesis turns  $Q(n)$  into  $Q(n + 1)$ . This is convenient for recurrences depending on the previous two stages. The same idea extends to any fixed finite number of preceding stages.

**Cauchy forward-backward induction.** This different pattern proves a statement at powers of two and then steps backward to every smaller index. Its classical model is the finite arithmetic-geometric mean inequality. Let  $P_n$  denote

$$\sqrt[n]{a_1 \cdots a_n} \leq \frac{a_1 + \cdots + a_n}{n}, \quad a_i \geq 0.$$

We prove every  $P_n$ , using the two-variable case [Proposition 5.6.6](#).

First,  $P_n \Rightarrow P_{2n}$ . Apply  $P_n$  to each block of  $n$  numbers and write their arithmetic means as  $A$  and  $B$ . Then

$$\sqrt[2n]{a_1 \cdots a_{2n}} \leq \sqrt{AB} \leq \frac{A + B}{2} = \frac{a_1 + \cdots + a_{2n}}{2n}.$$

The first inequality follows by multiplying the two block inequalities and taking the nonnegative square root; the second is two-variable AM-GM.

Second,  $P_n \Rightarrow P_{n-1}$  for  $n \geq 2$ . Given  $a_1, \dots, a_{n-1} \geq 0$ , put

$$A = \frac{a_1 + \cdots + a_{n-1}}{n - 1}$$

and append  $a_n = A$ . The assertion  $P_n$  gives

$$\sqrt[n]{a_1 \cdots a_{n-1} A} \leq A.$$

If  $A = 0$ , every  $a_i = 0$  and  $P_{n-1}$  is immediate. If  $A > 0$ , raise the inequality to the  $n$ th power and cancel  $A > 0$ :

$$a_1 \cdots a_{n-1} \leq A^{n-1}.$$

Taking the nonnegative  $(n - 1)$ st root proves  $P_{n-1}$ .

The forward steps give

$$P_2 \implies P_4 \implies P_8 \implies \cdots .$$

For a prescribed  $n$ , choose a power  $2^k \geq n$ , prove  $P_{2^k}$ , and apply the backward step until reaching  $P_n$ . This completes the equal-weight finite AM-GM proof. It is called Cauchy forward-backward induction, not two-step induction: its two implications are  $P_n \Rightarrow P_{2n}$  and  $P_n \Rightarrow P_{n-1}$ .

**Binomial coefficients and the binomial theorem**

Using the arithmetic developed in Chapter 2, define, for  $n \in \mathbb{N}_0$  and  $0 \leq k \leq n$ ,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad 0! = 1.$$

Adopt the extended convention

$$\binom{n}{k} = 0 \quad \text{if } k < 0 \text{ or } k > n.$$

In particular,  $n < k$  implies

$$\binom{n}{k} = 0.$$

This convention simplifies endpoint formulas and reindexing.

**Proposition A.2.7** (Pascal's identity). For  $n \in \mathbb{N}_0$  and every integer  $k$ ,

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}.$$

*Proof.* For  $1 \leq k \leq n$ ,

$$\begin{aligned} \binom{n}{k-1} + \binom{n}{k} &= \frac{n!}{(k-1)!(n-k+1)!} + \frac{n!}{k!(n-k)!} \\ &= \frac{n!k + n!(n-k+1)}{k!(n-k+1)!} \\ &= \frac{(n+1)!}{k!(n+1-k)!} \\ &= \binom{n+1}{k}. \end{aligned}$$

At the endpoints and outside the range, the zero convention gives the same identity without separate formulas.  $\square$

**Theorem A.2.8** (Binomial theorem). For  $x, y \in \mathbb{R}$  and  $n \in \mathbb{N}_0$ ,

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

*Proof.* The formula is immediate for  $n = 0$ . Assume it at  $n$ . Multiply by  $x + y$ , split the two

resulting sums, reindex, and use Pascal's identity:

$$\begin{aligned}
 (x + y)^{n+1} &= \sum_{k=0}^n \binom{n}{k} x^{n+1-k} y^k + \sum_{k=0}^n \binom{n}{k} x^{n-k} y^{k+1} \\
 &= \sum_{k=0}^{n+1} \binom{n}{k} x^{n+1-k} y^k + \sum_{k=0}^{n+1} \binom{n}{k-1} x^{n+1-k} y^k \\
 &= \sum_{k=0}^{n+1} \left( \binom{n}{k} + \binom{n}{k-1} \right) x^{n+1-k} y^k \\
 &= \sum_{k=0}^{n+1} \binom{n+1}{k} x^{n+1-k} y^k.
 \end{aligned}$$

Thus the mechanism is

multiply  $\rightarrow$  split  $\rightarrow$  reindex  $\rightarrow$  combine by Pascal.

□

The proof uses only commutativity and distributivity, so the same identity holds in any commutative ring. Ring theory is not needed for its use here.

### Natural-number well-ordering

**Proposition A.2.9** (Well-ordering principle for  $\mathbb{N}$ ). Every nonempty subset of  $\mathbb{N}$  has a least element. In elementary arithmetic this principle is equivalent to induction.

*Proof.* Assume induction and suppose  $E \subseteq \mathbb{N}$  has no least element. Prove by induction that  $E \cap \{1, \dots, n\}$  is empty. It is empty for  $n = 1$ ; if it is empty at  $n$ , membership of  $n + 1$  in  $E$  would make  $n + 1$  least. Thus  $E$  is empty, a contradiction.

Conversely, suppose a property holds at 1 and passes from  $n$  to  $n + 1$ . If the failures formed a nonempty set, its least element  $m$  could not be 1. Its predecessor  $m - 1$  would satisfy the property, forcing it at  $m$ , a contradiction. □

This concerns the usual, already specified order on  $\mathbb{N}$ , not the Choice-equivalent Well-Ordering Theorem for arbitrary sets. This completes the natural-number discussion; further arithmetic belongs to Chapter 2.

## A.3 The Axiom of Choice

Finite or recursively specified constructions supply a rule for each next step. Choice addresses a different problem: simultaneous selections from an arbitrary family when no rule has been supplied.

**Definition A.3.1** (Axiom of Choice). For every indexed family  $(X_i)_{i \in I}$  of nonempty sets, there is a function

$$c : I \longrightarrow \bigcup_{i \in I} X_i$$

with  $c(i) \in X_i$  for every  $i \in I$ .

The sets need not be disjoint. A choice function for a finite family can be constructed by induction, and an explicit rule needs no choice axiom. Countable choice and dependent choice are weaker principles sufficient for many familiar analytic selections.

## A.4 Choice, Zorn's Lemma, and Well-Ordering

Recall from Chapter 2 that a partially ordered set, or poset, is a set  $X$  equipped with a partial order  $\leq$ . Fix such a poset  $(X, \leq)$  throughout the following definitions. Two elements need not be comparable; this is exactly why “maximal” and “greatest” differ.

**Definition A.4.1** (Chains and upper bounds). A subset  $S \subseteq X$  is a **chain** if the restricted order is total: for every  $s, t \in S$ , either  $s \leq t$  or  $t \leq s$ . An element  $M \in X$  is an **upper bound** of  $S$  if  $s \leq M$  for every  $s \in S$ . The upper bound need not belong to  $S$ .

The empty set is a chain. If  $X$  is nonempty, every element of  $X$  vacuously upper-bounds it. This convention makes Zorn's statement uniform.

**Definition A.4.2** (Maximal and greatest elements). An element  $x \in X$  is **maximal** if

$$x \leq y \implies y = x.$$

An element  $g \in X$  is **greatest** if

$$x \leq g \quad \text{for every } x \in X.$$

In the inclusion poset  $X = \{\{1\}, \{2\}\}$ , both elements are maximal, but neither is greatest because they are incomparable. If a poset is totally ordered, however, a maximal element is greatest: comparability gives either  $x \leq g$  or  $g \leq x$ , and maximality of  $g$  turns the second possibility into equality. A maximal or greatest element need not exist.

**Theorem A.4.3** (Zorn's Lemma). If a nonempty partially ordered set has an upper bound for every chain, then it has a maximal element.

**Theorem A.4.4** (Well-Ordering Theorem). Every set admits a total order in which every nonempty subset has a least element.

We need a small amount of transfinite language to prove equivalence rather than merely announce it. A **well-order** is a total order in which every nonempty subset has a least element. Two well-ordered sets have the same **order type** if there is an order-preserving bijection between them. An **ordinal** is the canonical order type of a well-ordered set; its elements are precisely the earlier stages. At a successor stage one uses the preceding value, while at a limit stage one uses the entire family of earlier values.

**Theorem A.4.5** (Transfinite recursion). Let  $\alpha$  be an ordinal. If a typed rule assigns an element of a set  $Y$  to every function whose domain is an initial segment  $\beta < \alpha$ , then there is a unique function  $f : \alpha \rightarrow Y$  whose value at each  $\beta$  is the value prescribed from  $f|_\beta$ .

The full set-theoretic proof is not developed here; the following paragraph records its construction mechanism. The construction is the same “define from earlier stages” mechanism as ordinary recursion. Existence is proved by taking the union of all compatible partial solutions; uniqueness follows by considering the first stage at which two proposed solutions differ.

**Lemma A.4.6** (Hartogs). For every set  $X$  there is an ordinal  $h(X)$  for which no injection  $h(X) \rightarrow X$  exists.

*Proof.* Code a well-order on  $A \subseteq X$  by the pair  $(A, R)$ , where

$$R \subseteq A \times A \subseteq X \times X, \quad (A, R) \in \mathcal{P}(X) \times \mathcal{P}(X \times X).$$

The pairs that code well-orders therefore form a set. Replace each by its ordinal order type and let  $\gamma$  be an ordinal strictly larger than every type that occurs, for example the successor of their supremum. If  $\gamma$  injected into  $X$ , its image, transported with the ordinal order, would be a well-ordered subset of  $X$  of type  $\gamma$ , contradicting the construction. Take  $h(X) = \gamma$ .  $\square$

**Theorem A.4.7** (Equivalence of choice principles). Over the usual axioms of set theory without choice,

$$\text{AC} \iff \text{Well-Ordering Theorem} \iff \text{Zorn's Lemma}.$$

*Proof.* **AC  $\Rightarrow$  Well-Ordering.** Fix a set  $X$ . AC supplies a choice function

$$c : \mathcal{P}(X) \setminus \{\emptyset\} \longrightarrow X, \quad c(A) \in A.$$

To make stopping a typed recursion, adjoin a symbol  $\dagger \notin X$  and use  $Y = X \cup \{\dagger\}$  as target. Once  $\dagger$  has appeared, all later values are  $\dagger$ ; when no unused point remains, the next value is  $\dagger$ . Otherwise use the choice rule below. This defines a rule on every initial history (ignore any  $\dagger$  when forming its range). Use transfinite recursion through the Hartogs ordinal  $h(X)$ : at stage  $\alpha$ , if  $X \setminus \{x_\beta : \beta < \alpha\}$  is nonempty, choose

$$x_\alpha = c(X \setminus \{x_\beta : \beta < \alpha\}).$$

If the process never exhausted  $X$ , the distinct values  $x_\alpha$  would define an injection  $h(X) \rightarrow X$ , contrary to [Lemma A.4.6](#). Thus it has a first  $\dagger$  stage  $\lambda < h(X)$ , and  $\alpha \mapsto x_\alpha$  is a bijection from  $\lambda$  onto  $X$ . Transport the ordinal order to  $X$ ; this well-orders  $X$ .

**Well-Ordering  $\Rightarrow$  Zorn.** Let  $P$  satisfy Zorn's hypotheses, and well-order its underlying set. Build a chain by transfinite recursion through  $h(P)$ , using a stop symbol  $\dagger \notin P$  and target  $P \cup \{\dagger\}$  as above. For any history that is not a strictly increasing chain, prescribe  $\dagger$  as well; thus the rule is typed on every history. After selecting the chain  $C_\alpha = \{x_\beta : \beta < \alpha\}$ , choose the least element of the fixed well-order that is strictly above every member of  $C_\alpha$ , if such an element exists; otherwise stop. The selected elements are distinct and form a chain, so continuing without a stop would inject  $h(P)$  into  $P$ , contrary to Hartogs's lemma. At the stopping stage let  $C$  be the resulting chain. By hypothesis it has an upper bound  $u$ . If  $u \notin C$ , then  $u$  is strictly above every member of  $C$ : upper-bound status gives  $x \leq u$ , distinctness gives  $x \neq u$ , and antisymmetry rules out  $u \leq x$ . This contradicts the stopping rule; hence  $u \in C$ . If  $u < y$ , then  $y$  is strictly above every member of  $C$ , another contradiction. Therefore  $u$  is maximal.

**Zorn  $\Rightarrow$  AC.** Let  $(X_i)_{i \in I}$  be a family of nonempty sets. A partial choice function is a function

$$f : J \longrightarrow \bigcup_{i \in I} X_i, \quad J \subseteq I, \quad f(i) \in X_i.$$

Order partial choice functions by extension. This poset is nonempty because it contains the empty function. If  $\mathcal{C}$  is a chain, then

$$f = \bigcup_{g \in \mathcal{C}} g$$

is a function: two members that assign a value at the same index are comparable by extension and hence agree. It is again a partial choice function and upper-bounds the chain. Zorn gives a maximal partial choice function

$$f : J \rightarrow \bigcup_i X_i.$$

If  $J \neq I$ , choose one  $i_0 \in I \setminus J$  and one  $x_0 \in X_{i_0}$ . Then  $f \cup \{(i_0, x_0)\}$  is a larger partial choice function, a contradiction. Thus  $J = I$  and  $f$  is a choice function. Selecting a single element from the one already-given nonempty set  $X_{i_0}$  in this contradiction does not invoke the Axiom of Choice; the axiom concerns simultaneous choices from an arbitrary family.  $\square$

The well-ordering principle for the already ordered set  $\mathbb{N}$ , proved in [Proposition A.2.9](#), is equivalent to ordinary induction in this elementary setting. It is not the Well-Ordering Theorem for arbitrary sets, which the preceding theorem shows to be equivalent to AC.

## A.5 A Basis in Every Vector Space

Recall that spans use finite linear combinations, even in infinite dimensions.

**Theorem A.5.1** (Basis existence from Zorn's Lemma). Every vector space over a field has a basis.

*Proof.* Order the linearly independent subsets of  $V$  by inclusion. The poset is nonempty because it contains  $\emptyset$ . For a nonempty chain  $\mathcal{C}$ , let

$$U = \bigcup_{E \in \mathcal{C}} E.$$

Every finite collection of vectors of  $U$  lies in one member of the chain, so  $U$  is independent and bounds the chain; the empty chain is bounded by  $\emptyset$ . Zorn's Lemma gives a maximal independent set  $B$ . If  $v \notin \text{span } B$ , then  $B \cup \{v\}$  is independent, contradicting maximality. Hence  $B$  spans and is a basis.  $\square$

## Exercises

**Exercise 1.\*** For  $a \in \mathbb{R}$ , use recursion to define  $a^n$  for  $n \in \mathbb{N}_0$  and induction to prove  $a^{m+n} = a^m a^n$ . Identify which part is a definition and which is a proof.

**Exercise 2.** Prove by ordinary induction that

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

Mark the base case, induction hypothesis, and induction step.

**Exercise 3.\*** Use strong induction to prove that every integer  $n \geq 2$  is either prime or divisible by a prime. Explain why all smaller integers are useful.

**Exercise 4.** Let  $u_0 = 0$ ,  $u_1 = 1$ , and  $u_{n+2} = u_{n+1} + u_n$ . Use two-step induction to prove  $u_n < 2^n$  for every  $n \geq 1$ .

**Exercise 5.** Use Pascal's identity and the zero-outside-range convention to derive

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

State whether you invoke the binomial theorem or reprove this special case.

**Exercise 6.** In the poset of proper subsets of  $\{1, 2, 3\}$ , identify all maximal elements and explain why none is a greatest element.

**Exercise 7.\*** Modify the basis-existence argument to prove that any linearly independent subset extends to a basis. State exactly where Zorn's Lemma is used.



# Appendix B

## Categories and Functors

### Introduction

Category theory is used here as a language for organizing constructions that are already familiar, not as a separate prerequisite subject. The central questions are elementary: What kinds of objects are being studied? Which maps are allowed between them? Does a construction respect those maps? This appendix develops only the vocabulary needed elsewhere in these notes: categories, isomorphisms, opposite categories, functors, commutative diagrams, and natural transformations.

The pacing is intentionally concrete. Each abstract definition separates its typed data from its axioms, and examples explain what the symbols record. Yoneda’s lemma, adjunctions, representable-functor theory, limits and colimits, abelian categories, exact or derived functors, enriched categories, monads, and Kan extensions lie outside the present scope. The forms examples at the end may be read after Chapter 13; they organize results proved there and do not supply prerequisites for that chapter.

### B.1 Objects, Morphisms, and Composition

A category records objects, morphisms between them, and how those morphisms compose. Sets and functions give the first model, but the language is useful because the permitted morphisms can encode continuity, linearity, smoothness, an order relation, or some other structure.

**Definition B.1.1** (Category). A **category**  $\mathcal{C}$  consists of the following data.

1. A class of objects, denoted

$$\text{Obj}(\mathcal{C}).$$

We say “class” because the objects of an important category need not form a set.

2. For each ordered pair

$$A, B \in \text{Obj}(\mathcal{C}),$$

a set of morphisms

$$\text{Hom}_{\mathcal{C}}(A, B).$$

Writing  $f \in \text{Hom}_{\mathcal{C}}(A, B)$  is abbreviated to

$$f : A \longrightarrow B.$$

Here  $A$  is the source or domain of  $f$ , while  $B$  is its target or codomain. The actual range is a separate notion when morphisms happen to be functions.

3. For every triple of objects  $A, B, C$ , a composition map

$$\mathrm{Hom}_{\mathcal{C}}(B, C) \times \mathrm{Hom}_{\mathcal{C}}(A, B) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(A, C),$$

$$(g, f) \longmapsto g \circ f.$$

Thus

$$A \xrightarrow{f} B \xrightarrow{g} C$$

produces  $g \circ f : A \rightarrow C$ . The notation means first  $f$ , then  $g$ .

4. For every object  $A$ , an identity morphism

$$1_A : A \longrightarrow A.$$

These data satisfy two axioms.

**Associativity.** For composable arrows

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D,$$

one has

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

**Identity laws.** For every  $f : A \rightarrow B$ ,

$$1_B \circ f = f, \quad f \circ 1_A = f.$$

For ordinary functions, associativity says that both parenthesizations perform the same three evaluations in the same order. Identity functions do nothing. A category keeps precisely these compositional features even when its arrows are not presented as functions between displayed underlying sets.

## Four familiar categories

The following table gives the first examples to keep in mind.

| category                                       | objects                           | morphisms       |
|------------------------------------------------|-----------------------------------|-----------------|
| <b>Set</b>                                     | sets                              | functions       |
| <b>Vect</b> <sub><math>\mathbb{R}</math></sub> | real vector spaces                | linear maps     |
| <b>Top</b>                                     | topological spaces                | continuous maps |
| <b>Man</b>                                     | smooth manifolds without boundary | smooth maps     |

In each row, composition and identities are the ordinary ones. The defining property of the morphisms is preserved by composition: composites of linear, continuous, or smooth maps remain linear, continuous, or smooth.

For this appendix, **Man** uses smooth Hausdorff second-countable manifolds *without boundary*. Chapter 13 develops the broader boundary-capable setup. That broader class is important for Stokes' theorem, but it is not needed for the categorical examples here.

**Example B.1.2** (Divisibility as a category). Let the objects be the positive integers. Declare that there is one arrow

$$a \longrightarrow b \iff a \mid b,$$

and no arrow otherwise. Reflexivity  $a \mid a$  gives the identity arrow. If  $a \mid b$  and  $b \mid c$ , transitivity gives  $a \mid c$ , hence composition. For example,

$$2 \longrightarrow 6 \longrightarrow 30$$

composes to the arrow  $2 \rightarrow 30$ .

There is at most one arrow between any two objects. Therefore two composites with the same endpoints must agree, proving associativity and the identity laws. The arrow  $2 \rightarrow 6$  is not being treated as a function from one displayed integer to another; it records the divisibility relation.

More generally, every poset  $(P, \leq)$  gives a category with one arrow  $x \rightarrow y$  exactly when  $x \leq y$ . Reflexivity and transitivity become the identity and composition laws. Antisymmetry says that arrows in both directions force equality of objects. In the poset of subsets of a fixed set, the arrows are inclusions:  $A \subseteq B \subseteq C$  gives the composite inclusion  $A \rightarrow C$ .

## Isomorphism is not equality

**Definition B.1.3** (Isomorphism). A morphism

$$f : A \longrightarrow B$$

is an **isomorphism** if there exists a morphism

$$g : B \longrightarrow A$$

such that

$$g \circ f = 1_A, \quad f \circ g = 1_B.$$

The morphism  $g$  is called an inverse of  $f$ .

The inverse is unique. Indeed, if  $g$  and  $h$  are both inverses, then

$$g = g \circ 1_B = g \circ (f \circ h) = (g \circ f) \circ h = 1_A \circ h = h.$$

We may therefore write  $f^{-1}$  and write  $A \cong B$  when an isomorphism exists.

In the four familiar categories, the isomorphisms are respectively bijections, linear isomorphisms, homeomorphisms ([Definition 10.4.4](#)), and diffeomorphisms. A continuous bijection need not be an isomorphism in **Top**: the identity map from  $\mathbb{R}$  with the discrete topology to  $\mathbb{R}$  with its usual topology is continuous and bijective, but the inverse is not continuous. The category determines which structure the inverse must preserve.

Literal equality  $A = B$  and isomorphism  $A \cong B$  are different assertions. For example,

$$\mathbb{R}_{\leq 1}[t] := \{a + bt : a, b \in \mathbb{R}\}, \quad T : \mathbb{R}^2 \longrightarrow \mathbb{R}_{\leq 1}[t], \quad T(a, b) = a + bt,$$

is a linear isomorphism. The two vector spaces are nevertheless not literally the same set with the same presentation. Moreover, this particular isomorphism uses the displayed coordinates and polynomial basis.

**Remark B.1.4** (Canonical, natural, and noncanonical). The following working distinctions will be enough here:

canonical : determined by the stated data without arbitrary auxiliary choices,  
 natural : compatible with morphisms in the relevant variables,  
 noncanonical : requires additional choices.

The words are related but are not synonyms. Naturality is the formal compatibility condition defined later. Canonical is a working description of how an object or map is determined. For finite-dimensional  $V$ , the evaluation map  $V \rightarrow V^{**}$  is determined without choosing a basis and is natural; an isomorphism  $V \cong \mathbb{R}^n$  requires a basis and is noncanonical. Canonical does not mean “the only isomorphism,” and the bare existence of some isomorphism gives no preferred choice.

### The opposite category

Contravariant constructions reverse arrows. The opposite category turns that reversal into ordinary covariant bookkeeping.

**Definition B.1.5** (Opposite category). The **opposite category**  $\mathcal{C}^{\text{op}}$  has the same objects as  $\mathcal{C}$ :

$$\text{Obj}(\mathcal{C}^{\text{op}}) = \text{Obj}(\mathcal{C}).$$

For objects  $A, B$ , define

$$\text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) = \text{Hom}_{\mathcal{C}}(B, A).$$

Thus an arrow  $f : A \rightarrow B$  in  $\mathcal{C}$  gives a formally reversed arrow

$$f^{\text{op}} : B \longrightarrow A$$

in  $\mathcal{C}^{\text{op}}$ .

Suppose

$$A \xrightarrow{f^{\text{op}}} B \xrightarrow{g^{\text{op}}} C$$

in  $\mathcal{C}^{\text{op}}$ . The underlying arrows in  $\mathcal{C}$  are  $C \xrightarrow{g} B \xrightarrow{f} A$ , so composition is defined by

$$g^{\text{op}} \circ_{\mathcal{C}^{\text{op}}} f^{\text{op}} = (f \circ_{\mathcal{C}} g)^{\text{op}}.$$

The identity of  $A$  is the same underlying  $1_A$ .

The order reverses because the underlying arrows point the other way. The identity laws in  $\mathcal{C}^{\text{op}}$  are the identity laws in  $\mathcal{C}$  read backward. Likewise, for three composable opposite arrows, the two parenthesizations reduce to the two equal parenthesizations in  $\mathcal{C}$ ; hence associativity holds. A contravariant construction on  $\mathcal{C}$  is therefore an ordinary covariant functor from  $\mathcal{C}^{\text{op}}$ .

### A brief size convention

A category is **small** if its objects and all its morphisms form sets. It is **locally small** if

$$\text{Hom}_{\mathcal{C}}(A, B)$$

is a set for every fixed pair of objects  $A, B$ . Familiar categories such as **Set**, **Top**, and **Vect** $_{\mathbb{R}}$  are usually locally small rather than small: their collections of all objects are too large to be sets, but the maps between two fixed objects form a set. No further foundational size theory is needed in these notes.

## B.2 Functors and Variance

A construction becomes substantially more useful when it tells us not only what happens to objects, but also what happens to maps between those objects. For example, forgetting topology sends a topological space to its underlying set and sends a continuous map to the same underlying function. The action on maps is not extra decoration: it is what makes the construction compatible with changing the input.

Another elementary example assigns

$$P(X) = X \times X$$

to a set  $X$ . A function  $f : X \rightarrow Y$  determines

$$P(f) : X \times X \longrightarrow Y \times Y, \quad P(f)(x_1, x_2) = (f(x_1), f(x_2)).$$

Identity functions leave both coordinates unchanged. If  $X \xrightarrow{f} Y \xrightarrow{g} Z$ , then both  $P(g \circ f)$  and  $P(g) \circ P(f)$  send  $(x_1, x_2)$  to  $(g(f(x_1)), g(f(x_2)))$ . This is the pattern abstracted next.

**Definition B.2.1** (Covariant functor). A **covariant functor**

$$F : \mathcal{C} \longrightarrow \mathcal{D}$$

consists of two assignments.

1. To every object  $A$  of  $\mathcal{C}$ , it assigns an object  $F(A)$  of  $\mathcal{D}$ :

$$A \longmapsto F(A).$$

2. To every morphism  $f : A \rightarrow B$  in  $\mathcal{C}$ , it assigns a morphism in  $\mathcal{D}$  with the displayed domain and codomain:

$$f : A \rightarrow B \quad \longmapsto \quad F(f) : F(A) \rightarrow F(B).$$

These assignments satisfy

$$F(1_A) = 1_{F(A)}$$

for every object  $A$ , and

$$F(g \circ f) = F(g) \circ F(f)$$

for every composable pair  $A \xrightarrow{f} B \xrightarrow{g} C$ .

The forgetful construction  $U : \mathbf{Top} \rightarrow \mathbf{Set}$  satisfies the axioms because the underlying function of an identity is an identity and the underlying function of a composite is the composite of the underlying functions. The product construction  $P$  above satisfies them by the explicit coordinate calculation. Functors also preserve isomorphisms: applying  $F$  to the two inverse equations shows that  $F(f^{-1})$  is inverse to  $F(f)$ .

The direct-image construction gives another example. It sends a set  $X$  to its power set  $\mathcal{P}(X)$  and a function  $f : X \rightarrow Y$  to

$$f_* : \mathcal{P}(X) \longrightarrow \mathcal{P}(Y), \quad A \longmapsto f(A).$$

The identities

$$(1_X)_*(A) = A, \quad (g \circ f)_*(A) = g_*(f_*(A))$$

verify the functor laws elementwise.

### Contravariant functors

Variance records arrow direction. The two patterns are

$$\begin{array}{ccc}
 \text{covariant} & & \text{contravariant} \\
 f : A \rightarrow B & & f : A \rightarrow B \\
 \Downarrow & & \Downarrow \\
 F(f) : F(A) \rightarrow F(B) & & F(f) : F(B) \rightarrow F(A).
 \end{array}$$

**Definition B.2.2** (Contravariant functor). A **contravariant functor**  $F$  from  $\mathcal{C}$  to  $\mathcal{D}$  assigns an object  $F(A)$  to each object  $A$  and, to each arrow  $f : A \rightarrow B$ , an arrow

$$F(f) : F(B) \longrightarrow F(A).$$

It satisfies

$$F(1_A) = 1_{F(A)}$$

and, for  $A \xrightarrow{f} B \xrightarrow{g} C$ ,

$$\boxed{F(g \circ f) = F(f) \circ F(g).}$$

Equivalently, it is an ordinary covariant functor  $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ .

The algebraic dual is the basic example. A linear map  $T : V \rightarrow W$  induces

$$T^* : W^* \longrightarrow V^*, \quad T^*(\lambda) = \lambda \circ T.$$

The identity law follows from  $\lambda \circ 1_V = \lambda$ . If  $V \xrightarrow{T} W \xrightarrow{S} U$ , then for  $\mu \in U^*$ ,

$$(S \circ T)^*(\mu) = \mu \circ S \circ T = T^*(S^*(\mu)),$$

so

$$(S \circ T)^* = T^* \circ S^*.$$

The reversal is forced by precomposition. The pullback of differential forms will follow the same pattern.

## B.3 Commutative Diagrams and Natural Transformations

A diagram is **commutative** when composites along directed paths with the same starting and ending objects are equal. The diagram is a compact record of a typed equation.

For the triangle

$$\begin{array}{ccc}
 A & \xrightarrow{f} & B \\
 & \searrow h & \swarrow g \\
 & C &
 \end{array}$$

the two paths from  $A$  to  $C$  are the direct arrow  $h$  and the composite  $g \circ f$ . Commutativity means

$$g \circ f = h.$$

For the square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ u \downarrow & & \downarrow v \\ C & \xrightarrow{g} & D, \end{array}$$

the top-then-right path gives  $v \circ f : A \rightarrow D$ , while the left-then-bottom path gives  $g \circ u : A \rightarrow D$ . The square commutes exactly when

$$v \circ f = g \circ u.$$

A commutative diagram summarizes an equality of composites; it does not replace the proof of that equality.

For a concrete square, let  $P(X) = X \times X$  and define the diagonal

$$\Delta_X : X \rightarrow X \times X, \quad \Delta_X(x) = (x, x).$$

Given  $f : X \rightarrow Y$ , both routes around

$$\begin{array}{ccc} X & \xrightarrow{\Delta_X} & X \times X \\ f \downarrow & & \downarrow P(f) \\ Y & \xrightarrow{\Delta_Y} & Y \times Y \end{array}$$

send  $x$  to  $(f(x), f(x))$ . This is not one accidental square: it is a compatible comparison for every set and every function.

## Naturality

If  $F$  and  $G$  are two constructions on the same category, what does it mean to compare them compatibly with every morphism? A natural transformation answers this question.

**Definition B.3.1** (Natural transformation). Let

$$F, G : \mathcal{C} \rightarrow \mathcal{D}$$

be covariant functors. A **natural transformation**

$$\eta : F \Rightarrow G$$

assigns to each object  $A$  a component morphism

$$\eta_A : F(A) \rightarrow G(A).$$

For every morphism  $f : A \rightarrow B$ , these components satisfy the naturality equation

$$\boxed{G(f) \circ \eta_A = \eta_B \circ F(f).}$$

Equivalently, the square

$$\begin{array}{ccc} F(A) & \xrightarrow{\eta_A} & G(A) \\ F(f) \downarrow & & \downarrow G(f) \\ F(B) & \xrightarrow{\eta_B} & G(B) \end{array}$$

commutes.

In words: change the input first and then compare, or compare first and then change the input; the result is the same. The diagonal maps above give a natural transformation from the identity functor on **Set** to the product functor  $P$ . The elementwise calculation proves the required square for an arbitrary function  $f$ .

Naturality is stronger than choosing an unrelated morphism  $F(A) \rightarrow G(A)$  for each object. The components must cooperate with every allowed change of input. This is the precise sense in which “natural” is a compatibility property rather than merely a synonym for “familiar” or “canonical.”

## B.4 Differential Forms as the Analytic Payoff

The abstract language now packages constructions already proved in Chapter 13. For a smooth map

$$f : M \longrightarrow N,$$

pullback gives a linear map

$$f^* : \Omega^k(N) \longrightarrow \Omega^k(M).$$

The direction reverses. Identity and composition satisfy

$$(1_M)^* = 1_{\Omega^k(M)}, \quad (g \circ f)^* = f^* \circ g^*.$$

Consequently

$$\boxed{\Omega^k : \mathbf{Man}^{\text{op}} \longrightarrow \mathbf{Vect}_{\mathbb{R}}}$$

is a covariant functor. Equivalently,  $\Omega^k$  is contravariant on **Man**. In degree zero,  $f^*u = u \circ f$  is ordinary precomposition; the reversal on **Man** is already visible there.

Exterior differentiation supplies a comparison between two such functors:

$$d : \Omega^k \Rightarrow \Omega^{k+1}.$$

Here  $\Omega^k$  and  $\Omega^{k+1}$  are viewed as covariant functors on  $\mathbf{Man}^{\text{op}}$ , as required by the definition of a natural transformation. Its component at a manifold  $M$  is  $d_M : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ . For  $f : M \rightarrow N$ , the relevant naturality square, written for the underlying map in **Man**, is

$$\begin{array}{ccc} \Omega^k(N) & \xrightarrow{d_N} & \Omega^{k+1}(N) \\ f^* \downarrow & & \downarrow f^* \\ \Omega^k(M) & \xrightarrow{d_M} & \Omega^{k+1}(M). \end{array}$$

Starting with  $\omega \in \Omega^k(N)$ , the top-then-right route gives  $f^*(d\omega)$ , while the left-then-bottom route gives  $d(f^*\omega)$ . Thus commutativity is exactly the identity

$$\boxed{f^*(d\omega) = d(f^*\omega),}$$

proved in [Proposition 13.12.4](#).

Appendix H constructs de Rham cohomology and the maps induced by pullback. Thus  $H_{\text{dR}}^k$  is contravariant on **Man**, or equivalently the covariant functor

$$H_{\text{dR}}^k : \mathbf{Man}^{\text{op}} \longrightarrow \mathbf{Vect}_{\mathbb{R}}.$$

Appendix H, not the present appendix, supplies the cohomology construction and verifies that pullback descends to it. The role of the present language is organizational: one formula records the objects, the reversed arrows, identities, and composition laws together.



## Exercises

**Exercise 1.** Let  $P$  be a poset. Verify separately the object data, morphism sets, composition, identities, associativity, and identity laws for its category. Show that an arrow  $x \rightarrow y$  is an isomorphism exactly when  $x = y$ .

**Exercise 2.\*** In the divisibility category, determine every isomorphism. Then explain why the arrows  $2 \rightarrow 6$  and  $6 \rightarrow 30$  compose although they are not functions between the displayed integers.

**Exercise 3.\*** Suppose  $A \xrightarrow{f^{\text{op}}} B \xrightarrow{g^{\text{op}}} C$  in  $\mathcal{C}^{\text{op}}$ . Write the corresponding arrows in  $\mathcal{C}$  and compute their composite in  $\mathcal{C}^{\text{op}}$ , including its source and target.

**Exercise 4.** For the functor  $P(X) = X \times X$ , verify the identity and composition laws directly on an arbitrary ordered pair. Then prove that both coordinate projections  $P \Rightarrow \text{Id}_{\mathbf{Set}}$  are natural.

**Exercise 5.\*** Prove the two contravariant functor identities for the algebraic dual  $V \mapsto V^*$ . Track every domain and codomain before comparing the composites.

**Exercise 6.\*** Check the naturality square for the diagonal maps  $\Delta_X : X \rightarrow X \times X$  elementwise. State in words what the two routes do to an arbitrary  $x \in X$ .

**Exercise 7.** Let  $V$  be an  $n$ -dimensional real vector space. Explain why choosing a basis gives an isomorphism  $V \cong \mathbb{R}^n$  but does not give a canonical one. Contrast this with the evaluation map  $V \rightarrow V^{**}$ .

**Exercise 8.\*** For a smooth map  $f : M \rightarrow N$ , write the directions of the four arrows in the naturality square for exterior differentiation. Explain why a form on  $N$  pulls back to  $M$ , rather than the other way around.

## Appendix C

# Permutations and Determinants over a Field

### Introduction

This appendix supplements the self-contained determinant construction in Chapter 10 with a slower treatment of permutations and an arbitrary-field viewpoint. The inversion-based sign proof in the main text needs no appendix; the parity argument below is an alternative conceptual proof. A field  $F$  has addition, subtraction, multiplication, and division by nonzero elements obeying the familiar arithmetic laws, with  $0 \neq 1$ . No background in abstract algebra is assumed. Unless otherwise stated, vector spaces here are finite dimensional over  $F$ . Replace real scalars by elements of  $F$  in the vector-space axioms. Taking  $F = \mathbb{R}$  recovers the main text; neither this appendix nor the next is a prerequisite for Chapters 11–13. The arbitrary-field determinant will also supply the coordinate detectors for exterior bases in Appendix D; its value lies in the field even when no notion of positive orientation is available. The permutation exposition is influenced by the author’s algebra notes [4], specialized here to the elementary facts needed for determinants.

### C.1 What Survives over an Arbitrary Field?

Span, independence, exchange, bases, dimension, linear maps, matrices, dual spaces, and multilinear maps use only field arithmetic. The Chapter 10 proofs therefore work over  $F$ . Determinants, tensor products, and exterior powers also belong to this algebraic theory, as the constructions below show.

A compatible total order, positivity, absolute value or norm, metric, topology, convergence, and orientation via determinant sign are additional structures, not consequences canonically supplied by field axioms. Order, positivity, norms, topology, and convergence must be supplied separately. For instance, a symmetric bilinear form over a field need not be positive definite; the field axioms do not even specify a meaning for “positive.” The algebraic constructions here need none of these additions.

Integer coefficients encode repeated addition:  $m$  in  $F$  means  $m \cdot 1_F$  for  $m \geq 0$ , and negative integers mean additive inverses. The arithmetic laws survive, but distinct integers can have the same value. For a first example  $\mathbb{F}_2 = \{0, 1\}$  has  $1 + 1 = 0$ ,  $1 \cdot 1 = 1$ , and  $1^{-1} = 1$ . These tables obey the field laws. Here  $3 = 1$  and  $-2 = 0$ .

**Example C.1.1** (Finite fields and characteristic two). For prime  $p$ , the residues  $\mathbb{F}_p$  with arithmetic modulo  $p$  form a field, and  $\mathbb{F}_p^n$  has the usual standard basis. For completeness, elementary integer arithmetic supplies the inverse: if  $p \nmid a$ , Bezout's identity gives integers  $u, v$  with  $ua + vp = 1$ , so the residue of  $u$  inverts that of  $a$ . Bezout's identity is used here as elementary arithmetic background; no quotient-ring theory is needed. The **characteristic** of a field is the least positive integer  $p$  such that  $p \cdot 1 = 0$ , or zero if there is no such integer. A positive characteristic is prime: if  $p = ab$  with  $1 < a, b < p$ , then  $(a \cdot 1)(b \cdot 1) = 0$  contradicts invertibility of nonzero factors. An ordered field has  $1 > 0$ ; repeated addition would give  $p \cdot 1 > 0$ , contradicting  $p \cdot 1 = 0$ . Thus positive-characteristic fields cannot be ordered fields, and ordinary Euclidean positivity is unavailable. Over  $\mathbb{F}_2^2$ , the symmetric bilinear expression  $B(x, y) = x_1y_1 + x_2y_2$  has  $B((1, 1), (1, 1)) = 1 + 1 = 0$  although  $(1, 1) \neq 0$ .

alternating means vanishing on repeated inputs.

This is the field-independent definition. Expansion of two equal inputs  $u + v$  still proves the interchange identity  $A(u, v) = -A(v, u)$  (with any other slots fixed). In characteristic two,  $-1 = 1$ , so an alternating bilinear form is also symmetric. Symmetry alone does not force zero diagonal:  $B(x, y) = xy$  on  $\mathbb{F}_2$  has  $B(1, 1) = 1$ . The same warning applies to exterior products:  $v \wedge v = 0$  is fundamental; the consequence  $v \wedge w = -w \wedge v$  becomes  $v \wedge w = w \wedge v$ .

## C.2 Cycles, Transpositions, and Parity

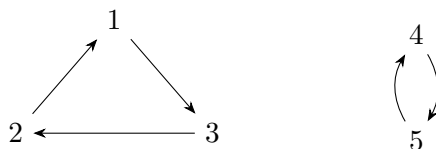
A permutation of  $\{1, \dots, n\}$  is a bijection of this set to itself. Their set is denoted  $S_n$ . Composition is composition of bijections:  $(\sigma\tau)(j) = \sigma(\tau(j))$ . We read the right-hand permutation first. The identity fixes every entry, and  $\sigma^{-1}$  reverses all arrows of  $\sigma$ . No formal group theory is needed. A **cycle**  $(a_1 \ a_2 \ \dots \ a_k)$  sends each displayed entry to the next and the last back to the first, fixing all others.

### Reading and composing bijections

A two-row display places each input above its output:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 5 & 4 \end{pmatrix}.$$

Its output list  $(3, 1, 2, 5, 4)$  is shorter, but a cycle uses a different convention:  $(1 \ 3 \ 2)$  means  $1 \mapsto 3 \mapsto 2 \mapsto 1$ . Thus  $\sigma = (1 \ 3 \ 2)(4 \ 5)$ . Starting the same cycle elsewhere does not change it:  $(1 \ 3 \ 2) = (3 \ 2 \ 1) = (2 \ 1 \ 3)$ . Reversing it does change the map and gives its inverse  $(1 \ 2 \ 3)$ . An omitted entry is fixed; it is not absent from the domain.



**Example C.2.1** (Why order matters). Let  $\alpha = (1\ 2\ 3)$  and  $\beta = (1\ 2)$  in  $S_3$ . Follow each input through two arrows:

| $j$ | $\beta(j)$ | $\alpha(\beta(j))$ | $\alpha(j)$ | $\beta(\alpha(j))$ |
|-----|------------|--------------------|-------------|--------------------|
| 1   | 2          | 3                  | 2           | 1                  |
| 2   | 1          | 2                  | 3           | 3                  |
| 3   | 3          | 1                  | 1           | 2                  |

Thus  $\alpha\beta = (1\ 3)$ , but  $\beta\alpha = (2\ 3)$ . By contrast  $(1\ 3\ 2)$  and  $(4\ 5)$  act on disjoint sets of entries. For an entry in the first set, the second cycle does nothing, and conversely; every other entry is fixed by both. Hence disjoint cycles commute.

### Following a permutation until it closes

Why should cycles describe every bijection? Start at one entry and keep following its image. There are finitely many entries, so eventually one repeats. Injectivity prevents the path from joining itself anywhere except its starting point: if  $\sigma^r(a) = \sigma^s(a)$  with  $0 < r < s$ , applying the inverse  $r$  times gives  $a = \sigma^{s-r}(a)$ , an earlier return. After closing one cycle, begin at an unused entry.

**Proposition C.2.2** (Cycle decomposition). Every permutation decomposes into disjoint cycles, uniquely up to the order of the cycles and cyclic rotation of their entries. Every permutation is a product of transpositions, where a transposition is a two-entry cycle.

*Proof.* Following successive images of an entry closes a cycle: finiteness forces a repetition, and injectivity makes the first repetition the starting entry. Starting again outside the entries already used gives disjoint cycles and exhausts the finite set. Each cycle is the orbit of any one of its entries, with its order prescribed by the permutation, proving the stated uniqueness. Finally

$$(a_1\ a_2\ \cdots\ a_k) = (a_1\ a_k)(a_1\ a_{k-1}) \cdots (a_1\ a_2).$$

Checking the images of the displayed entries proves this identity; applying it to each disjoint cycle gives a transposition decomposition.  $\square$

For example the permutation with list  $(2, 3, 1, 5, 4)$  is  $(1\ 2\ 3)(4\ 5)$ , and equals  $(1\ 3)(1\ 2)(4\ 5)$ .

**Example C.2.3** (A full decomposition and its inverse). The list  $(4, 3, 2, 5, 1, 6)$  sends  $1 \rightarrow 4 \rightarrow 5 \rightarrow 1$ ,  $2 \rightarrow 3 \rightarrow 2$ , and fixes 6. Hence

$$\sigma = (1\ 4\ 5)(2\ 3) = (1\ 5)(1\ 4)(2\ 3), \quad \sigma^{-1} = (1\ 5\ 4)(2\ 3).$$

For a cycle  $(a_1 \cdots a_k)$ , the transposition formula can be verified without multiplying symbols formally. The rightmost swap takes  $a_1$  to  $a_2$ , which later swaps leave alone. For  $2 \leq j < k$ , the swap  $(a_1\ a_j)$  takes  $a_j$  to  $a_1$ , and the next takes it to  $a_{j+1}$ . The final entry  $a_k$  goes to  $a_1$ . Every unlisted entry stays fixed.

### Adjacent swaps and the parity question

An **adjacent transposition** is  $(i\ i+1)$ . Swapping positions  $i < j$  can be accomplished by moving the entry at  $i$  rightward to  $j$  and moving the displaced entry leftward back to  $i$ . Intermediate entries return to their original places. This uses  $2(j-i) - 1$  adjacent swaps:

$$(i\ j) = (i\ i+1) \cdots (j-2\ j-1)(j-1\ j)(j-2\ j-1) \cdots (i\ i+1).$$

For example  $(1\ 3) = (1\ 2)(2\ 3)(1\ 2)$ . The same permutation therefore has a one-swap and a three-swap expression. Inserting  $(1\ 2)(1\ 2)$  produces still more expressions. Length is not intrinsic. We must determine whether its parity is intrinsic: could the same permutation have both a two-swap and a three-swap expression?

We need parity to be independent of the chosen transpositions. The next proof uses formal polynomials with integer coefficients only as finite expressions in indeterminates; it does not assume any polynomial theory.

**Theorem C.2.4** (Parity and sign). Any two transposition expressions for a permutation have the same parity of length. Define  $\text{sgn}(\sigma) = (-1)^r$  using that parity. Then sign is multiplicative under composition.

*Proof.* We seek a nonzero expression whose value reacts predictably when two positions are exchanged. The Vandermonde expression has exactly this property: every transposition negates it. It can therefore determine whether a transposition decomposition has even or odd length. Consider the integer polynomial

$$\Delta(t_1, \dots, t_n) = \prod_{i < j} (t_j - t_i).$$

Swapping adjacent variables  $t_i, t_{i+1}$  negates their mutual factor. For a third index  $k < i$ , the pair of factors is  $(t_i - t_k)(t_{i+1} - t_k)$ ; for  $k > i + 1$  it is  $(t_k - t_i)(t_k - t_{i+1})$ . Each pair merely exchanges its factors. Thus only the mutual factor changes sign and the whole expression is negated. Swapping positions  $i < j$  can be done in  $2(j - i) - 1$  adjacent swaps, hence also negates  $\Delta$ . Thus a decomposition of length  $r$  changes  $\Delta$  to  $(-1)^r \Delta$ . Two decompositions of the same permutation perform the same substitution, so yield equal expressions. They cannot yield opposite signs: evaluating at the explicit integer tuple gives

$$\Delta(1, \dots, n) = \prod_{i < j} (j - i) > 0$$

(the empty product is 1), so  $\Delta$  is nonzero and not its negative. Parity is therefore well defined. Concatenating decompositions proves  $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau)$ .  $\square$

Call permutations with sign 1 **even** and sign  $-1$  **odd**. A  $k$ -cycle has sign  $(-1)^{k-1}$ . The parity is defined over the integers even when coefficients later lie in characteristic two; interpreting its signs in that field sends both  $+1$  and  $-1$  to 1 without changing the combinatorial parity theorem.

The identity uses zero swaps and has sign 1. Reversing a swap expression gives an expression of the same length for the inverse, since each swap undoes itself. Therefore  $\text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma)$ . A  $k$ -cycle uses  $k - 1$  swaps; a permutation with disjoint cycles of lengths  $k_1, \dots, k_r$  has sign

$$(-1)^{\sum_j (k_j - 1)}.$$

Fixed points contribute zero to this exponent.

### Counting inversions instead of choosing swaps

The Vandermonde argument explains why parity is intrinsic; inversion counting now supplies an efficient way to compute it. Parity is known to be independent of a swap decomposition. To compute it without finding such a decomposition, count the pairs that appear in the wrong order. Each adjacent correction removes exactly one inversion.

**Proposition C.2.5** (Inversions compute the same sign). An inversion of  $\sigma$  is a pair  $i < j$  with  $\sigma(i) > \sigma(j)$ . Then  $\text{sgn}(\sigma) = (-1)^{\text{inv}(\sigma)}$ .

*Proof.* If the list is not increasing, it has an adjacent inverted pair. Exchanging that pair decreases the inversion count by one: their mutual inversion disappears and comparisons with all other entries have the same total count. After exactly  $\text{inv}(\sigma)$  exchanges the list is increasing, hence is the identity. Parity now gives the formula.  $\square$

**Example C.2.6** (Inversions in a complete list). For  $(3, 1, 2, 5, 4)$  the inverted position pairs are  $(1, 2), (1, 3), (4, 5)$ . There are three, so the permutation is odd. Sorting by adjacent exchanges makes the count visible:

$$31254 \longrightarrow 13254 \longrightarrow 12354 \longrightarrow 12345.$$

Its cycle expression  $(1\ 3\ 2)(4\ 5)$  also gives  $\text{sign}(-1)^2(-1) = -1$ . The two calculations agree for a proved reason, not because a preferred decomposition was selected.

In the inversion proof, a third entry to the left compares with both exchanged entries before and after; the number of these two inversions is unchanged. A third entry to the right has the same property. Only the comparison between the adjacent entries reverses. Moreover an unsorted list must have adjacent entries out of order: if every neighbor increased, transitivity would make the entire list increasing.

### Permutation matrices connect the notation to linear maps

The **permutation matrix**  $P_\sigma$  has column  $j$  equal to  $e_{\sigma(j)}$ . Applying it to basis vectors shows  $P_\sigma P_\tau = P_{\sigma\tau}$  and  $P_\sigma^{-1} = P_{\sigma^{-1}}$ .

**Example C.2.7** (Which way does a permutation matrix move coordinates?). For  $\sigma = (1\ 2\ 3)$  the columns are  $e_2, e_3, e_1$ , hence

$$P_\sigma = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad P_\sigma \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_3 \\ x_1 \\ x_2 \end{pmatrix}.$$

The old coordinate  $x_j$  moves to position  $\sigma(j)$ ; the new coordinate in position  $i$  is  $x_{\sigma^{-1}(i)}$ . In the Leibniz sum only the selection  $(2, 3, 1)$  has a nonzero product. Its sign is  $+1$ , so  $\det P_\sigma = 1$ . This example also explains why a permutation and its inverse can appear in different coordinate conventions without changing the determinant sign.

## C.3 Constructing the Determinant Form

We have constructed a sign for each permutation and proved how signs behave under composition. This is the missing coefficient in the determinant formula: multilinearity chooses one basis vector in each slot, alternation removes repeated choices, and permutation sign records the orientation of the surviving orderings.

Let  $(v_1, \dots, v_n)$  be a basis of  $V$  and write

$$x_j = \sum_i a_{ij} v_i.$$

The slot-by-slot uniqueness argument of [Theorem 10.17.3](#) uses only multilinearity and alternation, so works over  $F$ : each alternating form is determined by its value on this basis. We now give the field-valid existence argument in full.

**Theorem C.3.1** (Existence over an arbitrary field). The formula

$$\omega(x_1, \dots, x_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j), j}$$

defines the unique alternating  $n$ -linear form with  $\omega(v_1, \dots, v_n) = 1$ .

*Proof.* Each product uses one coordinate from each input, so distributing sums or scalar multiples in one column proves multilinearity. If  $x_r = x_s$  for  $r \neq s$ , pair  $\sigma$  with  $\sigma \circ (r \ s)$ . This is a fixed-point-free pairing. The products agree since the two coordinate columns agree, and the signs are opposite by parity. Their sum is zero in every field, including characteristic two where it is twice the same element. On the given basis, all terms except the identity term vanish, and that term is 1. For uniqueness, expanding an arbitrary form gives the same sum: repeated tuples vanish, and the surviving tuples are permutations with their forced signs.  $\square$

In the standard basis this is  $\det A$ . It agrees with the main-text determinant when  $F = \mathbb{R}$  by uniqueness, and gives  $\det P_\sigma = \operatorname{sgn}(\sigma)$  interpreted in  $F$ . The uniqueness proof of multiplicativity, the invertibility criterion, the transpose formula, and basis independence in Chapter 10 remain valid over  $F$ . No order or positivity was used in them.

**Proposition C.3.2** (Laplace expansion from the permutation sum). The minor and cofactor definitions and both expansions in [Theorem 10.17.12](#) are valid over  $F$ .

*Proof.* Group the Leibniz terms by the row  $i = \sigma(j)$  selected in the fixed column  $j$ . Delete row  $i$  and column  $j$ , preserving the order of all remaining rows and columns. The residual choices are precisely the permutations of this minor. Moving row  $i$  and column  $j$  to the first positions uses  $(i-1) + (j-1)$  adjacent swaps; their sign is  $(-1)^{i+j}$ . The contribution of this group is therefore  $a_{ij}(-1)^{i+j} M_{ij}$ . Summing over  $i$  gives column expansion, and transpose invariance gives row expansion. This is the permutation counterpart of the alternating-form proof of [Theorem 10.17.12](#), valid over every field.  $\square$

For example the sparse matrix of Chapter 10 has determinant  $-6(8-3) = -30$ , interpreted in  $F$ . The same calculation remains valid if this field element becomes zero:  $-30 = 0$  in characteristics 2, 3, 5, whereas  $-30 = 5$  in characteristic 7. The matrix is singular in the first three cases and invertible in the last. Interpret the scalar in  $F$  before using the determinant criterion.

**A normalized form versus an endomorphism determinant.** Let  $\omega_e$  be normalized to 1 on a basis  $e$ , and let the columns of  $P$  express a new basis  $e'$  in  $e$  coordinates. Then  $\omega_e(e'_1, \dots, e'_n) = \det P$ , so uniqueness of the normalized form gives

$$\omega_{e'} = (\det P)^{-1} \omega_e.$$

For a fixed endomorphism  $T : V \rightarrow V$ , its matrix changes instead from  $A$  to  $P^{-1}AP$ ; multiplicativity makes their determinants equal. For a map between distinct spaces there are separate source and target basis changes, so there is no independent scalar determinant without normalizations. The top exterior-power action in [Theorem D.9.10](#) will give a fourth equivalent viewpoint alongside normalized alternation, Leibniz, and recursive Laplace expansion.

## Exercises

**Exercise 1.** Find disjoint cycles, a transposition decomposition, sign, inversion count, and determinant of the permutation matrix for the list  $(2, 5, 4, 1, 3)$ .

**Exercise 2.\*** Define the adjugate by  $(\text{adj } A)_{ji} = C_{ij}(A)$ . Using cofactor expansion of a matrix with one replaced row, prove  $A \text{adj } A = (\det A)I$ . Derive the inverse formula when  $\det A \neq 0$ . For  $Ax = b$ , use column multilinearity to derive

$$x_j = \det(A_1, \dots, b, \dots, A_n) / \det A$$

(Cramer's rule).

**Exercise 3.** In  $S_5$ , compute both  $(1\ 4\ 2)(2\ 3\ 5)$  and  $(2\ 3\ 5)(1\ 4\ 2)$  by making a table of all five images. Give inverse lists and cycle expressions. Which entries witness noncommutation?

**Exercise 4.** Give three distinct transposition expressions for  $(1\ 4)$  and count their lengths. Explain what parity says, and what it does not say, about the smallest possible length of an expression.

**Exercise 5.\*** For the reversing list  $(n, n-1, \dots, 1)$  find the inversion count and sign. Compute its permutation matrix determinant for  $n = 4, 5, 6$ .

**Exercise 6.\*** Over  $\mathbb{F}_2$ , compute the determinant of  $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ . Explain why the answer gives an invertibility test but no positive or negative orientation. Distinguish permutation parity from its field-valued sign.



## Appendix D

# Tensor Products and Exterior Powers

### Introduction

Multilinear rules become easier to organize when they are represented by linear maps on a single vector space. This appendix develops that advantage over a field  $F$ , following Chapter 8 of the author's algebra notes. The route is construction, associativity, symmetry, direct sums, finite multilinear products, induced maps and calculations, Tensor–Hom, exterior powers, and finally the exterior algebra and alternating forms. A quotient-space preparation makes the constructions self-contained. No topology or ordering of  $F$  is needed;  $F = \mathbb{R}$  connects the final constructions to differential forms.

There are finitely many tensor factors and direct summands. Quotient and tensor constructions can use arbitrary vector spaces; the formal-symbol spaces in their proofs are usually infinite dimensional. Dimension formulas and duality identifications carry their finite-dimensional hypotheses explicitly. The exterior algebra below is formed for a finite-dimensional space, so only finitely many of its homogeneous components are nonzero.

The recurring task is to construct a map without choosing a representation of its input. First define a map on independent vectors, verify its slot identities, and factor it through the relevant universal property. Construct an inverse by the same method when an isomorphism is claimed. Once both maps exist, agreement on spanning elementary elements can prove the inverse identities; the uniqueness clause often proves the same equality more directly. Chapter 13 remains independent of tensor quotients, while the final duality and shuffle comparison explain why the two constructions agree. This order of proof will remain visible at each layer.

For a selective comparison of proof architecture, see Dummit–Foote [25], Sections 10.4 and 11.4–11.5. The displayed expansions and explicit factorization checks here retain a slower pace and specialize throughout to vector spaces over fields.

### D.1 Quotient Vector Spaces and Universal Factorization

In  $E = F^2$ , suppose horizontal displacements are to be ignored. Then  $(x, y)$  and  $(x + t, y)$  should describe the same object, a horizontal line. The second coordinate survives, whereas the first does not. This resembles the equivalence-class constructions of integers and rationals in Chapter 2 (Propositions 2.4.2 and 2.4.4): different representatives describe a single element. Here the differences we ignore form a vector subspace.

**Definition D.1.1** (Cosets). For a subspace  $R \subseteq E$ , declare

$$x \sim y \quad \text{if} \quad x - y \in R.$$

The class of  $x$  is the coset  $x + R = \{x + r : r \in R\}$ . Write  $E/R$  for the set of these classes.

The subspace laws give reflexivity, symmetry, and transitivity: differences are zero, negated, or added. Moreover  $x + R = y + R$  if and only if  $x - y \in R$ , since adding a vector of  $R$  leaves its coset unchanged.

**Theorem D.1.2** (The quotient vector space). The operations

$$(x + R) + (y + R) = (x + y) + R, \quad a(x + R) = ax + R$$

make  $E/R$  a vector space. The quotient map  $\pi : E \rightarrow E/R$ ,  $\pi(x) = x + R$ , is linear and surjective, with kernel  $R$ .

*Proof.* Changing representatives to  $x + r, y + s$  changes their sum by  $r + s \in R$  and a scalar multiple by  $ar \in R$ . Thus the operations are well defined. The vector-space identities descend from  $E$ ; zero is  $R$  and the inverse of  $x + R$  is  $-x + R$ . The displayed operations make  $\pi$  linear. Every class has a representative, and  $\pi(x) = R$  exactly when  $x \in R$ .  $\square$

**Example D.1.3** (A legal rule and an illegal rule). For  $R = \{(t, 0) : t \in F\} \subseteq F^2$ , the rule  $\mu((x, y) + R) = y$  is independent of representative. Its reverse is  $\eta(y) = (0, y) + R$ . Both are linear by the quotient operations;  $\mu\eta(y) = y$ , and  $\eta\mu((x, y) + R) = (0, y) + R = (x, y) + R$  because their difference is horizontal. Thus

$$F^2/R \cong F$$

by these specific maps. The rule  $(x, y) + R \mapsto x$  is not a function on classes:  $(0, 0)$  and  $(1, 0)$  represent the same class but would give different outputs.

**Theorem D.1.4** (Quotient universal factorization). For a linear  $L : E \rightarrow U$ , a linear map  $\tilde{L} : E/R \rightarrow U$  satisfying  $L = \tilde{L} \circ \pi$  exists if and only if  $R \subseteq \ker L$ . When it exists it is unique and is given by

$$\tilde{L}(x + R) = L(x).$$

$$\begin{array}{ccc} E & \xrightarrow{\pi} & E/R \\ & \searrow L & \downarrow \tilde{L} \\ & & U \end{array}$$

The diagram says both routes starting at  $x \in E$  give  $L(x) \in U$ . All three arrows here are linear; the dashed one is the map to be constructed.

*Proof.* If  $L = \tilde{L}\pi$  and  $r \in R$ , then  $\pi(r) = 0$  and  $L(r) = \tilde{L}(0) = 0$ , proving necessity. For sufficiency suppose  $R \subseteq \ker L$ . If  $x + R = y + R$ , then  $x - y \in R$ , so  $L(x) - L(y) = L(x - y) = 0$ . The displayed rule is therefore independent of representative and defines  $\tilde{L}$ . For classes  $x + R, y + R$  and scalars  $a, b$ ,

$$\tilde{L}(a(x + R) + b(y + R)) = L(ax + by) = a\tilde{L}(x + R) + b\tilde{L}(y + R).$$

Thus it has the required type, a linear map on the quotient, and  $\tilde{L}(\pi(x)) = L(x)$ . For uniqueness, any map  $H$  with  $H\pi = L$  has  $H(x + R) = H(\pi(x)) = L(x)$  on every class, since  $\pi$  is surjective. Hence  $H = \tilde{L}$ .  $\square$

The phrase “ $L$  descends to the quotient” abbreviates this construction under the relation-killing hypothesis.

**Example D.1.5** (First isomorphism theorem). For linear  $T : V \rightarrow W$ , quotient factorization gives

$$V/\ker T \longrightarrow \operatorname{im} T, \quad v + \ker T \longmapsto Tv.$$

Its kernel is zero and it is onto its stated target, so it is an isomorphism. Its inverse sends  $Tv$  to  $v + \ker T$ : any two preimages differ by a kernel vector. For  $T(x, y, z) = (x - y, y - z)$  the kernel consists of  $(t, t, t)$ ; the class of  $(a, 0, -b)$  corresponds to  $(a, b)$ .

**Proposition D.1.6** (Quotient bases and chosen complements). Let  $E$  be finite dimensional and  $R \subseteq E$  a subspace. Extend a basis  $r_1, \dots, r_d$  of  $R$  to  $r_1, \dots, r_d, s_1, \dots, s_m$  of  $E$ . Then  $s_1 + R, \dots, s_m + R$  form a basis of  $E/R$ , and

$$\dim(E/R) = \dim E - \dim R.$$

For any chosen complement  $E = R \oplus S$ , restriction gives an isomorphism  $\pi|_S : S \rightarrow E/R$ .

*Proof.* Expand  $x \in E$  in the extended basis. Its  $r$ -part disappears in the quotient, proving spanning of the surviving cosets. If

$$\sum_j a_j(s_j + R) = 0,$$

then

$$\sum_j a_j s_j \in R.$$

Expanding that vector in the  $r_i$  and using independence of the full basis gives every  $a_j = 0$ . This proves independence and the dimension formula separately. For a complement,  $\pi|_S$  is linear, has kernel  $S \cap R = 0$ , and is onto since  $\pi(r + s) = \pi(s)$ . Its reverse sends  $x + R$  to the  $S$ -component of  $x$ : adding an element of  $R$  does not change that component. Componentwise linearity makes the reverse linear; the composites return  $s$  and  $x + R$ .  $\square$

The quotient is not literally its chosen complement. Different complements give different representative choices; the quotient map chooses none canonically.

## D.2 Bilinear Maps and Binary Tensor Products

A bilinear rule accepts two independent vectors. Scalar multiplication  $(a, v) \mapsto av$ , evaluation  $(\lambda, v) \mapsto \lambda(v)$ , and the coordinate table  $(x, y) \mapsto (x_i y_j)_{ij}$  are examples. The word “independent” describes how the arguments may be varied, not linear independence of the vectors. Fixing either argument leaves a linear map in the other; varying both together need not do so. For example  $b : F \times F \rightarrow F$ ,  $b(x, y) = xy$ , has  $b((1, 0) + (0, 1)) = 1$  and  $b(1, 0) + b(0, 1) = 0$ . Thus the Cartesian product, with its usual vector addition, is not the source that linearizes this rule. We seek a vector space representing *all* bilinear rules on the pair.

**Definition D.2.1** (Bilinearity). A map  $b : V \times W \rightarrow U$  is bilinear if, for all vectors and scalars,

$$\begin{aligned} b(v + v', w) &= b(v, w) + b(v', w), \\ b(av, w) &= ab(v, w), \\ b(v, w + w') &= b(v, w) + b(v, w'), \\ b(v, aw) &= ab(v, w). \end{aligned}$$

These are two independent linearity requirements, one per slot.

A tensor product retains the common slot identities of these examples, without imposing special identities of one particular rule such as a dot product.

**Definition D.2.2** (Tensor product). A tensor product of  $V, W$  consists of a vector space  $T$  and a bilinear map  $\tau : V \times W \rightarrow T$  with the following property:

- (a) For every vector space  $U$  over  $F$ ,
- (b) and every bilinear map  $b : V \times W \rightarrow U$ ,
- (c) there exists a unique linear map  $\tilde{b} : T \rightarrow U$  such that  $\tilde{b} \circ \tau = b$ .

Write  $T = V \otimes_F W$  and  $v \otimes w = \tau(v, w)$ ; the latter is an elementary tensor. We omit the subscript  $F$  when the field is understood.

$$\begin{array}{ccc} V \times W & \xrightarrow{\tau} & V \otimes W \\ & \searrow b & \downarrow \tilde{b} \\ & & U \end{array}$$

“Every” lets us choose  $U$  and  $b$  for the problem at hand. “Exists” gives an actual linear map on tensors, not just a formula on a pair. “Unique” says two linear maps with this same composite with  $\tau$  must agree. The horizontal and diagonal arrows are bilinear; the vertical one is linear. The equation means both routes send  $(v, w)$  to  $\tilde{b}(v, w)$ .

### Construction from formal symbols

Let  $E$  consist of finitely supported coefficient families indexed by pairs  $(v, w) \in V \times W$ . Write such a family as

$$\sum c_{v,w} [v, w].$$

Here finite support means the set  $\{(v, w) : c_{v,w} \neq 0\}$  is finite; no closure or topology is involved. The pair  $(v, w)$  is a label,  $[v, w]$  its basis vector,  $c_{v,w}$  a scalar, and the full sum an element of  $E$ . The symbol  $[v, w]$  has coefficient 1 at that pair and zero elsewhere. These symbols form a basis by the definition of coefficient families; addition and scaling act on their coefficients. In particular  $[v + v', w]$  is initially a single symbol, not the sum  $[v, w] + [v', w]$ . Even  $[0, w]$  is a nonzero basis symbol at this stage.

Let  $R$  be the subspace spanned by all vectors of the following four types:

$$\begin{aligned} [v + v', w] - [v, w] - [v', w], \\ [av, w] - a[v, w], \\ [v, w + w'] - [v, w] - [v, w'], \\ [v, aw] - a[v, w]. \end{aligned}$$

Define  $T = E/R$  and  $\tau(v, w) = [v, w] + R$ . Each displayed difference becomes zero in the quotient, so  $\tau$  satisfies the four bilinear identities. No other types of relations have been imposed.

**Theorem D.2.3** (Existence of the tensor product). The pair  $(E/R, \tau)$  satisfies the tensor universal property.

*Proof.* Fix an arbitrary target  $U$  and bilinear  $b : V \times W \rightarrow U$ . First extend from the formal basis. Define

$$\widehat{b} : E \rightarrow U, \quad \widehat{b}\left(\sum c_{v,w}[v, w]\right) = \sum c_{v,w}b(v, w).$$

Each sum is finite, and its coefficients are uniquely specified in  $E$ . The formula is linear by distributing coefficients. At this stage bilinearity of  $b$  has not been used: any assignment of values to formal basis symbols extends linearly in this way.

Then check the relations. Bilinearity now gives all four checks:

$$\begin{aligned} \widehat{b}([v + v', w] - [v, w] - [v', w]) &= b(v + v', w) - b(v, w) - b(v', w) = 0, \\ \widehat{b}([av, w] - a[v, w]) &= b(av, w) - ab(v, w) = 0, \\ \widehat{b}([v, w + w'] - [v, w] - [v, w']) &= b(v, w + w') - b(v, w) - b(v, w') = 0, \\ \widehat{b}([v, aw] - a[v, w]) &= b(v, aw) - ab(v, w) = 0. \end{aligned}$$

Linearity makes  $\widehat{b}$  vanish on their whole span:  $R \subseteq \ker \widehat{b}$ . The quotient factorization theorem [Theorem D.1.4](#) therefore gives the linear map  $\widetilde{b} : E/R \rightarrow U$ , with

$$\widetilde{b}(e + R) = \widehat{b}(e), \quad \widetilde{b}(v \otimes w) = b(v, w).$$

Thus  $\widetilde{b}\tau = b$ . Any other linear  $H$  with this property has  $H\pi([v, w]) = b(v, w) = \widehat{b}([v, w])$  on the formal basis, hence  $H\pi = \widehat{b}$ . Quotient uniqueness gives  $H = \widetilde{b}$ .  $\square$

The two stages solve different problems: basis extension gives a map before identifications; killing the relations allows it to survive identifications.

$$\begin{array}{ccc} E & \xrightarrow{\pi} & E/R \\ & \searrow \widehat{b} & \downarrow \widetilde{b} \\ & & U \end{array}$$

**Working rule.** To construct a map from a tensor product: define a concrete map on the independent variables, verify bilinearity, invoke the universal property, and obtain the induced linear map.

To prove two existing linear maps equal: first establish both maps, compare their composites with the bilinear input map, then use uniqueness (or, equivalently here, agreement on the spanning elementary tensors). A formula on elementary tensors alone is not a definition of a linear map.

**Example D.2.4** (Construct before evaluating). On  $F^2 \times F^2$  let  $b(v, w) = v_1 w_1 + 2v_1 w_2 - v_2 w_1$ . For fixed  $w$  it is  $(w_1 + 2w_2)v_1 - w_1 v_2$ , linear in  $v$ ; for fixed  $v$  it is  $(v_1 - v_2)w_1 + 2v_1 w_2$ , linear in  $w$ . Thus a unique linear  $L : F^2 \otimes F^2 \rightarrow F$  exists with these input values. It sends  $3e_1 \otimes e_2 + e_2 \otimes e_1$  to  $6 - 1 = 5$ , interpreted in  $F$ . By contrast the proposed rule  $v \otimes w \mapsto v$  fails:  $v \otimes 0 = 0$  for every  $v$ , whereas the proposed outputs vary. Its independent-variable map is not linear in  $w$ , so the working rule detects the problem in advance.

**Theorem D.2.5** (Uniqueness up to canonical isomorphism). If  $(T, \tau)$  and  $(T', \tau')$  are tensor products of  $V, W$ , there is a unique linear isomorphism  $A : T \rightarrow T'$  satisfying  $A\tau = \tau'$ .

*Proof. Construct  $A$ .* Use the universal property of  $T$  with target  $U = T'$  and input bilinear map  $b = \tau'$ . It gives a unique linear  $A : T \rightarrow T'$  such that  $A\tau = \tau'$ . Temporarily distinguishing the spaces, this reads  $A(v \otimes_T w) = v \otimes_{T'} w$ .

*Construct the reverse.* Use the universal property of  $T'$  with target  $T$  and bilinear map  $\tau$ . This gives a linear  $C : T' \rightarrow T$  with  $C\tau' = \tau$ , or  $C(v \otimes_{T'} w) = v \otimes_T w$ .

*First composite.* Both  $CA$  and  $\text{id}_T$  are existing linear maps  $T \rightarrow T$ . Compute

$$(CA)\tau = C(A\tau) = C\tau' = \tau = \text{id}_T \tau.$$

The uniqueness clause for  $T$ , with target  $T$  and input map  $\tau$ , says only one linear map has this composite. Therefore  $CA = \text{id}_T$ .

*Second composite.* Both  $AC$  and  $\text{id}_{T'}$  are linear maps  $T' \rightarrow T'$ , and

$$(AC)\tau' = A(C\tau') = A\tau = \tau' = \text{id}_{T'} \tau'.$$

Uniqueness for  $T'$ , now with target  $T'$  and input map  $\tau'$ , gives  $AC = \text{id}_{T'}$ . Hence  $A, C$  are mutual inverses. The same uniqueness used to construct  $A$  proves that no other isomorphism satisfying the stated input condition exists.  $\square$

The map  $A$  is canonical: its behavior on every pair uniquely determines it, without any basis choice. Two constructions need not be literally equal as sets to be identified by this particular isomorphism.

$$\begin{array}{ccc} T & \xrightarrow{\quad A \quad} & T' \\ & \xleftarrow{\quad C \quad} & \\ & \tau \quad \quad \quad \tau' & \\ & V \times W & \end{array}$$

**Working facts: elementary tensor arithmetic.** Bilinearity gives

$$0 \otimes w = v \otimes 0 = 0, \quad (av) \otimes w = a(v \otimes w) = v \otimes (aw).$$

Expanding  $(0 + 0) \otimes w$  and subtracting  $0 \otimes w$  gives zero; the other slot is identical. The scalar equalities are the two scalar bilinearity identities.

### Spanning is not surjectivity of the input map

Every quotient class of a finite sum of symbols is a finite sum of elementary tensors. Therefore elementary tensors span. The universal map  $\tau : V \times W \rightarrow V \otimes W$  need not be surjective: being a finite sum of elementary tensors is weaker than being one elementary tensor.

**Proposition D.2.6** (Universality itself forces spanning). For any tensor product  $(T, \tau)$ ,  $\text{span } \tau(V \times W) = T$ .

*Proof.* Let  $S$  be that span. The quotient map  $\pi : T \rightarrow T/S$  has  $\pi\tau = 0$ . The zero linear map  $Z : T \rightarrow T/S$  also has  $Z\tau = 0$ . Tensor uniqueness with target  $T/S$  and zero bilinear input forces  $\pi = Z$ . But  $\pi$  is surjective, so  $T/S$  is the zero space. Hence every  $t \in T$  belongs to  $S$ , as claimed.  $\square$

Representations as sums need not be unique. For instance

$$(v + v') \otimes w = v \otimes w + v' \otimes w, \quad (av) \otimes w = v \otimes (aw).$$

The elementary factors and even the number of summands can change. A rule on a chosen representation therefore requires justification.

### D.3 Finite Associativity

Both parenthesizations solve the same problem: linearize a trilinear map on  $U \times V \times W$ . This is the reason for associativity.

**Theorem D.3.1** (Associativity). There is a canonical isomorphism

$$A : (U \otimes V) \otimes W \longrightarrow U \otimes (V \otimes W), \quad A((u \otimes v) \otimes w) = u \otimes (v \otimes w).$$

*Proof.* We first verify the common universal problem. Given trilinear  $b : U \times V \times W \rightarrow Z$ , fixing  $w$  gives a bilinear map and hence a unique linear  $L_w : U \otimes V \rightarrow Z$  with  $L_w(u \otimes v) = b(u, v, w)$ . Uniqueness in binary universality implies  $L_{aw+cw'} = aL_w + cL_{w'}$ : both sides factor the same bilinear input map. Consequently  $(t, w) \mapsto L_w(t)$  is bilinear on  $(U \otimes V) \times W$  and factors uniquely through  $(U \otimes V) \otimes W$ . Fixing  $u$  instead proves the identical trilinear universal property for  $U \otimes (V \otimes W)$ .

Apply the first property to the trilinear input  $(u, v, w) \mapsto u \otimes (v \otimes w)$  to construct  $A$ ; apply the second to  $(u, v, w) \mapsto (u \otimes v) \otimes w$  to construct its reverse  $C$ . The composites  $CA$  and  $AC$  factor their respective original tuple maps, as do the identity maps. Uniqueness in the two trilinear properties gives  $CA = \text{id}$  and  $AC = \text{id}$ .  $\square$

The finite version below uses the same argument inductively; no basis or dimension calculation is involved.

### D.4 Symmetry over a Field

The two inputs have occupied ordered slots, but we may wish to exchange their roles. Exchanging slots changes the target tensor product; it does not assert equality of tensors within the original space. The universal property turns this exchange into an isomorphism.

**Proposition D.4.1** (Symmetry). The rule  $v \otimes w \mapsto w \otimes v$  determines a canonical isomorphism  $S : V \otimes W \rightarrow W \otimes V$ .

*Proof.* The concrete map  $(v, w) \mapsto w \otimes v$  is bilinear by the two tensor-slot identities. The working rule constructs  $S$ ; interchanging the slots again constructs  $R : W \otimes V \rightarrow V \otimes W$ . Now  $RS(v \otimes w) = v \otimes w$  and  $SR(w \otimes v) = w \otimes v$ . Each composite and the relevant identity factor the same bilinear input map; uniqueness proves both inverse identities. No choice of representation has been made.  $\square$

In a space with basis  $e_1, e_2$ , symmetry interchanges  $e_1 \otimes e_2$  and  $e_2 \otimes e_1$ . The tensor-basis theorem below will prove that these are distinct. The field scalars may be moved from either tensor slot to the other. Swapping the inputs therefore still gives a bilinear map over the same field. Symmetry changes their order; associativity only changes grouping.

## D.5 Tensor Products and Finite Direct Sums

For a finite family  $(V_i)_{i \in I}$ , the direct sum  $\bigoplus_{i \in I} V_i$  is the vector space of tuples  $(v_i)$ , with componentwise addition and scalar multiplication. The injection  $\iota_i : V_i \rightarrow \bigoplus_i V_i$  places a vector in its  $i$ th component. Every tuple is

$$\sum_i \iota_i(v_i).$$

Given linear maps  $L_i : V_i \rightarrow X$ , Chapter 10's direct-sum construction gives the unique linear map  $L : \bigoplus_i V_i \rightarrow X$  with  $L\iota_i = L_i$ :

$$L((v_i)) = \sum_i L_i(v_i).$$

All index sets in the direct-sum formulas below are finite.

**Theorem D.5.1** (Finite direct sums). For finite index sets  $I, J$  there are canonical isomorphisms

$$\begin{aligned} \Phi : \left( \bigoplus_{i \in I} V_i \right) \otimes W &\longrightarrow \bigoplus_{i \in I} (V_i \otimes W), \\ \Theta : V \otimes \left( \bigoplus_{j \in J} W_j \right) &\longrightarrow \bigoplus_{j \in J} (V \otimes W_j). \end{aligned}$$

Their elementary values are  $\Phi((v_i) \otimes w) = (v_i \otimes w)_i$  and  $\Theta(v \otimes (w_j)) = (v \otimes w_j)_j$ .

*Proof. Construct the first forward map.* The proposed independent-variable map is  $b((v_i), w) = (v_i \otimes w)_i$ . Its output is a tuple in the stated finite direct sum. Tensor linearity in each component gives

$$b(a\mathbf{v} + c\mathbf{v}', w) = ab(\mathbf{v}, w) + cb(\mathbf{v}', w), \quad b(\mathbf{v}, aw + cw') = ab(\mathbf{v}, w) + cb(\mathbf{v}, w').$$

It is bilinear and hence induces  $\Phi$ .

*Construct its reverse.* For each  $i$ , the map  $(v, w) \mapsto \iota_i(v) \otimes w$  is bilinear, because  $\iota_i$  is linear and the tensor map is bilinear. It induces  $j_i : V_i \otimes W \rightarrow (\bigoplus_i V_i) \otimes W$ . The direct-sum construction just proved gives the linear map

$$\Psi : \bigoplus_i (V_i \otimes W) \longrightarrow \left( \bigoplus_i V_i \right) \otimes W, \quad \Psi((t_i)) = \sum_i j_i(t_i).$$

For an elementary tensor in the source of  $\Phi$ ,

$$\Psi\Phi((v_i) \otimes w) = \sum_i \iota_i(v_i) \otimes w = \left( \sum_i \iota_i(v_i) \right) \otimes w = (v_i) \otimes w.$$

For an elementary tensor injected into summand  $i$  of the other space,  $\Phi\Psi$  returns that same injected tensor. Such vectors span the direct sum, and elementary tensors span the first space. Both composites are therefore identities.

*Use symmetry for the other factor.* Compose the symmetry  $V \otimes (\bigoplus_j W_j) \cong (\bigoplus_j W_j) \otimes V$ , the proved first-factor distributivity, and symmetry in every resulting summand. The composite sends  $v \otimes (w_j)$  to  $(v \otimes w_j)_j$ , which is exactly  $\Theta$ . Its inverse is the reverse composite.  $\square$



**Corollary D.5.2** (Binary distributivity). In particular,

$$\begin{aligned}(U \oplus V) \otimes W &\cong (U \otimes W) \oplus (V \otimes W), \\ W \otimes (U \oplus V) &\cong (W \otimes U) \oplus (W \otimes V).\end{aligned}$$

*Proof.* Take a two-element index set in the two constructions above. Their inverses add the maps induced by the two component injections.  $\square$

**Corollary D.5.3** (Distributivity in both factors). For finite index sets  $I, J$ ,

$$\left(\bigoplus_i V_i\right) \otimes \left(\bigoplus_j W_j\right) \cong \bigoplus_{(i,j) \in I \times J} (V_i \otimes W_j).$$

*Proof.* Apply the first-factor theorem with  $W = \bigoplus_j W_j$ , then apply the second-factor theorem inside each summand. Reindex the finite tuple of tuples by  $(i, j)$ . The composite sends  $(v_i) \otimes (w_j)$  to  $(v_i \otimes w_j)_{i,j}$ ; all constituent maps are already isomorphisms.  $\square$

## D.6 Finite Multilinear Tensor Products

Three different input spaces arise when a trilinear rule accepts three different kinds of data. Repeated copies of one space are a special case. We first state the input problem, then solve it by recursively using the binary construction. A direct tuple-symbol quotient is an alternative realization, not a second theory to develop.

**Definition D.6.1** (Multilinearity). For  $n \geq 1$ , a map  $b : V_1 \times \cdots \times V_n \rightarrow U$  is multilinear if it is linear in each slot with every other input fixed. In slot  $i$ ,

$$\begin{aligned}b(v_1, \dots, av_i + cv'_i, \dots, v_n) &= ab(v_1, \dots, v_i, \dots, v_n) \\ &\quad + cb(v_1, \dots, v'_i, \dots, v_n).\end{aligned}$$

Here  $n$  is the number of factors and  $i$  specifies the varying slot.

**Definition D.6.2** (Finite tensor-product universal property). A tensor product of  $V_1, \dots, V_n$  consists of a vector space  $T$  and a multilinear map  $\tau : V_1 \times \cdots \times V_n \rightarrow T$  such that every multilinear map  $b : V_1 \times \cdots \times V_n \rightarrow U$  factors uniquely as  $b = \tilde{b} \circ \tau$  with  $\tilde{b} : T \rightarrow U$  linear. The value  $\tau(v_1, \dots, v_n)$  is an elementary tensor.

$$\begin{array}{ccc} V_1 \times \cdots \times V_n & \xrightarrow{\tau} & T \\ & \searrow b & \downarrow \tilde{b} \\ & & U \end{array}$$

The horizontal and slant arrows are multilinear; the vertical arrow is linear. We will build the upper-right space so that this factorization always exists and is unique.

**Recursive existence from the binary product**

Set

$$P_1 = V_1, \quad P_n = P_{n-1} \otimes V_n \quad (n \geq 2),$$

and define the nested tuple maps by

$$\tau_1(v_1) = v_1, \quad \tau_n(v_1, \dots, v_n) = \tau_{n-1}(v_1, \dots, v_{n-1}) \otimes v_n.$$

Tensor-slot linearity and induction make  $\tau_n$  multilinear. Its elementary values span  $P_n$ : elementary  $t \otimes v_n$  span the binary product, and expanding  $t$  in nested elementary tensors from  $P_{n-1}$  expands each such tensor into values of  $\tau_n$ . This argument uses no bases or dimension formula.

**Theorem D.6.3** (Recursive construction and universality). For every finite  $n \geq 1$ ,  $(P_n, \tau_n)$  represents  $n$ -linear maps.

*Proof.* Use  $(n-1)$ -linear universality, prove linear dependence on the last variable, then apply binary universality. For  $n = 1$ , the given linear map itself is the unique factor through  $\tau_1 = \text{id}_{V_1}$ . Assume the result at  $n-1$  and take an arbitrary target vector space  $U$  and multilinear  $b : V_1 \times \dots \times V_n \rightarrow U$ .

*Fix the last input.* For  $v_n \in V_n$ , define  $b_{v_n}(v_1, \dots, v_{n-1}) = b(v_1, \dots, v_n)$ . It is  $(n-1)$ -linear. The induction hypothesis constructs a linear map

$$L_{v_n} : P_{n-1} \longrightarrow U, \quad L_{v_n}(\tau_{n-1}(v_1, \dots, v_{n-1})) = b(v_1, \dots, v_n).$$

*Let that input vary.* For  $v, w \in V_n$  and  $a, c \in F$ , compare the already existing linear maps  $L_{av+cw}$  and  $aL_v + cL_w$ . On each spanning elementary value  $t = \tau_{n-1}(v_1, \dots, v_{n-1})$ , linearity of the last slot of  $b$  gives

$$L_{av+cw}(t) = aL_v(t) + cL_w(t).$$

Spanning proves equality on all of  $P_{n-1}$ . Thus  $v \mapsto L_v$  is a linear map  $V_n \rightarrow \mathcal{L}(P_{n-1}, U)$ .

*Apply binary universality.* Now define

$$\beta : P_{n-1} \times V_n \longrightarrow U, \quad \beta(t, v) = L_v(t).$$

For fixed  $v$ , it is linear in  $t$  by construction of  $L_v$ . For fixed  $t$ , it is linear in  $v$  by the map equality just proved. Hence  $\beta$  is bilinear and induces  $\tilde{b} : P_n \rightarrow U$ . Its value on  $\tau_n(v_1, \dots, v_n)$  is  $b(v_1, \dots, v_n)$ . Any two such factors agree on the spanning values of  $\tau_n$ , so are equal. This proves existence and uniqueness at  $n$ .  $\square$

**Remark D.6.4** (An alternative realization). One may alternatively start from tuple symbols modulo the slot-linearity relations. The same relation check as in the binary construction gives the finite multilinear universal property. Thus this realization is canonically isomorphic to  $P_n$ : factor each object's tuple map through the other and use uniqueness to identify both composites with identities. There is no need to repeat the formal-symbol construction.

**Corollary D.6.5** (Every finite parenthesization). Every recursively binary parenthesization of the ordered factors  $V_1, \dots, V_n$  represents the same multilinear problem as  $P_n$ . Every canonical comparison preserves the elementary tuple values, and any two canonical reassociation routes with the same endpoints agree.

*Proof.* Induct on the number of factors. At the outermost binary split, fix the inputs on the right, factor the multilinear inputs on the left, and then factor on the right. The uniqueness clause on the left ensures that the first factor depends multilinearly on the right inputs, just as in the proof of [Theorem D.6.3](#). This gives a bilinear map on the two parenthesized blocks, and binary universality gives the required linear factor on their tensor product. The same successive uniqueness clauses prove its uniqueness. Hence every parenthesization represents the same problem. Factoring their tuple maps constructs comparisons in both directions. Uniqueness proves they are inverse and that all reassociation routes with the same endpoints agree.  $\square$

For  $n = 3$ , the left-associated tuple map sends  $(u, v, w)$  to  $(u \otimes v) \otimes w$ ; the associativity map sends this to  $u \otimes (v \otimes w)$ . Both represent the same trilinear input rule. We may consequently write  $V_1 \otimes \cdots \otimes V_n$  without parentheses, using these particular comparisons.

**Remark D.6.6** (Finite-factor scope). All tensor products in this appendix have finitely many factors, although the individual vector spaces need not be finite dimensional. The finite induction justifies finite regrouping. Infinitely many tensor factors require a separate construction; the argument above neither defines such a product nor justifies infinite regrouping. That specialized subject is outside this appendix.

**Definition D.6.7** (Tensor powers). Write  $V^{\otimes n}$  for  $n$  repeated factors. Set  $V^{\otimes 0} = F$  and identify the one-factor product with  $V$  using  $P_1 = V$ . For exterior powers below,  $k$  will denote the degree, and  $V^{\otimes k}$  has  $k$  factors.

## D.7 Tensoring Linear Maps and Concrete Calculations

If each input vector is transformed by a linear map, the combined tensor data should transform as well. We must construct that transformation on the whole tensor product, so its value does not depend on a decomposition into elementary tensors.

**Proposition D.7.1** (Induced binary tensor homomorphisms). Linear maps  $T : V \rightarrow V'$  and  $S : W \rightarrow W'$  induce a unique linear map

$$T \otimes S : V \otimes W \longrightarrow V' \otimes W', \quad (T \otimes S)(v \otimes w) = T(v) \otimes S(w).$$

This construction preserves identities and composition, and takes pairs of isomorphisms to isomorphisms.

*Proof.* The concrete map  $(v, w) \mapsto T(v) \otimes S(w)$  is bilinear: linearity of  $T$  or  $S$  followed by the corresponding tensor-slot identity verifies each slot. The working rule constructs  $T \otimes S$  with the stated type. For further linear  $T' : V' \rightarrow V''$ ,  $S' : W' \rightarrow W''$ , both composites  $(T' \otimes S')(T \otimes S)$  and  $(T'T) \otimes (S'S)$  send  $v \otimes w$  to  $T'T(v) \otimes S'S(w)$ , hence factor the same bilinear map and agree by uniqueness. Identities act as identities. If  $T, S$  are invertible, the already justified construction of  $T^{-1} \otimes S^{-1}$  and the composition rule give both inverse identities.  $\square$

### Induced maps in all the factors

For linear maps  $T_i : V_i \rightarrow W_i$ , consider directly

$$b : V_1 \times \cdots \times V_n \longrightarrow \bigotimes_i W_i, \quad b(v_1, \dots, v_n) = T_1 v_1 \otimes \cdots \otimes T_n v_n.$$

Linearity of  $T_i$  and of the  $i$ th tensor slot proves multilinearity. The finite universal property therefore constructs the unique linear map

$$\bigotimes_i T_i : \bigotimes_i V_i \longrightarrow \bigotimes_i W_i$$

with these values. For  $S_i : W_i \rightarrow X_i$ , the two maps below factor the same multilinear rule  $(v_i) \mapsto \bigotimes_i S_i T_i v_i$ ; uniqueness gives

$$\bigotimes_i (S_i T_i) = (\bigotimes_i S_i) (\bigotimes_i T_i).$$

The identity map factors the original tuple map, so tensoring identities gives the identity. For invertible  $T_i$ , tensoring their inverses and using composition gives both inverse laws. The recursive binary construction has the same tuple values and is therefore compatible with this direct construction. For repeated  $T$  write  $T^{\otimes n}$ , and set  $T^{\otimes 0} = \text{id}_F$ .

### The scalar unit

The bilinear operation  $(a, v) \mapsto av$  already has its values in  $V$ . Linearizing it should recover  $V$ , with the scalar 1 providing the reverse map. This identifies a specific isomorphism, rather than merely comparing dimensions.

**Proposition D.7.2** (Scalar multiplication linearized). There is a canonical isomorphism

$$\mu : F \otimes V \longrightarrow V, \quad \mu(a \otimes v) = av.$$

*Proof.* The vector-space axioms make  $b : F \times V \rightarrow V$ ,  $b(a, v) = av$ , bilinear. For example  $b(a, cv + dw) = cb(a, v) + db(a, w)$ . This bilinear map induces the linear  $\mu$ . Define the reverse directly:

$$\eta : V \longrightarrow F \otimes V, \quad \eta(v) = 1 \otimes v.$$

The second tensor slot is linear, so  $\eta$  is linear. For every  $v$ ,  $\mu\eta(v) = 1v = v$ . For an elementary tensor,

$$\eta\mu(a \otimes v) = 1 \otimes (av) = a(1 \otimes v) = a \otimes v.$$

The two existing linear maps  $\eta\mu$  and the identity agree on a spanning set, hence agree everywhere. Thus both composites are identities.  $\square$

In particular  $F \otimes_F F$  is one-dimensional although the formal space has one independent symbol for every pair of field elements.

Composing the already constructed symmetry  $V \otimes F \rightarrow F \otimes V$  with  $\mu$  gives the right unit  $v \otimes a \mapsto av$ . Its inverse is  $v \mapsto v \otimes 1$ : composing in either order fixes vectors or elementary tensors, respectively.

### Tensor bases and coordinates

The construction has used no basis. With bases chosen, it should produce a finite table of coefficients indexed by pairs of basis vectors. Expansion shows that these pairs span; independent coordinate detectors must still show that none of those coefficients is redundant.

**Theorem D.7.3** (Tensor basis). If  $(e_i)_{i=1}^m$  and  $(f_j)_{j=1}^n$  are bases of  $V, W$ , the tensors  $e_i \otimes f_j$  form a basis of  $V \otimes W$ . Its dimension is  $mn$ .

*Proof. Spanning.* Write

$$v = \sum_i x_i e_i, \quad w = \sum_j y_j f_j.$$

Bilinearity expands

$$v \otimes w = \sum_{i,j} x_i y_j (e_i \otimes f_j).$$

Since elementary tensors span, so does the displayed finite list.

*Independence.* The coordinate maps  $e^r : V \rightarrow F$  and  $f^s : W \rightarrow F$  are already linear by uniqueness of coordinates in the two given bases. Define the scalar-valued input map  $b_{rs}(v, w) = e^r(v)f^s(w)$ . For example

$$b_{rs}(av + bw', w) = (ae^r(v) + be^r(w'))f^s(w) = ab_{rs}(v, w) + bb_{rs}(w', w),$$

and  $b_{rs}(v, aw + bw') = ab_{rs}(v, w) + bb_{rs}(v, w')$  follows by expanding the second coordinate functional. Thus it is bilinear. Universality gives a linear  $\ell_{rs} : V \otimes W \rightarrow F$ , for which

$$\ell_{rs}(e_i \otimes f_j) = \delta_{ri}\delta_{sj}.$$

Applying this existing map to

$$\sum c_{ij} e_i \otimes f_j = 0$$

gives  $c_{rs} = 0$ . All coefficients vanish, proving independence.  $\square$

We did not define  $\ell_{rs}$  as “the coefficient extractor”: uniqueness of tensor coefficients was precisely what the proof had to establish. The bilinear detector constructed it without assuming the conclusion.

**Proposition D.7.4** (Nonzero elementary tensors). For finite-dimensional  $V, W$ ,  $v \otimes w = 0$  if and only if  $v = 0$  or  $w = 0$ .

*Proof.* A zero factor gives zero by elementary arithmetic. Otherwise extend  $v$  and  $w$  separately to bases. Their tensor is one of the tensor-basis vectors, so is nonzero.  $\square$

**Example D.7.5** (A table, and a tensor that is not elementary). Choosing bases identifies tensors with rectangular coefficient tables. A rectangular coefficient table alone determines no canonical map  $V \rightarrow W$ ; later, evaluation of a covector factor in  $V^* \otimes W$  will define such a map. The maps sending

$$\sum c_{ij} e_i \otimes f_j$$

to  $(c_{ij})$  and sending  $(c_{ij})$  back to that sum are linear and inverse: each composite restores every coefficient. An elementary tensor has table  $(x_i y_j)$ . For example  $(e_1 + 2e_2) \otimes (f_1 - f_2)$  in  $F^2 \otimes F^2$  has table  $\begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix}$ . The tensor  $e_1 \otimes f_1 + e_2 \otimes f_2$  is not elementary: an elementary representation would require  $x_1 y_1 = x_2 y_2 = 1$  and  $x_1 y_2 = x_2 y_1 = 0$ . The first two equations make all four factors nonzero, contradicting the last two. This proves that  $\tau$  need not be surjective.

**Example D.7.6** (Six coordinates). In  $F^2 \otimes F^3$ , order the basis by  $(e_1 \otimes f_1, e_1 \otimes f_2, e_1 \otimes f_3, e_2 \otimes f_1, e_2 \otimes f_2, e_2 \otimes f_3)$ . Then  $(e_1 + 2e_2) \otimes (f_1 - f_2 + 3f_3)$  has coordinates  $(1, -1, 3, 2, -2, 6)$ . Integers are interpreted in  $F$ . The vector-space dimension is six, although a single elementary tensor's coordinates are constrained to be the products  $x_i y_j$ .

**Corollary D.7.7** (Finite tensor basis). Let  $n \geq 1$  and let  $\mathcal{B}_i = (e_1^{(i)}, \dots, e_{d_i}^{(i)})$  be a basis of  $V_i$ , where  $d_i = \dim V_i$ . Then

$$e_{j_1}^{(1)} \otimes \cdots \otimes e_{j_n}^{(n)}, \quad 1 \leq j_i \leq d_i,$$

form a basis of the finite tensor product, and its dimension is

$$\prod_{i=1}^n d_i.$$

*Proof.* For  $P_1 = V_1$ , this is the given basis. If the nested basis is known for  $P_{n-1}$ , apply the binary tensor-basis theorem ([Theorem D.7.3](#)) to  $P_{n-1} \otimes V_n$ . It appends  $e_{j_n}^{(n)}$  to each previous basis tensor and proves both spanning and independence. Transport this basis through the canonical associativity maps of [Corollary D.6.5](#). It becomes exactly the displayed list in any parenthesized realization. Counting gives the product dimension. If a factor is zero dimensional, the binary theorem gives the zero space at that stage and thereafter, in agreement with the empty list.  $\square$

There is only one independent tensor-basis proof here, the binary one. The recursive comparison was established without this corollary and therefore legitimately transports the basis to any finite realization.

**Example D.7.8** (A three-factor rule becomes a functional). Consider three factor spaces with dimensions 2, 2, 3:

$$b : F^2 \times F^2 \times F^3 \rightarrow F, \quad b(x, y, z) = x_1 y_1 z_1 + x_2 y_1 z_2 + x_1 y_2 z_3.$$

Hold two inputs fixed. As a function of each remaining input the rule is

$$\begin{aligned} x &\longmapsto (y_1 z_1 + y_2 z_3)x_1 + (y_1 z_2)x_2, \\ y &\longmapsto (x_1 z_1 + x_2 z_2)y_1 + (x_1 z_3)y_2, \\ z &\longmapsto (x_1 y_1)z_1 + (x_2 y_1)z_2 + (x_1 y_2)z_3. \end{aligned}$$

Each is a fixed linear combination of coordinate functions; substituting  $au + cv$  in its variable distributes into  $a$  times the value at  $u$  and  $c$  times the value at  $v$ . Thus all three linearity requirements hold. With target  $F$  and this input  $b$ , [Theorem D.6.3](#) constructs the unique linear functional

$$\tilde{b} : F^2 \otimes F^2 \otimes F^3 \rightarrow F, \quad \tilde{b}(x \otimes y \otimes z) = b(x, y, z).$$

Write  $e_i, f_j, g_k$  for the three standard bases. Its twelve basis values are

$$\tilde{b}(e_i \otimes f_j \otimes g_k) = \delta_{i1}\delta_{j1}\delta_{k1} + \delta_{i2}\delta_{j1}\delta_{k2} + \delta_{i1}\delta_{j2}\delta_{k3}.$$

Exactly the tensors indexed by  $(1, 1, 1), (2, 1, 2), (1, 2, 3)$  have value 1; the other nine have value zero. For the finite sum

$$t = e_1 \otimes f_1 \otimes g_1 + e_2 \otimes f_2 \otimes g_2, \quad \tilde{b}(t) = 1 + 0 = 1.$$

This  $t$  is not elementary. In an elementary tensor the coefficients  $c_{ijk} = x_i y_j z_k$  satisfy  $c_{111}c_{222} = c_{122}c_{211}$ . For  $t$  the left side is 1 and the right side is zero. Thus the new functional evaluates an input that cannot be supplied as one triple to  $b$ ; its value on a sum is independent of the chosen representation.

**Example D.7.9** (Four components of a tensor). Take  $A = F^2 \oplus F$  and  $B = F \oplus F^2$ , with injections  $i : F^2 \rightarrow A$ ,  $j : F \rightarrow A$ ,  $r : F \rightarrow B$ , and  $s : F^2 \rightarrow B$ . Name the forward isomorphism in [Corollary D.5.3](#)  $D$  and its constructed inverse  $E$ . Thus  $D$  has target

$$(F^2 \otimes F) \oplus (F^2 \otimes F^2) \oplus (F \otimes F) \oplus (F \otimes F^2),$$

in that order. Let  $e_1, e_2$  be the basis of the first  $F^2$  and  $f_1, f_2$  that of the second. On one elementary tensor it gives

$$D((e_1 + e_2, 1) \otimes (1, f_1)) = ((e_1 + e_2) \otimes 1, (e_1 + e_2) \otimes f_1, 1 \otimes 1, 1 \otimes f_1).$$

The inverse  $E$  is the already constructed sum of maps induced by the four pairs of injections. For the component tuple

$$z = (e_1 \otimes 1, e_2 \otimes f_1 + e_1 \otimes f_2, 1 \otimes 1, 1 \otimes f_2)$$

its value is the nontrivial sum

$$\begin{aligned} E(z) &= i(e_1) \otimes r(1) + i(e_2) \otimes s(f_1) + i(e_1) \otimes s(f_2) \\ &\quad + j(1) \otimes r(1) + j(1) \otimes s(f_2). \end{aligned}$$

Applying  $D$  to the five terms gives, respectively,

$$\begin{aligned} &(e_1 \otimes 1, 0, 0, 0), (0, e_2 \otimes f_1, 0, 0), (0, e_1 \otimes f_2, 0, 0), \\ &(0, 0, 1 \otimes 1, 0), (0, 0, 0, 1 \otimes f_2). \end{aligned}$$

Their sum is  $z$ , checking  $DE$  on this input. For the first displayed elementary tensor,  $ED$  expands its four terms and recombines them as  $(i(e_1 + e_2) + j(1)) \otimes (r(1) + s(f_1))$ , the original tensor. Dimension counting then gives  $3 \cdot 3 = 2 + 4 + 1 + 2$ . The count checks the construction but does not specify it.

## D.8 Tensor–Hom Correspondences

### A map whose values are maps

Recall from [Definition 10.10.2](#) and [Proposition 10.13.4](#) that  $\mathcal{L}(V, W)$  is the vector space of linear maps with pointwise addition and scalar multiplication. A map into this space returns a whole function, which can then be evaluated at a vector.

**Proposition D.8.1** (Currying). There are inverse linear maps

$$C : \mathcal{L}(V \otimes W, U) \rightarrow \mathcal{L}(V, \mathcal{L}(W, U)), \quad D : \mathcal{L}(V, \mathcal{L}(W, U)) \rightarrow \mathcal{L}(V \otimes W, U).$$

The forward rule is  $C(L)(v)(w) = L(v \otimes w)$ .

*Proof.* First check the types in the forward rule:

$$\begin{aligned} C(L) &\in \mathcal{L}(V, \mathcal{L}(W, U)), \\ C(L)(v) &\in \mathcal{L}(W, U), \\ C(L)(v)(w) &\in U. \end{aligned}$$

For fixed  $v$ ,  $w \mapsto L(v \otimes w)$  is linear by second-slot linearity and linearity of  $L$ . Thus  $C(L)(v)$  has the stated inner type. For all  $w$ , first-slot linearity gives  $C(L)(av + bw')(w) = aC(L)(v)(w) +$

$bC(L)(v')(w)$ . Equality at every  $w$  proves equality of maps, so  $v \mapsto C(L)(v)$  is linear. Finally  $C(aL + bL')(v)(w) = aC(L)(v)(w) + bC(L')(v)(w)$ ; hence  $C$  itself is linear.

For the inverse, take a linear  $A : V \rightarrow \mathcal{L}(W, U)$  and define  $b_A(v, w) = A(v)(w)$ . Since  $A$  is linear,  $b_A(av + bv', w) = aA(v)(w) + bA(v')(w)$ . Since  $A(v)$  is a linear map,  $b_A(v, aw + bw') = aA(v)(w) + bA(v)(w')$ . Thus  $b_A$  is bilinear and induces the linear  $D(A) : V \otimes W \rightarrow U$ . Comparing existing maps on elementary tensors shows  $D(aA + bA') = aD(A) + bD(A')$ , so  $D$  is linear as well.

For every  $v, w$ ,  $C(D(A))(v)(w) = D(A)(v \otimes w) = A(v)(w)$ , proving  $CD(A) = A$ . For every elementary tensor,  $D(C(L))(v \otimes w) = C(L)(v)(w) = L(v \otimes w)$ . Spanning proves  $DC(L) = L$ . These are both inverse identities.  $\square$

### A covector and a vector make a linear map

A covector can measure an input and use the resulting scalar to rescale a fixed output vector. Thus  $v \mapsto \lambda(v)w$  is a linear map with image in the line spanned by  $w$ . In finite dimension, basis coordinates let every linear map be assembled from finitely many such pieces.

**Proposition D.8.2** (Finite-dimensional tensor-map identification). If  $V$  is finite dimensional, there is a canonical isomorphism

$$J : V^* \otimes W \longrightarrow \mathcal{L}(V, W), \quad J(\lambda \otimes w)(v) = \lambda(v)w.$$

*Proof.* Begin with  $B(\lambda, w)$  equal to the map  $v \mapsto \lambda(v)w$ . This is a linear map  $V \rightarrow W$  because  $\lambda$  is linear. Evaluation at each  $v$  shows  $B(a\lambda + b\mu, w) = aB(\lambda, w) + bB(\mu, w)$  and  $B(\lambda, aw + bw') = aB(\lambda, w) + bB(\lambda, w')$  as equalities of maps. So  $B$  is bilinear with target  $\mathcal{L}(V, W)$  and induces  $J$ .

Choose a basis  $(e_i)$  of  $V$  and its dual basis  $(e^i)$  as in [Definition 10.16.2](#) (the same coordinate argument works over  $F$ ). Define the reverse by the finite sum

$$K(T) = \sum_i e^i \otimes T(e_i).$$

Pointwise linearity in  $T$  and tensor linearity show that  $K$  is linear. For

$$v = \sum_i e^i(v)e_i,$$

$$J(K(T))(v) = \sum_i e^i(v)T(e_i) = T(v),$$

so  $JK = \text{id}$ . On an elementary tensor the other composite is

$$K(J(\lambda \otimes w)) = \sum_i e^i \otimes \lambda(e_i)w = \left( \sum_i \lambda(e_i)e^i \right) \otimes w = \lambda \otimes w.$$

The last covector equality follows by evaluation on every  $e_j$ . Spanning now proves  $KJ = \text{id}$ . Although a basis was used to write  $K$ , it is the unique inverse of the basis-free map  $J$ . Any other basis therefore writes the same inverse, not another isomorphism.  $\square$

Finite dimension is used in the finite expansion of the identity on  $V$ . With a target basis  $(f_i)$ ,  $J(e^j \otimes f_i)$  sends  $e_j$  to  $f_i$  and the other basis vectors to zero: its matrix unit is in row  $i$ , column  $j$ . For example  $e^1 \otimes w_1 + e^2 \otimes w_2$  represents the map  $x_1e_1 + x_2e_2 \mapsto x_1w_1 + x_2w_2$ , the basis-image rule of Chapter 10.



**Naturality of currying.** For linear  $R : V' \rightarrow V$ ,  $S : W' \rightarrow W$ ,  $T : U \rightarrow U'$  and  $L : V \otimes W \rightarrow U$ , the already constructed maps satisfy

$$C(T \circ L \circ (R \otimes S))(v')(w') = T(C(L)(R(v'))(S(w'))).$$

Both sides equal  $T(L(R(v') \otimes S(w')))$ . This checks compatibility with composition by evaluating at the two independent inputs; it also shows why the displayed correspondence requires no bases.

## D.9 Exterior Powers

### Begin with two inputs

Signed area is bilinear in its two edge vectors and vanishes when they agree. Ordinary tensors retain  $v \otimes v$ ; to model area-like rules we must impose a further relation. The problem is to find a vector space with an alternating bilinear input map such that every alternating bilinear rule  $V \times V \rightarrow U$  factors uniquely to a linear map from it. The quotient will solve this problem. Define the subspace:

$$R_2 = \text{span}\{v \otimes v : v \in V\} \subseteq V \otimes V.$$

Then set

$$\Lambda^2 V = (V \otimes V) / R_2$$

and let  $u \wedge v$  denote the class of  $u \otimes v$ . The two-input wedge map is bilinear, as a composite of a bilinear tensor map and a linear quotient map, and  $v \wedge v = 0$ . Expanding the repeated input  $u + v$  gives

$$0 = (u + v) \wedge (u + v) = u \wedge v + v \wedge u,$$

so  $u \wedge v = -v \wedge u$ . In characteristic two the minus sign is a plus sign, but the repeated-input-zero relation still must be imposed.

$$\boxed{(ae_1 + be_2) \wedge (ce_1 + de_2) = (ad - bc)e_1 \wedge e_2.}$$

For example  $(e_1 + 2e_2) \wedge (3e_1 + e_2) = -5e_1 \wedge e_2$ . Over  $\mathbb{R}$  the coefficient records signed area. This is the two-dimensional calculation to keep in mind when we reach the determinant line.

The relation must use all vectors, not only basis vectors. In  $F^2$ , killing just  $e_1 \otimes e_1, e_2 \otimes e_2$  leaves the nonzero class of  $(e_1 + e_2) \otimes (e_1 + e_2)$ , since its two off-diagonal coordinates are 1. The correct  $R_2$  has basis

$$e_1 \otimes e_1, \quad e_2 \otimes e_2, \quad e_1 \otimes e_2 + e_2 \otimes e_1.$$

The third vector is obtained by subtracting the first two from the repeated tensor of  $e_1 + e_2$ . Conversely every repeated tensor has coordinates  $a^2, b^2, ab$  in this list. The list is independent by its tensor coordinates. Thus  $\dim R_2 = 3$  and its quotient has dimension  $4 - 3 = 1$ , even in characteristic two.

The determinant is already alternating bilinear, so its linear tensor factor kills  $R_2$  and factors once more to a linear map  $\Lambda^2 F^2 \rightarrow F$ . That map sends  $e_1 \wedge e_2$  to 1, proving this wedge is nonzero. The general version of this two-stage argument follows next.

**Definition D.9.1** (Alternating multilinear map). A multilinear  $A : V^k \rightarrow U$  is alternating if

$$v_i = v_j \text{ for some } i \neq j \implies A(v_1, \dots, v_k) = 0.$$

For  $k = 1$  the extra condition is empty. For  $k \geq 2$ , expansion with  $u + v$  in two equal slots proves the interchange sign rule, just as above.

For general  $k \geq 1$ , we seek an alternating multilinear map from  $V^k$  whose linear factors represent every alternating rule into every vector space. The following quotient is the solution, verified immediately below.

**Definition D.9.2** (Exterior power). For  $k \geq 1$ , let  $R_k \subseteq V^{\otimes k}$  be the span of elementary tensors having equal vectors in two distinct positions (so  $R_1 = \{0\}$ ). The space  $R_k$  is the smallest linear subspace containing all elementary tensors with repeated inputs. Every alternating multilinear rule must kill these tensors, so quotienting by  $R_k$  imposes exactly the relations forced by alternation. Define

$$\Lambda^k V = V^{\otimes k} / R_k, \quad \iota_k(v_1, \dots, v_k) = \pi(v_1 \otimes \dots \otimes v_k).$$

Write this value  $v_1 \wedge \dots \wedge v_k$ , an elementary wedge. We identify  $\Lambda^1 V$  with  $V$ , and set  $\Lambda^0 V = F$ .

Notice that  $R_k$  is a span. A relation is not required to be one repeated elementary tensor; it may be any finite linear combination of such tensors.

**Theorem D.9.3** (Exterior universal property). The map  $\iota_k$  is alternating multilinear. For every vector space  $U$  and every alternating multilinear  $A : V^k \rightarrow U$ , there exists a unique linear  $\tilde{A} : \Lambda^k V \rightarrow U$  with  $\tilde{A}\iota_k = A$ .

$$\begin{array}{ccccc} V^k & \xrightarrow{\tau_k} & V^{\otimes k} & \xrightarrow{\pi} & \Lambda^k V \\ & \searrow A & \downarrow L & \swarrow \tilde{A} & \\ & & U & & \end{array}$$

*Proof.* The composite  $\pi\tau_k$  is multilinear because  $\tau_k$  is and  $\pi$  is linear. A repeated input puts  $\tau_k(v_1, \dots, v_k)$  in  $R_k$ , so its quotient class is zero. Thus  $\iota_k$  has the claimed type.

*Tensor stage.* Given  $A$ , use its multilinearity and [Theorem D.6.3](#) to construct a unique linear  $L : V^{\otimes k} \rightarrow U$  with  $L\tau_k = A$ .

*Relation check.* On a generator  $v_1 \otimes \dots \otimes v_k$  of  $R_k$ , two inputs agree. Therefore  $L(v_1 \otimes \dots \otimes v_k) = A(v_1, \dots, v_k) = 0$  by alternation. Linearity makes  $L$  vanish on every finite combination of these generators, so  $R_k \subseteq \ker L$ .

*Quotient stage.* Apply [Theorem D.1.4](#) to obtain  $\tilde{A} : \Lambda^k V \rightarrow U$ , given by  $\tilde{A}(t + R_k) = L(t)$ . The relation check is precisely its representative-independence proof. Now  $\tilde{A}\iota_k = \tilde{A}\pi\tau_k = L\tau_k = A$ .

*Two uniqueness clauses.* If  $H : \Lambda^k V \rightarrow U$  is linear with  $H\iota_k = A$ , then  $H\pi$  and  $L$  are linear maps on  $V^{\otimes k}$  with the same composite  $A$  with  $\tau_k$ . Tensor uniqueness gives  $H\pi = L$ . Quotient uniqueness then gives  $H = \tilde{A}$ . Existence used first multilinearity and then alternation; uniqueness uses the two stages in the same order.  $\square$

Elementary wedges span because tensor elements are finite sums of elementary tensors and  $\pi$  is surjective. If another space with an alternating input map has the same universal property, construct  $A$  from its input map and  $C$  from the original one, as in [Theorem D.2.5](#). Both  $CA\iota_k = \iota_k$  and  $AC\iota'_k = \iota'_k$  hold, and the corresponding uniqueness clauses force both composites to be identities. Thus exterior powers too are determined up to the unique input-preserving isomorphism.

**Working rule for exterior powers.** Start with a concrete map on  $V^k$ , check multilinearity and repeated-input vanishing, then invoke the exterior universal property. Elementary wedges suggest a formula; this construction makes it well defined. Once maps exist, equality on elementary wedges suffices, because they span.

**Proposition D.9.4** (Elementary wedge calculus). The elementary wedge is linear in each vector slot, is zero on repeated inputs, changes sign on an interchange, and satisfies

$$v_{\sigma(1)} \wedge \cdots \wedge v_{\sigma(k)} = \text{sgn}(\sigma) v_1 \wedge \cdots \wedge v_k.$$

*Proof.* Write  $W(v_1, \dots, v_k) = \iota_k(v_1, \dots, v_k)$ . Multilinearity gives, with every unshown slot fixed,

$$W(\dots, au + bv, \dots) = aW(\dots, u, \dots) + bW(\dots, v, \dots).$$

Taking  $a = b = 0$  proves zero-slot vanishing. Alternation gives zero whenever two inputs agree. In any two chosen slots it consequently gives

$$\begin{aligned} 0 &= W(\dots, u + v, \dots, u + v, \dots) \\ &= W(\dots, u, \dots, v, \dots) + W(\dots, v, \dots, u, \dots), \end{aligned}$$

because the equal- $u$  and equal- $v$  terms vanish. Moving one term to the other side proves the interchange rule, with no division by 2. Decompose  $\sigma$  into transpositions; each contributes a minus sign. The parity theorem in Appendix C identifies their product with  $\text{sgn}(\sigma)$ , independently of the decomposition.  $\square$

### Expand, remove repeats, sort, and collect

For  $n = 3, k = 2$ , the full expansion is

$$\begin{aligned} &(a_1e_1 + a_2e_2 + a_3e_3) \wedge (b_1e_1 + b_2e_2 + b_3e_3) \\ &= a_1b_1e_1 \wedge e_1 + a_1b_2e_1 \wedge e_2 + a_1b_3e_1 \wedge e_3 \\ &\quad + a_2b_1e_2 \wedge e_1 + a_2b_2e_2 \wedge e_2 + a_2b_3e_2 \wedge e_3 \\ &\quad + a_3b_1e_3 \wedge e_1 + a_3b_2e_3 \wedge e_2 + a_3b_3e_3 \wedge e_3 \\ &= (a_1b_2 - a_2b_1)e_1 \wedge e_2 + (a_1b_3 - a_3b_1)e_1 \wedge e_3 + (a_2b_3 - a_3b_2)e_2 \wedge e_3. \end{aligned}$$

The three diagonal terms vanish. Sorting  $e_2 \wedge e_1$ ,  $e_3 \wedge e_1$ , and  $e_3 \wedge e_2$  changes each sign. Collecting the remaining pairs gives the three displayed coefficients. This proves spanning by the three increasing wedges; independence is a separate task.

**Theorem D.9.5** (Exterior basis). For a basis  $(e_1, \dots, e_n)$ , the wedges  $e_I = e_{i_1} \wedge \cdots \wedge e_{i_k}$  with  $i_1 < \cdots < i_k$  form a basis of  $\Lambda^k V$ . Thus

$$\dim \Lambda^k V = \binom{n}{k},$$

and the space is zero for  $k > n$ .

*Proof. Spanning.* By Corollary D.7.7, the tensors  $e_{i_1} \otimes \cdots \otimes e_{i_k}$  span  $V^{\otimes k}$ . The surjective quotient map  $\pi : V^{\otimes k} \rightarrow \Lambda^k V$  sends them to  $e_{i_1} \wedge \cdots \wedge e_{i_k}$ , so these wedges span the quotient. Repeated indices vanish. For distinct indices, sort them into  $j_1 < \cdots < j_k$ , writing  $i_s = j_{\sigma(s)}$ ; the interchange

law gives  $e_{i_1} \wedge \cdots \wedge e_{i_k} = \text{sgn}(\sigma) e_{j_1} \wedge \cdots \wedge e_{j_k}$ . Thus only increasing wedges are needed. The preceding three-dimensional expansion illustrates this general quotient-and-sorting mechanism.

*Independence.* For an increasing tuple  $I = (i_1, \dots, i_k)$  define the concrete map

$$A_I(v_1, \dots, v_k) = \det(e^{i_r}(v_s))_{r,s=1}^k.$$

Changing  $v_s$  linearly changes column  $s$  linearly, so determinant multilinearity gives linearity in that slot. Equal inputs give equal columns and determinant zero. The determinant theory of Chapter 10 and Appendix C therefore proves alternating multilinearity over  $F$ . The exterior universal property gives a unique linear map  $\tilde{A}_I : \Lambda^k V \rightarrow F$ . For  $J = (j_1 < \cdots < j_k)$ , its value is

$$\tilde{A}_I(e_J) = A_I(e_{j_1}, \dots, e_{j_k}) = \det(\delta_{i_r j_s})_{r,s=1}^k.$$

If  $I = J$  the matrix is the identity, so  $\tilde{A}_I(e_I) = 1$ . If  $I \neq J$ , choose  $j_s \in J \setminus I$ , which exists since the two different sets have the same size. The dual-basis rule (Definition 10.16.2) gives

$$e^{i_r}(e_{j_s}) = \delta_{i_r j_s} = 0 \quad (1 \leq r \leq k).$$

Thus column  $s$  of the determinant is zero and  $\tilde{A}_I(e_J) = 0$ . We have therefore proved  $\tilde{A}_I(e_J) = \delta_{IJ}$ . For a relation

$$\sum_J c_J e_J = 0,$$

fix  $I$  and apply this constructed map:

$$0 = \tilde{A}_I\left(\sum_J c_J e_J\right) = \sum_J c_J \tilde{A}_I(e_J) = c_I.$$

All coefficients vanish. There are

$$\binom{n}{k}$$

increasing index sets; for  $k > n$  none remain and the spanning space is zero. In degree zero the empty wedge is  $1 \in F$ , giving dimension one.  $\square$

For instance  $A_{13}(u, v) = u_1 v_3 - u_3 v_1$  induces values 0, 1, 0 on  $e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3$ . The induced linear map extracts a coefficient; its construction above ensures that it respects every presentation of an exterior element.

**Coordinates are minors.** Write

$$v_j = \sum_{i=1}^n a_{ij} e_i$$

for  $1 \leq j \leq k$ . The linear maps just constructed give the full coordinate expansion

$$v_1 \wedge \cdots \wedge v_k = \sum_{i_1 < \cdots < i_k} \det(a_{i_r j})_{r,j=1}^k e_{i_1} \wedge \cdots \wedge e_{i_k}.$$

Indeed applying  $\tilde{A}_I$  to the left yields its displayed coefficient; the exterior basis makes these coefficients unique. For two inputs these are exactly the 2 by 2 minors obtained by expanding and collecting above.

**Proposition D.9.6** (Nonvanishing of an elementary wedge). For finite-dimensional  $V$ , the wedge  $v_1 \wedge \cdots \wedge v_k$  is nonzero exactly when its inputs are linearly independent.

*Proof.* For a dependent nonempty list, choose a relation with nonzero coefficient at position  $j$  and solve it as

$$v_j = \sum_{i \neq j} c_i v_i.$$

Then

$$v_1 \wedge \cdots \wedge v_k = \sum_{i \neq j} c_i v_1 \wedge \cdots \wedge \underset{\text{slot } j}{v_i} \wedge \cdots \wedge v_k = 0.$$

Each summand repeats  $v_i$  in slots  $i, j$ ; for a one-vector dependent list the vector is zero and linearity gives the same conclusion. For an independent list, extend it to a basis; the wedge is an exterior-basis vector, hence nonzero. For  $k = 0$  the empty list is independent and its wedge is 1; for  $k > \dim V$  no independent list exists.  $\square$

Unlike  $v \otimes v$  for  $v \neq 0$ ,  $v \wedge v$  vanishes. The criterion concerns elementary wedges, not arbitrary sums of them.

### Representations need not be unique

Even an elementary wedge has many presentations:  $u \wedge (v + cu) = u \wedge v$ , and for  $c \neq 0$ ,  $(cu) \wedge (c^{-1}v) = u \wedge v$ . A general sum of elementary wedges need not reduce to one. In  $F^4$  put  $\omega = e_1 \wedge e_2 + e_3 \wedge e_4$ . For an elementary  $u \wedge v$  its coefficients  $p_{ij} = u_i v_j - u_j v_i$  satisfy  $p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0$ . To verify this without assuming any extra algebra, expand:

$$\begin{aligned} p_{12}p_{34} &= u_1 u_3 v_2 v_4 - u_1 u_4 v_2 v_3 - u_2 u_3 v_1 v_4 + u_2 u_4 v_1 v_3, \\ -p_{13}p_{24} &= -u_1 u_2 v_3 v_4 + u_1 u_4 v_3 v_2 + u_3 u_2 v_1 v_4 - u_3 u_4 v_1 v_2, \\ p_{14}p_{23} &= u_1 u_2 v_4 v_3 - u_1 u_3 v_4 v_2 - u_4 u_2 v_1 v_3 + u_4 u_3 v_1 v_2. \end{aligned}$$

Commutativity pairs each term with its negative. For  $\omega$ , the only nonzero increasing coefficients are  $p_{12} = p_{34} = 1$ , making that expression 1. Thus  $\omega$  is not elementary, in any characteristic. Coordinates in a fixed wedge basis are unique; arbitrary elementary decompositions are not. The product construction below must therefore respect relations among different decompositions.

### Exterior maps, minors, and determinants

A linear transformation sends a list of edge vectors to another list. We want its action on their oriented area or volume data, including sums of wedges with nonunique presentations. The exterior universal property turns an alternating rule on vector tuples directly into a well-defined linear map on exterior elements.

**Proposition D.9.7** (Induced exterior homomorphisms). Let  $T : V \rightarrow W$  be linear. For  $k \geq 1$  there is a unique linear map

$$\Lambda^k T : \Lambda^k V \longrightarrow \Lambda^k W,$$

characterized by

$$\Lambda^k T(v_1 \wedge \cdots \wedge v_k) = T v_1 \wedge \cdots \wedge T v_k.$$

Set  $\Lambda^0 T = \text{id}_F$ . These induced maps commute with the tensor quotient maps and preserve identities:

$$\Lambda^k(\text{id}_V) = \text{id}_{\Lambda^k V}.$$

For linear  $S : W \rightarrow X$ , they preserve composition:

$$\Lambda^k(S \circ T) = (\Lambda^k S) \circ (\Lambda^k T).$$

If  $T$  is invertible, its induced map is invertible, with

$$(\Lambda^k T)^{-1} = \Lambda^k(T^{-1}).$$

*Proof.* We first construct a map on the direct product  $V^k$ , then use the exterior universal property; no tensor representative needs to be chosen. For  $k \geq 1$  put

$$A_T : V^k \rightarrow \Lambda^k W, \quad A_T(v_1, \dots, v_k) = Tv_1 \wedge \dots \wedge Tv_k.$$

For a representative slot  $j$ , linearity of  $T$  and that wedge slot gives

$$\begin{aligned} A_T(v_1, \dots, av_j + bv'_j, \dots, v_k) \\ &= Tv_1 \wedge \dots \wedge T(av_j + bv'_j) \wedge \dots \wedge Tv_k \\ &= Tv_1 \wedge \dots \wedge (aTv_j + bTv'_j) \wedge \dots \wedge Tv_k \\ &= aA_T(v_1, \dots, v_j, \dots, v_k) + bA_T(v_1, \dots, v'_j, \dots, v_k). \end{aligned}$$

If  $v_i = v_j$ , then  $Tv_i = Tv_j$ , so the output wedge is zero. Thus  $A_T$  is alternating multilinear. By [Theorem D.9.3](#) there is a unique linear  $\Lambda^k T : \Lambda^k V \rightarrow \Lambda^k W$  with

$$\Lambda^k T(v_1 \wedge \dots \wedge v_k) = A_T(v_1, \dots, v_k).$$

The construction establishes the map before we compare its values. For  $\xi = v_1 \wedge \dots \wedge v_k$ ,

$$\Lambda^k(\text{id}_V)(\xi) = v_1 \wedge \dots \wedge v_k = \xi,$$

and

$$\begin{aligned} \Lambda^k(S \circ T)(\xi) &= STv_1 \wedge \dots \wedge STv_k \\ &= (\Lambda^k S)(Tv_1 \wedge \dots \wedge Tv_k) \\ &= ((\Lambda^k S) \circ (\Lambda^k T))(\xi). \end{aligned}$$

Each pair consists of linear maps factoring the same alternating input map on  $V^k$ . Uniqueness in the exterior universal property therefore proves both laws. If  $T$  is invertible,

$$(\Lambda^k(T^{-1}))(\Lambda^k T) = \Lambda^k(T^{-1}T) = \text{id},$$

and reversing the factors gives the identity on  $\Lambda^k W$ . Hence  $\Lambda^k(T^{-1})$  is the inverse. In degree zero all maps are  $\text{id}_F$ , so these statements hold directly.  $\square$

**Compatibility with the quotient realization.** We now compare the constructed map with tensors, rather than construct it a second time. The tensor map  $T^{\otimes k}$  sends a repeated-input generator to a repeated-input tensor, since  $v_i = v_j$  implies  $Tv_i = Tv_j$ . For a finite linear combination of such generators  $r_\ell$ ,

$$T^{\otimes k}\left(\sum_\ell c_\ell r_\ell\right) = \sum_\ell c_\ell T^{\otimes k}(r_\ell) \in R_k(W).$$

Thus  $T^{\otimes k}(R_k(V)) \subseteq R_k(W)$ . More explicitly, on every elementary tensor  $t = v_1 \otimes \dots \otimes v_k$  both routes give

$$(\Lambda^k T)\pi_V(t) = Tv_1 \wedge \dots \wedge Tv_k = \pi_W T^{\otimes k}(t).$$

Both routes are linear and elementary tensors span, so the equality holds on all tensors. This proves the compatibility asserted in the proposition:

$$\begin{array}{ccc} V^{\otimes k} & \xrightarrow{T^{\otimes k}} & W^{\otimes k} \\ \pi_V \downarrow & & \downarrow \pi_W \\ \Lambda^k V & \xrightarrow{\Lambda^k T} & \Lambda^k W. \end{array}$$

The square says that the direct universal-property construction agrees with the concrete quotient realization.

**Proposition D.9.8** (Minors of an exterior homomorphism). Let  $V, W$  have bases  $(e_j)_{j=1}^n, (f_i)_{i=1}^m$ , and let the possibly rectangular  $m$ -by- $n$  matrix  $A$  represent  $T : V \rightarrow W$ . For increasing  $k$ -tuples  $J$  in the source and  $I$  in the target,

$$\Lambda^k T(e_J) = \sum_I \det(A_{I,J}) f_I.$$

*Proof.* We extract each coefficient by the linear map  $\tilde{A}_I$  constructed in the exterior basis proof, now using the target dual basis  $(f^i)$ . The column convention for  $A$  says

$$Te_{j_s} = \sum_r a_{rj_s} f_r, \quad f^{i_r}(Te_{j_s}) = a_{i_r j_s}.$$

Consequently the evaluation matrix is exactly the indicated minor:

$$(f^{i_r}(Te_{j_s}))_{r,s} = (a_{i_r j_s})_{r,s} = A_{I,J}.$$

Writing

$$\Lambda^k T(e_J) = \sum_I c_I f_I,$$

we obtain

$$c_I = \tilde{A}_I(\Lambda^k T(e_J)) = \det(f^{i_r}(Te_{j_s}))_{r,s} = \det A_{I,J}.$$

Substituting these coefficients proves the formula. For  $k = 0$  the empty minor is 1; above either dimension the relevant source or target is zero.  $\square$

For  $k = 1$  this is the usual basis-image formula

$$T(e_j) = \sum_i a_{ij} f_i.$$

For example the columns  $(1, 0, 1)$  and  $(0, 1, 1)$  define  $T : F^2 \rightarrow F^3$ . Its three two-row minors are  $1, 1, -1$ , so  $\Lambda^2 T(e_1 \wedge e_2) = f_1 \wedge f_2 + f_1 \wedge f_3 - f_2 \wedge f_3$ .

**Example D.9.9** (Basis images still produce the matrix). For  $T(e_1) = e_1 + e_2$ ,  $T(e_2) = e_2$ ,  $T(e_3) = 2e_3$ , the images of  $e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3$  are respectively  $e_1 \wedge e_2, 2e_1 \wedge e_3 + 2e_2 \wedge e_3$ , and  $2e_2 \wedge e_3$ . The matrix is

$$[\Lambda^2 T] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 2 \end{pmatrix}.$$

This is the Chapter 10 basis-image procedure applied to an exterior basis.

The coefficient  $ad - bc$  from our first two-input calculation now has an intrinsic interpretation: the top exterior power is a line, and an endomorphism acts on that line by one scalar.

**Theorem D.9.10** (Fourth characterization of the determinant). *Top exterior-power action.* If  $\dim V = n$  and  $T : V \rightarrow V$  is linear, then  $\Lambda^n V$  is one-dimensional and

$$\Lambda^n T = (\det T) \operatorname{id}_{\Lambda^n V}.$$

Equivalently, for every  $v_1, \dots, v_n \in V$ ,

$$Tv_1 \wedge \dots \wedge Tv_n = (\det T)(v_1 \wedge \dots \wedge v_n).$$

*Proof.* The top exterior power is a line. We identify its scalar action by applying the normalized determinant form, whose value on the chosen generator is one. Let  $\mathcal{B} = (e_1, \dots, e_n)$  and let  $\det_{\mathcal{B}} : V^n \rightarrow F$  be the already established alternating  $n$ -linear map with  $\det_{\mathcal{B}}(e_1, \dots, e_n) = 1$ . For  $n \geq 1$ , the exterior universal property gives the unique linear map

$$\widetilde{\det}_{\mathcal{B}} : \Lambda^n V \rightarrow F, \quad \widetilde{\det}_{\mathcal{B}}(v_1 \wedge \dots \wedge v_n) = \det_{\mathcal{B}}(v_1, \dots, v_n).$$

By [Theorem D.9.5](#),  $g = e_1 \wedge \dots \wedge e_n$  is nonzero and  $\Lambda^n V = Fg$ . Hence the endomorphism  $\Lambda^n T$  satisfies  $\Lambda^n T(g) = \lambda g$  for a unique  $\lambda \in F$ . Now every step of the scalar identification is visible:

$$\begin{aligned} \lambda &= \lambda \widetilde{\det}_{\mathcal{B}}(g) \\ &= \widetilde{\det}_{\mathcal{B}}(\lambda g) \\ &= \widetilde{\det}_{\mathcal{B}}(\Lambda^n T(g)) \\ &= \widetilde{\det}_{\mathcal{B}}(Te_1 \wedge \dots \wedge Te_n) \\ &= \det_{\mathcal{B}}(Te_1, \dots, Te_n) \\ &= \det[T]_{\mathcal{B}} = \det T. \end{aligned}$$

The first equality uses normalization, the second linearity, the next two the defining action, and the fifth the factorization property. The last line is the matrix expression and basis independence of the determinant already established earlier. Both linear maps  $\Lambda^n T$  and  $(\det T) \operatorname{id}$  agree on the basis element  $g$  of this one-dimensional space, so they agree on every  $cg$ . Only now apply this map identity to arbitrary vectors:

$$\begin{aligned} Tv_1 \wedge \dots \wedge Tv_n &= \Lambda^n T(v_1 \wedge \dots \wedge v_n) \\ &= (\det T)(v_1 \wedge \dots \wedge v_n). \end{aligned}$$

Replacing the generator by  $g' = cg$ ,  $c \neq 0$ , does not change the scalar:

$$\Lambda^n T(g') = c\Lambda^n T(g) = c(\det T)g = (\det T)g'.$$

Any generator from another ordered basis differs in this way, because both are nonzero vectors in the same line. As a geometric consistency check, if  $T$  is singular its basis images are dependent, and

$$Te_1 \wedge \dots \wedge Te_n = 0$$

by [Proposition D.9.6](#), matching  $\det T = 0$ . For  $n = 0$ ,  $\Lambda^0 V = F$ ,  $\Lambda^0 T = \operatorname{id}_F$ , and  $\det T = 1$ , so the statement remains valid.  $\square$



For a coordinate confirmation, the minors formula has only one increasing  $n$ -tuple in top degree. The matrix of  $\Lambda^n T$  is therefore the 1-by-1 matrix  $(\det[T]_{\mathcal{B}})$ .

There are now four equivalent descriptions of the determinant: normalized alternating multilinearity determines it; the Leibniz formula constructs it; recursive Laplace expansion, starting with the empty determinant 1, computes that formula; and top exterior action records the same scalar. These are established equivalences, not new axioms for arbitrary computational algorithms. The exterior basis proof already uses determinants, so this is a retrospective characterization. In particular, the earlier multiplicativity theorem can now be understood retrospectively through composition:

$$\begin{aligned}\Lambda^n(S \circ T) &= \det(S \circ T) \text{id}, \\ (\Lambda^n S)(\Lambda^n T) &= (\det S) \text{id} \circ (\det T) \text{id} \\ &= (\det S)(\det T) \text{id}.\end{aligned}$$

Functoriality gives equality of these two maps. Evaluating at the nonzero generator  $g$  forces equality of their scalar coefficients, hence  $\det(S \circ T) = \det S \det T$ . This explains the earlier result; it cannot serve as its foundational proof because the exterior basis argument already used determinant theory. For  $T : V \rightarrow W$  between distinct  $n$ -dimensional spaces, the top map has different source and target lines; there is no scalar without separate basis or volume normalizations.

### Wedge products between exterior powers

The intended elementary rule is

$$(v_1 \wedge \cdots \wedge v_p) \wedge (w_1 \wedge \cdots \wedge w_q) = v_1 \wedge \cdots \wedge v_p \wedge w_1 \wedge \cdots \wedge w_q.$$

This is not yet an operation on general exterior elements: their expressions as sums of elementary wedges are nonunique.

We want to concatenate two wedge lists, even when neither element is given by a unique list. For  $p, q \geq 1$  the intended output lies in  $X = \Lambda^{p+q}V$ . We will first construct a map from one exterior power, then a map with values in a space of linear maps, and only then evaluate to obtain a bilinear product.

**Lemma D.9.11** (Equality on spanning tuples). If two existing multilinear maps  $A, B : V_1 \times \cdots \times V_r \rightarrow U$  agree on tuples from spanning sets  $S_1, \dots, S_r$ , they agree everywhere.

*Proof.* Write

$$v_i = \sum_j a_{ij} s_{ij}$$

with  $s_{ij} \in S_i$ . Expanding each slot gives for either map the finite sum over index tuples with coefficient

$$\prod_i a_{ij_i}$$

and value at  $(s_{1j_1}, \dots, s_{rj_r})$ . Those values agree by hypothesis, so the sums agree. The lemma compares maps that already exist; it cannot prove that a proposed rule is a map.  $\square$

**Theorem D.9.12** (Well-defined exterior product). There is a unique bilinear map  $m_{p,q} : \Lambda^p V \times \Lambda^q V \rightarrow \Lambda^{p+q} V$  whose value on two elementary wedges is their concatenated wedge.

*Proof.* First factor the  $u$ -variables. Fix the tuple  $\mathbf{v} = (v_1, \dots, v_q)$  and define

$$b_{\mathbf{v}}(u_1, \dots, u_p) = u_1 \wedge \dots \wedge u_p \wedge v_1 \wedge \dots \wedge v_q \in X.$$

The universal  $(p+q)$ -wedge map is linear in each  $u$  slot, and equal  $u$  entries make its value zero. Thus  $b_{\mathbf{v}}$  is alternating  $p$ -linear. The exterior universal property constructs a linear  $L_{\mathbf{v}} : \Lambda^p V \rightarrow X$  with this value on elementary wedges.

The map

$$B : V^q \rightarrow \mathcal{L}(\Lambda^p V, X), \quad B(\mathbf{v}) = L_{\mathbf{v}},$$

is alternating multilinear. Each value has the stated type by the preceding construction. Replace one slot  $v_j$  by  $av_j + bv'_j$ . There are now two existing linear maps  $L_{(\dots, av_j + bv'_j, \dots)}$  and  $aL_{(\dots, v_j, \dots)} + bL_{(\dots, v'_j, \dots)}$ . On an elementary  $u_1 \wedge \dots \wedge u_p$ , they agree by the linearity of slot  $p+j$  in the long wedge. Since elementary  $p$ -wedges span, the two maps agree on every  $\alpha \in \Lambda^p V$ . This proves linearity in slot  $j$  of  $B$ , not just linearity of an isolated scalar value. If two  $v$  entries agree,  $L_{\mathbf{v}}$  sends every elementary  $p$ -wedge to zero by alternation of the long wedge. It is then the zero linear map by spanning. Thus  $B$  is alternating multilinear with values in  $\mathcal{L}(\Lambda^p V, X)$ .

Factor the  $v$ -variables. Apply the exterior universal property again to obtain a linear map

$$H : \Lambda^q V \rightarrow \mathcal{L}(\Lambda^p V, X), \quad H(v_1 \wedge \dots \wedge v_q) = L_{\mathbf{v}}.$$

Define  $m_{p,q}(\alpha, \beta) = H(\beta)(\alpha)$ . For fixed  $\beta$ ,  $H(\beta)$  is linear, giving linearity in  $\alpha$ . For fixed  $\alpha$ , linearity of  $H$  gives

$$H(a\beta + b\beta')(\alpha) = (aH(\beta) + bH(\beta'))(\alpha) = aH(\beta)(\alpha) + bH(\beta')(\alpha).$$

Thus  $m_{p,q}$  is bilinear and has the prescribed concatenation values. The construction does not require a unique elementary decomposition of a general wedge. For uniqueness, any other bilinear product with those values agrees on pairs from the spanning elementary wedges; [Lemma D.9.11](#) gives equality everywhere.  $\square$

Write  $\alpha \wedge \beta = m_{p,q}(\alpha, \beta)$ . When a degree is zero, define the product by scalar multiplication in  $F = \Lambda^0 V$ . The construction can be tracked by the types

$$V^q \longrightarrow \mathcal{L}(\Lambda^p V, \Lambda^{p+q} V) \longleftarrow \Lambda^q V,$$

where the left arrow is alternating multilinear and the right arrow is the linear factor  $H$ , evaluated at  $\alpha$  only at the final stage.

**Proposition D.9.13** (Associativity and graded commutativity). For homogeneous elements of degrees  $p, q, r$ ,

$$(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma), \quad \alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

*Proof.* Both expressions in associativity are existing trilinear maps of  $\alpha, \beta, \gamma$ , since the products just constructed are bilinear. On elementary wedges both give the same concatenation of the three lists. The spanning-tuple lemma proves equality everywhere. For commutativity, moving a block of  $p$  vectors past a block of  $q$  requires  $pq$  interchanges, giving sign  $(-1)^{pq}$  on elementary pairs. Both sides are bilinear, so the same lemma extends this identity to arbitrary homogeneous elements. A zero degree is governed by scalar multiplication, for which associativity and the sign 1 hold directly.  $\square$

## D.10 The Exterior Algebra and Alternating Forms

Exterior multiplication has been constructed between separate degrees. For a finite-dimensional space, only finitely many degrees are nonzero, so we can collect them in one finite direct sum and multiply by distribution. Afterward we relate these vector wedges to alternating scalar measurements.

**Definition D.10.1** (Exterior algebra). For a finite-dimensional vector space  $V$  with  $\dim V = n$ , its **exterior algebra** is the finite direct sum

$$\Lambda V = \bigoplus_{k=0}^n \Lambda^k V,$$

with multiplication obtained by distributing the constructed wedge products between homogeneous components. The scalar  $1 \in \Lambda^0 V = F$  is its unit.

The exterior basis theorem makes degrees above  $n$  zero, so this finite sum contains every nonzero homogeneous component. An element  $\omega = \omega_0 + \cdots + \omega_n$  has a unique component in each degree, by the direct-sum construction. It is homogeneous of degree  $k$  if only that component can be nonzero. Extend the products above by

$$\left( \sum_p \alpha_p \right) \wedge \left( \sum_q \beta_q \right) = \sum_{p,q} \alpha_p \wedge \beta_q;$$

degrees above  $n$  are zero. The degree- $k$  component of the product is

$$\sum_{p+q=k} \alpha_p \wedge \beta_q.$$

This finite sum is well defined because the homogeneous components are unique. Associativity on homogeneous elements and finite distribution prove associativity for arbitrary elements. The scalar 1 is an identity. “Algebra” here means a vector space with this bilinear associative multiplication; no additional algebra course is needed to use the construction.

**Example D.10.2** (The complete exterior algebra of  $F^3$ ). For the standard basis, the degree table is

| degree | basis                                            | dimension |
|--------|--------------------------------------------------|-----------|
| 0      | 1                                                | 1         |
| 1      | $e_1, e_2, e_3$                                  | 3         |
| 2      | $e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3$ | 3         |
| 3      | $e_1 \wedge e_2 \wedge e_3$                      | 1         |

Thus the total dimension is 8. Products are calculated by distribution:

$$\begin{aligned} (1 + e_1) \wedge (1 + e_2) &= 1 + e_1 + e_2 + e_1 \wedge e_2, \\ (e_1 \wedge e_2) \wedge e_3 &= e_1 \wedge e_2 \wedge e_3, \\ e_2 \wedge (e_1 \wedge e_3) &= -e_1 \wedge e_2 \wedge e_3, \\ (e_1 \wedge e_2) \wedge (e_2 \wedge e_3) &= 0. \end{aligned}$$

The first result has mixed degrees; the next two are homogeneous of degree 3. Graded commutativity applies directly to homogeneous factors. Zero belongs to every homogeneous subspace and has no unique degree.

**Example D.10.3** (Mixed degrees in dimension two). For scalars  $a, b$ , vectors  $u, v$ , and degree-two elements  $\omega_2, \eta_2$ ,

$$(a + u + \omega_2) \wedge (b + v + \eta_2) = ab + (av + bu) + (a\eta_2 + b\omega_2 + u \wedge v).$$

The three groups have degrees 0, 1, 2. Every omitted product has degree above 2 and vanishes. A mixed sum has no single homogeneous degree to insert into the graded commutativity sign.

**Proposition D.10.4** (The assembled exterior homomorphism). Let  $T : V \rightarrow W$  be linear, with  $V, W$  finite dimensional. The direct sum of the maps  $\Lambda^k T$  defines  $\Lambda T : \Lambda V \rightarrow \Lambda W$ , preserving degrees, the unit, and wedge multiplication.

*Proof.* Each component map is linear and above the target dimension is zero; thus the finite component rule defines a linear map. Degree zero fixes 1. For elementary wedges, applying  $T$  before or after concatenation gives the same list of images. The two existing bilinear maps agree on spanning pairs, so agree on all homogeneous pairs. Finite distribution proves the identity on mixed sums. This assembled product-preserving map is a graded algebra homomorphism; each fixed-degree exterior homomorphism is a linear map.  $\square$

**Example D.10.5** (Compute with basis wedges). Write  $e_I$  with its indices increasing. If  $I, J$  overlap,  $e_I \wedge e_J = 0$ ; otherwise sort their concatenation and attach its sign. Thus

$$\begin{aligned} (2e_1 + e_3) \wedge (e_1 \wedge e_2 + 3e_2 \wedge e_4) \\ &= 6e_1 \wedge e_2 \wedge e_4 + e_3 \wedge e_1 \wedge e_2 + 3e_3 \wedge e_2 \wedge e_4 \\ &= e_1 \wedge e_2 \wedge e_3 + 6e_1 \wedge e_2 \wedge e_4 - 3e_2 \wedge e_3 \wedge e_4. \end{aligned}$$

The middle term needs two swaps, the last one swap. Over  $\mathbb{R}$ , the bivector  $\omega = e_1 \wedge e_2 + e_3 \wedge e_4$  has  $\omega \wedge \omega = 2e_1 \wedge e_2 \wedge e_3 \wedge e_4 \neq 0$ : the two cross terms add because  $(-1)^{2 \cdot 2} = 1$ . In characteristic two this square is zero; the earlier coefficient obstruction, not this square test, proves non-elementarity over every field.

## From alternating rules to linear maps into the field.

### First identification: factor an alternating map

Write  $\text{Alt}^k(V; F)$  for the vector space of alternating  $k$ -linear scalar-valued maps, with pointwise operations; set  $\text{Alt}^0(V; F) = F$ .

**Proposition D.10.6.** There is a canonical isomorphism  $\text{Alt}^k(V; F) \cong (\Lambda^k V)^*$ .

*Proof.* For  $A \in \text{Alt}^k(V; F)$  use [Theorem D.9.3](#) to construct its unique linear factor  $F(A) = \tilde{A} : \Lambda^k V \rightarrow F$ . For the reverse define  $G(\ell)(v_1, \dots, v_k) = \ell(v_1 \wedge \dots \wedge v_k)$ . Linearity of  $\ell$  preserves each slot's linearity and the zero value on repeated inputs, so  $G(\ell)$  is indeed in  $\text{Alt}^k(V; F)$ . Pointwise evaluation shows  $G$  is linear. To check  $F$ , let  $\xi = v_1 \wedge \dots \wedge v_k$ . Then

$$F(aA + bB)(\xi) = (aA + bB)(v_1, \dots, v_k) = aF(A)(\xi) + bF(B)(\xi).$$

The two sides are existing linear maps of  $\xi$  agreeing on elementary wedges; expansion of a finite sum of those wedges proves equality everywhere. This proves linearity of  $F$ . For each tuple,  $G(F(A))(v_1, \dots, v_k) = A(v_1, \dots, v_k)$ , so  $GF$  is the identity. On each elementary wedge,  $F(G(\ell))(v_1 \wedge \dots \wedge v_k) = \ell(v_1 \wedge \dots \wedge v_k)$ . Both are linear maps into  $F$ ; spanning proves  $FG$  is the identity. In degree zero both spaces are  $F$ , with the identity identification.  $\square$

**Second identification: pair covector wedges with vector wedges**

The previous identification used a linear map from vector wedges to  $F$ . Chapter 13 instead writes alternating forms as wedges of covectors. To connect these two objects, evaluate a list of covectors on a list of vectors and take the determinant. We must factor in each list separately; finite-dimensional dual bases will then give the inverse.

**Theorem D.10.7** (Exterior duality). For finite-dimensional  $V$ , there is a canonical isomorphism  $P : \Lambda^k(V^*) \rightarrow (\Lambda^k V)^*$  characterized by

$$P(\lambda_1 \wedge \cdots \wedge \lambda_k)(v_1 \wedge \cdots \wedge v_k) = \det(\lambda_i(v_j)).$$

*Proof. First factor in the vector variables.* Fix the covector list  $\lambda = (\lambda_1, \dots, \lambda_k)$ . The map  $(v_1, \dots, v_k) \mapsto \det(\lambda_i(v_j))$  is multilinear by the column laws of the determinant and vanishes when two vector entries agree. It therefore induces  $L_\lambda \in (\Lambda^k V)^*$ .

*Then factor in the covector variables.* Regard  $\lambda \mapsto L_\lambda$  as a map  $(V^*)^k \rightarrow (\Lambda^k V)^*$ . For  $\xi = v_1 \wedge \cdots \wedge v_k$ , the determinant row laws give

$$L_{(\dots, a\lambda_j + b\mu_j, \dots)}(\xi) = aL_{(\dots, \lambda_j, \dots)}(\xi) + bL_{(\dots, \mu_j, \dots)}(\xi).$$

Equal covector entries give equal rows, hence zero on every such  $\xi$ . The resulting linear maps already exist. Their equality on those spanning wedges proves these identities as equalities of linear maps. Thus this map is alternating multilinear, and the exterior universal property gives the linear  $P$  with the asserted value.

*Construct the inverse.* Choose a basis  $(e_i)$  of  $V$ , with dual  $(e^i)$ . Let  $e_I = e_{i_1} \wedge \cdots \wedge e_{i_k}$  and  $e^I = e^{i_1} \wedge \cdots \wedge e^{i_k}$  for increasing  $I$ . The determinant pairing gives  $P(e^I)(e_J) = \delta_{IJ}$ : matching lists give the identity matrix; for distinct lists choose  $j_s \in J \setminus I$ , so  $e^{i_r}(e_{j_s}) = 0$  for every  $r$  and column  $s$  is zero, exactly as in [Theorem D.9.5](#). Define

$$Q : (\Lambda^k V)^* \rightarrow \Lambda^k(V^*), \quad Q(\ell) = \sum_I \ell(e_I) e^I.$$

This finite sum is linear in  $\ell$ . For every basis wedge  $e_J$ ,

$$P(Q(\ell))(e_J) = \sum_I \ell(e_I) \delta_{IJ} = \ell(e_J).$$

Thus  $PQ(\ell) = \ell$  by equality of linear maps on a basis. In the reverse direction

$$Q(P(e^I)) = \sum_J P(e^I)(e_J) e^J = e^I.$$

Both sides are linear in the covector wedge, and the  $e^I$  form a basis, so  $QP = \text{id}$ . The basis writes the inverse explicitly, but the inverse is independent of the basis because it is the unique inverse of the canonical  $P$ . In degree zero the pairing is multiplication of scalars; if  $k > n$  both spaces are zero and the empty sums give their unique maps.  $\square$

Combining the two identifications recovers the alternating covectors called  $\Lambda^k(V^*)$  in Chapter 13. A vector wedge is made from vectors; a covector wedge evaluates on them and returns a scalar. For example  $(dx \wedge dz)((1, 2, 3), (0, 1, 4)) = 4$ . The determinant pairing relates these roles without confusing them or dividing by a factorial.

**Proposition D.10.8** (Duality commutes with maps). For linear  $T : V \rightarrow W$  between finite-dimensional spaces, the determinant pairings satisfy

$$P_V \circ \Lambda^k(T^*) = (\Lambda^k T)^* \circ P_W.$$

*Proof.* All four maps have been constructed. For an elementary covector wedge  $\lambda_1 \wedge \cdots \wedge \lambda_k$  and an elementary vector wedge  $v_1 \wedge \cdots \wedge v_k$ , both sides evaluate to  $\det(\lambda_i(Tv_j))$ . Spanning first in vectors, then in covectors proves equality of the linear maps. In degree zero both sides are the identity.  $\square$

In words: pull covectors back before pairing, or push vectors forward before pairing; the scalar result is the same. Over  $\mathbb{R}$  with  $n > 0$ , a nonzero top covector  $\omega$  is a volume form. It selects the orientation of bases on which its value is positive. Replacing it by a positive multiple preserves that orientation, whereas changing its magnitude changes the normalization of volume. Evaluation on a basis, now for an endomorphism  $T : V \rightarrow V$ , and the preceding determinant computation give  $T^*\omega = (\det T)\omega$ . This joins the constructions: a linear map acts on vectors, its top exterior map records the signed scaling, and a top covector evaluates that scaling. The scalar identity works over every field; the language of positive orientation uses real order.

**Proposition D.10.9** (Compatibility with the shuffle product). Let  $V$  be finite dimensional over  $\mathbb{R}$ . Denote by

$$J_k : \Lambda^k(V^*)_{\text{abstract}} \rightarrow \text{Alt}^k(V; \mathbb{R})$$

the determinant-pairing identification obtained above. For degrees  $p, q$ ,

$$J_{p+q}(\alpha \wedge \beta) = J_p(\alpha) \wedge_{\text{shuffle}} J_q(\beta).$$

*Proof.* Both sides are existing bilinear maps in  $\alpha, \beta$ : the abstract product was constructed in “Wedge products between exterior powers,” the maps  $J_k$  by duality, and the shuffle product in Section 13.6. It suffices by Lemma D.9.11 to take  $\alpha = \lambda_1 \wedge \cdots \wedge \lambda_p$  and  $\beta = \mu_1 \wedge \cdots \wedge \mu_q$ . Evaluate on  $v_1, \dots, v_{p+q}$ . The left side is the determinant whose rows are the values of  $\lambda_1, \dots, \lambda_p, \mu_1, \dots, \mu_q$  on these vectors. Expand by the columns assigned to the first  $p$  rows. For each  $(p, q)$ -shuffle  $\sigma$  the contribution is

$$\text{sgn}(\sigma) \det(\lambda_i(v_{\sigma(j)}))_{i,j=1}^p \det(\mu_i(v_{\sigma(p+j)}))_{i,j=1}^q.$$

Summing gives precisely the shuffle definition on the right. This is the determinant/concatenation calculation already explained in Chapter 13. Equality on every vector tuple proves equality of these alternating forms; spanning now extends it to all  $\alpha, \beta$ . Degree zero uses scalar multiplication and the empty determinant 1; degrees above  $\dim V$  give zero.  $\square$

Appendix D constructs an exterior algebra through quotients, whereas Chapter 13 constructs it through alternating covectors and shuffles. The canonical maps  $J_k$  identify their products as well as their vector spaces, so they give the same exterior algebra after this identification.

## Exercises

**Exercise 1.\*** Let  $R = \text{span}(1, 1)$  in  $F^2$ . Determine which of  $x + y$  and  $x - y$  factor through

$$F^2/R,$$

including characteristic two. Construct an isomorphism of the quotient with  $F$ , its reverse, and both composites.

**Exercise 2.\*** Let  $R \subseteq E$ ,  $S \subseteq H$  be subspaces and  $L : E \rightarrow H$  linear with  $L(R) \subseteq S$ . Construct  $\tilde{L} : E/R \rightarrow H/S$  by using the quotient universal property on  $\pi_H L$ . Verify the commutative square and prove uniqueness. Explain why the hypothesis is necessary.

**Exercise 3.** Construct the map  $F^2 \otimes F^2 \rightarrow F^2$  from  $b(x, y) = (x_1 y_2, x_2 y_1)$ , checking both slots first. Evaluate it on  $(e_1 - e_2) \otimes (e_1 + 2e_2)$  using two different expansions.

**Exercise 4.** A proposed real-valued rule on  $\mathbb{R}^2 \otimes \mathbb{R}^2$  sends  $v \otimes w$  to  $\|v\|_2 \|w\|_2$ . Compare the equal tensors  $(-v) \otimes w = v \otimes (-w)$  and the additivity requirement on  $(v + (-v)) \otimes w$ . Decide whether the proposed values can be those of a linear functional, and identify the failed slot-linearity condition.

**Exercise 5.\*** Suppose an extra relation  $v \otimes w = w \otimes v$  is imposed on  $F^2 \otimes F^2$ . Use  $b(v, w) = v_1 w_2$  to show the resulting quotient does not satisfy the ordinary tensor universal property.

**Exercise 6.** For  $V_1 = F$ ,  $V_2 = F^2$ , and  $W = F^2$ , construct both directions of  $(V_1 \oplus V_2) \otimes W \cong (V_1 \otimes W) \oplus (V_2 \otimes W)$ . Evaluate them on  $((1, (1, -1)) \otimes (2, 1))$  and on a sum of two injected elementary tensors. Verify both composites on these inputs.

**Exercise 7.\*** In [Example D.7.9](#), consider the component tuple  $(0, e_1 \otimes f_1 + e_2 \otimes f_2, 0, 0)$ . Send it backward explicitly and show that the result is not an elementary tensor of  $A \otimes B$ . Use coordinate projections to isolate a two-by-two coefficient table and its nonzero determinant. Explain why a dimension count alone cannot detect this.

**Exercise 8.** For  $A : V \rightarrow \mathcal{L}(W, U)$ , distinguish the types of  $A$ ,  $A(v)$ , and  $A(v)(w)$ . Reconstruct its tensor factor and prove the two currying inverse identities.

**Exercise 9.** Expand  $(e_1 + e_2) \wedge (e_2 + e_3)$  in  $F^3$ , and recover the three coefficients with the induced determinant maps constructed from independent inputs. Compute its wedge with  $e_1 + e_3$ . Interpret the final coefficient 2 in characteristic two and relate its vanishing to dependence of the three vectors.

**Exercise 10.** For  $T(e_1) = e_1 - e_3$ ,  $T(e_2) = 2e_2$ ,  $T(e_3) = e_1 + e_3$  over  $\mathbb{R}$ , compute  $\Lambda^2 T$  by its basis images and compare  $\Lambda^3 T$  with  $\det T$ . Verify the quotient compatibility square on an elementary tensor. Also compute  $(1 + e_1 + e_1 \wedge e_2) \wedge (2 + e_2)$  in  $\Lambda \mathbb{R}^2$ .

**Exercise 11.\*** In the construction of exterior multiplication, prove explicitly that the map  $\mathbf{v} \mapsto L_{\mathbf{v}}$  takes equal  $v$  entries to the zero linear map. State where spanning, existence, and alternation are used.

**Exercise 12.** For  $\alpha = e^1 \wedge e^3$  and  $\xi = (e_1 + e_2) \wedge (e_2 + 2e_3)$ , compute the determinant pairing and identify the spaces containing  $\alpha$ ,  $\xi$ , and the answer. Construct the inverse duality map for this basis.

## Appendix E

# Quotient Spaces, Gluing, and Classical Manifolds

### Introduction

This appendix makes the gluing pictures of Chapter 13 precise. It uses Chapter 10's open-set, product, compactness, and Hausdorff language, and Chapter 13's definitions of smooth manifolds and manifolds with boundary. The analysis-to-Stokes route uses the product, subspace, and chart constructions already given and does not depend on this appendix. The construction begins by specifying which points represent the same point, then determines continuity and local coordinates from that specification. These examples prepare the global questions of geometric topology and provide spaces on which the cohomological ideas of Appendix H can be tested. The quotient tools are structural constructions used repeatedly, not additional hypotheses in the main integration theorems.

### E.1 Quotient Topology and Quotient Maps

Appendix D forms quotient vector spaces by declaring vectors equivalent. Here we retain that idea but allow any equivalence relation on any topological space. The extra question is which subsets of the resulting set should be open. The progression is from quotient vector spaces to topological-group quotients to arbitrary equivalence-relation quotients: increasing generality does not mean that every quotient has a vector-space or group structure.

**Definition E.1.1** (Quotient topology). Let  $\sim$  be an equivalence relation on a topological space  $X$ . Its quotient set  $X/\sim$  consists of the classes  $[x]$ . The **canonical projection** and the **quotient topology** are

$$\pi : X \longrightarrow X/\sim, \quad \pi(x) = [x], \quad U \text{ open in } X/\sim \iff \pi^{-1}(U) \text{ open in } X.$$

Here  $\pi^{-1}(U)$  denotes an inverse image, not an inverse function.



**Proposition E.1.2** (The open-set rule defines a topology). The quotient topology is the finest topology making  $\pi$  continuous.

*Proof.* Inverse images take the empty set and whole quotient to  $\emptyset$  and  $X$ , and preserve arbitrary unions and finite intersections. The open-set axioms in  $X$  therefore imply the three topology axioms on the quotient. The rule makes  $\pi$  continuous. Any topology making  $\pi$  continuous can contain only sets whose inverse images are open, so it is contained in this one.  $\square$

**Definition E.1.3** (Quotient map). A surjection  $q : X \rightarrow Q$  between topological spaces is a **quotient map** if  $U \subseteq Q$  is open exactly when  $q^{-1}(U)$  is open in  $X$ .

We reserve  $\pi$  for canonical projections onto equivalence classes and  $q$  for general quotient maps;  $Q$  need not literally be written  $X/\sim$ .

**Proposition E.1.4** (The universal property). For any topological space  $Y$ , a map  $h : X/\sim \rightarrow Y$  is continuous if and only if  $h \circ \pi$  is continuous. Consequently, every continuous  $f : X \rightarrow Y$  constant on equivalence classes has a unique continuous factor

$$\tilde{f} : X/\sim \rightarrow Y, \quad \tilde{f}([x]) = f(x), \quad f = \tilde{f} \circ \pi.$$

*Proof.* For open  $V \subseteq Y$ ,  $(h \circ \pi)^{-1}(V) = \pi^{-1}(h^{-1}(V))$ . The quotient rule makes this open exactly when  $h^{-1}(V)$  is open. For the factorization, constancy on classes makes the displayed rule well defined; surjectivity of  $\pi$  gives uniqueness, and the equivalence just proved gives continuity.  $\square$

$$\begin{array}{ccc} X & \xrightarrow{\pi} & X/\sim \\ & \searrow f & \downarrow \tilde{f} \\ & & Y \end{array}$$

Construct  $f$  before gluing and check that its values agree at glued points; the dashed arrow is the uniquely induced map. We use a tilde for a map produced by such a factorization, while retaining structural names such as  $T_1 \otimes T_2$  and  $\Lambda^k T$  in Appendix D. The same inverse-image proof works for every quotient map. The factor need not be injective: a constant map may collapse distinct classes. It is injective exactly when  $f(x) = f(y)$  implies  $x \sim y$ , so that the classes are precisely the fibers of  $f$ .

**Example E.1.5** (Coset quotients). Let  $G$  be an abelian group written additively, equipped with a topology for which addition and negation are continuous, and let  $H \leq G$  be a subgroup. Then

$$x \sim y \iff x - y \in H, \quad [x] = x + H, \quad \pi : G \rightarrow G/H.$$

The subgroup laws give an equivalence relation; equip its classes with the quotient topology above. This is a special case of the same definition. If  $V$  is a topological vector space and  $W$  a linear subspace, it equips Appendix D's quotient vector space  $V/W$  with a topology. For the torus,  $\mathbb{Z}^2$  is an additive subgroup of  $\mathbb{R}^2$ , but it is not an  $\mathbb{R}$ -linear subspace (scalar multiplication by

$$1/2$$

does not preserve it).

## E.2 Quotient-Space Tools

**Proposition E.2.1** (Compact quotients). A quotient of a compact space is compact.

*Proof.* Its canonical projection is continuous and onto, so the continuous-image compactness theorem applies.  $\square$

**Separation can fail.** The quotient need not be Hausdorff even when the original space is. For example, in  $\mathbb{R}$  collapse all of  $(0, 1)$  to a single class and leave other points separate. If  $[0]$  and the collapsed class had disjoint open neighborhoods, the inverse image of the first would be an open set containing 0 and hence meeting  $(0, 1)$ , whereas the inverse image of the second contains all of  $(0, 1)$ . This is impossible.

Recall [Definition 10.5.5](#): in a Hausdorff space, any two distinct points have disjoint open neighborhoods. Closed points alone do not ensure this separation, as [Example 10.5.8](#) shows. The following criterion uses only the compactness results of Chapter 10.

**Proposition E.2.2** (Compact-to-Hausdorff criterion). A continuous bijection  $h : K \rightarrow H$  from a compact space to a Hausdorff space is a homeomorphism. No Hausdorff hypothesis on  $K$  is needed.

*Proof.* A closed subset  $F$  of  $K$  is compact: express a relatively open cover of  $F$  using open subsets of  $K$ , adjoin  $K \setminus F$ , use compactness of  $K$ , and discard that extra set. Its image  $h(F)$  is compact and hence closed in  $H$  by [Theorem 10.5.10](#). Thus  $h$  is a closed bijection, so  $(h^{-1})^{-1}(F) = h(F)$  is closed for every closed  $F$ . This proves continuity of  $h^{-1}$ .  $\square$

For the examples below, use this five-step strategy:

- (a) Build a continuous map on the original space.
- (b) Check that it is constant on the prescribed equivalence classes.
- (c) Descend it by [Proposition E.1.4](#).
- (d) Check that the descended map is bijective, by identifying its fibers.
- (e) Prove the source compact and target Hausdorff, then apply [Proposition E.2.2](#).

Compactness may come from a compact set of representatives even when the original space is not compact. This is the key to the next two examples.

**Proposition E.2.3** (A metric for separated isometric identifications). Let  $Y \subseteq \mathbb{R}^d$  have its relative Euclidean metric. Let  $\mathcal{G}$  be a family of bijective isometries of  $Y$ , containing the identity and closed under composition and inverse. Suppose that for every  $p, q \in Y$  and  $R > 0$  the set

$$\{g \in \mathcal{G} : \|p - g(q)\| \leq R\}$$

is finite. Identify  $p$  and  $q$  when  $p = g(q)$  for some  $g$ . Then

$$d([p], [q]) = \min_{g \in \mathcal{G}} \|p - g(q)\|$$

is a metric inducing the quotient topology. If in addition no nonidentity map fixes a point, every point has a sufficiently small relative ball that projects homeomorphically onto an open quotient neighborhood.

*Proof.* The idea is to find a nearest representative and then compare small balls. Only finitely many maps can improve the identity's distance, so the minimum exists. Replacing  $p, q$  by  $a(p), b(q)$  reindexes it by the bijection  $g \mapsto a^{-1}gb$  of  $\mathcal{G}$ . An attained zero means equal classes; inverses give symmetry. If  $g, h$  minimize for  $(p, q)$  and  $(q, r)$ , then

$$d([p], [r]) \leq \|p - gh(r)\| \leq \|p - g(q)\| + \|q - h(r)\|.$$

This proves the triangle inequality.

The inverse image of a metric ball about  $[p]$  is the union of the relative balls of the same radius about  $g(p)$ , hence is open in  $Y$ . Conversely, if  $\pi^{-1}(U)$  is open and contains  $p$ , choose a relative ball  $B_Y(p, \rho) \subseteq \pi^{-1}(U)$ . Every class at distance less than  $\rho$  has a representative in this ball and belongs to  $U$ . Thus the two topologies agree. Also  $\pi$  is open: for open  $A \subseteq Y$ ,

$$\pi^{-1}(\pi(A)) = \bigcup_g g(A)$$

is open.

For the last assertion, finiteness with  $p = q$  and  $R = 1$ , together with the absence of fixed points, gives  $\delta_p > 0$  such that  $\|p - g(p)\| \geq \delta_p$  for every nonidentity  $g$ . If  $0 < 2\rho < \delta_p$ , the balls  $B_Y(p, \rho)$  and  $g(B_Y(p, \rho))$  are disjoint: an intersection would put their centers at distance less than  $2\rho$ . The projection is consequently injective on this ball, and its continuity and openness make the restriction a homeomorphism onto an open neighborhood.  $\square$

**Lemma E.2.4** (Separated translates give quotient neighborhoods). Let a family  $\mathcal{G}$  of isometries of a Euclidean subspace  $Y$  contain the identity and be closed under composition and inverse. Identify points carried to each other by these maps, and give  $Y/\sim$  its quotient topology. If a relatively open ball  $D$  satisfies  $D \cap g(D) = \emptyset$  for every nonidentity  $g \in \mathcal{G}$ , then  $\pi|_D : D \rightarrow \pi(D)$  is a homeomorphism and  $\pi(D)$  is open.

*Proof.* If  $x, y \in D$  have the same class, then  $y = g(x)$  for some  $g$ . Disjointness forces  $g$  to be the identity, hence  $x = y$ . This proves injectivity. For any relatively open  $A \subseteq D$ , the set  $A$  is open in  $Y$  because  $D$  is, and the saturation has the exact description

$$\pi^{-1}(\pi(A)) = \bigcup_{g \in \mathcal{G}} g(A).$$

Both inclusions follow by writing equality of classes as  $y = g(a)$ . Every  $g(A)$  is open since an isometry with inverse is a homeomorphism. Thus  $\pi(A)$  is open by the quotient definition (Section E.1). The restriction is continuous and bijective onto its image; its openness proves continuity of its inverse. Taking  $A = D$  proves that its image is an open neighborhood.  $\square$

### E.3 Circles and Spheres as Quotients

**Construction and descent.** For  $s, t \in \mathbb{R}$ , write  $s \sim t$  if  $s - t \in \mathbb{Z}$ . The continuous map

$$e(t) = (\cos 2\pi t, \sin 2\pi t)$$

is onto  $S^1$ , and  $e(s) = e(t)$  exactly when  $s - t \in \mathbb{Z}$ . It therefore descends to a continuous bijection  $\tilde{e} : \mathbb{R}/\mathbb{Z} \rightarrow S^1$ .

**Topology.** Every class has a representative in  $[0, 1]$ , so the restriction of the canonical projection to  $[0, 1]$  is a continuous surjection. Thus  $\mathbb{R}/\mathbb{Z}$  is compact. The circle is Hausdorff as a subspace of  $\mathbb{R}^2$ . The criterion proves

$$\mathbb{R}/\mathbb{Z} \cong S^1.$$

**The finite interval model.** Restricting  $e$  to  $[0, 1]$  gives exactly the endpoint relation  $0 \sim 1$ , with every other class a singleton. The same factorization and compact-to-Hausdorff argument give

$$[0, 1]/(0 \sim 1) \cong S^1.$$

Joining endpoints is therefore a precise topological construction, illustrated in Figure E.1. The resulting circle has the local arc charts already used in Chapter 13; the whole closed interval is not one chart.

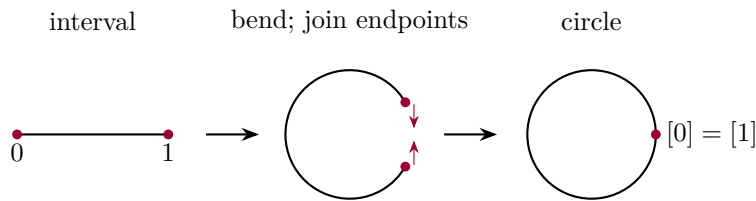


Figure E.1: Joining the interval's two purple endpoints gives one circle point. A neighborhood of that point joins short intervals near both endpoints. This explains the seam without mistaking the whole closed interval for a single circle chart.

### Collapsing the boundary of a disk

Closing an interval joins its two endpoints. In higher dimensions we can instead collapse the entire boundary to a single point.

**Proposition E.3.1** (Sphere as a disk quotient). For  $n \geq 1$ , collapsing  $\partial D^n$  to one class gives

$$D^n/\partial D^n \cong S^n.$$

*Proof.* Write  $r = \|x\|$  and define the continuous map  $f : D^n \rightarrow S^n$  by

$$f(x) = \left( \frac{\sin(\pi r)}{r} x, \cos(\pi r) \right) \quad (r > 0), \quad f(0) = (0, \dots, 0, 1).$$

Continuity at zero follows from

$$\sin(\pi r)/r \rightarrow \pi.$$

The whole boundary maps to the south pole. For  $0 < r < 1$  the last coordinate determines  $r$  uniquely, and the first  $n$  coordinates determine the direction of  $x$ . To see surjectivity explicitly, write a sphere point as  $(u, t)$  with  $\|u\|^2 + t^2 = 1$ . If  $-1 < t < 1$ , let

$$r = \arccos(t)/\pi \in (0, 1), \quad x = ru/\sin(\pi r).$$

Since  $\|u\| = \sin(\pi r)$ , this has  $\|x\| = r$  and  $f(x) = (u, t)$ . The north pole comes only from  $x = 0$ ; the south pole comes exactly from the boundary. Thus the fibers are precisely the prescribed classes. The universal property gives a continuous bijection

$$\tilde{f} : D^n/\partial D^n \rightarrow S^n.$$

The domain is compact and the sphere Hausdorff, so it is a homeomorphism.  $\square$

Figure E.2 shows the two-dimensional case. This construction also explains the sphere's one 0-cell and one  $n$ -cell in the optional CW discussion of [Section 13.17](#).

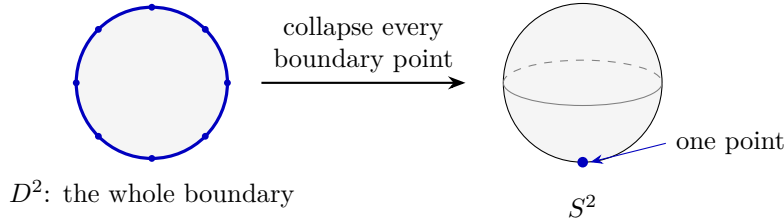


Figure E.2: The rule for  $D^2/\partial D^2$ : every marked boundary point belongs to one class, shown as the south pole; interior points remain distinct.

## E.4 The Torus as a Quotient

The formal smooth torus of Chapter 13 is  $T^2 = S^1 \times S^1$ .

**Construction and fibers.** To identify the topology of the gluing model, define

$$E(x, y) = ((\cos 2\pi x, \sin 2\pi x), (\cos 2\pi y, \sin 2\pi y)).$$

This map is continuous and onto. Equality of its first circle coordinates means the first parameters differ by an integer; equality of the second means the same for the second parameters. Its fibers are therefore exactly

$$(x, y) \sim (x + m, y + n), \quad (m, n) \in \mathbb{Z}^2.$$

The universal property gives a continuous bijection  $\tilde{E} : \mathbb{R}^2/\mathbb{Z}^2 \rightarrow S^1 \times S^1$ .

**Topology.** Every class has a representative in  $[0, 1]^2$ , so the continuous image of this compact square is the whole quotient. The target is Hausdorff: if two pairs differ, disjoint neighborhoods in a differing factor, together with the entire other factor, separate them. Thus [Proposition E.2.2](#) proves

$$\mathbb{R}^2/\mathbb{Z}^2 \cong S^1 \times S^1.$$

**The square model.** On the square, the fibers of  $E$  are precisely the equivalence relation generated by  $(0, y) \sim (1, y)$  and  $(x, 0) \sim (x, 1)$ . In particular the four corners form one class. Descending the restricted map gives a continuous bijection from the compact square quotient to the same Hausdorff product, hence a homeomorphism. This formally justifies [Figure 13.15](#), including both stages of gluing.

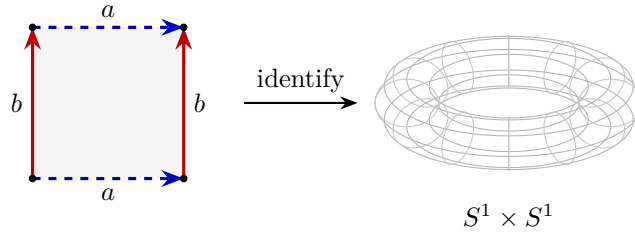


Figure E.3: Torus quotient:  $(0, y) \sim (1, y)$  and  $(x, 0) \sim (x, 1)$ . The dashed  $a$ -edges and solid  $b$ -edges distinguish the two pairs. The arrows agree on each paired edge, and all four vertices form one class. Figure 13.15 gives the full geometric sequence.

**Coordinate conventions.** The linear rescaling

$$(\theta, \varphi) \mapsto (\theta/(2\pi), \varphi/(2\pi))$$

and its inverse respect the respective relations; both descend continuously by the universal property. Hence  $\mathbb{R}^2/(2\pi\mathbb{Z})^2 \cong \mathbb{R}^2/\mathbb{Z}^2$ . These homeomorphisms identify the topological models. The smooth structure is the product-circle structure already constructed in Chapter 13, transported through the homeomorphism if the quotient notation is used. No general smooth-quotient theorem is needed.

## E.5 Cylinder and Möbius Strip

Gluing two edges leaves boundary; reversing that gluing changes the way in which the remaining edges join.

**Definition E.5.1** (Finite rectangle models). For  $\sigma \in \{1, -1\}$ , put

$$Q_\sigma = ([0, 1] \times [-1, 1])/\sim, \quad (0, t) \sim (1, \sigma t),$$

with only the equivalence relation generated by these edge identifications. The **cylinder** is  $Q_1$  and the **Möbius strip** is  $Q_{-1}$ . Both have the quotient topology and are compact.

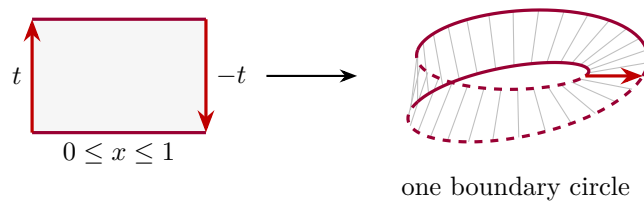


Figure E.4: Möbius quotient:  $(0, t) \sim (1, -t)$ . Opposite arrows encode the reversal. The two long edges join into the single emphasized boundary; dashes indicate its rear portions. Figure 13.17 shows the deformation, while Figure E.5 checks local charts.

For the cylinder,  $(x, t) \mapsto (e(x), t)$  has exactly these fibers. The compact-to-Hausdorff criterion therefore identifies  $Q_1$  with  $S^1 \times [-1, 1]$ .

### An infinite strip for local coordinates

The finite rectangle shows the global gluing; an infinite strip lets a neighborhood continue across a seam without cutting its coordinates.

**Construction E.5.2** (Infinite strip model). Set  $B = \mathbb{R} \times [-1, 1]$  and define

$$f_k(x, t) = (x + k, \sigma^k t), \quad k \in \mathbb{Z}.$$

These Euclidean isometries satisfy  $f_k f_l = f_{k+l}$  and  $f_k^{-1} = f_{-k}$ . Declare  $p \sim r$  when  $p = f_k(r)$  for some  $k$ ; the identities verify equivalence. Write  $\hat{Q}_\sigma = B/\sim$  and  $\pi : B \rightarrow \hat{Q}_\sigma$  for its canonical projection.

To compare the two models by compact-to-Hausdorff, we first establish Hausdorffness of the infinite model directly.

**Lemma E.5.3** (Distance between strip classes). The formula

$$d([p], [r]) = \min_{k \in \mathbb{Z}} \|p - f_k(r)\|$$

defines a metric on  $\hat{Q}_\sigma$ .

*Proof.* A bound on  $\|p - f_k(r)\|$  bounds the horizontal displacement and hence allows only finitely many integers  $k$ . No  $f_k$  with  $k \neq 0$  fixes a point, and  $\|p - f_k(p)\| \geq |k| \geq 1$ . Thus [Proposition E.2.3](#) applies. It also shows that this metric induces the quotient topology and that relative balls of radius less than

$$1/2$$

project to quotient neighborhoods. In particular the quotient is Hausdorff.  $\square$

**Proposition E.5.4** (The finite and infinite models agree). The rectangle inclusion induces a homeomorphism  $Q_\sigma \cong \hat{Q}_\sigma$ .

*Proof. Construct the map.* Compose inclusion with  $\pi$ . It respects the edge relation because  $f_1(0, t) = (1, \sigma t)$ , so  $\pi(0, t) = \pi(1, \sigma t)$ . Constancy on these generating pairs implies constancy on every equivalence class. The quotient universal property therefore gives the continuous map  $[x, t] \mapsto \pi(x, t)$ .

*Check the fibers.* Reducing the first coordinate modulo integers gives a representative in the rectangle: for  $x \in \mathbb{R}$  choose integer  $k$  with  $x - k \in [0, 1]$  and apply  $f_{-k}$ . If rectangle points satisfy  $(x, t) = f_k(y, u)$ , then  $x - y = k$  with  $x, y \in [0, 1]$ . Thus  $k = 0$  gives identical points, while  $k = \pm 1$  forces the paired endpoints 0, 1 and the specified sign change in the second coordinate. There are no other possibilities. These are exactly the finite-model fibers, proving bijectivity.

*Compare the topologies.* Its source is compact and its target is Hausdorff by [Lemma E.5.3](#). Apply [Proposition E.2.2](#).  $\square$

## Charts across seams and at boundary points

**Proposition E.5.5** (Small strip neighborhoods are quotient charts). For  $p \in B$  and

$$0 < \rho < 1/4,$$

let  $D = B \cap B_{\mathbb{R}^2}(p, \rho)$ . Then  $\pi|_D : D \rightarrow \pi(D)$  is a homeomorphism onto an open quotient neighborhood.

*Proof.* For  $k \neq 0$ , the centers  $p$  and  $f_k(p)$  have horizontal separation at least 1. If  $z \in D \cap f_k(D)$ , then

$$1 \leq \|p - f_k(p)\| \leq \|p - z\| + \|z - f_k(p)\| < 2\rho < 1,$$

a contradiction. Apply [Lemma E.2.4](#) to the relative ball and the maps  $f_k$ .  $\square$

At an interior point  $p = (x_0, t_0)$  choose also  $\rho < 1 - |t_0|$ ;  $D$  is an ordinary disk. At  $t_0 = 1$  use  $(x - x_0, 1 - t)$ , and at  $t_0 = -1$  use  $(x - x_0, t + 1)$ ; the image is an open half-disk relative to  $v \geq 0$ . These points give genuine boundary charts.

For a seam chart centered at  $(0, t_0)$ , let  $D$  be such a relative disk. In the finite rectangle use precisely the pieces

$$D_0 = D \cap \{x \geq 0\}, \quad D_1 = f_1(D \cap \{x \leq 0\}).$$

The coordinate map into  $D$  is

$$\chi([x, t]) = \begin{cases} (x, t), & (x, t) \in D_0, \\ (x - 1, \sigma t), & (x, t) \in D_1. \end{cases}$$

At the identified edges  $(0, t) \sim (1, \sigma t)$  both formulas give  $(0, t)$ . Thus the map is well defined; the small-disk proposition proves it is a homeomorphism. Interior seams assemble two half-disks into a disk; seam endpoints assemble two quarter-disks into a half-disk.

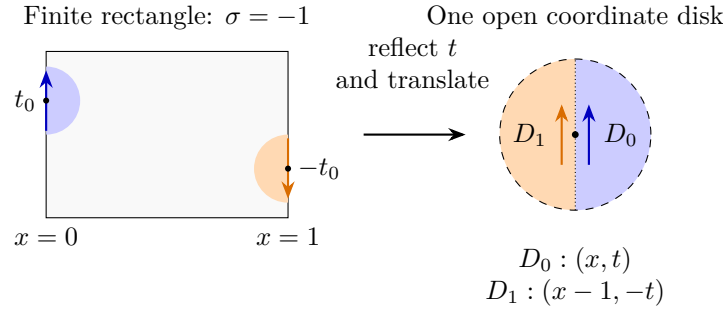


Figure E.5: A Möbius seam chart. The right-edge piece is centered at  $(1, -t_0)$  and is reflected transversely before being placed to the left of the piece at  $(0, t_0)$ . Both arrows then point in the same chart direction. For the cylinder use  $(x - 1, t)$  instead, without reflection. At a boundary endpoint the corresponding quarter-disks assemble into a half-disk. Dashed arcs indicate excluded disk boundaries.

**Smoothness and second countability.** On an overlap, lifts differ locally by one  $f_k$ ; the integer is locally constant: disjoint translated small balls make the chosen lift unique on a neighborhood. It is constant on each connected overlap component by [Proposition 10.7.2](#) and [Definition 10.7.10](#). Transitions, including the boundary coordinates, are locally affine and therefore smooth. Their maximal compatible atlas is the smooth structure. The projection is open by the saturation argument above. Second countability means having a countable topological basis ([Definition 10.4.10](#)). By [Example 10.4.7](#) and [Proposition 10.4.11](#),  $B$  has a countable relative Euclidean basis. Its images form a countable basis: lift a point of an open quotient set and choose a basis element inside its inverse image. Both spaces are consequently Hausdorff, second countable smooth surfaces with boundary.

**Proposition E.5.6** (Boundary components).

$$\partial Q_1 \cong S^1 \sqcup S^1, \quad \partial Q_{-1} \cong S^1.$$

*Proof.* The charts identify the boundary as the image of  $t = \pm 1$ . For  $\sigma = 1$  the two levels remain separate; on each, parameters are identified modulo  $\mathbb{Z}$ , giving two circles. For  $\sigma = -1$ , every



boundary class has a representative  $(x, 1)$ , and two such representatives agree exactly when their parameters differ by an even integer. Thus the map  $x \mapsto [x, 1]$  descends to a continuous bijection  $\mathbb{R}/(2\mathbb{Z}) \rightarrow \partial Q_{-1}$ . Its source is compact and its target is Hausdorff, so it is a homeomorphism. Rescaling  $x$  by

$$1/2$$

identifies the source with  $\mathbb{R}/\mathbb{Z} \cong S^1$ . □

Following the top rectangle edge continues along the bottom edge before returning to the start, as traced in Figure 13.17.

## E.6 The Klein Bottle

**Definition E.6.1** (Square model). With exactly the convention of Figure 13.18, set

$$K = [0, 1]^2 / \sim, \quad (0, y) \sim (1, y), \quad (x, 0) \sim (1 - x, 1).$$

The equivalence relation is generated by these edge identifications. All four corners form one class, and  $K$  is compact.

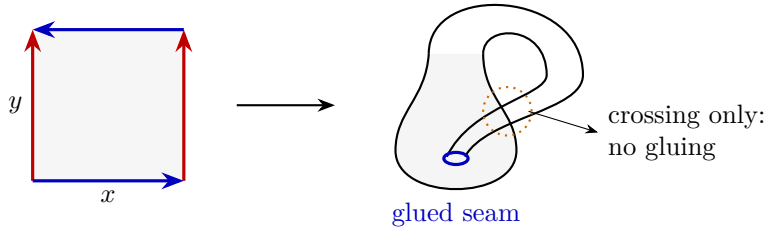


Figure E.6: Klein quotient:  $(0, y) \sim (1, y)$  and  $(x, 0) \sim (1 - x, 1)$ . The solid blue seam represents the joined circles; the dotted callout marks an incidental crossing of distinct sheets. The drawing in  $\mathbb{R}^3$  is not an embedding. See Figure 13.18 for the deformation and Figure E.7 for the local disk at the corner class.

### The plane model and its topology

As for the strip, extending coordinates past the seams makes neighborhoods easier to construct. On  $\mathbb{R}^2$  introduce the isometries

$$G_{m,n}(x, y) = ((-1)^n x + m, y + n), \quad m, n \in \mathbb{Z}.$$

They satisfy

$$G_{m,n} G_{a,b} = G_{m+(-1)^n a, n+b}, \quad G_{m,n}^{-1} = G_{-(-1)^n m, -n}.$$

Their orbits are equivalence classes; denote the quotient by  $\widehat{K}$  and its projection by  $\pi : \mathbb{R}^2 \rightarrow \widehat{K}$ .

**Proposition E.6.2** (Quotient metric and Hausdorffness). The formula

$$d([p], [r]) = \min_{m,n \in \mathbb{Z}} \|p - G_{m,n}(r)\|$$

is a metric inducing the quotient topology on  $\widehat{K}$ . The square inclusion induces a homeomorphism  $K \cong \widehat{K}$ .

*Proof. Apply the common metric criterion.* A distance bound first bounds the vertical integer  $n$  and then the horizontal integer  $m$ , so only finitely many pairs occur. For a nonidentity  $G_{m,n}$ , displacement is at least 1: use the vertical coordinate if  $n \neq 0$ , and the horizontal coordinate otherwise. The family therefore satisfies [Proposition E.2.3](#), proving the metric and topology claims and providing small quotient disks. No metric axioms need to be reconstructed for this model.

*Finite model.* Reduce  $y$  and then  $x$  to place any representative in the square. In the square the only duplications are the prescribed edges and corners:  $G_{1,0}$  identifies vertical edges and  $G_{1,1}$  identifies  $(x, 0)$  with  $(1 - x, 1)$ . Inclusion therefore descends to a continuous bijection from compact  $K$  onto Hausdorff  $\widehat{K}$ , a homeomorphism.  $\square$

### Interior, edge, and corner neighborhoods

**Proposition E.6.3** (Local Euclidean structure). Every point of  $K$  has an ordinary open-disk neighborhood, including the single class represented by all four square corners.

*Proof.* For  $(m, n) \neq (0, 0)$  the displacement of a point under  $G_{m,n}$  is at least 1: use the vertical displacement when  $n \neq 0$  and the horizontal one otherwise. A disk of radius

$$\rho < 1/4$$

therefore has disjoint translates. The triangle estimate in the strip proof shows the disk misses every nontrivial translate. By [Lemma E.2.4](#), its projection is a homeomorphism onto an open neighborhood.

*Interior.* A sufficiently small disk remains inside the square and already supplies the desired neighborhood.

*Edges.* A disk centered on an edge away from the corners has two half-disk pieces, represented at the paired edges. The edge rule fits them into the full disk, not a half-disk.

*Corner class.* Center the plane disk at  $(0, 0)$ . Write its coordinates as  $(u, v)$ . Its four quadrant pieces have square representatives

| quadrant             | representative in the square |
|----------------------|------------------------------|
| $u \geq 0, v \geq 0$ | $(u, v)$                     |
| $u \leq 0, v \geq 0$ | $(1 + u, v)$                 |
| $u \geq 0, v \leq 0$ | $(1 - u, 1 + v)$             |
| $u \leq 0, v \leq 0$ | $(-u, 1 + v)$                |

On the axes these choices agree as classes by the edge relations. Thus four quarter-disks assemble into one disk. There is no boundary or corner singularity.  $\square$

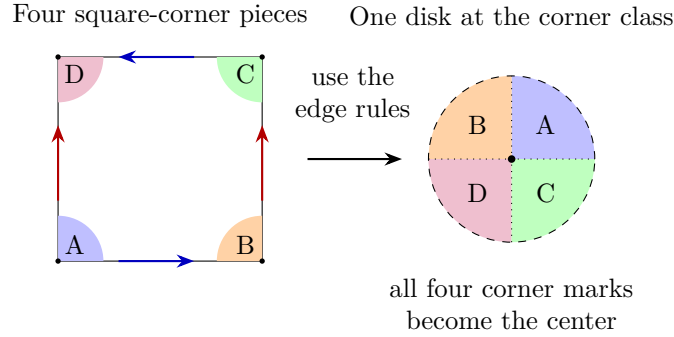


Figure E.7: The Klein-bottle corner chart. Bottom-left A stays in the positive quadrant; bottom-right B translates left. Top-right C and top-left D reflect horizontally and translate down, as in the displayed representative table. The four pieces fill a disk with no missing quadrant, boundary, or singular corner. The outer dashed circle is excluded.

**Smooth structure.** Lifted charts have transitions locally given by one  $G_{m,n}$ . These are affine maps and hence smooth; their maximal extension is a smooth atlas.

**Second countability.** Projection is open since saturations are unions of isometric translates. Images of the rational disks of [Example 10.4.7](#) form a countable quotient basis, by the same lift-and-shrink argument as for the strip. This checks the remaining manifold hypothesis independently of the pictures.

**The three-dimensional picture.** The familiar bottle-like drawing in  $\mathbb{R}^3$  self-intersects and is not an embedding. Its crossing represents distinct sheets, not an additional identification in  $K$ . The abstract compact smooth surface has the ordinary disk neighborhoods just proved.

## E.7 Real Projective Space

To record a direction without choosing a length or a sign, replace each nonzero vector by the line it spans. Ratios of coordinates will then give ordinary local coordinates on the resulting space. The proof architecture selectively follows the supplied smooth-manifolds notes [24], with explicit quotient factorizations and chart inverses in the notation used here.

**Definition E.7.1** (Scaling quotient and homogeneous coordinates). For  $n \geq 0$ , define

$$\mathbb{RP}^n = (\mathbb{R}^{n+1} \setminus \{0\}) / \sim, \quad x \sim y \iff y = \lambda x \text{ for some } \lambda \neq 0,$$

with the quotient topology and canonical projection  $\pi : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n$ . Write  $[x_0 : \cdots : x_n]$  for the class of  $x$ .

The scalars 1,  $\lambda^{-1}$ , and products of nonzero scalars verify reflexivity, symmetry, and transitivity. Each class consists of the nonzero vectors in exactly one one-dimensional linear subspace of  $\mathbb{R}^{n+1}$ . Thus a projective point is a line through the origin. Homogeneous coordinates are not ordinary global coordinates: common nonzero rescaling describes the same projective point.

### The antipodal sphere model

Every line meets  $S^n$  in exactly two antipodal points. To identify the topologies, rather than just the sets, we use the universal property twice.

**Proposition E.7.2** (Scaling and antipodal quotients). There is a natural homeomorphism

$$\mathbb{RP}^n \cong S^n / (x \sim -x).$$

*Proof.* Let

$$\pi_S : S^n \rightarrow S^n / \pm 1$$

be the antipodal projection and let

$$N(x) = x / \|x\|.$$

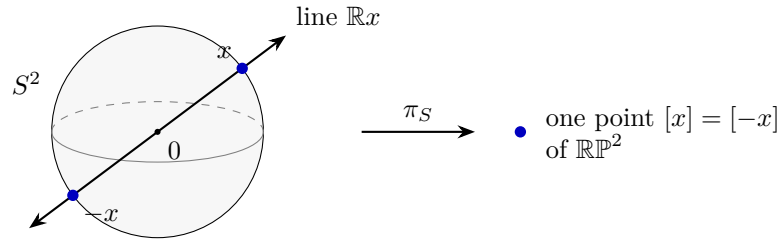
The continuous composite  $\pi_S \circ N$  is unchanged by nonzero scaling, including negative scaling, so it induces a continuous map

$$\tilde{N} : \mathbb{RP}^n \rightarrow S^n / \pm 1.$$

The inclusion  $\iota : S^n \rightarrow \mathbb{R}^{n+1} \setminus \{0\}$  followed by  $\pi$  is constant on antipodal pairs, so it induces a continuous map

$$\tilde{\iota} : S^n / \pm 1 \rightarrow \mathbb{RP}^n.$$

Normalizing a unit vector changes nothing, and normalizing a nonzero vector leaves its line unchanged. Thus both composites are identities.  $\square$



a line through the origin = an antipodal pair = a projective point

Figure E.8: The sphere model of projective space pairs  $x$  only with  $-x$ . The matching filled markers lie on the same line through the origin. The final dot represents one class, not a drawing of the whole quotient.

**Example: the projective line is a circle.** For  $n = 1$ , the map  $(x, y) \mapsto (x^2 - y^2, 2xy)$  from  $S^1$  onto  $S^1$  has exactly antipodal fibers (it doubles the angle). Its continuous factor is a bijection from a compact space to a Hausdorff space, proving  $\mathbb{RP}^1 \cong S^1$ .

### Affine patches and their coordinates

A line with  $x_i \neq 0$  meets the affine plane  $x_i = 1$  in exactly one point. The remaining coordinates of that intersection give the desired chart.

**Definition E.7.3** (Standard affine patches). For  $i = 0, \dots, n$ , define

$$U_i = \{[x_0 : \dots : x_n] : x_i \neq 0\}, \quad \varphi_i : U_i \rightarrow \mathbb{R}^n, \quad \varphi_i([x]) = (x_j/x_i)_{j \neq i},$$

with the retained indices in increasing order.

**Proposition E.7.4** (Affine charts). The  $U_i$  are open, cover  $\mathbb{RP}^n$ , and each  $\varphi_i$  is a homeomorphism onto  $\mathbb{R}^n$ .

*Proof. Domains and representative independence.* The preimage  $\pi^{-1}(U_i) = \{x : x_i \neq 0\}$  is open in  $\mathbb{R}^{n+1} \setminus \{0\}$ , so  $U_i$  is open. Every nonzero vector has a nonzero coordinate, proving the cover assertion. Rescaling multiplies both numerator and denominator by the same nonzero scalar, so the ratios and their domains are independent of the representative.

*Continuity by descent.* On  $\{x_i \neq 0\}$  define the continuous ratio map

$$\gamma_i(x) = (x_j/x_i)_{j \neq i}.$$

The restriction of  $\pi$  over  $U_i$  is quotient: a preimage open in this restricted source is open in the full source, since  $\pi^{-1}(U_i)$  is open; the quotient criterion then applies. The universal property gives  $\tilde{\gamma}_i = \varphi_i$ , proving continuity.

*Continuous inverse.* For  $u = (u_j)_{j \neq i} \in \mathbb{R}^n$ , let  $s_i(u)$  insert 1 in position  $i$ , keeping all other entries in increasing index order. This is a continuous map into  $\mathbb{R}^{n+1} \setminus \{0\}$ , and

$$\varphi_i^{-1} = \pi \circ s_i.$$

Indeed, taking ratios of  $s_i(u)$  returns  $u$ , while inserting 1 into the ratios of  $x$  returns

$$x/x_i,$$

representing the same line as  $x$ . This verifies both inverse identities and continuity of the inverse.  $\square$

**Example E.7.5** (The first overlap). For  $n \geq 1$ , if  $\varphi_0([x]) = (u_1, \dots, u_n)$ , then  $U_0 \cap U_1$  corresponds to  $u_1 \neq 0$ . Dividing instead by  $x_1$  gives

$$\varphi_1 \circ \varphi_0^{-1}(u_1, \dots, u_n) = \left( \frac{1}{u_1}, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1} \right).$$

For  $n = 1$ , only the first entry remains.

**General transitions.** For  $i \neq j$ , the domain of  $\varphi_j \circ \varphi_i^{-1}$  is  $\{u : u_j \neq 0\}$ . Its output has indices  $k \neq j$ , in increasing order:

$$v_i = 1/u_j, \quad v_k = u_k/u_j \quad (k \neq i, j).$$

These rational maps are smooth where their denominators are nonzero; the inverse exchanges  $i$  and  $j$ .

**Proposition E.7.6** (Hausdorffness from the antipodal metric). The space  $\mathbb{RP}^n$  is Hausdorff.

*Proof.* Apply [Proposition E.2.3](#) to  $Y = S^n$  and the two isometries  $\text{id}$  and  $-\text{id}$ . The finiteness hypothesis is automatic, and  $\|x - (-x)\| = 2$ . The metric

$$d([x], [y]) = \min\{\|x - y\|, \|x + y\|\} \quad (x, y \in S^n)$$

induces the antipodal quotient topology, already identified with the scaling quotient. Every metric space is Hausdorff.  $\square$

**Second countability, compactness, and smooth structure.** Pull back the countable Euclidean basis of [Example 10.4.7](#) in each of the  $n + 1$  open charts. Their finite union is countable and is a basis: intersect any open neighborhood with a chart containing its point and choose a basis member there. Thus the space is second countable. It is compact because the continuous map

$$S^n \rightarrow S^n / \pm 1 \cong \mathbb{RP}^n$$

is onto and  $S^n$  is compact. The smooth transitions give an atlas; extending it maximally proves that  $\mathbb{RP}^n$  is a compact smooth  $n$ -manifold without boundary. For  $n = 0$  the space is a point with a chart into  $\mathbb{R}^0$ .

**Example E.7.7** (The disk model of  $\mathbb{RP}^2$ ). Every antipodal pair on  $S^2$  has a representative in the closed upper hemisphere; the representative is unique in its interior, while equatorial points remain paired antipodally. Projection of the hemisphere onto its first two coordinates is a homeomorphism onto  $D^2$ , with inverse  $u \mapsto (u, \sqrt{1 - \|u\|^2})$ . The induced map from the disk quotient to  $\mathbb{RP}^2$  is a continuous bijection. Compactness of the disk quotient and the Hausdorff property just proved make it a homeomorphism. Thus  $\mathbb{RP}^2$  identifies only antipodal boundary pairs, whereas

$$D^2 / \partial D^2 \cong S^2$$

collapses the entire boundary to one point.

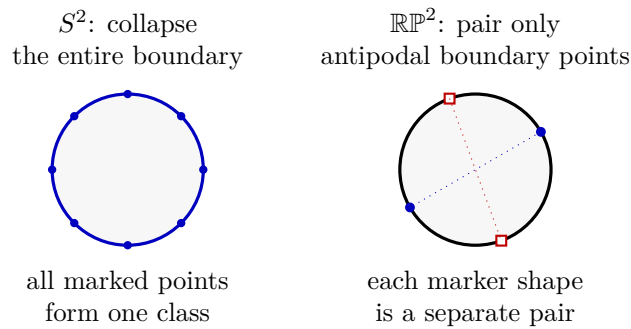


Figure E.9: Two different disk quotients. For the sphere, all of  $\partial D^2$  collapses together. For the projective plane,  $u \sim -u$  only on the boundary; interior points stay distinct. Dotted chords indicate pairings, not curves collapsed in the disk.

## Exercises

**Exercise 1.\*** Show that  $g(t) = (\cos 2\pi t, \sin 2\pi t)$  descends to  $\mathbb{R}/\mathbb{Z}$ . Explain separately why the descended map is well defined, continuous, and a homeomorphism.

**Exercise 2.** Descend  $(x, t) \mapsto (e(x), t)$  from the rectangle to  $Q_1$  and prove  $Q_1 \cong S^1 \times [-1, 1]$ . Identify its two boundary components.

**Exercise 3.\*** For  $\sigma = -1$ , write the seam chart centered at

$$[0, 1/2]$$

using a disk of radius

$$1/8.$$

State its two finite-rectangle domains and its inverse. Repeat at  $[0, 1]$ , using half-disk coordinates.

**Exercise 4.** Trace the Möbius boundary from  $(0, 1)$  through both horizontal edges. Write a parametrization with period 2 and prove that its classes have no smaller positive period.

**Exercise 5.** For  $\mathbb{RP}^2$ , compute  $\varphi_1 \varphi_0^{-1}(u_1, u_2)$ , state its domain, and check its inverse explicitly.

**Exercise 6.\*** Verify that  $(x, y) \mapsto (x^2 - y^2, 2xy)$  on  $S^1$  has exactly antipodal fibers and descends to a homeomorphism  $\mathbb{RP}^1 \rightarrow S^1$ .

# Appendix F

## Submanifolds

### Introduction

The inverse, implicit, and constant-rank theorems describe how nonlinear equations can be straightened locally. Chapter 13 supplies the intrinsic language in which that straightening becomes geometry:

$$\begin{array}{c} \text{inverse/implicit/constant-rank theory} + \text{smooth manifolds} \\ \longrightarrow \text{submanifold geometry.} \end{array}$$

This appendix joins those two parts of the book. It distinguishes a smooth parametrization from a well-behaved subset, identifies the tangent directions allowed by constraints, and prepares the matrix groups of Appendix G. It is a continuation of Chapters 11 and 13, not a prerequisite for Stokes.

Throughout this appendix manifolds are smooth, Hausdorff, second countable, and without boundary; maps are smooth unless stated otherwise. Boundary versions require additional hypotheses and are not implicit in our statements. The primary comparisons are Tu [15, §§9,11] and Lee [5, Chapters 4–6]. We use the local slice viewpoint for embedded subsets and reduce rank statements to Chapter 11, keeping the topological requirement in an embedding separate from the rank requirement.

### F.1 Embedded Submanifolds as Local Straightening

A curved subset need not be a linear subspace. It can nevertheless have exactly  $k$  independent local coordinates if an ambient coordinate change makes it a coordinate plane.

**Definition F.1.1** (Embedded submanifold). A subset  $S \subset M^n$ , with the subspace topology, is an **embedded  $k$ -submanifold** if every  $p \in S$  has a chart  $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$  such that

$$\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^k \times \{0\}).$$

Such a chart is called adapted to  $S$ . The induced coordinates on  $S$  are the first  $k$  components of  $\varphi|_{U \cap S}$ .



**Proposition F.1.2** (The induced smooth structure). These induced coordinates make  $S$  a smooth  $k$ -manifold, and the inclusion  $i : S \rightarrow M$  is smooth with injective differential.

*Proof.* Identify the slice with  $\mathbb{R}^k$ . Its intersection with the open set  $\varphi(U)$  is open in that slice, so the induced coordinates are charts for the subspace topology. An overlap map is obtained by inserting zeros, applying an ambient smooth transition, and retaining the first  $k$  components. It and its inverse are smooth. Hausdorffness and second countability pass to subspaces: separating neighborhoods restrict to the subspace (Definition 10.5.5), and a countable basis restricts as in Proposition 10.4.11. In these charts inclusion is  $x \mapsto (x, 0)$ ; its derivative  $v \mapsto (v, 0)$  is injective. The induced maximal atlas is independent of the chosen adapted charts, because the same overlap argument compares any two of them.  $\square$

We henceforth identify  $T_p S$  with the linear subspace  $di_p(T_p S)$  of  $T_p M$ . The identification is through this injective map, not an equality of the original chart-equivalence-class constructions.

**Example F.1.3** (Planes, open subsets, and graphs). A coordinate plane is already straight. Every open subset of  $M$  is an embedded submanifold of dimension  $\dim M$ . For smooth  $f : U \subseteq \mathbb{R}^k \rightarrow \mathbb{R}^q$  with  $U$  open, the graph  $\Gamma_f = \{(x, f(x)) : x \in U\}$  is straightened by

$$\varphi : U \times \mathbb{R}^q \rightarrow U \times \mathbb{R}^q, \quad \varphi(x, y) = (x, y - f(x)).$$

Its inverse is  $(x, z) \mapsto (x, z + f(x))$ , so it is a diffeomorphism. Graph coordinates give

$$T_{(x, f(x))} \Gamma_f = \{(v, Df(x)v) : v \in \mathbb{R}^k\}.$$

Here  $D$  is the Euclidean derivative; the displayed vector is the image under the intrinsic differential of the graph parametrization. The implicit function theorem, Theorem 11.5.8, produces just such graphs when an appropriate block of derivatives is invertible.

$$\begin{array}{ccc} \Gamma_f \cap (U \times \mathbb{R}^q) & \hookrightarrow & U \times \mathbb{R}^q \\ \varphi| \downarrow & & \downarrow \varphi \\ U \times \{0\} & \hookrightarrow & U \times \mathbb{R}^q \end{array}$$

The diagram records what straightening preserves: points of the subset become exactly the zero-normal-coordinate points, while nearby ambient points still have all  $k + q$  coordinates.

**Proposition F.1.4** (Testing a map with values in a submanifold). For an embedded  $S \subset M$ , a map  $h : P \rightarrow S$  is smooth if and only if  $i \circ h : P \rightarrow M$  is smooth.

*Proof.* One implication is composition. For the other, continuity into the subspace  $S$  follows from continuity into  $M$ . Near each image point use an adapted chart. The coordinate expression of  $h$  consists of the first  $k$  components of that of  $i \circ h$ , and is therefore smooth.  $\square$

This elementary test will verify the group operations in Appendix G without reconstructing their smoothness in every submanifold chart.

## F.2 Rank, Immersions, Submersions, and Embeddings

For  $f : M^m \rightarrow N^n$ , the intrinsic differential  $df_p : T_p M \rightarrow T_{f(p)} N$  has the chart representative  $D(\psi \circ f \circ \varphi^{-1})(\varphi(p))$ . The coordinate conversions are isomorphisms, so rank is intrinsic.

**Definition F.2.1.** The map  $f$  is an **immersion at  $p$**  if  $df_p$  is injective, and a **submersion at  $p$**  if it is surjective. It is an immersion or submersion if the condition holds at every point. A **smooth embedding** is an immersion that is a homeomorphism from  $M$  onto  $f(M)$  with its subspace topology in  $N$ .

An immersion preserves independent directions locally. An embedding also requires the global subset topology to recover the topology of the source.

**Theorem F.2.2** (Manifold constant-rank theorem). If  $f : M^m \rightarrow N^n$  has constant rank  $r$  on a neighborhood of  $p$ , there are charts centered at  $p$  and  $f(p)$  in which its expression is

$$(x^1, \dots, x^m) \mapsto (x^1, \dots, x^r, 0, \dots, 0).$$

*Proof.* The proof has two steps: express the intrinsic rank in coordinates, then absorb Euclidean coordinate changes into the charts. Choose charts  $\varphi, \psi$  and shrink the source so that its image is in the target chart. The Euclidean map  $h = \psi \circ f \circ \varphi^{-1}$  has rank  $r$  throughout this neighborhood, since its derivative represents  $df$ . Apply [Theorem 11.6.3](#) to obtain local Euclidean diffeomorphisms straightening  $h$ . Composing them with  $\varphi$  and  $\psi$  gives the required manifold charts. No new analytic assertion is needed.  $\square$

**Corollary F.2.3** (The three maximal-rank models). Near a point where  $df_p$  is invertible,  $f$  is a local diffeomorphism. Near a submersion point it is a coordinate projection, and near an immersion point it is the coordinate inclusion  $x \mapsto (x, 0)$ .

*Proof.* A nonzero maximal minor remains nonzero in a neighborhood by continuity. Rank there is therefore maximal and constant. Apply the theorem. In the equal-dimensional case the coordinate model is the identity; the other two cases are projection and inclusion.  $\square$

**Proposition F.2.4** (Embeddings and embedded images). A smooth embedding has an embedded image, and is a diffeomorphism onto that image. Conversely the inclusion of an embedded submanifold is a smooth embedding. In particular an injective immersion from a compact manifold into a Hausdorff manifold is an embedding.

*Proof.* The issue is to exclude other pieces of the image from a local slice. The immersion normal form makes  $f(U)$  a slice near  $f(p)$  for a source neighborhood  $U$ . Since  $f$  is a homeomorphism onto its image,  $f(U)$  is relatively open in  $f(M)$ . Intersect the target chart with an ambient open set whose intersection with  $f(M)$  lies in  $f(U)$ , and shrink again to a coordinate box. The whole image in this box is now exactly its coordinate slice. The inverse in slice coordinates is a coordinate projection and is smooth. The converse follows from the induced-chart proposition. Finally a continuous bijection from compact  $M$  to its image in a Hausdorff space is a homeomorphism by Chapter 10's compact-to-Hausdorff criterion; combine this with the assumed immersion.  $\square$

**Example F.2.5** (An immersed curve with a crossing). The map  $\gamma : S^1 \rightarrow \mathbb{R}^2$  given in angle coordinates by  $\gamma(t) = (\sin t, \sin 2t)$  is well defined and smooth. Its derivative is  $(\cos t, 2 \cos 2t)$ , never zero: when  $\cos t = 0$ ,  $\cos 2t = -1$ . Thus it is an immersion. The distinct circle points  $t = 0, \pi$  both map to the origin, so it is not an embedding. The two tangent lines there have slopes 2 and  $-2$ . The image itself is not an embedded curve: a small punctured neighborhood of the crossing has four arms, whereas a punctured coordinate interval has two.

**Example F.2.6** (Injectivity alone still does not suffice). Restrict  $\gamma(t) = (\sin t, \sin 2t)$  to  $(-\pi, \pi)$ . If two inputs have the same sine, either they agree or their cosines are opposite. Equality of  $\sin 2t = 2 \sin t \cos t$  then forces either the inputs to agree or  $\sin t = 0$ ; in this interval the latter gives  $t = 0$ . Thus the restriction is an injective immersion. But  $t_j \rightarrow \pi$  from below gives  $\gamma(t_j) \rightarrow \gamma(0)$  without  $t_j \rightarrow 0$ . Its inverse on the image is not continuous, so it is not an embedding.

**Remark F.2.7** (Immersion, injective immersion, embedding).

|                     |                                               |
|---------------------|-----------------------------------------------|
| immersion           | $df_p$ is injective at every point            |
| injective immersion | also distinct points have distinct images     |
| embedding           | also the inverse onto the image is continuous |

An immersion is a local differential condition. Injectivity removes crossings, but an embedding also requires the source topology to agree with the subspace topology on the image. The crossing and accumulation examples above separate these requirements. For compact sources and Hausdorff targets, [Proposition F.2.4](#) supplies the last condition.

### F.3 Regular Values and Tangent Kernels

Equations select subsets; their differentials select the allowed directions. The rank hypothesis ensures that these two selections fit together smoothly.

**Definition F.3.1.** A **regular point** of  $f : M \rightarrow N$  is a point where  $df_p$  is surjective. A **regular value**  $c \in N$  is a value for which every point of  $f^{-1}(c)$  is regular. An empty inverse image makes this condition vacuously true. Points that are not regular are called critical points.

**Theorem F.3.2** (Regular level sets). Let  $c$  be a regular value of  $f : M^n \rightarrow N^m$ . If  $f^{-1}(c)$  is nonempty, it is an embedded  $(n - m)$ -submanifold  $S$ , and, under the inclusion,

$$T_p S = \ker df_p \quad (p \in S).$$

An empty level set is an embedded empty submanifold; no assertion of a negative dimension is intended when  $n < m$ .

*Proof.* Near  $p \in S$ , surjectivity persists by a nonzero  $m$ -rowed minor. The constant-rank theorem makes  $f$  the projection onto its first  $m$  coordinates, with  $c$  represented by zero. Its fiber is the slice  $x^1 = \cdots = x^m = 0$ ; permuting coordinates puts this in the form of [Definition F.1.1](#). Differentiating  $f \circ i = c$  gives  $df_p \circ di_p = 0$ , hence  $T_p S \subseteq \ker df_p$ . Both spaces have dimension  $n - m$ , the latter by rank-nullity, so they are equal.  $\square$

For Euclidean  $f$ , the identified kernel is  $\ker Df(p)$ , exactly the formula of [Corollary 11.6.6](#). The intrinsic theorem transports that computation across charts rather than changing it.

**Example F.3.3** (Spheres and cylinders). For  $h : \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $h(x) = \|x\|^2$ , we have  $Dh(p)v = 2p \cdot v$ . At  $h(p) = 1$  this functional is nonzero. Consequently

$$S^{n-1} = h^{-1}(1), \quad T_p S^{n-1} = \{v : p \cdot v = 0\}.$$

Similarly  $h(x, y, z) = x^2 + y^2$  gives the cylinder  $h^{-1}(1)$  and tangent condition  $xv_1 + yv_2 = 0$ , leaving the vertical direction free. A rank failure is a failure of the criterion, not a proof of singularity: the zero set of  $x \mapsto x^2$  is still a point.

**Example F.3.4** (An orthogonality constraint). Let  $M_n(\mathbb{R})$  be the  $n^2$ -dimensional matrix space and let  $\text{Sym}_n$  be its subspace of symmetric matrices, of dimension

$$n(n+1)/2.$$

For  $h : M_n(\mathbb{R}) \rightarrow \text{Sym}_n$ ,  $h(A) = A^T A$ , expanding  $(A + tH)^T(A + tH)$  gives

$$Dh(A)H = A^T H + H^T A.$$

If  $A^T A = I$ , every symmetric  $K$  is the image of

$$H = AK/2.$$

Thus  $I$  is a regular value with this specified target, and

$$O(n) = h^{-1}(I), \quad \dim O(n) = \frac{n(n-1)}{2}, \quad T_A O(n) = \{H : A^T H + H^T A = 0\}.$$

Taking all matrices as the target would incorrectly demand surjectivity onto  $n^2$  coordinates. Symmetry removes the redundant equations.

**Remark F.3.5** (Three local pictures of the same submanifold). The preceding results fit together as

$$\text{coordinate slice} \longleftrightarrow \text{local graph} \longleftrightarrow \text{regular level set}.$$

For an embedded  $k$ -submanifold  $S \subseteq M^n$ , choose an adapted chart  $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ . Projection onto the last  $n - k$  coordinates gives a submersion  $h = \text{pr}_{\text{last}} \circ \varphi$ , with

$$S \cap U = h^{-1}(0), \quad T_p S = \ker dh_p \quad (p \in S \cap U).$$

Conversely, [Theorem F.3.2](#) supplies an adapted slice for a submersion's level set. The implicit function theorem, after selecting an invertible derivative block in coordinates, expresses that level set as a graph. A graph is straightened by  $(x, y) \mapsto (x, y - f(x))$ . These are local descriptions in suitable coordinates, not a claim that an entire submanifold is a single graph or a single global level set.

## F.4 Constrained Extrema and a Global Perspective

**Proposition F.4.1** (Intrinsic multipliers). If  $f : M \rightarrow \mathbb{R}$  has a local extremum on an embedded  $S$  at  $p$ , then  $df_p$  vanishes on  $T_p S$ . If near  $p$  the constraint is  $S = h^{-1}(c)$  for a submersion  $h : M \rightarrow \mathbb{R}^m$ , then there are scalars  $\lambda_1, \dots, \lambda_m$  such that

$$df_p = \sum_{j=1}^m \lambda_j dh_p^j.$$

These multipliers are unique for this independent constraint list.

*Proof.* In coordinates on  $S$ , the ordinary necessary derivative condition for a local extremum gives  $d(f|_S)_p = 0$ . The chain rule says this is  $df_p$  restricted to the image of  $di_p$ . For the multiplier statement put  $L = dh_p$ . The kernel formula shows that  $df_p$  vanishes on  $\ker L$ . Since  $L$  is onto, define a functional  $\ell : \mathbb{R}^m \rightarrow \mathbb{R}$  by  $\ell(Lv) = df_p(v)$ . Equal values of  $Lv$  differ by a kernel vector, so this definition is independent of the choice of  $v$  and is linear. Thus  $df_p = \ell L$ . Expanding  $\ell$  in the standard dual basis gives the displayed formula. Surjectivity of  $L$  proves uniqueness. This is quotient factorization in its simplest linear form.  $\square$

For  $f(x) = a \cdot x$  on the unit sphere, the equation is  $a = 2\lambda p$ . For  $a \neq 0$  it gives

$$p = \pm a/\|a\|,$$

the maximum and minimum found by Cauchy–Schwarz. The first-order equation supplies candidates; it does not by itself decide which candidate is a maximum or a minimum.

**Theorem F.4.2** (Whitney embedding theorem). Every smooth  $n$ -manifold without boundary, for  $n \geq 1$ , admits a smooth embedding into  $\mathbb{R}^{2n}$ .

See Lee [5, Theorem 6.19]; Tu [15, §11.3] discusses the embedding perspective. The theorem connects abstract and embedded manifolds globally. Our local slice and rank proofs do not prove it. Sard’s theorem, transversality, tubular neighborhoods, Frobenius theory, and Morse theory belong to subsequent study, signposted in Section 13.17.

## Exercises

**Exercise 1.\*** Straighten the graph of  $f(x, y) = x^2 - y^2$  explicitly and compute its tangent space at  $(a, b, a^2 - b^2)$ . Compare the graph and regular-level calculations.

**Exercise 2.\*** For  $h(x, y) = xy$ , determine all regular values and describe their level sets. Explain precisely why the zero fiber fails to be an embedded curve at the origin.

**Exercise 3.\*** Check the ranks of the projection  $S^2 \rightarrow \mathbb{R}$ ,  $(x, y, z) \mapsto z$ . Compute the tangent line of a nonempty regular fiber as an intersection of kernels.

**Exercise 4.\*** Verify the injectivity claim for the restricted crossing curve above and exhibit the sequence witnessing failure of continuity of its inverse. Explain why compactness prevents this phenomenon for an injective immersion.

**Exercise 5.\*** For  $1 \leq k \leq n$ , show that  $h(A) = A^T A$  from real  $n$  by  $k$  matrices to symmetric  $k$  by  $k$  matrices has  $I_k$  as a regular value. Find the dimension and tangent space of the space of orthonormal  $k$ -frames.

**Exercise 6.** Use intrinsic multipliers to find candidates for extrema of  $f(x, y, z) = xyz$  on the sphere. Explain why a necessary condition alone does not classify all these candidates.

**Exercise 7.\*** Prove that the product of two embedded submanifolds is embedded by using adapted product charts. Identify its tangent space and apply the result to the formal torus  $S^1 \times S^1$ .

# Appendix G

## Lie Groups and Lie Algebras

### Introduction

A translation, a rotation, or an invertible change of coordinates can be composed with another transformation of the same kind. These transformations also vary smoothly. A Lie group brings these two observations together:

a Lie group is simultaneously a smooth manifold and a group.

The compatibility of these structures transports a tangent vector at the identity to every point. Commutators of the resulting vector fields then encode infinitesimal noncommutativity in a finite-dimensional Lie algebra. This is a bridge from calculus to symmetry, not a development of representation theory or the classification of Lie groups.

We use Appendix F and only the vector-field theory constructed below. Compare Tu [15, §§14–16] with Lee [5, Chapters 7–8,20]. The matrix examples use regular level sets; the exponential is proved from its convergent series, avoiding the flow-existence machinery needed for a general Lie-group exponential. All Lie groups here are finite-dimensional real smooth manifolds without boundary.

### G.1 Smooth Groups and Concrete Examples

The additive space  $\mathbb{R}^n$  has smooth operations  $(x, y) \mapsto x + y$  and  $x \mapsto -x$ . The circle has complex multiplication  $(x + iy)(u + iv) = (xu - yv) + i(xv + yu)$  and inverse  $x - iy$  on  $x^2 + y^2 = 1$ . These are restrictions of polynomial maps, and their images lie in the circle. Finally invertible matrices form the open set

$$GL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\},$$

because the determinant is a continuous polynomial.

**Definition G.1.1** (Lie group). A group is a set with an associative multiplication, an identity  $e$ , and an inverse for each element. A **Lie group** is a smooth manifold  $G$  with a group structure for which both maps

$$m : G \times G \rightarrow G, \quad m(g, h) = gh, \quad \iota : G \rightarrow G, \quad \iota(g) = g^{-1}$$

are smooth. A **Lie-group homomorphism** is a smooth map preserving multiplication (and therefore identity and inverses).

Thus  $(\mathbb{R}^n, +)$  and  $S^1$  are Lie groups. For  $GL_n(\mathbb{R})$  multiplication has entries

$$\sum_k a_{ik} b_{kj},$$

and inversion has entries given by cofactors divided by  $\det A$ . They are smooth on the open set in question. Products of Lie groups use componentwise multiplication; in particular  $T^2 = S^1 \times S^1$  is a Lie group with its main-text product structure.

**Proposition G.1.2** (Special linear and orthogonal groups). The sets  $SL_n(\mathbb{R}) = (\det)^{-1}(\{1\})$ ,  $O(n) = \{A : A^T A = I\}$ , and  $SO(n) = \{A \in O(n) : \det A = 1\}$  are embedded submanifolds of matrix space and Lie groups under matrix multiplication. Their dimensions are  $n^2 - 1$ ,

$$n(n-1)/2, \quad n(n-1)/2,$$

respectively, for  $n \geq 1$ .

*Proof.* We first verify the rank hypotheses and then restrict the operations. Multilinearity of the determinant gives

$$\det(I + tH) = 1 + t \sum_i h_{ii} + O(t^2), \quad D(\det)(I)H = \operatorname{tr} H,$$

where

$$\operatorname{tr} H = \sum_i h_{ii}.$$

Indeed the linear terms choose one column from  $tH$  and all other columns from  $I$ ; only that column's diagonal entry survives. Factoring  $A + tH = A(I + tA^{-1}H)$  gives, for invertible  $A$ ,

$$D(\det)(A)H = \det A \operatorname{tr}(A^{-1}H).$$

At determinant 1, choosing

$$H = A/n$$

makes this derivative 1. The regular-level theorem therefore gives the stated dimension of  $SL_n$ . The orthogonal computation, including its symmetric-matrix target and surjectivity, is already proved in [Example F.3.4](#). For  $A \in O(n)$ ,  $(\det A)^2 = 1$ , so  $SO(n)$  is the open subset  $\{\det A > 0\}$  of  $O(n)$  and has the same local dimension.

The determinant product law and  $(AB)^T(AB) = B^T A^T A B$  show that these sets are groups; inverses remain in the same sets. The smooth operations of  $GL_n$  restrict smoothly to them by the submanifold map test in [Appendix F](#). This also justifies smoothness with the induced manifold structures, rather than merely with the ambient coordinates.  $\square$

We have proved these examples individually. No general closed-subgroup theorem is being assumed.

**Proposition G.1.3** (Group operations to first order). Let  $m : G \times G \rightarrow G$  be multiplication and  $\iota : G \rightarrow G$  inversion. Under  $T_{(e,e)}(G \times G) \cong T_e G \oplus T_e G$ ,

$$dm_{(e,e)}(v, w) = v + w, \quad d\iota_e(v) = -v.$$

*Proof.* Restricting  $m$  to either axis gives the identity:  $m(g, e) = g = m(e, g)$ . Thus its differential sends  $(v, 0)$  to  $v$  and  $(0, w)$  to  $w$ ; linearity gives their sum. Differentiate  $m(g, \iota(g)) = e$  at  $e$  to obtain  $v + d\iota_e(v) = 0$ .  $\square$

To first order, every Lie group looks like its tangent vector space with ordinary addition. This linearization cannot detect noncommutativity. Transporting vectors by left translation and then taking commutators of the resulting fields supplies that extra infinitesimal information:

$$\begin{aligned} \text{multiplication} &\longrightarrow \text{linearization} \longrightarrow \text{left-invariant fields} \\ &\longrightarrow \text{commutator} \longrightarrow \text{Lie algebra.} \end{aligned}$$

## G.2 Translation Turns One Tangent Space into Vector Fields

For  $g \in G$ , left translation is  $L_g(h) = gh$ . It is smooth, and  $L_{g^{-1}}$  is its smooth inverse. Hence

$$T_e G \xrightarrow{d(L_g)_e} T_g G$$

is a linear isomorphism. One tangent space supplies directions everywhere.

A smooth vector field  $X$  assigns  $X_p \in T_p M$  with smooth coordinate components. It acts on smooth scalar functions by  $Xf(p) = df_p(X_p)$ . In a chart this is

$$Xf = \sum_i X^i \partial_i f,$$

so  $Xf$  is smooth and obeys the product rule.

**Definition G.2.1.** A vector field  $X$  on  $G$  is **left invariant** if  $d(L_g)_h X_h = X_{gh}$  for all  $g, h \in G$ . Set  $\mathfrak{g} = T_e G$ .

**Proposition G.2.2** (Fields determined at the identity). Evaluation at  $e$  is a linear bijection between smooth left-invariant vector fields and  $\mathfrak{g}$ . Its inverse sends  $v$  to the field  $X_g^v = d(L_g)_e v$ .

*Proof.* Left invariance forces the formula, so it proves uniqueness. To check existence including smoothness, choose a smooth curve  $\gamma$  with  $\gamma(0) = e$ ,  $\gamma'(0) = v$ , using a coordinate line in a chart. For a local target coordinate  $x^j$  near  $g_0$ , the component of  $X_g^v$  is

$$\left. \frac{\partial}{\partial t} \right|_{t=0} x^j(g\gamma(t)).$$

For  $g$  near  $g_0$  and sufficiently small  $t$  the expression is defined and smooth jointly in  $(g, t)$ , because multiplication is smooth. Its  $t$ -derivative is smooth in  $g$ . Thus the field is smooth. The chain rule and  $L_g \circ L_h = L_{gh}$  give

$$d(L_g)_h X_h^v = d(L_g)_h d(L_h)_e v = d(L_{gh})_e v = X_{gh}^v.$$

Finally  $X_e^v = v$ , and linearity follows from that of each differential.  $\square$

For a matrix group the transported field is  $X_g^A = gA$ . For the additive group it is the constant field  $X_x^v = v$ . These formulas are concrete realizations of the same transport mechanism.

## G.3 The Bracket and the Lie Algebra

Composition of vector fields acting on functions generally has second derivatives. Subtracting the reverse composition cancels them, leaving a first-order operation and hence another vector field.



**Proposition G.3.1** (Vector-field commutator). For smooth vector fields  $X, Y$ , the rule

$$[X, Y]f = X(Yf) - Y(Xf)$$

defines a smooth vector field, with chart components

$$[X, Y]^j = \sum_i (X^i \partial_i Y^j - Y^i \partial_i X^j).$$

The bracket is bilinear, antisymmetric, and satisfies the Jacobi identity  $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ .

*Proof.* Expanding the first composition gives

$$X(Yf) = \sum_{i,j} X^i (\partial_i Y^j) \partial_j f + \sum_{i,j} X^i Y^j \partial_i \partial_j f.$$

Interchange  $X, Y$  in the second composition. Reindex  $i, j$  and use equality of mixed partials to cancel the second-derivative terms. The remaining coefficient is the stated smooth vector field. This construction is intrinsic because its action on every local scalar function is intrinsic; coordinate functions determine each tangent vector. Bilinearity and antisymmetry follow from the subtraction. For Jacobi, the three expansions as operators are

$$\begin{aligned} [X, [Y, Z]] &= XYZ - XZY - YZX + ZYX, \\ [Y, [Z, X]] &= YZX - YXZ - ZXY + XZY, \\ [Z, [X, Y]] &= ZXY - ZYX - XYZ + YXZ. \end{aligned}$$

Every term cancels with its opposite; evaluation on local coordinates then gives the zero vector field.  $\square$

If  $h : M \rightarrow N$  is a diffeomorphism, define  $(h_*X)_{h(p)} = dh_p X_p$ . The chain rule gives  $((h_*X)f) \circ h = X(f \circ h)$ . Applying this identity twice and subtracting proves

$$h_*[X, Y] = [h_*X, h_*Y].$$

In particular the bracket of left-invariant fields is left invariant: take  $h = L_g$  and use the defining invariances.

**Definition G.3.2** (Lie algebra of a Lie group). A real **Lie algebra** is a real vector space with a bilinear antisymmetric bracket satisfying the Jacobi identity. On  $\mathfrak{g} = T_e G$  define

$$[v, w]_{\mathfrak{g}} = [X^v, X^w]_e.$$

By [Proposition G.2.2](#), the left-invariant extension of this bracket is  $[X^v, X^w]$ . Thus all three identities pass from vector fields to  $\mathfrak{g}$ , proving that this definition gives a Lie algebra. Its underlying vector space has dimension  $\dim G$ .

**Proposition G.3.3** (Matrix Lie algebras). With tangent spaces identified in matrix space,

$$\begin{aligned}\mathfrak{gl}_n &= M_n(\mathbb{R}), \\ \mathfrak{sl}_n &= \{A : \operatorname{tr} A = 0\}, \\ \mathfrak{so}(n) &= \{A : A^T = -A\}.\end{aligned}$$

Both  $O(n)$  and  $SO(n)$  have the last tangent space at  $I$ . For these matrix groups the bracket is  $[A, B] = AB - BA$ .

*Proof.* The first group is open. For  $SL_n$ , use the kernel of  $D(\det)(I)H = \operatorname{tr} H$ . For  $O(n)$  use the kernel of  $H \mapsto H + H^T$ ; the open subset  $SO(n)$  has the same tangent space. For the bracket, the left-invariant fields are  $X^A(g) = gA$  and  $X^B(g) = gB$ . The derivatives of these ambient linear expressions are  $H \mapsto HA$  and  $H \mapsto HB$ . The coordinate bracket formula gives

$$[X^A, X^B](g) = (gA)B - (gB)A = g(AB - BA).$$

The calculation is also valid after restriction to an embedded matrix group: on every ambient matrix coordinate the restricted fields act by the same derivative, and these coordinates determine the included tangent vector. Evaluation at  $I$  proves the claim. Closure can also be checked directly:  $\operatorname{tr}(AB) = \operatorname{tr}(BA)$  by interchanging the two summation indices, and for skew  $A, B$ ,  $(AB - BA)^T = BA - AB$ .  $\square$

**Example G.3.4** (A nonzero bracket). In  $\mathfrak{sl}_2$ , take

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then  $HE = E$  and  $EH = -E$ , so the bracket from [Proposition G.3.3](#) is  $[H, E] = 2E \neq 0$ . Equivalently the left-invariant fields satisfy  $[X^H, X^E](g) = 2gE$ : their commutator, as defined in [Proposition G.3.1](#), need not vanish.

**Remark G.3.5** (The principal examples together). The tangent descriptions above give the following comparison. The circle and torus use their usual angle coordinates at the identity.

| $G$                 | $T_e G$                           | bracket   |
|---------------------|-----------------------------------|-----------|
| $(\mathbb{R}^n, +)$ | $\mathbb{R}^n$                    | 0         |
| $S^1$               | $\mathbb{R}$                      | 0         |
| $T^n$               | $\mathbb{R}^n$                    | 0         |
| $GL_n(\mathbb{R})$  | $M_n(\mathbb{R})$                 | $AB - BA$ |
| $SL_n(\mathbb{R})$  | $\{A : \operatorname{tr} A = 0\}$ | $AB - BA$ |
| $O(n)$              | $\{A : A^T = -A\}$                | $AB - BA$ |
| $SO(n)$             | $\{A : A^T = -A\}$                | $AB - BA$ |

For the additive group the invariant fields are constant, so their bracket vanishes. Angle coordinates give the same local calculation for circles and tori. Equal Lie algebras need not mean equal global groups.

## G.4 Matrix Exponentials and One-Parameter Subgroups

A tangent matrix describes initial velocity. Its exponential integrates that velocity into a family of transformations. We can prove this directly for matrices with the convergence tools already available.

**Definition G.4.1.** For  $A \in M_n(\mathbb{R})$  set

$$\exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!}, \quad A^0 = I.$$

The notation  $e^A$  is a standard alternative for this series-defined matrix exponential; it is not an application of real-power laws to matrices. A **one-parameter subgroup** of a Lie group  $G$  is a smooth map  $\gamma : \mathbb{R} \rightarrow G$  with  $\gamma(t+s) = \gamma(t)\gamma(s)$ .

**Proposition G.4.2** (The exponential laws needed here). The series converges, and  $\gamma(t) = \exp(tA)$  is smooth with

$$\begin{aligned} \gamma(0) &= I, & \gamma'(t) &= A \exp(tA) = \exp(tA)A, \\ \exp(tA) \exp(sA) &= \exp((t+s)A), & \exp(tA)^{-1} &= \exp(-tA). \end{aligned}$$

Every one-parameter subgroup of  $GL_n(\mathbb{R})$  has this form, with  $A = \gamma'(0)$ .

*Proof.* Use an operator norm, so  $\|A^k\| \leq \|A\|^k$ . The scalar exponential series gives absolute convergence and uniform convergence for bounded  $t$ . For each derivative order the differentiated series is bounded on compact  $t$ -intervals by a constant times an exponential series. The termwise differentiation theorem therefore gives smoothness and the derivative formula. For the product, absolute convergence permits the Cauchy product:

$$\begin{aligned} \exp(tA) \exp(sA) &= \sum_{m=0}^{\infty} \left( \sum_{k=0}^m \frac{t^k s^{m-k}}{k!(m-k)!} \right) A^m \\ &= \sum_{m=0}^{\infty} \frac{(t+s)^m A^m}{m!}. \end{aligned}$$

Taking  $s = -t$  proves invertibility. Conversely differentiate  $\gamma(t+s) = \gamma(t)\gamma(s)$  in  $s$  at zero to get  $\gamma'(t) = \gamma(t)A$ . The product rule gives  $(\gamma(t) \exp(-tA))' = 0$ , hence  $\gamma(t) \exp(-tA) = I$  by its value at zero. This proves the converse without an ODE existence theorem.  $\square$

The same exponential law is not asserted for  $\exp(A) \exp(B)$  with arbitrary noncommuting  $A, B$ .

**Proposition G.4.3** (Exponentiating the displayed Lie algebras). If  $\text{tr } A = 0$ , then  $\exp(tA) \in SL_n(\mathbb{R})$ . If  $A^T = -A$ , then  $\exp(tA) \in SO(n)$ . Thus the exponential supplies one-parameter subgroups with every initial tangent vector for the matrix groups above.

*Proof.* Set  $u(t) = \det(\exp(tA))$ . The determinant derivative and the preceding proposition give  $u'(t) = u(t) \text{tr } A$ ,  $u(0) = 1$ . Differentiating  $e^{-t \text{tr } A} u(t)$  proves  $u(t) = e^{t \text{tr } A}$ . For skew  $A$ , transposing the series gives  $(\exp(tA))^T = \exp(-tA)$ , so the matrix is orthogonal. Its determinant is 1, either by the trace formula or by continuity from  $t = 0$  into  $\{1, -1\}$ .  $\square$

**Example G.4.4** (The whole computation for  $SO(2)$ ). Every skew 2 by 2 matrix is  $A = \theta J$ , where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I.$$

Separate the even and odd powers in the absolutely convergent series:

$$\exp(\theta J) = \cos \theta I + \sin \theta J = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The one-parameter subgroup  $t \mapsto \exp(t\theta J)$  rotates at constant angular speed  $\theta$ . Acting on  $(1, 0)$  traces the unit circle; the initial velocity is  $\theta J(1, 0) = (0, \theta)$ , its tangent vector at that point. More generally  $J(\cos t, \sin t) = (-\sin t, \cos t)$  is the counterclockwise unit tangent. Thus the matrix generator, circle tangent, and exponential describe the same motion. Here  $\mathfrak{so}(2) = \mathbb{R}J$  has zero bracket, and exponentiation repeats after  $2\pi J$ . Infinitesimal data do not by themselves distinguish the additive line from the circle's periodic global geometry. For general Lie groups an exponential is constructed by integral curves of left-invariant fields; that existence theory is beyond this appendix.

## Exercises

**Exercise 1.\*** Check the group and smoothness axioms for  $S^1$  using real coordinates. Identify  $T_{(1,0)}S^1$  and its left-invariant fields.

**Exercise 2.\*** Compute  $T_A SL_n$  and  $T_A O(n)$  at a general point and show they are obtained from their identity tangent spaces by  $H \mapsto AH$ .

**Exercise 3.\*** For  $X = x\partial_y$  and  $Y = y\partial_x$  on  $\mathbb{R}^2$ , compute  $[X, Y]$ . Verify its action directly on  $x$ ,  $y$ , and  $xy$ .

**Exercise 4.\*** In  $\mathfrak{sl}_2$ , put  $H = \text{diag}(1, -1)$ ,  $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , and  $F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ . Compute all pairwise brackets and verify one Jacobi identity directly.

**Exercise 5.\*** Compute  $\exp(tA)$  for a diagonal matrix and for  $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Check the subgroup law and the initial velocity in each case.

**Exercise 6.\*** Show that a smooth homomorphism  $\gamma : \mathbb{R} \rightarrow S^1$  has the form  $\gamma(t) = (\cos at, \sin at)$  by embedding the circle as  $SO(2)$  and using the one-parameter subgroup proposition.

**Exercise 7.** Show that  $O(n)$  and  $SO(n)$  have identical Lie algebras although they are different groups for  $n \geq 1$ . Identify explicitly which global information their tangent spaces at the identity fail to record.

# Appendix H

## de Rham Cohomology

### Introduction

Chapter 13 proves  $d^2 = 0$ . Thus a form that is a derivative is automatically closed. The converse question is more subtle:

$$d\omega = 0 \quad \stackrel{?}{\implies} \quad \omega = d\eta.$$

Locally the answer is yes in positive degree. Globally a hole can prevent local primitives from fitting together. De Rham cohomology organizes that failure of global exactness, and integration provides a way to detect it. Here closed means  $d\omega = 0$  and exact means  $\omega = d\eta$ ; [Definition H.2.1](#) will collect the precise quotient definition.

This appendix starts from forms, pullbacks, and Stokes in Chapter 13. Appendix [E](#) enriches its geometric examples, while Appendices [F](#) and [G](#) are not prerequisites. Compare Tu [[15](#), §§24–29] and Lee [[5](#), Chapters 17–18]. We use the coefficientwise homotopy calculation to prove the star-shaped Poincaré lemma without flows or Lie derivatives, then compute circle and degree-one torus classes directly. Chain and cochain language first explains the quotient construction. After the concrete computations, exact sequences lead to Mayer–Vietoris and the de Rham comparison theorem, explicitly stated without full proofs. General homotopy invariance and larger computations remain later directions. Manifolds here are smooth and without boundary unless an integration domain is explicitly given a boundary.

**A first-reading route.** A first pass may follow the geometric core:

$$\text{closed/exact forms} \longrightarrow \text{Poincaré lemma} \longrightarrow S^1 \longrightarrow T^2 \longrightarrow \text{periods}.$$

The chain/cochain language organizes these constructions; it need not be mastered before reading the geometric examples. Exact sequences, Mayer–Vietoris, and the de Rham comparison theorem form a structural continuation. The Snake and Five Lemmas are landmarks along that route, not prerequisites for the direct circle and torus calculations.

### H.1 Chains, Cochains, and Induced Maps

We first isolate what the identity “two successive derivatives give zero” permits. All complexes in this appendix are complexes of real vector spaces; the algebraic statements also hold for abelian groups.

**Definition H.1.1** (Chain and cochain complexes). A **chain complex** consists of vector spaces and linear maps

$$\cdots \longrightarrow C_{k+1} \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \longrightarrow \cdots, \quad \partial_k \partial_{k+1} = 0.$$

Its cycles, boundaries, and homology are

$$Z_k = \ker \partial_k, \quad B_k = \operatorname{im} \partial_{k+1}, \quad H_k = Z_k / B_k.$$

A **cochain complex** has arrows raising degree,

$$\cdots \longrightarrow C^{k-1} \xrightarrow{d^{k-1}} C^k \xrightarrow{d^k} C^{k+1} \longrightarrow \cdots, \quad d^k d^{k-1} = 0.$$

Its cocycles, coboundaries, and cohomology are

$$Z^k = \ker d^k, \quad B^k = \operatorname{im} d^{k-1}, \quad H^k = Z^k / B^k.$$

The zero-composite condition says that every boundary is a cycle, and every coboundary is a cocycle, so the quotients are meaningful. A cycle has no boundary; its homology class ignores changes by boundaries of higher-dimensional chains. For cochains the arrows run in the other direction:

$$\begin{array}{c|c} \text{chains} & \partial : C_k \rightarrow C_{k-1} \\ \text{cochains} & d : C^k \rightarrow C^{k+1} \end{array}$$

Chapter 13 gives our central cochain complex immediately:

$$\Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots.$$

Its cocycles are closed forms and its coboundaries are exact forms. For de Rham and singular complexes, spaces in negative degrees are zero. The next section retains the concrete quotient notation for this example.

**Definition H.1.2** (Cochain map). A cochain map  $F : C^\bullet \rightarrow D^\bullet$  is a family of linear maps  $F^k : C^k \rightarrow D^k$  satisfying  $F^{k+1} d_C^k = d_D^k F^k$  for every  $k$ .

**Proposition H.1.3.** A cochain map induces  $H^k(F) : H^k(C) \rightarrow H^k(D)$  by  $[z] \mapsto [F^k z]$ . Identities and compositions pass to the induced maps.

*Proof.* If  $d_C z = 0$ , then  $d_D Fz = F d_C z = 0$ . If  $z = d_C b$ , then  $Fz = d_D Fb$ . Thus cocycles go to cocycles and coboundaries to coboundaries, so the quotient map is well defined and linear. Evaluation on representatives proves the identity and composition rules.  $\square$

For a smooth map  $f : M \rightarrow N$ , pullback is a cochain map  $f^* : \Omega^\bullet(N) \rightarrow \Omega^\bullet(M)$  because  $f^* d = d f^*$  (Proposition 13.12.4). In the language of Appendix B, this compatibility is the naturality of  $d$  (Definition B.3.1). The concrete induced pullback below is exactly Proposition H.1.3 applied to forms.

## H.2 Closed Forms, Exact Forms, and the Quotient

**Definition H.2.1.** Let  $\Omega^k(M)$  be the real vector space of smooth  $k$ -forms, with  $\Omega^0(M) = C^\infty(M)$  and  $\Omega^{-1}(M) = \{0\}$ . Put

$$\begin{aligned} Z^k(M) &= \ker(d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)), \\ B^k(M) &= \operatorname{im}(d : \Omega^{k-1}(M) \rightarrow \Omega^k(M)). \end{aligned}$$

Members of  $Z^k$  are **closed**, and members of  $B^k$  are **exact**. Since  $d^2 = 0$ ,  $B^k(M) \subseteq Z^k(M)$ . The **de Rham cohomology** is the vector space

$$H_{\text{dR}}^k(M) = Z^k(M) / B^k(M).$$

Concretely,  $[\omega]$  consists of all closed forms differing from  $\omega$  by an exact form. The operations are  $[\omega] + [\eta] = [\omega + \eta]$  and  $a[\omega] = [a\omega]$ ; changing representatives adds another exact form, so these are well defined. In particular

$$[\omega] = 0 \iff \omega \text{ is exact.}$$

These elementary quotient rules suffice here; the systematic linear factorization theory is in [Section D.1](#). This is a quotient of spaces of forms, not a quotient topology on  $M$ .

**Proposition H.2.2** (Pullback on cohomology). A smooth map  $f : M \rightarrow N$  induces a linear map

$$f^* : H_{\text{dR}}^k(N) \rightarrow H_{\text{dR}}^k(M), \quad f^*[\omega] = [f^*\omega].$$

It satisfies  $\operatorname{id}^* = \operatorname{id}$  and  $(g \circ f)^* = f^*g^*$  for composable smooth maps.

*Proof.* The identity  $d(f^*\omega) = f^*(d\omega)$ , proved in [Proposition 13.12.4](#), sends closed forms to closed forms. It sends  $d\eta$  to  $d(f^*\eta)$ , so it also sends exact forms to exact forms. Compose the linear map  $f^* : Z^k(N) \rightarrow Z^k(M)$  with the quotient map to  $H_{\text{dR}}^k(M)$ . This composite kills  $B^k(N)$ , and therefore factors uniquely through

$$Z^k(N) / B^k(N).$$

Equivalently, replacing  $\omega$  by  $\omega + d\eta$  changes its image by  $d(f^*\eta)$ , proving representative independence directly. The pullback identities on forms descend to the quotient and give the stated identities. As with forms, measurements move backward.  $\square$

## H.3 Degree Zero and Connected Components

**Proposition H.3.1.**  $H_{\text{dR}}^0(M)$  is the space of locally constant real functions. If  $M$  is nonempty and connected it is canonically  $\mathbb{R}$ . More generally, if  $\mathcal{C}$  is the set of connected components, it is

$$\prod_{C \in \mathcal{C}} \mathbb{R}.$$

*Proof.* There are no nonzero exact 0-forms, since  $\Omega^{-1} = 0$ . If  $df = 0$ , its expression in a sufficiently small coordinate ball has zero Euclidean derivative. Integrating along the segment between two points proves it is constant there. Hence  $f$  is locally constant; the converse follows by differentiating a constant expression. The connectedness criterion in [Proposition 10.7.2](#) makes a locally constant function constant on each connected component ([Definition 10.7.10](#) and [Proposition 10.7.11](#)).

Each component of a manifold is open: a sufficiently small coordinate ball about a point is connected, hence lies in that point's component. Thus any assignment of a real constant to every component is locally constant and smooth. There is no finite-support restriction, explaining the product rather than the direct sum. The empty manifold gives the zero vector space, while a nonempty connected manifold has one component.  $\square$

## H.4 The Poincaré Lemma by Contracting Segments

If  $U \subseteq \mathbb{R}^n$  is star shaped about 0, every point can be joined to 0 by a segment staying in  $U$ . We pull a form along these segments, extract the part measuring motion in the segment parameter, and integrate that part. The fundamental theorem of calculus measures the difference between the two endpoint pullbacks.

**Example H.4.1** (The one-form prototype). For a closed 1-form

$$\omega = \sum_i a_i(x) dx^i,$$

seek a primitive by integrating along the radial segment from 0 to  $x$ :

$$F(x) = \int_0^1 \omega_{tx}(x) dt = \int_0^1 \sum_i a_i(tx) x^i dt.$$

Closedness means  $\partial_j a_i = \partial_i a_j$ . Differentiating under the integral and using that symmetry gives

$$\begin{aligned} \partial_j F(x) &= \int_0^1 \left( a_j(tx) + t \sum_i x^i \partial_j a_i(tx) \right) dt \\ &= \int_0^1 \frac{\partial}{\partial t} (t a_j(tx)) dt = a_j(x). \end{aligned}$$

Thus  $dF = \omega$ . Smoothness along a compact segment justifies passage of derivatives through the integral; the coefficientwise argument in the next proof supplies the details. The general  $K$  below is this same radial integration with additional tangent-vector slots carried along. Each such slot is scaled by  $t$ , explaining the factor  $t^{k-1}$ .

**Lemma H.4.2** (The segment homotopy identity). For star-shaped open  $U$  about 0, let  $h(t, x) = tx$ ,  $0 \leq t \leq 1$ . There are linear maps  $K : \Omega^k(U) \rightarrow \Omega^{k-1}(U)$ , with  $K = 0$  in degree zero, such that

$$dK\omega + Kd\omega = \omega - h_0^*\omega, \quad h_0(x) = 0.$$

For  $k \geq 1$  their explicit value is

$$(K\omega)_x(v_1, \dots, v_{k-1}) = \int_0^1 t^{k-1} \omega_{tx}(x, v_1, \dots, v_{k-1}) dt.$$

*Proof.* We derive the identity before relying on the formula. Every pulled-back  $k$ -form has a unique coefficient decomposition

$$h^*\omega = dt \wedge \alpha(t, x) + \beta(t, x),$$

where  $\alpha$  has spatial degree  $k-1$  and  $\beta$  has spatial degree  $k$ , and neither contains  $dt$ . In degree zero the  $\alpha$  term is absent. Define

$$K\omega = \int_0^1 \alpha(t, \cdot) dt$$

coefficientwise. The integral has smooth coefficients: near a fixed  $x$ , compactness of  $[0, 1]$  and openness of  $U$  allow the smooth expression  $tx$  on a common neighborhood of the whole segment. Every coefficient derivative can be passed under this compact-interval integral: the fundamental theorem writes a spatial difference quotient of a smooth coefficient  $a$  as



$$\int_0^1 \partial_i a(t, x + sue_i) ds,$$

where  $u$  is the increment. Uniform continuity on a compact neighborhood makes this converge uniformly in  $t$  to  $\partial_i a(t, x)$  as  $u \rightarrow 0$ . The uniform integral-limit theorem (Theorem 9.3.1) permits passage through the  $t$ -integral. Repeating for each derivative proves smoothness. Thus  $K\omega$  really is a smooth form.

Write  $d_x$  for differentiation only in the spatial variables. The graded product rule, with the  $dt$  placed first, gives

$$d(h^*\omega) = dt \wedge (\partial_t \beta - d_x \alpha) + d_x \beta.$$

For clarity,  $d(dt \wedge \alpha) = -dt \wedge d_x \alpha$  and  $d\beta = dt \wedge \partial_t \beta + d_x \beta$ . Pullback commutes with  $d$ , so the coefficient of  $dt$  in  $h^*(d\omega)$  is  $\partial_t \beta - d_x \alpha$ . Consequently

$$\begin{aligned} Kd\omega + dK\omega &= \int_0^1 (\partial_t \beta - d_x \alpha) dt + d_x \int_0^1 \alpha dt \\ &= \beta(1, \cdot) - \beta(0, \cdot) = h_1^* \omega - h_0^* \omega. \end{aligned}$$

Here  $h_1$  is the identity. Finally the Euclidean derivative of  $h$  sends  $\partial_t$  to  $x$  and a spatial vector  $v$  to  $tv$ . Evaluating  $h^*\omega$  on  $(\partial_t, v_1, \dots, v_{k-1})$  therefore gives  $\alpha(t, x)(v_1, \dots) = t^{k-1} \omega_{tx}(x, v_1, \dots)$ , which proves the explicit formula. There is no singularity at  $t = 0$ .  $\square$

**Theorem H.4.3** (Star-shaped Poincaré lemma). On a star-shaped open subset of  $\mathbb{R}^n$ , every closed form of positive degree is exact. Every closed positive-degree form on a manifold is locally exact.

*Proof.* Translate the star center to zero. For  $k > 0$  the pullback by the constant map  $h_0$  is zero, because its derivative is zero in every input. If  $d\omega = 0$ , the homotopy identity gives  $\omega = d(K\omega)$ . For the manifold assertion choose a chart ball about the point, transfer the form there, apply this construction, and pull the primitive back.  $\square$

In degree one the primitive is especially concrete: if

$$\omega = \sum_i a_i(x) dx^i$$

is closed, then

$$u(x) = \int_0^1 \sum_i a_i(tx) x^i dt$$

satisfies  $du = \omega$ . The positive-degree restriction matters: in degree zero the homotopy identity reads  $Kdf = f - f(0)$ , and nonzero constant functions are not exact. Thus for nonempty star-shaped  $U$ , and in particular for  $\mathbb{R}^n$ ,

$$H_{\text{dR}}^0(U) = \mathbb{R}, \quad H_{\text{dR}}^k(U) = 0 \quad (k > 0).$$

## H.5 A Puncture and the Circle

**Example H.5.1** (Local angle, global obstruction). On  $\mathbb{R}^2 \setminus \{0\}$  set

$$\eta = \frac{-y dx + x dy}{x^2 + y^2}.$$

For

$$P = -y/(x^2 + y^2), \quad Q = x/(x^2 + y^2),$$

$$Q_x = \frac{y^2 - x^2}{(x^2 + y^2)^2} = P_y, \quad d\eta = (Q_x - P_y) dx \wedge dy = 0.$$

But  $\gamma(t) = (\cos t, \sin t)$  satisfies  $\gamma^*\eta = dt$ , so

$$\int_{\gamma} \eta = 2\pi.$$

An exact form would have integral zero around this closed curve by the line-integral fundamental theorem. In any polar angle chart, direct substitution gives  $\eta = d\theta$ . These local angles differ by constants on overlap components but cannot be chosen as one global real-valued primitive. The circle cannot be filled by its usual disk inside the punctured domain, so Stokes is not contradicted.

**Theorem H.5.2** (Cohomology of the circle). Let  $\alpha$  be the restriction of  $-y dx + x dy$  to the unit circle with its counterclockwise orientation. Then

$$H_{\text{dR}}^0(S^1) \cong \mathbb{R}, \quad H_{\text{dR}}^1(S^1) = \mathbb{R}[\alpha], \quad H_{\text{dR}}^k(S^1) = 0 \quad (k \geq 2).$$

Integration divided by  $2\pi$  identifies  $H_{\text{dR}}^1(S^1)$  with  $\mathbb{R}$ .

*Proof.* Only the degree-one assertion needs a new argument. Pull a 1-form  $\omega$  back by  $\gamma: \mathbb{R} \rightarrow S^1$ ,  $\gamma(t) = (\cos t, \sin t)$ :  $\gamma^*\omega = a(t) dt$  with  $a$  smooth and  $2\pi$ -periodic. Every 1-form on the circle is closed by dimension. If

$$\int_0^{2\pi} a(t) dt = 0,$$

the function

$$u(t) = \int_0^t a(s) ds$$

satisfies

$$u(t + 2\pi) - u(t) = \int_t^{t+2\pi} a(s) ds = 0.$$

It therefore defines  $f: S^1 \rightarrow \mathbb{R}$  by  $f(\gamma(t)) = u(t)$ . This is well defined because equal circle points differ by an integral multiple of  $2\pi$ , and it is smooth by local angle charts. The identity  $\gamma^*df = du = \gamma^*\omega$  implies  $df = \omega$ , since  $\gamma$  is locally a diffeomorphism. Conversely exact forms have zero integral. For arbitrary  $\omega$ , set

$$c = (2\pi)^{-1} \int_{S^1} \omega;$$

then  $\omega - c\alpha$  has integral zero and is exact. Finally

$$\int_{S^1} \alpha = 2\pi,$$

so  $[\alpha] \neq 0$ . Connectedness gives degree zero, and all forms of degree at least two vanish.  $\square$

The integral is not merely a nonexactness test here: the zero-period primitive construction proves it is a complete degree-one invariant.

## H.6 Two Independent Directions on the Torus

Keep  $T^2 = S^1 \times S^1$ , with projections  $\pi_1, \pi_2$ . Put  $\alpha_1 = \pi_1^* \alpha$  and  $\alpha_2 = \pi_2^* \alpha$ . They are closed by pullback compatibility. The loops moving once around one circle with the other fixed give the period table

|               | $\alpha_1$ | $\alpha_2$ |
|---------------|------------|------------|
| first circle  | $2\pi$     | 0          |
| second circle | 0          | $2\pi$     |

Thus their classes are independent. We can also prove they span without invoking a product theorem for cohomology.

**Proposition H.6.1.**  $H_{\text{dR}}^1(T^2) = \mathbb{R}[\alpha_1] \oplus \mathbb{R}[\alpha_2]$ .

*Proof.* Pull a closed form back by  $q(x, y) = ((\cos x, \sin x), (\cos y, \sin y))$  to obtain  $a(x, y) dx + b(x, y) dy$ . Both coefficients are periodic with period  $2\pi$  in each variable, and closedness says  $a_y = b_x$ . The horizontal period

$$a_0(y) = \int_0^{2\pi} a(x, y) dx$$

is constant, since

$$a'_0(y) = \int_0^{2\pi} b_x(x, y) dx = b(2\pi, y) - b(0, y) = 0.$$

The vertical period is constant by the same calculation with  $a_y = b_x$ . Subtract multiples of  $\alpha_1, \alpha_2$  to make both periods zero; write  $a, b$  for the adjusted coefficients. Define

$$u(x, y) = \int_0^x a(s, 0) ds + \int_0^y b(x, t) dt.$$

Then  $u_y = b(x, y)$  and

$$u_x = a(x, 0) + \int_0^y b_x(x, t) dt = a(x, 0) + \int_0^y a_y(x, t) dt = a(x, y).$$

The vertical increment of  $u$  over one period is the zero vertical period. Its horizontal increment is the zero horizontal period at  $y = 0$ , plus a difference of integrals of the periodic function  $b$ , also zero. Hence  $u$  is periodic in both variables and descends to a smooth function on  $T^2$ , using local angle charts. Its differential is the adjusted form. The period table proves uniqueness of the two coefficients and independence.  $\square$

The torus is connected, so  $H_{\text{dR}}^0(T^2) = \mathbb{R}$  as well. This calculation develops only the two degree-one directions; a general product formula and higher-dimensional torus computations are not needed.

## H.7 Integration Detects Classes

For a compact oriented  $k$ -manifold  $P$  without boundary and a smooth map  $f : P \rightarrow M$ , Stokes immediately gives

$$\omega = d\eta \implies \int_P f^* \omega = \int_P d(f^* \eta) = 0.$$

Thus a nonzero period of a closed form proves its class is nonzero. More general cycles express the same cancellation with pieces.

Cubes are used here only as a concrete integration device. The later singular cohomology section uses the standard simplicial model. This appendix does not need or prove a general equivalence between cubical and simplicial chain theories.

For a concrete elementary convention, a smooth singular  $k$ -cube is a smooth map from a neighborhood of  $[0, 1]^k$  to  $M$ . A finite real linear combination

$$c = \sum_j a_j \sigma_j$$

is a chain. Its boundary is the signed sum of its face maps: the faces  $x^i = 1$  and  $x^i = 0$ , with remaining coordinates in their original order, have coefficients  $(-1)^{i-1}$  and  $-(-1)^{i-1}$ , respectively. A cycle satisfies  $\partial c = 0$  as a formal sum of these maps. Set

$$\int_c \omega = \sum_j a_j \int_{[0,1]^k} \sigma_j^* \omega.$$

For  $k \geq 1$ , the coordinate proof of local Stokes, or the iterated one-variable fundamental theorem on the cube, gives

$$\int_{[0,1]^k} d(\sigma_j^* \eta) = \int_{\partial[0,1]^k} \sigma_j^* \eta.$$

Opposite face signs are exactly the signs obtained when the omitted coordinate is moved to the first position. By linearity and pullback compatibility,

$$\boxed{\omega = d\eta, \quad \partial c = 0 \implies \int_c \omega = \int_{\partial c} \eta = 0.}$$

This argument proves the needed cancellation directly and does not assume a general theory of manifolds with corners or singular homology.

## H.8 Exact Sequences and Connecting Maps

Start with a subspace  $U \subseteq V$ . Inclusion  $i$  and the quotient map  $q$  give the elementary pattern

$$0 \longrightarrow U \xrightarrow{i} V \xrightarrow{q} V/U \longrightarrow 0, \quad \ker q = U = \operatorname{im} i.$$

Indeed,  $q(v) = [0]$  exactly when  $v \in U$ ; every coset has a representative, and inclusion loses no information. Exactness names this matching of what one arrow produces with what the next arrow kills.

**Definition H.8.1** (Exactness). A sequence  $A \xrightarrow{i} B \xrightarrow{p} C$  of linear maps is **exact at  $B$**  if  $\text{im } i = \ker p$ . A sequence is exact if this holds at every term having an incoming and outgoing map. A **short exact sequence** is an exact sequence

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0.$$

For a linear map  $f : A \rightarrow B$ , its cokernel is  $\text{coker } f = B / \text{im } f$ .

The initial zero says  $\ker i = 0$ , hence  $i$  is injective; the final zero says  $\text{im } p = C$ , hence  $p$  is surjective. The middle condition identifies  $C$  with  $B/i(A)$ . Thus exactness records precisely which elements disappear and where they came from. A complex only requires  $\text{im } d \subseteq \ker d$ ; its cohomology measures the failure of equality.

For the later Mayer–Vietoris computation, keep the minimal route in view:

$$\begin{aligned} \text{short exact sequence of complexes} &\longrightarrow \text{connecting homomorphism} \\ &\longrightarrow \text{long exact sequence} \longrightarrow \text{Mayer–Vietoris.} \end{aligned}$$

The Snake and Five Lemmas give useful structural context. The calculation below uses the stated long exact sequence theorem and the explicit connecting-map mechanism, without requiring proofs of those two lemmas.

**Theorem H.8.2** (Snake Lemma). For a commutative diagram of vector spaces with short exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C & \longrightarrow & 0 \\ & & f \downarrow & & g \downarrow & & h \downarrow & & \\ 0 & \longrightarrow & A' & \xrightarrow{i'} & B' & \xrightarrow{p'} & C' & \longrightarrow & 0, \end{array}$$

there is a natural connecting map  $\delta$  making the following sequence exact (read the two lines consecutively):

$$\begin{aligned} 0 \longrightarrow \ker f &\longrightarrow \ker g \longrightarrow \ker h \xrightarrow{\delta} \text{coker } f, \\ \text{coker } f &\longrightarrow \text{coker } g \longrightarrow \text{coker } h \longrightarrow 0. \end{aligned}$$

The repeated cokernel is one term, displayed twice only to break the line.

The Snake Lemma extracts exact information about kernels and cokernels from a commutative diagram. We use its perspective rather than its proof. The next consequence is more directly relevant to differential forms.

**Theorem H.8.3** (Long exact sequences). A short exact sequence of chain complexes  $0 \rightarrow A_\bullet \rightarrow B_\bullet \rightarrow C_\bullet \rightarrow 0$ , exact in every degree and with maps commuting with boundaries, gives a natural exact sequence

$$\cdots \rightarrow H_n(A) \rightarrow H_n(B) \rightarrow H_n(C) \xrightarrow{\partial_*} H_{n-1}(A) \rightarrow \cdots.$$

A degree-wise short exact sequence of cochain complexes  $0 \rightarrow A^\bullet \xrightarrow{i} B^\bullet \xrightarrow{p} C^\bullet \rightarrow 0$  gives a natural exact sequence

$$\begin{aligned} \cdots \rightarrow H^{k-1}(C) &\xrightarrow{\delta} H^k(A) \xrightarrow{i_*} H^k(B) \xrightarrow{p_*} H^k(C), \\ H^k(C) &\xrightarrow{\delta} H^{k+1}(A) \rightarrow H^{k+1}(B) \rightarrow \cdots. \end{aligned}$$

Again the repeated term is a continuation, not an additional arrow. Naturality means that a commuting map between short exact sequences induces a commuting map between these long exact sequences, including the connecting arrows.

**Where does the connecting map come from?** Represent  $[c] \in H^k(C)$  by a cocycle. Lift  $c$  to  $b \in B^k$ , using surjectivity of  $p$ . Although  $c$  is closed,  $b$  need not be. However,  $p(db) = d(pb) = dc = 0$ . Exactness gives a unique  $a \in A^{k+1}$  with  $i(a) = db$ . Since  $i(da) = d^2b = 0$  and  $i$  is injective,  $da = 0$ . Define

$$\delta[c] = [a].$$

A different lift  $b + i(u)$  changes  $a$  to  $a + du$ . If  $c$  is replaced by  $c + dv$ , lift  $v$  to  $w$  and replace  $b$  by  $b + dw$ ; this leaves  $db$  unchanged. These checks explain representative independence. They do not constitute a proof of the entire long exact sequence.

The class  $\delta[c]$  measures the obstruction to lifting  $[c]$  by a closed element: if  $a = du$ , then  $b - i(u)$  is a closed lift. Conversely a closed lift makes the obstruction zero. This explains exactness at  $H^k(C)$  and why a new degree appears. For chains, lifting a cycle and taking its boundary yields the same construction one degree lower.

**Theorem H.8.4** (Five Lemma, isomorphism form). Suppose the following diagram commutes and both rows are exact:

$$\begin{array}{ccccccccc} A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 & \longrightarrow & A_5 \\ f_1 \downarrow & & f_2 \downarrow & & f_3 \downarrow & & f_4 \downarrow & & \downarrow f_5 \\ B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 & \longrightarrow & B_5. \end{array}$$

If  $f_1, f_2, f_4, f_5$  are isomorphisms, then  $f_3$  is an isomorphism. More generally it suffices that  $f_1$  is surjective,  $f_2, f_4$  are isomorphisms, and  $f_5$  is injective.

Exactness lets information in neighboring terms control the middle term. This is why long exact sequences can support comparisons as well as computations. The algebraic lemmas above are structural landmarks, not additional prerequisites for Chapters 1–13.

## H.9 Mayer–Vietoris: Assembling Overlapping Pieces

Let  $M = U \cup V$ , where  $U, V$  are open in a smooth manifold. Restricting forms and comparing their restrictions on the overlap gives

$$0 \longrightarrow \Omega^\bullet(M) \xrightarrow{r} \Omega^\bullet(U) \oplus \Omega^\bullet(V) \xrightarrow{s} \Omega^\bullet(U \cap V) \longrightarrow 0,$$

where  $r(\omega) = (\omega|_U, \omega|_V)$  and  $s(\alpha, \beta) = \alpha|_{U \cap V} - \beta|_{U \cap V}$ . Both are cochain maps (Definition H.1.2). Agreement on overlaps explains  $\text{im } r = \ker s$ , and a global form is determined by its two restrictions.

Surjectivity of  $s$  uses a smooth partition of unity  $\rho_U + \rho_V = 1$  subordinate to  $U, V$  (Theorem 13.14.7). For a form  $\eta$  on  $U \cap V$ , extend  $\rho_V \eta$  by zero to  $U$  and  $-\rho_U \eta$  by zero to  $V$ . The support conditions (Definitions 13.14.3 and 13.12.5) make these extensions smooth: near a point outside the corresponding open set the multiplier vanishes. Their difference restricts to  $\eta$ . This explains the short exact sequence without proving its full cohomological consequences.

**Theorem H.9.1** (De Rham Mayer–Vietoris; stated without full proof). The preceding short exact sequence induces

$$\begin{aligned} \cdots &\rightarrow H_{\text{dR}}^{k-1}(U \cap V) \xrightarrow{\delta} H_{\text{dR}}^k(M) \xrightarrow{r_*} H_{\text{dR}}^k(U) \oplus H_{\text{dR}}^k(V), \\ H_{\text{dR}}^k(U) \oplus H_{\text{dR}}^k(V) &\xrightarrow{s_*} H_{\text{dR}}^k(U \cap V) \xrightarrow{\delta} H_{\text{dR}}^{k+1}(M) \rightarrow \cdots. \end{aligned}$$

The repeated direct sum joins the two lines. The sequence is exact.

This is [Theorem H.8.3](#) applied to the displayed complex; compare Tu [15, §§25–26] for the exact-sequence construction and its application to forms. It expresses the same local-to-global strategy as integration in Chapter 13:

Global cohomology can be assembled from overlapping local pieces.

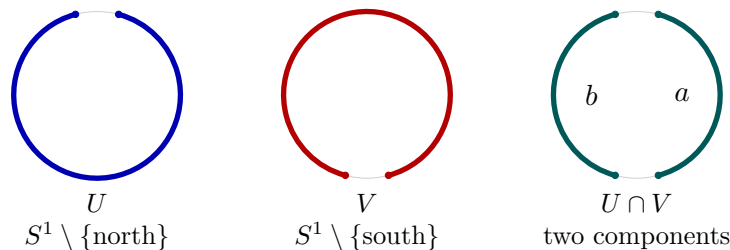


Figure H.1: Two punctured-circle open arcs cover  $S^1$ . Gaps are enlarged schematically to show the omitted poles. Their overlap has a left and a right component; a locally constant function chooses two independent values.

For example, cover a circle by two open arcs whose intersection has two components, as in [Figure H.1](#). A locally constant function on the overlap chooses one independent constant on each component, so  $H_{\text{dR}}^0(U \cap V) \cong \mathbb{R}^2$ . The arcs have degree-zero cohomology  $\mathbb{R}$  and higher cohomology zero; the intersection has degree-zero cohomology  $\mathbb{R}^2$ . Here the arc computations follow directly from one-variable primitives. The difference map sends  $(a, b)$  to  $(a - b, a - b)$ . Its cokernel is one dimensional, so the sequence recovers the one-dimensional  $H_{\text{dR}}^1(S^1)$  computed directly in [Theorem H.5.2](#). The direct proof also identifies the class by its period, whereas the sequence organizes the overlap obstruction.

## H.10 Singular Cohomology and de Rham's Theorem

A singular  $k$ -simplex is a continuous map  $\sigma : \Delta^k \rightarrow M$ , where  $\Delta^k$  is the standard simplex with ordered vertices. Finite real linear combinations form the singular chain space  $C_k(M; \mathbb{R})$ . The boundary is the alternating sum of restrictions to faces, with the  $i$ th face given sign  $(-1)^i$ , starting at  $i = 0$ . On 0-chains the boundary is zero. Each codimension-two face occurs twice with opposite signs, so  $\partial^2 = 0$ . This is a chain complex in the sense of [Definition H.1.1](#). Its dual spaces  $C^k(M; \mathbb{R}) = \text{Hom}(C_k(M; \mathbb{R}), \mathbb{R})$  form a cochain complex with  $(\delta\varphi)(c) = \varphi(\partial c)$ . Their cohomology is real singular cohomology  $H_{\text{sing}}^k(M; \mathbb{R})$ . A continuous map induces a chain map by composing simplices, and precomposition then gives the contravariant map on cohomology.

Continuous simplices need not be differentiable, so we do not integrate forms directly over an arbitrary continuous simplex. For a smooth simplex (smooth on a neighborhood in its affine span), define

$$I(\omega)(c) = \int_c \omega = \sum_j a_j \int_{\Delta^k} \sigma_j^* \omega, \quad c = \sum_j a_j \sigma_j.$$

The simplex version of Stokes, used here as part of the comparison theorem with the same oriented-face convention, gives

$$\int_{\partial c} \omega = \int_c d\omega, \quad \delta I(\omega) = I(d\omega).$$

The first identity is understood with  $c$  of dimension one larger than the degree of  $\omega$ ; it is the simplex version of the cubical cancellation already verified in [Section H.7](#). Thus integration is a cochain

map on smooth singular chains. Closed forms annihilate boundaries, and exact forms annihilate cycles. Passing from smooth to continuous singular theory and proving that this comparison is an isomorphism are substantive parts of the following external theorem, not consequences of Stokes alone.

**Theorem H.10.1** (de Rham theorem). For every smooth manifold  $M$  without boundary and  $k \geq 0$ , there is a natural isomorphism of real vector spaces

$$H_{\text{dR}}^k(M) \cong H_{\text{sing}}^k(M; \mathbb{R}).$$

The right-hand side is real singular cohomology, previewed in Section 13.17; it is defined from continuous maps of simplices and their boundaries. The theorem says that smooth differential forms recover that topological invariant. Naturality means that pullback by a smooth map commutes with the isomorphisms. Integration over smooth simplices supplies the comparison, with Stokes providing the compatibility with boundaries. Comparing smooth and continuous chains and proving that the resulting map is an isomorphism are substantial further steps, not consequences of the nonzero-period tests alone. See Lee [5, Theorem 18.14] and the cohomological route in Tu [15, §§24–29].

An invariant built analytically from differential forms  
agrees with a topological invariant.

For a smooth map  $f : M \rightarrow N$ , naturality is the commuting square

$$\begin{array}{ccc} H_{\text{dR}}^k(N) & \xrightarrow{\cong} & H_{\text{sing}}^k(N; \mathbb{R}) \\ f^* \downarrow & & \downarrow f^* \\ H_{\text{dR}}^k(M) & \xrightarrow{\cong} & H_{\text{sing}}^k(M; \mathbb{R}). \end{array}$$

Either compare a class first and then pull it back, or pull it back first and then compare: the result is the same. In the language of Definition B.3.1, these comparison isomorphisms form a natural isomorphism between the two cohomology functors, viewed as contravariant functors on smooth manifolds, equivalently as covariant functors on  $\mathbf{Man}^{\text{op}}$ . This is the structural meaning of the period computations, not a new prerequisite for the main course.

## Exercises

**Exercise 1.\*** Decide which of  $x \, dy$ ,  $y \, dx + x \, dy$ , and  $x \, dx + y \, dy$  are closed on  $\mathbb{R}^2$ . Find primitives whenever possible and identify the obstruction otherwise.

**Exercise 2.\*** Apply the operator  $K$  to  $dx \wedge dy$  on  $\mathbb{R}^2$  and verify by direct differentiation that the resulting 1-form is a primitive. Track the signs.



**Exercise 3.\*** Describe  $H_{\text{dR}}^0(\mathbb{R} \setminus \mathbb{Z})$  and explain why a direct sum over the components would give the wrong answer.

**Exercise 4.\*** For an integer  $m$ , let  $f_m : S^1 \rightarrow S^1$  send the angle  $t$  to  $mt$ . Compute  $f_m^*$  on  $H^0$  and  $H^1$  using the angular form and its periods.

**Exercise 5.\*** On an annulus centered at the origin, prove that the angular form is closed and not exact. Explain why every point nevertheless has a neighborhood on which it has a primitive.

**Exercise 6.\*** Use the torus period table to compute the induced map on  $H^1$  for  $f(z, w) = (z^a w^b, z^c w^d)$ , where  $a, b, c, d \in \mathbb{Z}$  and the circle uses complex multiplication. Specify which basis determines your matrix convention.

**Exercise 7.\*** Give the product torus its product orientation. Show that

$$\int_{T^2} \alpha_1 \wedge \alpha_2 = (2\pi)^2,$$

and conclude that this closed 2-form is not exact. Explain why this alone is not a computation of the dimension of  $H_{\text{dR}}^2(T^2)$ .

# Appendix I

## Hints for Selected Exercises

These optional hints suggest a starting point, not a completed solution. Try the exercise first; return to its hypotheses after reading the hint. The exercise number in each entry links back to the original problem. Numbers restart within each chapter or appendix, exactly as in the exercise sets. A superscript asterisk means *hint available*, not “hard exercise.” The optional appendices have more selected hints to support independent reading.

### Chapter 2

**Exercise 1.** Group strings by their length. Each group is finite; arrange the groups by increasing length and list each group in a fixed order.

**Exercise 2.** For each inclusion, start with an arbitrary element and unpack image and inverse image separately. For failure of equality, try a constant function on a two-element set.

**Exercise 5.** Write the sum through  $n + 1$  as the sum through  $n$  plus its last term. Use the induction hypothesis only on the first part.

### Chapter 3

**Exercise 1.** Compare a constant rational sequence with that constant plus

$$1/n.$$

Equivalence concerns the limit of a difference, not equality of individual terms.

**Exercise 4.** Insert  $x_j$  and  $x_k$  between  $r_j$  and  $r_k$  in the triangle inequality. Give each of the three errors a share of the tolerance.

## Chapter 4

**Exercise 1.** Simplify the error first. Choose a natural number strictly larger than your sufficient bound; verify the estimate for every  $n \geq N$ .

**Exercise 2.** Divide by the leading powers and bound the normalized denominator away from zero before applying a quotient law.

**Exercise 3.** Use Bernoulli on

$$4/3$$

for decay and on

$$3/2$$

for growth. For the negative base separate odd and even indices.

**Exercise 4.** Multiply by the conjugate, then divide the denominator by  $k$ .

**Exercise 5.** Let  $e_n$  be the error after  $n$  corrections, starting at  $n = 0$ . Inductively bound  $|e_n|$  by a geometric sequence; equality is not assumed.

**Exercise 6.** For the first use one fixed positive error gap. For the second, either exclude every candidate or exhibit two distinct subsequential limits.

**Exercise 9.** The number  $s - \varepsilon$  cannot be an upper bound. Find one term above it and use monotonicity for all subsequent indices.

**Exercise 11.** Let the candidate be the infimum of the range. What does the definition of infimum say about a term below that candidate plus  $\varepsilon$ ?

## Chapter 5

**Exercise 1.** For the square first restrict  $|x - 3| < 1$ ; for the reciprocal keep  $|x + 1|$  above a fixed positive number.

**Exercise 2.** Normalize by the degrees; for the negative end set  $x = -t$ . Retain the parity of the degree difference.

**Exercise 3.** Compute the punctured limits before examining the assigned values.

**Exercise 4.** For positive results estimate differences. For negative results use input pairs approaching each other while the outputs retain a fixed gap.

**Exercise 5.** Continuity and the endpoint readings give existence. No monotonicity assumption has been supplied.

**Exercise 6.** Factor the output difference and bound the sum of the two inputs over the entire domain.

**Exercise 13.** At the junction compare the sequences

$$1/n, \quad -1/n$$

and the assigned value. Away from the junction choose a neighborhood lying in one branch.

## Chapter 6

**Exercise 1.** Inspect  $-1, 0, 1$  separately. On  $[0, r]$  use  $r^n$ ; near 1 use moving inputs.

**Exercise 3.** Complete the square in  $x(1-x)$  to locate its largest possible value.

**Exercise 8.** For nonuniformity choose the input so that  $nx$  stays equal to 1. On  $[\delta, 1]$  use  $1 + n^2x^2 \geq n^2x^2$ .

## Chapter 7

**Exercise 1.** Start with the term test; then look for cancellation, a power benchmark, a ratio, or an  $n$ th root, in that order.

**Exercise 3.** Index the first contribution carefully. Apply the finite geometric identity before passing to the limiting total.

**Exercise 4.** Write finite upper bounds  $b$  on the integrals, evaluate, and then let  $b \rightarrow \infty$ . Keep this calculation separate from the condensation proof.

**Exercise 12.** Test each endpoint independently. For uniform convergence on the closed interval, a summable bound independent of  $x$  is useful; at a divergent endpoint uniform convergence is impossible.

## Chapter 8

**Exercise 2.** Keep

$$|h| < |a|/2$$

in the reciprocal quotient at  $a \neq 0$ . For the piecewise function compare the two one-sided quotients.

**Exercise 5.** Use the perimeter condition to express one side in terms of the other. Compare the interior critical point with the boundary values.

**Exercise 12.** Bound the third derivative on the whole interval from 0 to 0.1, then use the Lagrange remainder. A decimal guess is not an error bound.

## Chapter 9

**Exercise 1.** For the linear function, sum the oscillation on each cell times its width. For the other function use density in every cell.

**Exercise 3.** Accumulate the given rate over elapsed time. Divide the total accumulation by the duration to obtain its average rate.

**Exercise 17.** Place a triangle on

$$[1/(2n), 1/n]$$

and choose its height to keep its area fixed. Check the value at zero separately.

## Appendix A

**Exercise 1.** Fix  $a$  and recurse from 1 by multiplication by  $a$ . For the addition law, fix  $m$  and induct on  $n$ .

**Exercise 3.** If  $n$  is not prime, choose a proper divisor between 2 and  $n-1$  and apply the induction hypothesis to that divisor.

**Exercise 7.** Order the independent sets containing the prescribed set by inclusion. For a chain, test a dependence in its union using only the finitely many vectors involved.

## Appendix B

**Exercise 2.** In a poset category, arrows in both directions force equality. Apply this to divisibility, then use transitivity rather than inventing underlying functions.

**Exercise 3.** Reverse the displayed arrows first: in  $\mathcal{C}$  they have the form  $C \rightarrow B \rightarrow A$ . The composite in the opposite category is the formal opposite of their composite in  $\mathcal{C}$ .

**Exercise 5.** Evaluate each proposed composite on a covector and then on a vector. Precomposition determines the reversed arrow order.

**Exercise 6.** Write both composites explicitly. One route applies  $f$  before taking the diagonal; the other applies  $f$  to each coordinate after taking it.

**Exercise 8.** Start with  $\omega \in \Omega^k(N)$ . Both routes must end in  $\Omega^{k+1}(M)$ ; this requirement fixes every vertical arrow direction.

## Appendix C

**Exercise 2.** The  $(i, j)$  entry of the matrix product is a cofactor expansion. For  $i \neq j$ , identify a matrix with two equal rows.

**Exercise 5.** Every pair of positions is an inversion in the reversing list. Count pairs before evaluating the sign.

**Exercise 6.** Reduce the integer determinant in the field. The field equality  $-1 = 1$  does not change the parity of a permutation.

## Appendix D

**Exercise 1.** A linear map descends exactly when it vanishes on the relation subspace. Evaluate both candidates on  $(1, 1)$ , including in characteristic two.

**Exercise 2.** Check that  $\pi_H \circ L$  kills  $R$ , then invoke the quotient universal property.

**Exercise 5.** Compare the values of the proposed bilinear map on  $(e_1, e_2)$  and  $(e_2, e_1)$ .

**Exercise 7.** Project to the indicated two-dimensional factors. An elementary tensor has a coefficient table whose columns are proportional.

**Exercise 11.** First test  $L_v$  on elementary wedges in the remaining slots; then use that those wedges span.

## Appendix E

**Exercise 1.** First prove constancy on integer translates. For the topology, factor through the compact interval model and use the compact-to-Hausdorff criterion.

**Exercise 3.** Use a signed horizontal coordinate on the two sides of the seam. The negative side must apply the edge reversal.

**Exercise 6.** Identify the map with angle doubling. After checking its fibers, use compactness and the Hausdorff target.

## Appendix F

**Exercise 1.** Try  $(x, y, z) \mapsto (x, y, z - f(x, y))$ ; its inverse adds  $f$  back.

**Exercise 2.** Compute  $Dh$  and inspect the two axes in the zero fiber near their crossing.

**Exercise 3.** Restrict  $dz$  to the tangent plane of the sphere. At a regular point, intersect the two tangent kernels.

**Exercise 4.** Use the sequence approaching the omitted parameter endpoint. For the compact case, combine immersion with the compact-to-Hausdorff criterion.

**Exercise 5.** Differentiate  $A^T A$  with symmetric matrices as target. To test surjectivity at an orthonormal frame, try a variation of the form  $H = AB$  with  $B$  symmetric.

**Exercise 7.** Take products of adapted charts and reorder the coordinates so that the free coordinates come first.

## Appendix G

**Exercise 1.** Differentiate  $x^2 + y^2 = 1$  at  $(1, 0)$  and then apply the derivative of left multiplication.

**Exercise 2.** Left multiplication by  $A$  is a diffeomorphism. Translate the already computed identity tangent space.

**Exercise 3.** Compute  $[X, Y]$  on the coordinate functions first; these determine its coefficients.

**Exercise 4.** Multiply the displayed matrices in both orders. Use bilinearity and antisymmetry to reduce repeated work.

**Exercise 5.** For the diagonal matrix compute powers entrywise; for the other matrix use  $A^2 = 0$  to truncate the series.

**Exercise 6.** In the  $SO(2)$  realization, the initial tangent is  $aJ$ . Apply the one-parameter subgroup result and  $J^2 = -I$ .

## Appendix H

**Exercise 1.** For  $P dx + Q dy$ , compare  $\partial_x Q$  with  $\partial_y P$ . A primitive must have these two partial derivatives.

**Exercise 2.** For  $k = 2$ , evaluate the two-form on  $(x, v)$  and include the factor  $t$  before integrating.

**Exercise 3.** Identify the interval components. A smooth locally constant function may take a nonzero value on every component, so use a product rather than a direct sum.

**Exercise 4.** Pull back the angular form along  $t \mapsto mt$  and compare its period. In degree zero use connectedness.

**Exercise 5.** Compute closedness directly, then integrate around a circle in the annulus. Use the local Poincaré lemma for the last claim.

**Exercise 6.** Pull back the two degree-one generators separately. Record their periods on the two coordinate circles before writing the matrix.

**Exercise 7.** Use the product parametrization and orientation. A nonzero integral rules out exactness, but does not show that every other closed two-form is a multiple modulo exact forms.



# References and Further Reading

- [1] Robert G. Bartle. *The Elements of Real Analysis*. Second edition, John Wiley & Sons, 1976. A broad classical companion for Cartesian-space topology, sequences and function convergence, asymptotic and order notation, Riemann integration and convergence, finite-dimensional differentiation, and multivariable integration and change of variables. The manifold and differential-form development instead uses Munkres, Spivak, Tu, and Lee as its geometric references.
- [2] Stephen H. Friedberg, Arnold J. Insel, and Lawrence E. Spence. *Linear Algebra*. Fifth edition, Pearson. Chapters 1–2 provide the organizational benchmark for vector spaces, replacement, bases, linear maps, and their matrix representations. The analysis main text selects only the finite-dimensional real foundations.
- [3] Tom M. Apostol. *Calculus, Volume 2*. Second edition, Wiley. Further reading for the integration of linear algebra with multivariable calculus and a structural approach to determinants.
- [4] Tommy Chen. *Algebra lecture notes*. Author’s local revision draft, consulted September 12, 2026. Theorem 5.12.3, “Existence and uniqueness of the determinant form,” supplies the slot-by-slot expansion pattern adapted in Chapter 10. Section 1.5 guides the concrete permutation discussion in Appendix C. Chapter 8 informs the treatment of universal properties in Appendix D, specialized here to vector spaces over fields.
- [5] John M. Lee. *Introduction to Smooth Manifolds*. Graduate Texts in Mathematics 218, second edition, Springer, 2013. Primary supplementary technical benchmark for Chapter 13: boundary charts, tangent spaces, orientation, partitions of unity, integration, and Stokes. Appendices F–H compare Chapters 4–8, 17–18, and 20: rank and slice theorems, embeddings, Lie groups and brackets, cohomology and integration, and one-parameter subgroups. The strong Whitney theorem (Theorem 6.19) and de Rham theorem (Theorem 18.14) are stated without proof. The earlier imported refinement is localized to Theorem 1.15 and the coordinate-neighborhood construction in the proof of Theorem 2.23, with the boundary adaptation in Exercise 2.24.
- [6] Terence Tao. *Analysis I*. Springer. Foundations and rational Cauchy constructions; useful for checking the order of logical dependencies in Chapters 2–4.
- [7] Stephen Abbott. *Understanding Analysis*. Second edition, Springer, 2015. A companion for motivation, counterexamples, and learning to write the elementary proofs in Part I.
- [8] Kenneth A. Ross. *Elementary Analysis: The Theory of Calculus*. Second edition, Springer, 2013. Elementary rigor and graduated practice with sequences, continuity, and one-variable calculus.

- [9] Charles C. Pugh. *Real Mathematical Analysis*. Second edition, Springer, 2015. Geometric viewpoints on analysis, finite-dimensional topology, multivariable differentiation, and the geometric volume viewpoint.
- [10] Anthony W. Knap. *Basic Real Analysis*. Cornerstones, first edition, Birkhäuser Boston, 2005. A systematic companion for metric and normed spaces, finite-dimensional differentiation, inverse and implicit functions, finite partitions of unity, and reduction of substitutions to elementary coordinate transformations. The author's digital second edition (2016) is also available from <https://www.math.stonybrook.edu/~aknapp/>.
- [11] Vladimir A. Zorich. *Mathematical Analysis I–II*. Springer. A broader analysis perspective, with geometric interpretations of multivariable derivatives, integration, and classical vector analysis.
- [12] John H. Hubbard and Barbara Burke Hubbard. *Vector Calculus, Linear Algebra, and Differential Forms: A Unified Approach*. Fifth edition, Matrix Editions, 2015. Geometric companion for interpreting forms as oriented measurements and connecting them to vector calculus. The publisher's Chapter 6 sample supplies a selective pedagogical benchmark; <https://matrixeditions.com/UnifiedApproach5thedSamples.html>.
- [13] James R. Munkres. *Analysis on Manifolds*. Addison–Wesley, 1991. A systematic companion for multivariable differentiation, primitive local diffeomorphisms, integration, and the route from forms to Stokes.
- [14] Michael Spivak. *Calculus on Manifolds*. W. A. Benjamin, 1965. A compact account of the unity of the classical integral theorems; best read after the more detailed developments here.
- [15] Loring W. Tu. *An Introduction to Manifolds*. Second edition, Springer, 2011. Primary pedagogical and organizational benchmark for Chapter 13's expanded manifold introduction: gradual chart and atlas definitions, tangent and cotangent spaces, differential forms, orientation, and integration. Sections 3.10 and 4.1–4.2 also benchmark the distinction between pointwise covectors, their coordinate basis, and varying form fields. Appendices F–H also compare Sections 9 and 11 (submanifolds and rank), 14–16 (vector fields, matrix groups, and Lie algebras), and 24–29 (cohomology, examples, and homotopy). The elementary circle and torus period proofs avoid importing the book's Mayer–Vietoris machinery. The exposition here is reconstructed in the manuscript's own notation and dependency order.
- [16] Walter Rudin. *Principles of Mathematical Analysis*. International Series in Pure and Applied Mathematics, third edition, McGraw–Hill, 1976. Further reading for real analysis.
- [17] James R. Munkres. *Topology*. Second edition, Prentice Hall, 2000. Further reading for the general-topology background used sparingly in Part II.
- [18] Michael Spivak. *Calculus*. Fourth edition, Publish or Perish, 2008. Further reading for rigorous single-variable calculus.
- [19] Ira M. Gessel. Lagrange inversion. *Journal of Combinatorial Theory, Series A* 144 (2016), 212–249. Coefficient formulations and formal power-series proofs of the optional analytic refinement; <https://arxiv.org/abs/1609.05988>.
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- [univie.ac.at/~slc/s/s61Asokal2.html](http://univie.ac.at/~slc/s/s61Asokal2.html). Analytic convergence and formal versions of Lagrange inversion.
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- [22] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002. Further reading and a benchmark for the optional fundamental-group, CW-complex, homology, and cohomology previews in Section 13.17; not a claim of wholesale adaptation. The author’s freely available edition is at <https://pi.math.cornell.edu/~hatcher/AT/ATpage.html>.
- [23] Luciena Xiao Xiao. *MAST31023 Introduction to Algebraic Topology Lecture Notes*. Supplied lecture-note PDF. Supplementary pedagogical benchmark and further reading for the optional algebraic-topology preview: spaces and maps, homotopy, fundamental groups, singular chains, and CW characteristic maps and examples. The manuscript keeps real coefficients and does not adopt the source’s full course.
- [24] Xuewang Ai Shuxue (author name romanized from the supplied Chinese notes). *Weifen Liuxing Biji* [Notes on Smooth Manifolds]. Chinese-language lecture notes; Section 1, example 7, printed page 4. Selective benchmark for Appendix E’s projective-space proof architecture: scaling quotient, affine ratio charts, and transitions. The present Hausdorff proof uses the common isometric-quotient criterion. This acknowledgment applies to that construction, not to Appendix E as a whole. No date or publication venue is supplied.
- [25] David S. Dummit and Richard M. Foote, *Abstract Algebra*, 3rd ed., Wiley, 2004. Sections 10.4 and 11.4–11.5 provide a selective comparison for tensor products, determinant expansions, and exterior powers.

**Attribution note.** The principal results used in the manuscript’s later development are proved internally, apart from the external refinement theorem explicitly cited in Chapter 13. The optional analytic Lagrange inversion theorem in Chapter 11 and the Lebesgue transformation-of-measure preview in Chapter 12 are also explicitly identified after their statements as unproved and are not prerequisites. Chapter 9 likewise identifies after their statements the unproved compatibility of proper Riemann and Lebesgue integration, Lipschitz/a.e. Newton–Leibniz theorem, Cantor-function existence, Lebesgue monotone and dominated convergence theorems, and Lebesgue Fundamental Theorem of Calculus; those results are not prerequisites for the later development and carry local references. The references listed here provide verification and further reading. Section 13.17 contains additional explicitly unproved preview results, including dominated convergence, Hilbert-space Riesz representation, Cauchy’s integral formula, fundamental-group computations, Euler-characteristic invariance, cellular–singular homology comparison, the homological Euler formula, and the maximum principle; none is a prerequisite. Appendix F states Whitney’s embedding theorem without proof. Appendix A states ordinary and transfinite recursion without proof, proves Hartogs’s lemma, proves the equivalence of the Axiom of Choice, the Well-Ordering Theorem, and Zorn’s Lemma from those foundations, and then applies Zorn’s Lemma to the vector-space basis theorem. Appendix H states the Snake and Five Lemmas, long exact sequence theorem, Mayer–Vietoris, and de Rham theorem without full proofs. The bibliography entries are not claims that all of the manuscript’s proofs or exposition were adapted from those sources. The author’s algebra notes [4] are acknowledged locally for the determinant expansion pattern, their influence

on the permutation exposition, and Appendix D’s proof architecture adapted from their Chapter 8. These specific adaptations do not describe the origin of the entire manuscript.

**A reading route.** For a second pass through foundations, pair Tao with Abbott or Ross. For Part II, use Pugh or Zorich for geometry and Munkres for a systematic calculus-on-manifolds treatment. Read Spivak for the compressed unifying picture after the details are familiar. Tu offers a gradual continuation into manifolds; Lee supplies a more extensive technical reference. None of their larger scopes is a prerequisite for these notes. For the algebraic foundations, compare Chapters 1–2 of Friedberg–Insel–Spence [2]; for determinants and their calculus role, compare Apostol [3] and Munkres. Appendices A and B collect foundational and categorical language. Appendices C–H develop algebraic extensions, quotient constructions, submanifolds, Lie theory, and de Rham cohomology along explicit dependency routes.